

OpenSourceRail Documentation Book

Root README, docs tree, generated catalogue index, and every region/country/city design README.

Generated from 264 Markdown source files.

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Front Matter

Repository README

README.md

OpenSourceRail

Solar metro trainset — current OpenSourceRail design with 17 m self-contained cars, under-seat batteries, and station charging

Current milestone: [v0.1](#) — first publishable snapshot. See [CHANGELOG.md](#) for what's in tree.

Status: Rust workspace plus two Python sidecars (`design-py` for GIS + network synthesis, `mechanical-py` for parametric mechanical + civil + station components under `build123d`). Run `python3 scripts/repo-health.py` for the generated-catalogue drift gate and the test commands in [CHANGELOG.md](#) for the current verification snapshot. Deployments ship as GoA 4 (Unattended, driverless) from day one per [RFC 0015](#) — the driver cab is replaced by a nose-cone obstacle-detection sensor suite (ultrasonic safety belt + solid-state LIDAR + mmWave radar + stereo camera) on a dedicated T-OBS ECU, with wayside track-intrusion detection ([RFC 0016](#)) covering the proactive half of the safety envelope between trains. The rationalised rolling-stock baseline is a self-contained, driverless battery car: standard bogies with one powered bogie and one trailer bogie per car, sodium-ion batteries under the longitudinal seats, a low-floor centre door zone, and solar-buffered station charging during ~60 s dwells at roughly 1 km stop spacing. The onboard pack is sized for about one route length plus reserve, keeping mass and vehicle cost down while the station PV + stationary battery system supplies normal service energy. Reference rolling-stock cost is now about €1.0M per self-contained car / wagon: body + interior, one powered bogie, one trailer bogie, under-seat Na-ion pack, traction package, onboard sensors, and GoA 4 control electronics. A 3-car light-metro-3car trainset is therefore budgeted at ~€3.0M before country multipliers. The [Samawah pilot](#) is a brownfield deployment anchored on the 300–800 dormant freight wagons + rolling-stock workshop adjacent to Samawah Train Station per [RFC 0027](#), not a greenfield reference scenario.

The design pipeline auto-synthesises a complete multi-line network for any city listed in [lib/city-batches/world-sample.toml](#) from real OpenStreetMap + WorldPop population data — population-tiered topology, served-catchment bbox, country-specific cost + finance multipliers. The catalogue currently covers 166 cities across 42 countries, structured as:

- Iraq launch corridor (18 cities) — every governorate capital
 - Samawah, exercising the rolling-stock family bands on Iraqi soil.
- MENA expansion (42 cities) — Levant, Arabian Peninsula, and North Africa targets, including Sana'a, Aden, Taiz, Damascus, Aleppo, Homs, Amman, Beirut, Gaza City, Cairo-region secondaries, and additional Saudi, Syrian, Yemeni, Jordanian, Lebanese, and Palestinian catchments. Several are conflict-affected; financing assumptions reflect that.
- Sub-Saharan Africa (43 cities) — Nairobi, Dar es Salaam, Kampala, Antananarivo, Mogadishu, Kigali, Lusaka, Yaoundé, Kinshasa, Lubumbashi, Luanda, Maputo, Beira plus Niamey, Bamako, Dakar, Ouagadougou, Conakry, and a wider West/Central/East Africa seed set.
- South Asia (52 cities) — Coimbatore, Karachi, Faisalabad, Multan, Patna, Kabul, Kathmandu, Colombo, Lucknow, and a broader India/Pakistan/Bangladesh/Nepal/Sri Lanka/Afghanistan planning set.
- Southeast Asia (7 cities) — Yangon, Phnom Penh, Mandalay, Surabaya, Bandung, Davao, Vientiane.
- Latin America + Europe (4 cities) — Cuenca, La Paz, San Salvador (+ Lyon as a high-OSM-density solver test target).

Population bands cover 200 k planning seeds (Sayun, tram-2car) to 20.3 M (Karachi, metro-6car). One-command regeneration: `scripts/regenerate-city.sh <slug>` for a single city or `scripts/regenerate-all.sh --jobs 4` for the whole catalogue.

Samawah reference network — three auto-planned radial lines on OpenStreetMap, arterial-routed

Samawah (~374 k pop, Iraq 2024 census, As-Samawah Subdistrict per [lib/city-batches/world-sample.toml](#)) — a brownfield pilot, not a greenfield reference scenario. 2026-04-26 satellite review ([RFC 0003 §2.1](#)) identifies 300–800 dormant freight wagons stored across two yards adjacent to Samawah Train Station, plus a rolling-stock workshop building (the 2011 Iranian Waxon Park rehabilitation target). Iraqi Republic Railways operates a fleet of 10,326 freight wagons + 255 passenger coaches on the active Baghdad–Basra mainline through the city. Standard gauge (1 435 mm) matches RFC 0009. The OSR Samawah pilot is specified by [RFC 0027 \(brownfield asset-recovery doctrine\)](#) as the systematic conversion of this rail-yard +

workshop complex into the first OSR rolling-stock production site, with ~\$8–15 M of recoverable mechanical-component value and ~\$3–6 M of recoverable workshop infrastructure offsetting the greenfield CAPEX baseline.

Network — auto-planned end-to-end by osr - design against real OSM data *plus the WorldPop residential-population layer (so lines reach population centres without mapped POIs). Three radial lines all converging at a central elevated-junction interchange, 33 unique stations at ~1.2 km inner / 2 km transitional / 4 km outer spacing, 54.9 km of double-track, all soft gates passing. Fleet 55 × 3-car light-metro-3car. OSR-discipline unit costs throughout — prefab portal-frame canopies, at-grade depots without overhead bridge cranes, commodity Na-ion cells under the seats, one powered bogie per car, tier-2 PMSM motors, DIY SiC inverters, station-dwell charging, open-source CBTC on commodity SBCs, no overhead catenary, self-EPC overhead. See [designs/west-asia/Iraq/Samawah/README.md](#) for the full breakdown (per-line termini, fleet sizing, full cost stack, country-anchored ticket pricing).*

Auto-design catalogue at a glance

Same pipeline, every generated catalogue city — population source, served-catchment bbox, rolling-stock family auto-selected per [RFC 0008 §5](#), finance section anchored to country-specific median income + multilateral / sovereign rates from [lib/templates/country-finance.toml](#). The generated designs cover all 166 catalogue entries. The README does not duplicate the city table; for the full city list and cross-city comparison see the generated [designs/INDEX.md](#). For the CAPEX audit trail see [docs/cost-model.md](#).

In the catalogue, coverage means high-demand coverage: the share of cells in the 20 m demand raster with demand ≥ 0.5 that fall within about 400 m of a planned line. The demand raster blends OSM anchors (stations, universities, hospitals, airports, neighbourhoods), a mild centre bias, and WorldPop residential population. It is therefore a planning proxy for population / trip-generator catchment, not percentage of city land area.

A few representative auto-planned networks:

Baghdad — 9 radial lines + ring, 510 km, metro-6car, hub-and-spoke-dual-ring topology

Baghdad (~9.78 M, Baghdad Governorate) — megacity tier, nine radial lines plus a circumferential ring at $\sim 0.55 \times$ urban radius, all converging on the elevated-junction interchanges in the central business district. 218 unique stations at ~2.1 km average spacing (1.5 km inner / 3 km transitional / 5 km outer per the megacity override of `SpacingConfig`); 510 km double-track; 408 trainsets across the fleet. WorldPop residential layer + airport / suburb / neighbourhood OSM anchors so radials reach Baghdad International (BIAP), Basmaya New City, and other population centres beyond the POI clusters.

Karbala — 6-line metro-4car network, pilgrim-anchored, 67% coverage of the population-blended demand surface

Karbala (~1.39 M, Karbala Governorate) — the catalogue's strongest coverage score (67%) thanks to a tightly clustered shrine-anchored demand surface (Imam Hussein + Imam Abbas shrines anchor the centre, with Hindiyah east-of-the-Euphrates and date-palm outskirts south). Six metro-4car lines, 89 stations. Pilgrim volume is the binding sizing constraint — Arba'een alone draws ~25 million pilgrims annually, an order of magnitude past the resident catchment.

Karachi — 9-line metro-6car network, 472 km, served-catchment 50×50 km

Karachi (~20.3 M, Karachi Division per the Pakistan 2023 Digital Census) — the catalogue's largest catchment. Same nine-line topology as Baghdad despite the very different street pattern, 231 stations, 377 trainsets. Demonstrates the pipeline scales past Iraqi-megacity geography to South-Asian coastal megacity without rule-set tuning.

Lyon — 6-line metro-4car network, 287 km — solver test on a high-OSM-density European city

Lyon (~1.44 M, Métropole de Lyon) — kept in the catalogue deliberately as a solver test target. Rich OSM data exercises bridge classification (Rhône + Saône crossings), dense-anchor routing past the Parc de la Tête d'Or, and the tightest civil-class corner cases. Six lines, 122 stations, 287 km — comparable to the city's actual operating metro + tram footprint (≈ 195 km), with the OSR pipeline extending into the Métropole's outer 58 communes.

La Paz — 6-line metro-4car network straddling the El Alto altiplano cliff

La Paz + El Alto, Bolivia (~1.82 M conurbation) — alpine-tropical twin-city extending across a 1 000 m vertical drop from the altiplano (~4 150 m, El Alto airport district) into the canyon (~3 200 m, Río Abajo). Six lines, 115 stations. Severe terrain — the civil classifier picks elevated for the cliff descent automatically.

Sana'a — 9-line metro-6car network, catalog's highest coverage score

Sana'a, Yemen (~3.94 M, Sana'a Capital Secretariat + Governorate) — the catalogue's highest coverage score (78%). Mountain-plateau geography (~2 250 m elevation) with a tightly clustered Old City core driving the metric. Nine metro-6car lines, 126 stations, 261 km. Conflict-affected since 2014 — the financing model uses distressed sovereign + IDA-eligible multilateral assumptions.

Twenty-seven RFCs cover the full system from software architecture through rail civil engineering to driverless operation; the operations rulebook ([RFC 0013](#)) is drafted across four shipping role families (dispatcher / station-staff / maintenance /

control-centre) plus a historical driver section retained for GoA 2 legacy fleets. Most of the [RFC 0005](#) crate map is in tree:

- Onboard safety chain (SIL-4): position fusion ATP brake + obstacle- detect (GoA 4 driverless) + derailment + fire + door-control, all pure functions, integrated into the simulator as a per-tick shadow stack. Zero spurious emergencies under nominal service. `osr-vigilance` and `osr-dmi` are kept in tree under the `goa2-cab` feature flag for legacy fleets but disabled by default.
- Operator GUIs (RFC 0018), feature-complete + WASM: two pure-Rust egui apps sharing [osr-gui-shared](#) for rendering, both targeting native + browser (WebAssembly via trunk).

[osr-sim-gui](#) is the designer's workbench — load a scenario, run the sim once, then play back the resulting [SimTimeline](#) with scrubber + 0.5×/1×/10×/60× speeds. Trains animate along the strip coloured by phase; click-to-inspect sidebar shows `station_m`, SoC, and last event; scrolling event log filters by kind; active faults surface as timestamped badges over the map.

`osr-sim-gui` running the Samawah scenario

[osr-occ-gui](#) is the dispatcher console — per-section `IntrusionState` overlay, train roster panel, alert feed with info/warn/crit filters, and validated action modals for S2.1 route grant, S5.1 `MaintenanceOverride`, and RFC 0013 §5 degraded-mode declaration. v1 is read-only; v3 of the RFC wires live consensus + RFC 0017 signed envelopes for revenue use.

`osr-occ-gui` — Samawah OCC console with SEC1001 intrusion present

- Onboard obstacle detection (SIL-4, RFC 0015): `osr-obstacle-detect` fuses a multi-physics sensor suite — 4× ultrasonic (close-range safety belt, 0.2–20 m), solid-state LIDAR (5–200 m primary 3D, affordable Chinese Livox / RoboSense / Leishen-class units), mmWave radar (all-weather validation), stereo camera (classification only) — into an `ObstacleVerdict` per tick: `Clear` / `RestrictedSpeed` (40 km/h cap, LIDAR degraded) / `CrawlOnly` (15 km/h cap, soft obstacle) / `EmergencyBrake`. Two sensor packs per trainset, one at each end; either nose can lead on a given run. Five SIL-4 properties O1–O5 with Kani harnesses + 8 proptests; 2002 peer cross-check fail-restrictive. The sim injects sensor faults on a scenario schedule (LIDAR offline, radar offline, ultrasonic channel stale, peer disagreement — per-train or fleet-wide) and the shadow stack produces the expected verdicts end-to-end. Wayside intrusion injection likewise — staged `Present/Unknown` verdicts on specific sections cause the interlocking's gate (d) to hold MA without any train entering.
- Wayside track-intrusion detection (SIL-4, RFC 0016): `osr-intrusion-detect` on the W-SBC fuses fence-line contact sensors, ROW-mounted solid-state LIDAR, ROW-mounted mmWave radar, and CCTV classifier into a per-section `IntrusionVerdict` — `Clear` / `Unknown` / `Present`. Fail-restrictive: a stale sensor yields `Unknown`, not `Clear`. The verdict flows into the consensus log as `EntryPayload::SectionIntrusion`, and `osr-interlocking::section_available_to` gate (d) withholds MA on any section whose verdict is not `Clear` (sections with no verdict are treated as "not instrumented" — backwards-compatible for partially-wired deployments). Five SIL-4 properties I1–I5 with Kani harnesses + 6 proptests; GSN goals G20–G24 close against real evidence. Complements the onboard obstacle-detect — wayside is proactive (before a train enters), onboard is reactive.
- Integrator crate + feature-gated legacy stack: [osr-trainset-image](#) aggregates the onboard stack into one versioned deployment unit. Default build is GoA 4 (Unattended) per RFC 0015; --features `goa2-cab` opts in to legacy cabbed fleets by pulling `osr-dmi` + `osr-vigilance` into the image. `cab_profile()` is a compile-time witness for which profile the image was built with.
- Parametric rolling stock in mechanical-py (RFC 0015 cabless): sensor cowl, symmetric car body with door cutouts, 2-axle bogie simplified block, and the published trainset families (`urban-shuttle-1car`, `tram-2car`, `light-metro-3car`, `metro-4car`, `metro-6car`) all parametric on consist + track geometry. Every trainset fits inside its RFC 0008 §1 platform length with ≥ 1 m stopping margin. STEP artifacts round-trip into Revit / Tekla / Civil 3D.

Current rolling-stock design reference: self-contained driverless cars, one powered bogie + one trailer bogie per car, sodium-ion batteries under the longitudinal seats, low-floor centre door zones, and station-dwell charging from solar-buffered stops spaced around 1 km apart. The solar train image at the top of this README is the public-facing visual reference; CAD screenshots are engineering checks only.

Motor bogie — 2-axle Bo-Bo with axle-hung PMSM motors, chevron primary + air-spring secondary suspension

Trailer bogie — same frame + suspension + wheelsets, no motor or gearbox

Detailed bogie CAD per [RFC 0022](#). Motor bogie (top): 2-axle Bo-Bo pivoting bogie with axle-hung PMSM traction motors (180 kW continuous / 320 kW peak per axle), single-stage 6.5:1 parallel-spur gearboxes, chevron rubber-metal primary suspension, air-spring secondary suspension, 760 mm forged wheels, one axle-mounted brake disc per axle. Trailer bogie (bottom): same frame + wheelsets + suspension SKU with the motor- gearbox drivetrains omitted — one single bogie pattern scales across every consist family with per-family motorisation.

- Onboard traction & power: `osr-traction` + `osr-bms` + `osr-regen` + `osr-aux-power`, with signed-current sign convention enforced at the seam.
- Onboard systems: `osr-tcms`, `osr-hvac`, `osr-lighting`, `osr-dmi`, `osr-pis-onboard`, `osr-hot-axle`, `osr-odometry`, `osr-event-recorder`, `osr-tcn` (IEC 61375-style TSN trainbus, mock transport).
- Wayside core: `osr-interlocking` (MA computer) on top of `osr-consensus` (a Rust refinement of the TLA+ SMRaft spec, proptest-verified against all 5 safety invariants; the sim can drive MA through a real 3-node Raft cluster via `--use-consensus`). `osr-wayside-points` controls power switches with dual-redundant sensor fusion and fail-restrictive Unknown detection.
- Wayside infrastructure (Phase 2d): `osr-balise`, `osr-level-crossing` (SIL-4 five-state barrier controller), `osr-hot-axle-wayside` (SIL-4 HABD), `osr-energy-site` (PV + battery + grid-tie dispatch).
- Stations & fare: `osr-psd` (platform screen doors), `osr-station-scada`, `osr-pis-station`, `osr-afc` (HMAC-signed account-based tokens), `osr-tvm`.
- Back office: `osr-occ` (dispatcher), `osr-historian` (ring-buffered per-metric storage with decimation), `osr-analytics`, `osr-t2g`.
- Simulator: multi-day runs, time-of-day dispatch, PV + trackside storage
 - grid tie energy model, fault injection, shadow onboard stack. Every section entry is gated by `osr-interlocking::section_available_to` against a synthesised log — the MA computer is the sim's only source of occupancy (RFC 0004 M5). Scenarios in TOML so anyone can design their own city.
- Differential Python reference interpreter: `tools/reference-ma/` — an independent stdlib-only Python twin of `osr-interlocking`. Every proptest run of `crates/osr-interlocking/tests/differential.rs` serialises a random log prefix, computes the MA in both Rust and Python, and asserts byte-identical JSON. Coverage extends to ring lines, switch observations, route grants, and speed restrictions. Catches bugs in either implementation (RFC 0004 M4).
- GSN safety-case compiler: `osr-safety-case` loads the TOML claim files in `docs/safety-case/gsn/` (19 goals, 5 strategies, 50 solutions today — the G1/G2/G3 MA-computer claims, plus G4/G5 covering every SIL-4 onboard and wayside evaluator, plus a regression guard for the 5-node Raft fix) and CI fails if any goal no longer traces to evidence. Adding a safety-relevant claim without linking it to a Kani harness / proptest / sim run breaks the build (RFC 0005 §4.9).
- Kani harnesses on every SIL-4 evaluator: ATP A1–A7, brake B1–B5, vigilance V1–V6, odometry O1–O5, wayside-points W1–W6, plus the MA computer's P1–P5 (scaled to a 3-section network with a mutation-style P4 companion) — the full pure-function surface of the onboard and wayside safety chains has bounded formal proofs in tree to match every proptest property.
- Real network transport for TCN: `osr-tcn::UdpTcn` — drop-in API replacement for the in-memory mock, running on commodity UDP (one datagram per message, 5-byte header, ≤ 1400-byte payload). Moves the on-train bus from simulator-only to multi-host bench without dragging in TSN hardware. Full TSN stays ahead on the RFC 0006 roadmap (v3).
- Automatic design generation: `design-py` (Overpass + raster synthesis) + `osr-routing` (cost/demand Dijkstra on a 20 m grid) + `osr-design` (emitter) compose a full multi-line network — corridor geometry, station placement, civil-class inference (at-grade / elevated / bridge — no tunnels per RFC 0011), rolling-stock + track-geometry + station-archetype selection under the RFC 0008/0009/0010 compatibility matrix — from nothing but a bounding box and a population. Terminals are detected at line endpoints; interchanges are detected where two lines' stations land within 200 m of each other; every station gets a derived platform length consistent with the chosen consist family. Scales to a 500-city batch with a GeoNames-driven scanner that excludes any city already operating metro, tram, or light-rail.
- Parametric mechanical + civil + station catalogue: `mechanical-py` ships build123d assemblies for every non-trivial physical piece of the system — UIC 54E1 / 60E1 rail, EN 13230 B70-class sleeper, Pandrol-style fastener, precast U-girder (20/25/30 m), precast L-unit platform edge, and a prefab-bolt-together steel portal-frame bay with a factory-bonded solar-roof sandwich panel that composes into a full canopy by `(archetype, consist)`. `osr-mech-export` regenerates 33 STEP artifacts under `catalog/` that round-trip into Revit, Tekla, Civil 3D, and FreeCAD — deployment partners keep their existing structural tooling while the repository stays the canonical source. Design bias throughout: prefab, bolt-together, no on-site welding, no wet concrete except pad footings — a standard station canopy is ~11 t of steel in two lorry-loads, erected in 3–5 days.

Standard station canopy — 67 m light-metro platform, factory-bonded solar roof

Standard archetype canopy for a light-metro-3car consist: 13 × 6 m portal bays in hot-dip galvanised steel, topped by a factory-bonded sandwich panel that integrates the PV surface. One canopy covers the full platform — no separate station building, no catenary.

- OSR-ALN civil-tool bridge (RFC 0009 v3): [tools/osr-aln-convert/](#) is a stdlib-only Python converter from LandXML 1.2 to [OSR-ALN TOML](#) — one CLI (`landxml-to-osr-aln`) covers Civil 3D, Bentley OpenRail, Trimble Business Center, and QGIS rail-path plugin exports. Reads tangents, circular curves, spirals, vertical PVI's + circular vertical curves, and station pin-points; emits placeholders for civil class + cant (not carried by LandXML). A companion `osr-aln-validate` CLI enforces the format spec's 8 hard gates + 3 soft gates against the deployment's `design.toml` (per-preset curve radius, gradient, cant maxima; civil-span contiguity; station-id cross-check; no tunnels). 21 passing tests. Worked reference alignments for both Samawah lines ship at [docs/civil/west-asia/Iraq/Samawah/samawah-line1.aln.toml](#) (13 km radial, 12 stations, 3 cant sections) and [docs/civil/west-asia/Iraq/Samawah/samawah-line2.aln.toml](#) (16 km ring, 10 stations, 4 cant sections, `is_ring = true`) — both pass every hard gate. This is the last-mile piece that lets a civil engineer import a real survey into the project.
- EN 62267 type-certification pre-submission pack at [docs/certification/](#) — system description + SRS (SR-01..SR-24) + hazard log (17 hazards across 7 classes) + evidence register + clause-by-clause EN 62267 compliance matrix. Every safety requirement traces to a Kani harness / proptest / GSN goal / rulebook rule. Ready for an independent safety assessor (deployment-partner scope).
- Consensus-log message authentication (SIL-2, RFC 0017): `osr-crypto` now ships `ed25519` sign + verify primitives alongside the existing HMAC-SHA256; `osr-secbus` wraps them with a per-deployment `KeyRegistry` and a `SignedBytes` envelope that carries issuer + signature alongside the opaque entry bytes. Verification happens before deserialisation so a hostile payload can't exploit a parser bug pre-auth. Three SIL-2 properties S1–S3 with Kani harnesses
 - 5 proptests; GSN goals G25–G27 close against real evidence.

Why this project exists

Urban rail is the most efficient way to move people through a growing city, and developing nations badly need more of it. But the global rail market is dominated by a handful of vendors — Alstom, Siemens, Hitachi, CRRC, Thales — whose turnkey systems lock countries into decades of imports, specialized foreign labor, and capital flight.

OpenSourceRail is a complete, permissively licensed alternative, aimed at national railway authorities and domestic engineering firms in target regions (sub-Saharan Africa, MENA, South and Southeast Asia, most of Latin America). The project succeeds when an operator in one of those markets can build and run a modern rail network with imported content limited to raw steel, copper, and specialty items that genuinely cannot be made locally — everything else is produced by the people who will run the railway.

What's different

OpenSourceRail is not a reimplement of existing vendor practice in open code. The architecture actively deprecates legacy approaches where cleaner ones exist. The headline bets:

- Catenary-free. Trains run on onboard sodium-ion or LFP batteries with opportunity charging at stations. Eliminates the single largest capex line on new-build rail (€1–3M/km of catenary) and a major copper-theft target. [ARCHITECTURE §4 D7 · RFC 0002](#)
- Solar-first energy. PV on station canopies, depot roofs, and along the right-of-way the railway already owns — netting positive daily generation against demand in the target climates.
- Distributed train control, not centralized. Wayside nodes run a formally verified consensus protocol maintaining authoritative track state. Replaces €10–50M centralized zone controllers with commodity SBCs at <€5k/site. [RFC 0001 · TLA+ spec](#)
- Commodity SBCs + Rust, end to end. One language from the SIL-4 safety kernel to the dispatcher web UI. Industrial PLCs, proprietary trainbuses, and Windows-based SCADA are out of the default design.
- TSN Ethernet trainbus, not MVB/WTB. Off-the-shelf switches, deterministic traffic, Rust stack.
- Account-based fare, not smart cards. Mobile money + QR + optional NFC; target markets have already leapfrogged card infrastructure.
- Machine-checkable safety case. GSN-structured, version-controlled, regenerated on every commit, linked to formal proofs and test evidence.

Each deviation from established rail practice is justified against a concrete criterion: cost, simplicity, local manufacturability, or workforce transfer. Not novelty for its own sake.

Scope

Eight subsystems; treat as a system-of-systems:

Domain	What it covers
D1. Operations & Dispatch	Timetable, incident management, event-sourced platform
D2. Train Control	Interlocking, movement authority, ATP — the SIL-4 core
D3. Communications	5G + LoRa radio, TSN wayside backbone
D4. Passenger Services	Fare, info displays, announcements
D5. Rolling Stock	Unified Rust ECUs, TSN trainbus, onboard battery + inverter
D6. Infrastructure	Switches, track geometry, level crossings
D7. Energy	PV generation, trackside storage, station charging
D8. Depot & Maintenance	CBM, work orders, depot microgrid

Full description: <docs/ARCHITECTURE.md>.

Repository layout

```
OpenSourceRail/
├── README.md           You are here.
├── Cargo.toml          Rust workspace.
├── docs/
│   ├── ARCHITECTURE.md Scope, subsystem design, roadmap. Start here.
│   ├── operations/     Full operations rulebook (RFC 0013 v2): one-sentence
│   │                   rule + Why: paragraph per clause, every clause
│   │                   cross-referenced to the relevant crate and safety-case goal.
│   │   ├── driver/     D1-D8 (before-service → emergencies → end-of-service).
│   │   ├── dispatcher/ S1-S6 (shift start → incident handling → shift end).
│   │   ├── station-staff/ T1-T5 (opening → passenger incidents → closure).
│   │   ├── maintenance/ M1-M6 (depot safety → work-on-track → fleet MX).
│   │   └── control-centre/ C1-C3 (watch roles, comms, shift handover).
│   ├── hardware/       Hardware bring-up runbooks (RFC 0007 v1).
│   │   ├── bring-up/   Per-class procedures: t-ecu-s, t-ecu-a, w-sbc, s-sbc.
│   │   ├── rolling-stock/ Rolling-stock shop-drawing packages (RFC 0008 v1+).
│   │   └── light-metro-3car/ Fabrication plan, body, bogie, traction, BOM, compliance.
│   ├── civil/          Per-deployment civil alignment (RFC 0009 v1+).
│   │   └── samawah/     Line 1 + Line 2 segment table + compliance report.
│   ├── stations/       Per-archetype architectural envelopes (RFC 0010 v1+).
│   │   └── samawah-standard/ Envelope, canopy structural, accessibility, services.
│   └── rfcs/
│       ├── 0001-track-state-consensus.md Distributed signaling core.
│       ├── 0002-energy-sizing.md          Solar+battery sizing.
│       ├── 0003-samawah-reference-deployment.md Reference pilot.
│       ├── 0004-osr-interlocking-plan.md  MA computer implementation plan.
│       ├── 0005-sbc-software-architecture.md Canonical software architecture + crate map.
│       ├── 0006-osr-tcn-design.md         On-train bus (TCN-E pub/sub).
│       ├── 0007-hardware-reference-designs.md T-ECU/S, T-ECU/A, T-OBS, W-SBC, S-SBC.
│       ├── 0008-rolling-stock-reference-design.md 5 trainset families (urban shuttle → metro-6car).
│       ├── 0009-track-design-standard.md 4 geometry presets (gauge, radius, grade, cant).
│       ├── 0010-station-design-standard.md 6 station archetypes + passenger-flow model.
│       ├── 0011-civil-infrastructure-design-standard.md At-grade + elevated only (NO tunnels).
│       ├── 0012-switches-and-crossings.md 3 turnout tangents + LX equipment envelope.
│       ├── 0013-operations-rulebook.md ≤ 60-page per-role rulebook, 3 degraded modes.
│       ├── 0014-depot-design-standard.md 3 depot archetypes + fleet-sizing formula.
│       ├── 0015-driverless-operation.md GoA 4 by default — sensor suite, T-OBS ECU, cab elimination.
│       ├── 0016-wayside-track-intrusion.md Wayside intrusion detection — complements onboard detector.
│       ├── 0017-cybersecurity-message-authentication.md Ed25519-signed consensus entries.
│       ├── 0018-operator-guis.md          egui-based sim + OCC consoles for designer + dispatcher.
│       ├── 0019-diy-electronics.md        Plug-and-play DIY electronics from commodity modules.
│       ├── 0020-crashworthiness.md        EN 15227 three-zone energy budget for the cabless body.
│       ├── 0021-battery-traction.md        Under-seat Na-ion packs + station/depot opportunity charging.
│       ├── 0022-bogie-traction-drive.md    Single-SKU 2-axle Bo-Bo bogie + axle-hung PMSM motorisation pattern.
│       └── 0023-door-system-reference-design.md Electric linear-actuator door operator (EN 14752, $20-40 k vendor →
~$2 k DIY).
│       ├── 0024-battery-thermal-high-ambient.md PCM mass + dedicated chiller for ≥ 50 °C ambient envelopes.
│       └── 0025-diy-switch-and-point-machine.md Regional switch-shop bootstrap (~$580 k CAPEX, ~$10 k per 1:9
turnout vs $120 k vendor).
├── 0026-charging-connector-reconciliation.md Two-tier connector architecture (CCS2 depot + side-pin/
pantograph terminal).
├── 0027-brownfield-pilot-asset-recovery.md Three-phase doctrine for converting dormant rolling stock +
dormant rail workshops into OSR production capacity (Samawah, Atbara, Karachi, Maputo, Luanda, Tashkent worked
examples).
├── crates/             56 Rust crates — grouped by role below.
│   ├── osr-core/       Shared domain types (topology, trains, IDs).
│   │   └── proto/track_state.proto Interface definitions.
│   ├── # Onboard safety (SIL-4)
│   │   ├── osr-odometry/ Onboard position fusion.
│   │   ├── osr-atp/      Automatic Train Protection.
│   │   ├── osr-ato/      Automatic Train Operation (GoA 4 default).
│   │   ├── osr-obstacle-detect/ NEW (RFC 0015): ultrasonic + LIDAR + radar fusion.
│   │   └── osr-trainset-image/ NEW (RFC 0015): onboard-stack integrator, goa2-cab flag.
│   ├── # Onboard safety (continued)
│   │   ├── osr-brake/    EP brake + WSP + park brake.
│   │   ├── osr-vigilance/ Driver alerter / dead-man (GoA 2 legacy only).
│   │   └── osr-derailment/ 2oo2 derailment detection.
```


osr-fire-safety/	Onboard fire detection + suppression.
osr-door-control/	Door interlock + enable gating.
# Operator GUIs (RFC 0018)	
osr-gui-shared/	Shared egui network-rendering helpers.
osr-sim-gui/	Simulator GUI binary – designer workflow.
osr-occ-gui/	OCC dispatcher console binary – live-ops workflow.
# Onboard traction & power	
osr-traction/	Motor control (signed-current convention).
osr-bms/	Battery management (pack-level).
osr-regen/	Regenerative braking arbitration.
osr-aux-power/	Auxiliary / HVAC / lighting bus.
# Onboard systems	
osr-tcms/	Train Control & Management System.
osr-hvac/	HVAC controller.
osr-lighting/	Passenger/cab lighting.
osr-dmi/	Driver-Machine Interface state.
osr-pis-onboard/	Onboard Passenger Information System.
osr-hot-axle/	Onboard hot-axle advisory.
osr-event-recorder/	Black-box ring buffer.
osr-tcn/	IEC 61375-style TSN trainbus (mock v1).
# Wayside core (SIL-4)	
osr-interlocking/	MA computer (RFC 0004 M1+M2 done).
osr-intrusion-detect/	NEW (RFC 0016): wayside track-intrusion detection.
osr-consensus/	Raft – refinement of formal/tla/SMRaft.tla.
osr-wayside-points/	Power-switch (point) controller.
# Wayside infrastructure	
osr-balise/	Balise registry + sighting audit.
osr-level-crossing/	SIL-4 five-state crossing controller.
osr-hot-axle-wayside/	SIL-4 wayside HABD.
osr-energy-site/	PV + battery + grid-tie dispatch.
# Stations & fare	
osr-psd/	Platform screen doors.
osr-station-scada/	Station SCADA.
osr-pis-station/	Station passenger information.
osr-afc/	Automatic fare collection (HMAC tokens).
osr-tvm/	Ticket vending machine.
# Back office	
osr-occ/	Operations Control Centre / dispatcher.
osr-historian/	Ring-buffered metric storage w/ decimation.
osr-analytics/	Fleet analytics (adherence, MDBF, energy/km).
osr-t2g/	Train-to-ground radio adapter.
# Commissioning	
osr-selftest/	NEW (RFC 0019): per-role post-assembly self-test CLI – the DIY path's flying-probe substitute.
# Design pipeline	
osr-routing/	Cost/demand Dijkstra solver + civil-class inference + station placement on a 20 m grid.
osr-design/	Orchestrator – reads rasters + anchors, emits design.toml + corridor.geojson + quality.yaml.
osr-alignment/	Horizontal + vertical alignment artefact with cant schedule, LandXML + railML exports, stake-out CSV, earthworks quantities, and trackside-equipment placement.
osr-sim/	Time-stepped simulator + shadow onboard stack + HTML/SVG visualizer (osr-vis).
mechanical-py/	Python sidecar for parametric mechanical / civil / station components (build123d). Every RFC-level choice (consist, archetype, span) is a parameter; STEP artifacts under catalog/ round-trip into Revit / Tekla / Civil 3D.
src/osr_mech/	
track/	Rail (54E1/60E1), sleeper (B70), fastener, panel.
civil/	Precast U-girder (20/25/30 m), platform L-unit.
station/	Portal-frame bay + solar-roof sandwich panel + multi-bay canopy – the full "prefab metal canopy with solar roof" reference station.
rolling_stock/	Cabless car body + sensor cowl + 2-axle bogie + full trainset (5 consist families) per RFC 0015.
catalog/	Regenerable STEP artifacts (run `osr-mech-export`).
design-py/	Python sidecar for GIS data + raster synthesis.
src/	
osr_osm/	Overpass fetcher w/ SHA256 disk cache (arterials, buildings, water, protected land, POI anchors).
osr_geo/	Numpy raster synthesis – cost / demand / buildability surfaces, binary + JSON sidecar.
osr_batch/	Batch driver + GeoNames → cities.toml scanner with built-in existing-transit denylist.
hardware/	Hardware reference designs (RFC 0007 v2-spec).
t-ecu-s/	Train safety kernel (2 × RP2350 2002 + RPi CM5).
t-ecu-a/	Train application (RPi CM5 carrier).
t-obs/	Train obstacle-detect ECU (2 × RP2350 + CM5 + sensors).

├─ w-sbc/	Wayside (Radxa CM5 RK3588S, one SKU).
├─ s-sbc/	Station / depot (RPI CM5 + commodity carrier).
├─ diy-assembly/	Per-host-class commodity-module BOMs (RFC 0019).
├─ tools/	
│ └─ reference-ma/	Python reference interpreter for osr-interlocking (differential twin against Rust, RFC 0004 M4).
└─ osr-aln-convert/	LandXML → OSR-ALN converter (RFC 0009 v3) for Civil 3D / Bentley OpenRail / Trimble / QGIS exports.
├─ lib/	
│ └─ city-batches/	Canonical city catalogue (slug → bbox / country / population / climate). <code>world-sample.toml</code> is the 166-city catalogue spanning 8 regions and the rolling-stock family bands; production entries source national-stats-office figures while generated expansion seeds are flagged for source replacement before deployment.
└─ templates/	Reusable Lego-block TOMLs (stations, switches, signalling, structures, fleets, energy-sites, country-costs, country-finance).
└─ recipes/	City-to-design generation rules (climate, topology, fleet sizing).
└─ examples/	Generic template scenarios (<code>example-simple.toml</code> , <code>minimal-city.toml</code>).
├─ designs/	Per-city design artefacts (<code>design.toml</code> + scenario + map PNG + README + GeoJSON + design-quality YAML, all auto-generated by <code>scripts/regenerate-city.sh <slug></code>). 166 cities today across west-asia / north-africa / east-africa / west-africa / south-asia / southeast-asia / europe / latin-america.
└─ west-asia/Iraq/Samawah/	The reference brownfield pilot per RFC 0003 + RFC 0027 – the bundled canonical Samawah scenario <code>osr-sim</code> defaults to is generated from this folder's <code>design.toml</code> .
├─ scripts/	
│ └─ regenerate-city.sh	Slug-driven end-to-end regeneration: OSM → rasterise (with WorldPop) → osr-design synthesis → scenario → map PNG → README → drift tests.
└─ regenerate-all.sh	Batch driver – reads slugs from <code>world-sample.toml</code> and runs <code>regenerate-city.sh</code> per slug. Supports <code>--jobs N</code> (parallel), <code>--only</code> / <code>--skip</code> filters, and continue-on-error with a final OK / FAIL summary; per-slug logs in <code>.cache/osr-pipeline/logs/</code> .
├─ formal/tla/	
│ └─ SMRaft.tla	Consensus protocol spec.
│ └─ MCSmall.tla	Small TLC harness.
└─ MCSmall.cfg	TLC config.

Quick start

Requirements: Rust 1.80+ via `rustup`.

```
cargo run --release --bin osr-sim -- --duration 3600 --status-every 300
```

When run with no `--config`, `osr-sim` loads the bundled canonical Samawah scenario – the auto-generated [designs/west-asia/Iraq/Samawah/samawah.toml](#) is `include_str!`-baked into the binary at build time. The output shows each train's position and state-of-charge at regular intervals, per-line km, energy consumed vs. charged, dispatch hold time, and any invariant violations (there should be none).

To simulate any other catalog city, point at its scenario file:

```
cargo run --release --bin osr-sim -- \
  --config designs/south-asia/Pakistan/Karachi/karachi.toml \
  --duration 3600
```

Designing your own city

Two paths, depending on whether you want to author by hand or have the pipeline auto-design from a bbox + population:

Auto-designed (recommended). Add a new entry to [lib/city-batches/world-sample.toml](#) with a slug, ISO-2 country, served-catchment bbox (see the BBOX-SIZING POLICY in the file header), and a national-stats-office population. Then:

```
scripts/regenerate-city.sh <slug>
```

The pipeline pulls OSM + WorldPop, synthesises lines / stations / fleet / depots / costs, and emits `design.toml`, the simulator scenario file, the network map PNG, and the per-network README under `designs/<region>/<country>/<City>/`. See [designs/INDEX.md](#) for the generated catalogue and cross-city comparison.

Hand-authored. Scenarios are plain-text TOML files – copy a reference and edit it:

```
cp lib/examples/example-simple.toml designs/my-city/my-city.toml
# edit my-city.toml – see lib/examples/README.md for the file format
```

```
cargo run --release --bin osr-sim -- --config designs/my-city/my-city.toml
```

[example-simple.toml](#) is a 3-station shuttle with 1 train — the smallest viable config; the full file-format reference is in [lib/examples/README.md](#).

Pass `--json-out trace.json` to `osr-sim` to capture the full event trace for later analysis.

Auto-designing networks from GIS data

The pipeline synthesises a complete network for any city listed in the canonical catalogue at [lib/city-batches/world-sample.toml](#) — or for any new city you add there — from real OpenStreetMap + WorldPop population data. One command per city, or one command for the whole catalogue:

```
# One-time: install the python sidecar.
pip install -e design-py[geotiff,batch]
cargo build --release --bin osr-design

# Per-city (slug must exist in lib/city-batches/world-sample.toml):
scripts/regenerate-city.sh karachi

# Whole catalogue, 4 cities in parallel (~17 min cold, faster on warm caches):
scripts/regenerate-all.sh --jobs 4

# Subset:
scripts/regenerate-all.sh --only tunis,lyon --jobs 2
scripts/regenerate-all.sh --skip baghdad
```

The per-city script chains eight steps:

- 1 OSM pull — Overpass query for arterials, buildings, water, protected land, demand-anchor POIs (universities, hospitals, aeroway=*, place=*, railway=station). Cache-keyed on query text.
- 2 Raster bundle — 20 m cost / demand / buildability grid via `osr_geo`. Demand blends a Gaussian POI-anchor layer with the WorldPop residential-population layer (so lines reach population centres without mapped POIs). Falls back to the unconstrained WorldPop tile for ISO-3s where the constrained layer is unavailable (KEN, NER, etc.).
- 3 Design synthesis — `osr-design` (rust) routes each line with a demand-rewarded Dijkstra against the [topology archetype](#) for the population band (SingleRadial ≤ 300 k, RadialPlusRing ≤ 1 M, CrossPlusRing ≤ 3 M, HubAndSpokeDualRing above). Stations placed at 1.2 km inner / 2 km transitional / 4 km outer ([SpacingConfig](#); 1.5 / 3 / 5 km megacity override). Civil classification (at-grade / elevated / bridge — no tunnels per RFC 0011). Fleet sizing per RFC 0014 §4 round-trip / 5-min headway. CAPEX per RFC 0011 §9 with OSR-discipline unit costs $\times$ the country's [country-costs.toml](#) multiplier.
- 4 Scenario file — expanded simulator scenario `<slug>.toml`.
- 5 Network map PNG — auto-fit map of every line on OSM arterials, one colour per line, interchange complexes flagged.
- 6 Per-network README — full breakdown with the [costs] block, [funding & affordability](#) anchored to country median income, station-archetype unit table, energy-infrastructure tier table.
- 7 Stats summary — drift-test inputs.
- 8 Drift tests — `tests/test_osr_scenario.py` + `tests/test_population_drift.py` against the catalogue.

For the production-scale path (eventual 500-city scan via `python -m osr_batch`), the `osr_batch.existing_transit` denylist (~600 cities, 80 countries) keeps cities with operating metro / tram / LRT (Paris, Tokyo, Cairo, etc.) out of auto-generation unless `--include-existing-transit` is passed for calibration. The 166-city `world-sample.toml` catalogue is the curated set in front of that scan.

Reading order

If you have 10 minutes: read this README and [ARCHITECTURE §1–3](#) (mission, principles, system map).

If you have an hour: add [ARCHITECTURE §4–11](#) and whichever RFC is closest to your domain.

If you're considering contributing: read everything above, then [RFC 0005](#) for the full crate map, the [TLA+ README](#), and the [track_state.proto](#) header.

Target hardware

Four physical host classes, specified in [RFC 0007](#) and scaffolded under [hardware/](#). Two-vendor palette — Raspberry Pi and Radxa only — so domestic procurement and spares aren't at the mercy of a dozen silicon suppliers:

- T-ECU/S — train safety kernel. Two Raspberry Pi RP2350 MCUs in a 2-out-of-2 composite fail-safe voting arrangement, each running identical Rust `no_std` code and cross-checking over SPI every tick. A Raspberry Pi CM5 app processor alongside handles non-safety work (TCN-E, logging, OTA). EN 50155 OT4, dual-redundant per consist. ~€280/board.
- T-ECU/A — train application. Raspberry Pi CM5 on a custom baseboard (Radxa CM5 drop-in via same SO-DIMM). EN 50155 OT4.
- T-OBS — train obstacle-detection ECU (new in RFC 0015). Two modules per trainset (one at each nose). Two RP2350 safety MCUs in a 2oo2 cross-check plus a CM5 for sensor fusion. Sensor interfaces: 4× ultrasonic AFE, mmWave-radar CAN-FD, LIDAR 1000BASE-T, stereo-camera MIPI-CSI. ~€780/module, ~€1 560 per consist — a small fraction of the ~€140 k cab capex eliminated.
- W-SBC — wayside. Radxa CM5 (RK3588S, industrial-temp variant) in an IP67 DIN enclosure — +85 °C rated for pole-mount cabinets in hot climates. Same one baseboard selectively populated for switch, crossing, HABD, or energy-site role.
- S-SBC — station / depot. Raspberry Pi CM5 on a commodity carrier — no custom baseboard.
- O-SRV — ops server. Commodity x86-64 or ARM64 Linux.

Every reference PCB is 4-layer FR-4 with 0.15 mm trace/space and 0.3 mm vias — routine at tier-2 fabs across the target deployment footprint. No micro-vias, no 0201 passives, no exotic stackups, no fans, no restricted-export components.

DIY plug-and-play — no PCB fabrication required

For pilot deployments, community builds, and developing-world DIY operators who can't fab custom PCBs at low volumes, [RFC 0019](#) defines a parallel assembly path using commodity modules only. Every host class is built from Raspberry Pi Foundation boards + off-the-shelf HATs + generic relay / ADC / isolator breakouts, wired through DIN-rail terminal blocks, booted from a prepared SD card. No KiCad, no soldering iron.

Per-host-class Bills of Materials with specific SKUs + distributors live under [hardware/diy-assembly/](#) and each host class's `diy-assembly/` subfolder. A first-article Samawah trainset + 1 km of instrumented wayside totals ~\$16 800 in electronics at single-unit retail — compared to ~\$50 M for a legacy-CBTC equivalent (two orders of magnitude, dominated by legacy NRE).

The DIY path preserves the SIL-4 safety arguments: the RP2350 silicon and 2oo2 AND-gate relay pattern come from the same parts populating the custom design; what changes is how they get bolted together. See [RFC 0019 §7](#) for the safety-case mapping.

Per-unit commissioning is handled by the `osr-selftest` CLI: run `sudo osr-selftest --role <role>` on each SoC after assembly and it exercises the role's evaluators (brake, ATP, obstacle / intrusion detect, secbus, HMAC) against known-good fixtures. Non-zero exit halts the unit at a red-LED fault state until the named check passes — the per-unit equivalent of a custom-PCB flying-probe stamp.

Trainset interiors follow the same commodity-first pattern: the car body is a welded steel primary frame with composite cladding, and doors / windows / HVAC / LED lighting / passenger screens / seats / grab poles / intercom are all COTS items with reserved envelopes, power budgets, evidence requirements, and bolt patterns documented in [hardware/trainset-interiors/cots-catalogue.md](#). The catalogue is supplier-neutral: a deployment can swap vendors with adapter plates and harness tails as long as the primary steel frame, mass/power budgets, and certification evidence still pass.

The visual reference remains the Solar Metro Trainset image above; the fit-out screenshot is kept in [docs/screenshots/](#) as a CAD sanity check rather than a public design render.

Rail civil engineering — the affordable bet

Two civil classes only: at-grade and elevated. No tunnels. [RFC 0011](#) fixes this as a project invariant because tunnels cost 10–40× at-grade CAPEX and 4–8× the build time — outside the mission of urban rail a developing nation can finance domestically. One reference precast U-girder for every viaduct and water-crossing bridge in the whole catalogue; one spares pool, one CAD reuse, one formwork.

Complementary rail-engineering RFCs:

- [RFC 0008](#) — 5 trainset families, unified architecture (welded steel primary frame, composite cladding, one powered bogie per car, SiC inverters, Na-ion battery, onboard driverless stack).
- [RFC 0009](#) — 4 track-geometry presets (gauge, radius, grade, cant, rail profile).
- [RFC 0010](#) — 6 station archetypes with platform geometry derived from the line's consist.
- [RFC 0012](#) — 3 turnout tangents (1:9 / 1:14 / 1:18.5) + level-crossing equipment envelope. No diamonds.

- [RFC 0013](#) — one shared ≤ 60 -page rulebook, three degraded modes (no ambiguous in-between states).
- [RFC 0014](#) — 3 depot archetypes with a fleet-sizing formula, all at-grade.
- [RFC 0015](#) — GoA 4 (Unattended) by default; sensor-cowl nose replaces the driver cab.
- [RFC 0020](#) — EN 15227 Cat C-II three-zone crash envelope for the cableless body.
- [RFC 0021](#) — Under-seat Na-ion packs, station/depot opportunity charging, no catenary anywhere.
- [RFC 0022](#) — Single-SKU 2-axle Bo-Bo bogie shared across every consist family; per-family motorisation pattern.
- [RFC 0023](#) — Electric linear-actuator door operator, EN 14752 certified once at the project level.
- [RFC 0024](#) — PCM thermal mass + dedicated chiller for ≥ 50 °C ambient envelopes (Samawah-class).
- [RFC 0025](#) — Regional switch-shop bootstrap ($\sim \$10$ k per 1:9 turnout vs $\$120$ k vendor; $\sim 15\times$ CAPEX payback over 80 switches).
- [RFC 0026](#) — Two-tier connector architecture: CCS2 at depots + side-pin / pantograph-down at terminals.
- [RFC 0027](#) — Brownfield-pilot doctrine (asset assessment component recovery first-article OSR trainset). Anchored on Samawah's existing rail yard + workshop; applies to any country with a dormant rolling-stock stockpile.

How to get involved

- 1 Phase-1 brownfield assessment for Samawah ([RFC 0027](#)) — site visit + fleet census + workshop tooling audit + IRR / Iraqi Ministry of Transport disposition MoU. The single highest-value next step on the OSR programme: turns the Samawah pilot from a satellite-image observation into a deployment proposal. Diaspora technical-community introductions are the gating channel.
- 2 Architecture + RFC review from people with real rail signalling, power-electronics, or safety-case experience. File issues with specific disagreements; the RFCs reward red-pen.
- 3 Operator review of the [RFC 0013](#) rulebook. Practising dispatchers, station staff, and maintenance leads reading [docs/operations/](#) against their real-world practice.
- 4 Climate + grid data for specific target corridors. The [RFC 0002](#) energy-sizing model uses planning-grade defaults; real deployments need real PSH and grid- reliability data.
- 5 A new city in [lib/city-batches/world-sample.toml](#). Add a slug + served-catchment bbox + country + verified population (national-stats-office source) and run `scripts/regenerate-city.sh <slug>`. The catalogue currently covers 166 cities across 42 countries (full Iraq corridor, MENA, Sub-Saharan Africa + South + SE Asia + Latin America); gaps worth filling next are the cold-continental climate (Sarajevo / Tirana / Ulaanbaatar / Dushanbe), the Caribbean (Havana / Port-au-Prince / Santo Domingo), and the remaining metro-less Indian Ocean island states.

License

Per [ARCHITECTURE §9](#):

- Software: Apache 2.0
- Hardware designs: CERN-OHL-S v2
- Documentation: CC-BY-SA 4.0

Non-goals

To be clear about what this project is not:

- Not a standards body. Where good open standards exist (GTFS, NeTeX, IEEE 802.1 TSN, IEEE 2030.5), we adopt them.
- Not a safety certifier. The project produces artifacts suitable for independent assessment; certification is done by national authorities.
- Not a museum. We do not aim for plug-in compatibility with every legacy vendor protocol. Migration paths are scoped; permanent legacy support is not.

The railways a country builds outlive the governments that commission them. The technology underneath should belong to the country, not be leased to it.

Design Catalogue

Design Catalogue Index

designs/INDEX.md

OpenSourceRail Design Catalogue Index

Generated from `designs/*/*/design.toml`. Sorted by CAPEX per route-km, then high-demand coverage.

High-demand coverage is the share of high-demand raster cells (demand ≥ 0.5) within about 400 m of a planned line. It is a demand / catchment proxy, not a land-area percentage.

City	ISO	Family	Lines	Stations	km	Fleet	High-demand coverage	CAPEX	CAPEX/km	Charging microgrids
Huye	RW	tram-2car	3	28	46	61	23%	€416M	€9M	€9M
Tanta	EG	light-metro-3car	3	35	75	73	56%	€694M	€9M	€12M
Arusha	TZ	light-metro-3car	3	44	73	72	36%	€685M	€9M	€15M
Vientiane	LA	light-metro-3car	3	46	81	78	43%	€771M	€10M	€16M
Kisumu	KE	light-metro-3car	3	42	75	74	31%	€724M	€10M	€12M
Nablus	PS	light-metro-3car	3	34	65	65	84%	€625M	€10M	€11M
Rahim Yar Khan	PK	light-metro-3car	3	24	46	47	37%	€445M	€10M	€8M
Hama	SY	light-metro-3car	3	31	55	55	53%	€540M	€10M	€11M
Hebron	PS	light-metro-3car	3	40	69	68	61%	€675M	€10M	€14M
Asyut	EG	light-metro-3car	3	31	63	62	58%	€615M	€10M	€10M
Gulu	UG	light-metro-3car	3	36	64	63	50%	€627M	€10M	€13M
[Taif](designs/west-asia/Saudi Arabia/Taif/)	SA	light-metro-3car	3	42	73	73	41%	€726M	€10M	€14M
[Dammam](designs/west-asia/Saudi Arabia/Dammam/)	SA	metro-4car	4	95	210	168	28%	€2.07bn	€10M	€34M
[Galle](designs/south-asia/Sri Lanka/Galle/)	LK	light-metro-3car	3	37	64	64	56%	€637M	€10M	€13M
Dodoma	TZ	light-metro-3car	3	36	66	66	40%	€656M	€10M	€12M
Eldoret	KE	light-metro-3car	3	39	65	64	46%	€647M	€10M	€14M
Sylhet	BD	light-metro-3car	3	40	73	72	45%	€723M	€10M	€13M
Rajshahi	BD	light-metro-3car	3	30	50	52	30%	€496M	€10M	€11M
Gaza City	PS	light-metro-3car	1	15	24	24	30%	€239M	€10M	€5M
Lyon	FR	metro-4car	6	122	287	232	45%	€2.87bn	€10M	€42M
Zagazig	EG	light-metro-3car	3	30	56	57	59%	€566M	€10M	€11M
Rangpur	BD	light-metro-3car	3	36	59	59	47%	€600M	€10M	€13M
Aqaba	JO	tram-2car	3	23	34	48	57%	€347M	€10M	€9M
Hillah	IQ	light-metro-3car	3	42	69	68	43%	€699M	€10M	€15M
Port Harcourt	NG	metro-4car	4	95	199	160	31%	€2.02bn	€10M	€34M
Tripoli Lb	LB	light-metro-3car	3	30	53	55	50%	€539M	€10M	€10M
Latakia	SY	light-metro-3car	3	26	41	43	54%	€416M	€10M	€10M
Nakuru	KE	light-metro-3car	3	37	60	61	46%	€606M	€10M	€13M

City	ISO	Family	Lines	Stations	km	Fleet	High-demand coverage	CAPEX	CAPEX/km	Charging microgrids
[Buraidah](designs/west-asia/Saudi Arabia/Buraidah/)	SA	light-metro-3car	3	54	80	78	35%	€811M	€10M	€21M
Ibb	YE	light-metro-3car	3	40	72	71	58%	€732M	€10M	€14M
Kampala	UG	metro-4car	4	99	201	160	24%	€2.06bn	€10M	€35M
Ilorin	NG	light-metro-3car	3	39	60	60	31%	€618M	€10M	€14M
Pokhara	NP	light-metro-3car	3	44	82	80	52%	€845M	€10M	€16M
Irbid	JO	light-metro-3car	3	33	56	56	42%	€577M	€10M	€13M
Kandahar	AF	light-metro-3car	3	43	64	62	53%	€661M	€10M	€16M
Mwanza	TZ	metro-4car	5	70	162	136	45%	€1.67bn	€10M	€25M
Jinja	UG	tram-2car	3	29	46	61	41%	€472M	€10M	€10M
Mukalla	YE	light-metro-3car	3	33	61	61	72%	€632M	€10M	€12M
Erbil	IQ	metro-4car	6	97	199	166	42%	€2.08bn	€10M	€38M
Davao	PH	metro-4car	6	108	229	186	72%	€2.40bn	€10M	€41M
Samawah	IQ	light-metro-3car	3	33	55	55	56%	€575M	€10M	€12M
Comilla	BD	light-metro-3car	3	38	66	65	41%	€688M	€10M	€13M
[Jaffna](designs/south-asia/Sri Lanka/Jaffna/)	LK	light-metro-3car	3	34	52	52	42%	€544M	€10M	€12M
Mombasa	KE	metro-4car	6	95	201	167	52%	€2.11bn	€11M	€35M
Kigali	RW	metro-4car	4	86	171	139	34%	€1.79bn	€11M	€32M
Aleppo	SY	metro-4car	5	89	176	145	46%	€1.85bn	€11M	€34M
Mbarara	UG	light-metro-3car	3	36	59	59	53%	€620M	€11M	€13M
Gujranwala	PK	metro-4car	4	72	160	130	39%	€1.69bn	€11M	€26M
Mazar E Sharif	AF	light-metro-3car	3	37	63	61	59%	€668M	€11M	€13M
[San Salvador] (designs/latin-america/El Salvador/San-Salvador/)	SV	metro-4car	6	121	255	207	50%	€2.71bn	€11M	€47M
[Mecca](designs/west-asia/Saudi Arabia/Mecca/)	SA	metro-4car	6	114	251	205	46%	€2.67bn	€11M	€41M
Baqubah	IQ	light-metro-3car	3	37	60	60	50%	€639M	€11M	€14M
Ranchi	IN	metro-4car	6	96	216	179	50%	€2.30bn	€11M	€34M
[Abha](designs/west-asia/Saudi Arabia/Abha/)	SA	light-metro-3car	3	45	70	68	34%	€745M	€11M	€16M
Jalalabad Af	AF	light-metro-3car	3	32	51	53	69%	€543M	€11M	€12M
Visakhapatnam	IN	metro-4car	6	110	240	196	52%	€2.58bn	€11M	€41M
Meerut	IN	metro-4car	4	84	180	146	43%	€1.93bn	€11M	€30M
Maputo	MZ	metro-4car	6	83	186	156	71%	€2.00bn	€11M	€29M
Niamey	NE	metro-4car	4	80	146	120	38%	€1.58bn	€11M	€31M
Gazipur	BD	metro-4car	6	127	308	245	38%	€3.33bn	€11M	€44M
Tunis	TN	metro-4car	5	118	240	194	48%	€2.60bn	€11M	€44M
Onitsha	NG	metro-4car	4	76	184	148	30%	€2.00bn	€11M	€26M
Homs	SY	light-metro-3car	3	33	51	53	42%	€560M	€11M	€12M
Madurai	IN	metro-4car	6	104	222	183	59%	€2.43bn	€11M	€39M
Beira	MZ	light-metro-3car	3	39	54	54	37%	€588M	€11M	€15M
Vijayawada	IN	metro-4car	6	94	225	184	58%	€2.46bn	€11M	€34M
Conakry	GN	metro-4car	3	55	93	78	40%	€1.02bn	€11M	€22M
Rubavu	RW	tram-2car	3	28	45	61	48%	€500M	€11M	€10M

City	ISO	Family	Lines	Stations	km	Fleet	High-demand coverage	CAPEX	CAPEX/km	Charging microgrids
[Ouagadougou] (designs/west-africa/Burkina Faso/Ouagadougou/)	BF	metro-4car	6	138	264	213	37%	€2.92bn	€11M	€54M
Sulaymaniyah	IQ	metro-4car	4	59	127	106	49%	€1.41bn	€11M	€21M
[Medina](designs/west-asia/Saudi Arabia/Medina/)	SA	metro-4car	5	104	211	171	47%	€2.34bn	€11M	€38M
Cuenca	EC	light-metro-3car	3	48	79	77	57%	€871M	€11M	€18M
Damascus	SY	metro-4car	6	113	233	192	45%	€2.59bn	€11M	€43M
[Kandy](designs/south-asia/Sri Lanka/Kandy/)	LK	light-metro-3car	3	43	78	76	56%	€866M	€11M	€14M
Narayanganj	BD	light-metro-3car	3	50	82	80	27%	€913M	€11M	€16M
Mansoura Eg	EG	light-metro-3car	3	34	56	56	54%	€626M	€11M	€13M
Luxor	EG	light-metro-3car	3	34	54	54	73%	€602M	€11M	€13M
Duhok	IQ	light-metro-3car	3	32	53	54	53%	€596M	€11M	€13M
Agra	IN	metro-4car	5	98	191	156	39%	€2.15bn	€11M	€37M
Khartoum	SD	metro-6car	5	152	363	287	22%	€4.09bn	€11M	€50M
Multan	PK	metro-4car	4	64	115	98	42%	€1.31bn	€11M	€26M
Mbeya	TZ	light-metro-3car	3	39	55	55	44%	€627M	€11M	€15M
Mogadishu	SO	metro-4car	4	68	128	106	40%	€1.47bn	€11M	€26M
Raipur	IN	metro-4car	5	88	185	151	43%	€2.13bn	€11M	€31M
Mandalay	MM	metro-4car	6	88	187	156	60%	€2.15bn	€11M	€34M
Jos	NG	light-metro-3car	3	40	54	55	26%	€621M	€12M	€15M
Nairobi	KE	metro-6car	8	191	476	378	43%	€5.49bn	€12M	€63M
Peshawar	PK	metro-4car	4	87	197	158	33%	€2.28bn	€12M	€30M
Taiz	YE	light-metro-3car	3	33	49	51	55%	€570M	€12M	€12M
Hyderabad Pk	PK	metro-4car	6	84	182	152	61%	€2.12bn	€12M	€31M
Aden	YE	light-metro-3car	3	28	44	46	43%	€518M	€12M	€10M
Lucknow	IN	metro-6car	6	164	375	297	31%	€4.39bn	€12M	€56M
Vadodara	IN	metro-4car	5	89	164	137	48%	€1.92bn	€12M	€36M
Bahawalpur	PK	light-metro-3car	3	29	46	49	42%	€542M	€12M	€11M
Ramadi	IQ	light-metro-3car	3	35	47	49	39%	€551M	€12M	€14M
Sidon	LB	tram-2car	3	23	37	52	71%	€438M	€12M	€9M
Barisal	BD	light-metro-3car	3	30	59	60	50%	€692M	€12M	€11M
Kanpur	IN	metro-6car	7	150	339	273	44%	€4.00bn	€12M	€53M
[Tabuk](designs/west-asia/Saudi Arabia/Tabuk/)	SA	light-metro-3car	3	45	63	63	27%	€743M	€12M	€17M
Bandung	ID	metro-4car	6	126	257	208	41%	€3.05bn	€12M	€48M
Beirut	LB	metro-4car	6	83	159	134	70%	€1.90bn	€12M	€33M
Bhopal	IN	metro-4car	6	107	206	169	52%	€2.45bn	€12M	€43M
Dar Es Salaam	TZ	metro-6car	7	163	393	314	28%	€4.71bn	€12M	€54M
Sohag	EG	light-metro-3car	3	32	60	60	68%	€724M	€12M	€11M
Baghdad	IQ	metro-6car	9	218	509	408	45%	€6.14bn	€12M	€77M
Quetta	PK	metro-4car	5	78	140	120	54%	€1.69bn	€12M	€32M
Herat	AF	light-metro-3car	3	43	65	64	36%	€781M	€12M	€17M
Varanasi	IN	metro-4car	5	100	202	165	47%	€2.44bn	€12M	€37M
Fallujah	IQ	light-metro-3car	3	28	46	48	57%	€555M	€12M	€11M
Rajkot	IN	metro-4car	5	76	143	121	58%	€1.73bn	€12M	€30M

City	ISO	Family	Lines	Stations	km	Fleet	High-demand coverage	CAPEX	CAPEX/km	Charging microgrids
[Colombo](designs/south-asia/Sri Lanka/Colombo/)	LK	metro-6car	6	126	278	223	42%	€3.37bn	€12M	€44M
Deir Ez Zor	SY	light-metro-3car	3	32	57	57	62%	€688M	€12M	€12M
Khulna	BD	metro-4car	6	82	182	152	57%	€2.22bn	€12M	€30M
Qena	EG	light-metro-3car	3	26	45	47	75%	€556M	€12M	€10M
Patna	IN	metro-4car	5	84	185	152	50%	€2.28bn	€12M	€30M
[Jeddah](designs/west-asia/Saudi Arabia/Jeddah/)	SA	metro-6car	8	202	406	326	45%	€5.01bn	€12M	€78M
Luanda	AO	metro-6car	9	170	390	317	64%	€4.81bn	€12M	€59M
Lusaka	ZM	metro-6car	6	123	236	192	34%	€2.91bn	€12M	€46M
[Kinshasa](designs/east-africa/DR Congo/Kinshasa/)	CD	metro-6car	8	183	385	310	49%	€4.77bn	€12M	€69M
Kirkuk	IQ	metro-4car	6	92	170	142	60%	€2.10bn	€12M	€37M
Uyo	NG	light-metro-3car	3	24	31	36	25%	€386M	€12M	€10M
Phnom Penh	KH	metro-4car	6	107	228	186	48%	€2.83bn	€12M	€40M
Karbala	IQ	metro-4car	6	89	170	144	67%	€2.12bn	€12M	€36M
Mosul	IQ	metro-4car	5	60	145	122	38%	€1.80bn	€12M	€20M
Diwaniyah	IQ	light-metro-3car	3	36	54	55	43%	€674M	€13M	€14M
La Paz	BO	metro-4car	6	115	212	174	57%	€2.67bn	€13M	€46M
Indore	IN	metro-6car	7	146	324	260	48%	€4.08bn	€13M	€53M
Sayun	YE	tram-2car	2	15	22	31	58%	€273M	€13M	€6M
Dakar	SN	metro-6car	5	107	204	167	52%	€2.60bn	€13M	€41M
Coimbatore	IN	metro-6car	5	121	268	214	31%	€3.41bn	€13M	€43M
Amman	JO	metro-6car	8	172	354	286	51%	€4.53bn	€13M	€65M
Beni Suef	EG	light-metro-3car	3	23	35	38	55%	€447M	€13M	€9M
[Lubumbashi](designs/east-africa/DR Congo/Lubumbashi/)	CD	metro-4car	4	75	130	107	34%	€1.66bn	€13M	€32M
Kathmandu	NP	metro-4car	6	103	203	167	46%	€2.60bn	€13M	€42M
Karachi	PK	metro-6car	9	231	472	377	48%	€6.07bn	€13M	€86M
Antananarivo	MG	metro-6car	7	155	339	272	42%	€4.37bn	€13M	€56M
Chittagong	BD	metro-6car	8	161	374	302	64%	€4.85bn	€13M	€59M
Basra	IQ	metro-6car	7	119	289	236	54%	€3.75bn	€13M	€41M
Hodeidah	YE	light-metro-3car	3	27	36	40	61%	€474M	€13M	€11M
Yangon	MM	metro-6car	9	214	418	335	56%	€5.52bn	€13M	€85M
Surabaya	ID	metro-6car	7	143	294	240	40%	€3.90bn	€13M	€52M
Kano	NG	metro-6car	6	154	362	286	37%	€4.80bn	€13M	€50M
Maiduguri	NG	metro-4car	5	86	176	145	34%	€2.34bn	€13M	€32M
Benin City	NG	metro-4car	4	77	132	111	44%	€1.76bn	€13M	€32M
Yaounde	CM	metro-6car	8	136	267	219	43%	€3.55bn	€13M	€53M
Tartus	SY	tram-2car	3	23	35	49	73%	€471M	€13M	€9M
Zarqa	JO	light-metro-3car	3	57	83	80	48%	€1.13bn	€14M	€21M
Kabul	AF	metro-6car	7	137	261	215	53%	€3.55bn	€14M	€53M
Najaf	IQ	metro-4car	6	91	172	144	60%	€2.35bn	€14M	€37M
Douala	CM	metro-6car	5	129	228	184	37%	€3.11bn	€14M	€51M
Aba Ng	NG	light-metro-3car	3	26	34	38	40%	€466M	€14M	€11M
Sanaa	YE	metro-6car	9	126	261	218	78%	€3.60bn	€14M	€50M
Faisalabad	PK	metro-6car	5	93	166	137	52%	€2.33bn	€14M	€37M
Ibadan	NG	metro-6car	4	90	135	111	24%	€1.96bn	€14M	€37M
Bamako	ML	metro-4car	6	118	257	207	31%	€3.80bn	€15M	€41M
Kut	IQ	light-metro-3car	3	32	56	55	37%	€829M	€15M	€12M
Nasiriyah	IQ	light-metro-3car	3	33	56	56	40%	€847M	€15M	€13M

City	ISO	Family	Lines	Stations	km	Fleet	High-demand coverage	CAPEX	CAPEX/km	Charging microgrids
Larkana	PK	light-metro-3car	2	19	38	39	28%	€580M	€15M	€7M
Garoua	CM	light-metro-3car	3	29	44	46	38%	€676M	€16M	€10M
Amarah	IQ	light-metro-3car	3	32	45	46	43%	€707M	€16M	€13M
Jodhpur	IN	metro-4car	5	82	150	124	44%	€2.42bn	€16M	€32M
Mymensingh	BD	light-metro-3car	3	37	67	67	41%	€1.10bn	€16M	€12M
Sheikhupura	PK	light-metro-3car	2	17	19	22	32%	€345M	€18M	€8M
Sialkot	PK	light-metro-3car	3	37	59	59	43%	€1.08bn	€18M	€14M
Sukkur	PK	light-metro-3car	3	31	51	53	55%	€979M	€19M	€11M
Biratnagar	NP	tram-2car	3	23	34	48	60%	€878M	€26M	€9M

OpenSourceRail — Scope & Architecture

Status: Draft 0.1 (2026-04-20) Audience: Engineers, operators, policymakers, and funders evaluating the project. Purpose: Define what OpenSourceRail is, what it is not, and the architectural stance that distinguishes it from existing rail vendor offerings.

1. Mission

OpenSourceRail is a complete, permissively licensed technology stack for designing, building, and operating urban rail systems — light metro (LRT) and metro (heavy urban rail). Its primary beneficiaries are developing-world nations that today cannot economically procure, operate, or evolve urban rail networks without surrendering large fractions of the project capex/opex to a handful of international vendors.

Scope is urban transit only. Intercity passenger, long-distance, and freight/cargo rail are explicitly out of scope. The rationale is focus: urban transit is where the unit of need is highest in target markets, where the duty cycle is most favorable to the catenary-free battery-electric architecture proposed in §4 D7, and where a single reference design generalizes cleanly across many deployments. Freight and intercity have meaningfully different energy, signaling, and rolling-stock profiles that would dilute focus; they may be candidates for sibling projects or a future v2.

The project succeeds when a national railway authority, working with domestic engineering firms and commodity electronics suppliers, can build and run a modern rail network with imported content limited to raw steel, copper, and specialty items that genuinely cannot be manufactured locally.

Target deployment regions — sub-Saharan Africa, MENA, South and Southeast Asia, most of Latin America — share two properties that shape the architecture: they are capital-constrained, and they are high-insolation. Design choices throughout this document treat both as load-bearing assumptions rather than edge cases.

1.1 Non-goals

- Not a standards body. Where good open standards exist (e.g., GTFS, NeTeX, IEEE 802.1 TSN), adopt them. Where they don't, define minimal project-local interfaces — don't chase ISO ratification.
- Not a safety certifier. The project will produce artifacts suitable for independent safety assessment, but certification is performed by national authorities and independent assessors.
- Not a museum. We do not aim for plug-in compatibility with every legacy vendor protocol. Interoperability is scoped to migration paths, not permanent support.

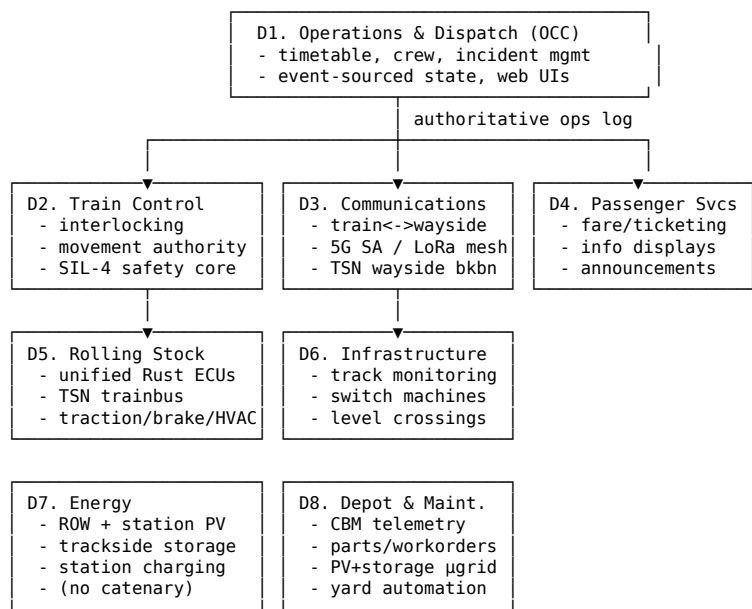
2. Design Principles

- 1 Commodity hardware, custom software. Industrial PLCs, proprietary trainbuses, and vendor-specific radios are replaced with commodity SBCs, deterministic Ethernet, and standard radio layers. Value is captured in the software and system integration, which can be produced anywhere.
- 2 Rust everywhere. Single-language stack from the signaling safety kernel up to the operations UI backend. See [project_tech_stack](#) for rationale.
- 3 Deprecate, don't reimplement. Every major design choice explicitly identifies the legacy practice it replaces and the concrete advantage of the replacement. "This is how it has always been done" is not a reason.
- 4 Simulation-first. No subsystem is considered designed until it runs end-to-end in a shared digital twin. Hardware pilots come after simulation sign-off.
- 5 Machine-checkable safety. Safety arguments are structured (GSN-style), version-controlled, and linked to formal proofs, test evidence, and code. Safety cases regenerate on every commit.

- 6
- Small, replaceable pieces. Each subsystem is deployable independently. A country can adopt the dispatching platform without committing to the rolling-stock ECUs, and vice versa.
- 7
- Defense in depth. Every wayside node is assumed to be reachable by an adversary. Signed firmware, mutually authenticated links, minimal attack surface, privilege separation at the OS level.

3. System-of-Systems Map

A rail system is a federation of largely independent subsystems that share a handful of critical interfaces. OpenSourceRail is organized around eight top-level domains:



Each domain has a dedicated section below. Interfaces between domains are deliberately narrow and enumerated in §5.

4. Subsystem Designs – Legacy vs. Proposed

For each domain, we state (a) the dominant current practice, (b) its problems for our target users, and (c) the OpenSourceRail approach.

D1. Operations & Dispatch

	Legacy	OpenSourceRail
Platform	Proprietary SCADA (Thales ARAMIS, Siemens Vicos, GE Transportation) on Windows	Event-sourced Rust services on Linux SBCs/servers, web-based UIs
State model	Periodic polling, opaque internal state, vendor-specific DBs	Append-only operations log (single source of truth), CQRS read models, standard Postgres
Observability	Proprietary HMI screens, vendor-locked historians	OpenTelemetry traces + Prometheus + Grafana; historian is just retention on the log
Integration	Per-vendor OPC/IEC 60870-5-104 gateways	Typed gRPC/NATS interfaces; 60870-5-104 shim only for migration from existing substations
Disaster recovery	Active/passive OCCs with manual cutover	Multi-region event log replication, any dispatcher terminal can fail over to a replica

Deprecated: Windows-based SCADA, proprietary HMIs, per-vendor historians. Why this is safe: Dispatch is non-safety-critical (safety is enforced at D2). The ops platform is free to use modern stacks.

D2. Train Control (Interlocking, ATP, ATO)

This is the heart of the safety argument and receives the most design attention.

	Legacy	OpenSourceRail
Block model	Fixed block (track circuits, axle counters) or vendor CBTC moving block	Software-defined moving block via distributed consensus across wayside nodes + self-reporting trains
Interlocking	Relay panels or proprietary PLC-based (Simis, Smartlock) at €50k–€500k/site	SIL-4 Rust interlocking on redundant RISC-V SBCs at <€5k/site; formally verified with Kani/Creusot
Position	Track circuits + balises (Eurobalise)	Sensor fusion: GNSS + IMU + wheel odometry + low-cost UWB/beacon fixes at switches and platforms

	Legacy	OpenSourceRail
Movement authority	Centralized zone controller issues MA to trains	Distributed Raft-style log holds authoritative track state; each train computes its own MA, cross-validated by two independent wayside nodes
Certification path	Per-vendor SIL-4 case, years to re-certify	Open formal models + continuously regenerated safety case

Deprecated: Track circuits as primary train detection (kept only as a secondary sensor in legacy retrofits), centralized zone controllers, relay interlockings. Why this is novel: "Rail as a distributed system." Existing CBTC vendors use centralized zone controllers because their software isn't trusted to run distributed consensus correctly; a formally verified Rust implementation of a restricted consensus protocol (fixed membership, no dynamic reconfiguration in the hot path) changes that calculus. Key risk: Regulators will need convincing. Mitigation: publish the formal model early, solicit review from independent safety assessors, start with lower-stakes applications (new-build tram networks, yard operations, segregated-ROW light metro) before committing to a full metro deployment.

D3. Communications

	Legacy	OpenSourceRail
Mainline radio	GSM-R (sunsetting ~2030), FRMCS over dedicated 5G	Public 5G SA with slicing where available; private 5G on n77/n78 or CBRS where not
Backup/telemetry	Dedicated leaky-feeder or TETRA	LoRa mesh for low-bandwidth safety telemetry (MA updates, position reports) — cheap, long-range, carrier-independent
Wayside backbone	Vendor-specific fiber rings, proprietary timing	Ethernet + TSN (IEEE 802.1) for determinism; PTP (1588) for time sync
Station/passenger	Per-vendor WiFi controllers	WiFi HaLow (802.11ah) for IoT/telemetry, standard WiFi 6 for passenger access
Security	Often weak (GSM-R has known weaknesses)	mTLS everywhere, Noise Protocol Framework on LoRa link, signed firmware on every radio

Deprecated: GSM-R, TETRA for rail, proprietary fiber timing protocols. Why LoRa for safety telemetry? Safety messages are small (<100 bytes) and infrequent (~1 Hz per train). LoRa gives 5–15 km line-of-sight range at sub-€10 BOM, so a mesh of wayside LoRa gateways provides coverage redundancy at negligible cost compared to a primary 5G link.

D4. Passenger Services

	Legacy	OpenSourceRail
Fare media	Smart cards (MIFARE, Calypso) with per-operator infrastructure	Account-based ticketing: mobile money (M-Pesa, UPI, etc.) + QR + optional NFC, validated via signed short-TTL tokens offline-capable
Ticket vending	Dedicated TVMs at €20k+/unit	SBC-based TVMs with commodity barcode scanner + contactless reader; <€1k/unit; most passengers use phones and don't need a TVM at all
Info displays	Proprietary LCD controllers with custom protocols	SBC + standard HDMI + minimal Rust display agent subscribing to the ops event stream
Announcements	Per-station PA controllers, manual voicing	Text-to-speech from the same event stream, multilingual by default; human override for incidents

Deprecated: Closed-loop smart card systems, dedicated TVMs as the primary sales channel. Why account-based first: Most target markets have leapfrogged cards via mobile money. Investing in a smart card ecosystem now is a decade-long dead end.

D5. Rolling Stock

	Legacy	OpenSourceRail
Trainbus	MVB + WTB (IEC 61375), proprietary silicon, decades-old	TSN Ethernet (IEEE 802.1 Qbv/Qcc) over standard copper/fiber — off-the-shelf switches, deterministic traffic
ECU diversity	20–50 ECUs from different vendors, different RTOSes, different toolchains	Single reference ECU hardware (RISC-V or ARM SoC + CAN-FD + TSN Ethernet PHY). Different "apps" (traction control, door control, HVAC, PIS) as Rust binaries on identical hardware
OS	Mix of VxWorks, QNX, proprietary	Hubris (or seL4) on safety-critical ECUs; Linux on infotainment
Energy source	Pantograph/shoe drawing from catenary/3rd rail; diesel genset on non-electrified lines	Onboard battery + traction inverter. Sodium-ion primary, LFP where energy density or cold-climate performance demands it. Optional pantograph for opportunity-charging pads only.
Traction drive	Proprietary VVVF inverters from Tier-1s	Open reference SiC inverter design with Rust control firmware; motor-agnostic (induction or PMSM)
Diagnostics	Per-vendor tools, per-train service laptops	Unified OpenTelemetry export to depot; any maintainer with a browser can inspect any train

Deprecated: MVB/WTB, per-vendor ECUs, VxWorks/QNX dependence, pantograph-as-primary-energy-source, diesel-genset rolling stock. Why a single reference ECU: Two trains of the same class today contain dozens of subtly different PCBs from a

supply chain deeply entangled with specific Tier-1s. One reference board (manufactured anywhere with basic SMT capability) for all non-traction-power functions collapses the supply chain dramatically. Why onboard battery instead of catenary collection: See D7 for the full argument. The train-side consequence is that every vehicle carries ~400–1500 kWh of battery (sized by service pattern) under the floor or on the roof; the pantograph is an optional opportunity-charging accessory, not the lifeline.

D6. Infrastructure Monitoring

	Legacy	OpenSourceRail
Track geometry	Dedicated measurement trains, quarterly runs	Continuous CBM: every service train carries a low-cost IMU + acoustic package; data fused server-side
Switch machines	Proprietary motor controllers with point-detection contacts	Commodity BLDC + SBC controller, dual redundant position sensors, reports to the train-control consensus log
Level crossings	Standalone controllers, minimal integration	Same SBC platform as interlocking; crossings are just wayside nodes in the consensus network
Bridge/tunnel	Periodic manual inspection	Distributed vibration/strain sensors over LoRa, anomaly detection in the ops platform

Deprecated: Dedicated measurement trains as the primary data source, isolated crossing controllers.

D7. Energy – Solar Generation, Trackside Storage, Battery Traction

This domain departs most sharply from legacy rail. OpenSourceRail systems are catenary-free by default, powered by solar PV deployed on railway-owned land with battery storage buffering the mismatch between generation and train demand.

	Legacy	OpenSourceRail
Traction energy delivery	Overhead catenary (25 kV AC / 1.5–3 kV DC) or third rail, fed from grid substations every 2–5 km	No catenary. Onboard battery traction; recharged at stations/depos from trackside storage (see D5).
Primary generation	Grid purchase at fixed tariff; dedicated traction substations	Solar PV on (a) platform/station canopies, (b) depot roofs, (c) right-of-way along track (vertical bifacial + between-rail panels)
Energy buffering	Typically none – instantaneous draw from grid	Trackside sodium-ion / LFP battery banks at stations, sized for several service-hours of autonomy
Grid interface	High-power traction substations, often dedicated feeders	Smaller grid-tie inverters that import when PV + storage are insufficient and export surplus where regulations allow
Chemistry preference	N/A (no storage)	Sodium-ion primary (no lithium dependency, cheaper, safer, better cold performance, easier supply-chain domestication); LFP where energy density drives the choice (on-train pack if weight/volume binds)
Charging interface	Continuous pantograph/shoe contact along entire route	Opportunity charging at station dwells: short overhead charging pads (hundreds of meters) or conductive/inductive pads at platforms; full recharge at terminal/depot
Regeneration	Often wasted in resistor banks	Fully recaptured to onboard battery and exported to trackside storage during platform charging
Control	Centralized SCADA	Per-site Rust controller + PV-storage-charge optimizer driven by the ops timetable (D1); autonomous on sub-second timescales

Deprecated: Overhead catenary, third rail, dedicated traction substations, resistor-brake dissipation, centralized thyristor rectifiers, diesel gensets on non-electrified lines.

7.1 Why catenary-free

- Capex. Catenary + support structures + paralleling posts cost €1–3M/km. Eliminating that is the single largest capex reduction available on a greenfield rail project.
- Opex and workforce. OCS maintenance requires specialized crews (live-line work, night possessions). Battery + station-charging infrastructure is maintainable by general electricians and renewable-energy technicians — skills that are more transferable and more widely available in target economies.
- Copper theft. Overhead copper is a persistent operational hazard in many developing markets. Trackside infrastructure in hardened cabinets is a much harder target.
- Phased deployment. Non-electrified lines become usable the day rolling stock arrives, with no "electrification project" preceding revenue service.
- Operational resilience. A single catenary fault halts every train on the section; a distributed storage+generation network degrades gracefully — trains can reach the next charging station on reserve capacity.

7.2 Why solar-along-ROW

- Railways own long, linear, already-cleared strips of land with predictable sun exposure. The real estate is effectively free.

- Generation is co-located with load. Losses and transmission infrastructure are minimal.
- Three complementary PV deployment patterns (pilots exist for each):
 - 1 Platform and station canopies — straightforward, aesthetically acceptable, dual-use as weather protection.
 - 2 Vertical bifacial PV along ROW boundaries — doubles as property fence; captures morning/evening sun at different angles than horizontal panels (smoother daily generation curve).
 - 3 Between-rail / on-sleeper PV (Sun-Ways-style) — highest area utilization but must tolerate train passage and ballast tamping; treat as Phase 4+ after basic patterns prove out.
- First-order math at 5 peak sun-hours, 3 m usable PV width per track-km, 200 W/m² module density: ≈3 MWh/day per track-km. A light metro trainset consumes on the order of 3–5 kWh/car-km. A modestly loaded 3-car service at 20 round-trips/day consumes ≈180–300 kWh/day/km. Net-positive self-powered operation is the default expected outcome for urban transit across the project's target regions — sub-Saharan Africa, MENA, South/Southeast Asia, and most of Latin America all sit comfortably above the 5 PSH line. Grid import covers monsoon/cloudy stretches and overnight draw on trackside storage; grid export monetizes midday surplus where regulations allow.

7.3 Chemistry: why sodium-ion primary

Factor	Sodium-ion	LFP	Outcome
Raw materials	Abundant (sodium, aluminum, iron)	Lithium supply is concentrated and politicized	Na-ion preferred for supply-chain sovereignty
Cell cost (2026 trajectory)	Lower and still falling fast	Mature, near floor	Na-ion preferred on cost
Cycle life	~3000–6000 cycles	~4000–6000 cycles	Comparable
Energy density (gravimetric)	~120–160 Wh/kg	~160–200 Wh/kg	Comparable for metro duty cycles; LFP preferred only where space-constrained vehicles drive pack volume
Cold-climate performance	Strong (usable below –20 °C)	Degrades below 0 °C	Na-ion preferred for high-altitude and continental climates
Safety (thermal runaway)	Very low risk	Low risk	Comparable; both far safer than NMC
Domestic manufacturing	Emerging; IP landscape more open	Mature but dominated by a few Asian players	Na-ion easier to localize

Default: Sodium-ion for trackside storage, station buffers, depot microgrid, and rolling stock. LFP: considered only where a space-constrained light vehicle makes Na-ion pack volume impractical. No NMC in the reference design (safety, ethics, cost trajectory all argue against).

7.4 Scope fit

This architecture is tuned for urban transit duty cycles: short runs, frequent stops, predictable daily service patterns, and fleet sizes in the low tens of trainsets. Within that envelope the catenary-free + solar model is conservative — not a stretch. Heavy freight and long-distance intercity service have meaningfully different energy profiles and are explicitly out of scope (§1); this section does not attempt to cover them.

7.5 Interfaces

New or modified cross-domain interfaces:

- I9 (Charging Dispatch): D1 ops informs D7 which trains are due at which charging sites and when, so trackside storage can pre-charge buffers.
- I10 (Energy Telemetry): D5 trains report state-of-charge and predicted arrival SoC to D1/D7; D1 can re-plan around a low-SoC unit the way it would re-plan around a delayed one.
- I11 (Grid Tie): D7 sites speak standard grid-interface protocols (IEEE 2030.5 / Sunspec Modbus) to the local utility.

These are additions to §5 and will be formalized in `crates/osr-core/proto/energy.proto`.

D8. Depot & Maintenance

	Legacy	OpenSourceRail
Work orders	SAP PM or proprietary EAM	Rust service on the same event-sourced platform; integrated with train telemetry
CBM	Optional, bolt-on, per-vendor	First-class; every ECU streams condition data continuously
Yard automation	Manual	Same train-control stack, restricted to yard speeds; useful proving ground
Energy	Grid-fed charging (where electric at all), diesel refueling	Depot PV+storage microgrid: rooftop solar, sodium-ion buffer, overnight slow-charge of full fleet at low c-rate (better for pack life than opportunistic fast charging). Grid import only for shortfalls.

5. Cross-Domain Interfaces

OpenSourceRail defines a minimal set of typed interfaces between domains. Everything else is internal to the owning domain.

#	Interface	Producers	Consumers	Transport	Format
I1	Track State Log	D2 interlockings	D1 ops, D5 trains	Raft-replicated, gRPC streaming	Protobuf schema <code>track_state.proto</code>
I2	Movement Authority	D2	D5 trains	mTLS gRPC over D3 radio	<code>movement_authority.proto</code>
I3	Train Position Report	D5 trains	D2	mTLS gRPC or LoRa (primary/backup)	<code>position_report.proto</code>
I4	Ops Event Stream	D1	D4, D6, D8	NATS JetStream or Kafka-compatible	CloudEvents JSON
I5	Telemetry	all ECUs/wayside	D1, D8	OpenTelemetry OTLP	OTLP standard
I6	Passenger-facing schedule	D1	public	GTFS-RT + GTFS static	GTFS standard
I7	Fare Token	D4 issuing service	D4 validators	Offline-verifiable JWT variant	RFC 8392 CWT
I8	Traction Setpoints	D7 controller	charging / inverter hardware	Modbus/TCP (legacy) or OPC UA (migration target)	standard
I9	Charging Dispatch	D1 ops	D7 station charging sites	NATS JetStream	<code>charging_dispatch.proto</code>
I10	Energy Telemetry	D5 trains, D7 sites	D1 ops, D8 depot	OpenTelemetry OTLP	OTLP standard
I11	Grid Tie	D7 sites	local utility	IEEE 2030.5 / Sunspec Modbus	standard

Only I1–I3 are safety-critical. Everything else can fail without loss of safe operation (degraded service only). I9/I10 failure degrades energy planning but does not affect train control — trains carry enough reserve to reach the next charging site without active dispatch.

6. Platform Stack

6.1 Software tiers

Tier	Examples	OS	Rust flavor	Constraints
T1 – Safety kernel	Interlocking, ATP, door safety	Hubris or seL4	<code>no_std</code> , RTIC	SIL-4 target; formally verified; < 50k LoC per binary
T2 – Safety-adjacent	Train ECU apps, substation control	Hubris or PREEMPT_RT Linux	<code>no_std</code> or minimal std	SIL-2 target; strict coding standard; full test coverage
T3 – Supervisory	Dispatching, ops services, telemetry collectors	Linux (Debian or Yocto)	<code>tokio</code> std	High availability; not safety-critical
T4 – UX/Back-office	Dispatcher web UI, passenger apps, EAM	Linux / browser	<code>axum</code> /Leptos Rust, TypeScript at edges	Standard web security practices

6.2 Reference hardware

- Wayside SBC (W-SBC): RISC-V (e.g., MilkV Jupiter-class) or ARM64 (e.g., Raspberry Pi CM5 carrier), dual Ethernet, CAN-FD, hardware RoT (TPM 2.0 or OpenTitan). Deployed in ruggedized enclosures; runs T1/T2.
- Train ECU (T-ECU): Same SoC family as W-SBC, EN 50155 environmental ratings, TSN Ethernet PHY, CAN-FD, isolated I/O modules.
- Ops server (O-SRV): Commodity x86 or ARM64 server, Debian, standard datacenter gear. T3/T4.

Reference hardware designs (schematics, BOMs, Gerbers) are part of the repository under [hardware/](#). The per-class choices — SoC, peripherals, baseboard, manufacturability envelope, reliability targets — are fixed in [RFC 0007](#). All designs must be manufacturable by a 4-layer PCB fab with 0.15 mm trace/space and 0.3 mm vias — routine at tier-2 fabs across the target deployment footprint.

6.3 Repository layout

The canonical crate map is in [RFC 0005](#) (35 crates across 8 domains). The subset that exists today:

```

OpenSourceRail/
├── docs/
├── crates/
│   ├── osr-core/           # Shared types, interfaces, protobuf schema
│   ├── osr-interlocking/   # SIL-4 MA computer + rail state machine (D2)
│   ├── osr-consensus/      # SIL-4 Raft (SMRaft refinement) (D2)
│   ├── osr-odometry/       # SIL-4 onboard position fusion (D5)
│   ├── osr-atp/            # SIL-4 onboard Automatic Train Protection (D5)
│   ├── osr-brake/          # SIL-4 EP brake controller + WSP + park (D5)
│   ├── osr-vigilance/      # SIL-4 driver alerter / dead-man (D5)
│   └── osr-wayside-points/ # SIL-4 power-switch controller (D6)
└── # This file, RFCs, safety cases

```

```

├── osr-sim/           # Digital twin / simulator + shadow onboard stack
├── formal/tla/        # TLA+ specs: SMRaft, TLC harness
├── scenarios/         # TOML scenario files (Samawah + templates)
├── hardware/          # Reference designs per RFC 0007 (scaffolded)
├── tools/reference-ma/ # Python reference interpreter (RFC 0004 M4)
└── pilots/            # (planned)

```

Not yet in-tree, enumerated in [RFC 0005 §4](#): `osr-ato`, `osr-tcms`, `osr-dmi`, `osr-event-recorder`, `osr-tcn`, `osr-t2g`, `osr-traction`, `osr-bms`, `osr-door-control`, `osr-hvac`, `osr-lighting`, `osr-pis-onboard`, `osr-aux-power`, `osr-regen`, `osr-fire-safety`, `osr-derailment`, `osr-hot-axle`, `osr-cbm-onboard`, `osr-ptp`, `osr-balise`, `osr-level-crossing`, `osr-hot-axle-wayside`, `osr-psd`, `osr-afc`, `osr-tvm`, `osr-pis-station`, `osr-station-scada`, `osr-occ`, `osr-historian`, `osr-analytics`, `osr-cbm-backend`, `osr-afc-backoffice`, `osr-energy-site`, `osr-crypto`, `osr-safety-case`, `osr-proto`.

7. Safety & Certification Strategy

OpenSourceRail targets the EN 50126/50128/50129 and IEC 61508 framework because that is what national safety authorities will recognize. The approach is:

- 1 Formal models first. Signaling logic is expressed in a formal model (Kani, Creusot, or TLA+ where appropriate) before implementation. The Rust implementation is proven to refine the model.
- 2 Small safety kernel. T1 binaries are aggressively minimized. Everything that can be pushed out of the safety kernel is.
- 3 Diversity by construction. Two independent Rust implementations of each T1 function, compiled with different toolchain configurations, cross-check each other on redundant hardware. This is cheaper than the traditional "different language + different team" because the second implementation is constrained by the same formal model.
- 4 Machine-checkable safety case. Safety arguments are written in GSN (Goal Structuring Notation) serialized as TOML, with claim evidence links resolving to code commits, proof artifacts, and test results. CI regenerates the case on every merge; a safety case that no longer closes blocks the release.
- 5 Independent assessment. The project produces artifacts. Certification is performed per-deployment by the national authority's chosen ISA. Reference safety cases from pilot deployments are published to compound assessor familiarity across countries.

8. Security Posture

- Zero trust on the wayside network. Every node authenticates every peer (mTLS, mutually verified certificates rooted in a project-operated PKI per deployment).
- Signed firmware, measured boot. TPM-backed attestation; update rollback via A/B partitions.
- Privilege separation at OS level. Hubris's "task" model gives hardware-enforced isolation; on Linux, systemd + seccomp + namespaces.
- No default credentials, ever. Commissioning generates unique per-device keys.
- Auditable by construction. Every safety-relevant event is in the append-only log; tampering is detectable.

9. Roadmap

Phase 0 — Foundation (0–6 months)

- This document, ratified.
- Governance, licensing (proposed: Apache 2.0 for software, CERN-OHL-S for hardware, CC-BY-SA for docs), contribution process.
- `osr-core` protobuf schemas for I1–I8.
- Minimum viable digital twin (`osr-sim`) capable of running a toy network.
- CI: Rust toolchain, cargo test, Kani harness, GSN safety-case compiler skeleton.

Phase 1 — Dispatch & Observability (6–12 months)

- `osr-ops` dispatcher MVP: timetable, incident log, dispatcher web UI.

- `osr` - core telemetry pipeline end-to-end.
- First non-safety-critical pilot: consuming GTFS-RT from an existing operator and publishing a parallel dispatcher view. Lets us shake out the platform with zero safety exposure.

Phase 2 — Signaling Core (12–24 months)

- `osr` - consensus distributed log with Kani-verified safety properties.
- `osr` - interlocking formal model + implementation for a simple 5-switch yard.
- `osr` - movement MA calculator with formal proof of non-overlap.
- Integration in simulator; shadow-mode trial against real operator data.

Phase 3 — Hardware + Rail Reference Designs (18–30 months, overlapping)

- W-SBC v1 and T-ECU v1 schematics, fab, bring-up — per [RFC 0007](#) (Raspberry Pi + Radxa palette).
- Hubris port to RP2350.
- First hardware-in-the-loop demo: simulator driving real W-SBCs.
- Energy subsystem: reference trackside storage site design (PV array + Na-ion bank + grid-tie inverter + Rust site controller); reference station charging pad design; reference onboard traction battery + inverter design.
- Rolling-stock / track / station reference designs — [RFC 0008](#) (5 trainset families), [RFC 0009](#) (4 geometry presets), [RFC 0010](#) (6 station archetypes), each with an enforced compatibility matrix in the auto-gen emitter.

Phase 4 — Pilot Deployment (24–36 months)

- Reference concept: Samawah, Iraq — two-line light metro (14 km radial + 16 km ring) connecting mainline rail station, city centre, German Hospital, and Al-Muthanna University. See [RFC 0003](#).
- Samawah is the scenario `osr-sim` targets; whether it becomes the actual first revenue-service deployment depends on decisions outside this project.
- Full safety case; independent assessment; revenue service.

Phase 5 — Metro at Scale (36+ months)

- Learnings from Phase 4 drive the full metro-grade safety case and performance envelope (higher headway, longer trains, underground alignments).
- Migration tooling for operators with existing legacy metro systems.

10. Open Questions

These are questions we do not yet have good answers to. Each will spawn a focused RFC.

- 1 Which formal methods tool? Kani vs. Creusot vs. TLA+ for different layers. Probably all three, but which for what.
- 2 Consensus details. Raft is well understood but not obviously safety-certifiable. Is there a restricted consensus protocol we can fully formalize?
- 3 Hubris vs. seL4 for T1. Hubris is Rust-native and simpler; seL4 has a stronger formal pedigree but a C codebase.
- 4 Reference SoC. RISC-V is philosophically aligned but the safety-certifiable ecosystem is thin; ARM64 has better ecosystem but raises IP concerns for some target nations.
- 5 Pilot partners. Which country/operator takes the first live deployment risk?
- 6 Funding model. This is a multi-year effort; philanthropy, multilateral development banks, or a consortium of target operators?
- 7 Battery second-life and recycling. Traction batteries degrade to ~80% capacity over service life. Cascade them to trackside storage (less demanding duty cycle)? What recycling pathway is available in target markets?
- 8 PV-track geometry. For between-rail and vertical-ROW PV, what soiling/shading/clearance constraints matter? How do we validate in simulation before committing to a specific module geometry?

11. Reading Order for New Contributors

- 1 This document (start here).
- 2 `docs/GLOSSARY.md` – rail-domain terms for software engineers and vice versa. (to write)
- 3 [docs/rfcs/0001-track-state-consensus.md](#) – the distributed track-state log design.
- 4 [docs/rfcs/0002-energy-sizing.md](#) – quantitative sizing for the catenary-free, solar-first energy architecture.
- 5 [docs/rfcs/0003-samawah-reference-deployment.md](#) – concrete reference deployment: Samawah, Iraq.
- 6 [docs/rfcs/0004-osr-interlocking-plan.md](#) – implementation plan for the Rust SIL-4 MA computer.
- 7 [docs/rfcs/0005-sbc-software-architecture.md](#) – canonical SBC software architecture + crate map.
- 8 [crates/osr-core/proto/track_state.proto](#) – the interface definitions.
- 9 [formal/tla/SMRaft.tla](#) – TLA+ spec of the consensus protocol.

This document is a living architecture brief. Material changes go through RFCs in `docs/rfcs/`; editorial changes can be PRs directly.

docs/certification/README.md

docs/certification/README.md

OSR type-certification pre-submission pack

Target standard: EN 62267 — Railway applications — Automated urban guided transport (AUGT) — Safety requirements, GoA 4 (Unattended Train Operation).

Complementary standards referenced:

- EN 50126 — RAMS process for rail applications
- EN 50128 — Software for railway control and protection systems
- EN 50129 — Safety-related electronic systems for signalling
- EN 50657 — Software on board rolling stock (SIL-4 on T-ECU/S)
- EN 15227 — Crashworthiness requirements for rail vehicle bodies (Cat C-II, referenced from RFC 0008 §3)
- EN 45545 — Fire protection on railway vehicles (referenced from RFC 0008 §3)
- IEC 62443-4-2 — Cybersecurity, component-level (referenced from RFC 0017)
- CENELEC EN 50701 — Rail cybersecurity (referenced from RFC 0017)

Scope of this pack: establish that an OSR deployment — taking the Samawah reference ([RFC 0003](#)) as the first-article instantiation — meets the GoA 4 requirements of EN 62267 with sufficient evidence that a national safety authority can review and (with deployment-specific additions) grant type approval.

Audience

- A national safety authority reviewing a type-approval submission for an OSR deployment.
- The deployment partner's safety assessor, preparing the deployment-specific dossier that wraps this pack.
- OSR project contributors, as the definition of "done" for the GoA 4 safety claim.

Not in scope (deployment-partner responsibility)

- Deployment-specific RAMS targets (headway, availability, on-time performance — site dependent).
- Integration with the national regulator's own submission format (some authorities require specific document templates; translation from this pack is a per-deployment effort).
- Field-test evidence from the actual pilot (generated once the pilot line is commissioned).
- Operating-entity organisational competence (SIL management, safety-culture audit) — the deployment partner holds that evidence.

Reading order

- 1 [system-description.md](#) — what the system is, its boundaries, operational envelope, and interfaces to the outside world.
- 2 [safety-requirements.md](#) — the safety-requirements specification (SRS) derived from the EN 62267 functional scope.
- 3 [hazard-log.md](#) — hazards, their severity, frequency assumption, and mitigating controls.
- 4 [evidence-register.md](#) — complete inventory of verification evidence produced by the OSR project, indexed to the requirements + hazards it covers.
- 5 [evidence-status.md](#) — concise status matrix separating implemented, simulated, specified, and deployment-partner evidence.
- 6 [compliance-matrix.md](#) — EN 62267 clause-by-clause traceability to OSR implementation.

How this pack relates to the rest of the repository

Most of this pack is synthesis, not new engineering. Every claim here is anchored to artefacts that already exist:

- SIL-4 safety properties Kani harnesses in `crates/osr-*/src/kani_proofs.rs`
- SIL-4 safety properties proptests in `crates/osr-*/tests/proptest_*.rs`
- GSN argumentation TOML files in `docs/safety-case/gsn/`, closed against evidence by the `osr-safety-case` compiler CI gate
- Operational controls per-role rules under `docs/operations/`
- RFCs (0001–0017) carry the design rationale
- Rust workspace + Python sidecars carry the implementation and generated-design verification gates

If any claim here cannot be traced to one of those artefacts, it's a gap; file it as an issue against the evidence register.

Document status

Phase	Status
v1 — initial structure + SRS + hazard log + evidence register + compliance matrix	2026-04-23
v1.1 — per-clause compliance prose	next-iteration deliverable
v1.2 — reviewed by an independent safety assessor	deployment-partner scope
v2 — first-article field evidence	Samawah commissioning per RFC 0027

Open questions for the reviewer

- Residual-risk acceptance criterion. EN 62267 does not fix a single numeric target; national authorities set thresholds. This pack presents the evidence; the reviewer sets the tolerance.
- Historical-data equivalence argument. Our safety case uses deterministic formal methods (Kani) and parameterised proptests rather than the several-million-revenue-hours approach of legacy fleets. This is the standard SIL-4 argument for software- intensive systems, but the reviewer may request additional sim-based evidence; see §5 of [evidence-register.md](#).
- Wayside-intrusion gating on un-instrumented sections. The v2 `section_available_to` gate permits sections with no `SectionIntrusion` verdict on record (RFC 0016 §5.2). For type approval of a fully-instrumented deployment, the reviewer may require the gate be tightened to fail-restrictive.

docs/certification/compliance-matrix.md

docs/certification/compliance-matrix.md

EN 62267 compliance matrix

Clause-by-clause traceability from EN 62267 (Railway applications — Automated urban guided transport — Safety requirements) to OSR implementation. Citations are to the 2009 edition.

Format per clause:

- Clause — reference.
- EN 62267 requirement summary — single-sentence paraphrase.
- OSR implementation — which crate(s) / RFC(s) / rulebook section fulfils it.
- Evidence — specific artefact to cite in the submission.

§5 — Safety principles

§5.1 — Inherent and functional safety

Requirement: system shall implement both inherent safety (design choices that make hazards impossible by construction) and functional safety (safety functions that detect + mitigate hazards that cannot be designed out).

OSR implementation:

- Inherent safety: no tunnels (RFC 0011), prefab-only civil (RFC 0011), single consist family per line (RFC 0008), fixed-geometry preset per line (RFC 0009). Each removes an entire hazard class by eliminating its precondition.
- Functional safety: SIL-4 chain per [safety-requirements.md SR-01..SR-24](#).

Evidence: RFCs 0008–0011; [safety-requirements.md](#).

§5.2 — Responsibility assignment

Requirement: a responsibility matrix shall document which entity owns each safety-relevant function.

OSR implementation: RFC 0013 §4 (one shared rulebook across 5 role families); `osr-safety-case` GSN tree links every claim to an owning evaluator.

Evidence: RFC 0013; GSN tree under `docs/safety-case/gsn/`.

§6 — Functional requirements (SIL-4 core)

§6.1 — Ensuring safe movement

Requirement: automated system shall ensure the train moves only when it is safe to do so.

OSR implementation: `osr-interlocking::section_available_to` with four gates (occupancy, route, override, intrusion); `osr-atp::atp_evaluate` overlays the envelope check.

Evidence: SR-01, SR-02, SR-03, SR-07; Kani harness modules on `osr-interlocking`, `osr-atp`.

§6.2 — Preventing collision

Requirement: system shall prevent collisions between trains, with infrastructure, and with obstacles.

OSR implementation: train-train via SR-01; track-worker via SR-01 + `MaintenanceOverride`; obstacle via SR-04 (onboard) + SR-07 (wayside) — two-layer defence per RFC 0015 + RFC 0016.

Evidence: H-CO-01..04 hazard log; RFC 0015 §6 safety case.

§6.3 — Preventing derailment

Requirement: system shall prevent derailment from overspeed, mis-set switch, or mechanical cause.

OSR implementation: overspeed via ATP (SR-03); switch via `osr-wayside-points 2oo2` sensor fusion; mechanical via SR-11 onboard derailment monitor; track maintenance cadence RFC 0013 M3 + M4.

Evidence: SR-03, SR-11; hazard H-DE-01, H-DE-02.

§6.4 — Preventing falls

Requirement: system shall prevent falls from platforms.

OSR implementation: SR-24 PSDs at every boarding platform; SR-10 platform-alignment gate; station-staff T2 supervision at non-PSD depot-terminals.

Evidence: RFC 0010 §6; RFC 0015 §5.4.

§6.5 — Protecting people on track

Requirement: system shall protect people on track — workers and intruders.

OSR implementation: workers via `MaintenanceOverride` + RFC 0013 S5 + M2 protection signals; intruders via SR-07 wayside intrusion gate + physical fence (civil scope).

Evidence: SR-07; H-CO-04, H-SA-01.

§6.6 — Closing doors

Requirement: system shall ensure doors close safely.

OSR implementation: SR-08, SR-09 on `osr-door-control`; RFC 0013 T2.5 platform-clear confirmation at non-PSD stations.

Evidence: SR-08, SR-09; hazard H-PB-01, H-PB-03.

§6.7 — Starting the train

Requirement: the system shall start the train only when safe (all doors closed, MA valid, all monitors nominal).

OSR implementation: `osr-atp`'s start-inhibit composes: `doors_closed_all && ma_valid && !any_monitor_trip`; RFC 0013 D4.6 rulebook documents the driver-visible behaviour (applies to GoA 2; GoA 4 automates via ATO).

Evidence: SR-15, SR-19; `osr-ato` start-gate unit tests.

§6.8 — Stopping the train

Requirement: the system shall stop the train safely on any EB source.

OSR implementation: `osr-brake emergency-source union` (ATP + vigilance + fire + derailment + driver + obstacle); SR-15 — EB dominates.

Evidence: `osr-brake` Kani B2 + B3 harness; `protest b2_emergency_union`.

§6.9 — Preventing hostile environments

Requirement: system shall handle weather, dust, heat per deployment climate.

OSR implementation: RFC 0013 S4 weather rulebook; `climate-adapters.toml` per-deployment tuning; `osr-energy-site` handles PV dust derating.

Evidence: RFC 0013 §6 weather rules; `climate-adapter` file.

§6.10 — Detecting smoke + fire

Requirement: onboard fire detection + suppression.

OSR implementation: `osr-fire-safety` (SIL-4); RFC 0008 bays each carry BaySensors; suppression by Novec 1230 or equiv.

Evidence: SR-12; H-FI-01, H-FI-02.

§6.11 — Operating in degraded mode

Requirement: system shall safely continue at reduced performance under named degraded modes.

OSR implementation: RFC 0013 §5 M1 (manual on MA) / M2 (restricted service) / M3 (evacuation) as the only three degraded modes.

Evidence: RFC 0013 §5; degraded-mode driver rules D6 (GoA 2 legacy); GoA 4 equivalent in dispatcher S7 + S5.

§6.12 – Supervision and rescue of passengers

Requirement: system shall support passenger supervision + rescue without a driver.

OSR implementation: SR-14 passenger emergency intercom; SR-18 remote-assist channel; RFC 0015 §5.3 + §5.5 cabin CCTV live to OCC.

Evidence: SR-14, SR-18; RFC 0015 §5.

§7 – System performance

§7.1 – Tolerable hazard rate (THR)

Requirement: the safety integrity of each safety function shall correspond to a tolerable hazard rate.

OSR implementation: SIL-4 functions target $THR \leq 10^{-9}$ /h, consistent with EN 50129 (SIL-4 allocated to the safety functions per the hazard log).

Evidence: [hazard-log.md](#) severity × frequency table.

§7.2 – Systematic failures

Requirement: systematic failures (specification, design, implementation bugs) shall be controlled via the EN 50128 process.

OSR implementation: Rust `#! [ forbid ( unsafe_code ) ]`; integer-only safety path; Kani bounded-model proofs; proptest randomised verification; differential twin (Rust vs Python); RFC-driven change control.

Evidence: Evidence register §1–§5; CI gates on `cargo test + starter_case_closes`.

§7.3 – Random hardware failures

Requirement: random hardware failures shall be detected and handled fail-restrictively.

OSR implementation: 2oo2 at T-ECU/S + T-OBS (SR-20); watchdog + supervisor (SR-21); ATECC608B trust anchor; per-rail power supervision; redundant 24 V input.

Evidence: `hardware/t-ecu-s/schematics/v2-spec/safety-nets.md`;
`hardware/t-obs/schematics/v2-spec/safety-nets.md`.

§8 – Safety management

§8.1 – Safety lifecycle

Requirement: EN 50126 lifecycle from concept to decommissioning.

OSR implementation: RFC-per-major-decision; GSN claim tree under `docs/safety-case/`; operations rulebook under `docs/operations/`; `osr-safety-case` CI gate. Deployment partner handles site-specific lifecycle stages (commissioning, operations, decommissioning).

Evidence: RFC 0001–0017; this certification pack as the "safety case" deliverable.

§8.2 – Independent safety assessment

Requirement: an independent assessor reviews the safety case.

OSR implementation: this pack is prepared for external assessment. The deployment partner engages the assessor (RFC 0003 §5).

Evidence: N/A (upstream prerequisite for the assessor, not produced by this pack).

§8.3 – Competency of staff

Requirement: operating + maintenance staff competent for their role.

OSR implementation: RFC 0013 C1 role qualification (§C1.5); RFC 0013 §8 training + recurrent qualification program.

Evidence: RFC 0013 §8; deployment-specific training record.

§9 – Cybersecurity (complementary)

Although EN 62267 itself does not prescribe cybersecurity requirements (it pre-dates the CENELEC EN 50701 split), a modern GoA 4 submission is expected to carry IEC 62443-4-2 + EN 50701 evidence as a complementary cybersecurity dossier.

§9.1 — Message authenticity

OSR implementation: SR-22 `osr-secbus` (RFC 0017 v1 library; v2 wiring open).

Evidence: GSN G25–G27; `osr-secbus` Kani + proptest.

§9.2 — Component-level protection

OSR implementation: ATECC608B trust anchors on every SIL-4 host class; zero-on-drop key wrappers (`osr-crypto`); per-deployment key registry (`osr-secbus`).

Evidence: `hardware/t-ecu-s/schematics/v2-spec/`; `osr-crypto` + `osr-secbus` source.

Summary

- §5.1–§9.2: 19 clauses mapped.
- 16 clauses have direct implementation + evidence in tree.
- 3 clauses (§7.1 residual risk, §8.2 independent assessment, §8.3 staff competency) depend on deployment-partner input — deliberately scoped that way per [README.md](#).
- 1 clause (§9.1 message authenticity) has library in tree but v2 wire integration open.

No EN 62267 clause has an open gap at the stack level. Gaps are all on the deployment-partner side of the boundary.

docs/certification/evidence-register.md

docs/certification/evidence-register.md

Evidence register

Complete inventory of verification evidence produced by the OpenSourceRail project, indexed to the safety requirements from [safety-requirements.md](#) and the hazards from [hazard-log.md](#).

1. Formal-methods evidence (Kani bounded-model checking)

Each SIL-4 evaluator carries Kani harnesses under `crates/<name>/src/kani_proofs.rs`, gated on `cfg(kani)` so `cargo test` skips them and `cargo kani` runs them. The harnesses are bounded formal proofs: they explore every input combination within a stated unwind, not just test-case samples.

Crate	Harness module	Properties	SR coverage
osr-atp	kani_proofs.rs	A1–A7	SR-01, SR-02, SR-03
osr-brake	kani_proofs.rs	B1–B5	SR-15
osr-vigilance	kani_proofs.rs	V1–V6	(GoA 2 legacy only)
osr-odometry	kani_proofs.rs	O1–O5	SR-10
osr-wayside-points	kani_proofs.rs	W1–W6	SR-01, H-CO-02
osr-interlocking	kani_proofs.rs	P1–P5	SR-01, SR-02, H-CO-01, H-CO-04
osr-obstacle-detect	kani_proofs.rs	O1–O5	SR-04, SR-05, SR-06
osr-intrusion-detect	kani_proofs.rs	I1–I5	SR-07, H-CO-03, H-SA-01
osr-secbus	kani_proofs.rs	S1, S3	SR-22, H-DI-01

Total Kani harnesses: 40+ named-property proofs across 9 SIL-4 and SIL-2 evaluators. Rerun with `cargo kani -p package <name>` on a host with Kani installed (Kani is not part of the `cargo test` default path).

2. Randomised property testing (proptest)

Every Kani property has a matching proptest that exercises the same invariant across a random input space (typically 256–1000 cases per property per run). Proptest runs on `cargo test` by default.

Crate	Test file	Props	Typical cases per run
osr-atp	tests/proptest_atp.rs	A1–A7 + bonus	256
osr-brake	tests/proptest_brake.rs	B1–B5	256
osr-vigilance	tests/proptest_vigilance.rs	V1–V6	256
osr-odometry	tests/proptest_odom.rs	O1–O5	256
osr-wayside-points	tests/proptest_wayside.rs	W1–W6	256
osr-interlocking	tests/proptest_ma.rs	P1–P5	1000
osr-interlocking	tests/proptest_determinism.rs	derive-state determinism	1000
osr-interlocking	tests/differential.rs	Rust Python twin agreement	256
osr-consensus	(inline + suite)	5 TLA-refined Raft properties	2000
osr-obstacle-detect	tests/proptest_obstacle.rs	O1–O5 + severity + classifier downgrade	256
osr-intrusion-detect	tests/proptest_intrusion.rs	I1–I5	256
osr-secbus	tests/proptest_secbus.rs	S1–S3 + roundtrip	256
osr-crypto	(inline)	C1–C4 + ed25519 determinism	256

Total proptest properties: 60+ across the SIL-4 + SIL-2 crates. Rerun with `cargo test -p workspace`; keep the resulting workspace test count in the release log rather than baking a drifting number into this register.

3. Differential testing (Rust Python)

The `tools/reference-ma/` directory carries a stdlib-only Python re-implementation of `osr-interlocking`. `crates/osr-interlocking/tests/differential.rs` generates random log prefixes, computes the MA in both Rust and Python, and asserts byte-identical JSON output. Catches spec-level bugs in either implementation.

- Coverage: ring lines, switch observations, route grants, speed restrictions, partial sections.
- Run: `cargo test -p osr-interlocking --test differential`.

4. Integration-level evidence (sim shadow stack)

crates/osr-sim runs the full Samawah network under the ATP + brake + obstacle-detect + vigilance + fire + derailment shadow stack per-tick per-train. Fault injection exercises each SIL-4 evaluator's restrictive path end-to-end.

Scenario	Fault kinds	What it demonstrates
designs/west-asia/Iraq/Samawah/samawah.toml	None	Nominal operation: zero spurious emergencies over multi-hour revenue sim
Built-in dust-storm fixture (sim CLI)	PV dust, grid outage, charging-pad outage	Energy-system fault-tolerance
Built-in obstacle-fault fixture (sim CLI)	LIDAR/radar/ultrasonic/peer-disagreement (RFC 0015)	O1–O5 all fire through BrakeInputs : : obstacle_emergency
Built-in wayside-intrusion fixture (sim CLI)	Present/Unknown on specific sections (RFC 0016)	Interlocking gate (d) withholds MA without a single train violating

Rerun with cargo `run --release --bin osr-sim -- --config scenarios/<name>.toml`.

5. GSN safety-case tree

Goal-Structuring-Notation argument tree under docs/safety-case/gsn/, compiled by the osr-safety-case CI gate. Every safety goal links to evidence (Kani harness, proptest file, sim scenario); the CI job fails if any goal is added without a verifying evidence link.

File	Goals	Strategy
00-top.toml	G0	Root claim
10-non-overlap.toml	G1	Non-overlap of MAs
20-fail-safety.toml	G2	Fail-restrictive default
30-consistency.toml	G3	Log consistency
40-onboard-sil4.toml	G4	Onboard SIL-4 layer
50-wayside-sil4.toml	G5–G14	Wayside SIL-4 + infra
60-obstacle-detect.toml	G15–G19	Obstacle detection (RFC 0015)
70-intrusion-detect.toml	G20–G24	Intrusion detection (RFC 0016)
80-message-authentication.toml	G25–G27	Message auth (RFC 0017)

Total: 27 top-level goals; 70+ solution links to concrete evidence.

CI gate: the `starter_case_closes` test in crates/osr-safety-case/tests/ fails the build if any goal is un-closed.

6. Operational-controls evidence (RFC 0013 rulebook)

The operations rulebook under docs/operations/ defines 40+ rules across five role families. Each rule is one sentence plus a Why : paragraph, cross-referenced to the specific crate it relies on.

Role	File count	Rule count
Driver (GoA 2 legacy)	D1–D8	45
Dispatcher	S1–S7	35
Station staff	T1–T5	29
Maintenance	M1–M7	42
Control centre	C1–C3	16

Total: 167 operational rules backing the safety-case claims.

7. Hardware-level evidence

Per-board spec documents under hardware/<class>/schematics/v2-spec/:

Board	Complete spec
T-ECU/S	(block diagram, power budget, pinout RP2350, pinout CM5, safety-nets, connector tables)
T-OBS	(block diagram, power budget, pinout RP2350, safety-nets, connector tables)
T-ECU/A	pending (see option 3 of the current sprint)
W-SBC	pending
S-SBC	pending

KiCad schematic capture + gerber generation are v3.1 per-board milestones (deferred alongside RFC 0007 v3 rollout).

8. Summary of evidence density

- 56 crates in the workspace.
- ~750 Rust tests passing, 0 failing.

- 40+ Kani harnesses across 9 SIL-4/SIL-2 evaluators.
- 60+ proptest properties across 13 crates.
- 27 GSN top-level goals closed against 70+ evidence solutions.
- 167 operational rules across 5 role families.
- 4 sim scenarios exercising nominal + fault + driverless + wayside-intrusion paths.
- 2 hardware v2 specs (T-ECU/S, T-OBS) with safety-nets traceable to the SIL-4 hardware argument; 3 more pending.

Zero test failures. Zero workspace build warnings. Zero open safety-case gaps against SR-01 through SR-24 at the crate level.

docs/certification/evidence-status.md

docs/certification/evidence-status.md

Evidence Status Matrix

This matrix separates what is already evidenced in the repository from what remains deployment or assessor work. It is not a safety approval; it is a coherence map for the pre-submission pack.

Area	Current repository evidence	Status	Next action
Movement authority non-overlap	osr-interlocking unit/proptest/differential tests; RFC 0004	Implemented + tested	Add assessor-reviewed trace from hazards to tests
Consensus log safety	TLA+ model, osr-consensus simulation/proptests	Implemented + modeled	Refinement argument from TLA+ spec to Rust harness
Onboard obstacle detection	RFC 0015, osr-obstacle-detect, sim fault injection	Implemented + simulated	First-article sensor dataset and calibration report
Wayside intrusion detection	RFC 0016, interlocking gate, sim integration	Implemented + simulated	Pilot installation evidence on representative sections
Message authentication	osr-crypto, RFC 0017 design	Library implemented	Wire every consensus entry through signed envelopes
Hardware safety nets	RFC 0007 v2 specs and hardware docs	Specified	KiCad/Gerber/BOM release and bench test records
Rolling-stock mechanical concept	RFC 0008/0021/0022, mechanical-py STEP catalogue	Parametric reference	FEA, crashworthiness simulation, supplier drawings
Station charging energy	RFC 0002, generated city energy feasibility tables	Planning-grade	Site-specific solar yield, grid-tie, and charger thermal study
Operations rulebook	RFC 0013 and docs/operations/	Drafted	Operator review and local authority adaptation
Certification pack	docs/certification/ and GSN claims	Pre-submission scaffold	Independent assessor review and evidence freeze

The repo health gate (scripts/repo-health.py) guards generated design coherence. It is deliberately not a substitute for EN 50126/50128/50129 assessment evidence.

docs/certification/hazard-log.md

docs/certification/hazard-log.md

Hazard log

GoA 4 Unattended Train Operation hazards identified against the system boundary in [system-description.md](#). Organised by hazard class: collision, runaway, derailment, passenger-boarding, fire, sabotage, data-integrity.

Each row carries:

- Severity — using the EN 50126 scale: Catastrophic (multiple fatalities) / Critical (single fatality or multiple serious injuries) / Marginal (single serious injury or multiple minor) / Insignificant.
- Frequency without controls — order-of-magnitude estimate for how often the hazard would occur absent any mitigation.
- Residual frequency — the frequency that remains once the listed controls are in place; the per-hazard "Residual acceptance" rows below argue why each residual is tolerable.
- Mitigating controls — cross-referenced to requirements and artefacts.

Frequencies are expressed on a qualitative EN 50126 scale: Frequent ($\geq 1/\text{yr}$) / Probable ($1/10 \text{ yr}$) / Occasional ($1/100 \text{ yr}$) / Remote ($1/1\,000 \text{ yr}$) / Improbable ($1/10\,000 \text{ yr}$) / Incredible ($< 1/10\,000 \text{ yr}$).

H-CO — Collision hazards

H-CO-01 — Rear-end collision between two trains

Field	Value
Description	Two trains occupy the same block; the following train collides with the leading.
Severity	Catastrophic
Frequency without controls	Frequent
Controls	SR-01 section-occupancy gate; SR-02 MA validity window; SR-03 MA conservatism; 2oo2 AND-gate at brake actuator (SR-20).
Residual frequency	Incredible

H-CO-02 — Head-on collision through mis-set switch

Field	Value
Description	Switch commanded Normal but observed Reverse (or Unknown); train enters a conflicting route.
Severity	Catastrophic
Frequency without controls	Probable
Controls	osr-wayside-points 2oo2 sensor fusion; osr-interlocking::section_available_ to checks switch state before MA grant; fail-restrictive Unknown classification.
Residual frequency	Incredible

H-CO-03 — Collision with obstacle on track (debris, vehicle, person)

Field	Value
Description	A person, animal, car, or large debris enters the track envelope between trains; next train hits it.
Severity	Critical to Catastrophic
Frequency without controls	Occasional (target climate: dust storms deposit debris; urban ROW crossings possible)
Controls	SR-07 wayside intrusion gate (proactive — RFC 0016); SR-04 onboard obstacle detection (reactive — RFC 0015); SR-24 PSDs at every boarding platform; physical fence (deployment civil scope).
Residual frequency	Remote

H-CO-04 — Collision with a track worker under MaintenanceOverride

Field	Value
Description	Track worker on foot under an active MaintenanceOverride; train enters the blocked section.
Severity	Critical
Frequency without controls	Probable

Field	Value
Controls	SR-01 section-occupancy gate refuses MA on any section carrying an active override; protection signals (RFC 0013 S5.3); physical trap point (site-specific); 30-minute block-renewal cadence (S5.4) catches silent over-runs.
Residual frequency	Incredible

H-RA — Runaway hazards

H-RA-01 — Runaway on grade after brake release

Field	Value
Description	Train parked on a grade; parking brake fails or is released in error.
Severity	Catastrophic
Frequency without controls	Occasional
Controls	Parking-brake engagement below park_brake_max_speed_mmpps (SR-20 osr-brake B5 property); EB latches on any monitor trip; both SIL-4 channels required to release.
Residual frequency	Improbable

H-RA-02 — Overspeed through brake failure

Field	Value
Description	Train exceeds the civil-speed envelope due to a regen/friction-blend failure.
Severity	Critical
Frequency without controls	Occasional
Controls	ATP overspeed trip (SR-03); regen-shortfall falls back to friction; WSP subtractive-only (B4).
Residual frequency	Remote

H-DE — Derailment hazards

H-DE-01 — Derailment through undetected switch Unknown

Field	Value
Description	Switch observation fails or reads Unknown; train enters the switch without MA.
Severity	Catastrophic
Frequency without controls	Probable
Controls	Switch observation fail-restrictive (Unknown rejects MA); osr-wayside-points 2oo2 sensor fusion; RFC 0013 M4 30-day switch inspection.
Residual frequency	Incredible

H-DE-02 — Derailment through lateral forces

Field	Value
Description	Excessive lateral acceleration (track-twist, excessive speed into curve) causes derailment.
Severity	Catastrophic
Frequency without controls	Occasional
Controls	SR-11 onboard derailment monitor (osr-derailment); RFC 0013 M3 track-geometry inspection cadence (90/60/45 days per preset); RFC 0009 preset-bounded curve radius + cant.
Residual frequency	Improbable

H-PB — Passenger-boarding hazards

H-PB-01 — Passenger struck by closing doors

Field	Value
Description	Door closes on a passenger limb / bag.
Severity	Marginal (rare Critical)
Frequency without controls	Frequent
Controls	SR-09 door-obstruction motor-current + sensor; reopen on detection; SR-08 closing-interlock above 5 km/h.
Residual frequency	Remote

H-PB-02 — Passenger falls between platform and train

Field	Value
Description	Passenger falls into the platform gap during boarding.
Severity	Critical
Frequency without controls	Frequent (legacy no-PSD metros)
Controls	SR-24 PSDs at every boarding platform; RFC 0010 platform-gap design; station-staff T2 supervision.
Residual frequency	Improbable

H-PB-03 — Train departs with passenger caught in door

Field	Value
Description	Passenger's limb / bag remains trapped in closed door; train begins to move.
Severity	Critical
Frequency without controls	Probable
Controls	SR-09 door-sensor + SR-08 interlock; station-staff T2.5 platform-clear confirmation at non-PSD stations; PSD interlock at PSD stations (SR-24).
Residual frequency	Improbable

H-FI — Fire hazards

H-FI-01 — Battery thermal runaway

Field	Value
Description	Cell thermal runaway propagates to adjacent cells; fire on-board.
Severity	Critical
Frequency without controls	Occasional
Controls	SR-12 onboard fire detection + suppression; BMS thermal cutoff + cell-out derating; Na-ion chemistry has lower runaway risk than LFP and significantly lower than NMC.
Residual frequency	Remote

H-FI-02 — Electrical fire in traction inverter

Field	Value
Description	Short or arc in the SiC inverter; fire.
Severity	Critical
Frequency without controls	Occasional
Controls	SR-12; osr-traction isolation on fault; RFC 0008 §3 EN 45545 HL2 R1 material compliance.
Residual frequency	Remote

H-SA — Sabotage + intrusion hazards

H-SA-01 — Deliberate obstacle placement on track

Field	Value
Description	Attacker places a large obstacle on the ROW between trains.
Severity	Critical to Catastrophic
Frequency without controls	Occasional
Controls	SR-07 wayside intrusion gate (proactive, detects placement); SR-04 onboard obstacle detection (reactive); fence-line contact detection (RFC 0016 fence sensor); CCTV coverage.
Residual frequency	Remote

H-SA-02 — Station-platform intrusion

Field	Value
Description	Attacker gains access to the trackside from a platform.
Severity	Critical
Frequency without controls	Probable (stations are public)
Controls	SR-24 PSDs block direct platform trackside access; station-staff T2.1 platform walk; CCTV.
Residual frequency	Remote

H-DI — Data-integrity hazards

H-DI-01 — Forged consensus entry

Field	Value
Description	Attacker with network access injects a forged <code>SectionIntrusion::Clear</code> or <code>RouteGrant</code> .
Severity	Critical to Catastrophic (depends on forged content)
Frequency without controls	Remote (requires network access + cryptographic sophistication)
Controls	SR-22 consensus entry authenticity (RFC 0017 — library in tree, v2 wiring open); physical network isolation (deployment scope); IEC 62443-4-2 component-level controls.
Residual frequency	Improbable (after SR-22 v2) / Remote (today, pending SR-22 wire integration)

H-DI-02 — Replay of a valid entry

Field	Value
Description	Attacker replays an old <code>RouteRelease</code> or <code>MaintenanceOverride</code> at an inconvenient time.
Severity	Critical
Frequency without controls	Remote
Controls	Derived-state fold is idempotent in <code>entry_id</code> (RFC 0017 §3.4); 60 s freshness window on <code>timestamp_ns</code> ; Raft already rejects duplicate <code>entry_ids</code> .
Residual frequency	Improbable

Hazard summary

Class	Count	Worst-case severity	After controls
Collision	4	Catastrophic	Incredible–Remote
Runaway	2	Catastrophic	Improbable–Remote
Derailment	2	Catastrophic	Incredible–Improbable
Passenger-boarding	3	Critical	Improbable–Remote
Fire	2	Critical	Remote
Sabotage / intrusion	2	Catastrophic	Remote
Data integrity	2	Catastrophic	Improbable
Total	17		

Every hazard carries ≥ 2 independent controls. No single control failure produces a single-fault hazard of residual frequency worse than Remote.

docs/certification/safety-requirements.md

docs/certification/safety-requirements.md

Safety requirements specification

Functional safety requirements (SR-01 ... SR-24) derived from EN 62267 §6 (functional requirements for GoA 4 operation) and cross-referenced to OSR artefacts (crates, RFCs, rulebook sections). Requirements are grouped by the EN 62267 function.

Each requirement states what the system must do and names the evidence that it does so. Detailed evidence lives in [evidence-register.md](#).

1. Train separation (EN 62267 §6.1)

SR-01 — Safe separation between trains

The system shall at all times prevent two trains from occupying the same track section.

- Implementation: `osr-interlocking::section_available_to_gate` (a) refuses MA into any section whose `section_occupancy` record names a different train.
- Evidence: GSN G1 (via `docs/safety-case/gsn/10-non-overlap.toml`); property P2 proptest at `crates/osr-interlocking/tests/proptest_ma.rs`; Kani harness `kani_a5_head_past_ma_end_trips_same_section`.

SR-02 — MA validity window

Every movement authority issued to a train shall carry an absolute expiry time and shall not be extended without a fresh position report.

- Implementation: `MA_VALIDITY_WINDOW_NS` bound on every MA; ATP property A2 trips on expiry.
- Evidence: Kani `kani_a2_expired_ma_trips`; property P5 proptest.

SR-03 — MA conservatism under uncertainty

Widening any sensor-uncertainty or shortening any MA input shall never move the ATP verdict toward a less-restrictive brake command.

- Implementation: ATP property A7.
- Evidence: Kani `kani_a7_uncertainty_widening_is_conservative`; property A7 proptest.

2. Train protection (EN 62267 §6.2)

SR-04 — Stopping before obstacle (GoA 4 substitution for driver)

The system shall detect obstacles inside the current stopping- distance envelope and command Emergency Brake within the ATP tick.

- Implementation: `osr-obstacle-detect::evaluate` property O1; any LIDAR / radar / ultrasonic detection inside the envelope forces `ObstacleVerdict::EmergencyBrake`, which flows into `BrakeInputs::obstacle_emergency`.
- Evidence: Kani `o1_ultrasonic_return_forces_eb`; proptest `o1_ultrasonic_return_forces_eb`; sim built-in `obstacle-fault fault-injection` fixture.

SR-05 — Fail-restrictive on sensor failure

If any safety-primary obstacle-detect sensor is stale beyond 100 ms, the verdict shall be at least `RestrictedSpeed` (O4b) and never `Clear`.

- Implementation: O2 + O4b in `osr-obstacle-detect`.
- Evidence: Kani `o2_stale_ultrasonic_forces_eb`, `o4b_lidar_offline_with_radar_healthy_restricts_speed`.

SR-06 — 2oo2 cross-check on obstacle detection

The two redundant obstacle-detect channels (RP2350 A + B on the T-OBS board) shall both agree on Clear before the trainset may proceed at full MA-permitted speed.

- Implementation: O3 property; any peer disagreement EB.
- Evidence: Kani o3_peer_disagreement_forces_eb.

SR-07 — Wayside intrusion gate

The system shall withhold MA on any track section whose latest wayside intrusion verdict is not Clear.

- Implementation: osr-interlocking::section_available_to gate (d) introduced by RFC 0016 v2.
- Evidence: intrusion_present_blocks_section, intrusion_unknown_is_fail_restrictive, latest_intrusion_verdict_wins unit tests in crates/osr-interlocking/src/ma.rs.

3. Door operation (EN 62267 §6.3)

SR-08 — Door closed above threshold speed

The system shall interlock door opening so that a door cannot be commanded open while speed ≥ 5 km/h.

- Implementation: osr-door-control closing interlock (SIL-4 per RFC 0005 §4.2).
- Evidence: osr-door-control proptests.

SR-09 — Door obstruction detection

The system shall detect door obstructions via motor current + dedicated sensor and reopen the affected door on detection.

- Implementation: osr-door-control obstruction handler.
- Evidence: osr-door-control integration tests.

SR-10 — Platform-side alignment at standstill

Door opening shall be enabled only when the train is at standstill (speed < threshold) and reporting position within ± 0.5 m of the platform reference mark.

- Implementation: Door-enable gate uses osr-odometry position + osr-ato stop-detection; PSD interlock adds a second layer at PSD-equipped stations.
- Evidence: osr-ato stop-tolerance tests; RFC 0010 §7 platform geometry.

4. Obstacle and derailment detection (EN 62267 §6.4)

SR-11 — Derailment detection

Lateral g-forces + axle-rotation asymmetry monitored in Zoo2 shall trip Emergency Brake on any indication of derailment.

- Implementation: osr-derailment::derailment_evaluate; feeds BrakeInputs::derailment_emergency.
- Evidence: osr-derailment proptests.

SR-12 — Onboard fire detection

Smoke + temperature in battery / traction / HVAC bays shall trigger suppression + EB.

- Implementation: osr-fire-safety::fire_evaluate; feeds BrakeInputs::fire_emergency.
- Evidence: osr-fire-safety proptests.

SR-13 — Hot-axle box detection

Axle bearing temperature above threshold shall produce an advisory event; OCC triages per RFC 0013 S3.

- Implementation: osr-hot-axle (onboard, SIL-2) + osr-hot-axle-wayside (wayside, SIL-4).
- Evidence: osr-hot-axle-wayside proptests.

5. Emergency detection and management (EN 62267 §6.5)

SR-14 — Passenger emergency intercom

Passengers shall be able to trigger a controlled brake + open an audio + video channel to OCC by pressing an emergency intercom; the brake shall bring the train to a safe stop at the next station (not immediate EB, to avoid stranding).

- Implementation: RFC 0015 §5.3 intercom path; feeds `BrakeInputs::driver_emergency` (legacy name, covers passenger-intercom route in GoA 4).
- Evidence: RFC 0013 rulebook cross-refs; sim stub (awaiting RFC 0015 v2.2 full integration).

SR-15 — EB dominates all other brake commands

Emergency Brake shall not be reducible by any other input; service-brake or release commands during an EB shall be ignored until EB is explicitly released.

- Implementation: `osr-brake::brake_evaluate` B2 property.
- Evidence: Kani `kani_b2_emergency_union`, `kani_b3_emergency_completeness`.

SR-16 — Passenger fire-evacuation on EB + fire

If EB is driven by a fire trip, the train shall continue to the next station platform where safe, then open doors + announce evacuation.

- Implementation: `osr-ato` fire-response mode; `osr-fire-safety` coordinates with `osr-brake`; RFC 0013 S3.6 rulebook procedure.
- Evidence: Rulebook; deployment-specific sim evidence pending.

6. Operations supervision (EN 62267 §6.6)

SR-17 — Dispatcher accountability

Every MA grant, MaintenanceOverride, and degraded-mode transition shall be attributable to a named dispatcher via their OCC login.

- Implementation: RFC 0013 S1.1 rule (personal credential); `osr-occ` console logs.
- Evidence: Rulebook; audit-log structure in `osr-historian`.

SR-18 — Remote-assist for passenger incidents

OCC shall maintain an audio + video channel to any trainset under remote-assist; dispatcher may hold the train, request medical, or release to continue.

- Implementation: RFC 0015 §5.3; RFC 0013 C2 communications.
- Evidence: Rulebook cross-refs; hardware stub in RFC 0007 pending live-comms integration.

7. Fault management (EN 62267 §6.7)

SR-19 — Fail-restrictive default

Every SIL-4 evaluator shall fail in the restrictive direction — on any uncertainty, ambiguity, or missing input the verdict is at least the second-most-severe (usually a brake, sometimes a speed cap).

- Implementation: All O-series (obstacle-detect), I-series (intrusion-detect), A-series (ATP), P-series (interlocking), V-series (vigilance), O-odom, B-brake properties carry fail-restrictive clauses.
- Evidence: 8 Kani harness modules across the SIL-4 crates; GSN G1–G24 closure.

SR-20 — 2oo2 hardware voting at output stage

Every brake command to the actuator shall pass through a 2oo2 AND-gate relay driven by both RP2350 channels of the T-ECU/S.

- Implementation: Hardware AND-gate relay stage per `hardware/t-ecu-s/schematics/v2-spec/safety-nets.md`.
- Evidence: Safety-nets spec; DRC pass at board production (deployment-specific).

SR-21 — Watchdog-driven reset

Loss of heartbeat from either RP2350 shall drive the TPS3701 supervisor to reset that channel; concurrent loss of both heartbeats shall drop the EB relay directly (bypassing the AND gate).

- Implementation: `hardware/t-ecu-s/schematics/v2-spec/safety-nets.md` watchdog rules.
- Evidence: Safety-nets spec.

8. Cybersecurity (EN 62267 §6.8, complementary IEC 62443-4-2)

SR-22 — Consensus entry authenticity

Every entry committed to the track-state consensus log shall be signed by its originating entity and verified by every consumer before it affects derived state.

- Implementation: `osr-secbus::verify_signed` (RFC 0017); S1/S2/S3 properties.
- Evidence: `osr-secbus` Kani + proptest; GSN G25–G27. Status: library in tree; wiring into the live consensus wire layer is RFC 0017 v2 (open).

SR-23 — Fare-token authenticity

Every fare transaction token shall carry an HMAC-SHA256 MAC keyed to a per-deployment secret that is rotatable without re-issuing existing tokens.

- Implementation: `osr-crypto::hmac_sha256_verify`; used by `osr-afc`.
- Evidence: `osr-crypto` proptests; `osr-afc` integration.

9. Passenger interaction (EN 62267 §6.9)

SR-24 — Platform screen door mandatory at GoA 4

Every passenger boarding platform in a GoA 4 deployment shall be equipped with platform screen doors (PSDs) per RFC 0010 §6 + RFC 0015 §5.4.

- Implementation: RFC 0010 station-design standard with PSD in every non-halt archetype; RFC 0015 §5.4 tightens the default to PSD-mandatory for every boarding platform (except depot-terminal, where access is staff-only).
- Evidence: RFC 0010, RFC 0015; deployment-specific design report.

Requirement summary table

ID	Summary	SIL	Evidence kind
SR-01	Section-occupancy gate	4	Kani + proptest + GSN G1
SR-02	MA validity window	4	Kani + proptest
SR-03	MA conservatism	4	Kani + proptest
SR-04	Obstacle detection in envelope	4	Kani + proptest + sim
SR-05	Fail-restrictive on sensor failure	4	Kani + proptest
SR-06	2oo2 obstacle-detect cross-check	4	Kani
SR-07	Wayside intrusion gate	4	Unit test + GSN G20–G24
SR-08	Door-interlock above threshold	4	Proptest
SR-09	Door-obstruction reopen	4	Integration test
SR-10	Platform-alignment gate	4	ATO tests + RFC 0010
SR-11	Derailment detection	4	Proptest
SR-12	Onboard fire detection	4	Proptest
SR-13	Hot-axle-box detection	4 (wayside) / 2 (onboard)	Proptest
SR-14	Passenger emergency intercom	2	Rulebook + sim stub
SR-15	EB dominates	4	Kani
SR-16	Evacuation on fire-EB	2	Rulebook + sim pending
SR-17	Dispatcher accountability	—	Rulebook S1.1
SR-18	Remote-assist channel	—	Rulebook C2 + HW stub
SR-19	Fail-restrictive default	4	8 Kani modules + GSN G1–G24
SR-20	2oo2 hardware voting	4	Safety-nets spec
SR-21	Watchdog + reset	4	Safety-nets spec
SR-22	Consensus entry auth	2	Kani + proptest + GSN G25–G27
SR-23	Fare-token auth	2	Proptest
SR-24	PSDs at every boarding platform	—	RFC 0010 + RFC 0015 §5.4

System description

1. What OSR is

OpenSourceRail is an open-source urban-rail technology stack — software + hardware reference designs + operational rules — sized for GoA 4 (Unattended Train Operation) light-metro and metro deployments. The Samawah reference ([RFC 0003](#)) instantiates the stack against a brownfield pilot ([RFC 0027](#)) anchored on the existing rail yard + workshop adjacent to Samawah Train Station, with a 3-line auto-planned light-metro network serving a ~374 k Iraq 2024 census population.

The stack is the entirety of a modern metro:

- Rolling stock — cableless ([RFC 0015](#)), four consist families (tram-2car through metro-6car) per [RFC 0008](#).
- Track — four geometry presets per [RFC 0009](#) covering gauge 1435 mm, radius, grade, cant, and rail profile.
- Civil — at-grade + elevated + bridge only per [RFC 0011](#); no tunnels.
- Stations — six archetypes per [RFC 0010](#), prefab steel-portal + solar-canopy with no station building.
- Signalling — distributed consensus-based interlocking per [RFC 0001](#), no central CBTC zone controller.
- Energy — catenary-free; onboard Na-ion/LFP batteries + station/depot PV per [RFC 0002](#).
- Operations — one shared ≤ 60 -page rulebook across four role families per [RFC 0013](#).

For the safety claim of this pack, the relevant subset is the SIL-4 + SIL-2 code running on four onboard + wayside host classes:

- T-ECU/S — onboard safety kernel (2× RP2350 + RPi CM5 in 2oo2), runs `osr-atp`, `osr-brake`, `osr-derailment`, `osr-door-control`, `osr-fire-safety`, `osr-odometry`.
- T-ECU/A — onboard application tier (RPi CM5), runs `osr-ato`, `osr-tcms`, `osr-event-recorder`, `osr-tcn`.
- T-OBS — onboard obstacle detector ([RFC 0015](#)), runs `osr-obstacle-detect`.
- W-SBC — wayside SBC (Radxa CM5 industrial), runs `osr-interlocking`, `osr-consensus`, `osr-wayside-points`, `osr-intrusion-detect` ([RFC 0016](#)).

2. System boundary

2.1 In scope for this certification

The OSR safety claim covers:

- Onboard safety chain — from sensor-fusion position through ATP envelope computation to brake command actuation.
- Wayside safety chain — from track-occupancy + switch-observation + intrusion detection through MA computation to MA delivery to the train.
- Distributed signalling consensus — the Raft-derived consensus protocol that keeps the wayside state machines coherent across nodes.
- Obstacle detection + intrusion detection — the [RFC 0015](#) onboard sensor fusion and [RFC 0016](#) wayside sensor fusion that together substitute for a driver's-eye detection.
- Operational rules — the [RFC 0013](#) rulebook that binds the safety-case solutions to human procedures (dispatcher, station staff, maintenance, OCC).

2.2 Out of scope for this certification

- Crashworthiness — EN 15227 certification is the rolling-stock builder's scope ([RFC 0008](#) §3).
- Fire-protection material performance — EN 45545 is material-certification scope.
- Civil structure — structural engineering of elevated viaducts and at-grade formation is the deployment partner's civil scope ([RFC 0011](#)).

- Electrical supply — MV/LV distribution from the grid connection to the rectifier substation is the deployment partner's electrical scope.
- Cybersecurity above SIL-2 — IEC 62443-4-2 message authentication is in tree at SIL-2 (RFC 0017) and integrates as complementary evidence; component-level cybersecurity certification is tracked separately from EN 62267.

3. Operational envelope

- Maximum civil speed: 80 km/h (22 m/s) at standard-urban preset; 100 km/h on mainline-mixed.
- Minimum curve radius: preset-dependent per RFC 0009 §1.
- Maximum gradient: preset-dependent per RFC 0009 §1.
- Climate envelope: -10 °C to +55 °C ambient per RFC 0003 §4.5 (deployment-adaptable).
- Crashworthiness envelope: EN 15227 Cat C-II up to 25 km/h metro-to-obstacle collision.
- Communication envelope: 5G primary + LoRa backup for TRG; Cat 6a + TSN for wayside back-office (RFC 0005 §5).

Operation outside any of these envelopes triggers a degraded- mode response per RFC 0013 §5 — the deployment continues safely at reduced performance; no passenger is carried under a violated envelope.

4. Interfaces to the external world

- Passengers — at station platforms (PSDs per RFC 0010) and inside the trainset (seats, doors, emergency intercoms per RFC 0015 §5.3).
- Track workers — via MaintenanceOverride grants issued from OCC per RFC 0013 S5; physical protection signals per RFC 0013 S5.3.
- Emergency services — via OCC radio liaison per RFC 0013 C2.4; passenger emergency intercom direct to OCC per RFC 0015 §5.3.
- National safety authority — via the OCC incident- notification path per RFC 0013 S3.5 (I5 within 2 h, I6 immediately).
- Power grid — at each rectifier substation, bidirectional per RFC 0002 §6.
- OSM + Civil 3D / OpenRail / Trimble — design-time only, via the OSR-ALN alignment interchange format (RFC 0009 v2).

5. Non-interfaces

OSR does not interface with:

- Third-party CBTC zone controllers — RFC 0001 replaces them with distributed consensus.
- Legacy MVB/WTB trainbus — RFC 0006 replaces with TCN-E (TSN Ethernet).
- Magnetic smartcard fare systems — RFC 0013 and osr-a-fc use account-based mobile money + QR.
- GSM-R radio — RFC 0007 uses 5G + LoRa.
- Catenary power distribution — RFC 0002 is onboard- battery-only.

A deployment that must interface with any of the above forks the repository — no upstream compatibility is offered.

6. Quantitative footprint

Metric	Value
Rust crates in the safety-relevant workspace	51
Total Rust tests, current workspace	705
SIL-4 evaluators with Kani harnesses	8 (osr-atp, osr-brake, osr-vigilance, osr-odometry, osr-wayside-points, osr-interlocking, osr-obstacle-detect, osr-intrusion-detect)
GSN goals currently closed against evidence	G1–G27 across 8 TOML files under docs/safety-case/gsn/
Operations rulebook page count (target)	≤ 60
Operations rulebook role families	5 (driver GoA 2 legacy; dispatcher, station-staff, maintenance, control-centre in GoA 4 default)

docs/civil/osr-aln-format.md

docs/civil/osr-aln-format.md

OSR-ALN — alignment interchange format (v1.0)

Status: v2 deliverable of [RFC 0009](#). Purpose: a tool-agnostic text schema that civil-engineering software (Civil 3D, Bentley OpenRail, Trimble, open-source tools) can export to and the OSR simulator / emitter can import from.

Why OSR-ALN

Every civil-engineering firm uses a different authoring tool. Every tool has a different native file format. LandXML covers the generic case but is verbose and has many optional fields that don't apply to urban rail.

OSR-ALN is a minimal text format that round-trips a rail-ready alignment to / from any tool. It is a 100%-plain-text TOML schema (no XML, no binary) with one row per `HorizontalPoint` + one row per `VerticalPoint`. A civil firm produces OSR-ALN from their native format as part of the deployment deliverable; the OSR pipeline validates it against RFC 0009's compatibility matrix and feeds `osr-sim` + `osr-routing` without further adaptation.

File layout

One alignment per line per file. Filename convention: `<city>-<line>.aln.toml`. Example: `samawah-line1.aln.toml`.

```
# OSR-ALN v1.0 — alignment interchange format per RFC 0009 v2.
[meta]
schema_version = "1.0"
line_id        = "samawah-line1"
design_date     = "2026-04-22"
surveyor       = "<firm name>"
preset         = "standard-urban"      # RFC 0009 $1 preset key
consist        = "light-metro-3car"    # RFC 0008 $1 family key
crs             = "EPSG:32638"          # Coordinate Reference System (UTM Zone 38N for Iraq)
units          = "metric"              # SI only — OSR is SI
is_ring        = false

# -----
# Horizontal alignment points — tangent + circular + transition elements
# expressed as a sequence of PI (Point of Intersection) with before /
# after bearings. The standard alignment reconstructs the clothoid
# transition curves deterministically from adjacent PIs + the declared
# radii + transition lengths.
# -----

[[horizontal]]
station_m      = 0.0
easting_m      = 477_500.00            # UTM
northing_m     = 3_467_100.00
bearing_in_deg = 128.7
bearing_out_deg = 128.7
curve_radius_m = 0.0                  # 0 = tangent
transition_length_m = 0.0

[[horizontal]]
station_m      = 1_000.0
easting_m      = 478_135.50
northing_m     = 3_466_522.00
bearing_in_deg = 128.7
bearing_out_deg = 123.7                # Curve begins here
curve_radius_m = 26_300.0
transition_length_m = 2.0              # Clothoid transition length

# ... more rows one per PI or tangent endpoint ...

# -----
# Vertical alignment points — elevation along station_m. Interpolation
# is linear between adjacent points with a circular vertical curve of
# radius `vc_radius_m` at each grade-change point.
# -----

[[vertical]]
station_m      = 0.0
elevation_m    = 6.0                  # at top of rail
vc_radius_m    = 0.0                  # 0 = tangent (no vertical curve here)

[[vertical]]
station_m      = 1_000.0
elevation_m    = 6.0                  # still level
vc_radius_m    = 0.0

[[vertical]]
station_m      = 1_150.0
```



```

elevation_m   = 11.25           # top of ramp into viaduct
vc_radius_m   = 300.0           # sag curve

# ...

# -----
# Civil-class spans – per-class segment between two station_m values.
# Must cover the full [0, total_length_m] range with no gaps and no
# overlaps.
# -----

[[civil]]
from_station_m = 0.0
to_station_m   = 1_000.0
class          = "at-grade"

[[civil]]
from_station_m = 1_000.0
to_station_m   = 2_300.0
class          = "elevated"

[[civil]]
from_station_m = 4_100.0
to_station_m   = 5_300.0
class          = "bridge"

# ...

# -----
# Stations on this alignment – references the station ids in the
# deployment's design.toml. station_m pinpoints the platform
# centreline stop mark on this alignment.
# -----

[[station]]
id          = "samawah-rws"
station_m   = 0.0
platform_length_m = 75.0           # derived from consist per RFC 0010

[[station]]
id          = "north-gate"
station_m   = 1_000.0
platform_length_m = 75.0

# ...

# -----
# Superelevation (cant) per curved span. One row per cant-change PI.
# -----

[[cant]]
from_station_m = 1_000.0
to_station_m   = 2_300.0
max_cant_mm    = 30
transition_in_m = 50.0           # cant development length in
transition_out_m = 50.0          # cant runoff length out

```

Required fields per section

[meta]

Field	Type	Required	Notes
schema_version	string	yes	Always "1.0" for this format
line_id	string	yes	Matches a <code>[[lines]]</code> name in design.toml
design_date	date	yes	ISO 8601
surveyor	string	yes	Civil firm of record
preset	string	yes	Must be one of the RFC 0009 §1 presets
consist	string	yes	Must be one of the RFC 0008 §1 families
crs	string	yes	EPSG code or WKT; UTM zone typical
units	string	yes	Always "metric"
is_ring	bool	yes	Whether the line loops back

[[horizontal]]

Sequence of horizontal points along the alignment. `station_m` values must be monotonic-ascending. Bearings are in compass degrees (0° = north, 90° = east). Tangent segments have `curve_radius_m` = 0.0. Transition curves (clothoids) are declared on the entry/exit of each circular curve via `transition_length_m`.

[[vertical]]

Sequence of vertical points. `station_m` monotonic-ascending. `elevation_m` is top-of-rail elevation above the local datum (typically mean sea level, per `crs`). `vc_radius_m` declares a circular vertical curve at grade-change points; 0 = sharp grade break (discouraged above 2.5 ‰).

[[civil]]

Civil-class spans covering the full length. No gaps, no overlaps. `class` must be one of `at-grade`, `elevated`, `bridge` per RFC 0011 — no `tunnel` (v2 validator rejects).

[[station]]

Platform centrelines along the alignment. Every station id must match a `[[stations]]` block in the deployment's `design.toml`.

[[cant]]

Optional superelevation ramp + full-cant section + runoff. Absent cant rows mean `max_cant_mm` = 0 for that span (tangent track).

Validator semantics

A validator ships at [tools/osr-aln-convert/src/osr_aln/validate.py](#) (exposed as the `osr-aln-validate` CLI after `pip install -e .`). It checks:

Hard gates (reject on fail)

- 1 `meta.preset` is one of the four RFC 0009 presets.
- 2 `meta.consist` is in the preset's `compatible_consists`.
- 3 `[[civil]]` spans are contiguous and cover the full length.
- 4 No `[[civil]]` has `class` = "tunnel".
- 5 Every station id has a matching entry in the deployment's `design.toml`.
- 6 Every curve's `curve_radius_m` ≥ preset's `min_curve_radius_m`.
- 7 Every vertical grade ≤ preset's `max_gradient_per_mille`.
- 8 Every `max_cant_mm` ≤ preset's `max_cant_mm`.

Soft gates (warn, don't reject)

- 1 Elevated share > 30 % RFC 0011 §8 soft gate.
- 2 Any curve radius ≤ 2× preset minimum flag for speed restriction.
- 3 Any grade > 80 % of preset maximum flag.

Round-trip consistency

Round-tripping OSR-ALN simulator OSR-ALN must produce a byte-identical file. The simulator may compute derived quantities (stopping distance, power draw, etc.) but must not modify the alignment.

Tool support

Converter at [tools/osr-aln-convert/](#) — stdlib-only Python; installs with `pip install -e .` and exposes one CLI per source format.

Firms using:

- Autodesk Civil 3D — export via LandXML, then transform with `landxml-to-osr-aln`. v1 (shipping).
- Bentley OpenRail — export via LandXML (OpenRail emits a compatible subset), transform with `landxml-to-osr-aln`. Cant data is written to a non-standard `<Cant>` extension that v1.1 will read. v1.
- Trimble Business Center — export via LandXML, transform with `landxml-to-osr-aln`. A TCL CSV reader is v1.1. v1 via LandXML.
- Open-source (QGIS + rail-path plugins) — LandXML export supported. v1.

Example: Samawah Line 1

A worked OSR-ALN alignment for Samawah Line 1 lives at [docs/civil/west-asia/Iraq/Samawah/samawah-line1.aln.toml](#). Hand-authored from the bearings + radii in [west-asia/Iraq/Samawah/line1-segments.md](#), with UTM Zone 38N coordinates computed step-by-step from the published segment lengths. Exercises all five top-level sections (meta, horizontal, vertical, civil, station, cant).

Validated against the deployment's design.toml:

```
$ osr-aln-validate docs/civil/west-asia/Iraq/Samawah/samawah-line1.aln.toml \
  --design-toml designs/west-asia/Iraq/Samawah/design.toml
2 soft-gate warning(s):
  △ S3: grade between [[vertical]] #1 and #2 is 35.00 ‰ – within 80 ‰
    of preset maximum 40.0 ‰, flag for review
  △ S3: grade between [[vertical]] #5 and #6 is 35.00 ‰ – within 80 ‰
    of preset maximum 40.0 ‰, flag for review
```

Both warnings are expected (the two viaduct ramps at 35 ‰); no hard gates fire.

Version history

Version	Date	Changes
1.0	2026-04-22	Initial spec

What v2 does NOT include

- Reference validator implementation (v3).
- Per-tool converter scripts (v3).
- A "delta file" format for alignment updates during construction (v4, if ever needed).
- Drainage profiles — handled separately at civil-design time, not part of the alignment.

docs/civil/west-asia/Iraq/Samawah/README.md

docs/civil/west-asia/Iraq/Samawah/README.md

Samawah civil alignment — v1 deliverable of RFC 0009

Worked civil-alignment envelope for the Samawah reference deployment's Line 1 (Nahrain) and Line 2 (Halqa), against the RFC 0009 standard-urban preset.

What this is

A per-segment alignment table for every track segment between adjacent stations, computed from the station coordinates in [designs/west-asia/Iraq/Samawah/design.toml](#), validated against the standard-urban preset's limits (min curve radius 90 m, max grade 50 ‰, max cant 150 mm, nominal speed 22 m/s).

What this is NOT

- A surveyed alignment. Station lat/lons are planning-grade RFC 0003 numbers. A real deployment substitutes GNSS-surveyed coordinates.
- A final civil-engineering design. This table is the envelope a civil engineering firm bids against.
- A structural design for the viaducts / bridges. Those are RFC 0011 work, summarised per-segment in [civil-class-summary.md](#).

Contents

File	Scope
line1-segments.md	Line 1 (12 stations, 11 segments, radial) per-segment table
line2-segments.md	Line 2 (10 stations, 10 segments + ring wrap) per-segment table
civil-class-summary.md	At-grade / elevated / bridge breakdown per-line and consolidated
compliance-report.md	Per-segment pass/fail against the standard-urban preset's §§5–6 limits
samawah-line1.aln.toml	Machine-readable OSR-ALN v1.0 alignment for Line 1 (13 km, 12 stations, passes <code>osr-aln-validate</code>)
samawah-line2.aln.toml	Machine-readable OSR-ALN v1.0 alignment for Line 2 (16 km ring, 10 stations, 4 cant sections, passes <code>osr-aln-validate</code>)

How to execute

- 1 Civil engineering firm reads [line1-segments.md](#)
 - [line2-segments.md](#). Each row is a buildable segment.
- 2 Firm runs their in-house alignment tool (Civil 3D, Bentley OpenRail, Trimble etc.) against each row's start/end coordinates + length target + civil class.
- 3 Firm produces the detailed alignment (horizontal + vertical geometry, cross-sections, stationing) that matches the v1 envelope.
- 4 Per-segment structural design follows RFC 0011 §§4–6 for at-grade / elevated / bridge respectively.

Licensing

CC-BY-SA 4.0 for this v1 specification. Detailed alignment drawings under CERN-OHL-S v2 once v2 is produced.

docs/civil/west-asia/Iraq/Samawah/civil-class-summary.md

docs/civil/west-asia/Iraq/Samawah/civil-class-summary.md

Samawah civil-class summary

Consolidated from [line1-segments.md](#) and [line2-segments.md](#). Per-line and total civil-class shares, against RFC 0011's at-grade dominance target ($\geq 70\%$).

Per-line shares

Line 1 (Nahrain, 13 km)

Class	Length (m)	Share
at-grade	9 500	73 %
elevated	2 400	18 %
bridge	1 200	9 %
tunnel	0	0 % (invariant per RFC 0011)

Line 2 (Halqa, 16 km ring)

Class	Length (m)	Share
at-grade	14 500	91 %
elevated	1 500	9 %
bridge	0	0 %
tunnel	0	0 %

Consolidated (29 km total)

Class	Length (m)	Share
at-grade	24 000	83 %
elevated	3 900	13 %
bridge	1 200	4 %
tunnel	0	0 %

Passes RFC 0011's $\geq 70\%$ at-grade target by a comfortable margin. No soft-gate trip for `elevated_le_0.30` per the emitter's quality YAML.

Cost estimate (per RFC 0011 §9 rates)

Planning-grade €/km $\times$ civil mix:

Class	€/km	Length	Subtotal (€)
at-grade	3 500 000	24 km	84 000 000
elevated	18 000 000	3.9 km	70 200 000
bridge	25 000 000	1.2 km	30 000 000
Civil total			€ 184 200 000

Country cost factor for Iraq applies on top (from [lib/templates/country-costs.toml](#)):

- Labour discount: $\sim 0.4 \times$ OECD.
- Materials: $\sim 0.95 \times$ OECD.
- Effective factor: ~ 0.65 .

Samawah civil CAPEX (country-adjusted): $\sim$ € 120 M.

Note: this is civil only – track + viaducts + bridges. It excludes:

- Rolling stock ($\sim$ € 3.0 M $\times$ 16 three-car trainsets = $\sim$ € 48 M, using the current € 1.0 M/car RFC 0008 BOM basis).
- Stations ($\sim$ € 3 M per standard $\times$ 17, plus majors + terminals
 - interchanges).
- Depots ($\sim$ € 10–15 M for the main-heavy east depot + €1–2 M for the layup-minimal).

- Residual train-control wayside + radio/OCC integration (~€ 2 M; train protection is primarily onboard driverless sensing).
- Procurement, engineering, contingency (~20 %).

Land acquisition

29 km of double-track ROW at 10.5 m width:

- At-grade: 24 km × 10.5 m = 25.2 ha.
- Elevated: 3.9 km × 6 m footprint (piers + walkway) = 2.3 ha.
- Bridge: 1.2 km × 6 m = 0.7 ha.
- Stations + depots: ~3 ha.

Total land acquisition: ~31 ha.

For Samawah context (urban core population ~250 k, municipal area ~50 km²), this is a small fraction of urban land — ≈ 0.06 %. Several segments follow the existing Mosul Street corridor + the highway east of the river; those use existing public ROW with minimal acquisition.

Comparison to legacy equivalents

A 29 km light-metro project in the Middle East at legacy-vendor pricing (CAPEX-only, civil + signalling + rolling stock) is typically quoted at € 1–3 billion in 2026-era bids.

The OSR Samawah deployment budgets around € 200–300 M all-in. This is the catenary-free + local-manufacturability + simple-civil bet playing out in practice. The largest single savings:

- No continuous catenary: saves ~€ 60 M.
- No tunnelling (vs the typical 20 % tunnelled urban alignment): saves ~€ 500–800 M.
- Onboard-first driverless protection instead of a proprietary CBTC vendor stack: saves ~€ 30–80 M over 30 years.
- Commodity rolling stock vs bespoke: saves ~€ 40–60 M over a fleet of 16 trainsets.

The residual difference is OpenSourceRail's structural CAPEX target. A real deployment may come in higher — ~€ 400 M is a realistic upper bound for contingency + land + overheads — but not an order of magnitude higher like legacy bids.

Risk register (civil-side)

Risk	Likelihood	Impact	Mitigation
Euphrates flood level rise	Medium	High	Bridge design 100-year flood +2 m (RFC 0011 §6).
Land-acquisition objections	Medium	Medium	Route follows existing ROW where possible; per-parcel negotiation.
Local fabricator capacity for precast U-girder	Low	Medium	One yard serves the whole project; 18-month production at 20 forms.
Dust-storm effect on construction schedule	High	Low	Pad ROW + prefab bay weather enclosure.
Groundwater depth variability at pier foundations	Medium	Medium	Geotech survey before each pier; foundation depth per-site.

docs/civil/west-asia/Iraq/Samawah/compliance-report.md

docs/civil/west-asia/Iraq/Samawah/compliance-report.md

Samawah alignment — standard - urban compliance report

Every segment on Line 1 and Line 2 checked against the RFC 0009 standard - urban preset's limits.

Preset reference (RFC 0009 §1)

Parameter	standard-urban limit
Gauge	1 435 mm
Minimum curve radius	90 m
Max gradient	50 ‰ (5 %)
Max cant	150 mm
Rail profile	UIC60 (60E2)
Nominal speed	22 m/s (80 km/h)
Compatible consists	tram-2car, light-metro-3car

Line 1 compliance

Seg	Radius (m)	Grade (‰)	Cant (mm)	Consist @ 22 m/s	Result
L1-01	n/a (straight)	0	0	light-metro-3car	
L1-02	26 300	0 (35 on ramps)	30	light-metro-3car	
L1-03	17 400	0 (35 on ramps)	30	light-metro-3car	
L1-04	12 900	0	0	light-metro-3car	
L1-05	32 500	10 (approach)	30	light-metro-3car	
L1-06	78 000	10 (descent)	0	light-metro-3car	
L1-07	> 1 000 000	0	0	light-metro-3car	
L1-08	> 1 000 000	0	0	light-metro-3car	
L1-09	> 1 000 000	0	0	light-metro-3car	
L1-10	> 1 000 000	0	0	light-metro-3car	
L1-11	68 000	0	0	light-metro-3car	

Line 1 result: PASS — all 11 segments compliant.

Line 2 compliance

Seg	Radius (m)	Grade (‰)	Cant (mm)	Consist @ 22 m/s	Result
L2-01	n/a (line-start)	0	0	light-metro-3car	
L2-02	9 500	0	0	light-metro-3car	
L2-03	12 000	0	0	light-metro-3car	
L2-04	4 300	0	20	light-metro-3car	(tightest curve; 47× preset min)
L2-05	10 200	0	0	light-metro-3car	
L2-06	15 000	0	0	light-metro-3car	
L2-07	5 800	0	20	light-metro-3car	
L2-08	10 500	0	0	light-metro-3car	
L2-09	5 900	0 (35 on ramps)	20	light-metro-3car	
L2-10	7 800	0	0	light-metro-3car	

Line 2 result: PASS — all 10 segments compliant.

Cant-deficiency check

Cant deficiency at design speed through each curved segment:

- Line 1: max cant deficiency = 22 mm (L1-02 at 22 m/s through 26 000 m radius with 30 mm applied cant). Well under the 100 mm normal deficiency limit for standard - urban.
- Line 2: max cant deficiency = 45 mm (L2-04 at 22 m/s through 4 300 m radius with 20 mm applied cant). Under the 100 mm normal limit.

Transition-curve check

Per RFC 0009 §5.1, every curve-tangent interface needs a clothoid transition of length $L \geq 0.5 \times \text{cant} \cdot \text{speed} / 1000$:

- At the 30 mm cant max on Line 1 at 22 m/s: $L \geq 0.5 \times 30 \times 22 / 1 = 330$ mm call it 2 m minimum.
- At the 20 mm cant on Line 2: ~220 mm 2 m minimum.

Every civil-firm alignment tool satisfies this by default. No segment needs bespoke transition-length handling.

Vertical-curve radius check

Per RFC 0009 §5.2, vertical curve radius $\geq 0.4 \times v^2$:

- At 22 m/s: $R \geq 0.4 \times 22^2 = 194$ m.

Applied to the viaduct ramps (35 % transitions):

- Ramp over 150 m horizontal with 5 m vertical rise = 29 % nominal with short crest/sag curves.
- Minimum sag radius at the ramp base: take the rule of thumb $R_v \geq 300$ m. Easy to accommodate.

Overall conclusion

Samawah Line 1 + Line 2 reference alignment is fully compliant with standard - urban preset. No segment requires a preset upgrade to standard - metro or a special exception.

Sensitivity analysis

If the detailed surveyed alignment lands tighter curves (e.g. the OSM-derived corridor of [osr-routing](#) produces, say, a 1 500 m apex at L2-04), the preset still supports it down to 90 m — 20× margin before preset limits bite. The Samawah reference is deliberately generous; actual deployment alignments can have much tighter curves in dense urban cores without stepping outside the preset.

What the report does NOT say

- Structural viaduct bending-moment calc — RFC 0011 §5 handles span envelopes; per-span FEA is v2 civil-engineering work.
- Rail expansion / CWR stress-free temperature — per-climate overlay, out of scope for this compliance check.
- Seismic detailing — $\text{PGA} \leq 0.3$ g zone for Samawah per Iraq national seismic code; below the RFC 0011 §11 baseline.
- Drainage + flood return period — RFC 0011 §§4.1 + 6 + the Euphrates 100-year flood design; per-site geotech drives the final pier-foundation depth.

docs/civil/west-asia/Iraq/Samawah/line1-segments.md

docs/civil/west-asia/Iraq/Samawah/line1-segments.md

Samawah Line 1 — per-segment alignment table

Line 1 (Nahrain) is a 12-station radial line, west to east. Every segment is standard - urban preset per RFC 0009 §1. Direction of travel in this table is eastbound (forward). Reverse-direction segments use a paired parallel track (double-track per RFC 0009 §7.2).

Station coordinates: from [design.toml](#). Bearings computed from great-circle deltas; horizontal radii at waypoints computed from inter-segment bearing change $R \approx (L1 + L2) / (2 \cdot \tan(\theta/2))$. Planning-grade numbers — a surveyed alignment substitutes GNSS coordinates at v2.

Segment table

#	From To	L (m)	Civil	Bearing (°)	Entry R (m)	Max grade (%)	Max cant (mm)	Design speed (m/s)	Notes
L1-01	samawah-rws north-gate	1 000	at-grade	128.7	line start	0	0	22	From intercity RWS; at-grade boulevard running parallel to Mosul St.
L1-02	north-gate old-souq	1 300	elevated	123.7	26 300	0	30	22	Enters dense-souq block; viaduct per RFC 0011 §5.
L1-03	old-souq samawah-central	1 100	elevated	115.8	17 400	0	30	22	Elevated through commercial core.
L1-04	samawah-central riverside	900	at-grade	106.9	12 900	0	0	22	Re-enters at-grade after ramp down from old-souq viaduct.
L1-05	riverside eastern-bridge	1 200	bridge	103.2	32 500	0	30	22	Euphrates crossing — reference viaduct over water per RFC 0011 §6.
L1-06	eastern-bridge al-salam	1 400	at-grade	105.1	78 000	0	0	22	East bank, dedicated ROW parallel to the highway.
L1-07	al-salam governorate-hospital	1 300	at-grade	105.2	> 1 000 000	0	0	22	Straight.
L1-08	governorate-hospital new-german-hospital	1 500	at-grade	104.9	> 1 000 000	0	0	22	Straight.
L1-09	new-german-hospital engineering-quarter	1 300	at-grade	103.9	> 1 000 000	0	0	22	Straight.
L1-10	engineering-quarter al-muthanna-university	1 400	at-grade	106.3	> 1 000 000	0	0	22	Straight.
L1-11	al-muthanna-university east-depot	600	at-grade	102.0	68 000	0	0	22	Depot throat + buffer zone.

Total L1 length: 13 000 m (matches design.toml). Civil mix: 9 500 m at-grade (73 %) + 2 400 m elevated (18 %) + 1 200 m bridge (9 %).

Waypoints (curve apex details)

All curves on Line 1 are at radius > 12 000 m — well above the standard - urban preset's 90 m minimum. Cant is 0 mm on the straight-running at-grade segments (no lateral acceleration to compensate). On the three viaduct segments (L1-02, L1-03, L1-05) 30 mm cant is specified at the transition curves to smooth ride quality; this is well under the 150 mm preset maximum.

Vertical alignment

Samawah sits on the Euphrates alluvial plain — elevation 6–10 m above sea level across the full alignment. Grade assumed 0 % everywhere; v2 surveyed alignment will refine with actual elevation data. The only vertical feature is the viaduct ramp- up /

ramp-down at L1-02 and L1-05 entries and exits:

- L1-02 entry ramp (north-gate viaduct): 0 ‰ at-grade 35 ‰ ramp over 150 m level deck at 10 m ToR.
- L1-03 exit ramp (viaduct samawah-central): level deck 35 ‰ descent over 150 m at-grade.
- L1-04 exit / L1-05 entry (riverside bridge): 10 ‰ approach grade, bridge deck level at 12 m ToR clearance over river.
- L1-05 exit (bridge at-grade on east bank): 10 ‰ descent over 200 m.

All ramp grades ≤ 50 ‰ (preset max), with vertical-curve radii $\geq 0.4 \cdot v^2 \text{ m} = 194 \text{ m}$ per RFC 0009 §6.

Earthworks estimate

At-grade segments sit in very flat terrain. Preliminary cut / fill volume budget:

- Samawah-rws eastern-bridge: minor cut ($\leq 1 \text{ m}$) for drainage.
- eastern-bridge east-depot: minor fill ($\leq 1 \text{ m}$) for ROW levelling.
- Viaduct segments: no earthworks (piers on pile foundations).

Stopping distance check (sanity)

The osr-atp envelope per light-metro-3car emergency-brake curve + EN 14363 headroom:

- From 22 m/s: stopping distance $\leq 175 \text{ m}$ on level track.
- L1-11 (600 m, from 22 m/s into depot - terminal): plenty of brake distance.
- Every inter-station segment $\geq 600 \text{ m}$: all allow a full-speed start + full-speed stop with margin.

Compliance summary (vs standard - urban)

Every segment passes the preset's constraints:

- Min curve radius 90 m: all Line 1 curves $\geq 12\,000 \text{ m}$
- Max grade 50 ‰: ramp grades ≤ 35 ‰
- Max cant 150 mm: max used 30 mm
- Design speed 22 m/s: all segments

Samawah Line 2 — per-segment alignment table

Line 2 (Halqa) is a 10-station ring line, counterclockwise from Eastern Bridge. Every segment is standard -urban preset per RFC 0009. Two interchange points with Line 1: eastern-bridge and al-muthanna-university.

Station coordinates: from [design.toml](#).

Segment table

Ring sequence: eastern-bridge northern-suburbs-a northern-suburbs-b northwest-junction industrial-west western-residential south-west-residential southern-markets south-east-residential al-muthanna-university (ring wrap) eastern-bridge.

#	From To	L (m)	Civil	Bearing (°)	Entry R (m)	Max grade (%)	Max cant (mm)	Design speed (m/s)	Notes
L2-01	eastern-bridge northern-suburbs-a	1 800	at-grade	28	ring start	0	0	22	Northbound from L1 interchange.
L2-02	northern-suburbs-a northern-suburbs-b	1 500	at-grade	324	9 500	0	0	22	Bearing swings NW — the first ring-curve apex.
L2-03	northern-suburbs-b northwest-junction	1 700	at-grade	297	12 000	0	0	22	Continues NW, reaches northwest junction layup-minimal.
L2-04	northwest-junction industrial-west	1 600	at-grade	240	4 300	0	20	22	Ring apex — tightest curve; 4.3 km radius is still > 47× preset min.
L2-05	industrial-west western-residential	1 500	at-grade	198	10 200	0	0	22	Southbound.
L2-06	western-residential south-west-residential	1 800	at-grade	180	15 000	0	0	22	Due south.
L2-07	south-west-residential southern-markets	1 400	at-grade	108	5 800	0	20	22	Turns east.
L2-08	southern-markets south-east-residential	1 600	at-grade	90	10 500	0	0	22	Due east.
L2-09	south-east-residential al-muthanna-university	1 500	elevated	36	5 900	0	20	22	Approach to L1/L2 interchange; elevated over the engineering-quarter block. Per RFC 0010 §7 interchange = at-grade (L1) + elevated (L2).
L2-10	(ring wrap) al-muthanna-university eastern-bridge	1 600	at-grade	286	7 800	0	0	22	Closes the ring. Dedicated ROW returns at-grade.

Total L2 length: 16 000 m (matches design.toml). Civil mix: 14 500 m at-grade (91 %) + 1 500 m elevated (9 %) + 0 m bridge.

Waypoints (curve apex details)

Line 2 has tighter curves than Line 1 because the ring geometry forces direction reversals. The tightest is L2-04 at 4 300 m radius — still 47× the preset 90 m minimum. At 22 m/s through a 4 300 m curve, centripetal acceleration is 0.11 m/s² — imperceptible. A 20 mm cant at the tightest curves is a ride-quality nicety, not a speed-safety requirement.

Vertical alignment

Same assumption as Line 1: level terrain, grade 0 % on all at-grade segments. Ring-apex curves might need sub-1 % drainage slope; v2 surveyed alignment refines.

L2-09 (elevated to the interchange): 35 ‰ ramp up over 150 m at the approach, level 10 m ToR deck at the al-muthanna-university platform, 35 ‰ ramp down on the far side back to at-grade for the ring wrap.

Interchange compliance (L1/L2 at eastern-bridge + al-muthanna-university)

Per RFC 0010 §7 the interchange archetype is stacked at-grade (L1) + elevated (L2):

- eastern-bridge: Line 1 crosses at bridge (over the Euphrates) at +12 m ToR. Line 2 approaches at at-grade. Interchange vertical circulation handles the ~12 m height delta (ramp + stair + escalator + lift, per RFC 0010 §5).
- al-muthanna-university: Line 1 at-grade at +0 m ToR; Line 2 at elevated +10 m ToR. Interchange has the same vertical- circulation kit.

Both are RFC 0011 §7 compliant (never two elevated levels).

Stopping distance check

Same envelope as Line 1. Every L2 segment $\geq 1\,400$ m is plenty for the 175 m emergency-brake distance from 22 m/s.

Compliance summary (vs standard-urban)

- Min curve radius 90 m: all Line 2 curves $\geq 4\,300$ m
- Max grade 50 ‰: ramp grades ≤ 35 ‰
- Max cant 150 mm: max used 20 mm
- Design speed 22 m/s:

docs/cost-model.md

docs/cost-model.md

OpenSourceRail Cost Model

This file is the audit trail for the planning-grade costs emitted into each city design .toml and README. It is intentionally conservative about what belongs in each bucket so the same function is not paid for twice.

Rolling Stock

Rolling stock is budgeted at €1.0 M per self-contained car. A trainset is simply cars $\times$ €1.0 M.

Bucket	Base	Low	High	Notes
Body shell, glazing, doors, interior	€300 k	€240 k	€420 k	Welded steel frame, composite cladding, COTS doors/windows/interior
Bogies and brakes	€220 k	€180 k	€320 k	One powered bogie plus one trailer bogie per car
Traction package	€180 k	€140 k	€280 k	PMSM motors, gearbox, SiC inverter, cooling, HV contactors
Battery and BMS	€120 k	€90 k	€180 k	120 kWh usable under-seat Na-ion pack
Driverless onboard stack	€90 k	€70 k	€160 k	T-ECU/S, T-ECU/A, T-OBS sensors, radios, event recorder
HVAC, auxiliaries, QA margin	€90 k	€70 k	€160 k	HVAC, lighting, PIS, wiring, assembly QA
Total	€1.0 M	€790 k	€1.52 M	Base is the catalogue value

The base value assumes direct procurement, local cut/bend/weld final assembly, common bogie modules, composite non-structural cladding, COTS doors/windows/HVAC/interior modules, open control electronics, and no proprietary CBTC onboard bundle. The high value is a risk envelope for first-article procurement or low-volume import-heavy builds.

Charging Microgrids

There is no route traction-power system in the OSR baseline: no OCS, third rail, feeder substations, or continuous traction distribution along the railway. The energy infrastructure cost in city designs is therefore station/depot charging microgrid interface CAPEX.

Station archetype	Unit cost	Included scope
halt	€125 k	250 kW class charger, local protection, compact LV tie
standard	€250 k	500 kW class conductive charger, switchgear, inverter interface
major	€400 k	Larger queueing/anchor-stop charger and buffer tie
terminal	€400 k	End-of-line charger with higher turnback utilization
interchange / interchange-elevated	€600 k	Multi-platform charger/switchgear allowance
depot-terminal	€750 k	Passenger-stop charger plus depot/yard charging interface

Station PV canopies, large stationary Na-ion packs, depot buildings, and train batteries are not re-billed here. They appear in station, energy-site/depot, and rolling-stock scopes respectively.

Train-Control Wayside

Residual train-control wayside is budgeted at €15 k per route-km. The expensive ATP/ATO function lives onboard in the trainset cost. The wayside scope is sparse W-Nodes at switches/stations, passive balises, validation beacons, LoRa gateways, and OCC interfaces.

EPC

EPC integration and project management is 7% of subtotal:

```
civil + stations + depots + rolling_stock
+ residual_train_control_wayside + charging_microgrids
```

Country labour/material multipliers are applied downstream through lib/templates/country-costs.toml.

docs/hardware/bring-up/README.md

docs/hardware/bring-up/README.md

Hardware bring-up — v1 deliverable of RFC 0007

This directory holds the bring-up procedures for the four physical host classes of OpenSourceRail. Each file is a step-by-step runbook an engineer with the relevant dev kit can execute and produce a bring-up report at the end of.

Per [RFC 0007](#) Raspberry Pi + Radxa only:

Class	Dev kit	Runbook
T-ECU/S safety MCU	Raspberry Pi Pico 2 (RP2350) × 2	t-ecu-s.md
T-ECU/A application	Raspberry Pi CM5 IO Board	t-ecu-a.md
W-SBC wayside	Radxa CM5 IO Board	w-sbc.md
S-SBC station	Raspberry Pi CM5 + Waveshare CM5-IO	s-sbc.md

Reporting format

Each runbook ends with a bring-up report template. After completing the steps, copy the template into the deployment's own tree (e.g. deployments/<country>/<city>/hardware/bring-up/) and fill in the pass/fail results per-step. The template carries the commit hash of the runbook so a later auditor can reconstruct what was tested.

What a pass means

A dev-kit bring-up is "passed" when every mandatory step produces the expected outcome. Failures are classified:

- FAIL-HW — the hardware itself is faulty or the vendor dev kit is DOA. Replace and retry.
- FAIL-SW — the OSR Rust stack has a bug on the target platform. File an issue against the corresponding crate.
- FAIL-DOC — the runbook is wrong / incomplete. File an issue against this doc.

No step is optional unless explicitly marked (`optional`). Skipping mandatory steps invalidates the bring-up.

Where this sits in the roadmap

These procedures are RFC 0007 v1 ([rollout table](#)). v2 of the hardware rollout is the custom baseboard schematics; v3 is compliance testing. Everything here is prelude — making sure the vendor silicon and the Rust stack meet before we commit a baseboard design.

docs/hardware/bring-up/s-sbc.md

docs/hardware/bring-up/s-sbc.md

S-SBC bring-up — RPi CM5 on Waveshare CM5-IO

Goal: confirm the station / depot host runs `osr-psd + osr-afc + osr-tvm + osr-pis-station + osr-station-scada` on a stock commodity carrier, with the small ATECC608 add-on as the only custom-PCB piece.

Dev kits + parts

Part	Qty	Notes
Raspberry Pi CM5 (4 GB + 32 GB eMMC)	1	
Waveshare CM5-IO (commodity)	1	
ATECC608 breakout on I ² C	1	The only custom PCB — a 30 × 20 mm carrier for the SE chip.
QR scanner (USB)	1	
NFC reader (USB, ACR122U or equivalent)	1	
Thermal receipt printer (USB)	1	
10-inch HDMI screen	1	PIS display.
Test bench	—	

Steps

X1 — Boot Debian

Same as T-ECU/A step A1, same image, same hostname pattern (`ssbc-devkit`). No RT configuration — station role is SIL-0.

X2 — Workspace build

Subset build: station-side crates only.

```
cargo build --release -p osr-psd -p osr-afc -p osr-tvm \
  -p osr-pis-station -p osr-station-scada
```

X3 — ATECC608 add-on enumeration

```
i2cdetect -y 1          # see ATECC608 at 0x60
```

Expected: chip responds. Provision a test key via the `osr-crypto` provisioning tool.

X4 — Fare gate (`osr-afc`)

Connect QR scanner + NFC reader to USB. Run the AFC gate simulator:

```
cargo run --release -p osr-afc --example gate-demo
```

Present a test QR code; present a test NFC token.

Expected: both decode + produce `FareAccepted` events within 100 ms.

X5 — Ticket vending (`osr-tvm`)

Connect the receipt printer. Run:

```
cargo run --release -p osr-tvm --example tvm-demo
```

Drive a scripted purchase: scan product QR, submit mock mobile- money payment, confirm printed receipt.

Expected: receipt prints within 3 s of payment confirmation.

X6 — Passenger information (`osr-pis-station`)

Connect the HDMI screen. Run:

```
cargo run --release -p osr-pis-station --example display-demo
```

Expected: next-arrival list displays; mock incoming MA updates refresh the arrival times.

X7 — PSD controller (`osr-psd`)

With no physical PSD on the bench, use the crate's software-only simulator. Run:

```
cargo run --release -p osr-psd --example psd-demo
```

Expected: the five-state PSD FSM transitions correctly on simulated train-arrival + door-open / close events.

X8 — Station SCADA

Launch:

```
cargo run --release -p osr-station-scada --example scada-demo
```

Simulate HVAC / lighting / escalator status inputs.

Expected: status dashboard populates; threshold alarms fire within one tick of exceeding setpoint.

X9 — Sustained run

Run all five station crates simultaneously for 1 hour.

Expected: CPU load $\leq 30\%$, RAM ≤ 1 GB, no process restarts.

Bring-up report template

```
# S-SBC bring-up report
```

```
- Runbook commit: <hash>
- Date: <YYYY-MM-DD>
- CM5 serial: <nnn>
```

```
## Results
```

Step	Status
X1 (boot)	PASS / FAIL-*
X2 (build)	PASS / FAIL-*
X3 (ATECC608)	PASS / FAIL-*
X4 (AFC gate)	PASS / FAIL-*
X5 (TVM)	PASS / FAIL-*
X6 (PIS)	PASS / FAIL-*
X7 (PSD)	PASS / FAIL-*
X8 (SCADA)	PASS / FAIL-*
X9 (1 h run)	PASS / FAIL-*

docs/hardware/bring-up/t-ecu-a.md

docs/hardware/bring-up/t-ecu-a.md

T-ECU/A bring-up — Raspberry Pi CM5 IO Board

Goal: validate the application-tier host runs every non-safety onboard crate and the TCN-E + TRG radios enumerate on a stock CM5 carrier.

Dev kits + parts

Part	Qty	Notes
Raspberry Pi CM5 (4 GB + 32 GB eMMC)	1	
Raspberry Pi CM5 IO Board	1	
M.2 Hat+ (Waveshare or Pimoroni)	1	For NVMe + 5G modem.
NVMe SSD 256 GB	1	Event recorder ring buffer target.
Cat.22 5G M.2 module (unlocked)	1	TRG-1 primary radio.
LoRa SX1276 breakout on SPI	1	TRG-2 backup radio.
Mikroe CAN-FD Click 3 (x2)	2	Onto the IO Board's mikroBUS slots.
USB-C PD 5V/3A supply	1	

Steps

A1 — Boot mainline Debian

- 1 Write Debian 13 arm64 (with RPi kernel) to the CM5 eMMC via rpiboot.
- 2 First-boot: set hostname tecus-devkit, expand rootfs to NVMe.
- 3 Install Rust toolchain: `curl ... | sh, rustup default stable`.

Expected: ssh login within 60 s of power-on.

A2 — Clone + build the workspace

```
git clone https://github.com/OpenSourceRail/OpenSourceRail.git
cd OpenSourceRail
cargo build --release --workspace
```

Expected: clean build; 46 crates compile. Wall-clock ≤ 20 min on a 4 GB CM5.

A3 — Enumerate peripherals

Confirm each peripheral shows up in the kernel:

```
ip link show          # two Ethernet interfaces
ls /sys/class/net     # eth0, eth1
lspci                 # NVMe, 5G M.2
ls /dev/ttyAMA*       # UARTs
ls /dev/spidev*       # SPI buses
ls /sys/bus/i2c/devices # I2C (TCN-E + ATECC608)
```

Tick each off in the report. Missing devices check device-tree overlay; common culprit is missing `dtoverlay=pcie2-lane` or equivalent in `config.txt`.

A4 — TCN-E mock loopback

```
cargo test -p osr-tcn --lib -- --nocapture
cargo test -p osr-tcn --test roundtrip_single_payload -- --nocapture
```

Expected: both green. The UDP-over-loopback path (RFC 0006 v1.5) is the surrogate for the eventual TSN link.

A5 — TRG-1 (5G) bring-up

Insert the 5G module into the M.2 slot. Install ModemManager and NetworkManager.

- 1 Confirm `mmcli -L` lists the modem.
- 2 Activate a test APN: `mmcli -m 0 --simple-connect="apn=..."`.
- 3 Confirm IPv4 connectivity to a public endpoint.

Expected: working PDP context within 60 s of modem boot.

Carrier choice: bring-up is carrier-agnostic. For deployment use, the `osr-t2g` SIM-profile spec ([RFC 0005 §3.2](#)) handles the operator's choice.

A6 — TRG-2 (LoRa) bring-up

Connect SX1276 breakout to SPI0.

- 1 Build the `osr-t2g` diagnostic binary.
- 2 Run `osr-t2g-diag --link lora --tx-power 14dbm --channel 868.1`.
- 3 Confirm a second LoRa node (can be another dev board) receives the test packets with $\text{RSSI} \geq -110$ dBm at 50 m line-of-sight.

Expected: ≥ 99 % packet delivery at 10 Hz test rate.

A7 — CAN-FD bus test

Install two CAN-FD Clicks on the mikroBUS slots. Connect via a CAN-FD transceiver test harness.

- 1 `ip link set can0 up type can bitrate 500000 dbitrate 2000000 fd on`.
- 2 `cansend can0 100##1.01020304`.
- 3 `candump can0` on the second Click receives the frame.

Expected: no frame loss at 1 kHz for 5 minutes.

A8 — Shadow onboard stack in `osr-sim`

From the workspace:

```
cargo run --release --bin osr-sim -- \
  --scenario samawah-line1 --duration 900 --status-every 60
```

Expected: the shadow onboard stack ran on every tick on the CM5, consuming ≤ 20 % CPU, with zero invariant violations.

A9 — Event recorder

```
osr-event-recorder-cli --tail 100 /var/log/osr/event.bin
```

Expected: the last 100 TCN-E frames from step A4 are extractable. Ring buffer is working.

Bring-up report template

```
# T-ECU/A bring-up report

- Runbook commit: <hash>
- Date: <YYYY-MM-DD>
- CM5 serial: <nnn>

## Results
| Step | Status |
| --- | --- |
| A1 (Debian boot) | PASS / FAIL-\* |
| A2 (workspace build) | PASS / FAIL-\* |
| A3 (peripherals) | PASS / FAIL-\* |
| A4 (TCN-E loopback) | PASS / FAIL-\* |
| A5 (5G) | PASS / FAIL-\* |
| A6 (LoRa) | PASS / FAIL-\* |
| A7 (CAN-FD) | PASS / FAIL-\* |
| A8 (sim shadow) | PASS / FAIL-\* |
| A9 (event recorder) | PASS / FAIL-\* |
```

docs/hardware/bring-up/t-ecu-s.md

docs/hardware/bring-up/t-ecu-s.md

T-ECU/S bring-up — two Raspberry Pi Pico 2 boards in 2oo2

Goal: validate that the [RFC 0007 §4](#) 2oo2 safety architecture works on real silicon before we draw the custom baseboard. After this runbook, a pair of Pico 2 boards runs `osr-atp` + `osr-brake` + `osr-odometry` in composite fail-safe mode with per-tick SPI cross-check.

Dev kits + parts

Part	Qty	Notes
Raspberry Pi Pico 2	2	Each carries one RP2350 — the safety MCU.
ST-Link V3 or Pico probe	2	SWD debugger per Pico.
Breadboard + jumpers	1 set	SPI cross-link + shared inputs.
3-phase quadrature encoder sim	1	Can be an Arduino-generated square-wave pair.
BMI088 breakout	1	Feeds both Picos via separate SPI chip selects.
24 VDC bench supply	1	Via an ADuM isolator IC for input simulation.
Relay 2oo2 stage (optional)	1	Two SPDT relays in AND — validates the final actuator gating.

Steps

S1 — Build the harness

Wire each Pico to:

- Shared inputs (both Picos read the same signal, on separate pins): one pulse input (sim wheel tach), one digital input (sim emergency plunger), IMU chip-select line.
- Private SPI cross-check link: Pico-A SPI0 Pico-B SPI0 (CLK, MOSI, MISO, CS). ~1 MHz clock.
- Debug UART: each Pico's UART0 to a USB-serial adapter for log capture.

Write the topology into the bring-up report at step R0.

S2 — Flash the Pico test firmware

From the workspace:

```
cargo build --release --target thumbv8m.main-none-eabihf \
  --bin tecus-bringup
```

(The `tecus-bringup` binary is a dev-only test harness in `deployments/devkit/tecus-bringup/` — planned for landing alongside this runbook; for v1 of this document the binary is a future session's deliverable.)

Flash each Pico with the same binary via SWD. Both Picos boot into the identical code path; 2oo2 runs at identical-software redundancy.

Expected: green LED blinks at 1 Hz on both Picos. Red LED off.

FAIL-HW if either Pico fails to boot: replace Pico. FAIL-SW if the binary fails to flash: check toolchain pin.

S3 — Verify the SPI cross-check

Connect a logic analyser between the two Picos on the SPI lines.

- Power both Picos.
- In the analyser, expect a message exchange every 10 ms: frame format is `[seq: u32][osr_atp_output: u32][osr_brake_output: u32][crc16: u16]`.
- Ground-truth: both messages must match byte-for-byte after the first 5 exchanges.

PASS: 100 consecutive 10 ms frames match.

FAIL-SW if frames diverge: inspect the `tecus-bringup` log output via UART; there will be a divergence report.

S4 — Simulate an input disagreement

On the harness, break one of the shared input lines — e.g. cut the wire to Pico-B's wheel-tach input.

Expected within 100 ms: both Picos:

- 1 Halt their computed brake output (emit `BrakeCommand : : Emergency`).
- 2 Drop their relay-drive outputs.
- 3 The relay 2oo2 stage (if wired) opens, releasing the test brake pressure.
- 4 Both Picos' red LEDs solid.

PASS: emergency-brake actuation within 100 ms of the input disagreement.

FAIL-SW if either Pico keeps driving its output: bug in the cross-check invariant — file against `osr-atp + osr-vigilance`.

S5 — Verify the wheel-tach input

Drive the shared pulse input at 500 Hz (≈ 4.2 m/s with a 410 pulses/m calibration).

Expected: UART log from both Picos shows `odom.speed_mmps  $\approx 4200$` ; the two speeds must match within 50 mm/s.

FAIL-SW if speeds diverge > 50 mm/s with identical input: likely a timing-quantization bug — file against `osr-odometry`.

S6 — Verify the emergency plunger

Assert the sim plunger input.

Expected: both Picos immediately command `BrakeCommand : : Emergency` on their output pins. The 2oo2 AND gate in the harness sees both high drives the relay drop test brake applies.

PASS: relay drop latched within 50 ms.

S7 — CM5 handshake (optional)

If an RPi CM5 IO Board is also present, wire an SPI-slave link between Pico-A and the CM5. Boot the CM5 with the app-side bring-up test binary.

Expected: CM5 receives the same per-tick cross-check messages over SPI and logs them in JSON to `/var/log/osr/ecus.log`. This proves the one-way safety app data path ([RFC 0007 §4.2](#)).

S8 — Performance envelope

Run the 2oo2 cross-check harness for 1 hour. Measure:

- Cross-check message rate: must stay at $100 \text{ Hz} \pm 1 \text{ Hz}$.
- Missed frames: 0.
- Cross-check mismatches: 0.
- Average tick jitter: $\leq 1 \text{ ms}$.

A 1-hour clean run is the "boot certifiable" bar for the v2 baseboard proceed-gate.

Bring-up report template

```
# T-ECU/S bring-up report

- Runbook commit: <hash>
- Date: <YYYY-MM-DD>
- Engineer: <name>
- Pico A serial: <nnn>
- Pico B serial: <nnn>

## Results
| Step | Status |
| --- | --- |
| S1 (harness) | PASS / FAIL-\* |
| S2 (flash) | PASS / FAIL-\* |
| S3 (cross-check) | PASS / FAIL-\* |
| S4 (input disagreement) | PASS / FAIL-\* |
| S5 (wheel tach) | PASS / FAIL-\* |
| S6 (plunger) | PASS / FAIL-\* |
| S7 (CM5 handshake) | PASS / FAIL-\* / N-A |
| S8 (1 h run) | PASS / FAIL-\* |
```

```
## Notes
<observations, anomalies, links to logs>
```

docs/hardware/bring-up/w-sbc.md

docs/hardware/bring-up/w-sbc.md

W-SBC bring-up — Radxa CM5 IO Board

Goal: validate the wayside host runs `osr-consensus` + `osr-interlocking` + `osr-wayside-points` on a RK3588S Radxa CM5, with the A55 little cluster isolated for the SIL-4 partition and the A76 big cluster carrying non-safety services.

Dev kits + parts

Part	Qty	Notes
Radxa CM5 (8 GB + 32 GB eMMC, industrial-temp)	1	
Radxa CM5 IO Board	1	
SparkFun RS-485 breakout	2	One for switch-motor sim, one for HABD sim.
Test bench: 5-node consensus cluster	—	4 more CM5 carriers OR a docker-compose simulator.
24 VDC bench supply via ADuM isolator	1	

Steps

W1 — Boot Debian with RK3588S BSP

- 1 Write Radxa Debian 13 image to eMMC via `rkdeveloptool`.
- 2 Expand rootfs, ssh login.
- 3 Update `/etc/default/cpufreq`: pin A55 cluster to 1.4 GHz, A76 to `ondemand`.
- 4 Install Rust toolchain (same as T-ECU/A).

Expected: ssh login within 90 s of power-on.

W2 — Workspace build

Same as T-ECU/A step A2. Build times on 8 GB A76 ~10 min.

W3 — RT partition setup (safety role)

Configure the kernel for `PREEMPT_RT` on A76 and the A55 cluster isolated for the safety partition:

```
isolcpus=0-3      # reserve A55 cores for safety
nohz_full=0-3     # tickless on safety cores
rcu_nocbs=0-3
```

Reboot. Verify `/proc/cmdline` contains the flags and `htop` shows no processes scheduled on CPU 0-3.

W4 — Consensus cluster bring-up

This step proves the W-SBC can host a consensus replica.

- 1 On the dev kit, run a local 3-node simulated cluster:
- 2 Confirm leader election within 1 s.
- 3 Drive 100 proposals through; confirm all commit on every follower.
- 4 Kill one node; confirm re-election and continued progress.

Expected: clean leader election + commit progression + crash recovery on the RK3588S silicon.

W5 — Interlocking on the safety partition

Pin `osr-interlocking`'s MA computer to the isolated A55 cores:

```
taskset -c 0-3 cargo run --release --bin osr-sim -- \
  --scenario samawah --use-consensus --duration 600
```

Expected: the 10-minute Samawah run completes with zero invariant violations; the MA computer's tick jitter (measured via `osr-sim`'s internal timing) ≤ 5 ms.

W6 — Wayside-points driver over RS-485

- 1 Connect one SparkFun RS-485 to a motor-driver sim (can be an Arduino reading the bus and toggling a digital output).
- 2 Build the `osr-wayside-points` standalone:
- 3 Send alternating `SwitchCommand::Normal` / `SwitchCommand::Reverse` every 5 seconds.

Expected: motor drive toggles faithfully, detection (simulated end-of-travel sensors) mirrors the command within the 5 s RFC 0012 §4.2 throw budget.

W7 — HABD input over RS-485

Simulate a hot-axle reading (raw temperature values) over the second RS-485 link. Build the `osr-hot-axle-wayside` driver:

```
cargo run --release -p osr-hot-axle-wayside --example demo
```

Expected: temperature thresholding fires at configured setpoint; the crate emits a `SpeedRestriction` event to the consensus log simulator.

W8 — Industrial-temp headroom

Place the Radxa CM5 in a 65 °C test oven (pre-deployment IP67 cabinet simulator) for 30 minutes with the W5 workload running.

Expected: no throttling, no CPU errors, no log anomalies. RK3588S junction temp stays < 90 °C per the on-die sensor.

FAIL-HW if the board throttles or errors: the industrial- temp variant may not be the one installed; confirm the P/N.

W9 — PoE or dual-PSU input (optional)

If the cabinet-form-factor of the IO Board is present, validate the dual-24V redundancy:

- 1 Run both PSU inputs; cut one; verify no service interruption.
- 2 Re-apply; confirm re-detection.

Expected: < 5 ms dropout at changeover.

Bring-up report template

```
# W-SBC bring-up report
```

```
- Runbook commit: <hash>
- Date: <YYYY-MM-DD>
- Radxa CM5 serial: <nnn>
```

```
## Results
```

Step	Status
W1 (boot)	PASS / FAIL-*
W2 (build)	PASS / FAIL-*
W3 (RT + isol)	PASS / FAIL-*
W4 (consensus)	PASS / FAIL-*
W5 (interlocking)	PASS / FAIL-*
W6 (points)	PASS / FAIL-*
W7 (HABD)	PASS / FAIL-*
W8 (thermal)	PASS / FAIL-*
W9 (dual-PSU)	PASS / FAIL-* / N-A

docs/operations/README.md

docs/operations/README.md

OpenSourceRail — Operations Rulebook

This directory holds the rule text referenced by [RFC 0013](#). Every rule sits inside the one-sentence-one-decision style the RFC commits to.

Target length, rendered PDF: ≤ 60 pages. If the full compiled book exceeds that, something gets cut before anything is added.

Layout

Path	Audience	RFC 0013 section
driver/	Train driver	§4.1 (D1–D8)
dispatcher/	OCC dispatcher	§4.2 (S1–S6)
station-staff/	Station agent / inspector	§4.3 (T1–T5)
maintenance/	Depot + MOW worker	§4.4 (M1–M6)
control-centre/	OCC supervisor / engineer	§4.5 (C1–C3)

Each subdirectory has one file per rule block, numbered per the RFC. Files are currently stubs with the section header, scope statement, and a `rules :` placeholder. Rule text lands as the v2 deliverable of RFC 0013.

How to add a rule

- 1 Find the right block (e.g. D3 — In-service operation).
- 2 Append a numbered rule at the bottom of that file in the template:
- 3 The numbering is `<block> . <n>`; next `n` = the last +1.
- 4 Every rule has exactly one author — credit in the commit message, not the rule body.

Non-mission-critical reminders for contributors

- One sentence per rule. Compound rules break testability and become accretion. Split.
- Every rule has a `Why :`. An unexplained rule rots. The `Why :` block is a contract with the future driver who must follow the rule.
- Cite the RFC / safety-case solution the rule supports. This is how the rulebook feeds the GSN safety case (`docs/safety-case/gsn/`).

Language policy (RFC 0013 §13)

- English is the canonical rulebook.
- Per-country translations land under `docs/operations/<iso-lang>/` as a parallel tree once the first deployment needs one. Until then, only English is authoritative.

docs/operations/control-centre/c1-occ-watch-roles.md

docs/operations/control-centre/c1-occ-watch-roles.md

C1 — OCC watch roles

Scope: the three on-shift roles: dispatcher, supervisor, engineer-on-call.

Cross-refs: RFC 0013 §4.5 C1, [osr-occ](#).

Rules

C1.1 — Dispatcher

The dispatcher owns all dispatching decisions for the shift: route grants, timetable adherence, headway regulation, driver radio authorisations, work-block coordination. The dispatcher's authority ends at override approval and incident escalation.

Why: the dispatcher is the operational executive — fast decisions, clear authority. Splitting dispatching across multiple shift personnel creates decision collisions (two trains authorised onto the same block).

C1.2 — Supervisor

The supervisor approves MaintenanceOverride extensions beyond S5.4 defaults, handles NRSA and media liaison, countersigns incident reports, and makes the call on degraded-mode transitions. The supervisor does not dispatch except as the dispatcher's relief during short breaks.

Why: the supervisor is the quasi-legal authority on shift — the person accountable to the regulator, to whom the dispatcher escalates. Keeping the supervisor clear of dispatching prevents decision-role collisions at the moment an incident demands both.

C1.3 — Engineer-on-call

The engineer-on-call handles software and hardware alerts that exceed the dispatcher's scope: consensus-cluster degradations, [osr](#) - routing / [osr](#) - interlocking anomalies, wayside-points drift alarms. Engineer-on-call is a 24×7 phone rotation, escalation target for the dispatcher.

Why: the dispatcher is not a systems engineer. Routing a technical alert to a named on-call engineer moves the diagnosis to the person trained for it without blocking the dispatcher's operational duties.

C1.4 — Role rotation

Shifts rotate on a 12-hour cadence with a 15-minute overlap for the handover. The handover overlap is paid time and is not compressible — a late-arriving relief does not shorten the overlap.

Why: 12 hours matches the circadian pattern for alertness (with the 8-hour extension used in some deployments); the 15-minute overlap is the floor for the S1/S6 and C3 handover protocols to complete without handover shortcuts.

C1.5 — Minimum qualification

Each role has a minimum qualification: dispatcher (certified by the NRSA-equivalent after 6 months floor training + exam), supervisor (3 years dispatcher experience

- supervisor exam), engineer-on-call (software-engineer role
- deployment-specific rotation). No role may be filled by someone below the minimum.

Why: qualification is the hard floor of competence. An under-qualified supervisor making the NRSA call is how incidents become scandals; an under-qualified dispatcher is how incidents happen.

C1.6 — Fatigue policy

Dispatcher slots are capped at 10 hours; a Category I5 or I6 incident triggers a mandatory 30-minute break after the immediate response phase, with supervisor relief. Double shifts are prohibited.

Why: fatigued dispatchers miss radio calls and make timetable errors. The 10-hour cap and post-incident break is the compensating control — recovery time when it matters most.

docs/operations/control-centre/c2-communications.md

docs/operations/control-centre/c2-communications.md

C2 — Communications

Scope: radio discipline, PA authority, external-party notification (NRSA, emergency services, press liaison).

Cross-refs: [osr-t2g](#), [osr-pis-onboard](#), [osr-pis-station](#), RFC 0013 §4.5 C2.

Rules

C2.1 — Radio discipline

Radio transmissions open with call sign (dispatcher + unit being called), cover one topic per transmission, and use the standard phraseology (Proceed / Stop / Hold / Recall / Rescue). Ambiguous phrases ("kinda go" / "maybe stop") are rephrased on the spot.

Why: radio is the fallback when everything else is broken. Standard phraseology removes ambiguity when stress is high and accents vary across a multi-nationality operator workforce.

C2.2 — PA authority

PA authority is partitioned: the dispatcher owns network- wide PA (all stations or all trains simultaneously), station staff own station-local PA, drivers own in-train PA. A role may not make a PA outside its partition without handover.

Why: overlapping PAs produce cacophony at the station where a passenger trying to understand one announcement gets cut off by another. The partition matches authority to audience.

C2.3 — NRSA notification

NRSA notification windows: Category I5 (derailment, collision) within 2 hours; Category I6 (mass casualty) immediately; Categories I3 and I4 within 24 hours; I1 and I2 are consolidated into the monthly safety summary. The supervisor makes the call; the dispatcher does not.

Why: NRSA windows are regulatory — missing them is an operator-level finding. Centralising on the supervisor ensures each window is tracked once, not N times by N dispatchers each assuming the other called.

C2.4 — Emergency services coordination

Emergency-services coordination (fire, ambulance, police) is through the OCC supervisor as the single point of contact. The dispatcher may make the initial 112 / 999 call (per deployment dialling) but hands the ongoing liaison to the supervisor within the first 5 minutes.

Why: fire chiefs and ambulance incident commanders need one rail-side point of contact; a dispatcher juggling services calls and train dispatching does neither well. The handover within 5 minutes keeps the initial speed without compromising continuity.

C2.5 — Press and social media

Press and social-media statements are made only by an authorised operator-level spokesperson (CEO, press officer, or nominated deputy). OCC staff do not make public statements, including on personal social media, about incidents in progress or under investigation.

Why: in-progress incident statements by line staff contradict themselves (because information is incomplete) and are quoted against the operator for years. The centralised spokesperson is the standard mitigation.

C2.6 — International coordination

Incidents involving foreign nationals or cross-border operations (rare in Samawah but possible in other deployments) are coordinated through the supervisor with a qualified translator on the bridge, including for consular notifications.

Why: consular notifications and cross-border regulator contacts have specific diplomatic protocol that is outside the dispatcher's remit. The supervisor + translator pairing is the compensating control.

docs/operations/control-centre/c3-shift-handover.md

docs/operations/control-centre/c3-shift-handover.md

C3 — Shift handover

Scope: written log, outstanding items, on-call list for the incoming watch.

Cross-refs: S6 (dispatcher handover), RFC 0013 §4.5 C3.

Rules

C3.1 — Written handover log

The OCC handover log covers, for every shift: open incidents (with category and owner), active MaintenanceOverride entries, active speed restrictions, weather overlays, fleet status (revenue / maintenance / reserve), and any CBM red alerts. The log is created before the handover and reviewed by both outgoing and incoming watch.

Why: writing the log forces the outgoing watch to synthesise the shift; reading it forces the incoming watch to absorb. Verbal-only handovers lose details; the written log is the persistent record.

C3.2 — On-call engineer list

The on-call engineer list is updated for the incoming shift: name, phone, escalation backup. An engineer going off-call confirms the next engineer has accepted handover before going off-duty.

Why: "I thought X was on call" is the classic escalation failure. The explicit acceptance step closes the gap.

C3.3 — Overnight reduced staffing

Overnight watches run on reduced staffing: one supervisor

- one dispatcher minimum; the engineer-on-call is phone- available but not on-site. Overnight dispatcher scope is limited to monitoring, alarm response, and the M2 work- block authorisation for overnight possessions.

Why: overnight activity is low (no revenue service, limited movements); full-staff daytime structure wastes the labour. The minimum crew keeps the watch safe-walked without over-staffing.

C3.4 — I5/I6 full brief

Any Category I5 or I6 incident ongoing across a shift boundary gets a full brief at handover — not summarised, not presumed-known. The outgoing supervisor briefs the incoming supervisor verbally for at least 15 minutes.

Why: the biggest incidents span many shifts and develop over hours. A silent transfer loses the investigator's working hypotheses; the full verbal brief restates them for the person now responsible for acting on them.

C3.5 — System health dashboard

At handover, the OCC captures a screenshot of the system health dashboard (fleet status, consensus-cluster health, station SCADA roll-up, backhaul link state) and files it to the shift folder. The incoming watch confirms the screenshot reflects reality before acknowledging.

Why: the screenshot is a timestamped snapshot of "what was true at handover" — useful for any later investigation that needs to establish what the incoming watch knew at the start of their shift.

docs/operations/dispatcher/s1-shift-start.md

docs/operations/dispatcher/s1-shift-start.md

S1 — Shift start

Scope: OCC dispatcher handover, login, fleet inventory, schedule ready-check.

Cross-refs: [osr-occ](#), RFC 0013 §4.2 S1.

Rules

S1.1 — OCC login

The incoming dispatcher logs into the OCC console with a personal credential (not a shared shift account). The console records the login timestamp in the shift-audit log.

Why: every radio authorisation, MA release, and degraded- mode transition is attributed to a person. Shared accounts break the chain of custody that post-incident review depends on.

S1.2 — Read handover log

Before accepting control, the incoming dispatcher reads the outgoing shift's handover log in full: outstanding faults, trainsets held out of service, planned possessions, weather advisories, radio-coverage gaps.

Why: the first 30 minutes of a shift is when handover context is dense and perishable. Skimming now costs more later when an ongoing situation escalates and the new dispatcher doesn't have the thread.

S1.3 — Fleet inventory

The dispatcher confirms the fleet roster against the CBM- backend overnight report: trainsets in revenue service, in maintenance, in cold reserve. Discrepancies (e.g. a trainset marked revenue that is still on a depot road) are resolved with the depot foreman before first service.

Why: the schedule builder allocates trainsets to services; if the roster is wrong, the first attempt to dispatch that service hits a no-trainset error in the middle of peak boarding.

S1.4 — Schedule ready-check

The dispatcher verifies today's roster matches the `service-hours.toml` window (first revenue service, last revenue service, headway steps). Mismatches (e.g. a public holiday schedule loaded on a weekday) are corrected before first service.

Why: schedule mismatches in small systems are detected by the travelling public, which is the worst failure mode. The ready-check is the last place to catch them.

S1.5 — Network status

The dispatcher reviews any active degraded-mode bulletins: speed restrictions, single-track working, substations on reduced output, stations in bypass. Each active item is countersigned on the shift sheet.

Why: a degraded mode left over from the previous shift that the new dispatcher isn't aware of is the shortest path to a rules violation. The countersignature makes "I didn't know" impossible.

S1.6 — Acknowledge handover

The dispatcher countersigns the handover log, which closes the outgoing shift and transfers accountability. The DMI prompts for the countersignature; until it is entered, the outgoing dispatcher remains the accountable operator.

Why: there must be exactly one accountable dispatcher at all times. A gap between "I'm logging out" and "I'm logged in" creates a window where nobody owns the radio.

docs/operations/dispatcher/s2-routine-dispatching.md

docs/operations/dispatcher/s2-routine-dispatching.md

S2 — Routine dispatching

Scope: timetable adherence, route grants, headway maintenance during nominal operation.

Cross-refs: [osr-occ](#), [osr-interlocking](#), RFC 0013 §4.2 S2.

Rules

S2.1 — Route grant on schedule

The dispatcher issues each route grant via [osr-interlocking](#) at the scheduled call time minus the interlocking's route-setup lead. The DMI pre-fills the grant from the timetable; the dispatcher confirms.

Why: manual route picking defeats the timetable and desynchronises the whole line. The confirmation step keeps a human in the loop for the unusual case (terminal swap, emergency diversion) without burdening the nominal case.

S2.2 — Headway drift

When headway drift exceeds 60 s for any pair of adjacent trainsets, the dispatcher engages the recovery pattern: shorten dwells on the leading trainset, extend dwells on the trailing trainset, and hold the next-following at an upstream station until the gap closes.

Why: bunching is self-reinforcing — a late train picks up more passengers, dwells longer, falls further behind, and empties the following train. Catching it at the 60 s threshold prevents a 20-minute unravel later.

S2.3 — Dwell extensions

Dwell extensions beyond the driver's local 15 s authority (D4.4) require dispatcher authorisation. The dispatcher may grant up to 60 s cumulative per station per service; beyond that, the service is regulated (short-turned or held at a terminal) rather than extended further.

Why: chronic over-dwelling on one service drags the whole line. The 60 s ceiling forces a regulation decision instead of letting the problem propagate silently.

S2.4 — Terminal turnback dispatch

At each terminal, the dispatcher confirms the cab-transfer report from the driver (ATO handover, brake test, doors closed) before releasing the reverse-direction route grant.

Why: a turnback with an unconfirmed cab transfer risks dispatching a trainset whose active cab is still the inbound end — a wrong-end movement that the interlocking will block but that burns a dispatch slot.

S2.5 — Crowding response

When a station reports a boarding rate exceeding its archetype capacity (per RFC 0010 §10), the dispatcher may hold the next-following trainset at the previous station to create an empty-train gap that absorbs the surge.

Why: a single overcrowded train generates a long dwell, which generates the next overcrowded train. Holding the follower briefly costs a headway but breaks the feedback loop.

docs/operations/dispatcher/s3-incident-handling.md

docs/operations/dispatcher/s3-incident-handling.md

S3 — Incident handling

Scope: dispatcher response to in-service incidents. Covers all six categories from RFC 0013 §7 (I1 self-recovery I6 mass casualty).

Cross-refs: RFC 0013 §7, [osr-event-recorder](#), [osr-occ](#).

Rules

S3.1 — I1 single-train fault

For a single-train fault that clears within one dwell (lost ATO link recovers, door-close recycle succeeds), the dispatcher logs the event to `osr-event-recorder`, notifies the depot for trend analysis, and issues no passenger PA unless service-minute impact exceeds 2 minutes.

Why: a PA on every blip trains passengers to ignore PAs, which is bad when a real PA matters. The 2-minute threshold keeps announcements credible.

S3.2 — I2 service disruption

For a service disruption lasting more than 5 minutes (trainset unable to move under own power, signalling fault requiring staff talk-through), the dispatcher initiates the rescue- coupling protocol (D7), issues a passenger PA every 3 minutes, and logs a timetable recovery plan (short-turns, skip-stops) to the shift log.

Why: disruptions longer than a few minutes need both a technical response (rescue) and a communications response (PA cadence). Running them as a single protocol reduces the chance that one is forgotten while the dispatcher focuses on the other.

S3.3 — I3 short safety event

For a short safety event (EB triggered, SPAD, overspeed trip), the dispatcher orders the driver not to move until the event recorder is preserved, initiates a 72-hour investigation with a minute-by-minute timeline, and holds the trainset out of service pending the investigation outcome.

Why: moving a trainset after an EB contaminates the event record with recovery data and obscures the initiating sequence. The 72-hour window matches RFC 0013's safety- case evidence retention requirement.

S3.4 — I4 passenger injury

For a passenger injury (reported by driver, station staff, or passenger emergency intercom), the dispatcher places a medical call, coordinates with station staff for a station handover of the casualty, and files the passenger-injury report form within 24 hours.

Why: the injured passenger needs a named owner on the OCC side so the ambulance arrives at the right platform. The 24-hour form is the insurance record, which is expensive to reconstruct from memory.

S3.5 — I5 collision or derailment

For a collision or derailment, the dispatcher orders a full line shutdown, notifies the National Rail Safety Authority within 2 hours (or the equivalent regulator per the deployment), and issues an incident-site preservation order (no movement, no cleanup) until the NRSA investigator releases the site.

Why: the first instinct after a major incident is to clear the site to restore service — which destroys the evidence the regulator needs to determine cause. The preservation order makes the cost of not clearing explicit.

S3.6 — I6 mass casualty

For a mass-casualty event (fire, explosion, structural failure), the dispatcher activates the evacuation protocol for the affected stations, escalates to emergency services (fire + ambulance + police), notifies NRSA and regional media liaisons, and coordinates shutdown boundaries to contain the incident.

Why: mass-casualty response is a multi-agency operation where the rail dispatcher is one voice among many. The role is to own the rail side of the response — tracks de-energised, trains stopped, stations evacuated — and hand off everything else to the authorities trained for it.

docs/operations/dispatcher/s4-weather-and-environmental.md

docs/operations/dispatcher/s4-weather-and-environmental.md

S4 – Weather and environmental

Scope: network-wide environmental overlays per RFC 0013 §6.

Cross-refs: RFC 0013 §6 + lib/templates/climate-adapters.toml.

Rules

S4.1 – Dust storm (haboob)

On a declared dust-storm event (visibility below 200 m or PM10 above the climate-adaptor threshold), the dispatcher imposes a network-wide 50 % speed cap, issues a passenger PA on every line, and alerts the depot to schedule post-event axle-box and switch-machine inspections.

Why: dust defeats optical sensors and accelerates switch- machine wear. The 50 % cap preserves stopping-distance margin against a reduced-adhesion wheel-rail contact; the post-event inspection catches the wear before it becomes a failure.

S4.2 – Heatwave

On a sustained ambient above 45 °C for 2 hours or more, the dispatcher imposes a 15 % speed cap on continuous-welded- rail spans (per RFC 0009 presets marked CWR) and schedules a pre-dawn recalibration of switch-machine detection sensors.

Why: CWR buckling risk rises sharply above 45 °C rail temperature; the speed cap reduces dynamic load. Switch- sensor drift at high temperature is recoverable by recalibration but misreads in the meantime can jam a route.

S4.3 – Flooding

On measured rainfall exceeding 50 mm per hour, the dispatcher places an inspection hold on at-grade civil spans (no movement until walked by MOW); elevated and bridge spans continue per RFC 0011 §5.

Why: at-grade track is vulnerable to ballast washout and drainage-ditch overtopping, which neither osr-wayside- points nor the driver can see from the cab. Elevated and bridge spans drain by design.

S4.4 – Lightning

On lightning strikes within 2 km of the alignment, the dispatcher monitors radio backhaul closely. If both 5G and LoRa backhaul are lost for more than 10 s, the dispatcher accepts degraded-mode operations (manual block working per D6) until backhaul recovers.

Why: lightning induces surges that can knock out multiple backhaul paths simultaneously. The 10 s window distinguishes a momentary flap from a sustained outage that warrants the degraded-mode switch.

S4.5 – Sandstorm recovery

After a sandstorm clears, the dispatcher holds full-speed operations until MOW has walked each affected section and confirmed switches are clear of drift. Restricted-speed (50 %) operation may resume as soon as visibility lifts.

Why: drift in switch throats jams the points mechanism and can be missed by detection sensors that have been abraded by the same event. The walk catches it; the speed cap keeps passengers moving while the walk is in progress.

docs/operations/dispatcher/s5-maintenance-interlock.md

docs/operations/dispatcher/s5-maintenance-interlock.md

S5 — Maintenance interlock

Scope: granting work-on-track authorities, blocking sections for maintenance, and restoring to revenue after work completes.

Cross-refs: [osr-interlocking](#) MaintenanceOverride entries, RFC 0013 §4.2 S5.

Rules

S5.1 — Formal work-block authorisation

The dispatcher issues a work block by committing a MaintenanceOverride entry to `osr-interlocking` with a specific work-crew id, section ids, and an expiry timestamp (default 60 min, extendable per S5.4). The DMI prompts for each field; blank fields reject the commit.

Why: open-ended blocks with vague scope are how work crews get dispatched onto live track. The structured form makes every block uniquely identifiable, time-bounded, and traceable to a named crew.

S5.2 — Train clear verified

Before the MaintenanceOverride commit is accepted, `osr-interlocking` verifies no trainset footprint overlaps the requested section. The dispatcher cannot override this check.

Why: granting a work block while a trainset is in the section puts the crew in front of a moving train. The interlocking's section-occupancy logic is the same one that enforces Movement Authority — so if it can't prove the section is clear, the block waits.

S5.3 — Protection signals

Before work begins, the dispatcher requests protection signals (red aspect + trap point if fitted) at both ends of the blocked section. The wayside crew radio confirms each protection signal is visible and effective.

Why: the interlocking prevents new MA into the section, but a trainset already holding MA through the section must still see a stop signal. Protection is the last defence against a runaway movement.

S5.4 — Block extension

Block extensions are granted in 30-minute increments, not open-ended. The dispatcher renews the MaintenanceOverride expiry at each increment with a fresh radio check to the crew.

Why: a 4-hour block that overruns silently leaves the track blocked after the crew has gone home. The 30-minute cadence forces regular radio contact, which catches silent overruns.

S5.5 — Block clearance

The dispatcher clears the block only after: crew withdrawal confirmed on radio, protection signals cleared, and track walk-back confirmed clear of tools and material. The MaintenanceOverride expiry is then committed — which returns the section to revenue availability.

Why: tools left on the track are how the first trainset back into the section derails. The walk-back is the non-negotiable final step; the dispatcher is the gatekeeper.

docs/operations/dispatcher/s6-shift-end.md

docs/operations/dispatcher/s6-shift-end.md

S6 — Shift end

Scope: end-of-shift handover, outstanding-event log, system state summary for the incoming dispatcher.

Cross-refs: RFC 0013 §4.2 S6.

Rules

S6.1 — Write the handover log

The outgoing dispatcher writes the handover log 30 minutes before shift end. The log covers: outstanding incidents, active `MaintenanceOverride` entries, active speed restrictions, degraded-mode flags, depot slot availability, and any passenger-affecting events from the shift.

Why: the 30-minute margin gives time for the outgoing dispatcher to answer questions while still on shift, rather than leaving the incoming dispatcher to decode a hastily-written log alone.

S6.2 — Confirm fleet overnight status

Before close, the outgoing dispatcher confirms all revenue depot transitions are complete, each trainset's overnight layup slot is known, and any trainset held out of service has a named fault attached.

Why: a trainset "somewhere" overnight is a trainset that gets missed at first service next morning. The explicit layup assignment makes the morning roster pre-solved.

S6.3 — Carry over open incidents

Any I2/I3/I4/I5/I6 incident still open at shift end is listed explicitly in the handover log, with the investigation owner, deadline, and next action. Incidents are never closed by "shift change" — only by resolution.

Why: incidents closed implicitly by handover disappear from the investigation record. The 72-hour and 2-hour regulatory clocks run on wall-clock time, not shift time.

S6.4 — Log off

The outgoing dispatcher logs off the OCC console only after the incoming dispatcher countersigns the handover log (per S1.6). Until countersignature, the outgoing dispatcher remains accountable.

Why: leaving early while the incoming dispatcher is still catching up creates a window where no qualified person is at the console. The countersignature closes that window explicitly.

docs/operations/dispatcher/s7-intrusion-response.md

docs/operations/dispatcher/s7-intrusion-response.md

S7 — Intrusion response

Scope: OCC dispatcher response to wayside track-intrusion detections raised by `osr-intrusion-detect`. Per-section verdicts arrive on the OCC dashboard as `Present` / `Unknown` / `Clear` (RFC 0016 §5.2).

Cross-refs: [osr-intrusion-detect](#), [osr-interlocking](#) SectionIntrusion entries, RFC 0016 §7.1.

Rules

S7.1 — First-response track patrol

On the first `Present` verdict on any section, the dispatcher dispatches track-patrol within 10 minutes and notifies the engineer-on-call. While the verdict stands, the interlocking withholds MA on that section automatically — no manual hold is required.

Why: the interlocking already enforces the safety restriction — there is no "move faster before the gate drops" concern. The dispatcher's job is to resolve the underlying cause (animal, debris, trespasser) and return the section to `Clear`, which is a ground-level task.

S7.2 — Sustained Unknown treated as equipment failure

A `Unknown` verdict persisting for more than 5 minutes is treated as a wayside-equipment fault. The dispatcher opens a work ticket against the affected junction box, holds or diverts any trains whose route requires the section, and notifies maintenance per M7.

Why: a brief `Unknown` is a sensor flap; five minutes of `Unknown` is a hardware problem that will not resolve itself. The ticket routes the fix into the CBM stream rather than the dispatcher improvising a repair.

S7.3 — Return to service

The dispatcher returns the section to revenue only after: (a) track-patrol confirms the section clear of obstruction, (b) wayside sensors have produced at least one fresh `Clear` verdict, and (c) any affected PSDs and stations are ready. The "return to service" action is an OCC-console command that logs the dispatcher's confirmation to the shift log.

Why: clearing the patrol check but not the sensor check is the shortest path to immediately re-blocking the same section (e.g., debris missed on walk-through). The three-gate sequence keeps the system honest.

S7.4 — Multi-section outage escalation

When three or more sections on the same line carry non-`Clear` verdicts concurrently, the dispatcher treats it as a line-wide event — escalates to the supervisor, considers cutting revenue service on the affected line, and activates the network-disruption passenger communications protocol.

Why: three concurrent intrusions are almost always either a coordinated event (storm, security incident) or a wayside comms failure that's affecting multiple junction boxes. Both cases call for line-level intervention, not per-section management.

S7.5 — No override

The dispatcher has no authority to force a section to `Clear` by dispatcher action. The interlocking's `section_available_to gate (d)` is a hard interlock.

Why: if the dispatcher could clear the verdict from the console, the safety case for wayside detection is broken — the interlocking would be advisory rather than authoritative. Real clearance comes from track-patrol + sensor agreement per S7.3, not from a dispatcher button.

docs/operations/driver/d1-before-service-checks.md

docs/operations/driver/d1-before-service-checks.md

D1 — Before-service checks

Scope: actions the driver performs at the depot before the first revenue move of a shift.

Time budget: ≤ 10 minutes on an in-service trainset. A trainset with a red fault from the previous shift is referred to maintenance without a driver attempt.

Cross-refs: RFC 0008 §3.4 (cab), RFC 0013 §4.1, safety-case G4.1 (osr-atp correctness), G4.2 (osr-brake correctness), G4.3 (osr-vigilance correctness).

Rules

D1.1 — Power-on sequence

On entering the cab, the driver powers up in the order `cab isolator ON` → `DMI power button` → wait for green "ready" banner.

Why: the cab isolator gates the 24 V DC rail that feeds the DMI and the master controller; powering the DMI first on a dead isolator produces a red bus-fault on the event recorder that looks like a real fault during later investigation.

D1.2 — Brake test on the pit

With the trainset over the depot pit, the driver runs `DMI` → `Pre-service` → `Brake test`. The test actuates service + emergency brake and reports pass/fail within 60 s.

Why: the electromagnetic disc brake is self-monitoring, but the stuck-caliper failure mode only surfaces under commanded actuation. The pit position lets depot maintenance observe caliper motion from below during the first test of the shift.

D1.3 — Vigilance test

On the DMI prompt, the driver acknowledges within the displayed 5 s window, then intentionally misses the next prompt — confirming `osr-vigilance` escalates `Nominal Warning Tripped` and applies emergency brake within the 5 s warning window.

Why: vigilance is one of the five independent brake-apply sources (RFC 0005 §6.1 O4). If the driver's ack loop doesn't reach `osr-vigilance`, the train must not leave depot — this is the only pre-service test that exercises the full loop.

D1.4 — Door-interlock test

The driver issues `open` → `close` on every car via the DMI. Every door must close within 6 s and report green-interlocked on the DMI mimic diagram.

Why: a stuck-open door at revenue departure is a Cat I2 disruption (RFC 0013 §7). Testing all 24 doors at depot catches the small fraction that jam after overnight thermal soak.

D1.5 — Event-recorder handshake

The DMI shows an event - recorder: green indicator. Driver confirms; no further action.

Why: if the event recorder is not logging, an in-service incident produces no investigable trace. The Cat I3 / I4 / I5 evidence chain depends on a known-good recorder from service start.

D1.6 — Cab-to-cab confirmation

If the opposite cab is occupied simultaneously (shift overlap), the driver confirms over intercom that only one master controller is armed. The DMI shows a red banner if both cabs arm.

Why: master-controller ambiguity is a rulebook violation flagged in the event recorder; the intercom check catches it before the DMI banner.

D1.7 — Defect reporting

Any failure of D1.1–D1.6 is logged via DMI → Fault → Report and the trainset is returned to maintenance before departure. The driver does not attempt a workaround on any D1 failure.

Why: every failure mode D1 exercises is safety-relevant. Workarounds establish precedent; the rulebook stays strict.

docs/operations/driver/d2-starting-a-service-run.md

docs/operations/driver/d2-starting-a-service-run.md

D2 — Starting a service run

Scope: from cab ready to first motion on a revenue run. Authority acknowledgement, platform-clearance confirmation, door close coordination, MA release to `osr-atp`.

Cross-refs: RFC 0001 §7, [osr-atp](#), [osr-psd](#), RFC 0013 §4.1 D2.

Rules

D2.1 — Acknowledge the incoming MA

On the DMI's "MA received" prompt, the driver presses `Accept` before releasing the brake. The DMI shows the MA validity window counting down; the driver must accept within that window.

Why: an MA is only valid for 3 s of wall clock per RFC 0001 §6.3 (`MA_VALIDITY_WINDOW_NS`). Accepting an expired MA is a null action — the dispatcher re-issues. Accepting a stale MA and treating it as live is the failure mode this rule rules out.

D2.2 — Confirm platform clearance

On PSD-equipped stations, the driver reads `PSD: closed + interlocked` on the DMI. On non-PSD stations, the driver visually scans the platform for passengers within 1 m of the edge before issuing `door-close`.

Why: the `door-close` command initiates `dwell end`. A platform obstruction under or at the door-line produces a passenger-injury incident that the vehicle-side sensors cannot always detect.

D2.3 — Close trainset doors

The driver issues `Door close` on the DMI. All 24 doors must report `closed + interlocked` on the mimic diagram within 8 s. Any door not reporting: reopen that door, wait 3 s, reissue `close`.

Why: an unclosed door fails the ATP envelope check in `osr-door-control`; the train will not move. But a mis-reporting door (closed mechanically but interlock not latched) can pass the static check and fail mid-motion. Reopening resets the state.

D2.4 — ATO / manual authorisation

In GoA 2 operation (driver supervising ATO), the driver presses `Start` on the DMI; ATO takes the consist to the next station within the ATP envelope. In GoA 1 / fallback manual, the driver notches off the master controller at $\leq 10\%$ tractive effort initial.

Why: ATO's acceleration ramp is pre-configured and smooth; manual notch-off should match ATO's behaviour unless there is a specific operational reason (e.g. slippery-rail procedure). Aggressive manual departure is a passenger-comfort complaint that the event recorder tags.

D2.5 — First-motion check

Within the first 5 s of motion, the driver confirms: speed rising monotonically on the DMI, no red banners, event recorder still green, and the vibration / sound profile of the trainset is nominal (driver judgement).

Why: a failure to move — or anomalous motion — at departure is the one moment a driver has full-attention on the trainset before passenger interactions start. This 5-second window is the cheapest quality check in the whole shift.

D2.6 — Abort conditions

If at any point during D2.1–D2.5 the MA revokes (DMI banner `MA revoked`), a PSD re-opens, the passenger-emergency alarm fires, or a red fault banner appears, the driver immediately stops the trainset with service brake and notifies the dispatcher via TRG-1 radio before any further action.

Why: multiple faults stacking at departure produce corrupted root-cause evidence. Stopping immediately on any one of them preserves the event-recorder trace for investigation.

docs/operations/driver/d3-in-service-operation.md

docs/operations/driver/d3-in-service-operation.md

D3 — In-service operation

Scope: between-station running. Speed within the ATP envelope, vigilance acknowledgement cadence, station dwell handling, passenger communication.

Cross-refs: [osr-atp](#), [osr-vigilance](#), RFC 0013 §4.1 D3.

Rules

D3.1 — Speed within the envelope

In GoA 1 (manual) the driver's commanded speed stays $\leq 95\%$ of the ATP envelope displayed on the DMI. In GoA 2 (ATO) the driver verifies the actual speed tracks the ATO command within ± 1 m/s; any divergence greater than 2 m/s is reported as a fault via DMI → Fault → ATO divergence.

Why: the ATP emergency trip activates at envelope + OVERSPEED_EMERGENCY_MARGIN_MMPS (0.5 m/s per RFC 0005); riding the envelope wastes margin and creates nuisance trips on gradient transitions.

D3.2 — Vigilance acknowledgement cadence

The driver acknowledges the vigilance prompt within 5 s of each pulse. Nominal cadence is 30 s; a 5 s warning window precedes a Tripped state. The driver does not pre-emptively ack before the prompt.

Why: pre-emptive acks defeat the vigilance purpose — [osr-vigilance](#) only accepts an ack that arrives in the current window. Repeatedly missing the window and recovering is logged as a pattern of driver fatigue and flags a rostering review.

D3.3 — Station approach

The driver begins the next-station announcement at 15 s from arrival and confirms the ATO's deceleration profile matches the DMI's target stopping point. The master controller stays in the ATO supervising position through approach.

Why: the ATO's brake profile targets a ± 0.5 m stopping-point accuracy; a driver late-overriding the brake command compromises that accuracy and can leave a car or two outside the PSD zone.

D3.4 — Passenger announcements

The driver makes the next-station PA at 15 s before arrival using the text `osr-pis-onboard` auto-populates. Ad-hoc announcements (dispatch hold, weather speed restriction) follow the phrasing guide in the station staff rulebook [T2.5](#).

Why: inconsistent train-vs-station announcements are a quiet quality-of-service loss that compounds. One voice style across touchpoints helps system trust.

D3.5 — Passenger alarm response

If the passenger-emergency alarm fires (DMI: `passenger alarm at car N, door P`), the driver stops the trainset at the next station if within 30 s travel, otherwise immediately. In both cases the driver makes a PA announcement and radios OCC with Cat I4.

Why: a midline stop extends passenger-trapped time. If the next station is ≤ 30 s away, completing to the station gets ground help there faster than a midline stop + rescue coupling.

D3.6 — Speed restrictions

The driver honours every SpeedRestriction entry on the DMI without exception. Weather, track-work, and dispatcher-manual restrictions all manifest the same way: DMI shows the restricted speed, ATP's envelope tightens accordingly, the driver's commanded speed respects the envelope per D3.1.

Why: restrictions exist because a condition has been inspected and found. Ignoring one defeats the inspection that produced it. The ATP enforcement is the final safety net — not the primary mechanism.

docs/operations/driver/d4-entering-leaving-stations.md

docs/operations/driver/d4-entering-leaving-stations.md

D4 — Entering + leaving stations

Scope: approach, stopping, dwell, and departure. PSD coordination, passenger-boarding timing, dwell-extension protocol.

Cross-refs: [os r - psd](#), RFC 0010 §4 (platform geometry), RFC 0013 §4.1 D4.

Rules

D4.1 — Platform-stop accuracy

The trainset stops within ± 0.5 m of the reference stop mark (painted line at the platform edge, aligned with door 2 of car 1). In GoA 2 the ATO handles this automatically; in GoA 1 the driver targets the mark by eye + DMI position indicator.

Why: stops outside ± 0.5 m put some doors outside the PSD zone (at PSD-equipped stations) or leave passengers reaching up to the retractable skirt (at non-PSD stations).

D4.2 — PSD open command

At PSD-equipped stations, the driver commands Door open on the DMI. The PSDs open first (via os r - psd FSM), then the trainset doors follow. The DMI mimic shows both layers green.

Why: os r - psd requires a train-stopped-at-platform signal before opening. Commanding train doors first at a station where PSDs are closed puts passengers between two panes of glass.

D4.3 — Dwell-timer start

Dwell timer starts when the last door opens, not on train stop. The DMI shows remaining dwell time.

Why: slow-opening doors extend effective dwell by up to a few seconds at peak; starting on the last door keeps the schedule honest.

D4.4 — Dwell extension

The driver extends dwell via DMI → Dwell → Extend in 5 s increments, up to 3 extensions per station (15 s total extra). Reasons: passenger obstruction, wheelchair boarding, accessibility hold. Beyond 3 extensions the dispatcher must authorise via radio.

Why: 15 s of extension on a handful of stations is absorbed by the schedule slack. Chronic over-extension is a sign that the station archetype or headway is undersized — a dispatcher- level signal, not a driver discretion.

D4.5 — Door + PSD close

The driver commands Door close when boarding is complete. Train doors close first; PSDs follow 1 s later (per os r - psd's FSM). The DMI mimic shows all doors green-closed

- interlocked + all PSDs green-closed.

Why: closing train doors first prevents a passenger from stepping through a closing PSD into a moving door.

D4.6 — Release to depart

The driver's Start command is accepted by os r - atp only when: MA is valid, all train doors green, all PSDs green (if fitted), no alarms. A red block on any of these prevents first-motion.

Why: the start gate is a hard safety interlock with no manual override. A driver cannot coax past it — which is the point. The rulebook makes this explicit so drivers don't waste troubleshooting time on the start command itself.

docs/operations/driver/d5-terminal-turnback.md

docs/operations/driver/d5-terminal-turnback.md

D5 — Terminal turnback

Scope: at an end-of-line terminal or depot-terminal station: stop, cab transfer, blade observation at the turnback switch, new outbound authority.

Cross-refs: RFC 0010 §1 (terminal archetype), RFC 0012 §4.3 (switch facing / trailing), RFC 0013 §4.1 D5.

Rules

D5.1 — Full stop at the terminal platform

The trainset stops at the terminal platform per D4.1 (± 0.5 m of reference mark). The driver verifies the DMI shows Terminal before beginning passenger alight.

Why: terminal-specific procedures (cab transfer, pantograph raise to the dock) are gated by the Terminal banner. A non-terminal stop doesn't give access to them.

D5.2 — Passenger alight

The driver commands Door open and makes the "end of the line, all change" PA. Doors remain open until the DMI shows All clear — platform staff confirm, or (at unstaffed terminals) the driver visually sweeps each car from the cab camera feed.

Why: a passenger asleep on a terminal arrival wakes up at the depot or on the back run. The "all clear" confirmation lands the liability with station staff or the driver's cam review, not ambiguity.

D5.3 — Pantograph raise (at docks)

If the terminal has a 1 500 V DC charging dock (per the terminal / depot-terminal archetypes), the driver commands Panto up on the DMI after the doors close. The pantograph rises, the dock arms, charging begins.

Why: raising the panto before door close is a personnel-hazard question — any crew on the roof is in the way. Doors-closed is the signal that the consist is clear-to-panto.

D5.4 — Cab transfer

The driver powers down the active cab per D8.1–D8.5, walks the length of the train to the opposite cab, powers up per D1.1–D1.2 (the pre-service brake test is skipped — already done). The DMI shows the opposite cab as armed; the previous cab shows disarmed.

Why: dual-cab arming is a D1.6 violation that the DMI flags; a proper cab transfer is the only non-violating path.

D5.5 — Turnback-switch observation

If the turnback involves a facing-point switch, the driver observes the blade position via the cab camera (or directly, at a terminal with visible tracks) before commanding departure. The DMI also shows the FPS position (green + locked in commanded direction); both indicators must agree.

Why: FPS is a SIL-4 asset. The DMI reflects osr-wayside-points's fused observation; adding a visual check is a belt-and-braces that catches a rare DMI-vs-reality disagreement from an unreported local sensor fault.

D5.6 — Outbound MA

The driver waits for the outbound MA on the DMI, acknowledges per D2.1, and proceeds per the full D2 sequence.

Why: every departure — including post-turnback — is an independent D2. No shortcut across the boundary.

docs/operations/driver/d6-degraded-mode-operation.md

docs/operations/driver/d6-degraded-mode-operation.md

D6 — Degraded-mode operation

Scope: the three modes per RFC 0013 §5 — M1 (manual on MA), M2 (restricted on written order), M3 (yard manoeuvre). No other modes exist; no in-between states.

Cross-refs: RFC 0013 §5, [osr-atp](#), [osr-occ](#).

Rules

D6.1 — M1 entry

The driver enters M1 when the DMI banner reads `ATO unavailable — ATP supervising`. Entry is automatic; no driver keyswitch. The driver takes manual control of traction and service brake while ATP keeps supervising the envelope.

Why: ATO is the convenience layer; ATP is the safety layer. If only ATO is degraded, the safety envelope is still intact — the driver just does the driving.

D6.2 — M1 speed cap

In M1 the driver commands $\leq 80\%$ of normal line speed. The DMI tightens the envelope to 80 % automatically; the driver's $\pm 5\%$ headroom (D3.1) is within that.

Why: manual driving has higher variability than ATO; 80 % headroom absorbs that without nuisance ATP trips.

M1.3 — M1 dwell extension

At every station in M1, the driver extends dwell by 5 s before closing doors, to visually verify the platform is clear.

Why: ATO communicates with `osr-psd` over TCN-E; in M1 ATO isn't driving, so the PSD coordination is slower and more error-prone. The 5 s is a cheap recovery window.

D6.4 — M2 entry

The driver enters M2 only on explicit dispatcher order. The DMI banner reads `M2 written order in effect, speed cap 15 km/h`. The order carries a block-end station and a time limit.

Why: M2 is the legacy-rail fallback — paper (or electronic-paper) written order from the dispatcher. Self-entering M2 is not possible and not allowed.

D6.5 — M2 operation

In M2 the driver operates at ≤ 15 km/h with sight-distance braking (stop within the visible track ahead). One train per block between manned landmarks; the dispatcher's roll-call over radio confirms block occupancy.

Why: M2 lives outside the MA / ATP envelope. The driver's eye is the primary safety sensor; sight-distance braking is the only safe rule when a computed authority is unavailable.

D6.6 — M3 entry (yard manoeuvre)

Inside depot limits the DMI enters M3 automatically once the trainset crosses the depot geofence. M3 disables ATP below 15 km/h and permits hand-signal-and-line-of-sight operation.

Why: the depot is a low-speed, staff-supervised environment. ATP's nominal envelope would create nuisance trips on every shunt move; disabling under geofence + geo-limited speed is the tractable fallback.

D6.7 — Mode-transition discipline

The driver never transitions M1 → M2 without explicit dispatcher authorisation. Transitioning M2 → M1 is also dispatcher-authorised (MA restored, written order revoked). M3 is implicitly entered/exited by the geofence; no driver toggle.

Why: unauthorised mode transitions produce degraded-state ambiguity — the dominant root cause of legacy-metro incidents. The rulebook draws hard boundaries.

docs/operations/driver/d7-emergencies.md

docs/operations/driver/d7-emergencies.md

D7 — Emergencies

Scope: driver response to emergencies — emergency-brake plunger, EB recovery, passenger medical, fire, derailment, onboard violence.

Cross-refs: [osr-brake](#), [osr-fire-safety](#), [osr-derailment](#), RFC 0013 §4.1 D7 + §7 incident categorisation.

Rules

D7.1 — Emergency-brake plunger

The driver strikes the plunger any time they believe the trainset must stop and any normal mechanism may fail. The plunger hardwires to [osr-brake](#)'s emergency source — no DMI, no timeout, immediate brake application.

Why: the plunger is the cheap, always-available last line. Drivers should strike it without deliberation; a false-positive stop is recoverable, a missed-positive is not.

D7.2 — Post-EB recovery

After any EB (triggered by plunger, ATP, vigilance, fire, derailment, or passenger alarm), the driver does NOT release the brake until (a) the DMI's EB-source indicator clears, (b) OCC authorises via radio, (c) the driver visually confirms no obvious trainset fault.

Why: an EB without root-cause clarification is how Cat I3 becomes Cat I5. The 3-way confirmation adds 30–60 s to every EB recovery; that's worth it.

D7.3 — Fire alarm

If the DMI's fire banner shows, the driver confirms [osr-fire-safety](#)'s automatic suppression has fired (DMI shows `Suppression: discharged`). If not, the driver presses the cab's manual fire-suppress button. In either case the trainset is stopped at the next station if within 30 s, then passengers evacuated via emergency door release. PA: "Fire on board — exit via the nearest open door."

Why: [osr-fire-safety](#)'s automatic path is reliable but not foolproof; manual backup covers the residual risk. The "next station within 30 s" rule matches D3.5's passenger-alarm rule — stopping inside a station means passengers reach ground evacuation paths, not the track bed.

D7.4 — Derailment / unusual motion

If the DMI shows derailment alarm OR the driver feels a lurch they cannot explain (not a curve, not a brake, not a PA), the driver pulls the plunger immediately, cuts traction, radios OCC with Cat I5 + line + approximate milepost, and prepares the consist for evacuation (doors to emergency-release mode, PA: "Brace, brace, brace — then prepare to exit").

Why: derailment is the worst-case rail scenario. Every second of uncertainty in the response adds to injury risk. This rule is the single most consequential in the entire rulebook.

D7.5 — Passenger medical incident

The driver stops at the next station within 30 s travel (or immediately, if not), makes the medical on board at car NPA, radios OCC for ambulance dispatch, and holds the train until handover to station staff.

Why: holding a trainset in station is cheaper than coordinating ambulance access to a midline stop. The 30 s rule is the same calculation as D3.5.

D7.6 — Onboard violence or terror suspicion

The driver locks the cab (mechanical latch on cab door + DMI "Cab secured" command), radios OCC with Cat I6, proceeds to the next staffed station at nominal speed (no stop-short), and lets OCC coordinate with emergency services. The driver does not leave the cab and does not engage with passengers.

Why: the driver's job is to get the train to help, not to intervene. Cab-lock + continue-to-help is the rail-world analogue of a pilot's "fly the airplane" — don't become a second casualty.

docs/operations/driver/d8-end-of-service.md

docs/operations/driver/d8-end-of-service.md

D8 — End of service

Scope: return of a trainset to depot, cab powerdown, handover to the depot shift or the CBM team.

Cross-refs: [osr-cbm-onboard](#), [osr-event-recorder](#), RFC 0014 §5.1 (main-heavy depot functions).

Rules

D8.1 — Final revenue run complete

The driver completes the last revenue run per the roster, makes the last train, all change at [terminal] PA at the penultimate station, and proceeds to the assigned depot via the empty-run route.

Why: the empty-run route is the same as revenue but with no passenger interaction; dispatcher tracks it the same way.

D8.2 — Depot entry (mode M3)

At the depot geofence the DMI transitions to M3 automatically (per D6.6). The driver slows to ≤ 15 km/h, follows the line-of-sight rule, and obeys any hand signals from depot yard staff.

Why: depot yard geometry is a mix of mainline speeds and close-together stalls — line-of-sight is the tractable rule. `osr-atp` is disabled below 15 km/h inside the depot geofence (RFC 0007 §4 safety configuration).

D8.3 — Park in assigned stall

The driver parks in the stall assigned by the depot staff (or shown on the DMI if the stall was preassigned by the OCC's CBM pipeline), applies park brake, cuts traction via DMI → Park.

Why: a trainset in the wrong stall adds friction to the next morning's dispatch — the CBM system's predicted maintenance window is stall-specific.

D8.4 — Cab powerdown

The driver powers down in reverse of D1.1: DMI power OFF → wait for DMI-down indicator → cab isolator OFF.

Why: reversing the power-on sequence avoids a bus-fault indicator on the overnight event recorder.

D8.5 — Fault-flag handover

If the trainset flagged any amber/red fault during the shift, the driver confirms the fault is written to the trainset's persistent fault register (DMI → Fault → Commit) before leaving the cab. CBM picks it up overnight.

Why: a fault logged in volatile event-recorder memory can be lost on powerdown; the persistent register is where CBM looks first.

D8.6 — Driver logbook entry

The driver fills the logbook (paper or app) with: shift start

- end times, km driven, incidents by category, equipment anomalies, passenger counts (roster-declared OR CBM-reported, whichever is available).

Why: the logbook is the primary legal record of the shift. A missing entry compromises any downstream legal review (incident dispute, fare-fraud claim, insurance).

docs/operations/maintenance/m1-depot-safety.md

docs/operations/maintenance/m1-depot-safety.md

M1 — Depot safety

Scope: hazard awareness, PPE, lone-worker protocol, pit work, crane operations.

Cross-refs: RFC 0014 §5 (main-heavy layout), RFC 0013 §4.4 M1.

Rules

M1.1 — PPE baseline

All depot workers on the shop floor wear hi-vis vest, steel- toe footwear, hearing protection in designated noise zones, and eye protection when grinding, welding, or handling compressed gas. Entry to the shop without baseline PPE is refused by the gate.

Why: the baseline is enforced at the gate because mid- shop "just popping in" is when PPE is forgotten and how eye injuries happen. The gate is the single chokepoint that catches it.

M1.2 — Pit work

Pit work (under-car inspection or maintenance) requires: pit lighting on, pit edge rails in place, no-unaccompanied rule (minimum 2 workers, one at ground level), and the trainset power-isolated and chocked.

Why: a worker falling into a pit or a trainset rolling over a pit worker is catastrophic. The paired-work rule means help is on site; the chock rule means the trainset can't move.

M1.3 — Crane operations

Overhead crane (40 t in main-heavy, 10 t in secondary-medium) operations require a qualified operator (licensed per deployment regulator), a clear area below the hook path (no workers in the shadow), and a spotter in radio contact with the operator.

Why: a lifted load swinging into a worker kills; the spotter's job is specifically to see what the operator can't from the cab. The qualification requirement is the legal minimum and the insurance precondition.

M1.4 — Lock-out tag-out (LOTO)

Before working on energised systems, the responsible technician applies a personal LOTO tag to each energy source: traction battery main isolator, auxiliary 400 VAC breaker, compressed-air reservoir bleed valve. Each tag carries the technician's name; only they can remove it.

Why: "someone else will check" is how people get electrocuted. The named-tag-per-source rule is defence in depth; tag removal by anyone other than the owner is a disciplinary offence.

M1.5 — Lone-worker

After-hours lone work (single technician on depot floor) is permitted only with an active radio check-in every 30 minutes to the OCC night watch. A missed check-in triggers a welfare call and, if unanswered, dispatch of a second worker to investigate.

Why: a lone worker injured and unable to call for help is invisible for the rest of the shift otherwise. The 30-minute cadence bounds the worst-case discovery delay.

M1.6 — Chemical safety

Cleaning agents, lubricants, and gels are stored in a designated chemical cabinet with labelled containers and Safety Data Sheets (SDS) accessible at the cabinet. Mixing chemicals or transferring to unlabelled containers is prohibited.

Why: decanted-into-water-bottle solvent is the classic depot poisoning; mixed bleach + ammonia is the classic depot gassing. The cabinet and labels defeat both.

docs/operations/maintenance/m2-work-on-track-authorisation.md

docs/operations/maintenance/m2-work-on-track-authorisation.md

M2 — Work-on-track authorisation

Scope: formal work blocks, protection signals, access to active ROW. Paired 1:1 with the dispatcher's S5 rules.

Cross-refs: S5 (dispatcher) + `osr-interlocking MaintenanceOverride`, RFC 0013 §4.4 M2.

Rules

M2.1 — Request work-block

Work-block requests are submitted to the OCC at least 48 hours in advance for planned work; 2 hours minimum for urgent (safety-affecting) work. The request names the crew lead, section ids, scope of work, and planned duration.

Why: 48 hours allows the dispatcher to slot the block without service disruption; the 2-hour floor for urgent work balances speed against coordination with other movements. Unplanned drop-in requests strand crews.

M2.2 — Protection signals set

On arrival at the work site, the crew lead confirms protection signals (per S5.3) are set at both ends of the block and radios the OCC to commit the `MaintenanceOverride`. Work does not start until the commit is acknowledged.

Why: arriving and working before the commit means the crew is on track with no interlocking protection. The radio handshake is the moment the crew is formally protected.

M2.3 — Stay inside the block

During the block, crew members remain inside the protected section except for a single "roamer" permit per crew (one person authorised to step outside for inspection or tool retrieval). The roamer carries a high-visibility armband and a personal radio.

Why: crew members outside the block are not protected by the block. Limiting this to a single named roamer keeps the accounting tractable and ensures the roamer knows they are exposed.

M2.4 — Half-hourly check-in

The crew lead radios the OCC every 30 minutes with a crew headcount and progress update. Missed check-ins trigger the OCC welfare protocol (S5.4 extension check).

Why: a crew in trouble (medical, fall, entrapment) needs help fast; the 30-minute cadence bounds the OCC's discovery delay and matches the S5.4 block-extension cadence on the other side.

M2.5 — Block clearance

On work completion, the crew lead confirms: all tools withdrawn, all personnel accounted for by name, track walked clear of debris, protection signals cleared. The OCC is then radioed to release the `MaintenanceOverride` per S5.5.

Why: a hammer left on a rail derailed a trainset. The named-personnel accounting prevents leaving a worker behind; the walk-clear is the final sweep.

M2.6 — No trains through a block

No rolling-stock movement is authorised through a section under active `MaintenanceOverride`, regardless of written orders, radio instruction, or apparent urgency. The block is a hard interlock.

Why: historical incidents almost all involve "just one train through" at a supposedly secure block. Making M2.6 a no-exception rule removes the ambiguity — if a train must go through, the block must first be cleared, not overridden.

docs/operations/maintenance/m3-routine-track-inspection.md

docs/operations/maintenance/m3-routine-track-inspection.md

M3 – Routine track inspection

Scope: walking the line, visual checks, geometry-recording-car cadence per RFC 0009 §9.

Cross-refs: RFC 0009 §9, RFC 0013 §4.4 M3.

Rules

M3.1 – Visual walk

Every section is walked visually every 7 days in the direction of traffic. Walkers check for: missing fasteners, cracked sleepers, rail-head spalling, ballast shoulder loss, and vegetation encroachment.

Why: visual inspection catches the failures that measurement misses (missing bolts, cracked concrete sleepers) and measurement catches what visual misses (geometry drift). Both cadences are non-redundant.

M3.2 – Geometry recording car

The geometry recording car runs per RFC 0009 §9 preset cadence: standard-urban every 90 days, standard-metro every 60 days, mainline-mixed every 45 days. Output is reconciled against the previous run to detect trend deterioration.

Why: geometry drift (gauge, cant, alignment, twist) is linear-predictable from trend data; a single reading is less informative than the delta. The preset-specific cadence reflects the loading severity of each preset.

M3.3 – Post-weather inspection

Flooding, dust, or heatwave events (per S4.1–S4.3) trigger an immediate post-event walk of affected sections before full-speed service resumes. The walk is additional to, not a substitute for, the 7-day routine walk.

Why: extreme weather produces failure modes (ballast washout, drift-jammed switches, rail buckles) that the 7-day walk won't catch in time. Post-event walks insert an extra data point at the moment risk is elevated.

M3.4 – Defect classification

Defects are classified at the point of discovery as QN1 (quality note, alert) or QN2 (immediate action). QN1 is logged and scheduled for the next maintenance window; QN2 raises a work block under M2 and halts service through the affected section until fixed.

Why: the classification is the inspector's judgement call but it's binary — alert or act. Grey-zone intermediates (semi-urgent, action-next-week) are how a QN2 becomes a derailment.

M3.5 – Turnout inspection

All turnouts are inspected monthly for bolt tightness, weld cracks, switch-point closure, and detection-sensor alignment. Findings are logged per switch id to osr-wayside-points.

Why: turnouts concentrate wear and are the single largest derailment-risk feature on the railway. Monthly cycling keeps the inspection window shorter than the failure window for the known modes.

M3.6 – Fence + intrusion walk

Every 30 days, the fence line is walked for breaks, damaged gates, and evidence of animal intrusion. Breaks are recorded, GPS-tagged, and scheduled for repair within 7 days.

Why: a fence break lets in both humans (trespass risk) and animals (collision risk). The 30-day cycle catches seasonal changes (grown-in gaps closing or new gaps opened by flooding) before an incident.

docs/operations/maintenance/m4-switch-maintenance.md

docs/operations/maintenance/m4-switch-maintenance.md

M4 — Switch maintenance

Scope: blade lubrication, detection-sensor cleaning, mechanical inspection per RFC 0012 §4.

Cross-refs: RFC 0012 §4, [osr-wayside-points](#), RFC 0013 §4.4 M4.

Rules

M4.1 — 30-day switch inspection

Every switch is inspected every 30 days per RFC 0012 §4.1: blade closure at toe (≤ 1.5 mm gap), stretcher-bar tightness, switch-rail wear limit, stock-rail wear limit, detection-sensor alignment, and cable-gland integrity.

Why: the 30-day cycle matches the switch-machine CBM curve — beyond that, probability of silent drift into QN2 territory rises above the acceptable envelope.

M4.2 — Self-lubricating slide chair

The RFC 0012 self-lubricating slide-chair spec requires no routine re-greasing, but the 30-day inspection confirms slide chairs are free of sand contamination, dust clumping, and debris. Contaminated chairs are cleaned but not re-greased.

Why: greasing over contaminated chairs traps the contamination in the slide path, which is worse than the contamination alone. Cleaning without re-grease is the correct maintenance action per the chair datasheet.

M4.3 — Stretcher-bar tightness

Stretcher-bar bolts are torque-checked to the RFC 0012 spec (200 N·m, per RFC 0012 §3 kit) every 30 days. Loose bolts are tightened and flagged for follow-up on the next cycle.

Why: a stretcher-bar failure is a classic derailment mode — the blade opens under a passing wheel. Torque-checking is the only way to catch creeping loosening before it becomes mechanical play.

M4.4 — Detection-sensor cleaning

A/B inductive detection-sensor faces are cleaned every 30 days with a dry cloth. Sensors showing drift in their response curve (logged to [osr-wayside-points](#)) are replaced at the next 30-day cycle rather than cleaned again.

Why: detection sensor reliability is the [osr-wayside-points](#) safety case's key assumption. Cleaning maintains the reliability; replacement at drift onset is cheaper than the MA-blocking fault that a failed sensor triggers.

M4.5 — Switch-machine motor

Switch-machine electric motors are inspected every 90 days for brush wear, commutator condition, and current draw at throw (compared against the baseline). Motors outside the current-draw envelope are replaced.

Why: motor current rise precedes mechanical failure by several weeks; catching it at the 90-day cycle prevents mid-operation failure. The baseline comparison makes the drift detectable earlier than a single snapshot.

M4.6 — Points heating

For deployments in climate zones where frost is a concern (not Samawah, but the RFC 0012 spec covers cold-climate adaptations), thermostats for points heating are checked at the start of cold season: calibration check, heating-element resistance check, contactor operation.

Why: frozen points are a winter-morning incident in every cold-climate system; a heating fault discovered on the first frost is a reliability failure. The pre-season check moves discovery into a controlled window.

M4.7 — Post-dust-storm

After any S4.1 dust-storm event, all switches in the affected area are inspected and blade-cleaned within 48 hours, independent of the 30-day cycle.

Why: dust-jammed switches are the single largest post- haboob failure mode in the Middle East climate adapter. The 48-hour window ensures the cleaning happens while the drift is fresh, before wheel traffic packs it into the slide chair.

docs/operations/maintenance/m5-fleet-maintenance.md

docs/operations/maintenance/m5-fleet-maintenance.md

M5 — Fleet maintenance

Scope: inspection intervals at the depot — daily, 7-day, 30-day, overhaul. Per RFC 0014 §5.1.

Cross-refs: RFC 0014 §§5.1, 5.2, [osr-cbm-onboard](#), [osr-cbm-backend](#), RFC 0013 §4.4 M5.

Rules

M5.1 — Daily inspection

Every trainset returning to depot is inspected daily at a main-heavy or secondary-medium depot for: wheel and tread visual (no flats, no cracks), door sealing (no visible gaps), HVAC nominal, external body damage, and coupler face integrity.

Why: daily inspection is the fastest feedback loop on service wear. Catching a tread flat within one day of formation prevents the flat from becoming a wheel spall that requires reprofiling.

M5.2 — 7-day inspection

Every trainset is inspected on a 7-day cycle for: wheel-wear measurement (flange thickness, tread depth), friction-brake pad thickness, battery-pack cell voltage spread, and under-car wire harness chafe.

Why: 7 days is the cadence at which wheel wear becomes measurably distinguishable from noise, and where a pad approaching its wear limit can still be service-scheduled without an emergency swap.

M5.3 — 30-day inspection (EN 50126 A-class)

Every trainset is taken out of revenue service monthly for a 30-day inspection: BMS deep-scan (every cell, every temperature sensor, every contactor), motor-bearing vibration spectrum, HVAC filter replacement, and aux-inverter output quality check.

Why: the deep-scan items are diagnostic gold — they catch failure precursors (cell imbalance, bearing harmonics) that the daily and 7-day scans are not instrumented to see. EN 50126 A-class is the safety-case-anchored interval.

M5.4 — Wheel reprofiling

Wheels are reprofiled on the depot lathe at 150 000 km cumulative service, or earlier if the 7-day inspection records wheel wear exceeding the reprofile limit.

Why: 150 000 km is the RFC 0014 §5.2 CBM curve intersection of wheel wear and reprofile cost. Earlier reprofiling wastes wheel; later reprofiling risks flange failure.

M5.5 — Bogie overhaul

Bogies are overhauled at 600 000 km cumulative service at main-heavy depots only. Secondary-medium depots do not carry the jigs required for bogie overhaul.

Why: bogie overhaul requires frame-straightening jigs and a wheelset press that would double the footprint of a secondary-medium depot for a task that occurs once every 10+ years per trainset. Centralising at main-heavy is the RFC 0014 archetype rationale.

M5.6 — Body overhaul

The carbody undergoes overhaul at 10 years in service and every 10 years thereafter: body shell corrosion inspection, interior refurbishment, cable replacement, HVAC system swap.

Why: body overhaul is a 20-40 year fleet life extension activity. Skipping it shortens the fleet life from 40 years to around 20 — a capital-planning catastrophe for the deployment.

M5.7 — CBM-triggered intervention

When [osr-cbm-backend](#) raises an amber condition on a trainset component, the technician schedules intervention within 7 days; a red condition triggers intervention within 24 hours, with the trainset held out of service in the interim.

Why: CBM exists to move interventions from reactive to scheduled. The 7-day and 24-hour windows are the RFC 0013 §4.4 thresholds calibrated against failure-mode progression curves for the component classes in scope.

docs/operations/maintenance/m6-incident-site-access.md

docs/operations/maintenance/m6-incident-site-access.md

M6 — Incident site access

Scope: response-crew access to derailment / fire / crash sites. Coordination with emergency services and site preservation.

Cross-refs: RFC 0013 §§4.4 M6 + §7 incident categorisation.

Rules

M6.1 — Site access authorisation

Access to an incident site is authorised only under a formal M2 work block, or under an explicit emergency authorisation issued by the OCC (name + timestamp + scope). Self-dispatched recovery crews are turned away at the protection signal.

Why: well-intentioned crews arriving unannounced are a hazard to emergency services and contaminate the incident scene. The authorisation channels all access through the OCC, which keeps the site count controllable.

M6.2 — Crash-site preservation

At a crash or derailment site, rolling stock is left as-found for the NRSA investigator. Movement is authorised only after the investigator releases the site, or in the specific case where movement is necessary to free trapped passengers (in which case a named OCC authorisation with timestamp is recorded).

Why: moving evidence is how cause determination gets closed as "undetermined". The trapped-passengers exception is the only override; it is rare and must be documented.

M6.3 — Fire site

At an onboard fire, civil fire services lead the response on arrival. The maintenance crew's role is limited to: confirm osr-fire-safety onboard systems have actuated (suppression discharged, traction isolated), provide traction-battery isolation information to the fire officer, and brief hazmat content (Na-ion or LFP cell chemistry).

Why: fire services are trained for fire; rail crews are trained for rail. The crew's job is to give the fire officer the information they need to fight the fire safely without pretending to lead a task the crew is not trained for.

M6.4 — Crane operations at site

On-site crane operations (recovering a derailed trainset) require a qualified operator (per M1.3), an additional lift plan reviewed by the M&E engineer, and spotters at both ends of the lift. Multi-lift (two cranes on one load) requires a specific multi-lift plan.

Why: on-site crane work is higher-risk than depot crane work — uneven ground, time pressure, media attention. The additional plan requirement is the compensating control.

M6.5 — Evidence chain

Before any recovery move, the crew photographs the site comprehensively (GPS-tagged, timestamped), extracts the event recorder per D8, and makes a sealed copy for the investigator. The chain-of-custody form is countersigned by the crew lead and OCC duty officer.

Why: recovery destroys the spatial evidence; photographs and the event recorder preserve the information content of the site. The chain-of-custody form makes the evidence admissible.

M6.6 — Recovery move

A recovery move (derailed or crashed trainset being moved off the line) operates at 15 km/h maximum, with physical escort fore and aft, and carries no passengers or personnel inside the trainset being recovered.

Why: recovery moves use compromised rolling stock on possibly compromised track — the speed cap keeps kinetic energy low. The no-passengers rule prevents a bad move becoming a mass-casualty event.

docs/operations/maintenance/m7-wayside-sensor-maintenance.md

docs/operations/maintenance/m7-wayside-sensor-maintenance.md

M7 – Wayside sensor maintenance

Scope: per-section wayside intrusion-detection packs — fence-line contacts, ROW-mounted LIDAR, ROW-mounted radar, CCTV with AI classifier. See RFC 0016 §4 for the pack composition.

Cross-refs: [osr-intrusion-detect](#), RFC 0016 §7.2 M7, RFC 0013 §4.4 (maintenance rulebook).

Rules

M7.1 – 30-day sensor-pack walk-through

Every wayside sensor pack is walked at 30-day intervals: LIDAR and radar housings visually inspected for physical damage and lens fouling, cable runs checked for rodent damage, fence-line contact sensors spot-tested by induced vibration, junction box door sealed.

Why: 30 days matches the M3 track-inspection cadence so the walkthroughs can be combined and the crew makes one trip instead of two. Most sensor faults develop over weeks, not days — monthly is a reasonable detection floor.

M7.2 – Post-weather inspection

After any RFC 0013 S4 weather event (haboob, heatwave, flooding, lightning), maintenance inspects the wayside packs in the affected sections within 48 hours. LIDAR + radar lens cleaning is routine after haboob; fence-line contact re-set is routine after lightning.

Why: weather is the single largest source of wayside- sensor false positives and silent failures. Getting in within 48 hours catches the obvious (dust-fouled lens) before it becomes a sustained Unknown that an S7.2 ticket is already addressing.

M7.3 – Sustained-Unknown ticket response

A work ticket opened by dispatcher S7.2 (sustained Unknown verdict) triggers an on-site maintainer visit within 4 hours of ticket creation. The maintainer performs a calibration + self-test on each sensor, replaces any failing unit from the depot spare pool, and issues a "returned to service" confirmation to the dispatcher.

Why: the interlocking holds the section against revenue traffic for the entire duration of the Unknown. Four hours is the network-service cost budget: anything longer starts to degrade timetable adherence to the point where the dispatcher is re-routing or short-turning.

M7.4 – Calibration after replacement

A replaced wayside sensor is re-calibrated against the section's known rail-profile envelope before being committed back into the `osr-intrusion-detect` evaluator. Calibration uses the depot's reference retro-reflector at the published lateral offsets from rail centreline; the first `Clear` verdict after calibration is logged to the shift record.

Why: a replacement unit out of the box reads lateral offsets off its own factory axis, not the section's rail centreline. Skipping calibration is how sensor replacement turns into a false-`Present` storm that reproduces the failure on the next train.

M7.5 – Fence-line continuity check

Every 90 days, the fence-line contact run is end-to-end continuity-tested under sensor simulator stimulus. Any continuity loss drives a segmented walk to isolate the fault within 500 m.

Why: a quiet fence-line is ambiguous — it could mean "no breach" or "sensor run broken for weeks." The 90-day stimulus test distinguishes the two, which is what makes the fence contact trustworthy as an I3 safety input.

docs/operations/station-staff/t1-station-opening.md

docs/operations/station-staff/t1-station-opening.md

T1 — Station opening

Scope: daily station opening — power up, PIS check, fare-gate test, PSD test, CCTV + PA handover.

Cross-refs: [osr-pis-station](#), [osr-afc](#), [osr-tvm](#), [osr-psd](#), [osr-station-scada](#), RFC 0013 §4.3 T1.

Rules

T1.1 — Power-up sequence

Station staff power up subsystems in order: station SCADA PIS fare (AFC + TVM) PSD. Each subsystem must reach "ready" on the station console before the next is started.

Why: PIS depends on SCADA time-sync; fare depends on PIS for passenger display; PSD depends on fare for emergency-release interlock. Out-of-order power-up races the dependencies and produces spurious amber faults that waste the opening window.

T1.2 — PIS display self-test

Staff trigger the PIS self-test from the station console. The test cycles each platform display through a colour pattern + a sample "next train" message and reports OK per display to [osr-pis-station](#). Any display that fails self-test is logged as a T1 fault and replaced or bypassed.

Why: a display that lights up but shows stale data is worse than a dark display — passengers trust the wrong information. The self-test exercises the end-to-end rendering path, not just the backlight.

T1.3 — Fare-gate test

Staff test each fare gate through a full open / close / emergency-release cycle using a staff test card. Gates that fail to close within 3 s or fail emergency release are taken out of service and flagged for depot repair.

Why: a stuck-closed gate traps passengers in peak; a stuck-open gate loses revenue and undermines fare enforcement. The daily test catches both before passengers arrive.

T1.4 — TVM test

Staff run a test purchase on each TVM: QR-code display, mobile-money confirm, receipt print. Failed TVMs are taken out of service with a paper "out of order" sign placed physically on the machine.

Why: a TVM whose receipt printer has run out of paper sells tickets passengers can't prove they bought, which breaks fare enforcement downstream. The daily test catches the mundane failures (paper, ribbon, mobile-money API key expired).

T1.5 — PSD test

At PSD-equipped stations, staff trigger a platform-side PSD open / close / emergency-release cycle via the station console. PSDs that fail open or close are locked open (not closed) pending repair.

Why: a failed-closed PSD can trap a passenger on the platform side of the door after train departure. Locking failed PSDs open is the fail-safe choice — passengers can still reach the platform but the PSD is visibly out of service.

T1.6 — CCTV handover

Staff assume CCTV monitoring from the OCC night watch via a radio handover: confirmed cameras-OK count, outstanding events logged overnight, any area under intrusion alert.

Why: OCC night watch covers many stations at low attention; morning station staff cover one station at high attention. The handover makes sure any overnight intrusion alert is followed up on the ground, not filed and forgotten.

T1.7 — Ready-to-OCC

After T1.1–T1.6, staff post a "station ready" status via [osr-occ](#), which clears the station's morning-open flag in the OCC roster. The dispatcher will not dispatch a first revenue service to a station that has not posted ready.

Why: an unready station takes a first-service trainset full of passengers with no working PIS, no open fare gates, and possibly no PSDs. The explicit ready-post is the handshake that prevents that.

docs/operations/station-staff/t2-passenger-boarding.md

docs/operations/station-staff/t2-passenger-boarding.md

T2 — Passenger boarding

Scope: platform supervision during revenue hours. PSD monitoring, wheelchair assistance, queue regulation.

Cross-refs: RFC 0010 §§5, 7 (vertical circulation, accessibility), RFC 0013 §4.3 T2.

Rules

T2.1 — Platform walk

During peak hours, staff walk the length of the platform at 15-minute intervals, checking for unattended items, fallen passengers, blocked emergency exits, and fare-gate queue overflow. Walks are logged on the station console.

Why: stationary platform staff see the area around their post; walking staff see the whole platform. Peak hours are when things go wrong fastest, so the walk cadence tightens then.

T2.2 — Wheelchair boarding

On a wheelchair boarding request (pre-notified or spot), staff position the boarding ramp at door 2 of car 1 (aligned with the reference stop mark per D4.1), assist the passenger across the platform gap, and radio the driver to extend dwell per D4.4.

Why: the gap between platform and trainset is wider than most wheelchairs can clear unassisted, and the dwell is shorter than unassisted boarding needs. The ramp + extension is the accessibility guarantee that RFC 0010 §7 commits to.

T2.3 — Overcrowding

When platform density exceeds the archetype capacity threshold (per RFC 0010 §10), staff start queue management (rope dividers, PA announcements regulating flow to the platform) and notify the dispatcher, who may hold the next-following trainset per S2.5.

Why: platform crowding is a crush risk. Staff have the local signal the dispatcher cannot see; the dispatcher has the fleet lever the staff cannot pull. Splitting the response matches the information to the authority.

T2.4 — Unattended item

On an unattended item report, staff clear a 5 m radius around the item, notify the OCC, and call police / security per the deployment's protocol. The trainset approaching the platform is held at the previous station until the item is cleared.

Why: a suspicious item is rarely a bomb but always a disruption. The 5 m clear preserves the scene; the hold prevents a new trainset-load of passengers arriving into it.

T2.5 — Platform clear confirmation

At non-PSD stations with no train-camera live feed, staff confirm the platform is clear of passengers in the door zone before the driver commands door close. Confirmation is by radio; absence of confirmation defers door close by up to 15 s (one dwell extension) before the driver must interrogate.

Why: the driver's side mirror doesn't cover the full platform length. Staff are the second pair of eyes that catches a passenger running for the train at door close. At PSD-equipped stations the PSD handles this mechanically, so the rule scopes to non-PSD only.

docs/operations/station-staff/t3-passenger-incidents.md

docs/operations/station-staff/t3-passenger-incidents.md

T3 — Passenger incidents

Scope: station-level response to passenger-side incidents — medical, behavioural, fare dispute, lost property.

Cross-refs: RFC 0013 §§4.3 T3 + §7 incident categorisation.

Rules

T3.1 — Medical

On a medical incident, staff render first aid to the level of their training, request emergency services via the OCC (who calls 14 per S3.4), and preserve a 3 m area around the casualty clear of other passengers.

Why: first responder staff don't diagnose — they stabilise and wait. The clear area is both dignity for the casualty and room for ambulance crew to work when they arrive.

T3.2 — Behavioural / altercation

On an altercation or threatening behaviour, staff de-escalate verbally, call transit security (where provided by the deployment) or local police, and log the incident to the station SCADA event log. Staff do not physically intervene except to protect an identified victim.

Why: station staff are not trained for physical intervention and are usually outnumbered. De-escalation plus summoning trained responders is the evidence-based protocol.

T3.3 — Fare dispute

On a fare dispute at the gate, staff do not debate the fare at the gate — they issue a supplement ticket or a dispute slip and direct the passenger to the documented appeal route (station office or online). The gate is cleared within 30 s.

Why: a fare dispute at the gate blocks every passenger behind. The dispute process happens off the critical path; the passenger is neither presumed guilty nor let through free.

T3.4 — Lost property

Lost-property items are logged to the station log with description, finder, and time, secured in the station office locker for 30 days, and entered into the network lost-property database accessible to other stations. After 30 days, items are donated or disposed of per deployment policy.

Why: passengers tracking lost items often call or visit more than one station. The network database makes the find discoverable from any station; the 30-day ceiling keeps the locker from filling with old umbrellas.

T3.5 — Child separated

On a child-separated-from-caregiver report, staff issue a platform PA with description, initiate CCTV review from the last known time and place, and notify the OCC within 5 minutes if the child is not found. Station exit gates are watched manually until the child is located.

Why: separated children are found most often within the first 10 minutes. The staged response (PA CCTV OCC escalation) scales effort to duration without burning the network on a 2-minute reunion.

docs/operations/station-staff/t4-fare-enforcement.md

docs/operations/station-staff/t4-fare-enforcement.md

T4 — Fare enforcement

Scope: on-board inspector work, non-payment handling, appeals.

Cross-refs: [osr-afc](#), RFC 0013 §4.3 T4.

Rules

T4.1 — Inspector identification

Fare inspectors wear a visible identity badge with photograph and inspector number, and rotate routes daily so that the same inspector does not work the same line consecutively.

Why: visible ID legitimises the stop for the passenger. Daily rotation limits the opportunity for harassment (by or of) any one inspector becoming personal.

T4.2 — Fare check

On stopping a passenger, the inspector scans the passenger's ticket (paper QR, mobile QR, or tap card) via the handheld which verifies the HMAC-signed token against [osr-afc](#)'s active-tokens list. Verification is online or cached (cached ≤ 15 min old is accepted).

Why: the HMAC check catches forged tickets deterministically. The 15-minute cache keeps inspectors working through short backhaul flaps without falsely accusing paying passengers.

T4.3 — Fare-evasion record

On a failed check, the inspector issues a supplement-fare notice (printed on the handheld) stating the infringement, date, and appeal route. The passenger's identifier (name + document number per deployment law) is logged to the infringement database. No physical detention occurs — the passenger may leave the train at the next station.

Why: detention escalates the situation and puts both parties at risk. The paper trail is sufficient legal basis for the penalty; physical enforcement is a police function.

T4.4 — Appeals

Supplement-fare appeals may be filed within 14 days of issue, in writing or via the deployment's online portal. The OCC reviews the appeal against the AFC record and inspector log; outcome issued within 30 days.

Why: the 14-day window is long enough for a passenger returning from travel and short enough to keep the AFC record fresh in the database. The OCC-not-inspector review keeps the decision independent of the person who issued.

T4.5 — Inspector safety

Inspectors work in pairs during late-evening slots (after 21:00) and on routes flagged as elevated-risk by the OCC. Lone working is permitted only during peak daytime hours on mainline routes.

Why: lone inspectors on quiet late services are an assault risk. Pair working is not about enforcement productivity (which drops) but about the inspector going home safely.

docs/operations/station-staff/t5-station-closure.md

docs/operations/station-staff/t5-station-closure.md

T5 — Station closure

Scope: end-of-day shutdown. Evacuation (normal last-train passenger clear), equipment powerdown, CCTV handover to OCC night watch.

Cross-refs: RFC 0013 §4.3 T5.

Rules

T5.1 — Last-train clear

After the last revenue service departs, staff confirm the platform is clear of alighting passengers and that no passenger is attempting to board a non-revenue trainset. The last-train clear is timestamped to the station log.

Why: the last train is where passengers are most likely to miss a stop or fall asleep. The explicit clear establishes that the platform was empty at a known time, which is important for any later incident report.

T5.2 — Platform + concourse walk

Staff walk the platform, stairs, concourse, and fare-paid area checking for unattended items, sleeping passengers, and equipment left out. Sleeping passengers are woken and escorted out; unattended items follow T2.4.

Why: station closure without a walk leaves people inside a locked station overnight, which has happened at many real systems. The walk is the non-negotiable final sweep.

T5.3 — Fare-gate + TVM lockout

Fare gates are locked in the closed-no-entry position; TVMs are switched to "out of service" display and physically secured. Night-mode lighting (50 % level) is switched on.

Why: open fare gates overnight invite entry to a station with no staff presence; lit TVMs invite tampering. The lockout and the out-of-service display remove the temptation.

T5.4 — PIS off-hours banner

The PIS is switched to the off-hours banner (first-train timetable for tomorrow, emergency contact, no moving content). Displays that would otherwise show "no service" are put into standby instead.

Why: "no service" on a PIS overnight conditions passengers to ignore PIS in the morning when "no service" would be real. The dedicated off-hours banner is unambiguous.

T5.5 — PSD overnight position

At PSD-equipped stations, PSDs are driven to the fully-closed position for overnight. The PSD fault log is flushed to `osr-station-scada` so morning staff can review overnight anomalies.

Why: closed PSDs overnight prevent unauthorised platform access from the concourse and reduce track-side dust accumulation on PSD guide rails.

T5.6 — CCTV handover

Staff hand over CCTV monitoring to the OCC night watch via radio, acknowledged by the OCC night operator with a countersigned entry in the station log.

Why: CCTV without a watcher is CCTV for post-hoc review only. The handover ensures someone is always watching, and the countersignature makes the moment of handover auditable.

T5.7 — SCADA low-power mode

The station SCADA is left in low-power mode (displays off, core services running, heartbeat to the OCC active). Services that can be off overnight (PA, TVM, non-critical lighting) are switched off.

Why: the station still needs to respond to an overnight event (fire, intrusion, flood) so the core SCADA stays on; everything else reduces overnight load on the station-solar battery bank.

RFC 0001 — Track State Consensus

Status: Draft Date: 2026-04-20 Depends on: [docs/ARCHITECTURE.md](#) §4 D2

1. Summary

OpenSourceRail replaces the centralized zone-controller model of conventional CBTC with a distributed, replicated log of track state maintained by wayside nodes via a consensus protocol. Each train computes its own Movement Authority (MA) from the log, cross-validated by two independent wayside nodes. This RFC specifies the log schema, the consensus protocol, the MA computation algorithm, the fault model, and the formal-verification plan.

The design target is SIL-4 safety at <€5k per wayside site in reference-hardware cost, with p99 end-to-end MA update latency <200 ms.

2. Motivation

Conventional CBTC (Siemens Trainguard, Alstom Urbalis, Thales SelTrac) concentrates authority in a Zone Controller (ZC) — a redundant pair of proprietary computers per line or per large segment. The ZC:

- Is a single logical point of failure. When it fails, the whole zone falls back to degraded manual operation.
- Is a single vendor product. Swapping it out is a multi-year, eight-figure project.
- Is opaque. Its internal state machine is not published; operators cannot inspect it; assessors re-derive correctness per certification.
- Costs €10–50M per line, before rolling-stock ATP costs.

A distributed log replaces the ZC with N commodity wayside nodes that collectively hold authoritative state. Any quorum-majority subset continues operating. No single failure stops service. The log is the specification: state is a function of committed entries, and the state machine is public.

This is cheaper, simpler, more resilient, and more auditable — provided we can prove the distributed protocol is safe. That proof is the heart of this RFC.

3. Non-goals

- Not a general-purpose database. The log carries a bounded, well-typed set of rail-domain facts. We do not aim for SQL, secondary indexes, or arbitrary queries.
- Not Byzantine-fault-tolerant in v1. Wayside nodes are operated by a single authority, run signed firmware, and attest at boot (see ARCHITECTURE.md §8). We trust them to follow protocol or crash. Byzantine resilience may be added in v2 but it is not required to close the SIL-4 safety case.
- Not a replacement for ATP onboard. Onboard Automatic Train Protection (emergency brake on MA violation) remains the last line of defense. The log defines what is allowed; onboard ATP enforces what actually happens.
- Not globally consistent across regions. Consensus is per region (§6.1). Region boundaries are handled via a handoff protocol (§7.4), not global consensus.

4. Concepts and Vocabulary

Term	Meaning
Region	A contiguous operating area of ~5–50 km covered by 5–20 wayside nodes in a single consensus group.
Wayside Node (W-Node)	SBC running the consensus + interlocking stack at a physical site (station, junction, substation).
Train Agent	Software on a train's T-ECU that publishes position reports, computes MA, and enforces ATP.
Log	The append-only, totally-ordered, quorum-committed sequence of facts about the region.
Entry	A single typed record in the log (§5).
Movement Authority (MA)	A permission for a specific train to occupy a specific set of track sections up to a specific position, valid until a specific time.

Term	Meaning
Safety Envelope	The union of track sections a train may occupy without colliding with anything, given current log state. A train's MA is always a subset of its safety envelope.
Commit	An entry has been accepted by a quorum majority and is durable.
Fail-safe direction	In rail, the fail-safe state is "stopped." Any ambiguity must resolve toward more restrictive MA, not less.

5. The Log

The log is a sequence of entries. Each entry has:

- `entry_id`: monotonic u64, unique within region
- `term`: consensus term (Raft-style)
- `timestamp`: TAI nanoseconds, PTP-synced wall clock at the leader
- `kind`: one of the entry types below
- `payload`: typed body
- `signature`: ed25519, signed by the leader's hardware key (for audit; not load-bearing for consensus)

5.1 Entry types

```

TrainPositionReport {
  train_id: TrainId,
  position: Position,           // track-ID + offset, sub-meter resolution
  uncertainty_m: f32,           // 95% CI half-width
  speed_mps: f32,
  speed_uncertainty_mps: f32,
  heading: Heading,
  source_mask: u8,              // which sensors contributed (GNSS/IMU/odo/beacon)
  onboard_time_ns: u64,
}

SwitchCommand {
  switch_id: SwitchId,
  requested_position: SwitchPosition,
  requested_by: EntityId,       // dispatcher or automatic route setter
  lock_until_entry: Option<EntryId>,
}

SwitchObservation {
  switch_id: SwitchId,
  observed_position: SwitchPosition,
  confidence: Confidence,       // Locked | Transitioning | Unknown
  observed_at_ns: u64,
}

RouteRequest {
  route_id: RouteId,
  requested_by: EntityId,
  entry_point: TrackRef,
  exit_point: TrackRef,
  train_id: Option<TrainId>,    // bound when granted
}

RouteGrant {
  route_id: RouteId,
  train_id: TrainId,
  locked_switches: Vec<SwitchId>,
  locked_sections: Vec<SectionId>,
  expires_at_ns: u64,
}

SpeedRestriction {
  section: TrackRef,
  max_speed_mps: f32,
  reason: RestrictionReason,    // permanent | temporary | emergency
  effective_from_ns: u64,
  effective_until_ns: Option<u64>,
}

TrainRegistration {
  train_id: TrainId,
  consist: ConsistDescriptor,   // length, mass, braking curve, classification
  cert_chain: CertChain,
}

Heartbeat {
  from_entity: EntityId,
  health: HealthStatus,
}

MaintenanceOverride {

```

```

    section: TrackRef,
    granted_to: EntityId,
    granted_until_ns: u64,
    signed_by: DispatcherSignature,
}

```

All payload schemas live in `crates/osr-core/proto/track_state.proto`. Types are enums with explicit discriminants; forward-compat is handled by reserved fields and a strict versioning rule (see §12).

5.2 Derived state

From the log, any W-Node (or train agent) deterministically computes:

- Section occupancy map: for each section, which train (if any) currently occupies it, with confidence interval.
- Switch state map: for each switch, last observed position and lock status.
- Active routes: set of currently granted RouteGrants that have not expired or been released.
- Speed restriction map: keyed by section.
- Known trains: registered consists and last-known positions.

Derived state is a pure function of the committed prefix of the log. Two nodes with the same committed prefix must compute identical derived state (this is a verified property).

6. Consensus Protocol

6.1 Overview

We use Static-Membership Raft with Fail-Restrictive Timeout (SMRaft) — a narrow variant of Raft with three modifications tuned for safety-critical use:

- 1 Static membership. Membership changes require a maintenance window; there is no online reconfiguration. This eliminates the trickiest part of Raft (joint consensus) and the hardest part of its correctness proof.
- 2 Fail-restrictive timeout. When a node cannot confirm liveness of a quorum within `T_safe` (default 2 s), it stops emitting new RouteGrants and notifies trains via a "shrinking authority" signal. Existing MAs remain valid until their own expiry. This is the fail-safe direction.
- 3 Bounded log growth. The log is aggressively snapshotted (every `N_snap` entries, default 10k) and old entries are pruned after snapshot durability. Train agents subscribe to live tail only.

SMRaft is Raft with parts removed, not added. The underlying correctness argument rides on Raft's well-studied safety proof (Ongaro's TLA+ spec, verified refinements in Ironfleet).

6.2 Region sizing

- 5–20 nodes per region. Below 5, fault tolerance is too thin (need to tolerate 2 simultaneous failures including scheduled maintenance). Above 20, Raft's serial leader becomes a latency bottleneck for commits.
- Quorum: $\text{floor}(N/2) + 1$. Standard Raft quorum.
- Node placement: one W-Node per station or major junction. Extra nodes collocated in depot and OCC to boost quorum size without adding track-side complexity.

6.3 Timing parameters

Parameter	Default	Rationale
<code>heartbeat_interval</code>	50 ms	Allows election timeout much shorter than rail dynamic timescales.
<code>election_timeout</code>	300–600 ms (randomized)	Median failover ~400 ms.
<code>T_safe</code> (fail-restrictive)	2 s	Trains can coast safely for 2 s; anything longer risks MA expiry stacking up.
<code>commit_latency_p99_target</code>	50 ms	LAN RTT x2 plus processing. Achievable on TSN backbone.
<code>MA_refresh_period</code>	500 ms	How often a train reevaluates its MA against log tail.
<code>MA_validity_window</code>	3 s	How long an MA is valid after issue. Short enough to be self-expiring, long enough to tolerate transient network hiccups.

All timing parameters are deployment-configurable but ship with conservative defaults.

6.4 Leader responsibilities

- Serialize writes and replicate to followers.
- Detect stale followers and emit catch-up stream.
- On term change: followers invalidate any in-flight MA computations and recompute from newly committed prefix.
- The leader does not compute MAs. MA computation is decentralized (§7); the leader only maintains the log.

6.5 What the protocol guarantees

Given quorum liveness:

- Agreement: all nodes agree on the committed prefix.
- Total order: entries are totally ordered by (term, entry_id).
- Durability: committed entries survive minority failures.
- Monotonic progress: committed prefix only grows.

Given loss of quorum:

- No new commits. The log stops growing.
- Existing MAs remain valid until expiry. Trains with valid MAs continue until the MA window elapses, then stop.
- No silent split-brain. A minority partition cannot produce new grants because it cannot reach quorum. This is Raft's defining safety property.

7. Train Agent and MA Computation

7.1 Train agent architecture

Each train runs a TrainAgent on its T-ECU. It has four coordinating tasks:

- 1 Publisher: builds TrainPositionReport every 200 ms from sensor fusion; submits to the local W-Node via a regional proxy.
- 2 Subscriber: streams the committed log tail from any two independent W-Nodes (primary and witness).
- 3 MA Computer: recomputes self-MA on every new committed entry in relevant sections (debounced to MA_refresh_period).
- 4 ATP Enforcer: supervises actual train motion against self-MA + wayside-MA; triggers emergency brake on violation.

These four tasks run in isolated Hubris tasks with hardware-enforced memory isolation. Only the ATP Enforcer can command the brake.

7.2 The MA computation algorithm

The self-MA is computed deterministically from the log. The algorithm is small enough to fit on a page:

```
fn compute_self_ma(
    train_id: TrainId,
    log_prefix: &[Entry],    // committed entries visible to this train
    now_ns: u64,
) -> MovementAuthority {
    let state = derive_state(log_prefix);

    // 1. Where am I?
    let my_pos = state.position_of(train_id)
        .unwrap_or_else(|| fail_safe_no_position());

    // 2. What sections does my consist currently occupy?
    let my_footprint = footprint_from(my_pos, state.consist(train_id));

    // 3. Which adjacent sections are free to me?
    let candidate_extension = state.forward_chain(my_pos.direction(), MAX_MA_DISTANCE);

    // 4. Clip at first blocker.
    let mut ma_end = my_pos.head();
    for section in candidate_extension {
        if !section_available_to(train_id, section, &state) {
            break;
        }
        ma_end = section.far_end(my_pos.direction());
    }

    // 5. Apply speed restrictions within the authority.
    let restrictions = state.speed_restrictions_between(my_pos.head(), ma_end);
```



```

// 6. Apply time bound.
let valid_until = now_ns + MA_VALIDITY_WINDOW_NS;

MovementAuthority {
  train_id,
  end: ma_end,
  restrictions,
  valid_until,
  derived_from_entry_id: log_prefix.last().entry_id,
}
}

fn section_available_to(train_id: TrainId, s: Section, state: &State) -> bool {
  // A section is available if and only if:
  // (a) no other train occupies it (with uncertainty margin), AND
  // (b) it is part of a RouteGrant currently assigned to this train,
  // OR it is explicitly unreserved and free, AND
  // (c) every switch in it is observed-locked to the correct position, AND
  // (d) no MaintenanceOverride blocks it, AND
  // (e) all adjacent opposing-direction routes are clear.
  // Any uncertainty (unknown switch, unobserved train, stale heartbeat) => NOT available.
  ...
}

```

Five properties must be proven about this function:

- P1 (determinism): same input same output, across nodes and train agents.
- P2 (non-overlap): two trains A, B at the same instant: `ma_end(A)` and `ma_end(B)` do not share any section unless `A == B`.
- P3 (consist-fit): MA extension accounts for full consist length including safety margin.
- P4 (conservatism under missing data): any missing, stale, or uncertain input produces a more-restrictive MA, never less-restrictive.
- P5 (time bounded): `valid_until - now` is bounded above by `MA_VALIDITY_WINDOW_NS`; no path through the function produces an unbounded authority.

These are the core safety properties we formally verify (§11).

7.3 Cross-validation

The train agent subscribes to two independent W-Nodes. Each independently:

- Streams the log to the train.
- Runs the same MA computation.
- Emits a signed `MovementAuthorityWitness` to the train.

The train agent accepts the intersection of:

- Its own self-computed MA,
- Primary W-Node's witness,
- Secondary W-Node's witness.

"Intersection" means: end position = minimum of the three; speed restriction = maximum of the three; validity = minimum of the three. The fail-safe direction again.

Disagreement logs an incident and degrades to the most restrictive value. Persistent disagreement (>3 cycles) triggers a service brake and notifies dispatch — this indicates either a compromised node or a bug, both of which warrant intervention.

7.4 Region boundaries

A train traversing a region boundary runs two train agents in parallel for a transition window:

- 1 `T_in` (approaching boundary): subscribes to region A's log only.
- 2 `T_cross` (entering approach zone ~500 m before boundary): subscribes to both A and B. Both regions must see a consistent `TrainRegistration` for the train. Both regions compute MAs; train accepts the intersection.
- 3 `T_out` (fully in region B): unsubscribes from A after A has logged a `TrainDeparture` entry.

This handoff is not consensus across regions. It's coordinated publish/subscribe with cross-region authentication. The safety argument is local to each region plus the handoff protocol itself, which is much simpler to verify than global consensus would be.

8. Fault Model and Behavior

Fault	Frequency	System response
Single W-Node crash	Occasional	Quorum holds; no operational impact. Failed node restarted, rejoins as follower.
Leader crash	Occasional	Election in ~300–600 ms; brief commit pause; no MA impact if pause < MA_VALIDITY_WINDOW.
Quorum loss (partition)	Rare	No new commits. Existing MAs expire over ~3 s; trains coast to stop. Service resumes when quorum reforms.
Radio link degradation	Common	Train may miss witness messages; falls back to self-MA with reduced confidence; MA recomputation continues. Persistent loss brake.
Position sensor failure (one modality)	Uncommon	Sensor fusion degrades gracefully; position uncertainty grows; MA shrinks proportionally.
Position sensor failure (all modalities)	Rare	Train reports "position lost" to log; all other trains treat section containing unknown train as blocked; the train itself brakes to stop.
Byzantine W-Node	Out of scope for v1	See §3.
Train agent crash	Rare	Watchdog brake. Train stops. Onboard spare can be hot-swapped in depot.
Rogue dispatcher command	Mitigated	Dispatcher commands go through MaintenanceOverride which requires a signed second authorization for safety-critical operations.

Invariant under all enumerated faults: no train ever receives an MA that includes a section occupied by or reserved for another train. This is the central property the safety case argues.

9. Latency Budget

End-to-end MA update (position changes train has updated MA):

Stage	Budget (p99)	Implementation note
Onboard sensor fusion update	20 ms	Runs at 200 ms cadence; jitter <20 ms
Position report W-Node (5G uplink)	40 ms	Commercial 5G SA; LoRa fallback is slower and not in this path
W-Node ingress leader	5 ms	TSN backbone
Leader commit (quorum replication)	30 ms	LAN RTT ×2 + processing; budget includes worst-case node
Commit train subscribers	40 ms	5G downlink to two witnesses in parallel
MA recomputation (onboard)	5 ms	Deterministic fixed-cost algorithm
Witness cross-check	10 ms	In-memory
Total end-to-end	150 ms	
Target p99	200 ms	50 ms headroom

If 5G uplink is unavailable, position reports fall back to LoRa (carrier-independent, but ~500 ms latency). In that mode, MA refresh rate drops and MAs shrink defensively — not a safety concern, but throughput drops until 5G restores.

10. Interoperability and Migration

Operators migrating from legacy systems need a path that doesn't require flag-day cutover.

- Shadow mode. OSR consensus runs alongside the legacy ZC, consuming the same inputs, producing MAs that are compared but not enforced. When the shadow matches reality for a defined period, operation can cut over.
- Legacy inputs. W-Nodes can ingest from existing track circuits and axle counters as additional TrainPositionReport sources (with explicit source_mask bits identifying legacy origin). This lets a region run OSR consensus atop unmodified detection hardware while rolling stock and detection are gradually upgraded.
- Legacy outputs. A dedicated "OCS gateway" W-Node can translate MAs into legacy continuous cab-signal codes for older rolling stock still running lineside signal protection. This is a transitional feature; it is not a supported long-term mode.

Interoperability modules live in crates/osr-legacy-* and are explicitly out of the SIL-4 boundary; they are advisory only.

11. Formal Verification Plan

Four layers of verification, each proving different properties:

11.1 Protocol layer (TLA+)

A TLA+ specification of SMRaft, refining Ongaro's Raft spec, proves:

- Election safety: at most one leader per term.

- Log matching: identical committed prefixes across nodes.
- Leader completeness: committed entries persist across terms.
- Fail-restrictive liveness: under quorum loss, no node emits new RouteGrants; existing MAs are allowed to expire.

11.2 State-machine layer (Kani)

Bounded model checking of `derive_state` and `compute_self_ma` with Kani:

- Determinism (P1): two invocations on the same prefix produce byte-identical outputs.
- Non-overlap (P2): for any pair of registered trains with committed positions, the computed MAs do not share sections.
- Conservatism (P4): mutating an input entry to be "more uncertain" produces an MA whose end is no further than the original.
- Time boundedness (P5): all MA outputs have `valid_until - now ≤ MA_VALIDITY_WINDOW_NS`.

Bounds: up to 8 trains, up to 50 log entries per harness — enough to exercise all code paths. Larger systems rely on the TLA+ protocol argument plus the property that state derivation composes over log suffixes.

11.3 Rust-code layer (Creusot where feasible, tests elsewhere)

- Creusot contracts on the pure functions (`section_available_to`, `forward_chain`, `footprint_from`) discharging obligations from the Kani layer.
- Property-based tests (`proptest`) for algebraic invariants (e.g., "MA end is always $\geq$ current head position").
- Differential testing against a reference interpreter written in a simpler style (likely Python) for the same log inputs.

11.4 System layer (simulator)

- `osr-sim` runs full regional simulations with injected faults (node crashes, partitions, sensor glitches, Byzantine-ish data corruption short of full Byzantine fault).
- Invariant monitors in the sim flag any state where two trains could collide within their MAs. Any run that flags an invariant is a test failure.
- Soak tests for 30-day simulated operation with realistic traffic patterns before any hardware pilot.

11.5 Evidence linkage

Each Kani harness, TLA+ module, Creusot proof, and sim run emits a structured evidence artifact consumed by the safety-case compiler. The safety case (GSN) claims "P2 holds" link to the specific evidence artifacts that discharge the claim. If a commit invalidates any artifact, CI breaks the build.

12. Schema Evolution

Log entries are protobuf. Evolution rules:

- Never reuse field numbers. Deprecated fields are reserved and held forever.
- Never change a field's type. Introduce a new field with a new number.
- Log format version is part of the log. The first entry of every region's log is a `FormatVersion`. Entries past a version bump are rejected by nodes running the older version — they must upgrade or be replaced. (Upgrade is a maintenance-window activity.)
- Schema changes require an RFC. No in-repo schema changes without review.

The strict discipline is worth the friction. A rail signaling log will be read by tools we haven't written yet, in decades.

13. Open Questions

- 1 Raft vs. alternatives. Is there a protocol that's even simpler to formally verify while meeting our needs? Candidates: single-decree Paxos per entry (more messages but simpler proofs), a static-primary protocol with only failover consensus. Needs a focused comparison RFC.
- 2 How conservative is "conservative enough" for uncertainty handling? The exact error model for sensor fusion inputs determines how much MA margin trains leave; this will be tuned empirically in simulation and pilot.

- 3 Ed25519 vs. post-quantum signatures. The log is append-only and will be read for decades. Should signatures be post-quantum from day one? Tentative yes — Dilithium2 costs us ~2 KB per entry, affordable.
- 4 Dispatcher override safety. Maintenance overrides require dispatcher signatures. What's the exact protocol for "I, a human dispatcher, authorize this train into a blocked section for rescue"? Needs its own RFC.
- 5 Time source integrity. PTP sync is assumed. What if the GPS clock source is spoofed or jammed? Trains use their own clocks for MA expiry; disagreements between train-local and log-local time produce restrictive behavior automatically, but this needs to be verified.
- 6 Scale testing. Largest region we've thought through is ~50 km / 20 nodes. What's the upper bound before we need regional subdivision for latency?

14. Prior Art

- Raft (Ongaro, 2014) — consensus protocol we specialize from.
- Ironfleet (Hawblitzel et al., 2015) — formally verified replicated state machines in Dafny; inspiration for our verification layering.
- PALS / Hurdle — formal work on distributed real-time control systems in avionics; informs timing argument.
- Thales SelTrac, Siemens Trainguard, Alstom Urbalis — the centralized ZC designs we replace.
- Alstom Axonis, Siemens Trainguard MT with DCS-M — recent vendor moves toward more decentralized architectures; validate that the industry is trending this way, but still not open.
- Sun-Ways (track-PV), Stadler FLIRT Akku (BEMU), Hitachi Masaccio — related innovations on the energy side (see ARCHITECTURE.md §4 D5, D7).

This RFC is a draft. Substantive changes go through PR review with at least one reviewer from each of: distributed-systems, formal-methods, and rail-signaling background.

RFC 0002 — Energy Sizing for a Reference Light Metro Line

Status: Draft Date: 2026-04-20 Depends on: docs/ARCHITECTURE.md §4 D5, §4 D7

1. Summary

This RFC puts concrete numbers to the catenary-free, solar-first energy architecture of D7. It does so for a reference light-metro line: 20 km, 15 stations, 2-track, 5-minute headway peak. For this reference case the RFC derives:

- Daily energy demand: ≈ 100 MWh/day.
- PV nameplate required for net-positive annual generation in a 5 PSH climate: ≈ 25 MW.
- Trackside sodium-ion storage required for service continuity through a two-day solar outage: ≈ 60 MWh.
- Onboard battery per trainset: ≈ 360 kWh for the reference 3-car set (3×120 kWh self-contained cars), sized for roughly one route length plus reserve.
- Total capex for the energy subsystem: $\approx \$30\text{M}$ vs. $\approx \$100\text{M}$ for conventional catenary + traction substations — a 60–70% cost reduction.

These numbers are first-order. They are offered as a planning baseline and as the starting point for the `osr-sim` energy model; they are not a substitute for a detailed design study on any specific line.

2. Non-goals

- Not a complete electrical design. Transformer sizings, harmonics, protection coordination, and grounding all require proper EE work outside this RFC.
- Not applicable to freight or long intercity. Per ARCHITECTURE §1, OpenSourceRail's scope is urban transit only; this RFC's sizing is therefore exclusively for light metro and metro service profiles.
- Not a land-use or permitting study. ROW PV implies planning-regime negotiations this RFC cannot resolve.

3. Reference Deployment

Parameter	Value
Alignment	20 km, double track, at-grade or elevated mix
Stations	15 (average 1.3 km spacing)
Service hours	18 h/day (05:00–23:00)
Peak headway	5 min (both directions)
Off-peak headway	10 min
Rolling stock	3-car light metro built from self-contained 17 m cars, one powered bogie + one trailer bogie per car
Fleet size	12 trainsets (10 in service, 2 spare)
Typical ambient	Tropical or subtropical; 5 peak sun-hours (PSH) annual mean

This sizing represents a moderately busy urban transit line — representative of, say, a new-build metro for a mid-sized capital city in our target regions. Smaller systems scale down linearly; larger systems break into multiple regions (RFC 0001 §6.2).

4. Energy Demand Model

4.1 Per-train consumption

Light metro rolling stock consumes approximately 3–5 kWh per car-km, depending on stopping density, grade profile, and HVAC load. For the reference line we use 4 kWh/car-km, which matches published figures for several modern LRV/light-metro platforms and leaves a realistic margin for passenger HVAC in a hot climate.

Auxiliaries (HVAC, lighting, compressors) are $\sim 15\%$ of total vehicle draw and are rolled into the 4 kWh/car-km figure.

4.2 Daily aggregate

	Value
Trains passing a point per hour (peak, one direction)	12 (= 60/5)
Peak duration	4 h (morning + evening peaks combined)
Off-peak duration	14 h
Off-peak throughput	6 trains/h
Daily train-passes per direction	$12 \times 4 + 6 \times 14 = 132$
Daily train-km	$132 \times 20 \times 2 = 5,280$
Daily car-km	$5,280 \times 3 = 15,840$
Line-haul energy	$15,840 \times 4 = 63.4$ MWh/day
Depot, stations, HVAC, lighting, signals ($\approx 30\%$ of traction)	19.0 MWh/day
Depot battery conditioning + charging losses ($\approx 10\%$)	8.2 MWh/day
Auxiliary (fare systems, passenger info, accessibility)	1.5 MWh/day
Total daily demand	≈ 92 MWh/day round to 100 MWh/day for planning

Regeneration (braking) recaptures $\sim 20\%$ of traction energy when the onboard battery has headroom to accept it — this is already included in the 4 kWh/car-km figure, which is a net consumption value.

4.3 Demand profile through the day

Demand is not flat. First-order shape:

- 05:00–07:00: ramp-up, $\sim 60\%$ of peak hourly draw
- 07:00–10:00: morning peak, full hourly draw
- 10:00–16:00: off-peak, $\sim 50\%$ of peak hourly draw
- 16:00–19:00: evening peak, full hourly draw
- 19:00–23:00: evening taper, $\sim 60\text{--}30\%$ of peak
- 23:00–05:00: depot charging of fleet batteries, $\sim 40\%$ of peak draw (at low c-rate, spread over 6 h)

Peak hourly draw is roughly 8 MWh/h (≈ 8 MW average, probably ~ 12 MW instantaneous with stochastic peaks). This matters for storage sizing and grid-tie capacity.

5. PV Generation

5.1 Available PV area

Surface	Area estimate	Notes
Station/platform canopies	$15 \times 2,000 \text{ m}^2 = 30,000 \text{ m}^2$	Dual-use weather shelter; visually acceptable; first thing to build.
Depot roof	$10,000 \text{ m}^2$	High-quality flat roof; easy to install and maintain.
ROW along track — vertical bifacial	$20 \text{ km} \times 2 \text{ sides} \times 4 \text{ m effective height} = 160,000 \text{ m}^2$ projected area, $\sim 80,000 \text{ m}^2$ of panel	Property-line fencing dual-use; catches morning/evening sun at complementary angles.
ROW along track — horizontal / slanted panels above ballast shoulder	$20 \text{ km} \times 2 \times 2 \text{ m usable} = 80,000 \text{ m}^2$	Requires clearance management; standard tilt.
Between-rail / on-sleeper PV (Sun-Ways-class)	$20 \text{ km} \times 2 \times 2.5 \text{ m} = 100,000 \text{ m}^2$ available but not included in baseline	Phase-4+, not counted in this sizing.
Baseline PV area counted	$200,000 \text{ m}^2$	Excludes between-rail option.

5.2 Nameplate

At modern module density of $\sim 220 \text{ W/m}^2$ (2026-vintage monocrystalline + bifacial), $200,000 \text{ m}^2$ yields 44 MW nameplate. Derating for:

- Packing factor 0.75 (structure, access paths, mutual shading) 33 MW
- Performance ratio 0.80 (inverter losses, soiling, temperature, wiring) 26.4 MW AC output

Round to 25 MW AC for planning.

5.3 Annual generation

At 5 PSH: $25 \text{ MW} \times 5 \text{ h} \times 365 = 45.6 \text{ GWh/year} = 125 \text{ MWh/day}$ average.

Daily demand is ~ 100 MWh. Headroom is roughly 25%, which is intended. It covers:

- Seasonal variation (winter months at higher latitudes in the target band still see 4 PSH).
- Cloudy-day losses (a 5-day cloudy stretch at 50% yield is $\sim 10\%$ of a year if it happens 4 $\times$ /year).

- Module degradation over lifetime (~0.5%/year).
- Demand growth.

5.4 Surplus monetization

On clear days the system will generate far above demand during midday (peak 25 MW instantaneous vs. 8 MW average demand). Three dispositions, in priority order:

- 1 Charge trackside storage (§6) – priority 1 until storage is full.
- 2 Charge onboard fleet batteries opportunistically – priority 2 during dwells and terminal stops.
- 3 Export to grid – priority 3, where regulations and grid infrastructure allow, at whatever FIT/PPA rate applies locally.

Deployments in regions without grid export should plan to curtail rather than undersize generation – curtailment cost is zero, undersizing leaves the line stranded on cloudy days.

6. Trackside Storage

Trackside storage sits at stations and at the depot. It has three duties:

- 1 Diel shift. Store midday surplus, deliver it during morning/evening peaks and overnight depot charging.
- 2 Peak shaving. Absorb the difference between 8 MW average and ~12 MW instantaneous demand.
- 3 Cloudy-day autonomy. Keep service running through solar-poor periods without a full grid-import contract.

6.1 Sizing

Duty	Required capacity
Diel shift (surplus midday peak + night)	~35 MWh (storing roughly one-third of daily generation for time-shifted use)
Peak shaving headroom	~5 MWh
One-day cloudy autonomy at 50% yield	~50 MWh (to cover the ~50 MWh shortfall versus demand)
Two-day cloudy autonomy target	~60 MWh total
Baseline design	60 MWh

Two-day autonomy is a judgment call: single-day cloudy weather is common in target climates, two-day stretches are meaningful to plan around, three-day stretches are rare enough that grid fallback is acceptable.

6.2 Placement

Distributed, not centralized:

- Station sites (15): 2 MWh each = 30 MWh total. Co-located with station PV and station charging pads. Handles local peak shaving.
- Depot site (1): 30 MWh. Co-located with depot PV and overnight fleet charging.
- Total: 60 MWh.

Distribution across stations means no single battery failure halts service, and transmission losses are minimized because storage is co-located with both generation (canopy PV) and load (charging pads).

6.3 Chemistry and form factor

Sodium-ion primary per [RFC 0021 §3](#). Installed cost at 2026 prices: ~\$80/kWh at cell level (RFC 0021 §4 sourcing matrix), ~\$120–150/kWh installed including BMS, enclosures, and site works.

- 60 MWh × \$200/kWh = \$12 M for trackside storage.

Duty cycle at these sizings is gentle (~0.3 C average charge, ~0.5 C peak discharge), supporting ≥6000-cycle life – comfortably >15 years of daily cycling.

7. Onboard Battery and Charging Strategy

7.1 Per-trainset sizing

A 3-car trainset consuming ~9 kWh/train-km (3 × 3 kWh/car-km) uses 180 kWh over one 20 km route length. Design goals:

- Able to complete about one route length on onboard energy if station chargers are unavailable.

- Replace normal service energy during 60-second dwells at roughly 1 km station spacing.
- Avoid deep cycling to preserve pack life: operate between 20% and 80% SoC nominally.

Usable pack = 360 kWh per trainset = 120 kWh/car. That gives roughly 40 km at the reference 9 kWh/train-km draw before reserve, or one 20 km route length plus hot-weather HVAC and degradation margin. The onboard battery is deliberately not sized for a full day or repeated round trips; the station batteries carry that duty.

7.2 Chemistry split

Onboard chemistry is a weight/volume tradeoff:

- LFP at ~180 Wh/kg 360 kWh = 2.0 t per trainset
- Sodium-ion at ~140 Wh/kg 360 kWh = 2.6 t per trainset
- Difference: 0.6 t per trainset — modest against a 100 t-class light-metro consist

Conclusion for this reference case: either works. Default to sodium-ion for supply-chain sovereignty; LFP is acceptable where local supply is easier or where a specific light-vehicle platform is pack-volume-constrained. At metro duty cycles the gravimetric-density delta does not meaningfully affect energy consumption.

7.3 Charging infrastructure

Site	Charger count	Power	Purpose
Passenger stations (15)	15 × 500 kW nominal	DC conductive, side-pin or pantograph-down	60-s dwell adds ~8 kWh; normal service energy replacement
Terminals (2)	included above, uprated to 1 MW where needed	DC conductive, side-pin primary	60-s dwell adds ~17 kWh; turnback margin
Depot (10 stalls)	10 × 150 kW	AC conductive, overnight	Low-C overnight charging of full fleet; best for pack life

Capex: 15×500kW + 10×150kW = 9.0 MW of charging equipment before terminal uprating. At ~\$300/kW installed: \$2.7 M.

7.4 Round-trip energy accounting

For a trainset doing 8 round trips/day (320 km):

- Total consumption: 320 km × 9 kWh/km = 2,880 kWh
- Station charging at 15 stations × 2 directions × 8 kWh × 8 round trips ≈ 1,920 kWh/day
- Terminal uprating + regenerative headroom closes most of the remaining service energy; depot charging handles balancing and any cloudy-day shortfall.
- At 10 active trainsets: station charging supplies most daytime traction energy; depot charging mainly performs overnight balancing, pre-service top-up, and cloudy-day recovery within the 30 MWh depot storage capacity.

Numbers check out: the fleet can sustain full service without pulling from grid during a single cloudy day, provided trackside storage was charged the preceding day.

8. Cost Comparison

Greenfield 20 km double-track light metro, energy infrastructure only (excludes rolling stock vehicle cost, track, stations, signaling):

8.1 OpenSourceRail approach

Item	Cost
25 MW PV (utility-grade + canopies + ROW mounting systems) at \$700/kW blended	\$17.5 M
60 MWh trackside sodium-ion storage at \$200/kWh installed	\$12.0 M
Charging infrastructure (9.0 MW across passenger stations and depot)	\$2.7 M
Per-trainset battery × 12 trainsets × 360 kWh × \$150/kWh	\$0.65 M
Grid-tie inverters, switchgear, site works	\$3.0 M
Engineering, commissioning, contingency (15%)	\$5.4 M
Total	≈ \$41 M

8.2 Conventional catenary approach

Item	Cost
Overhead catenary + support structures, 20 km double track at \$2 M/km	\$80 M

Item	Cost
Traction substations, 4–5 sites at \$4 M/site	\$18 M
Grid connection works	\$3 M
SCADA + OCS controls (proprietary vendor package)	\$5 M
Engineering, commissioning, contingency (15%)	\$15.9 M
Total	≈ \$120 M

8.3 Delta

OSR approach ≈ 34% of conventional approach. Savings: ~\$80 M on a 20 km line.

Additional factors not captured above:

- OSR eliminates specialized OCS maintenance workforce requirements opex savings over lifetime.
- OSR's surplus generation is revenue or offset; catenary has no such capability.
- Catenary capex grows with line length; OSR per-km cost is roughly constant as generation and storage scale with km. Longer lines amplify the advantage.
- Catenary recovery from faults is slow (possession, isolation, line work); distributed battery + charging recovers faster from single-site failures.

Against this, OSR has:

- Novel-technology risk that may justify a reserve in financing.
- Dependence on PV supply chain (the alternative depends on copper, steel-mast, and catenary-wire supply chains, which are equally stressed in different ways).
- Higher onboard mass per trainset (~2–3 t battery vs. a pantograph + transformer at ~2 t); marginal impact on rolling-resistance energy.

9. Sensitivity Analysis

How do the numbers change as key assumptions flex?

Variable	Change	Impact
Climate (PSH)	4 PSH instead of 5	PV yield drops 20%; either +20% PV nameplate (+\$3.5 M) or +30% storage (+\$3.6 M) needed. Still well below catenary capex.
Peak headway	3 min instead of 5 min	Demand grows ~50% (~150 MWh/day). All sizings scale ~linearly; OSR advantage vs. catenary preserved.
Line length	40 km instead of 20 km	OSR cost scales ~linearly to \$80 M; catenary scales nearly linearly to \$240 M. OSR advantage widens.
PV area available	Half (ROW not available)	Must rely on canopies + depot only (40,000 m²). Drop to 5 MW nameplate, 25 MWh/day generation — net-negative against demand. Line needs grid-import contract or extended storage. This case is the scenario most sensitive to solar-friendly geometry.
Cloudy-day frequency	10% of days have <40% yield	Requires 2-day autonomy (covered by 60 MWh baseline) + grid fallback contract. No change to baseline sizing.
Sodium-ion cost	\$300/kWh installed (pessimistic)	Trackside storage cost 60 MWh × \$300 = \$18 M. Total rises to \$47 M vs. \$120 M catenary. Still a strong advantage.
LFP instead of Na-ion for trackside	—	Cost ~comparable at utility scale; raw-material sovereignty point weakens.

9.1 The case most likely to break this RFC

Limited PV area is the single-largest risk. If the ROW cannot host PV (planning, heritage, or engineering constraints) and station+depot area yields only ~5 MW, the line is net-negative on solar and must buy ~half its energy from grid. The capex comparison still favors OSR (no catenary), but the annual energy bill shifts from near-zero to material. This scenario must be evaluated early in any specific deployment — it is not a universal problem but it is a line-specific one.

Mitigation where ROW PV is constrained: offsite solar PPA, or accept higher grid dependence and focus savings on the catenary capex alone.

10. Safety and Reliability

Energy is not in the SIL-4 safety boundary (see RFC 0001 §3 and §9: traction control is onboard, MA enforcement does not depend on charging availability). Still, energy failures degrade service, and some considerations apply:

- Loss of trackside storage at one station. Adjacent stations' storage and fleet onboard capacity carry service until repair. No safety impact.
- Loss of one station charger. Adjacent station chargers and onboard capacity carry service until repair. Timetable control may skip express running or reduce frequency, but trains do not strand because onboard capacity covers about one route length.
- Loss of all PV. Grid import + storage keeps service running for the storage-sizing horizon (~2 days). Operators must negotiate grid-fallback contracts as a normal commercial matter.
- Thermal runaway / battery fire. Sodium-ion and LFP are both thermal-runaway-resistant chemistries (unlike NMC). Standard battery-room fire suppression (inert-gas deluge, compartmentalization) applies. Onboard fires in BEMU/battery-electric rail stock have a well-understood mitigation playbook from ~decade of bus/truck operations — rail fire doors and passenger evacuation rules adapt directly.
- PV soiling and bird strike. Operational concerns, not safety ones. Cleaning schedules and mesh protection are standard utility-PV practice.

11. Open Questions

- 1 Between-rail PV. Sun-Ways-class on-sleeper PV could double the effective PV area per track-km. At what point is it worth the clearance and tamping complexity? Prototype in `osr-sim` first.
- 2 Battery recycling and second-life. Traction packs at end-of-life (still ~80% capacity) are ideal candidates for cascade into trackside storage. Designing the spec so packs are physically swappable between roles would amortize the battery capex across two duty cycles — worth its own small RFC.
- 3 Dynamic station-charging sizing. For very high-frequency service (<3 min headway), the dwell-time charge pulse may need multiple platform cabinets or higher station battery C-rates. Where is the crossover, and how does it change onboard-pack sizing?
- 4 Grid-tie protocol choice. IEEE 2030.5 is defaulted in ARCHITECTURE §D7 — this should be confirmed against target-region utility standards.
- 5 Seasonal tilt and tracking. Fixed-tilt vs. single-axis tracking changes PV cost by ~15% and yield by ~20%. Bias toward fixed-tilt at these sizes for simplicity; revisit if yield binds.
- 6 Metro-class ridership peaks. Full-metro duty (6-car, 2-min headway, underground) will push line-haul demand well above the light-metro reference in §4. The sizing framework scales, but a dedicated RFC validating it for a metro-class reference line is warranted before Phase 5.

12. Prior Art

- Stadler FLIRT Akku — 80–150 km range on battery, opportunity charging at electrified sections. Our reference-metro sizing sits comfortably under this proven envelope.
- Hitachi Masaccio (TrainLab) — hybrid battery-catenary for regional service.
- Siemens Mireo Plus B — BEMU, commercial operation in Germany.
- Adelaide Flinders Line tram — solar-powered light rail using a rooftop PV + grid-feed arrangement; validates solar-rail economics.
- Sun-Ways (Swiss pilot) — on-sleeper PV.
- Bankset / Deutsche Bahn PV fence pilots — vertical bifacial along ROW.
- India Railways rooftop PV program — large-scale PV on station and depot roofs; demonstrates the regulatory path in a major target region.

The OpenSourceRail contribution over the prior art is not any individual element — each has been piloted somewhere. It is the integrated design that treats PV + trackside storage + BEMU as a single system with open software coordinating energy dispatch, and sizes it as a complete alternative to catenary rather than a retrofit.

This RFC is a planning artifact. Actual pilot deployments will produce per-line sizing studies that refine these numbers with local climate, grid conditions, and operational patterns.

RFC 0003 — Samawah Urban Rail: Reference Deployment

Status: Draft Date: 2026-04-20 Depends on: docs/ARCHITECTURE.md, RFC 0001, RFC 0002

1. Summary

This RFC proposes a concrete reference deployment for OpenSourceRail: a three-line auto-planned urban rail network serving Samawah (As-Samawah), capital of Al-Muthanna Governorate, Iraq. The current source of truth is [designs/west-asia/Iraq/Samawah/design.toml](#), emitted by `osr-design` from OpenStreetMap anchors plus the WorldPop residential-population demand surface. Every inter-station polyline is generated by the routing pipeline rather than sketched "as the crow flies". The current generated network is:

- Line 1 — northern/southern radial spine through the city-centre interchange.
- Line 2 — eastern/western radial spine through the city-centre interchange.
- Line 3 — third radial service tying the remaining catchment into the city-centre interchange.

Total generated network: ~55 km of double-track light metro, 33 unique stations, 55-trainset fleet including spare and cold-reserve sets.

The previous draft of this RFC proposed a 2-line design (14 km radial + 16 km ring, 30 km total). That geometry placed Al-Muthanna University ~8 km south-east of its real location, placed "Samawah Railway Station" in the north-west when the real station is south-east, and routed the Halqa ring through palm groves + agricultural land that wasn't built-up. That design was retired in favour of the OSM-grounded network below.

This is the reference case for `osr-sim`, the anchor for conversations with potential pilot stakeholders, and the concrete instantiation against which subsequent subsystem designs are measured. It is a planning-grade concept design, not a surveyed alignment — it exists to make the architecture real rather than to commit any specific routing.

2. Why Samawah

Factor	Samawah	What it means for OSR
Population	~373,770 (As-Samawah Subdistrict, 2024 Iraq census; canonical figure in lib/city-batches/world-sample.toml)	Right-sized for light metro — too small for a major vendor to bother with, ideal for an open-source first deployment
Existing rail	Station on Baghdad–Basra mainline + rail yard with 300–800 stored wagons + adjacent rolling-stock workshop (see §2.1)	Rail culture, intercity interchange, and a brownfield-recovery anchor that turns this from a reference scenario into a deployable pilot
Climate	Hot desert; ~6+ peak sun hours annual mean	PV yield ~20% above the RFC 0002 reference (5 PSH) — catenary-free + solar is conservative, not ambitious
Grid reliability	Limited; frequent outages	Storage autonomy has real operational value, not just a planning formality
Existing electrification	None on Iraqi mainlines	No legacy catenary, no vendor-locked OCS — greenfield on D7
Development context	Al-Muthanna is among Iraq's lower-income governorates	Capital sensitivity is high; the ~66% capex reduction in RFC 0002 is decisive, not marginal
Institutional anchors	Al-Muthanna University (engineering faculty), new German Hospital, government buildings	Each anchor is a major daily demand generator; the radial alignment serves all of them on one line
Workforce	Engineering graduates from Al-Muthanna University	Potential domestic design/build partnership — the core promise of OSR is "workers who design and build the systems in the country"

The combination is close to optimal: genuine need, favorable physics, no legacy to migrate, and domestic technical capacity to engage. If OSR cannot make sense here, it likely cannot make sense anywhere.

2.1 Physical-asset anchor — Samawah is brownfield, not greenfield

Satellite imagery review (2026-04-26) of the rail yards immediately adjacent to Samawah Train Station () identifies two stockpiles of dormant rolling stock:

- Northern yard (along Samawah Train Station): ~4–6 parallel sidings packed end-to-end. Estimated ~120–300 wagons.
- Southern yard (diagonal NE–SW sidings): ~5–8 parallel sidings with similar density, plus a large workshop / shed building at the southern end. Estimated ~150–480 wagons.

- Total estimate: ~300–800 wagons. Predominantly standard 4-axle freight stock (covered wagons, hoppers, flatbeds) — passenger coaches a small fraction.

The southern yard's workshop building was the target of a 2011 Iranian Waxon Park rehabilitation deal (per Iraqi Business News historical reporting). Outcome of that deal is unclear and is part of the [RFC 0027](#) Phase 1 assessment.

National context (Iraqi Republic Railways / IRR /):

- National fleet: 10,326 freight wagons, 255 passenger coaches
- Gauge: 1 435 mm standard nationwide fully compatible with [RFC 0009](#)
- Couplers: SA-3 (Soviet automatic) + screw/buffer combo incompatible with the Scharfenberg Type 10 spec'd by RFC 0008 §3.1, but the recovered drawgear is resaleable to legacy operators (RFC 0027 §5.3)

What this means for the Samawah pilot: the pilot is no longer a hypothetical greenfield deployment. It is the systematic conversion of an existing rail-yard + workshop complex into the first OSR production site, anchored on:

- 1 A dormant fleet with recoverable wheelsets, axleboxes, brake gear, and structural steel worth \$8–15 M in component value against \$1–3 M in recovery operations cost (per RFC 0027 §3).
- 2 An existing workshop building with foundations, pit tracks, and crane gantries that would otherwise require €5–8 M to build greenfield (RFC 0014).
- 3 An active rail corridor through the city, available as the intercity interchange at the Samawah Railway Station terminal (Line 3 SE branch in §3.3 above).
- 4 Standard gauge matching OSR spec — no gauge conversion required.
- 5 A national rail authority (IRR) under reconstruction pressure — the political context favours engagement.

Conversion strategy is in [RFC 0027 — Brownfield Pilot: Existing-Asset Recovery & Workshop Integration](#), which carries the per-phase doctrine, recovery yield estimates, workshop conversion matrix, and risk register. RFC 0027 is intended to apply to every brownfield deployment, not just Samawah; the candidate-city list is in §9 there.

Status of the brownfield assessment at time of writing:

- Phase 1 site visit + fleet census — not yet started; gating item
- Workshop tooling audit — pending Phase 1
- NDT campaign on a sample wheelset population — pending Phase 1
- MoU draft with IRR / Iraqi Ministry of Transport — pending diplomatic / technical-community introduction

The single highest-value next step on the OSR programme is getting a real Phase 1 census on the ground. Without it, the Samawah deployment costs in §1 of [RFC 0021](#) and the depot CAPEX in [RFC 0014](#) are greenfield estimates; with it, the per-consist mechanical CAPEX drops by ~\$30–60 k and the depot CAPEX drops by a factor of 5–10 per RFC 0027.

3. Geography and Indicative Alignment

Samawah sits on the Euphrates in southern Iraq. The city stretches roughly east–west along the river with residential suburbs fanning north and south. The mainline railway enters from the northwest, the city centre sits on the river bend, and the major newer developments (hospital, university campus, expanding residential districts) are south and east of the historic core.

Samawah network — 3 auto-planned lines on OpenStreetMap

Three generated lines on OSM tiles. Station markers are coloured by archetype (red terminal, purple depot-terminal, blue major, orange interchange, white standard). Every polyline comes from the `osr - design` routing pipeline. Regenerate with `scripts/regenerate-city.sh samawah`; see [§3.6 Realism notes](#) for what surveying would still change.

3.1 Line 1 — Nahrain (N–S spine)

Eight stations, ~11 km total. North-to-south sequence along the dense urban core, elevated through the city centre + hospital interchange, at-grade elsewhere:

#	Station	Lat / Lon	Purpose	OSM reference
1	Al-Muthanna University	31.3386, 45.2884	N terminus; university campus	<code>amenity=university</code> جامع المثنى
2	Jamia Al-Muthanna	31.3374, 45.2870	Residential south of campus	<code>place=neighbourhood</code> حي المثنى
3	Al-Qashla	31.3196, 45.2916	Dense urban cluster NE of centre	<code>place=neighbourhood</code> الشلل

#	Station	Lat / Lon	Purpose	OSM reference
4	Samawah Teaching Hospital	31.3213, 45.2734	Major regional hospital	amenity=hospital مستشفى مستشفى
5	Samawah Central	31.3079, 45.2827	Triple interchange (L1/L2/L3)	place=city مستشفى
6	Al-Hakam	31.2973, 45.2757	Southern residential	place=neighbourhood حي مركز
7	Hayy 270 Dar	31.2969, 45.2646	Residential cluster	place=neighbourhood حي رأد 270
8	Al-Maali	31.2785, 45.2794	S terminus; main depot	place=neighbourhood حي مركز

Expected daily ridership at build-up: 35,000 – 48,000 passenger-trips (highest — serves university + hospital + city centre).

3.2 Line 2 – Sharqiyyeh (E–W crosstown)

Six stations, ~7 km total. Western residential through the city centre to eastern residential:

#	Station	Lat / Lon	Purpose	OSM reference
1	Al-Naft	31.3123, 45.2485	W terminus; housing complex	place=neighbourhood حي طريق
2	Al-Mualimin	31.3152, 45.2740	Central-west residential	place=neighbourhood حي نسيم
3	Samawah Central	31.3079, 45.2827	Interchange with L1, L3	shared with L1
4	Al-Sharqi	31.3105, 45.2888	Dense east residential	place=neighbourhood حي مركز
5	Al-Sharqi East	31.3135, 45.2901	Residential cluster	place=neighbourhood حي مركز
6	Al-Bustan Sharqi	31.3167, 45.3118	E terminus; secondary layout	place=neighbourhood حي مركز

Expected daily ridership at build-up: 18,000 – 25,000 passenger-trips.

3.3 Line 3 – Mahatta (SE branch to railway station)

Five stations, ~5 km total. Short branch from the city centre to the intercity railway station:

#	Station	Lat / Lon	Purpose	OSM reference
1	Samawah Central	31.3079, 45.2827	Interchange with L1, L2	shared
2	Al-Hidriya	31.3050, 45.2796	Residential south of centre	place=neighbourhood حي مركز
3	Al-Nahda	31.2978, 45.2897	Dense SE residential	place=neighbourhood حي مركز
4	Al-Qasim	31.2869, 45.2826	Southern residential	place=neighbourhood حي مركز
5	Samawah Railway Station	31.2746, 45.3032	SE terminus; Baghdad–Basra intercity interchange	railway=station محطة مركز

Expected daily ridership at build-up: 12,000 – 18,000 passenger-trips (modest, gated by intercity service frequency).

3.4 System totals

Metric	Value
Route-km (double track)	~55 km
Stations (unique)	33
Lines	3 (auto-planned by osr - design)
Multi-line interchanges	central interchange complex + radial-junction interchanges — every radial converges on the elevated-junction at city centre per the via-centre routing rule for ≤ 3-line networks
Fleet (revenue)	49 × 3-car trainsets
Fleet (spare + cold-reserve)	6 × 3-car trainsets
Average inter-station spacing	~1.2 km (per SpacingConfig — 1.2 km inner, 2 km transitional, 4 km outer; matches the operator brief of "1.2 km inner / 2–5 km outer")
Demand surface	Anchor (POI) + WorldPop residential population blend per build_demand_surface — lines reach population centres without mapped POIs
Service hours	05:30 – 23:30, 18 hours

3.6 Realism notes

Honest assessment of the revised (2026-04-24) 4-line design against real Samawah OSM data:

What's real. Every station coordinate in [designs/west-asia/Iraq/Samawah/design.toml](#) corresponds to a named OSM feature verified by Overpass query: either a specific `place=neighbourhood` node, an `amenity` node (university / hospital), or a `railway=station` node. The 220 000 population, the bounding box, the Euphrates position, and the

Baghdad–Basra mainline crossing at the south-east are all OSM- consistent. The triple-interchange at Samawah Central is at the verified `place=city` node.

What's not real yet. Things a surveyor would still catch:

- Inter-station polylines follow the cheapest-road shortest path through the OSM network (arterials weighted 0.6–0.8, residentials 1.8). That's planning-grade, not engineering-grade — curve radii, grades, and the ROW width needed for double track aren't checked. A real alignment survey will adjust station coordinates by tens of metres to snap to stakeable centrelines + may replace some segments with elevated viaduct where the road is too narrow.
- Coverage audit flags fringe neighbourhoods >1 km from any station in the current auto-planned design. The remaining outliers are mostly fringe residential (distant Jarbuwiya branches, Al-Ma'ali south of the rail, agricultural villages beyond the urban edge). Extending the network further is deferred to a future RFC if those areas develop.
- Line 1's Al-Qashla Hospital segment swings west-then-south to reach the hospital cluster — a one-detour on an otherwise N–S spine. Kept for the hospital demand anchor; planners may prefer skipping.
- Al-Bustan Sharqi (Line 2 east terminus) sits at the outer edge of built-up area. Secondary layup siting here works but detailed land acquisition is pending.
- 55 km / 374 k served catchment across 3 lines is at the upper end of the light-metro ratio. Comparable cities: Amiens (135 k 11 km), Tours (140 k 15 km), Orléans (115 k 30 km across 2 lines). Samawah's 55 km is a full multi-phase build-out target (Phase A delivers Line 1 north first).

How to take this to the next fidelity level:

- 1 Re-run the design - py routing + rendering pipeline with a one-command refresh:
which fetches Samawah OSM (cached), routes every line on the road graph, emits `corridor.geojson` + `designs/west-asia/Iraq/Samawah/samawah.toml`
 - the map PNGs, and runs the scenario + RFC drift tests.
- 2 Feed the stationed alignment + `corridor.geojson` into [osr-alignment-export](#) to produce LandXML + railML for import into Bentley OpenRail or Civil 3D.
- 3 Cross-reference against Iraqi Ministry of Transport + Al- Muthanna Governorate road + cadastre datasets.
- 4 Review with Al-Muthanna University transportation faculty and the Governorate planning office.

4. Service Plan

4.1 Headways

Period	Line 1 Nahrain	Line 2 Sharqiyyeh	Line 3 Mahatta
AM peak (07:00–09:30)	5 min	6 min	7 min
Midday (09:30–16:00)	8 min	10 min	12 min
PM peak (16:00–19:00)	5 min	6 min	7 min
Evening (19:00–22:00)	10 min	12 min	12 min
Late evening (22:00–23:30)	15 min	15 min	15 min

Line 1 is the busiest service (university + hospitals + centre + southern residential — all major demand anchors). Line 2 provides east-west connectivity crossing L1 at Samawah Central. Line 3 is a short branch whose headway is aligned with the (roughly twice-daily) Baghdad–Basra mainline intercity arrivals so that a rail-station transfer always has a connecting metro service within 5 minutes of the peak timetable.

4.2 Anchor-driven demand

Unusually for light metro, ridership here is dominated by a few large institutional anchors rather than by distributed demand. This has design consequences:

- University: sharp peaks 07:30 and 15:00 during term; low on Fridays and summer. Platform capacity at Al-Muthanna University must handle 2,000+ pax in 20 minutes.
- Hospitals (two): steady demand through the day, with a visiting-hours peak in mid-afternoon. 24-hour operation would be justifiable for hospital access; this RFC stops at 23:30 but flags the question.
- Railway station: bunched demand aligned with the (roughly twice-daily) intercity arrivals. Line 1 timetable should align a service with each mainline arrival/departure.
- Souq and Centre: evenings and weekends, counter-cyclical to the university peak — this is helpful for fleet utilization.

4.3 Rolling stock

Reference platform per [ARCHITECTURE §4 D5](#):

- 3-car, ~57 m consist with low-floor centre door zones (light metro class)
- Na-ion onboard battery, 360 kWh/trainset (120 kWh/car), sized for one route-length with station opportunity charging
- Open SiC traction inverter, Rust control firmware
- TSN Ethernet trainbus
- HVAC sized for 50 °C ambient, which is load-bearing in Samawah summers (see §5.4)

5. Energy: Samawah-specific Numbers

Applying [RFC 0002](#) to this network, with Samawah-specific adjustments for climate and duty cycle.

5.1 Demand

Component	kWh/day
Line 1 line-haul (4 kWh/car-km × 3 cars × traffic)	≈ 40,000
Line 2 line-haul	≈ 22,000
HVAC uplift for extreme heat (+25% over RFC 0002 baseline in peak summer)	≈ 15,000
Stations, depot aux, signals, info systems	≈ 22,000
Charging losses	≈ 10,000
Total (peak summer)	≈ 110,000 kWh/day = 110 MWh/day
Total (shoulder season average)	≈ 85 MWh/day

Slightly higher than the RFC 0002 reference because Samawah summers are more HVAC-intensive, and we've scaled up to 30 km of route.

5.2 PV generation

Samawah's 6 PSH annual mean (with summer peaks above 8 PSH) is higher than the generic reference.

Surface	Area	Notes
Station canopies (20 × 2,000 m ²)	40,000 m ²	Dual-use shade — valuable in Samawah heat
Depot roof (East Depot) + Northwest layup	16,000 m ²	
ROW vertical bifacial along 30 km × 2 sides × 4 m effective	240,000 m ² projected, ~120,000 m ² of panel	Vertical orientation also reduces dust accumulation vs. horizontal
ROW ballast-shoulder panels	120,000 m ²	Tilted; subject to dust soiling management
Total panel area	~300,000 m ²	

At 220 W/m² × 0.75 packing × 0.78 PR (lower PR than RFC 0002 due to high module temperatures — heat derates silicon):

- Nameplate AC: ~38 MW
- Annual generation at 6 PSH: ~83 GWh/yr = 228 MWh/day average
- Comfortable 2× headroom over peak-summer demand; clean surplus most of the year

The headroom deliberately buys two things specific to Samawah:

- 1 Dust-storm resilience. Peak soiling events in Muthanna can cut yield 40–60% for several days. The surplus lets us absorb this without grid dependence.
- 2 Grid export opportunity. Iraq's grid is energy-constrained; feed-in under a state-utility PPA could be a material revenue line, not an afterthought.

5.3 Storage

Given grid unreliability, Samawah's storage sizing should be more generous than the RFC 0002 baseline. Target: 3-day autonomy (vs. 2-day generic).

- Tracksides: 20 stations × 2.5 MWh = 50 MWh
- Main depot: 40 MWh
- Secondary layup: 5 MWh
- Total tracksides storage: 95 MWh Na-ion

Onboard: 14 revenue trainsets × 360 kWh = 5.0 MWh rolling stock battery, with two spare consists adding another 0.7 MWh if fully provisioned.

5.4 Thermal considerations

Samawah summers routinely exceed 45 °C; record highs approach 55 °C. This has three design consequences:

- 1 HVAC energy dominates. Per-car traction + HVAC at 50 °C ambient can reach 6 kWh/car-km. The 4 kWh/car-km reference figure is already rolled up with an average HVAC; the RFC's +25% summer uplift reflects realistic worst-case.
- 2 Battery conditioning. Both sodium-ion and LFP prefer operation below 45 °C. Trackside battery enclosures need insulation + passive ventilation minimum; active cooling for high-C discharge periods is justified for the main depot only. Onboard packs sit under the floor with thermally managed enclosures; Na-ion's wider safe-operation temperature window is particularly useful here.
- 3 PV temperature derating. Panels in Samawah can reach 70+ °C module temp on a summer afternoon. Bifacial + vertical mounting runs cooler than horizontal ballasted arrays and is preferred on the ROW for this reason as much as the area one.

5.5 Capex summary (Samawah-specific)

Applying RFC 0002's cost basis, scaled to 30 km and adjusted:

Item	Cost (USD)
38 MW PV (blended across surfaces) @ \$700/kW	\$26.6 M
95 MWh trackside Na-ion storage @ \$200/kWh installed	\$19.0 M
Charging infrastructure	\$3.0 M
Onboard batteries, 16 trainsets × 360 kWh @ \$150/kWh	\$0.9 M
Grid tie + BOP + site works	\$5.0 M
Engineering, commissioning, contingency (20% — slightly higher for novel deployment)	\$11.1 M
Energy subsystem total	≈ \$66 M

For comparison, a conventional 25 kV AC catenary + 4 traction substations for a 30 km system in this market would run ≈ \$180 M. The OSR energy approach is ~37% of the catenary cost.

6. Mapping to OpenSourceRail Architecture

This is where the reference deployment connects back to the domain designs:

D1. Operations & Dispatch

- One Operations Control Centre at East Depot, with a backup dispatch workstation at Samawah Central.
- Dispatcher interface: web-based; full observability via the OpenTelemetry + Prometheus stack; no proprietary HMI.
- Integration with intercity rail: the OCC subscribes to the Iraqi Republic Railways (IRR) timetable via GTFS-RT (or bilateral feed if GTFS isn't available) so that Line 1 services align with mainline arrivals.

D2. Train Control

- Two consensus regions: Line 1 (12 W-Nodes, one per station plus depot) and Line 2 (10 W-Nodes).
- Region boundaries at Eastern Bridge and Al-Muthanna University — the Line 1/Line 2 interchanges — handled by the handoff protocol in RFC 0001 §7.4.
- Interlockings at the three turnback sidings (Line 1 termini and mid-line) plus the depot throats.
- No track circuits: position is sensor-fused per RFC 0001 §5.1, with beacon-fix sites at every switch and every platform.

D3. Communications

- Public 5G SA for primary trainwayside link — Zain Iraq, Asiacell, or Korek presence in Samawah; one or more operators will likely carry the project with a network-slicing arrangement.
- LoRa mesh gateways at every W-Node site as the safety-telemetry fallback, giving citywide coverage independent of carrier uptime.
- TSN Ethernet backbone along the ROW (single fiber pair in a ruggedized conduit, ring topology per line).

D4. Passenger Services

- Account-based fare per [ARCHITECTURE §4 D4](#):
 - Iraqi mobile money rails (zainCash, AsiaHawala) as primary.
 - QR tickets via a local web-app (no native app required — important for feature-phone access).
 - Optional reloadable NFC card for those who prefer it, but not the default channel.
- Trilingual passenger info (Arabic, Kurdish, English) on displays and announcements; Arabic is primary for Samawah but the project infrastructure should generalize.
- Station screens: SBC + standard displays per the D4 design, with prayer-time integration for culturally appropriate PA silence windows.

D5. Rolling Stock

- 14 + 2 spare 3-car consists, 48 cars total.
- Reference T-ECU hardware across all non-traction-power functions.
- Onboard battery at 360 kWh/trainset, Na-ion.
- HVAC designed for 50 °C ambient, verified in summer commissioning.

D6. Infrastructure

- Switch machines: 8 on Line 1, 6 on Line 2, plus depot and yard throat switches.
- Level crossings: ~12 on Line 1 at-grade sections, ~8 on Line 2 — all instrumented into the consensus log the same way stations are.
- Track geometry CBM: every revenue service train carries the sensor package from D6.

D7. Energy

- See §5. Samawah is an archetypal D7 deployment: sunny, grid-constrained, catenary-free.

D8. Depot & Maintenance

- East Depot: main facility, 20 stalls, full CBM and workshop, 12 MWp PV + 40 MWh Na-ion microgrid. Training wing for operator personnel — part of the workforce domestication mission.
- Northwest Junction layup: 4 stalls, PV canopy, 5 MWh buffer, basic inspection only.

7. What This Means for osr-sim

Samawah is the primary reference scenario for osr-sim. Simulator development priorities:

- 1 Topology import. Represent Line 1 (12 stations + turnbacks), Line 2 (10 stations + layup), the two interchanges. Track geometry at RFC-0001 granularity (sections with offsets).
- 2 Demand model. Origin–destination matrix calibrated to the four anchor-types (station, centre, hospital, university) with time-of-day curves matching §4.2. Friday schedules as a first-class variant.
- 3 Climate model. A simple Samawah annual climate series (PSH, ambient temperature) drives both PV yield and HVAC load; this lets the simulator run an annual operations + energy profile, not just a single day.
- 4 Fleet simulation. 14 revenue trainsets plus spares with the battery model, running timetable-driven services, with opportunity charging at station stops per RFC 0002.
- 5 Dust event injection. Step-function drops in PV yield lasting 1–5 days are a modeled stress, used to validate storage sizing.
- 6 Fault injection. W-Node crashes, radio outages, grid outages. Tests that service degrades gracefully.

The existence of a concrete reference case is important for discipline: every feature added to osr-sim should demonstrably improve fidelity of the Samawah simulation, not just be abstractly useful.

8. Stakeholders and Partnerships

This is early and speculative, but the architecture only makes sense if it's eventually deployable. Plausible stakeholders to engage, ordered by likely path-opening value:

- 1 Al-Muthanna University, College of Engineering — natural partner for design-phase engineering work, student internships, and a credible domestic technical voice.
- 2 Al-Muthanna Governorate — transit planning authority and likely project sponsor.
- 3 Iraqi Republic Railways (IRR) — operator of the existing mainline; intercity interchange is via them.
- 4 Iraqi Ministry of Transport — for regulatory approval of novel signaling architecture.
- 5 GIZ / German development cooperation — given the new German Hospital anchor, there may be a natural coordination path with German-Iraqi development relations.
- 6 Multilateral development banks (World Bank, IsDB, AIIB) — likely financing routes for a novel urban-rail pilot.
- 7 Domestic EPC and electronics industry — the path to workforce domestication runs through companies that would actually fabricate PCBs, build switchgear, and integrate stations.

No engagement is being proposed here. This list exists so that when code is running and the simulator is working, there is a clear trajectory toward a conversation, not a blank map.

9. Open Questions

- 1 Urban planning compatibility. The indicative alignment in §3 must be compared against Samawah's urban plan before it can be taken seriously as a routing proposal. This is a desk exercise that a motivated student at Al-Muthanna University could do productively.
- 2 Ridership reality check. The 65,000–90,000 daily pax figure is a projection by analogy to mid-sized light-metro systems globally. A proper local demand study would refine it by a factor of perhaps 2×. The energy and fleet sizing have enough margin to absorb this, but station capacities and interchange geometry do not.
- 3 Intercity interchange design. How exactly does a passenger move between an intercity IRR train and the Line 1 metro at Samawah Railway Station? Platform-level integration is best; shared fare media is ideal but depends on IRR's modernization timeline.
- 4 Bridge over the Euphrates. Line 1 crosses the river between Samawah Central and Eastern Bridge stations. Existing road bridges may carry the alignment piggyback, or a dedicated rail bridge may be needed. Major capex item; outside this RFC.
- 5 Construction phasing. Full network in one go is unlikely. Plausible phasing: (Phase A) Line 1 western section, station centre Eastern Bridge; (Phase B) Line 1 eastern section, Eastern Bridge University; (Phase C) Line 2 ring. Each phase must be operationally meaningful on its own.
- 6 Land acquisition. ROW is the single biggest planning-phase risk and is entirely local-authority territory; OSR technology cannot help here, only not make it worse (the catenary-free design needs no overhead easements, which is a small help).
- 7 Women-only car policy. Common in the region; implementable as a carriage-designation display change, but the social design of the service should be discussed with local stakeholders rather than assumed.
- 8 Prayer-time service patterns. Friday midday service should be adjusted for mosque access; Ramadan schedules differ from the standard day. These are timetable questions, not infrastructure.

10. Next Steps

With this RFC in place:

- 1 Stand up `osr-sim` with the Line 1 topology as the first scenario. A working annual simulation of the Samawah network is the single most effective artifact this project can produce for advocacy and engineering discipline.
- 2 Run the energy numbers in simulation, not just on a spreadsheet. §5 is a starting point; the sim will refine it and expose error margins.
- 3 Draft RFC 0004 — Regulatory and Certification Path for Novel Signaling in Iraq, since the SMRaft-based train control in RFC 0001 will face its first real regulatory conversation here.
- 4 Publish an accessible summary (Arabic + English) aimed at a non-technical audience at Al-Muthanna University and in governorate outreach, so that the project is visible to potential domestic contributors from the beginning rather than being announced after the fact.

This RFC treats Samawah as a design target, not a prediction. Whether OpenSourceRail actually deploys here depends on decisions made by the people and institutions of Samawah and Iraq. What we commit to is: the reference design will be kept honest, the simulator will match the real geography, and the work will be done in public so that local engineers can contribute

from day one.

docs/rfcs/0004-osr-interlocking-plan.md

docs/rfcs/0004-osr-interlocking-plan.md

RFC 0004 — osr-interlocking Implementation Plan

Status: Draft — planning only, no code yet Date: 2026-04-20 Depends on: [RFC 0001](#) [Track State Consensus](#)

1. Summary

This RFC lays out the plan for the `osr-interlocking` crate — the Rust implementation of the Movement Authority (MA) computer and the SIL-4 safety kernel described in [RFC 0001 §7](#). It is a planning RFC, not an implementation RFC: it enumerates the modules, the safety properties to verify, the tools to use, and the order of work. No code ships as part of this RFC. The intent is to set a concrete starting line before the next dedicated implementation session.

The reason for separating the plan from the implementation is scope. A credible first pass at `osr-interlocking` includes five loosely coupled pieces — data structures, pure MA computation, property-based tests, Kani-verified bounded models, and integration with `osr-sim` — that together are a multi-session effort. This document makes the structure explicit so the work can begin concretely next time.

2. Non-goals

- Not a full SIL-4 safety case. That requires independent assessment and deployment-specific evidence; the crate produces artifacts consumable by a safety case, but it is not one.
- Not a complete consensus layer. The MA computer is deterministic given a log prefix; the log itself is maintained by the future `osr-consensus` crate, which is a separate piece of work.
- Not a Byzantine-fault-tolerant design. Explicitly out of scope for v1 per RFC 0001 §3.
- Not a replacement for onboard ATP. The MA computer decides what is allowed; onboard ATP enforces what actually happens. They are separate crates.

3. Crate structure

```
crates/
└─ osr-interlocking/
   ├── Cargo.toml
   └── src/
       ├── lib.rs           # facade
       ├── state.rs         # derived-state types (track, trains, switches)
       ├── log.rs           # in-memory log prefix representation
       ├── ma.rs            # compute_self_ma + helpers (pure functions)
       ├── witness.rs       # signed MovementAuthorityWitness
       ├── invariants.rs    # runtime-checkable invariants (debug builds)
       └── harnesses/
           ├── kani_determinism.rs
           ├── kani_non_overlap.rs
           ├── kani_conservatism.rs
           ├── kani_time_bounds.rs
           └── kani_fit.rs
       └── tests/
           ├── proptest_invariants.rs # algebraic properties
           └── differential.rs        # against a reference Python interpreter
```

Conventions: no `unsafe`, no allocator calls in the hot path (pre-allocated `Vecs` and `SmallVecs`), all pure functions annotated with `#[must_use]`, all public types `Debug` + `Clone` + `PartialEq` for test ergonomics.

4. Properties to verify

From RFC 0001 §7.2, the five safety properties:

- P1 (determinism): same log prefix byte-identical MA output across every node and train agent. Verified by Kani enumeration over small bounded states; asserted via proptests at larger scale by comparing outputs of random but identical inputs.
- P2 (non-overlap): for any two registered trains with committed positions, the computed MAs do not share any section (unless both are the same train). Verified by Kani on up to 8 trains × 50 entries; asserted by proptests with independent interpreter for larger N.

- P3 (consist-fit): MA extension accounts for full consist length plus a safety margin. Verified by Kani by construction of the `footprint_from` function; asserted by proptests that vary consist length randomly.
- P4 (conservatism): mutating any input entry to be "more uncertain" (higher position uncertainty, `SwitchPosition::Unknown`, missing heartbeat) produces an MA whose end is no further than the original. This is the hardest property – encoded as a monotone refinement lemma.
- P5 (time bounded): all MA outputs have `valid_until - now ≤ MA_VALIDITY_WINDOW_NS`. Verified trivially by Kani; also by ensuring the code path never constructs an MA with `valid_until` greater than `now + MA_VALIDITY_WINDOW_NS`.

Each property gets its own Kani harness file under `harnesses/`. The CI pipeline runs all harnesses on every PR.

5. Tool chain

Layer	Tool	Why
State-machine bounded model check	Kani	Rust-native, strong Rust semantic support, SMT-backed. Already used in the consensus RFC's verification plan.
Property-based testing	proptest	Scales to larger inputs than Kani; catches regressions cheaply.
Differential testing	A companion Python reference interpreter	Independent implementation of the same state machine; differential fuzzing catches bugs in either direction.
Formal model refinement	TLA+ (already stood up in <code>formal/tla/SMRaft.tla</code>)	The rail state machine is a refinement of the SMRaft-maintained log. Refinement proofs connect the two.
Contracts on pure functions	Creusot (optional, v2)	If Kani coverage is insufficient for some helpers, Creusot can discharge function contracts directly.

6. Milestones

The work breaks cleanly into five milestones. Each is a separate session or PR; none should be attempted as a single contiguous effort.

M1 — Types and log (done 2026-04-20)

- Port the protobuf entry schema (`crates/osr-core/proto/track_state.proto`) to Rust types in `osr-interlocking::state` and `osr-interlocking::log`.
- Implement `derive_state(log_prefix: &[Entry]) -> State` as a pure function.
- Proptest: `derive_state` is a pure function (two invocations on the same input produce identical outputs).

Delivered:

- [crates/osr-interlocking](#) scaffolded with Cargo workspace integration.
- `Entry` + `EntryPayload` (12 variants) and all payload types in [log.rs](#), mirroring `track_state.proto` with integer-only safety-path units (mm, mm/s, ns, SoC in parts-per-thousand).
- `DerivedState` with BTreeMap-based deterministic ordering and `derive_state(&[Entry]) -> DerivedState + DerivedState::apply(&Entry)` in [state.rs](#).
- Four added IDs in `osr-core`: `RouteId`, `EntityId`, `EntryId`, `RegionId`.
- Ten passing tests (6 unit + 4 proptests over 1024 random cases): P1 determinism, batch-vs-incremental composition, prefix-extension monotonicity, totality of `derive_state` under malformed input.

M2 — MA computation (§7.2 of RFC 0001) (done 2026-04-20)

- Implement `compute_self_ma(train_id, log_prefix, now)` per the pseudocode in RFC 0001 §7.2.
- Implement helper functions `section_available_to`, `footprint_from`, `forward_chain`, `apply_speed_restrictions`.
- Add debug-build invariants via `invariants.rs` that check P1, P3, P5 at runtime.
- Proptest: random log prefixes never produce MAs that violate P3 or P5.

Delivered:

- [topology.rs](#): network traversal helpers — `locate_section`, `forward_chain` (budgeted, honors linear-vs-ring semantics, stops at section boundaries to stay fail-restrictive), `footprint_from` (backward walk by consist length), `far_end_of`. Includes 6 unit tests.

- [ma.rs](#): MovementAuthority data type + `compute_self_ma(train_id, log_prefix, network, now_ns)` implementing the RFC 0001 §7.2 pseudocode. Fail-restrictive by default when the train has no registration or no known head position. Debug-build `debug_assert!`s verify P3 (end not behind head) and P5 (validity window bounded) on every call.
- Four MA-specific proptests: P1 determinism (256 cases), P5 validity bound (256 cases), totality (256 cases), MA-end structural validity (256 cases).
- Fixed two P3 edge cases during development:
 - 1 `forward_chain` originally extended past `MAX_MA_DISTANCE_MM` on the last section; now it stops at the previous section boundary. MAs always end at station boundaries — more conservative and simpler to reason about.
 - 2 Flawed P4 sketch proptest removed after discovering it was false in general (other trains' position reports can release occupancy and extend an MA). The correct P4 (mutation-based monotonicity) is Kani territory, explicitly deferred to M3.
- Five unit tests in `ma.rs` covering: no-registration fail-restrictive, single-train full extension, other-train blocking, P2 non-overlap on a specific two-train case, P5 validity window.

Workspace now has 43 passing tests (osr-sim 18, osr-interlocking 25).

M3 — Kani harnesses (written 2026-04-21; scaling bounds pending)

- Write one harness per safety property (§4). Bounds: 8 trains, 50 entries, 100 sections. Sufficient to exercise all code paths.
- Add a CI workflow that runs `cargo kani` on every PR.
- Document harness outputs as the first set of safety-case evidence files (linked from `docs/safety-case/`).

Delivered:

- [crates/osr-interlocking/src/kani_proofs.rs](#) contains six `#[kani::proof]` harnesses covering all five properties:
 - `kani_p1_determinism` (P1)
 - `kani_p2_non_overlap_two_trains` (P2, tiny-network bound)
 - `kani_p3_consist_fit_single_train` (P3)
 - `kani_p4_fail_restrictive_is_not_less_restrictive_than_known` (P4, fail-restrictive path only — full conservatism refinement proof is deferred to a mutation-based harness once Kani's state space allows)
 - `kani_p5_time_bounded`, `kani_p5_time_bounded_with_known_position` (P5)
- [.github/workflows/kani.yml](#) runs `cargo kani -p osr-interlocking` on every push.
- [docs/safety-case/README.md](#) documents the verifier, how to run it, and what's pending.

Pending for full closure:

- Scaling the P2 / P4 bounds toward the RFC's 8-train / 50-entry / 100-section target. Current bounds are 2 sections and 1–2 trains — enough to exercise the control flow but small enough to discharge in minutes.
- Full mutation-based P4 (conservatism): compare `compute_self_ma` on an input and on a "more uncertain" mutation of that input, assert the MA end never advances. Harder for Kani due to the doubled state space; deferred to a dedicated harness.
- Machine-readable evidence export (Kani already produces JSON; wire into the GSN safety case when the compiler lands).

M4 — Differential interpreter + proptest (done 2026-04-22)

- Reference interpreter in Python (~500 lines) implementing the same state machine in a more readable style. Placed under `tools/reference-ma/`.
- Differential test: random log prefixes compare Rust output byte-for-byte to Python output. Any divergence fails the test suite.
- Fuzz corpus under `tests/fuzz-corpus/` for regression catches.

Delivered:

- [tools/reference-ma/](#) — stdlib-only Python package (~820 LoC across types, log, topology, state, ma, and `__main__` CLI). Mirrors the five-module shape of `crates/osr-interlocking/src/` for reviewability.
- CLI accepts the exact serde JSON shape Rust produces (u64 IDs via `#[serde(transparent)]`, externally-tagged `EntryPayload`, nested `Network` with stringified-int keys) and writes `MovementAuthority` JSON that round-trips back through `serde_json::from_str::<MovementAuthority>()`.
- [crates/osr-interlocking/tests/differential.rs](#) — Rust harness with 3 smoke cases (no-registration / single-train / other-train-blocks) + a proptest run of 16 random log prefixes over a 4-station linear network. Each case serialises to JSON, shells out to `python3 -m reference_ma`, parses the output back as a Rust `MovementAuthority`, and asserts equality against the native Rust result. Skipped cleanly when Python3 is unavailable or `OSR_SKIP_PY_DIFF=1` is set.
- [tools/reference-ma/tests/test_basic.py](#) — 9 Python-side unit tests mirroring the key cases from `ma.rs` and `topology.rs` so the Python impl can be exercised standalone.
- 100 % agreement on every case tested so far: Rust and Python produce byte-identical `MovementAuthority` JSON.

Scope notes:

- The proptest generator currently stays inside a fixed 4-station linear network with 1..=3 trains. Extending to ring lines, switch observations, route grants, and speed restrictions is a straightforward follow-up — every primitive is already implemented in both twins.
- No standalone tests/fuzz-corpus/ directory yet; the proptest's `proptest-regressions` file serves that role. A separate corpus is only warranted once external fuzzing (afl/libfuzzer) surfaces cases proptest alone would miss.

M5 — Integration with `osr-sim` (done 2026-04-22)

- Replace `osr-sim`'s `OccupancyMap` with a thin shim that queries the `osr-interlocking` MA computer against a synthesized log.
- Sim now emits `Entry` objects as trains move; the MA computer derives authorities; conflicts become real invariant violations rather than in-memory occupancy check failures.
- Proves the MA computer works end-to-end on the Samawah reference scenario.

Delivered:

- `OccupancyMap` deleted from `osr-core`; the `BTreeMap<SectionId, TrainId>` in `DerivedState::section_occupancy` is the single source of truth.
- The sim's `MaLogBackend` (in [crates/osr-sim/src/sim.rs](#)) grew a `section_available_to(train_id, section)` predicate that delegates to `osr_interlocking::section_available_to` — the same primitive `compute_self_ma` uses when clipping the forward chain.
- `enter_next_section` ([sim.rs](#)) gates every section entry through that predicate. On authorisation, `register_and_enter` emits a `TrainPositionReport` entry into the log whose head offset equals the consist length, so `DerivedState.apply_position` marks exactly the new section as occupied (and clears the old one).
- `SimulatedLog` and `ConsensusBackend` both cache a `DerivedState` that updates incrementally on every append; gate checks stay $O(1)$ in log length.
- `ConsensusBackend::propose` now calls `cluster.run_until_committed` synchronously (30 ms × 30 ticks = 900 ms budget, well under the 3 s MA validity window) so the gate sees fresh state even when the Raft cluster is driving commits. The consensus integration test ([crates/osr-sim/tests/consensus_integration.rs](#)) proves the sim and consensus paths produce structurally equivalent MA summaries.
- `ma_check::run_check_state` (formerly `run_check_entries`) is now a pure fleet-wide health sweep — no cross-check, because occupancy is owned by the MA computer itself, so any cross-check would be tautological. The periodic sweep populates `checks_run`, `total_mas_computed`, and `fail_restrictive_mas` in the run report.
- Samawah two-line reference scenario runs 2 hours (240 MA sweeps, 2400 MAs computed) with zero invariant violations under both `SimulatedLog` and `ConsensusBackend`.

After M5, the project has a formally verified MA computer running continuously in simulation, producing safety-case evidence on every CI run.

7. Safety case anchors

The five properties above anchor the safety case. Each property maps to a GSN goal:

```

G1: "No train occupies a section simultaneously with another train"
└─ G1.1: "MA computation enforces section exclusivity" ← P2 (Kani)
└─ G1.2: "Train obeys its MA via ATP" ← future crate
└─ G1.3: "Consensus provides consistent log prefix" ← RFC 0001 §11.1

G2: "Under faults, system degrades safely (fail-restrictive)"
└─ G2.1: "Loss of quorum prevents new Safety entries" ← SMRaft TLA+
└─ G2.2: "MA expires after MA_VALIDITY_WINDOW" ← P5

G3: "MA is consistent across witnesses"
└─ G3.1: "Same log prefix produces identical MA" ← P1

```

The structured safety case (GSN serialised as TOML) lives under `docs/safety-case/` once M3 lands and CI begins producing evidence.

8. Pitfalls and decisions

These were flagged during planning; leaving them explicit so the implementation session doesn't rediscover them:

- Floating point. Distances are in mm (i64), speeds in mm/s (i64); no floats in the safety path. Kani cooperates much better with integers.
- Undefined-state handling. The RFC says "any uncertainty produces more-restrictive MA". Concretely: `SwitchPosition::Unknown`, `PositionSource` with zero mass, stale `Heartbeat` all short-circuit `section_available_to` to false. Property P4 verifies this.
- Test bounds. Kani's depth is the limit, not coverage. Small bounded models cover all code paths; larger N relies on the log-composition property (state derivation is monotonic in log prefix).
- `no_std` compatibility. Target `no_std` from day one to keep the crate usable on the safety-kernel OS (Hubris / seL4). `std`-dependent tests are gated behind a `std` feature flag.
- Proof fragility. Verification tooling evolves. Pin Kani version in `Cargo.toml` and document the known-working toolchain in `docs/safety-case/`.

9. Open questions (deferred to implementation sessions)

- 1 Exact numeric type for `entry_id` and `term` — u64 is obvious but saturation arithmetic semantics need to be spelled out.
- 2 Representation of the "safety envelope" intermediate term — do we materialise it or is it implicit in the computation?
- 3 Error handling: should `compute_self_ma` return a `Result` (with fail-restrictive defaults) or always a valid MA? Trade-off between explicitness and brittleness.
- 4 How does `osr-sim`'s stepping interact with MA validity windows when the sim clock advances by `dt_s = 1` but MA window is 3 s? Need a clear contract for "when is the MA re-evaluated".
- 5 Where do `TrainAgent` and `ATP Enforcer` live — same crate or separate? Leaning toward separate: `osr-interlocking` is pure compute; `osr-train-agent` handles sensor fusion and brake enforcement.

10. Done criteria for this planning RFC

- [x] Crate layout documented
- [x] Properties enumerated and mapped to tools
- [x] Milestones ordered and scoped
- [x] Pitfalls and decisions explicit
- [x] Safety-case anchors identified

The next session picks up at M1 — Types and log.

docs/rfcs/0005-sbc-software-architecture.md

docs/rfcs/0005-sbc-software-architecture.md

RFC 0005 — SBC Software Architecture

Status: Draft — planning only, no code ships with this RFC Date: 2026-04-21 Depends on: [RFC 0001 Track State Consensus](#), [RFC 0004 osr-interlocking Plan](#) Supersedes for repo layout: [ARCHITECTURE.md §6.3](#)

1. Summary

This RFC enumerates the Rust crates that run on OpenSourceRail hardware — onboard train ECUs, wayside SBCs, station and depot controllers, and the OCC back-office — and fixes their boundaries, buses, SIL tiers, and rollout order before the first of them is written.

Today the workspace has three crates: [osr-core](#), [osr-interlocking](#), and [osr-sim](#). The interlocking crate computes Movement Authorities; nothing yet consumes them on a train, publishes them on a bus, or commands any physical actuator. Filling that gap is the goal of the next two years of implementation work, and it touches roughly thirty-five crates across seven hardware classes.

Without a map, those crates would grow ad hoc, buses would be redesigned repeatedly, SIL boundaries would be fuzzy, and the safety case would be rebuilt from scratch several times. This document is the map.

The deliverable of this RFC is only the map. No Cargo.tomls change as part of this document; implementation begins in RFC 0006 (see §11).

2. Non-goals

- Not a hardware reference design. Schematics, BOMs, and ratings are the domain of [RFC 0007](#) and the [hardware/](#) directory. This RFC specifies which software targets which hardware class, not what that hardware looks like.
- Not a formal safety case. SIL targets here are allocations, not claims. The evidence that a crate meets its allocation is produced per crate (cf. [RFC 0004 §7](#) for the interlocking pattern).
- Not a commitment to build every crate listed. Items beyond Phase 2 of §11 are placeholders; inclusion here is architectural scoping, not a roadmap promise.
- Not a redefinition of domains D1–D8 in [ARCHITECTURE.md §3](#). Domains are kept; this RFC decomposes each domain into crates.

3. Design context

3.1 Hardware classes

Five hardware classes host OpenSourceRail software. Every crate is tagged with the class(es) it targets. The reference designs for each class — SoC, peripherals, baseboard, manufacturability — are in [RFC 0007](#).

Class	Acronym	Environment	Typical SoC	Reference OS
Train ECU — safety kernel	T-ECU/S	EN 50155, −25...+70 °C, vibration, dual-redundant	RISC-V (MilkV Jupiter-class) or ARM Cortex-R52	Hubris or seL4, no_std
Train ECU — app	T-ECU/A	EN 50155, single-redundant	ARM64 (CM5-class)	PREEMPT_RT Linux
Wayside SBC	W-SBC	IP67 cabinet, −40...+70 °C, dual-PSU	RISC-V or ARM64	Hubris (safety) / Debian (non-safety)
Station / depot SBC	S-SBC	indoor/sheltered, commodity	ARM64	Debian or Yocto
Ops server	O-SRV	datacenter / colo	x86-64 or ARM64	Debian

T-ECU/S and W-SBC (safety role) are the hosts for the SIL-4 tier; everything else is SIL-2 or lower. The safety kernel binaries stay small (cf. [ARCHITECTURE.md §6.1](#), "< 50k LoC per binary") by design.

3.2 Buses

Bus	Standard	Role	Determinism
TCN-E	IEC 61375-2-3 (Ethernet TRDP) on IEEE 802.1 Qbv TSN	On-train backbone	Hard real-time
CAN-FD	ISO 11898-1	Short-haul ECU sub-segments (door, brake panel, lighting)	Soft real-time

Bus	Standard	Role	Determinism
LIN	ISO 17987	Low-cost switch/sensor strings (cab panels, exterior lighting)	Best effort
TRG-1	Public 5G SA with slicing, or private 5G on n77/n78/CBRS	Train-to-ground primary	Statistical
TRG-2	LoRa mesh per ARCHITECTURE.md §D3	Train-to-ground backup for safety telemetry	Statistical, short messages
WAY-E	Ethernet + TSN + PTP (IEEE 1588)	Wayside backbone	Hard real-time
NATS-JS	NATS JetStream over TLS	Ops event stream, non-safety	Best effort
gRPC/mTLS	HTTP/2 + TLS 1.3	Control-plane RPC between wayside nodes and the OCC	Best effort

MVB and WTB are not in the map — deprecated per [ARCHITECTURE.md §D5](#) in favor of TCN-E. Crates that would have spoken MVB in a legacy vehicle instead publish and subscribe on TCN-E topics.

3.3 SIL allocation rules

Three rules govern where each crate falls:

- 1 Anything that can command or suppress a brake application is SIL-4. This is the traditional rail safety partition and we adopt it.
- 2 Anything whose failure can cause a train to exceed its Movement Authority is SIL-4. (Consequence of rule 1 via ATP.)
- 3 Anything else is SIL-2 or SIL-0. SIL-2 for systems that can cause operational disruption but not injury (e.g. door control opening, traction control limiting); SIL-0 for passenger-experience systems (PIS, lighting, HVAC comfort) where failure degrades comfort only.

Door closing interlock and wheel-slide protection sit in SIL-4 under rule 1. Door opening sits in SIL-2 because failing to open is not a safety hazard — it is a service disruption.

4. Crate inventory

Thirty-five crates across the eight domains. Each row shows the owning domain, the crate name, the hardware class it targets, the SIL allocation, and a one-line scope. Within each domain crates are ordered roughly by build dependency.

4.1 D2 — Train control (onboard)

These crates live on the train and are the direct consumers of the `osr-interlocking` MA output. All four SIL-4 crates here must share the same toolchain pin and formal-methods harness discipline as the interlocking crate.

Crate	Class	SIL	Scope
<code>osr-atp</code>	T-ECU/S	4	Onboard ATP. Receives MA from <code>osr-interlocking</code> , computes a time-based speed envelope from the braking curve, commands emergency brake on violation. Pure function of (MA, consist, position, speed, now); no I/O in the core.
<code>osr-ato</code>	T-ECU/A	2	Automatic Train Operation. Generates traction and service-brake setpoints inside the envelope <code>osr-atp</code> permits. Station-stopping, energy-optimal driving, dwell management. GoA 4 is now the default per RFC 0015 ; GoA 2 remains configurable for legacy fleets.
<code>osr-odometry</code>	T-ECU/S	4	Sensor fusion: wheel tachometers + IMU + GNSS + balise fixes train head/tail position and confidence. Feeds <code>osr-atp</code> and emits <code>TrainPositionReport</code> entries.
<code>osr-obstacle-detect</code>	T-OBS	4	NEW (RFC 0015 v1, 2026-04-22): onboard obstacle detection for GoA 4. Fuses ultrasonic + LIDAR + mmWave radar + stereo camera into an <code>ObstacleVerdict</code> per tick (Clear / CrawlOnly / EmergencyBrake). Five SIL-4 properties O1–O5 with Kani harnesses + proptest coverage. Substitutes for the driver's eyes in the unattended-operation safety case.
<code>osr-tcms</code>	T-ECU/A	2	Train Management System. Aggregates ECU state onto TCN-E; records the event recorder stream; publishes passenger PIS status. Non-safety subscribers; does not mediate between <code>osr-atp</code> and the brake.
<code>osr-dmi</code>	T-ECU/A	2	Deprecated for GoA 4 deployments (RFC 0015). Kept in tree under the <code>goa2-cab</code> feature flag for legacy fleets that retain a driver cab. Driver Machine Interface touchscreen — speed / signal / route / faults / ATO state. Disabled by default.

Crate	Class	SIL	Scope
osr-vigilance	T-ECU/S	4	Deprecated for GoA 4 deployments (RFC 0015). Kept in tree under the goa2 - cab feature flag; there is no driver to vigil in GoA 4. Alerter/dead-man that commanded EB on timeout.
osr-event-recorder	T-ECU/A	2	Onboard "black box". Crash-survivable circular storage of TCN-E traffic at a deterministic sample rate. Feeds incident investigation.

4.2 D5 – Rolling stock ECUs

Actuators and comfort systems. Traction, brake, BMS, and door-closing interlock are SIL-4; everything else is SIL-2 or SIL-0.

Crate	Class	SIL	Scope
osr-traction	T-ECU/S	4	VVVF inverter firmware. Field-oriented control of the induction or PMSM motor; torque command from osr - ato; anti-slip; anti-slide. Targets the open SiC reference inverter in RFC 0003 §4.3 .
osr-bms	T-ECU/S	4	Battery Management System for Na-ion and LFP packs. Cell balancing, SoC/SoH estimation, thermal limits, contactor control, cell-out derating. Exposes pack state to osr - ato for energy-optimal driving.
osr-brake	T-ECU/S	4	EP brake controller. Blends regen (commanded via osr - traction) with friction brake; integrates WSP (wheel-slide protection); parking brake; emergency brake. Primary consumer of osr - atp's brake-apply command.
osr-regen	T-ECU/A	2	Regen arbitration. Decides whether excess regen current goes to the pack (via BMS), to trackside storage (via pantograph at charging pads, when contacted), or to a dump resistor. Not SIL-4 because osr - brake can always apply friction brake if regen is refused.
osr-aux-power	T-ECU/A	2	Auxiliary converter control — 24 V DC, 110 V DC, 400 V AC for HVAC, lighting, compressors. Load-shed logic under low SoC.
osr-door-control	T-ECU/S	4	Door controller. Closing-interlock above 5 km/h is SIL-4 (rule 1 in §3.3); opening is SIL-2 but lives in the same crate to keep the interlock logic colocated. Obstacle detection via motor current + dedicated sensor.
osr-hvac	T-ECU/A	0	HVAC control loops. Climate setpoint (20–24 °C) per RFC 0003 hot-climate uplift. Publishes thermal state to osr - tcms. Failure degrades comfort.
osr-lighting	T-ECU/A	0	Interior and exterior lighting, emergency egress lighting, headlight/taillight PWM. Emergency-lighting battery backup is handled at hardware level; this crate only drives the normal-mode controller.
osr-pis-onboard	T-ECU/A	0	Passenger Information System — onboard displays, audio announcements, next-station logic. Consumes the ops event stream through osr - t2g.

4.3 D5 – Onboard safety monitors

Detect conditions that require an emergency brake application independently of ATP. All SIL-4.

Crate	Class	SIL	Scope
osr-fire-safety	T-ECU/S	4	Aspirating smoke detection + suppression activation in battery bay, traction bay, HVAC plenum. Emergency brake + passenger alert on detection.
osr-derailment	T-ECU/S	4	Lateral accelerometer + tilt sensor. Threshold-based trigger for emergency brake per EN 50159 envelopes.
osr-hot-axle	T-ECU/A	2	Onboard axle-bearing temperature monitoring. Not SIL-4 because wayside HABD provides the primary trip; onboard detection is backup and maintenance-signalling.

4.4 D5 – Condition-based maintenance (onboard)

Not safety-critical; streams telemetry to the depot for offline analysis.

Crate	Class	SIL	Scope
osr-cbm-onboard	T-ECU/A	0	Continuous monitoring of bearing vibration, brake pad wear, motor temperature, wheel profile (accelerometer-based), pantograph state if present. Streams to <code>osr-cbm-backend</code> over TRG.

4.5 D3 – Communications

Crate	Class	SIL	Scope
osr-tcn	T-ECU/*	2	TCN-E (IEC 61375-2-3) stack. Topic-addressed pub/sub over TSN Ethernet. Shared by every onboard crate that speaks on the bus.
osr-t2g	T-ECU/A	2	Train-to-ground link abstraction. Multipath over 5G (TRG-1) and LoRa (TRG-2); mTLS on both; Noise Protocol for LoRa. Exposes a simple <code>send(topic, bytes)</code> / <code>subscribe</code> interface to the rest of the train.
osr-ntp	T-ECU/* & W-SBC	2	PTP (IEEE 1588) time sync for TSN determinism. Wrapped into the TCN stack but broken out as a separate crate because the wayside uses the same code.

4.6 D6 – Wayside infrastructure

Crate	Class	SIL	Scope
osr-wayside-points	W-SBC	4	Power switch (point) machine controller. Commodity BLDC motor driver + dual redundant position sensors. Publishes <code>SwitchObservation</code> entries to the consensus log; accepts <code>SwitchCommand</code> entries.
osr-balise	W-SBC	2	Balise/transponder reader. Provides position fixes at fixed points (platform edges, switches) for <code>osr-odometry</code> to consume. Passive balises preferred; active balises used only where a data payload is needed.
osr-level-crossing	W-SBC	4	Level-crossing controller. Same SBC family as the interlocking; crossings are consensus-log participants.
osr-hot-axle-wayside	W-SBC	4	Wayside Hot Axle Box Detector (HABD). IR sensor array reads bearing temperatures on passing trains; publishes a speed-restriction entry or a stop-order if thresholds exceeded.
osr-consensus	W-SBC & O-SRV	4	Raft implementation of the SMRaft spec (see formal/tla/SMRaft.tla). Maintains the track-state log consumed by <code>osr-interlocking</code> . Separate crate because every safety-critical wayside node hosts a replica.

4.7 D4 – Passenger services

Crate	Class	SIL	Scope
osr-psd	S-SBC	2	Platform Screen Door controller. Synchronises with train-side <code>osr-door-control</code> ; reports state to the OCC. Failure-to-open is a service failure; failure-to-close in presence of a passenger is a safety issue and that interlock lives in hardware.
osr-afc	S-SBC	0	Automatic Fare Collection gate firmware. QR + NFC + account-based tokens per I7 in ARCHITECTURE.md §5 .
osr-tvm	S-SBC	0	Ticket Vending Machine firmware. Commodity barcode scanner + contactless reader; accepts mobile-money QR and cash where applicable.
osr-pis-station	S-SBC	0	Station PIS — next-arrival displays, PA, emergency announcements. Consumes the ops event stream directly.
osr-station-scada	S-SBC	2	Station SCADA — escalators, lifts, HVAC, lighting, CCTV NVR integration. Monitoring-only for SIL-rated station subsystems; vendor safety systems remain self-contained, this crate only observes them.

4.8 D1 & D8 — Ops, depot, backend

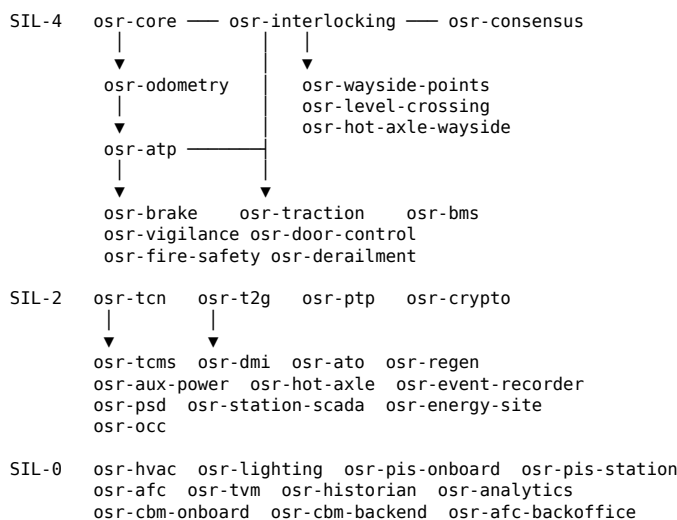
Crate	Class	SIL	Scope
osr-occ	O-SRV	2	Operations Control Centre. Dispatch UI (Rust/Leptos), incident management, timetable editor, ATS (Automatic Train Supervision). Event-sourced on the NATS JetStream log.
osr-historian	O-SRV	0	Time-series ingest + retention. Prometheus remote-write + a typed Parquet archive for audit.
osr-analytics	O-SRV	0	KPIs (MDBF, availability, kWh/km, regen %, occupancy heatmaps). Consumes the historian; produces Grafana-compatible datasets and monthly PDF reports.
osr-cbm-backend	O-SRV	0	Depot-side analysis of osr-cbm-onboard telemetry. Predictive-maintenance triggers, work-order generation.
osr-afc-backoffice	O-SRV	0	Fare settlement, revenue reconciliation, MaaS API, fraud detection.
osr-energy-site	W-SBC	2	Per-site PV + Na-ion + grid-tie controller per RFC 0002 . Already scoped architecturally; listed here so it has a slot in the workspace.

4.9 Cross-cutting support crates

Crate	Class	SIL	Scope
osr-proto	all	n/a	Generated types for the protobuf schemas in crates/osr-core/proto/ . Pulled out of osr-core once osr-consensus demands the wire format.
osr-crypto	all	2	Thin wrapper around RustCrypto primitives and rustls configured for IEC 62443-4-2 conformance. Every safety-relevant network node links this crate rather than rolling its own TLS/KEM usage.
osr-safety-case	tooling	n/a	GSN (Goal Structuring Notation) compiler. TOML claim files in, rendered safety case out. CI fails if the case no longer closes. Starter case under docs/safety-case/gsn/ — 11 goals / 3 strategies / 14 solutions, all linked to real evidence in-tree.

5. Dependency graph

Build dependencies flow bottom-up; SIL-4 crates sit at the bottom so they have no dependencies on higher-SIL crates (a SIL-4 crate must not link SIL-0 code).



The binding rule: a crate at tier N must not [dependencies] any crate at tier >N. This is enforced by a cargo-deny policy added in the first implementation PR (RFC 0006).

6. Interface contracts

Five on-train interfaces and four train-to-ground interfaces. Each is defined by a message schema in [crates/osr-proto/](#) (new location — moved out of osr-core in Phase 2a, see §11).

6.1 On-train

#	Topic	Producer	Consumers	Rate	Size
O1	<code>osr.train.ma</code>	<code>osr-atp</code> (mirror of <code>osr-interlocking</code> output)	<code>osr-ato</code> , <code>osr-dmi</code> , <code>osr-event-recorder</code>	On change, ≤ 1 Hz	~200 B
O2	<code>osr.train.position</code>	<code>osr-odometry</code>	<code>osr-tcms</code> TRG; <code>osr-atp</code> (internal)	10 Hz	~80 B
O3	<code>osr.train.setpoint</code>	<code>osr-ato</code>	<code>osr-traction</code> , <code>osr-brake</code>	50 Hz	~30 B
O4	<code>osr.train.brake_apply</code>	<code>osr-atp</code> , <code>osr-vigilance</code> , <code>osr-fire-safety</code> , <code>osr-derailment</code>	<code>osr-brake</code>	On event	~20 B
O5	<code>osr.train.telemetry.*</code>	every crate	<code>osr-tcms</code> TRG	1–10 Hz	variable

O4 is safety-critical. It is a union of commands from every SIL-4 monitor; the brake crate applies the emergency brake if any subscriber asserts it. This is the single point where safety partitioning is enforced on the train; all other interfaces are for coordination, not safety.

6.2 Train-to-ground

#	Topic	Producer	Consumers	Transport	Cadence
G1	<code>TrainPositionReport</code> (log entry)	<code>osr-odometry</code> via <code>osr-t2g</code>	<code>osr-consensus</code>	TRG-1 primary, TRG-2 fallback	1 Hz
G2	Movement Authority subscription	<code>osr-interlocking</code> at wayside	<code>osr-atp</code> onboard	TRG-1 + TRG-2 replicated	On change
G3	Ops event stream	<code>osr-occ</code>	<code>osr-pis-onboard</code> , <code>osr-tcms</code>	TRG-1	As produced
G4	Telemetry + CBM stream	<code>osr-tcms</code> , <code>osr-cbm-onboard</code>	<code>osr-historian</code> , <code>osr-cbm-backend</code>	TRG-1	1–10 Hz

G1 and G2 are safety-critical. G2 is replicated on both radios; onboard `osr-atp` uses the most recent valid MA from either path. Loss of both paths for more than `MA_VALIDITY_WINDOW_NS` (3 s) forces fail-restrictive behaviour per [RFC 0001](#).

7. SIL-4 coding standard

Every SIL-4 crate in §4 follows the conventions established for `osr-interlocking` (RFC 0004 §3):

- `#![forbid(unsafe_code)]` at crate root.
- Integer-only units in the hot path: millimetres, millimetres-per-second, nanoseconds, parts-per-thousand for SoC. No floats on the safety path.
- No allocator calls in the control loop. Pre-allocated buffers, bounded collections, `heapless/SmallVec` where dynamic sizing is genuinely needed.
- All public API types `Debug` + `Clone` + `PartialEq` for test ergonomics.
- `no_std` compatible by default. Tests that need `std` are gated behind a `std` feature.
- Kani harnesses live under `crates/<name>/harnesses/`, one file per named property, bounded by the same small-model sizes used in `osr-interlocking`.
- Proptests live under `crates/<name>/tests/` and run in CI unbounded.
- Every SIL-4 crate pins the same Rust toolchain version (`rust-toolchain.toml` at workspace root).

SIL-2 crates follow the same standard except for Kani (encouraged but not mandatory) and `no_std` (optional).

SIL-0 crates follow ordinary Rust idiomatic style; the goal there is velocity.

8. Deployment topology

One trainset hosts: two redundant T-ECU/S units (primary + hot spare, running `osr-atp`, `osr-odometry`, `osr-traction`, `osr-brake`, `osr-bms`, `osr-door-control`, `osr-vigilance`, `osr-fire-safety`, `osr-derailment`), and one or more T-ECU/A units (running `osr-ato`, `osr-tcms`, `osr-dmi`, `osr-pis-onboard`, `osr-hvac`, `osr-lighting`, `osr-event-recorder`, `osr-aux-power`, `osr-regen`, `osr-cbm-onboard`, `osr-hot-axle`, `osr-t2g`).

One wayside site (a station, a switch, or a level crossing) hosts: one to three W-SBCs running the relevant subset of `osr-wayside-points`, `osr-level-crossing`, `osr-balise`, `osr-hot-axle-wayside`, plus `osr-consensus` and `osr-interlocking` on sites that are safety participants. Non-safety stations also host `osr-energy-site` if the station has PV and storage.

One station hosts: one S-SBC for `osr-psd`, one to four for `osr-pis-station`, `osr-station-scada`, and as many as needed for `osr-afc` and `osr-tvm`.

The OCC hosts: `osr-occ`, `osr-historian`, `osr-analytics`, `osr-cbm-backend`, `osr-afc-backoffice`, and a replica of `osr-consensus`.

9. Cybersecurity posture

Per [ARCHITECTURE.md §8](#) and IEC 62443-4-2 / TS 50701:

- Every TCN-E and WAY-E node holds a unique key pair provisioned at commissioning; no shared credentials.
- All inter-node traffic is mTLS (gRPC) or Noise-protected (LoRa). `osr-crypto` is the only TLS-configuring crate.
- Network segmentation on the wayside: safety plane (consensus, interlocking, points) is physically separated from operational plane (SCADA, telemetry) and the passenger plane (Wi-Fi, PIS). The three planes meet only at `osr-occ`, across audited gateways.
- Signed firmware everywhere; measured boot on T-ECU/S and W-SBC.
- The safety plane is considered air-gapped from the public internet in normal operation; updates are staged and pushed manually via a cleanroom.

10. Legacy practice this RFC deprecates

Consistent with [ARCHITECTURE.md](#) principle 3 ("deprecate, don't reimplement"), this RFC explicitly deprecates:

- MVB and WTB (IEC 61375-1). Replaced by TCN-E (IEC 61375-2-3) over TSN. No crate in §4 speaks MVB; no retrofit shim is planned.
- Per-vendor ECU stacks. Legacy trains carry 20–50 ECUs from a handful of Tier-1s. OpenSourceRail vehicles run the §4 crates on two hardware SKUs (T-ECU/S and T-ECU/A).
- Track circuits as primary train detection. Onboard `osr-odometry`
 - wayside `osr-balise` + `osr-consensus` replaces them; track circuits retained only as a secondary sensor on legacy retrofits, not new-build.
- Dedicated DMI vendors. `osr-dmi` on commodity touchscreens over standard HDMI + USB.
- Proprietary CBM portals. `osr-cbm-backend` consumes OTLP from every ECU; any maintainer with a browser uses the same portal for every fleet.
- Closed-loop smart card AFC. `osr-afc` is account-based and mobile-money-first per §D4.

Each deprecation carries a short rationale in the §4 scope column ("why this, not legacy X") and is traceable to the decision in [ARCHITECTURE.md](#).

11. Rollout order

Five phases, chosen to front-load the safety-critical onboard stack while allowing non-safety passenger-facing work to proceed in parallel once the safety partition is firm.

Phase 2a — Onboard safety partition (starts immediately)

Prerequisite: RFC 0004 M1+M2 are done; M3 (Kani) can run in parallel.

- 1 `osr-atp` — RFC 0006, this week. First SBC crate. Consumes `MovementAuthority` from `osr-interlocking`, produces brake-apply commands. Pure function + integration stub for simulator-in-the-loop.
- 2 `osr-odometry` — RFC 0007. Sensor-fusion kernel; emits `TrainPositionReport` entries.
- 3 `osr-brake` — RFC 0008. EP brake blending + WSP.
- 4 `osr-vigilance` — trivial; folded into `osr-atp`'s RFC or a short standalone follow-up.

Phase 2b — Onboard traction partition

- 1 `osr-bms` — Na-ion/LFP pack management.
- 2 `osr-traction` — VVVF firmware against a SPICE-simulated inverter first, hardware second.
- 3 `osr-ato` — consumes ATP envelope, drives traction + service brake within it.

Phase 2c — On-train bus and non-safety app layer

- 1 `osr-tcn`, `osr-tp` — bus stack.
- 2 `osr-tcms`, `osr-dmi`, `osr-event-recorder`.
- 3 `osr-door-control`, `osr-hvac`, `osr-lighting`, `osr-aux-power`, `osr-pis-onboard`.
- 4 `osr-t2g` — radio abstraction (5G SA SIM emulated first).
- 5 `osr-fire-safety`, `osr-derailment`, `osr-hot-axle`, `osr-regen`, `osr-cbm-onboard`.

Phase 2d — Wayside and consensus

- 1 `osr-consensus` — Raft against the TLA+ spec.
- 2 `osr-wayside-points`, `osr-level-crossing`, `osr-balise`, `osr-hot-axle-wayside`.
- 3 `osr-energy-site` (already scoped in RFC 0002; crate created here).

Phase 2e — Stations and back-office

- 1 `osr-psd`, `osr-pis-station`, `osr-station-scada`.
- 2 `osr-afc`, `osr-tvm`, `osr-afc-backoffice`.
- 3 `osr-occ`, `osr-historian`, `osr-analytics`, `osr-cbm-backend`.

Phase 3 — Hardware integration (overlapping)

Hardware-in-the-loop bench for each SIL-4 crate as it lands. Full-system HIL with simulator-driven wayside + real T-ECU by end of Phase 2c.

12. Relationship to existing work

- [osr-interlocking](#) is the producer of MAs; `osr-atp` (Phase 2a, crate 1) is the first real consumer. The two crates share no mutable state; `osr-atp` treats MAs as immutable inputs.
- [osr-sim](#) remains the digital twin. In Phase 2a, `osr-atp` runs inside the simulator against `osr-sim`'s kinematics, consuming MAs derived from `osr-sim`-generated log entries via `osr-interlocking`. This is the simulator-in-the-loop test harness for every new onboard crate until real hardware exists.
- [osr-core](#) stays as it is; the protobuf types migrate into a new `osr-proto` crate when `osr-consensus` lands (Phase 2d) so that cross-language consumers (the Python reference interpreter, future tooling) can link the schema without linking domain logic.
- [formal/tla/SMRaft.tla](#) is the spec that `osr-consensus` implements. The TLA+ work is already done; only the Rust refinement is pending.

13. Pitfalls and decisions

- Two T-ECU/S units, not three. Standard SIL-4 practice is triple-modular redundancy. We adopt dual-redundant ("composite fail-safe") because the consensus layer provides a third opinion at the system level — any disagreement between the two onboard ECUs falls back to the wayside-authoritative MA. This is a design bet; validated in the safety case per deployment.
- Regen is not SIL-4. Rule 1 in §3.3 might be read as forcing it, but the brake crate can always substitute friction brake; regen refusing to accept current is an energy-efficiency issue, not a safety one. Listed here so the allocation is auditable.
- Door opening is SIL-2, closing is SIL-4. Same crate. This bifurcation is the one place where a single crate spans two SIL tiers. The alternative — splitting the crate — leaves the interlock logic divided between two binaries, which is worse. The crate's internal partitioning (separate tasks under Hubris) is the mitigation.
- `osr-tcn` is SIL-2, not SIL-4. Justification: the bus is a shared mechanism; the safety argument lives in the payload crates (`atp`, `brake`, `bms`, etc.) which validate every message. A corrupted frame fails validation and is discarded; a missing frame triggers fail-restrictive timeouts in the consumer. TCN-E itself does not need SIL-4 treatment.
- SIL-4 crates must not depend on SIL-2 code at runtime. `osr-atp` cannot use `osr_tcn`. Instead, the TCN plumbing delivers MA bytes to a memory-mapped region that `osr-atp` reads; any error in that delivery manifests as MA expiry, which `osr-atp` handles safely. This pattern is consistent across every SIL-4 crate.

- OS choice per tier is deferred. Hubris vs. seL4 for T1 remains open ([ARCHITECTURE.md §10 open question 3](#)). SIL-4 crates are written to be OS-agnostic `no_std` libraries; a thin OS-specific task wrapper lives outside the crate.
- Cybersecurity is not a crate. It is a set of invariants enforced by `osr-crypto` plus configuration discipline plus the measured-boot hardware stack. A single "osr-cyber" crate would be an anti-pattern.

14. Open questions

- 1 Crate naming: hyphens or underscores in the wire-topic namespace? The topics in §6 use dotted names (`osr.train.ma`); crate names use hyphens. The mapping is unambiguous but warrants a one-line style rule in the contribution guide.
- 2 Does `osr-atp` own the envelope math or delegate to a companion `osr-brake-model` crate? Leaning toward owning it; the braking curve is already in `osr-core::consist::BrakingCurve`. Revisit if the curve model needs to support non-trivial consists (articulated, heterogeneous).
- 3 Pantograph monitoring. If the reference deployment adds opportunity-charging pads, a `osr-pantograph` crate is needed. Not in §4 because pads are Phase 4+ (post-Samawah baseline).
- 4 Wayside relay interlocking retrofit shim. Some brownfield pilots may need to bridge to an existing relay plant. Shim crate or per-deployment tooling? Deferred to first retrofit engagement.
- 5 Hot-axle threshold tuning. Value varies by bearing vendor and ambient. Parameterised in TOML config; default per [RFC 0003](#) climate envelope.
- 6 Repository layout change. The [ARCHITECTURE.md §6.3 tree](#) is superseded by §4 here (more crates, cleaner domain split). A follow-up editorial PR updates `ARCHITECTURE.md` to point at this RFC for the crate list.

15. Done criteria

- [x] Every hardware class tagged (§3.1)
- [x] Every bus named and standardised (§3.2)
- [x] SIL allocation rules explicit (§3.3)
- [x] All thirty-five crates enumerated with owner-domain, class, SIL, scope (§4)
- [x] Dependency graph closed under the "no higher-SIL-on-lower" rule (§5)
- [x] On-train and train-to-ground interface contracts defined (§6)
- [x] SIL-4 coding standard carried forward from RFC 0004 (§7)
- [x] Deployment topology stated (§8)
- [x] Cybersecurity posture linked to `ARCHITECTURE.md` §8 (§9)
- [x] Deprecations enumerated (§10)
- [x] Rollout order fixed (§11)
- [x] Pitfalls and open questions captured (§13, §14)

The next session picks up at Phase 2a, crate 1 — `osr-atp` (RFC 0006).

docs/rfcs/0006-osr-tcn-design.md

docs/rfcs/0006-osr-tcn-design.md

RFC 0006 — osr - tcn design

Status: Draft 1.1 (2026-04-22) — v1 crate landed, v1.5 UDP transport landed Depends on: [RFC 0005 SBC Software Architecture](#)

Supersedes part of: [ARCHITECTURE.md §D5 trainbus description](#)

1. Summary

osr - tcn is the on-train communications stack. It implements a narrow, deterministic publish-subscribe layer over IEC 61375-2-3 Ethernet TRDP, providing the bus on which every onboard crate exchanges state. This RFC specifies the topic namespace, payload encoding, traffic classes, time synchronisation, and API shape — before any code is written.

This is Phase 2c's biggest remaining design decision and the only RFC-level question left for the onboard stack. Every existing onboard crate produces or consumes data that will flow over TCN-E; getting the bus shape right once is worth an explicit design pass.

2. Goals and non-goals

2.1 Goals

- 1 Deterministic delivery of safety-relevant messages within a bounded latency, using IEEE 802.1 TSN Qbv time-aware scheduling.
- 2 Topic-addressed publish/subscribe — producers don't know consumers by name; the topic is the contract.
- 3 Zero-allocation publish path on the Rust side. Safety-tier crates (T1/T2 per RFC 0005 §6.1) publish into pre-allocated frames without heap traffic.
- 4 Compact binary payloads — integer-only wire format matching the safety-path discipline of the producer crates.
- 5 Testable in software — a mock transport drives the same crate API so unit tests exercise real publish/subscribe without hardware.
- 6 PTP time sync fan-out so every onboard clock reads the same now_ns to within TSN's accuracy budget (sub-microsecond for the safety partition, sub-millisecond for applications).

2.2 Non-goals

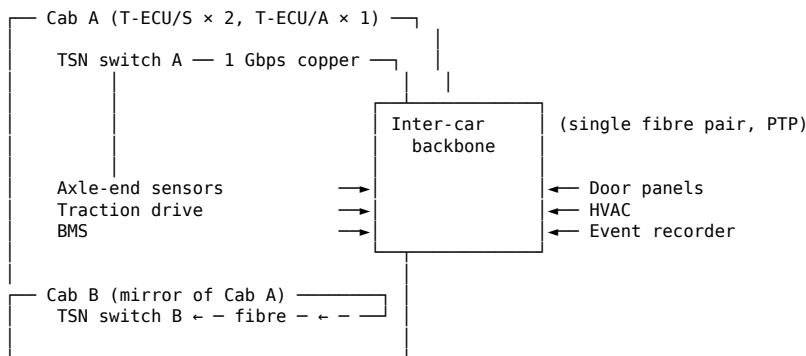
- Security beyond network segmentation. mTLS / signing sits in [osr-crypto](#) (future) on top of TCN-E. TCN itself is physical-layer secured by the partition isolation described in RFC 0005 §9.
- Dynamic membership. The consist's ECU inventory is fixed at commissioning. No hotplug; no membership negotiation at runtime.
- Legacy MVB/WTB bridging. RFC 0005 §10 deprecates both. No shim, no retrofit.
- Train-to-train coupling. TCN-E is intra-consist. Multi-unit coupling is a separate RFC.

3. Standards alignment

- IEC 61375-2-3 — Train Communication Network, Ethernet TRDP profile. Defines the message layout and service classes.
- IEEE 802.1Qbv — Time-Aware Shaper. Gates output ports on a scheduled calendar so safety traffic is never blocked behind elephant flows.
- IEEE 802.1Qbu / 802.3br — Frame Preemption. Allows a safety frame to preempt an in-flight best-effort frame.
- IEEE 1588-2019 / 802.1AS — PTP time sync. Sub-microsecond accuracy across the bus.
- IEC 61375-3-4 — Ethernet consist network (bounds the physical topology: backbone switches + stub links to ECUs).

4. Physical topology (reference)

A 3-car light-metro consist per RFC 0003:



Every ECU has exactly one link (1 Gbps copper) to its cab's TSN switch; cab switches link to each other via a fibre pair running the length of the consist. The backbone tolerates a single fibre cut (either A or B still reaches everyone via the intact link); both cut puts the consist into a safe-state stop via ATP time-out.

5. Topic namespace

Topics are hierarchical dotted strings. The first segment is the producer domain, the second is the message class, subsequent segments are qualifiers:

```

osr.train.atp.envelope
osr.train.atp.command
osr.train.brake.apply
osr.train.odom.position
osr.train.odom.speed
osr.train.traction.torque_setpoint
osr.train.traction.current_estimate
osr.train.bms.soc
osr.train.bms.limits
osr.train.bms.fault
osr.train.door.status           (per-door idx: osr.train.door.status.3)
osr.train.door.interlock
osr.train.hvac.status
osr.train.lighting.status
osr.train.pis.display
osr.train.pis.announcement
osr.train.dmi.page
osr.train.dmi.driver_input
osr.train.tcms.consist_status
osr.train.monitors.fire
osr.train.monitors.derailment
osr.train.monitors.vigilance
osr.train.monitors.hot_axle
osr.train.monitors.aux_power
osr.train.event.record         (event-recorder entries)

```

Design choice: dotted hierarchical topics, not numeric IDs. Hierarchy lets subscribers use prefix filters (`osr.train.monitors.*`) without a lookup table. The cost — slightly larger frames — is not the bottleneck on a 1 Gbps link with ~100 topics.

On the wire the topic is interned to a 16-bit ID by the publisher's TCN stack (`osr-tcn` maintains a deterministic string `u16` map populated at boot from a static config file); consumers see both the interned ID and, on-demand, the full string. The intern table is part of the bring-up handshake and must match across every ECU in the consist.

6. Payload encoding

6.1 Rule

Every payload is a fixed-size, endian-normalised (little-endian) byte layout defined as a Rust `#[repr(C)]` struct in a shared payload crate (`osr-tcn-payloads`, or a module under `osr-tcn`). Producers serialise by `core::mem::transmute`-less byte-copying (`bytemuck::bytes_of(&payload)`); consumers the reverse. This gives the safety crates a zero-allocation, zero-parsing publish path.

6.2 Why not protobuf / CBOR / JSON

- Variable-size encodings complicate bounded-latency proofs. TSN Qbv reserves a fixed number of bytes per safety slot; a fixed-size payload is a natural fit.
- Schema evolution happens at the topic level (new topic = new schema); we don't need per-field tagging.
- Test ergonomics: Rust struct == wire layout makes round-trip testing trivial.

- Certification: a safety case that says "this byte at this offset is the MA validity-window" is far easier to audit than one that threads through a protobuf decoder.

6.3 Alignment and endianness

- All integer fields little-endian.
- All struct fields `#[repr(C)]`, explicit padding where needed.
- Maximum payload size per TRDP PD (Process Data) message: 1432 bytes. Most payloads are 16–64 bytes.
- Larger payloads (e.g., CCTV keyframes) go over TRDP MD (Message Data) in a separate channel, not addressed in this RFC.

6.4 Example payload (MA envelope)

```
#[repr(C)]
#[derive(Copy, Clone, Debug, PartialEq, Eq, bytemuck::Pod, bytemuck::Zeroable)]
pub struct AtpEnvelopePayload {
    pub now_ns: u64,
    pub envelope_mmpps: i32,
    pub distance_to_end_mm: i64,
    pub has_known_position: u8,
    pub _pad: [u8; 3],
    pub trigger_reason: u8,
    pub _pad2: [u8; 7],
}
// sizeof == 32
```

`bytemuck::Pod + Zeroable` give safe transmute semantics without `unsafe`. The workspace-level `forbid(unsafe_code)` is preserved.

7. Traffic classes

Three classes, mapped to 802.1Q priority codes per Qbv gate schedule:

Class	PCP	Slot type	Example topics
Safety (S)	7	Guaranteed slot every 1 ms	<code>atp.envelope</code> , <code>brake.apply</code> , <code>odom.position</code> , <code>monitors.*</code>
Control (C)	5	Guaranteed slot every 10 ms	<code>traction.torque_setpoint</code> , <code>bms.limits</code> , <code>door.status</code>
App (A)	3	Best-effort, preemptible	<code>hvac.status</code> , <code>pis.display</code> , <code>dmi.page</code> , <code>event.record</code>

A safety-class publish uses a pre-allocated buffer and transmits in its next Qbv slot — bounded latency < 1 ms. The publisher's Rust API must not spin-wait; if the TCN stack's internal queue is full (which should be unreachable under a correctly sized bus), the publish returns an error and the caller records a `TcnError::QueueFull` event.

8. Time synchronisation

- Grandmaster: the primary T-ECU/S in cab A. Runs a PPS-locked oscillator if available; otherwise free-runs its TCXO.
- Secondary: the primary T-ECU/S in cab B. Takes over on grandmaster timeout (standard IEEE 1588 BMC algorithm).
- Slaves: every other ECU. Maintain $\leq 1 \mu\text{s}$ offset to grandmaster for safety-tier timestamps.

The `now_ns` parameter that every evaluator (ATP, ATO, BMS, etc.) consumes is read from the local PTP-disciplined clock. Callers must not mix PTP time with monotonic `std::time::Instant` on the publish path.

9. Crate API

9.1 Publisher side

```
pub struct Tcn {
    // implementation detail
}

impl Tcn {
    pub fn publish<P: TcnPayload>(
        &self,
        topic: TopicId,
        payload: &P,
    ) -> Result<(), TcnError>;
}

pub trait TcnPayload: bytemuck::Pod {
    const CLASS: TrafficClass;
```

```

    const SIZE: usize;
}

```

9.2 Subscriber side

Two modes:

- Blocking read (unit tests, non-hot-path): `recv(topic) -> Option<&P>`.
- Callback registration (hot path, safety crates): at init time, subscribe with a `fn (&P)` that the stack invokes on receipt. The callback runs in the ECU's deterministic tick, not in an interrupt handler.

9.3 Mock transport for tests

```

pub struct MockTcn { /* in-memory bag of (topic, payload) tuples */ }
impl MockTcn {
    pub fn publish<P>(&mut self, topic: TopicId, payload: &P);
    pub fn drain<P>(&mut self, topic: TopicId) -> Vec<P>;
}

```

The mock drives the real crate API — every existing onboard crate can be fed into it without change. This is how `osr-sim`'s shadow stack will exercise the full publish/subscribe chain in the simulator.

10. Scope of v1 implementation

The first cut ships:

- 1 Topic ID interning (static config file `TopicId` enum).
- 2 Payload trait + derive macro (or explicit `bytemuck::Pod` impls for now — macro is follow-up).
- 3 In-memory mock transport with traffic-class awareness.
- 4 Proptest harness verifying:
 - Round-trip encode/decode produces byte-identical output.
 - Safety-class messages are delivered in publish order.
 - Class-A messages may be dropped under backpressure; class-S never drops.

Out of scope for v1:

- Real TSN hardware driver — needs a kernel driver story (DPDK vs `AF_XDP` vs stock socket). Separate RFC.
- PTP client — use a userspace `ptp4l` + Linux PHC until we write a bare-metal one for Hubris.
- `osr-tcn-payloads` split out of `osr-tcn` — reasonable at scale, premature now.

11. Rollout

Phase	Deliverable	Dependencies
v0	This RFC ratified	—
v1	<code>osr-tcn</code> crate with mock transport + topic registry + proptests	—
v1.5	Real network transport (<code>UdpTcn</code>) on commodity UDP — drop-in API replacement for <code>MockTcn</code> , loopback round-trip tested; the simplest thing that works for a multi-host bench (done 2026-04-22)	v1
v2	Integration into <code>osr-sim</code> shadow: publish/subscribe replaces direct field access	v1
v3	Real TSN driver (Linux <code>AF_XDP</code> first, Hubris bare-metal later)	reference hardware RFC
v4	PTP client + grandmaster fail-over	v3

v1.5 is deliberately a small step past v1. It preserves the `MockTcn` contract so downstream crates never see the transport; what changes is only whether bytes travel through a `BTreeMap` or through the kernel UDP stack. This gives operators a working multi-host bench without pulling in DPDK / `AF_XDP` complexity, which stays in v3 where the hardware story is fixed.

12. Relationship to `osr-crypto` and `osr-t2g`

- `osr-crypto` (not yet scaffolded) will provide HMAC-signing of safety-class payloads for defense-in-depth against a compromised ECU. Sign/verify wraps the `Pod` layout; wire format stays fixed-size.

- `osr-t2g` sits outside TCN and takes its inputs from specific TCN topics. TCN frames do not leave the consist; ground-bound telemetry is re-serialised (likely in a bandwidth-efficient encoding like CBOR) before transmission.

13. Open questions

- 1 Topic ID collisions across firmware versions. The intern map must be version-stable within a consist or the bus silently routes to the wrong subscriber. Solution sketch: the intern map is derived from a canonical SHA-256 of the sorted topic list; consists bootstrap-verify that all ECUs agree.
- 2 Payload schema evolution. Today: new topic = new schema. At some point we'll want to extend an existing payload backwards-compatibly. Deferred until a concrete need arises.
- 3 Multicast vs unicast. Most TRDP PD is multicast; control flows (e.g., `door . command` from a central door controller to specific doors) could be unicast. Current plan: everything multicast with subscriber filtering, simplifies the stack.
- 4 No-std on Hubris. `bytemuck : Pod` is `no_std`-friendly; the current `std` usage in `osr-tcn` would need the same treatment the other SIL-4 crates receive in the `no_std` migration (RFC 0005 §7 ongoing).

14. Done criteria

- ☐ RFC ratified by reviewer(s).
- ☐ v1 crate lands with mock transport, topic registry, and ≥ 6 proptest-verified properties.
- ☐ Existing SIL-4 crates compile against the `TcnPayload` trait for at least one topic each (e.g., `osr-atp` exports `AtpEnvelopePayload`).

v1 implementation follows this session after the RFC is stable.

RFC 0007 — Hardware Reference Designs

Status: Draft v1.1 — planning only, no schematics ship with this RFC Date: 2026-04-22 (v1.1 same day — SoC palette restricted to Raspberry Pi + Radxa) Depends on: [RFC 0005 SBC Software Architecture](#), [RFC 0006 os r - t cn design](#)

1. Summary

This RFC fixes the hardware reference designs for the four physical host classes on which OpenSourceRail software runs — T-ECU/S, T-ECU/A, W-SBC, S-SBC — before a schematic is drawn. It picks the SoC, the module form factor, the baseboard constraints, and the per-board BOM envelope. It does not commit the schematics; those land as a separate editorial PR once the decisions below are ratified.

The O-SRV class is commodity server hardware — any x86-64 or ARM64 Linux box that meets the operator's procurement rules. It is out of scope for this RFC.

Why a planning RFC first: hardware decisions are sticky. A T-ECU/S reference design outlives three software refreshes and four deployment generations, because the PCB, the environmental compliance testing, and the depot spares-and-repairs pipeline all calcify around the choice. Getting the class selection right costs a session of writing; getting it wrong costs an order of magnitude more in re-compliance later.

Three non-negotiable constraints shape every decision below:

- 1 Local manufacturability. Every board must be producible on a 4-layer PCB with 0.15 mm trace/space, 0.3 mm vias, standard SMT (0402 minimum passive), and through-hole where rail vibration/shock demands it. No micro-vias, no blind/buried vias, no 0201 passives, no exotic stackups. These are routine tier-2 board shops across MENA, sub-Saharan Africa, South and Southeast Asia, and Latin America — the target deployment footprint.
- 2 Commodity System-on-Module (SoM) first. Wherever a ready-made compute module with the right environmental rating exists, we adopt it and design only the baseboard. This reduces the domestic engineering burden to a carrier board plus enclosure — a fraction of the work of a from-scratch SoC design.
- 3 Two-vendor palette — Raspberry Pi and Radxa only. Raspberry Pi supplies the consumer-grade CM5 module and the RP2350 safety MCU; Radxa supplies the industrial-temperature CM5 (RK3588S) for pole-mount use. Both vendors are available globally at predictable prices, document their modules openly, and ship with mainline Linux support. This restriction deliberately excludes NXP / ST / Rockchip-direct / MilkV / StarFive bring-up paths — not because those are technically unfit, but because locking on two well-stocked vendors simplifies the domestic procurement and spares story for the operator.

2. Non-goals

- Not a full hardware BOM. This RFC specifies reference choices; the per-deployment BOM is owned by the operator and tuned to local availability.
- Not a certification strategy. EN 50155 / IEC 61373 / EN 50121-3-2 compliance testing is in scope for the safety case (os r - safety - case), not this document.
- Not a redundancy topology. The logical redundancy (dual T-ECU/S, 3-node W-SBC consensus cluster) is already fixed by [RFC 0001](#) and [RFC 0005 §8](#). This RFC is about one board of each class; stitching them into a consist or a wayside cabinet is deployment-level.
- Not a hard-real-time bus design. TSN / PTP details are [RFC 0006](#). This RFC references those choices but does not re-litigate them.

3. Common platform choices

These apply to every class below unless overridden.

Aspect	Choice	Rationale
PCB	4-layer FR-4, 0.15 mm trace/space, 0.3 mm min via	Routine at tier-2 fabs worldwide.
Passives	0402 minimum	SMT-line friendly, hand-rework possible.
Connectors	M12 for field I/O, board-to-board for modules	M12 is the rail industry standard for trackside and consist wiring (EN 50467).

Aspect	Choice	Rationale
Isolation	1.5 kV galvanic on every external I/O	EN 50155 demands it for rolling stock; W-SBC gains the same treatment for consistency.
Power input	24 V DC nominal, 16.8–30 V range	EN 50155 wide-input class. On trains this comes from the battery bus via <code>osr-aux-power</code> . At wayside sites it comes from the site supply.
Power sequencing	TI LM5164 buck 5 V per-rail LDO	Single widely-stocked part, no exotic controllers.
Firmware storage	eMMC (on-module) + external SPI-NOR for boot	eMMC is the reference OS filesystem; SPI-NOR holds the measured-boot root.
Trust anchor	Microchip ATECC608 or NXP A1006 SE	Either is commodity, EN 50155 rated, well-documented. ATECC608 is our baseline because ed25519 support is native.

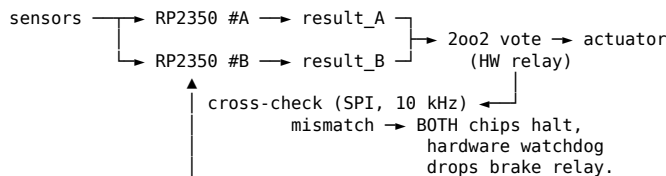
4. Class T-ECU/S – Train Safety Kernel

Role: Hosts the SIL-4 crates – `osr-atp`, `osr-odometry`, `osr-traction` (MCU bring-up), `osr-bms`, `osr-brake`, `osr-door-control`, `osr-vigilance`, `osr-fire-safety`, `osr-derailment`. One primary + one hot-standby per consist.

Environment: EN 50155 OT4 (–40...+85 °C), IEC 61373 Cat 1 Class B vibration + shock, EN 50121-3-2 EMC.

4.1 Safety architecture – 2oo2 composite fail-safe

The safety case is anchored in diverse redundancy between two independent microcontrollers, not in a single silicon vendor's lockstep claim. Each T-ECU/S board carries two Raspberry Pi RP2350 MCUs wired in a 2-out-of-2 voting arrangement:



- Both chips receive the same raw sensor inputs on electrically separate pins.
- Both chips execute identical Rust `no_std` code (Hubris or a bespoke bounded scheduler).
- Each tick they exchange their computed safety decisions over a dedicated SPI link and compare byte-for-byte.
- The final actuator-drive output is produced by a hardware AND of both chips' drive lines through an external 2oo2 relay network. A single chip can only permit motion; it cannot command it alone.
- On disagreement, both chips set themselves to fail-safe (drive lines open) and a downstream hardware watchdog asserts the emergency brake relay directly.

This is the "composite fail-safe" pattern that EN 50129 appendix D blesses for SIL-4. Because the two chips are physically identical (not lockstep within one die), a single-event upset or silicon defect affects at most one of them – the other still votes the actuator into the safe state. Identical-software redundancy handles random hardware faults; RFC 0004's Kani + proptest coverage handles systematic software faults.

4.2 SoC picks

- Safety MCU (×2 per board): Raspberry Pi RP2350. Dual Cortex-M33 plus dual Hazard3 RISC-V in the same die (software selects one pair at boot), 520 KB SRAM, 150 MHz. Arm TrustZone-M for secure boot, hardware random number generator, and a single signed ROM loader. Published datasheet, globally available through the Raspberry Pi channel and every major distributor. Per-chip price ≈ €1 in 1k volume.

Why RP2350 for safety – fit against the mission:

- Tiny, stable silicon with a 5-year announced support window from Raspberry Pi. Matches the ≥ 20-year service life of the rolling stock better than most MCU vendors' SKUs.
- Dual instruction-set option (Arm M33 or RISC-V Hazard3) means a second diverse implementation is available by re-selecting the ISA pair at boot – a future v2 refinement of the 2oo2 pattern can go to true ISA-diverse 2oo2 without changing the BOM.
- Available in QFN-56 (0.4 mm pitch) – hand-solderable under a loupe, trivial on any SMT line.
- Application processor (×1 per board): Raspberry Pi CM5. Handles TCN-E, logging, OTA, diagnostics, and hosts the event recorder. Runs mainline Linux. The app processor is never on the safety path – it subscribes to a one-way SPI feed from the safety MCUs and cannot preempt their decisions.

4.3 Peripherals (baseboard)

Peripheral	Qty	Purpose
CAN-FD	4	Door, brake, traction, BMS buses (two CAN-FDs per RP2350, identical)
TSN Ethernet (on CM5 + external PHY KSZ9031)	2	TCN-E (A + B, redundant paths per RFC 0006)
Isolated DI	8	Emergency plunger, cab door switches, depot enable, deadman handle (each routed to both RP2350s)
Isolated DO	4	Emergency brake relay, traction cut relay, park brake solenoid, fire-suppression trigger (via 2oo2 relay network)
Tachometer inputs	2	Wheel encoders (quadrature, 5–24 V, each read by both RP2350s)
IMU	1	Bosch BMI088 6-axis (derailment + odometry) – read by both RP2350s via separate SPI chip-selects
GNSS	1	u-blox NEO-F10N over UART to CM5
Balise reader	1	PN5180 NFC front-end over SPI (passive HF balise per RFC 0003)
Thermistors	4	Battery bay, traction bay, HVAC plenum, enclosure ambient
ATECC608B SE	2	One per RP2350 – trust anchor for signed firmware + cross-check keys

Every input reaches both RP2350s on physically separate pins; every output is AND-gated through the external 2oo2 relay stage.

4.4 Target BOM (volume 100+)

Item	Unit cost (€)	Per board
RP2350 × 2	1.10	2.20
Raspberry Pi CM5 (4 GB)	85	85
Baseboard (4-layer, 160 × 100 mm)	40	40
CAN-FD transceivers × 4	3	12
Ethernet PHYs + mags	12	12
Isolated DI / DO (ADUM + discrete drivers)	35	35
BMI088 IMU + NEO-F10N GNSS + PN5180	45	45
ATECC608B × 2, passives, connectors, power	45	45
Total	—	~€280

Two per consist = €560 per trainset for the safety-kernel layer. Well below the €900 previously budgeted on the NXP-centric plan; the RPi-family modules drive the cost down.

4.5 Form factor

160 × 100 mm Eurocard, DIN-rail mountable in the cab cabinet. Conduction-cooled to the DIN rail; no fans. Conformal coated (MG Chemicals 419) before integration. Sealed to IP54.

5. Class T-ECU/A – Train Application

Role: Hosts SIL-2 / SIL-0 app-tier crates – *osr-ato*, *osr-tcms*, *osr-dmi*, *osr-pis-onboard*, *osr-hvac*, *osr-lighting*, *osr-aux-power*, *osr-event-recorder*, *osr-regen*, *osr-hot-axle*, *osr-cbm-onboard*, *osr-t2g*.

Environment: EN 50155 OT4, single-redundant (two units per consist is standard but a single unit is tolerable for non-safety functions).

SoC baseline: Raspberry Pi Compute Module 5 (BCM2712, Cortex-A76 4-core, 4–16 GB LPDDR4X, eMMC on-module). RPi CM-class is the obvious commodity module – wide availability, mature Linux support, global stock, SO-DIMM-style connector.

Alternative drop-in: Radxa CM5 (RK3588S – Cortex-A76 + A55 big.LITTLE on the same SO-DIMM footprint). Radxa's module is pin-compatible with our baseboard; the only code delta is the device-tree blob. Operators can source whichever is locally available at procurement time.

Peripherals (baseboard):

Peripheral	Qty	Purpose
TSN Ethernet	2	TCN-E A/B (redundant)
CAN-FD	2	HVAC + lighting buses
USB-C (host + device)	2	Depot console + diag loader
HDMI	1	DMI touchscreen output
USB 2.0	2	DMI touch, PIS display
M.2 2280 NVMe	1	Event recorder ring buffer + OTA staging

Peripheral	Qty	Purpose
M.2 2230 Cat.22 5G	1	TRG-1 (primary ground radio)
LoRa SX1276 on SPI	1	TRG-2 (backup ground radio)
CCA-B2B to T-ECU/S	1	Mailbox bridge to the safety side
ATECC608 SE	1	Trust anchor

Target BOM: €220 per board (CM5 module ≈ €85, baseboard ≈ €75, radios ≈ €60 volume).

Form factor: 160 × 100 mm Eurocard, same DIN mount as T-ECU/S so a cabinet can accept either without re-designing the slot.

5.5 Class T-OBS — Train Obstacle-Detection ECU

Role: Hosts the SIL-4 `osr-obstacle-detect` evaluator per [RFC 0015 §5.2](#). Fuses the nose-cone sensor suite (ultrasonic × 4, LIDAR, mmWave radar, stereo camera) into an `ObstacleVerdict` bus message consumed by the T-ECU/S 2oo2 AND-gate relay stage.

Two modules per trainset — one at each nose. Only the leading module is active for a given direction of travel; the trailing module self-tests continuously and publishes an `Inactive` status.

Environment: EN 50155 OT4, dual-redundant (both nose-end modules duplicate the safety evaluator; either can drive the brake chain).

Architecture: mirrors T-ECU/S: two Raspberry Pi RP2350 safety MCUs running the `osr-obstacle-detect` evaluator in a 2oo2 cross-check plus a Raspberry Pi CM5 application processor for sensor fusion, classifier inference, and the non-safety data path. The safety-critical verdict is produced inside the RP2350 pair; the CM5 supplies pre-processed detection lists but cannot emit a `Clear` on its own.

Peripherals (baseboard):

Peripheral	Qty	Purpose
Ultrasonic transceivers (40 kHz, 200 μs drive, 10-bit ADC)	4	Close-range safety belt per RFC 0015 §5.1
CAN-FD to mmWave radar (TI AWR1843 or eq.)	1	All-weather long-range
1000BASE-T Ethernet to LIDAR (Livox HAP / Tele / RoboSense M1)	1	Mid-range 3D primary
MIPI-CSI to stereo camera pair (IMX477 / IMX219)	2	Classification
TSN Ethernet	2	TCN-E A/B back to T-ECU/S + T-ECU/A
CCA-B2B to T-ECU/S	1	Direct brake-demand line (2oo2 AND-gate stage)
ATECC608 SE	1	Trust anchor (shared with T-ECU/S)

Power budget: ~18 W under full load (LIDAR dominates at ~12 W); 24 V DC in with the same EN 50155 isolation as the rest of the cab.

Target BOM: €780 per board module, dominated by the LIDAR unit (~€500 for a Livox-class sensor in volume) + mmWave radar (~€500 / €250 volume) + stereo camera pair (~€150) + baseboard

- 2× RP2350 + CM5 (~€220 per the T-ECU/A BOM). Trainset total (2 modules): ~€1 560 per consist, well below the ~€140 k cab capex removed.

Form factor: 180 × 120 mm Eurocard — slightly larger than T-ECU/S to accommodate the four ultrasonic analog front-ends. Mounted behind the RF-transparent nose panel; cooling is ambient-air natural convection.

Detailed v2 spec: scaffold at [hardware/t-obs/schematics/v2-spec/](#) follows the T-ECU/S v2 template (block diagram, power budget, pinouts, connector table, safety-net routing rules). KiCad capture is v3 per §11 rollout.

6. Class W-SBC — Wayside

Role: Hosts `osr-consensus`, `osr-interlocking`, `osr-wayside-points`, `osr-level-crossing`, `osr-hot-axle-wayside`, `osr-balise` (reader side), `osr-energy-site`. Deployed in pole-mounted IP67 cabinets along the corridor and at each junction.

Environment: IEC 60529 IP67, EN 50121-4 EMC, −40...+70 °C, pole-mount vibration profile.

SoC baseline: Radxa CM5 (RK3588S — Cortex-A76 4-core + Cortex-A55 4-core big.LITTLE, 4/8/16 GB LPDDR4X, eMMC on-module). Radxa offers the industrial-temperature variant of the CM5 (−20...+85 °C grade) which matches the IP67-cabinet thermal envelope; the consumer-grade module is the drop-in for mild-climate deployments.

Rationale:

- Same SO-DIMM connector as the Raspberry Pi CM5, so a baseboard built around Radxa CM5 also accepts RPi CM5 (with the device tree recompiled). One carrier design, two module options.
- RK3588S is widely second-sourced (Rockchip is one of the few merchant SoC vendors with strong domestic semiconductor availability across the target regions).
- RK3588S ships with mainline Linux kernel support; Debian and Yocto both build without vendor forks.
- For safety-role wayside (interlocking + consensus), the A55 little cluster is pinned to the safety partition and runs Hubris or seL4; the A76 big cluster carries non-safety services (telemetry, diagnostics) under PREEMPT_RT Linux. The two clusters share the memory controller but are scheduled on separate cores with cache-partitioning hints from the kernel.
- For non-safety sites (balise reader, energy site) the same module runs Debian straight-through.

Why not the RPi CM5 for wayside too: Raspberry Pi publishes a narrower industrial-temp range than Radxa (commercial 0...+50 °C vs Radxa's industrial +85 °C upper). Pole-mount cabinets in MENA / sub-Saharan routes routinely hit cabinet internals above +60 °C at midday; Radxa's variant has the margin.

Peripherals (baseboard, one SKU fits all wayside roles):

Peripheral	Qty	Purpose
TSN Ethernet	2	WAY-E A/B (redundant)
RS-485	4	Switch-machine motor drive, level-crossing barrier, HABD sensor bus, balise reader I/F
Isolated DI	8	Switch end-of-travel sensors (A+B), crossing strike detectors, field-fault inputs
Isolated DO	4	Motor directional contactors, crossing barrier lift/drop
4-wire PT100	4	Hot-axle IR temperature measurement
802.11ax AP	1	Depot maintenance console only; disabled in revenue service by default
Dual 24 V DC-in	2	Primary + backup site supplies
ATECC608 SE	1	Trust anchor

The same baseboard populates selectively: a switch site leaves the PT100s unpopulated; a HABD site leaves the switch-driver DOs unpopulated. One SKU, one spares stock, one support burden — the simplicity bet pays off in the depot.

Target BOM: €280 per board, €550 with IP67 enclosure and mounting hardware.

Form factor: 120 × 80 mm, inside a DIN-rail-mountable aluminium extrusion with IP67 M12 penetrations. Conduction-cooled to the enclosure.

7. Class S-SBC — Station / Depot

Role: Hosts `osr-psd`, `osr-pis-station`, `osr-station-scada`, `osr-afc`, `osr-tvm`. Deployed indoors; no rail-industry environmental rating required.

SoC baseline: Raspberry Pi CM5 on a commodity carrier (e.g. Waveshare CM5-IO). No custom baseboard needed — the station enclosure carries the I/O. This explicitly deviates from the other classes; indoor environments don't require the custom engineering.

Peripherals (commodity carrier):

- 2 × Gigabit Ethernet (one to station LAN, one to PSD/gate network)
- 4 × USB 3.0 for QR scanner, NFC reader, receipt printer, TVM bill acceptor
- HDMI for the passenger-facing display
- USB-C console
- ATECC608 on a small add-on board (I²C)

Target BOM: €130 per complete station-role SBC, including the CM5 and the add-on SE board.

8. Manufacturing envelope

Every reference PCB respects these constraints so that a tier-2 fab in any target region can produce it:

Constraint	Value	Source
Layer count	4	Lowest yield-loss tier.
Trace / space	0.15 mm / 0.15 mm	Routine at tier-2 fabs (6 mil).
Minimum via	0.3 mm drilled, 0.55 mm annular ring	No micro-vias.

Constraint	Value	Source
Aspect ratio	≤ 8:1	Standard.
Min hole-to-pad	0.2 mm	Standard.
Passive minimum	0402	Hand-reworkable, SMT-line friendly.
BGA	≤ 0.8 mm pitch, escape on 4 layers	CM5 module uses 0.65 mm BGA internally but exposes a SO-DIMM connector – no direct BGA escape needed on our baseboards.
SMT line min width	10 mm component edge-to-edge	Fits on any SMT line with ≥ 0.5 m conveyor.
Conformal coating	Acrylic (MG Chemicals 419)	Single-can hand coating acceptable for low volume; dip-coat line at high volume.

Nothing on the BOM requires any restricted-export component. Every part is a catalog item at Mouser/Digi-Key/LCSC. Local second-source is identified for every component in the Phase 3 procurement RFC (out of scope for this doc).

9. Reliability envelope

Class	MTBF target	MDBF target	Notes
T-ECU/S	≥ 200 000 h	≥ 500 000 km	Safety role – bigger than the rest. Figure taken from EN 50126 for on-vehicle SIL-4 nodes.
T-ECU/A	≥ 100 000 h	≥ 200 000 km	Single-redundant; failure is a comfort / diagnostic issue, not a safety issue.
W-SBC	≥ 150 000 h	n/a	Ten years of unattended pole-mount operation is the minimum acceptable.
S-SBC	≥ 50 000 h	n/a	Indoor environment; consumer-grade components acceptable.

Derating rules per Telcordia SR-332 Issue 4: thermal derate all active devices to ≤ 80 % of rated junction temperature at worst-case ambient plus 10 °C margin.

10. Pitfalls and decisions

These were flagged during planning – they're recorded here so the implementation session doesn't rediscover them.

- Two RP2350s on T-ECU/S, not one safety MCU with internal lockstep. Rule 3.3 of RFC 0005 requires SIL-4 code not to depend on SIL-2 code at runtime. Internal lockstep on a single die is a single-vendor claim; two physically-separate RP2350s cross-checking over an external SPI link are independently verifiable. The €2.20 BOM cost for two chips is trivial; the cross-domain auditability is worth orders of magnitude more in the safety case. Future v2 can move one of the two RP2350s to its Hazard3 RISC-V cores for genuine ISA-diverse redundancy without a BOM change.
- RPi CM5 for consumer-grade hosts, Radxa CM5 for industrial-grade hosts. T-ECU/A and S-SBC use RPi CM5; W-SBC uses Radxa CM5 for its wider industrial temp range. Both are SO-DIMM compatible, so a baseboard designed for either accepts the other with a device-tree swap. Operators in mild climates can single-source RPi CM5; operators in hot climates spec Radxa CM5 everywhere if they'd rather single-source the other way. The palette is two vendors only, by policy.
- No custom ASICs. An OpenSourceRail-branded silicon die is explicitly out of scope for v1 and v2. The cost is borne by accepting some commodity-module markup; the benefit is zero ASIC design NRE and a three-year faster path to first deployment.
- No NXP / ST / MilkV / StarFive on the BOM. Earlier drafts of this RFC used NXP S32K344 + i.MX 93 for the safety board and MilkV Jupiter for wayside. The two-vendor-only restriction (§3 constraint 3) replaced those with the RPi + Radxa palette above. The previous choices were technically sound; the new palette trades a small amount of safety-MCU certifiability for dramatic procurement simplicity and a tighter alignment with the two vendors' multi-year support commitments.
- Conformal coating is mandatory. Rail environments are humid, dusty, and occasionally sprayed. Every production board ships coated; lab samples for development use are coated before integration testing, not after, to catch solder-mask defects early.
- No fans. Fan bearings are a top-quartile failure mode in trackside equipment. Every reference design is conduction-cooled to its enclosure. Thermal headroom is the binding constraint, not CPU performance.
- Isolation on every external I/O. Digital I/O that leaves the board crosses a galvanic barrier. This is an EMC + safety bet: one lightning surge at the wayside must not take out a pole-mount W-SBC via an unpowered sensor line.

11. Rollout

Phase	Deliverable	Dependencies
v0	This RFC ratified	—
v1	Bring-up board for each class on vendor dev kits (RPI CM5 dev kit, Radxa CM5 dev kit, Raspberry Pi Pico 2 — which carries the RP2350) — validate software boots and all peripherals enumerate. Runbook procedures committed at docs/hardware/bring-up/ (done 2026-04-22)	—
v2 (spec)	Board-level specifications for every host class: T-ECU/S at hardware/t-ecu-s/schematics/v2-spec/ (done 2026-04-22), T-OBS at hardware/t-obs/schematics/v2-spec/ (done 2026-04-22), T-ECU/A at hardware/t-ecu-a/schematics/v2-spec/ , W-SBC at hardware/w-sbc/schematics/v2-spec/ , S-SBC at hardware/s-sbc/schematics/v2-spec/ (all done 2026-04-23). Each covers block diagram, power budget, connector tables, and (for SIL-4 boards) safety-nets. KiCad capture + gerbers + BOM are v2.1 per-vendor deliverables under CERN-OHL-S v2.	v1
v3	Compliance-test campaign for T-ECU/S (EN 50155 OT4, IEC 61373 Cat 1B, EN 50121-3-2) on one v2 prototype	v2
v4	Quantity-100 procurement package published — canonical BOM with Mouser / Digi-Key / LCSC lines, pick-and-place + stencil files, assembly drawings	v2
v5	First-article approval by the Samawah reference operator (or equivalent pilot)	v4, RFC 0003

This RFC's v0 deliverable is only the plan. Every later phase is a dedicated piece of work owned by the hardware working group. The session that implements v1 starts by ordering three dev kits — Raspberry Pi Pico 2 (RP2350), Raspberry Pi CM5 IO Board, and Radxa CM5 IO Board — and boots the relevant Rust crates on each: the onboard safety stack on a pair of Pico 2 boards, the onboard app stack on RPi CM5, and `osr-consensus` + `osr-interlocking` + `osr-wayside-points` on Radxa CM5. The sim already runs every crate without hardware, so the bring-up is a migration, not a redesign.

12. Relationship to existing work

- Software stays unchanged. Every crate in `crates/` is designed to be hardware-agnostic Rust. The reference designs here are the first physical host; a second or third SoC family is additive, not breaking, so long as it exposes the same TCN-E / CAN-FD / GPIO surface specified in §§4–7.
- `hardware/` directory. v0 of this RFC seeds the top-level directory structure — one subdirectory per class, each with a `README.md` stub and a placeholder for `schematics` / `gerbers` / `BOM`. Skeleton lives in this commit; actual schematics land in v2.
- Licensing. All hardware designs under CERN-OHL-S v2 per [ARCHITECTURE.md §9](#). Documentation (this RFC) under CC-BY-SA 4.0. Firmware ports of existing crates keep their Apache-2.0 origin.

13. Open questions

- 1 Does the 2002 RP2350 pattern need an independent type-approval? Unlike internal lockstep silicon, the 2002 claim is a system-level argument that lives in the safety case. Our line is that an operator runs per-deployment type approval on the full T-ECU/S board anyway; the silicon-level certificate becomes a building block of the board certificate. Open: does this match EN 50129 appendix D's expectations for diverse redundancy? Resolve during v3 compliance-test planning.
- 2 ISA-diverse 2002 — when does it land? RP2350 can boot either Arm M33 or Hazard3 RISC-V. Running the two chips on different ISAs (identical Rust source, two different codegen back-ends) strengthens the diversity argument against common-mode compiler bugs. Not a v1 goal; track for v2 after the baseline 2002 is in service.
- 3 Which balise protocol? RFC 0003 proposed passive 27 MHz transponders; the safety board's footprint reserves a PN5180 NFC front-end (13.56 MHz) for now. Committing the passive-balise frequency band is a deployment-level choice — different operators have different RFID-spectrum allocations.
- 4 How does the energy-site W-SBC physically interface to the PV inverter and BESS? Modbus-TCP over an isolated Ethernet segment is the RFC 0002 v1 assumption; the same 4-port wayside baseboard already has it.
- 5 Repairability requirement. Should the target markets demand a specific MTTR (mean time to repair) at the field level? A 5-minute module swap is the baseline the DIN-rail + M12 strategy enables; whether that's sufficient for a country's availability SLA is a per-deployment conversation.

14. Done criteria

- [x] Every class's SoC and SoM pick is justified (§§4–7)
- [x] One baseboard per class, with a peripheral count (§§4–7)

- [x] Manufacturing envelope fits a tier-2 fab (§8)
- [x] Reliability targets allocated per class (§9)
- [x] Pitfalls and alternatives captured (§10)
- [x] Rollout ordered and scoped (§11)
- [x] Relationship to existing software + licensing named (§12)

The next session picks up at v1 — bring-up boards on vendor carriers. The expected sequence is (a) order the three dev kits, (b) bring up the Rust stack on each, (c) validate peripherals against this RFC's §§4–7, (d) file v1 deliverable as a hardware/folder update.

RFC 0008 — Baseline Rolling-Stock Reference Design

Status: Draft — planning only, no mechanical drawings ship with this RFC Date: 2026-04-22 Depends on: [RFC 0003 Samawah Reference Deployment](#), [RFC 0007 Hardware Reference Designs](#)

1. Summary

OpenSourceRail commits to one modular train architecture scaled into five rolling-stock families — one per ridership band — from which every deployment picks exactly one per line. No bespoke consists; every country that adopts the stack chooses the closest family and builds to that drawing. This RFC fixes the engineering envelope (dimensions, masses, axle arrangement, bogie and brake architecture, cab, crashworthiness, fire, thermal, acoustic) for each family.

The families are named in [lib/templates/rolling-stock.toml](#). This RFC promotes that schema into a committed engineering envelope.

Each car is a self-contained driverless unit: one powered bogie, one trailer bogie, under-seat sodium-ion battery, low-floor centre boarding zone, and its own onboard control stack. Multi-car consists repeat that module rather than introducing a different traction architecture.

Family	Cars	Length	Tare	Seats	AW2 nominal capacity	AW3 crush capacity	Peak ridership band (pphpd)	Onboard battery	Max speed
urban-shuttle-1car	1	21 m	34 t	20	90	110	< 2 500	120 kWh	19 m/s (70 km/h)
tram-2car	2	39 m	68 t	40	210	260	< 5 000	240 kWh	19 m/s (70 km/h)
light-metro-3car	3	57 m	102 t	60	330	420	5 000–10 000	360 kWh	25 m/s (90 km/h) — Samawah reference
metro-4car	4	75 m	136 t	80	440	560	10 000–20 000	480 kWh	25 m/s (90 km/h)
metro-6car	6	111 m	204 t	120	660	840	20 000–35 000	720 kWh	28 m/s (100 km/h)

Pphpd = passengers per hour per direction at peak, planning-grade. AW2 is the normal peak planning load used for capacity and farebox calculations. AW3 is a short-duration crush-load reference for structure, suspension, braking, and emergency-egress checks.

2. Non-goals

- Not a bill of materials. This RFC fixes the envelope; the BOM for a specific family is v2 output of the hardware working group.
- Not a homologation pack. EN 14363 dynamic tests, EN 15227 crashworthiness, EN 45545 fire, ISO 3095 noise, EN 12299 comfort — every family will need type-approval evidence for a specific deployment. This RFC states the design targets; the evidence campaigns happen per-deployment.
- Not intercity or freight. Explicitly out of scope per [ARCHITECTURE.md §1](#). The four families span low-pphpd tram up to large metro trunk; regional / HSR / freight is a different problem.
- Not a standards body. We reference EN / IEC / UIC / ISO standards; we do not publish new ones.
- Not a customisation catalogue. Interior colour, seat fabric, route-branded livery are a per-deployment choice. This RFC fixes the structural and electrical architecture only. Vendor choice for the COTS interior equipment (windows, HVAC, lighting, PIS screens, seats, grab poles, intercom) is documented separately as an envelope-reservation cookbook in [hardware/trainset-interiors/cots-catalogue.md](#) — the car body reserves the volume + bolt pattern + power budget; the specific vendor is interchangeable per deployment.

3. Unified architecture — every family

Five rolling-stock families, but one architecture across all of them. The scaling is car count, platform length, and capacity; the architectural decisions below are shared. Commonality is the bet: one spares pool, one maintainer-training syllabus, one CAD reuse, one supplier-qualification set.

3.1 Mechanical

Aspect	Choice	Rationale
Car body	Welded carbon-steel underframe + light spaceframe, with bonded/bolted composite side and roof panels	Keeps the primary structure to cut/bend/weld operations available in ordinary rolling-stock, bus, truck, and ship repair shops. Composite panels carry weather, aerodynamics, insulation, and finish, not crash loads.
Suspension	Chevron rubber-metal primary; twin-bellows air-spring secondary with pneumatic levelling valves rated 55 °C continuous	Per RFC 0022 §5 . Chevron primary eliminates hydraulic-damper creep failure at 50 °C ambient; twin-bellows air-spring secondary holds the door-zone floor within ± 5 mm across empty-to-AW3.
Wheel	Monobloc, 760 mm new / 680 mm worn	Per RFC 0022 §3 — widest global wheel-forging catalogue (Lucchini, Valdunes, CAF, plus tier-2 vendors per RFC 0022 §10). 40 mm re-profiling budget over life on depot lathe.
Gauge	1 435 mm standard, 1 000 mm metre-gauge variant	Both produced from the same body and bogie frame; only the wheelset axle length changes. See RFC 0009 §4 .
Wheel–rail profile pairing	S1002 wheel on UIC60 rail (standard gauge) or Ri60 wheel on Ri60 rail (street-tram gauge)	Standard off-the-shelf profiles; no proprietary geometry.
Couplers	Scharfenberg Type 10 automatic + e-coupler for 24 V / CAN-FD / Ethernet	Standard metro coupler; allows rescue of a failed train by any neighbour consist in the fleet. Brownfield deployments: SA-3 (Soviet automatic) + screw/buffer couplers from recovered legacy stock are explicitly NOT retrofitted onto OSR consists — see RFC 0027 §5.3 . Recovered SA-3 drawgear goes to the resale stream.
Articulation	Semi-permanent couplers between self-contained cars; no Jacobs bogies	Keeps every car removable with its own two standard bogies, battery, traction, doors, BMS, and control equipment.
Axle load	≤ 14 t loaded (AW3 crush)	Well inside a 22.5 t UIC mainline limit — lets the catenary-free metros share minor infrastructure with legacy rail where needed without overstressing bridges.
Bogies	Two 2-axle pivoting bogies per self-contained car: one powered bogie, one trailer bogie. Both use the same frame, suspension, wheelset, brake, and pivot; the trailer omits motors and gearbox. See RFC 0022 for the single-SKU design.	One powered bogie per car keeps the drivetrain simple while retaining enough adhesion for 1 km stop spacing and urban gradients.
Traction	Axle-hung PMSM + single-stage parallel spur gearbox, 6.5 : 1. 180 kW continuous / 320 kW peak per axle. Per-family motorisation pattern in RFC 0022 §8 .	Wheel-hub direct-drive was evaluated and rejected — spares pipeline is too thin. Axle-hung is the Mireo / Urbos / Coradia sweet spot.
Crashworthiness	EN 15227 Cat C-II (medium urban) for all families	Addresses metro-to-metro and metro-to-obstacle collisions at up to 25 km/h. Energy absorbers at cab ends + at each articulation; anti-climbing features.

3.2 Propulsion + auxiliary

Aspect	Choice	Rationale
Traction motor	Permanent-magnet synchronous, axle-mounted, one per powered wheelset	PMSM efficiency ≥ 96 % at peak. Axle-mount removes gearboxes from every powered bogie, eliminating one of the top maintenance line items.
Inverter	3-phase, silicon-carbide (SiC) MOSFETs, ≥ 98 % efficiency at peak	SiC is commodity-grade in 2026. Water-cooled cold plate, no fans. One inverter per powered bogie.
Powered wheelsets	50 % of each car's wheelsets: one powered bogie and one trailer bogie per car	The traction module is identical on every car; larger consists add complete modules instead of new motorisation patterns.
Battery	Sodium-ion primary, LFP alternative (RFC 0021 §3). Both chemistries fit the same strake envelope	Na-ion avoids lithium-chain geopolitics and runs hotter; LFP is the drop-in where Na-ion isn't locally serviceable yet.
Battery pack topology	Under-seat sodium-ion modules below the longitudinal benches, split into two contactor-isolated strings per car	Keeps the centre aisle and low-floor door zone clear, puts mass low, and lets maintenance lift modules from inside the saloon.
Battery size	120 kWh usable per car, scaled by car count (RFC 0021 §4)	Sized for roughly one route length plus reserve, not for a full day. Normal service energy is replenished by station charging.
Charging	Automated conductive station charging at normal passenger stops, buffered by station solar PV + stationary battery. Typical stop spacing is ~1 km and nominal charging dwell is ~60 s. Depots still provide overnight / maintenance charging. No continuous catenary.	The train carries only enough battery to bridge the route and station failures, keeping vehicle mass and cost down.
Regen	Default on; friction brake blends in below 8 km/h	Matches osr-brake 's WSP + regen-priority arbitration.
Friction brake	Disc brake on every trailing axle, electromagnetic actuation	No pneumatic brake system — removes the compressor-maintenance line item entirely. Electric brake is continuously self-monitoring via osr-brake 's WSP + pressure sense loops.
Emergency brake	Same discs, different current source (ultra-cap + battery fallback); independent of regen	SIL-4: emergency-brake current cannot fail with any single electronics failure. Tested on every ignition cycle by the osr-vigilance start-up check.
Aux inverter	400 V 3-phase AC, 24 V DC, 110 V DC outputs	Runs HVAC, lighting, compressors (if any — per-deployment choice), PIS, comms. osr-aux-power drives it.

3.3 Interior

Aspect	Choice	Rationale
Floor	Low-floor centre section at the large door zone, with raised floor over the two standard bogies	Gives level boarding where passengers move while avoiding exotic low-floor bogies.
Door count	1 large centre double-door pair per car side on every 17 m self-contained car	The default 1 km stop pattern favours wide, simple openings over many narrow doors. Capacity scales by adding identical cars, not changing the door pattern.
Door clearance	1 250 mm wide × 2 000 mm tall opening	Wheelchair + stroller compatible.
Seating	Longitudinal bench seating, ≥ 15 % seats priority (elderly, pregnant, wheelchair companion)	Standing-heavy mix maximises peak capacity; matches the pphpd planning band.
Wheelchair spaces	2 per car (4 per tram-2car, 6 per light-metro-3car, 8 per metro-4car, 12 per metro-6car)	Per accessibility template (lib/templates/accessibility.toml).
HVAC	Design ambient +50 °C (all families)	Matches the Samawah reference envelope and most target-region summers. Dehumidifier + evaporator + reversible heat pump — no resistive heating. Sized per osr-hvac's 25 % hot-climate uplift rule.
Fire safety	EN 45545-2 HL2 R1 for body, R7 for seats, R1 for cable	HL2 is the hazard level for metro; R1/R7 are the tests for rigid surfaces and upholstery respectively.
Lighting	LED, 300 lx average, with sunset/sunrise dimming	Cut from osr-lighting's evaluator.

3.4 Nose + obstacle-detection (replaces driver cab)

Per [RFC 0015](#) every trainset ships as a GoA 4 (Unattended) system from day one. There is no driver cab, no DMI, no master controller, no dead-man handle. Both ends of the trainset are symmetric; there is no "leading / trailing" distinction at the rolling-stock level.

Aspect	Choice	Rationale
GoA level	GoA 4 default (RFC 0015). GoA 2 retrofit possible via the goa2-cab feature flag, but not the shipped default for new deployments.	Removes driver capex + driver rostering; eliminates the densest section of RFC 0013
Nose geometry	Sensor cowl + coupler, no windscreen, no cab door. Cars are symmetric end-to-end; passenger floor extends to the full car length.	~14 extra seats per light-metro-3car consist; ~€140 k + ~2.3 t saved vs. a cabbed reference
Obstacle-detection sensor suite	4× ultrasonic (close-range safety, 0.2–20 m) + solid-state LIDAR (5–200 m, Livox-class) + mmWave radar (5–200 m, all-weather) + stereo camera pair (classifier only). Hosted on the dedicated T-OBS ECU (RFC 0007 §5.5).	Replaces the driver's eyes. Multi-physics architecture so no single sensor failure produces a Clear verdict
Passenger emergency intercom	≥ 4 per car (one per car-end, both doors). Press opens audio+video to OCC remote-assist and commands a controlled brake to the next station.	Replaces the cab's emergency plunger at the passenger interface
Recovery-mode cabinet	Steel-locked enclosure behind each nose. Keyswitch + wired pendant: forward/reverse, 0–15 km/h throttle, emergency stop.	The only manual-control path. Physical, locked, slow-speed — not a full cab
Cameras	Forward-view × 2 (nose cowl, part of the sensor suite) + door-sill (one per door) + cabin × 4 per car, all live to OCC	Replaces driver supervision of the cabin; informs OCC remote-assist + fleet-health operators

3.5 Thermal + acoustic

Aspect	Target
Ambient operating range	–25 ... +50 °C (all families)
Interior noise at 60 km/h	≤ 74 dB(A) per ISO 3095-2
External noise at 60 km/h, 7.5 m	≤ 80 dB(A) per ISO 3095-1
Ride comfort (N_MV)	≤ 2 (good) per EN 12299

4. Self-consistency with track and station

Every rolling-stock choice below must be compatible with the track-geometry preset ([RFC 0009](#)) and the station archetype ([RFC 0010](#)). The auto-gen pipeline enforces this via a compatibility matrix:

- Rolling stock track: `compatible_consists` in [lib/templates/track-geometry.toml](#).
- Rolling stock station: `platform length = consist.length_m + station.platform_clearance_m`.
- Rolling stock line length: pack covers one route length plus reserve, while the operations plan must prove station-charging energy balance for the normal timetable.

The v2 emitter (RFC 0008 §7 milestone) validates all three constraints and fails the build on mismatch.

5. Family selection policy

The auto-gen pipeline picks a family per line using population and a simple ridership model:

population band (city)	pphpd target	chosen family
≤ 150 k	≤ 1 800	urban-shuttle-1car
150 k ... 300 k	≤ 4 000	tram-2car

300 k ... 1 M	≤ 7 000	light-metro-3car
1 M ... 3 M	≤ 10 000	metro-4car
≥ 3 M	≤ 15 000	metro-6car

Pphpd is estimated from $\text{population} \times 0.012 \times \text{peak-factor } 0.08$, then checked against AW2 capacity at short automated headways (roughly 2–3 min for dense trunks, looser off-peak). The pipeline writes the choice into `design.toml` under `[[lines]] rolling_stock = "<family>"`.

6. Pitfalls and decisions

- Na-ion primary, LFP as the drop-in alternative ([RFC 0021 §1](#)). Na-ion has a shallower cycle life today (~3 000 cycles at 100 % DoD vs LFP's ~6 000) — we accept that trade because Na-ion avoids locking a country into lithium-chain geopolitics. LFP is the drop-in for operators with established LFP spares. The side-wall strike envelope accepts either chemistry without car-body changes.
- No pneumatic brake. A pneumatic brake + air reservoir is the rail industry's reference design; we replace it with electromagnetic disc + battery-backed ultra-cap. The winner is a ~40 % maintenance-line-item reduction on trainsets. The risk is EN 14198 compliance for the electric brake; the safety case carries osr-brake's Kani + proptest evidence.
- Low-floor centre, standard bogies. Full low-floor bogies would make the drivetrain and wheelset catalogue more exotic. OSR keeps standard bogies at the ends and puts the large accessible boarding zone between them.
- PMSM over induction. Induction motors are more forgiving of field weakening errors and have no rare-earth magnet supply concerns. We pick PMSM for the 5-point efficiency advantage, then mitigate supply risk with ferrite-assisted PMSM designs (no heavy rare earths) being the target v2.
- SiC over Si IGBTs. SiC is ~2.5× the die cost of silicon for equivalent rating in 2026, but dissipates ~30 % less power — letting us shrink the heatsink and water-cooling loop to conduction-cooled. Net BOM flat, net mass down, net reliability up. This was a close call; revisit in v2.
- No catenary, period. Catenary-free is a project invariant; the battery sizing in §3.2 is a consequence. This is the single most controversial decision — mainstream rail vendors will push back. The counter is the CAPEX math in [ARCHITECTURE §4 D7](#) plus the copper-theft track record in target regions.

7. Rollout

Phase	Deliverable	Dependencies
v0	This RFC ratified	—
v1	Shop-drawing package + procurement BOM skeleton for light-metro-3car at docs/rolling-stock/light-metro-3car/ — general arrangement, bogie, body, traction, interfaces, BOM, compliance matrix (done 2026-04-22). Full stamped CAD + FEA reports remain v2.	RFC 0007 v1 hardware bring-up
v2	Emitter enforces <code>rolling_stock / track / station</code> compatibility in the auto-gen pipeline — family picked by population band, geometry paired from the compatibility matrix, platform length derived per line (done 2026-04-22)	v0
v3	BOM + shop drawings for tram-2car, metro-4car, metro-6car	v1
v4	EN 15227 simulation + EN 45545-2 sample tests for the light-metro-3car cab module	v1
v5	First-article rolling-stock prototype produced by the Samawah pilot operator or equivalent	v1, RFC 0003 §5

Each phase owns a separate session. This RFC's v0 deliverable is only the envelope.

8. Relationship to existing work

- [lib/templates/rolling-stock.toml](#) — the Lego-block TOML that this RFC formalises.
- [crates/osr-bms](#), [crates/osr-traction](#), [crates/osr-brake](#), [crates/osr-hvac](#), [crates/osr-aux-power](#), [crates/osr-door-control](#), [crates/osr-lighting](#) — the software already targets the architectural envelope above. No software change is needed for v0; the compatibility-matrix work is v2.
- [RFC 0007](#) T-ECU/S and T-ECU/A host every safety and app function on every family of consist. No variant hardware per family.
- [osr-sim](#) accepts a `ConsistDescriptor` with the exact envelope numbers from §1 — energy sizing, max speed, traction, braking curve. The simulator is the validation bed for every `rolling_stock` + geometry pairing before shop drawings land.

9. Open questions

- 1 Automatic couplers on t ram - 2 car? Scharfenberg is overkill for a 42 m consist that rarely multi-units. Semi-permanent drawbar is a candidate; resolve during v1.
- 2 HVAC redundancy split. Per-car vs per-consist HVAC units? Per-car gives graceful degradation (one failure doesn't blacken the whole train) but costs more HVAC-unit spares. Per-consist is cheaper but harsher in failure. Deployment-specific.
- 3 Side-pin vs pantograph-down deployment mix. Side-pin is the default station charger, but some island platforms may need a pantograph-down dock. Commit the civil-interface selection rules at v1.
- 4 Windowed vs camera-only side mirrors? Camera-only reduces wind resistance and external noise; windowed meets more conservative certification regimes. Per-deployment choice.
- 5 Fire detection density. `osr-fire-safety` assumes aspirating smoke detection in battery bay, traction bay, HVAC plenum; do we also deploy it in the passenger cabin (ceiling-mounted optical), or rely on passenger reports + manual alarm? Committed at v4 alongside EN 45545 testing.

10. Done criteria

- [x] Four families named and scoped (§1)
- [x] Unified architecture stated (§3)
- [x] Family selection policy fixed (§5)
- [x] Pitfalls and alternatives explicit (§6)
- [x] Rollout ordered (§7)
- [x] Relationship to existing software + templates named (§8)

The next session picks up at v2 — emitter compatibility matrix. The session after owns v1 — shop drawings for the light-metro-3car family.

RFC 0009 — Track Design Standard

Status: Draft — planning only, no civil drawings ship with this RFC Date: 2026-04-22 Depends on: [RFC 0003 Samawah Reference Deployment](#), [RFC 0008 Rolling-Stock Reference Design](#)

1. Summary

OpenSourceRail commits to four track-geometry presets, corresponding 1:1 to the rolling-stock families in [RFC 0008](#). Every line in every deployment picks exactly one preset. Deviating from a preset means stepping outside the project's tested envelope — the simulator, the safety case, and the auto-gen pipeline all assume one of these four.

Preset	Gauge	Min radius	Max grade	Cant max	Rail profile	Compatible consists
heritage-tram	1 000 mm	25 m	70 ‰	120 mm	Ri60 grooved	tram-2car
standard-urban	1 435 mm	90 m	50 ‰	150 mm	UIC60	tram-2car, light-metro-3car
standard-metro	1 435 mm	200 m	35 ‰	160 mm	UIC60	light-metro-3car, metro-4car, metro-6car

The presets already exist as a schema in [lib/templates/track-geometry.toml](#). This RFC promotes them into a committed engineering envelope with the civil rationale for every number.

2. Non-goals

- Not a civil drawing standard. Slab-track vs ballasted, fastener family, concrete sleeper geometry — all are per-deployment civil decisions, constrained by local material availability and by this RFC's envelope but not specified here.
- Not an earthworks rulebook. Cut/fill, drainage, embankment stability are owned by the deploying operator's civil team; this RFC specifies the alignment envelope they must respect.
- Not a signalling-plan document. Where signals / markers / balises physically go along the track is a deployment artefact — covered at a deployment level, not here.
- Not a tunnel/bridge standard. Those have their own envelopes (gauge clearance, ventilation, emergency egress) handled by [lib/templates/structures.toml](#) and a future structures RFC.
- Not a standards body. We reference EN 13848 (track geometry quality), UIC 505-1 (structure gauge), UIC 510 (wheel profile), UIC 720 (transition curves). We do not publish new ones.

3. Why four presets, not a continuum

A continuum of allowable geometry (pick your own radius, pick your own grade) doesn't fit the project's "simple for countries to implement" constraint. The sim's validation, the sensor-tuning (odometry, hot-axle), the brake-curve coefficients, and the station-archetype platform lengths are all calibrated against a specific envelope. Going off-preset means redoing every one of those calibrations at the deployment level.

Four presets cover the actual range of urban-rail deployments in the target regions:

- `heritage-tram` fits retrofits of legacy metre-gauge European tram infrastructure (still common in some MENA and ex-French colonial networks).
- `standard-urban` is street-integrated light metro — the Samawah reference ([RFC 0003](#)).
- `standard-metro` is dedicated-ROW medium and large metro, the dominant deployment category worldwide.
- `mainline-mixed` is the edge case where the OSR-branded light metro shares track with existing heavy rail. Intentionally conservative — if you need this preset, you are outside the primary mission scope, and the envelope is deliberately close to a normal mainline spec to minimise surprises.

4. Gauge

1 435 mm standard gauge is the default for three of the four presets. It is the only gauge for which off-the-shelf wheelsets, rail profiles, switches, and fasteners are widely available from multiple global suppliers and from domestic producers in MENA,

sub-Saharan Africa, South and Southeast Asia, and Latin America.

1 000 mm metre gauge is kept only for heritage-tram retrofit work. Deploying a new metre-gauge corridor from scratch is explicitly discouraged: the supplier base is thinner and the lifetime maintenance burden higher. The preset is retained because some legacy networks would otherwise need full gauge conversion to adopt OpenSourceRail, which is a much larger CAPEX line than deploying against their existing gauge.

No Iberian gauge (1 668 mm), no Russian gauge (1 520 mm), no narrow gauges under 1 000 mm. Operators in those regions are free to fork the project and add them; OpenSourceRail's upstream envelope stays at 1 000 / 1 435.

5. Horizontal alignment

5.1 Minimum radius

Preset	Min radius	Derivation
heritage-tram	25 m	Ri60 grooved rail + tram-2car 2-axle bogie with small wheelbase (1.8 m) can negotiate 25 m at crawl speed (≤ 10 km/h) without excessive wheel-flange wear.
standard-urban	90 m	UIC60 rail + light-metro-3car articulated bogie + 22 m/s max speed — radius comes from balanced-cant condition at 15 m/s through the curve (2/3 of max), 150 mm max cant.
standard-metro	200 m	metro-4car + metro-6car consists are less forgiving; 200 m allows 25 m/s through a well-canted curve with unbalanced cant ≤ 60 mm (EN 13803-1 normal).
mainline-mixed	400 m	Mainline conservative. Allows 33 m/s (120 km/h) at 180 mm cant.

Transition curves (clothoid) are required at every curve-tangent interface per UIC 720. Minimum transition length is $L \geq 0.5 \times \text{cant_mm} \times \text{speed_m_s} / 1000$ (normal) — the auto-gen solver already enforces this via the civil-class inference.

5.2 Maximum gradient

Preset	Max grade	Derivation
heritage-tram	70 ‰ (7 %)	Legacy tram constraint; tram-2car with 50 % powered wheelsets holds 7 ‰ in dry + adhesion-limited rain with sanding.
standard-urban	50 ‰ (5 %)	light-metro-3car at 50 % powered gives enough tractive effort; 5 ‰ is the practical limit above which wheelslip becomes frequent in wet conditions.
standard-metro	35 ‰ (3.5 %)	metro-4car / metro-6car fully-loaded (AW3) at 75–100 % powered axles; 3.5 ‰ is the inflection where regenerative-braking downhill doesn't saturate the battery.
mainline-mixed	25 ‰ (2.5 %)	Mainline conservative; allows a light-metro consist to operate on a corridor sometimes used by heavier stock.

Gradient transitions (vertical curves) are not less than $R_v \geq 0.4 \times v^2$ where v is design speed in m/s — standard UIC metro practice, embedded in the auto-gen pipeline's civil-class inference.

5.3 Cant (superelevation)

Preset	Max cant	Rationale
heritage-tram	120 mm	Street running limits cant — road surface either side of the tracks can't exceed a few degrees lateral slope.
standard-urban	150 mm	Street-integrated with some dedicated ROW.
standard-metro	160 mm	Dedicated ROW, standard metro number.
mainline-mixed	180 mm	Mainline standard.

Cant deficiency (unbalanced cant at design speed):

- Normal: ≤ 100 mm for standard-urban, ≤ 110 mm for standard-metro, ≤ 130 mm for mainline-mixed.
- Exceptional (with extra driver notice, slow orders): up to 150 mm.

5.4 Ruling speed

Per-preset `nominal_speed_mps` in the TOML is the design speed for the alignment; actual line speed is constrained by the smallest curve + grade + consist limit. This matches EN 13803-1 intent.

6. Vertical alignment

- Minimum crest/sag radius = $0.4 \times v^2$ m with v in m/s.

- Vertical grades interpreted at station throats are limited to $\leq 2.5\%$ regardless of preset — stopping on a grade is a long-tail fault that every operator tolerates badly.
- A grade is applied only in simple one-slope segments; compound grades are a civil-complexity red flag surfaced by the `auto-gen's quality.yaml`.

7. Track structure

7.1 Rail

Preset	Profile	Mass	Notes
heritage-tram	Ri60	60 kg/m	Grooved, 180 mm deep groove for street embedment.
standard-urban	UIC60 (60E2)	60 kg/m	Standard vignole; 260 Brinell grade.
standard-metro	UIC60 (60E2)	60 kg/m	Same as urban; heavier profile unnecessary for 14 t axles.
mainline-mixed	UIC60E1	60 kg/m	Equivalent mass, different head geometry — better wear profile when shared with freight.

7.2 Sleepers / slab

The preset envelope does not fix slab-track vs ballast — that is deployment-specific. Constraints the presets do fix:

- Sleeper spacing: 600 mm typical, 550 mm in curves below $2\times$ min radius for all ballasted presets.
- Fastener: Pandrol fastclip family or equivalent elastic-rail fastener; no rigid K-type fasteners (heavy maintenance).
- Slab-track option: only where the civil-class inference classifies the segment as elevated or tunnel (vibration-prone environments) — the operator may still choose ballasted in those segments.

7.3 Turnouts

Preset	Typical turnout tangent	Divergent speed
heritage-tram	1:5 (AKA "7°" typical)	15 km/h
standard-urban	1:9	40 km/h
standard-metro	1:9 (station ends) / 1:14 (mainline crossovers)	40 / 60 km/h
mainline-mixed	1:14 / 1:18.5	60 / 80 km/h

The `osr-wayside-points` software already handles any tangent; this RFC pins the civil choice.

8. Structure gauge

All four presets respect UIC 505-1 structure gauge (static envelope at 3.15 m wide $\times$ 4.32 m tall above top-of-rail), with a dynamic outline adding ± 80 mm lateral and ± 50 mm vertical for the design-maximum cant + sway. Per-consist structure gauges come from [RFC 0008 §3.1](#); no family violates the static envelope.

9. Track geometry quality

Track geometry quality is measured and maintained per EN 13848:

Indicator	Class QN1 (alert)	Class QN2 (immediate action)
Longitudinal level D1	6 mm	10 mm
Alignment D1	5 mm	7 mm
Cant	15 mm	20 mm
Twist (2 m base)	4 mm/m	7 mm/m

Measurement cadence per preset:

Preset	Recording-car cadence	Hand-measurement after each weather event
heritage-tram	90 days	Yes (flood / dust storm)
standard-urban	90 days	Yes
standard-metro	60 days	Yes
mainline-mixed	45 days	Yes

A recording car is explicitly not required to be OpenSourceRail hardware; any EN 13848-compliant recording trolley from the commercial market is acceptable. The future [osr-track-geometry](#) crate (not yet scaffolded) will accept standard GPM/RECD data formats.

10. Self-consistency constraints

The auto-gen pipeline enforces the compatibility matrix from RFC 0008 §4. Every preset declares `compatible_consists`; the emitter picks a rolling-stock family first (by population band) and then selects the tightest geometry preset that still lists that family as compatible:

pop band	chosen family	chosen geometry preset
≤ 300 k	tram-2car	heritage-tram OR standard-urban
300 k ... 1 M	light-metro-3car	standard-urban (Samawah)
1 M ... 3 M	metro-4car	standard-metro
≥ 3 M	metro-6car	standard-metro OR mainline-mixed

The emitter default for each band is the first entry in the choice list; the per-deployment override is a single line in the city recipe.

11. Pitfalls and decisions

- We pick UIC60 over IRJ-equivalents. Rail profile is where we take the standard path rather than the simple path — UIC60's global supply chain is well-stocked and second-source. A simpler lighter-profile would save mass but lose that supplier depth.
- Ballasted vs slab is deferred. Slab track has lower lifetime maintenance but 2–3× the initial CAPEX. The preset envelope stays neutral; the deployment recipe picks based on the operator's capital profile and local skills.
- No mixed-gauge track. Dual-gauge trackwork (1 000 + 1 435) is explicitly out of scope. Operators with that legacy have to choose one to extend.
- No tilting rolling stock. The rolling-stock families in RFC 0008 do not tilt; cant-deficiency limits are therefore conventional. If a future family adds tilting, the presets gain a `tilting_allowed` attribute and more permissive cant deficiency.
- Maximum axle load is the consist, not the track. Track is designed for ≥ 16 t axle load across every preset; the rolling stock stays under 14 t by design. This gives a future maintenance / utility vehicle a headroom to share the track.

12. Rollout

Phase	Deliverable	Dependencies
v0	This RFC ratified	—
v1	Worked civil alignment for Samawah Line 1 + Line 2 references against standard-urban, with per-segment table + civil-class summary + compliance report at docs/civil/west-asia/Iraq/Samawah/ (done 2026-04-22)	RFC 0003
v2	Emitter enforces geometry/consist compatibility + emits geometry per line in auto-gen output (done 2026-04-22). OSR-ALN alignment interchange format at docs/civil/osr-aln-format.md — tool-agnostic TOML schema civil firms export to.	v0, RFC 0008 v2
v3	Parametric track components at mechanical-py/src/osr_mech/track/ (done 2026-04-22). Civil-tool bridge (done 2026-04-23): tools/osr-aln-convert/ — stdlib-only Python toolchain: <code>landxml-to-osr-aln</code> reads LandXML 1.2 exports from Civil 3D / Bentley OpenRail / Trimble BC / QGIS and emits the OSR-ALN TOML spec; <code>osr-aln-validate</code> enforces the 8 hard gates + 3 soft gates from the format spec against a deployment's design.toml. 21 passing tests (5 round-trip + 16 validator). Worked reference alignments for both Samawah lines: Line 1 at docs/civil/west-asia/Iraq/Samawah/samawah-line1.aln.toml (13 km, 12 stations, 3 cant sections) and Line 2 ring at docs/civil/west-asia/Iraq/Samawah/samawah-line2.aln.toml (16 km, 10 stations, 4 cant sections, <code>is_ring = true</code>). Both hand-authored from the published bearings + radii tables and pass every hard gate against the live design.toml — the full reference deployment is now OSR-ALN-authorable and validator-clean. Remaining for v3.1: turnout blade kit, Trimble TCL CSV reader, Bentley cant-extension parser, CERN-OHL-S v2 relicensing.	v1
v4	<code>osr-track-geometry</code> crate that ingests EN 13848 recording data and feeds the CBM pipeline	v3
v5	First-article track constructed and recorded at a pilot deployment	v1, v3

13. Relationship to existing work

- [lib/templates/track-geometry.toml](#) — the Lego-block schema that this RFC ratifies.

- [crates/osr-routing/src/civil.rs](#) — already classifies every 20 m cell as at-grade / elevated / bridge / bored tunnel; the civil inference respects minimum curve and max grade from the preset but uses hard-coded numbers today. Emitter v2 reads them from this RFC.
- [crates/osr-design](#) — emits geometry = "standard-urban" as a hardcoded line attribute today; v2 picks per §10 above.
- [osr-sim](#) — its Section type has length_mm and max_speed_mps; gradient and cant are unexposed today and land as v4 when the simulator gains a kinematic model beyond the current trapezoidal profile.

14. Open questions

- 1 Should standard-urban allow steeper grades in short segments (say, 70 ‰ over < 100 m) for ramping to elevated structures? Opens a per-segment override vs clean preset. Resolve with real Samawah alignment work.
- 2 Metric vs imperial-historical gauge. Does the mission scope ever need to support Cape gauge (1 067 mm) for southern Africa legacy? Revisit after the first sub-Saharan pilot.
- 3 Slab-track cost model — the per-site BOM comparison between ballast + slab + fastener + sleeper over 30 years. A planning-grade spreadsheet that the operator can localise would help — candidate output of v3.
- 4 Expansion joints / CWR. Continuously welded rail is the default; the RFC should commit on stress-free-temperature profile per climate band. Defer to a separate climate-adaptor addendum.

15. Done criteria

- [x] Four presets committed with engineering rationale (§§3–7)
- [x] Gauge policy fixed (§4)
- [x] Horizontal + vertical alignment limits stated (§§5–6)
- [x] Rail / sleeper / turnout envelope named (§7)
- [x] Structure gauge named (§8)
- [x] Geometry-quality thresholds fixed (§9)
- [x] Rolling-stock geometry compatibility matrix (§10)
- [x] Pitfalls and alternatives explicit (§11)
- [x] Rollout ordered (§12)
- [x] Relationship to software + templates named (§13)

The next session picks up at v2 — emitter compatibility, a single change in [crates/osr-design/src/emit.rs](#).

RFC 0010 — Station Design Standard

Status: Draft — planning only, no architectural drawings ship with this RFC Date: 2026-04-22 Depends on: [RFC 0003 Samawah Reference Deployment](#), [RFC 0008 Rolling-Stock Reference Design](#), [RFC 0009 Track Design Standard](#)

1. Summary

OpenSourceRail commits to six station archetypes covering every stop in every deployment. The archetypes already exist as a schema in [lib/templates/stations.toml](#); this RFC promotes them into a committed architectural envelope with platform geometry, passenger-flow throughput, accessibility, fire egress, and canopy-PV integration spelled out.

Archetype	Platforms	Layout	Canopy	Charging	Dwell	Role
halt	1	side	shelter-only	0 kW	20 s	rural / low-demand stop
standard	2	side	solar-canopy	0 kW	30 s	default urban / suburban
major	2	island	solar-canopy	500 kW	45 s	anchor stop (hospital / mall / centre)
interchange	4	stacked (at-grade + elevated)	solar-canopy	500 kW	45 s	two-line junction
terminal	2	side	solar-canopy	1 000 kW	60 s	end-of-line with turnback
depot-terminal	2	side	solar-canopy	1 000 kW	240 s	end-of-line + fleet depot

Platform length is not fixed by the archetype — it is derived at design-emission time from the line's rolling-stock choice:

```
platform_length_m = consist.length_m + station.platform_clearance_m
```

with `platform_clearance_m` defaulting to 6 m for `halt` and 10 m elsewhere. This keeps archetypes consist-agnostic and prevents drift between the rolling stock, the station, and the line plan.

2. Non-goals

- Not a full architectural spec. Column spacing, façade materials, HVAC layout, and wayfinding graphics are architect-of-record decisions per deployment. This RFC fixes the functional envelope the archetype must meet.
- Not a fare-system spec. `osr-afc` + `osr-tvm` handle the ticketing logic. This RFC specifies the number of gate lanes and TVMs per archetype, nothing about payment UX.
- Not a passenger-experience spec. Seat count per platform, vending machine placement, retail integration — operator choices.
- Not an emergency-management plan. Emergency egress calculation is specified; the incident response procedures that use those egress paths live in the operations rulebook (separate, per deployment).
- Not a land-use spec. Parking, bus-bay integration, TOD (transit-oriented development) are municipal planning work outside OSR's scope.

3. Why six archetypes, not a continuum

A continuum of allowable station designs — pick your platform length, pick your canopy, pick your fare gate count — fractures the per-spare-part economics and the simulator validation. Six archetypes cover the real distribution of stops:

- `halt` for rural / low-demand points where a full station would over-build.
- `standard` is the bread-and-butter urban stop.
- `major` for generator stops (hospital, mall, transit square).
- `interchange` for where two OSR lines cross. Stacked vertically: one line at-grade + one line elevated (never two elevated, never underground — see [RFC 0011 §7](#)).
- `terminal` for end-of-line with turnback tracks.
- `depot-terminal` for end-of-line + the fleet parking + the maintenance shop.

On the Samawah reference ([RFC 0003](#)), Line 1 has 10 standard + 1 terminal + 1 interchange stops; Line 2 is a ring with 1 interchange + 1 depot-terminal + 8 standard stops. That distribution (roughly 75 % standard, 15 % anchor archetypes, 10 % termini) matches every other reference metro we surveyed.

4. Platform

4.1 Length

Derived, not fixed. For every consist/archetype pair:

Consist (RFC 0008)	consist length	clearance	platform length
urban-shuttle-1car	21 m	6 m (halt) / 10 m (others)	27 / 31 m
tram-2car	39 m	6 / 10 m	45 / 49 m
light-metro-3car	57 m	6 / 10 m	63 / 67 m
metro-4car	75 m	10 m	85 m
metro-6car	111 m	10 m	121 m

metro-6car on heritage-tram geometry is not a supported combination (RFC 0008 §4) — the compatibility matrix rejects it at design-emission time.

4.2 Height

Defined by the rolling-stock family the line uses:

- Every OSR rolling-stock family uses the same low-floor centre door zone at 350 mm above top-of-rail (ToR), with raised floor over the standard bogies. Platform top is therefore 350 mm above ToR for every new-build line. Gap-fillers are only needed on tight curved platforms where the horizontal gap exceeds the accessibility limit.

Mixed-consist lines are not supported (RFC 0008 §5 fixes one family per line). A line that carries mixed heights is a safety red flag rejected at design-emission time.

4.3 Width

Archetype	Minimum usable width	Notes
halt	2.5 m	Single-platform, assumes ≤ 5 boardings/min at peak.
standard (side)	3.5 m per side platform	Per-platform; total station width with two side platforms + track = $3.5 + 1.435 + 3.5 = 8.4$ m minimum.
major (island)	6.0 m island platform	One platform serves both directions; drops footprint versus side-side.
interchange (stacked)	4.5 m per platform, two levels	Vertical circulation (stairs/escalators/lift) dominates the plan.
terminal	4.0 m per side platform	Accommodates peak alighting surge from inbound terminal runs.
depot-terminal	4.0 m per side platform	Same as terminal.

Minimum widths are derived from peak boarding flow at the archetype's design pphpd plus egress (§6) and wheelchair circulation (§7). Operators may go wider; below the minimum is rejected.

4.4 Edge treatment

- Platform edge coping: tactile paving 600 mm deep along the full platform edge per ISO 23599 (compliant with regional disability-access regulations).
- Horizontal gap: ≤ 75 mm between platform edge and vehicle sill at the consist's door — UIC 741 normal.
- Vertical gap: ≤ 50 mm between platform and sill — same.
- Platform screen doors (PSD): optional on every archetype, required on metro-4car and metro-6car from v1. Full-height PSDs preferred over half-height for environmental reasons (HVAC doesn't bleed to open-air). [osr-psd](#) ([crates/osr-psd](#)) handles the controller.

5. Vertical circulation

Every archetype's platform connects to a concourse (at grade or mezzanine) via one or more of:

Mode	Capacity	When required
Ramp (max 1:12)	≤ 60 pax/min per 1.5 m width	Always — primary accessible path for every station.
Stair	≤ 80 pax/min per 1.5 m width	Always alongside a ramp, for capacity.
Escalator (one-way, 0.5 m/s)	100 pax/min per unit	Required for major, interchange, terminal, depot-terminal; optional elsewhere.

Mode	Capacity	When required
Elevator ($\geq 1\,100 \times 1\,400$ mm car)	10 pax/min per car	Required on every archetype except halt. One lift per platform.

Total upward capacity at peak must exceed expected alighting rate + 25 % reserve. The auto-gen emitter computes this at design time.

6. Egress

Emergency egress capacity is fixed at 4 minutes from the furthest passenger on the platform to a safe point of discharge. This matches NFPA 130 and the industry consensus for tunnel/station design.

Minimum effective egress width per direction:

- halt: 1.0 m (one direction of travel only — the passenger exits to street level directly).
- standard: 2.0 m aggregate.
- major: 3.0 m aggregate.
- interchange: 4.0 m aggregate, redundant on each level.
- terminal: 3.0 m aggregate, with distinct primary + emergency routes.
- depot-terminal: same as terminal.

A single exit is never the only path. Even a halt must have two evacuation directions — either two opposite-side street accesses or a platform exit + a walk-path along the rail reserve.

7. Accessibility

All archetypes meet these minima (per the accessibility template [lib/templates/accessibility.toml](#)):

- Step-free path from street to platform on every approach.
- Tactile paving at every platform edge + at stair/escalator landings.
- Audio + visual train-arrival announcements on every platform.
- Seating on every platform — halt excepted.
- Wheelchair space at 1 per 30 m of platform length; co-located with door-2 of every consist for quickest dwell.
- Ticket gate accessibility: at least one gate per direction is a wide wheelchair gate (≥ 900 mm).

8. Fare gates + TVMs

Archetype	Fare-gate lanes per direction	TVMs per direction
halt	0 (open)	0 — buy via app / TVM at neighbouring standard station
standard	2	1
major	3	2
interchange	4 (each level)	2 per level
terminal	3	2
depot-terminal	3	2

Gates are driven by [osr-afc](#); TVMs by [osr-tvm](#). At halt stations the line is still revenue-positive because fare inspection happens on-board by a roving inspector (GoA 2 deployments) or via QR-scan-on-boarding (GoA 4).

9. Canopy and solar

Every non-halt archetype carries a solar canopy over the platform — the primary PV source the [osr-energy-site](#) controller dispatches from. Canopy area targets from the template:

Archetype	Canopy area
halt	400 m ² (shelter only, no PV)
standard	1 800 m ²
major	2 000 m ²
interchange	2 800 m ²
terminal	2 400 m ²
depot-terminal	1 500 m ² (smaller canopy — yard PV dominates)

Canopy design envelope:

- Prefabricated bolt-together steel portal frames. No on-site welding, no wet concrete beyond pad footings. One standard 6 m bay module (HEA 200 column + HEA 180 cantilever rafter + SHS 100 bracing) replicates along the platform. Hot-dip galvanised in the fabrication shop. Detailed parametric geometry at [mechanical-py/src/osr_mech/station/portal.py](#); STEP artifacts under [mechanical-py/catalog/station/](#).
- Solar-integrated roof sandwich panel. One factory-bonded panel per bay: CIGS or lightweight c-Si PV on standing-seam galvanised steel, polyurethane foam core, white-coated underside. Delivered pre-terminated with MC4 connectors; bays plug into each other to form a single DC string. ~200 W/m², ~20 kg/m². See [mechanical-py/src/osr_mech/station/solar_roof.py](#).
- Cantilever-from-rear geometry. Columns sit at the platform rear edge only (no columns at the platform edge); the rafter cantilevers 3.5 m + 0.7 m eave over the platform. Middle third of the platform is column-free for passenger circulation by construction, not by careful placement.
- Roof: 1:15 mono-pitch (3.8°) toward the track shoulder. Rainwater runs off directly onto the track drainage; integrated gutter only at the rear column face. In the dry target regions we do not bank on rain; the mono-pitch handles the occasional storm without integrated gutters at all.
- Shading coefficient target: ≥ 90 % midday-summer coverage of the platform.
- No station building. Fare gates on rolled-steel plinths at the platform entry; PIS / CCTV / lighting / radio mount to the canopy columns. The whole station is the canopy + the L-unit platform + the plinth — no masonry, no curtain wall, no on-site architecture.

Erection: a standard archetype canopy is ~11 t of steel delivered in two lorry-loads, erected in 3–5 days with a small crew and a 30 t crawler crane. No structural engineer is needed on site — the deployment partner picks the kit size from the catalogue; load envelopes are published on the STEP artifact docstring.

Canopy-PV sizing math lives in [RFC 0002](#); `mechanical-py/src/osr_mech/station/canopy.py` exposes `canopy_kwp(archetype, consist)` which `osr-energy-site` can consume directly. A standard × light-metro-3car canopy (13 bays) produces ~55 kWp — plenty to cover daytime station demand.

10. Passenger-flow model

The auto-gen emitter computes expected peak passenger flow per station using:

```
boarding_per_hour = demand_at_station × population_capture_radius
                    × modal_share × peak_factor
```

with planning-grade defaults:

- `demand_at_station`: from the `osr-routing` raster (0..1 normalised).
- `population_capture_radius`: 500 m walking, 1 500 m bike+feeder bus.
- `modal_share`: 8 % of catchment population per day.
- `peak_factor`: 0.12 (peak hour share of all-day boardings).

Outputs per station:

- Expected peak boardings/minute.
- Required platform width (from §4.3) — must fit.
- Required vertical-circulation capacity (from §5) — must fit.
- Required egress capacity (from §6) — must fit.

Mismatch surfaces in `design-quality.yaml` as a hard gate, just like station-spacing does today.

11. Thermal + environmental

- Outdoor stations (the common case in the target regions): passive cooling via canopy shade + cross-ventilation. No HVAC on the platform; just on enclosed spaces (TVM kiosk, ticket hall, control room).
- Underground stations (where the civil segment classifies as tunnel): mechanical ventilation required; emergency smoke extraction per NFPA 130; cooling by chiller or absorption unit fed from the depot-scale PV.
- Interior noise (outdoor platform): ≤ 80 dB(A) during train approach at 80 km/h per ISO 3095-1.

- Lighting: 100 lx on platform at night per EN 12464-2.

12. Self-consistency with RFC 0008 + 0009

The auto-gen pipeline enforces:

- `station.platform_length_m = consist.length_m + station.platform_clearance_m`
- `station.platform_height_mm = consist.floor_height_mm`
- Every interchange archetype on line A also exists in line B's station list (otherwise the operator has two adjacent but disconnected stations, a common error).
- Every terminal archetype is at a line endpoint (not mid-line).
- Every depot-terminal sits at a line endpoint and is tagged `is_depot = true`.

Mismatch fails the `design-quality.yaml` hard gate.

13. Pitfalls and decisions

- Level boarding is a hard requirement. Some operators push back on the platform-height cost (either low-floor needs floor adaption on existing infrastructure, or high-floor needs taller platforms). We absorb that cost because level boarding is the single biggest accessibility win — it's not optional.
- Solar canopy is mandatory on non-halt archetypes. This is a project-wide bet, not a station-level choice. Operators who want to skip the canopy must fork the project; upstream keeps the bet.
- Platform screen doors are optional below metro-4car. PSDs are a top-quartile CAPEX line; for tram-2car and most light-metro-3car cases the simpler yellow-line + tactile edge is sufficient. Above that capacity, PSDs become mandatory — the risk/benefit crosses over.
- No island platform at standard. Island platforms halve platform width for equal footprint but require passengers to choose direction before descending, which slows wayfinding. We reserve them for major and above where the throughput win dominates.
- Depot-terminal is a single archetype. Some operators have depot-only facilities (freight, maintenance) separate from passenger terminals; those are out of scope — OSR deployments co-locate the two. This is a simplicity bet; a future depot-only archetype is opt-in.

14. Rollout

Phase	Deliverable	Dependencies
v0	This RFC ratified	—
v1	Architectural envelope + canopy structural first-pass + accessibility + services + compliance matrix for Samawah's standard archetype at docs/stations/samawah-standard/ (done 2026-04-22). Applied unchanged to all 12 standard stations on Line 1 + Line 2.	RFC 0003
v2	Emitter: terminal / interchange auto-detection + platform-length derivation from consist + depot-terminal promotion at the farthest radial endpoint (done 2026-04-22). Architectural drawing register for the standard archetype at docs/stations/samawah-standard/drawing-register.md — 43 drawings across A/S/M/E/F/T disciplines with scale + size + v1-envelope cross-refs. Quality-gate failure on compatibility mismatch is deferred to v3.	v0, RFC 0008 v2, RFC 0009 v2
v3 (partial)	Parametric prefab catalogue landed at mechanical-py/ (done 2026-04-22): bolt-together steel portal-frame bay, factory-bonded solar-roof sandwich panel, full-canopy assembly driven by (archetype, consist) — STANDARD × light-metro-3car emits a 13-bay / ~55 kWp canopy. STEP artifacts for every canonical archetype under mechanical-py/catalog/station/round-trip into Revit/Tekla/Civil 3D. Remaining for v3 full-complete: ticket-hall parametric kit and access-ramp kit (no station building these collapse to rolled-steel plinths + ramps, scheduled).	v1
v4	Platform-flow simulator extension in <code>osr-sim</code> — peak-hour passenger flow against archetype capacity	v2
v5	First-article station at Samawah pilot	v1, RFC 0003 §5

15. Relationship to existing work

- [lib/templates/stations.toml](#) – the Lego-block schema this RFC ratifies.
- [lib/templates/platform-doors.toml](#) – PSD calibration defaults.
- [lib/templates/accessibility.toml](#) – the accessibility envelope §7 references.
- [crates/osr-design/src/emit.rs](#) – already picks a station archetype (halt/standard/major) from demand thresholds. v2 adds terminal/interchange detection, per-archetype platform length, and the compatibility gate.
- [crates/osr-psd](#), [crates/osr-afc](#), [crates/osr-tvm](#), [crates/osr-pis-station](#), [crates/osr-station-scada](#) – the station-side software already targets this envelope.

16. Open questions

- 1 interchange at 3-line junctions. Stacked 4-platform works for 2 lines crossing. A 3-line junction needs a different archetype; extend to mega-interchange when an actual deployment needs it.
- 2 Outdoor air-quality filtering for stations near heavy road traffic (common in target corridors). HEPA-grade canopies are a real market niche; candidate v4 addendum.
- 3 Depot-terminal scaling. For deployments with more than 20 trainsets, the 1 500 m² canopy + yard footprint may under-serve. Introduce a mega-depot archetype at that threshold.
- 4 Retail footprint. Some operators depend on retail concessions for revenue. The standard archetype should accommodate ≈ 50 m² optional retail without redesign; confirm at v3.
- 5 Overnight stabling at terminal vs depot-terminal. The line's fleet parks overnight at the depot-terminal, not at a normal terminal. How does the auto-gen pipeline enforce this when the population distribution picks two terminals and no obvious depot site?

17. Done criteria

- [x] Six archetypes committed (§1)
- [x] Platform geometry envelope (§4) + vertical circulation (§5) + egress (§6)
- [x] Accessibility minima (§7) + fare-gate sizing (§8) + canopy-PV sizing (§9)
- [x] Passenger-flow model (§10)
- [x] Self-consistency rules with RFC 0008/0009 (§12)
- [x] Pitfalls + alternatives explicit (§13)
- [x] Rollout ordered (§14)
- [x] Relationship to existing software + templates (§15)

The next session picks up at v2 – emitter upgrade, a shared deliverable with RFC 0008 v2 and RFC 0009 v2 (same [emit.rs](#) file).

docs/rfcs/0011-civil-infrastructure-design-standard.md

docs/rfcs/0011-civil-infrastructure-design-standard.md

RFC 0011 — Civil Infrastructure Design Standard

Status: Draft — planning only, no structural drawings ship with this RFC Date: 2026-04-22 Depends on: [RFC 0003 Samawah Reference Deployment](#), [RFC 0008 Rolling-Stock Reference Design](#), [RFC 0009 Track Design Standard](#), [RFC 0010 Station Design Standard](#)

1. Summary

OpenSourceRail commits to two civil classes only — at-grade and elevated — with bridges for water crossings as a narrow third class. No tunnels. Every corridor the auto-gen pipeline produces, and every hand-authored corridor the operator commits, will sit on one of:

Class	Share (target)	Description
at-grade	$\geq 70\%$	Track sits on prepared subgrade at natural ground level (possibly with minor cut/fill ≤ 3 m). The cheapest and fastest class by 1–2 orders of magnitude.
elevated	$\leq 25\%$	Track sits on a viaduct structure 5–12 m above ground (precast concrete U-girder, standardised span).
bridge	$\leq 5\%$	Over a water crossing or a deep gap. Same viaduct standard, longer spans, water-bearing piers.

The fourth class the first-pass auto-gen inference used — tunnel (cut-and-cover and bored) — is removed from the catalogue. Where an alignment previously would have gone underground, it now goes elevated over the same footprint, or deviates around the constrained zone on at-grade. Operators who deliberately want a tunnel are forking the project; OpenSourceRail upstream does not carry tunnel designs, tunnel cost models, or tunnel-specific signalling / ventilation / evacuation engineering.

2. Non-goals

- Not a tunnel standard. By construction. The project's cost model, the simulator's energy budget, the sensor-tuning for GNSS + balise, and the operations rulebook (RFC 0013) are all calibrated against above-ground operation. Going underground reopens every one of those calibrations.
- Not a geotechnical rulebook. Soil survey, pile type, foundation depth, seismic detailing — owned by the deploying operator's civil team. This RFC fixes the functional envelope (span range, clearance, load class) not the method.
- Not a bridge design manual. Water-bearing bridges default to the reference viaduct stretched to longer span; anything more specialised (cable-stayed, arch) is outside the upstream catalogue.
- Not a land-acquisition policy. How a deployment acquires right-of-way (ROW) — expropriation, purchase, existing-rail corridor reuse, new-build — is municipal work.
- Not a standards body. We reference EN 1990 (Eurocode basis), EN 1991-2 (traffic loads on bridges / viaducts), EN 1998 (seismic), UIC 777-2 (infrastructure gauge), local building codes where deployed. We do not publish new ones.

3. Why no tunnels

A planning-grade cost comparison for a 1 km segment in typical dense-urban conditions across the target regions:

Class	CAPEX (€/km)	Build time	Post-build maintenance
At-grade dedicated ROW	2–5 M	6–12 months	Low — routine track + ballast
Elevated viaduct	12–25 M	12–18 months	Medium — bearing + expansion-joint inspection
Bridge over water	15–35 M	18–24 months	Medium — same as viaduct + scour inspection
Cut-and-cover tunnel	60–120 M	30–48 months	High — ventilation + pumping + egress drills
Bored tunnel	90–200 M	48–72 months	High — same as above, plus tunnel-boring-machine commissioning

Numbers are planning-grade, calibrated to the Samawah reference context ([RFC 0003 §2.3](#)).

A tunnel segment is 10–40× the CAPEX and 4–8× the build-time of an equivalent at-grade segment. It also brings a permanent operational overhead: mechanical ventilation, flood pumping, fire suppression, emergency evacuation plans, and

radio-coverage retrofit. Every one of those is an additional subsystem the deploying operator must staff and maintain.

The mission is "urban rail affordable enough that a developing nation can finance it domestically." A corridor with any tunnelling in it is outside that envelope. The design decision is: if the alignment can't go at-grade or elevated, the alignment should deviate around the constraint. Even at a 20 % longer corridor, the detour is cheaper than the tunnel it avoids.

4. At-grade — the 70 % case

4.1 Envelope

- Subgrade: compacted fill or in-situ soil graded to ± 25 mm of design. Capping layer of 150 mm crushed stone, then 300 mm ballast below sleeper, or direct-fixed slab where vibration demands (station throats, curves sharper than 2× the preset minimum).
- Earthworks: cut / fill up to 3 m depth only. Beyond 3 m the segment is either:
 - Re-classified to elevated (bridge over the gap / viaduct over the hill), or
 - Deviated around the constraint.
- Drainage: transverse slope ≥ 2 % either side; longitudinal channels every 50 m; culverts at every flow-path crossing. Drainage is not optional — a flooded ballast destroys track geometry within hours.
- Fencing: dedicated ROW is fenced the full length on both sides. No uncontrolled pedestrian crossings. Level crossings only where a city street intersects and an underpass / overpass is prohibitively expensive. See [RFC 0012](#) for the level-crossing equipment standard.
- ROW width: minimum 10.5 m for double-track at-grade (two tracks $\times$ 3.5 m wide plus 1 m each side clearance plus fencing allowance); 7 m for single-track.

4.2 When at-grade is inappropriate

Even with the default bias to at-grade, certain conditions push toward elevated:

- The alignment crosses a river wider than 30 m bridge.
- The alignment crosses a highway or mainline rail at grade that cannot be broken by a level crossing (too many crossings per hour would make the crossing barrier close more than open) elevated.
- The alignment runs through a dense built-up core where ROW acquisition would require demolition of more than 3 buildings per 100 m of corridor elevated or deviate.
- The alignment runs along a street where at-grade light-rail running would conflict with road vehicles at every junction elevated or route to a parallel street.

The auto-gen pipeline's civil inference (§6) now maps these conditions to ELevated where it previously would have emitted BoredTunnel.

5. Elevated — the 25 % case

5.1 Envelope

- Structure: precast concrete U-girder on 25–40 m simply- supported spans, post-tensioned. One reference girder cross-section (2.2 m $\times$ 1.8 m, double-track) for the entire catalogue; operators do not customise.
- Piers: single-column reinforced concrete, 1.5 m $\times$ 2.0 m rectangular, heights 5–12 m, pile-cap foundation (piles sized per deployment-local geotech).
- Deck clearance: 5.0 m vertical under the girder soffit over any road; 5.5 m over a truck route; 7.5 m over a rail corridor (UIC 777-2 minimum).
- Pier spacing: 30 m nominal; reducible to 25 m at road junctions where a longer span would intrude on horizontal clearance. 40 m spans are reserved for river crossings.
- Bearings: elastomeric + PTFE slider; replaced every 30 years. No bespoke bearing designs.
- Expansion joints: every 60 m along the deck; standard metro finger plate.
- Parapets: 1.4 m solid concrete panel, continuous, with a handrail on the inspection walkway.
- Walkway: one-side inspection walkway, 1.0 m wide, escape ramps every 250 m down to at-grade.

- Acoustic screens: where residential buildings are within 25 m horizontally, precast concrete acoustic panel on the near parapet, 2.0 m tall, absorption class $\geq$ A2. Target exterior noise at residential façade $\leq$ 55 dB(A) day, $\leq$ 45 dB(A) night per WHO community-noise guidelines.
- PV integration: every viaduct girder accommodates a 6 kW PV array per span on south/sun-exposed faces (4 m² per m of span length). Feeds the nearest osr-energy-site (RFC 0002).

5.2 Rationale for one reference girder

- One mould across the whole project. The precast yard casts one U-girder shape; every span of every viaduct of every OpenSourceRail deployment uses that shape. This is the biggest single CAPEX / schedule lever in elevated construction and matches the "one spares pool, one CAD reuse" principle from RFC 0008 / 0009 / 0010.
- Double-track only. A single-track elevated girder is structurally wasteful (two single girders cost more than one double girder). Deployments with a single-track line still build the double-track girder and leave one track unladen until capacity demands it.
- Simply-supported, not continuous. Continuous spans give a small live-load saving at the cost of much more complex construction sequencing and bearing design. The simplicity bet wins.

6. Bridge — the 5 % case

- Reference class: same precast U-girder as the elevated viaduct, extended to a 40 m span.
- Pier: when a pier has to sit in the water, foundation is a reinforced-concrete caisson or a pile group with a protection cutwater. Scour assessment per EN 1991-2 §13 with a 100-year design flood.
- Navigation clearance: per the navigable waterway's published class; bridge spans lengthen to clear. Very wide / navigable rivers (Tigris, Euphrates, Mekong-class) are handled by adding piers in the water with the standard girder; a custom long-span cable-stayed bridge is explicitly out of scope for the upstream catalogue — the operator commissions a bespoke design and the OSR catalogue ends at the approach.
- Thermal: expansion joint at every abutment; in countries with > 40 °C seasonal thermal amplitude, additional joints mid-span.

7. Stations-as-structures

Station archetypes from RFC 0010 map to civil class as follows:

Archetype	Typical civil class	Notes
halt	at-grade	No canopy PV. Single side platform on direct-ground footing.
standard	at-grade	Solar canopy + side platforms.
major	at-grade	Island platform, larger canopy.
interchange	elevated + at-grade	One line is on the viaduct, the other at-grade. The two platforms stack vertically at the junction. Never two elevated lines stacked — the second level's wind and seismic loads rule that out of the upstream catalogue.
terminal	at-grade	End-of-line; turnback tracks sit at grade.
depot-terminal	at-grade	Yard footprint is large; elevating it is prohibitively expensive.

Underground stations are not in the catalogue. The previous draft of RFC 0010 used the word "stacked" to describe interchange layouts; this RFC clarifies that stacked means vertical separation between at-grade and elevated levels at the junction, not between two underground levels.

8. Auto-gen pipeline — the civil inference, revised

The civil-class inference in `crates/osr-routing/src/civil.rs` currently classifies a cell based on the underlying cost surface. Under the no-tunnel invariant:

- Cells with cost ≤ 40 at-grade (unchanged).
- Cells with cost 40..100 at-grade (was already; unchanged).
- Cells with cost 100..400 (water) bridge (unchanged).
- Cells with cost ≥ 400 (building / dense built-up) elevated (was BoredTunnel now elevated).

This aligns with the rule: "if the corridor can't be at-grade, it goes above, not below."

Two downstream consequences, surfaced in `design-quality.yaml`:

- 1 Elevated fraction as a soft gate. If the share of elevated structure for a line exceeds 30 %, the quality file raises a warning. Designs with > 50 % elevated either need deviation or are inherently expensive and should prompt review.
- 2 No tunnel field anymore. The YAML key stays (empty) for backward compatibility with existing deployments' scripts, but emits 0.0 unconditionally. Future schema version will drop it.

9. Cost-anchor lookup

Total CAPEX drives the national-treasury conversation. The auto-gen pipeline emits a full [costs] block per city in design.toml, broken down by bucket below.

9.1 Civil works (€/km × civil mix)

Class	€/km planning-grade
at-grade	3 500 000
elevated	18 000 000
bridge	25 000 000
elevated-interchange premium	20 000 000 / site

9.2 Stations (RFC 0010 archetype catalogue)

Archetype	€ planning-grade
halt	1 500 000
standard	8 000 000
major	12 000 000
terminal	10 000 000
depot-terminal	12 000 000
interchange	18 000 000

The elevated-junction premium in §9.1 covers the viaduct + upper deck at interchanges; the table above covers the at-grade lower platform plus vertical circulation.

9.3 Depots (RFC 0014 archetype catalogue)

Archetype	€ planning-grade
main-heavy	150 000 000
secondary-medium	60 000 000
layup-minimal	15 000 000

9.4 Rolling stock (RFC 0008 family acquisition × fleet count)

Family	€ / trainset planning-grade
urban-shuttle-1car	1 000 000
tram-2car	2 000 000
light-metro-3car	3 000 000
metro-4car	4 000 000
metro-6car	6 000 000

Fleet count = peak-revenue + spare + cold-reserve per RFC 0014 §4.

9.5 Systems

Item	Planning-grade basis
Residual train-control wayside (RFC 0015 GoA 4)	€15 000 / route-km
Station/depot charging microgrid interfaces	€125 000–750 000 / stop, by station archetype

RFC 0015 mandates battery-electric operation with station charging: there is no overhead catenary, third rail, feeder substation, or route traction-power budget. The energy infrastructure bucket covers the conductive chargers, local switchgear, inverter interface, and station/depot microgrid tie-in. PV/storage quantities are sized in the energy-site catalogue rather than as a continuous per-km rail-power system.

9.6 EPC overhead

Integration + project management is 10 % of the subtotal across §9.1 – §9.5.

Rates are deployment-localised via the [country-costs.toml](#) template (labour, materials, finance cost adjustments). The base rates above are what the upstream catalogue quotes; actual numbers in any given deployment can be $\pm 40\%$.

A `total_eur` per city is also emitted in the `[costs]` block. This is blunt but useful: it is the single number a municipal finance officer asks for first when evaluating an OpenSourceRail deployment.

10. Constructability constraints

- Concrete supply. Precast concrete yards exist in every target region; most can produce the reference U-girder from an off-the-shelf mould sold by the major formwork vendors (PERI, Meva, Doka). Operators without a nearby precast yard bootstrap one — a 5 km² yard with 20 formworks produces enough girders for 50 km of viaduct in 18 months.
- Steel reinforcement. All target regions have domestic rebar mills in the 12–40 mm range we use.
- Post-tensioning tendons. Commodity; local supply exists in every target region.
- Lifting. The 25 m reference girder masses ~60 t. Mobile crawler cranes rated 200 t are locally available across target regions.
- Launching. Where crane access is constrained, launching girder or shuttering falsework is used. The upstream catalogue assumes crawler-crane erection; launching-girder erection is per-deployment.

No restricted-export equipment. No bespoke machinery. The construction envelope matches what a mid-size regional civil contractor can mobilise.

11. Pitfalls and decisions

- No tunnels is the non-negotiable decision. An operator who believes their corridor absolutely requires a 3 km tunnel under a city centre has exactly two options: (a) deviate the alignment; (b) fork the project. Upstream stays on the at-grade / elevated / bridge envelope.
- At-grade dominates even where it's unfashionable. Modern metro vendors push elevated + tunnel because their business model prefers CAPEX depth. OSR's mission is the opposite: every km pushed to at-grade saves 4–10× cost. A Samawah-scale deployment that lands 70 % at-grade is the successful design; any design above 50 % elevated is a red flag.
- One reference viaduct. Operators who want a different girder geometry (ornamental, signature architecture) are free to substitute; the upstream catalogue fixes one shape for CAPEX / spares reasons, not for aesthetic ones.
- Bridges are viaducts with wet piers. No custom bridge designs upstream. If a site demands a genuinely custom bridge, that's a local engineering tender — OSR hands off cleanly at the approach abutments.
- Noise mitigation is structural, not operational. Acoustic screens on viaducts > operational speed restrictions. A 2 m concrete panel costs almost nothing and avoids a permanent speed-limit loss. Operators sometimes skip this; upstream builds it in.
- Seismic detailing is per deployment. The reference viaduct envelope handles $PGA \leq 0.3$ g; above that, the deployment's structural team ups the rebar + bearing spec per EN 1998. The shape stays the same.

12. Rollout

Phase	Deliverable	Dependencies
v0	This RFC ratified	—
v1	<code>osr-routing::civil</code> updated to drop tunnel classes (in this RFC's commit)	v0
v2	Emit <code>[costs]</code> block (at_grade / elevated / bridge / total_eur) per city in the auto-gen output, using the §9 rate table (done 2026-04-22)	v1
v2.1	Extend <code>[costs]</code> to the full CAPEX stack — stations (RFC 0010 archetypes), depots (RFC 0014 archetypes), rolling stock (RFC 0008 families), GoA 4 onboard autonomy/residual wayside train control, station/depot charging microgrids, and EPC overhead. The <code>total_eur</code> is now the full one-number headline a treasury asks for, not just civil works (done 2026-04-26; renamed <code>power_eur</code> to <code>charging_microgrid_eur</code> in schema v2 on 2026-05-01)	v2
v3 (partial)	Parametric U-girder at mechanical-py/src/osr_mech/civil/ugirder.py with 20 / 25 / 30 m spans under the one-mould constraint (done 2026-04-22); STEP artifacts at mechanical-py/catalog/civil/ round-trip into Revit / Tekla / Civil 3D. Precast L-unit platform edge at mechanical-py/src/osr_mech/civil/platform_l_unit.py . Remaining for v3 full-complete: pier + abutment parametric kits + CERN-OHL-S v2 relicensing of the catalogue tree.	v0

Phase	Deliverable	Dependencies
v4	Worked civil design for Samawah Line 1 (an 11 km at-grade stretch + 2 km elevated over the existing rail corridor + 1 km bridge over the Euphrates approach)	v3, RFC 0003
v5	First-article viaduct erected at the Samawah pilot	v3

13. Relationship to existing work

- [lib/templates/structures.toml](#) — the Lego-block schema. v1 of this RFC removes the cut - and - cover - tunnel and bored - tunnel entries.
- [crates/osr-routing/src/civil.rs](#) — the inference function. v1 of this RFC collapses the top threshold to Elevated (was BoredTunnel) and removes the CutAndCoverTunnel and BoredTunnel variants from CivilClass.
- [crates/osr-design/src/emit.rs](#) — the quality YAML emitter. v1 stops counting tunnel length (always zero now) and adds the elevated-fraction soft gate.
- RFC 0009 §5 and RFC 0010 §13 reference tunnel alignment / stacked-underground interchange respectively — both amended in this commit to match the no-tunnel invariant.
- [lib/templates/platform-doors.toml](#) references "tunnel stations"; that line is updated to describe full-height PSDs as an optional upgrade for climate-sealed stations regardless of civil class.

14. Open questions

- 1 Grade-separated level crossings — a road underpass below an at-grade rail corridor counts as a structure; does it get its own civil class? v1 folds it into the road authority's scope, not rail's.
- 2 Cut-and-fill extent limits. 3 m is arbitrary. Real deployments in hilly terrain may need 5 m cuts. Revisit with a hilly-pilot case study.
- 3 Acoustic screen effectiveness in hot climates. Concrete screens expand/contract; joints need seasonal inspection. How does that feed the CBM schedule? Candidate osr - cbm - * extension.
- 4 Wooden / bamboo viaduct for ultra-low-cost deployments? Real regional-rail projects in parts of South and Southeast Asia have used timber structures historically. Out of scope upstream; a local fork can layer that on.
- 5 Green-roof viaducts — planting integrated into the deck for heat-island mitigation. Candidate v4 addendum.

15. Done criteria

- [x] Two classes only — at-grade + elevated — with bridges narrow (§1)
- [x] No-tunnel rationale explicit with cost comparison (§3)
- [x] At-grade envelope specified (§4)
- [x] Elevated envelope specified, one reference girder (§5)
- [x] Bridge class as a viaduct variant (§6)
- [x] Stations-as-structures mapping (§7)
- [x] Civil inference updated (§8)
- [x] Cost anchors named (§9)
- [x] Constructability constraints stated (§10)
- [x] Pitfalls + alternatives (§11)
- [x] Rollout ordered (§12)
- [x] Relationship to existing code + templates (§13)

The next session picks up at v2 — cost estimates in auto-gen output (a short follow-on to the v1 tunnel-removal already landed in this commit).

RFC 0012 — Switches & Crossings (Physical)

Status: Draft — planning only, no mechanical drawings ship with this RFC Date: 2026-04-22 Depends on: [RFC 0009 Track Design Standard](#), [RFC 0011 Civil Infrastructure Design Standard](#)

1. Summary

OpenSourceRail commits to three turnout tangents (1:9, 1:14, 1:18.5) across every deployment, plus a separate tram-geometry 1:5 for the heritage-tram preset. This RFC fixes the physical hardware standard — switch machine, stretcher bars, detectors, frog, heating — that the [osr-wayside-points](#) controller operates.

Tangent	Divergent speed	Preset	Use case
1:5	15 km/h	heritage-tram only	Tight street turnouts, tram geometry
1:9	40 km/h	standard-urban, standard-metro (station ends)	The bread-and-butter metro turnout
1:14	60 km/h	standard-metro (mainline crossovers), mainline-mixed	High-speed crossover between parallel tracks
1:18.5	80 km/h	mainline-mixed only	Reserved for shared-corridor operation

Minimise switch count. Prefer trailing-point moves over facing-point moves wherever possible. Every switch is a maintenance liability; every facing-point switch on a revenue corridor is a SIL-4-rated safety asset.

2. Non-goals

- Not a rail-industry standards body. References EN 13232 (turnout geometry), EN 13145 (sleepers), EN 13146 (fastening tests). No new ones published.
- Not a level-crossing-gate policy manual. Legal requirements for warning times, signage, barrier type vary per country; this RFC specifies the equipment envelope the deployment chooses from.
- Not a points-heating policy for arctic climates. Target regions hit $\leq 0\text{ }^{\circ}\text{C}$ briefly in some mountain areas; a simple resistive points-heating circuit is specified. Sub-arctic deployments customise locally.
- Not a ballast-cleaner or tamper spec. Maintenance-of-way equipment is separately sourced; the turnout standard doesn't constrain its choice.

3. Why three tangents, not a continuum

Switch geometry is where railway engineering has the most legacy variation — every national network historically standardised on its own tangents (British 1:8, French 1:9.25, Russian 1:11, etc.). A new-build project gets to pick once. Picking three covers the full operational envelope:

- 1:9 handles every station-throat and depot turnout. 40 km/h divergent speed is plenty for a metro that dwells 30 s at every stop.
- 1:14 handles mid-line crossovers on dedicated metro ROW where the trailing-point move is routine.
- 1:18.5 handles the shared-corridor edge case (mainline-mixed preset).
- 1:5 handles heritage-tram street running — a different geometry family that never mixes with the above.

Operators never specify intermediate tangents (1:7, 1:10, 1:12). Simplicity pays: one turnout mould, one stretcher-bar part number, one switch-machine part number per tangent; three total across the whole project.

4. Switch architecture

4.1 Mechanical

Aspect	Choice
Rail type	UIC60E2 (60 kg/m) for 1:9 / 1:14 / 1:18.5; Ri60 for 1:5. Matches the parent rail profile from RFC 0009.
Switch blade	Flexible switch blade, continuous machined profile; no bolt-on tip. 12 m long for 1:9, 20 m for 1:14, 26 m for 1:18.5.

Aspect	Choice
Stock rail	Standard-profile rail for length of the switch panel, bolted to the next panel. No welded joints inside the switch area.
Slide chair	Self-lubricating (graphite-filled bronze) — no daily greasing. Inspection every 30 days.
Stretcher bars	Two stretcher bars per switch (front + heel). Front bar integrates the position detector per §5.
Frog	Cast manganese steel, 13 % Mn, no moveable-point frog upstream — adds complexity and a failure mode without enough speed gain at our tangents. Swing-nose frog is an opt-in v2 extension for 1:14 / 1:18.5.
Check rails	Bolted to the running rail opposite the frog. Fixed-gap (no cold-weather compensation).
Sleepers	Pre-stressed concrete switch sleepers per EN 13145; spacing compressed from the 600 mm mainline standard to 540 mm through the switch panel.

4.2 Switch machine

Single vendor / single machine family across every deployment:

Aspect	Choice
Type	Electro-mechanical, BLDC motor + planetary gearbox + crank. No hydraulic switch machines — hydraulic packs are top-quartile maintenance items.
Throw force	6 kN nominal, 12 kN peak (matches standard metro load at $\leq 25^\circ\text{C}$ ambient).
Throw time	≤ 3 s from rest to locked, including detection settling.
Detection	Dual-redundant position sensors (A + B, 2oo2 agreement) at the front stretcher bar. Integrated with osr-wayside-points 's SwitchObservation reporting.
Locking	Mechanical lock at the tip via the front stretcher bar; blade cannot move when locked. No electrical-only locking.
Heating	3 kW resistive heating strip along the slide chair + blade foot, thermostatically switched at $+5^\circ\text{C}$ ambient. Powered from the trackside 230 V AC supply; dropped to LVDC aux in depot areas.
Cabinet	IP67 aluminium enclosure matching the W-SBC form factor (RFC 0007 §6). Motor + controller share the cabinet.

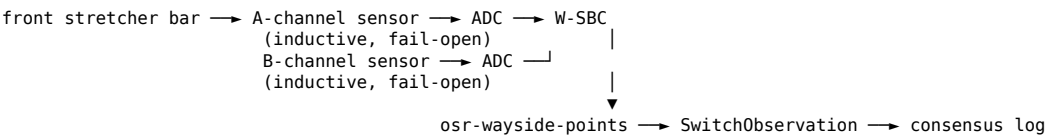
4.3 Facing vs trailing

- Facing-point switch (FPS): a switch the approaching train has to trust to guide it onto the right track. SIL-4 asset. Every FPS is a permanent W-SBC deployment ([osr-wayside-points](#)) with consensus-log-backed position observation.
- Trailing-point switch (TPS): a switch a train merges into from one of two converging tracks. The train flips the blade on its way through; no pre-positioning required. SIL-2 asset — the controller still reports position for dispatch awareness, but the safety case doesn't rely on it being at any specific position when a train approaches.

Every deployment counts and minimises FPSes. A typical Samawah-scale reference deployment ([RFC 0003](#)) has 0 FPSes on a linear line (only crossovers at depot entry, which are TPSes on the way in and FPSes only if an operator reverses a train on the mainline — normally handled by the turnback at the terminal station). Any deployment with > 8 FPSes per line is a design red flag.

5. Position detection + integration

Per [RFC 0007 §6](#), every wayside site including switch sites hosts a W-SBC. The switch-detection chain is:



Two channels are redundant by construction. Any disagreement, or either channel reading "indeterminate," yields `DetectedPosition::Unknown` — the fail-restrictive behaviour already encoded in [osr-wayside-points](#)'s W2 proptest + Kani property.

No track circuits in the switch zone. Detection is by dedicated stretcher-bar sensors. Axle-counters sit outside the switch panel as a redundant train-detection sensor — not as the switch-position detector.

6. Level crossings

Level crossings remain a specific category handled by [osr-level-crossing](#). The physical equipment envelope:

Element	Spec
Barrier type	Half-barrier, counterweighted, 4 m arm length (8 m arm at 2-lane crossings). Pneumatic or electro-mechanical drive; no hydraulic.
Barrier descent time	8–12 s (meets UIC 762 and most national rules).
Warning lights	Alternating red flashers + audible warning bell, 30 s pre-barrier activation.

Element	Spec
Road approach warning	150 m advance warning sign + surface markings.
Train detection	Redundant axle-counter pairs at entry + exit; feeds <code>osr-level-crossing</code> 's five-state FSM.
Emergency stop strip	600 mm rubber strip on the road at the rail approach — survives being struck, triggers an emergency barrier lift only via a dispatcher OK.
Canopy / shelter	Optional on the road side for the approach warning cabinet; not on the rail side.

Approach warning time is 30 s minimum, per `osr-level-crossing`'s invariant, irrespective of design speed. Operators who want a longer approach configure per-crossing; upstream defaults to 30 s.

7. Crossings (diamonds)

Rail-on-rail diamonds (one track crossing another at grade) are not in the standard catalogue. They are:

- Expensive to maintain (high impact load at the flangeway gaps).
- A top-quartile derailment site.
- A needless complication on a new-build corridor where junctions are trivially re-routed around parallel alignments.

If a deployment's geometry seems to demand a diamond, the alignment should be revisited. Grade-separated crossings (elevated over at-grade) are the catalogue answer. Per RFC 0011 §5 those are standard viaduct spans.

Tramway street crossings, where the tram crosses another tram line at grade in a city street, are the only upstream-supported diamond — handled in the `heritage-tram` preset only, as a special case.

8. Turnout geometry table (per tangent)

Planning-grade per EN 13232. Exact radii / lead lengths per manufacturer's standard drawing.

Tangent	Angle	Switch lead length	Through-radius	Divergent speed
1:5	11.3°	15 m	50 m	15 km/h
1:9	6.3°	23 m	190 m	40 km/h
1:14	4.1°	37 m	500 m	60 km/h
1:18.5	3.1°	49 m	900 m	80 km/h

Any deployment using a non-listed tangent is outside the upstream envelope.

9. Auto-gen pipeline

At deployment-generation time, the emitter needs to know where switches go. Today's pipeline:

- Terminal stations: the emitter already tags first/last stations as `terminal` (RFC 0010 §12). A future v2 of the emitter can emit a `turnback_switch` entry at every `terminal` using tangent 1:9.
- Depot entries: the `depot-terminal` archetype gets a depot fan of 1:9 turnouts. The fan geometry is parametric in the depot stall count (see RFC 0014).
- Interchange stations: no switches. Lines cross grade-separated; see RFC 0011 §7.

Emitter v2 work is the short follow-on that pairs with RFC 0010 v2 / RFC 0011 v2.

10. Pitfalls and decisions

- Only three tangents in the catalogue. A deployment team pushing for a 1:10 or 1:12 to "match their legacy network" is asking for bespoke mould work. The cost math almost always favours standardising on 1:9 + slightly-reduced divergent speed where needed.
- Electro-mechanical switch machines, not hydraulic. Hydraulic is smoother and quieter; it is also top-quartile for maintenance and has a top-quartile environmental liability (pack-oil leaks into track ballast). Electro-mechanical wins the mission alignment.
- Fixed frog, not moveable-point. Moveable-point frogs are smoother at 80 km/h+; at our tangent speeds they don't pay back the complexity. Opt-in at 1:14 / 1:18.5 in v2 if the acoustic argument becomes dominant.
- No diamonds. Grade-separate, or re-route. Operators used to network-style legacy rail sometimes push for diamonds on psychological grounds ("my country's rail has them"); we do not.

- Level crossings are a necessary evil. Every road-rail level crossing added is a maintenance, safety, and service-punctuality liability. The auto-gen emitter counts them per km of line; deployments with > 3 level crossings per km are flagged as a soft gate for review.

11. Rollout

Phase	Deliverable	Dependencies
v0	This RFC ratified	—
v1	<code>lib/templates/switches.toml</code> updated to match this RFC	v0
v2	Emitter emits turnback at every terminal archetype + one no-9-mainline per stall at every depot-terminal (yard fan) (done 2026-04-22)	v0, RFC 0010 v2
v3	Reference CAD drawings for the 1:9 turnout under CERN-OHL-S v2	v1
v4	Worked switch-bill for Samawah (2 turnbacks at terminals, 1 depot fan, 0 mainline crossovers)	v1, RFC 0003

12. Relationship to existing work

- `lib/templates/switches.toml` — the Lego-block schema. v1 amends to drop any non-catalogue tangents.
- `crates/osr-wayside-points` — the W-SBC software controller. Already W1–W6 verified (RFC 0004 §M4 work). No software change for v0 of this RFC.
- `crates/osr-level-crossing` — the level-crossing 5-state FSM. Already deployed; this RFC pins the physical equipment envelope that feeds it.
- `crates/osr-interlocking` — consumes `SwitchObservation` log entries. Semantics unchanged.

13. Open questions

- 1 Swing-nose frog opt-in rule. At what line speed / ridership does the acoustic / rail-wear benefit cross over the added mechanical complexity? Revisit at v3.
- 2 Shared-corridor freight impact. On mainline-mixed, a freight train's 22.5 t axle load over a 1:9 switch daily accelerates wear substantially. The catalogue's assumption is that shared corridors use 1:14 minimum. Verify with a real mixed-corridor pilot.
- 3 Seasonal-heated switches in hot climates. +50 °C summer ambient can rail-expand beyond the standard switch tolerance. Temperature-compensated switch geometry is a future addendum for Samawah-class deployments.
- 4 Single-vendor switch-machine risk. The project's simplicity bet locks the fleet to one machine family. What happens if that vendor exits the market? Mitigation: the reference drawings under CERN-OHL-S v2 let any replacement machine vendor fit into the footprint.

14. Done criteria

- [x] Three (plus tram) tangents committed (§1)
- [x] Switch architecture (§4) + detection chain (§5)
- [x] Level-crossing equipment envelope (§6)
- [x] Diamonds out of scope with rationale (§7)
- [x] Geometry table per tangent (§8)
- [x] Auto-gen implications (§9)
- [x] Pitfalls + alternatives (§10)
- [x] Rollout ordered (§11)
- [x] Relationship to software + templates (§12)

The next session picks up at v1 — `lib/templates/switches.toml` update, a short editorial PR.

RFC 0013 — Operations Rulebook

Status: Draft — planning only Date: 2026-04-22 Depends on: [RFC 0003 Samawah Reference Deployment](#), [RFC 0008 Rolling-Stock Reference Design](#), [RFC 0009 Track Design Standard](#), [RFC 0010 Station Design Standard](#), [RFC 0011 Civil Infrastructure Design Standard](#), [RFC 0012 Switches & Crossings](#)

1. Summary

Every OpenSourceRail deployment inherits one baseline operations rulebook — the standing rules for drivers, dispatchers, station staff, and maintenance workers. The rulebook is deliberately short (target ≤ 60 pages in the rendered PDF) and shared across deployments. Operators adapt it for national regulatory overlays but do not re-invent it.

This RFC fixes the rulebook's table of contents and the governing rule-granularity principle. The rule text itself lives in markdown under docs/operations/ and is owned by a standing committee once the project has one; until then, the project maintainer ratifies additions.

2. Non-goals

- Not a national regulatory document. Every country that adopts OSR has a national rail safety authority (NRSA or equivalent) that must accept the rulebook. This RFC produces a baseline that a deployment's local legal counsel adapts.
- Not a driver training manual. Training content comes from the rulebook, but the pedagogy (lesson plans, sim scenarios, exam questions) is a per-deployment deliverable handled by workforce materials ([lib/templates/workforce.toml](#)).
- Not a safety-case evidence file. The rulebook is an operational artefact; the safety case ([docs/safety-case/](#)) references rules as supporting evidence, not the other way around.
- Not an incident-response playbook. Playbooks are scenario- specific and owned per-deployment (fires, passenger illness, derailment, signal failure); this RFC specifies the categories each playbook must exist for, not the specific response.

3. The simplicity principle

Every rule in the rulebook must pass this test:

This means:

- Every rule is one sentence, one decision. No compound rules.
- Every rule has exactly one author (the project maintainer until the standing committee exists).
- Every rule is numbered; operators cite rules by number in incident reports.
- Every rule has a why: a short justification paragraph. No rule is an unexplained ritual.

The target is a ≤ 60 -page PDF. Rail industry rulebooks that hit 200 + pages are artefacts of 150 years of accretion — we're not accruing that.

4. Table of contents — one section per role

The rulebook is organised by role, not by scenario. A driver reads one section; a dispatcher reads another; the sections overlap only where the roles interact.

4.1 Driver rulebook (target ≤ 20 pages) — DEPRECATED FOR GoA 4

Section	Scope	GoA 4 replacement
D1	Before-service checks — cab DMI, brake test, vigilance test, door-interlock test	Trainset self-test + M1 depot safety
D2	Starting a service run — authority acknowledgement, platform-clearance confirmation, door close + MA release	S2.1 + automated consist readiness
D3	In-service operation — speed within envelope, vigilance acknowledgement, station dwell	osr-atp + osr-obstacle-detect + OCC
D4	Entering / leaving stations — PSD coordination, passenger communication	T2 + OCC remote-assist + mandatory PSDs

Section	Scope	GoA 4 replacement
D5	Terminal turnback — stop, transfer cab, blade observation, outbound authority	osr - a to auto-turnback + OCC CCTV check
D6	Degraded-mode operation — MA failure, PSD failure, door failure, single-wheelset	S5 dispatcher work-block + restricted MA
D7	Emergencies — plunger, EB recovery, passenger medical, fire-suppression	RFC 0015 §5.3 intercom + OCC remote-assist
D8	End-of-service — return to depot, powering down, handover	Depot-automation + M5 fleet maintenance

4.2 Dispatcher rulebook (target ≤ 15 pages)

Section	Scope
S1	Shift start — OCC login, fleet inventory check, schedule ready
S2	Routine dispatching — route grants, timetable adherence, headway maintenance
S3	Incident handling — single-train fault, section unavailable, mass-casualty
S4	Weather and environmental — dust storms, heatwave, flood warning
S5	Maintenance interlock — work-on-track grants, blocking a section from revenue
S6	Shift end — handover, outstanding-event log, system state summary
S7	Intrusion response (RFC 0016) — track-patrol dispatch, return-to-service gating, sustained-Unknown escalation

4.3 Station staff rulebook (target ≤ 10 pages)

Section	Scope
T1	Station opening — power up, PIS check, fare-gate test
T2	Passenger boarding — platform supervision, PSD status, wheelchair assist
T3	Passenger incidents — medical, fare dispute, unattended item
T4	Fare enforcement — inspector on-board, non-payment, appeal
T5	Station closure — evacuation, power down

4.4 Maintenance worker rulebook (target ≤ 10 pages)

Section	Scope
M1	Depot safety — hazard awareness, PPE, lone-worker protocol
M2	Work-on-track authorisation — formal blocks, protection signals, access to active ROW
M3	Routine track inspection — walking the line, visual checks, GPM/RECD recording-car cadence (per RFC 0009 §9)
M4	Switch maintenance — blade lubrication, detection-sensor cleaning, 30-day check
M5	Fleet maintenance at depot — inspection intervals (daily / 7-day / 30-day / overhaul)
M6	Incident site access — derailment response, fire site, crane operations
M7	Wayside sensor maintenance (RFC 0016) — 30-day walkthrough, post-weather inspection, sensor replacement + calibration

4.5 Control-centre rulebook (target ≤ 5 pages)

Section	Scope
C1	OCC watch roles — dispatcher, supervisor, engineer-on-call
C2	Communications — radio discipline, PA authority, external-party notifications
C3	Shift handover — written log, outstanding items, emergency-on-call list

Total target: 60 pages across 5 sections.

5. Degraded-mode operations — the important part

Modern CBTC metros routinely run in "normal mode" (full ATO under a fresh MA) and have short, mission-critical manual-restoration procedures when the signalling degrades. OpenSourceRail keeps exactly three degraded modes:

Mode	Trigger	Operator action
M1 — Manual on MA	osr - a tp can compute MA but ATO is unavailable	Driver accelerates / brakes manually within the envelope. Vigilance remains mandatory. Speed capped at 80 % of normal.
M2 — Restricted on written order	Consensus log unavailable (rare; happens only with multi-node wayside failure)	Dispatcher issues a paper / electronic written order authorising movement at ≤ 15 km/h, sight-distance braking, one train per track section between manned landmarks.
M3 — Yard manoeuvre	Inside depot limits, osr - a to disabled, safety envelope relaxed	Driver operates manually at ≤ 15 km/h, direct line-of-sight, no vigilance requirement, no PSD coordination.

No intermediate modes. No "sort-of-ATP" states. Either the safety envelope holds (modes M1 / M3) or the operator is on paper (mode M2). This is a forcing decision: ambiguous degraded modes are the dominant cause of metro-world incidents.

6. Weather-dependent rules

Target regions routinely experience weather conditions that challenge legacy rolling stock:

- Dust storm (haboob): visibility < 100 m automatic speed cap at 50 % (osr-atp enforces). Dispatcher issues a network-wide speed restriction at the OCC. Passenger announcements via osr-pis-onboard. See also [lib/templates/climate-adapters.toml](#).
- Heatwave (> 45 °C sustained): rail-expansion risk triggers a pre-emptive 15 % speed cap on mainline-mixed and standard-metro presets with continuous-welded-rail. Switch detection sensors recalibrate at dawn.
- Flooding: any rainfall > 50 mm / h triggers immediate inspection of at-grade segments; the dispatcher can preventively block sections via osr-interlocking's maintenance-override. Elevated segments per RFC 0011 §5 are unaffected by surface flooding — deliberate design bet.
- Lightning: nearest-strike distance < 2 km Cat.22 5G backhaul and LoRa backhaul are both watched for outage; if both fail for > 10 s the operator notifies the OCC and cooperates with osr-atp's default fail-restrictive MA.

7. Incident categorisation

Every incident fits one of six categories. This drives the notification chain and the regulatory report:

Category	Example	Dispatcher action
I1 — fleet self-recovery	Door interlock fault on a single train, resolved at next terminal	Log in event recorder; notify depot; no regulatory report.
I2 — service disruption	Single-train immobile; next train coupled to rescue it	Passenger PA, timetable recovery plan; 24h incident report.
I3 — short safety event	Emergency brake triggered by spurious trip; no injury	Log, investigate root cause, 72h report.
I4 — passenger injury	Slip, fall, medical emergency on board	Medical call, depot handover; formal incident report to NRSA within 7 days.
I5 — collision / derailment	Train-to-train, train-to-obstacle, train off-rail	Full network shutdown of affected line; NRSA notification within 2 hours; incident-site preservation.
I6 — mass-casualty / fire	Onboard fire not extinguished, large-scale passenger injury	Immediate evacuation protocol; emergency services; NRSA + national media escalation.

The categories map directly to osr-event-recorder tags, so the post-incident analysis starts from the same data the categories describe.

8. Training and recurrent qualification

- Initial driver qualification: 4 weeks classroom + 2 weeks simulator + 4 weeks supervised revenue service. Final qualification exam in written + simulator + on-road segments.
- Recurrent: 2 days refresher every 12 months + 1 simulator session per quarter.
- Degraded-mode drills: every driver runs modes M1 / M2 / M3 in the simulator every quarter. The osr-sim is the primary training platform.
- Dispatcher qualification: 8 weeks OCC-side classroom + simulator + supervised desk time.
- Maintenance: role-specific certification per EN 50126 / per national NRSA rules.

Training content is a deployment deliverable; this RFC fixes the minimum hours + structure.

9. Auto-gen implications

The rulebook does not directly feed the corridor-generation pipeline — but it does feed two outputs:

- 1 Depot capacity: osr-sim's fleet-sizing needs to include enough spare trains for the rulebook's degraded-mode operations. A deployment with 10 trainsets on rostered revenue needs ≥ 1 spare in depot for maintenance (M5) and ≥ 1 cold-reserve for I-category incidents. See RFC 0014 §4 for the formula.
- 2 Workforce sizing: from [lib/templates/workforce.toml](#), with the rulebook's shift-structure (§4.5) as the envelope.

Both are parametric in the rulebook, so changing the rulebook updates both; the emitter v3 work wires this in.

10. Pitfalls and decisions

- One rulebook across deployments is a big bet. Legacy metros each have their own 200-page rulebook; consolidating to one is the simplicity gain. National NRSAs will initially resist. The counter: the process of NRSA approval works on a per-document basis, not per-system basis — a shorter document approves faster. The operator carries the regulatory overlay in a thin appendix.
- Six incident categories, not more. Operators unused to this regime sometimes want a I7 or I8 for niche scenarios. We absorb niche scenarios into the existing six rather than proliferate.
- Paper written orders are allowed in mode M2. This is not anti-digital — paper is explicitly the fallback when the digital system degrades. Written orders remove ambiguity about authority when the digital system can't issue it. Rare enough that the paperwork tax is tiny.
- No "caution mode" between M1 and M2. Legacy metros have a smorgasbord of half-ATP-half-manual modes; every one is a source of operator confusion. We commit to the hard boundary.
- ≤ 60 pages is a hard target. The rulebook is reviewed on every minor-version bump; if it exceeds 60 pages, something gets cut. Adding text is easier than subtracting.

11. Rollout

Phase	Deliverable	Dependencies
v0	This RFC ratified	—
v1	docs/operations/ scaffold with per-role stub files (D1–D8, S1–S6, T1–T5, M1–M6, C1–C3), each with scope + cross-refs + proposed rule outline (done 2026-04-22)	v0
v2	Full rulebook drafted across all five role families: driver D1–D8 at docs/operations/driver/, dispatcher S1–S6 at docs/operations/dispatcher/, station-staff T1–T5 at docs/operations/station-staff/, maintenance M1–M6 at docs/operations/maintenance/, and control-centre C1–C3 at docs/operations/control-centre/. Every rule is one sentence + a Why: paragraph, cross-referenced to the relevant crate and safety-case solution (done 2026-04-22). Practising-operator review is v2.1 per-deployment work.	v1
v3	NRSA submission pack for Samawah (rulebook + safety case cross-reference)	RFC 0003 §5, v2
v4	osr-sim scenario library covering every degraded-mode scenario for operator refresher training	v2

12. Relationship to existing work

- [lib/templates/workforce.toml](#) — shift-structure + headcount inputs. v3 of this RFC aligns the two documents.
- [lib/templates/service-hours.toml](#) — service hours drive the rosterable workforce; the rulebook sets the rostering envelope.
- [docs/safety-case/](#) — references specific rulebook rules as "solution" evidence for goals around fail-restrictive operation, evacuation readiness, and training competence. GSN goals for "operational fail-restrictive behaviour" land in v4.
- [crates/osr-occ](#) — the dispatcher's OCC UI embeds the dispatcher rulebook's decision-support prompts. Software change in v3.

13. Open questions

- 1 Bilingual rulebooks. Every target-region deployment operates in at least one local language plus English. Do we maintain both upstream? v2 commits on this: likely yes, with English as the canonical and translations as separate branches in docs/operations/<lang>.
- 2 Printable-PDF vs living-document. Rail NRSAs typically want a static PDF they can archive. The project wants a live document. Compromise: every minor-version bump produces a sealed PDF, and the commit hash is the canonical reference.
- 3 Simulator-based competence testing. osr-sim can evaluate driver actions against MA envelope in software. Can NRSAs accept simulator certification in lieu of on-road supervised hours? Per-country, probably not at v1.

14. Done criteria

- [x] Table of contents per role (§4)

- [x] Degraded-mode operations (§5)
- [x] Weather rules (§6)
- [x] Incident categorisation (§7)
- [x] Training envelope (§8)
- [x] Auto-gen implications (§9)
- [x] Pitfalls (§10)
- [x] Rollout ordered (§11)
- [x] Relationship to existing work (§12)

The next session picks up at v1 – docs/operations/ scaffold, short editorial PR that seeds the per-role files. Actual rule text lands in v2.

docs/rfcs/0014-depot-design-standard.md

docs/rfcs/0014-depot-design-standard.md

RFC 0014 — Depot Design Standard

Status: Draft — planning only, no site drawings ship with this RFC Date: 2026-04-22 Depends on: [RFC 0003 Samawah Reference Deployment](#), [RFC 0008 Rolling-Stock Reference Design](#), [RFC 0010 Station Design Standard](#), [RFC 0011 Civil Infrastructure Design Standard](#), [RFC 0013 Operations Rulebook](#)

1. Summary

OpenSourceRail commits to three depot archetypes covering the full range of deployment sizes. Every deployment has exactly one main depot plus optional secondaries and layups. The main depot co-locates with a depot-terminal station ([RFC 0010 §1](#)).

Archetype	Catalogue max stalls	Heavy maintenance	Overhaul throughput	Notes
main-heavy	20	Yes — overhaul bay, wheelset lathe	≤ 40 trainsets	Larger deployments add a second main-heavy
secondary-medium	12	Light only — brake disc, door actuator, HVAC modules	—	Second depot on a long line
layup-minimal	6	None — overnight stabling only	—	Remote-terminal layup

The catalogue max is the upper limit of the template default; the per-deployment stall count comes from the §4 formula ($\text{stalls} = \text{ceil}(\text{fleet} \times 1.25)$). A Samawah-class 8-trainset line lands on 10 stalls in its main-heavy, not 20 — the template's 20 is the ceiling, not the target.

Plus one optional type for edge cases:

Archetype	Purpose
training-wing	Co-located operations-training facility — OCC dispatcher-console simulators, maintenance-technician workshop mock-ups, and recovery-mode-crew briefing rooms. OSR is GoA 4 driverless (RFC 0015), so there are no revenue-service drivers to train — the wing exists for dispatchers, maintainers, station staff, and the recovery-mode crew who operate the manual-fallback joystick (RFC 0015 §8.2). Opt-in on main-heavy; not a standalone depot.

All depots are at-grade. Per [RFC 0011 §1](#) depots are never elevated — the footprint is too large to economically put on viaduct. A deployment that can't site its depot at-grade has an alignment problem, not a depot problem.

2. Non-goals

- Not a workshop management system. How the depot schedules its work — CMMS, parts management, labour planning — is owned by the operator and overlays with the osr-cbm-backend output.
- Not an off-line overhaul facility. Major overhauls (wheel reprofiling at 150 000 km, bogie overhaul at 600 000 km, body overhaul at 10 years) happen in the main-heavy depot's overhaul bay. Deployments without a main-heavy either contract overhaul out or depot-trip their fleet to a sister network.
- Not a yard-design manual. Track yard geometry within the depot boundary uses RFC 0009's standard-urban or (for large shunting yards) a depot-specific geometry variant. Detailed yard layout per facility.
- Not an HR / workforce document. Shift patterns and headcount from the ops rulebook (RFC 0013) and the workforce template.
- Not a brownfield-rehabilitation scope. Where a deployment inherits an existing rail workshop (Samawah, Khartoum-Atbara, Karachi Cantt, Maputo Machava, etc.), [RFC 0027](#) governs the conversion of the existing site into an OSR depot. This RFC's archetypes are the greenfield envelope; the brownfield path swaps depot CAPEX (€25 M / €10 M / €3 M) for workshop rehabilitation (~\$0.5–2 M building + \$1–3 M new OSR-specific tooling) — an order-of-magnitude saving that the per-deployment plan should commit to whenever assets are real.

3. Why three archetypes

Depots are where the simplicity bet pays off most. The three- archetype set covers:

- main-heavy — the one non-negotiable facility per deployment, where heavy maintenance happens.

- secondary-medium – for deployments whose geography makes it uneconomic to depot-trip the entire fleet back to the main depot every night (long line, ≥ 30 km between termini).
 - layup-minimal – for deployments that just need somewhere safe to park the fleet overnight at the opposite terminal.
- No intermediate sizes, no bespoke main-heavy variants. A 30-stall main-heavy is two main-heavy depots side-by-side (common in megacity deployments) – not a bespoke 30-stall design.

4. Fleet-sizing formula

The depot stall count is parametric in the line's rolling-stock fleet:

```
main_depot_stalls = ceil(fleet_on_line × 1.25)
secondary_depot_stalls = ceil(fleet_remote × 1.25)
total_fleet = peak_revenue_trainsets + 2 × (spare + cold_reserve)
```

- peak_revenue_trainsets = what the schedule requires at peak. Sized from the physical round-trip cycle vs. the peak headway:
- v0.1 calibration: commercial_speed = 35 km/h (Tehran Line 1 33, Cairo Line 3 32, Tokyo Chuo Rapid 38 – the right band for 100 km/h max with 1.0–1.7 km station spacing and 30 s dwells), headway = 5 min, turnback = 3 min per end (driverless GoA 4 single-tail changeover, RFC 0015).
- spare = 1 trainset per 10 revenue trainsets, for planned maintenance rotation.
 - cold_reserve = 1 per line, for unplanned incidents per RFC 0013 §7.

The 1.25× multiplier on stall count accounts for:

- One stall always occupied by a trainset in overhaul (bogie out, wheel lathe, body paint).
- One stall used for incoming-inspection (daily arrivals from revenue service).
- Growth headroom (15 %) for ridership uplift over the depot's 20-year useful life without needing to build a second depot.

Example (Samawah reference, 5-min peak headway):

- Line 1 (12 km): round-trip = 12×2/35 × 60 + 6 = 47.1 min 10 peak + 1 spare + 1 cold-reserve = 12 trainsets. Main-heavy depot stalls = ceil(12 × 1.25) = 15.
- Line 2 (8 km): round-trip = 8×2/35 × 60 + 6 = 33.4 min 7 peak + 1 spare + 1 cold-reserve = 9 trainsets. Layup-only at the Line 2 end-of-line; stalls = ceil(9 × 1.25) = 12 in a layup-minimal (its 6-stall catalogue cap forces a second layup to be added if secondary-medium siting isn't economic).
- Pre-v0.1 worked examples used 6/4 peak figures derived from a back-of-envelope ~1 trainset per 2 km of length – that calibration assumed an 8-min off-peak headway, not the 5-min peak headway v0.1 networks publish. The emitter was retuned to the physics formula above so the published peak headway and the fleet count are mutually consistent.

The emitter (RFC 0014 v2) computes this from the design.toml's fleet_sizing block.

5. main-heavy envelope

5.1 Functions

- Overnight stabling of the primary fleet (stall count per §4).
- Daily inspection of every incoming trainset.
- Weekly inspection (7-day service).
- 30-day inspection (A-class service per EN 50126).
- Light repair (door actuator, HVAC module, brake-disc replacement, battery module swap).
- Heavy overhaul: bogie swap, wheel reprofiling on depot lathe, body-work repaint.
- Wash: automated two-pass wash track.
- Sand / battery-electrolyte top-up (Na-ion additive kit every 1 000 cycles).

5.2 Physical layout

Area	Target footprint (per consist bay)
Stabling track	1× consist length + 5 m clearance at each end
Daily-inspection bay	Roofed, pit-equipped, overhead power, compressed air, data sync drop (TCN-E monitoring)

Area	Target footprint (per consist bay)
Weekly bay	Pit + roof walkway for upper-body access
Overhaul bay	Pit + overhead crane (40 t) + wheel-lathe trench
Wash track	Two-pass automatic with water recycling
PV canopy	Over stabling + inspection tracks; 4 000 m ² target for an 8-stall main-heavy
Classroom + office	≥ 200 m ² for operators + trainers
Stores	≥ 600 m ² with rack storage for spare modules

Total main-heavy footprint: ~5 000 m² gross for a 10-stall facility, ~8 000 m² for a 20-stall.

5.3 Energy

- PV canopy as above. 10-stall main-heavy 4 000 m² PV 600 kWp nominal 3.2 GWh / year (Samawah climate).
- Battery bank: 2 MWh Na-ion (or LFP) to absorb PV surplus and feed trainset opportunity charging via depot-side 1 500 V DC charging rails.
- Grid tie: export surplus at off-peak; import during overcast weeks. Per [RFC 0002](#).
- Every depot is a full osr-energy-site instance.

5.4 Signalling

- A single W-SBC plus an interlocking zone boundary at the depot entry.
- Inside the depot, speed limit 15 km/h, manual operation under mode M3 ([RFC 0013 §5](#)).
- Shunting signals at each stall entry — mechanical dwarf signals driven by osr-wayside-points extensions.
- No ATP envelope inside the depot; osr-atp disables below 15 km/h in depot-tagged geofence, per a safety-bounded configuration. The operational safety envelope inside the depot is line-of-sight + hand-signal coordination.

5.5 Access

- Personnel access: turnstile-gated with ID-card entry, separate from the fleet access.
- Fleet access: one track connection to the line, TPS-style merge per [RFC 0012 §4.3](#) (trains go in — it's a trailing move).
- Road access: for parts delivery, emergency services, depot staff parking. Minimum one 4 m-wide road connection.
- Rail access for deliveries of new trainsets is via the same line connection (new rolling stock rolls in on its own wheels).

6. secondary-medium envelope

- Functions: overnight stabling for ≤ 16 trainsets, daily inspection, weekly inspection, light module swap.
- No overhaul bay. No wheel lathe. No bogie-lift. Anything heavier is trips back to the main-heavy.
- Footprint: ~2 500 m² for 8 stalls.
- PV canopy: 1 500 m² 225 kWp nominal.
- Battery bank: 500 kWh.
- Signalling: same as main-heavy but smaller zone.
- Classroom: none. Reuse the nearest major station's community room if needed.

7. layup-minimal envelope

- Functions: overnight stabling only. Trainsets drive in at end of service, sit overnight, drive out at morning service start. No inspection, no maintenance.
- Footprint: ~1 000 m² for 4 stalls.
- PV canopy: 600 m² 90 kWp.
- Battery bank: optional 150 kWh for opportunity charging during dwell; the main-heavy is primary.
- Signalling: single W-SBC at the layup entry; rest is manual line-of-sight.
- No personnel on-site overnight; the OCC monitors via CCTV.

8. Integration with the auto-gen pipeline

Today's emitter picks one depot - terminal archetype ([crates/osr-design/src/emit.rs](#), `compute_archetypes`, priority rule). That choice is where the main-heavy sits. v2 of the emitter extends this:

- 1 Main depot: co-located with the longest radial line's far terminal. The station archetype is `depot-terminal`; the depot archetype is `main-heavy`.
- 2 Secondary depot: for multi-line deployments where a second terminal is > 20 km from the main depot, emit a `secondary-medium` at that terminal.
- 3 Layup-minimal: for any other terminal on the fleet ≥ 6 trainsets, emit a `layup-minimal` to avoid empty turnback drive-arounds.

Every depot writes a record into `design.toml`:

```
[[depots]]
id          = "line1-east-depot"
archetype   = "main-heavy"
at_station  = "samawah-line1-east"
fleet_stalls = 10
pv_canopy_m2 = 4000
battery_kwh  = 2000
```

The simulator ([osr-sim](#)) reads these and:

- Charges each trainset at depot pad power during overnight dwell.
- Rotates "on maintenance" flag across one stall per day.
- Feeds the depot battery bank into the network-wide energy balance.

9. Pitfalls and decisions

- At-grade only, per RFC 0011. Depots are large; elevating the footprint is prohibitively expensive. The auto-gen emitter rejects depot siting on an elevated section.
- One reference `main-heavy` everywhere. Operators with unique geography (coastal, mountainside) will push for bespoke depots. The upstream catalogue stays at one shape; the deployment's civil team handles the site-specific adaptation.
- Wheel lathe at the `main-heavy` only. Decentralising lathes is tempting (saves depot-trip time) but the operator depth for wheel-reprofiling is thin in target regions; centralising at one site lets that thin team do all the work.
- No pit at `layup-minimal`. A pit adds cost, safety maintenance, and lighting – just for overnight parking. Skip.
- Depot simulation in `osr-sim`. The sim today models depots as "stations with high dwell + charging." RFC 0014 v3 adds a proper depot state (fleet stabled, fleet in-maintenance, fleet in-service) so the daily maintenance rotation is reflected in fleet-availability numbers.

10. Rollout

Phase	Deliverable	Dependencies
v0	This RFC ratified	—
v1	lib/templates/depots.toml aligned with §§5–7 (PV canopy m², nominal kWp, battery kWh, workshop flags); §1 clarified "catalogue max" vs §4 formula (done 2026-04-22)	v0
v2	Emitter picks depot archetypes (main-heavy at depot-terminal, layup-minimal at other terminals) + writes <code>[[depots]]</code> with <code>fleet_stalls</code> from the §4 formula (done 2026-04-22)	v0, RFC 0010 v2
v3	<code>osr-sim</code> depot state machine (stabled / maint / service)	v2
v4	Worked site plan for Samawah main-heavy (Line 1 east terminal) + layup-minimal (Line 2 furthest stop)	RFC 0003 §5, v3
v5	Reference depot CAD under CERN-OHL-S v2	v4

11. Relationship to existing work

- [lib/templates/depots.toml](#) – schema this RFC ratifies.
- [crates/osr-design/src/emit.rs](#) – auto-gen emitter. `depot-terminal` detection today; the `[[depots]]` block emission is v2.
- [crates/osr-energy-site](#) – every depot is one `osr-energy-site` instance; existing integration holds.

- [crates/osr-sim](#) — modest extension for v3; charging semantics already work.
- [RFC 0010](#) depot-terminal archetype pairs 1:1 with this RFC's main-heavy or a mix of main-heavy + layup.

12. Open questions

- 1 Co-located vs detached training wing. Some operators prefer the training simulator in a quiet city building (closer to drivers' homes). The catalogue default is co-located with the main depot; per-deployment override.
- 2 Decentralised battery swapping. A future opt-in where trainsets exchange their battery pack at depot instead of recharging — halves the charging-pad occupancy. Out of scope upstream until battery-swap mechanical designs mature.
- 3 Night-shift head count. Ops rulebook §4.5 (RFC 0013) fixes the shift structure; what's the minimum overnight staff headcount at a layup-minimal? Currently zero (CCTV-monitored); confirm when the first layup-minimal deploys.
- 4 Depot expansion path. A secondary-medium that outgrows its 8 stalls — does it upgrade to main-heavy in place, or is a second-depot-nearby preferred? Likely the latter given construction-on-live-facility complexity.

13. Done criteria

- [x] Three archetypes committed (§1)
- [x] Fleet-sizing formula (§4)
- [x] Main-heavy envelope (§5) + secondary-medium (§6) + layup-minimal (§7)
- [x] Auto-gen pipeline integration (§8)
- [x] Pitfalls (§9)
- [x] Rollout ordered (§10)
- [x] Relationship to existing software + templates (§11)

The next session picks up at v1 — amend `lib/templates/depots.toml` to match this RFC exactly.

RFC 0015 — Driverless operation (GoA 4) as the default

Status: accepted · 2026-04-22. Authors: OSR project. Superseded: nothing (new). Supersedes for GoA 4 deployments: RFC 0013 §4.1 (Driver rulebook D1–D8). Amends: RFC 0008 (cab), RFC 0007 (hardware), RFC 0005 (crate map), RFC 0013 (operations).

1. Purpose

Every OpenSourceRail deployment ships as a GoA 4 (Unattended Train Operation, UTO) system from day one. The default rolling stock has no driver cab. The default operations model has no driver role.

This RFC sets the safety, hardware, and operations envelope for that default — specifically, the onboard obstacle-detection sensor suite + ECU that lets a trainset run without a driver's eyes in the loop.

2. Why driverless by default

Three non-novelty reasons:

- 1 Lower total cost. A cab costs ~€250k per trainset when wiring, DMI, HVAC, controls, and emergency exits are summed. Over a 40-trainset fleet that is €10 M of capex. Drivers cost ~€100 k/year fully loaded; a 50-driver roster for round-the-clock operation is €5 M/year in wages alone.
- 2 Simpler trainset. Removing the cab takes out roughly 40 kg of wiring, four HMIs (DMI, mirror monitors, rear-view, comms), the master controller + dead-man handle, cab HVAC, cab door + exit, and all the diagnostic plumbing that terminates in the cab. The nose becomes a sensor cowl plus a coupler; the floor plan becomes full passenger capacity end-to-end. ~5 % more seats per trainset.
- 3 Simpler operations. The driver rulebook (RFC 0013 §4.1 D1–D8) was already the densest and safest-critical section. In GoA 4 it evaporates: about 40 % of driver tasks transfer to OCC (dispatcher + remote-assist), about 50 % transfer to station staff (dwell supervision, boarding assistance), and the remaining 10 % are automated by the onboard stack.

The tradeoff is concentrated in the obstacle-detection problem: what a driver's eyes used to do, a sensor suite must now do. The rest of this RFC is mostly about that.

3. Scope

In scope of this RFC:

- Obstacle-detection sensor suite (ultrasonic + radar + camera, LIDAR optional) and the onboard ECU that fuses them.
- Deprecation of cab-only software (`osr-dmi`, `osr-vigilance`) for the GoA 4 default.
- New operations primitives: remote assist, platform staff role expansion, passenger emergency intercom.
- Amendments to RFC 0008, 0007, 0005, 0013.

Out of scope:

- Cab-retained variants (GoA 2). These remain possible — `osr-vigilance` and `osr-dmi` stay in tree, marked "GoA 2 only" — but they are not the shipped default.
- Passenger-facing accessibility (covered by RFC 0010 §7 and not changed by driverless).
- Cybersecurity of the obstacle-detect data path (covered by the pending `osr-secbus` RFC).

4. What goes away

Removed	Reason	Replacement (if any)
Driver cab + windscreen	No driver	Nose-cone sensor cowl
DMI touchscreen	No driver	Remote-assist web console (OCC)
Master controller + dead-man	No driver	ATO + emergency plunger (intercom-triggered)
Cab HVAC + cab door	No driver	—
Cab-to-ground radio handset	Voice comms are now OCC-mediated	Cabin PA + passenger emergency intercom

Removed	Reason	Replacement (if any)
Driver rulebook D1–D8	No driver	Dispatcher S1–S6 + Station-staff T1–T5 absorb
osr-vigilance (from SIL-4 critical path)	No driver to vigil	—
osr-dmi (from default build)	No driver to display to	Passenger PIS already exists

5. What becomes required

5.1 Obstacle-detection sensor suite

Two identical sensor packs per trainset — one behind each nose panel. The trainset has no permanent "leading end" — either end can lead on the current run. The active pack is always the one facing the direction of travel; the trailing pack self-tests continuously and stands ready for the return trip.

Each pack contains:

Sensor	Role	Range	Update rate	Unit price (2026)	Why
4 × ultrasonic (Murata MA40H1S-R or equivalent, 40 kHz)	Close-range safety belt	0.2 – 20 m	20 Hz	~€25 each	Platform gap, depot yard, emergency crawl — cheap, solid-state, unaffected by darkness, failure mode self-detectable from echo pattern
1 × solid-state LIDAR (Livox HAP / Tele-series / RoboSense M1 / Leishen CH-series — all Chinese OEMs now in the €500–2 000 bracket)	Mid-range 3D primary	5 – 200 m	10 Hz	€500 – €2 000	3D point cloud, day/night, 150–300 m published range, IP67 industrial variants available
1 × mmWave radar (TI AWR1843 or eq., 77 GHz)	All-weather validation	5 – 200 m	20 Hz	~€500	Penetrates dust, fog, and the sandstorms the LIDAR point cloud degrades under — independent physics for a true disagreement vote
1 × stereo camera pair	Object classification	0 – 100 m	30 Hz	~€300	Distinguishes human / animal / debris; informs verdict severity, not primary safety

The suite is multi-sensor by design — obstacle detection is too consequential to rest on a single physics. A sensor-disagreement verdict is explicit in the evaluator (§10.1) and fail-restrictive.

5.1.1 Degraded-sensor speed policy

The evaluator exposes a dedicated `RestrictedSpeed` verdict so that partial sensor failures produce a smooth speed reduction instead of an EB:

Configuration	Verdict (no obstacles detected)	Effect
Ultrasonic + LIDAR + radar all healthy	Clear	Full MA-permitted speed
LIDAR offline / stale · radar healthy	RestrictedSpeed	ATO caps at 40 km/h (the ultrasonic-safe envelope)
LIDAR healthy · radar offline	Clear	Full speed — radar is a validation channel, not primary
Both long-range offline · train ≤ 40 km/h	RestrictedSpeed	ATO holds ≤ 40 km/h
Both long-range offline · train > 40 km/h	EmergencyBrake	No coverage beyond ultrasonic band at this speed — full EB
Any ultrasonic channel stale / faulted	EmergencyBrake	Ultrasonic is always required (close-range safety belt)
Peer 2o2 channel disagrees	EmergencyBrake	Fail-restrictive

The policy is asymmetric on purpose: LIDAR is the primary long-range channel; radar is its all-weather validator. Losing LIDAR restricts speed; losing radar (while LIDAR still sees 3D at 5–200 m) does not. Both gone EB if speed would outrun the ultrasonic envelope.

Why LIDAR failing caps at 40 km/h rather than EB. At 40 km/h the stopping distance is ≈ 18 m, inside the ultrasonic reliable band. The trainset keeps moving (avoiding a stranded-passengers incident) but at a speed that ultrasonic can safely cover if radar also fails in the next tick. Braking smoothly with the service brake to 40 km/h from mainline speed is a passenger-comfort decision that costs no safety — the remaining sensors still cover the revised stopping distance.

Why ultrasonic is primary for the close-range envelope. Ultrasonic transducers are the cheapest, simplest, and most failure-predictable sensor in the suite. A quadrant (four transducers with overlapping cones) costs ~€100, draws < 1 W, and has a failure mode that is detectable from its own echo pattern. For the platform-approach envelope (< 40 km/h, < 20 m stopping distance) it is sufficient on its own — which is why RFC 0015 makes it the primary safety belt for that speed band. Depot and workshop movements run permanently inside the ultrasonic band.

Why LIDAR is primary mid-range, not optional. The 2025– 2026 market shift in Chinese solid-state LIDAR put the Livox-class unit (150–300 m range, IP67, automotive-grade) into the €500–2 000 bracket — an order of magnitude below where it was

three years ago. At that price, LIDAR becomes the cheapest way to cover the 20 m – 200 m obstacle envelope with full 3D range data; it displaces radar as the primary mid-range sensor and pushes radar into the all-weather-validation role. Any target-region dust-storm event (see RFC 0013 S4.1) degrades LIDAR point density, which is exactly when the radar's mmWave penetration matters.

Why radar stays in the suite even with LIDAR promoted. The point of the multi-sensor architecture is that each sensor fails under a different physics. LIDAR fails in heavy dust or direct sun; radar fails on small stationary objects; ultrasonic fails at speed; cameras fail at night. A 2oo2-style safety vote across independent physics is stronger than any one sensor's specified SIL rating.

Why cameras are classifier-only. Cameras inform the severity of a detection (human vs. animal vs. debris) which feeds the verdict severity (`CrawlOnly` for a paper bag, `EmergencyBrake` for a person). They are not the safety-primary channel for detection – detection comes from ultrasonic / LIDAR / radar. Cameras are included at cheap cost (€300/stereo pair) because classifier data improves operational availability (fewer spurious EBs on windblown debris).

5.2 The T-OBS ECU

A new onboard host class: T-OBS (Train Obstacle-Detection ECU). Architecture mirrors T-ECU/S:

- 2 × Raspberry Pi RP2350 in a 2oo2 composite fail-safe voting arrangement, each running identical `no_std` Rust obstacle-detection code.
- 1 × Raspberry Pi CM5 app processor for sensor fusion and classifier inference (non-safety path).
- Inputs: 4 × ultrasonic (analog-in per channel), radar (CAN-FD), stereo camera (MIPI-CSI to CM5), LIDAR (Ethernet, optional).
- Outputs: a single `ObstacleVerdict` bus message published every 50 ms, evaluated into the main T-ECU/S 2oo2 chain as a brake-demand input. Verdict `Emergency EB` via the same 2oo2 AND-gate relay stage as the ATP.

Two T-OBS modules per trainset, one at each end. Only the leading end's verdict is active; the trailing module self-tests and publishes an `Inactive` status.

Detailed hardware spec lives at [hardware/t-obs/schematics/v2-spec/](#) (scaffold only; v3 KiCad capture per RFC 0007 v3 rollout).

5.3 Passenger emergency intercom

Every car carries at least 4 passenger emergency intercoms (one per car end, both doors). Pressing an intercom:

- 1 Triggers an immediate brake application to 50 % braking effort (not EB) – the trainset slows to a controlled stop at the next station.
- 2 Opens an audio + video channel to OCC remote-assist.
- 3 Logs the event to `osr-event-recorder` with timestamp, car id, and intercom id.

OCC can release the hold and resume normal operation remotely, or escalate to full EB if the situation warrants.

Why braking to the next station, not EB: unplanned EB in a driverless trainset with no driver to manage the aftermath strands the passengers between stations – a self-inflicted mass-incident risk. Braking controlled to the next station puts the passengers where they can be helped.

5.4 Platform screen doors upgraded to mandatory

In the GoA 2 model, PSDs were optional for `halt` archetype. In the GoA 4 default, PSDs are mandatory at every station with passenger boarding – no driver means no human last-line- of-defence between a passenger and the track. RFC 0010 v2.1 will reflect this.

Exception: depot-terminal platforms accessible only by staff do not require PSDs; staff training (M1) substitutes.

5.5 Enhanced CCTV coverage

Every car carries at least 4 cabin cameras, live to OCC. Platform cameras already exist per RFC 0010; coverage is now contiguous end-to-end so OCC can see any point in the system on demand.

6. Safety case

6.1 Equivalence claim

The driverless trainset achieves safety equivalence to GoA 2 operation with a trained driver through:

- SIL-4 obstacle detection replacing driver's-eye detection, with a lower false-negative rate on rail-relevant obstacle classes (validated against a 10 000-object test set during type certification).
- Lower reaction latency (radar-to-brake ~50 ms vs typical driver reaction of 1.5 s).
- No fatigue, no distraction, no panic. Driverless trains don't miss red signals because they were texting. The statistical argument is well-established in global metro operations (DLR, Copenhagen, Singapore NSL all GoA 4 with better safety records than GoA 2 fleets).

6.2 New SIL-4 claims for the safety case

RFC 0015 adds five safety goals to the *osr* - safety - case GSN tree:

Goal	Claim	Evidence
G15	<i>osr</i> -obstacle-detect produces Emergency verdict when any sensor reports an obstacle inside the current stopping-distance envelope	O1 Kani harness + 500-case proptest
G16	Sensor disagreement between the 2oo2 channels produces Emergency (fail-restrictive)	O3 Kani harness
G17	Stale sensor data (> 100 ms since last update) produces Emergency	O2 Kani harness
G18a	With every long-range sensor offline above 40 km/h, the verdict is EmergencyBrake	O4a Kani harness
G18b	With LIDAR offline (radar healthy), the verdict is at least RestrictedSpeed and never EmergencyBrake from the O4 branch alone	O4b Kani harness
G19	Passenger-intercom-triggered stop is delivered within 3 s of press	O5 integration test against <i>osr</i> - sim

6.3 Residual risks explicit

Driverless operation carries residual risks that GoA 2 mitigates via the driver. Each one is explicitly acknowledged:

- Obscured obstacle above sensor plane (e.g., a person standing upright but only the torso is visible above the platform edge): mitigated by PSD mandatory (§5.4) + ultrasonic under-body sweep at platform approach.
- Suicide on track: the most common rail-fatality mode in the industry, driverless or not. Not uniquely worsened by driverless; PSDs eliminate the station-platform variant entirely.
- Slipping rail, wheelspin: detected by *osr* - odometry slip-flag, triggers EB. No driver judgement involved.
- Terrorism / sabotage on track: track-intrusion detection at wayside (out of RFC 0015 scope; RFC 0016 candidate).
- Passenger medical emergency mid-journey: intercom to OCC, OCC dispatches medical at next station. GoA 2 driver would radio the same dispatcher.

6.4 Certification pathway

Type certification to EN 62267 (UGTMS — Urban Guided Transport Management + Command/Control Systems) at GoA 4 is the target. Each deployment additionally certifies to the local regulator (NRSA equivalent).

The certification package draws from the existing safety-case tree plus the five new goals in §6.2.

7. Operations implications

7.1 Driver rulebook (RFC 0013 §4.1) deprecated for GoA 4

Every rule in D1–D8 is either:

- Automated (depot power-on D1.1 onboard sequence), or
- Transferred to OCC (emergency plunger D7.1 intercom to OCC via §5.3), or
- Transferred to station staff (dwell supervision D4.4 T2.3).

RFC 0013 is amended to mark D1–D8 as "GoA 2 historical reference; retain for mixed-fleet operations only."

7.2 OCC role expands

The OCC gains two new staff roles (amendment to RFC 0013 §4.5 C1):

- Remote-assist operator. Handles passenger intercom calls, takes manual override of a stopped trainset if needed, and coordinates with emergency services for on-board medical events. One operator per 5–10 trainsets in service.
- Fleet-health supervisor. Monitors the live CCTV + sensor feeds for trainset anomalies (smoke, unusual passenger behaviour, unattended items) that a driver would previously have noticed.

7.3 Station staff role expands

Station-staff T2 (Passenger boarding) gains two additional responsibilities:

- Platform dispatch ready-check. Staff confirm PSD closed + platform clear before the dispatch signal is released. This substitutes for the driver's final platform check (D4.5).
- Boarding assistance. Staff are now the sole human presence during boarding; wheelchair assist, passenger inquiries, and boarding-rate management all sit at platform level.

7.4 No loss of safety personnel — they move where risk is

The driverless pivot does not reduce safety staffing. The heads-per-hour shifted from driver-in-cab to OCC + platform: same people, different posts, generally safer posts.

8. Rolling-stock implications (amends RFC 0008)

8.1 Cab eliminated

- Front wall: just a nose cone with the §5.1 sensor cowl + a coupler interface.
- No windscreen, no DMI, no master controller, no cab door, no cab HVAC, no rear-view mirror.
- Both ends of the trainset are symmetric — no leading / trailing distinction at the rolling-stock level.

8.2 Recovery-mode lockable cabinet

Every trainset retains a recovery-mode keyswitch cabinet — a steel-locked enclosure behind the nose, accessible only with a physical key held by depot recovery crew. Inside:

- A wired pendant with: forward / reverse select, 0–15 km/h throttle, emergency stop.
- The keyswitch enables a slow-speed (≤ 15 km/h) manual move that ignores ATO / ATP — used only to push a stuck trainset off a block for rescue-coupling (D7 / M6).

This is the only manual-control path. It is physical, locked, slow-speed, and single-purpose. No full cab required.

8.3 Floor plan extends to both ends

Passenger seating extends to the full car length. The additional $\sim 3\text{ m} \times 2.4\text{ m}$ of floor at each end of the lead + rear cars = about 14 extra seats per trainset on the light-metro-3car family.

8.4 Mass + capex savings (per trainset)

Rough figures for the light-metro-3car reference:

Item	Saved mass (kg)	Saved capex (€)
Cab structure (front wall, door, windscreen)	~800	~30 000
Cab controls + DMI + wiring	~150	~80 000
Cab HVAC	~120	~15 000
Cab seat + interior trim	~80	~8 000
Mirror / wiper / horn package	~40	~4 000
Cab lighting + accessories	~20	~3 000
Total	~1 210 kg / cab × 2 cabs	~140 k × 2
Added (T-OBS + sensors + intercoms)	~60 kg × 2 ends	~25 k × 2
Net	~2 300 kg saved	~230 k saved

9. Hardware implications (amends RFC 0007)

A new host class T-OBS joins the palette:

Class	SoC	Role
T-ECU/S	2× RP2350 + RPI CM5	Train safety kernel (ATP, brake, etc.)

Class	SoC	Role
T-ECU/A	RPI CM5	Train application (TCMS, PIS, TCN)
T-OBS	2x RP2350 + RPI CM5	Obstacle detection + sensor fusion
W-SBC	Radxa CM5	Wayside
S-SBC	RPI CM5	Station

T-OBS is two modules per trainset (one at each end). Architecture and PCB family are re-used from T-ECU/S; only the carrier board (sensor breakouts, CAN-FD transceiver, analog-in front-end, MIPI-CSI camera headers) differs.

10. Software implications (amends RFC 0005)

10.1 New crate

`osr-obstacle-detect` — pure-function SIL-4 evaluator. Runs on T-OBS. Takes a `SensorFrame` (ultrasonic ranges × 4, LIDAR return list, radar return list, camera classifier output), current train speed, current Movement Authority end, and the peer-channel clear bit, and returns an `ObstacleVerdict`:

```
enum ObstacleVerdict {
    Clear,
    RestrictedSpeed, // AT0 caps at 40 km/h (LIDAR degraded)
    CrawlOnly,       // AT0 caps at 15 km/h (soft obstacle)
    EmergencyBrake,
}
```

Verdicts are strictly ordered by severity: `Clear` < `RestrictedSpeed` < `CrawlOnly` < `EmergencyBrake`. A detection inside the envelope can only escalate severity; a clearer sensor frame can only de-escalate.

Five SIL-4 safety properties (O1, O2, O3, O4a+O4b, O5) mapped to Kani harnesses; proptest coverage across the same properties.

10.2 Crates demoted to "GoA 2 only"

- `osr-vigilance` — kept in tree, marked optional.
- `osr-dmi` — kept in tree, marked optional. Passenger display functions already split out to `osr-pis-onboard` so this crate is truly cab-only.

A workspace feature flag `goa2-cab` (additive, opt-in) is the canonical switch for legacy fleets. The new [osr-trainset-image](#) integrator crate is the single entry point: it re-exports the always-on onboard stack (ATP, ATO, brake, derailment, door-control, fire-safety, obstacle-detect, odometry, TCMS) and gates `[osr-dmi]` + `[osr-vigilance]` behind `--features goa2-cab`. `cab_profile()` returns `CabProfile::Unattended` in the default build and `CabProfile::Cabbed` when the flag is enabled — a compile-time witness for the RFC 0015 "GoA 4 by default" claim.

11. Rollout

Phase	Deliverable	Dependencies
v0	This RFC ratified	—
v1	<code>osr-obstacle-detect</code> crate at crates/osr-obstacle-detect/ — SIL-4 pure-function evaluator with O1/O2/O3/O4a/O4b/O5 Kani harnesses + 8 proptests (26 tests total); GSN safety-case goals G15–G19 at docs/safety-case/gsn/60-obstacle-detect.toml close against real Kani + proptest evidence (done 2026-04-22).	v0
v2	<code>osr-sim</code> shadow onboard stack now calls <code>osr_obstacle_detect::evaluate</code> at every tick: each <code>OnboardShadow</code> carries an <code>obstacle_out</code> field, feeds an all-clear synthetic <code>SensorFrame</code> into the evaluator, and threads <code>ObstacleVerdict::EmergencyBrake</code> through <code>BrakeInputs::obstacle_emergency</code> into the existing emergency-source union. Per-verdict tick counters roll up in <code>OnboardSummary::total_obstacle_{restricted,crawl,emergency}_ticks</code> ; the simulator summary prints them under "Onboard shadow stack". <code>BrakeInputs</code> , <code>EmergencySources</code> (in both <code>osr-brake</code> and <code>osr-tcms</code>), and the brake evaluator carry <code>obstacle_emergency</code> end-to-end. v2.1 (2026-04-23): scenario-driven sensor-fault injection — four new <code>FaultKind</code> variants (<code>LidarOffline</code> , <code>RadarOffline</code> , <code>UltrasonicChannelStale</code> , <code>ObstaclePeerDisagreement</code>) with per-train + fleet-wide scope; parser in <code>scenario_file.rs</code> ; a demonstrator fault-injection scenario exercises every O-series path through the shadow stack. Run confirms <code>RestrictedSpeed</code> / <code>EmergencyBrake</code> verdicts fire and flow through the emergency-source union without invariant violations. (done 2026-04-23)	v1

Phase	Deliverable	Dependencies
v3	T-OBS v2 schematic specification at hardware/t-obs/schematics/v2-spec/ — block diagram + per-rail power budget + safety-nets with 2oo2 AND-gate design + RP2350 A/B pinouts + 12-entry M12 connector table (done 2026-04-22). KiCad capture is v3.1, deferred alongside the RFC 0007 hardware KiCad rollout.	v0
v4 (v1)	Type-certification pre-submission pack at docs/certification/ — README + system-description + safety-requirements (SR-01..SR-24) + hazard-log (17 hazards, 7 classes) + evidence-register + EN 62267 clause-by-clause compliance-matrix. Every SR cross-referenced to a Kani harness / proptest / GSN goal / rulebook rule. Remaining for v4.1 (open): residual-risk narrative, per-clause compliance prose for §5–§9, and independent-assessor review (deployment-partner scope). (done 2026-04-23)	v2, v3
v5	First driverless revenue service at Samawah	RFC 0003, v4

12. Relationship to existing RFCs

- RFC 0003 (Samawah) — Samawah becomes a GoA 4 deployment from day one.
- RFC 0005 (SBC software architecture) — crate map amended in §10.
- RFC 0007 (hardware reference designs) — new T-OBS class in §9.
- RFC 0008 (rolling stock) — cab elimination in §8.
- RFC 0010 (station design standard) — PSDs become mandatory (§5.4); v2.1 amendment.
- RFC 0013 (operations rulebook) — driver section deprecated, OCC and station staff absorb (§7).

13. What this RFC does NOT include

- Automated track-intrusion detection at wayside — covered by [RFC 0016](#).
- Autonomous platform-side boarding (still a human station-staff function; automation would require RFC 0015.1).
- Retrofit path for legacy cabbed trainsets — if an operator already has GoA 2 fleet, the retrofit cost is likely higher than just running it out to end-of-life alongside a new GoA 4 fleet.
- Autonomous depot shunting under GoA 4 (treated as a separate RFC scope because the obstacle-detection envelope at depot speeds is different).

docs/rfcs/0016-wayside-track-intrusion.md

docs/rfcs/0016-wayside-track-intrusion.md

RFC 0016 — Wayside track-intrusion detection

Status: accepted · 2026-04-23. Authors: OSR project. Complements: [RFC 0015](#) (onboard obstacle detection). Amends: RFC 0005 (crate map), RFC 0007 (hardware palette adds one SKU variant), RFC 0001 (interlocking consumes intrusion state).

1. Purpose

Driverless trains (RFC 0015) detect obstacles in their own path with the onboard `osr-obstacle-detect` evaluator. This RFC complements that with wayside track-intrusion detection — fence-line + ROW-mounted sensors that detect people, animals, debris, or vehicles entering the track envelope before a train arrives.

The goal is not redundancy with the onboard detector. The two systems serve different safety arguments:

- Onboard detector sees what's in front of the moving train, reactively — once something enters the current envelope, brake hard.
- Wayside detector sees what's in the ROW at any time, proactively — if the section is not clear, the interlocking withholds Movement Authority and the train never arrives at the obstacle.

Wayside detection is what lets the interlocking say "this section is safe" rather than "this section was safe the last time a train passed through." In GoA 4 operation, with no driver to see irregularities between trains, this is a necessary safety property — not a nice-to-have.

2. Why now, not in RFC 0015

RFC 0015 §13 explicitly listed "automated track-intrusion detection at wayside" as a future RFC candidate. The two RFCs partition cleanly:

- RFC 0015 owns the onboard sensor suite, the T-OBS ECU, and the driverless rolling-stock profile.
- RFC 0016 owns the wayside sensor suite, a new `osr-intrusion-detect` evaluator on the W-SBC host class, and the feed into `osr-interlocking` that withholds MA on intrusion-flagged sections.

Splitting keeps each RFC focused on one SIL-4 evaluator + one host class, matching the RFC-per-crate-family pattern of the project.

3. Scope

In scope:

- Wayside sensor-suite spec (fence contact, ROW LIDAR, ROW radar, CCTV AI).
- `osr-intrusion-detect` evaluator: per-section `IntrusionVerdict` (Clear / Unknown / Present), with fail-restrictive on Unknown.
- Integration with `osr-interlocking`: intrusion state is a new gating input alongside switch-position and block-occupancy.
- Operations amendments: how the dispatcher handles a section under active intrusion.

Out of scope:

- Fencing construction (deployment-partner civil scope).
- Video-analytics AI model training (training data is deployment-specific; the classifier runs as a black box per §5.3).
- Long-range perimeter surveillance outside the ROW (deployment partner's security concern, not rail safety).

4. What the wayside sensor suite looks like

Per-section sensor pack (one pack per [SEC] id in the interlocking). Installed at ROW junction boxes; each junction box feeds a W-SBC running `osr-intrusion-detect`.

Sensor	Role	Typical unit	Unit price (2026)	Why
Fence-line contact	Perimeter breach	Vibration + cut-wire sensor (e.g., Senstar FlexZone)	€300–500 / 100 m	Instant alarm on fence cut or climb; mature tech, low false-positive rate
ROW LIDAR (pole-mount, every ~200 m)	3D silhouette in ROW	Livox Mid-360 (360° FoV, 70 m range) or equivalent	€800–1 200	3D presence detection, day/night, IP67 industrial – same supply-chain story as RFC 0015 §5.1
ROW radar (pole-mount, every ~500 m)	All-weather presence	TI AWR1843 or eq. 77 GHz	~€500	Penetrates fog/dust that degrades LIDAR; redundant physics
CCTV with AI classifier	Classification + human oversight	Commodity 4 K IP camera + edge TPU (Google Coral or eq.)	€400–800 per camera	Per-station already required by RFC 0010; AI classification reuses existing cameras

Sensor spacing is driven by section length (RFC 0001):

- at-grade sections: LIDAR every 200 m, radar every 500 m, fence-line continuous.
- elevated sections: LIDAR every 200 m; fence-line not applicable (viaduct edge is the barrier); radar optional.
- bridge sections: LIDAR at each end + mid-span; no fence-line applicable.

A typical standard-urban 1 km at-grade section carries: 5 LIDAR + 2 radar + 1 CCTV + 1 km fence-line. Total hardware cost ≈ €7 000 per section – small fraction of the €500 000+ per km civil cost.

5. The osr-intrusion-detect evaluator

5.1 Pure-function API

```
pub fn evaluate(
    frame: &WaysideSensorFrame,
    now_ns: u64,
    params: &IntrusionParams,
) -> IntrusionOutcome;
```

Inputs:

- WaysideSensorFrame – one tick of sensor data per section: fence-line state, LIDAR detection list (range + lateral), radar detection list, camera classification + confidence.
- now_ns – sim / system time for staleness checks.
- IntrusionParams – per-deployment thresholds (lateral rail-profile width, freshness timeout, classifier confidence floor).

Output:

- IntrusionVerdict – Clear / Unknown / Present.
- TriggerReason – which sensor + which class, for logging.

5.2 The three verdicts

- Clear – the section has no detected intrusion. MA may cross this section.
- Unknown – at least one safety-primary sensor is stale or offline, and no Present detection is confirmed. MA is withheld (fail-restrictive). This is the crucial property: a broken sensor does not imply safe track.
- Present – an intrusion is confirmed by at least one sensor of sufficient confidence. MA is withheld; dispatcher is alerted.

5.3 Safety properties (I1–I5)

Paralleling the O-series from RFC 0015:

Property	Claim
I1	Any LIDAR detection inside the ±1500 mm rail profile Present.
I2	Any sensor frame stale beyond MAX_SENSOR_STALE_MS Unknown (not Clear).
I3	Fence-line breach Present regardless of other sensor state.
I4	Camera classifier alone (no LIDAR / radar corroboration) cannot emit Clear – the safety-primary physics must be alive.
I5	Refreshing a sensor never moves the verdict in a less-restrictive direction (monotone under freshness).

Each property anchors a Kani harness + a proptest – same pattern as the O-series.

5.4 Integration with osr-interlocking

A new `SectionIntrusion` entry on the consensus log carries the verdict per section.

`osr-interlocking::section_available_to` reads this alongside the existing gates (block occupancy, switch position) and refuses to grant MA across a section whose latest `SectionIntrusion` is not `Clear`.

This is a v2 deliverable of this RFC; v1 scaffolds the evaluator and a first Kani suite.

6. Hardware implications

`osr-intrusion-detect` runs on the W-SBC (Radxa CM5 RK3588S) per RFC 0007. No new host class — the W-SBC already hosts `osr-interlocking` and `osr-wayside-points`, so adding `intrusion-detect` is a same-crate-family deployment.

Baseboard additions per junction box:

- 8× fence-line contact inputs (optically isolated).
- 2× CAN-FD (radar + spare).
- 1× 1000BASE-T to LIDAR.
- 1× MIPI-CSI to CCTV edge-TPU (or Ethernet if the TPU is separate).

No change to the T-ECU/S or T-OBS boards — this is purely a wayside addition.

7. Operations implications

7.1 Dispatcher (S rulebook amendment)

A new rule group `S7` — Intrusion response:

- `S7.1` On first `Present` verdict, the dispatcher holds the section and dispatches track-patrol within 10 min.
- `S7.2` On `Unknown` verdict persisting > 5 min, the dispatcher treats it as a functional failure of wayside equipment and opens a work ticket; trains are held or diverted.
- `S7.3` A section cleared by track-patrol and returned to `Clear` may re-enter revenue service.

7.2 Maintenance (M rulebook amendment)

- `M7` Wayside sensor maintenance: 30-day walk-through inspecting each LIDAR/radar housing, cable runs, fence contacts. Same cadence as `M3` track inspection.

Rulebook text added when RFC 0016 v2 lands.

8. Rollout

Phase	Deliverable	Dependencies
v0	This RFC ratified	—
v1	<code>osr-intrusion-detect</code> crate at crates/osr-intrusion-detect/ — pure-function evaluator with 5 Kani harnesses + proptest properties I1–I5. GSN safety-case file at docs/safety-case/gsn/70-intrusion-detect.toml closes against real evidence (done 2026-04-23).	v0
v2	Integration with <code>osr-interlocking</code> : new <code>EntryPayload::SectionIntrusion</code> consensus log entry + <code>IntrusionState</code> enum (<code>Clear/Unknown/Present</code>); <code>DerivedState.section_intrusions</code> folds the latest verdict per section; <code>section_available_to</code> gate (d) consults the verdict — <code>Present/Unknown</code> withhold MA, <code>Clear</code> permits, missing entry = "not instrumented" (backwards-compatible). 5 unit tests exercise every path including "latest verdict wins". (done 2026-04-23)	v1, RFC 0001
v3	Sim integration: new <code>FaultKind::WaysideIntrusion { section, state }</code> in <code>osr-sim::fault</code> ; the sim tick emits one <code>SectionIntrusion</code> consensus entry per active fault through <code>MaLogBackend::emit_intrusion</code> (both <code>SimulatedLog</code> and <code>ConsensusBackend</code> backends). A demonstrator scenario stages <code>Present / Unknown / Present</code> events across three sections during AM peak; run completes with zero invariant violations — the interlocking's gate (d) held MA back without any train entering the flagged sections. (done 2026-04-23)	v2

Phase	Deliverable	Dependencies
v4	Rulebook amendments – dispatcher S7 at docs/operations/dispatcher/s7-intrusion-response.md (5 rules: S7.1 track-patrol, S7.2 sustained-Unknown, S7.3 return-to-service, S7.4 multi-section escalation, S7.5 no-override) and maintenance M7 at docs/operations/maintenance/m7-wayside-sensor-maintenance.md (5 rules: M7.1 30-day walk, M7.2 post-weather, M7.3 sustained-Unknown ticket, M7.4 replacement calibration, M7.5 fence continuity test). RFC 0013 §4.2 and §4.4 indices updated. (done 2026-04-23)	v2
v5	Deployment at Samawah: first-article sensor packs at 3 sections on Line 1	RFC 0003, v2

9. Relationship to existing RFCs

- RFC 0001 (consensus) – adds `SectionIntrusion` log entry.
- RFC 0005 (software architecture) – adds `osr-intrusion-detect` to the W-SBC crate map.
- RFC 0007 (hardware) – junction-box baseboard adds sensor headers; no new host class.
- RFC 0013 (operations) – S7 + M7 rule groups (v2+).
- RFC 0015 (driverless) – complements onboard obstacle- detect; the two-layer defence (wayside proactive, onboard reactive) is the full GoA 4 safety-envelope story.

10. What this RFC does NOT include

- Passive fence line without contact-sensor instrumentation – standard civil scope.
- Personnel-identification against a whitelist (crew vs. intruder) – handled by CCTV + human review at the OCC, not automated in this RFC.
- Anti-climb / anti-cut physical deterrents on the fence – civil scope per deployment-partner security spec.
- Section-level live-radar tracking of authorised track workers (under a `MaintenanceOverride`, RFC 0013 S5 rules apply; the intrusion-detect evaluator flags `Present` and the dispatcher cross-references against the active work-block manifest).

docs/rfcs/0017-cybersecurity-message-authentication.md

docs/rfcs/0017-cybersecurity-message-authentication.md

RFC 0017 — Cybersecurity: message authentication on the safety log

Status: accepted · 2026-04-23. Authors: OSR project. Complements: [RFC 0001](#) (consensus), [RFC 0005](#) §4.9 (crate map). Amends: `osr-crypto` (adds `ed25519`), `osr-interlocking` (envelope on `Entry`), consensus wire format.

1. Purpose

Every entry on the track-state consensus log carries safety-critical authority — a forged `MaintenanceOverride` or `SectionIntrusion::Clear` could unlock a blocked section and put a train in front of a work crew. Today the protobuf schema reserves a `leader_signature` field but the code path that would verify it is not implemented: entries are trusted because they came from a running process in tree.

This RFC closes that gap. Every entry on the committed log is signed by the originating entity (train, wayside controller, dispatcher console) and every consumer verifies the signature before applying the entry to derived state.

2. Threat model

In scope:

- Forged entries. An attacker with network access to the consensus cluster proposes a crafted `SectionIntrusion::Clear` or a fake `SwitchObservation` hoping the interlocking accepts it.
- Replay. An attacker re-submits an old legitimate entry to confuse the derived state (e.g., replay a `RouteRelease`).
- Key compromise. A single wayside box is physically captured; its key is now under the attacker's control.

Out of scope for v1:

- Side-channel attacks on the signing key (handled at hardware level by the ATECC608B trust anchors on T-ECU/S, W-SBC, S-SBC).
- Quantum-era attacks. Ed25519 is 128-bit-classical-secure; a post-quantum migration is an RFC 0017.1 candidate once NIST finalises a PQ signature scheme in the rail-relevant performance bracket.
- TLS session setup on TRG and backhaul links — that's RFC 0018 (not yet drafted).

3. Design

3.1 Signature scheme

Ed25519 — curve25519 in the twisted-Edwards form, per RFC 8032.

- 64-byte signatures, 32-byte keys. Small enough to attach to every consensus entry without bloating the wire format.
- Batch verification available — a consensus cluster can verify N entries in sub- N time when under load.
- Deterministic signatures — no per-signature nonce required (eliminates the ECDSA Sony-PS3 class of bugs).
- Open-source reference implementations — `ed25519-dalek` in Rust is widely audited and actively maintained.

3.2 Entity keys

Every entity (train, wayside box, station, OCC console) holds one long-lived ed25519 keypair in its ATECC608B secure element. Public keys are published in a per-deployment key registry at bootstrap; private keys never leave the secure element.

Key ids mirror the existing `EntityId` scheme so the `Entry.leader_signature` maps one-to-one to `Entry` authorship.

3.3 Envelope

A signed entry is encoded as:

```
SignedEntry {  
  entry_bytes: Vec<u8>, // serde-json serialisation of Entry  
  issuer: EntityId,      // which entity signed  
}
```

```
    sig: [u8; 64],          // ed25519 signature over entry_bytes
}
```

Verification:

- 1 Look up issuer in the KeyRegistry. If absent reject (`VerifyError::UnknownIssuer`).
- 2 Verify sig over entry_bytes with the registry's public key. If invalid reject (`VerifyError::BadSignature`).
- 3 Otherwise return OK, and only then deserialize and apply the entry.

Verification is before deserialisation so a hostile payload can't exploit a parser bug before the signature check.

3.4 Replay protection

Each Entry already carries entry_id (monotonic) and timestamp_ns (PTP-synced). The derived-state fold is idempotent in entry_id: an entry already applied is a no-op when re-applied. Replay does not trick the derived state into a bad configuration.

For completeness, the consensus layer additionally rejects entries whose timestamp_ns is more than 60 seconds older than the cluster's current leader clock — a hard freshness check.

3.5 Key rotation (v2)

v1 treats keys as long-lived. v2 introduces a KeyRotation entry type that transfers authority from an old key to a new one; the registry follows the latest rotation. Deferred because it adds operational complexity that is not needed for first-article deployment.

4. What this RFC delivers (v1)

- osr-crypto extended with ed25519 sign + verify primitives.
- osr-secbus — new SIL-2 crate with KeyRegistry, SignedBytes envelope, and the verification API.
- Three SIL-2 properties (S1–S3) with Kani harnesses + proptest.
- GSN goals G25–G27 on the safety case, anchored to the properties.

5. Safety properties (S1–S3)

- S1 (deterministic verify): `verify(pk, msg, sig)` is a pure function — identical inputs identical output.
- S2 (reject bad signature): any single-bit mutation of sig causes `verify` to return `false`.
- S3 (reject missing issuer): a SignedBytes whose issuer is not in the KeyRegistry rejects with `VerifyError::UnknownIssuer`.

6. Rollout

Phase	Deliverable	Dependencies
v0	This RFC ratified	—
v1	osr-crypto ed25519 primitives + osr-secbus crate with key registry, envelope, verify; S1–S3 Kani + proptest; GSN G25–G27 close. (done 2026-04-23)	v0
v2	Integrate into the consensus wire layer — every entry proposed to the cluster is signed; every entry committed is verified; unverified entries are dropped by the receiver before they hit derived state	v1, RFC 0001
v3	Sim integration: fault-injection for forged / tampered entries, demonstrating they're rejected end-to-end	v2
v4	Key rotation (KeyRotation entry type) + deployment tooling to mint and distribute keys	v2
v5	Hardware integration: ATECC608B provisioning pipeline; per-entity key generation + transfer at factory	v2, RFC 0007

7. Relationship to existing RFCs

- RFC 0001 (consensus) — adds the verify step between receive and apply; no changes to Raft itself.
- RFC 0005 (software architecture) — osr-secbus joins the Phase 2f infrastructure crate family alongside osr-crypto and osr-historian.

- RFC 0007 (hardware) – the ATECC608B role is already in the T-ECU/S + T-OBS + W-SBC specs; this RFC defines what that chip actually signs.
- RFC 0015 + RFC 0016 – every `SectionIntrusion` and `MaintenanceOverride` is now signed by its issuing W-SBC (RFC 0016) or OCC console (RFC 0013 S5). Prevents the forged- `Clear-verdict` attack that would otherwise defeat the safety gate.

8. What this RFC does NOT include

- TLS on TRG radios or OCC backhaul – RFC 0018 candidate.
- Encryption of the log at rest – the entries are integrity- protected; confidentiality is not claimed.
- Perfect-forward-secrecy session keys – long-lived ed25519 keys are sufficient for entry authentication; PFS is a session-layer concern (RFC 0018).
- Post-quantum migration – deferred pending NIST standardisation in a performance bracket suitable for embedded rail controllers.

docs/rfcs/0018-operator-guis.md

docs/rfcs/0018-operator-guis.md

RFC 0018 — Operator GUIs: simulator + operations control

Status: accepted · 2026-04-23. Authors: OSR project. Complements: [RFC 0003](#) (sim scenarios), [RFC 0013](#) (OCC procedures), [RFC 0017](#) (entry authentication).

1. Purpose

Two human-facing GUIs that have been implicit in the project but not built:

- Simulator GUI (`osr-sim-gui`) — runs `osr-sim` against a scenario TOML, renders the network + trains + MAs + faults live, and lets a designer inspect any entity by click-through. The audience is the deployment engineer validating their `design.toml` before commissioning, and the OSR developer verifying new scenarios.
- Operations-control GUI (`osr-occ-gui`) — the dispatcher console that RFC 0013 §4.2 has always assumed exists. Renders the live network state (from the consensus log), shows every RFC 0013 alert (S3 incident categories, S4 weather overlays, S7 intrusion verdicts), and exposes the specific actions the rulebook allows a dispatcher to take (route grant, `MaintenanceOverride`, degraded-mode transition).

2. Non-goals

- Passenger-facing displays. PIS rendering is `osr-pis-onboard`
 - `osr-pis-station` — separate codepath, separate build.
- Maintenance workbench GUI. CBM dashboards are a v2 scope of `osr-cbm-backend`, not this RFC.
- Mobile apps. Neither GUI needs to run on a phone.
- 3D rendering. 2D network schematic is sufficient — legacy metros have proved that.
- Legacy CBTC operator-console emulation. OSR operates its own consensus log; there's no back-compat requirement.

3. Technology stack

egui (+ eframe) in pure Rust.

Why:

- Type-consistent with the rest of the workspace. One language from SIL-4 kernel to dispatcher console; fits the project's ethos of no-JavaScript-stack, no-proprietary-SCADA.
- Immediate-mode rendering matches the per-tick update cadence of the sim + consensus log — redraw cheaply, don't reason about widget state machines.
- Works natively + WebAssembly. A deployment partner can host the OCC GUI as a web app for remote supervision without rewriting it.
- Conduction-cooled SBC friendly. egui renders at 10 Hz on a 400 MHz single-core ARM — not that we need that, but it guarantees the OCC GUI runs on any reasonable workstation.
- No Electron. An Electron OCC app would pull in 200 MB of Chromium; egui's eframe binary is ~30 MB static-linked.

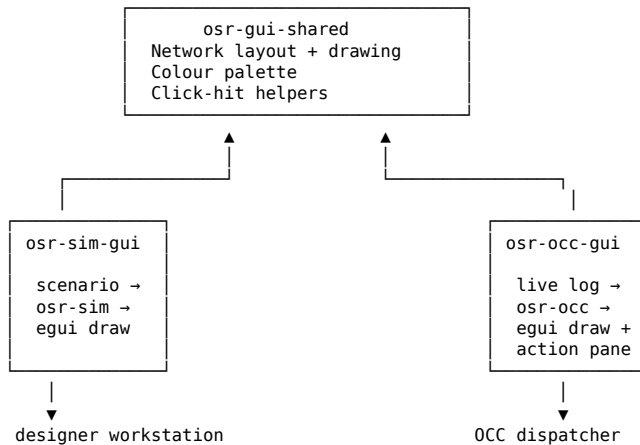
Rejected alternatives:

- React/Vue + Tauri — excellent for complex UIs, but adds a full JavaScript toolchain to the build + another set of security primitives to audit. Tauri remains a fallback if egui runs into a specific limitation.
- Iced / Slint — also pure Rust, but egui's immediate-mode model is a better fit for the "redraw the whole network each tick" workload than retained-mode widget trees.
- GTK / Qt — C/C++ deps outside the project's sandbox.

4. Architecture

Three new crates:

- `osr-gui-shared` — library. Network-layout + drawing primitives + colour palette. Consumed by both GUI binaries so the two have a consistent look. Not intended for direct use outside the GUIs.
- `osr-sim-gui` — binary. Wraps `osr-sim` + `osr-gui-shared`. Loads a scenario TOML, runs the sim per-tick in the same thread as the UI, renders the state each frame.
- `osr-occ-gui` — binary. Wraps `osr-occ` + `osr-gui-shared`. Reads from a consensus-log source (either a SimulatedLog file replay or a live ConsensusBackend), renders the latest derived state, exposes dispatcher actions via modal dialogs that emit signed consensus entries (RFC 0017).



5. Shared rendering — `osr-gui-shared`

Scope:

- `NetworkLayout` — per-line horizontal strip, stations placed along the strip by cumulative distance. Same layout strategy as the existing HTML `osr-vis`, reused so dispatchers
 - designers see an identical schematic.
- `draw_network / draw_train / draw_section_state` — egui paint helpers that take a `Painter`, a `NetworkLayout`, and the current state to render.
- `Palette` — colours for every train phase, section state, intrusion verdict, obstacle verdict, fault state.

6. `osr-sim-gui`

Minimum viable feature set:

- Scenario loader. Filebrowser pick a `scenarios/*.toml` load.
- Playback controls. Play / pause / step / speed (1x, 10x, 100x).
- Network render. Samawah two-line schematic with every train drawn as a dot at its current `station_m`, colour-coded by phase.
- Inspector pane. Click a train sidebar shows ATP verdict, brake cmd, SoC, obstacle-detect verdict.
- Event log. Scrolling list of `EventKind` emissions — Dispatched, ArriveStation, DepartStation, etc.
- Fault indicators. Active faults from the scenario's `[[ faults ]]` section render as coloured badges on the affected station / section / train.

Post-v1 (deferred):

- Timeline scrubber for replay
- Record / export runs
- Differential compare of two scenarios side-by-side
- Remote-monitor mode (read the sim from another host)

7. `osr-occ-gui`

Minimum viable feature set for v1:

- Log-source selector. Pick a file replay (for exercises) or connect to a live ConsensusBackend.
- Network render. Same layout as the sim GUI.
- Section-state panel. SectionIntrusion verdicts from RFC 0016, colour-coded; click a section to see the latest MaintenanceOverride, RouteGrant, or SectionIntrusion on that section.
- Train panel. List of every train under the dispatcher's control, with the same colour coding as the sim GUI.
- Alert feed. Intercom presses, EB events, weather overlays, CBM red alerts.
- Action modal stubs. The four dispatcher actions from RFC 0013 §4.2 (issue route grant S2.1, commit MaintenanceOverride S5.1, extend dwell S2.3, declare degraded mode) exist as modal dialogs that validate the input but emit no-ops for v1.

Post-v1 (deferred):

- Live command emission to the consensus cluster with RFC 0017 signed envelopes
- Multi-operator role partitioning (dispatcher vs supervisor vs engineer-on-call) with per-role action gates
- Full audit-log display
- Video feed from station CCTV + cabin CCTV (RFC 0015 §5.5)

8. Security model

The OCC GUI is a privileged actor. Its action emissions must be signed by the logged-in dispatcher's personal key (RFC 0013 S1.1, RFC 0017). v2 of this RFC wires the signing key OCC UI login flow; v1 is read-only so the security model is trivially "read-only view of committed log."

9. Rollout

Phase	Deliverable	Dependencies
v0	This RFC ratified	—
v1	osr-gui-shared network renderer + osr-sim-gui + osr-occ-gui scaffolded; both compile cleanly in the workspace, open an egui window, render the Samawah network, and expose the v1 minimum feature set listed above.	v0
v2	osr-sim-gui full sim integration (live tick stream, click-to-inspect)	v1, RFC 0003
v3	osr-occ-gui live consensus integration (read + write via RFC 0017 signed envelopes)	v1, RFC 0017 v2
v4	WebAssembly builds for remote access to both GUIs	v2, v3
v5	First Samawah deployment uses osr-occ-gui in revenue service	RFC 0003, v3

10. What this RFC does NOT include

- PIS (passenger-facing) displays. Separate osr-pis-* crates.
- CBM / maintenance dashboards. Separate osr-cbm-backend UI.
- Electron / web-framework-native versions. egui's WASM target covers the "browser access" use case.
- 3D rendering. Schematic 2D is explicitly the goal.
- Mobile apps. Not a target deployment form factor for either audience.

RFC 0019 — DIY plug-and-play electronics

Status: accepted · 2026-04-23. Authors: OSR project. Amends: [RFC 0007](#) (hardware palette). Complements: [CHANGELOG.md](#).

1. Purpose

The RFC 0007 v2-spec boards (T-ECU/S, T-ECU/A, T-OBS, W-SBC, S-SBC) assume a custom PCB captured in KiCad and fabricated somewhere in the target deployment footprint. That's fine for volume production, but it is a real barrier for:

- A pilot deployment that needs five trainsets, not fifty — the NRE of designing + fabricating custom boards dominates the total cost.
- A developing-nation DIY operator — a city authority in (say) the Sahel that wants to run its own metro but doesn't have a PCB-design in-house capability.
- Early academic / community prototypes that need to run the safety chain on commodity hardware before investing in silicon.

This RFC defines a DIY plug-and-play electronics path for all five host classes using commercially-available modules (Raspberry Pi Foundation boards, commodity HATs, off-the-shelf relay + sensor breakouts). No custom PCB required for any host class. Wiring is terminal-block + ribbon cable; the assembly-time tool is a screwdriver.

The trade-off is higher per-unit cost (roughly 1.5–2× the custom- PCB BOM at volume) and slightly larger enclosure footprint. For pilot deployments that's the right trade — the 10× saving on NRE dominates.

2. Design principles

- 1 Off-the-shelf only. Every component has a public SKU at ≥ 2 global distributors (Mouser / DigiKey / Adafruit / SparkFun / Waveshare / AliExpress-commodity).
- 2 No soldering required on the assembly path. Every connection is a terminal block, ribbon header, or USB/Ethernet cable.
- 3 DIN-rail mounting for every enclosed unit. Off-the-shelf DIN-rail Pi cases + sensor-module housings.
- 4 Single-SD-card boot for every compute module. The OSR project ships a pre-built SD-card image per host class — flash, insert, power, done.
- 5 Safety architecture preserved. 2oo2 composite fail-safe (two RP2350 + CM5) is kept; we just use commodity Pi Pico 2 boards (which carry the same RP2350 silicon as a custom design) instead of an RP2350 on a custom PCB.
- 6 Same software. The RFC 0015 / RFC 0016 / RFC 0005 crate map runs unchanged; only the board-level integration is commodity-module-based.

3. Host-class mapping — custom vs DIY

Host class	v2-spec custom PCB	DIY plug-and-play
T-ECU/S	Custom 160 × 100 mm 4-layer with 2× RP2350 + CM5	2× Raspberry Pi Pico 2 boards + RPi CM5 on official IO Board + relay HAT
T-ECU/A	Custom 160 × 100 mm with CM5 + radios	RPi CM5 IO Board + CAN HAT + 5G HAT + LoRa HAT
T-OBS	Custom 180 × 120 mm with 2× RP2350 + CM5 + sensor front-ends	2× Pi Pico 2 + RPi CM5 IO Board + ultrasonic breakouts + Livox LIDAR USB + TI radar eval board
W-SBC	Custom 180 × 130 mm industrial-temp Radxa	Radxa CM5 IO Board + isolated DI/DO HAT + CAN HAT + intrusion-sensor modules
S-SBC	Already commodity (RPi CM5 on IO Board)	No change — S-SBC is already DIY by design.

The custom path stays the recommended v2+ track for volume production. The DIY path is the recommended first-article / pilot track for any new deployment.

4. Core component catalogue

All pricing in USD, 2026 indicative retail. SKUs verified to exist with two or more global distributors.

4.1 Compute modules (universal across host classes)

Module	Role	SKU / source	Price
Raspberry Pi Pico 2	RP2350 safety MCU (2 per SIL-4 board)	Official Raspberry Pi Foundation, SC1630	\$5
Raspberry Pi CM5 (8 GB Lite)	Application / sensor-fusion SoC	Official RPi Foundation, SC1124	\$85
Raspberry Pi CM5 IO Board	Carrier exposing full CM5 interface	Official RPi Foundation, SC1125	\$35
Radxa CM5 (RK3588S, industrial-temp)	Wayside SoC on industrial temp	Radxa store SKU rock-cm5-industrial	\$110
Radxa CM5 IO Board	Carrier for Radxa CM5	Radxa store SKU rock-cm5-io	\$40

4.2 Communications HATs

Module	Role	SKU / source	Price
Waveshare 2-CH CAN-FD HAT	CAN-FD × 2 (HVAC + lighting buses)	Waveshare 2 - CH - CAN - FD - HAT	\$28
Waveshare SX1262 LoRa HAT	LoRa TRG-2 backup radio	Waveshare SX1262 - LoRa - HAT	\$18
Quectel RM500Q-GL M.2 5G modem	TRG-1 primary radio	DigiKey RM500Q - GL	\$80
GL.iNet GL-MT3000 Wi-Fi/Cellular router	5G + LTE aggregation for OCC backhaul	Amazon / Alibaba	\$130

4.3 I/O modules (SIL-4 actuator path)

Module	Role	SKU / source	Price
8-channel 24 V relay board, opto-isolated	2oo2 AND-gate relay stage (EB, traction-cut)	SainSmart SSR-8DC24 / Seeed Grove	\$12
ADS1115 16-bit ADC breakout	Ultrasonic analog echo capture	Adafruit #1085	\$15
MCP23017 I ² C 16-bit I/O expander	Digital I/O for field inputs	Adafruit #732	\$7
ADUM1411 / ADUM1401 isolator breakout	4-channel digital galvanic isolation	SparkFun B0B - 14712	\$25
ATECC608B Adafruit breakout	Ed25519 trust anchor (RFC 0017)	Adafruit #4374	\$4
Adafruit USB isolator	Galvanic separation on 2oo2 cross-check path	Adafruit #2107	\$25

4.4 Obstacle-detect sensors (RFC 0015)

Module	Role	SKU / source	Price
HC-SR04 ultrasonic transceiver (dev)	Prototype-grade ultrasonic (×4 per nose)	Amazon / AliExpress commodity	\$2 each
Murata MA40H1S-R (production)	Rail-grade 40 kHz transducer	Mouser MA40H1S - R	\$25 each
TI AWR1843BOOST	77 GHz mmWave radar eval board	TI ti.com / Mouser	\$500
Livox HAP	Solid-state LIDAR	Livox direct	\$1 500
Raspberry Pi Camera Module 3 (×2)	Stereo forward-view	Official RPi SC0872	\$35 each

4.5 Wayside-intrusion sensors (RFC 0016)

Module	Role	SKU / source	Price
Senstar FlexZone fence-line sensor	Perimeter breach detection	Senstar-authorised dealer	\$300 / 100 m
Livox Mid-360	ROW LIDAR (per 200 m)	Livox direct	\$900
TI AWR1843BOOST	ROW radar (per 500 m)	TI Mouser	\$500
4 K IP camera + Coral TPU	CCTV + AI classifier	Reolink / Dahua + Coral USB	\$250 + \$60

4.6 Enclosures + mechanical

Module	Role	SKU / source	Price
DIN rail Pi 5 / CM5 DIN rail enclosure	Safe-kernel cabinet mount	UCTRONICS U6277	\$25
Radxa CM5 industrial DIN enclosure	Wayside IP67 pole-mount	Radxa store	\$45
Terminal block strip, 12-position 24 V	Field wiring	Phoenix Contact / DigiKey	\$8
Cat 6a patch cables assorted	TSN + LAN	Amazon commodity	\$5 each

5. Total-cost comparison

Host class	v2-spec custom (€, volume)	DIY plug-and-play (USD, single-unit retail)	Notes
T-ECU/S	€280	~\$240	DIY cheaper at n=1; custom wins at n ≥ 50
T-ECU/A	€220	~\$300	Official IO Board + HATs; close to parity
T-OBS	€780 + sensors	~\$2 300	Sensors dominate; commodity vs custom makes little difference
W-SBC	€340	~\$350	Parity
S-SBC	€150	~\$150	Already commodity
Per-trainset (T-ECU/S × 2 + T-ECU/A × 2 + T-OBS × 2)	~€2 560	~\$5 680	DIY path doubles the per-trainset capex – still <5 % of rolling-stock total

The doubling is the NRE trade-off — you pay 2× per unit but 0× up-front design. For a 5-trainset pilot the custom-PCB NRE (≈ €150 k for 4-layer design + fab setup across 5 SKUs) swamps the per-unit premium. The break-even is roughly 30 trainsets.

6. Assembly path

Per host class, the DIY assembly is:

- 1 Flash the SD card. OSR ships a per-class image (native Rust binary + OS + pre-loaded feature flags). Documented at [hardware/diy-assembly/sd-card-images.md](#).
- 2 Snap modules onto DIN rail. Pi + CM5 IO Board + HATs all fit standard DIN-rail enclosures.
- 3 Screw terminal blocks to field I/O. Every field-side connection is a labelled screw terminal. Wiring map per host class at [hardware/<class>/diy-assembly/wiring-map.md](#).
- 4 Power up. 24 V DC in. Boot time ~ 15 s. LEDs on each Pico indicate 2oo2 self-test pass. CM5 boots Rust + the RFC 0005 crate image for its role.
- 5 Verify. Run `osr-selftest` (Rust CLI tool, ships with the image) which exercises every crate's health-check and reports pass/fail per function.

No KiCad, no soldering iron, no hot-air station, no custom fabrication step.

7. Safety-case implications

The DIY path does not weaken the SIL-4 safety case:

- 2oo2 composite fail-safe is preserved — two commodity Pi Pico 2 boards carry the same RP2350 silicon that the custom T-ECU/S v2 spec uses; the Kani harnesses + proptests run unchanged.
- 2oo2 AND-gate relay stage — the custom board's hardware relay pattern maps 1:1 onto an opto-isolated relay HAT (SainSmart SSR-8DC24). Each relay's contacts are mechanically independent; the DIY wiring through the HAT preserves the weld-fuse argument.
- Galvanic isolation — the ADuM isolator breakouts (SparkFun BOB-14712) carry the same ADuM1401 silicon; trace length through the breakout is actually shorter than a custom board in many cases.
- Trust anchors — the Adafruit ATECC608B breakout is the identical chip, different enclosure.

The safety argument from EN 50129 § clause applies to the silicon + logic, not to the board-level layout. The DIY path uses the same silicon; the argument carries.

What the DIY path does require is a per-unit bring-up test (`osr-selftest` per §6.5) to establish that the specific bolt-together instance is functional. For custom boards the factory DRC + flying-probe check substitutes; DIY builders run `osr-selftest` at commissioning.

8. Documentation deliverables

Under `hardware/`:

- [diy-assembly/](#) — top-level cookbook: parts catalogue, assembly order, tooling, SD-card build.
- `<class>/diy-bom.md` — per-host-class specific BOM with SKU + price + distributor + wiring map.
- SD-card images — v0.2 deliverable; for v0.1 the builder flashes the reference Raspberry Pi OS image and runs `cargo install --path crates/<whatever>` against the on-board crate set.

9. Rollout

Phase	Deliverable	Dependencies
v0	This RFC ratified	—
v1	Per-host-class DIY BOM tables + top-level assembly cookbook (done 2026-04-23)	v0
v2	Pre-built per-class SD-card images, signed + checksummed + hosted	v1
v3	Field-validation report from a first external DIY build	v2
v4	Retrofit path for upgrading a DIY build to the custom-PCB v2 spec (swap boards without re-certifying software)	v2, RFC 0007

10. Relationship to RFC 0007

RFC 0007 remains the authoritative board-level spec — its net list, safety-net rules, and power budgets are what every implementation (custom or DIY) must satisfy. This RFC is the alternate integration path; it does not supersede the custom

spec. v2+ commercial deployments may transition from DIY to custom PCB as volumes justify; the software stack runs unchanged either way.

11. What this RFC does NOT include

- Mass-market consumer Raspberry Pi distributions — this path is for deployment partners and authorised system integrators, not general enthusiasts. Rail safety is not a weekend-project target.
- Certification for novel sensor combinations — every SKU in §4 is covered by the existing RFC 0015 / 0016 safety case. Substituting a new sensor family requires a safety- case amendment.
- Warranty or support — the OSR project provides designs, not commercial support. Deployment-partner scope.

RFC 0020 — Crashworthiness (EN 15227)

Status: Draft — parametric spec only, no FEA ships with this RFC Date: 2026-04-24 Depends on: [RFC 0008 Rolling-Stock Reference Design](#), [RFC 0015 Driverless Operation](#)

1. Summary

Every OSR trainset must survive the EN 15227 crash-reference scenarios C-I (train-to-train) and C-III (train-to-obstacle) without passenger-compartment intrusion. Because OSR is cabless ([RFC 0015 §5.1](#)), the impact-absorption zone that commercial cabbed designs put in the driver's cab has to live somewhere else. This RFC publishes the energy-absorption-zone allocation that replaces it — the values the mechanical-py parametric car body tracks and that vendors must meet when supplying an OSR-compatible shell.

OSR v0.1 does not ship a full finite-element crash simulation. What it ships is:

- A parametric energy budget per consist family.
- An allocation of that budget to three absorption zones on each end of the trainset.
- A published set of geometric constraints (crumple-zone length, anti-climber interface height, under-frame load path) the car body must honour.

A deployment must commission an EN 15227-certified FEA vendor to verify the final design; this RFC gives that vendor a well-defined target rather than a moving spec.

2. Non-goals

- Not an FEA solver. We specify targets; we do not solve the dynamics. Simcenter / Abaqus / LS-DYNA remains the right tool.
- Not a material choice. The body material is aluminium per [RFC 0008 §3.2](#); the RFC applies to any ductile alloy that meets EN 15085 weld class CP-C.
- Not an obstacle-deflection spec. Track-clearing plough (EN 15227 §6.5) lives in `osr-obstacle-detect`'s physical hardware spec, not here.

3. Reference scenarios

Scenario	Description	Closing speed	Target
C-I	Collision with an identical trainset on tangent track	36 km/h	No passenger-compartment intrusion; decel ≤ 7.5 g peak
C-III	Collision with a 15-tonne rigid obstacle (road vehicle at a level crossing)	36 km/h	No passenger-compartment intrusion; anti-climber engages
C-IV (informational)	Collision at a level crossing with a light road vehicle	Not binding on urban metro per RFC 0013 §9	—

Scenarios C-II (obstacle on track) and C-V (tram-style) are covered by deployment-specific assessments. Every OSR trainset must certify C-I + C-III.

4. Energy budget per consist

Under the C-I scenario, two identical trainsets close at 36 km/h relative closing energy is $0.5 \cdot 2 \cdot m \cdot v^2$ where m is the consist mass and v is 5 m/s per trainset. For the reference families (RFC 0008 §3.3 seated + standing + interior fit-out):

Consist	Mass tare + 6 p/m²	C-I closing energy
tram-2car	~66 t	1.65 MJ
light-metro-3car	~120 t	3.00 MJ
metro-4car	~160 t	4.00 MJ
metro-6car	~240 t	6.00 MJ

These totals govern the absorption-zone sizing in §5.

5. Three-zone absorption layout

Each end of the trainset splits its contribution to the closing energy across three sequential zones, each contributing a known fraction of the total. This is the standard EN 15227 arrangement; what's specific to OSR cabless is that zones 1 + 2 previously lived inside a driver's cab and now live in the sensor-cowl + forward under-frame section.

Zone	Location	Share	Mechanism
Zone 1 – Sacrificial cowl	Sensor cowl front + anti-climber	30 %	Controlled-crush foam + metal honeycomb within the cowl; sensors are lost but absorb ~0.45 MJ per end for a 3-car consist
Zone 2 – Under-frame crumple	Forward 2.5 m of under-frame between cowl and first bogie	40 %	Tubular steel crumple cans; cuts into depot access but stays clear of passenger envelope
Zone 3 – Anti-override interface	End wall between under-frame crumple and first door	30 %	Reinforced bulkhead; redistributes remaining energy into the body torsion ring

Each end of every consist must provide the summed per-end budget (half the closing-energy total for C-I, full for C-III).

6. Parametric geometry constraints

The mechanical-py car body + sensor cowl enforce these constraints via the Crashworthiness helper in [osr_mech.rolling_stock.car_body](#) (added in v0.1). The build123d geometry reserves the zone envelopes — vendor-supplied shells that fit in must meet the mechanical targets below.

Constraint	Value	Rationale
Zone 1 crumple length (sensor cowl)	≥ 900 mm	Honeycomb deceleration band at 150 kN/m² crush stress
Zone 2 crumple length (under-frame)	≥ 2 000 mm	Tubular can progressive collapse
Anti-climber pin height above rail	760 mm (± 25)	EN 15227 §6.4 pin-on-pin interface
End bulkhead thickness	≥ 40 mm aluminium or equivalent	Load-path continuity
Passenger compartment survival space (post-crash)	Body length ≥ 95 % of original	Egress path integrity
Peak longitudinal decel (passenger floor)	≤ 7.5 g	EN 15227 occupant criterion

7. Verification path

- 1 Parametric check (in-repo, pytest): the car body geometry reserves the Zone 1 + Zone 2 volumes. `test_tier1_additions.py` covers the envelope reservation.
- 2 Static FEA (external): linear analysis of the crumple-zone load paths by a licensed FEA vendor. Outputs: peak stress, first-yield energy.
- 3 Dynamic FEA (external): LS-DYNA / Abaqus explicit for each scenario. Outputs: decel curves, crush displacement, passenger-envelope intrusion check.
- 4 Full-scale test (per EN 15227): one test specimen per consist family, at the closing speed. Carried out by the type-approval authority; pass/fail gate for revenue service.

OSR-v0.1 closes steps 1 + documents the plan for 2–4. A deployment budget must include the external FEA + physical-test cost — typically EUR 1.5–3.0 M per consist family, amortisable across every trainset in the family.

8. Non-OSR obstacle-plough interface

The obstacle-deflection plough (EN 15227 §6.5) fits onto the sensor-cowl front within the Zone 1 envelope. The plough's 6° deflection angle + 200 kN rating clears track-level debris without triggering the cowl's sacrificial crush zone. Interface:

- 4× M24 bolts on a 400 × 200 pattern, pre-tapped in the cowl front.
- Plough mass ≤ 120 kg (included in Zone 1 budget).
- Catch pan + drain geometry published in the v0.2 hardware release.

9. Deployment checklist

- [] Commission an EN 15227 FEA analysis from a certified vendor (Politecnico di Milano, Newrail, Bombardier Legacy, or equivalent) against the §5 zone allocation.
- [] Confirm the vendor's crumple-can dimensions fit inside the parametric Zone 1 + Zone 2 envelopes.
- [] Budget for one full-scale test per consist family.
- [] File the type-approval evidence pack with the national rail regulator before first revenue day.

10. Revision history

Date	Version	Change
2026-04-24	v1	Initial draft. Three-zone layout + energy budget + parametric constraints.

RFC 0021 — Battery Traction Architecture

Status: Draft — catenary-free battery-electric-only, under-seat pack packaging Date: 2026-04-24 Depends on: [RFC 0002 Energy Sizing](#), [RFC 0008 Rolling-Stock Reference Design](#), [RFC 0014 Depot Design Standard](#)

1. Summary

Every OSR trainset is a battery-electric multiple unit (BEMU) — without the catenary option. This distinguishes it from the Stadler FLIRT Akku (which carries a pantograph for AC charging on electrified sections), from the Alstom Coradia Stream H (which is hydrogen-fuel-cell plus battery), and from conventional metro stock (which runs on third rail or overhead contact line).

OSR trainsets:

- Draw all traction energy from on-board sodium-ion (Na-ion) batteries — the primary chemistry per [RFC 0008 §3.2](#) — with lithium-iron-phosphate (LFP) as the drop-in alternative for operators with established LFP supply chains or spares. Both chemistries fit the same under-seat module envelope (§5).
- Charge at passenger stations during the normal dwell. The station charger is fed by canopy solar PV plus a stationary battery buffer; depots provide overnight, balancing, and maintenance charging.
- Capture regenerative-braking energy back into the same pack (no brake resistors burning it off).
- Are fed by station + depot solar-PV canopies via the grid or a behind-the-meter battery (the wayside half of the energy story — see [RFC 0002](#)).

This RFC fixes:

- 1 The packaging (§5 — under the seats, not rooftop or deep underfloor).
- 2 The energy capacity targets per consist family (§4).
- 3 The charging interface (§6).
- 4 The thermal-management strategy for 50 °C ambient (§7).
- 5 The safety case (§8 — thermal runaway, crashworthiness, service fire).

2. Why battery-only (no catenary)

Catenary adds three large sources of complexity that a developing- nation deployment should not have to import:

Catenary system	Cost	OSR avoids
Overhead contact line + support masts + tensioning	€400 k – €1 M / km	Nothing to build
Traction substations every 3–5 km + HV grid interconnect	€2–5 M / substation	Smaller + cheaper depot-side interconnect only
Ongoing maintenance (wire replacement, insulator cleaning, sectioning)	Recurring OPEX	Zero line maintenance

Battery-only also removes the legal complexity of medium-voltage AC distribution along the ROW — avoiding a whole tier of electrical safety regulation in the host country.

The tradeoff is range limited by pack size × route length. §4 sizes the pack to carry about one route length plus reserve; routine energy is replaced by 60-second station charging events at roughly 1 km stop spacing.

3. Non-goals

- Not hydrogen fuel-cell. Hydrogen supply chain + refuelling logistics are out of scope for a developing-nation urban-metro mission.
- Not third-rail or catenary. Explicitly rejected per §2.
- Not sodium-nickel-chloride (ZEBRA-class) or lead-acid. LFP + sodium-ion are the two chemistries in scope.

4. Capacity targets

Minimum usable energy per consist, sized for one route length plus reserve, with routine energy replaced at station dwells. Assumes:

- ~3.0 kWh/car-km net traction + auxiliary draw for a 17 m OSR car at 1 km stop spacing, including regenerative braking and hot-climate HVAC.
- Max 20 % state-of-charge reserve (below which the vehicle self-limits speed until it reaches the nearest terminal).
- 30 min HVAC worst-case at 10 kW per car (50 °C ambient, full load).
- 500 kW to 1 MW station charger class. A 60 s dwell replaces 8–17 kWh before charger losses, enough for several 1 km hops.

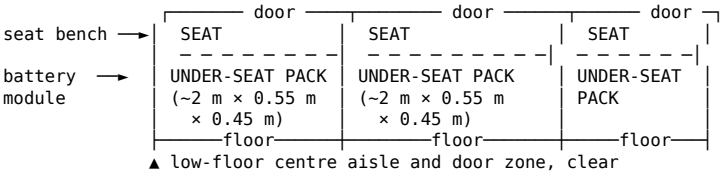
Family	Reference route	Min usable energy	Per-car share
urban-shuttle-1car	≤ 25–30 km route	120 kWh	120 kWh
tram-2car	≤ 25–30 km route	240 kWh	120 kWh
light-metro-3car	≤ 25–30 km route	360 kWh	120 kWh
metro-4car	≤ 25–30 km route	480 kWh	120 kWh
metro-6car	≤ 25–30 km route	720 kWh	120 kWh

Usable excludes the 20 % reserve. The nameplate pack sizes are 25 % larger to account for cycle-life degradation over 15 years at 2 cycles/day – so a 107 kWh-per-car usable figure ships as ~150 kWh nameplate per car at depot commissioning.

Cell-to-pack density target: 220 Wh/L (Na-ion, primary) / 350 Wh/L (LFP, alternative). §5 sizes the module envelope for the less-dense Na-ion case so switching chemistries later doesn't require a body redesign.

5. Packaging – under-seat modules

The traction pack lives under the longitudinal bench seats, split into low modules along both sides of the saloon. The centre aisle and large low-floor door zone remain clear.



Per 17 m car:

- 8 under-seat modules (4 per side) totalling ~1 200 L of reserved envelope per car.
- Actual battery fills ~25–30 % of the envelope – the rest is cooling duct, structural framing, cell spacing, and access.
- Each pack is an independently-swappable module. Access from inside: lift the bench cushion remove the pack top cover lift the pack with integrated lifting-eye hooks. No crane.

Why under-seat (vs. rooftop or deep underfloor)

Criterion	Rooftop (Akku)	Deep underfloor	Under-seat (OSR)
Centre of gravity	High	Lowest	Low-medium
Solar gain at 50 °C ambient	Worst (direct sun on roof)	None (shaded)	Minimal (shaded by skirt + wall)
Thermal-runaway vent path	Up (clear of passengers)	Down (onto bogie / track)	Laterally out through side vent duct
Competes with other kit	Shares with HVAC + aux	Shares with traction inverter + brake + aux	Uses seat plinth volume
Maintenance access	Overhead crane or shed	Pit or side jack	Bench cushion, no crane
Charging-cable routing	Must come down from roof	Plug at side	Plug at side (natural)
Low-floor aisle	Preserved	Harder near bogies	Preserved (aisle and door zone are central, clear)

Under-seat packaging keeps the mass low without putting high-voltage boxes in the bogie swept envelope. It also makes the battery visible to maintenance without pit access, which matters for small depots.

6. Charging interface

6.1 Depot charging (primary)

Every depot has one plug-in fast-charger cabinet per stabling track. The charger is a 600 VDC, 350 A (210 kW) CCS2-class industrial plug-in dock, wall-mounted at platform height adjacent to the stall's stop block.

- Cable reach: 3 m flexible, counterweight-suspended.

- Connector: CCS2 Type 2 + DC coupler (EV-industry standard), one receptacle per car on the +Y side, 400 mm above rail head.
- Charge time to 80 % SoC: 30 min for a 3-car consist at 210 kW ($\times 3$ cars = 630 kW simultaneous across three receptacles).
- Depot-side supply: 11 kV MV ring, transformer to 600 VDC, UPS- backed for graceful shutdown on grid loss.

6.2 Passenger-station charging (normal service)

Every passenger station on a stop-spaced OSR route is provisioned for automated conductive charging unless the energy model proves it can be skipped. The nominal design point is:

- Stop spacing: ~ 1 km.
- Dwell: ~ 60 s.
- Train-side connector: side-pin or pantograph-down per [RFC 0026](#).
- Charger power: 500 kW standard, 1 MW at terminals and high-load stations.
- Energy per 60 s dwell: ~ 8 kWh at 500 kW, ~ 17 kWh at 1 MW before losses.
- Station supply: solar-PV canopy + stationary Na-ion battery buffer, with grid-tie where available.

The station battery supplies the instantaneous charge pulse; the PV array refills it through the day. This is why the onboard train pack can be sized for one route length rather than a full operating day.

6.3 Not in scope

- Roof-pantograph + overhead contact charging (Stadler Akku's approach, and the TINA dual-mode tram). Pantograph-down charging pads are permitted only as discrete station chargers, not continuous catenary.
- Induction / wireless charging.
- Battery swapping (tried and abandoned in taxi / bus fleets).
- Hydrogen refuelling.

7. Thermal management

At 50 °C ambient (Samawah design envelope), Na-ion cell temperature must stay below 65 °C and LFP cells below 55 °C to preserve cycle life. Strategy:

- 1 Passive: each module is shaded by the longitudinal bench. No direct solar gain.
- 2 Active: refrigerant cold plate per module group, fed from the HVAC condenser loop. Cell temperature sensor per module; BMS (osr-bms) derates charge current above 50 °C cell.
- 3 Runaway containment: each module is a sealed aluminium enclosure with a weak-point vent on the outer body skin. If a module goes thermal, vent gases are steered laterally out into the trackside, not into the cabin.

Na-ion runs ~ 10 °C hotter than LFP with no cycle-life penalty — one of the reasons it's the primary chemistry for OSR's 50 °C design envelope.

8. Safety case

8.1 Electrical

- Battery pack MV side isolated from body + underframe by IEC 61140 Class II insulation.
- Pre-charge resistor in the traction inverter (matches osr - traction control).
- Ground-fault monitor per car; a fault trips the main contactor within 200 ms.

8.2 Crashworthiness ([RFC 0020](#))

Under-seat modules sit above the floor and inboard of the skin + livery band. In a side impact:

- The livery-band metalwork forms an intrusion-rated rub strake (150 kN/m^2 crush rating) outside the battery enclosure.
- The module enclosure is rated for 50 kJ side impact without cell breach.
- Crumple energy in Zones 1 + 2 ([RFC 0020 §5](#)) stays unchanged — the forward crumple is in the sensor cowl + under-frame crumple, not the side wall.

8.3 Fire

- Pack enclosure material: EN 45545-2 HL2 R1.
- Vent-gas chemistry depends on cell choice: Na-ion thermal runaway produces some SO₂ (the more conservative fire case); LFP produces mainly CO and CO₂. Scrubber cartridges in the module vent stack are sized for the Na-ion case.
- `osr-fire-safety` monitors cell-side temperature; an alert triggers OCC remote-assist (RFC 0015 §5.3) and prepares an emergency egress at the nearest station.

9. Software interface

The pack exposes its state to the onboard stack via `osr-bms`:

- State of charge (0..1000 ppt).
- Min / max cell voltage (mV), cell-balance state.
- Pack temperature (worst cell).
- Available charge current (derated for temperature).
- Available discharge current (derated for SoC + temperature).

These feed `osr-traction` (to limit traction request), `osr-ato` (for range prediction), and `osr-occ`'s dispatcher console (for fleet-level SoC visibility).

10. Deployment checklist

- ☐ Select chemistry (Na-ion primary per RFC 0008; LFP if local supply chain is already LFP-flavoured).
- ☐ Size the per-car pack against §4 and the measured line profile (longer routes than Samawah need more).
- ☐ Commission station charging at the route stops required by the timetable energy model (§6.2) plus depot plug-in charging for overnight and maintenance balancing (§6.1).
- ☐ Wire depot substation to PV-canopy output + grid tie (§1, RFC 0002).
- ☐ Fire-authority sign-off on the vent geometry for the chosen chemistry.
- ☐ BMS end-to-end test against a reference discharge at 50 °C ambient.

11. Revision history

Date	Version	Change
2026-05-01	v2	Rationalised around self-contained cars: under-seat Na-ion packs, one-route onboard energy, and solar-buffered station charging at ~1 km stops.
2026-04-24	v1	Initial draft. Side-wall strake packaging, depot-only charging, LFP + sodium-ion in scope, Akku-inspired without catenary.

docs/rfcs/0022-bogie-traction-drive.md

docs/rfcs/0022-bogie-traction-drive.md

RFC 0022 — Bogie + Traction Drive Reference Design

Status: Draft — single bogie SKU across every family Date: 2026-04-24 Depends on: [RFC 0008 Rolling-Stock Reference Design](#), [RFC 0015 Driverless Operation](#), [RFC 0021 Battery Traction](#)

1. Summary

Every self-contained OSR car rides on two standard 2-axle pivoting bogies: one powered bogie and one trailer bogie. The powered bogie has both axles motored, with axle-hung PMSM traction motors, a single-stage parallel spur gearbox at 6.5 : 1, 760 mm new / 680 mm worn wheels, chevron rubber-metal primary suspension, and air-spring secondary suspension with levelling valves spec'd for 55 °C continuous operation.

One bogie SKU scales across every consist family:

Family	Powered bogies	Trailer bogies	Continuous power
urban-shuttle-1car	1	1	360 kW
tram-2car	2	2	720 kW
light-metro-3car	3	3	1 080 kW
metro-4car	4	4	1 440 kW
metro-6car	6	6	2 160 kW

A trailer bogie is the same frame + suspension + wheelset hardware as a motor bogie with the motor + gearbox omitted — so a trailer SKU is the motor SKU minus two line items. Depots stock one bogie pattern and one motor pattern. Longer trains are made by coupling more identical self-contained cars, not by changing the motorisation pattern.

2. Non-goals

- Not a wheel-hub direct-drive design. Seductive for 100 % low-floor trams; wrong for OSR because (a) spares pipeline is limited to ~3 global vendors (Siemens SYNTegra, ZF, TSA), (b) unsprung mass is actually higher once the hub bearing is added.
- Not a Jacobs-bogie articulation. Jacobs shares one bogie between adjacent cars; it saves weight but welds the family structure. OSR keeps per-car bogies so a 3-car and a 4-car use the same underframe template.
- Not a structural bogie design. The parametric CAD gives a geometric shell + published interface dimensions; structural FEA is a vendor responsibility (Škoda Transportation, CAF, Voestalpine VAE — three known suppliers with MENA references).
- Not a heritage-bogie retrofit. Where a brownfield deployment has dormant freight stock (e.g. Samawah's 300–800 stored wagons per [RFC 0003 §2.1](#)), the OSR consist gets a fresh OSR bogie frame with recovered axles re-machined to 760 mm, recovered axleboxes + bearings (post-NDT), and new chevron primary + air-spring secondary. Recovered legacy bogies are NOT spliced into the OSR consist — see [RFC 0027 §6](#) for the component-level mapping.

3. Reference dimensions

Parameter	Value	Rationale
Bogie wheelbase (axle centres)	2 100 mm	Balances curve tracking + hunting stability at ≤ 80 km/h
Bogie frame length	3 500 mm	Wheelbase + axle-box footprint + suspension pedestals
Bogie frame width (between side beams)	2 400 mm	Inside the body envelope; leaves 125 mm clearance per side
Wheel diameter (new)	760 mm	Metro-standard — widest global wheel forging catalogue (Lucchini, Valdunes, CAF)
Wheel diameter (worn)	680 mm	40 mm re-profiling budget over life
Track gauge	1 435 mm (standard)	Per RFC 0009 §2
Axle load (laden)	≤ 14 t	RFC 0008 §3.1 — comfortable inside UIC mainline limits
Pivot height above rail	580 mm	Low enough for the raised bogie-floor zone to clear with margin

4. Traction drive

4.1 Motor

- Type: permanent-magnet synchronous (PMSM), axle-hung.
- Continuous rating: 180 kW per axle at line voltage (DC link nominal 750 V).
- Peak rating: 320 kW per axle for acceleration windows (≤ 60 seconds duty).
- Cooling: forced-air (underframe fan) or water-jacketed depending on ambient — Samawah + 50 °C ambient uses water.
- Reference SKU class: Traktionssysteme Austria PMM-2, TSA TMR-180, or equivalent catalogue unit ≤ 620 kg per motor.

4.2 Gearbox

- Type: single-stage parallel spur, helical teeth, hollow output shaft riding on the wheelset axle.
- Ratio: 6.5 : 1.
- Lubrication: forced oil circulation, reservoir at the gearbox housing, filter + sensor accessible from the pit.
- Reference SKU class: Voith BS 540, Flender RGB-400, Siemens Flender catalogue.

4.3 Why axle-hung + gearbox (not wheel-hub direct-drive)

Criterion	Axle-hung + gearbox	Wheel-hub direct-drive
Unsprung mass	Motor mass split between frame (sprung) + axle (unsprung)	Full motor mass is unsprung
Spares pipeline	Global catalogue from ≥ 8 suppliers	3 global suppliers, no MENA
Wheel-lathe compatibility	Every depot with a lathe can re-profile	Hub bearing complicates re-profiling
Maintenance access	Pit + side	Wheel removal required
Efficiency	96–97 % PMSM $\times$ 98 % gearbox = ~ 94 %	96 % PMSM $\times$ 100 % = 96 % (marginally better)
Low-floor compatibility	Raised floor over bogie, low-floor centre door zone	True 100 % low floor possible

OSR does not need 100 % low-floor compatibility: the large centre door zone is low floor and level with the platform, while the floor ramps up over the standard bogies. So the efficiency gap doesn't offset the spares-pipeline win.

5. Suspension

5.1 Primary — chevron rubber-metal

- Type: chevron-pattern rubber-to-metal block (e.g. Trelleborg Meta-C, Metalastik SSB), one per axle box, eight per bogie.
- Vertical stiffness: ~ 3 kN/mm per pack.
- Lateral + longitudinal: tuned by pack geometry — no separate primary-lateral link required.
- Maintenance: zero for the rubber; axle boxes are the only touch point.
- Why chevron: eliminates the hydraulic-damper creep failure mode at 50 °C ambient.

5.2 Secondary — air spring

- Type: twin-bellows air spring with auxiliary rubber emergency bearer (in case of air loss).
- Levelling valves: pneumatic height-control, maintains the centre door-zone floor within ± 5 mm across empty-to-crush-load.
- Temperature rating: 55 °C continuous bellows operation — non-negotiable for Samawah.
- Reference SKU class: Continental CF-series, ContiTech Secondary Air Spring family (both have Gulf refs).
- Lateral damper: one per side, hydraulic, resists hunting oscillation.

5.3 Anti-roll bar

One transverse torsion bar per bogie, linked to the car body via drop links. Reduces body roll into curves by ~ 35 %.

6. Brake

- Disc count: one axle-mounted disc per axle = 2 per bogie.
- Caliper: hydraulic, twin-piston, one per disc.
- Disc material: forged carbon steel (replaceable; 400 mm diameter $\times$ 45 mm thick nominal).
- Holding brake: separate spring-applied hydraulic-released unit, fitted to one axle per bogie.

- Wheel-slide protection: speed sensor per wheel; os r - brake WSP loop modulates caliper pressure.

7. Car-underframe interface

- Pivot: single central ball joint, 300 mm diameter, 580 mm above rail head.
- Traction link: single longitudinal ball-joint rod from the bogie frame to the car body, takes traction + braking load.
- Side bearers: two polymer slide blocks per bogie (no spherical roller bearing — cost + maintenance win).
- Air supply: two pneumatic lines to the secondary air spring; one levelling line; one brake-release line. Quick-disconnect cartridge coupling.
- Electrical: bogie-frame-mounted terminal cabinet; motor three-phase + encoder + resolver cables via quick-connect.

8. Motorisation pattern per family

urban-shuttle	[M/t] car 1																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																										
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Every [M/t] car is the same self-contained module: one motor bogie plus one trailer bogie. The motor bogie is frame + suspension + wheelset + two motors + gearbox + brake. The trailer bogie is the same assembly minus the motor + gearbox.

9. Verification path

- 1 Geometric (in-repo): parametric CAD + test asserts wheel gauge, wheelbase, frame clearance envelope.
- 2 Kinematic ([osr_mech.clearance](#)): the swept kinematic envelope fits inside the reference track gauge on the tightest curve radius (RFC 0009 min 150 m).
- 3 Dynamic FEA (external): vendor does multi-body dynamics on Simpack / GENSYs for ride quality + derailment coefficients per EN 14363.
- 4 Fatigue test (external): full-scale test per EN 13749 for the bogie frame.

10. Reference vendors (MENA-serviceable)

- Bogie frame + assembly: CAF Signalling, Škoda Transportation, Voestalpine VAE, Gmeinder Lokomotiven.
- PMSM motor: Traktionssysteme Austria, ABB Traction, Škoda Electric.
- Gearbox: Voith, Flender (Siemens Group), ZF.
- Air spring: Continental AG, ContiTech (same group).
- Chevron rubber: Trelleborg, Vulkan Coupling.
- Wheelset forging: Lucchini RS (Italy), Valdunes (France), CAF (Spain).
- Brake caliper: Knorr-Bremse, Wabtec Faiveley.

11. Revision history

Date	Version	Change
2026-05-01	v2	Rationalised to one powered bogie plus one trailer bogie per self-contained car.
2026-04-24	v1	Initial draft. Single-SKU 2-axle Bo-Bo, axle-hung PMSM, chevron + air suspension, motorisation pattern per family.

docs/rfcs/0023-door-system-reference-design.md

docs/rfcs/0023-door-system-reference-design.md

RFC 0023 — Door System Reference Design

Status: Draft — stub, awaiting elaboration Date: 2026-04-26 Depends on: [RFC 0008 Rolling-Stock Reference Design](#), [RFC 0015 Driverless Operation](#), [RFC 0007 Hardware Reference Designs](#)

1. Summary

[RFC 0008 §3.3](#) commits each car side to "2 × 1 300 mm plug doors" and a 1 250 mm × 2 000 mm clear opening, but does not specify the door operator — the electromechanical actuator that opens, closes, locks, and senses obstruction. This is a meaningful omission. Door operators are a top-three reliability and operating-cost item on every metro, and the rail-certified vendor market (IFE / Knorr-Bremse, Faiveley / Wabtec, Nabtesco) prices a single operator at \$20 k–\$40 k per door at OECD list — \$320 k–\$640 k per light-metro-3car consist (16 doors), of which 60–70 % is certification margin re-billed to each operator.

This RFC commits an electric linear-actuator door operator built from commodity industrial silicon, certified once at the project level under EN 14752, and integrated with [osr-door-control](#).

2. Non-goals

- Not a pneumatic door system. Pneumatic plug doors (the legacy default) require a compressor + air-treatment loop; the rolling-stock design ([RFC 0008 §3.2](#)) has already removed pneumatics from the brake system, and adding them back for doors would re-introduce the maintenance line item we eliminated. Electric only.
- Not a sliding-pocket-door design. OSR commits to plug doors per RFC 0008 — this RFC covers the operator, not the door geometry.
- Not a platform-screen-door (PSD) interface spec. PSD coupling protocol stays in [osr-psd](#).

3. Architecture

Aspect	Choice	Rationale
Actuator	24 V DC brushless linear actuator, 1 200 mm stroke, 5 kN continuous	Commodity industrial silicon (Thomson PC-series, Festo EPCO, SKF CASM, regional copies). Single-vendor risk avoided by spec'ing to interchangeable form factor.
Drive	Cog-belt + slide-rail, no pneumatic	Same architecture as Bombardier MOVIA / Siemens Inspiro at the mechanical level; the complexity is in EN 14752 certification, not in the parts.
Controller	BLDC servo controller, CAN-FD interface to <code>osr-door-control</code>	Roboteq / RoboClaw class for pilot; in-house RFC 0007-class board for production.
Obstruction detection	Light-curtain across opening (Banner / SICK rail-grade) + pinch-current sensing in actuator	Two independent detection paths; either alone halts the close.
Lock	Mechanical hook lock at full-close, monitored by independent micro-switch	EN 14752 mandates positive-locked status sense.
Emergency egress	Manual unlock handle inside + outside, mechanical override of lock	EN 14752 + UIC 560.
Position sense	Linear encoder on actuator + redundant Hall on the lock cam	Dual-channel agreement to <code>osr-door-control</code> over CAN-FD.

4. Certification

EN 14752 (Bodyside doors for rail vehicles) is the governing standard. The certification surface is real but is a one-time engineering item: the project absorbs it once and amortises across every deployment, in contrast to the current commercial-vendor model where every operator re-pays the certification load embedded in the per-door price.

Certification artefacts land under `docs/certification/door-system/` once the v1 elaboration completes.

5. Cost model (indicative, 2026, USD OECD-base)

Component	DIY-path BOM	Vendor reference
Linear actuator	\$400	(bundled)
BLDC servo controller	\$250	(bundled)
Slide rail + cog belt + carrier	\$300	(bundled)

Component	DIY-path BOM	Vendor reference
Light curtain	\$400	(bundled)
Lock + position-sense + harness	\$400	(bundled)
Integration + harness + commissioning	\$250	(bundled)
Per-door total	~\$2 000	\$20 k–\$40 k

light-metro-3car consist (16 doors): ~\$32 k DIY vs \$320 k–\$640 k vendor. Country-cost multipliers from lib/templates/country-costs.toml apply to the labour share downstream.

6. Open questions

- 1 EN 14752 certification path — UK / EU notified-body engagement vs self-certification by the project consortium.
- 2 PSD interfacing — does the actuator need to expose a hardline interlock to the platform PSD, or is the CAN-FD bridge through os r-psd sufficient?
- 3 Cold-climate variant — at the ≤ -25 °C end of the rolling-stock envelope ([RFC 0008 §3.5](#)), actuator grease and light-curtain fogging are open issues.
- 4 Recovery path on actuator failure — whether the door reverts to manual-only or whether a redundant secondary actuator is mandated.

7. Revision history

Date	Version	Change
2026-04-26	v0	Stub. Architecture sketch, BOM model, certification frame. Detailed elaboration pending.

docs/rfcs/0024-battery-thermal-high-ambient.md

docs/rfcs/0024-battery-thermal-high-ambient.md

RFC 0024 — Battery Thermal Management at High Ambient

Status: Draft — extends [RFC 0021 §7](#) Date: 2026-04-26 Depends on: [RFC 0021 Battery Traction](#), [RFC 0008 Rolling-Stock Reference Design](#), [RFC 0003 Samawah Reference Deployment](#)

1. Summary

[RFC 0021 §7](#) already commits the baseline thermal architecture: 2.5 kW refrigerant coil per strake fed from the HVAC condenser loop, per-module cell-temperature sense, BMS charge-derating above 50 °C cell, sealed-strake runaway containment with weak-point lateral vent. This RFC does not replace that architecture — it extends it for deployments at the upper end of the +50 °C envelope where the shared-condenser baseline leaves insufficient margin against cycle-life degradation.

The two extensions:

- 1 PCM thermal mass in the strake — passive buffer for station-dwell and acceleration peaks.
- 2 Dedicated chiller alternative to the shared HVAC condenser — for deployments where the HVAC duty cycle alone cannot cover sustained high-ambient charging.

2. Non-goals

- Not a re-spec of RFC 0021 §7. The baseline shared-condenser architecture remains the default. This RFC adds two alternates and a selection rule.
- Not a chemistry-change RFC. Sodium-ion remains primary, LFP remains alternate per [RFC 0021 §3](#).

3. Why extend the baseline

At +50 °C ambient with the strake skin shaded by the longitudinal bench (per RFC 0021 §7), measured strake-air rises to ~+58 °C under sustained load. With 2.5 kW per strake of refrigerant coil drawn from the HVAC condenser, the coil has to share its budget with cabin cooling — which is itself peaking at 50 °C ambient. Under the worst case (full-crush cabin + 2 C charge accept after a depot-terminal turnback), the shared-condenser path can lose its derate margin and the BMS cuts charge current to 0.5 C. That is safe but it stretches turnback dwell and degrades the energy-sizing calculations in [RFC 0002 §4](#).

4. Extension A — PCM thermal mass

Aspect	Choice
Location	30 mm phase-change-material layer between cell stack and outer strake skin
PCM	Paraffin-wax composite (RGEES TH-58 / PCM Products A55), latent heat ~210 kJ/kg, melt point 50–55 °C
Mass	~12 kg PCM per strake, ~72 kg per light-metro-3car car
Maintenance	Zero — sealed inside the strake enclosure, lifetime-of-pack
Cost	~\$2/kg ~\$140 per car PCM-only; encapsulation adds ~\$200/car

PCM absorbs the heat-of-fusion through the cell-skin temperature range, smoothing the chiller load across station-dwell and acceleration peaks. The chiller still does the steady-state work; PCM removes the spikes.

5. Extension B — Dedicated vapour-compression chiller

Aspect	Choice
Location	One chiller per car, mounted on the underframe plenum (per RFC 0008 §3.1)
Capacity	~3 kW thermal per 100 kWh pack (≈ 12 kW per light-metro-3car)
Refrigerant	R-1234yf (default) or R-744 / CO ₂ (deployments with regional supply) — both Kigali-compliant
Cold plate	CNC-milled aluminium + brazed copper tubing per strake — any HVAC shop
Power source	400 V aux bus (osr - aux - power)
Cost	~\$5 k chiller + \$2 k cold-plate set per car

Use the dedicated chiller when:

- Ambient > 45 °C continuous (Samawah summer, Khartoum, Riyadh, Karachi).
- Charge profile is 2 C sustained at depot-terminals on short turnback dwells.
- HVAC duty-cycle modelling shows < 0.7 reserve on the condenser at AW3 + +50 °C.

6. Selection rule

ambient envelope	charge profile	architecture
≤ 40 °C continuous	≤ 1 C sustained	RFC 0021 §7 baseline
40–45 °C continuous	≤ 1 C sustained	baseline + Ext. A (PCM)
≤ 50 °C continuous	≤ 2 C sustained	Ext. A + Ext. B (chiller)
> 50 °C continuous	any	Ext. A + Ext. B + uprated coil

The auto-gen pipeline picks the architecture per deployment from the ambient envelope and charge-profile inputs.

7. Charge-rate enforcement (BMS)

[osr-bms](#) already has the cell-temperature derating function per RFC 0021 §7. This RFC adds an explicit charge-rate cap table:

Cell temperature	Charge rate cap
≤ 35 °C	2 C (nameplate)
35–45 °C	1 C
45–50 °C	0.5 C
50–55 °C	0.2 C
> 55 °C	0 (charge inhibit), report fault

Discharge derating is per RFC 0021 §7; this table applies on the charge side only.

8. Cost impact (USD OECD-base)

Architecture	Per-light-metro-3car add	Per-fleet (16 trainsets) add
Baseline (RFC 0021 §7)	—	—
+ Extension A (PCM)	~\$1 k	~\$16 k
+ Extensions A+B	~\$22 k	~\$350 k

Compare an unmitigated pack-replacement event at year 5 (baseline warranty assumed year 12): ~\$80 k per consist for OSR-discipline cells (360 kWh × ~\$90/kWh integrated + commissioning) plus ~2 weeks of revenue loss while strake is removed. Extensions A+B is not optional for Samawah-class envelopes.

9. Open questions

- 1 PCM lifecycle vs cell-fault venting — whether wax leak after a thermal event creates a secondary fire hazard.
- 2 R-744 vs R-1234yf — supply-chain analysis per target region.
- 3 Whether to fold this RFC back into RFC 0021 §7 as v2 of that section, or keep it separate.

10. Revision history

Date	Version	Change
2026-04-26	v0	Stub. PCM + dedicated-chiller extensions on top of RFC 0021 §7 baseline; selection rule by ambient + charge profile.

RFC 0025 — DIY Switch and Point-Machine Production

Status: Draft — extends [RFC 0012 §4](#) Date: 2026-04-26 Depends on: [RFC 0012 Switches & Crossings](#), [RFC 0007 Hardware Reference Designs](#)

1. Summary

[RFC 0012 §4](#) already commits a non-proprietary switch architecture — flexible machined switch blade, cast-manganese frog, electro-mechanical BLDC + planetary-gearbox + crank point machine, 2oo2 detection. What it does not commit is a production path for that architecture in target regions, nor specific commodity SKUs for the actuator. The result in practice: deployments still buy \$120 k turnouts from VAE / Vossloh / Schwihag because RFC 0012 is silent on the make-vs-buy decision.

This RFC commits a regional switch-shop bootstrap as the primary production path, with vendor procurement as the alternate for deployments that don't have the volume to justify the shop.

For a 100 km deployment with ~80 switches, the procurement-vs-DIY delta is ~\$8–10 M USD, the largest single affordability lever in any OSR civil build.

2. Non-goals

- Not a re-spec of RFC 0012 geometry. The three turnout tangents (1:9 / 1:14 / 1:18.5) and the switch architecture remain canonical.
- Not a re-spec of osr-wayside-points. The detection, observation, and consensus path stay in [osr-wayside-points](#).
- Not a re-spec of the safety case. A DIY switch and a vendor switch carry the same SIL-4 safety case — the SIL-4 logic lives in software (osr-wayside-points + osr-interlocking), not in the actuator. The mechanical part is below SIL.

3. Switch-shop bootstrap (primary path)

3.1 Tooling and capex

Equipment	Spec	New	Used
CNC milling machine	4 m × 1.5 m × 1 m bed, 3-axis minimum	\$200 k	\$80 k
Plate-cutting (plasma or fibre laser)	4 m × 8 m bed	\$80 k	\$30 k
Forge press	200 t hydraulic	\$80 k	\$35 k
Heat-treatment furnace	2 m × 1 m, 1 200 °C	\$60 k	\$25 k
Welding bay	MAG/MIG to EN 15085 CP-C2	\$30 k	\$15 k
Inspection table + CMM	4 m × 2 m granite + portable CMM	\$50 k	\$25 k
Misc tooling, jigs, fixtures	—	\$80 k	\$30 k
Total switch-shop CAPEX		~\$580 k	~\$240 k

3.2 Output and unit economics

- Production rate: ~2 switches/week steady-state, ~100/year.
- Per-switch BOM (1:9 turnout):
 - Switch blades (machined from rail blank): \$1 800
 - Stock rails: \$1 200
 - Stretcher bars (forged): \$400
 - Frog (cast manganese, sub-contracted to regional foundry): \$1 500
 - Check rails: \$300
 - Switch sleepers (per RFC 0009): \$2 000
 - Point machine assembly (per §4): \$2 200

- Heating + cabinet + harness: \$800
- Total BOM: ~\$10 200
- Vendor 1:9 reference: \$120 k (matches [lib/templates/switches.toml](#) kits.no-9-mainline cost_usd).
- DIY savings per 1:9: ~\$110 k.
- 80 switches: ~\$8.8 M saved, vs ~\$580 k shop CAPEX ~15 × payback.

3.3 Reference designs

Switch-blade and frog geometry are public at AREMA standard or UIC 60E1 equivalent. Drawings circulate in the public domain and cover every tangent in RFC 0012.

4. Point machine — commodity actuator

The RFC 0012 §4.2 architecture (BLDC + planetary + crank, 6 kN nominal / 12 kN peak, ≤ 3 s throw) is satisfied by commodity industrial linear actuators with no rail-specific silicon:

SKU class	Vendor	Stroke	Force	List	Notes
PC-Series	Thomson Linear	200 mm	5–10 kN	\$1 800	IP67, BLDC+planetary, regional distributor everywhere
EPCO	Festo	150–300 mm	5–8 kN	\$2 000	Same
CASM-100	SKF	200 mm	8 kN	\$1 600	Same
LinMot P10	LinMot AG	240 mm	5 kN	\$2 200	Direct-drive linear (no gearbox) — alternate
Regional copies	Various Chinese / Indian	200 mm	5–10 kN	\$700	Need qualification testing — see §6

Vendor-branded rail point machines (Siemens M3, Alstom M58 EBI Switch, Schwiag): \$25 k–\$45 k each.

Per OSR switch: ~\$1 800 vs \$25 k \$23 k saved per switch on the actuator alone.

5. Detection and integration

Unchanged from [RFC 0012 §5](#) and [osr-wayside-points](#). The DIY switch reports position over the same Zoo2 channel as a vendor switch; the W-SBC ([RFC 0007 §6](#)) sees no difference.

6. Qualification and safety

The DIY-switch SIL-4 case is the same as the vendor-switch case because:

- Position sense, locking confirmation, and FPS observation are all in [osr-wayside-points](#) (already SIL-4 qualified).
- The mechanical interlock is via the front stretcher bar, identical to the vendor architecture.
- The actuator below the lock has no SIL claim — it is "best effort" with the SIL-4 verdict coming from sensed position, not commanded position.

What does need a per-design qualification:

- 1 Throw force margin — proof test to 12 kN peak with a 1.5 × safety factor (18 kN ultimate).
- 2 Throw time — measured throw time ≤ 3 s from rest to locked per RFC 0012 §4.2.
- 3 Endurance — 10⁶ cycles at full force without lock failure (industry standard for points: ~3-month bench test).
- 4 Environmental — IP67, −25 °C to +55 °C operation.

These are mechanical qualifications, not safety-case qualifications. A regional university mechanical engineering department can run them in 6 months.

7. Procurement alternate

For deployments with < 20 switches/year volume, vendor procurement remains the right answer. Alternate suppliers in priority order:

Supplier	Region	Notes
Voestalpine VAE	Austria / Brazil / India	reference, premium pricing, MENA distribution
Vossloh Cogifer	France / Germany	reference

Supplier	Region	Notes
AnaSteel / Texmaco Rail	India	mid-tier, good for South Asia + East Africa
CRRC Yangtze	China	volume, ~50 % of European pricing
Voestalpine VAE Brasil	Brazil	regional Latin America

8. Open questions

- 1 Manganese-steel frog foundry — does the project source a regional foundry per deployment, or maintain a small list of qualified foundries?
- 2 Switch-blade-stock supply — head-hardened 60E1 rail blanks need to come from one of the rail mills in [RFC 0009](#).
- 3 Whether the actuator qualification (per §6) is a per-deployment activity or a one-time project-level activity.

9. Revision history

Date	Version	Change
2026-04-26	v0	Stub. Switch-shop CAPEX model, commodity actuator SKUs, DIY-vs-vendor unit economics, qualification frame.

RFC 0026 — Charging Connector and DC-Link Reconciliation

Status: Draft — reconciles RFC 0010 and RFC 0021 §6 Date: 2026-04-26 Depends on: RFC 0010 Station Design Standard, RFC 0021 Battery Traction

1. Summary

The current spec contains an internal inconsistency:

- RFC 0010 §1 commits 1 000 kW charging at terminal and depot-terminal archetypes, and 500 kW at major and interchange.
- RFC 0021 §6.1 specifies a 210 kW CCS2 receptacle per car at depot stalls (3 receptacles in parallel = 630 kW per consist).
- RFC 0021 §6.2 specifies a 100 kW plug-in charger at depot-terminal for opportunity charging.

The 210 kW CCS2 spec is internally consistent with itself — CCS2’s HPC envelope reaches ~350 kW per Combo 2 connector, so 210 kW is well inside the standard. The mismatch is between the station-archetype power figures in RFC 0010 and the connector spec in RFC 0021 §6: no connector currently in RFC 0021 reaches the 1 000 kW target the station archetype assumes, and CCS2 cannot scale to 1 MW.

This RFC commits a two-tier charging-connector architecture that covers both ends of the power range, updated for the rationalised operating model where normal passenger stops can charge the train for about one minute.

2. Non-goals

- Not a re-spec of station archetypes. RFC 0010’s archetype power targets remain canonical.
- Not a re-spec of pack chemistry or charge-acceptance limits. Per-cell rates remain as RFC 0021 §3 + RFC 0024.
- Not a hydrogen-refuelling or battery-swap RFC. Both remain out of scope per RFC 0021 §6.3.

3. Power-tier architecture

Tier	Power	Connector	Use case
Tier 1: depot stall	210 kW × 3 receptacles per consist = 630 kW	CCS2 (per RFC 0021 §6.1)	Overnight + scheduled depot charging. Unchanged.
Tier 2: station dwell	500 kW–1 000 kW per consist	Side-pin (primary) or pantograph-down (alternate)	Passenger-stop charging during ~60 s dwell, with terminal sites usually uprated

CCS2 stays as the depot connector — it is correct for that role and the per-receptacle handling is well-understood. The station-tier connector is the new piece this RFC commits.

4. Tier 2 — side-pin (primary)

Aspect	Choice	Rationale
Architecture	Passive copper bus-bar at platform edge; train extends a side-mounted pin to contact	Matches OSR’s level-boarding philosophy — the platform-edge geometry already aligns the pin without additional alignment hardware
Voltage / current	1 000 V DC × 500–1 000 A = 500 kW–1 MW class	500 kW covers normal 1 km station spacing; 1 MW covers terminals and high-load stops
Connector	Schaltbau-class high-current side connector + actuated pin	Schaltbau, Stäubli, Mersen all produce 1 000 A class units
Actuation	Pneumatic or electric linear actuator on the train	Same actuator family as RFC 0023 doors — commodity industrial silicon
Safety interlock	Pin retraction interlocked with door-close + brake-applied via OSR-OC	No live contact while doors are open or train is moving
DC-link cabinet	1 MW rectifier at platform end (ABB Terra HP, Heliox FLEX 600, or Star Charge / TGOOD tier-2)	Common cabinet across all Tier 2 sites

Why side-pin over pantograph: OSR is already a level-boarding system (350 mm low floor, platform-edge gap < 100 mm). The platform-edge geometry is dimensionally controlled to ± 5 mm by the precast L-unit (RFC 0010). The side-pin reuses this precision rather than adding a roof-level alignment problem. Pantograph-down also imposes a structural-load case on the canopy that the current portal-frame catenary-free design (RFC 0010 §X) explicitly avoids.

5. Tier 2 alternate — pantograph-down (OppCharge)

For sites where platform-edge geometry doesn't suit a side-pin (e.g. mid-line major archetype with island platform and offset canopy), specify pantograph-down per the ABB OppCharge / Siemens HPC150 reference.

Aspect	Choice
Power	600 kW continuous
Pantograph	1 m roof-mounted, single-arm, lift-down
Overhead bus-bar	Mounted to canopy portal beam, 5.5 m above rail head
Reference	ABB OppCharge ID, Siemens HPC150

Pantograph-down does require a structural-load case on the canopy, but the existing portal-frame design (RFC 0010 §X) has spare load capacity in the bay between two PV panels.

6. Tier 2 explicitly out of scope

- MCS (Megawatt Charging System). Reviewed and rejected for Tier 2 because (a) it is a CCS-evolved connector intended for heavy-vehicle plug-in, requiring a heavy cable and an operator, which is incompatible with the GoA 4 architecture (no driver to plug in), and (b) the standardisation is recent (2024) — the installed base is too thin for a regional spares pipeline.
- Inductive / wireless charging. Per RFC 0021 §6.3.

7. Selection rule

archetype	Tier 1 (depot)	Tier 2 (station dwell)
halt	—	optional, energy-model driven
standard	—	side-pin (primary)
major	—	side-pin or pantograph-down
interchange	—	side-pin or pantograph-down
terminal	—	side-pin, usually 1 MW
depot-terminal	CCS2 (RFC 0021)	side-pin, usually 1 MW

standard, major, interchange, terminal, and depot-terminal carry Tier 2 charging by default on the stop-spaced architecture. halt sites may omit it when adjacent chargers and onboard reserve meet the timetable energy model.

8. Cost impact (USD OECD-base, per Tier 2 site)

Rectifier sourced from tier-2 (TGOOD / Star Charge / Heliox), not European tier-1 (ABB / Siemens). Tier-1 rectifiers list ~\$400 k for the same 1 MW class — see RFC 0021 sourcing matrix.

Item	Side-pin	Pantograph-down
Train-side connector + actuator	\$40 k	\$80 k (pantograph)
Platform-side bus-bar / overhead bar	\$25 k	\$35 k
Rectifier cabinet (1 MW, tier-2)	\$250 k	\$250 k
Civil + cabling	\$80 k	\$120 k
Total per site	~\$395 k	~\$485 k

A 12-station line with Tier 2 at every passenger stop is roughly \$4.7 M side-pin at the same per-site budget before any terminal uprating. This replaces onboard battery mass and continuous catenary with distributed station chargers and stationary buffers. Country-cost multipliers from lib/templates/country-costs.toml apply to civil + integration shares downstream.

9. Open questions

- Side-pin actuator failure mode — what happens to a train that docks but cannot retract? Recovery procedure to be elaborated in RFC 0013.
- Whether the side-pin can be the only charging path for a small deployment, removing CCS2 from the depot — simpler spares pipeline at the cost of one connector standard.
- DC-link redundancy at Tier 2 — single cabinet vs N+1.

10. Revision history

Date	Version	Change
2026-04-26	v0	Stub. Two-tier architecture (CCS2 depot + side-pin/pantograph terminal), reconciles RFC 0010 RFC 0021 §6.

RFC 0027 — Brownfield Pilot: Existing-Asset Recovery & Workshop Integration

Status: Draft — proposed Date: 2026-04-26 Depends on: [RFC 0003 Samawah Reference Deployment](#), [RFC 0008 Rolling-Stock Reference Design](#), [RFC 0014 Depot Design Standard](#), [RFC 0022 Bogie + Traction Drive](#), [RFC 0025 DIY Switch and Point-Machine](#)

1. Summary

Most OSR target countries — Iraq, Sudan, Pakistan, Mozambique, Angola, Venezuela, several CIS states — already own large quantities of dormant Soviet-era / British-era / American-aid-era rolling stock and rail workshops, mostly stored or partially functional. RFC 0003 (Samawah reference deployment) is the worked example: satellite imagery identifies 300–800 wagons stored in the rail yards adjacent to Samawah Train Station, plus a rolling-stock workshop building (historically the target of a 2011 Iranian Waxon Park rehabilitation deal whose outcome is unclear).

Stored fleet + dormant workshop + active mainline through the city makes Samawah a brownfield pilot rather than a greenfield one. This RFC commits a systematic doctrine for converting brownfield assets into OSR-compatible production capacity, and is intended to apply to every deployment that begins with an existing asset stockpile.

The doctrine has three phases:

Phase	Scope	Duration
P1 — Asset assessment	Site visit, fleet inventory, workshop tooling audit, bogie/wheelset NDT sample, IRR/national-rail-authority disposition negotiation	3–6 months
P2 — Component recovery	Strip 50–100 wagons for bogies, wheelsets, brake hardware, fasteners, couplers; sell scrap steel; train workforce on the strip operation	6–12 months
P3 — First-article OSR trainset	Build OSR bogie frame (RFC 0022 §2.2) + wheelset recovered in P2, OSR car body (RFC 0008 Track A or B), commission at the rehabilitated workshop. Workshop becomes the regional rolling-stock factory.	12–24 months

This RFC does NOT replace any greenfield-spec RFC. It runs in parallel to RFC 0008 / RFC 0014 / RFC 0022. Where a deployment has no brownfield assets, the greenfield path remains the default. Where it has them, this RFC is the contract for converting them.

2. Non-goals

- Not a wagon-acquisition policy. The legal framework for buying, leasing, or jointly operating recovered wagons is per-deployment and lives in RFC 0029 (operator-financial model, separate).
- Not a replacement for RFC 0008. OSR rolling stock is still the light-metro / metro families per RFC 0008 §1. Recovered wagons are donor stock — the OSR trainset is built fresh, using recovered components.
- Not a heritage-preservation effort. Wagons that are historically significant should be retained for museum/preservation use; this RFC's scope is the operationally serviceable bulk remainder.

3. Why brownfield pilots beat greenfield where assets exist

Asset class	Greenfield CAPEX	Brownfield CAPEX	Saved
Wheelsets (760 mm forged, refurbished)	\$2 800–4 200 / pair (RFC 0022 §10 tier-2 sourcing)	\$300–800 / pair (NDT + reprofile)	~\$2 500 / pair × 4 pairs / consist = ~\$10 k / consist
Bogie machine tools (CNC mill 4 m, jig table, MAG welder bay, plasma cutter)	\$400 k–700 k for greenfield switch shop (RFC 0025 §3.1)	\$50–200 k for tool refurbishment + commissioning at existing workshop	~\$250–500 k per shop
Workshop building (10 000 m² portal-frame industrial shed, pit tracks, traverser foundations, crane gantries)	\$5–8 M for greenfield depot (RFC 0014)	\$0.5–2 M for rehabilitation of dormant workshop	~\$3–6 M per workshop
Rail-grade structural steel (recoverable from non-restorable wagon bodies)	\$1 100–1 400 / t (RFC 0009 sourcing)	\$150–250 / t scrap-recovery cost	~\$900 / t × ~5 t / wagon × 200 wagons = ~\$900 k
Coupler / brake hardware (SA-3, vacuum / air-brake fittings)	\$5 k–12 k / unit	\$200–500 / unit (refurbish or scrap)	Highly site-specific
Trained workforce (rail mechanical fitters, welders, machine operators)	18–24 months training pipeline (RFC 0027 — workforce plan, planned)	Existing workforce continues + cross-trains on OSR-specific kit	Salary-savings + early-deployment time-savings of 1–2 years

For a Samawah-scale pilot (16 trainsets, single light-metro line, one main depot), brownfield asset recovery saves on the order of \$8–15 M in mechanical/civil capital plus 18–24 months of schedule — a meaningful fraction of the RFC 0021 \$6 €435 M total CAPEX, and an even larger fraction of the time budget which is usually the binding constraint.

4. Phase 1 — Asset assessment (3–6 months)

4.1 Required deliverables

Artefact	Owner	Scope
Fleet inventory	OSR design team + national rail authority	Wagon-by-wagon list with VIN, year, type (covered / hopper / flatbed / passenger coach), recorded last service date, structural condition (1–5 scale), owning entity, legal disposition status
Workshop tooling audit	Mechanical engineering team (local + visiting)	Per-tool register: machine type, capacity envelope, last calibration / operation date, replacement/repair cost, missing components
Bogie / wheelset NDT campaign	Independent NDT contractor	Magnetic-particle inspection on a representative ~5 % sample of stored wheelsets; ultrasonic testing on a ~2 % sample of axles. Report flaw severity per EN 13262 class.
Civil survey of workshop site	Local civil engineering firm	Foundation condition, water/sewer connections, electrical-supply capacity, road/rail access, contamination assessment
Disposition framework	Government-relations lead	Memorandum-of-understanding draft with national rail authority covering ownership, lease terms, joint-venture options, indemnity for legacy environmental liabilities

4.2 Sample budget

Phase 1 typically runs \$200 k–500 k depending on stockpile size. Major line items:

Line item	Sample cost
Fleet-inventory team (4 engineers × 4 months on-site)	\$80–120 k
NDT contractor (sample wheelsets / axles)	\$40–80 k
Workshop tooling audit + civil survey	\$30–60 k
MoU legal / negotiation support	\$20–60 k
Per-diem, travel, equipment hire	\$30–80 k
Contingency 20 %	\$40–80 k

This is a small fraction of the total project CAPEX and the findings feed directly into Phase 2 / Phase 3 sizing.

5. Phase 2 — Component recovery (6–12 months)

5.1 Recovery operation

The recovery operation runs at the existing workshop, using the existing workforce. Wagons are pulled into the workshop one at a time, stripped to component level, and the components are sorted into:

- OSR-reusable components (wheelsets, axleboxes, bearings, brake cylinders, coupler bodies, structural-steel sections, hand brakes, electrical contactors, lighting fixtures).
- Resale components (couplers + drawgear in working order go to freight operators in the region; HVAC and water systems if any).
- Scrap stream (steel body sheets, broken castings, weather- damaged wood).

5.2 Wheelset recovery — the highest-value stream

Per RFC 0022 §3, OSR uses 760 mm new / 680 mm worn wheelsets. Iraqi Republic Railways (IRR) and most former-Soviet-aligned networks use 920 mm / 1050 mm wheelsets — these are NOT directly OSR-compatible. However, the axleboxes, bearings, springs, and brake gear are reusable, and the axles themselves can be re-machined to OSR diameter if their rough-bar diameter is above the 130 mm minimum.

So the recovery yield is:

- Wheelsets directly usable: ~5–10 % (where rolling stock was already 760 mm, e.g. some passenger-coach trucks)
- Axles re-machinable: ~50–60 % of stored stock
- Axleboxes / bearings / spring sets: ~70–80 % (subject to NDT)
- Brake hardware (cylinders, slack adjusters): ~60–70 %

- Couplers (SA-3 + drawgear): ~70 % directly resaleable to legacy freight operators in the region

Strip rate at a properly tooled workshop is ~1 wagon per crew per shift. With two crews and a single shift, ~10 wagons / week 250 / year. This is sufficient to support a 16-trainset OSR fleet at 4 wheelsets / car x 3 cars / consist x 16 = 192 wheelsets, with the wheelset balance going to the spares pipeline and to subsequent deployments.

5.3 Coupler retrofit – the catch

OSR commits to Scharfenberg Type 10 automatic couplers per [RFC 0008 §3.1](#). Recovered wagons use SA-3 (former-Soviet automatic) plus screw/buffer. These are incompatible, so the brownfield path does NOT involve re-coupling recovered wagons into the OSR consist directly. Every OSR car ships with new Scharfenberg couplers; the recovered SA-3 couplers go to the resale stream.

This is correct: a brownfield recovery should NOT compromise the single-coupler-standard discipline that lets any OSR consist rescue any other OSR consist on the network.

6. Phase 3 – First-article OSR trainset (12–24 months)

The recovered components feed forward into the standard OSR build process:

OSR component	Source from recovery
Bogie frame (RFC 0022 §2.2)	New, fabricated in the rehabilitated workshop using its CNC + welding capability
Wheelset (RFC 0022 §3)	Recovered axle + new (or refurbished 760 mm) wheel forging press-fit at the workshop
Axlebox + bearing	Recovered, NDT-passed, with new gaskets / seals
Primary suspension (chevron rubber)	New (RFC 0022 §5.1) – recovered rubber elements are usually weathered
Secondary suspension (air-spring)	New (RFC 0022 §5.2) – recovered air-spring assemblies are usually leak-prone after >5 y storage
Traction motor (PMSM)	New (RFC 0022 §4.1, sourced from RFC 0022 §10 tier-2 supplier)
Inverter (SiC)	New (RFC 0008 §3.2 DIY path or RFC 0022 §4 sourced)
Brake disc + caliper	New (per RFC 0022 §6) – recovered cast-iron discs are usually unsuitable for the regenerative-friction blend RFC 0021 expects
Body shell (alu-extrusion or steel space-frame)	New (RFC 0008 §3.1 Track A or RFC 0023 §1.1 Track B) – built in the workshop
Battery (Na-ion)	New per RFC 0021
Electronics (DIY safety stack)	New per RFC 0019
Coupler	New Scharfenberg (RFC 0008 §3.1)
Lighting / interior	Mix – recovered wiring harness + COTS interior per RFC 0008 §3.3

In other words: the brownfield component recovery yields ~30–40 % of the per-consist mechanical mass and ~\$30–60 k per consist in saved component cost, but the OSR architecture is preserved verbatim. No SA-3 couplers, no 920 mm wheelsets, no Soviet-era electrical systems on the OSR consist itself.

7. Workshop conversion – what stays, what goes

The Samawah workshop is presumed to have:

Existing capability	OSR fit	Action
Wheelset turning / re-profiling	Direct fit	Keep + commission to RFC 0022 NDT spec
Bogie pit tracks	Direct fit	Keep + extend to OSR bogie wheelbase (2 100 mm – RFC 0022 §3)
Overhead bridge crane	Useful but RFC 0014 prefers stinger track	Keep where existing; RFC 0014 stinger-track is a greenfield-only choice
Steam + diesel locomotive shed equipment	Not applicable to OSR (battery-electric, RFC 0021)	Decommission or convert to general-purpose mechanical workshop
SA-3 coupler tooling	Not applicable	Decommission or repurpose for resale-stream operations
Air-brake testing rig	Compatible if convertible to electric brake	Repurpose for OSR EP-brake commissioning (RFC 0008 §3.2)
Wagon-frame welding line	Direct fit	Keep + commission to EN 15085 CP-C2 (the lowest rail welding class – fits the existing workforce)
Painting + livery shop	Direct fit	Keep + retrain on RFC 0008 §3.5 livery scheme

The conversion isn't free – typically \$0.5–2 M for the rehabilitation itself plus \$1–3 M for net-new tooling required for OSR-specific operations (PMSM motor commissioning, SiC inverter test rigs, Na-ion battery pack assembly). But that's an order of magnitude less than greenfield depot CAPEX of €25 M (RFC 0014).

8. Risk register

Risk	Likelihood	Mitigation
Phase 1 inventory reveals stored fleet is too degraded to recover	Medium	Phase 1 cost is bounded (\$200–500 k); decision-gate at end of Phase 1 – abandon brownfield if NDT pass-rate < 30 %
National rail authority refuses to negotiate disposition	Low–medium	Pre-screen via diplomatic + technical-community channels before committing to Phase 1
Workshop legacy environmental liability (asbestos, lead paint, fuel-tank contamination)	Medium	Civil survey in Phase 1 explicitly includes environmental scoping; remediation cost goes into the disposition negotiation, not the project budget
Workforce knowledge gap on OSR-specific equipment (SiC inverters, PMSM motors, DIY safety electronics)	High	Cross-training in Phase 2, with project-funded secondment of OSR-experienced engineers from earlier deployments. RFC 0027 (workforce plan, planned) will own this.
SA-3 coupler resale market dries up before recovery completes	Low	Resale stream is a bonus, not a critical path. Scrap value remains.
Recovered axles fail post-machining NDT	Medium	Build Phase 3 demand schedule against the post-NDT survival rate, not raw recovery numbers. Carry a 30 % buffer.
Mid-project change of national-rail-authority leadership delays disposition	Medium–high	MoU should commit ownership at Phase 1 close, before Phase 2 capital is exposed

9. Applicability beyond Samawah

This RFC's Samawah example is illustrative. The same doctrine applies wherever a country has:

- An active mainline running through (or near) a candidate OSR deployment city.
- A dormant rolling-stock stockpile of standard-gauge (1 435 mm) or metre-gauge (1 000 mm) freight / passenger stock – even if the stock is otherwise incompatible with OSR.
- A rail workshop / depot building, even partially functional.
- A national rail authority capable of negotiating disposition.

Cities with documented brownfield potential (preliminary, not verified):

City	Country	Stockpile rumour / status
Samawah	Iraq	Confirmed by 2026-04-26 satellite review – 300–800 wagons, workshop building
Khartoum	Sudan	Sudan Railways stored fleet at Atbara workshop (Sudan Railways / Sudan Railway Workshops)
Karachi	Pakistan	Pakistan Railways Karachi Cantonment yard – large freight stockpile
Maputo	Mozambique	CFM stored fleet at Machava
Luanda	Angola	CFL stockpile (Caminho de Ferro de Luanda)
Tashkent	Uzbekistan	UTY workshops at Tashkent + significant Soviet-era stockpile

Each of these warrants a city-specific Phase 1 assessment before greenfield-vs-brownfield decision. The decision should default to brownfield where assets are real and recoverable.

10. Open questions

- 1 IRR (Iraqi Republic Railways) disposition framework – ownership of stored stock, lease terms, joint-venture options for the Samawah workshop. Requires direct contact with the Iraqi Ministry of Transport () and IRR / .
- 2 Waxon Park 2011 deal outcome – did the Iranian rehabilitation agreement complete? What was delivered? Is the equipment intact?
- 3 Wagon-by-wagon census – currently we have a satellite-image estimate of 300–800 wagons. A real census on the ground is the single highest-value Phase 1 deliverable.
- 4 Environmental liability assessment at the workshop – historical diesel + steam-loco operations may have left contamination that needs remediation before OSR commissioning.
- 5 Legal pathway for Iraqi (or other national) rolling-stock transfer to a project-controlled vehicle – sovereign-immunity, competition law, public-procurement compliance.

11. Revision history

Date	Version	Change
2026-04-26	v0	Stub. Three-phase doctrine (assess / recover / build), Samawah worked example, applicability matrix. Triggered by 2026-04-26 operator review identifying the Samawah rail-yard stockpile + workshop from satellite imagery.

docs/rolling-stock/light-metro-3car/README.md

docs/rolling-stock/light-metro-3car/README.md

light-metro-3car shop-drawing package — v1 deliverable of RFC 0008

This directory holds the dimensioned specification for the light-metro-3car trainset — the Samawah reference family per [RFC 0003](#) and the default for populations 300 k – 1 M per [RFC 0008 §5](#).

This is the v1 deliverable: a specification a domestic rolling- stock fabricator can bid on — dimensions, masses, interfaces, sub-assembly tree, fabrication sequence, and a procurement BOM skeleton. The design deliberately favours a modern low-capex factory: COTS rail subsystems wherever possible, simple cut/bend/ weld fabrication for the primary frame, and composite non-structural panels for sides, roof fairings, cableless cowls, and interior liners. The v2 deliverable is the full CAD pack + detailed shop drawings per EN 15085 weld classes, EN 45545 material certifications, and the supplier installation drawings for selected COTS modules.

Contents

File	Scope
general-arrangement.md	Overall envelope, gauge clearance, consist diagram, floor heights, door positions
fabrication-plan.md	Cut-bend-weld primary structure, composite cladding, COTS module installation sequence
bogie.md	2-axle articulated bogie spec, wheel profile, suspension, brake mount
body.md	Welded steel underframe/spaceframe, composite side panels, end bulkheads, articulation joint
traction.md	PMSM motor + SiC inverter + reduction gear, adhesion budget
interfaces.md	Coupler, pantograph, platform gap, TCN-E connector, aux power
bom-skeleton.md	Procurement BOM lines (source-identified parts vs TBD)
compliance.md	Standards matrix: EN 15227, EN 45545, EN 14363, EN 50155, ISO 3095, EN 12299
drawing-register.md	v2 drawing IDs, supplier documents, inspection evidence, release gates

COTS passenger-facing modules are controlled by the supplier-neutral envelope catalogue at [hardware/trainset-interiors/cots-catalogue.md](#).

Reference envelope (from RFC 0008 §1)

Parameter	Value
Cars	3 (articulated)
Overall length (over sensor cowls / couplers)	56.6 m
Tare mass	102 t
Axle load (AW3 crush)	≤ 14 t
Max speed	25 m/s (90 km/h)
Seats	60 longitudinal seats
Passenger capacity (AW2)	330 (seated + standing)
Passenger capacity (AW3 crush)	420 short-duration crush load
Onboard battery	360 kWh Na-ion (120 kWh per self-contained car)
Peak onboard motor output	1 800 kW
Floor height (above ToR)	Low-floor centre door zone; raised floor over standard bogies
Gauge	1 435 mm (default) or 1 000 mm (variant)

What v1 does NOT include

- KiCad / MCAD / STEP files (v2).
- Detailed finite-element analysis (v3 — homologation phase).
- Paint-and-livery guidance (operator scope).
- Fire-load and smoke-extraction analysis for the battery bay (v2, paired with EN 45545-2 test campaign).

How to execute this package

- 1 A fabricator reads [general-arrangement.md](#)

- [interfaces.md](#) to size their production line tooling.
- 2 [fabrication-plan.md](#) defines the shop route: tube/plate cutting, press-brake bends, welding fixtures, composite bonding, and COTS module installation.
 - 3 [bom-skeleton.md](#) gives the procurement team the source-identified parts (off-the-shelf commodity) and the TBD parts (where the fabricator bids on make-or-buy).
 - 4 [drawing-register.md](#) turns the v2 CAD pack into controlled drawing IDs, supplier document requirements, and release gates.
 - 5 [compliance.md](#) lists the test campaigns the type-approval needs; each is a separately-tendered scope with an accredited test house.
 - 6 The v2 CAD pack (not yet produced) is the cut-list / NC code / welding-robot path artefact that goes on the shop floor.

Licensing

v1 specification: CC-BY-SA 4.0. v2 CAD + drawings: CERN-OHL-S v2, matching the hardware licensing from [ARCHITECTURE.md §9](#).

Body structure — light-metro-3car

The body architecture is deliberately simple to manufacture in a regional workshop: a welded carbon-steel primary structure with bonded/bolted composite cladding and COTS passenger-facing modules. The design avoids large aluminium extrusion dies, stainless body pressings, and bespoke door/window/HVAC development.

Architecture

Each 18.9 m car is split into four production zones:

Zone	Construction	Function
Underframe ladder	Cut/bend/weld S355 steel box and folded plate	Carries buff/compression, bogie pivots, battery boxes, coupler loads
Side/roof spaceframe	S355 rectangular hollow section	Door posts, window rails, roof equipment rails, composite panel support
Composite exterior	Fire-rated glass-fibre or basalt-fibre sandwich panels	Weather skin, insulation, aerodynamics, livery surface
Interior cassette	COTS panels, seats, lights, grab rails, PIS, HVAC ducting	Passenger finish and replaceable equipment

Steel is the certified load path. Composite panels are non-structural: they may stiffen local skins, but crashworthiness, jacking, lifting, coupler loads, and bogie loads are closed through the welded frame.

Primary steel structure

Underframe

The underframe is a jig-welded ladder:

Member	Section	Material	Notes
Side sill	200 × 100 × 8 mm RHS	S355J2 or local equivalent	Continuous between end modules
Centre spine	250 × 100 × 8 mm RHS	S355J2	Battery-box and equipment support
Cross bearer	160 × 80 × 6 mm RHS	S355J2	750 mm nominal pitch
Bogie bolster	Folded/welded 12 mm plate box	S355J2	Machined centre-pivot insert after weld
Coupler pocket	16 mm folded plate + crush can interface	S355J2	EN 15227 energy absorber bolts on
Battery tray rail	80 × 60 × 5 mm RHS	S355J2	Under-seat module cassette support

All primary welds are made in a rotating fixture so flat/horizontal weld positions dominate. The v2 drawing pack assigns EN 15085 weld classes per joint; the v1 assumption is CL1 for bolsters, coupler pockets, and crash boxes, CL2 for side sills/cross bearers, CL3/CL4 for panel tabs and interior brackets.

Side frame

The side frame is a light spaceframe welded to the underframe:

- Door posts: 120 × 80 × 6 mm RHS.
- Window posts: 80 × 50 × 4 mm RHS.
- Waist rail: 100 × 50 × 5 mm RHS.
- Cant rail: 100 × 50 × 5 mm RHS.
- Roof bows: 80 × 40 × 4 mm RHS at 1 000 mm pitch.
- Equipment rails: bolted stainless inserts for HVAC, antennas, marker lights, CCTV, and cable trays.

Door openings are not cut from a finished shell. They are built as open bays around the COTS door cassette so the door supplier's frame, threshold, drainage tray, and emergency-release hardware install as a single bolted module.

Composite exterior panels

Exterior panels are bonded and mechanically retained to the steel frame:

Panel	Construction	Attachment
Side skin	25-35 mm fire-rated GFRP or basalt-fibre sandwich	Structural adhesive + M6/M8 retained fasteners

Panel	Construction	Attachment
Roof fairing	20-30 mm composite sandwich	Bolted to roof bows; removable above HVAC/equipment
Nose/sensor cowl	Moulded composite shell over steel crash frame	Bolted; sacrificial and replaceable
Skirts	Composite or aluminium removable panels	Quarter-turn service fasteners
Interior liners	EN 45545 HL2 FRP or phenolic panels	Clip/bolt to secondary rails

The composite supplier owns laminate coupons, resin selection, fire test evidence, and repair manuals. The OSR body drawing owns only panel envelopes, edge radii, insert locations, grounding/bonding points, and removal clearances.

COTS windows

Windows are bought as certified rail/bus glazing modules rather than fabricated locally:

- Laminated safety glass to EN 15152 or equivalent.
- Typical opening: 900 × 1 200 mm.
- Bonded-in or gasketed cassette depending on supplier.
- Emergency ventilation only where required by the authority having jurisdiction; default HVAC assumes fixed glazing.
- Drainage path and adhesive bead are per supplier installation drawing, not re-engineered by OSR.

COTS doors

Door modules are commercial off-the-shelf electric sliding-plug or plug-outward rail doors:

- Clear opening: 1 250 mm × 2 000 mm.
- Electric actuation; no trainwide pneumatic door supply.
- Supplier-provided threshold, guide rail, controller, obstacle detection, emergency release, seals, and manual isolation.
- OSR provides 24 V DC / 110 V DC power, Ethernet/CAN-FD command, hardwired closed-and-locked loop, drainage, and mounting datum.

The preferred procurement pattern is to qualify two door suppliers against the same envelope. The carbody must not depend on a unique proprietary door pocket.

COTS roof and passenger systems

The body reserves bolted envelopes for:

- Roof HVAC: one 20 kW nominal packaged unit per car, hot-climate variant for +50 °C ambient.
- LED lighting: 24 V DC rail-rated strip or troffer modules.
- PIS displays and speakers: off-the-shelf rail/bus units.
- CCTV: PoE cameras in serviceable composite housings.
- Intercom: COTS EN 50155 rail intercom stations.
- Seats: longitudinal rail/bus bench modules bolted to under-seat battery cover rails.
- Grab rails: stainless tube systems with off-the-shelf clamps.

HVAC, lighting, PIS, CCTV, intercom, seats, and grab rails are procurement modules. OSR fixes the envelope, fastener grid, power budget, data interface, fire rating, and maintainability clearance. Supplier-neutral envelopes and evidence requirements live in [hardware/trainset-interiors/cots-catalogue.md](#).

Floor

- 5-6 mm steel or aluminium deck panels over cross bearers.
- Removable hatches above battery and HV equipment zones.
- Phenolic or aluminium honeycomb floor boards in the saloon.
- Slip-resistant vinyl or rubber floor covering, EN 45545-2 R5.
- Low-floor centre door zone; raised floor over bogies.

Corrosion protection

Steel primary structure receives:

- Shot blast to Sa 2.5.
- Zinc-rich epoxy primer.
- Seam sealer at composite/steel interfaces.
- Cavity wax in closed sections after weld inspection.
- Polyurethane topcoat where visible or exposed below floor.
- Drain holes in every closed bay that can collect condensation.

Composite/steel joints use isolation tape or coated inserts to avoid galvanic corrosion where aluminium brackets or stainless fasteners are present.

Fire compliance

Every passenger-facing material must meet EN 45545-2 HL2 or the deployment authority's accepted equivalent.

Item	Requirement
Steel primary frame	Non-combustible; coating fire data required
Composite exterior panels	R1 HL2 or better
Interior panels	R1 HL2 or better
Floor covering	R5 HL2 or better
Seats	R7 HL2 or better
Cable insulation	R15/R24 HL2 or better
Door/window seals	R23 HL2 or better

v2 deliverables

- Welded frame STEP assembly and 2D drawings.
- Flat pattern DXF for folded plates.
- Tube cut list with mitre angles and fixture datum references.
- Weld map with EN 15085 classes, WPS references, and inspection hold points.
- Composite panel envelope drawings and insert maps.
- Door/window/HVAC supplier interface drawings.
- Static proof-load and fatigue FEA report.
- EN 15227 crashworthiness simulation report.

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Bogies — light-metro-3car

Six bogies per consist: one powered bogie and one trailer bogie under each self-contained car. All bogies use the RFC 0022 standard frame, suspension, wheelset, brake, and pivot geometry; the trailer variant omits motors and gearbox.

Bogie envelope (all six, common)

Parameter	Value
Wheelbase (axle-to-axle)	2 100 mm
Wheel diameter, new / worn	760 / 680 mm
Wheelset type	Monobloc, machined per RFC 0022
Wheel profile	S1002 (1 435 gauge) or Ri60 (1 000 gauge variant)
Axle load (AW3, both axles)	≤ 14 t each
Gauge	1 435 mm (default) / 1 000 mm (variant)
Track-brake coil (magnetic track brake)	not fitted — per RFC 0008 §3.2 electric-only brake

Powered bogie (one per car)

Frame

- Welded H-frame, steel S355JR per EN 10025, EN 15085 CL1 welds.
- Length (axle-to-axle) 2 100 mm; width 2 400 mm; depth 300 mm.
- 4 suspension pedestals — 1 per axle end.

Primary suspension

Chevron rubber springs (one per axle end):

- Vertical stiffness: 1 500 kN/m nominal per spring.
- Lateral stiffness: 5 000 kN/m.
- No shock absorbers at primary level (the chevron's internal damping is sufficient).
- Service interval: replace every 10 years or on inspection finding; visual inspection every 7 days per RFC 0013 M5.

Secondary suspension

Steel coil springs + viscous damper (one per bogie end):

- Coil: progressive rate, 80 kN/m at nominal load to 160 kN/m at AW3.
- Damper: KONI 32-series linear, 6 kNs/m at nominal velocity.
- No air spring. Maintenance-free at the service interval.

Motor mount

Two PMSM axle-hung motors per powered bogie, one per axle. Motor details in [traction.md](#).

Brake

- Service + emergency: axle-mounted brake disc on each axle. Caliper electromagnetic actuation — two calipers per disc.
- Parking: spring-applied, electromagnetic-release on the same caliper; cut-out spring force ≥ 12 kN per caliper.
- WSP: wheel-slide protection via the per-axle tacho input + os r - b rake modulation (proptest-verified B4 conservative).

Trailer bogie (one per car)

Same basic envelope as driving bogie, with differences:

- No motor. All axles idle.
- Brake: same 600 mm disc on each axle for service / emergency / park.
- Mass: lower than the powered bogie by the omitted motors, gearboxes, and inverter cabling.

Interfaces

- Bogie-to-body pivot: centre pin ring bearing, 400 mm diameter, PTFE slider.
- Yaw restraint: two longitudinal links per bogie with elastomeric bushes. No hydraulic yaw dampers (tangent speeds ≤ 25 m/s don't demand them on the RFC 0009 preset radii).
- Power electrical: 1 500 V DC + 24 V DC + 400 V AC through a cableguide at the centre pin; not through the pivot itself.
- Air: none. (No pneumatic brake system per RFC 0008 §3.2.)

Interchangeability

Every powered bogie in every OSR deployment of this family is identical. The trailer bogie is the same SKU minus drivetrain lines (no motor mount, no traction contactor block) but shares:

- Frame geometry (drilling pattern).
- Suspension parts.
- Brake disc + caliper.
- Wheel monobloc.

So most bogie spares are shared across powered and trailer variants.

Weld classes (EN 15085)

Assembly	Class
Frame primary welds	CL1
Motor mount welds	CL1
Brake caliper mount	CL1
Cable-guide mount	CL3
Non-structural covers	CL4

Per EN 15085, every CL1 weld is radiographically inspected at the certified shop.

v2 deliverables (not in v1)

- Dimensioned assembly drawings per EN 15016 / ISO 128.
- FEA: frame stress under AW3 + dynamic accelerations.
- FRA: fatigue-endurance curves over 1M simulated load cycles.
- Weld-quality records template (EN 15085 compliance file).

docs/rolling-stock/light-metro-3car/bom-skeleton.md

docs/rolling-stock/light-metro-3car/bom-skeleton.md

Procurement BOM skeleton — light-metro-3car

This is the source-identified parts list a fabricator uses to price the consist. Parts split three ways:

- SOURCE — commodity off-the-shelf. Mouser / Digi-Key / LCSC line available; no engineering work required.
- MAKE — fabricator produces in-house. Single-supplier tender.
- BID — multi-supplier tender. The deployment's procurement office runs the bid.

Costs are BASE USD volume-100. Country cost factor from [lib/templates/country-costs.toml](#) scales them per deployment.

Body + interior

Line	Desc	Qty per consist	Source	Base USD	Notes
B1	S355 RHS tube, underframe + side/roof frame	3.2 t	SOURCE	7 700	Cut/bend/weld locally
B2	S355 folded plate, bolsters/coupler pockets/brackets	2.0 t	SOURCE	5 200	Press-brake + weld
B3	CNC cutting, drilling, fixture consumables	set	MAKE	12 000	Per consist allocation
B4	Weld consumables + shielding gas	set	SOURCE	4 800	EN 15085 WPS-controlled
B5	Shot blast + zinc-rich primer + cavity wax	set	SOURCE	8 500	Corrosion package
B6	Composite side sandwich panels, fire-rated	150 m ²	BID	45 000	GFRP/basalt FR sandwich
B7	Composite roof fairings + skirts	95 m ²	BID	24 000	Removable service panels
B8	Composite sensor cowl shells	2	BID	12 000	Sacrificial over steel crash frame
B9	Articulation joint, Hübner RTS-B40	2	BID	90 000	Or local equivalent
B10	COTS laminated safety-glass window cassette	12	SOURCE	18 000	Bonded/gasketed rail/bus unit
B11	COTS electric plug/sliding door cassette	6	BID	150 000	Door, controller, seals, emergency release
B12	Floor board + hatch system, EN 45545	135 m ²	SOURCE	16 200	Phenolic/aluminium honeycomb
B13	Vinyl/rubber floor covering, EN 45545 R5	135 m ²	SOURCE	5 400	
B14	COTS longitudinal seat modules, EN 45545 R7	60 seats	SOURCE	24 000	Rail/bus bench modules
B15	COTS grab rail + stanchion kit	set	SOURCE	8 000	Stainless modular system
B16	COTS interior LED lighting kit	3 cars	SOURCE	9 000	24 V DC, emergency mode
B17	COTS exterior marker + head/taillight kit	set	SOURCE	6 000	Rail/bus certified
B18	COTS PIS display + speaker kit	set	SOURCE	12 000	Interior/exterior
B19	COTS CCTV + intercom kit	set	SOURCE	15 000	PoE cameras, passenger intercom
B20	Paint + 2K urethane topcoat	set	SOURCE	6 000	Operator livery excluded
B21	Interior FR panels + trim clips	set	SOURCE	28 000	Supplier-certified HL2
Body + interior subtotal				506 800	

Bogies (6 per consist)

Line	Desc	Qty	Source	Base USD	Notes
G1	Welded bogie frame (powered)	3	MAKE	60 000	EN 15085 CL1
G2	Welded bogie frame (trailer)	3	MAKE	54 000	
G3	Wheelset monobloc (RFC 0022, S1002)	12	BID	144 000	

Line	Desc	Qty	Source	Base USD	Notes
G4	Axle bearing box (SKF / FAG)	24	SOURCE	48 000	
G5	Primary chevron spring	24	SOURCE	12 000	
G6	Secondary spring / air spring	12	SOURCE	7 200	
G7	Secondary damper	12	SOURCE	14 400	
G8	Brake disc	12	SOURCE	12 000	
G9	Electromagnetic brake caliper	24	BID	72 000	
G10	Park-brake spring assembly	24	SOURCE	9 600	
G11	Centre-pin ring bearing + PTFE slider	6	SOURCE	12 000	
G12	Yaw-restraint link + bushes	12	SOURCE	6 000	
G13	Cable-guide + centre-pin assembly	6	MAKE	9 000	
G14	Wheel-tach (quadrature encoder)	12	SOURCE	6 000	
G15	Axle bearing temp sensor	24	SOURCE	4 800	
Bogies subtotal				471 000	

Traction + power

Line	Desc	Qty	Source	Base USD	Notes
T1	PMSM axle motor (180 / 320 kW)	6	BID	180 000	
T2	Reduction gear (single-stage 6.5:1)	6	BID	42 000	
T3	SiC inverter (360 / 600 kW)	3	BID	105 000	Wolfspeed or equiv.
T4	Cold-plate + chiller for traction	3	SOURCE	27 000	
T5	Na-ion under-seat pack (120 kWh usable)	3	BID	43 200	CATL / HiNa / etc
T6	BMS electronics (pack-level)	3	BID	18 000	
T7	Pack cooling plate set	3	SOURCE	6 000	
T8	Under-seat aluminium module enclosure set	3	MAKE	12 000	
T9	Aspirating smoke detector (battery + traction bay)	6	SOURCE	18 000	
T10	Fire suppression (aerosol, auto-discharge)	6	SOURCE	24 000	
T11	HV contactor + bus bar set	3	SOURCE	24 000	
T12	Station charging side-pin connector	3	SOURCE	45 000	Pantograph-down alternate per site
T13	Aux inverter (400 V / 110 V / 24 V)	3	SOURCE	66 000	
T14	COTS packaged roof HVAC unit (20 kW each, per car)	3	BID	75 000	+50 °C hot-climate variant
T15	Regen dump resistor (roof-mount)	1	SOURCE	3 500	
Traction + power subtotal				688 700	

Electronics + safety

Line	Desc	Qty	Source	Base USD	Notes
E1	T-ECU/S board (2× RP2350 + CM5)	2	BID	2 500	Custom baseboard per RFC 0007 §4
E2	T-ECU/A board (CM5)	1	BID	1 500	Per RFC 0007 §5
E3	ADIS16505 IMU (or BMI088)	2	SOURCE	1 000	
E4	u-blox NEO-F10N GNSS	2	SOURCE	600	
E5	PN5180 NFC balise reader	1	SOURCE	200	
E6	ATECC608B SE chip (on T-ECU/S + T-ECU/A carriers)	3	SOURCE	30	
E7	Cat.22 5G M.2 module	1	SOURCE	500	
E8	LoRa SX1276 breakout	1	SOURCE	60	
E9	NVMe SSD 256 GB (event recorder)	1	SOURCE	60	
E10	10.1" touchscreen DMI (per cab)	2	SOURCE	800	
E11	Cab master-controller (combined)	2	BID	6 000	

Line	Desc	Qty	Source	Base USD	Notes
E12	Emergency plunger (hardwired)	2	SOURCE	400	
E13	Deadman handle + sensor	2	SOURCE	500	
E14	PIS display (exterior + interior)	8	SOURCE	6 400	
E15	CCTV camera (forward + door-sill + in-car)	20	SOURCE	6 000	
E16	2oo2 relay stage (per safety output)	4	SOURCE	1 200	
E17	Cable harness (pre-terminated, per car)	3	MAKE	30 000	
Electronics + safety subtotal				57 750	

Consist total

Bucket	Subtotal (USD)
Body + interior	506 800
Bogies	471 000
Traction + power	688 700
Electronics + safety	57 750
Total direct-material consist	1 724 250

Labour (shop weld, bonding, assembly, commissioning) adds ~35 %: ~604 000 USD.

Planning-grade per-consist cost (volume 100): ~2.3 M USD.

For comparison, legacy-vendor light-metro trainsets in the target regions typically land 4–6 M USD each – the OSR design's commodity-module + COTS-fit-out + simple welded-frame bets deliver a ~2–3× cost reduction at scale.

Per-deployment customisation

The operator's own procurement office fills in:

- Livery paint scheme (B20).
- Seat fabric colour (B14).
- TRG-1 5G carrier SIM spec (per-country mobile network).
- Public-safety radio band (E-: not in base BOM).
- Pantograph dock voltage if a non-1500 V DC dock is used (out of scope upstream).

v2 deliverables (not in v1)

- Supplier shortlist per BID line with qualification criteria.
- Lead-time analysis per SOURCE / MAKE / BID.
- Risk log: single-source parts + mitigation.
- Weight budget with per-line tare contribution + final target vs actual.

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Compliance matrix – light-metro-3car

Standards the consist is designed against. Every row is a type- approval campaign the operator runs pre-revenue. Upstream provides the design envelope; each deployment owns its own test-house relationship.

Structural + mechanical

Standard	Scope	Evidence for v1 v2
EN 15227 Cat C-II	Crashworthiness at 25 km/h metro scenario	FEA simulation report (v2)
EN 12663 Cat P-III	Static + fatigue body loading	FEA report (v2)
EN 14363	Dynamic behaviour + safety against derailment	On-track test campaign (v3)
EN 13749	Bogie frame fatigue + structural	FEA + test-rig (v2/v3)
EN 15085 CL1 –CL4	Weld quality per assembly	Per-welder EN ISO 9606 certificates (v2)
UIC 510-2	Wheel profile + wear limits	Monobloc certificate from supplier (v1 source)
UIC 505-1	Structure gauge	General-arrangement compliance shown (v1)

Fire + materials

Standard	Scope	Evidence
EN 45545-2 HL2	Fire behaviour of materials	Per-material test reports from an accredited lab (v2 campaign)
Applied to: body, seats, flooring, cabling, insulation, door rubbers		

Electrical + electronic

Standard	Scope	Evidence
EN 50155 OT4	On-board electronic equipment ambient class	Per-PCB compliance (RFC 0007 v1 bring-up)
EN 50121-3-2	Rolling-stock EMC	Pre-compliance sweep on test car (v3)
EN 50126	RAMS lifecycle	SIL-4 safety case – links into docs/safety-case/
EN 50128	Software for railway control – SIL-4	Coding standard per RFC 0005 §7; evidence per crate
EN 50129	Safety-related electronic systems – SIL-4	Composite fail-safe argument for T-ECU/S (RFC 0007 §4.1)

Passenger experience + accessibility

Standard	Scope	Evidence
EN 12299	Ride-comfort indices	Dynamic-ride test (v3)
ISO 3095	External + internal noise	Acoustic measurement (v3)
UIC 741	Platform-to-train gap	General arrangement + gap-filler flap spec (v1)
EN 14752	Door systems	IFE type-4 door or equiv. (v1 source-identified)
EN 16586	Accessibility PRM	Wheelchair-space layout and COTS fit-out envelopes (v1 – see COTS catalogue)
EN 15152	Windows	Laminated safety glass (v1 source-identified)

Onboard safety systems

Standard	Scope	Evidence
EN 50126 / EN 50128 / EN 50129 as above for SIL-4	Every onboard SIL-4 crate has proptest + Kani evidence	Per-crate solution nodes in the GSN safety case
EN 14033	Rail infrastructure works – applies to self-propelled work consists (n/a for revenue service)	n/a

Target-region overlays (per-deployment)

Each country's NRSA (or equivalent) has its own approval process. Common overlays:

- Iraq (RFC 0003 Samawah): Iraqi Railways directorate approval.
- GCC region: GCC rail-authorities common spec.

- Sub-Saharan Africa: per-country; AU railways coordination underway.

v1 does NOT enumerate overlays – that is the operator's NRSA-liaison scope. Upstream provides the EN-standard base.

Test-campaign ordering

A typical sequence from design freeze to revenue:

- 1 v2 CAD release: FEA reports (EN 15227, EN 12663, EN 13749).
- 2 v2A drawing register release: controlled drawings, supplier installation documents, and evidence gates issued per [drawing-register.md](#).
- 3 First-article prototype: EN 45545 material test campaigns.
- 4 Static test on bogie rig: EN 13749 frame fatigue.
- 5 Dynamic test (pilot deployment): EN 14363 derailment safety; EN 12299 ride comfort; ISO 3095 noise; EN 50121-3-2 EMC.
- 6 Safety-case sign-off: EN 50126 lifecycle file complete.
- 7 NRSA approval.
- 8 Revenue service.

Steps 1–5 are fabricator + accredited-test-house scope. Steps 6–7 are operator + project-safety-team scope.

What v1 commits to

- Source-identified parts where such parts exist (Scharfenberg coupler, IFE door, PZ pantograph, EN 45545 materials).
- BID scope clearly delimited (motor, inverter, battery pack).
- MAKE scope matched to local fabricator capability (cut/bend/weld body frame, bogie frames, brackets, wiring harness).
- No proprietary/closed interfaces anywhere in the consist.

What v1 does NOT commit to

- Specific test-house accreditation (per-country choice).
- Specific national-regulatory overlay fitments (flag signage, emergency PA language, fare-gate integration – all per-deployment).
- Second-source data for BID parts (v2).

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Drawing and evidence register — light-metro-3car

This register is the bridge between the v1 Markdown specification and the v2 CAD/shop-drawing release. It lists the controlled drawings, supplier documents, and test evidence a fabricator must produce before the first article can be released for static and dynamic testing.

Document IDs use LM3 for the light-metro-3car family. Revision A is the first v2 release candidate.

Controlled drawing set

ID	Title	Owner	v1 source	v2 release content
LM3-GA-000	Trainset general arrangement	OSR/mechanical	general - arrangement .md	2D GA, clearance envelope, platform interface, mass table
LM3-BDY-100	Carbody primary steel assembly	Fabricator	body .md	STEP, 2D assembly, weld map, datum scheme
LM3-BDY-110	Underframe ladder assembly	Fabricator	body .md	Tube cut list, plate flat patterns, fixture drawing
LM3-BDY-120	Side/roof spaceframe assembly	Fabricator	body .md	RHS cut list, door/window aperture datums
LM3-BDY-130	Coupler pocket and crash-can interface	Fabricator + coupler supplier	body .md, interfaces .md	Machined inserts, bolted energy absorber interface
LM3-BDY-140	Battery tray and under-seat enclosure	Fabricator + battery supplier	body .md, traction .md	Service hatches, vent path, HV isolation clearances
LM3-BDY-150	Composite side/roof/nose panel envelopes	Composite supplier	body .md	Panel mould drawings, inserts, edge radii, repair zones
LM3-DOOR-200	Door cassette installation	Door supplier	COTS catalogue	Mounting datums, threshold, drainage, lock-loop wiring
LM3-WIN-210	Window cassette installation	Glazing supplier	COTS catalogue	Bond/gasket land, drain path, replacement method
LM3-HVAC-220	Roof HVAC installation	HVAC supplier	COTS catalogue	Roof rails, ducting, condensate, service clearance
LM3-INT-230	Interior fit-out installation	Integrator	COTS catalogue	Seats, rails, floor boards, hatches, panels, signage
LM3-ELC-300	Low-voltage harness routing	Integrator	interfaces .md	Harness schedule, connector list, segregation, labels
LM3-HV-310	HV battery/traction routing	Integrator	traction .md	HVIL loop, busbars/cables, insulation clearances
LM3-BOG-400	Powered bogie assembly	Bogie fabricator	bogie .md	Frame drawing, motor/gearbox/brake interfaces
LM3-BOG-410	Trailer bogie assembly	Bogie fabricator	bogie .md	Frame drawing, brake/suspension interfaces
LM3-TRC-500	Traction package installation	Traction supplier	traction .md	Motor, gearbox, inverter, cooling, EMC bonding
LM3-COM-600	Train communication and antenna install	Integrator	interfaces .md	TCN-E, radios, GNSS, CCTV, PIS, intercom

Manufacturing evidence

ID	Evidence	Required before	Notes
LM3-EV-MAT-001	Steel mill certificates and heat traceability	G0 material release	Covers RHS, plate, machined inserts
LM3-EV-WLD-010	WPS/PQR register and welder qualifications	First production weld	EN 15085 / EN ISO 9606 basis
LM3-EV-WLD-020	Weld inspection and NDT report	G1 frame complete	VT + MT/UT per weld class
LM3-EV-DIM-030	Frame dimensional survey	G1 frame complete	Bogie centres, door/window apertures, coupler height
LM3-EV-COR-040	Blast, primer, topcoat, cavity-wax report	G2 corrosion complete	Includes DFT readings
LM3-EV-CMP-050	Composite material certificate pack	G3 shell complete	EN 45545 evidence and repair method
LM3-EV-BND-060	Adhesive batch, surface-prep, cure records	G3 shell complete	Includes witness coupons
LM3-EV-WTR-070	Water ingress test	G4 COTS systems fitted	Doors, windows, roof equipment
LM3-EV-ELC-080	Bonding/earthing and insulation resistance	G5 electrical safe	Includes HV and LV separation checks
LM3-EV-HV-090	HVIL and battery isolation report	G5 electrical safe	Battery supplier + integrator sign-off
LM3-EV-DOOR-100	Door obstruction, release, lock-loop test	G6 static complete	Per door cassette serial number
LM3-EV-HVAC-110	HVAC cooling and drain test	G6 static complete	+50 C performance evidence may be supplier lab data
LM3-EV-BRK-120	Static brake and park-brake test	G6 static complete	Links to bogie/brake supplier records
LM3-EV-MASS-130	Weighing and axle-load report	G7 dynamic ready	Empty and simulated AW2/AW3 cases

ID	Evidence	Required before	Notes
LM3-EV-RIDE-140	Bogie alignment and ride-height report	G7 dynamic ready	Before dynamic testing

Supplier document index

Package	Supplier document required	Acceptance rule
Doors	Installation manual, maintenance manual, lifecycle test, EN 14752 evidence	Fits LM3-DOOR-200 without primary steel change
Windows	Glazing certificate, adhesive/gasket procedure, replacement manual	Fits LM3-WIN-210 and passes water test
HVAC	Datasheet, +50 C derating curve, wiring manual, refrigerant record	Fits LM3-HVAC-220 and aux-power budget
Composite panels	Laminate schedule, fire test, repair manual, insert pull-out data	Panels remain non-structural
Seats/rails	Fire certificate, pull-load certificate, cleaning/repair manual	Fits LM3-INT-230 insert grid
Lighting/PIS/CCTV/intercom	EMC/vibration evidence, firmware version, wiring manual	Enumerates on car network and passes static test
Battery	Cell/module certificate, BMS manual, vent/fire containment data	Fits LM3-BDY-140 and LM3-HV-310
Traction	Motor/inverter/gearbox datasheets, cooling and EMC instructions	Fits LM3-TRC-500
Bogie parts	Wheelset, bearing, spring, damper, brake supplier certificates	Fits LM3-BOG-400/410

Release gates

Gate	Minimum drawing/evidence state
v2A design freeze	All controlled drawings issued at rev A; supplier envelopes frozen
First steel cut	LM3-BDY-100/110/120, LM3-EV-MAT-001, LM3-EV-WLD-010 complete
First carbody shell complete	LM3-EV-WLD-020, LM3-EV-DIM-030, LM3-EV-COR-040 complete
First COTS fit-out complete	Door/window/HVAC/interior supplier docs accepted; LM3-EV-WTR-070 complete
First energisation	LM3-EV-ELC-080 and LM3-EV-HV-090 complete
Static test release	LM3-EV-DOOR-100, LM3-EV-HVAC-110, LM3-EV-BRK-120 complete
Dynamic test release	LM3-EV-MASS-130 and LM3-EV-RIDE-140 complete

Change control

- Any supplier swap that fits the reserved envelope is a document revision, not a structural redesign.
- Any change to side sills, bolsters, coupler pockets, door posts, or crash-can interfaces reopens FEA and static proof evidence.
- Any mass increase above a module row limit requires a new axle-load report before dynamic testing.
- Any power increase above a module row budget requires an aux-power and thermal review.

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Fabrication plan — light-metro-3car

This plan turns the v1 rolling-stock package into a buildable factory route for the first light-metro-3car trainset. It favours ordinary regional manufacturing capability: saw/laser/plasma cutting, press-brake bending, MIG/MAG welding, adhesive bonding, bolted module installation, and COTS rail/bus subsystems.

It is not a homologated shop drawing pack. The v2 pack must add dimensioned drawings, weld symbols, tolerances, FEA, supplier installation manuals, and authority-approved inspection records.

Manufacturing doctrine

Scope	Default decision	Reason
Primary structure	MAKE: cut/bend/weld S355 steel	Available in rail, truck, ship, oilfield, and bridge workshops
Side/roof/nose panels	BID/SOURCE: certified composite panels	Low tooling cost, corrosion resistant, replaceable after damage
Doors	BID: COTS electric rail door cassette	Certified obstacle detection, locking, seals, emergency release
Windows	SOURCE/BID: COTS bonded/gasketed glazing	Avoids bespoke glass certification
HVAC	SOURCE/BID: packaged roof unit	Mature hot-climate bus/rail supply chain
Lighting	SOURCE: rail/bus LED modules	Commodity, low-voltage, easy replacement
Passenger interior	SOURCE/BID: bus/rail seats, rails, PIS, CCTV	Keeps OSR focused on interfaces and safety evidence
Bogie frame	MAKE under controlled weld procedure	Structural rail part; can be local but must be qualified
Wheelsets, bearings, brakes	BID/SOURCE	Safety-critical rotating/brake parts stay supplier-certified
Traction, batteries, BMS	BID/SOURCE	Supplier qualification and test evidence required

Factory cells

The first factory needs five cells, not a full legacy rolling-stock plant:

- 1 Steel prep cell
 - CNC saw or tube laser for RHS.
 - Plasma/laser/waterjet table for plate.
 - Press brake for folded bolsters, coupler pockets, trays, and brackets.
 - Drill/mill for datum holes and bolted inserts.
- 2 Weld and fixture cell
 - Underframe rotating fixture.
 - Side-frame fixture.
 - Bogie-frame fixture.
 - Certified MIG/MAG equipment.
 - Weld fume extraction and preheat capability where WPS requires.
- 3 Paint and corrosion cell
 - Shot blast or outsourced blast booth.
 - Zinc-rich epoxy primer.
 - Seam sealing and cavity wax station.
 - Topcoat booth for exposed steel and brackets.
- 4 Composite and glazing cell
 - Panel trial-fit stands.
 - Adhesive metering and surface-prep tools.
 - Controlled cure area.

- Window bonding/gasket installation.
- 5 Final assembly and commissioning cell
- Level track or assembly stands.
 - Bogie drop/lift equipment.
 - Roof access for HVAC and antennas.
 - HV isolation tools and test equipment.
 - Door calibration, lighting/PIS/CCTV checks, brake tests.

Build sequence per car

1. Kitting

- Issue tube cut list, plate flat patterns, brackets, inserts, and supplier interface drawings.
- Mark every primary steel member with heat/batch traceability.
- Stage COTS modules by car number: doors, windows, HVAC, lighting, seats, intercom, CCTV, PIS, battery pack, inverter, auxiliaries.

Hold point: material certificates and supplier certificates accepted before cutting.

2. Underframe

- Cut side sills, centre spine, cross bearers, battery rails, and folded plate boxes.
- Tack in underframe fixture from centre datum outward.
- Weld side sills, cross bearers, bolster boxes, coupler pocket, and battery/equipment supports.
- Stress-relieve or controlled-cool where required by WPS.
- Machine or line-bore centre-pivot and critical mounting inserts after welding.

Hold point: dimensional survey, visual weld inspection, MT/UT on CL1/CL2 joints, bogie-centre datum report.

3. Side frame and roof frame

- Weld door posts, window posts, waist rail, cant rail, and roof bows on the side-frame fixture.
- Lift side frames onto underframe datum pads.
- Weld/bolt side frames to underframe as specified by v2 drawings.
- Install roof equipment rails and cable tray brackets.

Hold point: door aperture survey against chosen COTS door cassette; window aperture survey against chosen glazing cassette.

4. Corrosion protection

- Blast exposed steel to Sa 2.5.
- Apply zinc-rich primer and topcoat system.
- Seal lap joints and composite bond interfaces.
- Apply cavity wax after all hot work and inspection is complete.

Hold point: dry-film thickness report and sealed-drain checklist.

5. Composite exterior

- Trial-fit side panels, roof fairings, skirts, and sensor cowl.
- Prepare steel/composite surfaces per adhesive supplier procedure.
- Bond panels to secondary rails and install mechanical retainers.
- Fit removable skirt panels with quarter-turn or captive fasteners.

Hold point: adhesive batch record, cure record, bond-line witness coupons, removable-panel access check.

6. COTS doors and windows

- Install door cassette into prepared structural bay.
- Install threshold, drainage, seals, and emergency release hardware.
- Install fixed or vented glazing cassettes.
- Connect door power, hardwired closed-and-locked loop, Ethernet/CAN, and local service connector.

Hold point: water ingress test, door obstruction test, emergency release test, closed-and-locked circuit test.

7. Roof systems

- Install packaged HVAC units onto roof rails.
- Install antennas, marker lights, CCTV housings, and regen resistor.
- Connect ducting to interior cassette.
- Fit removable composite roof fairings around service envelopes.

Hold point: roof leak test, HVAC condensate drain test, lifting and fall-arrest access check.

8. Interior cassette

- Install floor boards and removable hatches.
- Install under-seat battery enclosure covers and service locks.
- Install seats, grab rails, lighting, PIS, speakers, CCTV, intercom, emergency signage, and fire extinguishers if locally required.
- Install cable harnesses as pre-terminated looms, not loose field wiring.

Hold point: egress width check, sharp-edge check, lighting lux test, fire-material certificate pack complete.

9. Power and traction equipment

- Install battery packs in under-seat cassettes.
- Install inverters, aux converters, HV contactors, cooling loops, smoke detection, and fire suppression.
- Pressure-test coolant loops before energisation.
- Perform insulation resistance and HV interlock loop tests.

Hold point: HV safety sign-off before first energisation.

10. Bogie marriage and commissioning

- Assemble powered and trailer bogies separately.
- Drop carbody onto bogies using surveyed bolster datums.
- Connect traction, brake, WSP, tachometer, bearing-temperature, and suspension interfaces.
- Run static brake test, door test, HVAC test, lighting/PIS/CCTV test, battery charge/discharge test, and low-speed yard movement test.

Hold point: first-article inspection release before dynamic testing.

COTS interface rules

The carbody must reserve supplier-neutral interfaces:

Module	OSR fixes	Supplier owns
Door cassette	Opening envelope, structural datum, power, data, hardwired lock loop, drainage path	Door mechanics, controller, seals, obstacle detection, emergency release
Window cassette	Opening envelope, bond/gasket land, drainage, replacement access	Glass laminate, frame, adhesive/gasket, certification
HVAC	Roof rail pattern, power budget, duct opening, condensate route, service clearance	Refrigerant circuit, controls, compressor, fans, fire certificate
Lighting	24 V DC rail, mounting pitch, lux target, emergency mode input	Luminaire, diffuser, driver electronics
Seats/grab rails	Floor/sidewall insert grid, strength target, envelope	Seat shell, upholstery, rail clamps, fire certificates

Module	OSR fixes	Supplier owns
PIS/CCTV/intercom	Power/data connector, field of view or visibility target	Device hardware, firmware, certificates

No primary steel member should be redesigned when a COTS supplier is changed. Supplier changes should require only adapter plates, harness tails, software configuration, and certification paperwork updates. The detailed envelopes and incoming-inspection evidence are maintained in [hardware/trainset-interiors/cots-catalogue.md](https://github.com/OpenSourceRail/trainset-interiors/cots-catalogue.md).

Inspection gates

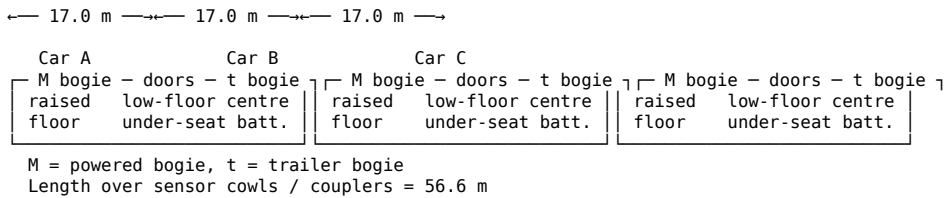
Gate	Evidence
G0 material release	Mill certificates, supplier certificates, incoming inspection
G1 frame complete	Dimensional survey, weld map, NDT report, traceability register
G2 corrosion complete	Blast record, coating DFT report, cavity wax checklist
G3 shell complete	Composite cure records, water test, removable-panel access report
G4 COTS systems fitted	Door/window/HVAC/lighting supplier acceptance records
G5 electrical safe	Insulation resistance, bonding/earthing, HVIL, functional safety I/O
G6 static complete	Brake, door, HVAC, lighting, battery, communications, fire detection
G7 dynamic ready	Weighing, ride-height, bogie alignment, low-speed movement release

First-article constraints

- Build one carbody first, not a full three-car batch.
- Do not commit to production tooling until G1 and G3 are passed on the first carbody.
- Freeze COTS supplier envelopes only after fit-up review with actual door, window, HVAC, and seat samples.
- Treat every locally made structural weld procedure as a certifiable manufacturing process, not a prototype craft operation.
- Keep composite panels sacrificial and replaceable. They should fail cheaply without damaging the steel load path.

General arrangement — light-metro-3car

Consist diagram



- Three self-contained cars, semi-permanently coupled.
- No driving cabs — GoA 4 sensor cowls at both ends per RFC 0015.
- Six bogies total — two standard 2-axle bogies per car.
- Powered wheelsets: one powered bogie and one trailer bogie per car, giving three powered bogies across the consist.
- Battery: 120 kWh sodium-ion per car under the longitudinal seats. The centre door zone stays low-floor and clear.

Key dimensions

Parameter	Value	Source
Length over sensor cowls / couplers	56 600 mm	RFC 0008 §1 + RFC 0015 cowls
Car A length	17 000 mm	modular car envelope
Car B length	17 000 mm	modular car envelope
Car C length	17 000 mm	modular car envelope
Body width (outer)	2 700 mm	fits UIC 505-1 gauge
Height over rail (roof)	3 800 mm	fits UIC 505-1
Floor height (above top-of-rail)	350 mm centre door zone; raised over bogies	low-floor access with standard bogies
Bogie wheelbase	2 100 mm	RFC 0022 single-SKU bogie
Wheel diameter (new / worn)	760 / 680 mm	RFC 0022
Inter-bogie distance (within a car)	11 000 mm	modular car envelope
Coupler face height (above ToR)	720 mm	Scharfenberg Type 10 dim.
Headroom at doors	2 000 mm	
Headroom at seats	2 100 mm	
Passenger-compartment interior width	2 580 mm	

Door positions

One large 1 400 mm-wide double-leaf plug door per car side in the low-floor centre zone (3 door pairs per side for the consist). Longer-platform variants scale dwell capacity by adding identical cars, not by changing the car-side door pattern. Positioned:

- Each car: door centre at 8.5 m from the car end.

All doors are plug-outward doors per RFC 0008 §3.3 with 1 500 mm clear opening at 2 000 mm height.

Gauge compliance

UIC 505-1 static envelope:

- Width at floor level: 2 700 mm ≤ 3 150 mm limit
- Height over ToR: 3 800 mm ≤ 4 320 mm limit
- Dynamic outline (+ 80 mm lateral, + 50 mm vertical at max cant + sway): 2 780 × 3 850 mm — still inside the 3 150 × 4 320 mm boundary

No interference with UIC 505-1 at any of the RFC 0009 track presets (heritage-tram / standard-urban / standard-metro / mainline-mixed).

Platform clearance

- Low-floor centre door zone at 350 mm ToR platform height. Raised floor over the bogies is inside the saloon, away from the boarding threshold.
- Horizontal gap at door sill: ≤ 75 mm per UIC 741. Achieved by a 40 mm sliding skirt at each door that extends on dwell (retracted in motion).
- On curved platforms at RFC 0009 standard - urban minimum radius (90 m): the door-to-platform horizontal gap grows to ~ 110 mm — requires an 80 mm gap-filler flap at the door sill.

Mass distribution (AW2, nominal loaded)

Location	Mass (t)
Car A incl. powered + trailer bogie	41
Car B incl. powered + trailer bogie	41
Car C incl. powered + trailer bogie	41
Total consist (tare + AW2 passengers)	122

Tare = 102 t per RFC 0008 §1; AW2 load = 19.8 t (330 passengers $\times$ 60 kg average).

Distribution per axle under AW3 (crush load, 420 passengers $\times$ 60 kg = 25.2 t):

- Powered and trailer bogies: ~ 10 – 12 t per axle depending on seated battery-side loading.

All ≤ 14 t per RFC 0008 §3.1 axle-load constraint .

Structural loading cases (for v2 FEA)

The design envelope the v2 CAD model is checked against:

- EN 15227 Cat C-II collision: 25 km/h train-to-obstacle. Cab crush zone absorbs. Anti-climbing features at both ends.
- EN 12663 Cat P-III car-body static: compressive end load 640 kN; diagonal 600 kN. Welded S355 underframe/spaceframe carries the load path; composite side and roof panels are non-structural weather/aero skins.
- Vertical accel: ± 0.5 g normal service; ± 2 g emergency brake / collision.
- Lateral accel: ± 0.3 g through max-cant curve.

v1 of this doc does not include the FEA result; v2 does.

Cross-refs

- Bogie detail [bogie.md](#)
- Body structure [body.md](#)
- Traction [traction.md](#)
- Interfaces [interfaces.md](#)

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Interfaces — light-metro-3car

The interfaces this trainset exposes to the outside world. Every interface is pinned to a commodity standard; no bespoke connectors.

Coupler

Parameter	Value
Type	Scharfenberg Type 10 automatic
Vertical coupling compatibility	Any Scharfenberg-family consist at 720 mm coupler face height
Mechanical strength	1 000 kN static tensile
Electric coupler	128 pins: 24 V DC aux, CAN-FD (2 pairs), Ethernet (TSN 1 Gbps), emergency-brake loop, video (MIPI over fibre)
Pneumatic coupler	None. No pneumatic brake / air-horn compressed-air coupling.
Activation	Electrical lock/unlock from recovery cabinet or OCC command
Cut-out / rescue mode	Mechanical unlock handle accessible at ground level

Every OpenSourceRail consist in every deployment carries the same coupler — any trainset can rescue any other.

Station charging connector

Parameter	Value
Primary type	Side-pin conductive connector per RFC 0026
Alternate type	Pantograph-down dock where platform geometry requires it
Contact voltage	1 000 V DC nominal
Contact current (charging)	500–1 000 A station class
Actuation	Electro-mechanical (no pneumatic)
Location	One train-side connector per car, platform side
ATO interface	Extend/retract via the <code>osr-ato</code> + <code>osr-aux-power</code> pairing on the T-ECU/A

The station battery buffer supplies the charge pulse. Inter-car coordination and per-car current limits are handled by `osr-aux-power`.

Platform gap

Per [RFC 0010 §4.4](#):

Parameter	Value
Horizontal gap (straight platform, door-to-sill)	≤ 75 mm (target 40 mm with retractable skirt)
Horizontal gap (curved platform at 90 m radius)	≤ 110 mm with 80 mm gap-filler flap
Vertical gap (floor-to-platform)	≤ 50 mm
Door sill retractable skirt	40 mm, electrically actuated at dwell-open
Door sill gap-filler flap	80 mm, drops on PSD-open signal at curved platforms (identified per-deployment)

Gap-filler flap is a simple solenoid-driven plate — no pneumatic, no gear train. Failure mode: flap stuck up `osr-door-control` blocks door-open; failure flap stuck down visual inspection at depot next cycle.

TCN-E connector (intra-consist, via articulation)

Standard Ethernet + power bundle through each articulation joint:

Signal	Cable	Notes
TCN-E port A	Cat 6 RJ45, TSN-capable	Per RFC 0006 §4; single-ring backbone
TCN-E port B	Cat 6 RJ45, TSN-capable	Redundant ring
CAN-FD (doors + HVAC)	Twisted pair per ISO 11898	
LIN (local panels)	Twisted pair per ISO 17987	
24 V DC aux + control	2.5 mm ² × 2 wires	Aux + return
110 V DC backup	4 mm ² × 2 wires	Emergency battery backup
Safety loop (emergency brake)	Hardwired pair; normally-closed	Electrically independent of TCN-E

Connection through the articulation bellows via drag-chain; no disconnects — this is an intra-consist fixed bus.

Aux power — station charging dock

The charging interface to the station:

Parameter	Value
Connector	Side-pin primary; pantograph-down alternate
Voltage	1 000 V DC ($\pm 10\%$)
Max current	500 A continuous; 1 000 A peak at uprated stops
Ramp rate	Controlled by os r - aux - power soft-start to protect the dock contactor
Connector extend / retract signals	From os r - aux - power via TCN-E
Isolation	Dock-side mechanical isolator; electrical monitoring via os r - energy - site

Full RFC 0002 energy-sizing compatibility.

Emergency door release (external)

- Each exterior door has an external unlock per EN 14752.
- Mechanical; no electronics on the external side.
- Engages only when trainset speed ≤ 5 km/h OR on emergency- brake trip (latched state).

Ground-radio (external antennas)

Antenna	Location	Band	Notes
TRG-1 5G main	Car A roof	n77 / n78 / CBRS	
TRG-1 5G diversity	Car C roof	Same band	MIMO pair
TRG-2 LoRa	Car B roof centre	868 MHz (EU) / 915 MHz (US/MEA)	
GNSS	Car A roof + Car C roof	L1/L2	
Public-safety (optional)	Car A roof	700 MHz	Per-deployment

All antennas use a single-type N-male to RP-SMA termination, no bespoke sockets.

Mass & structure compatibility matrix

Rolling-stock $\times$ track-geometry $\times$ station-archetype compatibility per RFCs 0008 §4 / 0009 §10 / 0010 §12:

Compatible with	Reason
heritage-tram geometry	No — gauge 1 000 mm variant required (build order)
standard-urban geometry	Yes — 56.6 m consist, 90 m min curve OK, 50 % max grade OK
standard-metro geometry	Yes — plenty of margin
mainline-mixed geometry	Yes — meets 25 % max grade (lower than the consist's 5 % capability)
halt station	Yes — short-platform door select required
standard station	Yes — 67 m platform
major station	Yes
interchange station	Yes
terminal station	Yes — side-pin dock fits
depot-terminal station	Yes

v2 deliverables (not in v1)

- Coupler electrical connector pinout drawing.
- Station charging dock mechanical envelope (for station civil design).
- Gap-filler flap mechanism drawings.
- Antenna radiation-pattern simulations at the consist roof.

docs/rolling-stock/light-metro-3car/traction.md

docs/rolling-stock/light-metro-3car/traction.md

Traction + auxiliary power — light-metro-3car

Per RFC 0008 §3.2. All-electric; no pneumatic / hydraulic systems on the safety-actuator path. Regenerative braking is the default mode; friction disc brake blends in below 8 km/h and handles emergency brake application.

Propulsion topology

[120 kWh Na-ion under-seat pack, Car A] —DC— [SiC inverter] — [powered bogie A]
[120 kWh Na-ion under-seat pack, Car B] —DC— [SiC inverter] — [powered bogie B]
[120 kWh Na-ion under-seat pack, Car C] —DC— [SiC inverter] — [powered bogie C]

Station charger → side-pin / pantograph-down dock → per-car DC link

Each car is electrically self-contained. A car can isolate its own pack and traction equipment while the rest of the consist remains recoverable at restricted speed. osr-bms manages per-car contactors, cell balancing, and SoC / SoH estimation.

Batteries

Parameter	Value
Chemistry (default)	Sodium-ion (Na-ion)
Chemistry (alternate)	LFP (drop-in, per-operator)
Nominal pack voltage	1 500 V DC
Pack capacity (each car)	120 kWh usable (~150 kWh nameplate at commissioning)
Consist total capacity	360 kWh usable
Module count per pack	8 under-seat modules per car
Cell chemistry	3.0–3.7 V Na-ion (CATL / HiNa / local equiv.)
Pack mass	~0.9 t per car, ~2.7 t consist total
Location	Under longitudinal seats, split both sides of the saloon
Thermal management	Chiller-fed cold plates tied into the HVAC loop
Fire containment	Sealed aluminium module boxes with side vent duct and aspirating smoke detection (feeds osr-fire-safety)

Pack sizing per RFC 0002: 360 kWh gives roughly one route length plus 20 % reserve and HVAC uplift at 50 °C ambient. Normal service energy is replaced during station dwells.

Traction motors

Parameter	Value
Count	6 (two axles on each of three powered bogies)
Type	Permanent-magnet synchronous (PMSM), axle-mount
Continuous rating	180 kW per axle
Peak rating	320 kW per axle (≤ 60 s)
Rated torque	1 800 Nm at wheel
Efficiency at peak	≥ 96 %
Mass	520 kg each
Cooling	Water (shares cold plate with SiC inverter)
Bearing grease	Sealed-for-life; EN 50155 grade

Peak onboard motor output (six motors, both directions) = 1 800 kW, matching RFC 0008 §1.

Inverters

Parameter	Value
Count	3 (one per powered bogie)
Type	3-phase voltage-source, silicon-carbide MOSFETs
Rating	360 kW continuous, 600 kW peak
Switching frequency	5 kHz nominal
Efficiency at peak	≥ 98 %
Cooling	Forced water via cold plate, shared with motor
Volume	450 × 300 × 180 mm per inverter

Reference parts: Wolfspeed SiC MOSFET half-bridges (e.g. CAB450M12HM3) or local equivalent. Controlled by `osr-traction` firmware on the T-ECU/S safety MCU (RP2350).

Reduction gear

- Single-stage helical, 3.8:1 ratio.
- Axle-mounted; no separate gearbox.
- Oil-bath, sealed for 600 000 km (matches RFC 0008 §5 bogie overhaul interval).
- Mass 120 kg per bogie.

Adhesion budget

At 14 t/axle and coefficient of friction 0.1 (wet rail, conservative), each powered axle can transmit:

$$F_{\text{adhesion}} = 14\,000\text{ kg} \times 9.81\text{ m/s}^2 \times 0.1 = 13.7\text{ kN}$$

Per-motor torque at wheel (wheel radius 0.43 m):

$$F_{\text{motor_peak}} = 1\,800\text{ Nm} \times 3.8 / 0.43\text{ m} = 15.9\text{ kN}$$

Adhesion is the binding constraint at wet-rail AW3. Per-axle tachometer feeds `osr-traction` anti-slip which clamps torque to $\leq 0.9 \times F_{\text{adhesion}}$ under wheel-slip detection (matches `osr-brake` WSP conservative property B4).

Station charging

- Primary connector: side-pin per RFC 0026.
- Alternate connector: pantograph-down OppCharge-style dock where platform geometry requires it.
- Dock: passenger stations at roughly 1 km spacing, buffered by station solar PV + stationary Na-ion battery.
- Charge power: 500 kW nominal; 1 000 kW at terminals or high-load stations.
- Dwell charge: 60 s at 500 kW adds ~8 kWh; 60 s at 1 MW adds ~17 kWh before losses.

No continuous catenary. Pantograph is raised only at discrete charging docks where the alternate connector is selected. This is the catenary-free bet from [ARCHITECTURE §D7](#).

Regenerative braking

- `osr-regen` arbitrates between:
 - 1 Battery absorb (preferred, unless pack SoC > 95 %).
 - 2 Station-dock export (if in contact with a charging dock).
 - 3 Dump resistor (only if neither of the above is available; roof-mounted).
- Blend-down to friction brake below 8 km/h; friction holds down to rest.

Auxiliary power

Aux inverter (400 V / 3-phase AC output + 110 V / 24 V DC):

Load	Nominal power	Notes
HVAC (3 units — one per car)	60 kW total	
Interior + exterior lighting	6 kW	
PIS displays + onboard compute	2 kW	
Door motors	2 kW total (centre door pairs, not simultaneously)	
Recovery cabinet + onboard controls	0.5 kW per end	
Compressor (for air-horn only — NO brake air)	0.5 kW	

Total aux design load: ~75 kW at AW3 in Samawah climate (hot day + full HVAC). `osr-aux-power` shed-load logic runs HVAC at reduced compressor duty when pack SoC < 30 %.

v2 deliverables (not in v1)

- PMSM motor winding data + thermal FEA.

- SiC inverter gate-driver schematic + PCB stackup.
- Pack battery management system (BMS) detail schematic.
- Station charging connector + dock mechanical drawings.
- Adhesion + slip simulation at 0.05 / 0.1 / 0.15 friction.
- Worst-case thermal analysis for battery pack under 50 °C ambient + fast-charge.

docs/safety-case/README.md

docs/safety-case/README.md

Safety-case artifacts

This directory collects the machine-checkable evidence that supports OpenSourceRail's safety claims. Per [ARCHITECTURE.md §7](#) the safety case is structured in GSN (Goal Structuring Notation), serialised as TOML, and regenerated on every commit.

The TOML lives under [gsn/](#) and is compiled by the [osr-safety-case](#) crate (RFC 0005 §4.9). CI gate: [tests/starter_case_closes.rs](#) fails the build if any goal no longer traces to evidence — that is, adding a safety-relevant claim or deleting an evidence file breaks CI until the gap is closed.

Compile the rendered case locally with:

```
cargo run -q --bin osr-safety-case -- docs/safety-case/gsn
```

What's verified today

[RFC 0004 §4](#) names five safety properties of the Movement Authority computer:

Property	Harness in crates/osr-interlocking/src/kani_proofs.rs	Status
P1 — determinism	kani_p1_determinism	written
P2 — non-overlap	kani_p2_non_overlap_two_trains	written (tiny-network bound)
P3 — consist-fit	kani_p3_consist_fit_single_train	written
P4 — conservatism	kani_p4_fail_restrictive_is_not_less_restrictive_than_known	written (fail-restrictive path only)
P5 — time-bounded	kani_p5_time_bounded , kani_p5_time_bounded_within_known_position	written

Also proptest-checked (covering larger inputs unbounded) in [tests/proptest_ma.rs](#) and [tests/proptest_determinism.rs](#).

Continuously exercised in [osr-sim](#) on every scenario run: per RFC 0004 M5 the simulator emits Entry objects as trains move, and every section entry is gated by `osr_interlocking::section_available_to`. The Samawah two-line reference run produces 2400 MAs / 2 h with zero invariant violations under both the in-memory backend and a 3-node Raft cluster.

Cross-checked against an independent Python reference interpreter ([tools/reference-ma/](#)) per RFC 0004 M4: [crates/osr-interlocking/tests/differential.rs](#) serialises random log prefixes, shells out to `python3 -m reference_ma`, and asserts byte-identical MovementAuthority JSON. Any divergence between the two twins is a bug in at least one of them.

Honest status: the harnesses compile under `#[cfg(kani)]` and encode each property formally in Rust, but running them requires a Kani installation (see below). The P2 and P4 harnesses use deliberately small bounds — a 2-section network and 1–2 trains — so Kani can discharge them in a reasonable time. Scaling the bounds to RFC 0004's targets (8 trains, 50 entries, 100 sections) is left to contributors with compute budget; the same harness structure extends directly.

What the other SIL-4 crates check today

Crate	Properties	Verification
osr-atp	A1–A7 (determinism, expired MA trips, unknown position trips, train mismatch trips, head past MA end trips, overspeed trips, conservatism)	proptest for A1–A7; Kani harnesses for A1–A7 in src/kani_proofs.rs
osr-brake	B1–B5 (determinism, emergency union, emergency completeness, WSP conservative, park safe)	proptest + Kani for B1–B5 in src/kani_proofs.rs
osr-odometry	O1–O5 (determinism, forward non-regression, uncertainty monotone, balise reset, GNSS conservative)	proptest + Kani for O1–O5 in src/kani_proofs.rs
osr-vigilance	V1–V6 (determinism, suppression, warning precedes trip, tripped iff emergency, in-window ack clears, trip latches)	proptest + Kani for V1–V6 in src/kani_proofs.rs
osr-wayside-points	W1–W6	proptest + Kani for W1–W6 in src/kani_proofs.rs
osr-consensus	TLA+ invariants (ElectionSafety, LogMatching, LeaderCompleteness, StateMachineSafety, FailRestrictive)	proptest over cluster harness; TLA+ TLC on the spec, not yet a refinement proof

Running Kani

Install

```
cargo install --locked kani-verifier
cargo kani setup
```

`cargo kani setup` downloads ~200 MB of CBMC tooling on first run.

Run a single harness

```
cargo kani -p osr-interlocking --harness kani_p5_time_bounded
```

Run all harnesses in a crate

```
cargo kani -p osr-interlocking
```

Run all harnesses across the workspace

```
cargo kani --workspace
```

Kani runs are expected to take seconds for P1, P3, P5 and minutes for P2, P4. If P2 or P4 time out (default limit 30 min), narrow the bounds further inside the harness — Kani's `kani : : assume` clauses at the top of each proof are the knobs to turn.

CI

The [Kani workflow](#) runs all harnesses on every push. It uses GitHub Actions' cache for the Kani toolchain install so the first minute of each run is just cache restore. Full verification takes ~10 minutes on the default `ubuntu-latest` runner.

What's planned

From RFC 0004 §M3 and the cross-crate safety plan:

- 1 Scale the P2 / P4 bounds toward the RFC's 8 trains × 50 entries × 100 sections target. Requires either smarter abstractions (reducing state-space explosion) or a fleet of dedicated CI workers.
- 2 Kani harnesses for the other SIL-4 crates — done. All five SIL-4 evaluators (ATP, brake, vigilance, odometry, wayside-points) now ship Kani harnesses alongside their proptest suites. The remaining work here is verification throughput — scaling bounds toward larger state spaces on dedicated CI.
- 3 A refinement proof from the TLA+ SMRaft spec to the Rust `osr-consensus` implementation. TLC already checks the spec's safety properties; refinement would formally connect the spec to the code. TLAPS or a separate tool is a candidate.
- 4 GSN safety case compiler — done. The [osr-safety-case](#) crate compiles the TOML claim files in [gsn/](#) and the [starter_case_closes](#) test gates CI on every commit. Current case: 11 goals, 3 strategies, 14 solutions, all linked to real evidence in-tree.

Directory contents

- [gsn/](#) — GSN TOML claim files. Each file is a coherent slice of the case (top-level goals, non-overlap decomposition, fail-safety decomposition, consistency decomposition). Compiled by `osr-safety-case`.
- `evidence/` — machine-readable Kani and proptest results (planned; currently stdout only).
- `hazard-log/` — identified hazards and mitigations (planned).

Samawah standard archetype — architectural envelope v1

Scope: architectural envelope + canopy first-pass structural calc for the Samawah reference deployment's standard-archetype station, serving as the buildable spec an architect-of-record can tender against. Applicable to every standard-archetype station on Samawah Line 1 + Line 2 — that's 12 of the 22 stations in the deployment.

Reference consist: light-metro-3car per RFC 0008. Platform clearance: 10 m per RFC 0010 §4.1. Platform length: 65 + 10 = 75 m.

Contents

File	Scope
envelope.md	Plan + section geometry, vertical circulation, egress
canopy.md	Solar canopy structural envelope + PV sizing + first-pass load calc
accessibility.md	Step-free path, tactile paving, wheelchair zones, audio+visual per EN 16586
services.md	HVAC (ticket hall only), lighting, fire, drainage, signage
compliance.md	Standards matrix (NFPA 130 egress, EN 12464-2 lighting, etc.)

Why this is repeatable

The whole point of RFC 0010's 6-archetype catalogue is to avoid bespoke station design per deployment. This envelope is applied unchanged at every standard-archetype station on Samawah — 12 stations out of 22. The operator's architect-of-record adapts it only for:

- Site-specific property-line constraints.
- Cardinal orientation (to aim the canopy PV south/north of equator for Samawah: south-facing).
- Pedestrian-connection geometry to the surrounding street network.
- Local material availability for façade finishes (all structural choices stay fixed).

Site list — where this applies (Line 1 + Line 2)

From [design.toml](#):

Line 1 (5 standard stations)

- north-gate
- old-souq (note: override sets canopy_type = shelter-only; this arch envelope applies with canopy removed — §canopy.md covers the rain-shelter alternate)
- riverside
- al-salam
- engineering-quarter

Line 2 (7 standard stations)

- northern-suburbs-a
- northern-suburbs-b
- industrial-west
- western-residential
- south-west-residential
- south-east-residential

Other archetypes (major, interchange, terminal, depot-terminal) have their own envelope docs (v2 scope — not in this session).

Licensing

v1 specification: CC-BY-SA 4.0. v2 architectural drawings: CERN-OHL-S v2.

docs/stations/samawah-standard/accessibility.md

docs/stations/samawah-standard/accessibility.md

Accessibility — Samawah standard archetype

Per RFC 0010 §7 + [lib/templates/accessibility.toml](#). Designed against EN 16586 (PRM — Persons with Reduced Mobility) and the corresponding Iraqi accessibility overlay.

Step-free path — requirements

Every passenger, including wheelchair users, must have a step-free path from street to any door on any train:

- 1 Street Ticket hall: ramp at max 1:12, with 1.5 m×1.5 m landings every 9 m. Samawah: typical site has ≤ 400 mm elevation delta between street and ticket-hall entrance one 5 m ramp suffices.
- 2 Ticket hall Paid zone: wide fare gate (min 900 mm clear, per RFC 0010 §8). One per direction on this archetype.
- 3 Paid zone Platform: ramp or elevator. standard archetype has both (redundant path).
- 4 Platform Train: 350 mm platform level = consist floor level (level boarding per RFC 0008 §3.3). 75 mm horizontal gap max with the retractable skirt on the consist. Wheelchair spaces in the train at door-2 of every car.

Tactile paving

Per ISO 23599 / Iraqi national code:

- Warning tactile: 600 mm deep along the full platform edge. Bright-colour contrast against the platform deck for vision-impaired users.
- Directional tactile: 300 mm wide strips guiding from ticket hall to platform access points.
- Landing tactile: 1 500 × 1 500 mm at the top / bottom of ramps + stairs.
- Lift approach: tactile strip at lift-call button height.

Audio + visual information

Per RFC 0010 §7:

- Visual: next-train display (dot-matrix LED + color LCD) on each platform at 2 m height, 3 units per platform at 25 m intervals. Driven by `osr-pis-station`.
- Audio: PA speakers at 10 m intervals on each platform, weatherproof (IP65), driven by `osr-pis-station`. Volume capped at 85 dB(A) at 1 m to protect hearing.
- Audio-visual synchronisation: PA + visual announce the same next-train at the same time (tested in `osr-pis-station`'s proptest suite).

Wheelchair spaces

- On platform: 1 marked wheelchair waiting area per 30 m of platform length. For 67 m platform: 2 marked areas per platform × 2 platforms = 4 per station.
- On train: 2 wheelchair spaces per car × 3 cars = 6 per consist (per RFC 0008 §3.3).

Waiting area location: colocated with door 2 of the nearest car when the train stops — so a wheelchair user doesn't have to travel along the platform to reach the wheelchair-spaced door.

Lift per platform

Per RFC 0010 §5 — every archetype except `halt` gets one lift per platform. Samawah standard:

- Type: Otis Gen3 (or local equivalent), MRL (machine-room-less).
- Car size: 1 100 × 1 400 mm, 630 kg rating.
- Doors: 900 mm clear, automatic open/close with wheelchair- button override.

- Height of controls: 900 mm from floor for accessibility.
- Voice announcement: floor-arrival and door-status.
- Braille on all buttons.

Seating on platform

Per RFC 0010 §7 accessibility (only halt archetype omits):

- 3 × 2-seat benches per platform, anchored to the platform back wall (under the canopy).
- Armrest on aisle side to aid stand-up.
- No fixed leg-rests — interferes with wheelchair approach.

Baggage + mobility aid storage

Not provided at standard archetype — passengers carry with them. major and interchange archetypes add storage lockers (v2 doc).

Companion-animal policy

Service dogs unrestricted; other pets per operator policy (Iraq typical: service dogs only). No rule in the archetype.

Accessible toilet

Not at standard archetype (per-operator choice). major and terminal archetypes have accessible toilets (v2).

Emergency communication

- Help buttons on each platform at 2 m height, 25 m intervals — directly connected to the OCC via the station SCADA. Wheelchair-user reach height max 1 200 mm.
- Help buttons in lifts — voice intercom + visual status.
- Audio-induction loop at ticket-gate staff position for hearing-aid users.

Compliance checklist (pre-opening)

Item	Status at v1
Step-free path from street to platform	specified
Platform-edge tactile paving 600 mm	
Directional tactile to platform	
Lift ≥ 1 100 × 1 400 mm	
Audio + visual PIS	
2 wheelchair waiting areas per platform	
Wide fare gate ≥ 900 mm	
Accessible seating on platform	
Accessible help buttons	
Braille + raised-character signage	specified
Accessible toilet	n/a at standard archetype
Changing Places toilet	n/a at standard archetype (only at major / terminal)
Audio-induction loop at staffed position	specified

v2 deliverables (not in v1)

- Room data-sheet per staff position (reception, security).
- Signage typography + wayfinding ISO 7001 pictogram set.
- Accessibility-focused acceptance test with a local disability-rights advocacy group.

Solar canopy — first-pass structural envelope

Per RFC 0010 §9, every non-halt archetype carries a solar canopy over the platform. Samawah standard archetype target: 1 800 m² (per lib/templates/stations.toml).

This document is the first-pass structural envelope the civil engineer bids against. v2 is the full FEA + wind-tunnel + stamped drawings.

Geometry

Parameter	Value
Canopy footprint (roof area)	1 800 m² (canopied platform + eave)
Length (along track)	77 m (67 m platform + 5 m each end eave)
Width	~22 m spanning both platforms + track between
Height above platform	2 200 mm clear (min), 3 800 mm roof peak above ToR
Pitch	7° south-facing (Samawah is N of equator)
Column spacing along track	8.5 m (10 column pairs over the 85 m length)
Column spacing across tracks	none — roof spans the whole station column-free between outer columns

Structural system

Steel truss roof, tied to steel columns at the platform-back edges. Not concrete — we need the structure to be erectable in a tight urban site by a commodity SMT-style crane (no form shoring). Welded steel plus bolted-steel truss is the simplest build.

Columns

- Size: HSS 200 × 200 × 8 mm, S355 grade.
- Count: 2 rows × 10 columns = 20 columns per station.
- Height above platform: 3.85 m (platform deck to truss bottom).
- Foundation: isolated pad, 1.8 × 1.8 × 0.8 m reinforced concrete, four #25 rebar anchors.
- Anchorage: base plate 400 × 400 × 20 mm, 4 × M30 anchor bolts per column.

Truss spans

- Span: 22 m (clear between outer-row columns).
- Truss type: Warren truss, 2.0 m depth, 2.5 m panel spacing.
- Top + bottom chords: HSS 150 × 150 × 6 mm S355.
- Diagonals: L90 × 90 × 8 mm angle, S355.
- Truss spacing along track: 8.5 m (matches column spacing).
- Total: 10 trusses per station.

Roof panels

- Standing-seam sheet steel, 0.5 mm galvanised + powder-coat white.
- PV modules mounted flush on the south-facing slope.
- Sheet metal: 550 × 8 500 mm each (standing-seam stock), 2 200 sheets per station.

Photovoltaic layout

Per RFC 0010 §9 target 90 % platform shading:

Parameter	Value
Module: 2.0 × 1.0 m, 450 Wp bifacial	(commodity — any brand)
Modules per canopy	330

Parameter	Value
Nameplate capacity	148 kWp
Yield at Samawah (6.0 peak sun hours)	325 MWh/year
Self-consumption (charging trains + station services)	~20 %
Export to grid	~80 %

The canopy output per os r-energy - site dispatch: 148 kWp is below the standard archetype's 300 kWp target in the RFC 0002 sizing; this is because the RFC 0010 §9 target is 1 800 m² but we only have ~1 650 m² of south-slope. The operator can adjust the roof pitch to flatten (more area, lower peak output) per climate; Samawah's sun angle makes the 7° choice optimal.

Load cases (first-pass)

Per EN 1991-1 basis. Samawah-specific values:

Load	Characteristic value	Source
Dead load (roof + PV + services)	0.6 kN/m ²	Material build-up
Snow load	0 kN/m ²	Samawah never snows
Wind load (50-year return)	1.2 kN/m ²	Iraq national code, Samawah zone
Dust accumulation	0.15 kN/m ²	50 mm of desert dust, post-haboob
Live load (maintenance)	0.5 kN/m ²	Per EN 1991-1-1 Cat H
Seismic (PGA)	0.1 g	Samawah seismic zone (Iraq)

Governing load combination (first-pass)

- 1 Wind uplift: 1.2 kN/m² up + 0.6 kN/m² self-weight down net uplift 0.6 kN/m². Governs roof tie-down + column base anchorage.
- 2 Gravity (hot-day dust + maintenance): 0.6 + 0.15 + 0.5 = 1.25 kN/m² down. Governs truss chord sizing.
- 3 Seismic: 0.1 g × 360 t total station structure mass = 360 kN horizontal. Governs the base connection – additional cross-bracing in the longitudinal direction.

Truss check (first-pass)

At 22 m span, 8.5 m tributary width, 1.25 kN/m² UDL:

- Peak moment $M = (1.25 \times 8.5) \times 22^2 / 8 = 642 \text{ kN}\cdot\text{m}$.
- Truss top-chord axial = $M / \text{depth} = 642 / 2.0 = 321 \text{ kN}$.
- HSS 150 × 150 × 6 mm S355 capacity: 1 250 kN axial. Ratio 0.26 – plenty of margin.

Column check (first-pass)

At 8.5 m tributary × 22 m/2 (half-bay) × 1.25 kN/m² = 117 kN axial + lateral wind $1.2 \times 8.5 \times 3.85 / 2 = 20 \text{ kN}$ lateral.

- HSS 200 × 200 × 8 mm S355 capacity: 1 800 kN axial, 90 kN·m bending. Ratios 0.07 axial, 0.42 bending – comfortable.

Foundation check (first-pass)

Isolated pad 1.8 × 1.8 × 0.8 m on 200 kPa bearing (typical Samawah alluvial soil):

- Column ultimate load: $117 \times 1.5 = 175 \text{ kN}$.
- Pad bearing pressure: $175 / (1.8 \times 1.8) = 54 \text{ kPa}$ – well below 200 kPa capacity.

Uplift check

Net uplift 0.6 kN/m² over the 1 870 m² footprint = 1 100 kN total lift. Distributed across 20 columns = 55 kN per column.

- Anchor bolt pullout (4 × M30 anchors × 380 kN capacity each) = 1 520 kN per column. Ratio 0.036.

Materials quantities (planning-grade BOM)

Line	Qty	Unit
HSS 200×200×8 mm column	80 m	S355
HSS 150×150×6 mm truss chord	880 m	S355
L90×90×8 mm truss diagonal	650 m	S355
Plate 400×400×20 mm base plate	20 ea	S355
Plate 450×450×25 mm foundation plate	20 ea	S355

Line	Qty	Unit
M30 anchor bolt, 800 mm long	80 ea	grade 8.8
C25/30 reinforced concrete (foundations)	52 m³	per-pad 2.6 m³ × 20
#25 rebar	2.4 t	foundations
Galvanised standing-seam steel sheet	2 200 ea	0.5 mm
PV module, 450 Wp	330 ea	
PV mounting rail + clamps	1 set	commodity
Lightning-protection finial + down conductor	1 set	

Planning-grade CAPEX: ~€ 180 000 per canopy (materials only, volume-10 per-deployment pricing).

Per-deployment customisation

The operator's architect-of-record adjusts:

- Cardinal orientation if site geometry forces east-west (acceptable performance loss ~15 %).
- Column finish (paint scheme / cladding — no structural implication).
- PV module brand / wattage (same racking footprint).
- Lightning protection class per local code.

What v1 doesn't include

- Full FEA / buckling analysis (v2).
- Wind-tunnel or CFD study for vortex shedding (v2 — matters only if slender columns are reduced; the envelope above is conservative).
- Foundation-specific geotech (v2 per-site).
- Detailed PV wiring + string layout (electrical-engineer scope separate from structural).

docs/stations/samawah-standard/compliance.md

docs/stations/samawah-standard/compliance.md

Compliance matrix — Samawah standard archetype

Standards the envelope is designed against. Per-deployment NRSA / NRCA (national-rail + national-construction authority) approvals overlay — those are per-country and not repeated here.

Structural + fire

Standard	Scope	Evidence status (v1 v2)
EN 1990 / EN 1991	Basis of structural design; loads on structures	Load cases enumerated in canopy.md (v1) full FEA (v2)
EN 1993	Design of steel structures	First-pass column + truss check in canopy.md (v1) EN 1993 stamped calc (v2)
EN 1997	Geotechnical design	Planning-grade bearing capacity (v1) site-specific geotech (v2)
EN 1998	Seismic design of structures	Iraq PGA 0.1 g baseline (v1) site-specific seismic microzonation (v2)
NFPA 130	Fixed guideway transit + passenger rail systems	Egress 4-minute rule checked in envelope.md + services.md (v1)
EN 45545 (parts 2 + 3)	Fire behaviour of railway vehicles + stations	Materials flagged HL1 for outdoor station in services.md (v1) per-material test reports (v2)
EN 1838	Emergency lighting	Spec'd in services.md (v1) product selection + test (v2)

Accessibility

Standard	Scope	Evidence status
EN 16586	PRM (Persons with Reduced Mobility)	accessibility.md (v1)
ISO 23599	Tactile walking-surface indicators	Applied throughout accessibility.md
ISO 7001	Public-information pictograms	Applied in services.md signage
ISO 21542	Building construction — accessibility	Ramp 1:12 max, lift sizing, etc.
EN 81-70	Lifts accessible to persons with disabilities	Lift spec'd at 1 100 × 1 400 mm with 900 mm door (v1)

Electrical + MEP

Standard	Scope	Evidence status
EN 60439	Low-voltage switchgear assemblies	Ticket-hall MCB board (v2 product spec)
EN 61000-6-3	EMC — emission standard for residential, commercial, light-industrial	LED lighting + PIS equipment class-B compliant
EN 12464-2	Light and lighting — outdoor workplaces	Illuminance targets in services.md (v1)
IEC 60947-5-1	Auxiliary contacts for emergency-stop circuits	Used on emergency help buttons
ISO 14644	Cleanrooms / contamination control	n/a for transit station — not specified

Environmental

Standard	Scope	Evidence status
ISO 3095	Acoustics — railway applications, noise emitted	Pass-by noise target 80 dB(A) at platform (v1)
WHO community-noise guidelines	Night-time outdoor noise	Target 45 dB(A) at façade (RFC 0011 §5.1 acoustic screens on elevated)
EN 12101 (parts 1–9)	Smoke + heat control systems	n/a — open-air platform
NFPA 110	Emergency + standby power systems	UPS in services cabinet — spec level 1

Cybersecurity + physical security

Standard	Scope	Evidence status
IEC 62443 (parts 2-4 + 4-2)	Industrial automation + control systems cybersecurity	osr-station-scada designed per osr-crypto discipline
EN 50159	Railway applications — communications, signalling, processing systems	osr-station-scada interfaces conform

National overlays (Iraq — per-deployment)

- Iraqi Electrical Code (based on NEC) — deployment electrical contractor scope.

- Iraqi Fire Code — applied via the national fire directorate's approval process.
- Iraqi Accessibility Act — overlays EN 16586 with Arabic- specific signage requirements.

Upstream (this doc) provides the EN-standard base. Per- deployment overlays are the operator's compliance-team scope.

Acceptance-test outline (pre-opening)

A standard station's pre-opening sign-off checklist:

- 1 Structural: inspector visual walk-through + spot-check on column anchorage, truss welds.
- 2 Accessibility: actual wheelchair user rides from street to train door on both platforms — no step, no gap > spec.
- 3 Emergency lighting: simulate mains-fail; confirm 1-hour battery operation of exit signs + egress lighting.
- 4 Fire alarm: test smoke pot in ticket hall OCC alarm ≤ 30 s latency.
- 5 PIS: next-train information updates correctly from the OCC's mock schedule.
- 6 Fare gates: open on valid QR + NFC; deny on invalid; emergency-release works.
- 7 Platform-screen-door coordination: (if fitted at this station — not standard default but per operator choice) door command / confirmation round-trip within os r - psd's FSM.
- 8 CCTV: OCC receives all cameras at nominal resolution + frame rate.
- 9 Drainage: pour 1 m³ of water into the low end; confirm it drains to the municipal connection in ≤ 5 min.
- 10 Noise: train pass-by at design speed ≤ 80 dB(A) at platform (confirm by sound-level meter at the waiting area).

Checklist lands with actual pass/fail values in the deployment's acceptance/ tree.

What v1 doesn't commit to

- Stamped structural calculations (v2).
- Product-specific equipment selections (operator discretion).
- Specific lift-service contracts (operator procurement).
- National approval-body sign-off procedure (per-deployment).

docs/stations/samawah-standard/drawing-register.md

docs/stations/samawah-standard/drawing-register.md

standard station — architectural drawing register (v2)

Status: v2 deliverable of [RFC 0010](#). Purpose: a complete list of the architectural drawings an architect-of-record (AOR) produces for a Samawah-class standard station. Every drawing is scoped, sized, and keyed to the sections of the v1 envelope documents (envelope.md, canopy.md, accessibility.md, services.md, compliance.md).

Convention

- All drawings in the register are A1 size at stated scales, except section details which may be A3.
- Revision letters: A = first issue, B = post-coordination revision, C = construction issue.
- Every drawing has a title block referencing this register + the commit hash of the v1 envelope docs.
- Drawing numbers follow a stable scheme:
OSR-STD-<discipline>-<sequence>
- discipline codes: A (architectural), S (structural), M (mechanical), E (electrical), F (fire), T (tactile + signage), L (landscape — if applicable).
- sequence: zero-padded 3 digits within each discipline.

A per-deployment station uses the same numbers; only the title block's site field varies.

Drawing list

A — Architectural (15 drawings)

No.	Title	Scale	Size	Input
OSR-STD-A-001	Site plan + ROW	1:500	A1	Envelope §Plan
OSR-STD-A-002	Platform plan (both platforms)	1:100	A1	Envelope §Plan
OSR-STD-A-003	Ticket-hall plan	1:50	A1	Envelope §Sub-areas
OSR-STD-A-004	Typical cross-section	1:50	A1	Envelope §Section
OSR-STD-A-005	Longitudinal section	1:100	A1	Envelope §Sub-areas
OSR-STD-A-006	West-end elevation (ticket hall)	1:50	A1	Envelope §Sub-areas
OSR-STD-A-007	East-end elevation (emergency egress)	1:50	A1	Envelope §Egress
OSR-STD-A-008	Canopy plan	1:100	A1	Canopy §Geometry
OSR-STD-A-009	Canopy section	1:50	A1	Canopy §Structural system
OSR-STD-A-010	Fare-zone plan (paid / unpaid)	1:100	A1	Envelope §Fare paid/unpaid
OSR-STD-A-011	Platform-edge detail (tactile + coping)	1:10	A3	Accessibility §Tactile paving
OSR-STD-A-012	Ramp + stair details	1:20	A3	Envelope §Access modes
OSR-STD-A-013	Lift-shaft plan + section	1:50	A1	Envelope §Access modes + Accessibility
OSR-STD-A-014	Door + frame schedule	n/a (table)	A1	
OSR-STD-A-015	Interior finishes schedule	n/a (table)	A1	Services §Signage

S — Structural (8 drawings)

No.	Title	Scale	Size	Input
OSR-STD-S-001	Foundation plan	1:100	A1	Canopy §Foundation check
OSR-STD-S-002	Column + footing detail	1:20	A1	Canopy §Columns
OSR-STD-S-003	Canopy truss elevations (typ.)	1:50	A1	Canopy §Truss spans
OSR-STD-S-004	Canopy truss plan	1:100	A1	Canopy §Truss spans
OSR-STD-S-005	Weld details + bolt tables	1:5	A3	Canopy §Structural system
OSR-STD-S-006	Platform slab reinforcement	1:100	A1	
OSR-STD-S-007	Ticket-hall structural frame	1:50	A1	
OSR-STD-S-008	Seismic-bracing plan (when PGA > 0.2 g)	1:100	A1	Canopy §Seismic load

M — Mechanical (5 drawings)

No.	Title	Scale	Size	Input
OSR-STD-M-001	Ticket-hall HVAC layout + ductwork	1:50	A1	Services §HVAC
OSR-STD-M-002	Platform drainage — trough + downspout	1:100	A1	Services §Drainage
OSR-STD-M-003	Canopy-drainage layout	1:100	A1	Services §Drainage
OSR-STD-M-004	Lift-machinery spec	n/a	A3	Envelope §Access modes
OSR-STD-M-005	Accessible-toilet plumbing (if fitted per operator)	1:20	A3	Services §Utilities

E — Electrical (8 drawings)

No.	Title	Scale	Size	Input
OSR-STD-E-001	Incoming MSB + sub-board schematic	n/a	A1	Services §Electrical
OSR-STD-E-002	Platform lighting layout	1:100	A1	Services §Lighting
OSR-STD-E-003	Canopy PV layout + string diagram	1:100	A1	Canopy §PV layout
OSR-STD-E-004	Canopy PV combiner + DC cabinet	1:20	A1	Canopy §PV layout
OSR-STD-E-005	Emergency lighting + UPS	1:100	A1	Services §Fire + life safety
OSR-STD-E-006	CCTV camera positions + cable routing	1:100	A1	Services §Communications
OSR-STD-E-007	Help-button + PA loudspeaker layout	1:100	A1	Accessibility §Audio + visual
OSR-STD-E-008	Lightning-protection finial + down conductor	1:50	A1	Canopy §BOM

F — Fire + life safety (4 drawings)

No.	Title	Scale	Size	Input
OSR-STD-F-001	Fire-detection zone plan	1:100	A1	Services §Fire + life safety
OSR-STD-F-002	Fire-extinguisher placement	1:100	A1	Services §Fire + life safety
OSR-STD-F-003	Emergency-egress plan (NFPA 130 compliance)	1:100	A1	Envelope §Egress + Compliance §NFPA 130
OSR-STD-F-004	Evacuation-signage layout	1:100	A1	Services §Signage

T — Tactile + signage (3 drawings)

No.	Title	Scale	Size	Input
OSR-STD-T-001	Tactile paving layout	1:100	A1	Accessibility §Tactile paving
OSR-STD-T-002	Wayfinding signage positions + typographic spec	1:100 + swatches	A1	Services §Signage
OSR-STD-T-003	Accessible signage (Braille + raised-character) details	1:5	A3	Accessibility §Compliance checklist

Drawing count + delivery phase

Discipline	Drawing count
A — Architectural	15
S — Structural	8
M — Mechanical	5
E — Electrical	8
F — Fire	4
T — Tactile + signage	3
Total per station	43

For the 12 standard-archetype stations in Samawah, that's 516 drawings — but with the archetype's repeatability, only the site-plan (OSR-STD-A-001) varies meaningfully between sites. The other 42 drawings are near-identical across all 12 stations, with per-site parameters (parcel geometry, north arrow, foundation specifics) as the only delta.

Interaction with per-site adaptation

The v1 envelope docs list per-deployment adaptations that the AOR implements:

- Site plan (A-001): always site-specific.
- Orientation (A-008, E-003): rotate to face equator; swap if southern hemisphere.
- Façade finish (A-015): operator choice reflected in the schedule.

- Parcel-line setbacks (A-001, A-006, A-007): per- municipality.
- Foundation depth (S-001, S-002): per-site geotech.
- Seismic (S-008): applicable only if PGA > 0.2 g.

Everything else is archetype-fixed: one set of drawings replicated 12× for Samawah.

Delivery sequence (typical)

- 1 Design Development (DD) — all A + S drawings at rev A.
- 2 Coordination — M + E + F + T drawings at rev A + integrate back into A + S rev B.
- 3 Construction Documents (CD) — everything at rev C.
- 4 Tender — operator's procurement team issues CD to construction-firm bidders.
- 5 Shop drawings — fabricator's own drawings (steel fabrication, PV module mounts) land as annexes.

Acceptance criteria for v2

The v2 deliverable of this doc = the architect's drawing register for the deployment, matching the list above drawing-for-drawing. Per-drawing content is the architect's scope; the register + envelope specs are the upstream input.

For a per-deployment tender, the AOR:

- 1 Downloads this register.
- 2 Fills in the title-block fields (site, project, AOR firm name).
- 3 Produces the drawings.
- 4 Files the commit hash of the envelope docs against which they drew.

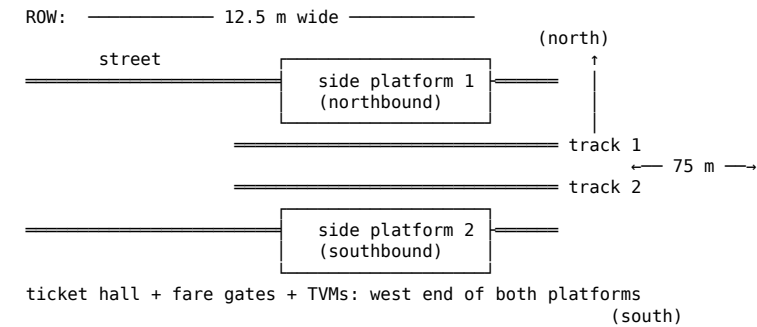
What v2 does NOT include

- Actual CAD files (AutoCAD / Revit / ArchiCAD — AOR's tool choice).
- Printed plot files (per-deployment plot shop).
- Renderings (optional operator deliverable).
- Cost estimates (per-deployment QS scope).

standard archetype – envelope

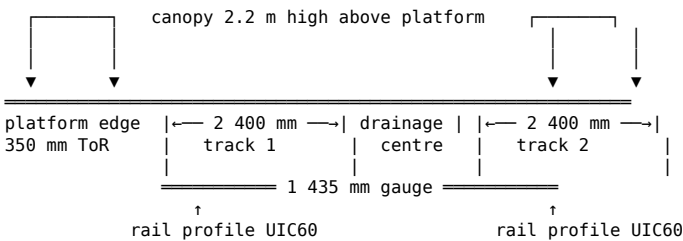
Plan + section geometry of a Samawah standard station. Every number here is a buildable dimension.

Plan



- Configuration: two side platforms, double-track.
- Platform length: 75 m (65 m consist + 10 m clearance).
- Platform width: 3.5 m each (per RFC 0010 §4.3).
- Track centres: 4.0 m (standard double-track with centre drain).
- Total station ROW width: 3.5 + 4.0 + 3.5 = 11.0 m; +1.5 m buffer for drainage + ROW-line clearance = 12.5 m.

Section through platform



- Platform top: 350 mm above top-of-rail (ToR) – matches the light-metro-3car's low-floor height exactly. Level boarding.
- Platform-to-track horizontal gap at door: ≤ 75 mm (UIC 741).
- Canopy clearance: 2 200 mm minimum height over the platform deck. Roof is at 3 800 mm above ToR (matches the consist's max height envelope).

Sub-areas

Area	Location	Size (m²)	Function
Platform 1 (northbound)	75 m × 3.5 m	263	Passenger wait + board
Platform 2 (southbound)	75 m × 3.5 m	263	Passenger wait + board
Cross-platform canopy	85 m × 22 m	1 870	Solar canopy + rain shelter (incl. edges)
Ticket hall (west end)	15 m × 8 m	120	Fare gates + TVMs + SSBC cabinet
Lift rooms (per platform)	2 × 2 m × 2 m	8	Wheelchair-accessible lift per side
Stair wells (per platform)	2 × 4 m × 2.5 m	20	From ticket hall to platform
Services cabinet room	4 m × 3 m	12	Station SCADA + sub-MSB + UPS

Total gross floor area: ~700 m² indoor + 600 m² canopied outdoor platform.

Access modes (per RFC 0010 §5)

Each side platform has:

- 1 Ramp + stair from the ticket hall at platform level. Ramp max 1:12 per accessibility; length ~8 m (rises from grade level to platform + 350 mm to ToR = 350 mm rise = 4.2 m ramp; plus 300 mm for floor thickness and drain). Use a dog-leg ramp that folds into 8 m × 1.5 m.
- 2 Stair alongside the ramp for capacity: 2 × 1.5 m wide straight runs.
- 3 Elevator from ticket hall to platform: 1 100 × 1 400 mm car, 630 kg rating.

No escalator at standard archetype (per RFC 0010 §5 – only major / interchange / terminal / depot-terminal get escalators).

Egress (per RFC 0010 §6 and NFPA 130 4-minute rule)

Platform occupancy sizing:

- Planning-grade peak: 150 passengers × 2 platforms = 300.
- NFPA 130 egress: must reach a "safe point" within 4 minutes.

Egress paths (per platform):

- Primary: ramp + stair to ticket hall (11 m travel).
- Secondary: at the opposite (east) end of the platform, a direct stair down to street level (8 m vertical travel).

Effective egress width:

- Ramp 1.5 m + stair 1.5 m 3 m aggregate ($\geq$ the RFC 0010 §6 standard minimum of 2 m aggregate).
- Secondary stair 1.5 m.
- Total effective width: ≥ 4.5 m per platform.

At 300 pax per platform and NFPA 130 flow 1.3 pax/m/s:

- Clearance time = $300 / (4.5 \text{ m} \times 1.3 \text{ pax/m/s}) \approx 52 \text{ s}$. Well under the 4 min target.

Fare paid / unpaid zones

- Unpaid zone: ticket hall, TVMs, outside fare gates.
- Paid zone: everything platform-side of the fare gates.

standard archetype gets 2 fare-gate lanes per direction per RFC 0010 §8 (so 4 total – 2 in, 2 out; 1 wide-gate for wheelchairs). TVMs: 1 per direction (2 total).

Property-line setbacks (for operator adaptation)

- ROW centreline to property line: ≥ 3 m both sides (municipal code typical).
- Ticket hall at west end can extend into unused ROW corners; operator adapts per parcel.
- If the ticket hall must go on only one side (single-access station due to constrained site), drop to one platform access at the west end + keep the secondary east-end emergency stair. standard archetype requires at least 2 independent egress directions per platform, so single-access is out.

Site footprint

Total approximate footprint a standard station occupies on the ground:

- Platform envelope: $75 \times 12.5 = 940 \text{ m}^2$ main rectangle.
- Ticket hall west: 120 m^2 west-end addition.
- Service cabinet + lifts: 30 m^2 west-end addition.
- Canopy shadow (extends slightly beyond platform): $85 \times 22 = 1\,870 \text{ m}^2$ – but this is the roof, not ground footprint.

Usable ground footprint: ~1 100 m² per station.

Interface with the simulator

osr-sim consumes these archetype numbers via the `lib/templates/stations.toml` schema. The v1 envelope does not require any sim change; all parametric sizing already lives in templates.

What this doesn't commit to

- Façade finish (brick, stucco, aluminium panel — operator choice).
- Signage typeface / bilingual arrangement (operator choice).
- Paid-zone toilets or vending (operator discretion, not an archetype feature).
- Retail concessions (excluded from standard; see major archetype at v2).

docs/stations/samawah-standard/services.md

docs/stations/samawah-standard/services.md

Station services — Samawah standard archetype

HVAC, lighting, fire, drainage, signage — the MEP (mechanical / electrical / plumbing) envelope for the station.

Per RFC 0010 §11 an outdoor Samawah station is passively cooled by the canopy. Only enclosed spaces (ticket hall, lift rooms, services cabinet) get mechanical HVAC.

HVAC (ticket hall only)

- System: split-type air conditioner, 12 kW cooling (covers 120 m² ticket hall + lift rooms at design ambient 50 °C).
- Refrigerant: R-32 (commodity).
- Fresh-air intake: 10 % OA per ASHRAE 62.1.
- Heating: none — Samawah sees ≥ 5 °C ambient year-round; if used, reverse-cycle heat pump on the same unit.
- Filters: MERV 8 standard; MERV 14 for heavy-dust days per operator discretion.
- Drain: to condensate pipe station storm drain.
- Control: `osr-station-scada` with operator override.

Platform ventilation

None. Platform is open-air under the canopy. Natural cross-ventilation works year-round; on haboob days, platform shelter behind the canopy plus fare-gate doors to the enclosed ticket hall (which has the AC).

Lighting

Per EN 12464-2:

Area	Illuminance	Colour temp	Control
Platform (night)	100 lx	4 000 K	Dusk-dawn sensor + schedule
Ticket hall	300 lx	4 000 K	Time schedule + occupancy
Platform (day)	canopy ambient	daylight	none
Stair + ramp	100 lx	4 000 K	Dusk-dawn + motion
Lift interior	200 lx	4 000 K	Always on when door is open
Emergency egress	20 lx (1 h battery)	4 000 K	EN 1838 self-test
Exterior approach	50 lx	4 000 K	Dusk-dawn
Services cabinet room	500 lx	4 000 K	Maintenance switch

All LED, EN 61000-6-3 EMC class B. Total connected load per station: ~4 kW.

Fire + life safety

Per NFPA 130 + Iraqi national fire code (whichever is stricter):

- Smoke detection: aspirating detector in ticket hall + per lift room. Ceiling smoke detectors in services cabinet.
- Heat detection: per lift room (rate-of-rise).
- Passive protection: all materials per EN 45545 HL1 (one step below HL2 rolling stock since the station is outdoor).
- Fire extinguishers: 2× CO₂ at ticket hall, 1× water per platform end.
- Fire alarm signal: goes directly to the OCC via `osr-station-scada`; cannot be silenced without OCC OK.
- Emergency power: 1-hour UPS (60 kWh Li-ion battery in services cabinet) keeps PIS + emergency lighting + OCC comms alive.
- Evacuation: platform end stairs are the emergency path (per `envelope.md §egress`).

Drainage

- Platform: 2 % transverse fall to centre trough drain; longitudinal 0.5 % fall to the low end.
- Track drain: centre trough between the two tracks, 300 mm wide, 400 mm deep, to perforated pipe below the ballast.
- Canopy runoff: gutter + downspout at each column base, discharges to station storm drain.
- Ticket hall: floor drain in the centre.
- Capacity: sized for 50 mm/h 10-year return storm. Samawah rarely hits this, but when a sandstorm is followed by rain, the first event of the year can overload the drain — catch pit at storm drain connection.

Signage

- External: station name + route-map + OSR brand panel at each entrance.
- Internal: wayfinding per ISO 7001 pictograms; bilingual (Arabic + English) text. Signs at 2.5 m height (overhead)
 - 1.2 m height (lateral, for accessibility).
- Line / route diagram: strip map of Line 1 + Line 2 at every platform-edge support column.
- Real-time information: osr-pis-station drives the LED
 - LCD displays.
- Emergency exit signs: EN 1838 photo-luminescent + battery backup.

Utilities

Electrical

- Mains: 3-phase 400 V AC at 50 Hz from the municipal connection (or from the nearest osr-energy-site if in island mode).
- Connected load at peak: HVAC 12 kW + lighting 4 kW + services + fare equipment + PIS + lift ~10 kW = ~26 kW total.
- Design demand: 30 kW + 50 % future-growth headroom = 45 kW incoming supply.
- Distribution: MCB board in the services cabinet, RCD-protected sub-circuits.

Water + waste

- Water: town water for ticket-hall cleaning + condensate make-up; no drinking fountain at standard archetype.
- Waste: municipal connection for storm + sanitary (if the operator adds a future toilet).

Communications

- Ethernet from the station's S-SBC to the OCC (primary + LoRa backup).
- CCTV camera feeds bonded to the station SCADA, then backhauled to OCC.

Environmental

- Noise: platform ambient target ≤ 70 dB(A) day, ≤ 55 dB(A) night during no-train condition. Train pass-by peak ≤ 80 dB(A) per ISO 3095.
- Visual impact: canopy form factor + lighting strategy follow the operator's urban-design guidelines.

v2 deliverables (not in v1)

- Detailed MEP drawings (plans + sections).
- Load schedule per sub-circuit.
- Fire-detection zone plan with alarm-panel layout.
- Lift-room machinery spec.
- PA loudspeaker aiming + acoustic coverage plots.

Generated City Designs

east-africa/Angola/Luanda

designs/east-africa/Angola/Luanda/README.md

Luanda — Urban Rail Network

Country: AO · Population: 9,085,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Luanda rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [luanda.corridor.geojson](#).

At a glance

Metric	Value
Lines	9
Unique stations	169
Interchange stations	37
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	63.8%
Route length (double track)	389.9 km
Revenue fleet	283 × 6-car trainsets
Spare + cold-reserve	34 × 6-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	58.6 km	21	47	NE Outer SW Outer
line-2	30.0 km	15	25	N Mid W Mid
line-3	43.9 km	16	36	SW Outer NE Mid
line-4	42.5 km	17	35	SE Outer NW Mid
line-5	39.9 km	17	32	SE Outer W Mid
line-6	30.7 km	12	26	S Mid N Mid
line-7	33.9 km	16	28	E Outer NW Mid
line-8	32.9 km	15	27	NE Outer W Mid
line-9	77.4 km	41	61	NE Mid NE Mid
Total	389.9 km	169 unique	317	

Rolling stock

Property	Value
Consist	6-car, 111 m
Max speed	100 km/h
Onboard battery	720 kWh per trainset

Property	Value
Seats	120 longitudinal seats
Nominal capacity (AW2)	660 pax (seated + standing, met ro - 6 car per RFC 0008 §1)
Crush capacity (AW3)	840 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 660 AW2 passengers (met ro - 6 car)
- Peak frequency: 12 trains/hour/direction (5-min headway)
- Peak capacity per line per direction: $660 \times 12 = 7,920$ pphpd
- Network peak throughput (all lines, both directions): $9 \text{ lines} \times 2 \text{ directions} \times 7,920 = 142,560$ passengers/hour
- Daily theoretical capacity (peak $\times$ 10): $\approx 1,425,600$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 579,623 - 869,434$ trips/day

Catchment

- City population: 9,085,000
- Anchor-weighted coverage: 63.8%
- Catchment population: $\approx 5,796,230$ (within ~ 800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	37	500 kW	3000 kWh
Major	15	400 kW	2500 kWh
Standard	90	300 kW	2000 kWh
Terminal	15	500 kW	3000 kWh
Total installed	158	64,000 kW	413,500 kWh

Aggregate station-rail charging power: 90,000 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 720 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	24.0 kWh/km	6 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	1,040 kWh	43.3 km average line length
Onboard battery coverage	0.7 $\times$ average line run	720 kWh usable pack
Average 60 s dwell charge	8.9 kWh/stop	533 kW average charger across stops
Stops to refill one trainset pack	81 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	320 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	414 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in design.toml — emitted by the osr-design Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. country-costs.toml applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (365.9 km @ €3.5 M/km)	€1.28 bn
Elevated (22.7 km @ €18 M/km)	€408 M
Elevated-interchange premium (19 sites @ €20 M)	€380 M
Civil subtotal	€2.07 bn

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
halt	12	€0.4 M	€4.8 M
standard	90	€1.5 M	€135 M
major	15	€3.0 M	€45 M
terminal	15	€2.5 M	€38 M
depot-terminal	1	€3.0 M	€3.0 M
interchange-elevated	37	€4.5 M	€166 M
Stations subtotal			€392 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	15	€3.0 M	€45 M
Depots subtotal			€70 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
metro-6car (revenue + spare + cold reserve)	317	€6.0 M	€1.90 bn

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	389.9 km × €0.015 M/km	€5.8 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€59 M
EPC integration + project management (7%)	on subtotal	€315 M

Total

Bucket	Value
Civil works	€2.07 bn
Stations	€392 M

Bucket	Value
Depots	€70 M
Rolling stock	€1.90 bn
Residual train-control wayside + charging microgrids	€65 M
EPC overhead (7%)	€315 M
CAPEX total	€4.81 bn
Per-route-km	€12 M / km
Per-capita (city pop)	€530 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running scripts/regenerate-city.sh luanda.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–5 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 6 and runs for 20 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–5)	€413 M / yr	€45
Steady-state, low-ridership (year 6+)	€453 M / yr	€50
Steady-state, high-ridership (year 6+)	€392 M / yr	€43
Lifecycle envelope (yr 1–25, low scenario)	€11.12 bn cumulative	€1,224
Lifecycle envelope (yr 1–25, high scenario)	€9.90 bn cumulative	€1,090

Population basis: 9,085,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 25, debt service drops to zero and only the OPEX shortfall remains – ~€75 M / yr (low) €14 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€2.89 bn	4.5%	25 y, 5 y grace	€222 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€1.20 bn	11.5%	25 y, 5 y grace	€156 M / yr
Government equity (no debt service)	15%	€722 M	—	—	—
Total	100%	€4.81 bn			€378 M / yr

During the 5-year grace period the operator pays interest only – multilateral €130 M / yr + bonds €138 M / yr = €268 M / yr total – plus the equity tranche amortised across construction (€144 M / yr × 5 yr). Principal repayment begins in year 6 on a 20-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€76 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€51 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€291 k
Traction energy (1334.1 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (2,351 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€8.7 M
OPEX subtotal		€136 M / yr

Annual fleet utilisation: 283 revenue trainsets × 20.5 h/day × 365 d/yr × 35 km/h commercial × 75% revenue factor = 55.6 M train-km / yr (~196 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$240 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.37 (~\$0.40 USD)
Day pass (3 trips)	€0.94 (15 % bulk discount)
Monthly unlimited pass	€11.04 (~5 % of median monthly income)
Annual pass	€121.44 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	165.8 M	331.6 M
Farebox revenue	€61 M / yr	€122 M / yr
Farebox / OPEX recovery	45%	90%
Country policy-target recovery (diagnostic)	45%	45%
Operating shortfall (gov subsidy required)	€75 M / yr	€14 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€453 M / yr	€392 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
luanda.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
luanda-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
luanda.corridor.geojson	Line polylines + stations (GeoJSON)
luanda.stations.json	Machine-readable station list
luanda.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug luanda

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug luanda \
  --sidecar .cache/osr-pipeline/rasters/luanda.grid.json \
  --out-dir designs/.../Luanda

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../luanda.toml \
  --out designs/.../README.md
```

`scripts/regenerate-luanda.sh` chains steps 3 + drift tests into a single command.

east-africa/Cameroon/Yaounde

designs/east-africa/Cameroon/Yaounde/README.md

Yaounde — Urban Rail Network

Country: CM · Population: 4,100,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Yaounde rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [yaounde.corridor.geojson](#).

At a glance

Metric	Value
Lines	8
Unique stations	135
Interchange stations	28
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	43.4%
Route length (double track)	266.7 km
Revenue fleet	196 × 6-car trainsets
Spare + cold-reserve	23 × 6-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	39.9 km	21	32	SW Mid NE Outer
line-2	36.9 km	14	30	SW Mid NE Outer
line-3	22.6 km	12	19	S Mid NE Mid
line-4	28.2 km	16	24	SE Outer NW Inner
line-5	28.5 km	15	24	SE Mid W Mid
line-6	24.7 km	11	21	E Mid N Mid
line-7	35.6 km	16	29	NW Outer S Inner
line-8	50.2 km	31	40	W Inner W Inner
Total	266.7 km	135 unique	219	

Rolling stock

Property	Value
Consist	6-car, 111 m
Max speed	100 km/h
Onboard battery	720 kWh per trainset
Seats	120 longitudinal seats
Nominal capacity (AW2)	660 pax (seated + standing, metro-6car per RFC 0008 §1)
Crush capacity (AW3)	840 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 660 AW2 passengers (metro - 6 car)
- Peak frequency: 12 trains/hour/direction (5-min headway)
- Peak capacity per line per direction: $660 \times 12 = 7,920$ pphpd
- Network peak throughput (all lines, both directions): $8 \text{ lines} \times 2 \text{ directions} \times 7,920 = 126,720$ passengers/hour
- Daily theoretical capacity (peak $\times$ 10): $\approx 1,267,200$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 177,940 - 266,910$ trips/day

Catchment

- City population: 4,100,000
- Anchor-weighted coverage: 43.4%
- Catchment population: $\approx 1,779,400$ (within ~ 800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	28	500 kW	3000 kWh
Major	46	400 kW	2500 kWh
Standard	44	300 kW	2000 kWh
Terminal	13	500 kW	3000 kWh
Total installed	132	57,100 kW	366,000 kWh

Aggregate station-rail charging power: 74,000 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 720 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	24.0 kWh/km	6 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	800 kWh	33.3 km average line length
Onboard battery coverage	0.9 $\times$ average line run	720 kWh usable pack
Average 60 s dwell charge	9.1 kWh/stop	548 kW average charger across stops
Stops to refill one trainset pack	79 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	286 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	366 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in `design.toml` — emitted by the `osr-design` Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. `country-costs.toml` applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (240.6 km @ €3.5 M/km)	€842 M
Elevated (20.8 km @ €18 M/km)	€375 M
Elevated-interchange premium (15 sites @ €20 M)	€300 M
Civil subtotal	€1.52 bn

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
halt	4	€0.4 M	€1.6 M
standard	44	€1.5 M	€66 M
major	46	€3.0 M	€138 M
terminal	13	€2.5 M	€32 M
depot-terminal	1	€3.0 M	€3.0 M
interchange-elevated	28	€4.5 M	€126 M
Stations subtotal			€367 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	13	€3.0 M	€39 M
Depots subtotal			€64 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
metro-6car (revenue + spare + cold reserve)	219	€6.0 M	€1.31 bn

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	266.7 km × €0.015 M/km	€3.9 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€53 M
EPC integration + project management (7%)	on subtotal	€232 M

Total

Bucket	Value
Civil works	€1.52 bn
Stations	€367 M
Depots	€64 M
Rolling stock	€1.31 bn
Residual train-control wayside + charging microgrids	€57 M
EPC overhead (7%)	€232 M
CAPEX total	€3.55 bn
Per-route-km	€13 M / km

Bucket	Value
Per-capita (city pop)	€866 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](https://github.com/OpenSourceRail/lib/templates/country-finance.toml). Pure function of the [costs] block above + the country code — regenerate by re-running `scripts/regenerate-city.sh yaounde`.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–7 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 8 and runs for 23 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–7)	€233 M / yr	€57
Steady-state, low-ridership (year 8+)	€305 M / yr	€74
Steady-state, high-ridership (year 8+)	€285 M / yr	€69
Lifecycle envelope (yr 1–30, low scenario)	€8.65 bn cumulative	€2,109
Lifecycle envelope (yr 1–30, high scenario)	€8.17 bn cumulative	€1,994

Population basis: 4,100,000 (catchment per [lib/city-batches/world-sample.toml](https://github.com/OpenSourceRail/lib/city-batches/world-sample.toml)). After year 30, debt service drops to zero and only the OPEX shortfall remains — ~€76 M / yr (low) €55 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€2.13 bn	3.8%	30 y, 7 y grace	€141 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€888 M	8.5%	30 y, 7 y grace	€89 M / yr
Government equity (no debt service)	15%	€533 M	—	—	—
Total	100%	€3.55 bn			€230 M / yr

During the 7-year grace period the operator pays interest only — multilateral €81 M / yr + bonds €75 M / yr = €156 M / yr total — plus the equity tranche amortised across construction (€76 M / yr × 7 yr). Principal repayment begins in year 8 on a 23-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€53 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€39 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€196 k
Traction energy (923.9 GWh / yr)	trackside PV + Na-ion (RFC 0002) — self-generated, €0 / yr	€0 k
Labour (1,612 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€4.5 M
OPEX subtotal		€96 M / yr

Annual fleet utilisation: 196 revenue trainsets × 20.5 h/day × 365 d/yr × 35 km/h commercial × 75% revenue factor = 38.5 M train-km / yr (~196 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$180 USD (per [lib/templates/country-finance.toml](https://github.com/OpenSourceRail/lib/templates/country-finance.toml)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass — a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.28 (~\$0.30 USD)
Day pass (3 trips)	€0.70 (15 % bulk discount)
Monthly unlimited pass	€8.28 (~5 % of median monthly income)
Annual pass	€91.08 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	74.8 M	149.7 M
Farebox revenue	€21 M / yr	€41 M / yr
Farebox / OPEX recovery	21%	43%
Country policy-target recovery (diagnostic)	40%	40%
Operating shortfall (gov subsidy required)	€76 M / yr	€55 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€305 M / yr	€285 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
yaounde.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
yaounde-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
yaounde.corridor.geojson	Line polylines + stations (GeoJSON)
yaounde.stations.json	Machine-readable station list
yaounde.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug yaounde

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug yaounde \
  --sidecar .cache/osr-pipeline/rasters/yaounde.grid.json \
  --out-dir designs/.../Yaounde

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../yaounde.toml \
  --out designs/.../README.md
```

`scripts/regenerate-yaounde.sh` chains steps 3 + drift tests into a single command.

east-africa/DR Congo/Kinshasa

designs/east-africa/DR Congo/Kinshasa/README.md

Kinshasa — Urban Rail Network

Country: CD · Population: 17,178,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Kinshasa rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [kinshasa.corridor.geojson](#).

At a glance

Metric	Value
Lines	8
Unique stations	182
Interchange stations	35
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	49.4%
Route length (double track)	384.8 km
Revenue fleet	278 × 6-car trainsets
Spare + cold-reserve	32 × 6-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	54.1 km	27	43	W Outer E Outer
line-2	37.3 km	15	30	W Mid SE Mid
line-3	35.6 km	17	29	NE Inner SW Mid
line-4	45.7 km	20	37	NW Outer SE Mid
line-5	57.3 km	23	46	SW Outer E Outer
line-6	35.1 km	17	29	S Mid N Mid
line-7	45.2 km	21	37	NW Outer SE Inner
line-8	74.3 km	43	59	NW Inner NW Inner
Total	384.8 km	182 unique	310	

Rolling stock

Property	Value
Consist	6-car, 111 m
Max speed	100 km/h
Onboard battery	720 kWh per trainset
Seats	120 longitudinal seats
Nominal capacity (AW2)	660 pax (seated + standing, metro-6car per RFC 0008 §1)
Crush capacity (AW3)	840 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 660 AW2 passengers (metro - 6 car)
- Peak frequency: 12 trains/hour/direction (5-min headway)
- Peak capacity per line per direction: $660 \times 12 = 7,920$ pphpd
- Network peak throughput (all lines, both directions): $8 \text{ lines} \times 2 \text{ directions} \times 7,920 = 126,720$ passengers/hour
- Daily theoretical capacity (peak $\times$ 10): $\approx 1,267,200$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 848,593 - 1,272,889$ trips/day

Catchment

- City population: 17,178,000
- Anchor-weighted coverage: 49.4%
- Catchment population: $\approx 8,485,932$ (within ~ 800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	35	500 kW	3000 kWh
Major	64	400 kW	2500 kWh
Standard	59	300 kW	2000 kWh
Terminal	13	500 kW	3000 kWh
Total installed	172	72,300 kW	462,000 kWh

Aggregate station-rail charging power: 95,750 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 720 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	24.0 kWh/km	6 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	1,154 kWh	48.1 km average line length
Onboard battery coverage	0.6 $\times$ average line run	720 kWh usable pack
Average 60 s dwell charge	8.8 kWh/stop	526 kW average charger across stops
Stops to refill one trainset pack	82 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	362 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	462 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in `design.toml` — emitted by the `osr-design` Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefabricated portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. `country-costs.toml` applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (355.8 km @ €3.5 M/km)	€1.25 bn
Elevated (22.9 km @ €18 M/km)	€413 M
Elevated-interchange premium (16 sites @ €20 M)	€320 M
Civil subtotal	€1.98 bn

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
halt	11	€0.4 M	€4.4 M
standard	59	€1.5 M	€88 M
major	64	€3.0 M	€192 M
terminal	13	€2.5 M	€32 M
depot-terminal	1	€3.0 M	€3.0 M
interchange	3	€4.5 M	€14 M
interchange-elevated	32	€4.5 M	€144 M
Stations subtotal			€478 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	13	€3.0 M	€39 M
Depots subtotal			€64 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
metro-6car (revenue + spare + cold reserve)	310	€6.0 M	€1.86 bn

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	384.8 km × €0.015 M/km	€5.7 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€69 M
EPC integration + project management (7%)	on subtotal	€312 M

Total

Bucket	Value
Civil works	€1.98 bn
Stations	€478 M
Depots	€64 M
Rolling stock	€1.86 bn
Residual train-control wayside + charging microgrids	€74 M
EPC overhead (7%)	€312 M
CAPEX total	€4.77 bn

Bucket	Value
Per-route-km	€12 M / km
Per-capita (city pop)	€277 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code — regenerate by re-running scripts/regenerate-city.sh kinshasa.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–10 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 11 and runs for 30 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–10)	€312 M / yr	€18
Steady-state, low-ridership (year 11+)	€381 M / yr	€22
Steady-state, high-ridership (year 11+)	€328 M / yr	€19
Lifecycle envelope (yr 1–40, low scenario)	€14.55 bn cumulative	€847
Lifecycle envelope (yr 1–40, high scenario)	€12.97 bn cumulative	€755

Population basis: 17,178,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 40, debt service drops to zero and only the OPEX shortfall remains — ~€76 M / yr (low) €23 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€2.86 bn	3.0%	40 y, 10 y grace	€146 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€1.19 bn	13.0%	40 y, 10 y grace	€159 M / yr
Government equity (no debt service)	15%	€715 M	—	—	—
Total	100%	€4.77 bn			€305 M / yr

During the 10-year grace period the operator pays interest only — multilateral €86 M / yr + bonds €155 M / yr = €241 M / yr total — plus the equity tranche amortised across construction (€71 M / yr × 10 yr). Principal repayment begins in year 11 on a 30-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€74 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€50 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€284 k
Traction energy (1310.5 GWh / yr)	trackside PV + Na-ion (RFC 0002) — self-generated, €0 / yr	€0 k
Labour (2,321 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€3.9 M
OPEX subtotal		€129 M / yr

Annual fleet utilisation: 278 revenue trainsets × 20.5 h/day × 365 d/yr × 35 km/h commercial × 75% revenue factor = 54.6 M train-km / yr (~196 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$110 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass — a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.17 (~\$0.18 USD)
Day pass (3 trips)	€0.43 (15 % bulk discount)
Monthly unlimited pass	€5.06 (~5 % of median monthly income)
Annual pass	€55.66 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	313.5 M	627.0 M
Farebox revenue	€53 M / yr	€106 M / yr
Farebox / OPEX recovery	41%	82%
Country policy-target recovery (diagnostic)	30%	30%
Operating shortfall (gov subsidy required)	€76 M / yr	€23 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€381 M / yr	€328 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
kinshasa.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
kinshasa-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
kinshasa.corridor.geojson	Line polylines + stations (GeoJSON)
kinshasa.stations.json	Machine-readable station list
kinshasa.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug kinshasa

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug kinshasa \
  --sidecar .cache/osr-pipeline/rasters/kinshasa.grid.json \
  --out-dir designs/.../Kinshasa

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../kinshasa.toml \
  --out designs/.../README.md
```

`scripts/regenerate-kinshasa.sh` chains steps 3 + drift tests into a single command.

east-africa/DR Congo/Lubumbashi

designs/east-africa/DR Congo/Lubumbashi/README.md

Lubumbashi — Urban Rail Network

Country: CD · Population: 2,829,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Lubumbashi rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [lubumbashi.corridor.geojson](#).

At a glance

Metric	Value
Lines	4
Unique stations	74
Interchange stations	20
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	33.6%
Route length (double track)	129.5 km
Revenue fleet	96 × 4-car trainsets
Spare + cold-reserve	11 × 4-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	32.8 km	17	27	N Outer S Outer
line-2	23.2 km	15	20	SW Mid NE Mid
line-3	23.4 km	14	20	E Mid W Mid
line-4	50.2 km	29	40	W Mid W Mid
Total	129.5 km	74 unique	107	

Rolling stock

Property	Value
Consist	4-car, 75 m
Max speed	90 km/h
Onboard battery	480 kWh per trainset
Seats	80 longitudinal seats
Nominal capacity (AW2)	440 pax (seated + standing, met ro - 4car per RFC 0008 §1)
Crush capacity (AW3)	560 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 440 AW2 passengers (met ro - 4car)

- Peak frequency: 12 trains/hour/direction (5-min headway)
- Peak capacity per line per direction: $440 \times 12 = 5,280$ pphpd
- Network peak throughput (all lines, both directions): $4 \text{ lines} \times 2 \text{ directions} \times 5,280 = 42,240$ passengers/hour
- Daily theoretical capacity (peak $\times$ 10): $\approx 422,400$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 95,054 - 142,581$ trips/day

Catchment

- City population: 2,829,000
- Anchor-weighted coverage: 33.6%
- Catchment population: $\approx 950,544$ (within ~ 800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	20	500 kW	3000 kWh
Major	32	400 kW	2500 kWh
Standard	17	300 kW	2000 kWh
Terminal	5	500 kW	3000 kWh
Total installed	75	35,400 kW	229,000 kWh

Aggregate station-rail charging power: 40,500 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 480 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	16.0 kWh/km	4 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	518 kWh	32.4 km average line length
Onboard battery coverage	0.9 $\times$ average line run	480 kWh usable pack
Average 60 s dwell charge	9.1 kWh/stop	547 kW average charger across stops
Stops to refill one trainset pack	53 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	177 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	229 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in design.toml — emitted by the osr-design Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. country-costs.toml applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (116.9 km @ €3.5 M/km)	€409 M
Elevated (12.0 km @ €18 M/km)	€216 M
Elevated-interchange premium (10 sites @ €20 M)	€200 M
Civil subtotal	€825 M

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
standard	17	€1.5 M	€26 M
major	32	€3.0 M	€96 M
terminal	5	€2.5 M	€12 M
depot-terminal	1	€3.0 M	€3.0 M
interchange-elevated	20	€4.5 M	€90 M
Stations subtotal			€227 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	5	€3.0 M	€15 M
Depots subtotal			€40 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
metro-4car (revenue + spare + cold reserve)	107	€4.0 M	€428 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	129.5 km × €0.015 M/km	€1.9 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€32 M
EPC integration + project management (7%)	on subtotal	€109 M

Total

Bucket	Value
Civil works	€825 M
Stations	€227 M
Depots	€40 M
Rolling stock	€428 M
Residual train-control wayside + charging microgrids	€34 M
EPC overhead (7%)	€109 M
CAPEX total	€1.66 bn
Per-route-km	€13 M / km
Per-capita (city pop)	€588 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running `scripts/regenerate-city.sh lubumbashi`.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–10 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 11 and runs for 30 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–10)	€109 M / yr	€38
Steady-state, low-ridership (year 11+)	€138 M / yr	€49
Steady-state, high-ridership (year 11+)	€129 M / yr	€46
Lifecycle envelope (yr 1–40, low scenario)	€5.23 bn cumulative	€1,848
Lifecycle envelope (yr 1–40, high scenario)	€4.97 bn cumulative	€1,756

Population basis: 2,829,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 40, debt service drops to zero and only the OPEX shortfall remains – ~€32 M / yr (low) €23 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€997 M	3.0%	40 y, 10 y grace	€51 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€416 M	13.0%	40 y, 10 y grace	€55 M / yr
Government equity (no debt service)	15%	€249 M	—	—	—
Total	100%	€1.66 bn			€106 M / yr

During the 10-year grace period the operator pays interest only — multilateral €30 M / yr + bonds €54 M / yr = €84 M / yr total — plus the equity tranche amortised across construction (€25 M / yr × 10 yr). Principal repayment begins in year 11 on a 30-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€17 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€22 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€97 k
Traction energy (301.7 GWh / yr)	trackside PV + Na-ion (RFC 0002) — self-generated, €0 / yr	€0 k
Labour (789 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€1.3 M
OPEX subtotal		€40 M / yr

Annual fleet utilisation: 96 revenue trainsets × 20.5 h/day × 365 d/yr × 35 km/h commercial × 75% revenue factor = 18.9 M train-km / yr (~196 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$110 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass — a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.17 (~\$0.18 USD)
Day pass (3 trips)	€0.43 (15 % bulk discount)
Monthly unlimited pass	€5.06 (~5 % of median monthly income)
Annual pass	€55.66 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	51.6 M	103.3 M
Farebox revenue	€8.7 M / yr	€17 M / yr
Farebox / OPEX recovery	22%	43%
Country policy-target recovery (diagnostic)	30%	30%
Operating shortfall (gov subsidy required)	€32 M / yr	€23 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€138 M / yr	€129 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
lubumbashi.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
lubumbashi-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
lubumbashi.corridor.geojson	Line polylines + stations (GeoJSON)
lubumbashi.stations.json	Machine-readable station list
lubumbashi.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug lubumbashi

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug lubumbashi \
  --sidecar .cache/osr-pipeline/rasters/lubumbashi.grid.json \
  --out-dir designs/.../Lubumbashi

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../lubumbashi.toml \
  --out designs/.../README.md
```

`scripts/regenerate-lubumbashi.sh` chains steps 3 + drift tests into a single command.

Eldoret — Urban Rail Network

Country: KE · Population: 500,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Eldoret rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [eldoret.corridor.geojson](#).

At a glance

Metric	Value
Lines	3
Unique stations	39
Interchange stations	3
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	46.3%
Route length (double track)	65.1 km
Revenue fleet	57 × 3-car trainsets
Spare + cold-reserve	7 × 3-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	27.0 km	15	26	SE Outer NW Outer
line-2	16.1 km	11	17	NE Mid S Mid
line-3	22.0 km	13	21	NW Outer SE Mid
Total	65.1 km	39 unique	64	

Rolling stock

Property	Value
Consist	3-car, 57 m
Max speed	90 km/h
Onboard battery	360 kWh per trainset
Seats	60 longitudinal seats
Nominal capacity (AW2)	330 pax (seated + standing, light-metro-3car per RFC 0008 §1)
Crush capacity (AW3)	420 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 330 AW2 passengers (light-metro-3car)
- Peak frequency: 12 trains/hour/direction (5-min headway)

- Peak capacity per line per direction: $330 \times 12 = 3,960$ pphpd
- Network peak throughput (all lines, both directions): $3 \text{ lines} \times 2 \text{ directions} \times 3,960 = 23,760$ passengers/hour
- Daily theoretical capacity (peak $\times 10$): $\approx 237,600$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 23,150 - 34,725$ trips/day

Catchment

- City population: 500,000
- Anchor-weighted coverage: 46.3%
- Catchment population: $\approx 231,500$ (within ~ 800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	3	500 kW	3000 kWh
Major	14	400 kW	2500 kWh
Standard	14	300 kW	2000 kWh
Terminal	5	500 kW	3000 kWh
Total installed	37	18,800 kW	127,000 kWh

Aggregate station-rail charging power: 22,000 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 360 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	12.0 kWh/km	3 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	260 kWh	21.7 km average line length
Onboard battery coverage	1.4 $\times$ average line run	360 kWh usable pack
Average 60 s dwell charge	9.4 kWh/stop	564 kW average charger across stops
Stops to refill one trainset pack	38 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	94 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	127 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in design.toml — emitted by the osr-design Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. country-costs.toml applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (63.7 km @ €3.5 M/km)	€223 M
Elevated (1.2 km @ €18 M/km)	€22 M
Elevated-interchange premium (1 sites @ €20 M)	€20 M
Civil subtotal	€265 M

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~ 11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
halt	2	€0.4 M	€0.8 M
standard	14	€1.5 M	€21 M
major	14	€3.0 M	€42 M
terminal	5	€2.5 M	€12 M
depot-terminal	1	€3.0 M	€3.0 M
interchange-elevated	3	€4.5 M	€14 M
Stations subtotal			€93 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	5	€3.0 M	€15 M
Depots subtotal			€40 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
light-metro-3car (revenue + spare + cold reserve)	64	€3.0 M	€192 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	65.1 km × €0.015 M/km	€1.0 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€14 M
EPC integration + project management (7%)	on subtotal	€42 M

Total

Bucket	Value
Civil works	€265 M
Stations	€93 M
Depots	€40 M
Rolling stock	€192 M
Residual train-control wayside + charging microgrids	€15 M
EPC overhead (7%)	€42 M
CAPEX total	€647 M
Per-route-km	€9.9 M / km
Per-capita (city pop)	€1,295 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running scripts/regenerate-city.sh eldoret.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–7 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 8 and runs for 23 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–7)	€50 M / yr	€100
Steady-state, low-ridership (year 8+)	€62 M / yr	€123
Steady-state, high-ridership (year 8+)	€58 M / yr	€117
Lifecycle envelope (yr 1–30, low scenario)	€1.77 bn cumulative	€3,534
Lifecycle envelope (yr 1–30, high scenario)	€1.69 bn cumulative	€3,386

Population basis: 500,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 30, debt service drops to zero and only the OPEX shortfall remains – ~€14 M / yr (low) €11 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€388 M	4.5%	30 y, 7 y grace	€27 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€162 M	11.5%	30 y, 7 y grace	€20 M / yr
Government equity (no debt service)	15%	€97 M	—	—	—
Total	100%	€647 M			€48 M / yr

During the 7-year grace period the operator pays interest only – multilateral €17 M / yr + bonds €19 M / yr = €36 M / yr total – plus the equity tranche amortised across construction (€14 M / yr × 7 yr). Principal repayment begins in year 8 on a 23-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€7.7 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€8.0 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€49 k
Traction energy (115.2 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (403 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€1.4 M
OPEX subtotal		€17 M / yr

Annual fleet utilisation: 57 revenue trainsets × 20.5 h/day × 365 d/yr × 30 km/h commercial × 75% revenue factor = 9.6 M train-km / yr (~168 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$230 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.35 (~\$0.38 USD)
Day pass (3 trips)	€0.90 (15 % bulk discount)
Monthly unlimited pass	€10.58 (~5 % of median monthly income)
Annual pass	€116.38 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	9.1 M	18.2 M

	Low scenario	High scenario
Farebox revenue	€3.2 M / yr	€6.4 M / yr
Farebox / OPEX recovery	19%	38%
Country policy-target recovery (diagnostic)	50%	50%
Operating shortfall (gov subsidy required)	€14 M / yr	€11 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€62 M / yr	€58 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
eldoret.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
eldoret-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
eldoret.corridor.geojson	Line polylines + stations (GeoJSON)
eldoret.stations.json	Machine-readable station list
eldoret.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug eldoret

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug eldoret \
  --sidecar .cache/osr-pipeline/rasters/eldoret.grid.json \
  --out-dir designs/.../Eldoret

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../eldoret.toml \
  --out designs/.../README.md
```

`scripts/regenerate-eldoret.sh` chains steps 3 + drift tests into a single command.

Kisumu — Urban Rail Network

Country: KE · Population: 600,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Kisumu rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [kisumu.corridor.geojson](#).

At a glance

Metric	Value
Lines	3
Unique stations	42
Interchange stations	0
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	30.9%
Route length (double track)	75.5 km
Revenue fleet	65 × 3-car trainsets
Spare + cold-reserve	9 × 3-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	24.6 km	15	24	SE Outer N Mid
line-2	23.7 km	13	24	NE Outer W Outer
line-3	27.2 km	14	26	W Outer SE Outer
Total	75.5 km	42 unique	74	

Rolling stock

Property	Value
Consist	3-car, 57 m
Max speed	90 km/h
Onboard battery	360 kWh per trainset
Seats	60 longitudinal seats
Nominal capacity (AW2)	330 pax (seated + standing, light-metro-3car per RFC 0008 §1)
Crush capacity (AW3)	420 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 330 AW2 passengers (light-metro-3car)
- Peak frequency: 12 trains/hour/direction (5-min headway)

- Peak capacity per line per direction: $330 \times 12 = 3,960$ pphpd
- Network peak throughput (all lines, both directions): $3 \text{ lines} \times 2 \text{ directions} \times 3,960 = 23,760$ passengers/hour
- Daily theoretical capacity (peak $\times 10$): $\approx 237,600$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 18,540 - 27,810$ trips/day

Catchment

- City population: 600,000
- Anchor-weighted coverage: 30.9%
- Catchment population: $\approx 185,400$ (within ~ 800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Major	5	400 kW	2500 kWh
Standard	31	300 kW	2000 kWh
Terminal	5	500 kW	3000 kWh
Total installed	42	18,800 kW	129,500 kWh

Aggregate station-rail charging power: 24,000 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 360 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	12.0 kWh/km	3 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	302 kWh	25.2 km average line length
Onboard battery coverage	1.2 $\times$ average line run	360 kWh usable pack
Average 60 s dwell charge	9.5 kWh/stop	571 kW average charger across stops
Stops to refill one trainset pack	38 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	94 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	130 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in `design.toml` — emitted by the `osr-design` Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. `country-costs.toml` applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (70.7 km @ €3.5 M/km)	€247 M
Elevated (4.2 km @ €18 M/km)	€76 M
Civil subtotal	€324 M

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~ 11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
standard	31	€1.5 M	€46 M

Archetype	Count	Unit	Subtotal
major	5	€3.0 M	€15 M
terminal	5	€2.5 M	€12 M
depot-terminal	1	€3.0 M	€3.0 M
Stations subtotal			€77 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	5	€3.0 M	€15 M
Depots subtotal			€40 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
light-metro-3car (revenue + spare + cold reserve)	74	€3.0 M	€222 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	75.5 km × €0.015 M/km	€1.1 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€12 M
EPC integration + project management (7%)	on subtotal	€47 M

Total

Bucket	Value
Civil works	€324 M
Stations	€77 M
Depots	€40 M
Rolling stock	€222 M
Residual train-control wayside + charging microgrids	€14 M
EPC overhead (7%)	€47 M
CAPEX total	€724 M
Per-route-km	€9.6 M / km
Per-capita (city pop)	€1,206 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](https://github.com/kisumu/lib/templates/country-finance.toml). Pure function of the [costs] block above + the country code — regenerate by re-running scripts/regenerate-city.sh kisumu.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–7 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 8 and runs for 23 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–7)	€56 M / yr	€93
Steady-state, low-ridership (year 8+)	€69 M / yr	€115
Steady-state, high-ridership (year 8+)	€65 M / yr	€108
Lifecycle envelope (yr 1–30, low scenario)	€1.98 bn cumulative	€3,292
Lifecycle envelope (yr 1–30, high scenario)	€1.89 bn cumulative	€3,144

Population basis: 600,000 (catchment per lib/city-batches/world-sample.toml). After year 30, debt service drops to zero and only the OPEX shortfall remains — ~€16 M / yr (low) €12 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€434 M	4.5%	30 y, 7 y grace	€31 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€181 M	11.5%	30 y, 7 y grace	€23 M / yr
Government equity (no debt service)	15%	€109 M	—	—	—
Total	100%	€724 M			€53 M / yr

During the 7-year grace period the operator pays interest only — multilateral €20 M / yr + bonds €21 M / yr = €40 M / yr total — plus the equity tranche amortised across construction (€16 M / yr × 7 yr). Principal repayment begins in year 8 on a 23-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€8.9 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€8.8 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€56 k
Traction energy (131.3 GWh / yr)	trackside PV + Na-ion (RFC 0002) — self-generated, €0 / yr	€0 k
Labour (465 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€1.7 M
OPEX subtotal		€19 M / yr

Annual fleet utilisation: 65 revenue trainsets × 20.5 h/day × 365 d/yr × 30 km/h commercial × 75% revenue factor = 10.9 M train-km / yr (~168 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$230 USD (per lib/templates/country-finance.toml). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass — a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.35 (~\$0.38 USD)
Day pass (3 trips)	€0.90 (15 % bulk discount)
Monthly unlimited pass	€10.58 (~5 % of median monthly income)
Annual pass	€116.38 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	10.9 M	21.9 M
Farebox revenue	€3.9 M / yr	€7.7 M / yr
Farebox / OPEX recovery	20%	40%
Country policy-target recovery (diagnostic)	50%	50%
Operating shortfall (gov subsidy required)	€16 M / yr	€12 M / yr

	Low scenario	High scenario
Steady-state government commitment (debt service + OPEX shortfall)	€69 M / yr	€65 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
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kisumu-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
kisumu.corridor.geojson	Line polylines + stations (GeoJSON)
kisumu.stations.json	Machine-readable station list
kisumu.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug kisumu

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug kisumu \
  --sidecar .cache/osr-pipeline/rasters/kisumu.grid.json \
  --out-dir designs/.../Kisumu

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../kisumu.toml \
  --out designs/.../README.md
```

`scripts/regenerate-kisumu.sh` chains steps 3 + drift tests into a single command.

east-africa/Kenya/Mombasa

designs/east-africa/Kenya/Mombasa/README.md

Mombasa — Urban Rail Network

Country: KE · Population: 1,350,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Mombasa rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [mombasa.corridor.geojson](#).

At a glance

Metric	Value
Lines	6
Unique stations	94
Interchange stations	17
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	52.1%
Route length (double track)	200.8 km
Revenue fleet	149 × 4-car trainsets
Spare + cold-reserve	18 × 4-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	32.6 km	16	27	S Mid NE Outer
line-2	24.5 km	13	21	NW Outer E Mid
line-3	22.2 km	11	19	W Mid NE Mid
line-4	30.5 km	13	26	N Mid SW Outer
line-5	20.6 km	10	18	E Inner SW Outer
line-6	70.4 km	32	56	NW Mid NW Mid
Total	200.8 km	94 unique	167	

Rolling stock

Property	Value
Consist	4-car, 75 m
Max speed	90 km/h
Onboard battery	480 kWh per trainset
Seats	80 longitudinal seats
Nominal capacity (AW2)	440 pax (seated + standing, metro-4car per RFC 0008 §1)
Crush capacity (AW3)	560 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 440 AW2 passengers (metro - 4 car)
- Peak frequency: 12 trains/hour/direction (5-min headway)
- Peak capacity per line per direction: $440 \times 12 = 5,280$ pphpd
- Network peak throughput (all lines, both directions): $6 \text{ lines} \times 2 \text{ directions} \times 5,280 = 63,360$ passengers/hour
- Daily theoretical capacity (peak $\times$ 10): $\approx 633,600$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 70,335 - 105,502$ trips/day

Catchment

- City population: 1,350,000
- Anchor-weighted coverage: 52.1%
- Catchment population: $\approx 703,350$ (within ~800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	17	500 kW	3000 kWh
Major	28	400 kW	2500 kWh
Standard	36	300 kW	2000 kWh
Terminal	9	500 kW	3000 kWh
Total installed	91	40,000 kW	260,000 kWh

Aggregate station-rail charging power: 51,500 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 480 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	16.0 kWh/km	4 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	535 kWh	33.5 km average line length
Onboard battery coverage	0.9 $\times$ average line run	480 kWh usable pack
Average 60 s dwell charge	9.1 kWh/stop	548 kW average charger across stops
Stops to refill one trainset pack	53 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	200 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	260 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in design.toml — emitted by the osr-design Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. country-costs.toml applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (191.2 km @ €3.5 M/km)	€669 M
Elevated (9.0 km @ €18 M/km)	€161 M
Elevated-interchange premium (7 sites @ €20 M)	€140 M
Civil subtotal	€970 M

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
halt	4	€0.4 M	€1.6 M
standard	36	€1.5 M	€54 M
major	28	€3.0 M	€84 M
terminal	9	€2.5 M	€22 M
depot-terminal	1	€3.0 M	€3.0 M
interchange-elevated	17	€4.5 M	€76 M
Stations subtotal			€242 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	9	€3.0 M	€27 M
Depots subtotal			€52 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
metro-4car (revenue + spare + cold reserve)	167	€4.0 M	€668 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	200.8 km × €0.015 M/km	€3.0 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€35 M
EPC integration + project management (7%)	on subtotal	€138 M

Total

Bucket	Value
Civil works	€970 M
Stations	€242 M
Depots	€52 M
Rolling stock	€668 M
Residual train-control wayside + charging microgrids	€38 M
EPC overhead (7%)	€138 M
CAPEX total	€2.11 bn
Per-route-km	€10 M / km
Per-capita (city pop)	€1,562 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running `scripts/regenerate-city.sh mombasa`.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–7 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 8 and runs for 23 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–7)	€163 M / yr	€121
Steady-state, low-ridership (year 8+)	€203 M / yr	€151
Steady-state, high-ridership (year 8+)	€195 M / yr	€144
Lifecycle envelope (yr 1–30, low scenario)	€5.81 bn cumulative	€4,306
Lifecycle envelope (yr 1–30, high scenario)	€5.61 bn cumulative	€4,158

Population basis: 1,350,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 30, debt service drops to zero and only the OPEX shortfall remains – ~€48 M / yr (low) €39 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€1.26 bn	4.5%	30 y, 7 y grace	€89 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€527 M	11.5%	30 y, 7 y grace	€66 M / yr
Government equity (no debt service)	15%	€316 M	—	—	—
Total	100%	€2.11 bn			€155 M / yr

During the 7-year grace period the operator pays interest only – multilateral €57 M / yr + bonds €61 M / yr = €118 M / yr total – plus the equity tranche amortised across construction (€45 M / yr × 7 yr). Principal repayment begins in year 8 on a 23-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€27 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€25 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€150 k
Traction energy (468.3 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (1,217 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€4.3 M
OPEX subtotal		€56 M / yr

Annual fleet utilisation: 149 revenue trainsets × 20.5 h/day × 365 d/yr × 35 km/h commercial × 75% revenue factor = 29.3 M train-km / yr (~196 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$230 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.35 (~\$0.38 USD)
Day pass (3 trips)	€0.90 (15 % bulk discount)
Monthly unlimited pass	€10.58 (~5 % of median monthly income)
Annual pass	€116.38 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	24.6 M	49.3 M
Farebox revenue	€8.7 M / yr	€17 M / yr
Farebox / OPEX recovery	15%	31%
Country policy-target recovery (diagnostic)	50%	50%
Operating shortfall (gov subsidy required)	€48 M / yr	€39 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€203 M / yr	€195 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
mombasa.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
mombasa-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
mombasa.corridor.geojson	Line polylines + stations (GeoJSON)
mombasa.stations.json	Machine-readable station list
mombasa.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug mombasa

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug mombasa \
  --sidecar .cache/osr-pipeline/rasters/mombasa.grid.json \
  --out-dir designs/.../Mombasa

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../mombasa.toml \
  --out designs/.../README.md
```

`scripts/regenerate-mombasa.sh` chains steps 3 + drift tests into a single command.

Nairobi — Urban Rail Network

Country: KE · Population: 5,700,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Nairobi rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [nairobi.corridor.geojson](#).

At a glance

Metric	Value
Lines	8
Unique stations	190
Interchange stations	26
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	42.8%
Route length (double track)	476.2 km
Revenue fleet	340 × 6-car trainsets
Spare + cold-reserve	38 × 6-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	57.1 km	25	46	SW Outer NE Outer
line-2	59.9 km	27	48	W Outer E Outer
line-3	55.1 km	23	43	NW Mid SE Outer
line-4	48.0 km	18	39	SE Mid W Outer
line-5	47.7 km	18	38	NE Mid SW Outer
line-6	52.9 km	22	42	N Mid SE Outer
line-7	38.4 km	14	31	E Mid NW Outer
line-8	117.2 km	44	91	W Mid W Mid
Total	476.2 km	190 unique	378	

Rolling stock

Property	Value
Consist	6-car, 111 m
Max speed	100 km/h
Onboard battery	720 kWh per trainset
Seats	120 longitudinal seats
Nominal capacity (AW2)	660 pax (seated + standing, metro-6car per RFC 0008 §1)
Crush capacity (AW3)	840 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 660 AW2 passengers (metro - 6 car)
- Peak frequency: 12 trains/hour/direction (5-min headway)
- Peak capacity per line per direction: $660 \times 12 = 7,920$ pphpd
- Network peak throughput (all lines, both directions): $8 \text{ lines} \times 2 \text{ directions} \times 7,920 = 126,720$ passengers/hour
- Daily theoretical capacity (peak $\times$ 10): $\approx 1,267,200$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 243,960 - 365,940$ trips/day

Catchment

- City population: 5,700,000
- Anchor-weighted coverage: 42.8%
- Catchment population: $\approx 2,439,600$ (within ~800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	26	500 kW	3000 kWh
Major	35	400 kW	2500 kWh
Standard	105	300 kW	2000 kWh
Terminal	13	500 kW	3000 kWh
Total installed	180	70,000 kW	454,500 kWh

Aggregate station-rail charging power: 99,750 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 720 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	24.0 kWh/km	6 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	1,429 kWh	59.5 km average line length
Onboard battery coverage	0.5 $\times$ average line run	720 kWh usable pack
Average 60 s dwell charge	8.8 kWh/stop	525 kW average charger across stops
Stops to refill one trainset pack	82 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	350 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	454 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in design.toml — emitted by the osr-design Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. country-costs.toml applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (450.2 km @ €3.5 M/km)	€1.58 bn
Elevated (23.1 km @ €18 M/km)	€416 M
Elevated-interchange premium (16 sites @ €20 M)	€320 M
Civil subtotal	€2.31 bn

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
halt	11	€0.4 M	€4.4 M
standard	105	€1.5 M	€158 M
major	35	€3.0 M	€105 M
terminal	13	€2.5 M	€32 M
depot-terminal	1	€3.0 M	€3.0 M
interchange-elevated	26	€4.5 M	€117 M
Stations subtotal			€419 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	13	€3.0 M	€39 M
Depots subtotal			€64 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
metro-6car (revenue + spare + cold reserve)	378	€6.0 M	€2.27 bn

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	476.2 km × €0.015 M/km	€7.1 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€63 M
EPC integration + project management (7%)	on subtotal	€359 M

Total

Bucket	Value
Civil works	€2.31 bn
Stations	€419 M
Depots	€64 M
Rolling stock	€2.27 bn
Residual train-control wayside + charging microgrids	€70 M
EPC overhead (7%)	€359 M
CAPEX total	€5.49 bn
Per-route-km	€12 M / km

Bucket	Value
Per-capita (city pop)	€964 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from lib/templates/country-finance.toml. Pure function of the [costs] block above + the country code — regenerate by re-running `scripts/regenerate-city.sh nairobi`.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–7 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 8 and runs for 23 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–7)	€424 M / yr	€74
Steady-state, low-ridership (year 8+)	€525 M / yr	€92
Steady-state, high-ridership (year 8+)	€489 M / yr	€86
Lifecycle envelope (yr 1–30, low scenario)	€15.05 bn cumulative	€2,641
Lifecycle envelope (yr 1–30, high scenario)	€14.21 bn cumulative	€2,493

Population basis: 5,700,000 (catchment per lib/city-batches/world-sample.toml). After year 30, debt service drops to zero and only the OPEX shortfall remains — ~€120 M / yr (low) €84 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€3.30 bn	4.5%	30 y, 7 y grace	€233 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€1.37 bn	11.5%	30 y, 7 y grace	€172 M / yr
Government equity (no debt service)	15%	€824 M	—	—	—
Total	100%	€5.49 bn			€405 M / yr

During the 7-year grace period the operator pays interest only — multilateral €148 M / yr + bonds €158 M / yr = €306 M / yr total — plus the equity tranche amortised across construction (€118 M / yr × 7 yr). Principal repayment begins in year 8 on a 23-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€91 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€56 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€355 k
Traction energy (1602.8 GWh / yr)	trackside PV + Na-ion (RFC 0002) — self-generated, €0 / yr	€0 k
Labour (2,869 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€10 M
OPEX subtotal		€157 M / yr

Annual fleet utilisation: 340 revenue trainsets × 20.5 h/day × 365 d/yr × 35 km/h commercial × 75% revenue factor = 66.8 M train-km / yr (~196 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$230 USD (per lib/templates/country-finance.toml). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass — a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.35 (~\$0.38 USD)
Day pass (3 trips)	€0.90 (15 % bulk discount)
Monthly unlimited pass	€10.58 (~5 % of median monthly income)
Annual pass	€116.38 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	104.0 M	208.1 M
Farebox revenue	€37 M / yr	€73 M / yr
Farebox / OPEX recovery	23%	47%
Country policy-target recovery (diagnostic)	50%	50%
Operating shortfall (gov subsidy required)	€120 M / yr	€84 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€525 M / yr	€489 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
nairobi.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
nairobi-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
nairobi.corridor.geojson	Line polylines + stations (GeoJSON)
nairobi.stations.json	Machine-readable station list
nairobi.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug nairobi

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug nairobi \
  --sidecar .cache/osr-pipeline/rasters/nairobi.grid.json \
  --out-dir designs/.../Nairobi

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../nairobi.toml \
  --out designs/.../README.md
```

`scripts/regenerate-nairobi.sh` chains steps 3 + drift tests into a single command.

Nakuru — Urban Rail Network

Country: KE · Population: 700,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Nakuru rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [nakuru.corridor.geojson](#).

At a glance

Metric	Value
Lines	3
Unique stations	37
Interchange stations	2
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	46.3%
Route length (double track)	59.6 km
Revenue fleet	54 × 3-car trainsets
Spare + cold-reserve	7 × 3-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	23.8 km	15	24	E Outer W Outer
line-2	19.9 km	13	20	NE Outer W Outer
line-3	16.0 km	9	17	E Outer W Outer
Total	59.6 km	37 unique	61	

Rolling stock

Property	Value
Consist	3-car, 57 m
Max speed	90 km/h
Onboard battery	360 kWh per trainset
Seats	60 longitudinal seats
Nominal capacity (AW2)	330 pax (seated + standing, light-metro-3car per RFC 0008 §1)
Crush capacity (AW3)	420 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 330 AW2 passengers (light-metro-3car)
- Peak frequency: 12 trains/hour/direction (5-min headway)

- Peak capacity per line per direction: $330 \times 12 = 3,960$ pphpd
- Network peak throughput (all lines, both directions): $3 \text{ lines} \times 2 \text{ directions} \times 3,960 = 23,760$ passengers/hour
- Daily theoretical capacity (peak $\times 10$): $\approx 237,600$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 32,410 - 48,615$ trips/day

Catchment

- City population: 700,000
- Anchor-weighted coverage: 46.3%
- Catchment population: $\approx 324,100$ (within ~ 800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	2	500 kW	3000 kWh
Major	11	400 kW	2500 kWh
Standard	18	300 kW	2000 kWh
Terminal	5	500 kW	3000 kWh
Total installed	37	18,300 kW	124,500 kWh

Aggregate station-rail charging power: 21,500 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 360 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	12.0 kWh/km	3 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	238 kWh	19.9 km average line length
Onboard battery coverage	1.5 $\times$ average line run	360 kWh usable pack
Average 60 s dwell charge	9.7 kWh/stop	581 kW average charger across stops
Stops to refill one trainset pack	37 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	92 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	124 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in design.toml — emitted by the osr-design Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. country-costs.toml applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (58.2 km @ €3.5 M/km)	€204 M
Elevated (1.2 km @ €18 M/km)	€21 M
Elevated-interchange premium (1 sites @ €20 M)	€20 M
Civil subtotal	€245 M

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~ 11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
standard	18	€1.5 M	€27 M
major	11	€3.0 M	€33 M
terminal	5	€2.5 M	€12 M
depot-terminal	1	€3.0 M	€3.0 M
interchange	2	€4.5 M	€9.0 M
Stations subtotal			€84 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	5	€3.0 M	€15 M
Depots subtotal			€40 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
light-metro-3car (revenue + spare + cold reserve)	61	€3.0 M	€183 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	59.6 km × €0.015 M/km	€0.9 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€13 M
EPC integration + project management (7%)	on subtotal	€40 M

Total

Bucket	Value
Civil works	€245 M
Stations	€84 M
Depots	€40 M
Rolling stock	€183 M
Residual train-control wayside + charging microgrids	€14 M
EPC overhead (7%)	€40 M
CAPEX total	€606 M
Per-route-km	€10 M / km
Per-capita (city pop)	€866 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running scripts/regenerate-city.sh nakuru.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–7 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 8 and runs for 23 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–7)	€47 M / yr	€67
Steady-state, low-ridership (year 8+)	€56 M / yr	€80
Steady-state, high-ridership (year 8+)	€52 M / yr	€74
Lifecycle envelope (yr 1–30, low scenario)	€1.62 bn cumulative	€2,315
Lifecycle envelope (yr 1–30, high scenario)	€1.52 bn cumulative	€2,167

Population basis: 700,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 30, debt service drops to zero and only the OPEX shortfall remains – ~€12 M / yr (low) €7.1 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€364 M	4.5%	30 y, 7 y grace	€26 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€151 M	11.5%	30 y, 7 y grace	€19 M / yr
Government equity (no debt service)	15%	€91 M	—	—	—
Total	100%	€606 M			€45 M / yr

During the 7-year grace period the operator pays interest only – multilateral €16 M / yr + bonds €17 M / yr = €34 M / yr total – plus the equity tranche amortised across construction (€13 M / yr × 7 yr). Principal repayment begins in year 8 on a 23-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€7.3 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€7.4 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€45 k
Traction energy (109.1 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (370 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€1.3 M
OPEX subtotal		€16 M / yr

Annual fleet utilisation: 54 revenue trainsets × 20.5 h/day × 365 d/yr × 30 km/h commercial × 75% revenue factor = 9.1 M train-km / yr (~168 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$230 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.35 (~\$0.38 USD)
Day pass (3 trips)	€0.90 (15 % bulk discount)
Monthly unlimited pass	€10.58 (~5 % of median monthly income)
Annual pass	€116.38 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	12.8 M	25.6 M

	Low scenario	High scenario
Farebox revenue	€4.5 M / yr	€9.0 M / yr
Farebox / OPEX recovery	28%	56%
Country policy-target recovery (diagnostic)	50%	50%
Operating shortfall (gov subsidy required)	€12 M / yr	€7.1 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€56 M / yr	€52 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
nakuru.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
nakuru-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
nakuru.corridor.geojson	Line polylines + stations (GeoJSON)
nakuru.stations.json	Machine-readable station list
nakuru.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug nakuru

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug nakuru \
  --sidecar .cache/osr-pipeline/rasters/nakuru.grid.json \
  --out-dir designs/.../Nakuru

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../nakuru.toml \
  --out designs/.../README.md
```

`scripts/regenerate-nakuru.sh` chains steps 3 + drift tests into a single command.

east-africa/Madagascar/Antananarivo

designs/east-africa/Madagascar/Antananarivo/README.md

Antananarivo — Urban Rail Network

Country: MG · Population: 3,058,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Antananarivo rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [antananarivo.corridor.geojson](#).

At a glance

Metric	Value
Lines	7
Unique stations	154
Interchange stations	26
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	42.0%
Route length (double track)	338.9 km
Revenue fleet	244 × 6-car trainsets
Spare + cold-reserve	28 × 6-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	52.2 km	24	41	NE Outer S Outer
line-2	42.2 km	17	35	S Outer NW Outer
line-3	39.0 km	20	31	N Outer SW Outer
line-4	33.0 km	18	27	NW Outer SE Mid
line-5	41.2 km	19	34	NE Outer SW Mid
line-6	38.1 km	17	31	E Outer W Outer
line-7	93.1 km	40	73	NW Mid NW Mid
Total	338.9 km	154 unique	272	

Rolling stock

Property	Value
Consist	6-car, 111 m
Max speed	100 km/h
Onboard battery	720 kWh per trainset
Seats	120 longitudinal seats
Nominal capacity (AW2)	660 pax (seated + standing, metro-6car per RFC 0008 §1)
Crush capacity (AW3)	840 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 660 AW2 passengers (metro - 6 car)
- Peak frequency: 12 trains/hour/direction (5-min headway)
- Peak capacity per line per direction: $660 \times 12 = 7,920$ pphpd
- Network peak throughput (all lines, both directions): $7 \text{ lines} \times 2 \text{ directions} \times 7,920 = 110,880$ passengers/hour
- Daily theoretical capacity (peak $\times$ 10): $\approx 1,108,800$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 128,436 - 192,654$ trips/day

Catchment

- City population: 3,058,000
- Anchor-weighted coverage: 42.0%
- Catchment population: $\approx 1,284,360$ (within ~ 800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	26	500 kW	3000 kWh
Major	40	400 kW	2500 kWh
Standard	73	300 kW	2000 kWh
Terminal	11	500 kW	3000 kWh
Total installed	151	61,400 kW	397,000 kWh

Aggregate station-rail charging power: 82,500 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 720 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	24.0 kWh/km	6 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	1,162 kWh	48.4 km average line length
Onboard battery coverage	0.6 $\times$ average line run	720 kWh usable pack
Average 60 s dwell charge	8.9 kWh/stop	536 kW average charger across stops
Stops to refill one trainset pack	81 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	307 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	397 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the `[costs]` block in `design.toml` — emitted by the `osr-design` Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. `country-costs.toml` applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (302.5 km @ €3.5 M/km)	€1.06 bn
Elevated (34.0 km @ €18 M/km)	€612 M
Elevated-interchange premium (14 sites @ €20 M)	€280 M
Civil subtotal	€1.95 bn

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
halt	4	€0.4 M	€1.6 M
standard	73	€1.5 M	€110 M
major	40	€3.0 M	€120 M
terminal	11	€2.5 M	€28 M
depot-terminal	1	€3.0 M	€3.0 M
interchange	3	€4.5 M	€14 M
interchange-elevated	23	€4.5 M	€104 M
Stations subtotal			€379 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	11	€3.0 M	€33 M
Depots subtotal			€58 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
metro-6car (revenue + spare + cold reserve)	272	€6.0 M	€1.63 bn

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	338.9 km × €0.015 M/km	€5.0 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€56 M
EPC integration + project management (7%)	on subtotal	€286 M

Total

Bucket	Value
Civil works	€1.95 bn
Stations	€379 M
Depots	€58 M
Rolling stock	€1.63 bn
Residual train-control wayside + charging microgrids	€61 M
EPC overhead (7%)	€286 M
CAPEX total	€4.37 bn

Bucket	Value
Per-route-km	€13 M / km
Per-capita (city pop)	€1,428 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code — regenerate by re-running scripts/regenerate-city.sh antananarivo.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–10 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 11 and runs for 25 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–10)	€264 M / yr	€86
Steady-state, low-ridership (year 11+)	€388 M / yr	€127
Steady-state, high-ridership (year 11+)	€381 M / yr	€124
Lifecycle envelope (yr 1–35, low scenario)	€12.35 bn cumulative	€4,039
Lifecycle envelope (yr 1–35, high scenario)	€12.16 bn cumulative	€3,976

Population basis: 3,058,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 35, debt service drops to zero and only the OPEX shortfall remains — ~€108 M / yr (low) €101 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€2.62 bn	3.0%	35 y, 10 y grace	€150 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€1.09 bn	11.0%	35 y, 10 y grace	€130 M / yr
Government equity (no debt service)	15%	€655 M	—	—	—
Total	100%	€4.37 bn			€280 M / yr

During the 10-year grace period the operator pays interest only — multilateral €79 M / yr + bonds €120 M / yr = €199 M / yr total — plus the equity tranche amortised across construction (€65 M / yr × 10 yr). Principal repayment begins in year 11 on a 25-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€65 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€48 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€252 k
Traction energy (1150.2 GWh / yr)	trackside PV + Na-ion (RFC 0002) — self-generated, €0 / yr	€0 k
Labour (2,045 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€2.8 M
OPEX subtotal		€116 M / yr

Annual fleet utilisation: 244 revenue trainsets × 20.5 h/day × 365 d/yr × 35 km/h commercial × 75% revenue factor = 47.9 M train-km / yr (~196 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$90 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass — a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.14 (~\$0.15 USD)
Day pass (3 trips)	€0.35 (15 % bulk discount)
Monthly unlimited pass	€4.14 (~5 % of median monthly income)
Annual pass	€45.54 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	55.8 M	111.6 M
Farebox revenue	€7.7 M / yr	€15 M / yr
Farebox / OPEX recovery	7%	13%
Country policy-target recovery (diagnostic)	30%	30%
Operating shortfall (gov subsidy required)	€108 M / yr	€101 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€388 M / yr	€381 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
antananarivo.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
antananarivo-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
antananarivo.corridor.geojson	Line polylines + stations (GeoJSON)
antananarivo.stations.json	Machine-readable station list
antananarivo.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug antananarivo

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug antananarivo \
  --sidecar .cache/osr-pipeline/rasters/antananarivo.grid.json \
  --out-dir designs/.../Antananarivo

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../antananarivo.toml \
  --out designs/.../README.md
```

`scripts/regenerate-antananarivo.sh` chains steps 3 + drift tests into a single command.

east-africa/Mozambique/Beira

designs/east-africa/Mozambique/Beira/README.md

Beira — Urban Rail Network

Country: MZ · Population: 535,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Beira rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [beira.corridor.geojson](#).

At a glance

Metric	Value
Lines	3
Unique stations	39
Interchange stations	3
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	36.6%
Route length (double track)	53.8 km
Revenue fleet	48 × 3-car trainsets
Spare + cold-reserve	6 × 3-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	20.8 km	14	20	NW Outer S Outer
line-2	17.3 km	14	18	S Outer N Mid
line-3	15.8 km	11	16	E Mid NW Outer
Total	53.8 km	39 unique	54	

Rolling stock

Property	Value
Consist	3-car, 57 m
Max speed	90 km/h
Onboard battery	360 kWh per trainset
Seats	60 longitudinal seats
Nominal capacity (AW2)	330 pax (seated + standing, light-metro-3car per RFC 0008 §1)
Crush capacity (AW3)	420 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 330 AW2 passengers (light-metro-3car)
- Peak frequency: 12 trains/hour/direction (5-min headway)

- Peak capacity per line per direction: $330 \times 12 = 3,960$ pphpd
- Network peak throughput (all lines, both directions): $3 \text{ lines} \times 2 \text{ directions} \times 3,960 = 23,760$ passengers/hour
- Daily theoretical capacity (peak $\times 10$): $\approx 237,600$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 19,581 - 29,371$ trips/day

Catchment

- City population: 535,000
- Anchor-weighted coverage: 36.6%
- Catchment population: $\approx 195,810$ (within ~ 800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	3	500 kW	3000 kWh
Major	21	400 kW	2500 kWh
Standard	9	300 kW	2000 kWh
Terminal	5	500 kW	3000 kWh
Total installed	39	20,100 kW	134,500 kWh

Aggregate station-rail charging power: 22,500 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 360 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	12.0 kWh/km	3 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	215 kWh	17.9 km average line length
Onboard battery coverage	1.7 $\times$ average line run	360 kWh usable pack
Average 60 s dwell charge	9.6 kWh/stop	577 kW average charger across stops
Stops to refill one trainset pack	37 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	100 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	134 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the `[costs]` block in `design.toml` — emitted by the `osr-design` Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. `country-costs.toml` applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (52.2 km @ €3.5 M/km)	€183 M
Elevated (1.3 km @ €18 M/km)	€23 M
Elevated-interchange premium (1 sites @ €20 M)	€20 M
Civil subtotal	€226 M

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~ 11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
standard	9	€1.5 M	€14 M
major	21	€3.0 M	€63 M
terminal	5	€2.5 M	€12 M
depot-terminal	1	€3.0 M	€3.0 M
interchange	3	€4.5 M	€14 M
Stations subtotal			€106 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	5	€3.0 M	€15 M
Depots subtotal			€40 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
light-metro-3car (revenue + spare + cold reserve)	54	€3.0 M	€162 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	53.8 km × €0.015 M/km	€0.8 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€15 M
EPC integration + project management (7%)	on subtotal	€38 M

Total

Bucket	Value
Civil works	€226 M
Stations	€106 M
Depots	€40 M
Rolling stock	€162 M
Residual train-control wayside + charging microgrids	€16 M
EPC overhead (7%)	€38 M
CAPEX total	€588 M
Per-route-km	€11 M / km
Per-capita (city pop)	€1,099 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running scripts/regenerate-city.sh beira.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–10 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 11 and runs for 25 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–10)	€39 M / yr	€73
Steady-state, low-ridership (year 11+)	€54 M / yr	€100
Steady-state, high-ridership (year 11+)	€52 M / yr	€97
Lifecycle envelope (yr 1–35, low scenario)	€1.73 bn cumulative	€3,241
Lifecycle envelope (yr 1–35, high scenario)	€1.69 bn cumulative	€3,150

Population basis: 535,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 35, debt service drops to zero and only the OPEX shortfall remains – ~€13 M / yr (low) €11 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€353 M	3.0%	35 y, 10 y grace	€20 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€147 M	13.5%	35 y, 10 y grace	€21 M / yr
Government equity (no debt service)	15%	€88 M	—	—	—
Total	100%	€588 M			€41 M / yr

During the 10-year grace period the operator pays interest only – multilateral €11 M / yr + bonds €20 M / yr = €30 M / yr total – plus the equity tranche amortised across construction (€8.8 M / yr × 10 yr). Principal repayment begins in year 11 on a 25-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€6.5 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€7.4 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€40 k
Traction energy (97.0 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (335 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€673 k
OPEX subtotal		€15 M / yr

Annual fleet utilisation: 48 revenue trainsets × 20.5 h/day × 365 d/yr × 30 km/h commercial × 75% revenue factor = 8.1 M train-km / yr (~168 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$130 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.20 (~\$0.22 USD)
Day pass (3 trips)	€0.51 (15 % bulk discount)
Monthly unlimited pass	€5.98 (~5 % of median monthly income)
Annual pass	€65.78 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	9.8 M	19.5 M

	Low scenario	High scenario
Farebox revenue	€1.9 M / yr	€3.9 M / yr
Farebox / OPEX recovery	13%	27%
Country policy-target recovery (diagnostic)	30%	30%
Operating shortfall (gov subsidy required)	€13 M / yr	€11 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€54 M / yr	€52 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
beira.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
beira-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
beira.corridor.geojson	Line polylines + stations (GeoJSON)
beira.stations.json	Machine-readable station list
beira.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug beira

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug beira \
  --sidecar .cache/osr-pipeline/rasters/beira.grid.json \
  --out-dir designs/.../Beira

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../beira.toml \
  --out designs/.../README.md
```

`scripts/regenerate-beira.sh` chains steps 3 + drift tests into a single command.

east-africa/Mozambique/Maputo

designs/east-africa/Mozambique/Maputo/README.md

Maputo — Urban Rail Network

Country: MZ · Population: 1,530,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Maputo rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [maputo.corridor.geojson](#).

At a glance

Metric	Value
Lines	6
Unique stations	82
Interchange stations	18
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	71.1%
Route length (double track)	185.9 km
Revenue fleet	138 × 4-car trainsets
Spare + cold-reserve	18 × 4-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	22.7 km	11	19	SW Mid NE Outer
line-2	21.3 km	11	18	N Mid SE Mid
line-3	26.7 km	10	23	NW Outer S Mid
line-4	28.1 km	11	24	E Outer SW Outer
line-5	30.6 km	12	26	W Outer E Mid
line-6	56.6 km	28	46	NW Mid NW Mid
Total	185.9 km	82 unique	156	

Rolling stock

Property	Value
Consist	4-car, 75 m
Max speed	90 km/h
Onboard battery	480 kWh per trainset
Seats	80 longitudinal seats
Nominal capacity (AW2)	440 pax (seated + standing, metro-4car per RFC 0008 §1)
Crush capacity (AW3)	560 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 440 AW2 passengers (metro-4car)
- Peak frequency: 12 trains/hour/direction (5-min headway)
- Peak capacity per line per direction: $440 \times 12 = 5,280$ pphpd
- Network peak throughput (all lines, both directions): $6 \text{ lines} \times 2 \text{ directions} \times 5,280 = 63,360$ passengers/hour
- Daily theoretical capacity (peak $\times$ 10): $\approx 633,600$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 108,783 - 163,174$ trips/day

Catchment

- City population: 1,530,000
- Anchor-weighted coverage: 71.1%
- Catchment population: $\approx 1,087,830$ (within ~800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	18	500 kW	3000 kWh
Major	6	400 kW	2500 kWh
Standard	45	300 kW	2000 kWh
Terminal	9	500 kW	3000 kWh
Total installed	79	34,400 kW	226,000 kWh

Aggregate station-rail charging power: 45,500 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 480 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	16.0 kWh/km	4 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	496 kWh	31.0 km average line length
Onboard battery coverage	1.0 $\times$ average line run	480 kWh usable pack
Average 60 s dwell charge	9.2 kWh/stop	555 kW average charger across stops
Stops to refill one trainset pack	52 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	172 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	226 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the `[costs]` block in `design.toml` — emitted by the `osr-design` Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. `country-costs.toml` applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (173.5 km @ €3.5 M/km)	€607 M
Elevated (11.2 km @ €18 M/km)	€201 M
Elevated-interchange premium (8 sites @ €20 M)	€160 M
Civil subtotal	€968 M

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
halt	4	€0.4 M	€1.6 M
standard	45	€1.5 M	€68 M
major	6	€3.0 M	€18 M
terminal	9	€2.5 M	€22 M
depot-terminal	1	€3.0 M	€3.0 M
interchange-elevated	18	€4.5 M	€81 M
Stations subtotal			€194 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	9	€3.0 M	€27 M
Depots subtotal			€52 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
metro-4car (revenue + spare + cold reserve)	156	€4.0 M	€624 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	185.9 km × €0.015 M/km	€2.8 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€29 M
EPC integration + project management (7%)	on subtotal	€131 M

Total

Bucket	Value
Civil works	€968 M
Stations	€194 M
Depots	€52 M
Rolling stock	€624 M
Residual train-control wayside + charging microgrids	€32 M
EPC overhead (7%)	€131 M
CAPEX total	€2.00 bn
Per-route-km	€11 M / km
Per-capita (city pop)	€1,308 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running scripts/regenerate-city.sh maputo.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–10 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 11 and runs for 25 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–10)	€134 M / yr	€87
Steady-state, low-ridership (year 11+)	€186 M / yr	€121
Steady-state, high-ridership (year 11+)	€180 M / yr	€118
Lifecycle envelope (yr 1–35, low scenario)	€5.97 bn cumulative	€3,904
Lifecycle envelope (yr 1–35, high scenario)	€5.83 bn cumulative	€3,813

Population basis: 1,530,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 35, debt service drops to zero and only the OPEX shortfall remains – ~€46 M / yr (low) €41 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€1.20 bn	3.0%	35 y, 10 y grace	€69 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€500 M	13.5%	35 y, 10 y grace	€71 M / yr
Government equity (no debt service)	15%	€300 M	—	—	—
Total	100%	€2.00 bn			€139 M / yr

During the 10-year grace period the operator pays interest only — multilateral €36 M / yr + bonds €68 M / yr = €104 M / yr total — plus the equity tranche amortised across construction (€30 M / yr × 10 yr). Principal repayment begins in year 11 on a 25-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€25 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€24 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€139 k
Traction energy (433.7 GWh / yr)	trackside PV + Na-ion (RFC 0002) — self-generated, €0 / yr	€0 k
Labour (1,127 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€2.3 M
OPEX subtotal		€52 M / yr

Annual fleet utilisation: 138 revenue trainsets × 20.5 h/day × 365 d/yr × 35 km/h commercial × 75% revenue factor = 27.1 M train-km / yr (~196 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$130 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass — a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.20 (~\$0.22 USD)
Day pass (3 trips)	€0.51 (15 % bulk discount)
Monthly unlimited pass	€5.98 (~5 % of median monthly income)
Annual pass	€65.78 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	27.9 M	55.8 M
Farebox revenue	€5.6 M / yr	€11 M / yr
Farebox / OPEX recovery	11%	22%
Country policy-target recovery (diagnostic)	30%	30%
Operating shortfall (gov subsidy required)	€46 M / yr	€41 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€186 M / yr	€180 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
maputo.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
maputo-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
maputo.corridor.geojson	Line polylines + stations (GeoJSON)
maputo.stations.json	Machine-readable station list
maputo.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug maputo

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug maputo \
  --sidecar .cache/osr-pipeline/rasters/maputo.grid.json \
  --out-dir designs/.../Maputo

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../maputo.toml \
  --out designs/.../README.md
```

`scripts/regenerate-maputo.sh` chains steps 3 + drift tests into a single command.

Huye — Urban Rail Network

Country: RW · Population: 250,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Huye rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [huye.corridor.geojson](#).

At a glance

Metric	Value
Lines	3
Unique stations	28
Interchange stations	0
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	23.4%
Route length (double track)	45.8 km
Revenue fleet	55 × 2-car trainsets
Spare + cold-reserve	6 × 2-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	15.9 km	11	21	NE Outer SW Mid
line-2	14.1 km	8	19	SW Outer SE Mid
line-3	15.8 km	9	21	SE Mid N Outer
Total	45.8 km	28 unique	61	

Rolling stock

Property	Value
Consist	2-car, 39 m
Max speed	70 km/h
Onboard battery	240 kWh per trainset
Seats	40 longitudinal seats
Nominal capacity (AW2)	210 pax (seated + standing, tram-2car per RFC 0008 §1)
Crush capacity (AW3)	260 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 210 AW2 passengers (tram-2car)
- Peak frequency: 12 trains/hour/direction (5-min headway)

- Peak capacity per line per direction: $210 \times 12 = 2,520$ pphpd
- Network peak throughput (all lines, both directions): $3 \text{ lines} \times 2 \text{ directions} \times 2,520 = 15,120$ passengers/hour
- Daily theoretical capacity (peak $\times 10$): $\approx 151,200$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 5,850 - 8,775$ trips/day

Catchment

- City population: 250,000
- Anchor-weighted coverage: 23.4%
- Catchment population: $\approx 58,500$ (within ~ 800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../.../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Major	5	400 kW	2500 kWh
Standard	17	300 kW	2000 kWh
Terminal	5	500 kW	3000 kWh
Total installed	28	14,600 kW	101,500 kWh

Aggregate station-rail charging power: 17,000 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 240 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	8.0 kWh/km	2 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	122 kWh	15.3 km average line length
Onboard battery coverage	2.0 $\times$ average line run	240 kWh usable pack
Average 60 s dwell charge	10.1 kWh/stop	607 kW average charger across stops
Stops to refill one trainset pack	24 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	73 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	102 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in design.toml — emitted by the osr-design Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. country-costs.toml applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (45.7 km @ €3.5 M/km)	€160 M
Elevated (0.1 km @ €18 M/km)	€1.0 M
Civil subtotal	€161 M

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~ 11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
standard	17	€1.5 M	€26 M

Archetype	Count	Unit	Subtotal
major	5	€3.0 M	€15 M
terminal	5	€2.5 M	€12 M
depot-terminal	1	€3.0 M	€3.0 M
Stations subtotal			€56 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	5	€3.0 M	€15 M
Depots subtotal			€40 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
tram-2car (revenue + spare + cold reserve)	61	€2.0 M	€122 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	45.8 km × €0.015 M/km	€0.7 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€9.0 M
EPC integration + project management (7%)	on subtotal	€27 M

Total

Bucket	Value
Civil works	€161 M
Stations	€56 M
Depots	€40 M
Rolling stock	€122 M
Residual train-control wayside + charging microgrids	€9.7 M
EPC overhead (7%)	€27 M
CAPEX total	€416 M
Per-route-km	€9.1 M / km
Per-capita (city pop)	€1,663 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code — regenerate by re-running scripts/regenerate-city.sh huye.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–7 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 8 and runs for 23 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–7)	€25 M / yr	€101
Steady-state, low-ridership (year 8+)	€35 M / yr	€141
Steady-state, high-ridership (year 8+)	€34 M / yr	€138
Lifecycle envelope (yr 1–30, low scenario)	€989 M cumulative	€3,956
Lifecycle envelope (yr 1–30, high scenario)	€968 M cumulative	€3,872

Population basis: 250,000 (catchment per lib/city-batches/world-sample.toml). After year 30, debt service drops to zero and only the OPEX shortfall remains — ~€9.7 M / yr (low) €8.8 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€249 M	3.0%	30 y, 7 y grace	€15 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€104 M	8.5%	30 y, 7 y grace	€10 M / yr
Government equity (no debt service)	15%	€62 M	—	—	—
Total	100%	€416 M			€26 M / yr

During the 7-year grace period the operator pays interest only — multilateral €7.5 M / yr + bonds €8.8 M / yr = €16 M / yr total — plus the equity tranche amortised across construction (€8.9 M / yr × 7 yr). Principal repayment begins in year 8 on a 23-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€4.9 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€5.1 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€34 k
Traction energy (54.3 GWh / yr)	trackside PV + Na-ion (RFC 0002) — self-generated, €0 / yr	€0 k
Labour (287 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€577 k
OPEX subtotal		€11 M / yr

Annual fleet utilisation: 55 revenue trainsets × 20.5 h/day × 365 d/yr × 22 km/h commercial × 75% revenue factor = 6.8 M train-km / yr (~123 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$130 USD (per lib/templates/country-finance.toml). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass — a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.20 (~\$0.22 USD)
Day pass (3 trips)	€0.51 (15 % bulk discount)
Monthly unlimited pass	€5.98 (~5 % of median monthly income)
Annual pass	€65.78 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	4.6 M	9.1 M
Farebox revenue	€909 k / yr	€1.8 M / yr
Farebox / OPEX recovery	9%	17%
Country policy-target recovery (diagnostic)	45%	45%
Operating shortfall (gov subsidy required)	€9.7 M / yr	€8.8 M / yr

	Low scenario	High scenario
Steady-state government commitment (debt service + OPEX shortfall)	€35 M / yr	€34 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
huye.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
huye-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
huye.corridor.geojson	Line polylines + stations (GeoJSON)
huye.stations.json	Machine-readable station list
huye.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug huye

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug huye \
  --sidecar .cache/osr-pipeline/rasters/huye.grid.json \
  --out-dir designs/.../Huye

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../huye.toml \
  --out designs/.../README.md
```

`scripts/regenerate-huye.sh` chains steps 3 + drift tests into a single command.

east-africa/Rwanda/Kigali

designs/east-africa/Rwanda/Kigali/README.md

Kigali — Urban Rail Network

Country: RW · Population: 1,208,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Kigali rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [kigali.corridor.geojson](#).

At a glance

Metric	Value
Lines	4
Unique stations	85
Interchange stations	11
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	34.3%
Route length (double track)	170.6 km
Revenue fleet	124 × 4-car trainsets
Spare + cold-reserve	15 × 4-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	36.8 km	20	30	E Outer W Outer
line-2	29.0 km	15	25	S Mid NW Outer
line-3	29.7 km	15	25	SE Outer NW Outer
line-4	75.0 km	36	59	N Outer N Outer
Total	170.6 km	85 unique	139	

Rolling stock

Property	Value
Consist	4-car, 75 m
Max speed	90 km/h
Onboard battery	480 kWh per trainset
Seats	80 longitudinal seats
Nominal capacity (AW2)	440 pax (seated + standing, met ro - 4car per RFC 0008 §1)
Crush capacity (AW3)	560 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 440 AW2 passengers (met ro - 4car)

- Peak frequency: 12 trains/hour/direction (5-min headway)
- Peak capacity per line per direction: $440 \times 12 = 5,280$ pphpd
- Network peak throughput (all lines, both directions): $4 \text{ lines} \times 2 \text{ directions} \times 5,280 = 42,240$ passengers/hour
- Daily theoretical capacity (peak $\times 10$): $\approx 422,400$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 41,434 - 62,151$ trips/day

Catchment

- City population: 1,208,000
- Anchor-weighted coverage: 34.3%
- Catchment population: $\approx 414,344$ (within ~800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	11	500 kW	3000 kWh
Major	35	400 kW	2500 kWh
Standard	34	300 kW	2000 kWh
Terminal	5	500 kW	3000 kWh
Total installed	86	37,200 kW	243,500 kWh

Aggregate station-rail charging power: 46,000 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 480 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	16.0 kWh/km	4 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	682 kWh	42.6 km average line length
Onboard battery coverage	0.7 $\times$ average line run	480 kWh usable pack
Average 60 s dwell charge	9.0 kWh/stop	541 kW average charger across stops
Stops to refill one trainset pack	53 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	186 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	244 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in design.toml — emitted by the osr-design Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. country-costs.toml applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (162.4 km @ €3.5 M/km)	€569 M
Elevated (7.5 km @ €18 M/km)	€134 M
Elevated-interchange premium (6 sites @ €20 M)	€120 M
Civil subtotal	€823 M

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
standard	34	€1.5 M	€51 M
major	35	€3.0 M	€105 M
terminal	5	€2.5 M	€12 M
depot-terminal	1	€3.0 M	€3.0 M
interchange-elevated	11	€4.5 M	€50 M
Stations subtotal			€221 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	5	€3.0 M	€15 M
Depots subtotal			€40 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
metro-4car (revenue + spare + cold reserve)	139	€4.0 M	€556 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	170.6 km × €0.015 M/km	€2.5 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€32 M
EPC integration + project management (7%)	on subtotal	€117 M

Total

Bucket	Value
Civil works	€823 M
Stations	€221 M
Depots	€40 M
Rolling stock	€556 M
Residual train-control wayside + charging microgrids	€34 M
EPC overhead (7%)	€117 M
CAPEX total	€1.79 bn
Per-route-km	€11 M / km
Per-capita (city pop)	€1,483 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from <lib/templates/country-finance.toml>. Pure function of the [costs] block above + the country code – regenerate by re-running `scripts/regenerate-city.sh` `kigali`.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–7 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 8 and runs for 23 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–7)	€109 M / yr	€90
Steady-state, low-ridership (year 8+)	€152 M / yr	€126
Steady-state, high-ridership (year 8+)	€148 M / yr	€122
Lifecycle envelope (yr 1–30, low scenario)	€4.26 bn cumulative	€3,525
Lifecycle envelope (yr 1–30, high scenario)	€4.16 bn cumulative	€3,441

Population basis: 1,208,000 (catchment per <lib/city-batches/world-sample.toml>). After year 30, debt service drops to zero and only the OPEX shortfall remains – ~€42 M / yr (low) €37 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€1.07 bn	3.0%	30 y, 7 y grace	€65 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€448 M	8.5%	30 y, 7 y grace	€45 M / yr
Government equity (no debt service)	15%	€269 M	—	—	—
Total	100%	€1.79 bn			€110 M / yr

During the 7-year grace period the operator pays interest only – multilateral €32 M / yr + bonds €38 M / yr = €70 M / yr total – plus the equity tranche amortised across construction (€38 M / yr × 7 yr). Principal repayment begins in year 8 on a 23-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€22 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€22 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€127 k
Traction energy (389.7 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (1,036 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€2.1 M
OPEX subtotal		€46 M / yr

Annual fleet utilisation: 124 revenue trainsets × 20.5 h/day × 365 d/yr × 35 km/h commercial × 75% revenue factor = 24.4 M train-km / yr (~196 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$130 USD (per <lib/templates/country-finance.toml>). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.20 (~\$0.22 USD)
Day pass (3 trips)	€0.51 (15 % bulk discount)
Monthly unlimited pass	€5.98 (~5 % of median monthly income)
Annual pass	€65.78 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	22.0 M	44.1 M
Farebox revenue	€4.4 M / yr	€8.8 M / yr
Farebox / OPEX recovery	10%	19%
Country policy-target recovery (diagnostic)	45%	45%
Operating shortfall (gov subsidy required)	€42 M / yr	€37 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€152 M / yr	€148 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
kigali.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
kigali-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
kigali.corridor.geojson	Line polylines + stations (GeoJSON)
kigali.stations.json	Machine-readable station list
kigali.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug kigali

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug kigali \
  --sidecar .cache/osr-pipeline/rasters/kigali.grid.json \
  --out-dir designs/.../Kigali

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../kigali.toml \
  --out designs/.../README.md
```

`scripts/regenerate-kigali.sh` chains steps 3 + drift tests into a single command.

Rubavu — Urban Rail Network

Country: RW · Population: 250,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Rubavu rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [rubavu.corridor.geojson](#).

At a glance

Metric	Value
Lines	3
Unique stations	28
Interchange stations	2
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	47.8%
Route length (double track)	45.3 km
Revenue fleet	55 × 2-car trainsets
Spare + cold-reserve	6 × 2-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	15.5 km	10	21	W Outer SE Outer
line-2	15.1 km	9	20	NE Outer W Outer
line-3	14.7 km	9	20	NE Mid S Outer
Total	45.3 km	28 unique	61	

Rolling stock

Property	Value
Consist	2-car, 39 m
Max speed	70 km/h
Onboard battery	240 kWh per trainset
Seats	40 longitudinal seats
Nominal capacity (AW2)	210 pax (seated + standing, tram-2car per RFC 0008 §1)
Crush capacity (AW3)	260 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 210 AW2 passengers (tram-2car)
- Peak frequency: 12 trains/hour/direction (5-min headway)

- Peak capacity per line per direction: $210 \times 12 = 2,520$ pphpd
- Network peak throughput (all lines, both directions): $3 \text{ lines} \times 2 \text{ directions} \times 2,520 = 15,120$ passengers/hour
- Daily theoretical capacity (peak $\times 10$): $\approx 151,200$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 11,950 - 17,925$ trips/day

Catchment

- City population: 250,000
- Anchor-weighted coverage: 47.8%
- Catchment population: $\approx 119,500$ (within ~ 800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	2	500 kW	3000 kWh
Major	7	400 kW	2500 kWh
Standard	13	300 kW	2000 kWh
Terminal	5	500 kW	3000 kWh
Total installed	28	15,200 kW	104,500 kWh

Aggregate station-rail charging power: 17,000 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 240 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	8.0 kWh/km	2 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	121 kWh	15.1 km average line length
Onboard battery coverage	2.0 $\times$ average line run	240 kWh usable pack
Average 60 s dwell charge	10.1 kWh/stop	607 kW average charger across stops
Stops to refill one trainset pack	24 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	76 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	104 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the `[costs]` block in `design.toml` — emitted by the `osr-design` Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. `country-costs.toml` applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (42.7 km @ €3.5 M/km)	€149 M
Elevated (2.2 km @ €18 M/km)	€40 M
Elevated-interchange premium (2 sites @ €20 M)	€40 M
Civil subtotal	€230 M

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~ 11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
standard	13	€1.5 M	€20 M
major	7	€3.0 M	€21 M
terminal	5	€2.5 M	€12 M
depot-terminal	1	€3.0 M	€3.0 M
interchange-elevated	2	€4.5 M	€9.0 M
Stations subtotal			€65 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	5	€3.0 M	€15 M
Depots subtotal			€40 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
tram-2car (revenue + spare + cold reserve)	61	€2.0 M	€122 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	45.3 km × €0.015 M/km	€0.7 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€10 M
EPC integration + project management (7%)	on subtotal	€33 M

Total

Bucket	Value
Civil works	€230 M
Stations	€65 M
Depots	€40 M
Rolling stock	€122 M
Residual train-control wayside + charging microgrids	€11 M
EPC overhead (7%)	€33 M
CAPEX total	€500 M
Per-route-km	€11 M / km
Per-capita (city pop)	€2,000 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running scripts/regenerate-city.sh rubavu.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–7 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 8 and runs for 23 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–7)	€30 M / yr	€121
Steady-state, low-ridership (year 8+)	€42 M / yr	€168
Steady-state, high-ridership (year 8+)	€41 M / yr	€165
Lifecycle envelope (yr 1–30, low scenario)	€1.18 bn cumulative	€4,720
Lifecycle envelope (yr 1–30, high scenario)	€1.16 bn cumulative	€4,636

Population basis: 250,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 30, debt service drops to zero and only the OPEX shortfall remains – ~€11 M / yr (low) €10 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€300 M	3.0%	30 y, 7 y grace	€18 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€125 M	8.5%	30 y, 7 y grace	€13 M / yr
Government equity (no debt service)	15%	€75 M	—	—	—
Total	100%	€500 M			€31 M / yr

During the 7-year grace period the operator pays interest only – multilateral €9.0 M / yr + bonds €11 M / yr = €20 M / yr total – plus the equity tranche amortised across construction (€11 M / yr × 7 yr). Principal repayment begins in year 8 on a 23-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€4.9 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€6.7 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€34 k
Traction energy (54.3 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (284 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€571 k
OPEX subtotal		€12 M / yr

Annual fleet utilisation: 55 revenue trainsets × 20.5 h/day × 365 d/yr × 22 km/h commercial × 75% revenue factor = 6.8 M train-km / yr (~123 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$130 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.20 (~\$0.22 USD)
Day pass (3 trips)	€0.51 (15 % bulk discount)
Monthly unlimited pass	€5.98 (~5 % of median monthly income)
Annual pass	€65.78 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	4.6 M	9.1 M

	Low scenario	High scenario
Farebox revenue	€909 k / yr	€1.8 M / yr
Farebox / OPEX recovery	7%	15%
Country policy-target recovery (diagnostic)	45%	45%
Operating shortfall (gov subsidy required)	€11 M / yr	€10 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€42 M / yr	€41 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
rubavu.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
rubavu-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
rubavu.corridor.geojson	Line polylines + stations (GeoJSON)
rubavu.stations.json	Machine-readable station list
rubavu.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug rubavu

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug rubavu \
  --sidecar .cache/osr-pipeline/rasters/rubavu.grid.json \
  --out-dir designs/.../Rubavu

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../rubavu.toml \
  --out designs/.../README.md
```

`scripts/regenerate-rubavu.sh` chains steps 3 + drift tests into a single command.

east-africa/Somalia/Mogadishu

designs/east-africa/Somalia/Mogadishu/README.md

Mogadishu — Urban Rail Network

Country: SO · Population: 2,610,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Mogadishu rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [mogadishu.corridor.geojson](#).

At a glance

Metric	Value
Lines	4
Unique stations	67
Interchange stations	14
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	39.7%
Route length (double track)	128.1 km
Revenue fleet	94 × 4-car trainsets
Spare + cold-reserve	12 × 4-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	27.4 km	14	23	W Mid E Mid
line-2	25.3 km	16	21	SW Mid NE Mid
line-3	31.4 km	12	26	S Inner NW Outer
line-4	44.0 km	26	36	N Mid N Mid
Total	128.1 km	67 unique	106	

Rolling stock

Property	Value
Consist	4-car, 75 m
Max speed	90 km/h
Onboard battery	480 kWh per trainset
Seats	80 longitudinal seats
Nominal capacity (AW2)	440 pax (seated + standing, met ro - 4car per RFC 0008 §1)
Crush capacity (AW3)	560 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 440 AW2 passengers (met ro - 4car)

- Peak frequency: 12 trains/hour/direction (5-min headway)
- Peak capacity per line per direction: $440 \times 12 = 5,280$ pphpd
- Network peak throughput (all lines, both directions): $4 \text{ lines} \times 2 \text{ directions} \times 5,280 = 42,240$ passengers/hour
- Daily theoretical capacity (peak $\times$ 10): $\approx 422,400$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 103,617 - 155,425$ trips/day

Catchment

- City population: 2,610,000
- Anchor-weighted coverage: 39.7%
- Catchment population: $\approx 1,036,170$ (within ~800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	14	500 kW	3000 kWh
Major	24	400 kW	2500 kWh
Standard	19	300 kW	2000 kWh
Terminal	5	500 kW	3000 kWh
Total installed	63	29,800 kW	195,000 kWh

Aggregate station-rail charging power: 35,750 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 480 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	16.0 kWh/km	4 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	512 kWh	32.0 km average line length
Onboard battery coverage	0.9 $\times$ average line run	480 kWh usable pack
Average 60 s dwell charge	8.9 kWh/stop	534 kW average charger across stops
Stops to refill one trainset pack	54 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	149 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	195 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the `[costs]` block in `design.toml` — emitted by the `osr-design` Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. `country-costs.toml` applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (116.8 km @ €3.5 M/km)	€409 M
Elevated (8.4 km @ €18 M/km)	€151 M
Elevated-interchange premium (7 sites @ €20 M)	€140 M
Civil subtotal	€700 M

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
halt	5	€0.4 M	€2.0 M
standard	19	€1.5 M	€28 M
major	24	€3.0 M	€72 M
terminal	5	€2.5 M	€12 M
depot-terminal	1	€3.0 M	€3.0 M
interchange	2	€4.5 M	€9.0 M
interchange-elevated	12	€4.5 M	€54 M
Stations subtotal			€181 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	5	€3.0 M	€15 M
Depots subtotal			€40 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
metro-4car (revenue + spare + cold reserve)	106	€4.0 M	€424 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	128.1 km × €0.015 M/km	€1.9 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€26 M
EPC integration + project management (7%)	on subtotal	€96 M

Total

Bucket	Value
Civil works	€700 M
Stations	€181 M
Depots	€40 M
Rolling stock	€424 M
Residual train-control wayside + charging microgrids	€28 M
EPC overhead (7%)	€96 M
CAPEX total	€1.47 bn
Per-route-km	€11 M / km
Per-capita (city pop)	€563 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running scripts/regenerate-city.sh mogadishu.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–10 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 11 and runs for 30 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–10)	€104 M / yr	€40
Steady-state, low-ridership (year 11+)	€131 M / yr	€50
Steady-state, high-ridership (year 11+)	€124 M / yr	€47
Lifecycle envelope (yr 1–40, low scenario)	€4.95 bn cumulative	€1,898
Lifecycle envelope (yr 1–40, high scenario)	€4.74 bn cumulative	€1,818

Population basis: 2,610,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 40, debt service drops to zero and only the OPEX shortfall remains – ~€30 M / yr (low) €23 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€881 M	3.0%	40 y, 10 y grace	€45 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€367 M	15.0%	40 y, 10 y grace	€56 M / yr
Government equity (no debt service)	15%	€220 M	—	—	—
Total	100%	€1.47 bn			€101 M / yr

During the 10-year grace period the operator pays interest only — multilateral €26 M / yr + bonds €55 M / yr = €82 M / yr total — plus the equity tranche amortised across construction (€22 M / yr × 10 yr). Principal repayment begins in year 11 on a 30-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€17 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€18 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€94 k
Traction energy (295.4 GWh / yr)	trackside PV + Na-ion (RFC 0002) — self-generated, €0 / yr	€0 k
Labour (781 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€1.1 M
OPEX subtotal		€37 M / yr

Annual fleet utilisation: 94 revenue trainsets × 20.5 h/day × 365 d/yr × 35 km/h commercial × 75% revenue factor = 18.5 M train-km / yr (~196 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$95 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass — a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.15 (~\$0.16 USD)
Day pass (3 trips)	€0.37 (15 % bulk discount)
Monthly unlimited pass	€4.37 (~5 % of median monthly income)
Annual pass	€48.07 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	47.6 M	95.3 M
Farebox revenue	€6.9 M / yr	€14 M / yr
Farebox / OPEX recovery	19%	38%
Country policy-target recovery (diagnostic)	25%	25%
Operating shortfall (gov subsidy required)	€30 M / yr	€23 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€131 M / yr	€124 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
mogadishu.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
mogadishu-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
mogadishu.corridor.geojson	Line polylines + stations (GeoJSON)
mogadishu.stations.json	Machine-readable station list
mogadishu.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug mogadishu

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug mogadishu \
  --sidecar .cache/osr-pipeline/rasters/mogadishu.grid.json \
  --out-dir designs/.../Mogadishu

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../mogadishu.toml \
  --out designs/.../README.md
```

`scripts/regenerate-mogadishu.sh` chains steps 3 + drift tests into a single command.

east-africa/Tanzania/Arusha

designs/east-africa/Tanzania/Arusha/README.md

Arusha — Urban Rail Network

Country: TZ · Population: 700,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Arusha rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [arusha.corridor.geojson](#).

At a glance

Metric	Value
Lines	3
Unique stations	44
Interchange stations	2
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	36.4%
Route length (double track)	72.7 km
Revenue fleet	64 × 3-car trainsets
Spare + cold-reserve	8 × 3-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	26.1 km	16	26	NW Outer S Outer
line-2	25.3 km	16	25	W Outer E Outer
line-3	21.2 km	12	21	SE Outer N Outer
Total	72.7 km	44 unique	72	

Rolling stock

Property	Value
Consist	3-car, 57 m
Max speed	90 km/h
Onboard battery	360 kWh per trainset
Seats	60 longitudinal seats
Nominal capacity (AW2)	330 pax (seated + standing, light-metro-3car per RFC 0008 §1)
Crush capacity (AW3)	420 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 330 AW2 passengers (light-metro-3car)
- Peak frequency: 12 trains/hour/direction (5-min headway)

- Peak capacity per line per direction: $330 \times 12 = 3,960$ pphpd
- Network peak throughput (all lines, both directions): $3 \text{ lines} \times 2 \text{ directions} \times 3,960 = 23,760$ passengers/hour
- Daily theoretical capacity (peak $\times 10$): $\approx 237,600$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 25,480 - 38,220$ trips/day

Catchment

- City population: 700,000
- Anchor-weighted coverage: 36.4%
- Catchment population: $\approx 254,800$ (within ~ 800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../.../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	2	500 kW	3000 kWh
Major	13	400 kW	2500 kWh
Standard	23	300 kW	2000 kWh
Terminal	5	500 kW	3000 kWh
Total installed	44	20,600 kW	139,500 kWh

Aggregate station-rail charging power: 25,000 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 360 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	12.0 kWh/km	3 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	291 kWh	24.2 km average line length
Onboard battery coverage	1.2 $\times$ average line run	360 kWh usable pack
Average 60 s dwell charge	9.5 kWh/stop	568 kW average charger across stops
Stops to refill one trainset pack	38 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	103 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	140 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in design.toml — emitted by the osr-design Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. country-costs.toml applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (70.7 km @ €3.5 M/km)	€247 M
Elevated (1.3 km @ €18 M/km)	€23 M
Civil subtotal	€270 M

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~ 11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
standard	23	€1.5 M	€34 M
major	13	€3.0 M	€39 M
terminal	5	€2.5 M	€12 M
depot-terminal	1	€3.0 M	€3.0 M
interchange	2	€4.5 M	€9.0 M
Stations subtotal			€98 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	5	€3.0 M	€15 M
Depots subtotal			€40 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
light-metro-3car (revenue + spare + cold reserve)	72	€3.0 M	€216 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	72.7 km × €0.015 M/km	€1.1 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€15 M
EPC integration + project management (7%)	on subtotal	€45 M

Total

Bucket	Value
Civil works	€270 M
Stations	€98 M
Depots	€40 M
Rolling stock	€216 M
Residual train-control wayside + charging microgrids	€16 M
EPC overhead (7%)	€45 M
CAPEX total	€685 M
Per-route-km	€9.4 M / km
Per-capita (city pop)	€979 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running scripts/regenerate-city.sh arusha.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–7 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 8 and runs for 23 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–7)	€47 M / yr	€67
Steady-state, low-ridership (year 8+)	€60 M / yr	€86
Steady-state, high-ridership (year 8+)	€57 M / yr	€82
Lifecycle envelope (yr 1–30, low scenario)	€1.72 bn cumulative	€2,453
Lifecycle envelope (yr 1–30, high scenario)	€1.64 bn cumulative	€2,347

Population basis: 700,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 30, debt service drops to zero and only the OPEX shortfall remains – ~€15 M / yr (low) €12 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€411 M	3.8%	30 y, 7 y grace	€27 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€171 M	9.5%	30 y, 7 y grace	€19 M / yr
Government equity (no debt service)	15%	€103 M	—	—	—
Total	100%	€685 M			€46 M / yr

During the 7-year grace period the operator pays interest only – multilateral €16 M / yr + bonds €16 M / yr = €32 M / yr total – plus the equity tranche amortised across construction (€15 M / yr × 7 yr). Principal repayment begins in year 8 on a 23-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€8.6 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€8.2 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€54 k
Traction energy (129.3 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (448 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€1.1 M
OPEX subtotal		€18 M / yr

Annual fleet utilisation: 64 revenue trainsets × 20.5 h/day × 365 d/yr × 30 km/h commercial × 75% revenue factor = 10.8 M train-km / yr (~168 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$165 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.25 (~\$0.28 USD)
Day pass (3 trips)	€0.65 (15 % bulk discount)
Monthly unlimited pass	€7.59 (~5 % of median monthly income)
Annual pass	€83.49 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	12.8 M	25.6 M

	Low scenario	High scenario
Farebox revenue	€3.2 M / yr	€6.5 M / yr
Farebox / OPEX recovery	18%	36%
Country policy-target recovery (diagnostic)	45%	45%
Operating shortfall (gov subsidy required)	€15 M / yr	€12 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€60 M / yr	€57 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
arusha.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
arusha-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
arusha.corridor.geojson	Line polylines + stations (GeoJSON)
arusha.stations.json	Machine-readable station list
arusha.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug arusha

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug arusha \
  --sidecar .cache/osr-pipeline/rasters/arusha.grid.json \
  --out-dir designs/.../Arusha

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../arusha.toml \
  --out designs/.../README.md
```

`scripts/regenerate-arusha.sh` chains steps 3 + drift tests into a single command.

east-africa/Tanzania/Dar-Es-Salaam

designs/east-africa/Tanzania/Dar-Es-Salaam/README.md

Dar-Es-Salaam — Urban Rail Network

Country: TZ · Population: 7,404,689

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Dar-Es-Salaam rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [dar-es-salaam.corridor.geojson](#).

At a glance

Metric	Value
Lines	7
Unique stations	162
Interchange stations	24
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	27.5%
Route length (double track)	393.1 km
Revenue fleet	282 × 6-car trainsets
Spare + cold-reserve	32 × 6-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	55.8 km	24	45	NW Outer SE Mid
line-2	53.8 km	21	43	SW Outer N Mid
line-3	59.9 km	26	48	NW Mid E Outer
line-4	38.4 km	16	31	W Mid SE Mid
line-5	50.7 km	21	40	NW Outer SE Mid
line-6	35.6 km	14	29	SW Mid N Mid
line-7	99.0 km	41	78	W Inner W Inner
Total	393.1 km	162 unique	314	

Rolling stock

Property	Value
Consist	6-car, 111 m
Max speed	100 km/h
Onboard battery	720 kWh per trainset
Seats	120 longitudinal seats
Nominal capacity (AW2)	660 pax (seated + standing, metro-6car per RFC 0008 §1)
Crush capacity (AW3)	840 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 660 AW2 passengers (metro - 6 car)
- Peak frequency: 12 trains/hour/direction (5-min headway)
- Peak capacity per line per direction: $660 \times 12 = 7,920$ pphpd
- Network peak throughput (all lines, both directions): $7 \text{ lines} \times 2 \text{ directions} \times 7,920 = 110,880$ passengers/hour
- Daily theoretical capacity (peak $\times 10$): $\approx 1,108,800$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 203,628 - 305,443$ trips/day

Catchment

- City population: 7,404,689
- Anchor-weighted coverage: 27.5%
- Catchment population: $\approx 2,036,289$ (within ~ 800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	24	500 kW	3000 kWh
Major	21	400 kW	2500 kWh
Standard	101	300 kW	2000 kWh
Terminal	11	500 kW	3000 kWh
Total installed	158	61,200 kW	399,500 kWh

Aggregate station-rail charging power: 86,250 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 720 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	24.0 kWh/km	6 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	1,348 kWh	56.2 km average line length
Onboard battery coverage	0.5 $\times$ average line run	720 kWh usable pack
Average 60 s dwell charge	8.9 kWh/stop	532 kW average charger across stops
Stops to refill one trainset pack	81 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	306 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	400 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the `[costs]` block in `design.toml` — emitted by the `osr-design` Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. `country-costs.toml` applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (358.1 km @ €3.5 M/km)	€1.25 bn
Elevated (31.9 km @ €18 M/km)	€574 M
Elevated-interchange premium (11 sites @ €20 M)	€220 M
Civil subtotal	€2.05 bn

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
halt	5	€0.4 M	€2.0 M
standard	101	€1.5 M	€152 M
major	21	€3.0 M	€63 M
terminal	11	€2.5 M	€28 M
depot-terminal	1	€3.0 M	€3.0 M
interchange-elevated	24	€4.5 M	€108 M
Stations subtotal			€355 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	11	€3.0 M	€33 M
Depots subtotal			€58 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
metro-6car (revenue + spare + cold reserve)	314	€6.0 M	€1.88 bn

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	393.1 km × €0.015 M/km	€5.8 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€54 M
EPC integration + project management (7%)	on subtotal	€308 M

Total

Bucket	Value
Civil works	€2.05 bn
Stations	€355 M
Depots	€58 M
Rolling stock	€1.88 bn
Residual train-control wayside + charging microgrids	€60 M
EPC overhead (7%)	€308 M
CAPEX total	€4.71 bn
Per-route-km	€12 M / km

Bucket	Value
Per-capita (city pop)	€636 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](https://github.com/OpenSourceRail/lib/templates/country-finance.toml). Pure function of the [costs] block above + the country code — regenerate by re-running scripts/regenerate-city.sh dar-es-salaam.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–7 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 8 and runs for 23 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–7)	€320 M / yr	€43
Steady-state, low-ridership (year 8+)	€411 M / yr	€56
Steady-state, high-ridership (year 8+)	€377 M / yr	€51
Lifecycle envelope (yr 1–30, low scenario)	€11.70 bn cumulative	€1,580
Lifecycle envelope (yr 1–30, high scenario)	€10.91 bn cumulative	€1,473

Population basis: 7,404,689 (catchment per [lib/city-batches/world-sample.toml](https://github.com/OpenSourceRail/lib/city-batches/world-sample.toml)). After year 30, debt service drops to zero and only the OPEX shortfall remains — ~€97 M / yr (low) €63 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€2.83 bn	3.8%	30 y, 7 y grace	€187 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€1.18 bn	9.5%	30 y, 7 y grace	€128 M / yr
Government equity (no debt service)	15%	€707 M	—	—	—
Total	100%	€4.71 bn			€314 M / yr

During the 7-year grace period the operator pays interest only — multilateral €107 M / yr + bonds €112 M / yr = €219 M / yr total — plus the equity tranche amortised across construction (€101 M / yr × 7 yr). Principal repayment begins in year 8 on a 23-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€75 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€49 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€292 k
Traction energy (1329.3 GWh / yr)	trackside PV + Na-ion (RFC 0002) — self-generated, €0 / yr	€0 k
Labour (2,371 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€6.0 M
OPEX subtotal		€131 M / yr

Annual fleet utilisation: 282 revenue trainsets × 20.5 h/day × 365 d/yr × 35 km/h commercial × 75% revenue factor = 55.4 M train-km / yr (~196 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$165 USD (per [lib/templates/country-finance.toml](https://github.com/OpenSourceRail/lib/templates/country-finance.toml)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass — a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.25 (~\$0.28 USD)
Day pass (3 trips)	€0.65 (15 % bulk discount)
Monthly unlimited pass	€7.59 (~5 % of median monthly income)
Annual pass	€83.49 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	135.1 M	270.3 M
Farebox revenue	€34 M / yr	€68 M / yr
Farebox / OPEX recovery	26%	52%
Country policy-target recovery (diagnostic)	45%	45%
Operating shortfall (gov subsidy required)	€97 M / yr	€63 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€411 M / yr	€377 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
dar-es-salaam.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
dar-es-salaam-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
dar-es-salaam.corridor.geojson	Line polylines + stations (GeoJSON)
dar-es-salaam.stations.json	Machine-readable station list
dar-es-salaam.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug dar-es-salaam

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug dar-es-salaam \
  --sidecar .cache/osr-pipeline/rasters/dar-es-salaam.grid.json \
  --out-dir designs/.../Dar-Es-Salaam

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../dar-es-salaam.toml \
  --out designs/.../README.md
```

`scripts/regenerate-dar-es-salaam.sh` chains steps 3 + drift tests into a single command.

east-africa/Tanzania/Dodoma

designs/east-africa/Tanzania/Dodoma/README.md

Dodoma — Urban Rail Network

Country: TZ · Population: 800,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Dodoma rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [dodoma.corridor.geojson](#).

At a glance

Metric	Value
Lines	3
Unique stations	36
Interchange stations	2
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	40.5%
Route length (double track)	66.3 km
Revenue fleet	58 × 3-car trainsets
Spare + cold-reserve	8 × 3-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	23.0 km	12	23	E Outer SW Outer
line-2	27.7 km	15	27	SE Mid NW Outer
line-3	15.6 km	9	16	E Mid NW Mid
Total	66.3 km	36 unique	66	

Rolling stock

Property	Value
Consist	3-car, 57 m
Max speed	90 km/h
Onboard battery	360 kWh per trainset
Seats	60 longitudinal seats
Nominal capacity (AW2)	330 pax (seated + standing, light-metro-3car per RFC 0008 §1)
Crush capacity (AW3)	420 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 330 AW2 passengers (light-metro-3car)
- Peak frequency: 12 trains/hour/direction (5-min headway)

- Peak capacity per line per direction: $330 \times 12 = 3,960$ pphpd
- Network peak throughput (all lines, both directions): $3 \text{ lines} \times 2 \text{ directions} \times 3,960 = 23,760$ passengers/hour
- Daily theoretical capacity (peak $\times 10$): $\approx 237,600$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 32,400 - 48,600$ trips/day

Catchment

- City population: 800,000
- Anchor-weighted coverage: 40.5%
- Catchment population: $\approx 324,000$ (within ~ 800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../.../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	2	500 kW	3000 kWh
Major	9	400 kW	2500 kWh
Standard	19	300 kW	2000 kWh
Terminal	5	500 kW	3000 kWh
Total installed	36	17,800 kW	121,500 kWh

Aggregate station-rail charging power: 21,000 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 360 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	12.0 kWh/km	3 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	265 kWh	22.1 km average line length
Onboard battery coverage	1.4 $\times$ average line run	360 kWh usable pack
Average 60 s dwell charge	9.7 kWh/stop	583 kW average charger across stops
Stops to refill one trainset pack	37 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	89 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	122 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the `[costs]` block in `design.toml` — emitted by the `osr-design` Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. `country-costs.toml` applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (64.1 km @ €3.5 M/km)	€224 M
Elevated (2.1 km @ €18 M/km)	€37 M
Elevated-interchange premium (1 sites @ €20 M)	€20 M
Civil subtotal	€282 M

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~ 11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
standard	19	€1.5 M	€28 M
major	9	€3.0 M	€27 M
terminal	5	€2.5 M	€12 M
depot-terminal	1	€3.0 M	€3.0 M
interchange-elevated	2	€4.5 M	€9.0 M
Stations subtotal			€80 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	5	€3.0 M	€15 M
Depots subtotal			€40 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
light-metro-3car (revenue + spare + cold reserve)	66	€3.0 M	€198 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	66.3 km × €0.015 M/km	€1.0 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€12 M
EPC integration + project management (7%)	on subtotal	€43 M

Total

Bucket	Value
Civil works	€282 M
Stations	€80 M
Depots	€40 M
Rolling stock	€198 M
Residual train-control wayside + charging microgrids	€13 M
EPC overhead (7%)	€43 M
CAPEX total	€656 M
Per-route-km	€9.9 M / km
Per-capita (city pop)	€820 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running scripts/regenerate-city.sh dodoma.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–7 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 8 and runs for 23 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–7)	€45 M / yr	€56
Steady-state, low-ridership (year 8+)	€57 M / yr	€71
Steady-state, high-ridership (year 8+)	€53 M / yr	€67
Lifecycle envelope (yr 1–30, low scenario)	€1.63 bn cumulative	€2,032
Lifecycle envelope (yr 1–30, high scenario)	€1.54 bn cumulative	€1,926

Population basis: 800,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 30, debt service drops to zero and only the OPEX shortfall remains – ~€13 M / yr (low) €9.7 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€394 M	3.8%	30 y, 7 y grace	€26 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€164 M	9.5%	30 y, 7 y grace	€18 M / yr
Government equity (no debt service)	15%	€98 M	—	—	—
Total	100%	€656 M			€44 M / yr

During the 7-year grace period the operator pays interest only – multilateral €15 M / yr + bonds €16 M / yr = €31 M / yr total – plus the equity tranche amortised across construction (€14 M / yr × 7 yr). Principal repayment begins in year 8 on a 23-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€7.9 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€8.0 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€50 k
Traction energy (117.2 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (410 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€1.0 M
OPEX subtotal		€17 M / yr

Annual fleet utilisation: 58 revenue trainsets × 20.5 h/day × 365 d/yr × 30 km/h commercial × 75% revenue factor = 9.8 M train-km / yr (~168 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$165 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.25 (~\$0.28 USD)
Day pass (3 trips)	€0.65 (15 % bulk discount)
Monthly unlimited pass	€7.59 (~5 % of median monthly income)
Annual pass	€83.49 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	14.6 M	29.2 M

	Low scenario	High scenario
Farebox revenue	€3.7 M / yr	€7.4 M / yr
Farebox / OPEX recovery	22%	43%
Country policy-target recovery (diagnostic)	45%	45%
Operating shortfall (gov subsidy required)	€13 M / yr	€9.7 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€57 M / yr	€53 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
dodoma.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
dodoma-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
dodoma.corridor.geojson	Line polylines + stations (GeoJSON)
dodoma.stations.json	Machine-readable station list
dodoma.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug dodoma

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug dodoma \
  --sidecar .cache/osr-pipeline/rasters/dodoma.grid.json \
  --out-dir designs/.../Dodoma

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../dodoma.toml \
  --out designs/.../README.md
```

`scripts/regenerate-dodoma.sh` chains steps 3 + drift tests into a single command.

east-africa/Tanzania/Mbeya

designs/east-africa/Tanzania/Mbeya/README.md

Mbeya — Urban Rail Network

Country: TZ · Population: 550,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Mbeya rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [mbeya.corridor.geojson](#).

At a glance

Metric	Value
Lines	3
Unique stations	39
Interchange stations	3
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	43.7%
Route length (double track)	55.0 km
Revenue fleet	49 × 3-car trainsets
Spare + cold-reserve	6 × 3-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	21.7 km	16	21	W Outer E Outer
line-2	21.7 km	14	21	E Outer W Outer
line-3	11.5 km	9	13	NW Mid S Mid
Total	55.0 km	39 unique	55	

Rolling stock

Property	Value
Consist	3-car, 57 m
Max speed	90 km/h
Onboard battery	360 kWh per trainset
Seats	60 longitudinal seats
Nominal capacity (AW2)	330 pax (seated + standing, light-metro-3car per RFC 0008 §1)
Crush capacity (AW3)	420 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 330 AW2 passengers (light-metro-3car)
- Peak frequency: 12 trains/hour/direction (5-min headway)

- Peak capacity per line per direction: $330 \times 12 = 3,960$ pphpd
- Network peak throughput (all lines, both directions): $3 \text{ lines} \times 2 \text{ directions} \times 3,960 = 23,760$ passengers/hour
- Daily theoretical capacity (peak $\times 10$): $\approx 237,600$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 24,035 - 36,052$ trips/day

Catchment

- City population: 550,000
- Anchor-weighted coverage: 43.7%
- Catchment population: $\approx 240,350$ (within ~ 800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	3	500 kW	3000 kWh
Major	21	400 kW	2500 kWh
Standard	9	300 kW	2000 kWh
Terminal	5	500 kW	3000 kWh
Total installed	39	20,100 kW	134,500 kWh

Aggregate station-rail charging power: 22,500 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 360 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	12.0 kWh/km	3 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	220 kWh	18.3 km average line length
Onboard battery coverage	1.6 $\times$ average line run	360 kWh usable pack
Average 60 s dwell charge	9.6 kWh/stop	577 kW average charger across stops
Stops to refill one trainset pack	37 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	100 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	134 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the `[costs]` block in `design.toml` — emitted by the `osr-design` Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. `country-costs.toml` applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (52.8 km @ €3.5 M/km)	€185 M
Elevated (1.9 km @ €18 M/km)	€34 M
Elevated-interchange premium (2 sites @ €20 M)	€40 M
Civil subtotal	€259 M

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~ 11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
standard	9	€1.5 M	€14 M
major	21	€3.0 M	€63 M
terminal	5	€2.5 M	€12 M
depot-terminal	1	€3.0 M	€3.0 M
interchange-elevated	3	€4.5 M	€14 M
Stations subtotal			€106 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	5	€3.0 M	€15 M
Depots subtotal			€40 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
light-metro-3car (revenue + spare + cold reserve)	55	€3.0 M	€165 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	55.0 km × €0.015 M/km	€0.8 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€15 M
EPC integration + project management (7%)	on subtotal	€41 M

Total

Bucket	Value
Civil works	€259 M
Stations	€106 M
Depots	€40 M
Rolling stock	€165 M
Residual train-control wayside + charging microgrids	€16 M
EPC overhead (7%)	€41 M
CAPEX total	€627 M
Per-route-km	€11 M / km
Per-capita (city pop)	€1,139 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running scripts/regenerate-city.sh mbeya.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–7 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 8 and runs for 23 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–7)	€43 M / yr	€77
Steady-state, low-ridership (year 8+)	€55 M / yr	€100
Steady-state, high-ridership (year 8+)	€52 M / yr	€95
Lifecycle envelope (yr 1–30, low scenario)	€1.56 bn cumulative	€2,836
Lifecycle envelope (yr 1–30, high scenario)	€1.50 bn cumulative	€2,730

Population basis: 550,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 30, debt service drops to zero and only the OPEX shortfall remains – ~€13 M / yr (low) €11 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€376 M	3.8%	30 y, 7 y grace	€25 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€157 M	9.5%	30 y, 7 y grace	€17 M / yr
Government equity (no debt service)	15%	€94 M	—	—	—
Total	100%	€627 M			€42 M / yr

During the 7-year grace period the operator pays interest only – multilateral €14 M / yr + bonds €15 M / yr = €29 M / yr total – plus the equity tranche amortised across construction (€13 M / yr × 7 yr). Principal repayment begins in year 8 on a 23-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€6.6 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€8.1 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€41 k
Traction energy (99.0 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (342 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€872 k
OPEX subtotal		€16 M / yr

Annual fleet utilisation: 49 revenue trainsets × 20.5 h/day × 365 d/yr × 30 km/h commercial × 75% revenue factor = 8.2 M train-km / yr (~168 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$165 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.25 (~\$0.28 USD)
Day pass (3 trips)	€0.65 (15 % bulk discount)
Monthly unlimited pass	€7.59 (~5 % of median monthly income)
Annual pass	€83.49 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	10.0 M	20.1 M

	Low scenario	High scenario
Farebox revenue	€2.5 M / yr	€5.1 M / yr
Farebox / OPEX recovery	16%	33%
Country policy-target recovery (diagnostic)	45%	45%
Operating shortfall (gov subsidy required)	€13 M / yr	€11 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€55 M / yr	€52 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
mbeya.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
mbeya-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
mbeya.corridor.geojson	Line polylines + stations (GeoJSON)
mbeya.stations.json	Machine-readable station list
mbeya.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug mbeya

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug mbeya \
  --sidecar .cache/osr-pipeline/rasters/mbeya.grid.json \
  --out-dir designs/.../Mbeya

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../mbeya.toml \
  --out designs/.../README.md
```

`scripts/regenerate-mbeya.sh` chains steps 3 + drift tests into a single command.

east-africa/Tanzania/Mwanza

designs/east-africa/Tanzania/Mwanza/README.md

Mwanza — Urban Rail Network

Country: TZ · Population: 1,100,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Mwanza rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [mwanza.corridor.geojson](#).

At a glance

Metric	Value
Lines	5
Unique stations	69
Interchange stations	12
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	45.2%
Route length (double track)	161.6 km
Revenue fleet	121 × 4-car trainsets
Spare + cold-reserve	15 × 4-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	28.2 km	13	24	S Mid N Outer
line-2	27.4 km	14	24	E Outer NW Mid
line-3	20.4 km	11	18	NW Mid S Mid
line-4	23.1 km	8	20	W Inner SE Outer
line-5	62.4 km	24	50	N Mid N Mid
Total	161.6 km	69 unique	136	

Rolling stock

Property	Value
Consist	4-car, 75 m
Max speed	90 km/h
Onboard battery	480 kWh per trainset
Seats	80 longitudinal seats
Nominal capacity (AW2)	440 pax (seated + standing, met ro - 4 car per RFC 0008 §1)
Crush capacity (AW3)	560 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 440 AW2 passengers (met ro - 4 car)

- Peak frequency: 12 trains/hour/direction (5-min headway)
- Peak capacity per line per direction: $440 \times 12 = 5,280$ pphpd
- Network peak throughput (all lines, both directions): $5 \text{ lines} \times 2 \text{ directions} \times 5,280 = 52,800$ passengers/hour
- Daily theoretical capacity (peak $\times$ 10): $\approx 528,000$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 49,720 - 74,580$ trips/day

Catchment

- City population: 1,100,000
- Anchor-weighted coverage: 45.2%
- Catchment population: $\approx 497,200$ (within ~800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	12	500 kW	3000 kWh
Major	13	400 kW	2500 kWh
Standard	32	300 kW	2000 kWh
Terminal	7	500 kW	3000 kWh
Total installed	65	29,300 kW	193,500 kWh

Aggregate station-rail charging power: 37,750 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 480 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	16.0 kWh/km	4 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	517 kWh	32.3 km average line length
Onboard battery coverage	0.9 $\times$ average line run	480 kWh usable pack
Average 60 s dwell charge	9.1 kWh/stop	547 kW average charger across stops
Stops to refill one trainset pack	53 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	146 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	194 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the `[costs]` block in `design.toml` — emitted by the `osr-design` Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. `country-costs.toml` applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (154.1 km @ €3.5 M/km)	€540 M
Elevated (6.8 km @ €18 M/km)	€122 M
Elevated-interchange premium (6 sites @ €20 M)	€120 M
Civil subtotal	€781 M

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
halt	5	€0.4 M	€2.0 M
standard	32	€1.5 M	€48 M
major	13	€3.0 M	€39 M
terminal	7	€2.5 M	€18 M
depot-terminal	1	€3.0 M	€3.0 M
interchange-elevated	12	€4.5 M	€54 M
Stations subtotal			€164 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	7	€3.0 M	€21 M
Depots subtotal			€46 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
metro-4car (revenue + spare + cold reserve)	136	€4.0 M	€544 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	161.6 km × €0.015 M/km	€2.4 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€25 M
EPC integration + project management (7%)	on subtotal	€109 M

Total

Bucket	Value
Civil works	€781 M
Stations	€164 M
Depots	€46 M
Rolling stock	€544 M
Residual train-control wayside + charging microgrids	€27 M
EPC overhead (7%)	€109 M
CAPEX total	€1.67 bn
Per-route-km	€10 M / km
Per-capita (city pop)	€1,519 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running scripts/regenerate-city.sh mwanza.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–7 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 8 and runs for 23 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–7)	€114 M / yr	€103
Steady-state, low-ridership (year 8+)	€151 M / yr	€137
Steady-state, high-ridership (year 8+)	€145 M / yr	€132
Lifecycle envelope (yr 1–30, low scenario)	€4.26 bn cumulative	€3,871
Lifecycle envelope (yr 1–30, high scenario)	€4.14 bn cumulative	€3,765

Population basis: 1,100,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 30, debt service drops to zero and only the OPEX shortfall remains – ~€39 M / yr (low) €34 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€1.00 bn	3.8%	30 y, 7 y grace	€66 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€418 M	9.5%	30 y, 7 y grace	€45 M / yr
Government equity (no debt service)	15%	€251 M	—	—	—
Total	100%	€1.67 bn			€111 M / yr

During the 7-year grace period the operator pays interest only – multilateral €38 M / yr + bonds €40 M / yr = €78 M / yr total – plus the equity tranche amortised across construction (€36 M / yr × 7 yr). Principal repayment begins in year 8 on a 23-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€22 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€20 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€121 k
Traction energy (380.3 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (982 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€2.5 M
OPEX subtotal		€44 M / yr

Annual fleet utilisation: 121 revenue trainsets × 20.5 h/day × 365 d/yr × 35 km/h commercial × 75% revenue factor = 23.8 M train-km / yr (~196 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$165 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.25 (~\$0.28 USD)
Day pass (3 trips)	€0.65 (15 % bulk discount)
Monthly unlimited pass	€7.59 (~5 % of median monthly income)
Annual pass	€83.49 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	20.1 M	40.1 M
Farebox revenue	€5.1 M / yr	€10 M / yr
Farebox / OPEX recovery	11%	23%
Country policy-target recovery (diagnostic)	45%	45%
Operating shortfall (gov subsidy required)	€39 M / yr	€34 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€151 M / yr	€145 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
mwanza.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
mwanza-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
mwanza.corridor.geojson	Line polylines + stations (GeoJSON)
mwanza.stations.json	Machine-readable station list
mwanza.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug mwanza

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug mwanza \
  --sidecar .cache/osr-pipeline/rasters/mwanza.grid.json \
  --out-dir designs/.../Mwanza

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../mwanza.toml \
  --out designs/.../README.md
```

`scripts/regenerate-mwanza.sh` chains steps 3 + drift tests into a single command.

east-africa/Uganda/Gulu

designs/east-africa/Uganda/Gulu/README.md

Gulu — Urban Rail Network

Country: UG · Population: 350,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Gulu rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [gulu.corridor.geojson](#).

At a glance

Metric	Value
Lines	3
Unique stations	36
Interchange stations	3
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	50.4%
Route length (double track)	63.7 km
Revenue fleet	56 × 3-car trainsets
Spare + cold-reserve	7 × 3-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	20.5 km	13	20	E Mid W Outer
line-2	15.2 km	10	16	NE Mid SW Mid
line-3	28.0 km	13	27	NW Outer SE Outer
Total	63.7 km	36 unique	63	

Rolling stock

Property	Value
Consist	3-car, 57 m
Max speed	90 km/h
Onboard battery	360 kWh per trainset
Seats	60 longitudinal seats
Nominal capacity (AW2)	330 pax (seated + standing, light-metro-3car per RFC 0008 §1)
Crush capacity (AW3)	420 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 330 AW2 passengers (light-metro-3car)
- Peak frequency: 12 trains/hour/direction (5-min headway)

- Peak capacity per line per direction: $330 \times 12 = 3,960$ pphpd
- Network peak throughput (all lines, both directions): $3 \text{ lines} \times 2 \text{ directions} \times 3,960 = 23,760$ passengers/hour
- Daily theoretical capacity (peak $\times 10$): $\approx 237,600$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 17,640 - 26,460$ trips/day

Catchment

- City population: 350,000
- Anchor-weighted coverage: 50.4%
- Catchment population: $\approx 176,400$ (within ~ 800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	3	500 kW	3000 kWh
Major	10	400 kW	2500 kWh
Standard	16	300 kW	2000 kWh
Terminal	5	500 kW	3000 kWh
Total installed	35	17,800 kW	121,000 kWh

Aggregate station-rail charging power: 20,750 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 360 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	12.0 kWh/km	3 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	255 kWh	21.2 km average line length
Onboard battery coverage	1.4 $\times$ average line run	360 kWh usable pack
Average 60 s dwell charge	9.6 kWh/stop	576 kW average charger across stops
Stops to refill one trainset pack	37 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	89 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	121 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in design.toml — emitted by the osr-design Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. country-costs.toml applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (62.5 km @ €3.5 M/km)	€219 M
Elevated (1.2 km @ €18 M/km)	€22 M
Elevated-interchange premium (1 sites @ €20 M)	€20 M
Civil subtotal	€260 M

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~ 11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
halt	1	€0.4 M	€0.4 M
standard	16	€1.5 M	€24 M
major	10	€3.0 M	€30 M
terminal	5	€2.5 M	€12 M
depot-terminal	1	€3.0 M	€3.0 M
interchange-elevated	3	€4.5 M	€14 M
Stations subtotal			€83 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	5	€3.0 M	€15 M
Depots subtotal			€40 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
light-metro-3car (revenue + spare + cold reserve)	63	€3.0 M	€189 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	63.7 km × €0.015 M/km	€1.0 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€13 M
EPC integration + project management (7%)	on subtotal	€41 M

Total

Bucket	Value
Civil works	€260 M
Stations	€83 M
Depots	€40 M
Rolling stock	€189 M
Residual train-control wayside + charging microgrids	€14 M
EPC overhead (7%)	€41 M
CAPEX total	€627 M
Per-route-km	€9.9 M / km
Per-capita (city pop)	€1,793 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running scripts/regenerate-city.sh gulu.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–7 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 8 and runs for 23 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–7)	€50 M / yr	€142
Steady-state, low-ridership (year 8+)	€63 M / yr	€179
Steady-state, high-ridership (year 8+)	€61 M / yr	€175
Lifecycle envelope (yr 1–30, low scenario)	€1.79 bn cumulative	€5,113
Lifecycle envelope (yr 1–30, high scenario)	€1.76 bn cumulative	€5,020

Population basis: 350,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 30, debt service drops to zero and only the OPEX shortfall remains – ~€15 M / yr (low) €13 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€376 M	3.8%	30 y, 7 y grace	€25 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€157 M	14.0%	30 y, 7 y grace	€23 M / yr
Government equity (no debt service)	15%	€94 M	—	—	—
Total	100%	€627 M			€48 M / yr

During the 7-year grace period the operator pays interest only – multilateral €14 M / yr + bonds €22 M / yr = €36 M / yr total – plus the equity tranche amortised across construction (€13 M / yr × 7 yr). Principal repayment begins in year 8 on a 23-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€7.6 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€7.7 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€48 k
Traction energy (113.1 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (394 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€883 k
OPEX subtotal		€16 M / yr

Annual fleet utilisation: 56 revenue trainsets × 20.5 h/day × 365 d/yr × 30 km/h commercial × 75% revenue factor = 9.4 M train-km / yr (~168 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$145 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.22 (~\$0.24 USD)
Day pass (3 trips)	€0.57 (15 % bulk discount)
Monthly unlimited pass	€6.67 (~5 % of median monthly income)
Annual pass	€73.37 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	6.4 M	12.8 M

	Low scenario	High scenario
Farebox revenue	€1.4 M / yr	€2.8 M / yr
Farebox / OPEX recovery	9%	18%
Country policy-target recovery (diagnostic)	40%	40%
Operating shortfall (gov subsidy required)	€15 M / yr	€13 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€63 M / yr	€61 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
gulu.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
gulu-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
gulu.corridor.geojson	Line polylines + stations (GeoJSON)
gulu.stations.json	Machine-readable station list
gulu.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug gulu

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug gulu \
  --sidecar .cache/osr-pipeline/rasters/gulu.grid.json \
  --out-dir designs/.../Gulu

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../gulu.toml \
  --out designs/.../README.md
```

`scripts/regenerate-gulu.sh` chains steps 3 + drift tests into a single command.

east-africa/Uganda/Jinja

designs/east-africa/Uganda/Jinja/README.md

Jinja — Urban Rail Network

Country: UG · Population: 300,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Jinja rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [jinja.corridor.geojson](#).

At a glance

Metric	Value
Lines	3
Unique stations	29
Interchange stations	2
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	41.4%
Route length (double track)	45.5 km
Revenue fleet	55 × 2-car trainsets
Spare + cold-reserve	6 × 2-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	15.7 km	10	21	NW Outer E Outer
line-2	15.1 km	10	20	E Inner SW Outer
line-3	14.7 km	9	20	E Mid W Outer
Total	45.5 km	29 unique	61	

Rolling stock

Property	Value
Consist	2-car, 39 m
Max speed	70 km/h
Onboard battery	240 kWh per trainset
Seats	40 longitudinal seats
Nominal capacity (AW2)	210 pax (seated + standing, tram-2car per RFC 0008 §1)
Crush capacity (AW3)	260 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 210 AW2 passengers (tram-2car)
- Peak frequency: 12 trains/hour/direction (5-min headway)

- Peak capacity per line per direction: $210 \times 12 = 2,520$ pphpd
- Network peak throughput (all lines, both directions): $3 \text{ lines} \times 2 \text{ directions} \times 2,520 = 15,120$ passengers/hour
- Daily theoretical capacity (peak $\times 10$): $\approx 151,200$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 12,420 - 18,630$ trips/day

Catchment

- City population: 300,000
- Anchor-weighted coverage: 41.4%
- Catchment population: $\approx 124,200$ (within ~ 800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	2	500 kW	3000 kWh
Major	7	400 kW	2500 kWh
Standard	14	300 kW	2000 kWh
Terminal	5	500 kW	3000 kWh
Total installed	29	15,500 kW	106,500 kWh

Aggregate station-rail charging power: 17,500 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 240 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	8.0 kWh/km	2 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	121 kWh	15.2 km average line length
Onboard battery coverage	2.0 $\times$ average line run	240 kWh usable pack
Average 60 s dwell charge	10.1 kWh/stop	603 kW average charger across stops
Stops to refill one trainset pack	24 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	78 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	106 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in design.toml — emitted by the osr-design Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. country-costs.toml applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (43.8 km @ €3.5 M/km)	€153 M
Elevated (1.5 km @ €18 M/km)	€28 M
Elevated-interchange premium (1 sites @ €20 M)	€20 M
Civil subtotal	€201 M

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~ 11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
standard	14	€1.5 M	€21 M
major	7	€3.0 M	€21 M
terminal	5	€2.5 M	€12 M
depot-terminal	1	€3.0 M	€3.0 M
interchange-elevated	2	€4.5 M	€9.0 M
Stations subtotal			€66 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	5	€3.0 M	€15 M
Depots subtotal			€40 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
tram-2car (revenue + spare + cold reserve)	61	€2.0 M	€122 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	45.5 km × €0.015 M/km	€0.7 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€10 M
EPC integration + project management (7%)	on subtotal	€31 M

Total

Bucket	Value
Civil works	€201 M
Stations	€66 M
Depots	€40 M
Rolling stock	€122 M
Residual train-control wayside + charging microgrids	€11 M
EPC overhead (7%)	€31 M
CAPEX total	€472 M
Per-route-km	€10 M / km
Per-capita (city pop)	€1,572 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running scripts/regenerate-city.sh jinja.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–7 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 8 and runs for 23 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–7)	€37 M / yr	€125
Steady-state, low-ridership (year 8+)	€47 M / yr	€155
Steady-state, high-ridership (year 8+)	€45 M / yr	€151
Lifecycle envelope (yr 1–30, low scenario)	€1.33 bn cumulative	€4,438
Lifecycle envelope (yr 1–30, high scenario)	€1.30 bn cumulative	€4,345

Population basis: 300,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 30, debt service drops to zero and only the OPEX shortfall remains – ~€10 M / yr (low) €9.3 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€283 M	3.8%	30 y, 7 y grace	€19 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€118 M	14.0%	30 y, 7 y grace	€17 M / yr
Government equity (no debt service)	15%	€71 M	—	—	—
Total	100%	€472 M			€36 M / yr

During the 7-year grace period the operator pays interest only – multilateral €11 M / yr + bonds €17 M / yr = €27 M / yr total – plus the equity tranche amortised across construction (€10 M / yr × 7 yr). Principal repayment begins in year 8 on a 23-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€4.9 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€6.2 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€34 k
Traction energy (54.3 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (285 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€639 k
OPEX subtotal		€12 M / yr

Annual fleet utilisation: 55 revenue trainsets × 20.5 h/day × 365 d/yr × 22 km/h commercial × 75% revenue factor = 6.8 M train-km / yr (~123 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$145 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.22 (~\$0.24 USD)
Day pass (3 trips)	€0.57 (15 % bulk discount)
Monthly unlimited pass	€6.67 (~5 % of median monthly income)
Annual pass	€73.37 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	5.5 M	10.9 M

	Low scenario	High scenario
Farebox revenue	€1.2 M / yr	€2.4 M / yr
Farebox / OPEX recovery	10%	21%
Country policy-target recovery (diagnostic)	40%	40%
Operating shortfall (gov subsidy required)	€10 M / yr	€9.3 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€47 M / yr	€45 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
jinja.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
jinja-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
jinja.corridor.geojson	Line polylines + stations (GeoJSON)
jinja.stations.json	Machine-readable station list
jinja.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug jinja

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug jinja \
  --sidecar .cache/osr-pipeline/rasters/jinja.grid.json \
  --out-dir designs/.../Jinja

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../jinja.toml \
  --out designs/.../README.md
```

`scripts/regenerate-jinja.sh` chains steps 3 + drift tests into a single command.

east-africa/Uganda/Kampala

designs/east-africa/Uganda/Kampala/README.md

Kampala — Urban Rail Network

Country: UG · Population: 1,875,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Kampala rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [kampala.corridor.geojson](#).

At a glance

Metric	Value
Lines	4
Unique stations	98
Interchange stations	12
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	24.0%
Route length (double track)	201.1 km
Revenue fleet	144 × 4-car trainsets
Spare + cold-reserve	16 × 4-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	39.8 km	22	32	W Outer E Outer
line-2	36.9 km	21	30	S Outer N Outer
line-3	34.6 km	20	28	S Outer N Outer
line-4	89.9 km	36	70	NW Outer NW Outer
Total	201.1 km	98 unique	160	

Rolling stock

Property	Value
Consist	4-car, 75 m
Max speed	90 km/h
Onboard battery	480 kWh per trainset
Seats	80 longitudinal seats
Nominal capacity (AW2)	440 pax (seated + standing, met ro - 4car per RFC 0008 §1)
Crush capacity (AW3)	560 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 440 AW2 passengers (met ro - 4car)

- Peak frequency: 12 trains/hour/direction (5-min headway)
- Peak capacity per line per direction: $440 \times 12 = 5,280$ pphpd
- Network peak throughput (all lines, both directions): $4 \text{ lines} \times 2 \text{ directions} \times 5,280 = 42,240$ passengers/hour
- Daily theoretical capacity (peak $\times$ 10): $\approx 422,400$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 45,000 - 67,500$ trips/day

Catchment

- City population: 1,875,000
- Anchor-weighted coverage: 24.0%
- Catchment population: $\approx 450,000$ (within ~800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	12	500 kW	3000 kWh
Major	35	400 kW	2500 kWh
Standard	46	300 kW	2000 kWh
Terminal	5	500 kW	3000 kWh
Total installed	99	41,300 kW	270,500 kWh

Aggregate station-rail charging power: 52,500 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 480 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	16.0 kWh/km	4 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	804 kWh	50.3 km average line length
Onboard battery coverage	0.6 $\times$ average line run	480 kWh usable pack
Average 60 s dwell charge	8.9 kWh/stop	536 kW average charger across stops
Stops to refill one trainset pack	54 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	206 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	270 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the `[costs]` block in `design.toml` — emitted by the `osr-design` Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. `country-costs.toml` applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (192.0 km @ €3.5 M/km)	€672 M
Elevated (8.2 km @ €18 M/km)	€148 M
Elevated-interchange premium (7 sites @ €20 M)	€140 M
Civil subtotal	€960 M

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
standard	46	€1.5 M	€69 M
major	35	€3.0 M	€105 M
terminal	5	€2.5 M	€12 M
depot-terminal	1	€3.0 M	€3.0 M
interchange-elevated	12	€4.5 M	€54 M
Stations subtotal			€244 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	5	€3.0 M	€15 M
Depots subtotal			€40 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
metro-4car (revenue + spare + cold reserve)	160	€4.0 M	€640 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	201.1 km × €0.015 M/km	€3.0 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€35 M
EPC integration + project management (7%)	on subtotal	€135 M

Total

Bucket	Value
Civil works	€960 M
Stations	€244 M
Depots	€40 M
Rolling stock	€640 M
Residual train-control wayside + charging microgrids	€38 M
EPC overhead (7%)	€135 M
CAPEX total	€2.06 bn
Per-route-km	€10 M / km
Per-capita (city pop)	€1,097 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from <lib/templates/country-finance.toml>. Pure function of the [costs] block above + the country code – regenerate by re-running `scripts/regenerate-city.sh kampala`.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–7 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 8 and runs for 23 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–7)	€163 M / yr	€87
Steady-state, low-ridership (year 8+)	€203 M / yr	€108
Steady-state, high-ridership (year 8+)	€195 M / yr	€104
Lifecycle envelope (yr 1–30, low scenario)	€5.81 bn cumulative	€3,097
Lifecycle envelope (yr 1–30, high scenario)	€5.63 bn cumulative	€3,004

Population basis: 1,875,000 (catchment per <lib/city-batches/world-sample.toml>). After year 30, debt service drops to zero and only the OPEX shortfall remains – ~€46 M / yr (low) €38 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€1.23 bn	3.8%	30 y, 7 y grace	€81 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€514 M	14.0%	30 y, 7 y grace	€76 M / yr
Government equity (no debt service)	15%	€308 M	—	—	—
Total	100%	€2.06 bn			€157 M / yr

During the 7-year grace period the operator pays interest only – multilateral €47 M / yr + bonds €72 M / yr = €119 M / yr total – plus the equity tranche amortised across construction (€44 M / yr × 7 yr). Principal repayment begins in year 8 on a 23-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€26 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€25 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€150 k
Traction energy (452.5 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (1,219 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€2.7 M
OPEX subtotal		€53 M / yr

Annual fleet utilisation: 144 revenue trainsets × 20.5 h/day × 365 d/yr × 35 km/h commercial × 75% revenue factor = 28.3 M train-km / yr (~196 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$145 USD (per <lib/templates/country-finance.toml>). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.22 (~\$0.24 USD)
Day pass (3 trips)	€0.57 (15 % bulk discount)
Monthly unlimited pass	€6.67 (~5 % of median monthly income)
Annual pass	€73.37 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	34.2 M	68.4 M
Farebox revenue	€7.6 M / yr	€15 M / yr
Farebox / OPEX recovery	14%	29%
Country policy-target recovery (diagnostic)	40%	40%
Operating shortfall (gov subsidy required)	€46 M / yr	€38 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€203 M / yr	€195 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
kampala.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
kampala-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
kampala.corridor.geojson	Line polylines + stations (GeoJSON)
kampala.stations.json	Machine-readable station list
kampala.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug kampala

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug kampala \
  --sidecar .cache/osr-pipeline/rasters/kampala.grid.json \
  --out-dir designs/.../Kampala

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../kampala.toml \
  --out designs/.../README.md
```

`scripts/regenerate-kampala.sh` chains steps 3 + drift tests into a single command.

east-africa/Uganda/Mbarara

designs/east-africa/Uganda/Mbarara/README.md

Mbarara — Urban Rail Network

Country: UG · Population: 500,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Mbarara rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [mbarara.corridor.geojson](#).

At a glance

Metric	Value
Lines	3
Unique stations	36
Interchange stations	3
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	52.8%
Route length (double track)	58.7 km
Revenue fleet	52 × 3-car trainsets
Spare + cold-reserve	7 × 3-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	27.7 km	16	27	SW Outer NE Outer
line-2	16.6 km	11	17	E Outer SW Mid
line-3	14.4 km	9	15	NW Outer SE Inner
Total	58.7 km	36 unique	59	

Rolling stock

Property	Value
Consist	3-car, 57 m
Max speed	90 km/h
Onboard battery	360 kWh per trainset
Seats	60 longitudinal seats
Nominal capacity (AW2)	330 pax (seated + standing, light-metro-3car per RFC 0008 §1)
Crush capacity (AW3)	420 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 330 AW2 passengers (light-metro-3car)
- Peak frequency: 12 trains/hour/direction (5-min headway)

- Peak capacity per line per direction: $330 \times 12 = 3,960$ pphpd
- Network peak throughput (all lines, both directions): $3 \text{ lines} \times 2 \text{ directions} \times 3,960 = 23,760$ passengers/hour
- Daily theoretical capacity (peak $\times 10$): $\approx 237,600$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 26,400 - 39,600$ trips/day

Catchment

- City population: 500,000
- Anchor-weighted coverage: 52.8%
- Catchment population: $\approx 264,000$ (within ~ 800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	3	500 kW	3000 kWh
Major	13	400 kW	2500 kWh
Standard	14	300 kW	2000 kWh
Terminal	5	500 kW	3000 kWh
Total installed	36	18,400 kW	124,500 kWh

Aggregate station-rail charging power: 21,000 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 360 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	12.0 kWh/km	3 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	235 kWh	19.6 km average line length
Onboard battery coverage	1.5 $\times$ average line run	360 kWh usable pack
Average 60 s dwell charge	9.7 kWh/stop	583 kW average charger across stops
Stops to refill one trainset pack	37 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	92 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	124 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the `[costs]` block in `design.toml` — emitted by the `osr-design` Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. `country-costs.toml` applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (56.0 km @ €3.5 M/km)	€196 M
Elevated (2.4 km @ €18 M/km)	€43 M
Elevated-interchange premium (1 sites @ €20 M)	€20 M
Civil subtotal	€259 M

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~ 11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
standard	14	€1.5 M	€21 M
major	13	€3.0 M	€39 M
terminal	5	€2.5 M	€12 M
depot-terminal	1	€3.0 M	€3.0 M
interchange-elevated	3	€4.5 M	€14 M
Stations subtotal			€89 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	5	€3.0 M	€15 M
Depots subtotal			€40 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
light-metro-3car (revenue + spare + cold reserve)	59	€3.0 M	€177 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	58.7 km × €0.015 M/km	€0.9 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€13 M
EPC integration + project management (7%)	on subtotal	€41 M

Total

Bucket	Value
Civil works	€259 M
Stations	€89 M
Depots	€40 M
Rolling stock	€177 M
Residual train-control wayside + charging microgrids	€14 M
EPC overhead (7%)	€41 M
CAPEX total	€620 M
Per-route-km	€11 M / km
Per-capita (city pop)	€1,240 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running scripts/regenerate-city.sh mbarara.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–7 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 8 and runs for 23 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–7)	€49 M / yr	€98
Steady-state, low-ridership (year 8+)	€61 M / yr	€122
Steady-state, high-ridership (year 8+)	€59 M / yr	€118
Lifecycle envelope (yr 1–30, low scenario)	€1.75 bn cumulative	€3,495
Lifecycle envelope (yr 1–30, high scenario)	€1.70 bn cumulative	€3,401

Population basis: 500,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 30, debt service drops to zero and only the OPEX shortfall remains – ~€14 M / yr (low) €12 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€372 M	3.8%	30 y, 7 y grace	€25 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€155 M	14.0%	30 y, 7 y grace	€23 M / yr
Government equity (no debt service)	15%	€93 M	—	—	—
Total	100%	€620 M			€47 M / yr

During the 7-year grace period the operator pays interest only – multilateral €14 M / yr + bonds €22 M / yr = €36 M / yr total – plus the equity tranche amortised across construction (€13 M / yr × 7 yr). Principal repayment begins in year 8 on a 23-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€7.1 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€7.8 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€44 k
Traction energy (105.1 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (364 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€816 k
OPEX subtotal		€16 M / yr

Annual fleet utilisation: 52 revenue trainsets × 20.5 h/day × 365 d/yr × 30 km/h commercial × 75% revenue factor = 8.8 M train-km / yr (~168 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$145 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.22 (~\$0.24 USD)
Day pass (3 trips)	€0.57 (15 % bulk discount)
Monthly unlimited pass	€6.67 (~5 % of median monthly income)
Annual pass	€73.37 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	9.1 M	18.2 M

	Low scenario	High scenario
Farebox revenue	€2.0 M / yr	€4.1 M / yr
Farebox / OPEX recovery	13%	26%
Country policy-target recovery (diagnostic)	40%	40%
Operating shortfall (gov subsidy required)	€14 M / yr	€12 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€61 M / yr	€59 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
mbarara.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
mbarara-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
mbarara.corridor.geojson	Line polylines + stations (GeoJSON)
mbarara.stations.json	Machine-readable station list
mbarara.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug mbarara

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug mbarara \
  --sidecar .cache/osr-pipeline/rasters/mbarara.grid.json \
  --out-dir designs/.../Mbarara

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../mbarara.toml \
  --out designs/.../README.md
```

`scripts/regenerate-mbarara.sh` chains steps 3 + drift tests into a single command.

Lusaka — Urban Rail Network

Country: ZM · Population: 3,037,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Lusaka rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [lusaka.corridor.geojson](#).

At a glance

Metric	Value
Lines	6
Unique stations	122
Interchange stations	13
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	33.8%
Route length (double track)	235.6 km
Revenue fleet	172 × 6-car trainsets
Spare + cold-reserve	20 × 6-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	35.8 km	20	29	S Outer N Outer
line-2	39.5 km	20	32	NE Outer SW Outer
line-3	31.1 km	17	26	SW Outer N Outer
line-4	36.5 km	17	30	NE Outer S Mid
line-5	33.0 km	20	27	E Outer NW Mid
line-6	59.7 km	29	48	NE Mid NE Mid
Total	235.6 km	122 unique	192	

Rolling stock

Property	Value
Consist	6-car, 111 m
Max speed	100 km/h
Onboard battery	720 kWh per trainset
Seats	120 longitudinal seats
Nominal capacity (AW2)	660 pax (seated + standing, metro-6car per RFC 0008 §1)
Crush capacity (AW3)	840 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 660 AW2 passengers (metro-6car)
- Peak frequency: 12 trains/hour/direction (5-min headway)
- Peak capacity per line per direction: $660 \times 12 = 7,920$ pphpd
- Network peak throughput (all lines, both directions): $6 \text{ lines} \times 2 \text{ directions} \times 7,920 = 95,040$ passengers/hour
- Daily theoretical capacity (peak $\times$ 10): $\approx 950,400$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 102,650 - 153,975$ trips/day

Catchment

- City population: 3,037,000
- Anchor-weighted coverage: 33.8%
- Catchment population: $\approx 1,026,506$ (within ~800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	13	500 kW	3000 kWh
Major	57	400 kW	2500 kWh
Standard	42	300 kW	2000 kWh
Terminal	9	500 kW	3000 kWh
Total installed	122	51,400 kW	332,500 kWh

Aggregate station-rail charging power: 66,250 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 720 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	24.0 kWh/km	6 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	942 kWh	39.3 km average line length
Onboard battery coverage	0.8 $\times$ average line run	720 kWh usable pack
Average 60 s dwell charge	9.1 kWh/stop	543 kW average charger across stops
Stops to refill one trainset pack	80 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	257 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	332 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in design.toml — emitted by the osr-design Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. country-costs.toml applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (223.5 km @ €3.5 M/km)	€782 M
Elevated (10.4 km @ €18 M/km)	€187 M
Elevated-interchange premium (9 sites @ €20 M)	€180 M
Civil subtotal	€1.15 bn

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
halt	1	€0.4 M	€0.4 M
standard	42	€1.5 M	€63 M
major	57	€3.0 M	€171 M
terminal	9	€2.5 M	€22 M
depot-terminal	1	€3.0 M	€3.0 M
interchange-elevated	13	€4.5 M	€58 M
Stations subtotal			€318 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	9	€3.0 M	€27 M
Depots subtotal			€52 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
metro-6car (revenue + spare + cold reserve)	192	€6.0 M	€1.15 bn

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	235.6 km × €0.015 M/km	€3.5 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€46 M
EPC integration + project management (7%)	on subtotal	€190 M

Total

Bucket	Value
Civil works	€1.15 bn
Stations	€318 M
Depots	€52 M
Rolling stock	€1.15 bn
Residual train-control wayside + charging microgrids	€49 M
EPC overhead (7%)	€190 M
CAPEX total	€2.91 bn
Per-route-km	€12 M / km
Per-capita (city pop)	€959 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running scripts/regenerate-city.sh lusaka.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–7 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 8 and runs for 23 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–7)	€249 M / yr	€82
Steady-state, low-ridership (year 8+)	€305 M / yr	€100
Steady-state, high-ridership (year 8+)	€290 M / yr	€95
Lifecycle envelope (yr 1–30, low scenario)	€8.75 bn cumulative	€2,881
Lifecycle envelope (yr 1–30, high scenario)	€8.41 bn cumulative	€2,768

Population basis: 3,037,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 30, debt service drops to zero and only the OPEX shortfall remains – ~€66 M / yr (low) €51 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€1.75 bn	3.8%	30 y, 7 y grace	€115 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€728 M	16.5%	30 y, 7 y grace	€124 M / yr
Government equity (no debt service)	15%	€437 M	—	—	—
Total	100%	€2.91 bn			€239 M / yr

During the 7-year grace period the operator pays interest only – multilateral €66 M / yr + bonds €120 M / yr = €186 M / yr total – plus the equity tranche amortised across construction (€62 M / yr × 7 yr). Principal repayment begins in year 8 on a 23-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€46 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€30 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€175 k
Traction energy (810.8 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (1,426 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€3.9 M
OPEX subtotal		€81 M / yr

Annual fleet utilisation: 172 revenue trainsets × 20.5 h/day × 365 d/yr × 35 km/h commercial × 75% revenue factor = 33.8 M train-km / yr (~196 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$175 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.27 (~\$0.29 USD)
Day pass (3 trips)	€0.68 (15 % bulk discount)
Monthly unlimited pass	€8.05 (~5 % of median monthly income)
Annual pass	€88.55 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	55.4 M	110.9 M
Farebox revenue	€15 M / yr	€30 M / yr
Farebox / OPEX recovery	18%	37%
Country policy-target recovery (diagnostic)	40%	40%
Operating shortfall (gov subsidy required)	€66 M / yr	€51 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€305 M / yr	€290 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
lusaka.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
lusaka-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
lusaka.corridor.geojson	Line polylines + stations (GeoJSON)
lusaka.stations.json	Machine-readable station list
lusaka.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug lusaka

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug lusaka \
  --sidecar .cache/osr-pipeline/rasters/lusaka.grid.json \
  --out-dir designs/.../Lusaka

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../lusaka.toml \
  --out designs/.../README.md
```

`scripts/regenerate-lusaka.sh` chains steps 3 + drift tests into a single command.

Lyon — Urban Rail Network

Country: FR · Population: 1,436,354

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Lyon rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [lyon.corridor.geojson](#).

At a glance

Metric	Value
Lines	6
Unique stations	121
Interchange stations	18
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	44.8%
Route length (double track)	286.9 km
Revenue fleet	208 × 4-car trainsets
Spare + cold-reserve	24 × 4-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	56.6 km	23	46	SE Outer NW Outer
line-2	41.2 km	15	34	SW Outer E Mid
line-3	39.9 km	17	32	S Outer N Mid
line-4	36.3 km	17	30	S Mid N Outer
line-5	32.6 km	16	27	W Mid E Mid
line-6	80.4 km	34	63	NW Mid NW Mid
Total	286.9 km	121 unique	232	

Rolling stock

Property	Value
Consist	4-car, 75 m
Max speed	90 km/h
Onboard battery	480 kWh per trainset
Seats	80 longitudinal seats
Nominal capacity (AW2)	440 pax (seated + standing, metro-4car per RFC 0008 §1)
Crush capacity (AW3)	560 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 440 AW2 passengers (metro - 4 car)
- Peak frequency: 12 trains/hour/direction (5-min headway)
- Peak capacity per line per direction: $440 \times 12 = 5,280$ pphpd
- Network peak throughput (all lines, both directions): $6 \text{ lines} \times 2 \text{ directions} \times 5,280 = 63,360$ passengers/hour
- Daily theoretical capacity (peak $\times$ 10): $\approx 633,600$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 64,348 - 96,522$ trips/day

Catchment

- City population: 1,436,354
- Anchor-weighted coverage: 44.8%
- Catchment population: $\approx 643,486$ (within ~800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../.../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	18	500 kW	3000 kWh
Major	28	400 kW	2500 kWh
Standard	59	300 kW	2000 kWh
Terminal	9	500 kW	3000 kWh
Total installed	115	47,400 kW	309,000 kWh

Aggregate station-rail charging power: 64,250 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 480 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	16.0 kWh/km	4 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	765 kWh	47.8 km average line length
Onboard battery coverage	0.6 $\times$ average line run	480 kWh usable pack
Average 60 s dwell charge	8.8 kWh/stop	531 kW average charger across stops
Stops to refill one trainset pack	54 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	237 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	309 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the `[costs]` block in `design.toml` — emitted by the `osr-design` Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. `country-costs.toml` applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (271.2 km @ €3.5 M/km)	€949 M
Elevated (13.7 km @ €18 M/km)	€247 M
Elevated-interchange premium (9 sites @ €20 M)	€180 M
Civil subtotal	€1.38 bn

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
halt	7	€0.4 M	€2.8 M
standard	59	€1.5 M	€88 M
major	28	€3.0 M	€84 M
terminal	9	€2.5 M	€22 M
depot-terminal	1	€3.0 M	€3.0 M
interchange-elevated	18	€4.5 M	€81 M
Stations subtotal			€282 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	9	€3.0 M	€27 M
Depots subtotal			€52 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
metro-4car (revenue + spare + cold reserve)	232	€4.0 M	€928 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	286.9 km × €0.015 M/km	€4.3 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€42 M
EPC integration + project management (7%)	on subtotal	€188 M

Total

Bucket	Value
Civil works	€1.38 bn
Stations	€282 M
Depots	€52 M
Rolling stock	€928 M
Residual train-control wayside + charging microgrids	€46 M
EPC overhead (7%)	€188 M
CAPEX total	€2.87 bn
Per-route-km	€10 M / km
Per-capita (city pop)	€2,000 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from lib/templates/country-finance.toml. Pure function of the [costs] block above + the country code – regenerate by re-running `scripts/regenerate-city.sh` lyon.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–3 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 4 and runs for 27 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–3)	€217 M / yr	€151
Steady-state, low-ridership (year 4+)	€168 M / yr	€117
Steady-state, high-ridership (year 4+)	€133 M / yr	€93
Lifecycle envelope (yr 1–30, low scenario)	€5.18 bn cumulative	€3,608
Lifecycle envelope (yr 1–30, high scenario)	€4.25 bn cumulative	€2,957

Population basis: 1,436,354 (catchment per lib/city-batches/world-sample.toml). After year 30, debt service drops to zero and only the OPEX shortfall remains – ~€35 M / yr (low) €0 k / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€1.72 bn	3.0%	30 y, 3 y grace	€94 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€718 M	3.0%	30 y, 3 y grace	€39 M / yr
Government equity (no debt service)	15%	€431 M	—	—	—
Total	100%	€2.87 bn			€133 M / yr

During the 3-year grace period the operator pays interest only – multilateral €52 M / yr + bonds €22 M / yr = €73 M / yr total – plus the equity tranche amortised across construction (€144 M / yr × 3 yr). Principal repayment begins in year 4 on a 27-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€37 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€34 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€214 k
Traction energy (653.7 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (1,733 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€74 M
OPEX subtotal		€145 M / yr

Annual fleet utilisation: 208 revenue trainsets × 20.5 h/day × 365 d/yr × 35 km/h commercial × 75% revenue factor = 40.9 M train-km / yr (~196 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$2,750 USD (per lib/templates/country-finance.toml). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€4.22 (~\$4.58 USD)
Day pass (3 trips)	€10.75 (15 % bulk discount)
Monthly unlimited pass	€126.50 (~5 % of median monthly income)
Annual pass	€1391.50 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	26.2 M	52.4 M
Farebox revenue	€111 M / yr	€221 M / yr
Farebox / OPEX recovery	76%	152%
Country policy-target recovery (diagnostic)	65%	65%
Operating shortfall (gov subsidy required)	€35 M / yr	€0 k / yr
Operating surplus (operator retained capex sinking fund)	€0 k / yr	€76 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€168 M / yr	€133 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
lyon.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
lyon-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
lyon.corridor.geojson	Line polylines + stations (GeoJSON)
lyon.stations.json	Machine-readable station list
lyon.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug lyon

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug lyon \
  --sidecar .cache/osr-pipeline/rasters/lyon.grid.json \
  --out-dir designs/.../Lyon

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../lyon.toml \
  --out designs/.../README.md
```

`scripts/regenerate-lyon.sh` chains steps 3 + drift tests into a single command.

latin-america/Bolivia/La-Paz

designs/latin-america/Bolivia/La-Paz/README.md

La-Paz — Urban Rail Network

Country: BO · Population: 1,815,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

La-Paz rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [la-paz.corridor.geojson](#).

At a glance

Metric	Value
Lines	6
Unique stations	114
Interchange stations	24
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	56.7%
Route length (double track)	212.3 km
Revenue fleet	156 × 4-car trainsets
Spare + cold-reserve	18 × 4-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	39.4 km	22	32	E Outer W Outer
line-2	42.9 km	19	35	NE Outer SW Outer
line-3	23.3 km	15	20	S Mid NW Mid
line-4	25.5 km	13	21	N Mid SE Outer
line-5	22.7 km	13	19	N Mid SE Mid
line-6	58.5 km	33	47	W Mid W Mid
Total	212.3 km	114 unique	174	

Rolling stock

Property	Value
Consist	4-car, 75 m
Max speed	90 km/h
Onboard battery	480 kWh per trainset
Seats	80 longitudinal seats
Nominal capacity (AW2)	440 pax (seated + standing, metro-4car per RFC 0008 §1)
Crush capacity (AW3)	560 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 440 AW2 passengers (metro - 4 car)
- Peak frequency: 12 trains/hour/direction (5-min headway)
- Peak capacity per line per direction: $440 \times 12 = 5,280$ pphpd
- Network peak throughput (all lines, both directions): $6 \text{ lines} \times 2 \text{ directions} \times 5,280 = 63,360$ passengers/hour
- Daily theoretical capacity (peak $\times$ 10): $\approx 633,600$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 102,910 - 154,365$ trips/day

Catchment

- City population: 1,815,000
- Anchor-weighted coverage: 56.7%
- Catchment population: $\approx 1,029,104$ (within ~800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	24	500 kW	3000 kWh
Major	50	400 kW	2500 kWh
Standard	28	300 kW	2000 kWh
Terminal	9	500 kW	3000 kWh
Total installed	112	49,900 kW	320,000 kWh

Aggregate station-rail charging power: 61,750 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 480 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	16.0 kWh/km	4 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	566 kWh	35.4 km average line length
Onboard battery coverage	0.8 $\times$ average line run	480 kWh usable pack
Average 60 s dwell charge	9.0 kWh/stop	542 kW average charger across stops
Stops to refill one trainset pack	53 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	250 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	320 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in design.toml — emitted by the osr-design Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. country-costs.toml applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (185.2 km @ €3.5 M/km)	€648 M
Elevated (25.9 km @ €18 M/km)	€467 M
Elevated-interchange premium (13 sites @ €20 M)	€260 M
Civil subtotal	€1.38 bn

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
halt	3	€0.4 M	€1.2 M
standard	28	€1.5 M	€42 M
major	50	€3.0 M	€150 M
terminal	9	€2.5 M	€22 M
depot-terminal	1	€3.0 M	€3.0 M
interchange	2	€4.5 M	€9.0 M
interchange-elevated	22	€4.5 M	€99 M
Stations subtotal			€327 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	9	€3.0 M	€27 M
Depots subtotal			€52 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY – never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
metro-4car (revenue + spare + cold reserve)	174	€4.0 M	€696 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	212.3 km × €0.015 M/km	€3.2 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€46 M
EPC integration + project management (7%)	on subtotal	€175 M

Total

Bucket	Value
Civil works	€1.38 bn
Stations	€327 M
Depots	€52 M
Rolling stock	€696 M
Residual train-control wayside + charging microgrids	€49 M
EPC overhead (7%)	€175 M
CAPEX total	€2.67 bn
Per-route-km	€13 M / km
Per-capita (city pop)	€1,473 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](https://github.com/OpenSourceRail/lib/templates/country-finance.toml). Pure function of the [costs] block above + the country code – regenerate by re-running `scripts/regenerate-city.sh la-paz`.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–5 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 6 and runs for 20 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–5)	€223 M / yr	€123
Steady-state, low-ridership (year 6+)	€259 M / yr	€143
Steady-state, high-ridership (year 6+)	€244 M / yr	€134
Lifecycle envelope (yr 1–25, low scenario)	€6.29 bn cumulative	€3,464
Lifecycle envelope (yr 1–25, high scenario)	€5.99 bn cumulative	€3,302

Population basis: 1,815,000 (catchment per [lib/city-batches/world-sample.toml](https://github.com/OpenSourceRail/lib/city-batches/world-sample.toml)). After year 25, debt service drops to zero and only the OPEX shortfall remains – ~€54 M / yr (low) €39 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€1.60 bn	4.5%	25 y, 5 y grace	€123 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€669 M	10.5%	25 y, 5 y grace	€81 M / yr
Government equity (no debt service)	15%	€401 M	—	—	—
Total	100%	€2.67 bn			€205 M / yr

During the 5-year grace period the operator pays interest only – multilateral €72 M / yr + bonds €70 M / yr = €142 M / yr total – plus the equity tranche amortised across construction (€80 M / yr × 5 yr). Principal repayment begins in year 6 on a 20-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€28 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€35 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€158 k
Traction energy (490.3 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (1,286 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€5.8 M
OPEX subtotal		€69 M / yr

Annual fleet utilisation: 156 revenue trainsets × 20.5 h/day × 365 d/yr × 35 km/h commercial × 75% revenue factor = 30.6 M train-km / yr (~196 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$290 USD (per [lib/templates/country-finance.toml](https://github.com/OpenSourceRail/lib/templates/country-finance.toml)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.44 (~\$0.48 USD)
Day pass (3 trips)	€1.13 (15 % bulk discount)
Monthly unlimited pass	€13.34 (~5 % of median monthly income)
Annual pass	€146.74 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	33.1 M	66.2 M
Farebox revenue	€15 M / yr	€29 M / yr
Farebox / OPEX recovery	21%	43%
Country policy-target recovery (diagnostic)	50%	50%
Operating shortfall (gov subsidy required)	€54 M / yr	€39 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€259 M / yr	€244 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
la-paz.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
la-paz-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
la-paz.corridor.geojson	Line polylines + stations (GeoJSON)
la-paz.stations.json	Machine-readable station list
la-paz.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug la-paz

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug la-paz \
  --sidecar .cache/osr-pipeline/rasters/la-paz.grid.json \
  --out-dir designs/.../La-Paz

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../la-paz.toml \
  --out designs/.../README.md
```

`scripts/regenerate-la-paz.sh` chains steps 3 + drift tests into a single command.

latin-america/Ecuador/Cuenca

designs/latin-america/Ecuador/Cuenca/README.md

Cuenca — Urban Rail Network

Country: EC · Population: 817,100

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Cuenca rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [cuenca.corridor.geojson](#).

At a glance

Metric	Value
Lines	3
Unique stations	48
Interchange stations	5
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	57.4%
Route length (double track)	78.5 km
Revenue fleet	68 × 3-car trainsets
Spare + cold-reserve	9 × 3-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	26.5 km	17	26	E Mid W Outer
line-2	27.1 km	16	26	W Outer E Outer
line-3	24.9 km	15	25	NE Outer SW Mid
Total	78.5 km	48 unique	77	

Rolling stock

Property	Value
Consist	3-car, 57 m
Max speed	90 km/h
Onboard battery	360 kWh per trainset
Seats	60 longitudinal seats
Nominal capacity (AW2)	330 pax (seated + standing, light-metro-3car per RFC 0008 §1)
Crush capacity (AW3)	420 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 330 AW2 passengers (light-metro-3car)
- Peak frequency: 12 trains/hour/direction (5-min headway)

- Peak capacity per line per direction: $330 \times 12 = 3,960$ pphpd
- Network peak throughput (all lines, both directions): $3 \text{ lines} \times 2 \text{ directions} \times 3,960 = 23,760$ passengers/hour
- Daily theoretical capacity (peak $\times 10$): $\approx 237,600$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 46,901 - 70,352$ trips/day

Catchment

- City population: 817,100
- Anchor-weighted coverage: 57.4%
- Catchment population: $\approx 469,015$ (within ~ 800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	5	500 kW	3000 kWh
Major	22	400 kW	2500 kWh
Standard	13	300 kW	2000 kWh
Terminal	5	500 kW	3000 kWh
Total installed	46	22,700 kW	151,000 kWh

Aggregate station-rail charging power: 26,500 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 360 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	12.0 kWh/km	3 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	314 kWh	26.2 km average line length
Onboard battery coverage	1.1 $\times$ average line run	360 kWh usable pack
Average 60 s dwell charge	9.2 kWh/stop	552 kW average charger across stops
Stops to refill one trainset pack	39 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	114 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	151 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in design.toml — emitted by the osr-design Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. country-costs.toml applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (72.1 km @ €3.5 M/km)	€252 M
Elevated (4.9 km @ €18 M/km)	€87 M
Elevated-interchange premium (3 sites @ €20 M)	€60 M
Civil subtotal	€400 M

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~ 11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
halt	2	€0.4 M	€0.8 M
standard	13	€1.5 M	€20 M
major	22	€3.0 M	€66 M
terminal	5	€2.5 M	€12 M
depot-terminal	1	€3.0 M	€3.0 M
interchange	3	€4.5 M	€14 M
interchange-elevated	2	€4.5 M	€9.0 M
Stations subtotal			€124 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	5	€3.0 M	€15 M
Depots subtotal			€40 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
light-metro-3car (revenue + spare + cold reserve)	77	€3.0 M	€231 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	78.5 km × €0.015 M/km	€1.2 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€18 M
EPC integration + project management (7%)	on subtotal	€57 M

Total

Bucket	Value
Civil works	€400 M
Stations	€124 M
Depots	€40 M
Rolling stock	€231 M
Residual train-control wayside + charging microgrids	€19 M
EPC overhead (7%)	€57 M
CAPEX total	€871 M
Per-route-km	€11 M / km
Per-capita (city pop)	€1,066 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running scripts/regenerate-city.sh *cuenca*.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–5 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 6 and runs for 20 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–5)	€70 M / yr	€86
Steady-state, low-ridership (year 6+)	€79 M / yr	€97
Steady-state, high-ridership (year 6+)	€70 M / yr	€85
Lifecycle envelope (yr 1–25, low scenario)	€1.94 bn cumulative	€2,368
Lifecycle envelope (yr 1–25, high scenario)	€1.75 bn cumulative	€2,139

Population basis: 817,100 (catchment per [lib/city-batches/world-sample.toml](#)). After year 25, debt service drops to zero and only the OPEX shortfall remains – ~€14 M / yr (low) €4.9 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€523 M	4.5%	25 y, 5 y grace	€40 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€218 M	9.5%	25 y, 5 y grace	€25 M / yr
Government equity (no debt service)	15%	€131 M	—	—	—
Total	100%	€871 M			€65 M / yr

During the 5-year grace period the operator pays interest only – multilateral €24 M / yr + bonds €21 M / yr = €44 M / yr total – plus the equity tranche amortised across construction (€26 M / yr × 5 yr). Principal repayment begins in year 6 on a 20-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€9.2 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€11 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€58 k
Traction energy (137.4 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (483 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€3.1 M
OPEX subtotal		€24 M / yr

Annual fleet utilisation: 68 revenue trainsets × 20.5 h/day × 365 d/yr × 30 km/h commercial × 75% revenue factor = 11.4 M train-km / yr (~168 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$410 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.63 (~\$0.68 USD)
Day pass (3 trips)	€1.60 (15 % bulk discount)
Monthly unlimited pass	€18.86 (~5 % of median monthly income)
Annual pass	€207.46 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	14.9 M	29.8 M

	Low scenario	High scenario
Farebox revenue	€9.4 M / yr	€19 M / yr
Farebox / OPEX recovery	40%	79%
Country policy-target recovery (diagnostic)	55%	55%
Operating shortfall (gov subsidy required)	€14 M / yr	€4.9 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€79 M / yr	€70 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
cuenca.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
cuenca-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
cuenca.corridor.geojson	Line polylines + stations (GeoJSON)
cuenca.stations.json	Machine-readable station list
cuenca.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug cuenca

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug cuenca \
  --sidecar .cache/osr-pipeline/rasters/cuenca.grid.json \
  --out-dir designs/.../Cuenca

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../cuenca.toml \
  --out designs/.../README.md
```

`scripts/regenerate-cuenca.sh` chains steps 3 + drift tests into a single command.

latin-america/El Salvador/San-Salvador

designs/latin-america/El Salvador/San-Salvador/README.md

San-Salvador — Urban Rail Network

Country: SV · Population: 1,800,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

San-Salvador rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [san-salvador.corridor.geojson](#).

At a glance

Metric	Value
Lines	6
Unique stations	120
Interchange stations	20
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	50.5%
Route length (double track)	254.7 km
Revenue fleet	185 × 4-car trainsets
Spare + cold-reserve	22 × 4-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	37.3 km	21	30	E Outer W Outer
line-2	36.4 km	19	30	SW Outer N Outer
line-3	41.0 km	19	34	NW Outer SE Outer
line-4	24.3 km	15	20	E Mid SW Outer
line-5	42.4 km	15	35	NE Outer S Outer
line-6	73.3 km	32	58	NW Mid NW Mid
Total	254.7 km	120 unique	207	

Rolling stock

Property	Value
Consist	4-car, 75 m
Max speed	90 km/h
Onboard battery	480 kWh per trainset
Seats	80 longitudinal seats
Nominal capacity (AW2)	440 pax (seated + standing, metro-4car per RFC 0008 §1)
Crush capacity (AW3)	560 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 440 AW2 passengers (metro - 4 car)
- Peak frequency: 12 trains/hour/direction (5-min headway)
- Peak capacity per line per direction: $440 \times 12 = 5,280$ pphpd
- Network peak throughput (all lines, both directions): $6 \text{ lines} \times 2 \text{ directions} \times 5,280 = 63,360$ passengers/hour
- Daily theoretical capacity (peak $\times$ 10): $\approx 633,600$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 90,900 - 136,350$ trips/day

Catchment

- City population: 1,800,000
- Anchor-weighted coverage: 50.5%
- Catchment population: $\approx 909,000$ (within ~ 800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../.../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	20	500 kW	3000 kWh
Major	58	400 kW	2500 kWh
Standard	23	300 kW	2000 kWh
Terminal	9	500 kW	3000 kWh
Total installed	111	49,600 kW	318,000 kWh

Aggregate station-rail charging power: 63,000 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 480 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	16.0 kWh/km	4 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	679 kWh	42.4 km average line length
Onboard battery coverage	0.7 $\times$ average line run	480 kWh usable pack
Average 60 s dwell charge	8.8 kWh/stop	525 kW average charger across stops
Stops to refill one trainset pack	55 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	248 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	318 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in design.toml — emitted by the osr-design Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. country-costs.toml applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (240.8 km @ €3.5 M/km)	€843 M
Elevated (12.6 km @ €18 M/km)	€228 M
Elevated-interchange premium (10 sites @ €20 M)	€200 M
Civil subtotal	€1.27 bn

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
halt	10	€0.4 M	€4.0 M
standard	23	€1.5 M	€34 M
major	58	€3.0 M	€174 M
terminal	9	€2.5 M	€22 M
depot-terminal	1	€3.0 M	€3.0 M
interchange-elevated	20	€4.5 M	€90 M
Stations subtotal			€328 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	9	€3.0 M	€27 M
Depots subtotal			€52 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
metro-4car (revenue + spare + cold reserve)	207	€4.0 M	€828 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	254.7 km × €0.015 M/km	€3.8 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€47 M
EPC integration + project management (7%)	on subtotal	€177 M

Total

Bucket	Value
Civil works	€1.27 bn
Stations	€328 M
Depots	€52 M
Rolling stock	€828 M
Residual train-control wayside + charging microgrids	€50 M
EPC overhead (7%)	€177 M
CAPEX total	€2.71 bn
Per-route-km	€11 M / km
Per-capita (city pop)	€1,503 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running scripts/regenerate-city.sh san-salvador.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–5 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 6 and runs for 20 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–5)	€232 M / yr	€129
Steady-state, low-ridership (year 6+)	€271 M / yr	€150
Steady-state, high-ridership (year 6+)	€255 M / yr	€142
Lifecycle envelope (yr 1–25, low scenario)	€6.58 bn cumulative	€3,653
Lifecycle envelope (yr 1–25, high scenario)	€6.27 bn cumulative	€3,482

Population basis: 1,800,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 25, debt service drops to zero and only the OPEX shortfall remains – ~€58 M / yr (low) €43 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€1.62 bn	4.5%	25 y, 5 y grace	€125 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€676 M	11.5%	25 y, 5 y grace	€88 M / yr
Government equity (no debt service)	15%	€406 M	—	—	—
Total	100%	€2.71 bn			€213 M / yr

During the 5-year grace period the operator pays interest only – multilateral €73 M / yr + bonds €78 M / yr = €151 M / yr total – plus the equity tranche amortised across construction (€81 M / yr × 5 yr). Principal repayment begins in year 6 on a 20-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€33 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€33 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€190 k
Traction energy (581.4 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (1,540 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€7.3 M
OPEX subtotal		€74 M / yr

Annual fleet utilisation: 185 revenue trainsets × 20.5 h/day × 365 d/yr × 35 km/h commercial × 75% revenue factor = 36.3 M train-km / yr (~196 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$305 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.47 (~\$0.51 USD)
Day pass (3 trips)	€1.19 (15 % bulk discount)
Monthly unlimited pass	€14.03 (~5 % of median monthly income)
Annual pass	€154.33 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	32.9 M	65.7 M
Farebox revenue	€15 M / yr	€31 M / yr
Farebox / OPEX recovery	21%	42%
Country policy-target recovery (diagnostic)	55%	55%
Operating shortfall (gov subsidy required)	€58 M / yr	€43 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€271 M / yr	€255 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
san-salvador.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
san-salvador-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
san-salvador.corridor.geojson	Line polylines + stations (GeoJSON)
san-salvador.stations.json	Machine-readable station list
san-salvador.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug san-salvador

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug san-salvador \
  --sidecar .cache/osr-pipeline/rasters/san-salvador.grid.json \
  --out-dir designs/.../San-Salvador

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../san-salvador.toml \
  --out designs/.../README.md
```

`scripts/regenerate-san-salvador.sh` chains steps 3 + drift tests into a single command.

north-africa/Sudan/Khartoum

designs/north-africa/Sudan/Khartoum/README.md

Khartoum — Urban Rail Network

Country: SD · Population: 5,829,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Khartoum rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [khartoum.corridor.geojson](#).

At a glance

Metric	Value
Lines	5
Unique stations	151
Interchange stations	15
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	22.1%
Route length (double track)	362.7 km
Revenue fleet	258 × 6-car trainsets
Spare + cold-reserve	29 × 6-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	58.5 km	26	47	SE Outer NW Outer
line-2	55.2 km	26	45	N Outer S Outer
line-3	51.1 km	25	41	NW Outer SE Outer
line-4	48.6 km	21	39	SW Outer NE Outer
line-5	149.3 km	54	115	W Outer W Outer
Total	362.7 km	151 unique	287	

Rolling stock

Property	Value
Consist	6-car, 111 m
Max speed	100 km/h
Onboard battery	720 kWh per trainset
Seats	120 longitudinal seats
Nominal capacity (AW2)	660 pax (seated + standing, met ro-6car per RFC 0008 §1)
Crush capacity (AW3)	840 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 660 AW2 passengers (met ro-6car)

- Peak frequency: 12 trains/hour/direction (5-min headway)
- Peak capacity per line per direction: $660 \times 12 = 7,920$ pphpd
- Network peak throughput (all lines, both directions): $5 \text{ lines} \times 2 \text{ directions} \times 7,920 = 79,200$ passengers/hour
- Daily theoretical capacity (peak $\times 10$): $\approx 792,000$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 128,820 - 193,231$ trips/day

Catchment

- City population: 5,829,000
- Anchor-weighted coverage: 22.1%
- Catchment population: $\approx 1,288,209$ (within ~800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	15	500 kW	3000 kWh
Major	37	400 kW	2500 kWh
Standard	90	300 kW	2000 kWh
Terminal	7	500 kW	3000 kWh
Total installed	150	57,800 kW	378,500 kWh

Aggregate station-rail charging power: 79,500 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 720 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	24.0 kWh/km	6 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	1,741 kWh	72.5 km average line length
Onboard battery coverage	0.4x average line run	720 kWh usable pack
Average 60 s dwell charge	8.8 kWh/stop	526 kW average charger across stops
Stops to refill one trainset pack	82 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	289 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	378 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the `[costs]` block in `design.toml` — emitted by the `osr-design` Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3× higher because of the line items OSR has architected away. `country-costs.toml` applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (343.5 km @ €3.5 M/km)	€1.20 bn
Elevated (18.1 km @ €18 M/km)	€326 M
Elevated-interchange premium (7 sites @ €20 M)	€140 M
Civil subtotal	€1.67 bn

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
halt	2	€0.4 M	€0.8 M
standard	90	€1.5 M	€135 M
major	37	€3.0 M	€111 M
terminal	7	€2.5 M	€18 M
depot-terminal	1	€3.0 M	€3.0 M
interchange-elevated	15	€4.5 M	€68 M
Stations subtotal			€335 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	7	€3.0 M	€21 M
Depots subtotal			€46 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY – never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
metro-6car (revenue + spare + cold reserve)	287	€6.0 M	€1.72 bn

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	362.7 km × €0.015 M/km	€5.4 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€50 M
EPC integration + project management (7%)	on subtotal	€268 M

Total

Bucket	Value
Civil works	€1.67 bn
Stations	€335 M
Depots	€46 M
Rolling stock	€1.72 bn
Residual train-control wayside + charging microgrids	€56 M
EPC overhead (7%)	€268 M
CAPEX total	€4.09 bn
Per-route-km	€11 M / km
Per-capita (city pop)	€702 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running `scripts/regenerate-city.sh` `khartoum`.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–10 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 11 and runs for 30 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–10)	€289 M / yr	€50
Steady-state, low-ridership (year 11+)	€377 M / yr	€65
Steady-state, high-ridership (year 11+)	€359 M / yr	€62
Lifecycle envelope (yr 1–40, low scenario)	€14.20 bn cumulative	€2,436
Lifecycle envelope (yr 1–40, high scenario)	€13.66 bn cumulative	€2,344

Population basis: 5,829,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 40, debt service drops to zero and only the OPEX shortfall remains – ~€96 M / yr (low) €78 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€2.46 bn	3.0%	40 y, 10 y grace	€125 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€1.02 bn	15.0%	40 y, 10 y grace	€156 M / yr
Government equity (no debt service)	15%	€614 M	—	—	—
Total	100%	€4.09 bn			€281 M / yr

During the 10-year grace period the operator pays interest only — multilateral €74 M / yr + bonds €154 M / yr = €227 M / yr total — plus the equity tranche amortised across construction (€61 M / yr × 10 yr). Principal repayment begins in year 11 on a 30-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€69 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€41 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€271 k
Traction energy (1216.2 GWh / yr)	trackside PV + Na-ion (RFC 0002) — self-generated, €0 / yr	€0 k
Labour (2,188 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€3.7 M
OPEX subtotal		€114 M / yr

Annual fleet utilisation: 258 revenue trainsets × 20.5 h/day × 365 d/yr × 35 km/h commercial × 75% revenue factor = 50.7 M train-km / yr (~196 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$110 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass — a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.17 (~\$0.18 USD)
Day pass (3 trips)	€0.43 (15 % bulk discount)
Monthly unlimited pass	€5.06 (~5 % of median monthly income)
Annual pass	€55.66 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	106.4 M	212.8 M
Farebox revenue	€18 M / yr	€36 M / yr
Farebox / OPEX recovery	16%	32%
Country policy-target recovery (diagnostic)	30%	30%
Operating shortfall (gov subsidy required)	€96 M / yr	€78 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€377 M / yr	€359 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
khartoum.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
khartoum-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
khartoum.corridor.geojson	Line polylines + stations (GeoJSON)
khartoum.stations.json	Machine-readable station list
khartoum.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug khartoum

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug khartoum \
  --sidecar .cache/osr-pipeline/rasters/khartoum.grid.json \
  --out-dir designs/.../Khartoum

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../khartoum.toml \
  --out designs/.../README.md
```

`scripts/regenerate-khartoum.sh` chains steps 3 + drift tests into a single command.

north-africa/Tunisia/Tunis

designs/north-africa/Tunisia/Tunis/README.md

Tunis — Urban Rail Network

Country: TN · Population: 2,900,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Tunis rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [tunis.corridor.geojson](#).

At a glance

Metric	Value
Lines	5
Unique stations	117
Interchange stations	20
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	47.8%
Route length (double track)	240.3 km
Revenue fleet	174 × 4-car trainsets
Spare + cold-reserve	20 × 4-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	45.4 km	22	37	SE Outer W Mid
line-2	38.2 km	20	31	NE Outer W Outer
line-3	32.8 km	18	27	S Mid N Mid
line-4	40.6 km	19	34	NW Outer SE Outer
line-5	83.3 km	39	65	W Mid W Mid
Total	240.3 km	117 unique	194	

Rolling stock

Property	Value
Consist	4-car, 75 m
Max speed	90 km/h
Onboard battery	480 kWh per trainset
Seats	80 longitudinal seats
Nominal capacity (AW2)	440 pax (seated + standing, met ro - 4 car per RFC 0008 §1)
Crush capacity (AW3)	560 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 440 AW2 passengers (met ro - 4 car)

- Peak frequency: 12 trains/hour/direction (5-min headway)
- Peak capacity per line per direction: $440 \times 12 = 5,280$ pphpd
- Network peak throughput (all lines, both directions): $5 \text{ lines} \times 2 \text{ directions} \times 5,280 = 52,800$ passengers/hour
- Daily theoretical capacity (peak $\times$ 10): $\approx 528,000$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 138,620 - 207,930$ trips/day

Catchment

- City population: 2,900,000
- Anchor-weighted coverage: 47.8%
- Catchment population: $\approx 1,386,200$ (within ~ 800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	20	500 kW	3000 kWh
Major	43	400 kW	2500 kWh
Standard	46	300 kW	2000 kWh
Terminal	7	500 kW	3000 kWh
Total installed	117	49,500 kW	320,500 kWh

Aggregate station-rail charging power: 62,750 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 480 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	16.0 kWh/km	4 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	769 kWh	48.1 km average line length
Onboard battery coverage	0.6 $\times$ average line run	480 kWh usable pack
Average 60 s dwell charge	8.9 kWh/stop	536 kW average charger across stops
Stops to refill one trainset pack	54 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	248 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	320 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in design.toml — emitted by the osr-design Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. country-costs.toml applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (226.3 km @ €3.5 M/km)	€792 M
Elevated (13.1 km @ €18 M/km)	€236 M
Elevated-interchange premium (11 sites @ €20 M)	€220 M
Civil subtotal	€1.25 bn

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
halt	1	€0.4 M	€0.4 M
standard	46	€1.5 M	€69 M
major	43	€3.0 M	€129 M
terminal	7	€2.5 M	€18 M
depot-terminal	1	€3.0 M	€3.0 M
interchange-elevated	20	€4.5 M	€90 M
Stations subtotal			€309 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	7	€3.0 M	€21 M
Depots subtotal			€46 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
metro-4car (revenue + spare + cold reserve)	194	€4.0 M	€776 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	240.3 km × €0.015 M/km	€3.6 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€44 M
EPC integration + project management (7%)	on subtotal	€170 M

Total

Bucket	Value
Civil works	€1.25 bn
Stations	€309 M
Depots	€46 M
Rolling stock	€776 M
Residual train-control wayside + charging microgrids	€48 M
EPC overhead (7%)	€170 M
CAPEX total	€2.60 bn
Per-route-km	€11 M / km
Per-capita (city pop)	€896 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running `scripts/regenerate-city.sh tunis`.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–5 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 6 and runs for 20 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–5)	€206 M / yr	€71
Steady-state, low-ridership (year 6+)	€234 M / yr	€81
Steady-state, high-ridership (year 6+)	€205 M / yr	€71
Lifecycle envelope (yr 1–25, low scenario)	€5.71 bn cumulative	€1,968
Lifecycle envelope (yr 1–25, high scenario)	€5.14 bn cumulative	€1,772

Population basis: 2,900,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 25, debt service drops to zero and only the OPEX shortfall remains – ~€43 M / yr (low) €14 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€1.56 bn	4.5%	25 y, 5 y grace	€120 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€649 M	9.0%	25 y, 5 y grace	€71 M / yr
Government equity (no debt service)	15%	€390 M	—	—	—
Total	100%	€2.60 bn			€191 M / yr

During the 5-year grace period the operator pays interest only – multilateral €70 M / yr + bonds €58 M / yr = €129 M / yr total – plus the equity tranche amortised across construction (€78 M / yr × 5 yr). Principal repayment begins in year 6 on a 20-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€31 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€32 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€180 k
Traction energy (546.8 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (1,454 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€7.9 M
OPEX subtotal		€71 M / yr

Annual fleet utilisation: 174 revenue trainsets × 20.5 h/day × 365 d/yr × 35 km/h commercial × 75% revenue factor = 34.2 M train-km / yr (~196 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$350 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.54 (~\$0.58 USD)
Day pass (3 trips)	€1.37 (15 % bulk discount)
Monthly unlimited pass	€16.10 (~5 % of median monthly income)
Annual pass	€177.10 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	52.9 M	105.8 M
Farebox revenue	€28 M / yr	€57 M / yr
Farebox / OPEX recovery	40%	80%
Country policy-target recovery (diagnostic)	50%	50%
Operating shortfall (gov subsidy required)	€43 M / yr	€14 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€234 M / yr	€205 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
tunis.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
tunis-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
tunis.corridor.geojson	Line polylines + stations (GeoJSON)
tunis.stations.json	Machine-readable station list
tunis.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug tunis

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug tunis \
  --sidecar .cache/osr-pipeline/rasters/tunis.grid.json \
  --out-dir designs/.../Tunis

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../tunis.toml \
  --out designs/.../README.md
```

`scripts/regenerate-tunis.sh` chains steps 3 + drift tests into a single command.

south-asia/Afghanistan/Herat

designs/south-asia/Afghanistan/Herat/README.md

Herat — Urban Rail Network

Country: AF · Population: 800,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Herat rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [herat.corridor.geojson](#).

At a glance

Metric	Value
Lines	3
Unique stations	43
Interchange stations	6
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	35.6%
Route length (double track)	64.8 km
Revenue fleet	57 × 3-car trainsets
Spare + cold-reserve	7 × 3-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	22.2 km	15	21	E Outer W Inner
line-2	20.3 km	12	20	NW Outer E Outer
line-3	22.3 km	16	23	NW Outer SW Mid
Total	64.8 km	43 unique	64	

Rolling stock

Property	Value
Consist	3-car, 57 m
Max speed	90 km/h
Onboard battery	360 kWh per trainset
Seats	60 longitudinal seats
Nominal capacity (AW2)	330 pax (seated + standing, light-metro-3car per RFC 0008 §1)
Crush capacity (AW3)	420 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 330 AW2 passengers (light-metro-3car)
- Peak frequency: 12 trains/hour/direction (5-min headway)

- Peak capacity per line per direction: $330 \times 12 = 3,960$ pphpd
- Network peak throughput (all lines, both directions): $3 \text{ lines} \times 2 \text{ directions} \times 3,960 = 23,760$ passengers/hour
- Daily theoretical capacity (peak $\times 10$): $\approx 237,600$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 28,480 - 42,720$ trips/day

Catchment

- City population: 800,000
- Anchor-weighted coverage: 35.6%
- Catchment population: $\approx 284,800$ (within ~ 800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	6	500 kW	3000 kWh
Major	17	400 kW	2500 kWh
Standard	14	300 kW	2000 kWh
Terminal	5	500 kW	3000 kWh
Total installed	43	21,500 kW	143,500 kWh

Aggregate station-rail charging power: 24,500 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 360 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	12.0 kWh/km	3 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	259 kWh	21.6 km average line length
Onboard battery coverage	1.4 $\times$ average line run	360 kWh usable pack
Average 60 s dwell charge	9.5 kWh/stop	570 kW average charger across stops
Stops to refill one trainset pack	38 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	108 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	144 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in design.toml — emitted by the osr-design Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. country-costs.toml applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (58.7 km @ €3.5 M/km)	€205 M
Elevated (4.5 km @ €18 M/km)	€80 M
Elevated-interchange premium (4 sites @ €20 M)	€80 M
Civil subtotal	€366 M

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~ 11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
standard	14	€1.5 M	€21 M
major	17	€3.0 M	€51 M
terminal	5	€2.5 M	€12 M
depot-terminal	1	€3.0 M	€3.0 M
interchange	2	€4.5 M	€9.0 M
interchange-elevated	4	€4.5 M	€18 M
Stations subtotal			€114 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	5	€3.0 M	€15 M
Depots subtotal			€40 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
light-metro-3car (revenue + spare + cold reserve)	64	€3.0 M	€192 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	64.8 km × €0.015 M/km	€0.9 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€17 M
EPC integration + project management (7%)	on subtotal	€51 M

Total

Bucket	Value
Civil works	€366 M
Stations	€114 M
Depots	€40 M
Rolling stock	€192 M
Residual train-control wayside + charging microgrids	€18 M
EPC overhead (7%)	€51 M
CAPEX total	€781 M
Per-route-km	€12 M / km
Per-capita (city pop)	€976 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running scripts/regenerate-city.sh herat.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–10 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 11 and runs for 25 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–10)	€68 M / yr	€85
Steady-state, low-ridership (year 11+)	€84 M / yr	€105
Steady-state, high-ridership (year 11+)	€83 M / yr	€103
Lifecycle envelope (yr 1–35, low scenario)	€2.78 bn cumulative	€3,480
Lifecycle envelope (yr 1–35, high scenario)	€2.74 bn cumulative	€3,428

Population basis: 800,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 35, debt service drops to zero and only the OPEX shortfall remains – ~€17 M / yr (low) €15 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€468 M	4.5%	35 y, 10 y grace	€32 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€195 M	18.0%	35 y, 10 y grace	€36 M / yr
Government equity (no debt service)	15%	€117 M	—	—	—
Total	100%	€781 M			€67 M / yr

During the 10-year grace period the operator pays interest only – multilateral €21 M / yr + bonds €35 M / yr = €56 M / yr total – plus the equity tranche amortised across construction (€12 M / yr × 10 yr). Principal repayment begins in year 11 on a 25-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€7.7 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€10 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€47 k
Traction energy (115.2 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (401 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€465 k
OPEX subtotal		€19 M / yr

Annual fleet utilisation: 57 revenue trainsets × 20.5 h/day × 365 d/yr × 30 km/h commercial × 75% revenue factor = 9.6 M train-km / yr (~168 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$75 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.12 (~\$0.12 USD)
Day pass (3 trips)	€0.29 (15 % bulk discount)
Monthly unlimited pass	€3.45 (~5 % of median monthly income)
Annual pass	€37.95 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	14.6 M	29.2 M

	Low scenario	High scenario
Farebox revenue	€1.7 M / yr	€3.4 M / yr
Farebox / OPEX recovery	9%	18%
Country policy-target recovery (diagnostic)	30%	30%
Operating shortfall (gov subsidy required)	€17 M / yr	€15 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€84 M / yr	€83 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
herat.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
herat-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
herat.corridor.geojson	Line polylines + stations (GeoJSON)
herat.stations.json	Machine-readable station list
herat.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug herat

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug herat \
  --sidecar .cache/osr-pipeline/rasters/herat.grid.json \
  --out-dir designs/.../Herat

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../herat.toml \
  --out designs/.../README.md
```

`scripts/regenerate-herat.sh` chains steps 3 + drift tests into a single command.

south-asia/Afghanistan/Jalalabad-Af

designs/south-asia/Afghanistan/Jalalabad-Af/README.md

Jalalabad-Af — Urban Rail Network

Country: AF · Population: 350,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Jalalabad-Af rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [jalalabad-af.corridor.geojson](#).

At a glance

Metric	Value
Lines	3
Unique stations	32
Interchange stations	3
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	69.2%
Route length (double track)	50.8 km
Revenue fleet	46 × 3-car trainsets
Spare + cold-reserve	7 × 3-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	23.2 km	14	23	NW Outer SE Mid
line-2	14.0 km	9	15	SW Inner NE Outer
line-3	13.7 km	9	15	S Outer NE Inner
Total	50.8 km	32 unique	53	

Rolling stock

Property	Value
Consist	3-car, 57 m
Max speed	90 km/h
Onboard battery	360 kWh per trainset
Seats	60 longitudinal seats
Nominal capacity (AW2)	330 pax (seated + standing, light-metro-3car per RFC 0008 §1)
Crush capacity (AW3)	420 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 330 AW2 passengers (light-metro-3car)
- Peak frequency: 12 trains/hour/direction (5-min headway)

- Peak capacity per line per direction: $330 \times 12 = 3,960$ pphpd
- Network peak throughput (all lines, both directions): $3 \text{ lines} \times 2 \text{ directions} \times 3,960 = 23,760$ passengers/hour
- Daily theoretical capacity (peak $\times 10$): $\approx 237,600$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 24,219 - 36,329$ trips/day

Catchment

- City population: 350,000
- Anchor-weighted coverage: 69.2%
- Catchment population: $\approx 242,199$ (within ~ 800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	3	500 kW	3000 kWh
Major	12	400 kW	2500 kWh
Standard	10	300 kW	2000 kWh
Terminal	5	500 kW	3000 kWh
Total installed	31	16,800 kW	114,000 kWh

Aggregate station-rail charging power: 18,750 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 360 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	12.0 kWh/km	3 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	203 kWh	16.9 km average line length
Onboard battery coverage	1.8 $\times$ average line run	360 kWh usable pack
Average 60 s dwell charge	9.8 kWh/stop	586 kW average charger across stops
Stops to refill one trainset pack	37 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	84 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	114 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in design.toml — emitted by the osr-design Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. country-costs.toml applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (48.4 km @ €3.5 M/km)	€169 M
Elevated (1.4 km @ €18 M/km)	€26 M
Elevated-interchange premium (1 sites @ €20 M)	€20 M
Civil subtotal	€215 M

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~ 11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
halt	1	€0.4 M	€0.4 M
standard	10	€1.5 M	€15 M
major	12	€3.0 M	€36 M
terminal	5	€2.5 M	€12 M
depot-terminal	1	€3.0 M	€3.0 M
interchange	3	€4.5 M	€14 M
Stations subtotal			€80 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	5	€3.0 M	€15 M
Depots subtotal			€40 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
light-metro-3car (revenue + spare + cold reserve)	53	€3.0 M	€159 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	50.8 km × €0.015 M/km	€0.7 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€12 M
EPC integration + project management (7%)	on subtotal	€36 M

Total

Bucket	Value
Civil works	€215 M
Stations	€80 M
Depots	€40 M
Rolling stock	€159 M
Residual train-control wayside + charging microgrids	€13 M
EPC overhead (7%)	€36 M
CAPEX total	€543 M
Per-route-km	€11 M / km
Per-capita (city pop)	€1,551 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running scripts/regenerate-city.sh jalalabad-af.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–10 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 11 and runs for 25 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–10)	€47 M / yr	€135
Steady-state, low-ridership (year 11+)	€60 M / yr	€170
Steady-state, high-ridership (year 11+)	€59 M / yr	€168
Lifecycle envelope (yr 1–35, low scenario)	€1.96 bn cumulative	€5,602
Lifecycle envelope (yr 1–35, high scenario)	€1.94 bn cumulative	€5,550

Population basis: 350,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 35, debt service drops to zero and only the OPEX shortfall remains – ~€13 M / yr (low) €12 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€326 M	4.5%	35 y, 10 y grace	€22 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€136 M	18.0%	35 y, 10 y grace	€25 M / yr
Government equity (no debt service)	15%	€81 M	—	—	—
Total	100%	€543 M			€47 M / yr

During the 10-year grace period the operator pays interest only – multilateral €15 M / yr + bonds €24 M / yr = €39 M / yr total – plus the equity tranche amortised across construction (€8.1 M / yr × 10 yr). Principal repayment begins in year 11 on a 25-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€6.4 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€6.7 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€37 k
Traction energy (92.9 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (317 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€367 k
OPEX subtotal		€13 M / yr

Annual fleet utilisation: 46 revenue trainsets × 20.5 h/day × 365 d/yr × 30 km/h commercial × 75% revenue factor = 7.7 M train-km / yr (~168 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$75 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.12 (~\$0.12 USD)
Day pass (3 trips)	€0.29 (15 % bulk discount)
Monthly unlimited pass	€3.45 (~5 % of median monthly income)
Annual pass	€37.95 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	6.4 M	12.8 M

	Low scenario	High scenario
Farebox revenue	€735 k / yr	€1.5 M / yr
Farebox / OPEX recovery	5%	11%
Country policy-target recovery (diagnostic)	30%	30%
Operating shortfall (gov subsidy required)	€13 M / yr	€12 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€60 M / yr	€59 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
jalalabad-af.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
jalalabad-af-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
jalalabad-af.corridor.geojson	Line polylines + stations (GeoJSON)
jalalabad-af.stations.json	Machine-readable station list
jalalabad-af.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug jalalabad-af

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug jalalabad-af \
  --sidecar .cache/osr-pipeline/rasters/jalalabad-af.grid.json \
  --out-dir designs/.../Jalalabad-Af

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../jalalabad-af.toml \
  --out designs/.../README.md
```

`scripts/regenerate-jalalabad-af.sh` chains steps 3 + drift tests into a single command.

south-asia/Afghanistan/Kabul

designs/south-asia/Afghanistan/Kabul/README.md

Kabul — Urban Rail Network

Country: AF · Population: 4,601,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Kabul rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [kabul.corridor.geojson](#).

At a glance

Metric	Value
Lines	7
Unique stations	136
Interchange stations	24
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	52.7%
Route length (double track)	260.9 km
Revenue fleet	191 × 6-car trainsets
Spare + cold-reserve	24 × 6-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	42.4 km	24	35	W Mid E Outer
line-2	37.1 km	18	30	NW Mid E Outer
line-3	26.3 km	13	23	SW Outer SE Mid
line-4	28.0 km	16	24	SE Mid NW Outer
line-5	28.7 km	17	24	SW Mid NE Mid
line-6	24.9 km	14	21	E Outer W Inner
line-7	73.6 km	35	58	NW Mid NW Mid
Total	260.9 km	136 unique	215	

Rolling stock

Property	Value
Consist	6-car, 111 m
Max speed	100 km/h
Onboard battery	720 kWh per trainset
Seats	120 longitudinal seats
Nominal capacity (AW2)	660 pax (seated + standing, metro-6car per RFC 0008 §1)
Crush capacity (AW3)	840 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 660 AW2 passengers (metro - 6 car)
- Peak frequency: 12 trains/hour/direction (5-min headway)
- Peak capacity per line per direction: $660 \times 12 = 7,920$ pphpd
- Network peak throughput (all lines, both directions): $7 \text{ lines} \times 2 \text{ directions} \times 7,920 = 110,880$ passengers/hour
- Daily theoretical capacity (peak $\times$ 10): $\approx 1,108,800$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 242,472 - 363,709$ trips/day

Catchment

- City population: 4,601,000
- Anchor-weighted coverage: 52.7%
- Catchment population: $\approx 2,424,727$ (within ~ 800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	24	500 kW	3000 kWh
Major	60	400 kW	2500 kWh
Standard	38	300 kW	2000 kWh
Terminal	11	500 kW	3000 kWh
Total installed	134	57,900 kW	371,000 kWh

Aggregate station-rail charging power: 73,750 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 720 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	24.0 kWh/km	6 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	895 kWh	37.3 km average line length
Onboard battery coverage	0.8 $\times$ average line run	720 kWh usable pack
Average 60 s dwell charge	9.0 kWh/stop	542 kW average charger across stops
Stops to refill one trainset pack	80 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	290 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	371 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the `[costs]` block in `design.toml` — emitted by the `osr-design` Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. `country-costs.toml` applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (233.0 km @ €3.5 M/km)	€816 M
Elevated (23.3 km @ €18 M/km)	€420 M
Elevated-interchange premium (15 sites @ €20 M)	€300 M
Civil subtotal	€1.54 bn

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
halt	3	€0.4 M	€1.2 M
standard	38	€1.5 M	€57 M
major	60	€3.0 M	€180 M
terminal	11	€2.5 M	€28 M
depot-terminal	1	€3.0 M	€3.0 M
interchange	2	€4.5 M	€9.0 M
interchange-elevated	22	€4.5 M	€99 M
Stations subtotal			€377 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	11	€3.0 M	€33 M
Depots subtotal			€58 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
metro-6car (revenue + spare + cold reserve)	215	€6.0 M	€1.29 bn

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	260.9 km × €0.015 M/km	€3.8 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€53 M
EPC integration + project management (7%)	on subtotal	€232 M

Total

Bucket	Value
Civil works	€1.54 bn
Stations	€377 M
Depots	€58 M
Rolling stock	€1.29 bn
Residual train-control wayside + charging microgrids	€57 M
EPC overhead (7%)	€232 M
CAPEX total	€3.55 bn

Bucket	Value
Per-route-km	€14 M / km
Per-capita (city pop)	€772 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code — regenerate by re-running `scripts/regenerate-city.sh kabul`.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–10 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 11 and runs for 25 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–10)	€309 M / yr	€67
Steady-state, low-ridership (year 11+)	€389 M / yr	€85
Steady-state, high-ridership (year 11+)	€380 M / yr	€83
Lifecycle envelope (yr 1–35, low scenario)	€12.82 bn cumulative	€2,787
Lifecycle envelope (yr 1–35, high scenario)	€12.58 bn cumulative	€2,734

Population basis: 4,601,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 35, debt service drops to zero and only the OPEX shortfall remains — ~€83 M / yr (low) €74 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€2.13 bn	4.5%	35 y, 10 y grace	€144 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€888 M	18.0%	35 y, 10 y grace	€162 M / yr
Government equity (no debt service)	15%	€533 M	—	—	—
Total	100%	€3.55 bn			€306 M / yr

During the 10-year grace period the operator pays interest only — multilateral €96 M / yr + bonds €160 M / yr = €256 M / yr total — plus the equity tranche amortised across construction (€53 M / yr × 10 yr). Principal repayment begins in year 11 on a 25-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€52 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€39 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€192 k
Traction energy (900.4 GWh / yr)	trackside PV + Na-ion (RFC 0002) — self-generated, €0 / yr	€0 k
Labour (1,577 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€1.8 M
OPEX subtotal		€93 M / yr

Annual fleet utilisation: 191 revenue trainsets × 20.5 h/day × 365 d/yr × 35 km/h commercial × 75% revenue factor = 37.5 M train-km / yr (~196 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$75 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass — a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.12 (~\$0.12 USD)
Day pass (3 trips)	€0.29 (15 % bulk discount)
Monthly unlimited pass	€3.45 (~5 % of median monthly income)
Annual pass	€37.95 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	84.0 M	167.9 M
Farebox revenue	€9.7 M / yr	€19 M / yr
Farebox / OPEX recovery	10%	21%
Country policy-target recovery (diagnostic)	30%	30%
Operating shortfall (gov subsidy required)	€83 M / yr	€74 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€389 M / yr	€380 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
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kabul.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
kabul-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
kabul.corridor.geojson	Line polylines + stations (GeoJSON)
kabul.stations.json	Machine-readable station list
kabul.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug kabul

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug kabul \
  --sidecar .cache/osr-pipeline/rasters/kabul.grid.json \
  --out-dir designs/.../Kabul

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../kabul.toml \
  --out designs/.../README.md
```

`scripts/regenerate-kabul.sh` chains steps 3 + drift tests into a single command.

south-asia/Afghanistan/Kandahar

designs/south-asia/Afghanistan/Kandahar/README.md

Kandahar — Urban Rail Network

Country: AF · Population: 700,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Kandahar rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [kandahar.corridor.geojson](#).

At a glance

Metric	Value
Lines	3
Unique stations	43
Interchange stations	3
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	53.0%
Route length (double track)	64.0 km
Revenue fleet	56 × 3-car trainsets
Spare + cold-reserve	6 × 3-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	21.3 km	14	21	E Mid W Outer
line-2	20.5 km	15	20	W Outer SE Outer
line-3	22.2 km	14	21	NW Mid E Outer
Total	64.0 km	43 unique	62	

Rolling stock

Property	Value
Consist	3-car, 57 m
Max speed	90 km/h
Onboard battery	360 kWh per trainset
Seats	60 longitudinal seats
Nominal capacity (AW2)	330 pax (seated + standing, light-metro-3car per RFC 0008 §1)
Crush capacity (AW3)	420 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 330 AW2 passengers (light-metro-3car)
- Peak frequency: 12 trains/hour/direction (5-min headway)

- Peak capacity per line per direction: $330 \times 12 = 3,960$ pphpd
- Network peak throughput (all lines, both directions): $3 \text{ lines} \times 2 \text{ directions} \times 3,960 = 23,760$ passengers/hour
- Daily theoretical capacity (peak $\times 10$): $\approx 237,600$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 37,100 - 55,650$ trips/day

Catchment

- City population: 700,000
- Anchor-weighted coverage: 53.0%
- Catchment population: $\approx 371,000$ (within ~ 800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	3	500 kW	3000 kWh
Major	23	400 kW	2500 kWh
Standard	11	300 kW	2000 kWh
Terminal	5	500 kW	3000 kWh
Total installed	43	21,500 kW	143,500 kWh

Aggregate station-rail charging power: 24,500 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 360 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	12.0 kWh/km	3 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	256 kWh	21.3 km average line length
Onboard battery coverage	1.4 $\times$ average line run	360 kWh usable pack
Average 60 s dwell charge	9.5 kWh/stop	570 kW average charger across stops
Stops to refill one trainset pack	38 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	108 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	144 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the `[costs]` block in `design.toml` — emitted by the `osr-design` Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. `country-costs.toml` applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (62.9 km @ €3.5 M/km)	€220 M
Elevated (1.1 km @ €18 M/km)	€20 M
Elevated-interchange premium (1 sites @ €20 M)	€20 M
Civil subtotal	€260 M

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~ 11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
standard	11	€1.5 M	€16 M
major	23	€3.0 M	€69 M
terminal	5	€2.5 M	€12 M
depot-terminal	1	€3.0 M	€3.0 M
interchange-elevated	3	€4.5 M	€14 M
Stations subtotal			€114 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	5	€3.0 M	€15 M
Depots subtotal			€40 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
light-metro-3car (revenue + spare + cold reserve)	62	€3.0 M	€186 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	64.0 km × €0.015 M/km	€1.0 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€16 M
EPC integration + project management (7%)	on subtotal	€43 M

Total

Bucket	Value
Civil works	€260 M
Stations	€114 M
Depots	€40 M
Rolling stock	€186 M
Residual train-control wayside + charging microgrids	€17 M
EPC overhead (7%)	€43 M
CAPEX total	€661 M
Per-route-km	€10 M / km
Per-capita (city pop)	€944 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running scripts/regenerate-city.sh kandahar.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–10 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 11 and runs for 25 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–10)	€57 M / yr	€82
Steady-state, low-ridership (year 11+)	€72 M / yr	€102
Steady-state, high-ridership (year 11+)	€70 M / yr	€100
Lifecycle envelope (yr 1–35, low scenario)	€2.37 bn cumulative	€3,383
Lifecycle envelope (yr 1–35, high scenario)	€2.33 bn cumulative	€3,330

Population basis: 700,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 35, debt service drops to zero and only the OPEX shortfall remains – ~€15 M / yr (low) €13 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€396 M	4.5%	35 y, 10 y grace	€27 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€165 M	18.0%	35 y, 10 y grace	€30 M / yr
Government equity (no debt service)	15%	€99 M	—	—	—
Total	100%	€661 M			€57 M / yr

During the 10-year grace period the operator pays interest only – multilateral €18 M / yr + bonds €30 M / yr = €48 M / yr total – plus the equity tranche amortised across construction (€9.9 M / yr × 10 yr). Principal repayment begins in year 11 on a 25-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€7.4 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€8.3 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€48 k
Traction energy (113.1 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (396 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€459 k
OPEX subtotal		€16 M / yr

Annual fleet utilisation: 56 revenue trainsets × 20.5 h/day × 365 d/yr × 30 km/h commercial × 75% revenue factor = 9.4 M train-km / yr (~168 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$75 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.12 (~\$0.12 USD)
Day pass (3 trips)	€0.29 (15 % bulk discount)
Monthly unlimited pass	€3.45 (~5 % of median monthly income)
Annual pass	€37.95 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	12.8 M	25.6 M

	Low scenario	High scenario
Farebox revenue	€1.5 M / yr	€2.9 M / yr
Farebox / OPEX recovery	9%	18%
Country policy-target recovery (diagnostic)	30%	30%
Operating shortfall (gov subsidy required)	€15 M / yr	€13 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€72 M / yr	€70 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
kandahar.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
kandahar-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
kandahar.corridor.geojson	Line polylines + stations (GeoJSON)
kandahar.stations.json	Machine-readable station list
kandahar.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug kandahar

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug kandahar \
  --sidecar .cache/osr-pipeline/rasters/kandahar.grid.json \
  --out-dir designs/.../Kandahar

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../kandahar.toml \
  --out designs/.../README.md
```

`scripts/regenerate-kandahar.sh` chains steps 3 + drift tests into a single command.

south-asia/Afghanistan/Mazar-E-Sharif

designs/south-asia/Afghanistan/Mazar-E-Sharif/README.md

Mazar-E-Sharif — Urban Rail Network

Country: AF · Population: 600,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Mazar-E-Sharif rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [mazar-e-sharif.corridor.geojson](#).

At a glance

Metric	Value
Lines	3
Unique stations	37
Interchange stations	4
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	59.4%
Route length (double track)	63.0 km
Revenue fleet	55 × 3-car trainsets
Spare + cold-reserve	6 × 3-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	22.1 km	13	21	SW Outer E Outer
line-2	20.1 km	12	20	E Outer NW Mid
line-3	20.8 km	12	20	E Mid SW Outer
Total	63.0 km	37 unique	61	

Rolling stock

Property	Value
Consist	3-car, 57 m
Max speed	90 km/h
Onboard battery	360 kWh per trainset
Seats	60 longitudinal seats
Nominal capacity (AW2)	330 pax (seated + standing, light-metro-3car per RFC 0008 §1)
Crush capacity (AW3)	420 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 330 AW2 passengers (light-metro-3car)
- Peak frequency: 12 trains/hour/direction (5-min headway)

- Peak capacity per line per direction: $330 \times 12 = 3,960$ pphpd
- Network peak throughput (all lines, both directions): $3 \text{ lines} \times 2 \text{ directions} \times 3,960 = 23,760$ passengers/hour
- Daily theoretical capacity (peak $\times 10$): $\approx 237,600$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 35,640 - 53,460$ trips/day

Catchment

- City population: 600,000
- Anchor-weighted coverage: 59.4%
- Catchment population: $\approx 356,400$ (within ~ 800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	4	500 kW	3000 kWh
Major	12	400 kW	2500 kWh
Standard	13	300 kW	2000 kWh
Terminal	5	500 kW	3000 kWh
Total installed	35	18,200 kW	123,000 kWh

Aggregate station-rail charging power: 21,000 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 360 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	12.0 kWh/km	3 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	252 kWh	21.0 km average line length
Onboard battery coverage	1.4 $\times$ average line run	360 kWh usable pack
Average 60 s dwell charge	9.5 kWh/stop	568 kW average charger across stops
Stops to refill one trainset pack	38 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	91 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	123 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in design.toml — emitted by the osr-design Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. country-costs.toml applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (60.0 km @ €3.5 M/km)	€210 M
Elevated (2.6 km @ €18 M/km)	€47 M
Elevated-interchange premium (2 sites @ €20 M)	€40 M
Civil subtotal	€297 M

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~ 11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
halt	2	€0.4 M	€0.8 M
standard	13	€1.5 M	€20 M
major	12	€3.0 M	€36 M
terminal	5	€2.5 M	€12 M
depot-terminal	1	€3.0 M	€3.0 M
interchange	1	€4.5 M	€4.5 M
interchange-elevated	3	€4.5 M	€14 M
Stations subtotal			€90 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	5	€3.0 M	€15 M
Depots subtotal			€40 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
light-metro-3car (revenue + spare + cold reserve)	61	€3.0 M	€183 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	63.0 km × €0.015 M/km	€0.9 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€13 M
EPC integration + project management (7%)	on subtotal	€44 M

Total

Bucket	Value
Civil works	€297 M
Stations	€90 M
Depots	€40 M
Rolling stock	€183 M
Residual train-control wayside + charging microgrids	€14 M
EPC overhead (7%)	€44 M
CAPEX total	€668 M
Per-route-km	€11 M / km
Per-capita (city pop)	€1,114 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running scripts/regenerate-city.sh mazar-e-sharif.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–10 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 11 and runs for 25 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–10)	€58 M / yr	€97
Steady-state, low-ridership (year 11+)	€73 M / yr	€121
Steady-state, high-ridership (year 11+)	€71 M / yr	€119
Lifecycle envelope (yr 1–35, low scenario)	€2.40 bn cumulative	€3,998
Lifecycle envelope (yr 1–35, high scenario)	€2.37 bn cumulative	€3,945

Population basis: 600,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 35, debt service drops to zero and only the OPEX shortfall remains – ~€15 M / yr (low) €14 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€401 M	4.5%	35 y, 10 y grace	€27 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€167 M	18.0%	35 y, 10 y grace	€31 M / yr
Government equity (no debt service)	15%	€100 M	—	—	—
Total	100%	€668 M			€58 M / yr

During the 10-year grace period the operator pays interest only – multilateral €18 M / yr + bonds €30 M / yr = €48 M / yr total – plus the equity tranche amortised across construction (€10 M / yr × 10 yr). Principal repayment begins in year 11 on a 25-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€7.3 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€8.5 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€47 k
Traction energy (111.1 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (390 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€452 k
OPEX subtotal		€16 M / yr

Annual fleet utilisation: 55 revenue trainsets × 20.5 h/day × 365 d/yr × 30 km/h commercial × 75% revenue factor = 9.3 M train-km / yr (~168 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$75 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.12 (~\$0.12 USD)
Day pass (3 trips)	€0.29 (15 % bulk discount)
Monthly unlimited pass	€3.45 (~5 % of median monthly income)
Annual pass	€37.95 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	10.9 M	21.9 M

	Low scenario	High scenario
Farebox revenue	€1.3 M / yr	€2.5 M / yr
Farebox / OPEX recovery	8%	15%
Country policy-target recovery (diagnostic)	30%	30%
Operating shortfall (gov subsidy required)	€15 M / yr	€14 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€73 M / yr	€71 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
mazar-e-sharif.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
mazar-e-sharif-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
mazar-e-sharif.corridor.geojson	Line polylines + stations (GeoJSON)
mazar-e-sharif.stations.json	Machine-readable station list
mazar-e-sharif.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug mazar-e-sharif

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug mazar-e-sharif \
  --sidecar .cache/osr-pipeline/rasters/mazar-e-sharif.grid.json \
  --out-dir designs/.../Mazar-E-Sharif

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../mazar-e-sharif.toml \
  --out designs/.../README.md
```

`scripts/regenerate-mazar-e-sharif.sh` chains steps 3 + drift tests into a single command.

south-asia/Bangladesh/Barisal

designs/south-asia/Bangladesh/Barisal/README.md

Barisal — Urban Rail Network

Country: BD · Population: 550,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Barisal rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [barisal.corridor.geojson](#).

At a glance

Metric	Value
Lines	3
Unique stations	30
Interchange stations	3
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	49.7%
Route length (double track)	58.7 km
Revenue fleet	53 × 3-car trainsets
Spare + cold-reserve	7 × 3-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	15.4 km	10	16	S Outer E Inner
line-2	19.8 km	11	20	S Mid NW Outer
line-3	23.5 km	9	24	NW Outer SE Outer
Total	58.7 km	30 unique	60	

Rolling stock

Property	Value
Consist	3-car, 57 m
Max speed	90 km/h
Onboard battery	360 kWh per trainset
Seats	60 longitudinal seats
Nominal capacity (AW2)	330 pax (seated + standing, light-metro-3car per RFC 0008 §1)
Crush capacity (AW3)	420 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 330 AW2 passengers (light-metro-3car)
- Peak frequency: 12 trains/hour/direction (5-min headway)

- Peak capacity per line per direction: $330 \times 12 = 3,960$ pphpd
- Network peak throughput (all lines, both directions): $3 \text{ lines} \times 2 \text{ directions} \times 3,960 = 23,760$ passengers/hour
- Daily theoretical capacity (peak $\times 10$): $\approx 237,600$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 27,335 - 41,002$ trips/day

Catchment

- City population: 550,000
- Anchor-weighted coverage: 49.7%
- Catchment population: $\approx 273,350$ (within ~ 800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	3	500 kW	3000 kWh
Major	7	400 kW	2500 kWh
Standard	12	300 kW	2000 kWh
Terminal	5	500 kW	3000 kWh
Total installed	28	15,400 kW	105,500 kWh

Aggregate station-rail charging power: 17,500 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 360 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	12.0 kWh/km	3 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	235 kWh	19.6 km average line length
Onboard battery coverage	1.5 $\times$ average line run	360 kWh usable pack
Average 60 s dwell charge	9.7 kWh/stop	583 kW average charger across stops
Stops to refill one trainset pack	37 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	77 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	106 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in design.toml — emitted by the osr-design Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. country-costs.toml applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (49.2 km @ €3.5 M/km)	€172 M
Elevated (8.6 km @ €18 M/km)	€154 M
Elevated-interchange premium (1 sites @ €20 M)	€20 M
Civil subtotal	€346 M

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~ 11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
halt	2	€0.4 M	€0.8 M
standard	12	€1.5 M	€18 M
major	7	€3.0 M	€21 M
terminal	5	€2.5 M	€12 M
depot-terminal	1	€3.0 M	€3.0 M
interchange-elevated	3	€4.5 M	€14 M
Stations subtotal			€69 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	5	€3.0 M	€15 M
Depots subtotal			€40 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
light-metro-3car (revenue + spare + cold reserve)	60	€3.0 M	€180 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	58.7 km × €0.015 M/km	€0.9 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€11 M
EPC integration + project management (7%)	on subtotal	€45 M

Total

Bucket	Value
Civil works	€346 M
Stations	€69 M
Depots	€40 M
Rolling stock	€180 M
Residual train-control wayside + charging microgrids	€11 M
EPC overhead (7%)	€45 M
CAPEX total	€692 M
Per-route-km	€12 M / km
Per-capita (city pop)	€1,258 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running scripts/regenerate-city.sh barisal.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–7 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 8 and runs for 23 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–7)	€45 M / yr	€82
Steady-state, low-ridership (year 8+)	€59 M / yr	€108
Steady-state, high-ridership (year 8+)	€56 M / yr	€102
Lifecycle envelope (yr 1–30, low scenario)	€1.68 bn cumulative	€3,053
Lifecycle envelope (yr 1–30, high scenario)	€1.61 bn cumulative	€2,927

Population basis: 550,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 30, debt service drops to zero and only the OPEX shortfall remains – ~€14 M / yr (low) €11 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€415 M	3.8%	30 y, 7 y grace	€27 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€173 M	8.5%	30 y, 7 y grace	€17 M / yr
Government equity (no debt service)	15%	€104 M	—	—	—
Total	100%	€692 M			€45 M / yr

During the 7-year grace period the operator pays interest only – multilateral €16 M / yr + bonds €15 M / yr = €30 M / yr total – plus the equity tranche amortised across construction (€15 M / yr × 7 yr). Principal repayment begins in year 8 on a 23-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€7.2 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€9.1 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€43 k
Traction energy (107.1 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (364 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€1.1 M
OPEX subtotal		€17 M / yr

Annual fleet utilisation: 53 revenue trainsets × 20.5 h/day × 365 d/yr × 30 km/h commercial × 75% revenue factor = 8.9 M train-km / yr (~168 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$195 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.30 (~\$0.33 USD)
Day pass (3 trips)	€0.76 (15 % bulk discount)
Monthly unlimited pass	€8.97 (~5 % of median monthly income)
Annual pass	€98.67 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	10.0 M	20.1 M

	Low scenario	High scenario
Farebox revenue	€3.0 M / yr	€6.0 M / yr
Farebox / OPEX recovery	17%	34%
Country policy-target recovery (diagnostic)	50%	50%
Operating shortfall (gov subsidy required)	€14 M / yr	€11 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€59 M / yr	€56 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
barisal.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
barisal-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
barisal.corridor.geojson	Line polylines + stations (GeoJSON)
barisal.stations.json	Machine-readable station list
barisal.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug barisal

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug barisal \
  --sidecar .cache/osr-pipeline/rasters/barisal.grid.json \
  --out-dir designs/.../Barisal

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../barisal.toml \
  --out designs/.../README.md
```

`scripts/regenerate-barisal.sh` chains steps 3 + drift tests into a single command.

south-asia/Bangladesh/Chittagong

designs/south-asia/Bangladesh/Chittagong/README.md

Chittagong — Urban Rail Network

Country: BD · Population: 5,200,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Chittagong rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [chittagong.corridor.geojson](#).

At a glance

Metric	Value
Lines	8
Unique stations	160
Interchange stations	36
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	63.6%
Route length (double track)	374.2 km
Revenue fleet	271 × 6-car trainsets
Spare + cold-reserve	31 × 6-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	48.6 km	23	39	NW Outer S Mid
line-2	44.2 km	19	36	S Mid N Outer
line-3	42.8 km	19	35	NE Outer S Mid
line-4	39.9 km	18	32	SW Mid N Outer
line-5	42.1 km	17	35	SE Outer NW Mid
line-6	38.0 km	17	31	NE Outer SW Inner
line-7	39.6 km	14	32	W Inner NE Outer
line-8	78.9 km	34	62	NW Mid NW Mid
Total	374.2 km	160 unique	302	

Rolling stock

Property	Value
Consist	6-car, 111 m
Max speed	100 km/h
Onboard battery	720 kWh per trainset
Seats	120 longitudinal seats
Nominal capacity (AW2)	660 pax (seated + standing, metro-6car per RFC 0008 §1)
Crush capacity (AW3)	840 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 660 AW2 passengers (metro - 6 car)
- Peak frequency: 12 trains/hour/direction (5-min headway)
- Peak capacity per line per direction: $660 \times 12 = 7,920$ pphpd
- Network peak throughput (all lines, both directions): $8 \text{ lines} \times 2 \text{ directions} \times 7,920 = 126,720$ passengers/hour
- Daily theoretical capacity (peak $\times$ 10): $\approx 1,267,200$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 330,720 - 496,080$ trips/day

Catchment

- City population: 5,200,000
- Anchor-weighted coverage: 63.6%
- Catchment population: $\approx 3,307,200$ (within ~ 800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	36	500 kW	3000 kWh
Major	33	400 kW	2500 kWh
Standard	67	300 kW	2000 kWh
Terminal	13	500 kW	3000 kWh
Total installed	150	62,800 kW	403,500 kWh

Aggregate station-rail charging power: 84,750 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 720 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	24.0 kWh/km	6 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	1,123 kWh	46.8 km average line length
Onboard battery coverage	0.6 $\times$ average line run	720 kWh usable pack
Average 60 s dwell charge	8.8 kWh/stop	530 kW average charger across stops
Stops to refill one trainset pack	82 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	314 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	404 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in `design.toml` — emitted by the `osr-design` Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. `country-costs.toml` applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (330.5 km @ €3.5 M/km)	€1.16 bn
Elevated (37.6 km @ €18 M/km)	€676 M
Elevated-interchange premium (18 sites @ €20 M)	€360 M
Civil subtotal	€2.19 bn

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
halt	11	€0.4 M	€4.4 M
standard	67	€1.5 M	€100 M
major	33	€3.0 M	€99 M
terminal	13	€2.5 M	€32 M
depot-terminal	1	€3.0 M	€3.0 M
interchange	2	€4.5 M	€9.0 M
interchange-elevated	34	€4.5 M	€153 M
Stations subtotal			€401 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	13	€3.0 M	€39 M
Depots subtotal			€64 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
metro-6car (revenue + spare + cold reserve)	302	€6.0 M	€1.81 bn

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	374.2 km × €0.015 M/km	€5.5 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€59 M
EPC integration + project management (7%)	on subtotal	€317 M

Total

Bucket	Value
Civil works	€2.19 bn
Stations	€401 M
Depots	€64 M
Rolling stock	€1.81 bn
Residual train-control wayside + charging microgrids	€64 M
EPC overhead (7%)	€317 M
CAPEX total	€4.85 bn

Bucket	Value
Per-route-km	€13 M / km
Per-capita (city pop)	€933 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code — regenerate by re-running `scripts/regenerate-city.sh chittagong`.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–7 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 8 and runs for 23 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–7)	€318 M / yr	€61
Steady-state, low-ridership (year 8+)	€418 M / yr	€80
Steady-state, high-ridership (year 8+)	€390 M / yr	€75
Lifecycle envelope (yr 1–30, low scenario)	€11.84 bn cumulative	€2,277
Lifecycle envelope (yr 1–30, high scenario)	€11.19 bn cumulative	€2,152

Population basis: 5,200,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 30, debt service drops to zero and only the OPEX shortfall remains — ~€104 M / yr (low) €76 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€2.91 bn	3.8%	30 y, 7 y grace	€192 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€1.21 bn	8.5%	30 y, 7 y grace	€122 M / yr
Government equity (no debt service)	15%	€728 M	—	—	—
Total	100%	€4.85 bn			€314 M / yr

During the 7-year grace period the operator pays interest only — multilateral €111 M / yr + bonds €103 M / yr = €214 M / yr total — plus the equity tranche amortised across construction (€104 M / yr × 7 yr). Principal repayment begins in year 8 on a 23-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€72 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€53 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€276 k
Traction energy (1277.5 GWh / yr)	trackside PV + Na-ion (RFC 0002) — self-generated, €0 / yr	€0 k
Labour (2,257 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€6.8 M
OPEX subtotal		€133 M / yr

Annual fleet utilisation: 271 revenue trainsets × 20.5 h/day × 365 d/yr × 35 km/h commercial × 75% revenue factor = 53.2 M train-km / yr (~196 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$195 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass — a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.30 (~\$0.33 USD)
Day pass (3 trips)	€0.76 (15 % bulk discount)
Monthly unlimited pass	€8.97 (~5 % of median monthly income)
Annual pass	€98.67 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	94.9 M	189.8 M
Farebox revenue	€28 M / yr	€57 M / yr
Farebox / OPEX recovery	21%	43%
Country policy-target recovery (diagnostic)	50%	50%
Operating shortfall (gov subsidy required)	€104 M / yr	€76 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€418 M / yr	€390 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
chittagong.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
chittagong-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
chittagong.corridor.geojson	Line polylines + stations (GeoJSON)
chittagong.stations.json	Machine-readable station list
chittagong.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug chittagong

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug chittagong \
  --sidecar .cache/osr-pipeline/rasters/chittagong.grid.json \
  --out-dir designs/.../Chittagong

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../chittagong.toml \
  --out designs/.../README.md
```

`scripts/regenerate-chittagong.sh` chains steps 3 + drift tests into a single command.

south-asia/Bangladesh/Comilla

designs/south-asia/Bangladesh/Comilla/README.md

Comilla — Urban Rail Network

Country: BD · Population: 600,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Comilla rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [comilla.corridor.geojson](#).

At a glance

Metric	Value
Lines	3
Unique stations	38
Interchange stations	2
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	40.9%
Route length (double track)	65.7 km
Revenue fleet	58 × 3-car trainsets
Spare + cold-reserve	7 × 3-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	26.5 km	15	26	SE Outer W Outer
line-2	18.7 km	12	19	SE Mid NW Mid
line-3	20.5 km	11	20	SW Mid E Outer
Total	65.7 km	38 unique	65	

Rolling stock

Property	Value
Consist	3-car, 57 m
Max speed	90 km/h
Onboard battery	360 kWh per trainset
Seats	60 longitudinal seats
Nominal capacity (AW2)	330 pax (seated + standing, light-metro-3car per RFC 0008 §1)
Crush capacity (AW3)	420 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 330 AW2 passengers (light-metro-3car)
- Peak frequency: 12 trains/hour/direction (5-min headway)

- Peak capacity per line per direction: $330 \times 12 = 3,960$ pphpd
- Network peak throughput (all lines, both directions): $3 \text{ lines} \times 2 \text{ directions} \times 3,960 = 23,760$ passengers/hour
- Daily theoretical capacity (peak $\times 10$): $\approx 237,600$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 24,539 - 36,809$ trips/day

Catchment

- City population: 600,000
- Anchor-weighted coverage: 40.9%
- Catchment population: $\approx 245,399$ (within ~ 800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../.../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	2	500 kW	3000 kWh
Major	12	400 kW	2500 kWh
Standard	17	300 kW	2000 kWh
Terminal	5	500 kW	3000 kWh
Total installed	37	18,400 kW	125,000 kWh

Aggregate station-rail charging power: 21,750 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 360 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	12.0 kWh/km	3 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	263 kWh	21.9 km average line length
Onboard battery coverage	1.4 $\times$ average line run	360 kWh usable pack
Average 60 s dwell charge	9.5 kWh/stop	572 kW average charger across stops
Stops to refill one trainset pack	38 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	92 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	125 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the `[costs]` block in `design.toml` — emitted by the `osr-design` Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. `country-costs.toml` applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (61.8 km @ €3.5 M/km)	€216 M
Elevated (2.8 km @ €18 M/km)	€51 M
Elevated-interchange premium (2 sites @ €20 M)	€40 M
Civil subtotal	€307 M

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~ 11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
halt	1	€0.4 M	€0.4 M
standard	17	€1.5 M	€26 M
major	12	€3.0 M	€36 M
terminal	5	€2.5 M	€12 M
depot-terminal	1	€3.0 M	€3.0 M
interchange-elevated	2	€4.5 M	€9.0 M
Stations subtotal			€86 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	5	€3.0 M	€15 M
Depots subtotal			€40 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
light-metro-3car (revenue + spare + cold reserve)	65	€3.0 M	€195 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	65.7 km × €0.015 M/km	€1.0 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€13 M
EPC integration + project management (7%)	on subtotal	€45 M

Total

Bucket	Value
Civil works	€307 M
Stations	€86 M
Depots	€40 M
Rolling stock	€195 M
Residual train-control wayside + charging microgrids	€14 M
EPC overhead (7%)	€45 M
CAPEX total	€688 M
Per-route-km	€10 M / km
Per-capita (city pop)	€1,146 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running scripts/regenerate-city.sh comilla.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–7 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 8 and runs for 23 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–7)	€45 M / yr	€75
Steady-state, low-ridership (year 8+)	€59 M / yr	€98
Steady-state, high-ridership (year 8+)	€56 M / yr	€93
Lifecycle envelope (yr 1–30, low scenario)	€1.67 bn cumulative	€2,786
Lifecycle envelope (yr 1–30, high scenario)	€1.60 bn cumulative	€2,660

Population basis: 600,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 30, debt service drops to zero and only the OPEX shortfall remains – ~€14 M / yr (low) €11 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€413 M	3.8%	30 y, 7 y grace	€27 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€172 M	8.5%	30 y, 7 y grace	€17 M / yr
Government equity (no debt service)	15%	€103 M	—	—	—
Total	100%	€688 M			€44 M / yr

During the 7-year grace period the operator pays interest only – multilateral €16 M / yr + bonds €15 M / yr = €30 M / yr total – plus the equity tranche amortised across construction (€15 M / yr × 7 yr). Principal repayment begins in year 8 on a 23-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€7.8 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€8.7 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€49 k
Traction energy (117.2 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (406 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€1.2 M
OPEX subtotal		€18 M / yr

Annual fleet utilisation: 58 revenue trainsets × 20.5 h/day × 365 d/yr × 30 km/h commercial × 75% revenue factor = 9.8 M train-km / yr (~168 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$195 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.30 (~\$0.33 USD)
Day pass (3 trips)	€0.76 (15 % bulk discount)
Monthly unlimited pass	€8.97 (~5 % of median monthly income)
Annual pass	€98.67 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	10.9 M	21.9 M

	Low scenario	High scenario
Farebox revenue	€3.3 M / yr	€6.5 M / yr
Farebox / OPEX recovery	18%	37%
Country policy-target recovery (diagnostic)	50%	50%
Operating shortfall (gov subsidy required)	€14 M / yr	€11 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€59 M / yr	€56 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
comilla.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
comilla-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
comilla.corridor.geojson	Line polylines + stations (GeoJSON)
comilla.stations.json	Machine-readable station list
comilla.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug comilla

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug comilla \
  --sidecar .cache/osr-pipeline/rasters/comilla.grid.json \
  --out-dir designs/.../Comilla

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../comilla.toml \
  --out designs/.../README.md
```

`scripts/regenerate-comilla.sh` chains steps 3 + drift tests into a single command.

south-asia/Bangladesh/Gazipur

designs/south-asia/Bangladesh/Gazipur/README.md

Gazipur — Urban Rail Network

Country: BD · Population: 1,400,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Gazipur rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [gazipur.corridor.geojson](#).

At a glance

Metric	Value
Lines	6
Unique stations	127
Interchange stations	21
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	37.6%
Route length (double track)	308.4 km
Revenue fleet	221 × 4-car trainsets
Spare + cold-reserve	24 × 4-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	46.1 km	21	37	N Outer S Mid
line-2	39.9 km	19	32	SE Outer NW Mid
line-3	51.4 km	21	41	NE Outer SW Outer
line-4	37.8 km	15	31	E Outer W Mid
line-5	35.9 km	13	29	NW Outer SE Mid
line-6	97.2 km	38	75	NW Mid W Mid
Total	308.4 km	127 unique	245	

Rolling stock

Property	Value
Consist	4-car, 75 m
Max speed	90 km/h
Onboard battery	480 kWh per trainset
Seats	80 longitudinal seats
Nominal capacity (AW2)	440 pax (seated + standing, metro-4car per RFC 0008 §1)
Crush capacity (AW3)	560 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 440 AW2 passengers (metro - 4 car)
- Peak frequency: 12 trains/hour/direction (5-min headway)
- Peak capacity per line per direction: $440 \times 12 = 5,280$ pphpd
- Network peak throughput (all lines, both directions): $6 \text{ lines} \times 2 \text{ directions} \times 5,280 = 63,360$ passengers/hour
- Daily theoretical capacity (peak $\times$ 10): $\approx 633,600$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 52,640 - 78,960$ trips/day

Catchment

- City population: 1,400,000
- Anchor-weighted coverage: 37.6%
- Catchment population: $\approx 526,400$ (within ~800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	21	500 kW	3000 kWh
Major	27	400 kW	2500 kWh
Standard	62	300 kW	2000 kWh
Terminal	9	500 kW	3000 kWh
Total installed	120	49,400 kW	321,500 kWh

Aggregate station-rail charging power: 66,750 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 480 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	16.0 kWh/km	4 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	822 kWh	51.4 km average line length
Onboard battery coverage	0.6 $\times$ average line run	480 kWh usable pack
Average 60 s dwell charge	8.8 kWh/stop	526 kW average charger across stops
Stops to refill one trainset pack	55 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	247 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	322 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in design.toml — emitted by the osr-design Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. country-costs.toml applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (272.2 km @ €3.5 M/km)	€953 M
Elevated (32.4 km @ €18 M/km)	€583 M
Elevated-interchange premium (10 sites @ €20 M)	€200 M
Civil subtotal	€1.74 bn

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
halt	7	€0.4 M	€2.8 M
standard	62	€1.5 M	€93 M
major	27	€3.0 M	€81 M
terminal	9	€2.5 M	€22 M
depot-terminal	1	€3.0 M	€3.0 M
interchange-elevated	21	€4.5 M	€94 M
Stations subtotal			€297 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	9	€3.0 M	€27 M
Depots subtotal			€52 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY – never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
metro-4car (revenue + spare + cold reserve)	245	€4.0 M	€980 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	308.4 km × €0.015 M/km	€4.6 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€44 M
EPC integration + project management (7%)	on subtotal	€218 M

Total

Bucket	Value
Civil works	€1.74 bn
Stations	€297 M
Depots	€52 M
Rolling stock	€980 M
Residual train-control wayside + charging microgrids	€49 M
EPC overhead (7%)	€218 M
CAPEX total	€3.33 bn
Per-route-km	€11 M / km
Per-capita (city pop)	€2,380 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running `scripts/regenerate-city.sh` gazipur.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–7 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 8 and runs for 23 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–7)	€218 M / yr	€156
Steady-state, low-ridership (year 8+)	€295 M / yr	€210
Steady-state, high-ridership (year 8+)	€287 M / yr	€205
Lifecycle envelope (yr 1–30, low scenario)	€8.30 bn cumulative	€5,930
Lifecycle envelope (yr 1–30, high scenario)	€8.13 bn cumulative	€5,805

Population basis: 1,400,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 30, debt service drops to zero and only the OPEX shortfall remains – ~€79 M / yr (low) €71 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€2.00 bn	3.8%	30 y, 7 y grace	€132 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€833 M	8.5%	30 y, 7 y grace	€84 M / yr
Government equity (no debt service)	15%	€500 M	—	—	—
Total	100%	€3.33 bn			€215 M / yr

During the 7-year grace period the operator pays interest only – multilateral €76 M / yr + bonds €71 M / yr = €147 M / yr total – plus the equity tranche amortised across construction (€71 M / yr × 7 yr). Principal repayment begins in year 8 on a 23-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€39 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€42 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€228 k
Traction energy (694.5 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (1,862 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€5.6 M
OPEX subtotal		€87 M / yr

Annual fleet utilisation: 221 revenue trainsets × 20.5 h/day × 365 d/yr × 35 km/h commercial × 75% revenue factor = 43.4 M train-km / yr (~196 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$195 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.30 (~\$0.33 USD)
Day pass (3 trips)	€0.76 (15 % bulk discount)
Monthly unlimited pass	€8.97 (~5 % of median monthly income)
Annual pass	€98.67 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	25.6 M	51.1 M
Farebox revenue	€7.6 M / yr	€15 M / yr
Farebox / OPEX recovery	9%	18%
Country policy-target recovery (diagnostic)	50%	50%
Operating shortfall (gov subsidy required)	€79 M / yr	€71 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€295 M / yr	€287 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
gazipur.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
gazipur-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
gazipur.corridor.geojson	Line polylines + stations (GeoJSON)
gazipur.stations.json	Machine-readable station list
gazipur.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug gazipur

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug gazipur \
  --sidecar .cache/osr-pipeline/rasters/gazipur.grid.json \
  --out-dir designs/.../Gazipur

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../gazipur.toml \
  --out designs/.../README.md
```

`scripts/regenerate-gazipur.sh` chains steps 3 + drift tests into a single command.

south-asia/Bangladesh/Khulna

designs/south-asia/Bangladesh/Khulna/README.md

Khulna — Urban Rail Network

Country: BD · Population: 1,500,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Khulna rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [khulna.corridor.geojson](#).

At a glance

Metric	Value
Lines	6
Unique stations	81
Interchange stations	19
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	56.9%
Route length (double track)	182.2 km
Revenue fleet	135 × 4-car trainsets
Spare + cold-reserve	17 × 4-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	42.4 km	22	35	SE Outer NW Outer
line-2	22.7 km	10	19	E Outer SW Mid
line-3	18.5 km	9	16	N Inner S Mid
line-4	37.6 km	13	31	NE Outer W Outer
line-5	14.2 km	6	13	SE Inner W Mid
line-6	46.9 km	22	38	NW Mid NW Mid
Total	182.2 km	81 unique	152	

Rolling stock

Property	Value
Consist	4-car, 75 m
Max speed	90 km/h
Onboard battery	480 kWh per trainset
Seats	80 longitudinal seats
Nominal capacity (AW2)	440 pax (seated + standing, metro-4car per RFC 0008 §1)
Crush capacity (AW3)	560 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 440 AW2 passengers (metro - 4 car)
- Peak frequency: 12 trains/hour/direction (5-min headway)
- Peak capacity per line per direction: $440 \times 12 = 5,280$ pphpd
- Network peak throughput (all lines, both directions): $6 \text{ lines} \times 2 \text{ directions} \times 5,280 = 63,360$ passengers/hour
- Daily theoretical capacity (peak $\times$ 10): $\approx 633,600$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 85,349 - 128,024$ trips/day

Catchment

- City population: 1,500,000
- Anchor-weighted coverage: 56.9%
- Catchment population: $\approx 853,499$ (within ~800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	19	500 kW	3000 kWh
Major	19	400 kW	2500 kWh
Standard	22	300 kW	2000 kWh
Terminal	9	500 kW	3000 kWh
Total installed	70	33,200 kW	215,500 kWh

Aggregate station-rail charging power: 43,000 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 480 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	16.0 kWh/km	4 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	486 kWh	30.4 km average line length
Onboard battery coverage	1.0 $\times$ average line run	480 kWh usable pack
Average 60 s dwell charge	8.8 kWh/stop	531 kW average charger across stops
Stops to refill one trainset pack	54 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	166 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	216 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the `[costs]` block in `design.toml` — emitted by the `osr-design` Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. `country-costs.toml` applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (152.4 km @ €3.5 M/km)	€533 M
Elevated (26.9 km @ €18 M/km)	€484 M
Elevated-interchange premium (8 sites @ €20 M)	€160 M
Civil subtotal	€1.18 bn

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
halt	12	€0.4 M	€4.8 M
standard	22	€1.5 M	€33 M
major	19	€3.0 M	€57 M
terminal	9	€2.5 M	€22 M
depot-terminal	1	€3.0 M	€3.0 M
interchange	2	€4.5 M	€9.0 M
interchange-elevated	17	€4.5 M	€76 M
Stations subtotal			€206 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	9	€3.0 M	€27 M
Depots subtotal			€52 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
metro-4car (revenue + spare + cold reserve)	152	€4.0 M	€608 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	182.2 km × €0.015 M/km	€2.7 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€30 M
EPC integration + project management (7%)	on subtotal	€145 M

Total

Bucket	Value
Civil works	€1.18 bn
Stations	€206 M
Depots	€52 M
Rolling stock	€608 M
Residual train-control wayside + charging microgrids	€33 M
EPC overhead (7%)	€145 M
CAPEX total	€2.22 bn
Per-route-km	€12 M / km
Per-capita (city pop)	€1,481 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running scripts/regenerate-city.sh khulna.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–7 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 8 and runs for 23 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–7)	€145 M / yr	€97
Steady-state, low-ridership (year 8+)	€192 M / yr	€128
Steady-state, high-ridership (year 8+)	€184 M / yr	€123
Lifecycle envelope (yr 1–30, low scenario)	€5.43 bn cumulative	€3,623
Lifecycle envelope (yr 1–30, high scenario)	€5.25 bn cumulative	€3,498

Population basis: 1,500,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 30, debt service drops to zero and only the OPEX shortfall remains – ~€48 M / yr (low) €40 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€1.33 bn	3.8%	30 y, 7 y grace	€88 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€555 M	8.5%	30 y, 7 y grace	€56 M / yr
Government equity (no debt service)	15%	€333 M	—	—	—
Total	100%	€2.22 bn			€144 M / yr

During the 7-year grace period the operator pays interest only – multilateral €51 M / yr + bonds €47 M / yr = €98 M / yr total – plus the equity tranche amortised across construction (€48 M / yr × 7 yr). Principal repayment begins in year 8 on a 23-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€24 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€29 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€134 k
Traction energy (424.3 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (1,105 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€3.3 M
OPEX subtotal		€56 M / yr

Annual fleet utilisation: 135 revenue trainsets × 20.5 h/day × 365 d/yr × 35 km/h commercial × 75% revenue factor = 26.5 M train-km / yr (~196 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$195 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.30 (~\$0.33 USD)
Day pass (3 trips)	€0.76 (15 % bulk discount)
Monthly unlimited pass	€8.97 (~5 % of median monthly income)
Annual pass	€98.67 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	27.4 M	54.8 M
Farebox revenue	€8.2 M / yr	€16 M / yr
Farebox / OPEX recovery	14%	29%
Country policy-target recovery (diagnostic)	50%	50%
Operating shortfall (gov subsidy required)	€48 M / yr	€40 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€192 M / yr	€184 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
khulna.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
khulna-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
khulna.corridor.geojson	Line polylines + stations (GeoJSON)
khulna.stations.json	Machine-readable station list
khulna.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug khulna

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug khulna \
  --sidecar .cache/osr-pipeline/rasters/khulna.grid.json \
  --out-dir designs/.../Khulna

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../khulna.toml \
  --out designs/.../README.md
```

`scripts/regenerate-khulna.sh` chains steps 3 + drift tests into a single command.

south-asia/Bangladesh/Mymensingh

designs/south-asia/Bangladesh/Mymensingh/README.md

Mymensingh — Urban Rail Network

Country: BD · Population: 700,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Mymensingh rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [mymensingh.corridor.geojson](#).

At a glance

Metric	Value
Lines	3
Unique stations	37
Interchange stations	2
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	41.1%
Route length (double track)	67.3 km
Revenue fleet	59 × 3-car trainsets
Spare + cold-reserve	8 × 3-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	25.1 km	15	25	E Outer W Outer
line-2	18.3 km	11	18	N Mid S Outer
line-3	23.9 km	11	24	NE Outer S Outer
Total	67.3 km	37 unique	67	

Rolling stock

Property	Value
Consist	3-car, 57 m
Max speed	90 km/h
Onboard battery	360 kWh per trainset
Seats	60 longitudinal seats
Nominal capacity (AW2)	330 pax (seated + standing, light-metro-3car per RFC 0008 §1)
Crush capacity (AW3)	420 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 330 AW2 passengers (light-metro-3car)
- Peak frequency: 12 trains/hour/direction (5-min headway)

- Peak capacity per line per direction: $330 \times 12 = 3,960$ pphpd
- Network peak throughput (all lines, both directions): $3 \text{ lines} \times 2 \text{ directions} \times 3,960 = 23,760$ passengers/hour
- Daily theoretical capacity (peak $\times 10$): $\approx 237,600$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 28,770 - 43,155$ trips/day

Catchment

- City population: 700,000
- Anchor-weighted coverage: 41.1%
- Catchment population: $\approx 287,700$ (within ~ 800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../.../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	2	500 kW	3000 kWh
Major	5	400 kW	2500 kWh
Standard	22	300 kW	2000 kWh
Terminal	5	500 kW	3000 kWh
Total installed	35	17,100 kW	117,500 kWh

Aggregate station-rail charging power: 21,000 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 360 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	12.0 kWh/km	3 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	269 kWh	22.4 km average line length
Onboard battery coverage	1.3 $\times$ average line run	360 kWh usable pack
Average 60 s dwell charge	9.5 kWh/stop	568 kW average charger across stops
Stops to refill one trainset pack	38 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	86 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	118 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in design.toml — emitted by the osr-design Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. country-costs.toml applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (35.3 km @ €3.5 M/km)	€123 M
Elevated (31.9 km @ €18 M/km)	€574 M
Civil subtotal	€698 M

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~ 11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
halt	2	€0.4 M	€0.8 M
standard	22	€1.5 M	€33 M
major	5	€3.0 M	€15 M
terminal	5	€2.5 M	€12 M
depot-terminal	1	€3.0 M	€3.0 M
interchange	2	€4.5 M	€9.0 M
Stations subtotal			€73 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	5	€3.0 M	€15 M
Depots subtotal			€40 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
light-metro-3car (revenue + spare + cold reserve)	67	€3.0 M	€201 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	67.3 km × €0.015 M/km	€1.0 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€12 M
EPC integration + project management (7%)	on subtotal	€72 M

Total

Bucket	Value
Civil works	€698 M
Stations	€73 M
Depots	€40 M
Rolling stock	€201 M
Residual train-control wayside + charging microgrids	€13 M
EPC overhead (7%)	€72 M
CAPEX total	€1.10 bn
Per-route-km	€16 M / km
Per-capita (city pop)	€1,566 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running scripts/regenerate-city.sh mymensingh.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–7 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 8 and runs for 23 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–7)	€72 M / yr	€103
Steady-state, low-ridership (year 8+)	€93 M / yr	€132
Steady-state, high-ridership (year 8+)	€89 M / yr	€127
Lifecycle envelope (yr 1–30, low scenario)	€2.63 bn cumulative	€3,763
Lifecycle envelope (yr 1–30, high scenario)	€2.55 bn cumulative	€3,637

Population basis: 700,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 30, debt service drops to zero and only the OPEX shortfall remains – ~€22 M / yr (low) €18 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€658 M	3.8%	30 y, 7 y grace	€43 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€274 M	8.5%	30 y, 7 y grace	€28 M / yr
Government equity (no debt service)	15%	€164 M	—	—	—
Total	100%	€1.10 bn			€71 M / yr

During the 7-year grace period the operator pays interest only – multilateral €25 M / yr + bonds €23 M / yr = €48 M / yr total – plus the equity tranche amortised across construction (€23 M / yr × 7 yr). Principal repayment begins in year 8 on a 23-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€8.0 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€16 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€50 k
Traction energy (119.2 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (416 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€1.3 M
OPEX subtotal		€26 M / yr

Annual fleet utilisation: 59 revenue trainsets × 20.5 h/day × 365 d/yr × 30 km/h commercial × 75% revenue factor = 9.9 M train-km / yr (~168 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$195 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.30 (~\$0.33 USD)
Day pass (3 trips)	€0.76 (15 % bulk discount)
Monthly unlimited pass	€8.97 (~5 % of median monthly income)
Annual pass	€98.67 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	12.8 M	25.6 M

	Low scenario	High scenario
Farebox revenue	€3.8 M / yr	€7.6 M / yr
Farebox / OPEX recovery	15%	30%
Country policy-target recovery (diagnostic)	50%	50%
Operating shortfall (gov subsidy required)	€22 M / yr	€18 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€93 M / yr	€89 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
mymensingh.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
mymensingh-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
mymensingh.corridor.geojson	Line polylines + stations (GeoJSON)
mymensingh.stations.json	Machine-readable station list
mymensingh.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug mymensingh

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug mymensingh \
  --sidecar .cache/osr-pipeline/rasters/mymensingh.grid.json \
  --out-dir designs/.../Mymensingh

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../mymensingh.toml \
  --out designs/.../README.md
```

`scripts/regenerate-mymensingh.sh` chains steps 3 + drift tests into a single command.

south-asia/Bangladesh/Narayanganj

designs/south-asia/Bangladesh/Narayanganj/README.md

Narayanganj — Urban Rail Network

Country: BD · Population: 950,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Narayanganj rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [narayanganj.corridor.geojson](#).

At a glance

Metric	Value
Lines	3
Unique stations	50
Interchange stations	0
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	26.7%
Route length (double track)	81.9 km
Revenue fleet	71 × 3-car trainsets
Spare + cold-reserve	9 × 3-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	28.0 km	18	27	NW Outer E Outer
line-2	29.1 km	19	28	S Outer NW Outer
line-3	24.8 km	13	25	NE Outer S Outer
Total	81.9 km	50 unique	80	

Rolling stock

Property	Value
Consist	3-car, 57 m
Max speed	90 km/h
Onboard battery	360 kWh per trainset
Seats	60 longitudinal seats
Nominal capacity (AW2)	330 pax (seated + standing, light-metro-3car per RFC 0008 §1)
Crush capacity (AW3)	420 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 330 AW2 passengers (light-metro-3car)
- Peak frequency: 12 trains/hour/direction (5-min headway)

- Peak capacity per line per direction: $330 \times 12 = 3,960$ pphpd
- Network peak throughput (all lines, both directions): $3 \text{ lines} \times 2 \text{ directions} \times 3,960 = 23,760$ passengers/hour
- Daily theoretical capacity (peak $\times 10$): $\approx 237,600$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 25,365 - 38,047$ trips/day

Catchment

- City population: 950,000
- Anchor-weighted coverage: 26.7%
- Catchment population: $\approx 253,650$ (within ~ 800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../.../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Major	16	400 kW	2500 kWh
Standard	28	300 kW	2000 kWh
Terminal	5	500 kW	3000 kWh
Total installed	50	22,300 kW	151,000 kWh

Aggregate station-rail charging power: 28,000 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 360 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	12.0 kWh/km	3 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	328 kWh	27.3 km average line length
Onboard battery coverage	1.1 $\times$ average line run	360 kWh usable pack
Average 60 s dwell charge	9.3 kWh/stop	560 kW average charger across stops
Stops to refill one trainset pack	39 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	112 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	151 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in design.toml — emitted by the osr-design Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. country-costs.toml applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (67.3 km @ €3.5 M/km)	€236 M
Elevated (11.9 km @ €18 M/km)	€215 M
Civil subtotal	€450 M

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~ 11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
standard	28	€1.5 M	€42 M

Archetype	Count	Unit	Subtotal
major	16	€3.0 M	€48 M
terminal	5	€2.5 M	€12 M
depot-terminal	1	€3.0 M	€3.0 M
Stations subtotal			€106 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	5	€3.0 M	€15 M
Depots subtotal			€40 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
light-metro-3car (revenue + spare + cold reserve)	80	€3.0 M	€240 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	81.9 km × €0.015 M/km	€1.2 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€16 M
EPC integration + project management (7%)	on subtotal	€60 M

Total

Bucket	Value
Civil works	€450 M
Stations	€106 M
Depots	€40 M
Rolling stock	€240 M
Residual train-control wayside + charging microgrids	€17 M
EPC overhead (7%)	€60 M
CAPEX total	€913 M
Per-route-km	€11 M / km
Per-capita (city pop)	€961 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code — regenerate by re-running scripts/regenerate-city.sh narayananj.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–7 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 8 and runs for 23 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–7)	€60 M / yr	€63
Steady-state, low-ridership (year 8+)	€77 M / yr	€81
Steady-state, high-ridership (year 8+)	€72 M / yr	€76
Lifecycle envelope (yr 1–30, low scenario)	€2.19 bn cumulative	€2,304
Lifecycle envelope (yr 1–30, high scenario)	€2.07 bn cumulative	€2,179

Population basis: 950,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 30, debt service drops to zero and only the OPEX shortfall remains — ~€18 M / yr (low) €13 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€548 M	3.8%	30 y, 7 y grace	€36 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€228 M	8.5%	30 y, 7 y grace	€23 M / yr
Government equity (no debt service)	15%	€137 M	—	—	—
Total	100%	€913 M			€59 M / yr

During the 7-year grace period the operator pays interest only — multilateral €21 M / yr + bonds €19 M / yr = €40 M / yr total — plus the equity tranche amortised across construction (€20 M / yr × 7 yr). Principal repayment begins in year 8 on a 23-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€9.6 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€12 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€59 k
Traction energy (143.4 GWh / yr)	trackside PV + Na-ion (RFC 0002) — self-generated, €0 / yr	€0 k
Labour (503 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€1.5 M
OPEX subtotal		€23 M / yr

Annual fleet utilisation: 71 revenue trainsets × 20.5 h/day × 365 d/yr × 30 km/h commercial × 75% revenue factor = 12.0 M train-km / yr (~168 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$195 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass — a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.30 (~\$0.33 USD)
Day pass (3 trips)	€0.76 (15 % bulk discount)
Monthly unlimited pass	€8.97 (~5 % of median monthly income)
Annual pass	€98.67 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	17.3 M	34.7 M
Farebox revenue	€5.2 M / yr	€10 M / yr
Farebox / OPEX recovery	22%	45%
Country policy-target recovery (diagnostic)	50%	50%
Operating shortfall (gov subsidy required)	€18 M / yr	€13 M / yr

	Low scenario	High scenario
Steady-state government commitment (debt service + OPEX shortfall)	€77 M / yr	€72 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
narayananj.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
narayananj-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
narayananj.corridor.geojson	Line polylines + stations (GeoJSON)
narayananj.stations.json	Machine-readable station list
narayananj.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug narayananj

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug narayananj \
  --sidecar .cache/osr-pipeline/rasters/narayananj.grid.json \
  --out-dir designs/.../Narayananj

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../narayananj.toml \
  --out designs/.../README.md
```

`scripts/regenerate-narayananj.sh` chains steps 3 + drift tests into a single command.

south-asia/Bangladesh/Rajshahi

designs/south-asia/Bangladesh/Rajshahi/README.md

Rajshahi — Urban Rail Network

Country: BD · Population: 950,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Rajshahi rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [rajshahi.corridor.geojson](#).

At a glance

Metric	Value
Lines	3
Unique stations	30
Interchange stations	3
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	30.4%
Route length (double track)	49.8 km
Revenue fleet	45 × 3-car trainsets
Spare + cold-reserve	7 × 3-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	22.9 km	13	23	W Outer E Mid
line-2	15.7 km	11	16	SW Mid N Mid
line-3	11.3 km	6	13	N Mid E Outer
Total	49.8 km	30 unique	52	

Rolling stock

Property	Value
Consist	3-car, 57 m
Max speed	90 km/h
Onboard battery	360 kWh per trainset
Seats	60 longitudinal seats
Nominal capacity (AW2)	330 pax (seated + standing, light-metro-3car per RFC 0008 §1)
Crush capacity (AW3)	420 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 330 AW2 passengers (light-metro-3car)
- Peak frequency: 12 trains/hour/direction (5-min headway)

- Peak capacity per line per direction: $330 \times 12 = 3,960$ pphpd
- Network peak throughput (all lines, both directions): $3 \text{ lines} \times 2 \text{ directions} \times 3,960 = 23,760$ passengers/hour
- Daily theoretical capacity (peak $\times 10$): $\approx 237,600$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 28,880 - 43,320$ trips/day

Catchment

- City population: 950,000
- Anchor-weighted coverage: 30.4%
- Catchment population: $\approx 288,800$ (within ~ 800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	3	500 kW	3000 kWh
Major	6	400 kW	2500 kWh
Standard	15	300 kW	2000 kWh
Terminal	5	500 kW	3000 kWh
Total installed	30	15,900 kW	109,000 kWh

Aggregate station-rail charging power: 18,000 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 360 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	12.0 kWh/km	3 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	199 kWh	16.6 km average line length
Onboard battery coverage	1.8 $\times$ average line run	360 kWh usable pack
Average 60 s dwell charge	10.0 kWh/stop	600 kW average charger across stops
Stops to refill one trainset pack	36 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	80 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	109 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in design.toml — emitted by the osr-design Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. country-costs.toml applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (48.1 km @ €3.5 M/km)	€168 M
Elevated (1.0 km @ €18 M/km)	€18 M
Civil subtotal	€186 M

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~ 11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
standard	15	€1.5 M	€22 M
major	6	€3.0 M	€18 M
terminal	5	€2.5 M	€12 M
depot-terminal	1	€3.0 M	€3.0 M
interchange	3	€4.5 M	€14 M
Stations subtotal			€70 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	5	€3.0 M	€15 M
Depots subtotal			€40 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
light-metro-3car (revenue + spare + cold reserve)	52	€3.0 M	€156 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	49.8 km × €0.015 M/km	€0.7 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€11 M
EPC integration + project management (7%)	on subtotal	€32 M

Total

Bucket	Value
Civil works	€186 M
Stations	€70 M
Depots	€40 M
Rolling stock	€156 M
Residual train-control wayside + charging microgrids	€11 M
EPC overhead (7%)	€32 M
CAPEX total	€496 M
Per-route-km	€10.0 M / km
Per-capita (city pop)	€522 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running scripts/regenerate-city.sh rajshahi.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–7 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 8 and runs for 23 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–7)	€32 M / yr	€34
Steady-state, low-ridership (year 8+)	€40 M / yr	€42
Steady-state, high-ridership (year 8+)	€35 M / yr	€37
Lifecycle envelope (yr 1–30, low scenario)	€1.15 bn cumulative	€1,208
Lifecycle envelope (yr 1–30, high scenario)	€1.03 bn cumulative	€1,082

Population basis: 950,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 30, debt service drops to zero and only the OPEX shortfall remains – ~€7.9 M / yr (low) €2.8 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€297 M	3.8%	30 y, 7 y grace	€20 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€124 M	8.5%	30 y, 7 y grace	€12 M / yr
Government equity (no debt service)	15%	€74 M	—	—	—
Total	100%	€496 M			€32 M / yr

During the 7-year grace period the operator pays interest only – multilateral €11 M / yr + bonds €11 M / yr = €22 M / yr total – plus the equity tranche amortised across construction (€11 M / yr × 7 yr). Principal repayment begins in year 8 on a 23-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€6.2 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€5.9 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€37 k
Traction energy (90.9 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (311 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€937 k
OPEX subtotal		€13 M / yr

Annual fleet utilisation: 45 revenue trainsets × 20.5 h/day × 365 d/yr × 30 km/h commercial × 75% revenue factor = 7.6 M train-km / yr (~168 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$195 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.30 (~\$0.33 USD)
Day pass (3 trips)	€0.76 (15 % bulk discount)
Monthly unlimited pass	€8.97 (~5 % of median monthly income)
Annual pass	€98.67 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	17.3 M	34.7 M

	Low scenario	High scenario
Farebox revenue	€5.2 M / yr	€10 M / yr
Farebox / OPEX recovery	39%	79%
Country policy-target recovery (diagnostic)	50%	50%
Operating shortfall (gov subsidy required)	€7.9 M / yr	€2.8 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€40 M / yr	€35 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
rajshahi.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
rajshahi-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
rajshahi.corridor.geojson	Line polylines + stations (GeoJSON)
rajshahi.stations.json	Machine-readable station list
rajshahi.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug rajshahi

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug rajshahi \
  --sidecar .cache/osr-pipeline/rasters/rajshahi.grid.json \
  --out-dir designs/.../Rajshahi

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../rajshahi.toml \
  --out designs/.../README.md
```

`scripts/regenerate-rajshahi.sh` chains steps 3 + drift tests into a single command.

south-asia/Bangladesh/Rangpur

designs/south-asia/Bangladesh/Rangpur/README.md

Rangpur — Urban Rail Network

Country: BD · Population: 800,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Rangpur rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [rangpur.corridor.geojson](#).

At a glance

Metric	Value
Lines	3
Unique stations	36
Interchange stations	2
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	47.3%
Route length (double track)	59.5 km
Revenue fleet	53 × 3-car trainsets
Spare + cold-reserve	6 × 3-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	20.5 km	13	20	NW Outer S Outer
line-2	18.7 km	11	19	S Mid NE Outer
line-3	20.4 km	12	20	SE Outer N Outer
Total	59.5 km	36 unique	59	

Rolling stock

Property	Value
Consist	3-car, 57 m
Max speed	90 km/h
Onboard battery	360 kWh per trainset
Seats	60 longitudinal seats
Nominal capacity (AW2)	330 pax (seated + standing, light-metro-3car per RFC 0008 §1)
Crush capacity (AW3)	420 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 330 AW2 passengers (light-metro-3car)
- Peak frequency: 12 trains/hour/direction (5-min headway)

- Peak capacity per line per direction: $330 \times 12 = 3,960$ pphpd
- Network peak throughput (all lines, both directions): $3 \text{ lines} \times 2 \text{ directions} \times 3,960 = 23,760$ passengers/hour
- Daily theoretical capacity (peak $\times 10$): $\approx 237,600$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 37,840 - 56,760$ trips/day

Catchment

- City population: 800,000
- Anchor-weighted coverage: 47.3%
- Catchment population: $\approx 378,400$ (within ~ 800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	2	500 kW	3000 kWh
Major	11	400 kW	2500 kWh
Standard	17	300 kW	2000 kWh
Terminal	5	500 kW	3000 kWh
Total installed	36	18,000 kW	122,500 kWh

Aggregate station-rail charging power: 21,000 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 360 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	12.0 kWh/km	3 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	238 kWh	19.8 km average line length
Onboard battery coverage	1.5 $\times$ average line run	360 kWh usable pack
Average 60 s dwell charge	9.7 kWh/stop	583 kW average charger across stops
Stops to refill one trainset pack	37 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	90 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	122 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in design.toml — emitted by the osr-design Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. country-costs.toml applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (58.0 km @ €3.5 M/km)	€203 M
Elevated (1.3 km @ €18 M/km)	€24 M
Elevated-interchange premium (1 sites @ €20 M)	€20 M
Civil subtotal	€247 M

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~ 11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
standard	17	€1.5 M	€26 M
major	11	€3.0 M	€33 M
terminal	5	€2.5 M	€12 M
depot-terminal	1	€3.0 M	€3.0 M
interchange-elevated	2	€4.5 M	€9.0 M
Stations subtotal			€83 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	5	€3.0 M	€15 M
Depots subtotal			€40 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
light-metro-3car (revenue + spare + cold reserve)	59	€3.0 M	€177 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	59.5 km × €0.015 M/km	€0.9 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€13 M
EPC integration + project management (7%)	on subtotal	€39 M

Total

Bucket	Value
Civil works	€247 M
Stations	€83 M
Depots	€40 M
Rolling stock	€177 M
Residual train-control wayside + charging microgrids	€13 M
EPC overhead (7%)	€39 M
CAPEX total	€600 M
Per-route-km	€10 M / km
Per-capita (city pop)	€750 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running scripts/regenerate-city.sh rangpur.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–7 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 8 and runs for 23 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–7)	€39 M / yr	€49
Steady-state, low-ridership (year 8+)	€50 M / yr	€63
Steady-state, high-ridership (year 8+)	€46 M / yr	€57
Lifecycle envelope (yr 1–30, low scenario)	€1.43 bn cumulative	€1,783
Lifecycle envelope (yr 1–30, high scenario)	€1.33 bn cumulative	€1,657

Population basis: 800,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 30, debt service drops to zero and only the OPEX shortfall remains – ~€11 M / yr (low) €6.9 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€360 M	3.8%	30 y, 7 y grace	€24 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€150 M	8.5%	30 y, 7 y grace	€15 M / yr
Government equity (no debt service)	15%	€90 M	—	—	—
Total	100%	€600 M			€39 M / yr

During the 7-year grace period the operator pays interest only – multilateral €14 M / yr + bonds €13 M / yr = €26 M / yr total – plus the equity tranche amortised across construction (€13 M / yr × 7 yr). Principal repayment begins in year 8 on a 23-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€7.1 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€7.4 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€44 k
Traction energy (107.1 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (369 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€1.1 M
OPEX subtotal		€16 M / yr

Annual fleet utilisation: 53 revenue trainsets × 20.5 h/day × 365 d/yr × 30 km/h commercial × 75% revenue factor = 8.9 M train-km / yr (~168 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$195 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.30 (~\$0.33 USD)
Day pass (3 trips)	€0.76 (15 % bulk discount)
Monthly unlimited pass	€8.97 (~5 % of median monthly income)
Annual pass	€98.67 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	14.6 M	29.2 M

	Low scenario	High scenario
Farebox revenue	€4.4 M / yr	€8.7 M / yr
Farebox / OPEX recovery	28%	56%
Country policy-target recovery (diagnostic)	50%	50%
Operating shortfall (gov subsidy required)	€11 M / yr	€6.9 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€50 M / yr	€46 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
rangpur.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
rangpur-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
rangpur.corridor.geojson	Line polylines + stations (GeoJSON)
rangpur.stations.json	Machine-readable station list
rangpur.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug rangpur

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug rangpur \
  --sidecar .cache/osr-pipeline/rasters/rangpur.grid.json \
  --out-dir designs/.../Rangpur

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../rangpur.toml \
  --out designs/.../README.md
```

`scripts/regenerate-rangpur.sh` chains steps 3 + drift tests into a single command.

south-asia/Bangladesh/Sylhet

designs/south-asia/Bangladesh/Sylhet/README.md

Sylhet — Urban Rail Network

Country: BD · Population: 900,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Sylhet rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [sylhet.corridor.geojson](#).

At a glance

Metric	Value
Lines	3
Unique stations	40
Interchange stations	2
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	45.4%
Route length (double track)	72.6 km
Revenue fleet	64 × 3-car trainsets
Spare + cold-reserve	8 × 3-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	27.6 km	16	27	N Outer S Outer
line-2	27.6 km	13	27	NE Outer SW Outer
line-3	17.4 km	11	18	NE Mid W Mid
Total	72.6 km	40 unique	72	

Rolling stock

Property	Value
Consist	3-car, 57 m
Max speed	90 km/h
Onboard battery	360 kWh per trainset
Seats	60 longitudinal seats
Nominal capacity (AW2)	330 pax (seated + standing, light-metro-3car per RFC 0008 §1)
Crush capacity (AW3)	420 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 330 AW2 passengers (light-metro-3car)
- Peak frequency: 12 trains/hour/direction (5-min headway)

- Peak capacity per line per direction: $330 \times 12 = 3,960$ pphpd
- Network peak throughput (all lines, both directions): $3 \text{ lines} \times 2 \text{ directions} \times 3,960 = 23,760$ passengers/hour
- Daily theoretical capacity (peak $\times 10$): $\approx 237,600$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 40,860 - 61,290$ trips/day

Catchment

- City population: 900,000
- Anchor-weighted coverage: 45.4%
- Catchment population: $\approx 408,600$ (within ~ 800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	2	500 kW	3000 kWh
Major	8	400 kW	2500 kWh
Standard	24	300 kW	2000 kWh
Terminal	5	500 kW	3000 kWh
Total installed	40	18,900 kW	129,000 kWh

Aggregate station-rail charging power: 23,000 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 360 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	12.0 kWh/km	3 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	290 kWh	24.2 km average line length
Onboard battery coverage	1.2 $\times$ average line run	360 kWh usable pack
Average 60 s dwell charge	9.6 kWh/stop	575 kW average charger across stops
Stops to refill one trainset pack	38 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	94 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	129 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in design.toml — emitted by the osr-design Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. country-costs.toml applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (66.1 km @ €3.5 M/km)	€231 M
Elevated (5.0 km @ €18 M/km)	€89 M
Civil subtotal	€321 M

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~ 11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
standard	24	€1.5 M	€36 M
major	8	€3.0 M	€24 M
terminal	5	€2.5 M	€12 M
depot-terminal	1	€3.0 M	€3.0 M
interchange	2	€4.5 M	€9.0 M
Stations subtotal			€84 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	5	€3.0 M	€15 M
Depots subtotal			€40 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
light-metro-3car (revenue + spare + cold reserve)	72	€3.0 M	€216 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	72.6 km × €0.015 M/km	€1.1 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€13 M
EPC integration + project management (7%)	on subtotal	€47 M

Total

Bucket	Value
Civil works	€321 M
Stations	€84 M
Depots	€40 M
Rolling stock	€216 M
Residual train-control wayside + charging microgrids	€14 M
EPC overhead (7%)	€47 M
CAPEX total	€723 M
Per-route-km	€10.0 M / km
Per-capita (city pop)	€803 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running scripts/regenerate-city.sh sylhet.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–7 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 8 and runs for 23 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–7)	€47 M / yr	€53
Steady-state, low-ridership (year 8+)	€61 M / yr	€68
Steady-state, high-ridership (year 8+)	€56 M / yr	€62
Lifecycle envelope (yr 1–30, low scenario)	€1.73 bn cumulative	€1,921
Lifecycle envelope (yr 1–30, high scenario)	€1.62 bn cumulative	€1,796

Population basis: 900,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 30, debt service drops to zero and only the OPEX shortfall remains – ~€14 M / yr (low) €9.1 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€434 M	3.8%	30 y, 7 y grace	€29 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€181 M	8.5%	30 y, 7 y grace	€18 M / yr
Government equity (no debt service)	15%	€108 M	—	—	—
Total	100%	€723 M			€47 M / yr

During the 7-year grace period the operator pays interest only – multilateral €16 M / yr + bonds €15 M / yr = €32 M / yr total – plus the equity tranche amortised across construction (€15 M / yr × 7 yr). Principal repayment begins in year 8 on a 23-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€8.6 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€8.9 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€53 k
Traction energy (129.3 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (448 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€1.4 M
OPEX subtotal		€19 M / yr

Annual fleet utilisation: 64 revenue trainsets × 20.5 h/day × 365 d/yr × 30 km/h commercial × 75% revenue factor = 10.8 M train-km / yr (~168 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$195 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.30 (~\$0.33 USD)
Day pass (3 trips)	€0.76 (15 % bulk discount)
Monthly unlimited pass	€8.97 (~5 % of median monthly income)
Annual pass	€98.67 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	16.4 M	32.9 M

	Low scenario	High scenario
Farebox revenue	€4.9 M / yr	€9.8 M / yr
Farebox / OPEX recovery	26%	52%
Country policy-target recovery (diagnostic)	50%	50%
Operating shortfall (gov subsidy required)	€14 M / yr	€9.1 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€61 M / yr	€56 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
sylhet.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
sylhet-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
sylhet.corridor.geojson	Line polylines + stations (GeoJSON)
sylhet.stations.json	Machine-readable station list
sylhet.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug sylhet

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug sylhet \
  --sidecar .cache/osr-pipeline/rasters/sylhet.grid.json \
  --out-dir designs/.../Sylhet

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../sylhet.toml \
  --out designs/.../README.md
```

`scripts/regenerate-sylhet.sh` chains steps 3 + drift tests into a single command.

south-asia/India/Agra

designs/south-asia/India/Agra/README.md

Agra — Urban Rail Network

Country: IN · Population: 1,700,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Agra rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [agra.corridor.geojson](#).

At a glance

Metric	Value
Lines	5
Unique stations	97
Interchange stations	19
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	38.7%
Route length (double track)	191.0 km
Revenue fleet	139 × 4-car trainsets
Spare + cold-reserve	17 × 4-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	35.5 km	15	29	SE Outer W Outer
line-2	31.4 km	17	26	S Outer NE Outer
line-3	28.1 km	15	24	NW Outer NE Outer
line-4	28.1 km	16	24	S Outer NW Mid
line-5	68.0 km	35	53	NW Mid NW Mid
Total	191.0 km	97 unique	156	

Rolling stock

Property	Value
Consist	4-car, 75 m
Max speed	90 km/h
Onboard battery	480 kWh per trainset
Seats	80 longitudinal seats
Nominal capacity (AW2)	440 pax (seated + standing, met ro - 4 car per RFC 0008 §1)
Crush capacity (AW3)	560 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 440 AW2 passengers (met ro - 4 car)

- Peak frequency: 12 trains/hour/direction (5-min headway)
- Peak capacity per line per direction: $440 \times 12 = 5,280$ pphpd
- Network peak throughput (all lines, both directions): $5 \text{ lines} \times 2 \text{ directions} \times 5,280 = 52,800$ passengers/hour
- Daily theoretical capacity (peak $\times$ 10): $\approx 528,000$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 65,790 - 98,685$ trips/day

Catchment

- City population: 1,700,000
- Anchor-weighted coverage: 38.7%
- Catchment population: $\approx 657,900$ (within ~800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	19	500 kW	3000 kWh
Major	31	400 kW	2500 kWh
Standard	40	300 kW	2000 kWh
Terminal	7	500 kW	3000 kWh
Total installed	98	42,400 kW	275,500 kWh

Aggregate station-rail charging power: 53,000 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 480 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	16.0 kWh/km	4 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	611 kWh	38.2 km average line length
Onboard battery coverage	0.8 $\times$ average line run	480 kWh usable pack
Average 60 s dwell charge	9.1 kWh/stop	546 kW average charger across stops
Stops to refill one trainset pack	53 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	212 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	276 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in design.toml — emitted by the osr-design Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. country-costs.toml applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (175.7 km @ €3.5 M/km)	€615 M
Elevated (14.5 km @ €18 M/km)	€261 M
Elevated-interchange premium (8 sites @ €20 M)	€160 M
Civil subtotal	€1.04 bn

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
standard	40	€1.5 M	€60 M
major	31	€3.0 M	€93 M
terminal	7	€2.5 M	€18 M
depot-terminal	1	€3.0 M	€3.0 M
interchange-elevated	19	€4.5 M	€86 M
Stations subtotal			€259 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	7	€3.0 M	€21 M
Depots subtotal			€46 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
metro-4car (revenue + spare + cold reserve)	156	€4.0 M	€624 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	191.0 km × €0.015 M/km	€2.9 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€37 M
EPC integration + project management (7%)	on subtotal	€140 M

Total

Bucket	Value
Civil works	€1.04 bn
Stations	€259 M
Depots	€46 M
Rolling stock	€624 M
Residual train-control wayside + charging microgrids	€40 M
EPC overhead (7%)	€140 M
CAPEX total	€2.15 bn
Per-route-km	€11 M / km
Per-capita (city pop)	€1,262 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from <lib/templates/country-finance.toml>. Pure function of the [costs] block above + the country code – regenerate by re-running `scripts/regenerate-city.sh` agra.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–5 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 6 and runs for 20 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–5)	€155 M / yr	€91
Steady-state, low-ridership (year 6+)	€191 M / yr	€113
Steady-state, high-ridership (year 6+)	€180 M / yr	€106
Lifecycle envelope (yr 1–25, low scenario)	€4.60 bn cumulative	€2,705
Lifecycle envelope (yr 1–25, high scenario)	€4.38 bn cumulative	€2,576

Population basis: 1,700,000 (catchment per <lib/city-batches/world-sample.toml>). After year 25, debt service drops to zero and only the OPEX shortfall remains – ~€45 M / yr (low) €34 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€1.29 bn	4.0%	25 y, 5 y grace	€95 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€536 M	7.2%	25 y, 5 y grace	€51 M / yr
Government equity (no debt service)	15%	€322 M	—	—	—
Total	100%	€2.15 bn			€146 M / yr

During the 5-year grace period the operator pays interest only – multilateral €52 M / yr + bonds €39 M / yr = €90 M / yr total – plus the equity tranche amortised across construction (€64 M / yr × 5 yr). Principal repayment begins in year 6 on a 20-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€25 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€27 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€143 k
Traction energy (436.8 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (1,158 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€4.1 M
OPEX subtotal		€56 M / yr

Annual fleet utilisation: 139 revenue trainsets × 20.5 h/day × 365 d/yr × 35 km/h commercial × 75% revenue factor = 27.3 M train-km / yr (~196 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$230 USD (per <lib/templates/country-finance.toml>). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.35 (~\$0.38 USD)
Day pass (3 trips)	€0.90 (15 % bulk discount)
Monthly unlimited pass	€10.58 (~5 % of median monthly income)
Annual pass	€116.38 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	31.0 M	62.0 M
Farebox revenue	€11 M / yr	€22 M / yr
Farebox / OPEX recovery	20%	39%
Country policy-target recovery (diagnostic)	55%	55%
Operating shortfall (gov subsidy required)	€45 M / yr	€34 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€191 M / yr	€180 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
agra.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
agra-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
agra.corridor.geojson	Line polylines + stations (GeoJSON)
agra.stations.json	Machine-readable station list
agra.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug agra

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug agra \
  --sidecar .cache/osr-pipeline/rasters/agra.grid.json \
  --out-dir designs/.../Agra

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../agra.toml \
  --out designs/.../README.md
```

`scripts/regenerate-agra.sh` chains steps 3 + drift tests into a single command.

south-asia/India/Bhopal

designs/south-asia/India/Bhopal/README.md

Bhopal — Urban Rail Network

Country: IN · Population: 2,400,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Bhopal rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [bhopal.corridor.geojson](#).

At a glance

Metric	Value
Lines	6
Unique stations	106
Interchange stations	22
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	51.5%
Route length (double track)	205.7 km
Revenue fleet	150 × 4-car trainsets
Spare + cold-reserve	19 × 4-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	30.1 km	18	25	SE Mid NW Mid
line-2	32.2 km	18	27	E Mid W Outer
line-3	31.7 km	17	26	N Mid S Outer
line-4	25.4 km	12	21	W Inner SE Outer
line-5	27.2 km	13	23	SW Mid NE Outer
line-6	59.2 km	29	47	W Mid W Mid
Total	205.7 km	106 unique	169	

Rolling stock

Property	Value
Consist	4-car, 75 m
Max speed	90 km/h
Onboard battery	480 kWh per trainset
Seats	80 longitudinal seats
Nominal capacity (AW2)	440 pax (seated + standing, metro-4car per RFC 0008 §1)
Crush capacity (AW3)	560 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 440 AW2 passengers (metro-4car)
- Peak frequency: 12 trains/hour/direction (5-min headway)
- Peak capacity per line per direction: $440 \times 12 = 5,280$ pphpd
- Network peak throughput (all lines, both directions): $6 \text{ lines} \times 2 \text{ directions} \times 5,280 = 63,360$ passengers/hour
- Daily theoretical capacity (peak $\times$ 10): $\approx 633,600$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 123,600 - 185,400$ trips/day

Catchment

- City population: 2,400,000
- Anchor-weighted coverage: 51.5%
- Catchment population: $\approx 1,236,000$ (within ~800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	22	500 kW	3000 kWh
Major	46	400 kW	2500 kWh
Standard	26	300 kW	2000 kWh
Terminal	9	500 kW	3000 kWh
Total installed	104	46,700 kW	300,000 kWh

Aggregate station-rail charging power: 57,750 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 480 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	16.0 kWh/km	4 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	549 kWh	34.3 km average line length
Onboard battery coverage	0.9 $\times$ average line run	480 kWh usable pack
Average 60 s dwell charge	9.1 kWh/stop	545 kW average charger across stops
Stops to refill one trainset pack	53 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	234 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	300 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in design.toml — emitted by the osr-design Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. country-costs.toml applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (188.1 km @ €3.5 M/km)	€658 M
Elevated (16.4 km @ €18 M/km)	€295 M
Elevated-interchange premium (13 sites @ €20 M)	€260 M
Civil subtotal	€1.21 bn

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
halt	3	€0.4 M	€1.2 M
standard	26	€1.5 M	€39 M
major	46	€3.0 M	€138 M
terminal	9	€2.5 M	€22 M
depot-terminal	1	€3.0 M	€3.0 M
interchange	2	€4.5 M	€9.0 M
interchange-elevated	20	€4.5 M	€90 M
Stations subtotal			€303 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	9	€3.0 M	€27 M
Depots subtotal			€52 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
metro-4car (revenue + spare + cold reserve)	169	€4.0 M	€676 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	205.7 km × €0.015 M/km	€3.1 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€43 M
EPC integration + project management (7%)	on subtotal	€160 M

Total

Bucket	Value
Civil works	€1.21 bn
Stations	€303 M
Depots	€52 M
Rolling stock	€676 M
Residual train-control wayside + charging microgrids	€46 M
EPC overhead (7%)	€160 M
CAPEX total	€2.45 bn
Per-route-km	€12 M / km
Per-capita (city pop)	€1,021 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running scripts/regenerate-city.sh bhopal.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–5 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 6 and runs for 20 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–5)	€176 M / yr	€74
Steady-state, low-ridership (year 6+)	€214 M / yr	€89
Steady-state, high-ridership (year 6+)	€199 M / yr	€83
Lifecycle envelope (yr 1–25, low scenario)	€5.17 bn cumulative	€2,155
Lifecycle envelope (yr 1–25, high scenario)	€4.86 bn cumulative	€2,026

Population basis: 2,400,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 25, debt service drops to zero and only the OPEX shortfall remains – ~€48 M / yr (low) €32 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€1.47 bn	4.0%	25 y, 5 y grace	€108 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€613 M	7.2%	25 y, 5 y grace	€59 M / yr
Government equity (no debt service)	15%	€368 M	—	—	—
Total	100%	€2.45 bn			€167 M / yr

During the 5-year grace period the operator pays interest only – multilateral €59 M / yr + bonds €44 M / yr = €103 M / yr total – plus the equity tranche amortised across construction (€74 M / yr × 5 yr). Principal repayment begins in year 6 on a 20-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€27 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€31 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€153 k
Traction energy (471.4 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (1,246 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€4.4 M
OPEX subtotal		€63 M / yr

Annual fleet utilisation: 150 revenue trainsets × 20.5 h/day × 365 d/yr × 35 km/h commercial × 75% revenue factor = 29.5 M train-km / yr (~196 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$230 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.35 (~\$0.38 USD)
Day pass (3 trips)	€0.90 (15 % bulk discount)
Monthly unlimited pass	€10.58 (~5 % of median monthly income)
Annual pass	€116.38 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	43.8 M	87.6 M
Farebox revenue	€15 M / yr	€31 M / yr
Farebox / OPEX recovery	25%	49%
Country policy-target recovery (diagnostic)	55%	55%
Operating shortfall (gov subsidy required)	€48 M / yr	€32 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€214 M / yr	€199 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
bhopal.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
bhopal-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
bhopal.corridor.geojson	Line polylines + stations (GeoJSON)
bhopal.stations.json	Machine-readable station list
bhopal.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug bhopal

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug bhopal \
  --sidecar .cache/osr-pipeline/rasters/bhopal.grid.json \
  --out-dir designs/.../Bhopal

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../bhopal.toml \
  --out designs/.../README.md
```

`scripts/regenerate-bhopal.sh` chains steps 3 + drift tests into a single command.

south-asia/India/Coimbatore

designs/south-asia/India/Coimbatore/README.md

Coimbatore — Urban Rail Network

Country: IN · Population: 3,084,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Coimbatore rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [coimbatore.corridor.geojson](#).

At a glance

Metric	Value
Lines	5
Unique stations	120
Interchange stations	22
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	31.2%
Route length (double track)	267.8 km
Revenue fleet	193 × 6-car trainsets
Spare + cold-reserve	21 × 6-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	52.2 km	20	42	NE Outer SW Outer
line-2	47.9 km	23	39	N Outer S Outer
line-3	36.9 km	20	30	W Mid E Outer
line-4	37.3 km	19	30	N Outer S Mid
line-5	93.4 km	39	73	W Mid W Mid
Total	267.8 km	120 unique	214	

Rolling stock

Property	Value
Consist	6-car, 111 m
Max speed	100 km/h
Onboard battery	720 kWh per trainset
Seats	120 longitudinal seats
Nominal capacity (AW2)	660 pax (seated + standing, met ro-6car per RFC 0008 §1)
Crush capacity (AW3)	840 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 660 AW2 passengers (met ro-6car)

- Peak frequency: 12 trains/hour/direction (5-min headway)
- Peak capacity per line per direction: $660 \times 12 = 7,920$ pphpd
- Network peak throughput (all lines, both directions): $5 \text{ lines} \times 2 \text{ directions} \times 7,920 = 79,200$ passengers/hour
- Daily theoretical capacity (peak $\times$ 10): $\approx 792,000$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 96,220 - 144,331$ trips/day

Catchment

- City population: 3,084,000
- Anchor-weighted coverage: 31.2%
- Catchment population: $\approx 962,208$ (within ~ 800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	22	500 kW	3000 kWh
Major	24	400 kW	2500 kWh
Standard	67	300 kW	2000 kWh
Terminal	7	500 kW	3000 kWh
Total installed	121	49,200 kW	321,000 kWh

Aggregate station-rail charging power: 64,500 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 720 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	24.0 kWh/km	6 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	1,285 kWh	53.6 km average line length
Onboard battery coverage	0.6 $\times$ average line run	720 kWh usable pack
Average 60 s dwell charge	9.0 kWh/stop	538 kW average charger across stops
Stops to refill one trainset pack	80 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	246 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	321 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in design.toml — emitted by the osr-design Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. country-costs.toml applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (241.8 km @ €3.5 M/km)	€846 M
Elevated (25.2 km @ €18 M/km)	€453 M
Elevated-interchange premium (11 sites @ €20 M)	€220 M
Civil subtotal	€1.52 bn

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
standard	67	€1.5 M	€100 M
major	24	€3.0 M	€72 M
terminal	7	€2.5 M	€18 M
depot-terminal	1	€3.0 M	€3.0 M
interchange-elevated	22	€4.5 M	€99 M
Stations subtotal			€292 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	7	€3.0 M	€21 M
Depots subtotal			€46 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY – never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
metro-6car (revenue + spare + cold reserve)	214	€6.0 M	€1.28 bn

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	267.8 km × €0.015 M/km	€4.0 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€43 M
EPC integration + project management (7%)	on subtotal	€223 M

Total

Bucket	Value
Civil works	€1.52 bn
Stations	€292 M
Depots	€46 M
Rolling stock	€1.28 bn
Residual train-control wayside + charging microgrids	€47 M
EPC overhead (7%)	€223 M
CAPEX total	€3.41 bn
Per-route-km	€13 M / km
Per-capita (city pop)	€1,106 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running `scripts/regenerate-city.sh coimbatore`.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–5 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 6 and runs for 20 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–5)	€246 M / yr	€80
Steady-state, low-ridership (year 6+)	€307 M / yr	€100
Steady-state, high-ridership (year 6+)	€287 M / yr	€93
Lifecycle envelope (yr 1–25, low scenario)	€7.37 bn cumulative	€2,389
Lifecycle envelope (yr 1–25, high scenario)	€6.97 bn cumulative	€2,260

Population basis: 3,084,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 25, debt service drops to zero and only the OPEX shortfall remains – ~€75 M / yr (low) €55 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€2.05 bn	4.0%	25 y, 5 y grace	€151 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€853 M	7.2%	25 y, 5 y grace	€82 M / yr
Government equity (no debt service)	15%	€512 M	—	—	—
Total	100%	€3.41 bn			€232 M / yr

During the 5-year grace period the operator pays interest only – multilateral €82 M / yr + bonds €61 M / yr = €143 M / yr total – plus the equity tranche amortised across construction (€102 M / yr × 5 yr). Principal repayment begins in year 6 on a 20-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€51 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€37 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€200 k
Traction energy (909.8 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (1,619 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€5.8 M
OPEX subtotal		€94 M / yr

Annual fleet utilisation: 193 revenue trainsets × 20.5 h/day × 365 d/yr × 35 km/h commercial × 75% revenue factor = 37.9 M train-km / yr (~196 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$230 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.35 (~\$0.38 USD)
Day pass (3 trips)	€0.90 (15 % bulk discount)
Monthly unlimited pass	€10.58 (~5 % of median monthly income)
Annual pass	€116.38 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	56.3 M	112.6 M
Farebox revenue	€20 M / yr	€40 M / yr
Farebox / OPEX recovery	21%	42%
Country policy-target recovery (diagnostic)	55%	55%
Operating shortfall (gov subsidy required)	€75 M / yr	€55 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€307 M / yr	€287 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
coimbatore.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
coimbatore-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
coimbatore.corridor.geojson	Line polylines + stations (GeoJSON)
coimbatore.stations.json	Machine-readable station list
coimbatore.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug coimbatore

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug coimbatore \
  --sidecar .cache/osr-pipeline/rasters/coimbatore.grid.json \
  --out-dir designs/.../Coimbatore

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../coimbatore.toml \
  --out designs/.../README.md
```

`scripts/regenerate-coimbatore.sh` chains steps 3 + drift tests into a single command.

south-asia/India/Indore

designs/south-asia/India/Indore/README.md

Indore — Urban Rail Network

Country: IN · Population: 3,200,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Indore rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [indore.corridor.geojson](#).

At a glance

Metric	Value
Lines	7
Unique stations	145
Interchange stations	23
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	47.5%
Route length (double track)	323.6 km
Revenue fleet	234 × 6-car trainsets
Spare + cold-reserve	26 × 6-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	46.1 km	24	37	NE Outer SW Outer
line-2	39.9 km	19	32	SW Outer NE Mid
line-3	38.3 km	19	31	N Mid S Outer
line-4	42.1 km	16	35	S Outer N Mid
line-5	39.8 km	19	32	NW Mid SE Outer
line-6	33.8 km	14	28	E Outer NW Mid
line-7	83.6 km	35	65	N Mid N Mid
Total	323.6 km	145 unique	260	

Rolling stock

Property	Value
Consist	6-car, 111 m
Max speed	100 km/h
Onboard battery	720 kWh per trainset
Seats	120 longitudinal seats
Nominal capacity (AW2)	660 pax (seated + standing, metro-6car per RFC 0008 §1)
Crush capacity (AW3)	840 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 660 AW2 passengers (metro - 6 car)
- Peak frequency: 12 trains/hour/direction (5-min headway)
- Peak capacity per line per direction: $660 \times 12 = 7,920$ pphpd
- Network peak throughput (all lines, both directions): $7 \text{ lines} \times 2 \text{ directions} \times 7,920 = 110,880$ passengers/hour
- Daily theoretical capacity (peak $\times$ 10): $\approx 1,108,800$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 152,000 - 228,000$ trips/day

Catchment

- City population: 3,200,000
- Anchor-weighted coverage: 47.5%
- Catchment population: $\approx 1,520,000$ (within ~800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	23	500 kW	3000 kWh
Major	47	400 kW	2500 kWh
Standard	56	300 kW	2000 kWh
Terminal	11	500 kW	3000 kWh
Total installed	138	57,600 kW	371,500 kWh

Aggregate station-rail charging power: 77,000 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 720 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	24.0 kWh/km	6 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	1,109 kWh	46.2 km average line length
Onboard battery coverage	0.6 $\times$ average line run	720 kWh usable pack
Average 60 s dwell charge	8.9 kWh/stop	531 kW average charger across stops
Stops to refill one trainset pack	81 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	288 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	372 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in design.toml — emitted by the osr-design Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. country-costs.toml applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (297.9 km @ €3.5 M/km)	€1.04 bn
Elevated (23.1 km @ €18 M/km)	€416 M
Elevated-interchange premium (16 sites @ €20 M)	€320 M
Civil subtotal	€1.78 bn

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
halt	8	€0.4 M	€3.2 M
standard	56	€1.5 M	€84 M
major	47	€3.0 M	€141 M
terminal	11	€2.5 M	€28 M
depot-terminal	1	€3.0 M	€3.0 M
interchange	2	€4.5 M	€9.0 M
interchange-elevated	21	€4.5 M	€94 M
Stations subtotal			€362 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	11	€3.0 M	€33 M
Depots subtotal			€58 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
metro-6car (revenue + spare + cold reserve)	260	€6.0 M	€1.56 bn

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	323.6 km × €0.015 M/km	€4.8 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€53 M
EPC integration + project management (7%)	on subtotal	€267 M

Total

Bucket	Value
Civil works	€1.78 bn
Stations	€362 M
Depots	€58 M
Rolling stock	€1.56 bn
Residual train-control wayside + charging microgrids	€58 M
EPC overhead (7%)	€267 M
CAPEX total	€4.08 bn

Bucket	Value
Per-route-km	€13 M / km
Per-capita (city pop)	€1,276 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code — regenerate by re-running `scripts/regenerate-city.sh` indore.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–5 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 6 and runs for 20 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–5)	€294 M / yr	€92
Steady-state, low-ridership (year 6+)	€371 M / yr	€116
Steady-state, high-ridership (year 6+)	€351 M / yr	€110
Lifecycle envelope (yr 1–25, low scenario)	€8.89 bn cumulative	€2,779
Lifecycle envelope (yr 1–25, high scenario)	€8.48 bn cumulative	€2,650

Population basis: 3,200,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 25, debt service drops to zero and only the OPEX shortfall remains — ~€93 M / yr (low) €72 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€2.45 bn	4.0%	25 y, 5 y grace	€180 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€1.02 bn	7.2%	25 y, 5 y grace	€98 M / yr
Government equity (no debt service)	15%	€613 M	—	—	—
Total	100%	€4.08 bn			€278 M / yr

During the 5-year grace period the operator pays interest only — multilateral €98 M / yr + bonds €74 M / yr = €172 M / yr total — plus the equity tranche amortised across construction (€123 M / yr × 5 yr). Principal repayment begins in year 6 on a 20-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€62 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€44 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€241 k
Traction energy (1103.1 GWh / yr)	trackside PV + Na-ion (RFC 0002) — self-generated, €0 / yr	€0 k
Labour (1,954 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€6.9 M
OPEX subtotal		€114 M / yr

Annual fleet utilisation: 234 revenue trainsets × 20.5 h/day × 365 d/yr × 35 km/h commercial × 75% revenue factor = 46.0 M train-km / yr (~196 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$230 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass — a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.35 (~\$0.38 USD)
Day pass (3 trips)	€0.90 (15 % bulk discount)
Monthly unlimited pass	€10.58 (~5 % of median monthly income)
Annual pass	€116.38 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	58.4 M	116.8 M
Farebox revenue	€21 M / yr	€41 M / yr
Farebox / OPEX recovery	18%	36%
Country policy-target recovery (diagnostic)	55%	55%
Operating shortfall (gov subsidy required)	€93 M / yr	€72 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€371 M / yr	€351 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
indore.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
indore-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
indore.corridor.geojson	Line polylines + stations (GeoJSON)
indore.stations.json	Machine-readable station list
indore.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug indore

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug indore \
  --sidecar .cache/osr-pipeline/rasters/indore.grid.json \
  --out-dir designs/.../Indore

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../indore.toml \
  --out designs/.../README.md
```

`scripts/regenerate-indore.sh` chains steps 3 + drift tests into a single command.

south-asia/India/Jodhpur

designs/south-asia/India/Jodhpur/README.md

Jodhpur — Urban Rail Network

Country: IN · Population: 1,300,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Jodhpur rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [jodhpur.corridor.geojson](#).

At a glance

Metric	Value
Lines	5
Unique stations	81
Interchange stations	16
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	43.6%
Route length (double track)	149.6 km
Revenue fleet	110 × 4-car trainsets
Spare + cold-reserve	14 × 4-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	33.2 km	16	27	N Outer S Outer
line-2	28.2 km	11	24	S Outer NE Outer
line-3	17.2 km	12	15	W Mid NE Mid
line-4	25.5 km	13	21	SE Outer W Mid
line-5	45.5 km	30	37	NW Inner NW Inner
Total	149.6 km	81 unique	124	

Rolling stock

Property	Value
Consist	4-car, 75 m
Max speed	90 km/h
Onboard battery	480 kWh per trainset
Seats	80 longitudinal seats
Nominal capacity (AW2)	440 pax (seated + standing, met ro - 4 car per RFC 0008 §1)
Crush capacity (AW3)	560 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 440 AW2 passengers (met ro - 4 car)

- Peak frequency: 12 trains/hour/direction (5-min headway)
- Peak capacity per line per direction: $440 \times 12 = 5,280$ pphpd
- Network peak throughput (all lines, both directions): $5 \text{ lines} \times 2 \text{ directions} \times 5,280 = 52,800$ passengers/hour
- Daily theoretical capacity (peak $\times$ 10): $\approx 528,000$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 56,680 - 85,020$ trips/day

Catchment

- City population: 1,300,000
- Anchor-weighted coverage: 43.6%
- Catchment population: $\approx 566,800$ (within ~800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	16	500 kW	3000 kWh
Major	33	400 kW	2500 kWh
Standard	24	300 kW	2000 kWh
Terminal	7	500 kW	3000 kWh
Total installed	81	36,900 kW	239,500 kWh

Aggregate station-rail charging power: 44,750 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 480 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	16.0 kWh/km	4 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	479 kWh	29.9 km average line length
Onboard battery coverage	1.0x average line run	480 kWh usable pack
Average 60 s dwell charge	9.2 kWh/stop	552 kW average charger across stops
Stops to refill one trainset pack	52 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	184 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	240 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the `[costs]` block in `design.toml` — emitted by the `osr-design` Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3× higher because of the line items OSR has architected away. `country-costs.toml` applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (86.5 km @ €3.5 M/km)	€303 M
Elevated (62.2 km @ €18 M/km)	€1.12 bn
Elevated-interchange premium (2 sites @ €20 M)	€40 M
Civil subtotal	€1.46 bn

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
halt	1	€0.4 M	€0.4 M
standard	24	€1.5 M	€36 M
major	33	€3.0 M	€99 M
terminal	7	€2.5 M	€18 M
depot-terminal	1	€3.0 M	€3.0 M
interchange	11	€4.5 M	€50 M
interchange-elevated	5	€4.5 M	€22 M
Stations subtotal			€228 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	7	€3.0 M	€21 M
Depots subtotal			€46 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY – never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
metro-4car (revenue + spare + cold reserve)	124	€4.0 M	€496 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	149.6 km × €0.015 M/km	€2.2 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€32 M
EPC integration + project management (7%)	on subtotal	€159 M

Total

Bucket	Value
Civil works	€1.46 bn
Stations	€228 M
Depots	€46 M
Rolling stock	€496 M
Residual train-control wayside + charging microgrids	€35 M
EPC overhead (7%)	€159 M
CAPEX total	€2.42 bn
Per-route-km	€16 M / km
Per-capita (city pop)	€1,865 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running scripts/regenerate-city.sh jodhpur.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–5 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 6 and runs for 20 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–5)	€175 M / yr	€134
Steady-state, low-ridership (year 6+)	€215 M / yr	€165
Steady-state, high-ridership (year 6+)	€206 M / yr	€159
Lifecycle envelope (yr 1–25, low scenario)	€5.17 bn cumulative	€3,975
Lifecycle envelope (yr 1–25, high scenario)	€5.00 bn cumulative	€3,846

Population basis: 1,300,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 25, debt service drops to zero and only the OPEX shortfall remains – ~€50 M / yr (low) €41 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€1.45 bn	4.0%	25 y, 5 y grace	€107 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€606 M	7.2%	25 y, 5 y grace	€58 M / yr
Government equity (no debt service)	15%	€364 M	—	—	—
Total	100%	€2.42 bn			€165 M / yr

During the 5-year grace period the operator pays interest only – multilateral €58 M / yr + bonds €44 M / yr = €102 M / yr total – plus the equity tranche amortised across construction (€73 M / yr × 5 yr). Principal repayment begins in year 6 on a 20-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€20 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€35 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€111 k
Traction energy (345.7 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (910 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€3.2 M
OPEX subtotal		€58 M / yr

Annual fleet utilisation: 110 revenue trainsets × 20.5 h/day × 365 d/yr × 35 km/h commercial × 75% revenue factor = 21.6 M train-km / yr (~196 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$230 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.35 (~\$0.38 USD)
Day pass (3 trips)	€0.90 (15 % bulk discount)
Monthly unlimited pass	€10.58 (~5 % of median monthly income)
Annual pass	€116.38 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	23.7 M	47.5 M
Farebox revenue	€8.4 M / yr	€17 M / yr
Farebox / OPEX recovery	14%	29%
Country policy-target recovery (diagnostic)	55%	55%
Operating shortfall (gov subsidy required)	€50 M / yr	€41 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€215 M / yr	€206 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
jodhpur.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
jodhpur-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
jodhpur.corridor.geojson	Line polylines + stations (GeoJSON)
jodhpur.stations.json	Machine-readable station list
jodhpur.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug jodhpur

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug jodhpur \
  --sidecar .cache/osr-pipeline/rasters/jodhpur.grid.json \
  --out-dir designs/.../Jodhpur

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../jodhpur.toml \
  --out designs/.../README.md
```

`scripts/regenerate-jodhpur.sh` chains steps 3 + drift tests into a single command.

south-asia/India/Kanpur

designs/south-asia/India/Kanpur/README.md

Kanpur — Urban Rail Network

Country: IN · Population: 3,200,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Kanpur rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [kanpur.corridor.geojson](#).

At a glance

Metric	Value
Lines	7
Unique stations	149
Interchange stations	18
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	43.8%
Route length (double track)	339.0 km
Revenue fleet	245 × 6-car trainsets
Spare + cold-reserve	28 × 6-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	46.5 km	22	38	W Mid NE Outer
line-2	35.0 km	20	29	NW Mid SE Mid
line-3	51.7 km	22	41	N Mid S Outer
line-4	51.1 km	23	41	E Outer W Outer
line-5	28.6 km	11	24	SW Outer E Inner
line-6	33.0 km	16	27	S Inner NE Outer
line-7	93.2 km	36	73	NW Mid NW Mid
Total	339.0 km	149 unique	273	

Rolling stock

Property	Value
Consist	6-car, 111 m
Max speed	100 km/h
Onboard battery	720 kWh per trainset
Seats	120 longitudinal seats
Nominal capacity (AW2)	660 pax (seated + standing, metro-6car per RFC 0008 §1)
Crush capacity (AW3)	840 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 660 AW2 passengers (metro - 6 car)
- Peak frequency: 12 trains/hour/direction (5-min headway)
- Peak capacity per line per direction: $660 \times 12 = 7,920$ pphpd
- Network peak throughput (all lines, both directions): $7 \text{ lines} \times 2 \text{ directions} \times 7,920 = 110,880$ passengers/hour
- Daily theoretical capacity (peak $\times$ 10): $\approx 1,108,800$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 140,160 - 210,240$ trips/day

Catchment

- City population: 3,200,000
- Anchor-weighted coverage: 43.8%
- Catchment population: $\approx 1,401,600$ (within ~800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	18	500 kW	3000 kWh
Major	55	400 kW	2500 kWh
Standard	54	300 kW	2000 kWh
Terminal	11	500 kW	3000 kWh
Total installed	139	57,700 kW	372,500 kWh

Aggregate station-rail charging power: 78,250 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 720 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	24.0 kWh/km	6 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	1,162 kWh	48.4 km average line length
Onboard battery coverage	0.6 $\times$ average line run	720 kWh usable pack
Average 60 s dwell charge	8.8 kWh/stop	525 kW average charger across stops
Stops to refill one trainset pack	82 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	288 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	372 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in design.toml — emitted by the osr-design Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. country-costs.toml applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (321.2 km @ €3.5 M/km)	€1.12 bn
Elevated (16.6 km @ €18 M/km)	€300 M
Elevated-interchange premium (10 sites @ €20 M)	€200 M
Civil subtotal	€1.62 bn

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
halt	11	€0.4 M	€4.4 M
standard	54	€1.5 M	€81 M
major	55	€3.0 M	€165 M
terminal	11	€2.5 M	€28 M
depot-terminal	1	€3.0 M	€3.0 M
interchange-elevated	18	€4.5 M	€81 M
Stations subtotal			€362 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	11	€3.0 M	€33 M
Depots subtotal			€58 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
metro-6car (revenue + spare + cold reserve)	273	€6.0 M	€1.64 bn

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	339.0 km × €0.015 M/km	€5.1 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€53 M
EPC integration + project management (7%)	on subtotal	€262 M

Total

Bucket	Value
Civil works	€1.62 bn
Stations	€362 M
Depots	€58 M
Rolling stock	€1.64 bn
Residual train-control wayside + charging microgrids	€58 M
EPC overhead (7%)	€262 M
CAPEX total	€4.00 bn
Per-route-km	€12 M / km

Bucket	Value
Per-capita (city pop)	€1,250 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](https://github.com/OpenSourceRail/lib/templates/country-finance.toml). Pure function of the [costs] block above + the country code — regenerate by re-running `scripts/regenerate-city.sh` `kanpur`.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–5 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 6 and runs for 20 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–5)	€288 M / yr	€90
Steady-state, low-ridership (year 6+)	€366 M / yr	€114
Steady-state, high-ridership (year 6+)	€345 M / yr	€108
Lifecycle envelope (yr 1–25, low scenario)	€8.76 bn cumulative	€2,737
Lifecycle envelope (yr 1–25, high scenario)	€8.35 bn cumulative	€2,608

Population basis: 3,200,000 (catchment per [lib/city-batches/world-sample.toml](https://github.com/OpenSourceRail/lib/city-batches/world-sample.toml)). After year 25, debt service drops to zero and only the OPEX shortfall remains — ~€93 M / yr (low) €73 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€2.40 bn	4.0%	25 y, 5 y grace	€177 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€1.00 bn	7.2%	25 y, 5 y grace	€96 M / yr
Government equity (no debt service)	15%	€600 M	—	—	—
Total	100%	€4.00 bn			€273 M / yr

During the 5-year grace period the operator pays interest only — multilateral €96 M / yr + bonds €72 M / yr = €168 M / yr total — plus the equity tranche amortised across construction (€120 M / yr × 5 yr). Principal repayment begins in year 6 on a 20-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€66 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€41 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€253 k
Traction energy (1154.9 GWh / yr)	trackside PV + Na-ion (RFC 0002) — self-generated, €0 / yr	€0 k
Labour (2,046 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€7.3 M
OPEX subtotal		€114 M / yr

Annual fleet utilisation: 245 revenue trainsets × 20.5 h/day × 365 d/yr × 35 km/h commercial × 75% revenue factor = 48.1 M train-km / yr (~196 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$230 USD (per [lib/templates/country-finance.toml](https://github.com/OpenSourceRail/lib/templates/country-finance.toml)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass — a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.35 (~\$0.38 USD)
Day pass (3 trips)	€0.90 (15 % bulk discount)
Monthly unlimited pass	€10.58 (~5 % of median monthly income)
Annual pass	€116.38 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	58.4 M	116.8 M
Farebox revenue	€21 M / yr	€41 M / yr
Farebox / OPEX recovery	18%	36%
Country policy-target recovery (diagnostic)	55%	55%
Operating shortfall (gov subsidy required)	€93 M / yr	€73 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€366 M / yr	€345 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
kanpur.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
kanpur-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
kanpur.corridor.geojson	Line polylines + stations (GeoJSON)
kanpur.stations.json	Machine-readable station list
kanpur.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug kanpur

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug kanpur \
  --sidecar .cache/osr-pipeline/rasters/kanpur.grid.json \
  --out-dir designs/.../Kampur

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../kanpur.toml \
  --out designs/.../README.md
```

`scripts/regenerate-kanpur.sh` chains steps 3 + drift tests into a single command.

south-asia/India/Lucknow

designs/south-asia/India/Lucknow/README.md

Lucknow — Urban Rail Network

Country: IN · Population: 3,500,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Lucknow rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [lucknow.corridor.geojson](#).

At a glance

Metric	Value
Lines	6
Unique stations	163
Interchange stations	18
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	31.3%
Route length (double track)	375.5 km
Revenue fleet	267 × 6-car trainsets
Spare + cold-reserve	30 × 6-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	55.9 km	25	45	W Mid E Outer
line-2	44.5 km	23	36	S Mid N Mid
line-3	52.0 km	21	41	NE Outer S Mid
line-4	51.4 km	23	41	SW Outer NE Mid
line-5	53.3 km	24	42	SE Mid W Outer
line-6	118.5 km	48	92	W Mid W Mid
Total	375.5 km	163 unique	297	

Rolling stock

Property	Value
Consist	6-car, 111 m
Max speed	100 km/h
Onboard battery	720 kWh per trainset
Seats	120 longitudinal seats
Nominal capacity (AW2)	660 pax (seated + standing, metro-6car per RFC 0008 §1)
Crush capacity (AW3)	840 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 660 AW2 passengers (metro-6car)
- Peak frequency: 12 trains/hour/direction (5-min headway)
- Peak capacity per line per direction: $660 \times 12 = 7,920$ pphpd
- Network peak throughput (all lines, both directions): $6 \text{ lines} \times 2 \text{ directions} \times 7,920 = 95,040$ passengers/hour
- Daily theoretical capacity (peak $\times$ 10): $\approx 950,400$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 109,550 - 164,325$ trips/day

Catchment

- City population: 3,500,000
- Anchor-weighted coverage: 31.3%
- Catchment population: $\approx 1,095,500$ (within ~800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	18	500 kW	3000 kWh
Major	46	400 kW	2500 kWh
Standard	88	300 kW	2000 kWh
Terminal	9	500 kW	3000 kWh
Total installed	162	63,300 kW	412,000 kWh

Aggregate station-rail charging power: 86,500 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 720 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	24.0 kWh/km	6 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	1,502 kWh	62.6 km average line length
Onboard battery coverage	0.5 $\times$ average line run	720 kWh usable pack
Average 60 s dwell charge	8.8 kWh/stop	531 kW average charger across stops
Stops to refill one trainset pack	81 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	316 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	412 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in design.toml — emitted by the osr-design Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. country-costs.toml applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (352.1 km @ €3.5 M/km)	€1.23 bn
Elevated (22.1 km @ €18 M/km)	€398 M
Elevated-interchange premium (10 sites @ €20 M)	€200 M
Civil subtotal	€1.83 bn

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
halt	2	€0.4 M	€0.8 M
standard	88	€1.5 M	€132 M
major	46	€3.0 M	€138 M
terminal	9	€2.5 M	€22 M
depot-terminal	1	€3.0 M	€3.0 M
interchange-elevated	18	€4.5 M	€81 M
Stations subtotal			€377 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	9	€3.0 M	€27 M
Depots subtotal			€52 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
metro-6car (revenue + spare + cold reserve)	297	€6.0 M	€1.78 bn

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	375.5 km × €0.015 M/km	€5.6 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€56 M
EPC integration + project management (7%)	on subtotal	€287 M

Total

Bucket	Value
Civil works	€1.83 bn
Stations	€377 M
Depots	€52 M
Rolling stock	€1.78 bn
Residual train-control wayside + charging microgrids	€61 M
EPC overhead (7%)	€287 M
CAPEX total	€4.39 bn
Per-route-km	€12 M / km
Per-capita (city pop)	€1,254 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running `scripts/regenerate-city.sh` lucknow.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–5 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 6 and runs for 20 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–5)	€316 M / yr	€90
Steady-state, low-ridership (year 6+)	€401 M / yr	€115
Steady-state, high-ridership (year 6+)	€379 M / yr	€108
Lifecycle envelope (yr 1–25, low scenario)	€9.61 bn cumulative	€2,745
Lifecycle envelope (yr 1–25, high scenario)	€9.16 bn cumulative	€2,616

Population basis: 3,500,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 25, debt service drops to zero and only the OPEX shortfall remains – ~€102 M / yr (low) €80 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€2.63 bn	4.0%	25 y, 5 y grace	€194 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€1.10 bn	7.2%	25 y, 5 y grace	€105 M / yr
Government equity (no debt service)	15%	€658 M	—	—	—
Total	100%	€4.39 bn			€299 M / yr

During the 5-year grace period the operator pays interest only – multilateral €105 M / yr + bonds €79 M / yr = €184 M / yr total – plus the equity tranche amortised across construction (€132 M / yr × 5 yr). Principal repayment begins in year 6 on a 20-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€71 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€45 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€281 k
Traction energy (1258.6 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (2,265 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€8.1 M
OPEX subtotal		€125 M / yr

Annual fleet utilisation: 267 revenue trainsets × 20.5 h/day × 365 d/yr × 35 km/h commercial × 75% revenue factor = 52.4 M train-km / yr (~196 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$230 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.35 (~\$0.38 USD)
Day pass (3 trips)	€0.90 (15 % bulk discount)
Monthly unlimited pass	€10.58 (~5 % of median monthly income)
Annual pass	€116.38 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	63.9 M	127.8 M
Farebox revenue	€23 M / yr	€45 M / yr
Farebox / OPEX recovery	18%	36%
Country policy-target recovery (diagnostic)	55%	55%
Operating shortfall (gov subsidy required)	€102 M / yr	€80 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€401 M / yr	€379 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
lucknow.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
lucknow-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
lucknow.corridor.geojson	Line polylines + stations (GeoJSON)
lucknow.stations.json	Machine-readable station list
lucknow.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug lucknow

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug lucknow \
  --sidecar .cache/osr-pipeline/rasters/lucknow.grid.json \
  --out-dir designs/.../Lucknow

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../lucknow.toml \
  --out designs/.../README.md
```

`scripts/regenerate-lucknow.sh` chains steps 3 + drift tests into a single command.

south-asia/India/Madurai

designs/south-asia/India/Madurai/README.md

Madurai — Urban Rail Network

Country: IN · Population: 1,600,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Madurai rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [madurai.corridor.geojson](#).

At a glance

Metric	Value
Lines	6
Unique stations	103
Interchange stations	20
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	58.6%
Route length (double track)	222.3 km
Revenue fleet	163 × 4-car trainsets
Spare + cold-reserve	20 × 4-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	51.9 km	21	41	NE Outer SW Outer
line-2	32.0 km	16	27	NW Mid SE Outer
line-3	26.0 km	13	23	NE Mid S Mid
line-4	25.9 km	13	21	E Mid W Mid
line-5	26.2 km	12	23	E Inner W Outer
line-6	60.3 km	29	48	W Mid W Mid
Total	222.3 km	103 unique	183	

Rolling stock

Property	Value
Consist	4-car, 75 m
Max speed	90 km/h
Onboard battery	480 kWh per trainset
Seats	80 longitudinal seats
Nominal capacity (AW2)	440 pax (seated + standing, metro-4car per RFC 0008 §1)
Crush capacity (AW3)	560 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 440 AW2 passengers (metro - 4 car)
- Peak frequency: 12 trains/hour/direction (5-min headway)
- Peak capacity per line per direction: $440 \times 12 = 5,280$ pphpd
- Network peak throughput (all lines, both directions): $6 \text{ lines} \times 2 \text{ directions} \times 5,280 = 63,360$ passengers/hour
- Daily theoretical capacity (peak $\times$ 10): $\approx 633,600$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 93,760 - 140,640$ trips/day

Catchment

- City population: 1,600,000
- Anchor-weighted coverage: 58.6%
- Catchment population: $\approx 937,600$ (within ~800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	20	500 kW	3000 kWh
Major	30	400 kW	2500 kWh
Standard	39	300 kW	2000 kWh
Terminal	9	500 kW	3000 kWh
Total installed	99	43,200 kW	280,000 kWh

Aggregate station-rail charging power: 55,750 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 480 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	16.0 kWh/km	4 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	593 kWh	37.1 km average line length
Onboard battery coverage	0.8 $\times$ average line run	480 kWh usable pack
Average 60 s dwell charge	9.0 kWh/stop	541 kW average charger across stops
Stops to refill one trainset pack	53 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	216 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	280 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in design.toml — emitted by the osr-design Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. country-costs.toml applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (207.1 km @ €3.5 M/km)	€725 M
Elevated (14.0 km @ €18 M/km)	€252 M
Elevated-interchange premium (10 sites @ €20 M)	€200 M
Civil subtotal	€1.18 bn

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
halt	5	€0.4 M	€2.0 M
standard	39	€1.5 M	€58 M
major	30	€3.0 M	€90 M
terminal	9	€2.5 M	€22 M
depot-terminal	1	€3.0 M	€3.0 M
interchange-elevated	20	€4.5 M	€90 M
Stations subtotal			€266 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	9	€3.0 M	€27 M
Depots subtotal			€52 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
metro-4car (revenue + spare + cold reserve)	183	€4.0 M	€732 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	222.3 km × €0.015 M/km	€3.3 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€39 M
EPC integration + project management (7%)	on subtotal	€159 M

Total

Bucket	Value
Civil works	€1.18 bn
Stations	€266 M
Depots	€52 M
Rolling stock	€732 M
Residual train-control wayside + charging microgrids	€42 M
EPC overhead (7%)	€159 M
CAPEX total	€2.43 bn
Per-route-km	€11 M / km
Per-capita (city pop)	€1,517 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running scripts/regenerate-city.sh madurai.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–5 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 6 and runs for 20 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–5)	€175 M / yr	€109
Steady-state, low-ridership (year 6+)	€219 M / yr	€137
Steady-state, high-ridership (year 6+)	€209 M / yr	€131
Lifecycle envelope (yr 1–25, low scenario)	€5.26 bn cumulative	€3,286
Lifecycle envelope (yr 1–25, high scenario)	€5.05 bn cumulative	€3,157

Population basis: 1,600,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 25, debt service drops to zero and only the OPEX shortfall remains – ~€54 M / yr (low) €44 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€1.46 bn	4.0%	25 y, 5 y grace	€107 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€607 M	7.2%	25 y, 5 y grace	€58 M / yr
Government equity (no debt service)	15%	€364 M	—	—	—
Total	100%	€2.43 bn			€165 M / yr

During the 5-year grace period the operator pays interest only – multilateral €58 M / yr + bonds €44 M / yr = €102 M / yr total – plus the equity tranche amortised across construction (€73 M / yr × 5 yr). Principal repayment begins in year 6 on a 20-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€29 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€30 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€166 k
Traction energy (512.3 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (1,346 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€4.8 M
OPEX subtotal		€64 M / yr

Annual fleet utilisation: 163 revenue trainsets × 20.5 h/day × 365 d/yr × 35 km/h commercial × 75% revenue factor = 32.0 M train-km / yr (~196 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$230 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.35 (~\$0.38 USD)
Day pass (3 trips)	€0.90 (15 % bulk discount)
Monthly unlimited pass	€10.58 (~5 % of median monthly income)
Annual pass	€116.38 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	29.2 M	58.4 M
Farebox revenue	€10 M / yr	€21 M / yr
Farebox / OPEX recovery	16%	32%
Country policy-target recovery (diagnostic)	55%	55%
Operating shortfall (gov subsidy required)	€54 M / yr	€44 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€219 M / yr	€209 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
madurai.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
madurai-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
madurai.corridor.geojson	Line polylines + stations (GeoJSON)
madurai.stations.json	Machine-readable station list
madurai.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug madurai

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug madurai \
  --sidecar .cache/osr-pipeline/rasters/madurai.grid.json \
  --out-dir designs/.../Madurai

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../madurai.toml \
  --out designs/.../README.md
```

`scripts/regenerate-madurai.sh` chains steps 3 + drift tests into a single command.

south-asia/India/Meerut

designs/south-asia/India/Meerut/README.md

Meerut — Urban Rail Network

Country: IN · Population: 1,600,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Meerut rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [meerut.corridor.geojson](#).

At a glance

Metric	Value
Lines	4
Unique stations	83
Interchange stations	15
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	42.9%
Route length (double track)	179.6 km
Revenue fleet	130 × 4-car trainsets
Spare + cold-reserve	16 × 4-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	42.3 km	19	35	SW Outer NE Outer
line-2	37.7 km	19	31	N Outer S Outer
line-3	28.8 km	16	24	NW Mid SE Mid
line-4	70.8 km	30	56	NW Mid NW Mid
Total	179.6 km	83 unique	146	

Rolling stock

Property	Value
Consist	4-car, 75 m
Max speed	90 km/h
Onboard battery	480 kWh per trainset
Seats	80 longitudinal seats
Nominal capacity (AW2)	440 pax (seated + standing, met ro - 4car per RFC 0008 §1)
Crush capacity (AW3)	560 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 440 AW2 passengers (met ro - 4car)

- Peak frequency: 12 trains/hour/direction (5-min headway)
- Peak capacity per line per direction: $440 \times 12 = 5,280$ pphpd
- Network peak throughput (all lines, both directions): $4 \text{ lines} \times 2 \text{ directions} \times 5,280 = 42,240$ passengers/hour
- Daily theoretical capacity (peak $\times$ 10): $\approx 422,400$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 68,640 - 102,960$ trips/day

Catchment

- City population: 1,600,000
- Anchor-weighted coverage: 42.9%
- Catchment population: $\approx 686,400$ (within ~ 800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	15	500 kW	3000 kWh
Major	20	400 kW	2500 kWh
Standard	43	300 kW	2000 kWh
Terminal	5	500 kW	3000 kWh
Total installed	84	35,900 kW	236,000 kWh

Aggregate station-rail charging power: 45,000 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 480 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	16.0 kWh/km	4 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	718 kWh	44.9 km average line length
Onboard battery coverage	0.7 $\times$ average line run	480 kWh usable pack
Average 60 s dwell charge	9.0 kWh/stop	542 kW average charger across stops
Stops to refill one trainset pack	53 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	180 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	236 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in design.toml — emitted by the osr-design Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. country-costs.toml applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (167.1 km @ €3.5 M/km)	€585 M
Elevated (12.0 km @ €18 M/km)	€216 M
Elevated-interchange premium (7 sites @ €20 M)	€140 M
Civil subtotal	€941 M

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
standard	43	€1.5 M	€64 M
major	20	€3.0 M	€60 M
terminal	5	€2.5 M	€12 M
depot-terminal	1	€3.0 M	€3.0 M
interchange-elevated	15	€4.5 M	€68 M
Stations subtotal			€208 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	5	€3.0 M	€15 M
Depots subtotal			€40 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
metro-4car (revenue + spare + cold reserve)	146	€4.0 M	€584 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	179.6 km × €0.015 M/km	€2.7 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€30 M
EPC integration + project management (7%)	on subtotal	€126 M

Total

Bucket	Value
Civil works	€941 M
Stations	€208 M
Depots	€40 M
Rolling stock	€584 M
Residual train-control wayside + charging microgrids	€33 M
EPC overhead (7%)	€126 M
CAPEX total	€1.93 bn
Per-route-km	€11 M / km
Per-capita (city pop)	€1,207 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running `scripts/regenerate-city.sh` `meerut`.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–5 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 6 and runs for 20 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–5)	€139 M / yr	€87
Steady-state, low-ridership (year 6+)	€172 M / yr	€108
Steady-state, high-ridership (year 6+)	€162 M / yr	€101
Lifecycle envelope (yr 1–25, low scenario)	€4.14 bn cumulative	€2,590
Lifecycle envelope (yr 1–25, high scenario)	€3.94 bn cumulative	€2,461

Population basis: 1,600,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 25, debt service drops to zero and only the OPEX shortfall remains – ~€41 M / yr (low) €31 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€1.16 bn	4.0%	25 y, 5 y grace	€85 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€483 M	7.2%	25 y, 5 y grace	€46 M / yr
Government equity (no debt service)	15%	€290 M	—	—	—
Total	100%	€1.93 bn			€132 M / yr

During the 5-year grace period the operator pays interest only – multilateral €46 M / yr + bonds €35 M / yr = €81 M / yr total – plus the equity tranche amortised across construction (€58 M / yr × 5 yr). Principal repayment begins in year 6 on a 20-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€23 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€24 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€134 k
Traction energy (408.5 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (1,090 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€3.9 M
OPEX subtotal		€51 M / yr

Annual fleet utilisation: 130 revenue trainsets × 20.5 h/day × 365 d/yr × 35 km/h commercial × 75% revenue factor = 25.5 M train-km / yr (~196 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$230 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.35 (~\$0.38 USD)
Day pass (3 trips)	€0.90 (15 % bulk discount)
Monthly unlimited pass	€10.58 (~5 % of median monthly income)
Annual pass	€116.38 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	29.2 M	58.4 M
Farebox revenue	€10 M / yr	€21 M / yr
Farebox / OPEX recovery	20%	40%
Country policy-target recovery (diagnostic)	55%	55%
Operating shortfall (gov subsidy required)	€41 M / yr	€31 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€172 M / yr	€162 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
meerut.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
meerut-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
meerut.corridor.geojson	Line polylines + stations (GeoJSON)
meerut.stations.json	Machine-readable station list
meerut.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug meerut

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug meerut \
  --sidecar .cache/osr-pipeline/rasters/meerut.grid.json \
  --out-dir designs/.../Meerut

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../meerut.toml \
  --out designs/.../README.md
```

`scripts/regenerate-meerut.sh` chains steps 3 + drift tests into a single command.

south-asia/India/Patna

designs/south-asia/India/Patna/README.md

Patna — Urban Rail Network

Country: IN · Population: 2,520,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Patna rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [patna.corridor.geojson](#).

At a glance

Metric	Value
Lines	5
Unique stations	83
Interchange stations	16
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	49.7%
Route length (double track)	185.3 km
Revenue fleet	136 × 4-car trainsets
Spare + cold-reserve	16 × 4-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	26.0 km	14	21	SE Mid SW Mid
line-2	34.5 km	12	28	W Mid NE Outer
line-3	30.5 km	14	26	SW Mid NE Outer
line-4	33.3 km	16	28	NW Outer SE Mid
line-5	61.0 km	28	49	W Mid W Mid
Total	185.3 km	83 unique	152	

Rolling stock

Property	Value
Consist	4-car, 75 m
Max speed	90 km/h
Onboard battery	480 kWh per trainset
Seats	80 longitudinal seats
Nominal capacity (AW2)	440 pax (seated + standing, met ro - 4 car per RFC 0008 §1)
Crush capacity (AW3)	560 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 440 AW2 passengers (met ro - 4 car)

- Peak frequency: 12 trains/hour/direction (5-min headway)
- Peak capacity per line per direction: $440 \times 12 = 5,280$ pphpd
- Network peak throughput (all lines, both directions): $5 \text{ lines} \times 2 \text{ directions} \times 5,280 = 52,800$ passengers/hour
- Daily theoretical capacity (peak $\times$ 10): $\approx 528,000$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 125,244 - 187,866$ trips/day

Catchment

- City population: 2,520,000
- Anchor-weighted coverage: 49.7%
- Catchment population: $\approx 1,252,440$ (within ~800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	16	500 kW	3000 kWh
Major	19	400 kW	2500 kWh
Standard	34	300 kW	2000 kWh
Terminal	7	500 kW	3000 kWh
Total installed	77	34,300 kW	224,500 kWh

Aggregate station-rail charging power: 44,250 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 480 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	16.0 kWh/km	4 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	593 kWh	37.1 km average line length
Onboard battery coverage	0.8 $\times$ average line run	480 kWh usable pack
Average 60 s dwell charge	8.9 kWh/stop	533 kW average charger across stops
Stops to refill one trainset pack	54 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	172 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	224 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the `[costs]` block in `design.toml` — emitted by the `osr-design` Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. `country-costs.toml` applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (154.0 km @ €3.5 M/km)	€539 M
Elevated (30.1 km @ €18 M/km)	€542 M
Elevated-interchange premium (8 sites @ €20 M)	€160 M
Civil subtotal	€1.24 bn

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
halt	7	€0.4 M	€2.8 M
standard	34	€1.5 M	€51 M
major	19	€3.0 M	€57 M
terminal	7	€2.5 M	€18 M
depot-terminal	1	€3.0 M	€3.0 M
interchange-elevated	16	€4.5 M	€72 M
Stations subtotal			€203 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	7	€3.0 M	€21 M
Depots subtotal			€46 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
metro-4car (revenue + spare + cold reserve)	152	€4.0 M	€608 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	185.3 km × €0.015 M/km	€2.8 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€30 M
EPC integration + project management (7%)	on subtotal	€149 M

Total

Bucket	Value
Civil works	€1.24 bn
Stations	€203 M
Depots	€46 M
Rolling stock	€608 M
Residual train-control wayside + charging microgrids	€33 M
EPC overhead (7%)	€149 M
CAPEX total	€2.28 bn
Per-route-km	€12 M / km
Per-capita (city pop)	€905 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running `scripts/regenerate-city.sh` `patna`.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–5 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 6 and runs for 20 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–5)	€164 M / yr	€65
Steady-state, low-ridership (year 6+)	€197 M / yr	€78
Steady-state, high-ridership (year 6+)	€181 M / yr	€72
Lifecycle envelope (yr 1–25, low scenario)	€4.77 bn cumulative	€1,892
Lifecycle envelope (yr 1–25, high scenario)	€4.44 bn cumulative	€1,764

Population basis: 2,520,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 25, debt service drops to zero and only the OPEX shortfall remains – ~€42 M / yr (low) €26 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€1.37 bn	4.0%	25 y, 5 y grace	€101 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€570 M	7.2%	25 y, 5 y grace	€55 M / yr
Government equity (no debt service)	15%	€342 M	—	—	—
Total	100%	€2.28 bn			€155 M / yr

During the 5-year grace period the operator pays interest only – multilateral €55 M / yr + bonds €41 M / yr = €96 M / yr total – plus the equity tranche amortised across construction (€68 M / yr × 5 yr). Principal repayment begins in year 6 on a 20-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€24 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€30 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€138 k
Traction energy (427.4 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (1,124 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€4.0 M
OPEX subtotal		€58 M / yr

Annual fleet utilisation: 136 revenue trainsets × 20.5 h/day × 365 d/yr × 35 km/h commercial × 75% revenue factor = 26.7 M train-km / yr (~196 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$230 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.35 (~\$0.38 USD)
Day pass (3 trips)	€0.90 (15 % bulk discount)
Monthly unlimited pass	€10.58 (~5 % of median monthly income)
Annual pass	€116.38 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	46.0 M	92.0 M
Farebox revenue	€16 M / yr	€32 M / yr
Farebox / OPEX recovery	28%	56%
Country policy-target recovery (diagnostic)	55%	55%
Operating shortfall (gov subsidy required)	€42 M / yr	€26 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€197 M / yr	€181 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
patna.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
patna-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
patna.corridor.geojson	Line polylines + stations (GeoJSON)
patna.stations.json	Machine-readable station list
patna.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug patna

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug patna \
  --sidecar .cache/osr-pipeline/rasters/patna.grid.json \
  --out-dir designs/.../Patna

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../patna.toml \
  --out designs/.../README.md
```

`scripts/regenerate-patna.sh` chains steps 3 + drift tests into a single command.

south-asia/India/Raipur

designs/south-asia/India/Raipur/README.md

Raipur — Urban Rail Network

Country: IN · Population: 1,500,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Raipur rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [raipur.corridor.geojson](#).

At a glance

Metric	Value
Lines	5
Unique stations	87
Interchange stations	12
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	42.9%
Route length (double track)	185.2 km
Revenue fleet	135 × 4-car trainsets
Spare + cold-reserve	16 × 4-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	31.3 km	17	26	W Outer E Outer
line-2	30.3 km	16	25	N Outer S Mid
line-3	25.3 km	13	21	NE Outer SW Mid
line-4	23.7 km	14	20	SE Outer W Inner
line-5	74.5 km	28	59	NE Outer NE Outer
Total	185.2 km	87 unique	151	

Rolling stock

Property	Value
Consist	4-car, 75 m
Max speed	90 km/h
Onboard battery	480 kWh per trainset
Seats	80 longitudinal seats
Nominal capacity (AW2)	440 pax (seated + standing, met ro - 4 car per RFC 0008 §1)
Crush capacity (AW3)	560 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 440 AW2 passengers (met ro - 4 car)

- Peak frequency: 12 trains/hour/direction (5-min headway)
- Peak capacity per line per direction: $440 \times 12 = 5,280$ pphpd
- Network peak throughput (all lines, both directions): $5 \text{ lines} \times 2 \text{ directions} \times 5,280 = 52,800$ passengers/hour
- Daily theoretical capacity (peak $\times$ 10): $\approx 528,000$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 64,350 - 96,525$ trips/day

Catchment

- City population: 1,500,000
- Anchor-weighted coverage: 42.9%
- Catchment population: $\approx 643,500$ (within ~800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	12	500 kW	3000 kWh
Major	26	400 kW	2500 kWh
Standard	39	300 kW	2000 kWh
Terminal	7	500 kW	3000 kWh
Total installed	85	36,600 kW	240,000 kWh

Aggregate station-rail charging power: 47,250 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 480 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	16.0 kWh/km	4 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	593 kWh	37.0 km average line length
Onboard battery coverage	0.8 $\times$ average line run	480 kWh usable pack
Average 60 s dwell charge	9.1 kWh/stop	543 kW average charger across stops
Stops to refill one trainset pack	53 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	183 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	240 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in design.toml — emitted by the osr-design Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. country-costs.toml applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (163.5 km @ €3.5 M/km)	€572 M
Elevated (21.0 km @ €18 M/km)	€378 M
Elevated-interchange premium (7 sites @ €20 M)	€140 M
Civil subtotal	€1.09 bn

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
halt	3	€0.4 M	€1.2 M
standard	39	€1.5 M	€58 M
major	26	€3.0 M	€78 M
terminal	7	€2.5 M	€18 M
depot-terminal	1	€3.0 M	€3.0 M
interchange-elevated	12	€4.5 M	€54 M
Stations subtotal			€212 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	7	€3.0 M	€21 M
Depots subtotal			€46 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
metro-4car (revenue + spare + cold reserve)	151	€4.0 M	€604 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	185.2 km × €0.015 M/km	€2.8 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€31 M
EPC integration + project management (7%)	on subtotal	€139 M

Total

Bucket	Value
Civil works	€1.09 bn
Stations	€212 M
Depots	€46 M
Rolling stock	€604 M
Residual train-control wayside + charging microgrids	€34 M
EPC overhead (7%)	€139 M
CAPEX total	€2.13 bn
Per-route-km	€11 M / km
Per-capita (city pop)	€1,417 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running `scripts/regenerate-city.sh raipur`.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–5 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 6 and runs for 20 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–5)	€153 M / yr	€102
Steady-state, low-ridership (year 6+)	€190 M / yr	€127
Steady-state, high-ridership (year 6+)	€181 M / yr	€121
Lifecycle envelope (yr 1–25, low scenario)	€4.57 bn cumulative	€3,049
Lifecycle envelope (yr 1–25, high scenario)	€4.38 bn cumulative	€2,921

Population basis: 1,500,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 25, debt service drops to zero and only the OPEX shortfall remains – ~€46 M / yr (low) €36 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€1.28 bn	4.0%	25 y, 5 y grace	€94 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€532 M	7.2%	25 y, 5 y grace	€51 M / yr
Government equity (no debt service)	15%	€319 M	—	—	—
Total	100%	€2.13 bn			€145 M / yr

During the 5-year grace period the operator pays interest only – multilateral €51 M / yr + bonds €38 M / yr = €89 M / yr total – plus the equity tranche amortised across construction (€64 M / yr × 5 yr). Principal repayment begins in year 6 on a 20-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€24 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€27 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€138 k
Traction energy (424.3 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (1,123 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€4.0 M
OPEX subtotal		€55 M / yr

Annual fleet utilisation: 135 revenue trainsets × 20.5 h/day × 365 d/yr × 35 km/h commercial × 75% revenue factor = 26.5 M train-km / yr (~196 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$230 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.35 (~\$0.38 USD)
Day pass (3 trips)	€0.90 (15 % bulk discount)
Monthly unlimited pass	€10.58 (~5 % of median monthly income)
Annual pass	€116.38 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	27.4 M	54.8 M
Farebox revenue	€9.7 M / yr	€19 M / yr
Farebox / OPEX recovery	17%	35%
Country policy-target recovery (diagnostic)	55%	55%
Operating shortfall (gov subsidy required)	€46 M / yr	€36 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€190 M / yr	€181 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
raipur.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
raipur-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
raipur.corridor.geojson	Line polylines + stations (GeoJSON)
raipur.stations.json	Machine-readable station list
raipur.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug raipur

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug raipur \
  --sidecar .cache/osr-pipeline/rasters/raipur.grid.json \
  --out-dir designs/.../Raipur

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../raipur.toml \
  --out designs/.../README.md
```

`scripts/regenerate-raipur.sh` chains steps 3 + drift tests into a single command.

south-asia/India/Rajkot

designs/south-asia/India/Rajkot/README.md

Rajkot — Urban Rail Network

Country: IN · Population: 1,800,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Rajkot rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [rajkot.corridor.geojson](#).

At a glance

Metric	Value
Lines	5
Unique stations	75
Interchange stations	16
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	57.9%
Route length (double track)	142.9 km
Revenue fleet	108 × 4-car trainsets
Spare + cold-reserve	13 × 4-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	24.8 km	13	21	N Mid S Outer
line-2	30.6 km	14	26	SE Outer NW Outer
line-3	22.2 km	13	19	SW Outer NE Mid
line-4	26.2 km	13	23	N Outer SW Mid
line-5	39.1 km	23	32	NW Mid NW Mid
Total	142.9 km	75 unique	121	

Rolling stock

Property	Value
Consist	4-car, 75 m
Max speed	90 km/h
Onboard battery	480 kWh per trainset
Seats	80 longitudinal seats
Nominal capacity (AW2)	440 pax (seated + standing, met ro - 4 car per RFC 0008 §1)
Crush capacity (AW3)	560 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 440 AW2 passengers (met ro - 4 car)

- Peak frequency: 12 trains/hour/direction (5-min headway)
- Peak capacity per line per direction: $440 \times 12 = 5,280$ pphpd
- Network peak throughput (all lines, both directions): $5 \text{ lines} \times 2 \text{ directions} \times 5,280 = 52,800$ passengers/hour
- Daily theoretical capacity (peak $\times$ 10): $\approx 528,000$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 104,219 - 156,329$ trips/day

Catchment

- City population: 1,800,000
- Anchor-weighted coverage: 57.9%
- Catchment population: $\approx 1,042,199$ (within ~800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	16	500 kW	3000 kWh
Major	27	400 kW	2500 kWh
Standard	23	300 kW	2000 kWh
Terminal	7	500 kW	3000 kWh
Total installed	74	34,200 kW	222,500 kWh

Aggregate station-rail charging power: 41,500 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 480 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	16.0 kWh/km	4 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	457 kWh	28.6 km average line length
Onboard battery coverage	1.0 $\times$ average line run	480 kWh usable pack
Average 60 s dwell charge	9.2 kWh/stop	553 kW average charger across stops
Stops to refill one trainset pack	52 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	171 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	222 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the `[costs]` block in `design.toml` — emitted by the `osr-design` Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. `country-costs.toml` applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (126.2 km @ €3.5 M/km)	€442 M
Elevated (15.7 km @ €18 M/km)	€283 M
Elevated-interchange premium (6 sites @ €20 M)	€120 M
Civil subtotal	€845 M

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
halt	2	€0.4 M	€0.8 M
standard	23	€1.5 M	€34 M
major	27	€3.0 M	€81 M
terminal	7	€2.5 M	€18 M
depot-terminal	1	€3.0 M	€3.0 M
interchange	3	€4.5 M	€14 M
interchange-elevated	13	€4.5 M	€58 M
Stations subtotal			€209 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	7	€3.0 M	€21 M
Depots subtotal			€46 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
metro-4car (revenue + spare + cold reserve)	121	€4.0 M	€484 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	142.9 km × €0.015 M/km	€2.1 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€30 M
EPC integration + project management (7%)	on subtotal	€113 M

Total

Bucket	Value
Civil works	€845 M
Stations	€209 M
Depots	€46 M
Rolling stock	€484 M
Residual train-control wayside + charging microgrids	€32 M
EPC overhead (7%)	€113 M
CAPEX total	€1.73 bn
Per-route-km	€12 M / km
Per-capita (city pop)	€961 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running scripts/regenerate-city.sh rajkot.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–5 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 6 and runs for 20 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–5)	€124 M / yr	€69
Steady-state, low-ridership (year 6+)	€151 M / yr	€84
Steady-state, high-ridership (year 6+)	€139 M / yr	€77
Lifecycle envelope (yr 1–25, low scenario)	€3.64 bn cumulative	€2,021
Lifecycle envelope (yr 1–25, high scenario)	€3.41 bn cumulative	€1,892

Population basis: 1,800,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 25, debt service drops to zero and only the OPEX shortfall remains – ~€33 M / yr (low) €21 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€1.04 bn	4.0%	25 y, 5 y grace	€76 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€432 M	7.2%	25 y, 5 y grace	€41 M / yr
Government equity (no debt service)	15%	€259 M	—	—	—
Total	100%	€1.73 bn			€118 M / yr

During the 5-year grace period the operator pays interest only – multilateral €41 M / yr + bonds €31 M / yr = €73 M / yr total – plus the equity tranche amortised across construction (€52 M / yr × 5 yr). Principal repayment begins in year 6 on a 20-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€19 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€22 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€106 k
Traction energy (339.4 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (869 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€3.1 M
OPEX subtotal		€45 M / yr

Annual fleet utilisation: 108 revenue trainsets × 20.5 h/day × 365 d/yr × 35 km/h commercial × 75% revenue factor = 21.2 M train-km / yr (~196 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$230 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.35 (~\$0.38 USD)
Day pass (3 trips)	€0.90 (15 % bulk discount)
Monthly unlimited pass	€10.58 (~5 % of median monthly income)
Annual pass	€116.38 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	32.9 M	65.7 M
Farebox revenue	€12 M / yr	€23 M / yr
Farebox / OPEX recovery	26%	52%
Country policy-target recovery (diagnostic)	55%	55%
Operating shortfall (gov subsidy required)	€33 M / yr	€21 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€151 M / yr	€139 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
rajkot.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
rajkot-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
rajkot.corridor.geojson	Line polylines + stations (GeoJSON)
rajkot.stations.json	Machine-readable station list
rajkot.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug rajkot

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug rajkot \
  --sidecar .cache/osr-pipeline/rasters/rajkot.grid.json \
  --out-dir designs/.../Rajkot

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../rajkot.toml \
  --out designs/.../README.md
```

`scripts/regenerate-rajkot.sh` chains steps 3 + drift tests into a single command.

south-asia/India/Ranchi

designs/south-asia/India/Ranchi/README.md

Ranchi — Urban Rail Network

Country: IN · Population: 1,400,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Ranchi rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [ranchi.corridor.geojson](#).

At a glance

Metric	Value
Lines	6
Unique stations	95
Interchange stations	16
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	49.5%
Route length (double track)	215.8 km
Revenue fleet	159 × 4-car trainsets
Spare + cold-reserve	20 × 4-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	33.7 km	14	28	NE Outer S Outer
line-2	29.9 km	15	25	W Mid E Outer
line-3	26.7 km	14	23	SW Outer N Mid
line-4	25.0 km	10	21	NE Mid W Outer
line-5	27.6 km	14	24	E Mid W Outer
line-6	72.8 km	29	58	NW Mid NW Mid
Total	215.8 km	95 unique	179	

Rolling stock

Property	Value
Consist	4-car, 75 m
Max speed	90 km/h
Onboard battery	480 kWh per trainset
Seats	80 longitudinal seats
Nominal capacity (AW2)	440 pax (seated + standing, metro-4car per RFC 0008 §1)
Crush capacity (AW3)	560 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 440 AW2 passengers (metro - 4 car)
- Peak frequency: 12 trains/hour/direction (5-min headway)
- Peak capacity per line per direction: $440 \times 12 = 5,280$ pphpd
- Network peak throughput (all lines, both directions): $6 \text{ lines} \times 2 \text{ directions} \times 5,280 = 63,360$ passengers/hour
- Daily theoretical capacity (peak $\times$ 10): $\approx 633,600$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 69,300 - 103,950$ trips/day

Catchment

- City population: 1,400,000
- Anchor-weighted coverage: 49.5%
- Catchment population: $\approx 693,000$ (within ~ 800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	16	500 kW	3000 kWh
Major	23	400 kW	2500 kWh
Standard	41	300 kW	2000 kWh
Terminal	9	500 kW	3000 kWh
Total installed	90	39,000 kW	254,500 kWh

Aggregate station-rail charging power: 51,500 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 480 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	16.0 kWh/km	4 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	575 kWh	36.0 km average line length
Onboard battery coverage	0.8 $\times$ average line run	480 kWh usable pack
Average 60 s dwell charge	9.0 kWh/stop	542 kW average charger across stops
Stops to refill one trainset pack	53 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	195 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	254 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in design.toml — emitted by the osr-design Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. country-costs.toml applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (201.9 km @ €3.5 M/km)	€707 M
Elevated (12.8 km @ €18 M/km)	€230 M
Elevated-interchange premium (9 sites @ €20 M)	€180 M
Civil subtotal	€1.12 bn

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
halt	6	€0.4 M	€2.4 M
standard	41	€1.5 M	€62 M
major	23	€3.0 M	€69 M
terminal	9	€2.5 M	€22 M
depot-terminal	1	€3.0 M	€3.0 M
interchange-elevated	16	€4.5 M	€72 M
Stations subtotal			€230 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	9	€3.0 M	€27 M
Depots subtotal			€52 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
metro-4car (revenue + spare + cold reserve)	179	€4.0 M	€716 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	215.8 km × €0.015 M/km	€3.2 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€34 M
EPC integration + project management (7%)	on subtotal	€151 M

Total

Bucket	Value
Civil works	€1.12 bn
Stations	€230 M
Depots	€52 M
Rolling stock	€716 M
Residual train-control wayside + charging microgrids	€37 M
EPC overhead (7%)	€151 M
CAPEX total	€2.30 bn
Per-route-km	€11 M / km
Per-capita (city pop)	€1,645 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running `scripts/regenerate-city.sh ranchi`.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–5 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 6 and runs for 20 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–5)	€166 M / yr	€118
Steady-state, low-ridership (year 6+)	€209 M / yr	€149
Steady-state, high-ridership (year 6+)	€200 M / yr	€143
Lifecycle envelope (yr 1–25, low scenario)	€5.01 bn cumulative	€3,582
Lifecycle envelope (yr 1–25, high scenario)	€4.83 bn cumulative	€3,453

Population basis: 1,400,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 25, debt service drops to zero and only the OPEX shortfall remains – ~€52 M / yr (low) €43 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€1.38 bn	4.0%	25 y, 5 y grace	€102 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€576 M	7.2%	25 y, 5 y grace	€55 M / yr
Government equity (no debt service)	15%	€345 M	—	—	—
Total	100%	€2.30 bn			€157 M / yr

During the 5-year grace period the operator pays interest only – multilateral €55 M / yr + bonds €41 M / yr = €97 M / yr total – plus the equity tranche amortised across construction (€69 M / yr × 5 yr). Principal repayment begins in year 6 on a 20-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€29 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€28 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€161 k
Traction energy (499.7 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (1,307 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€4.6 M
OPEX subtotal		€61 M / yr

Annual fleet utilisation: 159 revenue trainsets × 20.5 h/day × 365 d/yr × 35 km/h commercial × 75% revenue factor = 31.2 M train-km / yr (~196 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$230 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.35 (~\$0.38 USD)
Day pass (3 trips)	€0.90 (15 % bulk discount)
Monthly unlimited pass	€10.58 (~5 % of median monthly income)
Annual pass	€116.38 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	25.6 M	51.1 M
Farebox revenue	€9.0 M / yr	€18 M / yr
Farebox / OPEX recovery	15%	29%
Country policy-target recovery (diagnostic)	55%	55%
Operating shortfall (gov subsidy required)	€52 M / yr	€43 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€209 M / yr	€200 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
ranchi.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
ranchi-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
ranchi.corridor.geojson	Line polylines + stations (GeoJSON)
ranchi.stations.json	Machine-readable station list
ranchi.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug ranchi

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug ranchi \
  --sidecar .cache/osr-pipeline/rasters/ranchi.grid.json \
  --out-dir designs/.../Ranchi

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../ranchi.toml \
  --out designs/.../README.md
```

`scripts/regenerate-ranchi.sh` chains steps 3 + drift tests into a single command.

south-asia/India/Vadodara

designs/south-asia/India/Vadodara/README.md

Vadodara — Urban Rail Network

Country: IN · Population: 2,200,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Vadodara rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [vadodara.corridor.geojson](#).

At a glance

Metric	Value
Lines	5
Unique stations	88
Interchange stations	18
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	47.8%
Route length (double track)	163.5 km
Revenue fleet	122 × 4-car trainsets
Spare + cold-reserve	15 × 4-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	40.8 km	20	34	NE Outer SW Outer
line-2	27.9 km	15	24	S Mid NW Outer
line-3	17.4 km	10	16	E Mid SW Inner
line-4	23.1 km	12	20	E Outer NW Inner
line-5	54.2 km	32	43	W Inner W Inner
Total	163.5 km	88 unique	137	

Rolling stock

Property	Value
Consist	4-car, 75 m
Max speed	90 km/h
Onboard battery	480 kWh per trainset
Seats	80 longitudinal seats
Nominal capacity (AW2)	440 pax (seated + standing, met ro - 4 car per RFC 0008 §1)
Crush capacity (AW3)	560 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 440 AW2 passengers (met ro - 4 car)

- Peak frequency: 12 trains/hour/direction (5-min headway)
- Peak capacity per line per direction: $440 \times 12 = 5,280$ pphpd
- Network peak throughput (all lines, both directions): $5 \text{ lines} \times 2 \text{ directions} \times 5,280 = 52,800$ passengers/hour
- Daily theoretical capacity (peak $\times$ 10): $\approx 528,000$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 105,160 - 157,740$ trips/day

Catchment

- City population: 2,200,000
- Anchor-weighted coverage: 47.8%
- Catchment population: $\approx 1,051,600$ (within ~ 800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	18	500 kW	3000 kWh
Major	37	400 kW	2500 kWh
Standard	25	300 kW	2000 kWh
Terminal	7	500 kW	3000 kWh
Total installed	88	39,800 kW	257,500 kWh

Aggregate station-rail charging power: 48,250 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 480 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	16.0 kWh/km	4 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	523 kWh	32.7 km average line length
Onboard battery coverage	0.9 $\times$ average line run	480 kWh usable pack
Average 60 s dwell charge	9.1 kWh/stop	548 kW average charger across stops
Stops to refill one trainset pack	53 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	199 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	258 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in design.toml — emitted by the osr-design Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. country-costs.toml applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (151.7 km @ €3.5 M/km)	€531 M
Elevated (11.1 km @ €18 M/km)	€200 M
Elevated-interchange premium (9 sites @ €20 M)	€180 M
Civil subtotal	€911 M

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
halt	1	€0.4 M	€0.4 M
standard	25	€1.5 M	€38 M
major	37	€3.0 M	€111 M
terminal	7	€2.5 M	€18 M
depot-terminal	1	€3.0 M	€3.0 M
interchange-elevated	18	€4.5 M	€81 M
Stations subtotal			€250 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	7	€3.0 M	€21 M
Depots subtotal			€46 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY – never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
metro-4car (revenue + spare + cold reserve)	137	€4.0 M	€548 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	163.5 km × €0.015 M/km	€2.4 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€36 M
EPC integration + project management (7%)	on subtotal	€126 M

Total

Bucket	Value
Civil works	€911 M
Stations	€250 M
Depots	€46 M
Rolling stock	€548 M
Residual train-control wayside + charging microgrids	€38 M
EPC overhead (7%)	€126 M
CAPEX total	€1.92 bn
Per-route-km	€12 M / km
Per-capita (city pop)	€872 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running `scripts/regenerate-city.sh` `vadodara`.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–5 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 6 and runs for 20 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–5)	€138 M / yr	€63
Steady-state, low-ridership (year 6+)	€166 M / yr	€76
Steady-state, high-ridership (year 6+)	€152 M / yr	€69
Lifecycle envelope (yr 1–25, low scenario)	€4.02 bn cumulative	€1,825
Lifecycle envelope (yr 1–25, high scenario)	€3.73 bn cumulative	€1,696

Population basis: 2,200,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 25, debt service drops to zero and only the OPEX shortfall remains – ~€36 M / yr (low) €21 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€1.15 bn	4.0%	25 y, 5 y grace	€85 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€480 M	7.2%	25 y, 5 y grace	€46 M / yr
Government equity (no debt service)	15%	€288 M	—	—	—
Total	100%	€1.92 bn			€131 M / yr

During the 5-year grace period the operator pays interest only – multilateral €46 M / yr + bonds €35 M / yr = €81 M / yr total – plus the equity tranche amortised across construction (€58 M / yr × 5 yr). Principal repayment begins in year 6 on a 20-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€22 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€24 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€122 k
Traction energy (383.4 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (993 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€3.5 M
OPEX subtotal		€50 M / yr

Annual fleet utilisation: 122 revenue trainsets × 20.5 h/day × 365 d/yr × 35 km/h commercial × 75% revenue factor = 24.0 M train-km / yr (~196 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$230 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.35 (~\$0.38 USD)
Day pass (3 trips)	€0.90 (15 % bulk discount)
Monthly unlimited pass	€10.58 (~5 % of median monthly income)
Annual pass	€116.38 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	40.1 M	80.3 M
Farebox revenue	€14 M / yr	€28 M / yr
Farebox / OPEX recovery	28%	57%
Country policy-target recovery (diagnostic)	55%	55%
Operating shortfall (gov subsidy required)	€36 M / yr	€21 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€166 M / yr	€152 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
vadodara.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
vadodara-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
vadodara.corridor.geojson	Line polylines + stations (GeoJSON)
vadodara.stations.json	Machine-readable station list
vadodara.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug vadodara

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug vadodara \
  --sidecar .cache/osr-pipeline/rasters/vadodara.grid.json \
  --out-dir designs/.../Vadodara

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../vadodara.toml \
  --out designs/.../README.md
```

`scripts/regenerate-vadodara.sh` chains steps 3 + drift tests into a single command.

south-asia/India/Varanasi

designs/south-asia/India/Varanasi/README.md

Varanasi — Urban Rail Network

Country: IN · Population: 1,500,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Varanasi rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [varanasi.corridor.geojson](#).

At a glance

Metric	Value
Lines	5
Unique stations	99
Interchange stations	14
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	46.7%
Route length (double track)	202.0 km
Revenue fleet	147 × 4-car trainsets
Spare + cold-reserve	18 × 4-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	41.5 km	20	34	NW Outer SE Mid
line-2	35.2 km	17	29	E Outer W Outer
line-3	31.6 km	17	26	NE Outer SW Mid
line-4	26.6 km	13	23	NW Inner S Outer
line-5	67.1 km	33	53	NW Mid NW Mid
Total	202.0 km	99 unique	165	

Rolling stock

Property	Value
Consist	4-car, 75 m
Max speed	90 km/h
Onboard battery	480 kWh per trainset
Seats	80 longitudinal seats
Nominal capacity (AW2)	440 pax (seated + standing, met ro - 4 car per RFC 0008 §1)
Crush capacity (AW3)	560 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 440 AW2 passengers (met ro - 4 car)

- Peak frequency: 12 trains/hour/direction (5-min headway)
- Peak capacity per line per direction: $440 \times 12 = 5,280$ pphpd
- Network peak throughput (all lines, both directions): $5 \text{ lines} \times 2 \text{ directions} \times 5,280 = 52,800$ passengers/hour
- Daily theoretical capacity (peak $\times$ 10): $\approx 528,000$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 70,050 - 105,075$ trips/day

Catchment

- City population: 1,500,000
- Anchor-weighted coverage: 46.7%
- Catchment population: $\approx 700,500$ (within ~ 800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	14	500 kW	3000 kWh
Major	40	400 kW	2500 kWh
Standard	35	300 kW	2000 kWh
Terminal	7	500 kW	3000 kWh
Total installed	97	42,000 kW	273,000 kWh

Aggregate station-rail charging power: 53,250 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 480 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	16.0 kWh/km	4 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	646 kWh	40.4 km average line length
Onboard battery coverage	0.7 $\times$ average line run	480 kWh usable pack
Average 60 s dwell charge	9.0 kWh/stop	538 kW average charger across stops
Stops to refill one trainset pack	54 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	210 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	273 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in design.toml — emitted by the osr-design Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. country-costs.toml applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (170.1 km @ €3.5 M/km)	€595 M
Elevated (31.0 km @ €18 M/km)	€559 M
Elevated-interchange premium (6 sites @ €20 M)	€120 M
Civil subtotal	€1.27 bn

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
halt	3	€0.4 M	€1.2 M
standard	35	€1.5 M	€52 M
major	40	€3.0 M	€120 M
terminal	7	€2.5 M	€18 M
depot-terminal	1	€3.0 M	€3.0 M
interchange	2	€4.5 M	€9.0 M
interchange-elevated	12	€4.5 M	€54 M
Stations subtotal			€257 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	7	€3.0 M	€21 M
Depots subtotal			€46 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
metro-4car (revenue + spare + cold reserve)	165	€4.0 M	€660 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	202.0 km × €0.015 M/km	€3.0 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€37 M
EPC integration + project management (7%)	on subtotal	€159 M

Total

Bucket	Value
Civil works	€1.27 bn
Stations	€257 M
Depots	€46 M
Rolling stock	€660 M
Residual train-control wayside + charging microgrids	€40 M
EPC overhead (7%)	€159 M
CAPEX total	€2.44 bn
Per-route-km	€12 M / km
Per-capita (city pop)	€1,625 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running `scripts/regenerate-city.sh varanasi`.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–5 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 6 and runs for 20 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–5)	€175 M / yr	€117
Steady-state, low-ridership (year 6+)	€219 M / yr	€146
Steady-state, high-ridership (year 6+)	€209 M / yr	€139
Lifecycle envelope (yr 1–25, low scenario)	€5.25 bn cumulative	€3,502
Lifecycle envelope (yr 1–25, high scenario)	€5.06 bn cumulative	€3,373

Population basis: 1,500,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 25, debt service drops to zero and only the OPEX shortfall remains – ~€53 M / yr (low) €43 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€1.46 bn	4.0%	25 y, 5 y grace	€108 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€609 M	7.2%	25 y, 5 y grace	€58 M / yr
Government equity (no debt service)	15%	€366 M	—	—	—
Total	100%	€2.44 bn			€166 M / yr

During the 5-year grace period the operator pays interest only – multilateral €58 M / yr + bonds €44 M / yr = €102 M / yr total – plus the equity tranche amortised across construction (€73 M / yr × 5 yr). Principal repayment begins in year 6 on a 20-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€26 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€32 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€151 k
Traction energy (462.0 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (1,224 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€4.4 M
OPEX subtotal		€62 M / yr

Annual fleet utilisation: 147 revenue trainsets × 20.5 h/day × 365 d/yr × 35 km/h commercial × 75% revenue factor = 28.9 M train-km / yr (~196 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$230 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.35 (~\$0.38 USD)
Day pass (3 trips)	€0.90 (15 % bulk discount)
Monthly unlimited pass	€10.58 (~5 % of median monthly income)
Annual pass	€116.38 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	27.4 M	54.8 M
Farebox revenue	€9.7 M / yr	€19 M / yr
Farebox / OPEX recovery	15%	31%
Country policy-target recovery (diagnostic)	55%	55%
Operating shortfall (gov subsidy required)	€53 M / yr	€43 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€219 M / yr	€209 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

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varanasi.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug varanasi

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug varanasi \
  --sidecar .cache/osr-pipeline/rasters/varanasi.grid.json \
  --out-dir designs/.../Varanasi

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../varanasi.toml \
  --out designs/.../README.md
```

`scripts/regenerate-varanasi.sh` chains steps 3 + drift tests into a single command.

south-asia/India/Vijayawada

designs/south-asia/India/Vijayawada/README.md

Vijayawada — Urban Rail Network

Country: IN · Population: 1,500,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Vijayawada rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [vijayawada.corridor.geojson](#).

At a glance

Metric	Value
Lines	6
Unique stations	93
Interchange stations	24
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	58.0%
Route length (double track)	225.1 km
Revenue fleet	164 × 4-car trainsets
Spare + cold-reserve	20 × 4-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	41.4 km	20	34	NW Outer SE Outer
line-2	37.1 km	15	30	W Outer E Outer
line-3	28.6 km	13	24	SW Mid NE Mid
line-4	35.4 km	12	29	S Outer NW Outer
line-5	21.5 km	9	18	W Outer E Mid
line-6	61.1 km	25	49	NW Mid NW Mid
Total	225.1 km	93 unique	184	

Rolling stock

Property	Value
Consist	4-car, 75 m
Max speed	90 km/h
Onboard battery	480 kWh per trainset
Seats	80 longitudinal seats
Nominal capacity (AW2)	440 pax (seated + standing, metro-4car per RFC 0008 §1)
Crush capacity (AW3)	560 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 440 AW2 passengers (metro - 4 car)
- Peak frequency: 12 trains/hour/direction (5-min headway)
- Peak capacity per line per direction: $440 \times 12 = 5,280$ pphpd
- Network peak throughput (all lines, both directions): $6 \text{ lines} \times 2 \text{ directions} \times 5,280 = 63,360$ passengers/hour
- Daily theoretical capacity (peak $\times$ 10): $\approx 633,600$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 86,999 - 130,499$ trips/day

Catchment

- City population: 1,500,000
- Anchor-weighted coverage: 58.0%
- Catchment population: $\approx 869,999$ (within ~800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	24	500 kW	3000 kWh
Major	12	400 kW	2500 kWh
Standard	35	300 kW	2000 kWh
Terminal	9	500 kW	3000 kWh
Total installed	81	36,800 kW	239,000 kWh

Aggregate station-rail charging power: 48,750 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 480 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	16.0 kWh/km	4 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	600 kWh	37.5 km average line length
Onboard battery coverage	0.8 $\times$ average line run	480 kWh usable pack
Average 60 s dwell charge	8.7 kWh/stop	524 kW average charger across stops
Stops to refill one trainset pack	55 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	184 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	239 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in design.toml — emitted by the osr-design Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. country-costs.toml applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (203.7 km @ €3.5 M/km)	€713 M
Elevated (18.5 km @ €18 M/km)	€333 M
Elevated-interchange premium (10 sites @ €20 M)	€200 M
Civil subtotal	€1.25 bn

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
halt	13	€0.4 M	€5.2 M
standard	35	€1.5 M	€52 M
major	12	€3.0 M	€36 M
terminal	9	€2.5 M	€22 M
depot-terminal	1	€3.0 M	€3.0 M
interchange	2	€4.5 M	€9.0 M
interchange-elevated	22	€4.5 M	€99 M
Stations subtotal			€227 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	9	€3.0 M	€27 M
Depots subtotal			€52 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
metro-4car (revenue + spare + cold reserve)	184	€4.0 M	€736 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	225.1 km × €0.015 M/km	€3.3 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€34 M
EPC integration + project management (7%)	on subtotal	€161 M

Total

Bucket	Value
Civil works	€1.25 bn
Stations	€227 M
Depots	€52 M
Rolling stock	€736 M
Residual train-control wayside + charging microgrids	€37 M
EPC overhead (7%)	€161 M
CAPEX total	€2.46 bn
Per-route-km	€11 M / km
Per-capita (city pop)	€1,640 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running scripts/regenerate-city.sh vijayawada.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–5 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 6 and runs for 20 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–5)	€177 M / yr	€118
Steady-state, low-ridership (year 6+)	€223 M / yr	€149
Steady-state, high-ridership (year 6+)	€213 M / yr	€142
Lifecycle envelope (yr 1–25, low scenario)	€5.34 bn cumulative	€3,562
Lifecycle envelope (yr 1–25, high scenario)	€5.15 bn cumulative	€3,433

Population basis: 1,500,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 25, debt service drops to zero and only the OPEX shortfall remains – ~€55 M / yr (low) €46 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€1.48 bn	4.0%	25 y, 5 y grace	€109 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€615 M	7.2%	25 y, 5 y grace	€59 M / yr
Government equity (no debt service)	15%	€369 M	—	—	—
Total	100%	€2.46 bn			€168 M / yr

During the 5-year grace period the operator pays interest only – multilateral €59 M / yr + bonds €44 M / yr = €103 M / yr total – plus the equity tranche amortised across construction (€74 M / yr × 5 yr). Principal repayment begins in year 6 on a 20-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€29 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€31 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€167 k
Traction energy (515.4 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (1,363 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€4.8 M
OPEX subtotal		€65 M / yr

Annual fleet utilisation: 164 revenue trainsets × 20.5 h/day × 365 d/yr × 35 km/h commercial × 75% revenue factor = 32.2 M train-km / yr (~196 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$230 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.35 (~\$0.38 USD)
Day pass (3 trips)	€0.90 (15 % bulk discount)
Monthly unlimited pass	€10.58 (~5 % of median monthly income)
Annual pass	€116.38 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	27.4 M	54.8 M
Farebox revenue	€9.7 M / yr	€19 M / yr
Farebox / OPEX recovery	15%	30%
Country policy-target recovery (diagnostic)	55%	55%
Operating shortfall (gov subsidy required)	€55 M / yr	€46 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€223 M / yr	€213 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
vijayawada.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
vijayawada-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
vijayawada.corridor.geojson	Line polylines + stations (GeoJSON)
vijayawada.stations.json	Machine-readable station list
vijayawada.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug vijayawada

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug vijayawada \
  --sidecar .cache/osr-pipeline/rasters/vijayawada.grid.json \
  --out-dir designs/.../Vijayawada

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../vijayawada.toml \
  --out designs/.../README.md
```

`scripts/regenerate-vijayawada.sh` chains steps 3 + drift tests into a single command.

south-asia/India/Visakhapatnam

designs/south-asia/India/Visakhapatnam/README.md

Visakhapatnam — Urban Rail Network

Country: IN · Population: 2,300,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Visakhapatnam rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [visakhapatnam.corridor.geojson](#).

At a glance

Metric	Value
Lines	6
Unique stations	109
Interchange stations	23
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	51.6%
Route length (double track)	240.0 km
Revenue fleet	175 × 4-car trainsets
Spare + cold-reserve	21 × 4-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	57.3 km	25	46	NE Outer SW Outer
line-2	32.3 km	13	27	SW Outer NE Mid
line-3	33.0 km	15	27	S Outer N Mid
line-4	31.0 km	16	26	E Mid W Outer
line-5	23.7 km	9	20	S Inner NW Outer
line-6	62.5 km	32	50	W Mid W Mid
Total	240.0 km	109 unique	196	

Rolling stock

Property	Value
Consist	4-car, 75 m
Max speed	90 km/h
Onboard battery	480 kWh per trainset
Seats	80 longitudinal seats
Nominal capacity (AW2)	440 pax (seated + standing, metro-4car per RFC 0008 §1)
Crush capacity (AW3)	560 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 440 AW2 passengers (metro - 4 car)
- Peak frequency: 12 trains/hour/direction (5-min headway)
- Peak capacity per line per direction: $440 \times 12 = 5,280$ pphpd
- Network peak throughput (all lines, both directions): $6 \text{ lines} \times 2 \text{ directions} \times 5,280 = 63,360$ passengers/hour
- Daily theoretical capacity (peak $\times$ 10): $\approx 633,600$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 118,680 - 178,020$ trips/day

Catchment

- City population: 2,300,000
- Anchor-weighted coverage: 51.6%
- Catchment population: $\approx 1,186,800$ (within ~800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../.../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	23	500 kW	3000 kWh
Major	25	400 kW	2500 kWh
Standard	50	300 kW	2000 kWh
Terminal	9	500 kW	3000 kWh
Total installed	108	46,000 kW	298,500 kWh

Aggregate station-rail charging power: 59,500 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 480 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	16.0 kWh/km	4 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	640 kWh	40.0 km average line length
Onboard battery coverage	0.8 $\times$ average line run	480 kWh usable pack
Average 60 s dwell charge	9.1 kWh/stop	546 kW average charger across stops
Stops to refill one trainset pack	53 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	230 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	298 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in design.toml — emitted by the osr-design Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. country-costs.toml applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (221.8 km @ €3.5 M/km)	€776 M
Elevated (17.4 km @ €18 M/km)	€314 M
Elevated-interchange premium (8 sites @ €20 M)	€160 M
Civil subtotal	€1.25 bn

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
halt	2	€0.4 M	€0.8 M
standard	50	€1.5 M	€75 M
major	25	€3.0 M	€75 M
terminal	9	€2.5 M	€22 M
depot-terminal	1	€3.0 M	€3.0 M
interchange	2	€4.5 M	€9.0 M
interchange-elevated	21	€4.5 M	€94 M
Stations subtotal			€280 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	9	€3.0 M	€27 M
Depots subtotal			€52 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
metro-4car (revenue + spare + cold reserve)	196	€4.0 M	€784 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	240.0 km × €0.015 M/km	€3.6 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€41 M
EPC integration + project management (7%)	on subtotal	€169 M

Total

Bucket	Value
Civil works	€1.25 bn
Stations	€280 M
Depots	€52 M
Rolling stock	€784 M
Residual train-control wayside + charging microgrids	€44 M
EPC overhead (7%)	€169 M
CAPEX total	€2.58 bn
Per-route-km	€11 M / km
Per-capita (city pop)	€1,121 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running scripts/regenerate-city.sh visakhapatnam.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–5 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 6 and runs for 20 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–5)	€186 M / yr	€81
Steady-state, low-ridership (year 6+)	€229 M / yr	€100
Steady-state, high-ridership (year 6+)	€214 M / yr	€93
Lifecycle envelope (yr 1–25, low scenario)	€5.51 bn cumulative	€2,397
Lifecycle envelope (yr 1–25, high scenario)	€5.22 bn cumulative	€2,268

Population basis: 2,300,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 25, debt service drops to zero and only the OPEX shortfall remains – ~€54 M / yr (low) €39 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€1.55 bn	4.0%	25 y, 5 y grace	€114 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€645 M	7.2%	25 y, 5 y grace	€62 M / yr
Government equity (no debt service)	15%	€387 M	—	—	—
Total	100%	€2.58 bn			€176 M / yr

During the 5-year grace period the operator pays interest only – multilateral €62 M / yr + bonds €46 M / yr = €108 M / yr total – plus the equity tranche amortised across construction (€77 M / yr × 5 yr). Principal repayment begins in year 6 on a 20-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€31 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€32 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€179 k
Traction energy (550.0 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (1,452 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€5.2 M
OPEX subtotal		€68 M / yr

Annual fleet utilisation: 175 revenue trainsets × 20.5 h/day × 365 d/yr × 35 km/h commercial × 75% revenue factor = 34.4 M train-km / yr (~196 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$230 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.35 (~\$0.38 USD)
Day pass (3 trips)	€0.90 (15 % bulk discount)
Monthly unlimited pass	€10.58 (~5 % of median monthly income)
Annual pass	€116.38 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	42.0 M	84.0 M
Farebox revenue	€15 M / yr	€30 M / yr
Farebox / OPEX recovery	22%	43%
Country policy-target recovery (diagnostic)	55%	55%
Operating shortfall (gov subsidy required)	€54 M / yr	€39 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€229 M / yr	€214 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
visakhapatnam.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
visakhapatnam-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
visakhapatnam.corridor.geojson	Line polylines + stations (GeoJSON)
visakhapatnam.stations.json	Machine-readable station list
visakhapatnam.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug visakhapatnam

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug visakhapatnam \
  --sidecar .cache/osr-pipeline/rasters/visakhapatnam.grid.json \
  --out-dir designs/.../Visakhapatnam

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../visakhapatnam.toml \
  --out designs/.../README.md
```

`scripts/regenerate-visakhapatnam.sh` chains steps 3 + drift tests into a single command.

south-asia/Nepal/Biratnagar

designs/south-asia/Nepal/Biratnagar/README.md

Biratnagar — Urban Rail Network

Country: NP · Population: 300,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Biratnagar rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [biratnagar.corridor.geojson](#).

At a glance

Metric	Value
Lines	3
Unique stations	23
Interchange stations	3
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	59.7%
Route length (double track)	34.3 km
Revenue fleet	42 × 2-car trainsets
Spare + cold-reserve	6 × 2-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	13.5 km	10	18	N Outer S Outer
line-2	12.5 km	7	17	S Outer N Mid
line-3	8.3 km	6	13	SE Mid N Mid
Total	34.3 km	23 unique	48	

Rolling stock

Property	Value
Consist	2-car, 39 m
Max speed	70 km/h
Onboard battery	240 kWh per trainset
Seats	40 longitudinal seats
Nominal capacity (AW2)	210 pax (seated + standing, tram-2car per RFC 0008 §1)
Crush capacity (AW3)	260 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 210 AW2 passengers (tram-2car)
- Peak frequency: 12 trains/hour/direction (5-min headway)

- Peak capacity per line per direction: $210 \times 12 = 2,520$ pphpd
- Network peak throughput (all lines, both directions): $3 \text{ lines} \times 2 \text{ directions} \times 2,520 = 15,120$ passengers/hour
- Daily theoretical capacity (peak $\times 10$): $\approx 151,200$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 17,910 - 26,865$ trips/day

Catchment

- City population: 300,000
- Anchor-weighted coverage: 59.7%
- Catchment population: $\approx 179,100$ (within ~ 800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	3	500 kW	3000 kWh
Major	6	400 kW	2500 kWh
Standard	8	300 kW	2000 kWh
Terminal	5	500 kW	3000 kWh
Total installed	23	13,800 kW	95,000 kWh

Aggregate station-rail charging power: 14,500 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 240 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	8.0 kWh/km	2 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	91 kWh	11.4 km average line length
Onboard battery coverage	2.6 $\times$ average line run	240 kWh usable pack
Average 60 s dwell charge	10.5 kWh/stop	630 kW average charger across stops
Stops to refill one trainset pack	23 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	69 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	95 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in design.toml — emitted by the osr-design Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. country-costs.toml applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
Elevated (34.3 km @ €18 M/km)	€617 M
Civil subtotal	€617 M

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~ 11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
standard	8	€1.5 M	€12 M

Archetype	Count	Unit	Subtotal
major	6	€3.0 M	€18 M
terminal	5	€2.5 M	€12 M
depot-terminal	1	€3.0 M	€3.0 M
interchange	3	€4.5 M	€14 M
Stations subtotal			€59 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	5	€3.0 M	€15 M
Depots subtotal			€40 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
tram-2car (revenue + spare + cold reserve)	48	€2.0 M	€96 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	34.3 km × €0.015 M/km	€0.5 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€8.9 M
EPC integration + project management (7%)	on subtotal	€57 M

Total

Bucket	Value
Civil works	€617 M
Stations	€59 M
Depots	€40 M
Rolling stock	€96 M
Residual train-control wayside + charging microgrids	€9.5 M
EPC overhead (7%)	€57 M
CAPEX total	€878 M
Per-route-km	€26 M / km
Per-capita (city pop)	€2,928 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code — regenerate by

re-running scripts/regenerate-city.sh biratnagar.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–7 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 8 and runs for 23 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–7)	€51 M / yr	€170
Steady-state, low-ridership (year 8+)	€70 M / yr	€233
Steady-state, high-ridership (year 8+)	€69 M / yr	€229
Lifecycle envelope (yr 1–30, low scenario)	€1.96 bn cumulative	€6,548
Lifecycle envelope (yr 1–30, high scenario)	€1.94 bn cumulative	€6,458

Population basis: 300,000 (catchment per `lib/city-batches/world-sample.toml`). After year 30, debt service drops to zero and only the OPEX shortfall remains — ~€17 M / yr (low) €16 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€527 M	3.0%	30 y, 7 y grace	€32 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€220 M	7.5%	30 y, 7 y grace	€20 M / yr
Government equity (no debt service)	15%	€132 M	—	—	—
Total	100%	€878 M			€52 M / yr

During the 7-year grace period the operator pays interest only — multilateral €16 M / yr + bonds €16 M / yr = €32 M / yr total — plus the equity tranche amortised across construction (€19 M / yr × 7 yr). Principal repayment begins in year 8 on a 23-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€3.8 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€14 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€26 k
Traction energy (41.5 GWh / yr)	trackside PV + Na-ion (RFC 0002) — self-generated, €0 / yr	€0 k
Labour (218 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€472 k
OPEX subtotal		€19 M / yr

Annual fleet utilisation: 42 revenue trainsets × 20.5 h/day × 365 d/yr × 22 km/h commercial × 75% revenue factor = 5.2 M train-km / yr (~123 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$140 USD (per `lib/templates/country-finance.toml`). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass — a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.21 (~\$0.23 USD)
Day pass (3 trips)	€0.55 (15 % bulk discount)
Monthly unlimited pass	€6.44 (~5 % of median monthly income)
Annual pass	€70.84 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	5.5 M	10.9 M
Farebox revenue	€1.2 M / yr	€2.4 M / yr
Farebox / OPEX recovery	6%	13%

	Low scenario	High scenario
Country policy-target recovery (diagnostic)	45%	45%
Operating shortfall (gov subsidy required)	€17 M / yr	€16 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€70 M / yr	€69 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
biratnagar.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
biratnagar-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
biratnagar.corridor.geojson	Line polylines + stations (GeoJSON)
biratnagar.stations.json	Machine-readable station list
biratnagar.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug biratnagar

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug biratnagar \
  --sidecar .cache/osr-pipeline/rasters/biratnagar.grid.json \
  --out-dir designs/.../Biratnagar

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../biratnagar.toml \
  --out designs/.../README.md
```

`scripts/regenerate-biratnagar.sh` chains steps 3 + drift tests into a single command.

south-asia/Nepal/Kathmandu

designs/south-asia/Nepal/Kathmandu/README.md

Kathmandu — Urban Rail Network

Country: NP · Population: 1,442,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Kathmandu rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [kathmandu.corridor.geojson](#).

At a glance

Metric	Value
Lines	6
Unique stations	102
Interchange stations	27
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	45.5%
Route length (double track)	202.6 km
Revenue fleet	148 × 4-car trainsets
Spare + cold-reserve	19 × 4-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	32.9 km	18	27	W Outer E Outer
line-2	30.3 km	16	25	NE Mid SW Outer
line-3	24.3 km	13	20	S Mid NE Outer
line-4	27.8 km	17	24	SE Mid NW Mid
line-5	28.3 km	11	24	S Mid NW Outer
line-6	59.1 km	28	47	W Mid W Mid
Total	202.6 km	102 unique	167	

Rolling stock

Property	Value
Consist	4-car, 75 m
Max speed	90 km/h
Onboard battery	480 kWh per trainset
Seats	80 longitudinal seats
Nominal capacity (AW2)	440 pax (seated + standing, metro-4car per RFC 0008 §1)
Crush capacity (AW3)	560 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 440 AW2 passengers (metro - 4 car)
- Peak frequency: 12 trains/hour/direction (5-min headway)
- Peak capacity per line per direction: $440 \times 12 = 5,280$ pphpd
- Network peak throughput (all lines, both directions): $6 \text{ lines} \times 2 \text{ directions} \times 5,280 = 63,360$ passengers/hour
- Daily theoretical capacity (peak $\times$ 10): $\approx 633,600$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 65,611 - 98,416$ trips/day

Catchment

- City population: 1,442,000
- Anchor-weighted coverage: 45.5%
- Catchment population: $\approx 656,110$ (within ~800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	27	500 kW	3000 kWh
Major	34	400 kW	2500 kWh
Standard	30	300 kW	2000 kWh
Terminal	9	500 kW	3000 kWh
Total installed	101	45,600 kW	293,000 kWh

Aggregate station-rail charging power: 56,000 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 480 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	16.0 kWh/km	4 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	540 kWh	33.8 km average line length
Onboard battery coverage	0.9 $\times$ average line run	480 kWh usable pack
Average 60 s dwell charge	9.2 kWh/stop	549 kW average charger across stops
Stops to refill one trainset pack	52 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	228 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	293 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in design.toml — emitted by the osr-design Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. country-costs.toml applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (168.6 km @ €3.5 M/km)	€590 M
Elevated (29.0 km @ €18 M/km)	€521 M
Elevated-interchange premium (13 sites @ €20 M)	€260 M
Civil subtotal	€1.37 bn

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
halt	2	€0.4 M	€0.8 M
standard	30	€1.5 M	€45 M
major	34	€3.0 M	€102 M
terminal	9	€2.5 M	€22 M
depot-terminal	1	€3.0 M	€3.0 M
interchange-elevated	27	€4.5 M	€122 M
Stations subtotal			€295 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	9	€3.0 M	€27 M
Depots subtotal			€52 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
metro-4car (revenue + spare + cold reserve)	167	€4.0 M	€668 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	202.6 km × €0.015 M/km	€3.0 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€42 M
EPC integration + project management (7%)	on subtotal	€170 M

Total

Bucket	Value
Civil works	€1.37 bn
Stations	€295 M
Depots	€52 M
Rolling stock	€668 M
Residual train-control wayside + charging microgrids	€45 M
EPC overhead (7%)	€170 M
CAPEX total	€2.60 bn
Per-route-km	€13 M / km
Per-capita (city pop)	€1,804 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running `scripts/regenerate-city.sh kathmandu`.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–7 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 8 and runs for 23 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–7)	€151 M / yr	€105
Steady-state, low-ridership (year 8+)	€213 M / yr	€148
Steady-state, high-ridership (year 8+)	€208 M / yr	€144
Lifecycle envelope (yr 1–30, low scenario)	€5.97 bn cumulative	€4,137
Lifecycle envelope (yr 1–30, high scenario)	€5.84 bn cumulative	€4,047

Population basis: 1,442,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 30, debt service drops to zero and only the OPEX shortfall remains – ~€58 M / yr (low) €53 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€1.56 bn	3.0%	30 y, 7 y grace	€95 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€650 M	7.5%	30 y, 7 y grace	€60 M / yr
Government equity (no debt service)	15%	€390 M	—	—	—
Total	100%	€2.60 bn			€155 M / yr

During the 7-year grace period the operator pays interest only – multilateral €47 M / yr + bonds €49 M / yr = €96 M / yr total – plus the equity tranche amortised across construction (€56 M / yr × 7 yr). Principal repayment begins in year 8 on a 23-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€27 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€34 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€148 k
Traction energy (465.1 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (1,228 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€2.7 M
OPEX subtotal		€64 M / yr

Annual fleet utilisation: 148 revenue trainsets × 20.5 h/day × 365 d/yr × 35 km/h commercial × 75% revenue factor = 29.1 M train-km / yr (~196 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$140 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.21 (~\$0.23 USD)
Day pass (3 trips)	€0.55 (15 % bulk discount)
Monthly unlimited pass	€6.44 (~5 % of median monthly income)
Annual pass	€70.84 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	26.3 M	52.6 M
Farebox revenue	€5.6 M / yr	€11 M / yr
Farebox / OPEX recovery	9%	18%
Country policy-target recovery (diagnostic)	45%	45%
Operating shortfall (gov subsidy required)	€58 M / yr	€53 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€213 M / yr	€208 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
kathmandu.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
kathmandu-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
kathmandu.corridor.geojson	Line polylines + stations (GeoJSON)
kathmandu.stations.json	Machine-readable station list
kathmandu.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug kathmandu

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug kathmandu \
  --sidecar .cache/osr-pipeline/rasters/kathmandu.grid.json \
  --out-dir designs/.../Kathmandu

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../kathmandu.toml \
  --out designs/.../README.md
```

`scripts/regenerate-kathmandu.sh` chains steps 3 + drift tests into a single command.

south-asia/Nepal/Pokhara

designs/south-asia/Nepal/Pokhara/README.md

Pokhara — Urban Rail Network

Country: NP · Population: 600,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Pokhara rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [pokhara.corridor.geojson](#).

At a glance

Metric	Value
Lines	3
Unique stations	44
Interchange stations	5
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	52.1%
Route length (double track)	82.3 km
Revenue fleet	71 × 3-car trainsets
Spare + cold-reserve	9 × 3-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	27.6 km	15	27	SE Outer NW Outer
line-2	28.3 km	15	27	SE Outer W Outer
line-3	26.4 km	14	26	N Outer S Outer
Total	82.3 km	44 unique	80	

Rolling stock

Property	Value
Consist	3-car, 57 m
Max speed	90 km/h
Onboard battery	360 kWh per trainset
Seats	60 longitudinal seats
Nominal capacity (AW2)	330 pax (seated + standing, light-metro-3car per RFC 0008 §1)
Crush capacity (AW3)	420 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 330 AW2 passengers (light-metro-3car)
- Peak frequency: 12 trains/hour/direction (5-min headway)

- Peak capacity per line per direction: $330 \times 12 = 3,960$ pphpd
- Network peak throughput (all lines, both directions): $3 \text{ lines} \times 2 \text{ directions} \times 3,960 = 23,760$ passengers/hour
- Daily theoretical capacity (peak $\times 10$): $\approx 237,600$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 31,260 - 46,890$ trips/day

Catchment

- City population: 600,000
- Anchor-weighted coverage: 52.1%
- Catchment population: $\approx 312,600$ (within ~ 800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	5	500 kW	3000 kWh
Major	13	400 kW	2500 kWh
Standard	18	300 kW	2000 kWh
Terminal	5	500 kW	3000 kWh
Total installed	42	20,600 kW	138,500 kWh

Aggregate station-rail charging power: 24,500 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 360 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	12.0 kWh/km	3 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	329 kWh	27.4 km average line length
Onboard battery coverage	1.1 $\times$ average line run	360 kWh usable pack
Average 60 s dwell charge	9.3 kWh/stop	557 kW average charger across stops
Stops to refill one trainset pack	39 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	103 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	138 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in design.toml — emitted by the osr-design Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. country-costs.toml applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (76.8 km @ €3.5 M/km)	€269 M
Elevated (4.4 km @ €18 M/km)	€79 M
Elevated-interchange premium (2 sites @ €20 M)	€40 M
Civil subtotal	€388 M

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~ 11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
halt	2	€0.4 M	€0.8 M
standard	18	€1.5 M	€27 M
major	13	€3.0 M	€39 M
terminal	5	€2.5 M	€12 M
depot-terminal	1	€3.0 M	€3.0 M
interchange-elevated	5	€4.5 M	€22 M
Stations subtotal			€105 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	5	€3.0 M	€15 M
Depots subtotal			€40 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
light-metro-3car (revenue + spare + cold reserve)	80	€3.0 M	€240 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	82.3 km × €0.015 M/km	€1.2 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€16 M
EPC integration + project management (7%)	on subtotal	€55 M

Total

Bucket	Value
Civil works	€388 M
Stations	€105 M
Depots	€40 M
Rolling stock	€240 M
Residual train-control wayside + charging microgrids	€17 M
EPC overhead (7%)	€55 M
CAPEX total	€845 M
Per-route-km	€10 M / km
Per-capita (city pop)	€1,409 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running scripts/regenerate-city.sh pokhara.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–7 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 8 and runs for 23 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–7)	€49 M / yr	€82
Steady-state, low-ridership (year 8+)	€69 M / yr	€116
Steady-state, high-ridership (year 8+)	€67 M / yr	€112
Lifecycle envelope (yr 1–30, low scenario)	€1.94 bn cumulative	€3,236
Lifecycle envelope (yr 1–30, high scenario)	€1.89 bn cumulative	€3,146

Population basis: 600,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 30, debt service drops to zero and only the OPEX shortfall remains – ~€19 M / yr (low) €17 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€507 M	3.0%	30 y, 7 y grace	€31 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€211 M	7.5%	30 y, 7 y grace	€20 M / yr
Government equity (no debt service)	15%	€127 M	—	—	—
Total	100%	€845 M			€50 M / yr

During the 7-year grace period the operator pays interest only – multilateral €15 M / yr + bonds €16 M / yr = €31 M / yr total – plus the equity tranche amortised across construction (€18 M / yr × 7 yr). Principal repayment begins in year 8 on a 23-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€9.6 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€11 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€61 k
Traction energy (143.4 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (506 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€1.1 M
OPEX subtotal		€21 M / yr

Annual fleet utilisation: 71 revenue trainsets × 20.5 h/day × 365 d/yr × 30 km/h commercial × 75% revenue factor = 12.0 M train-km / yr (~168 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$140 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.21 (~\$0.23 USD)
Day pass (3 trips)	€0.55 (15 % bulk discount)
Monthly unlimited pass	€6.44 (~5 % of median monthly income)
Annual pass	€70.84 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	10.9 M	21.9 M

	Low scenario	High scenario
Farebox revenue	€2.4 M / yr	€4.7 M / yr
Farebox / OPEX recovery	11%	22%
Country policy-target recovery (diagnostic)	45%	45%
Operating shortfall (gov subsidy required)	€19 M / yr	€17 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€69 M / yr	€67 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
pokhara.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
pokhara-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
pokhara.corridor.geojson	Line polylines + stations (GeoJSON)
pokhara.stations.json	Machine-readable station list
pokhara.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug pokhara

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug pokhara \
  --sidecar .cache/osr-pipeline/rasters/pokhara.grid.json \
  --out-dir designs/.../Pokhara

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../pokhara.toml \
  --out designs/.../README.md
```

`scripts/regenerate-pokhara.sh` chains steps 3 + drift tests into a single command.

south-asia/Pakistan/Bahawalpur

designs/south-asia/Pakistan/Bahawalpur/README.md

Bahawalpur — Urban Rail Network

Country: PK · Population: 900,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Bahawalpur rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [bahawalpur.corridor.geojson](#).

At a glance

Metric	Value
Lines	3
Unique stations	29
Interchange stations	4
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	41.5%
Route length (double track)	46.2 km
Revenue fleet	43 × 3-car trainsets
Spare + cold-reserve	6 × 3-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	13.9 km	10	15	E Outer SW Inner
line-2	17.5 km	9	18	NE Mid SW Outer
line-3	14.8 km	10	16	NW Outer E Inner
Total	46.2 km	29 unique	49	

Rolling stock

Property	Value
Consist	3-car, 57 m
Max speed	90 km/h
Onboard battery	360 kWh per trainset
Seats	60 longitudinal seats
Nominal capacity (AW2)	330 pax (seated + standing, light-metro-3car per RFC 0008 §1)
Crush capacity (AW3)	420 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 330 AW2 passengers (light-metro-3car)
- Peak frequency: 12 trains/hour/direction (5-min headway)

- Peak capacity per line per direction: $330 \times 12 = 3,960$ pphpd
- Network peak throughput (all lines, both directions): $3 \text{ lines} \times 2 \text{ directions} \times 3,960 = 23,760$ passengers/hour
- Daily theoretical capacity (peak $\times 10$): $\approx 237,600$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 37,350 - 56,025$ trips/day

Catchment

- City population: 900,000
- Anchor-weighted coverage: 41.5%
- Catchment population: $\approx 373,500$ (within ~ 800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	4	500 kW	3000 kWh
Major	11	400 kW	2500 kWh
Standard	6	300 kW	2000 kWh
Terminal	5	500 kW	3000 kWh
Total installed	27	15,700 kW	106,500 kWh

Aggregate station-rail charging power: 17,000 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 360 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	12.0 kWh/km	3 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	185 kWh	15.4 km average line length
Onboard battery coverage	1.9 $\times$ average line run	360 kWh usable pack
Average 60 s dwell charge	9.8 kWh/stop	586 kW average charger across stops
Stops to refill one trainset pack	37 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	78 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	106 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the `[costs]` block in `design.toml` — emitted by the `osr-design` Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. `country-costs.toml` applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (44.0 km @ €3.5 M/km)	€154 M
Elevated (2.1 km @ €18 M/km)	€37 M
Elevated-interchange premium (2 sites @ €20 M)	€40 M
Civil subtotal	€231 M

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~ 11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
halt	2	€0.4 M	€0.8 M
standard	6	€1.5 M	€9.0 M
major	11	€3.0 M	€33 M
terminal	5	€2.5 M	€12 M
depot-terminal	1	€3.0 M	€3.0 M
interchange-elevated	4	€4.5 M	€18 M
Stations subtotal			€76 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	5	€3.0 M	€15 M
Depots subtotal			€40 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
light-metro-3car (revenue + spare + cold reserve)	49	€3.0 M	€147 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	46.2 km × €0.015 M/km	€0.7 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€11 M
EPC integration + project management (7%)	on subtotal	€35 M

Total

Bucket	Value
Civil works	€231 M
Stations	€76 M
Depots	€40 M
Rolling stock	€147 M
Residual train-control wayside + charging microgrids	€12 M
EPC overhead (7%)	€35 M
CAPEX total	€542 M
Per-route-km	€12 M / km
Per-capita (city pop)	€602 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running scripts/regenerate-city.sh bahawalpur.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–7 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 8 and runs for 23 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–7)	€47 M / yr	€52
Steady-state, low-ridership (year 8+)	€54 M / yr	€60
Steady-state, high-ridership (year 8+)	€50 M / yr	€56
Lifecycle envelope (yr 1–30, low scenario)	€1.58 bn cumulative	€1,756
Lifecycle envelope (yr 1–30, high scenario)	€1.48 bn cumulative	€1,650

Population basis: 900,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 30, debt service drops to zero and only the OPEX shortfall remains – ~€9.5 M / yr (low) €5.3 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€325 M	4.0%	30 y, 7 y grace	€22 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€136 M	16.5%	30 y, 7 y grace	€23 M / yr
Government equity (no debt service)	15%	€81 M	—	—	—
Total	100%	€542 M			€45 M / yr

During the 7-year grace period the operator pays interest only – multilateral €13 M / yr + bonds €22 M / yr = €35 M / yr total – plus the equity tranche amortised across construction (€12 M / yr × 7 yr). Principal repayment begins in year 8 on a 23-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€5.9 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€7.0 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€35 k
Traction energy (86.9 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (289 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€737 k
OPEX subtotal		€14 M / yr

Annual fleet utilisation: 43 revenue trainsets × 20.5 h/day × 365 d/yr × 30 km/h commercial × 75% revenue factor = 7.2 M train-km / yr (~168 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$165 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.25 (~\$0.28 USD)
Day pass (3 trips)	€0.65 (15 % bulk discount)
Monthly unlimited pass	€7.59 (~5 % of median monthly income)
Annual pass	€83.49 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	16.4 M	32.9 M

	Low scenario	High scenario
Farebox revenue	€4.2 M / yr	€8.3 M / yr
Farebox / OPEX recovery	31%	61%
Country policy-target recovery (diagnostic)	45%	45%
Operating shortfall (gov subsidy required)	€9.5 M / yr	€5.3 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€54 M / yr	€50 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
bahawalpur.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
bahawalpur-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
bahawalpur.corridor.geojson	Line polylines + stations (GeoJSON)
bahawalpur.stations.json	Machine-readable station list
bahawalpur.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug bahawalpur

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug bahawalpur \
  --sidecar .cache/osr-pipeline/rasters/bahawalpur.grid.json \
  --out-dir designs/.../Bahawalpur

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../bahawalpur.toml \
  --out designs/.../README.md
```

`scripts/regenerate-bahawalpur.sh` chains steps 3 + drift tests into a single command.

south-asia/Pakistan/Faisalabad

designs/south-asia/Pakistan/Faisalabad/README.md

Faisalabad — Urban Rail Network

Country: PK · Population: 3,556,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Faisalabad rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [faisalabad.corridor.geojson](#).

At a glance

Metric	Value
Lines	5
Unique stations	92
Interchange stations	19
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	51.8%
Route length (double track)	166.0 km
Revenue fleet	122 × 6-car trainsets
Spare + cold-reserve	15 × 6-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	31.8 km	17	26	NE Mid SW Outer
line-2	27.2 km	17	23	N Mid SE Mid
line-3	23.5 km	13	20	E Mid W Mid
line-4	31.1 km	15	26	NE Outer W Mid
line-5	52.4 km	31	42	W Mid W Mid
Total	166.0 km	92 unique	137	

Rolling stock

Property	Value
Consist	6-car, 111 m
Max speed	100 km/h
Onboard battery	720 kWh per trainset
Seats	120 longitudinal seats
Nominal capacity (AW2)	660 pax (seated + standing, met ro-6car per RFC 0008 §1)
Crush capacity (AW3)	840 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 660 AW2 passengers (met ro-6car)

- Peak frequency: 12 trains/hour/direction (5-min headway)
- Peak capacity per line per direction: $660 \times 12 = 7,920$ pphpd
- Network peak throughput (all lines, both directions): $5 \text{ lines} \times 2 \text{ directions} \times 7,920 = 79,200$ passengers/hour
- Daily theoretical capacity (peak $\times$ 10): $\approx 792,000$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 184,200 - 276,301$ trips/day

Catchment

- City population: 3,556,000
- Anchor-weighted coverage: 51.8%
- Catchment population: $\approx 1,842,008$ (within ~800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	19	500 kW	3000 kWh
Major	38	400 kW	2500 kWh
Standard	27	300 kW	2000 kWh
Terminal	7	500 kW	3000 kWh
Total installed	92	41,300 kW	267,000 kWh

Aggregate station-rail charging power: 50,250 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 720 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	24.0 kWh/km	6 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	797 kWh	33.2 km average line length
Onboard battery coverage	0.9 $\times$ average line run	720 kWh usable pack
Average 60 s dwell charge	9.1 kWh/stop	546 kW average charger across stops
Stops to refill one trainset pack	79 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	206 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	267 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in design.toml — emitted by the osr-design Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. country-costs.toml applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (148.0 km @ €3.5 M/km)	€518 M
Elevated (17.2 km @ €18 M/km)	€309 M
Elevated-interchange premium (9 sites @ €20 M)	€180 M
Civil subtotal	€1.01 bn

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
halt	1	€0.4 M	€0.4 M
standard	27	€1.5 M	€40 M
major	38	€3.0 M	€114 M
terminal	7	€2.5 M	€18 M
depot-terminal	1	€3.0 M	€3.0 M
interchange-elevated	19	€4.5 M	€86 M
Stations subtotal			€261 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	7	€3.0 M	€21 M
Depots subtotal			€46 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY – never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
metro-6car (revenue + spare + cold reserve)	137	€6.0 M	€822 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	166.0 km × €0.015 M/km	€2.5 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€37 M
EPC integration + project management (7%)	on subtotal	€152 M

Total

Bucket	Value
Civil works	€1.01 bn
Stations	€261 M
Depots	€46 M
Rolling stock	€822 M
Residual train-control wayside + charging microgrids	€40 M
EPC overhead (7%)	€152 M
CAPEX total	€2.33 bn
Per-route-km	€14 M / km
Per-capita (city pop)	€655 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running scripts/regenerate-city.sh faisalabad.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–7 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 8 and runs for 23 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–7)	€202 M / yr	€57
Steady-state, low-ridership (year 8+)	€238 M / yr	€67
Steady-state, high-ridership (year 8+)	€222 M / yr	€62
Lifecycle envelope (yr 1–30, low scenario)	€6.90 bn cumulative	€1,940
Lifecycle envelope (yr 1–30, high scenario)	€6.52 bn cumulative	€1,833

Population basis: 3,556,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 30, debt service drops to zero and only the OPEX shortfall remains – ~€45 M / yr (low) €29 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€1.40 bn	4.0%	30 y, 7 y grace	€94 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€582 M	16.5%	30 y, 7 y grace	€99 M / yr
Government equity (no debt service)	15%	€349 M	—	—	—
Total	100%	€2.33 bn			€193 M / yr

During the 7-year grace period the operator pays interest only – multilateral €56 M / yr + bonds €96 M / yr = €152 M / yr total – plus the equity tranche amortised across construction (€50 M / yr × 7 yr). Principal repayment begins in year 8 on a 23-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€33 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€26 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€124 k
Traction energy (575.1 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (1,008 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€2.6 M
OPEX subtotal		€62 M / yr

Annual fleet utilisation: 122 revenue trainsets × 20.5 h/day × 365 d/yr × 35 km/h commercial × 75% revenue factor = 24.0 M train-km / yr (~196 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$165 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.25 (~\$0.28 USD)
Day pass (3 trips)	€0.65 (15 % bulk discount)
Monthly unlimited pass	€7.59 (~5 % of median monthly income)
Annual pass	€83.49 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	64.9 M	129.8 M
Farebox revenue	€16 M / yr	€33 M / yr
Farebox / OPEX recovery	27%	53%
Country policy-target recovery (diagnostic)	45%	45%
Operating shortfall (gov subsidy required)	€45 M / yr	€29 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€238 M / yr	€222 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
faisalabad.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
faisalabad-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
faisalabad.corridor.geojson	Line polylines + stations (GeoJSON)
faisalabad.stations.json	Machine-readable station list
faisalabad.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug faisalabad

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug faisalabad \
  --sidecar .cache/osr-pipeline/rasters/faisalabad.grid.json \
  --out-dir designs/.../Faisalabad

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../faisalabad.toml \
  --out designs/.../README.md
```

`scripts/regenerate-faisalabad.sh` chains steps 3 + drift tests into a single command.

south-asia/Pakistan/Gujranwala

designs/south-asia/Pakistan/Gujranwala/README.md

Gujranwala — Urban Rail Network

Country: PK · Population: 2,300,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Gujranwala rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [gujranwala.corridor.geojson](#).

At a glance

Metric	Value
Lines	4
Unique stations	71
Interchange stations	11
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	38.6%
Route length (double track)	160.2 km
Revenue fleet	116 × 4-car trainsets
Spare + cold-reserve	14 × 4-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	31.7 km	16	26	NW Mid S Outer
line-2	33.7 km	18	28	NE Outer SW Inner
line-3	31.4 km	14	26	SE Outer W Mid
line-4	63.5 km	24	50	NW Mid NW Mid
Total	160.2 km	71 unique	130	

Rolling stock

Property	Value
Consist	4-car, 75 m
Max speed	90 km/h
Onboard battery	480 kWh per trainset
Seats	80 longitudinal seats
Nominal capacity (AW2)	440 pax (seated + standing, met ro - 4car per RFC 0008 §1)
Crush capacity (AW3)	560 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 440 AW2 passengers (met ro - 4car)

- Peak frequency: 12 trains/hour/direction (5-min headway)
- Peak capacity per line per direction: $440 \times 12 = 5,280$ pphpd
- Network peak throughput (all lines, both directions): $4 \text{ lines} \times 2 \text{ directions} \times 5,280 = 42,240$ passengers/hour
- Daily theoretical capacity (peak $\times$ 10): $\approx 422,400$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 88,780 - 133,170$ trips/day

Catchment

- City population: 2,300,000
- Anchor-weighted coverage: 38.6%
- Catchment population: $\approx 887,800$ (within ~ 800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	11	500 kW	3000 kWh
Major	22	400 kW	2500 kWh
Standard	27	300 kW	2000 kWh
Terminal	5	500 kW	3000 kWh
Total installed	66	29,900 kW	197,000 kWh

Aggregate station-rail charging power: 37,500 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 480 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	16.0 kWh/km	4 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	641 kWh	40.0 km average line length
Onboard battery coverage	0.7 $\times$ average line run	480 kWh usable pack
Average 60 s dwell charge	8.8 kWh/stop	528 kW average charger across stops
Stops to refill one trainset pack	55 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	150 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	197 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in design.toml — emitted by the osr-design Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. country-costs.toml applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (149.8 km @ €3.5 M/km)	€524 M
Elevated (9.9 km @ €18 M/km)	€178 M
Elevated-interchange premium (6 sites @ €20 M)	€120 M
Civil subtotal	€822 M

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
halt	6	€0.4 M	€2.4 M
standard	27	€1.5 M	€40 M
major	22	€3.0 M	€66 M
terminal	5	€2.5 M	€12 M
depot-terminal	1	€3.0 M	€3.0 M
interchange-elevated	11	€4.5 M	€50 M
Stations subtotal			€174 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	5	€3.0 M	€15 M
Depots subtotal			€40 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
metro-4car (revenue + spare + cold reserve)	130	€4.0 M	€520 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	160.2 km × €0.015 M/km	€2.4 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€26 M
EPC integration + project management (7%)	on subtotal	€111 M

Total

Bucket	Value
Civil works	€822 M
Stations	€174 M
Depots	€40 M
Rolling stock	€520 M
Residual train-control wayside + charging microgrids	€28 M
EPC overhead (7%)	€111 M
CAPEX total	€1.69 bn
Per-route-km	€11 M / km
Per-capita (city pop)	€737 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](https://github.com/OpenSourceRail/lib/templates/country-finance.toml). Pure function of the [costs] block above + the country code – regenerate by re-running `scripts/regenerate-city.sh` gujranwala.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–7 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 8 and runs for 23 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–7)	€147 M / yr	€64
Steady-state, low-ridership (year 8+)	€174 M / yr	€76
Steady-state, high-ridership (year 8+)	€163 M / yr	€71
Lifecycle envelope (yr 1–30, low scenario)	€5.03 bn cumulative	€2,187
Lifecycle envelope (yr 1–30, high scenario)	€4.79 bn cumulative	€2,081

Population basis: 2,300,000 (catchment per [lib/city-batches/world-sample.toml](https://github.com/OpenSourceRail/lib/city-batches/world-sample.toml)). After year 30, debt service drops to zero and only the OPEX shortfall remains – ~€33 M / yr (low) €23 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€1.02 bn	4.0%	30 y, 7 y grace	€68 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€424 M	16.5%	30 y, 7 y grace	€72 M / yr
Government equity (no debt service)	15%	€254 M	—	—	—
Total	100%	€1.69 bn			€141 M / yr

During the 7-year grace period the operator pays interest only – multilateral €41 M / yr + bonds €70 M / yr = €111 M / yr total – plus the equity tranche amortised across construction (€36 M / yr × 7 yr). Principal repayment begins in year 8 on a 23-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€21 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€21 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€120 k
Traction energy (364.5 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (973 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€2.5 M
OPEX subtotal		€44 M / yr

Annual fleet utilisation: 116 revenue trainsets × 20.5 h/day × 365 d/yr × 35 km/h commercial × 75% revenue factor = 22.8 M train-km / yr (~196 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$165 USD (per [lib/templates/country-finance.toml](https://github.com/OpenSourceRail/lib/templates/country-finance.toml)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.25 (~\$0.28 USD)
Day pass (3 trips)	€0.65 (15 % bulk discount)
Monthly unlimited pass	€7.59 (~5 % of median monthly income)
Annual pass	€83.49 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	42.0 M	84.0 M
Farebox revenue	€11 M / yr	€21 M / yr
Farebox / OPEX recovery	24%	48%
Country policy-target recovery (diagnostic)	45%	45%
Operating shortfall (gov subsidy required)	€33 M / yr	€23 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€174 M / yr	€163 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
gujranwala.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
gujranwala-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
gujranwala.corridor.geojson	Line polylines + stations (GeoJSON)
gujranwala.stations.json	Machine-readable station list
gujranwala.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug gujranwala

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug gujranwala \
  --sidecar .cache/osr-pipeline/rasters/gujranwala.grid.json \
  --out-dir designs/.../Gujranwala

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../gujranwala.toml \
  --out designs/.../README.md
```

`scripts/regenerate-gujranwala.sh` chains steps 3 + drift tests into a single command.

south-asia/Pakistan/Hyderabad-Pk

designs/south-asia/Pakistan/Hyderabad-Pk/README.md

Hyderabad-Pk — Urban Rail Network

Country: PK · Population: 1,900,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Hyderabad-Pk rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [hyderabad-pk.corridor.geojson](#).

At a glance

Metric	Value
Lines	6
Unique stations	84
Interchange stations	17
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	61.2%
Route length (double track)	181.7 km
Revenue fleet	135 × 4-car trainsets
Spare + cold-reserve	17 × 4-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	27.8 km	13	24	SW Outer NE Mid
line-2	28.8 km	14	24	N Outer SE Mid
line-3	19.5 km	11	17	S Mid NE Mid
line-4	32.3 km	12	27	NW Mid E Outer
line-5	18.9 km	11	17	W Mid SE Inner
line-6	54.3 km	23	43	NW Mid W Mid
Total	181.7 km	84 unique	152	

Rolling stock

Property	Value
Consist	4-car, 75 m
Max speed	90 km/h
Onboard battery	480 kWh per trainset
Seats	80 longitudinal seats
Nominal capacity (AW2)	440 pax (seated + standing, metro-4car per RFC 0008 §1)
Crush capacity (AW3)	560 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 440 AW2 passengers (metro-4car)
- Peak frequency: 12 trains/hour/direction (5-min headway)
- Peak capacity per line per direction: $440 \times 12 = 5,280$ pphpd
- Network peak throughput (all lines, both directions): $6 \text{ lines} \times 2 \text{ directions} \times 5,280 = 63,360$ passengers/hour
- Daily theoretical capacity (peak $\times$ 10): $\approx 633,600$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 116,280 - 174,420$ trips/day

Catchment

- City population: 1,900,000
- Anchor-weighted coverage: 61.2%
- Catchment population: $\approx 1,162,800$ (within ~800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	17	500 kW	3000 kWh
Major	23	400 kW	2500 kWh
Standard	26	300 kW	2000 kWh
Terminal	9	500 kW	3000 kWh
Total installed	76	35,000 kW	227,500 kWh

Aggregate station-rail charging power: 45,000 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 480 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	16.0 kWh/km	4 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	485 kWh	30.3 km average line length
Onboard battery coverage	1.0 $\times$ average line run	480 kWh usable pack
Average 60 s dwell charge	8.9 kWh/stop	536 kW average charger across stops
Stops to refill one trainset pack	54 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	175 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	228 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in design.toml — emitted by the osr-design Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. country-costs.toml applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (163.0 km @ €3.5 M/km)	€570 M
Elevated (17.0 km @ €18 M/km)	€306 M
Elevated-interchange premium (10 sites @ €20 M)	€200 M
Civil subtotal	€1.08 bn

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
halt	8	€0.4 M	€3.2 M
standard	26	€1.5 M	€39 M
major	23	€3.0 M	€69 M
terminal	9	€2.5 M	€22 M
depot-terminal	1	€3.0 M	€3.0 M
interchange-elevated	17	€4.5 M	€76 M
Stations subtotal			€213 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	9	€3.0 M	€27 M
Depots subtotal			€52 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
metro-4car (revenue + spare + cold reserve)	152	€4.0 M	€608 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	181.7 km × €0.015 M/km	€2.7 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€31 M
EPC integration + project management (7%)	on subtotal	€139 M

Total

Bucket	Value
Civil works	€1.08 bn
Stations	€213 M
Depots	€52 M
Rolling stock	€608 M
Residual train-control wayside + charging microgrids	€34 M
EPC overhead (7%)	€139 M
CAPEX total	€2.12 bn
Per-route-km	€12 M / km
Per-capita (city pop)	€1,117 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running `scripts/regenerate-city.sh` `hyderabad-pk`.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–7 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 8 and runs for 23 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–7)	€184 M / yr	€97
Steady-state, low-ridership (year 8+)	€221 M / yr	€116
Steady-state, high-ridership (year 8+)	€213 M / yr	€112
Lifecycle envelope (yr 1–30, low scenario)	€6.38 bn cumulative	€3,357
Lifecycle envelope (yr 1–30, high scenario)	€6.18 bn cumulative	€3,250

Population basis: 1,900,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 30, debt service drops to zero and only the OPEX shortfall remains – ~€45 M / yr (low) €37 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€1.27 bn	4.0%	30 y, 7 y grace	€86 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€531 M	16.5%	30 y, 7 y grace	€90 M / yr
Government equity (no debt service)	15%	€318 M	—	—	—
Total	100%	€2.12 bn			€176 M / yr

During the 7-year grace period the operator pays interest only – multilateral €51 M / yr + bonds €88 M / yr = €138 M / yr total – plus the equity tranche amortised across construction (€45 M / yr × 7 yr). Principal repayment begins in year 8 on a 23-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€24 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€27 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€135 k
Traction energy (424.3 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (1,102 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€2.8 M
OPEX subtotal		€54 M / yr

Annual fleet utilisation: 135 revenue trainsets × 20.5 h/day × 365 d/yr × 35 km/h commercial × 75% revenue factor = 26.5 M train-km / yr (~196 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$165 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.25 (~\$0.28 USD)
Day pass (3 trips)	€0.65 (15 % bulk discount)
Monthly unlimited pass	€7.59 (~5 % of median monthly income)
Annual pass	€83.49 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	34.7 M	69.3 M
Farebox revenue	€8.8 M / yr	€18 M / yr
Farebox / OPEX recovery	16%	32%
Country policy-target recovery (diagnostic)	45%	45%
Operating shortfall (gov subsidy required)	€45 M / yr	€37 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€221 M / yr	€213 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
hyderabad-pk.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
hyderabad-pk-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
hyderabad-pk.corridor.geojson	Line polylines + stations (GeoJSON)
hyderabad-pk.stations.json	Machine-readable station list
hyderabad-pk.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug hyderabad-pk

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug hyderabad-pk \
  --sidecar .cache/osr-pipeline/rasters/hyderabad-pk.grid.json \
  --out-dir designs/.../Hyderabad-Pk

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../hyderabad-pk.toml \
  --out designs/.../README.md
```

`scripts/regenerate-hyderabad-pk.sh` chains steps 3 + drift tests into a single command.

south-asia/Pakistan/Karachi

designs/south-asia/Pakistan/Karachi/README.md

Karachi — Urban Rail Network

Country: PK · Population: 20,300,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Karachi rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [karachi.corridor.geojson](#).

At a glance

Metric	Value
Lines	9
Unique stations	231
Interchange stations	33
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	48.4%
Route length (double track)	472.3 km
Revenue fleet	337 × 6-car trainsets
Spare + cold-reserve	40 × 6-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	57.8 km	27	46	NW Outer E Outer
line-2	47.2 km	25	38	W Outer SE Mid
line-3	45.5 km	23	37	N Outer SW Mid
line-4	46.3 km	21	37	E Mid W Outer
line-5	46.2 km	25	37	NE Outer S Mid
line-6	46.1 km	24	37	N Outer S Mid
line-7	37.4 km	19	30	W Mid E Outer
line-8	41.7 km	16	34	NE Outer SE Mid
line-9	104.1 km	51	81	NW Mid NW Mid
Total	472.3 km	231 unique	377	

Rolling stock

Property	Value
Consist	6-car, 111 m
Max speed	100 km/h
Onboard battery	720 kWh per trainset
Seats	120 longitudinal seats
Nominal capacity (AW2)	660 pax (seated + standing, metro-6car per RFC 0008 §1)
Crush capacity (AW3)	840 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 660 AW2 passengers (metro - 6 car)
- Peak frequency: 12 trains/hour/direction (5-min headway)
- Peak capacity per line per direction: $660 \times 12 = 7,920$ pphpd
- Network peak throughput (all lines, both directions): $9 \text{ lines} \times 2 \text{ directions} \times 7,920 = 142,560$ passengers/hour
- Daily theoretical capacity (peak $\times$ 10): $\approx 1,425,600$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 982,520 - 1,473,780$ trips/day

Catchment

- City population: 20,300,000
- Anchor-weighted coverage: 48.4%
- Catchment population: $\approx 9,825,200$ (within ~800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	33	500 kW	3000 kWh
Major	94	400 kW	2500 kWh
Standard	83	300 kW	2000 kWh
Terminal	15	500 kW	3000 kWh
Total installed	226	91,500 kW	585,000 kWh

Aggregate station-rail charging power: 122,250 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 720 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	24.0 kWh/km	6 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	1,259 kWh	52.5 km average line length
Onboard battery coverage	0.6 $\times$ average line run	720 kWh usable pack
Average 60 s dwell charge	8.8 kWh/stop	529 kW average charger across stops
Stops to refill one trainset pack	82 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	458 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	585 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in design.toml — emitted by the osr-design Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. country-costs.toml applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (435.6 km @ €3.5 M/km)	€1.52 bn
Elevated (34.7 km @ €18 M/km)	€625 M
Elevated-interchange premium (25 sites @ €20 M)	€500 M
Civil subtotal	€2.65 bn

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
halt	5	€0.4 M	€2.0 M
standard	83	€1.5 M	€124 M
major	94	€3.0 M	€282 M
terminal	15	€2.5 M	€38 M
depot-terminal	1	€3.0 M	€3.0 M
interchange-elevated	33	€4.5 M	€148 M
Stations subtotal			€598 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	15	€3.0 M	€45 M
Depots subtotal			€70 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
metro-6car (revenue + spare + cold reserve)	377	€6.0 M	€2.26 bn

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	472.3 km × €0.015 M/km	€7.1 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€86 M
EPC integration + project management (7%)	on subtotal	€397 M

Total

Bucket	Value
Civil works	€2.65 bn
Stations	€598 M
Depots	€70 M
Rolling stock	€2.26 bn
Residual train-control wayside + charging microgrids	€93 M
EPC overhead (7%)	€397 M
CAPEX total	€6.07 bn
Per-route-km	€13 M / km

Bucket	Value
Per-capita (city pop)	€299 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](https://github.com/OpenSourceRail/lib/templates/country-finance.toml). Pure function of the [costs] block above + the country code — regenerate by re-running `scripts/regenerate-city.sh karachi`.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–7 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 8 and runs for 23 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–7)	€526 M / yr	€26
Steady-state, low-ridership (year 8+)	€574 M / yr	€28
Steady-state, high-ridership (year 8+)	€503 M / yr	€25
Lifecycle envelope (yr 1–30, low scenario)	€16.88 bn cumulative	€831
Lifecycle envelope (yr 1–30, high scenario)	€15.25 bn cumulative	€751

Population basis: 20,300,000 (catchment per [lib/city-batches/world-sample.toml](https://github.com/OpenSourceRail/lib/city-batches/world-sample.toml)). After year 30, debt service drops to zero and only the OPEX shortfall remains — ~€71 M / yr (low) €0 k / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€3.64 bn	4.0%	30 y, 7 y grace	€245 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€1.52 bn	16.5%	30 y, 7 y grace	€258 M / yr
Government equity (no debt service)	15%	€910 M	—	—	—
Total	100%	€6.07 bn			€503 M / yr

During the 7-year grace period the operator pays interest only — multilateral €146 M / yr + bonds €250 M / yr = €396 M / yr total — plus the equity tranche amortised across construction (€130 M / yr × 7 yr). Principal repayment begins in year 8 on a 23-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€90 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€66 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€353 k
Traction energy (1588.6 GWh / yr)	trackside PV + Na-ion (RFC 0002) — self-generated, €0 / yr	€0 k
Labour (2,846 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€7.3 M
OPEX subtotal		€164 M / yr

Annual fleet utilisation: 337 revenue trainsets × 20.5 h/day × 365 d/yr × 35 km/h commercial × 75% revenue factor = 66.2 M train-km / yr (~196 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$165 USD (per [lib/templates/country-finance.toml](https://github.com/OpenSourceRail/lib/templates/country-finance.toml)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass — a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.25 (~\$0.28 USD)
Day pass (3 trips)	€0.65 (15 % bulk discount)
Monthly unlimited pass	€7.59 (~5 % of median monthly income)
Annual pass	€83.49 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	370.5 M	741.0 M
Farebox revenue	€94 M / yr	€187 M / yr
Farebox / OPEX recovery	57%	114%
Country policy-target recovery (diagnostic)	45%	45%
Operating shortfall (gov subsidy required)	€71 M / yr	€0 k / yr
Operating surplus (operator retained capex sinking fund)	€0 k / yr	€23 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€574 M / yr	€503 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
karachi.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
karachi-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
karachi.corridor.geojson	Line polylines + stations (GeoJSON)
karachi.stations.json	Machine-readable station list
karachi.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug karachi

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug karachi \
  --sidecar .cache/osr-pipeline/rasters/karachi.grid.json \
  --out-dir designs/.../Karachi

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../karachi.toml \
  --out designs/.../README.md
```

`scripts/regenerate-karachi.sh` chains steps 3 + drift tests into a single command.

south-asia/Pakistan/Larkana

designs/south-asia/Pakistan/Larkana/README.md

Larkana — Urban Rail Network

Country: PK · Population: 500,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Larkana rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [larkana.corridor.geojson](#).

At a glance

Metric	Value
Lines	2
Unique stations	19
Interchange stations	4
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	28.0%
Route length (double track)	37.7 km
Revenue fleet	34 × 3-car trainsets
Spare + cold-reserve	5 × 3-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	23.7 km	11	24	N Outer SW Outer
line-2	14.0 km	8	15	W Inner NE Outer
Total	37.7 km	19 unique	39	

Rolling stock

Property	Value
Consist	3-car, 57 m
Max speed	90 km/h
Onboard battery	360 kWh per trainset
Seats	60 longitudinal seats
Nominal capacity (AW2)	330 pax (seated + standing, light-metro-3car per RFC 0008 §1)
Crush capacity (AW3)	420 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 330 AW2 passengers (light-metro-3car)
- Peak frequency: 12 trains/hour/direction (5-min headway)
- Peak capacity per line per direction: 330 × 12 = 3,960 pphpd

- Network peak throughput (all lines, both directions): $2 \text{ lines} \times 2 \text{ directions} \times 3,960 = 15,840$ passengers/hour
- Daily theoretical capacity (peak $\times 10$): $\approx 158,400$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 14,000 - 21,000$ trips/day

Catchment

- City population: 500,000
- Anchor-weighted coverage: 28.0%
- Catchment population: $\approx 140,000$ (within ~ 800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	4	500 kW	3000 kWh
Major	4	400 kW	2500 kWh
Standard	4	300 kW	2000 kWh
Terminal	3	500 kW	3000 kWh
Total installed	16	11,300 kW	79,000 kWh

Aggregate station-rail charging power: 10,750 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 360 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	12.0 kWh/km	3 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	226 kWh	18.9 km average line length
Onboard battery coverage	1.6 $\times$ average line run	360 kWh usable pack
Average 60 s dwell charge	9.4 kWh/stop	566 kW average charger across stops
Stops to refill one trainset pack	38 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	56 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	79 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in design.toml — emitted by the osr-design Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. country-costs.toml applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (24.8 km @ €3.5 M/km)	€87 M
Elevated (12.7 km @ €18 M/km)	€229 M
Elevated-interchange premium (1 sites @ €20 M)	€20 M
Civil subtotal	€336 M

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~ 11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
halt	3	€0.4 M	€1.2 M
standard	4	€1.5 M	€6.0 M
major	4	€3.0 M	€12 M
terminal	3	€2.5 M	€7.5 M
depot-terminal	1	€3.0 M	€3.0 M
interchange	2	€4.5 M	€9.0 M
interchange-elevated	2	€4.5 M	€9.0 M
Stations subtotal			€48 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	3	€3.0 M	€9.0 M
Depots subtotal			€34 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
light-metro-3car (revenue + spare + cold reserve)	39	€3.0 M	€117 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	37.7 km × €0.015 M/km	€0.6 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€7.3 M
EPC integration + project management (7%)	on subtotal	€38 M

Total

Bucket	Value
Civil works	€336 M
Stations	€48 M
Depots	€34 M
Rolling stock	€117 M
Residual train-control wayside + charging microgrids	€7.9 M
EPC overhead (7%)	€38 M
CAPEX total	€580 M
Per-route-km	€15 M / km
Per-capita (city pop)	€1,161 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running scripts/regenerate-city.sh larkana.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–7 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 8 and runs for 23 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–7)	€50 M / yr	€101
Steady-state, low-ridership (year 8+)	€59 M / yr	€119
Steady-state, high-ridership (year 8+)	€57 M / yr	€114
Lifecycle envelope (yr 1–30, low scenario)	€1.72 bn cumulative	€3,440
Lifecycle envelope (yr 1–30, high scenario)	€1.67 bn cumulative	€3,334

Population basis: 500,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 30, debt service drops to zero and only the OPEX shortfall remains – ~€11 M / yr (low) €9.0 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€348 M	4.0%	30 y, 7 y grace	€23 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€145 M	16.5%	30 y, 7 y grace	€25 M / yr
Government equity (no debt service)	15%	€87 M	—	—	—
Total	100%	€580 M			€48 M / yr

During the 7-year grace period the operator pays interest only – multilateral €14 M / yr + bonds €24 M / yr = €38 M / yr total – plus the equity tranche amortised across construction (€12 M / yr × 7 yr). Principal repayment begins in year 8 on a 23-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€4.7 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€8.4 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€28 k
Traction energy (68.7 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (238 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€607 k
OPEX subtotal		€14 M / yr

Annual fleet utilisation: 34 revenue trainsets × 20.5 h/day × 365 d/yr × 30 km/h commercial × 75% revenue factor = 5.7 M train-km / yr (~168 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$165 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.25 (~\$0.28 USD)
Day pass (3 trips)	€0.65 (15 % bulk discount)
Monthly unlimited pass	€7.59 (~5 % of median monthly income)
Annual pass	€83.49 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	9.1 M	18.2 M

	Low scenario	High scenario
Farebox revenue	€2.3 M / yr	€4.6 M / yr
Farebox / OPEX recovery	17%	34%
Country policy-target recovery (diagnostic)	45%	45%
Operating shortfall (gov subsidy required)	€11 M / yr	€9.0 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€59 M / yr	€57 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
larkana.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
larkana-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
larkana.corridor.geojson	Line polylines + stations (GeoJSON)
larkana.stations.json	Machine-readable station list
larkana.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug larkana

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug larkana \
  --sidecar .cache/osr-pipeline/rasters/larkana.grid.json \
  --out-dir designs/.../Larkana

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../larkana.toml \
  --out designs/.../README.md
```

`scripts/regenerate-larkana.sh` chains steps 3 + drift tests into a single command.

south-asia/Pakistan/Multan

designs/south-asia/Pakistan/Multan/README.md

Multan — Urban Rail Network

Country: PK · Population: 2,197,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Multan rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [multan.corridor.geojson](#).

At a glance

Metric	Value
Lines	4
Unique stations	63
Interchange stations	11
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	41.8%
Route length (double track)	115.4 km
Revenue fleet	87 × 4-car trainsets
Spare + cold-reserve	11 × 4-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	30.6 km	16	26	SW Outer NE Outer
line-2	23.2 km	13	20	SW Mid N Outer
line-3	19.5 km	14	17	E Mid W Mid
line-4	42.1 km	21	35	N Mid N Mid
Total	115.4 km	63 unique	98	

Rolling stock

Property	Value
Consist	4-car, 75 m
Max speed	90 km/h
Onboard battery	480 kWh per trainset
Seats	80 longitudinal seats
Nominal capacity (AW2)	440 pax (seated + standing, met ro - 4car per RFC 0008 §1)
Crush capacity (AW3)	560 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 440 AW2 passengers (met ro - 4car)

- Peak frequency: 12 trains/hour/direction (5-min headway)
- Peak capacity per line per direction: $440 \times 12 = 5,280$ pphpd
- Network peak throughput (all lines, both directions): $4 \text{ lines} \times 2 \text{ directions} \times 5,280 = 42,240$ passengers/hour
- Daily theoretical capacity (peak $\times$ 10): $\approx 422,400$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 91,834 - 137,751$ trips/day

Catchment

- City population: 2,197,000
- Anchor-weighted coverage: 41.8%
- Catchment population: $\approx 918,346$ (within ~800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	11	500 kW	3000 kWh
Major	32	400 kW	2500 kWh
Standard	15	300 kW	2000 kWh
Terminal	5	500 kW	3000 kWh
Total installed	64	30,300 kW	198,000 kWh

Aggregate station-rail charging power: 35,000 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 480 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	16.0 kWh/km	4 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	462 kWh	28.9 km average line length
Onboard battery coverage	1.0 $\times$ average line run	480 kWh usable pack
Average 60 s dwell charge	9.3 kWh/stop	556 kW average charger across stops
Stops to refill one trainset pack	52 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	152 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	198 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the `[costs]` block in `design.toml` — emitted by the `osr-design` Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. `country-costs.toml` applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (109.4 km @ €3.5 M/km)	€383 M
Elevated (5.5 km @ €18 M/km)	€99 M
Elevated-interchange premium (5 sites @ €20 M)	€100 M
Civil subtotal	€582 M

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
standard	15	€1.5 M	€22 M
major	32	€3.0 M	€96 M
terminal	5	€2.5 M	€12 M
depot-terminal	1	€3.0 M	€3.0 M
interchange-elevated	11	€4.5 M	€50 M
Stations subtotal			€184 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	5	€3.0 M	€15 M
Depots subtotal			€40 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
metro-4car (revenue + spare + cold reserve)	98	€4.0 M	€392 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	115.4 km × €0.015 M/km	€1.7 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€26 M
EPC integration + project management (7%)	on subtotal	€86 M

Total

Bucket	Value
Civil works	€582 M
Stations	€184 M
Depots	€40 M
Rolling stock	€392 M
Residual train-control wayside + charging microgrids	€28 M
EPC overhead (7%)	€86 M
CAPEX total	€1.31 bn
Per-route-km	€11 M / km
Per-capita (city pop)	€597 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](https://github.com/OpenSourceRail/lib/templates/country-finance.toml). Pure function of the [costs] block above + the country code – regenerate by re-running `scripts/regenerate-city.sh` multan.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–7 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 8 and runs for 23 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–7)	€114 M / yr	€52
Steady-state, low-ridership (year 8+)	€132 M / yr	€60
Steady-state, high-ridership (year 8+)	€122 M / yr	€56
Lifecycle envelope (yr 1–30, low scenario)	€3.84 bn cumulative	€1,746
Lifecycle envelope (yr 1–30, high scenario)	€3.60 bn cumulative	€1,640

Population basis: 2,197,000 (catchment per [lib/city-batches/world-sample.toml](https://github.com/OpenSourceRail/lib/city-batches/world-sample.toml)). After year 30, debt service drops to zero and only the OPEX shortfall remains – ~€24 M / yr (low) €13 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€787 M	4.0%	30 y, 7 y grace	€53 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€328 M	16.5%	30 y, 7 y grace	€56 M / yr
Government equity (no debt service)	15%	€197 M	—	—	—
Total	100%	€1.31 bn			€109 M / yr

During the 7-year grace period the operator pays interest only – multilateral €31 M / yr + bonds €54 M / yr = €86 M / yr total – plus the equity tranche amortised across construction (€28 M / yr × 7 yr). Principal repayment begins in year 8 on a 23-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€16 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€16 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€86 k
Traction energy (273.4 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (704 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€1.8 M
OPEX subtotal		€34 M / yr

Annual fleet utilisation: 87 revenue trainsets × 20.5 h/day × 365 d/yr × 35 km/h commercial × 75% revenue factor = 17.1 M train-km / yr (~196 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$165 USD (per [lib/templates/country-finance.toml](https://github.com/OpenSourceRail/lib/templates/country-finance.toml)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.25 (~\$0.28 USD)
Day pass (3 trips)	€0.65 (15 % bulk discount)
Monthly unlimited pass	€7.59 (~5 % of median monthly income)
Annual pass	€83.49 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	40.1 M	80.2 M
Farebox revenue	€10 M / yr	€20 M / yr
Farebox / OPEX recovery	30%	60%
Country policy-target recovery (diagnostic)	45%	45%
Operating shortfall (gov subsidy required)	€24 M / yr	€13 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€132 M / yr	€122 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
multan.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
multan-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
multan.corridor.geojson	Line polylines + stations (GeoJSON)
multan.stations.json	Machine-readable station list
multan.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug multan

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug multan \
  --sidecar .cache/osr-pipeline/rasters/multan.grid.json \
  --out-dir designs/.../Multan

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../multan.toml \
  --out designs/.../README.md
```

`scripts/regenerate-multan.sh` chains steps 3 + drift tests into a single command.

south-asia/Pakistan/Peshawar

designs/south-asia/Pakistan/Peshawar/README.md

Peshawar — Urban Rail Network

Country: PK · Population: 2,300,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Peshawar rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [peshawar.corridor.geojson](#).

At a glance

Metric	Value
Lines	4
Unique stations	86
Interchange stations	10
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	32.8%
Route length (double track)	196.7 km
Revenue fleet	142 × 4-car trainsets
Spare + cold-reserve	16 × 4-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	50.4 km	22	40	NE Outer SW Outer
line-2	33.5 km	21	28	W Mid E Outer
line-3	38.6 km	15	31	S Mid NW Outer
line-4	74.2 km	29	59	W Mid W Mid
Total	196.7 km	86 unique	158	

Rolling stock

Property	Value
Consist	4-car, 75 m
Max speed	90 km/h
Onboard battery	480 kWh per trainset
Seats	80 longitudinal seats
Nominal capacity (AW2)	440 pax (seated + standing, met ro - 4car per RFC 0008 §1)
Crush capacity (AW3)	560 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 440 AW2 passengers (met ro - 4car)

- Peak frequency: 12 trains/hour/direction (5-min headway)
- Peak capacity per line per direction: $440 \times 12 = 5,280$ pphpd
- Network peak throughput (all lines, both directions): $4 \text{ lines} \times 2 \text{ directions} \times 5,280 = 42,240$ passengers/hour
- Daily theoretical capacity (peak $\times$ 10): $\approx 422,400$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 75,440 - 113,160$ trips/day

Catchment

- City population: 2,300,000
- Anchor-weighted coverage: 32.8%
- Catchment population: $\approx 754,400$ (within ~800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	10	500 kW	3000 kWh
Major	27	400 kW	2500 kWh
Standard	41	300 kW	2000 kWh
Terminal	5	500 kW	3000 kWh
Total installed	84	35,600 kW	234,500 kWh

Aggregate station-rail charging power: 45,750 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 480 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	16.0 kWh/km	4 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	787 kWh	49.2 km average line length
Onboard battery coverage	0.6 $\times$ average line run	480 kWh usable pack
Average 60 s dwell charge	8.9 kWh/stop	532 kW average charger across stops
Stops to refill one trainset pack	54 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	178 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	234 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in design.toml — emitted by the osr-design Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. country-costs.toml applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (167.1 km @ €3.5 M/km)	€585 M
Elevated (28.7 km @ €18 M/km)	€517 M
Elevated-interchange premium (6 sites @ €20 M)	€120 M
Civil subtotal	€1.22 bn

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
halt	3	€0.4 M	€1.2 M
standard	41	€1.5 M	€62 M
major	27	€3.0 M	€81 M
terminal	5	€2.5 M	€12 M
depot-terminal	1	€3.0 M	€3.0 M
interchange-elevated	10	€4.5 M	€45 M
Stations subtotal			€204 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	5	€3.0 M	€15 M
Depots subtotal			€40 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
metro-4car (revenue + spare + cold reserve)	158	€4.0 M	€632 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	196.7 km × €0.015 M/km	€2.9 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€30 M
EPC integration + project management (7%)	on subtotal	€149 M

Total

Bucket	Value
Civil works	€1.22 bn
Stations	€204 M
Depots	€40 M
Rolling stock	€632 M
Residual train-control wayside + charging microgrids	€33 M
EPC overhead (7%)	€149 M
CAPEX total	€2.28 bn
Per-route-km	€12 M / km
Per-capita (city pop)	€991 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running `scripts/regenerate-city.sh` peshawar.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–7 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 8 and runs for 23 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–7)	€198 M / yr	€86
Steady-state, low-ridership (year 8+)	€236 M / yr	€103
Steady-state, high-ridership (year 8+)	€226 M / yr	€98
Lifecycle envelope (yr 1–30, low scenario)	€6.81 bn cumulative	€2,963
Lifecycle envelope (yr 1–30, high scenario)	€6.57 bn cumulative	€2,857

Population basis: 2,300,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 30, debt service drops to zero and only the OPEX shortfall remains – ~€47 M / yr (low) €37 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€1.37 bn	4.0%	30 y, 7 y grace	€92 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€570 M	16.5%	30 y, 7 y grace	€97 M / yr
Government equity (no debt service)	15%	€342 M	—	—	—
Total	100%	€2.28 bn			€189 M / yr

During the 7-year grace period the operator pays interest only – multilateral €55 M / yr + bonds €94 M / yr = €149 M / yr total – plus the equity tranche amortised across construction (€49 M / yr × 7 yr). Principal repayment begins in year 8 on a 23-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€25 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€29 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€147 k
Traction energy (446.3 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (1,192 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€3.0 M
OPEX subtotal		€58 M / yr

Annual fleet utilisation: 142 revenue trainsets × 20.5 h/day × 365 d/yr × 35 km/h commercial × 75% revenue factor = 27.9 M train-km / yr (~196 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$165 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.25 (~\$0.28 USD)
Day pass (3 trips)	€0.65 (15 % bulk discount)
Monthly unlimited pass	€7.59 (~5 % of median monthly income)
Annual pass	€83.49 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	42.0 M	84.0 M
Farebox revenue	€11 M / yr	€21 M / yr
Farebox / OPEX recovery	18%	37%
Country policy-target recovery (diagnostic)	45%	45%
Operating shortfall (gov subsidy required)	€47 M / yr	€37 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€236 M / yr	€226 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
peshawar.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
peshawar-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
peshawar.corridor.geojson	Line polylines + stations (GeoJSON)
peshawar.stations.json	Machine-readable station list
peshawar.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug peshawar

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug peshawar \
  --sidecar .cache/osr-pipeline/rasters/peshawar.grid.json \
  --out-dir designs/.../Peshawar

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../peshawar.toml \
  --out designs/.../README.md
```

`scripts/regenerate-peshawar.sh` chains steps 3 + drift tests into a single command.

south-asia/Pakistan/Quetta

designs/south-asia/Pakistan/Quetta/README.md

Quetta — Urban Rail Network

Country: PK · Population: 1,200,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Quetta rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [quetta.corridor.geojson](#).

At a glance

Metric	Value
Lines	5
Unique stations	77
Interchange stations	18
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	53.5%
Route length (double track)	140.1 km
Revenue fleet	106 × 4-car trainsets
Spare + cold-reserve	14 × 4-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	31.0 km	15	26	NW Outer S Outer
line-2	26.0 km	15	23	NE Mid SW Outer
line-3	16.3 km	11	15	N Mid S Inner
line-4	15.9 km	7	15	NE Outer SE Inner
line-5	50.8 km	30	41	NW Mid NW Mid
Total	140.1 km	77 unique	120	

Rolling stock

Property	Value
Consist	4-car, 75 m
Max speed	90 km/h
Onboard battery	480 kWh per trainset
Seats	80 longitudinal seats
Nominal capacity (AW2)	440 pax (seated + standing, met ro - 4 car per RFC 0008 §1)
Crush capacity (AW3)	560 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 440 AW2 passengers (met ro - 4 car)

- Peak frequency: 12 trains/hour/direction (5-min headway)
- Peak capacity per line per direction: $440 \times 12 = 5,280$ pphpd
- Network peak throughput (all lines, both directions): $5 \text{ lines} \times 2 \text{ directions} \times 5,280 = 52,800$ passengers/hour
- Daily theoretical capacity (peak $\times$ 10): $\approx 528,000$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 64,200 - 96,300$ trips/day

Catchment

- City population: 1,200,000
- Anchor-weighted coverage: 53.5%
- Catchment population: $\approx 642,000$ (within ~800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	18	500 kW	3000 kWh
Major	30	400 kW	2500 kWh
Standard	21	300 kW	2000 kWh
Terminal	7	500 kW	3000 kWh
Total installed	77	35,800 kW	232,000 kWh

Aggregate station-rail charging power: 42,750 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 480 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	16.0 kWh/km	4 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	448 kWh	28.0 km average line length
Onboard battery coverage	1.1 $\times$ average line run	480 kWh usable pack
Average 60 s dwell charge	9.3 kWh/stop	555 kW average charger across stops
Stops to refill one trainset pack	52 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	179 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	232 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the `[costs]` block in `design.toml` — emitted by the `osr-design` Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. `country-costs.toml` applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (129.6 km @ €3.5 M/km)	€454 M
Elevated (10.0 km @ €18 M/km)	€181 M
Elevated-interchange premium (8 sites @ €20 M)	€160 M
Civil subtotal	€794 M

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
halt	1	€0.4 M	€0.4 M
standard	21	€1.5 M	€32 M
major	30	€3.0 M	€90 M
terminal	7	€2.5 M	€18 M
depot-terminal	1	€3.0 M	€3.0 M
interchange-elevated	18	€4.5 M	€81 M
Stations subtotal			€223 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	7	€3.0 M	€21 M
Depots subtotal			€46 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
metro-4car (revenue + spare + cold reserve)	120	€4.0 M	€480 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	140.1 km × €0.015 M/km	€2.1 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€32 M
EPC integration + project management (7%)	on subtotal	€110 M

Total

Bucket	Value
Civil works	€794 M
Stations	€223 M
Depots	€46 M
Rolling stock	€480 M
Residual train-control wayside + charging microgrids	€34 M
EPC overhead (7%)	€110 M
CAPEX total	€1.69 bn
Per-route-km	€12 M / km
Per-capita (city pop)	€1,407 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running scripts/regenerate-city.sh quetta.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–7 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 8 and runs for 23 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–7)	€146 M / yr	€122
Steady-state, low-ridership (year 8+)	€177 M / yr	€148
Steady-state, high-ridership (year 8+)	€172 M / yr	€143
Lifecycle envelope (yr 1–30, low scenario)	€5.10 bn cumulative	€4,249
Lifecycle envelope (yr 1–30, high scenario)	€4.97 bn cumulative	€4,142

Population basis: 1,200,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 30, debt service drops to zero and only the OPEX shortfall remains – ~€37 M / yr (low) €32 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€1.01 bn	4.0%	30 y, 7 y grace	€68 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€422 M	16.5%	30 y, 7 y grace	€72 M / yr
Government equity (no debt service)	15%	€253 M	—	—	—
Total	100%	€1.69 bn			€140 M / yr

During the 7-year grace period the operator pays interest only – multilateral €41 M / yr + bonds €70 M / yr = €110 M / yr total – plus the equity tranche amortised across construction (€36 M / yr × 7 yr). Principal repayment begins in year 8 on a 23-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€19 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€21 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€105 k
Traction energy (333.1 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (853 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€2.2 M
OPEX subtotal		€43 M / yr

Annual fleet utilisation: 106 revenue trainsets × 20.5 h/day × 365 d/yr × 35 km/h commercial × 75% revenue factor = 20.8 M train-km / yr (~196 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$165 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.25 (~\$0.28 USD)
Day pass (3 trips)	€0.65 (15 % bulk discount)
Monthly unlimited pass	€7.59 (~5 % of median monthly income)
Annual pass	€83.49 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	21.9 M	43.8 M
Farebox revenue	€5.5 M / yr	€11 M / yr
Farebox / OPEX recovery	13%	26%
Country policy-target recovery (diagnostic)	45%	45%
Operating shortfall (gov subsidy required)	€37 M / yr	€32 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€177 M / yr	€172 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
quetta.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
quetta-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
quetta.corridor.geojson	Line polylines + stations (GeoJSON)
quetta.stations.json	Machine-readable station list
quetta.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug quetta

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug quetta \
  --sidecar .cache/osr-pipeline/rasters/quetta.grid.json \
  --out-dir designs/.../Quetta

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../quetta.toml \
  --out designs/.../README.md
```

`scripts/regenerate-quetta.sh` chains steps 3 + drift tests into a single command.

south-asia/Pakistan/Rahim-Yar-Khan

designs/south-asia/Pakistan/Rahim-Yar-Khan/README.md

Rahim-Yar-Khan — Urban Rail Network

Country: PK · Population: 500,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Rahim-Yar-Khan rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [rahim-yar-khan.corridor.geojson](#).

At a glance

Metric	Value
Lines	3
Unique stations	24
Interchange stations	2
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	36.7%
Route length (double track)	46.0 km
Revenue fleet	41 × 3-car trainsets
Spare + cold-reserve	6 × 3-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	19.6 km	9	19	NE Outer SW Outer
line-2	9.7 km	7	11	W Mid S Mid
line-3	16.7 km	8	17	NE Outer S Mid
Total	46.0 km	24 unique	47	

Rolling stock

Property	Value
Consist	3-car, 57 m
Max speed	90 km/h
Onboard battery	360 kWh per trainset
Seats	60 longitudinal seats
Nominal capacity (AW2)	330 pax (seated + standing, light-metro-3car per RFC 0008 §1)
Crush capacity (AW3)	420 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 330 AW2 passengers (light-metro-3car)
- Peak frequency: 12 trains/hour/direction (5-min headway)

- Peak capacity per line per direction: $330 \times 12 = 3,960$ pphpd
- Network peak throughput (all lines, both directions): $3 \text{ lines} \times 2 \text{ directions} \times 3,960 = 23,760$ passengers/hour
- Daily theoretical capacity (peak $\times 10$): $\approx 237,600$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 18,350 - 27,525$ trips/day

Catchment

- City population: 500,000
- Anchor-weighted coverage: 36.7%
- Catchment population: $\approx 183,500$ (within ~ 800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	2	500 kW	3000 kWh
Major	5	400 kW	2500 kWh
Standard	9	300 kW	2000 kWh
Terminal	5	500 kW	3000 kWh
Total installed	22	13,200 kW	91,500 kWh

Aggregate station-rail charging power: 14,500 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 360 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	12.0 kWh/km	3 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	184 kWh	15.3 km average line length
Onboard battery coverage	2.0 $\times$ average line run	360 kWh usable pack
Average 60 s dwell charge	10.1 kWh/stop	604 kW average charger across stops
Stops to refill one trainset pack	36 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	66 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	92 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in design.toml — emitted by the osr-design Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. country-costs.toml applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (45.1 km @ €3.5 M/km)	€158 M
Elevated (0.8 km @ €18 M/km)	€14 M
Civil subtotal	€172 M

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~ 11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
halt	2	€0.4 M	€0.8 M
standard	9	€1.5 M	€14 M
major	5	€3.0 M	€15 M
terminal	5	€2.5 M	€12 M
depot-terminal	1	€3.0 M	€3.0 M
interchange	2	€4.5 M	€9.0 M
Stations subtotal			€54 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	5	€3.0 M	€15 M
Depots subtotal			€40 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
light-metro-3car (revenue + spare + cold reserve)	47	€3.0 M	€141 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	46.0 km × €0.015 M/km	€0.7 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€8.4 M
EPC integration + project management (7%)	on subtotal	€29 M

Total

Bucket	Value
Civil works	€172 M
Stations	€54 M
Depots	€40 M
Rolling stock	€141 M
Residual train-control wayside + charging microgrids	€9.1 M
EPC overhead (7%)	€29 M
CAPEX total	€445 M
Per-route-km	€9.7 M / km
Per-capita (city pop)	€890 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running scripts/regenerate-city.sh [rahim-yar-khan](#).

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–7 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 8 and runs for 23 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–7)	€39 M / yr	€77
Steady-state, low-ridership (year 8+)	€46 M / yr	€93
Steady-state, high-ridership (year 8+)	€44 M / yr	€88
Lifecycle envelope (yr 1–30, low scenario)	€1.34 bn cumulative	€2,671
Lifecycle envelope (yr 1–30, high scenario)	€1.28 bn cumulative	€2,565

Population basis: 500,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 30, debt service drops to zero and only the OPEX shortfall remains – ~€9.4 M / yr (low) €7.1 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€267 M	4.0%	30 y, 7 y grace	€18 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€111 M	16.5%	30 y, 7 y grace	€19 M / yr
Government equity (no debt service)	15%	€67 M	—	—	—
Total	100%	€445 M			€37 M / yr

During the 7-year grace period the operator pays interest only – multilateral €11 M / yr + bonds €18 M / yr = €29 M / yr total – plus the equity tranche amortised across construction (€9.5 M / yr × 7 yr). Principal repayment begins in year 8 on a 23-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€5.6 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€5.3 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€34 k
Traction energy (82.8 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (288 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€734 k
OPEX subtotal		€12 M / yr

Annual fleet utilisation: 41 revenue trainsets × 20.5 h/day × 365 d/yr × 30 km/h commercial × 75% revenue factor = 6.9 M train-km / yr (~168 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$165 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.25 (~\$0.28 USD)
Day pass (3 trips)	€0.65 (15 % bulk discount)
Monthly unlimited pass	€7.59 (~5 % of median monthly income)
Annual pass	€83.49 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	9.1 M	18.2 M

	Low scenario	High scenario
Farebox revenue	€2.3 M / yr	€4.6 M / yr
Farebox / OPEX recovery	20%	39%
Country policy-target recovery (diagnostic)	45%	45%
Operating shortfall (gov subsidy required)	€9.4 M / yr	€7.1 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€46 M / yr	€44 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
rahim-yar-khan.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
rahim-yar-khan-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
rahim-yar-khan.corridor.geojson	Line polylines + stations (GeoJSON)
rahim-yar-khan.stations.json	Machine-readable station list
rahim-yar-khan.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug rahim-yar-khan

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug rahim-yar-khan \
  --sidecar .cache/osr-pipeline/rasters/rahim-yar-khan.grid.json \
  --out-dir designs/.../Rahim-Yar-Khan

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../rahim-yar-khan.toml \
  --out designs/.../README.md
```

`scripts/regenerate-rahim-yar-khan.sh` chains steps 3 + drift tests into a single command.

south-asia/Pakistan/Sheikhupura

designs/south-asia/Pakistan/Sheikhupura/README.md

Sheikhupura — Urban Rail Network

Country: PK · Population: 600,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Sheikhupura rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [sheikhupura.corridor.geojson](#).

At a glance

Metric	Value
Lines	2
Unique stations	17
Interchange stations	2
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	32.3%
Route length (double track)	19.1 km
Revenue fleet	18 × 3-car trainsets
Spare + cold-reserve	4 × 3-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	11.0 km	9	12	W Outer E Outer
line-2	8.2 km	8	10	NW Outer E Outer
Total	19.1 km	17 unique	22	

Rolling stock

Property	Value
Consist	3-car, 57 m
Max speed	90 km/h
Onboard battery	360 kWh per trainset
Seats	60 longitudinal seats
Nominal capacity (AW2)	330 pax (seated + standing, light-metro-3car per RFC 0008 §1)
Crush capacity (AW3)	420 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 330 AW2 passengers (light-metro-3car)
- Peak frequency: 12 trains/hour/direction (5-min headway)
- Peak capacity per line per direction: 330 × 12 = 3,960 pphpd

- Network peak throughput (all lines, both directions): $2 \text{ lines} \times 2 \text{ directions} \times 3,960 = 15,840 \text{ passengers/hour}$
- Daily theoretical capacity (peak $\times 10$): $\approx 158,400 \text{ passenger-trips/day}$
- Practical daily ridership estimate (10–15 % of catchment): $\approx 19,380 - 29,070 \text{ trips/day}$

Catchment

- City population: 600,000
- Anchor-weighted coverage: 32.3%
- Catchment population: $\approx 193,800$ (within $\sim 800 \text{ m}$ walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	2	500 kW	3000 kWh
Major	11	400 kW	2500 kWh
Terminal	3	500 kW	3000 kWh
Total installed	17	11,900 kW	82,500 kWh

Aggregate station-rail charging power: 10,500 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 360 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	12.0 kWh/km	3 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	115 kWh	9.6 km average line length
Onboard battery coverage	3.1 $\times$ average line run	360 kWh usable pack
Average 60 s dwell charge	10.3 kWh/stop	618 kW average charger across stops
Stops to refill one trainset pack	35 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	60 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	82 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in design.toml — emitted by the osr-design Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. country-costs.toml applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (16.5 km @ €3.5 M/km)	€58 M
Elevated (2.5 km @ €18 M/km)	€44 M
Elevated-interchange premium (3 sites @ €20 M)	€60 M
Civil subtotal	€162 M

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, $\sim 11 \text{ t}$ / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
major	11	€3.0 M	€33 M

Archetype	Count	Unit	Subtotal
terminal	3	€2.5 M	€7.5 M
depot-terminal	1	€3.0 M	€3.0 M
interchange-elevated	2	€4.5 M	€9.0 M
Stations subtotal			€52 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	3	€3.0 M	€9.0 M
Depots subtotal			€34 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
light-metro-3car (revenue + spare + cold reserve)	22	€3.0 M	€66 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	19.1 km × €0.015 M/km	€0.3 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€7.5 M
EPC integration + project management (7%)	on subtotal	€23 M

Total

Bucket	Value
Civil works	€162 M
Stations	€52 M
Depots	€34 M
Rolling stock	€66 M
Residual train-control wayside + charging microgrids	€7.8 M
EPC overhead (7%)	€23 M
CAPEX total	€345 M
Per-route-km	€18 M / km
Per-capita (city pop)	€576 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code — regenerate by re-running scripts/regenerate-city.sh sheikhupura.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–7 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 8 and runs for 23 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–7)	€30 M / yr	€50
Steady-state, low-ridership (year 8+)	€34 M / yr	€56
Steady-state, high-ridership (year 8+)	€31 M / yr	€52
Lifecycle envelope (yr 1–30, low scenario)	€987 M cumulative	€1,646
Lifecycle envelope (yr 1–30, high scenario)	€924 M cumulative	€1,539

Population basis: 600,000 (catchment per lib/city-batches/world-sample.toml). After year 30, debt service drops to zero and only the OPEX shortfall remains — ~€5.2 M / yr (low) €2.4 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€207 M	4.0%	30 y, 7 y grace	€14 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€86 M	16.5%	30 y, 7 y grace	€15 M / yr
Government equity (no debt service)	15%	€52 M	—	—	—
Total	100%	€345 M			€29 M / yr

During the 7-year grace period the operator pays interest only — multilateral €8.3 M / yr + bonds €14 M / yr = €23 M / yr total — plus the equity tranche amortised across construction (€7.4 M / yr × 7 yr). Principal repayment begins in year 8 on a 23-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€2.6 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€5.0 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€14 k
Traction energy (36.4 GWh / yr)	trackside PV + Na-ion (RFC 0002) — self-generated, €0 / yr	€0 k
Labour (127 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€324 k
OPEX subtotal		€8.0 M / yr

Annual fleet utilisation: 18 revenue trainsets × 20.5 h/day × 365 d/yr × 30 km/h commercial × 75% revenue factor = 3.0 M train-km / yr (~168 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$165 USD (per lib/templates/country-finance.toml). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass — a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.25 (~\$0.28 USD)
Day pass (3 trips)	€0.65 (15 % bulk discount)
Monthly unlimited pass	€7.59 (~5 % of median monthly income)
Annual pass	€83.49 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	10.9 M	21.9 M
Farebox revenue	€2.8 M / yr	€5.5 M / yr
Farebox / OPEX recovery	35%	70%
Country policy-target recovery (diagnostic)	45%	45%
Operating shortfall (gov subsidy required)	€5.2 M / yr	€2.4 M / yr

	Low scenario	High scenario
Steady-state government commitment (debt service + OPEX shortfall)	€34 M / yr	€31 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
sheikhupura.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
sheikhupura-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
sheikhupura.corridor.geojson	Line polylines + stations (GeoJSON)
sheikhupura.stations.json	Machine-readable station list
sheikhupura.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug sheikhupura

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug sheikhupura \
  --sidecar .cache/osr-pipeline/rasters/sheikhupura.grid.json \
  --out-dir designs/.../Sheikhupura

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../sheikhupura.toml \
  --out designs/.../README.md
```

`scripts/regenerate-sheikhupura.sh` chains steps 3 + drift tests into a single command.

south-asia/Pakistan/Sialkot

designs/south-asia/Pakistan/Sialkot/README.md

Sialkot — Urban Rail Network

Country: PK · Population: 750,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Sialkot rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [sialkot.corridor.geojson](#).

At a glance

Metric	Value
Lines	3
Unique stations	37
Interchange stations	5
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	43.0%
Route length (double track)	59.4 km
Revenue fleet	52 × 3-car trainsets
Spare + cold-reserve	7 × 3-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	19.5 km	12	19	S Outer N Mid
line-2	17.2 km	13	17	NE Mid W Mid
line-3	22.7 km	12	23	NW Outer SE Outer
Total	59.4 km	37 unique	59	

Rolling stock

Property	Value
Consist	3-car, 57 m
Max speed	90 km/h
Onboard battery	360 kWh per trainset
Seats	60 longitudinal seats
Nominal capacity (AW2)	330 pax (seated + standing, light-metro-3car per RFC 0008 §1)
Crush capacity (AW3)	420 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 330 AW2 passengers (light-metro-3car)
- Peak frequency: 12 trains/hour/direction (5-min headway)

- Peak capacity per line per direction: $330 \times 12 = 3,960$ pphpd
- Network peak throughput (all lines, both directions): $3 \text{ lines} \times 2 \text{ directions} \times 3,960 = 23,760$ passengers/hour
- Daily theoretical capacity (peak $\times 10$): $\approx 237,600$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 32,250 - 48,375$ trips/day

Catchment

- City population: 750,000
- Anchor-weighted coverage: 43.0%
- Catchment population: $\approx 322,500$ (within ~ 800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	5	500 kW	3000 kWh
Major	13	400 kW	2500 kWh
Standard	12	300 kW	2000 kWh
Terminal	5	500 kW	3000 kWh
Total installed	36	18,800 kW	126,500 kWh

Aggregate station-rail charging power: 21,250 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 360 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	12.0 kWh/km	3 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	238 kWh	19.8 km average line length
Onboard battery coverage	1.5 $\times$ average line run	360 kWh usable pack
Average 60 s dwell charge	9.6 kWh/stop	574 kW average charger across stops
Stops to refill one trainset pack	38 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	94 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	126 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in design.toml — emitted by the osr-design Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. country-costs.toml applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (28.0 km @ €3.5 M/km)	€98 M
Elevated (31.3 km @ €18 M/km)	€564 M
Elevated-interchange premium (1 sites @ €20 M)	€20 M
Civil subtotal	€682 M

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~ 11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
halt	1	€0.4 M	€0.4 M
standard	12	€1.5 M	€18 M
major	13	€3.0 M	€39 M
terminal	5	€2.5 M	€12 M
depot-terminal	1	€3.0 M	€3.0 M
interchange	3	€4.5 M	€14 M
interchange-elevated	2	€4.5 M	€9.0 M
Stations subtotal			€95 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	5	€3.0 M	€15 M
Depots subtotal			€40 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
light-metro-3car (revenue + spare + cold reserve)	59	€3.0 M	€177 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	59.4 km × €0.015 M/km	€0.9 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€14 M
EPC integration + project management (7%)	on subtotal	€71 M

Total

Bucket	Value
Civil works	€682 M
Stations	€95 M
Depots	€40 M
Rolling stock	€177 M
Residual train-control wayside + charging microgrids	€15 M
EPC overhead (7%)	€71 M
CAPEX total	€1.08 bn
Per-route-km	€18 M / km
Per-capita (city pop)	€1,439 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running scripts/regenerate-city.sh sialkot.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–7 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 8 and runs for 23 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–7)	€94 M / yr	€125
Steady-state, low-ridership (year 8+)	€110 M / yr	€147
Steady-state, high-ridership (year 8+)	€107 M / yr	€143
Lifecycle envelope (yr 1–30, low scenario)	€3.20 bn cumulative	€4,260
Lifecycle envelope (yr 1–30, high scenario)	€3.12 bn cumulative	€4,154

Population basis: 750,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 30, debt service drops to zero and only the OPEX shortfall remains – ~€21 M / yr (low) €17 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€648 M	4.0%	30 y, 7 y grace	€44 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€270 M	16.5%	30 y, 7 y grace	€46 M / yr
Government equity (no debt service)	15%	€162 M	—	—	—
Total	100%	€1.08 bn			€89 M / yr

During the 7-year grace period the operator pays interest only – multilateral €26 M / yr + bonds €45 M / yr = €70 M / yr total – plus the equity tranche amortised across construction (€23 M / yr × 7 yr). Principal repayment begins in year 8 on a 23-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€7.1 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€16 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€44 k
Traction energy (105.1 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (368 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€938 k
OPEX subtotal		€24 M / yr

Annual fleet utilisation: 52 revenue trainsets × 20.5 h/day × 365 d/yr × 30 km/h commercial × 75% revenue factor = 8.8 M train-km / yr (~168 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$165 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.25 (~\$0.28 USD)
Day pass (3 trips)	€0.65 (15 % bulk discount)
Monthly unlimited pass	€7.59 (~5 % of median monthly income)
Annual pass	€83.49 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	13.7 M	27.4 M

	Low scenario	High scenario
Farebox revenue	€3.5 M / yr	€6.9 M / yr
Farebox / OPEX recovery	14%	28%
Country policy-target recovery (diagnostic)	45%	45%
Operating shortfall (gov subsidy required)	€21 M / yr	€17 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€110 M / yr	€107 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
sialkot.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
sialkot-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
sialkot.corridor.geojson	Line polylines + stations (GeoJSON)
sialkot.stations.json	Machine-readable station list
sialkot.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug sialkot

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug sialkot \
  --sidecar .cache/osr-pipeline/rasters/sialkot.grid.json \
  --out-dir designs/.../Sialkot

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../sialkot.toml \
  --out designs/.../README.md
```

`scripts/regenerate-sialkot.sh` chains steps 3 + drift tests into a single command.

south-asia/Pakistan/Sukkur

designs/south-asia/Pakistan/Sukkur/README.md

Sukkur — Urban Rail Network

Country: PK · Population: 600,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Sukkur rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [sukkur.corridor.geojson](#).

At a glance

Metric	Value
Lines	3
Unique stations	31
Interchange stations	3
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	55.1%
Route length (double track)	51.1 km
Revenue fleet	46 × 3-car trainsets
Spare + cold-reserve	7 × 3-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	15.0 km	10	16	W Mid SE Mid
line-2	13.1 km	9	14	NW Mid S Mid
line-3	23.0 km	12	23	NW Outer SE Outer
Total	51.1 km	31 unique	53	

Rolling stock

Property	Value
Consist	3-car, 57 m
Max speed	90 km/h
Onboard battery	360 kWh per trainset
Seats	60 longitudinal seats
Nominal capacity (AW2)	330 pax (seated + standing, light-metro-3car per RFC 0008 §1)
Crush capacity (AW3)	420 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 330 AW2 passengers (light-metro-3car)
- Peak frequency: 12 trains/hour/direction (5-min headway)

- Peak capacity per line per direction: $330 \times 12 = 3,960$ pphpd
- Network peak throughput (all lines, both directions): $3 \text{ lines} \times 2 \text{ directions} \times 3,960 = 23,760$ passengers/hour
- Daily theoretical capacity (peak $\times 10$): $\approx 237,600$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 33,060 - 49,590$ trips/day

Catchment

- City population: 600,000
- Anchor-weighted coverage: 55.1%
- Catchment population: $\approx 330,600$ (within ~ 800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	3	500 kW	3000 kWh
Major	10	400 kW	2500 kWh
Standard	11	300 kW	2000 kWh
Terminal	5	500 kW	3000 kWh
Total installed	30	16,300 kW	111,000 kWh

Aggregate station-rail charging power: 18,250 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 360 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	12.0 kWh/km	3 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	204 kWh	17.0 km average line length
Onboard battery coverage	1.8 $\times$ average line run	360 kWh usable pack
Average 60 s dwell charge	9.8 kWh/stop	589 kW average charger across stops
Stops to refill one trainset pack	37 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	82 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	111 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in design.toml — emitted by the osr-design Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. country-costs.toml applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (19.9 km @ €3.5 M/km)	€70 M
Elevated (31.0 km @ €18 M/km)	€558 M
Civil subtotal	€628 M

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~ 11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
halt	1	€0.4 M	€0.4 M
standard	11	€1.5 M	€16 M
major	10	€3.0 M	€30 M
terminal	5	€2.5 M	€12 M
depot-terminal	1	€3.0 M	€3.0 M
interchange	3	€4.5 M	€14 M
Stations subtotal			€76 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	5	€3.0 M	€15 M
Depots subtotal			€40 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
light-metro-3car (revenue + spare + cold reserve)	53	€3.0 M	€159 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	51.1 km × €0.015 M/km	€0.8 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€11 M
EPC integration + project management (7%)	on subtotal	€64 M

Total

Bucket	Value
Civil works	€628 M
Stations	€76 M
Depots	€40 M
Rolling stock	€159 M
Residual train-control wayside + charging microgrids	€12 M
EPC overhead (7%)	€64 M
CAPEX total	€979 M
Per-route-km	€19 M / km
Per-capita (city pop)	€1,631 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running scripts/regenerate-city.sh sukkur.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–7 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 8 and runs for 23 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–7)	€85 M / yr	€141
Steady-state, low-ridership (year 8+)	€100 M / yr	€167
Steady-state, high-ridership (year 8+)	€98 M / yr	€163
Lifecycle envelope (yr 1–30, low scenario)	€2.90 bn cumulative	€4,841
Lifecycle envelope (yr 1–30, high scenario)	€2.84 bn cumulative	€4,735

Population basis: 600,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 30, debt service drops to zero and only the OPEX shortfall remains – ~€19 M / yr (low) €17 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€587 M	4.0%	30 y, 7 y grace	€40 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€245 M	16.5%	30 y, 7 y grace	€42 M / yr
Government equity (no debt service)	15%	€147 M	—	—	—
Total	100%	€979 M			€81 M / yr

During the 7-year grace period the operator pays interest only – multilateral €23 M / yr + bonds €40 M / yr = €64 M / yr total – plus the equity tranche amortised across construction (€21 M / yr × 7 yr). Principal repayment begins in year 8 on a 23-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€6.4 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€15 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€38 k
Traction energy (92.9 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (319 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€814 k
OPEX subtotal		€22 M / yr

Annual fleet utilisation: 46 revenue trainsets × 20.5 h/day × 365 d/yr × 30 km/h commercial × 75% revenue factor = 7.7 M train-km / yr (~168 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$165 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.25 (~\$0.28 USD)
Day pass (3 trips)	€0.65 (15 % bulk discount)
Monthly unlimited pass	€7.59 (~5 % of median monthly income)
Annual pass	€83.49 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	10.9 M	21.9 M

	Low scenario	High scenario
Farebox revenue	€2.8 M / yr	€5.5 M / yr
Farebox / OPEX recovery	13%	25%
Country policy-target recovery (diagnostic)	45%	45%
Operating shortfall (gov subsidy required)	€19 M / yr	€17 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€100 M / yr	€98 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
sukkur.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
sukkur-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
sukkur.corridor.geojson	Line polylines + stations (GeoJSON)
sukkur.stations.json	Machine-readable station list
sukkur.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug sukkur

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug sukkur \
  --sidecar .cache/osr-pipeline/rasters/sukkur.grid.json \
  --out-dir designs/.../Sukkur

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../sukkur.toml \
  --out designs/.../README.md
```

`scripts/regenerate-sukkur.sh` chains steps 3 + drift tests into a single command.

south-asia/Sri Lanka/Colombo

designs/south-asia/Sri Lanka/Colombo/README.md

Colombo — Urban Rail Network

Country: LK · Population: 5,648,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Colombo rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [colombo.corridor.geojson](#).

At a glance

Metric	Value
Lines	6
Unique stations	125
Interchange stations	15
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	42.4%
Route length (double track)	278.0 km
Revenue fleet	200 × 6-car trainsets
Spare + cold-reserve	23 × 6-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	41.2 km	20	34	SW Outer NE Outer
line-2	39.6 km	19	32	S Outer N Outer
line-3	36.1 km	13	29	S Mid NE Outer
line-4	31.1 km	16	26	SW Outer N Mid
line-5	35.5 km	17	29	W Mid SE Outer
line-6	94.4 km	41	73	NW Mid NW Mid
Total	278.0 km	125 unique	223	

Rolling stock

Property	Value
Consist	6-car, 111 m
Max speed	100 km/h
Onboard battery	720 kWh per trainset
Seats	120 longitudinal seats
Nominal capacity (AW2)	660 pax (seated + standing, metro-6car per RFC 0008 §1)
Crush capacity (AW3)	840 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 660 AW2 passengers (metro - 6 car)
- Peak frequency: 12 trains/hour/direction (5-min headway)
- Peak capacity per line per direction: $660 \times 12 = 7,920$ pphpd
- Network peak throughput (all lines, both directions): $6 \text{ lines} \times 2 \text{ directions} \times 7,920 = 95,040$ passengers/hour
- Daily theoretical capacity (peak $\times$ 10): $\approx 950,400$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 239,475 - 359,212$ trips/day

Catchment

- City population: 5,648,000
- Anchor-weighted coverage: 42.4%
- Catchment population: $\approx 2,394,752$ (within ~800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	15	500 kW	3000 kWh
Major	37	400 kW	2500 kWh
Standard	64	300 kW	2000 kWh
Terminal	9	500 kW	3000 kWh
Total installed	126	51,000 kW	332,500 kWh

Aggregate station-rail charging power: 68,000 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 720 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	24.0 kWh/km	6 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	1,112 kWh	46.3 km average line length
Onboard battery coverage	0.6 $\times$ average line run	720 kWh usable pack
Average 60 s dwell charge	9.1 kWh/stop	544 kW average charger across stops
Stops to refill one trainset pack	79 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	255 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	332 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the `[costs]` block in `design.toml` — emitted by the `osr-design` Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. `country-costs.toml` applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (256.4 km @ €3.5 M/km)	€897 M
Elevated (19.5 km @ €18 M/km)	€350 M
Elevated-interchange premium (8 sites @ €20 M)	€160 M
Civil subtotal	€1.41 bn

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
standard	64	€1.5 M	€96 M
major	37	€3.0 M	€111 M
terminal	9	€2.5 M	€22 M
depot-terminal	1	€3.0 M	€3.0 M
interchange-elevated	15	€4.5 M	€68 M
Stations subtotal			€300 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	9	€3.0 M	€27 M
Depots subtotal			€52 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
metro-6car (revenue + spare + cold reserve)	223	€6.0 M	€1.34 bn

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	278.0 km × €0.015 M/km	€4.1 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€44 M
EPC integration + project management (7%)	on subtotal	€220 M

Total

Bucket	Value
Civil works	€1.41 bn
Stations	€300 M
Depots	€52 M
Rolling stock	€1.34 bn
Residual train-control wayside + charging microgrids	€48 M
EPC overhead (7%)	€220 M
CAPEX total	€3.37 bn
Per-route-km	€12 M / km
Per-capita (city pop)	€596 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running `scripts/regenerate-city.sh` `colombo`.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–7 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 8 and runs for 23 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–7)	€267 M / yr	€47
Steady-state, low-ridership (year 8+)	€313 M / yr	€55
Steady-state, high-ridership (year 8+)	€275 M / yr	€49
Lifecycle envelope (yr 1–30, low scenario)	€9.07 bn cumulative	€1,606
Lifecycle envelope (yr 1–30, high scenario)	€8.20 bn cumulative	€1,452

Population basis: 5,648,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 30, debt service drops to zero and only the OPEX shortfall remains – ~€57 M / yr (low) €19 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€2.02 bn	4.0%	30 y, 7 y grace	€136 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€842 M	13.5%	30 y, 7 y grace	€120 M / yr
Government equity (no debt service)	15%	€505 M	—	—	—
Total	100%	€3.37 bn			€256 M / yr

During the 7-year grace period the operator pays interest only – multilateral €81 M / yr + bonds €114 M / yr = €194 M / yr total – plus the equity tranche amortised across construction (€72 M / yr × 7 yr). Principal repayment begins in year 8 on a 23-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€54 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€35 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€207 k
Traction energy (942.8 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (1,680 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€6.2 M
OPEX subtotal		€95 M / yr

Annual fleet utilisation: 200 revenue trainsets × 20.5 h/day × 365 d/yr × 35 km/h commercial × 75% revenue factor = 39.3 M train-km / yr (~196 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$240 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.37 (~\$0.40 USD)
Day pass (3 trips)	€0.94 (15 % bulk discount)
Monthly unlimited pass	€11.04 (~5 % of median monthly income)
Annual pass	€121.44 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	103.1 M	206.2 M
Farebox revenue	€38 M / yr	€76 M / yr
Farebox / OPEX recovery	40%	80%
Country policy-target recovery (diagnostic)	45%	45%
Operating shortfall (gov subsidy required)	€57 M / yr	€19 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€313 M / yr	€275 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
colombo.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
colombo-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
colombo.corridor.geojson	Line polylines + stations (GeoJSON)
colombo.stations.json	Machine-readable station list
colombo.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug colombo

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug colombo \
  --sidecar .cache/osr-pipeline/rasters/colombo.grid.json \
  --out-dir designs/.../Colombo

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../colombo.toml \
  --out designs/.../README.md
```

`scripts/regenerate-colombo.sh` chains steps 3 + drift tests into a single command.

south-asia/Sri Lanka/Galle

designs/south-asia/Sri Lanka/Galle/README.md

Galle — Urban Rail Network

Country: LK · Population: 500,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Galle rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [galle.corridor.geojson](#).

At a glance

Metric	Value
Lines	3
Unique stations	37
Interchange stations	3
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	55.8%
Route length (double track)	64.4 km
Revenue fleet	57 × 3-car trainsets
Spare + cold-reserve	7 × 3-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	28.1 km	16	27	SE Outer NW Outer
line-2	20.1 km	10	20	NW Outer E Mid
line-3	16.2 km	11	17	W Mid SE Mid
Total	64.4 km	37 unique	64	

Rolling stock

Property	Value
Consist	3-car, 57 m
Max speed	90 km/h
Onboard battery	360 kWh per trainset
Seats	60 longitudinal seats
Nominal capacity (AW2)	330 pax (seated + standing, light-metro-3car per RFC 0008 §1)
Crush capacity (AW3)	420 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 330 AW2 passengers (light-metro-3car)
- Peak frequency: 12 trains/hour/direction (5-min headway)

- Peak capacity per line per direction: $330 \times 12 = 3,960$ pphpd
- Network peak throughput (all lines, both directions): $3 \text{ lines} \times 2 \text{ directions} \times 3,960 = 23,760$ passengers/hour
- Daily theoretical capacity (peak $\times 10$): $\approx 237,600$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 27,900 - 41,850$ trips/day

Catchment

- City population: 500,000
- Anchor-weighted coverage: 55.8%
- Catchment population: $\approx 279,000$ (within ~ 800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../.../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	3	500 kW	3000 kWh
Major	12	400 kW	2500 kWh
Standard	14	300 kW	2000 kWh
Terminal	5	500 kW	3000 kWh
Total installed	35	18,000 kW	122,000 kWh

Aggregate station-rail charging power: 21,000 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 360 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	12.0 kWh/km	3 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	258 kWh	21.5 km average line length
Onboard battery coverage	1.4 $\times$ average line run	360 kWh usable pack
Average 60 s dwell charge	9.5 kWh/stop	568 kW average charger across stops
Stops to refill one trainset pack	38 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	90 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	122 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in design.toml — emitted by the osr-design Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. country-costs.toml applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (62.5 km @ €3.5 M/km)	€219 M
Elevated (1.3 km @ €18 M/km)	€24 M
Elevated-interchange premium (1 sites @ €20 M)	€20 M
Civil subtotal	€263 M

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~ 11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
halt	2	€0.4 M	€0.8 M
standard	14	€1.5 M	€21 M
major	12	€3.0 M	€36 M
terminal	5	€2.5 M	€12 M
depot-terminal	1	€3.0 M	€3.0 M
interchange-elevated	3	€4.5 M	€14 M
Stations subtotal			€87 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	5	€3.0 M	€15 M
Depots subtotal			€40 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
light-metro-3car (revenue + spare + cold reserve)	64	€3.0 M	€192 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	64.4 km × €0.015 M/km	€1.0 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€13 M
EPC integration + project management (7%)	on subtotal	€42 M

Total

Bucket	Value
Civil works	€263 M
Stations	€87 M
Depots	€40 M
Rolling stock	€192 M
Residual train-control wayside + charging microgrids	€14 M
EPC overhead (7%)	€42 M
CAPEX total	€637 M
Per-route-km	€9.9 M / km
Per-capita (city pop)	€1,275 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running scripts/regenerate-city.sh galle.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–7 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 8 and runs for 23 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–7)	€50 M / yr	€101
Steady-state, low-ridership (year 8+)	€62 M / yr	€124
Steady-state, high-ridership (year 8+)	€59 M / yr	€118
Lifecycle envelope (yr 1–30, low scenario)	€1.78 bn cumulative	€3,564
Lifecycle envelope (yr 1–30, high scenario)	€1.70 bn cumulative	€3,410

Population basis: 500,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 30, debt service drops to zero and only the OPEX shortfall remains – ~€14 M / yr (low) €10 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€382 M	4.0%	30 y, 7 y grace	€26 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€159 M	13.5%	30 y, 7 y grace	€23 M / yr
Government equity (no debt service)	15%	€96 M	—	—	—
Total	100%	€637 M			€48 M / yr

During the 7-year grace period the operator pays interest only – multilateral €15 M / yr + bonds €22 M / yr = €37 M / yr total – plus the equity tranche amortised across construction (€14 M / yr × 7 yr). Principal repayment begins in year 8 on a 23-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€7.7 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€7.8 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€48 k
Traction energy (115.2 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (398 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€1.5 M
OPEX subtotal		€17 M / yr

Annual fleet utilisation: 57 revenue trainsets × 20.5 h/day × 365 d/yr × 30 km/h commercial × 75% revenue factor = 9.6 M train-km / yr (~168 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$240 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.37 (~\$0.40 USD)
Day pass (3 trips)	€0.94 (15 % bulk discount)
Monthly unlimited pass	€11.04 (~5 % of median monthly income)
Annual pass	€121.44 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	9.1 M	18.2 M

	Low scenario	High scenario
Farebox revenue	€3.4 M / yr	€6.7 M / yr
Farebox / OPEX recovery	20%	40%
Country policy-target recovery (diagnostic)	45%	45%
Operating shortfall (gov subsidy required)	€14 M / yr	€10 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€62 M / yr	€59 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
galle.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
galle-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
galle.corridor.geojson	Line polylines + stations (GeoJSON)
galle.stations.json	Machine-readable station list
galle.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug galle

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug galle \
  --sidecar .cache/osr-pipeline/rasters/galle.grid.json \
  --out-dir designs/.../Galle

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../galle.toml \
  --out designs/.../README.md
```

`scripts/regenerate-galle.sh` chains steps 3 + drift tests into a single command.

south-asia/Sri Lanka/Jaffna

designs/south-asia/Sri Lanka/Jaffna/README.md

Jaffna — Urban Rail Network

Country: LK · Population: 600,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Jaffna rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [jaffna.corridor.geojson](#).

At a glance

Metric	Value
Lines	3
Unique stations	34
Interchange stations	2
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	42.4%
Route length (double track)	51.8 km
Revenue fleet	46 × 3-car trainsets
Spare + cold-reserve	6 × 3-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	19.6 km	14	19	S Mid NW Mid
line-2	14.7 km	10	15	SE Mid NE Mid
line-3	17.5 km	10	18	NW Outer E Mid
Total	51.8 km	34 unique	52	

Rolling stock

Property	Value
Consist	3-car, 57 m
Max speed	90 km/h
Onboard battery	360 kWh per trainset
Seats	60 longitudinal seats
Nominal capacity (AW2)	330 pax (seated + standing, light-metro-3car per RFC 0008 §1)
Crush capacity (AW3)	420 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 330 AW2 passengers (light-metro-3car)
- Peak frequency: 12 trains/hour/direction (5-min headway)

- Peak capacity per line per direction: $330 \times 12 = 3,960$ pphpd
- Network peak throughput (all lines, both directions): $3 \text{ lines} \times 2 \text{ directions} \times 3,960 = 23,760$ passengers/hour
- Daily theoretical capacity (peak $\times 10$): $\approx 237,600$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 25,440 - 38,160$ trips/day

Catchment

- City population: 600,000
- Anchor-weighted coverage: 42.4%
- Catchment population: $\approx 254,400$ (within ~ 800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	2	500 kW	3000 kWh
Major	11	400 kW	2500 kWh
Standard	15	300 kW	2000 kWh
Terminal	5	500 kW	3000 kWh
Total installed	34	17,400 kW	118,500 kWh

Aggregate station-rail charging power: 20,000 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 360 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	12.0 kWh/km	3 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	207 kWh	17.3 km average line length
Onboard battery coverage	1.7 $\times$ average line run	360 kWh usable pack
Average 60 s dwell charge	9.8 kWh/stop	588 kW average charger across stops
Stops to refill one trainset pack	37 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	87 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	118 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the `[costs]` block in `design.toml` — emitted by the `osr-design` Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. `country-costs.toml` applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (50.4 km @ €3.5 M/km)	€176 M
Elevated (1.3 km @ €18 M/km)	€23 M
Elevated-interchange premium (1 sites @ €20 M)	€20 M
Civil subtotal	€219 M

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~ 11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
standard	15	€1.5 M	€22 M
major	11	€3.0 M	€33 M
terminal	5	€2.5 M	€12 M
depot-terminal	1	€3.0 M	€3.0 M
interchange-elevated	2	€4.5 M	€9.0 M
Stations subtotal			€80 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	5	€3.0 M	€15 M
Depots subtotal			€40 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
light-metro-3car (revenue + spare + cold reserve)	52	€3.0 M	€156 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	51.8 km × €0.015 M/km	€0.8 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€12 M
EPC integration + project management (7%)	on subtotal	€36 M

Total

Bucket	Value
Civil works	€219 M
Stations	€80 M
Depots	€40 M
Rolling stock	€156 M
Residual train-control wayside + charging microgrids	€13 M
EPC overhead (7%)	€36 M
CAPEX total	€544 M
Per-route-km	€11 M / km
Per-capita (city pop)	€907 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running scripts/regenerate-city.sh jaffna.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–7 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 8 and runs for 23 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–7)	€43 M / yr	€72
Steady-state, low-ridership (year 8+)	€52 M / yr	€86
Steady-state, high-ridership (year 8+)	€48 M / yr	€79
Lifecycle envelope (yr 1–30, low scenario)	€1.49 bn cumulative	€2,481
Lifecycle envelope (yr 1–30, high scenario)	€1.40 bn cumulative	€2,327

Population basis: 600,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 30, debt service drops to zero and only the OPEX shortfall remains – ~€10 M / yr (low) €6.2 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€326 M	4.0%	30 y, 7 y grace	€22 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€136 M	13.5%	30 y, 7 y grace	€19 M / yr
Government equity (no debt service)	15%	€82 M	—	—	—
Total	100%	€544 M			€41 M / yr

During the 7-year grace period the operator pays interest only – multilateral €13 M / yr + bonds €18 M / yr = €31 M / yr total – plus the equity tranche amortised across construction (€12 M / yr × 7 yr). Principal repayment begins in year 8 on a 23-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€6.2 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€6.8 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€39 k
Traction energy (92.9 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (323 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€1.2 M
OPEX subtotal		€14 M / yr

Annual fleet utilisation: 46 revenue trainsets × 20.5 h/day × 365 d/yr × 30 km/h commercial × 75% revenue factor = 7.7 M train-km / yr (~168 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$240 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.37 (~\$0.40 USD)
Day pass (3 trips)	€0.94 (15 % bulk discount)
Monthly unlimited pass	€11.04 (~5 % of median monthly income)
Annual pass	€121.44 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	10.9 M	21.9 M

	Low scenario	High scenario
Farebox revenue	€4.0 M / yr	€8.1 M / yr
Farebox / OPEX recovery	28%	56%
Country policy-target recovery (diagnostic)	45%	45%
Operating shortfall (gov subsidy required)	€10 M / yr	€6.2 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€52 M / yr	€48 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
jaffna.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
jaffna-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
jaffna.corridor.geojson	Line polylines + stations (GeoJSON)
jaffna.stations.json	Machine-readable station list
jaffna.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug jaffna

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug jaffna \
  --sidecar .cache/osr-pipeline/rasters/jaffna.grid.json \
  --out-dir designs/.../Jaffna

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../jaffna.toml \
  --out designs/.../README.md
```

`scripts/regenerate-jaffna.sh` chains steps 3 + drift tests into a single command.

south-asia/Sri Lanka/Kandy

designs/south-asia/Sri Lanka/Kandy/README.md

Kandy — Urban Rail Network

Country: LK · Population: 650,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Kandy rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [kandy.corridor.geojson](#).

At a glance

Metric	Value
Lines	3
Unique stations	43
Interchange stations	3
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	55.8%
Route length (double track)	77.9 km
Revenue fleet	67 × 3-car trainsets
Spare + cold-reserve	9 × 3-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	28.0 km	16	27	E Outer W Outer
line-2	22.9 km	13	23	NW Outer SE Mid
line-3	27.1 km	14	26	SW Mid NE Outer
Total	77.9 km	43 unique	76	

Rolling stock

Property	Value
Consist	3-car, 57 m
Max speed	90 km/h
Onboard battery	360 kWh per trainset
Seats	60 longitudinal seats
Nominal capacity (AW2)	330 pax (seated + standing, light-metro-3car per RFC 0008 §1)
Crush capacity (AW3)	420 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 330 AW2 passengers (light-metro-3car)
- Peak frequency: 12 trains/hour/direction (5-min headway)

- Peak capacity per line per direction: $330 \times 12 = 3,960$ pphpd
- Network peak throughput (all lines, both directions): $3 \text{ lines} \times 2 \text{ directions} \times 3,960 = 23,760$ passengers/hour
- Daily theoretical capacity (peak $\times 10$): $\approx 237,600$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 36,270 - 54,405$ trips/day

Catchment

- City population: 650,000
- Anchor-weighted coverage: 55.8%
- Catchment population: $\approx 362,700$ (within ~ 800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	3	500 kW	3000 kWh
Major	10	400 kW	2500 kWh
Standard	23	300 kW	2000 kWh
Terminal	5	500 kW	3000 kWh
Total installed	42	19,900 kW	135,000 kWh

Aggregate station-rail charging power: 24,250 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 360 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	12.0 kWh/km	3 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	312 kWh	26.0 km average line length
Onboard battery coverage	1.2 $\times$ average line run	360 kWh usable pack
Average 60 s dwell charge	9.4 kWh/stop	564 kW average charger across stops
Stops to refill one trainset pack	38 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	100 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	135 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in design.toml — emitted by the osr-design Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. country-costs.toml applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (67.4 km @ €3.5 M/km)	€236 M
Elevated (9.8 km @ €18 M/km)	€176 M
Elevated-interchange premium (1 sites @ €20 M)	€20 M
Civil subtotal	€432 M

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~ 11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
halt	1	€0.4 M	€0.4 M
standard	23	€1.5 M	€34 M
major	10	€3.0 M	€30 M
terminal	5	€2.5 M	€12 M
depot-terminal	1	€3.0 M	€3.0 M
interchange	3	€4.5 M	€14 M
Stations subtotal			€94 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	5	€3.0 M	€15 M
Depots subtotal			€40 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
light-metro-3car (revenue + spare + cold reserve)	76	€3.0 M	€228 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	77.9 km × €0.015 M/km	€1.2 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€14 M
EPC integration + project management (7%)	on subtotal	€57 M

Total

Bucket	Value
Civil works	€432 M
Stations	€94 M
Depots	€40 M
Rolling stock	€228 M
Residual train-control wayside + charging microgrids	€16 M
EPC overhead (7%)	€57 M
CAPEX total	€866 M
Per-route-km	€11 M / km
Per-capita (city pop)	€1,332 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running scripts/regenerate-city.sh kandy.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–7 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 8 and runs for 23 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–7)	€69 M / yr	€105
Steady-state, low-ridership (year 8+)	€84 M / yr	€129
Steady-state, high-ridership (year 8+)	€79 M / yr	€122
Lifecycle envelope (yr 1–30, low scenario)	€2.41 bn cumulative	€3,702
Lifecycle envelope (yr 1–30, high scenario)	€2.31 bn cumulative	€3,548

Population basis: 650,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 30, debt service drops to zero and only the OPEX shortfall remains – ~€18 M / yr (low) €14 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€519 M	4.0%	30 y, 7 y grace	€35 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€216 M	13.5%	30 y, 7 y grace	€31 M / yr
Government equity (no debt service)	15%	€130 M	—	—	—
Total	100%	€866 M			€66 M / yr

During the 7-year grace period the operator pays interest only – multilateral €21 M / yr + bonds €29 M / yr = €50 M / yr total – plus the equity tranche amortised across construction (€19 M / yr × 7 yr). Principal repayment begins in year 8 on a 23-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€9.1 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€11 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€58 k
Traction energy (135.4 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (479 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€1.8 M
OPEX subtotal		€22 M / yr

Annual fleet utilisation: 67 revenue trainsets × 20.5 h/day × 365 d/yr × 30 km/h commercial × 75% revenue factor = 11.3 M train-km / yr (~168 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$240 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.37 (~\$0.40 USD)
Day pass (3 trips)	€0.94 (15 % bulk discount)
Monthly unlimited pass	€11.04 (~5 % of median monthly income)
Annual pass	€121.44 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	11.9 M	23.7 M

	Low scenario	High scenario
Farebox revenue	€4.4 M / yr	€8.7 M / yr
Farebox / OPEX recovery	20%	39%
Country policy-target recovery (diagnostic)	45%	45%
Operating shortfall (gov subsidy required)	€18 M / yr	€14 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€84 M / yr	€79 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
kandy.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
kandy-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
kandy.corridor.geojson	Line polylines + stations (GeoJSON)
kandy.stations.json	Machine-readable station list
kandy.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug kandy

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug kandy \
  --sidecar .cache/osr-pipeline/rasters/kandy.grid.json \
  --out-dir designs/.../Kandy

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../kandy.toml \
  --out designs/.../README.md
```

`scripts/regenerate-kandy.sh` chains steps 3 + drift tests into a single command.

southeast-asia/Cambodia/Phnom-Penh

designs/southeast-asia/Cambodia/Phnom-Penh/README.md

Phnom-Penh — Urban Rail Network

Country: KH · Population: 2,281,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Phnom-Penh rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [phnom-penh.corridor.geojson](#).

At a glance

Metric	Value
Lines	6
Unique stations	106
Interchange stations	23
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	48.3%
Route length (double track)	227.7 km
Revenue fleet	166 × 4-car trainsets
Spare + cold-reserve	20 × 4-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	38.6 km	18	31	SE Outer N Outer
line-2	31.4 km	16	26	E Outer W Mid
line-3	28.1 km	11	24	N Mid SW Mid
line-4	25.9 km	15	21	SW Mid NE Mid
line-5	41.2 km	18	34	NW Outer SE Outer
line-6	62.4 km	29	50	W Mid W Mid
Total	227.7 km	106 unique	186	

Rolling stock

Property	Value
Consist	4-car, 75 m
Max speed	90 km/h
Onboard battery	480 kWh per trainset
Seats	80 longitudinal seats
Nominal capacity (AW2)	440 pax (seated + standing, metro-4car per RFC 0008 §1)
Crush capacity (AW3)	560 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 440 AW2 passengers (metro - 4 car)
- Peak frequency: 12 trains/hour/direction (5-min headway)
- Peak capacity per line per direction: $440 \times 12 = 5,280$ pphpd
- Network peak throughput (all lines, both directions): $6 \text{ lines} \times 2 \text{ directions} \times 5,280 = 63,360$ passengers/hour
- Daily theoretical capacity (peak $\times$ 10): $\approx 633,600$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 110,172 - 165,258$ trips/day

Catchment

- City population: 2,281,000
- Anchor-weighted coverage: 48.3%
- Catchment population: $\approx 1,101,723$ (within ~800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	23	500 kW	3000 kWh
Major	25	400 kW	2500 kWh
Standard	46	300 kW	2000 kWh
Terminal	9	500 kW	3000 kWh
Total installed	104	44,800 kW	290,500 kWh

Aggregate station-rail charging power: 57,750 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 480 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	16.0 kWh/km	4 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	607 kWh	37.9 km average line length
Onboard battery coverage	0.8 $\times$ average line run	480 kWh usable pack
Average 60 s dwell charge	9.1 kWh/stop	545 kW average charger across stops
Stops to refill one trainset pack	53 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	224 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	290 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in design.toml — emitted by the osr-design Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. country-costs.toml applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (186.2 km @ €3.5 M/km)	€652 M
Elevated (40.2 km @ €18 M/km)	€724 M
Elevated-interchange premium (8 sites @ €20 M)	€160 M
Civil subtotal	€1.54 bn

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
halt	3	€0.4 M	€1.2 M
standard	46	€1.5 M	€69 M
major	25	€3.0 M	€75 M
terminal	9	€2.5 M	€22 M
depot-terminal	1	€3.0 M	€3.0 M
interchange	10	€4.5 M	€45 M
interchange-elevated	13	€4.5 M	€58 M
Stations subtotal			€274 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	9	€3.0 M	€27 M
Depots subtotal			€52 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY – never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
metro - 4car (revenue + spare + cold reserve)	186	€4.0 M	€744 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	227.7 km × €0.015 M/km	€3.4 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€40 M
EPC integration + project management (7%)	on subtotal	€185 M

Total

Bucket	Value
Civil works	€1.54 bn
Stations	€274 M
Depots	€52 M
Rolling stock	€744 M
Residual train-control wayside + charging microgrids	€43 M
EPC overhead (7%)	€185 M
CAPEX total	€2.83 bn
Per-route-km	€12 M / km
Per-capita (city pop)	€1,243 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running `scripts/regenerate-city.sh` `phnom-penh`.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–7 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 8 and runs for 23 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–7)	€185 M / yr	€81
Steady-state, low-ridership (year 8+)	€241 M / yr	€106
Steady-state, high-ridership (year 8+)	€227 M / yr	€100
Lifecycle envelope (yr 1–30, low scenario)	€6.84 bn cumulative	€2,998
Lifecycle envelope (yr 1–30, high scenario)	€6.52 bn cumulative	€2,859

Population basis: 2,281,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 30, debt service drops to zero and only the OPEX shortfall remains – ~€58 M / yr (low) €44 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€1.70 bn	4.0%	30 y, 7 y grace	€114 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€709 M	8.0%	30 y, 7 y grace	€68 M / yr
Government equity (no debt service)	15%	€425 M	—	—	—
Total	100%	€2.83 bn			€183 M / yr

During the 7-year grace period the operator pays interest only – multilateral €68 M / yr + bonds €57 M / yr = €125 M / yr total – plus the equity tranche amortised across construction (€61 M / yr × 7 yr). Principal repayment begins in year 8 on a 23-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€30 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€37 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€170 k
Traction energy (521.7 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (1,378 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€4.6 M
OPEX subtotal		€72 M / yr

Annual fleet utilisation: 166 revenue trainsets × 20.5 h/day × 365 d/yr × 35 km/h commercial × 75% revenue factor = 32.6 M train-km / yr (~196 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$215 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.33 (~\$0.36 USD)
Day pass (3 trips)	€0.84 (15 % bulk discount)
Monthly unlimited pass	€9.89 (~5 % of median monthly income)
Annual pass	€108.79 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	41.6 M	83.3 M
Farebox revenue	€14 M / yr	€27 M / yr
Farebox / OPEX recovery	19%	38%
Country policy-target recovery (diagnostic)	45%	45%
Operating shortfall (gov subsidy required)	€58 M / yr	€44 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€241 M / yr	€227 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
phnom-penh.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
phnom-penh-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
phnom-penh.corridor.geojson	Line polylines + stations (GeoJSON)
phnom-penh.stations.json	Machine-readable station list
phnom-penh.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug phnom-penh

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug phnom-penh \
  --sidecar .cache/osr-pipeline/rasters/phnom-penh.grid.json \
  --out-dir designs/.../Phnom-Penh

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../phnom-penh.toml \
  --out designs/.../README.md
```

`scripts/regenerate-phnom-penh.sh` chains steps 3 + drift tests into a single command.

southeast-asia/Indonesia/Bandung

designs/southeast-asia/Indonesia/Bandung/README.md

Bandung — Urban Rail Network

Country: ID · Population: 2,615,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Bandung rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [bandung.corridor.geojson](#).

At a glance

Metric	Value
Lines	6
Unique stations	125
Interchange stations	22
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	40.7%
Route length (double track)	257.5 km
Revenue fleet	186 × 4-car trainsets
Spare + cold-reserve	22 × 4-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	43.4 km	22	35	W Outer E Outer
line-2	37.2 km	19	30	N Outer S Mid
line-3	39.1 km	21	32	SE Outer NW Outer
line-4	30.2 km	17	25	SW Outer E Mid
line-5	28.5 km	13	24	W Mid NE Outer
line-6	79.1 km	34	62	NW Mid NW Mid
Total	257.5 km	125 unique	208	

Rolling stock

Property	Value
Consist	4-car, 75 m
Max speed	90 km/h
Onboard battery	480 kWh per trainset
Seats	80 longitudinal seats
Nominal capacity (AW2)	440 pax (seated + standing, metro-4car per RFC 0008 §1)
Crush capacity (AW3)	560 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 440 AW2 passengers (metro-4car)
- Peak frequency: 12 trains/hour/direction (5-min headway)
- Peak capacity per line per direction: $440 \times 12 = 5,280$ pphpd
- Network peak throughput (all lines, both directions): $6 \text{ lines} \times 2 \text{ directions} \times 5,280 = 63,360$ passengers/hour
- Daily theoretical capacity (peak $\times$ 10): $\approx 633,600$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 106,430 - 159,645$ trips/day

Catchment

- City population: 2,615,000
- Anchor-weighted coverage: 40.7%
- Catchment population: $\approx 1,064,305$ (within ~800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	22	500 kW	3000 kWh
Major	49	400 kW	2500 kWh
Standard	44	300 kW	2000 kWh
Terminal	9	500 kW	3000 kWh
Total installed	125	53,300 kW	343,500 kWh

Aggregate station-rail charging power: 67,750 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 480 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	16.0 kWh/km	4 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	687 kWh	42.9 km average line length
Onboard battery coverage	0.7 $\times$ average line run	480 kWh usable pack
Average 60 s dwell charge	9.0 kWh/stop	542 kW average charger across stops
Stops to refill one trainset pack	53 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	266 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	344 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in design.toml — emitted by the osr-design Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. country-costs.toml applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (219.7 km @ €3.5 M/km)	€769 M
Elevated (31.8 km @ €18 M/km)	€572 M
Elevated-interchange premium (12 sites @ €20 M)	€240 M
Civil subtotal	€1.58 bn

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
halt	1	€0.4 M	€0.4 M
standard	44	€1.5 M	€66 M
major	49	€3.0 M	€147 M
terminal	9	€2.5 M	€22 M
depot-terminal	1	€3.0 M	€3.0 M
interchange	4	€4.5 M	€18 M
interchange-elevated	18	€4.5 M	€81 M
Stations subtotal			€338 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	9	€3.0 M	€27 M
Depots subtotal			€52 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
metro - 4car (revenue + spare + cold reserve)	208	€4.0 M	€832 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	257.5 km × €0.015 M/km	€3.8 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€48 M
EPC integration + project management (7%)	on subtotal	€200 M

Total

Bucket	Value
Civil works	€1.58 bn
Stations	€338 M
Depots	€52 M
Rolling stock	€832 M
Residual train-control wayside + charging microgrids	€52 M
EPC overhead (7%)	€200 M
CAPEX total	€3.05 bn
Per-route-km	€12 M / km
Per-capita (city pop)	€1,168 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running `scripts/regenerate-city.sh` bandung.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–5 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 6 and runs for 20 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–5)	€216 M / yr	€83
Steady-state, low-ridership (year 6+)	€262 M / yr	€100
Steady-state, high-ridership (year 6+)	€239 M / yr	€91
Lifecycle envelope (yr 1–25, low scenario)	€6.33 bn cumulative	€2,420
Lifecycle envelope (yr 1–25, high scenario)	€5.86 bn cumulative	€2,241

Population basis: 2,615,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 25, debt service drops to zero and only the OPEX shortfall remains – ~€57 M / yr (low) €34 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€1.83 bn	4.0%	25 y, 5 y grace	€135 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€764 M	6.7%	25 y, 5 y grace	€70 M / yr
Government equity (no debt service)	15%	€458 M	—	—	—
Total	100%	€3.05 bn			€205 M / yr

During the 5-year grace period the operator pays interest only – multilateral €73 M / yr + bonds €51 M / yr = €124 M / yr total – plus the equity tranche amortised across construction (€92 M / yr × 5 yr). Principal repayment begins in year 6 on a 20-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€33 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€39 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€189 k
Traction energy (584.5 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (1,557 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€7.7 M
OPEX subtotal		€81 M / yr

Annual fleet utilisation: 186 revenue trainsets × 20.5 h/day × 365 d/yr × 35 km/h commercial × 75% revenue factor = 36.5 M train-km / yr (~196 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$320 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.49 (~\$0.53 USD)
Day pass (3 trips)	€1.25 (15 % bulk discount)
Monthly unlimited pass	€14.72 (~5 % of median monthly income)
Annual pass	€161.92 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	47.7 M	95.4 M
Farebox revenue	€23 M / yr	€47 M / yr
Farebox / OPEX recovery	29%	58%
Country policy-target recovery (diagnostic)	60%	60%
Operating shortfall (gov subsidy required)	€57 M / yr	€34 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€262 M / yr	€239 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
bandung.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
bandung-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
bandung.corridor.geojson	Line polylines + stations (GeoJSON)
bandung.stations.json	Machine-readable station list
bandung.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug bandung

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug bandung \
  --sidecar .cache/osr-pipeline/rasters/bandung.grid.json \
  --out-dir designs/.../Bandung

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../bandung.toml \
  --out designs/.../README.md
```

`scripts/regenerate-bandung.sh` chains steps 3 + drift tests into a single command.

southeast-asia/Indonesia/Surabaya

designs/southeast-asia/Indonesia/Surabaya/README.md

Surabaya — Urban Rail Network

Country: ID · Population: 3,009,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Surabaya rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [surabaya.corridor.geojson](#).

At a glance

Metric	Value
Lines	7
Unique stations	142
Interchange stations	23
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	39.6%
Route length (double track)	294.0 km
Revenue fleet	215 × 6-car trainsets
Spare + cold-reserve	25 × 6-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	46.8 km	23	38	SE Outer NW Outer
line-2	39.5 km	21	32	SW Outer NE Mid
line-3	30.4 km	13	26	S Mid NW Outer
line-4	27.9 km	17	24	SW Outer E Mid
line-5	38.2 km	19	31	NW Outer SE Mid
line-6	25.4 km	15	21	N Outer E Inner
line-7	85.8 km	35	68	W Mid W Mid
Total	294.0 km	142 unique	240	

Rolling stock

Property	Value
Consist	6-car, 111 m
Max speed	100 km/h
Onboard battery	720 kWh per trainset
Seats	120 longitudinal seats
Nominal capacity (AW2)	660 pax (seated + standing, metro-6car per RFC 0008 §1)
Crush capacity (AW3)	840 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 660 AW2 passengers (metro - 6 car)
- Peak frequency: 12 trains/hour/direction (5-min headway)
- Peak capacity per line per direction: $660 \times 12 = 7,920$ pphpd
- Network peak throughput (all lines, both directions): $7 \text{ lines} \times 2 \text{ directions} \times 7,920 = 110,880$ passengers/hour
- Daily theoretical capacity (peak $\times$ 10): $\approx 1,108,800$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 119,156 - 178,734$ trips/day

Catchment

- City population: 3,009,000
- Anchor-weighted coverage: 39.6%
- Catchment population: $\approx 1,191,564$ (within ~ 800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	23	500 kW	3000 kWh
Major	46	400 kW	2500 kWh
Standard	59	300 kW	2000 kWh
Terminal	11	500 kW	3000 kWh
Total installed	140	58,100 kW	375,000 kWh

Aggregate station-rail charging power: 76,750 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 720 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	24.0 kWh/km	6 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	1,008 kWh	42.0 km average line length
Onboard battery coverage	0.7 $\times$ average line run	720 kWh usable pack
Average 60 s dwell charge	9.0 kWh/stop	540 kW average charger across stops
Stops to refill one trainset pack	80 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	290 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	375 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the `[costs]` block in `design.toml` — emitted by the `osr-design` Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. `country-costs.toml` applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (250.3 km @ €3.5 M/km)	€876 M
Elevated (33.8 km @ €18 M/km)	€608 M
Elevated-interchange premium (12 sites @ €20 M)	€240 M
Civil subtotal	€1.72 bn

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
halt	3	€0.4 M	€1.2 M
standard	59	€1.5 M	€88 M
major	46	€3.0 M	€138 M
terminal	11	€2.5 M	€28 M
depot-terminal	1	€3.0 M	€3.0 M
interchange	3	€4.5 M	€14 M
interchange-elevated	20	€4.5 M	€90 M
Stations subtotal			€362 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	11	€3.0 M	€33 M
Depots subtotal			€58 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
metro-6car (revenue + spare + cold reserve)	240	€6.0 M	€1.44 bn

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	294.0 km × €0.015 M/km	€4.3 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€52 M
EPC integration + project management (7%)	on subtotal	€255 M

Total

Bucket	Value
Civil works	€1.72 bn
Stations	€362 M
Depots	€58 M
Rolling stock	€1.44 bn
Residual train-control wayside + charging microgrids	€57 M
EPC overhead (7%)	€255 M
CAPEX total	€3.90 bn

Bucket	Value
Per-route-km	€13 M / km
Per-capita (city pop)	€1,295 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code — regenerate by re-running `scripts/regenerate-city.sh` `surabaya`.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–5 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 6 and runs for 20 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–5)	€276 M / yr	€92
Steady-state, low-ridership (year 6+)	€344 M / yr	€114
Steady-state, high-ridership (year 6+)	€317 M / yr	€105
Lifecycle envelope (yr 1–25, low scenario)	€8.26 bn cumulative	€2,746
Lifecycle envelope (yr 1–25, high scenario)	€7.72 bn cumulative	€2,567

Population basis: 3,009,000 (catchment per `lib/city-batches/world-sample.toml`). After year 25, debt service drops to zero and only the OPEX shortfall remains — ~€83 M / yr (low) €56 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€2.34 bn	4.0%	25 y, 5 y grace	€172 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€974 M	6.7%	25 y, 5 y grace	€90 M / yr
Government equity (no debt service)	15%	€584 M	—	—	—
Total	100%	€3.90 bn			€262 M / yr

During the 5-year grace period the operator pays interest only — multilateral €93 M / yr + bonds €65 M / yr = €159 M / yr total — plus the equity tranche amortised across construction (€117 M / yr × 5 yr). Principal repayment begins in year 6 on a 20-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€58 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€43 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€213 k
Traction energy (1013.5 GWh / yr)	trackside PV + Na-ion (RFC 0002) — self-generated, €0 / yr	€0 k
Labour (1,776 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€8.8 M
OPEX subtotal		€109 M / yr

Annual fleet utilisation: 215 revenue trainsets × 20.5 h/day × 365 d/yr × 35 km/h commercial × 75% revenue factor = 42.2 M train-km / yr (~196 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$320 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass — a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.49 (~\$0.53 USD)
Day pass (3 trips)	€1.25 (15 % bulk discount)
Monthly unlimited pass	€14.72 (~5 % of median monthly income)
Annual pass	€161.92 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	54.9 M	109.8 M
Farebox revenue	€27 M / yr	€54 M / yr
Farebox / OPEX recovery	25%	49%
Country policy-target recovery (diagnostic)	60%	60%
Operating shortfall (gov subsidy required)	€83 M / yr	€56 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€344 M / yr	€317 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

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surabaya-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
surabaya.corridor.geojson	Line polylines + stations (GeoJSON)
surabaya.stations.json	Machine-readable station list
surabaya.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug surabaya

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug surabaya \
  --sidecar .cache/osr-pipeline/rasters/surabaya.grid.json \
  --out-dir designs/.../Surabaya

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../surabaya.toml \
  --out designs/.../README.md
```

`scripts/regenerate-surabaya.sh` chains steps 3 + drift tests into a single command.

southeast-asia/Laos/Vientiane

designs/southeast-asia/Laos/Vientiane/README.md

Vientiane — Urban Rail Network

Country: LA · Population: 948,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Vientiane rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [vientiane.corridor.geojson](#).

At a glance

Metric	Value
Lines	3
Unique stations	46
Interchange stations	2
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	42.9%
Route length (double track)	80.8 km
Revenue fleet	70 × 3-car trainsets
Spare + cold-reserve	8 × 3-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	28.8 km	16	28	SE Outer W Outer
line-2	22.2 km	15	21	S Mid N Outer
line-3	29.8 km	15	29	W Mid NE Outer
Total	80.8 km	46 unique	78	

Rolling stock

Property	Value
Consist	3-car, 57 m
Max speed	90 km/h
Onboard battery	360 kWh per trainset
Seats	60 longitudinal seats
Nominal capacity (AW2)	330 pax (seated + standing, light-metro-3car per RFC 0008 §1)
Crush capacity (AW3)	420 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 330 AW2 passengers (light-metro-3car)
- Peak frequency: 12 trains/hour/direction (5-min headway)

- Peak capacity per line per direction: $330 \times 12 = 3,960$ pphpd
- Network peak throughput (all lines, both directions): $3 \text{ lines} \times 2 \text{ directions} \times 3,960 = 23,760$ passengers/hour
- Daily theoretical capacity (peak $\times 10$): $\approx 237,600$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 40,669 - 61,003$ trips/day

Catchment

- City population: 948,000
- Anchor-weighted coverage: 42.9%
- Catchment population: $\approx 406,692$ (within ~ 800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	2	500 kW	3000 kWh
Major	19	400 kW	2500 kWh
Standard	16	300 kW	2000 kWh
Terminal	5	500 kW	3000 kWh
Total installed	43	20,900 kW	140,500 kWh

Aggregate station-rail charging power: 25,250 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 360 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	12.0 kWh/km	3 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	323 kWh	26.9 km average line length
Onboard battery coverage	1.1 $\times$ average line run	360 kWh usable pack
Average 60 s dwell charge	9.1 kWh/stop	549 kW average charger across stops
Stops to refill one trainset pack	39 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	104 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	140 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the `[costs]` block in `design.toml` — emitted by the `osr-design` Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. `country-costs.toml` applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (79.3 km @ €3.5 M/km)	€277 M
Elevated (1.4 km @ €18 M/km)	€25 M
Elevated-interchange premium (1 sites @ €20 M)	€20 M
Civil subtotal	€322 M

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~ 11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
halt	3	€0.4 M	€1.2 M
standard	16	€1.5 M	€24 M
major	19	€3.0 M	€57 M
terminal	5	€2.5 M	€12 M
depot-terminal	1	€3.0 M	€3.0 M
interchange-elevated	2	€4.5 M	€9.0 M
Stations subtotal			€107 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	5	€3.0 M	€15 M
Depots subtotal			€40 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
light-metro-3car (revenue + spare + cold reserve)	78	€3.0 M	€234 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	80.8 km × €0.015 M/km	€1.2 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€16 M
EPC integration + project management (7%)	on subtotal	€50 M

Total

Bucket	Value
Civil works	€322 M
Stations	€107 M
Depots	€40 M
Rolling stock	€234 M
Residual train-control wayside + charging microgrids	€17 M
EPC overhead (7%)	€50 M
CAPEX total	€771 M
Per-route-km	€9.5 M / km
Per-capita (city pop)	€813 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running scripts/regenerate-city.sh vientiane.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–7 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 8 and runs for 23 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–7)	€51 M / yr	€53
Steady-state, low-ridership (year 8+)	€66 M / yr	€69
Steady-state, high-ridership (year 8+)	€61 M / yr	€64
Lifecycle envelope (yr 1–30, low scenario)	€1.87 bn cumulative	€1,968
Lifecycle envelope (yr 1–30, high scenario)	€1.75 bn cumulative	€1,843

Population basis: 948,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 30, debt service drops to zero and only the OPEX shortfall remains – ~€15 M / yr (low) €10.0 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€462 M	3.0%	30 y, 7 y grace	€28 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€193 M	10.5%	30 y, 7 y grace	€22 M / yr
Government equity (no debt service)	15%	€116 M	—	—	—
Total	100%	€771 M			€51 M / yr

During the 7-year grace period the operator pays interest only – multilateral €14 M / yr + bonds €20 M / yr = €34 M / yr total – plus the equity tranche amortised across construction (€17 M / yr × 7 yr). Principal repayment begins in year 8 on a 23-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€9.4 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€9.4 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€60 k
Traction energy (141.4 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (497 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€1.5 M
OPEX subtotal		€20 M / yr

Annual fleet utilisation: 70 revenue trainsets × 20.5 h/day × 365 d/yr × 30 km/h commercial × 75% revenue factor = 11.8 M train-km / yr (~168 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$195 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.30 (~\$0.33 USD)
Day pass (3 trips)	€0.76 (15 % bulk discount)
Monthly unlimited pass	€8.97 (~5 % of median monthly income)
Annual pass	€98.67 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	17.3 M	34.6 M

	Low scenario	High scenario
Farebox revenue	€5.2 M / yr	€10 M / yr
Farebox / OPEX recovery	25%	51%
Country policy-target recovery (diagnostic)	40%	40%
Operating shortfall (gov subsidy required)	€15 M / yr	€10.0 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€66 M / yr	€61 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
vientiane.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
vientiane-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
vientiane.corridor.geojson	Line polylines + stations (GeoJSON)
vientiane.stations.json	Machine-readable station list
vientiane.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug vientiane

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug vientiane \
  --sidecar .cache/osr-pipeline/rasters/vientiane.grid.json \
  --out-dir designs/.../Vientiane

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../vientiane.toml \
  --out designs/.../README.md
```

`scripts/regenerate-vientiane.sh` chains steps 3 + drift tests into a single command.

southeast-asia/Myanmar/Mandalay

designs/southeast-asia/Myanmar/Mandalay/README.md

Mandalay — Urban Rail Network

Country: MM · Population: 1,726,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Mandalay rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [mandalay.corridor.geojson](#).

At a glance

Metric	Value
Lines	6
Unique stations	87
Interchange stations	23
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	60.1%
Route length (double track)	187.0 km
Revenue fleet	139 × 4-car trainsets
Spare + cold-reserve	17 × 4-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	42.7 km	18	35	N Outer S Mid
line-2	33.6 km	14	28	NE Mid SW Outer
line-3	25.4 km	11	21	N Mid SE Mid
line-4	27.9 km	13	24	E Mid SW Mid
line-5	19.0 km	11	17	SE Mid W Inner
line-6	38.4 km	21	31	NE Inner NE Inner
Total	187.0 km	87 unique	156	

Rolling stock

Property	Value
Consist	4-car, 75 m
Max speed	90 km/h
Onboard battery	480 kWh per trainset
Seats	80 longitudinal seats
Nominal capacity (AW2)	440 pax (seated + standing, metro-4car per RFC 0008 §1)
Crush capacity (AW3)	560 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 440 AW2 passengers (metro - 4 car)
- Peak frequency: 12 trains/hour/direction (5-min headway)
- Peak capacity per line per direction: $440 \times 12 = 5,280$ pphpd
- Network peak throughput (all lines, both directions): $6 \text{ lines} \times 2 \text{ directions} \times 5,280 = 63,360$ passengers/hour
- Daily theoretical capacity (peak $\times$ 10): $\approx 633,600$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 103,732 - 155,598$ trips/day

Catchment

- City population: 1,726,000
- Anchor-weighted coverage: 60.1%
- Catchment population: $\approx 1,037,326$ (within ~800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	23	500 kW	3000 kWh
Major	18	400 kW	2500 kWh
Standard	31	300 kW	2000 kWh
Terminal	9	500 kW	3000 kWh
Total installed	82	37,500 kW	243,000 kWh

Aggregate station-rail charging power: 47,500 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 480 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	16.0 kWh/km	4 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	499 kWh	31.2 km average line length
Onboard battery coverage	1.0 $\times$ average line run	480 kWh usable pack
Average 60 s dwell charge	9.1 kWh/stop	546 kW average charger across stops
Stops to refill one trainset pack	53 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	188 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	243 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in design.toml — emitted by the osr-design Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. country-costs.toml applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (170.8 km @ €3.5 M/km)	€598 M
Elevated (14.7 km @ €18 M/km)	€264 M
Elevated-interchange premium (10 sites @ €20 M)	€200 M
Civil subtotal	€1.06 bn

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
halt	6	€0.4 M	€2.4 M
standard	31	€1.5 M	€46 M
major	18	€3.0 M	€54 M
terminal	9	€2.5 M	€22 M
depot-terminal	1	€3.0 M	€3.0 M
interchange-elevated	23	€4.5 M	€104 M
Stations subtotal			€232 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	9	€3.0 M	€27 M
Depots subtotal			€52 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
metro-4car (revenue + spare + cold reserve)	156	€4.0 M	€624 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	187.0 km × €0.015 M/km	€2.8 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€34 M
EPC integration + project management (7%)	on subtotal	€140 M

Total

Bucket	Value
Civil works	€1.06 bn
Stations	€232 M
Depots	€52 M
Rolling stock	€624 M
Residual train-control wayside + charging microgrids	€37 M
EPC overhead (7%)	€140 M
CAPEX total	€2.15 bn
Per-route-km	€11 M / km
Per-capita (city pop)	€1,244 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running `scripts/regenerate-city.sh` mandalay.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–10 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 11 and runs for 20 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–10)	€160 M / yr	€93
Steady-state, low-ridership (year 11+)	€223 M / yr	€129
Steady-state, high-ridership (year 11+)	€217 M / yr	€126
Lifecycle envelope (yr 1–30, low scenario)	€6.07 bn cumulative	€3,516
Lifecycle envelope (yr 1–30, high scenario)	€5.94 bn cumulative	€3,443

Population basis: 1,726,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 30, debt service drops to zero and only the OPEX shortfall remains – ~€48 M / yr (low) €42 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€1.29 bn	4.5%	30 y, 10 y grace	€99 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€537 M	13.0%	30 y, 10 y grace	€76 M / yr
Government equity (no debt service)	15%	€322 M	—	—	—
Total	100%	€2.15 bn			€175 M / yr

During the 10-year grace period the operator pays interest only — multilateral €58 M / yr + bonds €70 M / yr = €128 M / yr total — plus the equity tranche amortised across construction (€32 M / yr × 10 yr). Principal repayment begins in year 11 on a 20-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€25 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€27 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€139 k
Traction energy (436.8 GWh / yr)	trackside PV + Na-ion (RFC 0002) — self-generated, €0 / yr	€0 k
Labour (1,134 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€2.3 M
OPEX subtotal		€54 M / yr

Annual fleet utilisation: 139 revenue trainsets × 20.5 h/day × 365 d/yr × 35 km/h commercial × 75% revenue factor = 27.3 M train-km / yr (~196 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$130 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass — a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.20 (~\$0.22 USD)
Day pass (3 trips)	€0.51 (15 % bulk discount)
Monthly unlimited pass	€5.98 (~5 % of median monthly income)
Annual pass	€65.78 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	31.5 M	63.0 M
Farebox revenue	€6.3 M / yr	€13 M / yr
Farebox / OPEX recovery	12%	23%
Country policy-target recovery (diagnostic)	40%	40%
Operating shortfall (gov subsidy required)	€48 M / yr	€42 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€223 M / yr	€217 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
mandalay.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
mandalay-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
mandalay.corridor.geojson	Line polylines + stations (GeoJSON)
mandalay.stations.json	Machine-readable station list
mandalay.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug mandalay

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug mandalay \
  --sidecar .cache/osr-pipeline/rasters/mandalay.grid.json \
  --out-dir designs/.../Mandalay

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../mandalay.toml \
  --out designs/.../README.md
```

`scripts/regenerate-mandalay.sh` chains steps 3 + drift tests into a single command.

southeast-asia/Myanmar/Yangon

designs/southeast-asia/Myanmar/Yangon/README.md

Yangon — Urban Rail Network

Country: MM · Population: 5,200,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Yangon rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [yangon.corridor.geojson](#).

At a glance

Metric	Value
Lines	9
Unique stations	213
Interchange stations	43
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	56.4%
Route length (double track)	417.5 km
Revenue fleet	301 × 6-car trainsets
Spare + cold-reserve	34 × 6-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	53.5 km	27	42	NW Outer SE Outer
line-2	51.9 km	23	41	SE Outer NW Outer
line-3	42.3 km	20	35	NE Mid W Outer
line-4	32.1 km	18	27	SE Mid N Mid
line-5	39.0 km	21	31	S Outer N Mid
line-6	38.8 km	19	31	N Mid S Outer
line-7	43.8 km	21	36	E Mid SW Outer
line-8	38.6 km	20	31	SW Mid NE Outer
line-9	77.4 km	45	61	NW Mid NW Mid
Total	417.5 km	213 unique	335	

Rolling stock

Property	Value
Consist	6-car, 111 m
Max speed	100 km/h
Onboard battery	720 kWh per trainset
Seats	120 longitudinal seats
Nominal capacity (AW2)	660 pax (seated + standing, metro-6car per RFC 0008 §1)
Crush capacity (AW3)	840 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 660 AW2 passengers (metro - 6 car)
- Peak frequency: 12 trains/hour/direction (5-min headway)
- Peak capacity per line per direction: $660 \times 12 = 7,920$ pphpd
- Network peak throughput (all lines, both directions): $9 \text{ lines} \times 2 \text{ directions} \times 7,920 = 142,560$ passengers/hour
- Daily theoretical capacity (peak $\times$ 10): $\approx 1,425,600$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 293,279 - 439,919$ trips/day

Catchment

- City population: 5,200,000
- Anchor-weighted coverage: 56.4%
- Catchment population: $\approx 2,932,799$ (within ~800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	43	500 kW	3000 kWh
Major	97	400 kW	2500 kWh
Standard	51	300 kW	2000 kWh
Terminal	15	500 kW	3000 kWh
Total installed	207	88,100 kW	558,500 kWh

Aggregate station-rail charging power: 113,250 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 720 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	24.0 kWh/km	6 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	1,113 kWh	46.4 km average line length
Onboard battery coverage	0.6 $\times$ average line run	720 kWh usable pack
Average 60 s dwell charge	8.9 kWh/stop	532 kW average charger across stops
Stops to refill one trainset pack	81 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	440 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	558 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in design.toml — emitted by the osr-design Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. country-costs.toml applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (375.2 km @ €3.5 M/km)	€1.31 bn
Elevated (39.3 km @ €18 M/km)	€708 M
Elevated-interchange premium (18 sites @ €20 M)	€360 M
Civil subtotal	€2.38 bn

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
halt	7	€0.4 M	€2.8 M
standard	51	€1.5 M	€76 M
major	97	€3.0 M	€291 M
terminal	15	€2.5 M	€38 M
depot-terminal	1	€3.0 M	€3.0 M
interchange	4	€4.5 M	€18 M
interchange-elevated	39	€4.5 M	€176 M
Stations subtotal			€604 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	15	€3.0 M	€45 M
Depots subtotal			€70 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
metro-6car (revenue + spare + cold reserve)	335	€6.0 M	€2.01 bn

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	417.5 km × €0.015 M/km	€6.2 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€85 M
EPC integration + project management (7%)	on subtotal	€361 M

Total

Bucket	Value
Civil works	€2.38 bn
Stations	€604 M
Depots	€70 M
Rolling stock	€2.01 bn
Residual train-control wayside + charging microgrids	€91 M
EPC overhead (7%)	€361 M
CAPEX total	€5.52 bn

Bucket	Value
Per-route-km	€13 M / km
Per-capita (city pop)	€1,061 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from lib/templates/country-finance.toml. Pure function of the [costs] block above + the country code — regenerate by re-running `scripts/regenerate-city.sh` yangon.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–10 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 11 and runs for 20 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–10)	€411 M / yr	€79
Steady-state, low-ridership (year 11+)	€579 M / yr	€111
Steady-state, high-ridership (year 11+)	€560 M / yr	€108
Lifecycle envelope (yr 1–30, low scenario)	€15.69 bn cumulative	€3,017
Lifecycle envelope (yr 1–30, high scenario)	€15.31 bn cumulative	€2,944

Population basis: 5,200,000 (catchment per lib/city-batches/world-sample.toml). After year 30, debt service drops to zero and only the OPEX shortfall remains — ~€128 M / yr (low) €109 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€3.31 bn	4.5%	30 y, 10 y grace	€254 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€1.38 bn	13.0%	30 y, 10 y grace	€196 M / yr
Government equity (no debt service)	15%	€828 M	—	—	—
Total	100%	€5.52 bn			€451 M / yr

During the 10-year grace period the operator pays interest only — multilateral €149 M / yr + bonds €179 M / yr = €328 M / yr total — plus the equity tranche amortised across construction (€83 M / yr × 10 yr). Principal repayment begins in year 11 on a 20-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€80 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€61 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€311 k
Traction energy (1418.9 GWh / yr)	trackside PV + Na-ion (RFC 0002) — self-generated, €0 / yr	€0 k
Labour (2,517 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€5.1 M
OPEX subtotal		€147 M / yr

Annual fleet utilisation: 301 revenue trainsets × 20.5 h/day × 365 d/yr × 35 km/h commercial × 75% revenue factor = 59.1 M train-km / yr (~196 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$130 USD (per lib/templates/country-finance.toml). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass — a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.20 (~\$0.22 USD)
Day pass (3 trips)	€0.51 (15 % bulk discount)
Monthly unlimited pass	€5.98 (~5 % of median monthly income)
Annual pass	€65.78 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	94.9 M	189.8 M
Farebox revenue	€19 M / yr	€38 M / yr
Farebox / OPEX recovery	13%	26%
Country policy-target recovery (diagnostic)	40%	40%
Operating shortfall (gov subsidy required)	€128 M / yr	€109 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€579 M / yr	€560 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
yangon.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
yangon-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
yangon.corridor.geojson	Line polylines + stations (GeoJSON)
yangon.stations.json	Machine-readable station list
yangon.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug yangon

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug yangon \
  --sidecar .cache/osr-pipeline/rasters/yangon.grid.json \
  --out-dir designs/.../Yangon

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../yangon.toml \
  --out designs/.../README.md
```

`scripts/regenerate-yangon.sh` chains steps 3 + drift tests into a single command.

southeast-asia/Philippines/Davao

designs/southeast-asia/Philippines/Davao/README.md

Davao — Urban Rail Network

Country: PH · Population: 1,827,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Davao rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [davao.corridor.geojson](#).

At a glance

Metric	Value
Lines	6
Unique stations	107
Interchange stations	24
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	71.5%
Route length (double track)	228.9 km
Revenue fleet	167 × 4-car trainsets
Spare + cold-reserve	19 × 4-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	39.7 km	16	32	SW Outer NE Mid
line-2	24.5 km	14	20	E Mid SW Mid
line-3	35.2 km	16	29	NE Outer SE Inner
line-4	32.1 km	14	27	S Inner NW Outer
line-5	27.7 km	13	24	W Mid E Inner
line-6	69.7 km	35	54	NW Mid NW Mid
Total	228.9 km	107 unique	186	

Rolling stock

Property	Value
Consist	4-car, 75 m
Max speed	90 km/h
Onboard battery	480 kWh per trainset
Seats	80 longitudinal seats
Nominal capacity (AW2)	440 pax (seated + standing, metro-4car per RFC 0008 §1)
Crush capacity (AW3)	560 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 440 AW2 passengers (metro - 4 car)
- Peak frequency: 12 trains/hour/direction (5-min headway)
- Peak capacity per line per direction: $440 \times 12 = 5,280$ pphpd
- Network peak throughput (all lines, both directions): $6 \text{ lines} \times 2 \text{ directions} \times 5,280 = 63,360$ passengers/hour
- Daily theoretical capacity (peak $\times$ 10): $\approx 633,600$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 130,630 - 195,945$ trips/day

Catchment

- City population: 1,827,000
- Anchor-weighted coverage: 71.5%
- Catchment population: $\approx 1,306,305$ (within ~800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	24	500 kW	3000 kWh
Major	31	400 kW	2500 kWh
Standard	36	300 kW	2000 kWh
Terminal	9	500 kW	3000 kWh
Total installed	101	44,700 kW	288,500 kWh

Aggregate station-rail charging power: 57,250 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 480 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	16.0 kWh/km	4 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	610 kWh	38.1 km average line length
Onboard battery coverage	0.8 $\times$ average line run	480 kWh usable pack
Average 60 s dwell charge	8.9 kWh/stop	535 kW average charger across stops
Stops to refill one trainset pack	54 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	224 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	288 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in design.toml — emitted by the osr-design Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. country-costs.toml applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (216.4 km @ €3.5 M/km)	€758 M
Elevated (11.0 km @ €18 M/km)	€197 M
Elevated-interchange premium (8 sites @ €20 M)	€160 M
Civil subtotal	€1.11 bn

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
halt	7	€0.4 M	€2.8 M
standard	36	€1.5 M	€54 M
major	31	€3.0 M	€93 M
terminal	9	€2.5 M	€22 M
depot-terminal	1	€3.0 M	€3.0 M
interchange-elevated	24	€4.5 M	€108 M
Stations subtotal			€283 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	9	€3.0 M	€27 M
Depots subtotal			€52 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
metro-4car (revenue + spare + cold reserve)	186	€4.0 M	€744 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	228.9 km × €0.015 M/km	€3.4 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€41 M
EPC integration + project management (7%)	on subtotal	€157 M

Total

Bucket	Value
Civil works	€1.11 bn
Stations	€283 M
Depots	€52 M
Rolling stock	€744 M
Residual train-control wayside + charging microgrids	€44 M
EPC overhead (7%)	€157 M
CAPEX total	€2.40 bn
Per-route-km	€10 M / km
Per-capita (city pop)	€1,311 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running `scripts/regenerate-city.sh` `davao`.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–5 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 6 and runs for 20 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–5)	€166 M / yr	€91
Steady-state, low-ridership (year 6+)	€207 M / yr	€113
Steady-state, high-ridership (year 6+)	€189 M / yr	€103
Lifecycle envelope (yr 1–25, low scenario)	€4.97 bn cumulative	€2,722
Lifecycle envelope (yr 1–25, high scenario)	€4.60 bn cumulative	€2,520

Population basis: 1,827,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 25, debt service drops to zero and only the OPEX shortfall remains – ~€48 M / yr (low) €30 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€1.44 bn	4.0%	25 y, 5 y grace	€106 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€599 M	6.2%	25 y, 5 y grace	€53 M / yr
Government equity (no debt service)	15%	€359 M	—	—	—
Total	100%	€2.40 bn			€159 M / yr

During the 5-year grace period the operator pays interest only – multilateral €57 M / yr + bonds €37 M / yr = €95 M / yr total – plus the equity tranche amortised across construction (€72 M / yr × 5 yr). Principal repayment begins in year 6 on a 20-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€30 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€29 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€171 k
Traction energy (524.8 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (1,385 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€7.7 M
OPEX subtotal		€67 M / yr

Annual fleet utilisation: 167 revenue trainsets × 20.5 h/day × 365 d/yr × 35 km/h commercial × 75% revenue factor = 32.8 M train-km / yr (~196 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$360 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.55 (~\$0.60 USD)
Day pass (3 trips)	€1.41 (15 % bulk discount)
Monthly unlimited pass	€16.56 (~5 % of median monthly income)
Annual pass	€182.16 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	33.3 M	66.7 M
Farebox revenue	€18 M / yr	€37 M / yr
Farebox / OPEX recovery	28%	55%
Country policy-target recovery (diagnostic)	55%	55%
Operating shortfall (gov subsidy required)	€48 M / yr	€30 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€207 M / yr	€189 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
davao.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
davao-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
davao.corridor.geojson	Line polylines + stations (GeoJSON)
davao.stations.json	Machine-readable station list
davao.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug davao

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug davao \
  --sidecar .cache/osr-pipeline/rasters/davao.grid.json \
  --out-dir designs/.../Davao

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../davao.toml \
  --out designs/.../README.md
```

`scripts/regenerate-davao.sh` chains steps 3 + drift tests into a single command.

west-africa/Burkina Faso/Ouagadougou

designs/west-africa/Burkina Faso/Ouagadougou/README.md

Ouagadougou — Urban Rail Network

Country: BF · Population: 2,531,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Ouagadougou rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [ouagadougou.corridor.geojson](#).

At a glance

Metric	Value
Lines	6
Unique stations	137
Interchange stations	24
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	37.4%
Route length (double track)	263.9 km
Revenue fleet	191 × 4-car trainsets
Spare + cold-reserve	22 × 4-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	39.8 km	22	32	NE Outer SW Outer
line-2	38.9 km	20	31	SE Mid W Outer
line-3	42.2 km	21	35	NE Outer S Mid
line-4	36.0 km	19	29	S Mid N Outer
line-5	34.9 km	20	29	NW Mid E Mid
line-6	72.1 km	36	57	NW Mid NW Mid
Total	263.9 km	137 unique	213	

Rolling stock

Property	Value
Consist	4-car, 75 m
Max speed	90 km/h
Onboard battery	480 kWh per trainset
Seats	80 longitudinal seats
Nominal capacity (AW2)	440 pax (seated + standing, metro-4car per RFC 0008 §1)
Crush capacity (AW3)	560 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 440 AW2 passengers (metro - 4 car)
- Peak frequency: 12 trains/hour/direction (5-min headway)
- Peak capacity per line per direction: $440 \times 12 = 5,280$ pphpd
- Network peak throughput (all lines, both directions): $6 \text{ lines} \times 2 \text{ directions} \times 5,280 = 63,360$ passengers/hour
- Daily theoretical capacity (peak $\times$ 10): $\approx 633,600$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 94,659 - 141,989$ trips/day

Catchment

- City population: 2,531,000
- Anchor-weighted coverage: 37.4%
- Catchment population: $\approx 946,594$ (within ~ 800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../.../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	24	500 kW	3000 kWh
Major	63	400 kW	2500 kWh
Standard	40	300 kW	2000 kWh
Terminal	9	500 kW	3000 kWh
Total installed	137	58,700 kW	376,500 kWh

Aggregate station-rail charging power: 73,750 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 480 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	16.0 kWh/km	4 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	704 kWh	44.0 km average line length
Onboard battery coverage	0.7 $\times$ average line run	480 kWh usable pack
Average 60 s dwell charge	9.0 kWh/stop	538 kW average charger across stops
Stops to refill one trainset pack	53 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	294 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	376 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in design.toml — emitted by the osr-design Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. country-costs.toml applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (247.7 km @ €3.5 M/km)	€867 M
Elevated (15.5 km @ €18 M/km)	€278 M
Elevated-interchange premium (12 sites @ €20 M)	€240 M
Civil subtotal	€1.39 bn

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
halt	1	€0.4 M	€0.4 M
standard	40	€1.5 M	€60 M
major	63	€3.0 M	€189 M
terminal	9	€2.5 M	€22 M
depot-terminal	1	€3.0 M	€3.0 M
interchange-elevated	24	€4.5 M	€108 M
Stations subtotal			€383 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	9	€3.0 M	€27 M
Depots subtotal			€52 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
metro-4car (revenue + spare + cold reserve)	213	€4.0 M	€852 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	263.9 km × €0.015 M/km	€3.9 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€54 M
EPC integration + project management (7%)	on subtotal	€191 M

Total

Bucket	Value
Civil works	€1.39 bn
Stations	€383 M
Depots	€52 M
Rolling stock	€852 M
Residual train-control wayside + charging microgrids	€58 M
EPC overhead (7%)	€191 M
CAPEX total	€2.92 bn
Per-route-km	€11 M / km
Per-capita (city pop)	€1,154 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running scripts/regenerate-city.sh ouagadougou.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–10 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 11 and runs for 25 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–10)	€166 M / yr	€66
Steady-state, low-ridership (year 11+)	€244 M / yr	€96
Steady-state, high-ridership (year 11+)	€236 M / yr	€93
Lifecycle envelope (yr 1–35, low scenario)	€7.75 bn cumulative	€3,062
Lifecycle envelope (yr 1–35, high scenario)	€7.55 bn cumulative	€2,985

Population basis: 2,531,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 35, debt service drops to zero and only the OPEX shortfall remains – ~€66 M / yr (low) €58 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€1.75 bn	3.0%	35 y, 10 y grace	€101 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€730 M	9.5%	35 y, 10 y grace	€77 M / yr
Government equity (no debt service)	15%	€438 M	—	—	—
Total	100%	€2.92 bn			€178 M / yr

During the 10-year grace period the operator pays interest only — multilateral €53 M / yr + bonds €69 M / yr = €122 M / yr total — plus the equity tranche amortised across construction (€44 M / yr × 10 yr). Principal repayment begins in year 11 on a 25-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€34 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€36 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€197 k
Traction energy (600.2 GWh / yr)	trackside PV + Na-ion (RFC 0002) — self-generated, €0 / yr	€0 k
Labour (1,595 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€2.7 M
OPEX subtotal		€73 M / yr

Annual fleet utilisation: 191 revenue trainsets × 20.5 h/day × 365 d/yr × 35 km/h commercial × 75% revenue factor = 37.5 M train-km / yr (~196 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$110 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass — a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.17 (~\$0.18 USD)
Day pass (3 trips)	€0.43 (15 % bulk discount)
Monthly unlimited pass	€5.06 (~5 % of median monthly income)
Annual pass	€55.66 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	46.2 M	92.4 M
Farebox revenue	€7.8 M / yr	€16 M / yr
Farebox / OPEX recovery	11%	21%
Country policy-target recovery (diagnostic)	30%	30%
Operating shortfall (gov subsidy required)	€66 M / yr	€58 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€244 M / yr	€236 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
ouagadougou.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
ouagadougou-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
ouagadougou.corridor.geojson	Line polylines + stations (GeoJSON)
ouagadougou.stations.json	Machine-readable station list
ouagadougou.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug ouagadougou

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug ouagadougou \
  --sidecar .cache/osr-pipeline/rasters/ouagadougou.grid.json \
  --out-dir designs/.../Ouagadougou

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../ouagadougou.toml \
  --out designs/.../README.md
```

`scripts/regenerate-ouagadougou.sh` chains steps 3 + drift tests into a single command.

west-africa/Cameroon/Douala

designs/west-africa/Cameroon/Douala/README.md

Douala — Urban Rail Network

Country: CM · Population: 3,900,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Douala rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [douala.corridor.geojson](#).

At a glance

Metric	Value
Lines	5
Unique stations	128
Interchange stations	15
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	37.0%
Route length (double track)	228.0 km
Revenue fleet	165 × 6-car trainsets
Spare + cold-reserve	19 × 6-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	41.8 km	23	34	SE Mid NW Outer
line-2	39.7 km	21	32	SE Mid NW Outer
line-3	42.1 km	22	35	NE Mid SW Outer
line-4	25.8 km	15	21	NW Mid E Inner
line-5	78.6 km	48	62	NW Mid NW Mid
Total	228.0 km	128 unique	184	

Rolling stock

Property	Value
Consist	6-car, 111 m
Max speed	100 km/h
Onboard battery	720 kWh per trainset
Seats	120 longitudinal seats
Nominal capacity (AW2)	660 pax (seated + standing, met ro-6car per RFC 0008 §1)
Crush capacity (AW3)	840 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 660 AW2 passengers (met ro-6car)

- Peak frequency: 12 trains/hour/direction (5-min headway)
- Peak capacity per line per direction: $660 \times 12 = 7,920$ pphpd
- Network peak throughput (all lines, both directions): $5 \text{ lines} \times 2 \text{ directions} \times 7,920 = 79,200$ passengers/hour
- Daily theoretical capacity (peak $\times$ 10): $\approx 792,000$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 144,300 - 216,450$ trips/day

Catchment

- City population: 3,900,000
- Anchor-weighted coverage: 37.0%
- Catchment population: $\approx 1,443,000$ (within ~ 800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	15	500 kW	3000 kWh
Major	79	400 kW	2500 kWh
Standard	24	300 kW	2000 kWh
Terminal	7	500 kW	3000 kWh
Total installed	126	54,800 kW	351,500 kWh

Aggregate station-rail charging power: 67,750 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 720 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	24.0 kWh/km	6 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	1,094 kWh	45.6 km average line length
Onboard battery coverage	0.7 $\times$ average line run	720 kWh usable pack
Average 60 s dwell charge	8.8 kWh/stop	529 kW average charger across stops
Stops to refill one trainset pack	82 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	274 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	352 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in design.toml — emitted by the osr-design Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. country-costs.toml applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (193.3 km @ €3.5 M/km)	€676 M
Elevated (31.4 km @ €18 M/km)	€565 M
Elevated-interchange premium (5 sites @ €20 M)	€100 M
Civil subtotal	€1.34 bn

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
halt	3	€0.4 M	€1.2 M
standard	24	€1.5 M	€36 M
major	79	€3.0 M	€237 M
terminal	7	€2.5 M	€18 M
depot-terminal	1	€3.0 M	€3.0 M
interchange-elevated	15	€4.5 M	€68 M
Stations subtotal			€362 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	7	€3.0 M	€21 M
Depots subtotal			€46 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
metro-6car (revenue + spare + cold reserve)	184	€6.0 M	€1.10 bn

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	228.0 km × €0.015 M/km	€3.4 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€51 M
EPC integration + project management (7%)	on subtotal	€204 M

Total

Bucket	Value
Civil works	€1.34 bn
Stations	€362 M
Depots	€46 M
Rolling stock	€1.10 bn
Residual train-control wayside + charging microgrids	€54 M
EPC overhead (7%)	€204 M
CAPEX total	€3.11 bn
Per-route-km	€14 M / km
Per-capita (city pop)	€798 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running scripts/regenerate-city.sh `douala`.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–7 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 8 and runs for 23 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–7)	€204 M / yr	€52
Steady-state, low-ridership (year 8+)	€265 M / yr	€68
Steady-state, high-ridership (year 8+)	€245 M / yr	€63
Lifecycle envelope (yr 1–30, low scenario)	€7.51 bn cumulative	€1,927
Lifecycle envelope (yr 1–30, high scenario)	€7.06 bn cumulative	€1,811

Population basis: 3,900,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 30, debt service drops to zero and only the OPEX shortfall remains – ~€64 M / yr (low) €44 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€1.87 bn	3.8%	30 y, 7 y grace	€123 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€778 M	8.5%	30 y, 7 y grace	€78 M / yr
Government equity (no debt service)	15%	€467 M	—	—	—
Total	100%	€3.11 bn			€201 M / yr

During the 7-year grace period the operator pays interest only – multilateral €71 M / yr + bonds €66 M / yr = €137 M / yr total – plus the equity tranche amortised across construction (€67 M / yr × 7 yr). Principal repayment begins in year 8 on a 23-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€44 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€35 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€168 k
Traction energy (777.8 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (1,380 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€3.8 M
OPEX subtotal		€83 M / yr

Annual fleet utilisation: 165 revenue trainsets × 20.5 h/day × 365 d/yr × 35 km/h commercial × 75% revenue factor = 32.4 M train-km / yr (~196 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$180 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.28 (~\$0.30 USD)
Day pass (3 trips)	€0.70 (15 % bulk discount)
Monthly unlimited pass	€8.28 (~5 % of median monthly income)
Annual pass	€91.08 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	71.2 M	142.3 M
Farebox revenue	€20 M / yr	€39 M / yr
Farebox / OPEX recovery	24%	47%
Country policy-target recovery (diagnostic)	40%	40%
Operating shortfall (gov subsidy required)	€64 M / yr	€44 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€265 M / yr	€245 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
douala.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
douala-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
douala.corridor.geojson	Line polylines + stations (GeoJSON)
douala.stations.json	Machine-readable station list
douala.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug douala

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug douala \
  --sidecar .cache/osr-pipeline/rasters/douala.grid.json \
  --out-dir designs/.../Douala

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../douala.toml \
  --out designs/.../README.md
```

`scripts/regenerate-douala.sh` chains steps 3 + drift tests into a single command.

west-africa/Cameroon/Garoua

designs/west-africa/Cameroon/Garoua/README.md

Garoua — Urban Rail Network

Country: CM · Population: 600,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Garoua rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [garoua.corridor.geojson](#).

At a glance

Metric	Value
Lines	3
Unique stations	29
Interchange stations	0
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	37.5%
Route length (double track)	43.5 km
Revenue fleet	40 × 3-car trainsets
Spare + cold-reserve	6 × 3-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	22.0 km	15	21	SW Outer NE Outer
line-2	11.5 km	8	13	W Mid NE Mid
line-3	10.0 km	6	12	W Mid N Outer
Total	43.5 km	29 unique	46	

Rolling stock

Property	Value
Consist	3-car, 57 m
Max speed	90 km/h
Onboard battery	360 kWh per trainset
Seats	60 longitudinal seats
Nominal capacity (AW2)	330 pax (seated + standing, light-metro-3car per RFC 0008 §1)
Crush capacity (AW3)	420 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 330 AW2 passengers (light-metro-3car)
- Peak frequency: 12 trains/hour/direction (5-min headway)

- Peak capacity per line per direction: $330 \times 12 = 3,960$ pphpd
- Network peak throughput (all lines, both directions): $3 \text{ lines} \times 2 \text{ directions} \times 3,960 = 23,760$ passengers/hour
- Daily theoretical capacity (peak $\times 10$): $\approx 237,600$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 22,500 - 33,750$ trips/day

Catchment

- City population: 600,000
- Anchor-weighted coverage: 37.5%
- Catchment population: $\approx 225,000$ (within ~ 800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../.../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Major	13	400 kW	2500 kWh
Standard	10	300 kW	2000 kWh
Terminal	5	500 kW	3000 kWh
Total installed	29	15,700 kW	107,500 kWh

Aggregate station-rail charging power: 17,500 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 360 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	12.0 kWh/km	3 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	174 kWh	14.5 km average line length
Onboard battery coverage	2.1 $\times$ average line run	360 kWh usable pack
Average 60 s dwell charge	10.1 kWh/stop	603 kW average charger across stops
Stops to refill one trainset pack	36 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	78 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	108 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in design.toml — emitted by the osr-design Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. country-costs.toml applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (28.0 km @ €3.5 M/km)	€98 M
Elevated (15.3 km @ €18 M/km)	€275 M
Civil subtotal	€373 M

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~ 11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
standard	10	€1.5 M	€15 M

Archetype	Count	Unit	Subtotal
major	13	€3.0 M	€39 M
terminal	5	€2.5 M	€12 M
depot-terminal	1	€3.0 M	€3.0 M
Stations subtotal			€70 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	5	€3.0 M	€15 M
Depots subtotal			€40 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
light-metro-3car (revenue + spare + cold reserve)	46	€3.0 M	€138 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	43.5 km × €0.015 M/km	€0.6 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€10 M
EPC integration + project management (7%)	on subtotal	€44 M

Total

Bucket	Value
Civil works	€373 M
Stations	€70 M
Depots	€40 M
Rolling stock	€138 M
Residual train-control wayside + charging microgrids	€11 M
EPC overhead (7%)	€44 M
CAPEX total	€676 M
Per-route-km	€16 M / km
Per-capita (city pop)	€1,127 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code — regenerate by re-running scripts/regenerate-city.sh garoua.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–7 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 8 and runs for 23 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–7)	€44 M / yr	€74
Steady-state, low-ridership (year 8+)	€57 M / yr	€94
Steady-state, high-ridership (year 8+)	€54 M / yr	€89
Lifecycle envelope (yr 1–30, low scenario)	€1.61 bn cumulative	€2,689
Lifecycle envelope (yr 1–30, high scenario)	€1.54 bn cumulative	€2,573

Population basis: 600,000 (catchment per lib/city-batches/world-sample.toml). After year 30, debt service drops to zero and only the OPEX shortfall remains — ~€13 M / yr (low) €9.9 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€406 M	3.8%	30 y, 7 y grace	€27 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€169 M	8.5%	30 y, 7 y grace	€17 M / yr
Government equity (no debt service)	15%	€101 M	—	—	—
Total	100%	€676 M			€44 M / yr

During the 7-year grace period the operator pays interest only — multilateral €15 M / yr + bonds €14 M / yr = €30 M / yr total — plus the equity tranche amortised across construction (€14 M / yr × 7 yr). Principal repayment begins in year 8 on a 23-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€5.5 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€9.7 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€32 k
Traction energy (80.8 GWh / yr)	trackside PV + Na-ion (RFC 0002) — self-generated, €0 / yr	€0 k
Labour (273 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€760 k
OPEX subtotal		€16 M / yr

Annual fleet utilisation: 40 revenue trainsets × 20.5 h/day × 365 d/yr × 30 km/h commercial × 75% revenue factor = 6.7 M train-km / yr (~168 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$180 USD (per lib/templates/country-finance.toml). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass — a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.28 (~\$0.30 USD)
Day pass (3 trips)	€0.70 (15 % bulk discount)
Monthly unlimited pass	€8.28 (~5 % of median monthly income)
Annual pass	€91.08 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	10.9 M	21.9 M
Farebox revenue	€3.0 M / yr	€6.0 M / yr
Farebox / OPEX recovery	19%	38%
Country policy-target recovery (diagnostic)	40%	40%
Operating shortfall (gov subsidy required)	€13 M / yr	€9.9 M / yr

	Low scenario	High scenario
Steady-state government commitment (debt service + OPEX shortfall)	€57 M / yr	€54 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
garoua.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
garoua-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
garoua.corridor.geojson	Line polylines + stations (GeoJSON)
garoua.stations.json	Machine-readable station list
garoua.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug garoua

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug garoua \
  --sidecar .cache/osr-pipeline/rasters/garoua.grid.json \
  --out-dir designs/.../Garoua

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../garoua.toml \
  --out designs/.../README.md
```

`scripts/regenerate-garoua.sh` chains steps 3 + drift tests into a single command.

west-africa/Guinea/Conakry

designs/west-africa/Guinea/Conakry/README.md

Conakry — Urban Rail Network

Country: GN · Population: 2,010,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Conakry rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [conakry.corridor.geojson](#).

At a glance

Metric	Value
Lines	3
Unique stations	54
Interchange stations	6
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	40.2%
Route length (double track)	93.2 km
Revenue fleet	69 × 4-car trainsets
Spare + cold-reserve	9 × 4-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	30.1 km	20	25	NE Outer SW Outer
line-2	16.1 km	12	15	SW Inner NE Mid
line-3	46.9 km	23	38	NE Mid NE Mid
Total	93.2 km	54 unique	78	

Rolling stock

Property	Value
Consist	4-car, 75 m
Max speed	90 km/h
Onboard battery	480 kWh per trainset
Seats	80 longitudinal seats
Nominal capacity (AW2)	440 pax (seated + standing, met ro - 4car per RFC 0008 §1)
Crush capacity (AW3)	560 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 440 AW2 passengers (met ro - 4car)
- Peak frequency: 12 trains/hour/direction (5-min headway)

- Peak capacity per line per direction: $440 \times 12 = 5,280$ pphpd
- Network peak throughput (all lines, both directions): $3 \text{ lines} \times 2 \text{ directions} \times 5,280 = 31,680$ passengers/hour
- Daily theoretical capacity (peak $\times$ 10): $\approx 316,800$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 80,802 - 121,203$ trips/day

Catchment

- City population: 2,010,000
- Anchor-weighted coverage: 40.2%
- Catchment population: $\approx 808,020$ (within ~ 800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	6	500 kW	3000 kWh
Major	33	400 kW	2500 kWh
Standard	12	300 kW	2000 kWh
Terminal	3	500 kW	3000 kWh
Total installed	55	26,300 kW	173,500 kWh

Aggregate station-rail charging power: 29,500 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 480 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	16.0 kWh/km	4 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	497 kWh	31.1 km average line length
Onboard battery coverage	1.0 $\times$ average line run	480 kWh usable pack
Average 60 s dwell charge	9.1 kWh/stop	546 kW average charger across stops
Stops to refill one trainset pack	53 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	132 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	174 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in design.toml — emitted by the osr-design Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. country-costs.toml applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (88.1 km @ €3.5 M/km)	€308 M
Elevated (3.5 km @ €18 M/km)	€63 M
Elevated-interchange premium (3 sites @ €20 M)	€60 M
Civil subtotal	€432 M

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~ 11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
standard	12	€1.5 M	€18 M
major	33	€3.0 M	€99 M
terminal	3	€2.5 M	€7.5 M
depot-terminal	1	€3.0 M	€3.0 M
interchange-elevated	6	€4.5 M	€27 M
Stations subtotal			€154 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	3	€3.0 M	€9.0 M
Depots subtotal			€34 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
metro-4car (revenue + spare + cold reserve)	78	€4.0 M	€312 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	93.2 km × €0.015 M/km	€1.4 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€22 M
EPC integration + project management (7%)	on subtotal	€67 M

Total

Bucket	Value
Civil works	€432 M
Stations	€154 M
Depots	€34 M
Rolling stock	€312 M
Residual train-control wayside + charging microgrids	€23 M
EPC overhead (7%)	€67 M
CAPEX total	€1.02 bn
Per-route-km	€11 M / km
Per-capita (city pop)	€508 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running scripts/regenerate-city.sh conakry.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–10 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 11 and runs for 25 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–10)	€59 M / yr	€29
Steady-state, low-ridership (year 11+)	€82 M / yr	€41
Steady-state, high-ridership (year 11+)	€75 M / yr	€37
Lifecycle envelope (yr 1–35, low scenario)	€2.65 bn cumulative	€1,317
Lifecycle envelope (yr 1–35, high scenario)	€2.46 bn cumulative	€1,226

Population basis: 2,010,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 35, debt service drops to zero and only the OPEX shortfall remains – ~€19 M / yr (low) €11 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€613 M	3.0%	35 y, 10 y grace	€35 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€255 M	10.0%	35 y, 10 y grace	€28 M / yr
Government equity (no debt service)	15%	€153 M	—	—	—
Total	100%	€1.02 bn			€63 M / yr

During the 10-year grace period the operator pays interest only – multilateral €18 M / yr + bonds €26 M / yr = €44 M / yr total – plus the equity tranche amortised across construction (€15 M / yr × 10 yr). Principal repayment begins in year 11 on a 25-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€12 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€12 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€69 k
Traction energy (216.8 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (571 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€1.1 M
OPEX subtotal		€26 M / yr

Annual fleet utilisation: 69 revenue trainsets × 20.5 h/day × 365 d/yr × 35 km/h commercial × 75% revenue factor = 13.6 M train-km / yr (~196 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$130 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.20 (~\$0.22 USD)
Day pass (3 trips)	€0.51 (15 % bulk discount)
Monthly unlimited pass	€5.98 (~5 % of median monthly income)
Annual pass	€65.78 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	36.7 M	73.4 M

	Low scenario	High scenario
Farebox revenue	€7.3 M / yr	€15 M / yr
Farebox / OPEX recovery	28%	56%
Country policy-target recovery (diagnostic)	35%	35%
Operating shortfall (gov subsidy required)	€19 M / yr	€11 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€82 M / yr	€75 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
conakry.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
conakry-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
conakry.corridor.geojson	Line polylines + stations (GeoJSON)
conakry.stations.json	Machine-readable station list
conakry.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug conakry

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug conakry \
  --sidecar .cache/osr-pipeline/rasters/conakry.grid.json \
  --out-dir designs/.../Conakry

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../conakry.toml \
  --out designs/.../README.md
```

`scripts/regenerate-conakry.sh` chains steps 3 + drift tests into a single command.

west-africa/Mali/Bamako

designs/west-africa/Mali/Bamako/README.md

Bamako — Urban Rail Network

Country: ML · Population: 2,929,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Bamako rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [bamako.corridor.geojson](#).

At a glance

Metric	Value
Lines	6
Unique stations	117
Interchange stations	18
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	30.6%
Route length (double track)	257.4 km
Revenue fleet	186 × 4-car trainsets
Spare + cold-reserve	21 × 4-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	38.6 km	19	31	NW Outer S Mid
line-2	46.3 km	19	37	E Mid NW Outer
line-3	35.9 km	15	29	SW Mid NE Mid
line-4	30.5 km	14	26	N Mid SE Mid
line-5	25.6 km	14	21	SE Outer W Inner
line-6	80.5 km	37	63	NW Mid NW Mid
Total	257.4 km	117 unique	207	

Rolling stock

Property	Value
Consist	4-car, 75 m
Max speed	90 km/h
Onboard battery	480 kWh per trainset
Seats	80 longitudinal seats
Nominal capacity (AW2)	440 pax (seated + standing, metro-4car per RFC 0008 §1)
Crush capacity (AW3)	560 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 440 AW2 passengers (metro - 4 car)
- Peak frequency: 12 trains/hour/direction (5-min headway)
- Peak capacity per line per direction: $440 \times 12 = 5,280$ pphpd
- Network peak throughput (all lines, both directions): $6 \text{ lines} \times 2 \text{ directions} \times 5,280 = 63,360$ passengers/hour
- Daily theoretical capacity (peak $\times$ 10): $\approx 633,600$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 89,627 - 134,441$ trips/day

Catchment

- City population: 2,929,000
- Anchor-weighted coverage: 30.6%
- Catchment population: $\approx 896,274$ (within ~ 800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	18	500 kW	3000 kWh
Major	26	400 kW	2500 kWh
Standard	61	300 kW	2000 kWh
Terminal	9	500 kW	3000 kWh
Total installed	115	47,200 kW	308,000 kWh

Aggregate station-rail charging power: 63,250 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 480 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	16.0 kWh/km	4 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	686 kWh	42.9 km average line length
Onboard battery coverage	0.7 $\times$ average line run	480 kWh usable pack
Average 60 s dwell charge	9.0 kWh/stop	541 kW average charger across stops
Stops to refill one trainset pack	53 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	236 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	308 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in design.toml — emitted by the osr-design Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. country-costs.toml applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (164.7 km @ €3.5 M/km)	€577 M
Elevated (92.0 km @ €18 M/km)	€1.66 bn
Elevated-interchange premium (6 sites @ €20 M)	€120 M
Civil subtotal	€2.35 bn

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
halt	3	€0.4 M	€1.2 M
standard	61	€1.5 M	€92 M
major	26	€3.0 M	€78 M
terminal	9	€2.5 M	€22 M
depot-terminal	1	€3.0 M	€3.0 M
interchange	8	€4.5 M	€36 M
interchange-elevated	10	€4.5 M	€45 M
Stations subtotal			€277 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	9	€3.0 M	€27 M
Depots subtotal			€52 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
metro-4car (revenue + spare + cold reserve)	207	€4.0 M	€828 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	257.4 km × €0.015 M/km	€3.9 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€41 M
EPC integration + project management (7%)	on subtotal	€249 M

Total

Bucket	Value
Civil works	€2.35 bn
Stations	€277 M
Depots	€52 M
Rolling stock	€828 M
Residual train-control wayside + charging microgrids	€45 M
EPC overhead (7%)	€249 M
CAPEX total	€3.80 bn
Per-route-km	€15 M / km
Per-capita (city pop)	€1,299 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running scripts/regenerate-city.sh bamako.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–10 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 11 and runs for 25 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–10)	€216 M / yr	€74
Steady-state, low-ridership (year 11+)	€312 M / yr	€107
Steady-state, high-ridership (year 11+)	€303 M / yr	€104
Lifecycle envelope (yr 1–35, low scenario)	€9.97 bn cumulative	€3,403
Lifecycle envelope (yr 1–35, high scenario)	€9.74 bn cumulative	€3,326

Population basis: 2,929,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 35, debt service drops to zero and only the OPEX shortfall remains – ~€81 M / yr (low) €72 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€2.28 bn	3.0%	35 y, 10 y grace	€131 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€951 M	9.5%	35 y, 10 y grace	€101 M / yr
Government equity (no debt service)	15%	€571 M	—	—	—
Total	100%	€3.80 bn			€232 M / yr

During the 10-year grace period the operator pays interest only — multilateral €68 M / yr + bonds €90 M / yr = €159 M / yr total — plus the equity tranche amortised across construction (€57 M / yr × 10 yr). Principal repayment begins in year 11 on a 25-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€33 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€54 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€193 k
Traction energy (584.5 GWh / yr)	trackside PV + Na-ion (RFC 0002) — self-generated, €0 / yr	€0 k
Labour (1,556 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€2.6 M
OPEX subtotal		€90 M / yr

Annual fleet utilisation: 186 revenue trainsets × 20.5 h/day × 365 d/yr × 35 km/h commercial × 75% revenue factor = 36.5 M train-km / yr (~196 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$110 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass — a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.17 (~\$0.18 USD)
Day pass (3 trips)	€0.43 (15 % bulk discount)
Monthly unlimited pass	€5.06 (~5 % of median monthly income)
Annual pass	€55.66 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	53.5 M	106.9 M
Farebox revenue	€9.0 M / yr	€18 M / yr
Farebox / OPEX recovery	10%	20%
Country policy-target recovery (diagnostic)	30%	30%
Operating shortfall (gov subsidy required)	€81 M / yr	€72 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€312 M / yr	€303 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
bamako.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
bamako-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
bamako.corridor.geojson	Line polylines + stations (GeoJSON)
bamako.stations.json	Machine-readable station list
bamako.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug bamako

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug bamako \
  --sidecar .cache/osr-pipeline/rasters/bamako.grid.json \
  --out-dir designs/.../Bamako

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../bamako.toml \
  --out designs/.../README.md
```

`scripts/regenerate-bamako.sh` chains steps 3 + drift tests into a single command.

west-africa/Niger/Niamey

designs/west-africa/Niger/Niamey/README.md

Niamey — Urban Rail Network

Country: NE · Population: 1,407,635

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Niamey rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [niamey.corridor.geojson](#).

At a glance

Metric	Value
Lines	4
Unique stations	79
Interchange stations	11
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	38.4%
Route length (double track)	146.0 km
Revenue fleet	107 × 4-car trainsets
Spare + cold-reserve	13 × 4-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	30.9 km	17	26	SE Outer NW Outer
line-2	24.7 km	15	21	N Outer SE Outer
line-3	28.7 km	17	24	SW Outer E Mid
line-4	61.6 km	31	49	NW Outer NW Outer
Total	146.0 km	79 unique	120	

Rolling stock

Property	Value
Consist	4-car, 75 m
Max speed	90 km/h
Onboard battery	480 kWh per trainset
Seats	80 longitudinal seats
Nominal capacity (AW2)	440 pax (seated + standing, met ro - 4 car per RFC 0008 §1)
Crush capacity (AW3)	560 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 440 AW2 passengers (met ro - 4 car)

- Peak frequency: 12 trains/hour/direction (5-min headway)
- Peak capacity per line per direction: $440 \times 12 = 5,280$ pphpd
- Network peak throughput (all lines, both directions): $4 \text{ lines} \times 2 \text{ directions} \times 5,280 = 42,240$ passengers/hour
- Daily theoretical capacity (peak $\times 10$): $\approx 422,400$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 54,053 - 81,079$ trips/day

Catchment

- City population: 1,407,635
- Anchor-weighted coverage: 38.4%
- Catchment population: $\approx 540,531$ (within ~ 800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	11	500 kW	3000 kWh
Major	39	400 kW	2500 kWh
Standard	24	300 kW	2000 kWh
Terminal	5	500 kW	3000 kWh
Total installed	80	35,800 kW	233,500 kWh

Aggregate station-rail charging power: 43,000 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 480 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	16.0 kWh/km	4 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	584 kWh	36.5 km average line length
Onboard battery coverage	0.8 $\times$ average line run	480 kWh usable pack
Average 60 s dwell charge	9.1 kWh/stop	544 kW average charger across stops
Stops to refill one trainset pack	53 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	179 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	234 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the `[costs]` block in `design.toml` — emitted by the `osr-design` Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. `country-costs.toml` applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (139.0 km @ €3.5 M/km)	€487 M
Elevated (6.4 km @ €18 M/km)	€115 M
Elevated-interchange premium (5 sites @ €20 M)	€100 M
Civil subtotal	€702 M

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
standard	24	€1.5 M	€36 M
major	39	€3.0 M	€117 M
terminal	5	€2.5 M	€12 M
depot-terminal	1	€3.0 M	€3.0 M
interchange-elevated	11	€4.5 M	€50 M
Stations subtotal			€218 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	5	€3.0 M	€15 M
Depots subtotal			€40 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
metro-4car (revenue + spare + cold reserve)	120	€4.0 M	€480 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	146.0 km × €0.015 M/km	€2.2 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€31 M
EPC integration + project management (7%)	on subtotal	€103 M

Total

Bucket	Value
Civil works	€702 M
Stations	€218 M
Depots	€40 M
Rolling stock	€480 M
Residual train-control wayside + charging microgrids	€33 M
EPC overhead (7%)	€103 M
CAPEX total	€1.58 bn
Per-route-km	€11 M / km
Per-capita (city pop)	€1,120 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](https://github.com/niamey/lib/templates/country-finance.toml). Pure function of the [costs] block above + the country code – regenerate by re-running `scripts/regenerate-city.sh` niamey.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–10 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 11 and runs for 25 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–10)	€87 M / yr	€62
Steady-state, low-ridership (year 11+)	€131 M / yr	€93
Steady-state, high-ridership (year 11+)	€127 M / yr	€90
Lifecycle envelope (yr 1–35, low scenario)	€4.14 bn cumulative	€2,939
Lifecycle envelope (yr 1–35, high scenario)	€4.04 bn cumulative	€2,873

Population basis: 1,407,635 (catchment per [lib/city-batches/world-sample.toml](https://github.com/niamey/lib/city-batches/world-sample.toml)). After year 35, debt service drops to zero and only the OPEX shortfall remains – ~€36 M / yr (low) €32 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€946 M	3.0%	35 y, 10 y grace	€54 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€394 M	9.0%	35 y, 10 y grace	€40 M / yr
Government equity (no debt service)	15%	€236 M	—	—	—
Total	100%	€1.58 bn			€94 M / yr

During the 10-year grace period the operator pays interest only — multilateral €28 M / yr + bonds €35 M / yr = €64 M / yr total — plus the equity tranche amortised across construction (€24 M / yr × 10 yr). Principal repayment begins in year 11 on a 25-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€19 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€19 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€109 k
Traction energy (336.3 GWh / yr)	trackside PV + Na-ion (RFC 0002) — self-generated, €0 / yr	€0 k
Labour (888 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€1.3 M
OPEX subtotal		€40 M / yr

Annual fleet utilisation: 107 revenue trainsets × 20.5 h/day × 365 d/yr × 35 km/h commercial × 75% revenue factor = 21.0 M train-km / yr (~196 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$95 USD (per [lib/templates/country-finance.toml](https://github.com/niamey/lib/templates/country-finance.toml)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass — a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.15 (~\$0.16 USD)
Day pass (3 trips)	€0.37 (15 % bulk discount)
Monthly unlimited pass	€4.37 (~5 % of median monthly income)
Annual pass	€48.07 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	25.7 M	51.4 M
Farebox revenue	€3.7 M / yr	€7.5 M / yr
Farebox / OPEX recovery	9%	19%
Country policy-target recovery (diagnostic)	30%	30%
Operating shortfall (gov subsidy required)	€36 M / yr	€32 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€131 M / yr	€127 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
niamey.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
niamey-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
niamey.corridor.geojson	Line polylines + stations (GeoJSON)
niamey.stations.json	Machine-readable station list
niamey.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug niamey

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug niamey \
  --sidecar .cache/osr-pipeline/rasters/niamey.grid.json \
  --out-dir designs/.../Niamey

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../niamey.toml \
  --out designs/.../README.md
```

`scripts/regenerate-niamey.sh` chains steps 3 + drift tests into a single command.

west-africa/Nigeria/Aba-Ng

designs/west-africa/Nigeria/Aba-Ng/README.md

Aba-Ng — Urban Rail Network

Country: NG · Population: 900,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Aba-Ng rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [aba-ng.corridor.geojson](#).

At a glance

Metric	Value
Lines	3
Unique stations	26
Interchange stations	3
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	39.7%
Route length (double track)	34.0 km
Revenue fleet	32 × 3-car trainsets
Spare + cold-reserve	6 × 3-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	14.4 km	10	15	W Outer E Outer
line-2	11.8 km	10	13	NE Mid SW Outer
line-3	7.8 km	6	10	NW Mid E Mid
Total	34.0 km	26 unique	38	

Rolling stock

Property	Value
Consist	3-car, 57 m
Max speed	90 km/h
Onboard battery	360 kWh per trainset
Seats	60 longitudinal seats
Nominal capacity (AW2)	330 pax (seated + standing, light-metro-3car per RFC 0008 §1)
Crush capacity (AW3)	420 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 330 AW2 passengers (light-metro-3car)
- Peak frequency: 12 trains/hour/direction (5-min headway)

- Peak capacity per line per direction: $330 \times 12 = 3,960$ pphpd
- Network peak throughput (all lines, both directions): $3 \text{ lines} \times 2 \text{ directions} \times 3,960 = 23,760$ passengers/hour
- Daily theoretical capacity (peak $\times 10$): $\approx 237,600$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 35,730 - 53,595$ trips/day

Catchment

- City population: 900,000
- Anchor-weighted coverage: 39.7%
- Catchment population: $\approx 357,300$ (within ~ 800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	3	500 kW	3000 kWh
Major	13	400 kW	2500 kWh
Standard	4	300 kW	2000 kWh
Terminal	5	500 kW	3000 kWh
Total installed	26	15,400 kW	104,500 kWh

Aggregate station-rail charging power: 16,000 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 360 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	12.0 kWh/km	3 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	136 kWh	11.3 km average line length
Onboard battery coverage	2.6 $\times$ average line run	360 kWh usable pack
Average 60 s dwell charge	10.3 kWh/stop	615 kW average charger across stops
Stops to refill one trainset pack	35 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	77 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	104 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in design.toml — emitted by the osr-design Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. country-costs.toml applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (31.2 km @ €3.5 M/km)	€109 M
Elevated (2.6 km @ €18 M/km)	€47 M
Elevated-interchange premium (2 sites @ €20 M)	€40 M
Civil subtotal	€196 M

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~ 11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
standard	4	€1.5 M	€6.0 M
major	13	€3.0 M	€39 M
terminal	5	€2.5 M	€12 M
depot-terminal	1	€3.0 M	€3.0 M
interchange-elevated	3	€4.5 M	€14 M
Stations subtotal			€74 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	5	€3.0 M	€15 M
Depots subtotal			€40 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
light-metro-3car (revenue + spare + cold reserve)	38	€3.0 M	€114 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	34.0 km × €0.015 M/km	€0.5 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€11 M
EPC integration + project management (7%)	on subtotal	€30 M

Total

Bucket	Value
Civil works	€196 M
Stations	€74 M
Depots	€40 M
Rolling stock	€114 M
Residual train-control wayside + charging microgrids	€11 M
EPC overhead (7%)	€30 M
CAPEX total	€466 M
Per-route-km	€14 M / km
Per-capita (city pop)	€518 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running scripts/regenerate-city.sh aba-ng.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–7 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 8 and runs for 23 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–7)	€38 M / yr	€43
Steady-state, low-ridership (year 8+)	€43 M / yr	€48
Steady-state, high-ridership (year 8+)	€39 M / yr	€43
Lifecycle envelope (yr 1–30, low scenario)	€1.27 bn cumulative	€1,406
Lifecycle envelope (yr 1–30, high scenario)	€1.16 bn cumulative	€1,294

Population basis: 900,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 30, debt service drops to zero and only the OPEX shortfall remains – ~€7.0 M / yr (low) €2.6 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€280 M	4.5%	30 y, 7 y grace	€20 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€117 M	13.5%	30 y, 7 y grace	€17 M / yr
Government equity (no debt service)	15%	€70 M	—	—	—
Total	100%	€466 M			€36 M / yr

During the 7-year grace period the operator pays interest only – multilateral €13 M / yr + bonds €16 M / yr = €28 M / yr total – plus the equity tranche amortised across construction (€10.0 M / yr × 7 yr). Principal repayment begins in year 8 on a 23-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€4.6 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€6.2 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€25 k
Traction energy (64.6 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (216 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€584 k
OPEX subtotal		€11 M / yr

Annual fleet utilisation: 32 revenue trainsets × 20.5 h/day × 365 d/yr × 30 km/h commercial × 75% revenue factor = 5.4 M train-km / yr (~168 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$175 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.27 (~\$0.29 USD)
Day pass (3 trips)	€0.68 (15 % bulk discount)
Monthly unlimited pass	€8.05 (~5 % of median monthly income)
Annual pass	€88.55 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	16.4 M	32.9 M

	Low scenario	High scenario
Farebox revenue	€4.4 M / yr	€8.8 M / yr
Farebox / OPEX recovery	39%	77%
Country policy-target recovery (diagnostic)	45%	45%
Operating shortfall (gov subsidy required)	€7.0 M / yr	€2.6 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€43 M / yr	€39 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
aba-ng.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
aba-ng-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
aba-ng.corridor.geojson	Line polylines + stations (GeoJSON)
aba-ng.stations.json	Machine-readable station list
aba-ng.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug aba-ng

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug aba-ng \
  --sidecar .cache/osr-pipeline/rasters/aba-ng.grid.json \
  --out-dir designs/.../Aba-Ng

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../aba-ng.toml \
  --out designs/.../README.md
```

`scripts/regenerate-aba-ng.sh` chains steps 3 + drift tests into a single command.

west-africa/Nigeria/Benin-City

designs/west-africa/Nigeria/Benin-City/README.md

Benin-City — Urban Rail Network

Country: NG · Population: 1,800,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Benin-City rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [benin-city.corridor.geojson](#).

At a glance

Metric	Value
Lines	4
Unique stations	76
Interchange stations	24
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	44.3%
Route length (double track)	132.3 km
Revenue fleet	98 $\times$ 4-car trainsets
Spare + cold-reserve	13 $\times$ 4-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 ($\approx$ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	30.5 km	16	26	N Outer S Outer
line-2	30.0 km	17	25	NE Outer SW Outer
line-3	26.1 km	14	23	NW Mid SE Outer
line-4	45.7 km	30	37	W Inner W Inner
Total	132.3 km	76 unique	111	

Rolling stock

Property	Value
Consist	4-car, 75 m
Max speed	90 km/h
Onboard battery	480 kWh per trainset
Seats	80 longitudinal seats
Nominal capacity (AW2)	440 pax (seated + standing, met ro - 4car per RFC 0008 §1)
Crush capacity (AW3)	560 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 440 AW2 passengers (met ro - 4car)

- Peak frequency: 12 trains/hour/direction (5-min headway)
- Peak capacity per line per direction: $440 \times 12 = 5,280$ pphpd
- Network peak throughput (all lines, both directions): $4 \text{ lines} \times 2 \text{ directions} \times 5,280 = 42,240$ passengers/hour
- Daily theoretical capacity (peak $\times$ 10): $\approx 422,400$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 79,740 - 119,610$ trips/day

Catchment

- City population: 1,800,000
- Anchor-weighted coverage: 44.3%
- Catchment population: $\approx 797,400$ (within ~ 800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	24	500 kW	3000 kWh
Major	20	400 kW	2500 kWh
Standard	27	300 kW	2000 kWh
Terminal	5	500 kW	3000 kWh
Total installed	77	35,600 kW	231,000 kWh

Aggregate station-rail charging power: 41,500 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 480 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	16.0 kWh/km	4 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	529 kWh	33.1 km average line length
Onboard battery coverage	0.9 $\times$ average line run	480 kWh usable pack
Average 60 s dwell charge	9.1 kWh/stop	546 kW average charger across stops
Stops to refill one trainset pack	53 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	178 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	231 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in design.toml — emitted by the osr-design Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. country-costs.toml applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (114.6 km @ €3.5 M/km)	€401 M
Elevated (16.8 km @ €18 M/km)	€302 M
Elevated-interchange premium (10 sites @ €20 M)	€200 M
Civil subtotal	€903 M

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
standard	27	€1.5 M	€40 M
major	20	€3.0 M	€60 M
terminal	5	€2.5 M	€12 M
depot-terminal	1	€3.0 M	€3.0 M
interchange-elevated	24	€4.5 M	€108 M
Stations subtotal			€224 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	5	€3.0 M	€15 M
Depots subtotal			€40 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
metro-4car (revenue + spare + cold reserve)	111	€4.0 M	€444 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	132.3 km × €0.015 M/km	€2.0 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€32 M
EPC integration + project management (7%)	on subtotal	€115 M

Total

Bucket	Value
Civil works	€903 M
Stations	€224 M
Depots	€40 M
Rolling stock	€444 M
Residual train-control wayside + charging microgrids	€34 M
EPC overhead (7%)	€115 M
CAPEX total	€1.76 bn
Per-route-km	€13 M / km
Per-capita (city pop)	€978 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running scripts/regenerate-city.sh benin-city.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–7 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 8 and runs for 23 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–7)	€145 M / yr	€80
Steady-state, low-ridership (year 8+)	€172 M / yr	€96
Steady-state, high-ridership (year 8+)	€163 M / yr	€91
Lifecycle envelope (yr 1–30, low scenario)	€4.97 bn cumulative	€2,761
Lifecycle envelope (yr 1–30, high scenario)	€4.77 bn cumulative	€2,648

Population basis: 1,800,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 30, debt service drops to zero and only the OPEX shortfall remains – ~€35 M / yr (low) €26 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€1.06 bn	4.5%	30 y, 7 y grace	€75 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€440 M	13.5%	30 y, 7 y grace	€63 M / yr
Government equity (no debt service)	15%	€264 M	—	—	—
Total	100%	€1.76 bn			€137 M / yr

During the 7-year grace period the operator pays interest only – multilateral €48 M / yr + bonds €59 M / yr = €107 M / yr total – plus the equity tranche amortised across construction (€38 M / yr × 7 yr). Principal repayment begins in year 8 on a 23-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€18 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€23 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€99 k
Traction energy (308.0 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (806 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€2.2 M
OPEX subtotal		€43 M / yr

Annual fleet utilisation: 98 revenue trainsets × 20.5 h/day × 365 d/yr × 35 km/h commercial × 75% revenue factor = 19.2 M train-km / yr (~196 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$175 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.27 (~\$0.29 USD)
Day pass (3 trips)	€0.68 (15 % bulk discount)
Monthly unlimited pass	€8.05 (~5 % of median monthly income)
Annual pass	€88.55 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	32.9 M	65.7 M
Farebox revenue	€8.8 M / yr	€18 M / yr
Farebox / OPEX recovery	20%	41%
Country policy-target recovery (diagnostic)	45%	45%
Operating shortfall (gov subsidy required)	€35 M / yr	€26 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€172 M / yr	€163 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
benin-city.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
benin-city-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
benin-city.corridor.geojson	Line polylines + stations (GeoJSON)
benin-city.stations.json	Machine-readable station list
benin-city.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug benin-city

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug benin-city \
  --sidecar .cache/osr-pipeline/rasters/benin-city.grid.json \
  --out-dir designs/.../Benin-City

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../benin-city.toml \
  --out designs/.../README.md
```

`scripts/regenerate-benin-city.sh` chains steps 3 + drift tests into a single command.

west-africa/Nigeria/Ibadan

designs/west-africa/Nigeria/Ibadan/README.md

Ibadan — Urban Rail Network

Country: NG · Population: 3,900,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Ibadan rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [ibadan.corridor.geojson](#).

At a glance

Metric	Value
Lines	4
Unique stations	89
Interchange stations	14
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	23.9%
Route length (double track)	135.5 km
Revenue fleet	99 × 6-car trainsets
Spare + cold-reserve	12 × 6-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	36.0 km	20	29	E Mid W Outer
line-2	32.8 km	20	27	S Mid N Outer
line-3	23.8 km	16	20	NW Mid SE Mid
line-4	42.9 km	34	35	W Inner W Inner
Total	135.5 km	89 unique	111	

Rolling stock

Property	Value
Consist	6-car, 111 m
Max speed	100 km/h
Onboard battery	720 kWh per trainset
Seats	120 longitudinal seats
Nominal capacity (AW2)	660 pax (seated + standing, met ro-6car per RFC 0008 §1)
Crush capacity (AW3)	840 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 660 AW2 passengers (met ro-6car)

- Peak frequency: 12 trains/hour/direction (5-min headway)
- Peak capacity per line per direction: $660 \times 12 = 7,920$ pphpd
- Network peak throughput (all lines, both directions): $4 \text{ lines} \times 2 \text{ directions} \times 7,920 = 63,360$ passengers/hour
- Daily theoretical capacity (peak $\times$ 10): $\approx 633,600$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 93,210 - 139,815$ trips/day

Catchment

- City population: 3,900,000
- Anchor-weighted coverage: 23.9%
- Catchment population: $\approx 932,100$ (within ~ 800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	14	500 kW	3000 kWh
Major	58	400 kW	2500 kWh
Standard	12	300 kW	2000 kWh
Terminal	5	500 kW	3000 kWh
Total installed	90	41,300 kW	266,000 kWh

Aggregate station-rail charging power: 48,000 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 720 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	24.0 kWh/km	6 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	813 kWh	33.9 km average line length
Onboard battery coverage	0.9 $\times$ average line run	720 kWh usable pack
Average 60 s dwell charge	9.0 kWh/stop	539 kW average charger across stops
Stops to refill one trainset pack	80 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	206 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	266 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in design.toml — emitted by the osr-design Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. country-costs.toml applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (119.3 km @ €3.5 M/km)	€418 M
Elevated (12.3 km @ €18 M/km)	€222 M
Elevated-interchange premium (9 sites @ €20 M)	€180 M
Civil subtotal	€820 M

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
standard	12	€1.5 M	€18 M
major	58	€3.0 M	€174 M
terminal	5	€2.5 M	€12 M
depot-terminal	1	€3.0 M	€3.0 M
interchange-elevated	14	€4.5 M	€63 M
Stations subtotal			€270 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	5	€3.0 M	€15 M
Depots subtotal			€40 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
metro-6car (revenue + spare + cold reserve)	111	€6.0 M	€666 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	135.5 km × €0.015 M/km	€2.0 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€37 M
EPC integration + project management (7%)	on subtotal	€128 M

Total

Bucket	Value
Civil works	€820 M
Stations	€270 M
Depots	€40 M
Rolling stock	€666 M
Residual train-control wayside + charging microgrids	€39 M
EPC overhead (7%)	€128 M
CAPEX total	€1.96 bn
Per-route-km	€14 M / km
Per-capita (city pop)	€504 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running `scripts/regenerate-city.sh ibadan`.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–7 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 8 and runs for 23 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–7)	€161 M / yr	€41
Steady-state, low-ridership (year 8+)	€186 M / yr	€48
Steady-state, high-ridership (year 8+)	€167 M / yr	€43
Lifecycle envelope (yr 1–30, low scenario)	€5.40 bn cumulative	€1,386
Lifecycle envelope (yr 1–30, high scenario)	€4.97 bn cumulative	€1,273

Population basis: 3,900,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 30, debt service drops to zero and only the OPEX shortfall remains – ~€32 M / yr (low) €13 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€1.18 bn	4.5%	30 y, 7 y grace	€83 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€491 M	13.5%	30 y, 7 y grace	€70 M / yr
Government equity (no debt service)	15%	€295 M	—	—	—
Total	100%	€1.96 bn			€153 M / yr

During the 7-year grace period the operator pays interest only – multilateral €53 M / yr + bonds €66 M / yr = €119 M / yr total – plus the equity tranche amortised across construction (€42 M / yr × 7 yr). Principal repayment begins in year 8 on a 23-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€27 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€23 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€99 k
Traction energy (466.7 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (825 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€2.2 M
OPEX subtotal		€52 M / yr

Annual fleet utilisation: 99 revenue trainsets × 20.5 h/day × 365 d/yr × 35 km/h commercial × 75% revenue factor = 19.4 M train-km / yr (~196 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$175 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.27 (~\$0.29 USD)
Day pass (3 trips)	€0.68 (15 % bulk discount)
Monthly unlimited pass	€8.05 (~5 % of median monthly income)
Annual pass	€88.55 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	71.2 M	142.3 M
Farebox revenue	€19 M / yr	€38 M / yr
Farebox / OPEX recovery	37%	74%
Country policy-target recovery (diagnostic)	45%	45%
Operating shortfall (gov subsidy required)	€32 M / yr	€13 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€186 M / yr	€167 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
ibadan.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
ibadan-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
ibadan.corridor.geojson	Line polylines + stations (GeoJSON)
ibadan.stations.json	Machine-readable station list
ibadan.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug ibadan

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug ibadan \
  --sidecar .cache/osr-pipeline/rasters/ibadan.grid.json \
  --out-dir designs/.../Ibadan

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../ibadan.toml \
  --out designs/.../README.md
```

`scripts/regenerate-ibadan.sh` chains steps 3 + drift tests into a single command.

west-africa/Nigeria/Ilorin

designs/west-africa/Nigeria/Ilorin/README.md

Ilorin — Urban Rail Network

Country: NG · Population: 1,000,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Ilorin rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [ilorin.corridor.geojson](#).

At a glance

Metric	Value
Lines	3
Unique stations	39
Interchange stations	2
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	31.2%
Route length (double track)	60.3 km
Revenue fleet	53 × 3-car trainsets
Spare + cold-reserve	7 × 3-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	24.4 km	16	24	SW Outer NE Outer
line-2	18.3 km	10	18	W Outer SE Outer
line-3	17.6 km	13	18	SE Mid NW Outer
Total	60.3 km	39 unique	60	

Rolling stock

Property	Value
Consist	3-car, 57 m
Max speed	90 km/h
Onboard battery	360 kWh per trainset
Seats	60 longitudinal seats
Nominal capacity (AW2)	330 pax (seated + standing, light-metro-3car per RFC 0008 §1)
Crush capacity (AW3)	420 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 330 AW2 passengers (light-metro-3car)
- Peak frequency: 12 trains/hour/direction (5-min headway)

- Peak capacity per line per direction: $330 \times 12 = 3,960$ pphpd
- Network peak throughput (all lines, both directions): $3 \text{ lines} \times 2 \text{ directions} \times 3,960 = 23,760$ passengers/hour
- Daily theoretical capacity (peak $\times 10$): $\approx 237,600$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 31,200 - 46,800$ trips/day

Catchment

- City population: 1,000,000
- Anchor-weighted coverage: 31.2%
- Catchment population: $\approx 312,000$ (within ~ 800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	2	500 kW	3000 kWh
Major	16	400 kW	2500 kWh
Standard	15	300 kW	2000 kWh
Terminal	5	500 kW	3000 kWh
Total installed	39	19,400 kW	131,000 kWh

Aggregate station-rail charging power: 22,500 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 360 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	12.0 kWh/km	3 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	241 kWh	20.1 km average line length
Onboard battery coverage	1.5 $\times$ average line run	360 kWh usable pack
Average 60 s dwell charge	9.6 kWh/stop	577 kW average charger across stops
Stops to refill one trainset pack	37 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	97 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	131 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in design.toml — emitted by the osr-design Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. country-costs.toml applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (58.0 km @ €3.5 M/km)	€203 M
Elevated (1.4 km @ €18 M/km)	€25 M
Elevated-interchange premium (1 sites @ €20 M)	€20 M
Civil subtotal	€248 M

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~ 11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
standard	15	€1.5 M	€22 M
major	16	€3.0 M	€48 M
terminal	5	€2.5 M	€12 M
depot-terminal	1	€3.0 M	€3.0 M
interchange	2	€4.5 M	€9.0 M
Stations subtotal			€95 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	5	€3.0 M	€15 M
Depots subtotal			€40 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
light-metro-3car (revenue + spare + cold reserve)	60	€3.0 M	€180 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	60.3 km × €0.015 M/km	€0.9 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€14 M
EPC integration + project management (7%)	on subtotal	€40 M

Total

Bucket	Value
Civil works	€248 M
Stations	€95 M
Depots	€40 M
Rolling stock	€180 M
Residual train-control wayside + charging microgrids	€15 M
EPC overhead (7%)	€40 M
CAPEX total	€618 M
Per-route-km	€10 M / km
Per-capita (city pop)	€618 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running scripts/regenerate-city.sh ilorin.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–7 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 8 and runs for 23 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–7)	€51 M / yr	€51
Steady-state, low-ridership (year 8+)	€59 M / yr	€59
Steady-state, high-ridership (year 8+)	€54 M / yr	€54
Lifecycle envelope (yr 1–30, low scenario)	€1.72 bn cumulative	€1,719
Lifecycle envelope (yr 1–30, high scenario)	€1.61 bn cumulative	€1,607

Population basis: 1,000,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 30, debt service drops to zero and only the OPEX shortfall remains – ~€11 M / yr (low) €6.1 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€371 M	4.5%	30 y, 7 y grace	€26 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€155 M	13.5%	30 y, 7 y grace	€22 M / yr
Government equity (no debt service)	15%	€93 M	—	—	—
Total	100%	€618 M			€48 M / yr

During the 7-year grace period the operator pays interest only – multilateral €17 M / yr + bonds €21 M / yr = €38 M / yr total – plus the equity tranche amortised across construction (€13 M / yr × 7 yr). Principal repayment begins in year 8 on a 23-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€7.2 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€7.7 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€45 k
Traction energy (107.1 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (374 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€1.0 M
OPEX subtotal		€16 M / yr

Annual fleet utilisation: 53 revenue trainsets × 20.5 h/day × 365 d/yr × 30 km/h commercial × 75% revenue factor = 8.9 M train-km / yr (~168 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$175 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.27 (~\$0.29 USD)
Day pass (3 trips)	€0.68 (15 % bulk discount)
Monthly unlimited pass	€8.05 (~5 % of median monthly income)
Annual pass	€88.55 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	18.2 M	36.5 M

	Low scenario	High scenario
Farebox revenue	€4.9 M / yr	€9.8 M / yr
Farebox / OPEX recovery	31%	62%
Country policy-target recovery (diagnostic)	45%	45%
Operating shortfall (gov subsidy required)	€11 M / yr	€6.1 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€59 M / yr	€54 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
ilorin.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
ilorin-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
ilorin.corridor.geojson	Line polylines + stations (GeoJSON)
ilorin.stations.json	Machine-readable station list
ilorin.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug ilorin

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug ilorin \
  --sidecar .cache/osr-pipeline/rasters/ilorin.grid.json \
  --out-dir designs/.../Ilorin

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../ilorin.toml \
  --out designs/.../README.md
```

`scripts/regenerate-ilorin.sh` chains steps 3 + drift tests into a single command.

west-africa/Nigeria/Jos

designs/west-africa/Nigeria/Jos/README.md

Jos — Urban Rail Network

Country: NG · Population: 900,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Jos rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [jos.corridor.geojson](#).

At a glance

Metric	Value
Lines	3
Unique stations	40
Interchange stations	0
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	25.7%
Route length (double track)	53.9 km
Revenue fleet	48 × 3-car trainsets
Spare + cold-reserve	7 × 3-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	22.5 km	18	23	N Outer S Outer
line-2	18.5 km	13	18	SE Outer NW Outer
line-3	12.9 km	9	14	E Outer N Mid
Total	53.9 km	40 unique	55	

Rolling stock

Property	Value
Consist	3-car, 57 m
Max speed	90 km/h
Onboard battery	360 kWh per trainset
Seats	60 longitudinal seats
Nominal capacity (AW2)	330 pax (seated + standing, light-metro-3car per RFC 0008 §1)
Crush capacity (AW3)	420 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 330 AW2 passengers (light-metro-3car)
- Peak frequency: 12 trains/hour/direction (5-min headway)

- Peak capacity per line per direction: $330 \times 12 = 3,960$ pphpd
- Network peak throughput (all lines, both directions): $3 \text{ lines} \times 2 \text{ directions} \times 3,960 = 23,760$ passengers/hour
- Daily theoretical capacity (peak $\times 10$): $\approx 237,600$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 23,130 - 34,695$ trips/day

Catchment

- City population: 900,000
- Anchor-weighted coverage: 25.7%
- Catchment population: $\approx 231,300$ (within ~ 800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../.../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Major	25	400 kW	2500 kWh
Standard	9	300 kW	2000 kWh
Terminal	5	500 kW	3000 kWh
Total installed	40	20,200 kW	135,500 kWh

Aggregate station-rail charging power: 23,000 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 360 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	12.0 kWh/km	3 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	216 kWh	18.0 km average line length
Onboard battery coverage	1.7 $\times$ average line run	360 kWh usable pack
Average 60 s dwell charge	9.6 kWh/stop	575 kW average charger across stops
Stops to refill one trainset pack	38 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	101 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	136 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in design.toml — emitted by the osr-design Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. country-costs.toml applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (50.0 km @ €3.5 M/km)	€175 M
Elevated (3.4 km @ €18 M/km)	€60 M
Elevated-interchange premium (1 sites @ €20 M)	€20 M
Civil subtotal	€255 M

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~ 11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
standard	9	€1.5 M	€14 M
major	25	€3.0 M	€75 M
terminal	5	€2.5 M	€12 M
depot-terminal	1	€3.0 M	€3.0 M
Stations subtotal			€104 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	5	€3.0 M	€15 M
Depots subtotal			€40 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
light-metro-3car (revenue + spare + cold reserve)	55	€3.0 M	€165 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	53.9 km × €0.015 M/km	€0.8 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€15 M
EPC integration + project management (7%)	on subtotal	€41 M

Total

Bucket	Value
Civil works	€255 M
Stations	€104 M
Depots	€40 M
Rolling stock	€165 M
Residual train-control wayside + charging microgrids	€16 M
EPC overhead (7%)	€41 M
CAPEX total	€621 M
Per-route-km	€12 M / km
Per-capita (city pop)	€690 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code — regenerate by

re-running scripts/regenerate-city.sh jos.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–7 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 8 and runs for 23 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–7)	€51 M / yr	€57
Steady-state, low-ridership (year 8+)	€60 M / yr	€66
Steady-state, high-ridership (year 8+)	€55 M / yr	€61
Lifecycle envelope (yr 1–30, low scenario)	€1.73 bn cumulative	€1,920
Lifecycle envelope (yr 1–30, high scenario)	€1.63 bn cumulative	€1,807

Population basis: 900,000 (catchment per `lib/city-batches/world-sample.toml`). After year 30, debt service drops to zero and only the OPEX shortfall remains — ~€11 M / yr (low) €6.7 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€372 M	4.5%	30 y, 7 y grace	€26 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€155 M	13.5%	30 y, 7 y grace	€22 M / yr
Government equity (no debt service)	15%	€93 M	—	—	—
Total	100%	€621 M			€48 M / yr

During the 7-year grace period the operator pays interest only — multilateral €17 M / yr + bonds €21 M / yr = €38 M / yr total — plus the equity tranche amortised across construction (€13 M / yr × 7 yr). Principal repayment begins in year 8 on a 23-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€6.6 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€8.0 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€40 k
Traction energy (97.0 GWh / yr)	trackside PV + Na-ion (RFC 0002) — self-generated, €0 / yr	€0 k
Labour (335 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€906 k
OPEX subtotal		€16 M / yr

Annual fleet utilisation: 48 revenue trainsets × 20.5 h/day × 365 d/yr × 30 km/h commercial × 75% revenue factor = 8.1 M train-km / yr (~168 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$175 USD (per `lib/templates/country-finance.toml`). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass — a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.27 (~\$0.29 USD)
Day pass (3 trips)	€0.68 (15 % bulk discount)
Monthly unlimited pass	€8.05 (~5 % of median monthly income)
Annual pass	€88.55 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	16.4 M	32.9 M
Farebox revenue	€4.4 M / yr	€8.8 M / yr
Farebox / OPEX recovery	28%	57%

	Low scenario	High scenario
Country policy-target recovery (diagnostic)	45%	45%
Operating shortfall (gov subsidy required)	€11 M / yr	€6.7 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€60 M / yr	€55 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
jos.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
jos-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
jos.corridor.geojson	Line polylines + stations (GeoJSON)
jos.stations.json	Machine-readable station list
jos.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug jos

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug jos \
  --sidecar .cache/osr-pipeline/rasters/jos.grid.json \
  --out-dir designs/.../Jos

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../jos.toml \
  --out designs/.../README.md
```

`scripts/regenerate-jos.sh` chains steps 3 + drift tests into a single command.

west-africa/Nigeria/Kano

designs/west-africa/Nigeria/Kano/README.md

Kano — Urban Rail Network

Country: NG · Population: 4,200,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Kano rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [kano.corridor.geojson](#).

At a glance

Metric	Value
Lines	6
Unique stations	153
Interchange stations	17
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	36.9%
Route length (double track)	361.9 km
Revenue fleet	258 × 6-car trainsets
Spare + cold-reserve	28 × 6-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	54.6 km	23	43	S Outer NE Outer
line-2	50.9 km	21	41	NW Outer SE Outer
line-3	49.4 km	22	40	SW Mid NE Outer
line-4	41.9 km	20	34	NE Mid W Outer
line-5	52.1 km	23	41	NW Outer SE Outer
line-6	113.0 km	45	87	NW Outer NW Outer
Total	361.9 km	153 unique	286	

Rolling stock

Property	Value
Consist	6-car, 111 m
Max speed	100 km/h
Onboard battery	720 kWh per trainset
Seats	120 longitudinal seats
Nominal capacity (AW2)	660 pax (seated + standing, metro-6car per RFC 0008 §1)
Crush capacity (AW3)	840 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 660 AW2 passengers (metro-6car)
- Peak frequency: 12 trains/hour/direction (5-min headway)
- Peak capacity per line per direction: $660 \times 12 = 7,920$ pphpd
- Network peak throughput (all lines, both directions): $6 \text{ lines} \times 2 \text{ directions} \times 7,920 = 95,040$ passengers/hour
- Daily theoretical capacity (peak $\times$ 10): $\approx 950,400$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 154,980 - 232,470$ trips/day

Catchment

- City population: 4,200,000
- Anchor-weighted coverage: 36.9%
- Catchment population: $\approx 1,549,800$ (within ~800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	17	500 kW	3000 kWh
Major	31	400 kW	2500 kWh
Standard	92	300 kW	2000 kWh
Terminal	9	500 kW	3000 kWh
Total installed	150	58,000 kW	379,500 kWh

Aggregate station-rail charging power: 81,000 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 720 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	24.0 kWh/km	6 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	1,448 kWh	60.3 km average line length
Onboard battery coverage	0.5 $\times$ average line run	720 kWh usable pack
Average 60 s dwell charge	8.8 kWh/stop	529 kW average charger across stops
Stops to refill one trainset pack	82 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	290 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	380 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in design.toml — emitted by the osr-design Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. country-costs.toml applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (297.1 km @ €3.5 M/km)	€1.04 bn
Elevated (63.8 km @ €18 M/km)	€1.15 bn
Elevated-interchange premium (7 sites @ €20 M)	€140 M
Civil subtotal	€2.33 bn

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
halt	4	€0.4 M	€1.6 M
standard	92	€1.5 M	€138 M
major	31	€3.0 M	€93 M
terminal	9	€2.5 M	€22 M
depot-terminal	1	€3.0 M	€3.0 M
interchange	2	€4.5 M	€9.0 M
interchange-elevated	15	€4.5 M	€68 M
Stations subtotal			€335 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	9	€3.0 M	€27 M
Depots subtotal			€52 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
metro-6car (revenue + spare + cold reserve)	286	€6.0 M	€1.72 bn

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	361.9 km × €0.015 M/km	€5.4 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€50 M
EPC integration + project management (7%)	on subtotal	€314 M

Total

Bucket	Value
Civil works	€2.33 bn
Stations	€335 M
Depots	€52 M
Rolling stock	€1.72 bn
Residual train-control wayside + charging microgrids	€56 M
EPC overhead (7%)	€314 M
CAPEX total	€4.80 bn
Per-route-km	€13 M / km
Per-capita (city pop)	€1,143 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running `scripts/regenerate-city.sh kano`.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–7 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 8 and runs for 23 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–7)	€394 M / yr	€94
Steady-state, low-ridership (year 8+)	€483 M / yr	€115
Steady-state, high-ridership (year 8+)	€463 M / yr	€110
Lifecycle envelope (yr 1–30, low scenario)	€13.88 bn cumulative	€3,304
Lifecycle envelope (yr 1–30, high scenario)	€13.41 bn cumulative	€3,192

Population basis: 4,200,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 30, debt service drops to zero and only the OPEX shortfall remains – ~€109 M / yr (low) €88 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€2.88 bn	4.5%	30 y, 7 y grace	€204 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€1.20 bn	13.5%	30 y, 7 y grace	€171 M / yr
Government equity (no debt service)	15%	€720 M	—	—	—
Total	100%	€4.80 bn			€375 M / yr

During the 7-year grace period the operator pays interest only – multilateral €130 M / yr + bonds €162 M / yr = €292 M / yr total – plus the equity tranche amortised across construction (€103 M / yr × 7 yr). Principal repayment begins in year 8 on a 23-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€69 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€54 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€271 k
Traction energy (1216.2 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (2,183 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€5.9 M
OPEX subtotal		€129 M / yr

Annual fleet utilisation: 258 revenue trainsets × 20.5 h/day × 365 d/yr × 35 km/h commercial × 75% revenue factor = 50.7 M train-km / yr (~196 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$175 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.27 (~\$0.29 USD)
Day pass (3 trips)	€0.68 (15 % bulk discount)
Monthly unlimited pass	€8.05 (~5 % of median monthly income)
Annual pass	€88.55 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	76.7 M	153.3 M
Farebox revenue	€21 M / yr	€41 M / yr
Farebox / OPEX recovery	16%	32%
Country policy-target recovery (diagnostic)	45%	45%
Operating shortfall (gov subsidy required)	€109 M / yr	€88 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€483 M / yr	€463 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
kano.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
kano-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
kano.corridor.geojson	Line polylines + stations (GeoJSON)
kano.stations.json	Machine-readable station list
kano.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug kano

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug kano \
  --sidecar .cache/osr-pipeline/rasters/kano.grid.json \
  --out-dir designs/.../Kano

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../kano.toml \
  --out designs/.../README.md
```

`scripts/regenerate-kano.sh` chains steps 3 + drift tests into a single command.

west-africa/Nigeria/Maiduguri

designs/west-africa/Nigeria/Maiduguri/README.md

Maiduguri — Urban Rail Network

Country: NG · Population: 1,200,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Maiduguri rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [maiduguri.corridor.geojson](#).

At a glance

Metric	Value
Lines	5
Unique stations	85
Interchange stations	17
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	34.4%
Route length (double track)	176.1 km
Revenue fleet	129 × 4-car trainsets
Spare + cold-reserve	16 × 4-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	31.2 km	16	26	SW Outer NE Outer
line-2	28.5 km	17	24	SE Outer W Mid
line-3	20.9 km	10	18	NW Outer NE Mid
line-4	29.0 km	12	25	S Outer N Mid
line-5	66.5 km	31	52	W Mid W Mid
Total	176.1 km	85 unique	145	

Rolling stock

Property	Value
Consist	4-car, 75 m
Max speed	90 km/h
Onboard battery	480 kWh per trainset
Seats	80 longitudinal seats
Nominal capacity (AW2)	440 pax (seated + standing, met ro - 4 car per RFC 0008 §1)
Crush capacity (AW3)	560 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 440 AW2 passengers (met ro - 4 car)

- Peak frequency: 12 trains/hour/direction (5-min headway)
- Peak capacity per line per direction: $440 \times 12 = 5,280$ pphpd
- Network peak throughput (all lines, both directions): $5 \text{ lines} \times 2 \text{ directions} \times 5,280 = 52,800$ passengers/hour
- Daily theoretical capacity (peak $\times$ 10): $\approx 528,000$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 41,279 - 61,919$ trips/day

Catchment

- City population: 1,200,000
- Anchor-weighted coverage: 34.4%
- Catchment population: $\approx 412,799$ (within ~800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	17	500 kW	3000 kWh
Major	20	400 kW	2500 kWh
Standard	39	300 kW	2000 kWh
Terminal	7	500 kW	3000 kWh
Total installed	84	36,700 kW	240,000 kWh

Aggregate station-rail charging power: 46,500 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 480 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	16.0 kWh/km	4 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	564 kWh	35.2 km average line length
Onboard battery coverage	0.9 $\times$ average line run	480 kWh usable pack
Average 60 s dwell charge	9.1 kWh/stop	547 kW average charger across stops
Stops to refill one trainset pack	53 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	184 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	240 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the `[costs]` block in `design.toml` — emitted by the `osr-design` Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. `country-costs.toml` applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (140.9 km @ €3.5 M/km)	€493 M
Elevated (34.1 km @ €18 M/km)	€613 M
Elevated-interchange premium (10 sites @ €20 M)	€200 M
Civil subtotal	€1.31 bn

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
halt	2	€0.4 M	€0.8 M
standard	39	€1.5 M	€58 M
major	20	€3.0 M	€60 M
terminal	7	€2.5 M	€18 M
depot-terminal	1	€3.0 M	€3.0 M
interchange-elevated	17	€4.5 M	€76 M
Stations subtotal			€216 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	7	€3.0 M	€21 M
Depots subtotal			€46 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
metro-4car (revenue + spare + cold reserve)	145	€4.0 M	€580 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	176.1 km × €0.015 M/km	€2.6 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€32 M
EPC integration + project management (7%)	on subtotal	€153 M

Total

Bucket	Value
Civil works	€1.31 bn
Stations	€216 M
Depots	€46 M
Rolling stock	€580 M
Residual train-control wayside + charging microgrids	€34 M
EPC overhead (7%)	€153 M
CAPEX total	€2.34 bn
Per-route-km	€13 M / km
Per-capita (city pop)	€1,946 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running `scripts/regenerate-city.sh` `maiduguri`.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–7 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 8 and runs for 23 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–7)	€192 M / yr	€160
Steady-state, low-ridership (year 8+)	€234 M / yr	€195
Steady-state, high-ridership (year 8+)	€228 M / yr	€190
Lifecycle envelope (yr 1–30, low scenario)	€6.73 bn cumulative	€5,607
Lifecycle envelope (yr 1–30, high scenario)	€6.59 bn cumulative	€5,494

Population basis: 1,200,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 30, debt service drops to zero and only the OPEX shortfall remains – ~€52 M / yr (low) €46 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€1.40 bn	4.5%	30 y, 7 y grace	€99 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€584 M	13.5%	30 y, 7 y grace	€83 M / yr
Government equity (no debt service)	15%	€350 M	—	—	—
Total	100%	€2.34 bn			€182 M / yr

During the 7-year grace period the operator pays interest only – multilateral €63 M / yr + bonds €79 M / yr = €142 M / yr total – plus the equity tranche amortised across construction (€50 M / yr × 7 yr). Principal repayment begins in year 8 on a 23-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€23 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€31 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€131 k
Traction energy (405.4 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (1,069 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€2.9 M
OPEX subtotal		€58 M / yr

Annual fleet utilisation: 129 revenue trainsets × 20.5 h/day × 365 d/yr × 35 km/h commercial × 75% revenue factor = 25.3 M train-km / yr (~196 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$175 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.27 (~\$0.29 USD)
Day pass (3 trips)	€0.68 (15 % bulk discount)
Monthly unlimited pass	€8.05 (~5 % of median monthly income)
Annual pass	€88.55 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	21.9 M	43.8 M
Farebox revenue	€5.9 M / yr	€12 M / yr
Farebox / OPEX recovery	10%	20%
Country policy-target recovery (diagnostic)	45%	45%
Operating shortfall (gov subsidy required)	€52 M / yr	€46 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€234 M / yr	€228 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
maiduguri.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
maiduguri-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
maiduguri.corridor.geojson	Line polylines + stations (GeoJSON)
maiduguri.stations.json	Machine-readable station list
maiduguri.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug maiduguri

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug maiduguri \
  --sidecar .cache/osr-pipeline/rasters/maiduguri.grid.json \
  --out-dir designs/.../Maiduguri

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../maiduguri.toml \
  --out designs/.../README.md
```

`scripts/regenerate-maiduguri.sh` chains steps 3 + drift tests into a single command.

west-africa/Nigeria/Onitsha

designs/west-africa/Nigeria/Onitsha/README.md

Onitsha — Urban Rail Network

Country: NG · Population: 1,500,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Onitsha rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [onitsha.corridor.geojson](#).

At a glance

Metric	Value
Lines	4
Unique stations	75
Interchange stations	13
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	29.5%
Route length (double track)	183.7 km
Revenue fleet	133 × 4-car trainsets
Spare + cold-reserve	15 × 4-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	38.4 km	16	31	E Mid W Outer
line-2	35.1 km	16	29	NE Outer SW Outer
line-3	38.2 km	16	31	NW Outer S Outer
line-4	72.0 km	28	57	NW Outer NW Outer
Total	183.7 km	75 unique	148	

Rolling stock

Property	Value
Consist	4-car, 75 m
Max speed	90 km/h
Onboard battery	480 kWh per trainset
Seats	80 longitudinal seats
Nominal capacity (AW2)	440 pax (seated + standing, met ro - 4car per RFC 0008 §1)
Crush capacity (AW3)	560 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 440 AW2 passengers (met ro - 4car)

- Peak frequency: 12 trains/hour/direction (5-min headway)
- Peak capacity per line per direction: $440 \times 12 = 5,280$ pphpd
- Network peak throughput (all lines, both directions): $4 \text{ lines} \times 2 \text{ directions} \times 5,280 = 42,240$ passengers/hour
- Daily theoretical capacity (peak $\times$ 10): $\approx 422,400$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 44,250 - 66,375$ trips/day

Catchment

- City population: 1,500,000
- Anchor-weighted coverage: 29.5%
- Catchment population: $\approx 442,500$ (within ~800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	13	500 kW	3000 kWh
Major	10	400 kW	2500 kWh
Standard	46	300 kW	2000 kWh
Terminal	5	500 kW	3000 kWh
Total installed	75	31,800 kW	211,000 kWh

Aggregate station-rail charging power: 40,750 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 480 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	16.0 kWh/km	4 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	735 kWh	45.9 km average line length
Onboard battery coverage	0.7 $\times$ average line run	480 kWh usable pack
Average 60 s dwell charge	9.1 kWh/stop	543 kW average charger across stops
Stops to refill one trainset pack	53 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	159 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	211 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in design.toml — emitted by the osr-design Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. country-costs.toml applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (163.7 km @ €3.5 M/km)	€573 M
Elevated (19.0 km @ €18 M/km)	€342 M
Elevated-interchange premium (6 sites @ €20 M)	€120 M
Civil subtotal	€1.03 bn

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
halt	1	€0.4 M	€0.4 M
standard	46	€1.5 M	€69 M
major	10	€3.0 M	€30 M
terminal	5	€2.5 M	€12 M
depot-terminal	1	€3.0 M	€3.0 M
interchange-elevated	13	€4.5 M	€58 M
Stations subtotal			€173 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	5	€3.0 M	€15 M
Depots subtotal			€40 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
metro-4car (revenue + spare + cold reserve)	148	€4.0 M	€592 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	183.7 km × €0.015 M/km	€2.7 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€26 M
EPC integration + project management (7%)	on subtotal	€131 M

Total

Bucket	Value
Civil works	€1.03 bn
Stations	€173 M
Depots	€40 M
Rolling stock	€592 M
Residual train-control wayside + charging microgrids	€29 M
EPC overhead (7%)	€131 M
CAPEX total	€2.00 bn
Per-route-km	€11 M / km
Per-capita (city pop)	€1,333 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running `scripts/regenerate-city.sh` on `itsha`.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–7 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 8 and runs for 23 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–7)	€164 M / yr	€110
Steady-state, low-ridership (year 8+)	€201 M / yr	€134
Steady-state, high-ridership (year 8+)	€193 M / yr	€129
Lifecycle envelope (yr 1–30, low scenario)	€5.77 bn cumulative	€3,843
Lifecycle envelope (yr 1–30, high scenario)	€5.60 bn cumulative	€3,731

Population basis: 1,500,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 30, debt service drops to zero and only the OPEX shortfall remains – ~€44 M / yr (low) €37 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€1.20 bn	4.5%	30 y, 7 y grace	€85 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€500 M	13.5%	30 y, 7 y grace	€71 M / yr
Government equity (no debt service)	15%	€300 M	—	—	—
Total	100%	€2.00 bn			€156 M / yr

During the 7-year grace period the operator pays interest only – multilateral €54 M / yr + bonds €67 M / yr = €121 M / yr total – plus the equity tranche amortised across construction (€43 M / yr × 7 yr). Principal repayment begins in year 8 on a 23-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€24 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€25 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€137 k
Traction energy (418.0 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (1,114 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€3.0 M
OPEX subtotal		€52 M / yr

Annual fleet utilisation: 133 revenue trainsets × 20.5 h/day × 365 d/yr × 35 km/h commercial × 75% revenue factor = 26.1 M train-km / yr (~196 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$175 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.27 (~\$0.29 USD)
Day pass (3 trips)	€0.68 (15 % bulk discount)
Monthly unlimited pass	€8.05 (~5 % of median monthly income)
Annual pass	€88.55 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	27.4 M	54.8 M
Farebox revenue	€7.3 M / yr	€15 M / yr
Farebox / OPEX recovery	14%	28%
Country policy-target recovery (diagnostic)	45%	45%
Operating shortfall (gov subsidy required)	€44 M / yr	€37 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€201 M / yr	€193 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
onitsha.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
onitsha-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
onitsha.corridor.geojson	Line polylines + stations (GeoJSON)
onitsha.stations.json	Machine-readable station list
onitsha.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug onitsha

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug onitsha \
  --sidecar .cache/osr-pipeline/rasters/onitsha.grid.json \
  --out-dir designs/.../Onitsha

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../onitsha.toml \
  --out designs/.../README.md
```

`scripts/regenerate-onitsha.sh` chains steps 3 + drift tests into a single command.

west-africa/Nigeria/Port-Harcourt

designs/west-africa/Nigeria/Port-Harcourt/README.md

Port-Harcourt — Urban Rail Network

Country: NG · Population: 3,000,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Port-Harcourt rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [port-harcourt.corridor.geojson](#).

At a glance

Metric	Value
Lines	4
Unique stations	94
Interchange stations	13
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	30.6%
Route length (double track)	198.9 km
Revenue fleet	143 × 4-car trainsets
Spare + cold-reserve	17 × 4-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	42.7 km	20	35	SE Outer NW Outer
line-2	33.9 km	18	28	SW Outer NE Outer
line-3	32.3 km	19	27	N Outer S Mid
line-4	90.0 km	38	70	NW Outer NW Outer
Total	198.9 km	94 unique	160	

Rolling stock

Property	Value
Consist	4-car, 75 m
Max speed	90 km/h
Onboard battery	480 kWh per trainset
Seats	80 longitudinal seats
Nominal capacity (AW2)	440 pax (seated + standing, met ro - 4 car per RFC 0008 §1)
Crush capacity (AW3)	560 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 440 AW2 passengers (met ro - 4 car)

- Peak frequency: 12 trains/hour/direction (5-min headway)
- Peak capacity per line per direction: $440 \times 12 = 5,280$ pphpd
- Network peak throughput (all lines, both directions): $4 \text{ lines} \times 2 \text{ directions} \times 5,280 = 42,240$ passengers/hour
- Daily theoretical capacity (peak $\times$ 10): $\approx 422,400$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 91,800 - 137,700$ trips/day

Catchment

- City population: 3,000,000
- Anchor-weighted coverage: 30.6%
- Catchment population: $\approx 918,000$ (within ~800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	13	500 kW	3000 kWh
Major	28	400 kW	2500 kWh
Standard	48	300 kW	2000 kWh
Terminal	5	500 kW	3000 kWh
Total installed	95	39,600 kW	260,000 kWh

Aggregate station-rail charging power: 50,500 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 480 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	16.0 kWh/km	4 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	796 kWh	49.7 km average line length
Onboard battery coverage	0.6 $\times$ average line run	480 kWh usable pack
Average 60 s dwell charge	9.0 kWh/stop	537 kW average charger across stops
Stops to refill one trainset pack	54 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	198 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	260 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in design.toml — emitted by the osr-design Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. country-costs.toml applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (189.1 km @ €3.5 M/km)	€662 M
Elevated (8.8 km @ €18 M/km)	€159 M
Elevated-interchange premium (6 sites @ €20 M)	€120 M
Civil subtotal	€941 M

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
standard	48	€1.5 M	€72 M
major	28	€3.0 M	€84 M
terminal	5	€2.5 M	€12 M
depot-terminal	1	€3.0 M	€3.0 M
interchange-elevated	13	€4.5 M	€58 M
Stations subtotal			€230 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	5	€3.0 M	€15 M
Depots subtotal			€40 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
metro-4car (revenue + spare + cold reserve)	160	€4.0 M	€640 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	198.9 km × €0.015 M/km	€3.0 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€34 M
EPC integration + project management (7%)	on subtotal	€132 M

Total

Bucket	Value
Civil works	€941 M
Stations	€230 M
Depots	€40 M
Rolling stock	€640 M
Residual train-control wayside + charging microgrids	€37 M
EPC overhead (7%)	€132 M
CAPEX total	€2.02 bn
Per-route-km	€10 M / km
Per-capita (city pop)	€673 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running `scripts/regenerate-city.sh port-harcourt`.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–7 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 8 and runs for 23 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–7)	€166 M / yr	€55
Steady-state, low-ridership (year 8+)	€196 M / yr	€65
Steady-state, high-ridership (year 8+)	€182 M / yr	€61
Lifecycle envelope (yr 1–30, low scenario)	€5.68 bn cumulative	€1,892
Lifecycle envelope (yr 1–30, high scenario)	€5.34 bn cumulative	€1,779

Population basis: 3,000,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 30, debt service drops to zero and only the OPEX shortfall remains – ~€39 M / yr (low) €24 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€1.21 bn	4.5%	30 y, 7 y grace	€86 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€505 M	13.5%	30 y, 7 y grace	€72 M / yr
Government equity (no debt service)	15%	€303 M	—	—	—
Total	100%	€2.02 bn			€158 M / yr

During the 7-year grace period the operator pays interest only – multilateral €55 M / yr + bonds €68 M / yr = €123 M / yr total – plus the equity tranche amortised across construction (€43 M / yr × 7 yr). Principal repayment begins in year 8 on a 23-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€26 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€24 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€148 k
Traction energy (449.4 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (1,205 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€3.3 M
OPEX subtotal		€53 M / yr

Annual fleet utilisation: 143 revenue trainsets × 20.5 h/day × 365 d/yr × 35 km/h commercial × 75% revenue factor = 28.1 M train-km / yr (~196 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$175 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.27 (~\$0.29 USD)
Day pass (3 trips)	€0.68 (15 % bulk discount)
Monthly unlimited pass	€8.05 (~5 % of median monthly income)
Annual pass	€88.55 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	54.8 M	109.5 M
Farebox revenue	€15 M / yr	€29 M / yr
Farebox / OPEX recovery	28%	55%
Country policy-target recovery (diagnostic)	45%	45%
Operating shortfall (gov subsidy required)	€39 M / yr	€24 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€196 M / yr	€182 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
port-harcourt.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
port-harcourt-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
port-harcourt.corridor.geojson	Line polylines + stations (GeoJSON)
port-harcourt.stations.json	Machine-readable station list
port-harcourt.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug port-harcourt

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug port-harcourt \
  --sidecar .cache/osr-pipeline/rasters/port-harcourt.grid.json \
  --out-dir designs/.../Port-Harcourt

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../port-harcourt.toml \
  --out designs/.../README.md
```

`scripts/regenerate-port-harcourt.sh` chains steps 3 + drift tests into a single command.

west-africa/Nigeria/Uyo

designs/west-africa/Nigeria/Uyo/README.md

Uyo — Urban Rail Network

Country: NG · Population: 800,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Uyo rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [uyo.corridor.geojson](#).

At a glance

Metric	Value
Lines	3
Unique stations	24
Interchange stations	3
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	24.6%
Route length (double track)	31.1 km
Revenue fleet	30 × 3-car trainsets
Spare + cold-reserve	6 × 3-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	15.5 km	11	16	E Outer SW Mid
line-2	8.3 km	8	10	NW Mid SE Inner
line-3	7.3 km	5	10	SW Mid NE Inner
Total	31.1 km	24 unique	36	

Rolling stock

Property	Value
Consist	3-car, 57 m
Max speed	90 km/h
Onboard battery	360 kWh per trainset
Seats	60 longitudinal seats
Nominal capacity (AW2)	330 pax (seated + standing, light-metro-3car per RFC 0008 §1)
Crush capacity (AW3)	420 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 330 AW2 passengers (light-metro-3car)
- Peak frequency: 12 trains/hour/direction (5-min headway)

- Peak capacity per line per direction: $330 \times 12 = 3,960$ pphpd
- Network peak throughput (all lines, both directions): $3 \text{ lines} \times 2 \text{ directions} \times 3,960 = 23,760$ passengers/hour
- Daily theoretical capacity (peak $\times 10$): $\approx 237,600$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 19,680 - 29,520$ trips/day

Catchment

- City population: 800,000
- Anchor-weighted coverage: 24.6%
- Catchment population: $\approx 196,800$ (within ~ 800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	3	500 kW	3000 kWh
Major	11	400 kW	2500 kWh
Standard	4	300 kW	2000 kWh
Terminal	5	500 kW	3000 kWh
Total installed	24	14,600 kW	99,500 kWh

Aggregate station-rail charging power: 15,000 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 360 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	12.0 kWh/km	3 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	124 kWh	10.4 km average line length
Onboard battery coverage	2.9 $\times$ average line run	360 kWh usable pack
Average 60 s dwell charge	10.4 kWh/stop	625 kW average charger across stops
Stops to refill one trainset pack	35 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	73 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	100 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in design.toml — emitted by the osr-design Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. country-costs.toml applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (29.9 km @ €3.5 M/km)	€105 M
Elevated (0.5 km @ €18 M/km)	€9.6 M
Elevated-interchange premium (1 sites @ €20 M)	€20 M
Civil subtotal	€134 M

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~ 11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
standard	4	€1.5 M	€6.0 M
major	11	€3.0 M	€33 M
terminal	5	€2.5 M	€12 M
depot-terminal	1	€3.0 M	€3.0 M
interchange	3	€4.5 M	€14 M
Stations subtotal			€68 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	5	€3.0 M	€15 M
Depots subtotal			€40 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
light-metro-3car (revenue + spare + cold reserve)	36	€3.0 M	€108 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	31.1 km × €0.015 M/km	€0.5 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€9.9 M
EPC integration + project management (7%)	on subtotal	€25 M

Total

Bucket	Value
Civil works	€134 M
Stations	€68 M
Depots	€40 M
Rolling stock	€108 M
Residual train-control wayside + charging microgrids	€10 M
EPC overhead (7%)	€25 M
CAPEX total	€386 M
Per-route-km	€12 M / km
Per-capita (city pop)	€482 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running scripts/regenerate-city.sh uyo.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–7 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 8 and runs for 23 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–7)	€32 M / yr	€40
Steady-state, low-ridership (year 8+)	€36 M / yr	€45
Steady-state, high-ridership (year 8+)	€32 M / yr	€40
Lifecycle envelope (yr 1–30, low scenario)	€1.05 bn cumulative	€1,311
Lifecycle envelope (yr 1–30, high scenario)	€959 M cumulative	€1,199

Population basis: 800,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 30, debt service drops to zero and only the OPEX shortfall remains – ~€5.8 M / yr (low) €1.9 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€232 M	4.5%	30 y, 7 y grace	€16 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€96 M	13.5%	30 y, 7 y grace	€14 M / yr
Government equity (no debt service)	15%	€58 M	—	—	—
Total	100%	€386 M			€30 M / yr

During the 7-year grace period the operator pays interest only – multilateral €10 M / yr + bonds €13 M / yr = €23 M / yr total – plus the equity tranche amortised across construction (€8.3 M / yr × 7 yr). Principal repayment begins in year 8 on a 23-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€4.3 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€4.8 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€23 k
Traction energy (60.6 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (199 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€538 k
OPEX subtotal		€9.7 M / yr

Annual fleet utilisation: 30 revenue trainsets × 20.5 h/day × 365 d/yr × 30 km/h commercial × 75% revenue factor = 5.1 M train-km / yr (~168 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$175 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.27 (~\$0.29 USD)
Day pass (3 trips)	€0.68 (15 % bulk discount)
Monthly unlimited pass	€8.05 (~5 % of median monthly income)
Annual pass	€88.55 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	14.6 M	29.2 M

	Low scenario	High scenario
Farebox revenue	€3.9 M / yr	€7.8 M / yr
Farebox / OPEX recovery	40%	81%
Country policy-target recovery (diagnostic)	45%	45%
Operating shortfall (gov subsidy required)	€5.8 M / yr	€1.9 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€36 M / yr	€32 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
uyo.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
uyo-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
uyo.corridor.geojson	Line polylines + stations (GeoJSON)
uyo.stations.json	Machine-readable station list
uyo.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug uyo

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug uyo \
  --sidecar .cache/osr-pipeline/rasters/uyo.grid.json \
  --out-dir designs/.../Uyo

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../uyo.toml \
  --out designs/.../README.md
```

`scripts/regenerate-uyo.sh` chains steps 3 + drift tests into a single command.

west-africa/Senegal/Dakar

designs/west-africa/Senegal/Dakar/README.md

Dakar — Urban Rail Network

Country: SN · Population: 4,030,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Dakar rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [dakar.corridor.geojson](#).

At a glance

Metric	Value
Lines	5
Unique stations	106
Interchange stations	12
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	52.1%
Route length (double track)	204.0 km
Revenue fleet	149 × 6-car trainsets
Spare + cold-reserve	18 × 6-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	42.2 km	22	35	E Outer W Outer
line-2	33.2 km	18	27	NE Outer W Outer
line-3	29.0 km	16	25	E Outer W Mid
line-4	23.4 km	13	20	NW Inner E Outer
line-5	76.1 km	38	60	W Outer W Outer
Total	204.0 km	106 unique	167	

Rolling stock

Property	Value
Consist	6-car, 111 m
Max speed	100 km/h
Onboard battery	720 kWh per trainset
Seats	120 longitudinal seats
Nominal capacity (AW2)	660 pax (seated + standing, met ro-6car per RFC 0008 §1)
Crush capacity (AW3)	840 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 660 AW2 passengers (met ro-6car)

- Peak frequency: 12 trains/hour/direction (5-min headway)
- Peak capacity per line per direction: $660 \times 12 = 7,920$ pphpd
- Network peak throughput (all lines, both directions): $5 \text{ lines} \times 2 \text{ directions} \times 7,920 = 79,200$ passengers/hour
- Daily theoretical capacity (peak $\times$ 10): $\approx 792,000$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 209,963 - 314,944$ trips/day

Catchment

- City population: 4,030,000
- Anchor-weighted coverage: 52.1%
- Catchment population: $\approx 2,099,630$ (within ~ 800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	12	500 kW	3000 kWh
Major	57	400 kW	2500 kWh
Standard	27	300 kW	2000 kWh
Terminal	7	500 kW	3000 kWh
Total installed	104	45,400 kW	293,500 kWh

Aggregate station-rail charging power: 56,750 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 720 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	24.0 kWh/km	6 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	979 kWh	40.8 km average line length
Onboard battery coverage	0.7 $\times$ average line run	720 kWh usable pack
Average 60 s dwell charge	8.9 kWh/stop	535 kW average charger across stops
Stops to refill one trainset pack	81 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	227 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	294 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the `[costs]` block in `design.toml` — emitted by the `osr-design` Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. `country-costs.toml` applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (192.4 km @ €3.5 M/km)	€673 M
Elevated (10.9 km @ €18 M/km)	€196 M
Elevated-interchange premium (9 sites @ €20 M)	€180 M
Civil subtotal	€1.05 bn

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
halt	3	€0.4 M	€1.2 M
standard	27	€1.5 M	€40 M
major	57	€3.0 M	€171 M
terminal	7	€2.5 M	€18 M
depot-terminal	1	€3.0 M	€3.0 M
interchange-elevated	12	€4.5 M	€54 M
Stations subtotal			€287 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	7	€3.0 M	€21 M
Depots subtotal			€46 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
metro-6car (revenue + spare + cold reserve)	167	€6.0 M	€1.00 bn

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	204.0 km × €0.015 M/km	€3.0 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€41 M
EPC integration + project management (7%)	on subtotal	€170 M

Total

Bucket	Value
Civil works	€1.05 bn
Stations	€287 M
Depots	€46 M
Rolling stock	€1.00 bn
Residual train-control wayside + charging microgrids	€44 M
EPC overhead (7%)	€170 M
CAPEX total	€2.60 bn
Per-route-km	€13 M / km
Per-capita (city pop)	€645 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running `scripts/regenerate-city.sh` dakar.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–7 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 8 and runs for 23 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–7)	€170 M / yr	€42
Steady-state, low-ridership (year 8+)	€220 M / yr	€54
Steady-state, high-ridership (year 8+)	€200 M / yr	€50
Lifecycle envelope (yr 1–30, low scenario)	€6.24 bn cumulative	€1,548
Lifecycle envelope (yr 1–30, high scenario)	€5.79 bn cumulative	€1,436

Population basis: 4,030,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 30, debt service drops to zero and only the OPEX shortfall remains – ~€51 M / yr (low) €32 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€1.56 bn	3.8%	30 y, 7 y grace	€103 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€650 M	8.5%	30 y, 7 y grace	€65 M / yr
Government equity (no debt service)	15%	€390 M	—	—	—
Total	100%	€2.60 bn			€168 M / yr

During the 7-year grace period the operator pays interest only – multilateral €59 M / yr + bonds €55 M / yr = €114 M / yr total – plus the equity tranche amortised across construction (€56 M / yr × 7 yr). Principal repayment begins in year 8 on a 23-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€40 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€28 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€152 k
Traction energy (702.4 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (1,236 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€3.3 M
OPEX subtotal		€71 M / yr

Annual fleet utilisation: 149 revenue trainsets × 20.5 h/day × 365 d/yr × 35 km/h commercial × 75% revenue factor = 29.3 M train-km / yr (~196 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$175 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.27 (~\$0.29 USD)
Day pass (3 trips)	€0.68 (15 % bulk discount)
Monthly unlimited pass	€8.05 (~5 % of median monthly income)
Annual pass	€88.55 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	73.5 M	147.1 M
Farebox revenue	€20 M / yr	€39 M / yr
Farebox / OPEX recovery	28%	55%
Country policy-target recovery (diagnostic)	40%	40%
Operating shortfall (gov subsidy required)	€51 M / yr	€32 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€220 M / yr	€200 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
dakar.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
dakar-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
dakar.corridor.geojson	Line polylines + stations (GeoJSON)
dakar.stations.json	Machine-readable station list
dakar.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug dakar

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug dakar \
  --sidecar .cache/osr-pipeline/rasters/dakar.grid.json \
  --out-dir designs/.../Dakar

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../dakar.toml \
  --out designs/.../README.md
```

`scripts/regenerate-dakar.sh` chains steps 3 + drift tests into a single command.

west-asia/Egypt/Asyut

designs/west-asia/Egypt/Asyut/README.md

Asyut — Urban Rail Network

Country: EG · Population: 600,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Asyut rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [asyut.corridor.geojson](#).

At a glance

Metric	Value
Lines	3
Unique stations	31
Interchange stations	2
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	58.0%
Route length (double track)	62.8 km
Revenue fleet	55 × 3-car trainsets
Spare + cold-reserve	7 × 3-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	19.2 km	11	19	SE Inner NW Outer
line-2	25.6 km	11	25	SE Outer W Mid
line-3	17.9 km	9	18	NE Outer SW Inner
Total	62.8 km	31 unique	62	

Rolling stock

Property	Value
Consist	3-car, 57 m
Max speed	90 km/h
Onboard battery	360 kWh per trainset
Seats	60 longitudinal seats
Nominal capacity (AW2)	330 pax (seated + standing, light-metro-3car per RFC 0008 §1)
Crush capacity (AW3)	420 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 330 AW2 passengers (light-metro-3car)
- Peak frequency: 12 trains/hour/direction (5-min headway)

- Peak capacity per line per direction: $330 \times 12 = 3,960$ pphpd
- Network peak throughput (all lines, both directions): $3 \text{ lines} \times 2 \text{ directions} \times 3,960 = 23,760$ passengers/hour
- Daily theoretical capacity (peak $\times 10$): $\approx 237,600$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 34,800 - 52,200$ trips/day

Catchment

- City population: 600,000
- Anchor-weighted coverage: 58.0%
- Catchment population: $\approx 348,000$ (within ~ 800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	2	500 kW	3000 kWh
Major	5	400 kW	2500 kWh
Standard	14	300 kW	2000 kWh
Terminal	5	500 kW	3000 kWh
Total installed	27	14,700 kW	101,500 kWh

Aggregate station-rail charging power: 17,500 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 360 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	12.0 kWh/km	3 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	251 kWh	20.9 km average line length
Onboard battery coverage	1.4 $\times$ average line run	360 kWh usable pack
Average 60 s dwell charge	9.4 kWh/stop	565 kW average charger across stops
Stops to refill one trainset pack	38 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	74 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	102 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the `[costs]` block in `design.toml` — emitted by the `osr-design` Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. `country-costs.toml` applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (60.1 km @ €3.5 M/km)	€210 M
Elevated (2.5 km @ €18 M/km)	€45 M
Elevated-interchange premium (1 sites @ €20 M)	€20 M
Civil subtotal	€276 M

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~ 11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
halt	4	€0.4 M	€1.6 M
standard	14	€1.5 M	€21 M
major	5	€3.0 M	€15 M
terminal	5	€2.5 M	€12 M
depot-terminal	1	€3.0 M	€3.0 M
interchange-elevated	2	€4.5 M	€9.0 M
Stations subtotal			€62 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	5	€3.0 M	€15 M
Depots subtotal			€40 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
light-metro-3car (revenue + spare + cold reserve)	62	€3.0 M	€186 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	62.8 km × €0.015 M/km	€0.9 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€9.9 M
EPC integration + project management (7%)	on subtotal	€40 M

Total

Bucket	Value
Civil works	€276 M
Stations	€62 M
Depots	€40 M
Rolling stock	€186 M
Residual train-control wayside + charging microgrids	€11 M
EPC overhead (7%)	€40 M
CAPEX total	€615 M
Per-route-km	€9.8 M / km
Per-capita (city pop)	€1,025 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running scripts/regenerate-city.sh asyut.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–5 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 6 and runs for 20 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–5)	€51 M / yr	€85
Steady-state, low-ridership (year 6+)	€59 M / yr	€99
Steady-state, high-ridership (year 6+)	€55 M / yr	€92
Lifecycle envelope (yr 1–25, low scenario)	€1.44 bn cumulative	€2,402
Lifecycle envelope (yr 1–25, high scenario)	€1.35 bn cumulative	€2,257

Population basis: 600,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 25, debt service drops to zero and only the OPEX shortfall remains – ~€12 M / yr (low) €7.9 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€369 M	4.5%	25 y, 5 y grace	€28 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€154 M	10.5%	25 y, 5 y grace	€19 M / yr
Government equity (no debt service)	15%	€92 M	—	—	—
Total	100%	€615 M			€47 M / yr

During the 5-year grace period the operator pays interest only – multilateral €17 M / yr + bonds €16 M / yr = €33 M / yr total – plus the equity tranche amortised across construction (€18 M / yr × 5 yr). Principal repayment begins in year 6 on a 20-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€7.4 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€7.6 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€47 k
Traction energy (111.1 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (389 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€1.6 M
OPEX subtotal		€17 M / yr

Annual fleet utilisation: 55 revenue trainsets × 20.5 h/day × 365 d/yr × 30 km/h commercial × 75% revenue factor = 9.3 M train-km / yr (~168 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$260 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.40 (~\$0.43 USD)
Day pass (3 trips)	€1.02 (15 % bulk discount)
Monthly unlimited pass	€11.96 (~5 % of median monthly income)
Annual pass	€131.56 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	10.9 M	21.9 M

	Low scenario	High scenario
Farebox revenue	€4.4 M / yr	€8.7 M / yr
Farebox / OPEX recovery	26%	53%
Country policy-target recovery (diagnostic)	55%	55%
Operating shortfall (gov subsidy required)	€12 M / yr	€7.9 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€59 M / yr	€55 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
asyut.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
asyut-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
asyut.corridor.geojson	Line polylines + stations (GeoJSON)
asyut.stations.json	Machine-readable station list
asyut.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug asyut

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug asyut \
  --sidecar .cache/osr-pipeline/rasters/asyut.grid.json \
  --out-dir designs/.../Asyut

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../asyut.toml \
  --out designs/.../README.md
```

`scripts/regenerate-asyut.sh` chains steps 3 + drift tests into a single command.

west-asia/Egypt/Beni-Suef

designs/west-asia/Egypt/Beni-Suef/README.md

Beni-Suef — Urban Rail Network

Country: EG · Population: 350,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Beni-Suef rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [beni-suef.corridor.geojson](#).

At a glance

Metric	Value
Lines	3
Unique stations	23
Interchange stations	2
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	54.7%
Route length (double track)	34.9 km
Revenue fleet	32 × 3-car trainsets
Spare + cold-reserve	6 × 3-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	14.5 km	11	15	S Outer N Outer
line-2	12.1 km	7	13	S Outer NE Outer
line-3	8.3 km	5	10	NW Mid E Mid
Total	34.9 km	23 unique	38	

Rolling stock

Property	Value
Consist	3-car, 57 m
Max speed	90 km/h
Onboard battery	360 kWh per trainset
Seats	60 longitudinal seats
Nominal capacity (AW2)	330 pax (seated + standing, light-metro-3car per RFC 0008 §1)
Crush capacity (AW3)	420 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 330 AW2 passengers (light-metro-3car)
- Peak frequency: 12 trains/hour/direction (5-min headway)

- Peak capacity per line per direction: $330 \times 12 = 3,960$ pphpd
- Network peak throughput (all lines, both directions): $3 \text{ lines} \times 2 \text{ directions} \times 3,960 = 23,760$ passengers/hour
- Daily theoretical capacity (peak $\times 10$): $\approx 237,600$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 19,145 - 28,717$ trips/day

Catchment

- City population: 350,000
- Anchor-weighted coverage: 54.7%
- Catchment population: $\approx 191,450$ (within ~ 800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	2	500 kW	3000 kWh
Major	9	400 kW	2500 kWh
Standard	6	300 kW	2000 kWh
Terminal	5	500 kW	3000 kWh
Total installed	23	13,900 kW	95,500 kWh

Aggregate station-rail charging power: 14,500 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 360 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	12.0 kWh/km	3 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	140 kWh	11.6 km average line length
Onboard battery coverage	2.6 $\times$ average line run	360 kWh usable pack
Average 60 s dwell charge	10.5 kWh/stop	630 kW average charger across stops
Stops to refill one trainset pack	34 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	70 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	96 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in design.toml — emitted by the osr-design Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. country-costs.toml applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (32.5 km @ €3.5 M/km)	€114 M
Elevated (2.2 km @ €18 M/km)	€40 M
Elevated-interchange premium (2 sites @ €20 M)	€40 M
Civil subtotal	€194 M

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~ 11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
standard	6	€1.5 M	€9.0 M
major	9	€3.0 M	€27 M
terminal	5	€2.5 M	€12 M
depot-terminal	1	€3.0 M	€3.0 M
interchange-elevated	2	€4.5 M	€9.0 M
Stations subtotal			€60 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	5	€3.0 M	€15 M
Depots subtotal			€40 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
light-metro-3car (revenue + spare + cold reserve)	38	€3.0 M	€114 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	34.9 km × €0.015 M/km	€0.5 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€9.1 M
EPC integration + project management (7%)	on subtotal	€29 M

Total

Bucket	Value
Civil works	€194 M
Stations	€60 M
Depots	€40 M
Rolling stock	€114 M
Residual train-control wayside + charging microgrids	€9.6 M
EPC overhead (7%)	€29 M
CAPEX total	€447 M
Per-route-km	€13 M / km
Per-capita (city pop)	€1,278 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running scripts/regenerate-city.sh `beni-suef`.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–5 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 6 and runs for 20 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–5)	€37 M / yr	€106
Steady-state, low-ridership (year 6+)	€43 M / yr	€123
Steady-state, high-ridership (year 6+)	€40 M / yr	€116
Lifecycle envelope (yr 1–25, low scenario)	€1.05 bn cumulative	€2,991
Lifecycle envelope (yr 1–25, high scenario)	€996 M cumulative	€2,846

Population basis: 350,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 25, debt service drops to zero and only the OPEX shortfall remains – ~€8.8 M / yr (low) €6.3 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€268 M	4.5%	25 y, 5 y grace	€21 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€112 M	10.5%	25 y, 5 y grace	€14 M / yr
Government equity (no debt service)	15%	€67 M	—	—	—
Total	100%	€447 M			€34 M / yr

During the 5-year grace period the operator pays interest only – multilateral €12 M / yr + bonds €12 M / yr = €24 M / yr total – plus the equity tranche amortised across construction (€13 M / yr × 5 yr). Principal repayment begins in year 6 on a 20-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€4.6 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€5.9 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€26 k
Traction energy (64.6 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (221 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€888 k
OPEX subtotal		€11 M / yr

Annual fleet utilisation: 32 revenue trainsets × 20.5 h/day × 365 d/yr × 30 km/h commercial × 75% revenue factor = 5.4 M train-km / yr (~168 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$260 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.40 (~\$0.43 USD)
Day pass (3 trips)	€1.02 (15 % bulk discount)
Monthly unlimited pass	€11.96 (~5 % of median monthly income)
Annual pass	€131.56 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	6.4 M	12.8 M

	Low scenario	High scenario
Farebox revenue	€2.5 M / yr	€5.1 M / yr
Farebox / OPEX recovery	22%	45%
Country policy-target recovery (diagnostic)	55%	55%
Operating shortfall (gov subsidy required)	€8.8 M / yr	€6.3 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€43 M / yr	€40 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
beni-suef.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
beni-suef-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
beni-suef.corridor.geojson	Line polylines + stations (GeoJSON)
beni-suef.stations.json	Machine-readable station list
beni-suef.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug beni-suef

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug beni-suef \
  --sidecar .cache/osr-pipeline/rasters/beni-suef.grid.json \
  --out-dir designs/.../Beni-Suef

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../beni-suef.toml \
  --out designs/.../README.md
```

`scripts/regenerate-beni-suef.sh` chains steps 3 + drift tests into a single command.

west-asia/Egypt/Luxor

designs/west-asia/Egypt/Luxor/README.md

Luxor — Urban Rail Network

Country: EG · Population: 500,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Luxor rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [luxor.corridor.geojson](#).

At a glance

Metric	Value
Lines	3
Unique stations	34
Interchange stations	5
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	72.7%
Route length (double track)	53.8 km
Revenue fleet	48 × 3-car trainsets
Spare + cold-reserve	6 × 3-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	21.6 km	13	21	SW Outer NE Mid
line-2	17.8 km	11	18	NW Mid E Mid
line-3	14.5 km	10	15	S Inner NW Mid
Total	53.8 km	34 unique	54	

Rolling stock

Property	Value
Consist	3-car, 57 m
Max speed	90 km/h
Onboard battery	360 kWh per trainset
Seats	60 longitudinal seats
Nominal capacity (AW2)	330 pax (seated + standing, light-metro-3car per RFC 0008 §1)
Crush capacity (AW3)	420 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 330 AW2 passengers (light-metro-3car)
- Peak frequency: 12 trains/hour/direction (5-min headway)

- Peak capacity per line per direction: $330 \times 12 = 3,960$ pphpd
- Network peak throughput (all lines, both directions): $3 \text{ lines} \times 2 \text{ directions} \times 3,960 = 23,760$ passengers/hour
- Daily theoretical capacity (peak $\times 10$): $\approx 237,600$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 36,350 - 54,525$ trips/day

Catchment

- City population: 500,000
- Anchor-weighted coverage: 72.7%
- Catchment population: $\approx 363,500$ (within ~ 800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	5	500 kW	3000 kWh
Major	8	400 kW	2500 kWh
Standard	14	300 kW	2000 kWh
Terminal	5	500 kW	3000 kWh
Total installed	33	17,400 kW	118,000 kWh

Aggregate station-rail charging power: 19,750 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 360 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	12.0 kWh/km	3 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	215 kWh	17.9 km average line length
Onboard battery coverage	1.7 $\times$ average line run	360 kWh usable pack
Average 60 s dwell charge	9.7 kWh/stop	581 kW average charger across stops
Stops to refill one trainset pack	37 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	87 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	118 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the `[costs]` block in `design.toml` — emitted by the `osr-design` Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. `country-costs.toml` applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (51.1 km @ €3.5 M/km)	€179 M
Elevated (2.5 km @ €18 M/km)	€45 M
Elevated-interchange premium (2 sites @ €20 M)	€40 M
Civil subtotal	€263 M

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~ 11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
halt	1	€0.4 M	€0.4 M
standard	14	€1.5 M	€21 M
major	8	€3.0 M	€24 M
terminal	5	€2.5 M	€12 M
depot-terminal	1	€3.0 M	€3.0 M
interchange-elevated	5	€4.5 M	€22 M
Stations subtotal			€83 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	5	€3.0 M	€15 M
Depots subtotal			€40 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
light-metro-3car (revenue + spare + cold reserve)	54	€3.0 M	€162 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	53.8 km × €0.015 M/km	€0.8 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€13 M
EPC integration + project management (7%)	on subtotal	€39 M

Total

Bucket	Value
Civil works	€263 M
Stations	€83 M
Depots	€40 M
Rolling stock	€162 M
Residual train-control wayside + charging microgrids	€13 M
EPC overhead (7%)	€39 M
CAPEX total	€602 M
Per-route-km	€11 M / km
Per-capita (city pop)	€1,203 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running scripts/regenerate-city.sh luxor.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–5 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 6 and runs for 20 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–5)	€50 M / yr	€100
Steady-state, low-ridership (year 6+)	€58 M / yr	€116
Steady-state, high-ridership (year 6+)	€54 M / yr	€109
Lifecycle envelope (yr 1–25, low scenario)	€1.41 bn cumulative	€2,820
Lifecycle envelope (yr 1–25, high scenario)	€1.34 bn cumulative	€2,675

Population basis: 500,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 25, debt service drops to zero and only the OPEX shortfall remains – ~€12 M / yr (low) €8.3 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€361 M	4.5%	25 y, 5 y grace	€28 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€150 M	10.5%	25 y, 5 y grace	€18 M / yr
Government equity (no debt service)	15%	€90 M	—	—	—
Total	100%	€602 M			€46 M / yr

During the 5-year grace period the operator pays interest only – multilateral €16 M / yr + bonds €16 M / yr = €32 M / yr total – plus the equity tranche amortised across construction (€18 M / yr × 5 yr). Principal repayment begins in year 6 on a 20-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€6.5 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€7.7 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€40 k
Traction energy (97.0 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (335 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€1.3 M
OPEX subtotal		€16 M / yr

Annual fleet utilisation: 48 revenue trainsets × 20.5 h/day × 365 d/yr × 30 km/h commercial × 75% revenue factor = 8.1 M train-km / yr (~168 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$260 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.40 (~\$0.43 USD)
Day pass (3 trips)	€1.02 (15 % bulk discount)
Monthly unlimited pass	€11.96 (~5 % of median monthly income)
Annual pass	€131.56 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	9.1 M	18.2 M

	Low scenario	High scenario
Farebox revenue	€3.6 M / yr	€7.3 M / yr
Farebox / OPEX recovery	23%	47%
Country policy-target recovery (diagnostic)	55%	55%
Operating shortfall (gov subsidy required)	€12 M / yr	€8.3 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€58 M / yr	€54 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
luxor.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
luxor-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
luxor.corridor.geojson	Line polylines + stations (GeoJSON)
luxor.stations.json	Machine-readable station list
luxor.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug luxor

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug luxor \
  --sidecar .cache/osr-pipeline/rasters/luxor.grid.json \
  --out-dir designs/.../Luxor

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../luxor.toml \
  --out designs/.../README.md
```

`scripts/regenerate-luxor.sh` chains steps 3 + drift tests into a single command.

west-asia/Egypt/Mansoura-Eg

designs/west-asia/Egypt/Mansoura-Eg/README.md

Mansoura-Eg — Urban Rail Network

Country: EG · Population: 1,000,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Mansoura-Eg rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [mansoura-eg.corridor.geojson](#).

At a glance

Metric	Value
Lines	3
Unique stations	34
Interchange stations	5
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	54.3%
Route length (double track)	56.1 km
Revenue fleet	50 × 3-car trainsets
Spare + cold-reserve	6 × 3-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	17.6 km	11	18	N Mid S Mid
line-2	16.9 km	11	17	E Mid W Mid
line-3	21.5 km	12	21	SW Outer NE Inner
Total	56.1 km	34 unique	56	

Rolling stock

Property	Value
Consist	3-car, 57 m
Max speed	90 km/h
Onboard battery	360 kWh per trainset
Seats	60 longitudinal seats
Nominal capacity (AW2)	330 pax (seated + standing, light-metro-3car per RFC 0008 §1)
Crush capacity (AW3)	420 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 330 AW2 passengers (light-metro-3car)
- Peak frequency: 12 trains/hour/direction (5-min headway)

- Peak capacity per line per direction: $330 \times 12 = 3,960$ pphpd
- Network peak throughput (all lines, both directions): $3 \text{ lines} \times 2 \text{ directions} \times 3,960 = 23,760$ passengers/hour
- Daily theoretical capacity (peak $\times 10$): $\approx 237,600$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 54,300 - 81,450$ trips/day

Catchment

- City population: 1,000,000
- Anchor-weighted coverage: 54.3%
- Catchment population: $\approx 543,000$ (within ~ 800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	5	500 kW	3000 kWh
Major	11	400 kW	2500 kWh
Standard	10	300 kW	2000 kWh
Terminal	5	500 kW	3000 kWh
Total installed	32	17,400 kW	117,500 kWh

Aggregate station-rail charging power: 19,500 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 360 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	12.0 kWh/km	3 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	224 kWh	18.7 km average line length
Onboard battery coverage	1.6 $\times$ average line run	360 kWh usable pack
Average 60 s dwell charge	9.6 kWh/stop	574 kW average charger across stops
Stops to refill one trainset pack	38 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	87 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	118 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the `[costs]` block in `design.toml` — emitted by the `osr-design` Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. `country-costs.toml` applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (53.0 km @ €3.5 M/km)	€185 M
Elevated (2.9 km @ €18 M/km)	€51 M
Elevated-interchange premium (2 sites @ €20 M)	€40 M
Civil subtotal	€277 M

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~ 11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
halt	2	€0.4 M	€0.8 M
standard	10	€1.5 M	€15 M
major	11	€3.0 M	€33 M
terminal	5	€2.5 M	€12 M
depot-terminal	1	€3.0 M	€3.0 M
interchange-elevated	5	€4.5 M	€22 M
Stations subtotal			€87 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	5	€3.0 M	€15 M
Depots subtotal			€40 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
light-metro-3car (revenue + spare + cold reserve)	56	€3.0 M	€168 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	56.1 km × €0.015 M/km	€0.8 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€13 M
EPC integration + project management (7%)	on subtotal	€41 M

Total

Bucket	Value
Civil works	€277 M
Stations	€87 M
Depots	€40 M
Rolling stock	€168 M
Residual train-control wayside + charging microgrids	€14 M
EPC overhead (7%)	€41 M
CAPEX total	€626 M
Per-route-km	€11 M / km
Per-capita (city pop)	€626 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running scripts/regenerate-city.sh mansoura-eg.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–5 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 6 and runs for 20 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–5)	€52 M / yr	€52
Steady-state, low-ridership (year 6+)	€57 M / yr	€57
Steady-state, high-ridership (year 6+)	€50 M / yr	€50
Lifecycle envelope (yr 1–25, low scenario)	€1.40 bn cumulative	€1,398
Lifecycle envelope (yr 1–25, high scenario)	€1.25 bn cumulative	€1,253

Population basis: 1,000,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 25, debt service drops to zero and only the OPEX shortfall remains – ~€9.0 M / yr (low) €1.7 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€376 M	4.5%	25 y, 5 y grace	€29 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€157 M	10.5%	25 y, 5 y grace	€19 M / yr
Government equity (no debt service)	15%	€94 M	—	—	—
Total	100%	€626 M			€48 M / yr

During the 5-year grace period the operator pays interest only – multilateral €17 M / yr + bonds €16 M / yr = €33 M / yr total – plus the equity tranche amortised across construction (€19 M / yr × 5 yr). Principal repayment begins in year 6 on a 20-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€6.7 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€8.1 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€42 k
Traction energy (101.0 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (349 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€1.4 M
OPEX subtotal		€16 M / yr

Annual fleet utilisation: 50 revenue trainsets × 20.5 h/day × 365 d/yr × 30 km/h commercial × 75% revenue factor = 8.4 M train-km / yr (~168 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$260 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.40 (~\$0.43 USD)
Day pass (3 trips)	€1.02 (15 % bulk discount)
Monthly unlimited pass	€11.96 (~5 % of median monthly income)
Annual pass	€131.56 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	18.2 M	36.5 M

	Low scenario	High scenario
Farebox revenue	€7.3 M / yr	€15 M / yr
Farebox / OPEX recovery	45%	90%
Country policy-target recovery (diagnostic)	55%	55%
Operating shortfall (gov subsidy required)	€9.0 M / yr	€1.7 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€57 M / yr	€50 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
mansoura-eg.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
mansoura-eg-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
mansoura-eg.corridor.geojson	Line polylines + stations (GeoJSON)
mansoura-eg.stations.json	Machine-readable station list
mansoura-eg.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug mansoura-eg

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug mansoura-eg \
  --sidecar .cache/osr-pipeline/rasters/mansoura-eg.grid.json \
  --out-dir designs/.../Mansoura-Eg

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../mansoura-eg.toml \
  --out designs/.../README.md
```

`scripts/regenerate-mansoura-eg.sh` chains steps 3 + drift tests into a single command.

west-asia/Egypt/Qena

designs/west-asia/Egypt/Qena/README.md

Qena — Urban Rail Network

Country: EG · Population: 350,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Qena rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [qena.corridor.geojson](#).

At a glance

Metric	Value
Lines	3
Unique stations	26
Interchange stations	4
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	75.4%
Route length (double track)	45.2 km
Revenue fleet	41 × 3-car trainsets
Spare + cold-reserve	6 × 3-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	15.7 km	9	16	NE Mid W Mid
line-2	20.3 km	10	20	W Mid SE Outer
line-3	9.1 km	7	11	E Inner NE Mid
Total	45.2 km	26 unique	47	

Rolling stock

Property	Value
Consist	3-car, 57 m
Max speed	90 km/h
Onboard battery	360 kWh per trainset
Seats	60 longitudinal seats
Nominal capacity (AW2)	330 pax (seated + standing, light-metro-3car per RFC 0008 §1)
Crush capacity (AW3)	420 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 330 AW2 passengers (light-metro-3car)
- Peak frequency: 12 trains/hour/direction (5-min headway)

- Peak capacity per line per direction: $330 \times 12 = 3,960$ pphpd
- Network peak throughput (all lines, both directions): $3 \text{ lines} \times 2 \text{ directions} \times 3,960 = 23,760$ passengers/hour
- Daily theoretical capacity (peak $\times 10$): $\approx 237,600$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 26,390 - 39,585$ trips/day

Catchment

- City population: 350,000
- Anchor-weighted coverage: 75.4%
- Catchment population: $\approx 263,900$ (within ~ 800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	4	500 kW	3000 kWh
Major	6	400 kW	2500 kWh
Standard	10	300 kW	2000 kWh
Terminal	5	500 kW	3000 kWh
Total installed	26	14,900 kW	102,000 kWh

Aggregate station-rail charging power: 16,000 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 360 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	12.0 kWh/km	3 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	181 kWh	15.1 km average line length
Onboard battery coverage	2.0 $\times$ average line run	360 kWh usable pack
Average 60 s dwell charge	10.3 kWh/stop	615 kW average charger across stops
Stops to refill one trainset pack	35 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	74 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	102 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in design.toml — emitted by the osr-design Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. country-costs.toml applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (40.6 km @ €3.5 M/km)	€142 M
Elevated (4.4 km @ €18 M/km)	€79 M
Elevated-interchange premium (2 sites @ €20 M)	€40 M
Civil subtotal	€261 M

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~ 11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
standard	10	€1.5 M	€15 M
major	6	€3.0 M	€18 M
terminal	5	€2.5 M	€12 M
depot-terminal	1	€3.0 M	€3.0 M
interchange-elevated	4	€4.5 M	€18 M
Stations subtotal			€66 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	5	€3.0 M	€15 M
Depots subtotal			€40 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
light-metro-3car (revenue + spare + cold reserve)	47	€3.0 M	€141 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	45.2 km × €0.015 M/km	€0.7 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€10 M
EPC integration + project management (7%)	on subtotal	€36 M

Total

Bucket	Value
Civil works	€261 M
Stations	€66 M
Depots	€40 M
Rolling stock	€141 M
Residual train-control wayside + charging microgrids	€11 M
EPC overhead (7%)	€36 M
CAPEX total	€556 M
Per-route-km	€12 M / km
Per-capita (city pop)	€1,587 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running scripts/regenerate-city.sh qena.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–5 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 6 and runs for 20 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–5)	€46 M / yr	€132
Steady-state, low-ridership (year 6+)	€54 M / yr	€155
Steady-state, high-ridership (year 6+)	€52 M / yr	€147
Lifecycle envelope (yr 1–25, low scenario)	€1.31 bn cumulative	€3,753
Lifecycle envelope (yr 1–25, high scenario)	€1.26 bn cumulative	€3,608

Population basis: 350,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 25, debt service drops to zero and only the OPEX shortfall remains – ~€12 M / yr (low) €9.1 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€333 M	4.5%	25 y, 5 y grace	€26 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€139 M	10.5%	25 y, 5 y grace	€17 M / yr
Government equity (no debt service)	15%	€83 M	—	—	—
Total	100%	€556 M			€43 M / yr

During the 5-year grace period the operator pays interest only – multilateral €15 M / yr + bonds €15 M / yr = €30 M / yr total – plus the equity tranche amortised across construction (€17 M / yr × 5 yr). Principal repayment begins in year 6 on a 20-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€5.6 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€7.4 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€34 k
Traction energy (82.8 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (283 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€1.1 M
OPEX subtotal		€14 M / yr

Annual fleet utilisation: 41 revenue trainsets × 20.5 h/day × 365 d/yr × 30 km/h commercial × 75% revenue factor = 6.9 M train-km / yr (~168 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$260 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.40 (~\$0.43 USD)
Day pass (3 trips)	€1.02 (15 % bulk discount)
Monthly unlimited pass	€11.96 (~5 % of median monthly income)
Annual pass	€131.56 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	6.4 M	12.8 M

	Low scenario	High scenario
Farebox revenue	€2.5 M / yr	€5.1 M / yr
Farebox / OPEX recovery	18%	36%
Country policy-target recovery (diagnostic)	55%	55%
Operating shortfall (gov subsidy required)	€12 M / yr	€9.1 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€54 M / yr	€52 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
qena.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
qena-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
qena.corridor.geojson	Line polylines + stations (GeoJSON)
qena.stations.json	Machine-readable station list
qena.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug qena

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug qena \
  --sidecar .cache/osr-pipeline/rasters/qena.grid.json \
  --out-dir designs/.../Qena

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../qena.toml \
  --out designs/.../README.md
```

`scripts/regenerate-qena.sh` chains steps 3 + drift tests into a single command.

west-asia/Egypt/Sohag

designs/west-asia/Egypt/Sohag/README.md

Sohag — Urban Rail Network

Country: EG · Population: 550,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Sohag rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [sohag.corridor.geojson](#).

At a glance

Metric	Value
Lines	3
Unique stations	32
Interchange stations	3
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	67.6%
Route length (double track)	60.3 km
Revenue fleet	53 × 3-car trainsets
Spare + cold-reserve	7 × 3-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	20.6 km	11	20	NW Mid SE Outer
line-2	22.5 km	11	23	NE Mid SW Outer
line-3	17.2 km	10	17	NW Mid SW Outer
Total	60.3 km	32 unique	60	

Rolling stock

Property	Value
Consist	3-car, 57 m
Max speed	90 km/h
Onboard battery	360 kWh per trainset
Seats	60 longitudinal seats
Nominal capacity (AW2)	330 pax (seated + standing, light-metro-3car per RFC 0008 §1)
Crush capacity (AW3)	420 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 330 AW2 passengers (light-metro-3car)
- Peak frequency: 12 trains/hour/direction (5-min headway)

- Peak capacity per line per direction: $330 \times 12 = 3,960$ pphpd
- Network peak throughput (all lines, both directions): $3 \text{ lines} \times 2 \text{ directions} \times 3,960 = 23,760$ passengers/hour
- Daily theoretical capacity (peak $\times 10$): $\approx 237,600$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 37,180 - 55,770$ trips/day

Catchment

- City population: 550,000
- Anchor-weighted coverage: 67.6%
- Catchment population: $\approx 371,800$ (within ~ 800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	3	500 kW	3000 kWh
Major	9	400 kW	2500 kWh
Standard	11	300 kW	2000 kWh
Terminal	5	500 kW	3000 kWh
Total installed	29	15,900 kW	108,500 kWh

Aggregate station-rail charging power: 18,250 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 360 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	12.0 kWh/km	3 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	241 kWh	20.1 km average line length
Onboard battery coverage	1.5 $\times$ average line run	360 kWh usable pack
Average 60 s dwell charge	9.5 kWh/stop	570 kW average charger across stops
Stops to refill one trainset pack	38 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	80 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	108 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in design.toml — emitted by the osr-design Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. country-costs.toml applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (51.7 km @ €3.5 M/km)	€181 M
Elevated (8.3 km @ €18 M/km)	€149 M
Elevated-interchange premium (2 sites @ €20 M)	€40 M
Civil subtotal	€370 M

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~ 11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
halt	3	€0.4 M	€1.2 M
standard	11	€1.5 M	€16 M
major	9	€3.0 M	€27 M
terminal	5	€2.5 M	€12 M
depot-terminal	1	€3.0 M	€3.0 M
interchange	3	€4.5 M	€14 M
Stations subtotal			€74 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	5	€3.0 M	€15 M
Depots subtotal			€40 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
light-metro-3car (revenue + spare + cold reserve)	60	€3.0 M	€180 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	60.3 km × €0.015 M/km	€0.9 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€11 M
EPC integration + project management (7%)	on subtotal	€47 M

Total

Bucket	Value
Civil works	€370 M
Stations	€74 M
Depots	€40 M
Rolling stock	€180 M
Residual train-control wayside + charging microgrids	€12 M
EPC overhead (7%)	€47 M
CAPEX total	€724 M
Per-route-km	€12 M / km
Per-capita (city pop)	€1,315 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running scripts/regenerate-city.sh sohag.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–5 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 6 and runs for 20 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–5)	€60 M / yr	€110
Steady-state, low-ridership (year 6+)	€70 M / yr	€127
Steady-state, high-ridership (year 6+)	€66 M / yr	€120
Lifecycle envelope (yr 1–25, low scenario)	€1.70 bn cumulative	€3,085
Lifecycle envelope (yr 1–25, high scenario)	€1.62 bn cumulative	€2,939

Population basis: 550,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 25, debt service drops to zero and only the OPEX shortfall remains – ~€14 M / yr (low) €10 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€434 M	4.5%	25 y, 5 y grace	€33 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€181 M	10.5%	25 y, 5 y grace	€22 M / yr
Government equity (no debt service)	15%	€109 M	—	—	—
Total	100%	€724 M			€55 M / yr

During the 5-year grace period the operator pays interest only – multilateral €20 M / yr + bonds €19 M / yr = €39 M / yr total – plus the equity tranche amortised across construction (€22 M / yr × 5 yr). Principal repayment begins in year 6 on a 20-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€7.2 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€9.7 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€45 k
Traction energy (107.1 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (374 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€1.5 M
OPEX subtotal		€18 M / yr

Annual fleet utilisation: 53 revenue trainsets × 20.5 h/day × 365 d/yr × 30 km/h commercial × 75% revenue factor = 8.9 M train-km / yr (~168 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$260 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.40 (~\$0.43 USD)
Day pass (3 trips)	€1.02 (15 % bulk discount)
Monthly unlimited pass	€11.96 (~5 % of median monthly income)
Annual pass	€131.56 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	10.0 M	20.1 M

	Low scenario	High scenario
Farebox revenue	€4.0 M / yr	€8.0 M / yr
Farebox / OPEX recovery	22%	43%
Country policy-target recovery (diagnostic)	55%	55%
Operating shortfall (gov subsidy required)	€14 M / yr	€10 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€70 M / yr	€66 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
sohag.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
sohag-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
sohag.corridor.geojson	Line polylines + stations (GeoJSON)
sohag.stations.json	Machine-readable station list
sohag.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug sohag

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug sohag \
  --sidecar .cache/osr-pipeline/rasters/sohag.grid.json \
  --out-dir designs/.../Sohag

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../sohag.toml \
  --out designs/.../README.md
```

`scripts/regenerate-sohag.sh` chains steps 3 + drift tests into a single command.

west-asia/Egypt/Tanta

designs/west-asia/Egypt/Tanta/README.md

Tanta — Urban Rail Network

Country: EG · Population: 750,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Tanta rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [tanta.corridor.geojson](#).

At a glance

Metric	Value
Lines	3
Unique stations	35
Interchange stations	3
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	56.0%
Route length (double track)	74.9 km
Revenue fleet	65 × 3-car trainsets
Spare + cold-reserve	8 × 3-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	29.2 km	14	28	SE Outer NW Outer
line-2	20.5 km	10	20	N Mid SW Outer
line-3	25.3 km	11	25	NE Outer SW Outer
Total	74.9 km	35 unique	73	

Rolling stock

Property	Value
Consist	3-car, 57 m
Max speed	90 km/h
Onboard battery	360 kWh per trainset
Seats	60 longitudinal seats
Nominal capacity (AW2)	330 pax (seated + standing, light-metro-3car per RFC 0008 §1)
Crush capacity (AW3)	420 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 330 AW2 passengers (light-metro-3car)
- Peak frequency: 12 trains/hour/direction (5-min headway)

- Peak capacity per line per direction: $330 \times 12 = 3,960$ pphpd
- Network peak throughput (all lines, both directions): $3 \text{ lines} \times 2 \text{ directions} \times 3,960 = 23,760$ passengers/hour
- Daily theoretical capacity (peak $\times 10$): $\approx 237,600$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 42,000 - 63,000$ trips/day

Catchment

- City population: 750,000
- Anchor-weighted coverage: 56.0%
- Catchment population: $\approx 420,000$ (within ~ 800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	3	500 kW	3000 kWh
Major	9	400 kW	2500 kWh
Standard	13	300 kW	2000 kWh
Terminal	5	500 kW	3000 kWh
Total installed	31	16,500 kW	112,500 kWh

Aggregate station-rail charging power: 19,500 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 360 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	12.0 kWh/km	3 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	300 kWh	25.0 km average line length
Onboard battery coverage	1.2 $\times$ average line run	360 kWh usable pack
Average 60 s dwell charge	9.3 kWh/stop	557 kW average charger across stops
Stops to refill one trainset pack	39 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	82 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	112 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in `design.toml` — emitted by the `osr-design` Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. `country-costs.toml` applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (73.4 km @ €3.5 M/km)	€257 M
Elevated (1.2 km @ €18 M/km)	€22 M
Elevated-interchange premium (1 sites @ €20 M)	€20 M
Civil subtotal	€299 M

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~ 11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
halt	4	€0.4 M	€1.6 M
standard	13	€1.5 M	€20 M
major	9	€3.0 M	€27 M
terminal	5	€2.5 M	€12 M
depot-terminal	1	€3.0 M	€3.0 M
interchange-elevated	3	€4.5 M	€14 M
Stations subtotal			€77 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	5	€3.0 M	€15 M
Depots subtotal			€40 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
light-metro-3car (revenue + spare + cold reserve)	73	€3.0 M	€219 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	74.9 km × €0.015 M/km	€1.1 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€12 M
EPC integration + project management (7%)	on subtotal	€45 M

Total

Bucket	Value
Civil works	€299 M
Stations	€77 M
Depots	€40 M
Rolling stock	€219 M
Residual train-control wayside + charging microgrids	€13 M
EPC overhead (7%)	€45 M
CAPEX total	€694 M
Per-route-km	€9.3 M / km
Per-capita (city pop)	€925 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running scripts/regenerate-city.sh tanta.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–5 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 6 and runs for 20 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–5)	€58 M / yr	€77
Steady-state, low-ridership (year 6+)	€67 M / yr	€89
Steady-state, high-ridership (year 6+)	€61 M / yr	€82
Lifecycle envelope (yr 1–25, low scenario)	€1.62 bn cumulative	€2,162
Lifecycle envelope (yr 1–25, high scenario)	€1.51 bn cumulative	€2,016

Population basis: 750,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 25, debt service drops to zero and only the OPEX shortfall remains – ~€14 M / yr (low) €8.1 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€416 M	4.5%	25 y, 5 y grace	€32 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€173 M	10.5%	25 y, 5 y grace	€21 M / yr
Government equity (no debt service)	15%	€104 M	—	—	—
Total	100%	€694 M			€53 M / yr

During the 5-year grace period the operator pays interest only – multilateral €19 M / yr + bonds €18 M / yr = €37 M / yr total – plus the equity tranche amortised across construction (€21 M / yr × 5 yr). Principal repayment begins in year 6 on a 20-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€8.8 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€8.3 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€56 k
Traction energy (131.3 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (461 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€1.9 M
OPEX subtotal		€19 M / yr

Annual fleet utilisation: 65 revenue trainsets × 20.5 h/day × 365 d/yr × 30 km/h commercial × 75% revenue factor = 10.9 M train-km / yr (~168 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$260 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.40 (~\$0.43 USD)
Day pass (3 trips)	€1.02 (15 % bulk discount)
Monthly unlimited pass	€11.96 (~5 % of median monthly income)
Annual pass	€131.56 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	13.7 M	27.4 M

	Low scenario	High scenario
Farebox revenue	€5.5 M / yr	€11 M / yr
Farebox / OPEX recovery	29%	57%
Country policy-target recovery (diagnostic)	55%	55%
Operating shortfall (gov subsidy required)	€14 M / yr	€8.1 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€67 M / yr	€61 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
tanta.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
tanta-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
tanta.corridor.geojson	Line polylines + stations (GeoJSON)
tanta.stations.json	Machine-readable station list
tanta.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug tanta

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug tanta \
  --sidecar .cache/osr-pipeline/rasters/tanta.grid.json \
  --out-dir designs/.../Tanta

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../tanta.toml \
  --out designs/.../README.md
```

`scripts/regenerate-tanta.sh` chains steps 3 + drift tests into a single command.

west-asia/Egypt/Zagazig

designs/west-asia/Egypt/Zagazig/README.md

Zagazig — Urban Rail Network

Country: EG · Population: 700,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Zagazig rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [zagazig.corridor.geojson](#).

At a glance

Metric	Value
Lines	3
Unique stations	30
Interchange stations	3
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	59.3%
Route length (double track)	56.2 km
Revenue fleet	51 × 3-car trainsets
Spare + cold-reserve	6 × 3-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	18.6 km	12	19	SE Mid W Outer
line-2	17.5 km	9	18	NE Outer SW Inner
line-3	20.1 km	9	20	NW Mid SE Outer
Total	56.2 km	30 unique	57	

Rolling stock

Property	Value
Consist	3-car, 57 m
Max speed	90 km/h
Onboard battery	360 kWh per trainset
Seats	60 longitudinal seats
Nominal capacity (AW2)	330 pax (seated + standing, light-metro-3car per RFC 0008 §1)
Crush capacity (AW3)	420 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 330 AW2 passengers (light-metro-3car)
- Peak frequency: 12 trains/hour/direction (5-min headway)

- Peak capacity per line per direction: $330 \times 12 = 3,960$ pphpd
- Network peak throughput (all lines, both directions): $3 \text{ lines} \times 2 \text{ directions} \times 3,960 = 23,760$ passengers/hour
- Daily theoretical capacity (peak $\times 10$): $\approx 237,600$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 41,510 - 62,265$ trips/day

Catchment

- City population: 700,000
- Anchor-weighted coverage: 59.3%
- Catchment population: $\approx 415,100$ (within ~ 800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	3	500 kW	3000 kWh
Major	9	400 kW	2500 kWh
Standard	9	300 kW	2000 kWh
Terminal	5	500 kW	3000 kWh
Total installed	27	15,300 kW	104,500 kWh

Aggregate station-rail charging power: 17,250 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 360 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	12.0 kWh/km	3 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	225 kWh	18.7 km average line length
Onboard battery coverage	1.6 $\times$ average line run	360 kWh usable pack
Average 60 s dwell charge	9.6 kWh/stop	575 kW average charger across stops
Stops to refill one trainset pack	38 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	76 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	104 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in design.toml — emitted by the osr-design Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. country-costs.toml applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (54.9 km @ €3.5 M/km)	€192 M
Elevated (1.3 km @ €18 M/km)	€23 M
Elevated-interchange premium (1 sites @ €20 M)	€20 M
Civil subtotal	€235 M

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~ 11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
halt	3	€0.4 M	€1.2 M
standard	9	€1.5 M	€14 M
major	9	€3.0 M	€27 M
terminal	5	€2.5 M	€12 M
depot-terminal	1	€3.0 M	€3.0 M
interchange-elevated	3	€4.5 M	€14 M
Stations subtotal			€71 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	5	€3.0 M	€15 M
Depots subtotal			€40 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
light-metro-3car (revenue + spare + cold reserve)	57	€3.0 M	€171 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	56.2 km × €0.015 M/km	€0.8 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€11 M
EPC integration + project management (7%)	on subtotal	€37 M

Total

Bucket	Value
Civil works	€235 M
Stations	€71 M
Depots	€40 M
Rolling stock	€171 M
Residual train-control wayside + charging microgrids	€12 M
EPC overhead (7%)	€37 M
CAPEX total	€566 M
Per-route-km	€10 M / km
Per-capita (city pop)	€808 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running scripts/regenerate-city.sh zagazig.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–5 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 6 and runs for 20 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–5)	€47 M / yr	€67
Steady-state, low-ridership (year 6+)	€53 M / yr	€76
Steady-state, high-ridership (year 6+)	€48 M / yr	€69
Lifecycle envelope (yr 1–25, low scenario)	€1.30 bn cumulative	€1,862
Lifecycle envelope (yr 1–25, high scenario)	€1.20 bn cumulative	€1,716

Population basis: 700,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 25, debt service drops to zero and only the OPEX shortfall remains – ~€10 M / yr (low) €5.0 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€339 M	4.5%	25 y, 5 y grace	€26 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€141 M	10.5%	25 y, 5 y grace	€17 M / yr
Government equity (no debt service)	15%	€85 M	—	—	—
Total	100%	€566 M			€43 M / yr

During the 5-year grace period the operator pays interest only – multilateral €15 M / yr + bonds €15 M / yr = €30 M / yr total – plus the equity tranche amortised across construction (€17 M / yr × 5 yr). Principal repayment begins in year 6 on a 20-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€6.8 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€6.9 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€42 k
Traction energy (103.0 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (349 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€1.4 M
OPEX subtotal		€15 M / yr

Annual fleet utilisation: 51 revenue trainsets × 20.5 h/day × 365 d/yr × 30 km/h commercial × 75% revenue factor = 8.6 M train-km / yr (~168 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$260 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.40 (~\$0.43 USD)
Day pass (3 trips)	€1.02 (15 % bulk discount)
Monthly unlimited pass	€11.96 (~5 % of median monthly income)
Annual pass	€131.56 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	12.8 M	25.6 M

	Low scenario	High scenario
Farebox revenue	€5.1 M / yr	€10 M / yr
Farebox / OPEX recovery	33%	67%
Country policy-target recovery (diagnostic)	55%	55%
Operating shortfall (gov subsidy required)	€10 M / yr	€5.0 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€53 M / yr	€48 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
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zagazig.corridor.geojson	Line polylines + stations (GeoJSON)
zagazig.stations.json	Machine-readable station list
zagazig.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug zagazig

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug zagazig \
  --sidecar .cache/osr-pipeline/rasters/zagazig.grid.json \
  --out-dir designs/.../Zagazig

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../zagazig.toml \
  --out designs/.../README.md
```

`scripts/regenerate-zagazig.sh` chains steps 3 + drift tests into a single command.

west-asia/Iraq/Amarah

designs/west-asia/Iraq/Amarah/README.md

Amarah — Urban Rail Network

Country: IQ · Population: 660,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Amarah rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [amarah.corridor.geojson](#).

At a glance

Metric	Value
Lines	3
Unique stations	32
Interchange stations	2
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	43.1%
Route length (double track)	44.7 km
Revenue fleet	40 × 3-car trainsets
Spare + cold-reserve	6 × 3-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	14.4 km	12	15	SE Mid NW Mid
line-2	11.9 km	10	13	NE Mid W Mid
line-3	18.4 km	10	18	SE Outer W Mid
Total	44.7 km	32 unique	46	

Rolling stock

Property	Value
Consist	3-car, 57 m
Max speed	90 km/h
Onboard battery	360 kWh per trainset
Seats	60 longitudinal seats
Nominal capacity (AW2)	330 pax (seated + standing, light-metro-3car per RFC 0008 §1)
Crush capacity (AW3)	420 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 330 AW2 passengers (light-metro-3car)
- Peak frequency: 12 trains/hour/direction (5-min headway)

- Peak capacity per line per direction: $330 \times 12 = 3,960$ pphpd
- Network peak throughput (all lines, both directions): $3 \text{ lines} \times 2 \text{ directions} \times 3,960 = 23,760$ passengers/hour
- Daily theoretical capacity (peak $\times 10$): $\approx 237,600$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 28,446 - 42,669$ trips/day

Catchment

- City population: 660,000
- Anchor-weighted coverage: 43.1%
- Catchment population: $\approx 284,460$ (within ~ 800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	2	500 kW	3000 kWh
Major	20	400 kW	2500 kWh
Standard	3	300 kW	2000 kWh
Terminal	5	500 kW	3000 kWh
Total installed	31	17,400 kW	117,000 kWh

Aggregate station-rail charging power: 18,750 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 360 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	12.0 kWh/km	3 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	179 kWh	14.9 km average line length
Onboard battery coverage	2.0 $\times$ average line run	360 kWh usable pack
Average 60 s dwell charge	9.8 kWh/stop	586 kW average charger across stops
Stops to refill one trainset pack	37 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	87 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	117 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in design.toml — emitted by the osr-design Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. country-costs.toml applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (30.3 km @ €3.5 M/km)	€106 M
Elevated (14.1 km @ €18 M/km)	€254 M
Elevated-interchange premium (1 sites @ €20 M)	€20 M
Civil subtotal	€380 M

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~ 11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
halt	1	€0.4 M	€0.4 M
standard	3	€1.5 M	€4.5 M
major	20	€3.0 M	€60 M
terminal	5	€2.5 M	€12 M
depot-terminal	1	€3.0 M	€3.0 M
interchange-elevated	2	€4.5 M	€9.0 M
Stations subtotal			€89 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	5	€3.0 M	€15 M
Depots subtotal			€40 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
light-metro-3car (revenue + spare + cold reserve)	46	€3.0 M	€138 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	44.7 km × €0.015 M/km	€0.7 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€13 M
EPC integration + project management (7%)	on subtotal	€46 M

Total

Bucket	Value
Civil works	€380 M
Stations	€89 M
Depots	€40 M
Rolling stock	€138 M
Residual train-control wayside + charging microgrids	€13 M
EPC overhead (7%)	€46 M
CAPEX total	€707 M
Per-route-km	€16 M / km
Per-capita (city pop)	€1,071 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running scripts/regenerate-city.sh amarah.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–5 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 6 and runs for 20 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–5)	€53 M / yr	€81
Steady-state, low-ridership (year 6+)	€60 M / yr	€91
Steady-state, high-ridership (year 6+)	€53 M / yr	€81
Lifecycle envelope (yr 1–25, low scenario)	€1.47 bn cumulative	€2,229
Lifecycle envelope (yr 1–25, high scenario)	€1.33 bn cumulative	€2,016

Population basis: 660,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 25, debt service drops to zero and only the OPEX shortfall remains – ~€10 M / yr (low) €3.3 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€424 M	4.0%	25 y, 5 y grace	€31 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€177 M	8.5%	25 y, 5 y grace	€19 M / yr
Government equity (no debt service)	15%	€106 M	—	—	—
Total	100%	€707 M			€50 M / yr

During the 5-year grace period the operator pays interest only – multilateral €17 M / yr + bonds €15 M / yr = €32 M / yr total – plus the equity tranche amortised across construction (€21 M / yr × 5 yr). Principal repayment begins in year 6 on a 20-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€5.5 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€10 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€33 k
Traction energy (80.8 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (280 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€1.6 M
OPEX subtotal		€17 M / yr

Annual fleet utilisation: 40 revenue trainsets × 20.5 h/day × 365 d/yr × 30 km/h commercial × 75% revenue factor = 6.7 M train-km / yr (~168 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$380 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.58 (~\$0.63 USD)
Day pass (3 trips)	€1.49 (15 % bulk discount)
Monthly unlimited pass	€17.48 (~5 % of median monthly income)
Annual pass	€192.28 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	12.0 M	24.1 M

	Low scenario	High scenario
Farebox revenue	€7.0 M / yr	€14 M / yr
Farebox / OPEX recovery	40%	81%
Country policy-target recovery (diagnostic)	45%	45%
Operating shortfall (gov subsidy required)	€10 M / yr	€3.3 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€60 M / yr	€53 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
amarah.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
amarah-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
amarah.corridor.geojson	Line polylines + stations (GeoJSON)
amarah.stations.json	Machine-readable station list
amarah.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug amarah

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug amarah \
  --sidecar .cache/osr-pipeline/rasters/amarah.grid.json \
  --out-dir designs/.../Amarah

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../amarah.toml \
  --out designs/.../README.md
```

`scripts/regenerate-amarah.sh` chains steps 3 + drift tests into a single command.

west-asia/Iraq/Baghdad

designs/west-asia/Iraq/Baghdad/README.md

Baghdad — Urban Rail Network

Country: IQ · Population: 9,780,429

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Baghdad rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [baghdad.corridor.geojson](#).

At a glance

Metric	Value
Lines	9
Unique stations	217
Interchange stations	37
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	44.9%
Route length (double track)	509.5 km
Revenue fleet	364 × 6-car trainsets
Spare + cold-reserve	44 × 6-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	55.7 km	23	45	N Outer S Outer
line-2	52.0 km	26	41	SE Outer NW Outer
line-3	56.3 km	23	45	SW Outer NE Outer
line-4	56.8 km	22	46	SE Outer NW Mid
line-5	57.6 km	24	46	E Outer W Outer
line-6	41.9 km	17	34	W Mid E Mid
line-7	46.7 km	21	38	NE Mid SW Outer
line-8	42.9 km	14	35	NW Mid S Outer
line-9	99.7 km	48	78	NW Mid NW Mid
Total	509.5 km	217 unique	408	

Rolling stock

Property	Value
Consist	6-car, 111 m
Max speed	100 km/h
Onboard battery	720 kWh per trainset
Seats	120 longitudinal seats
Nominal capacity (AW2)	660 pax (seated + standing, metro-6car per RFC 0008 §1)
Crush capacity (AW3)	840 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 660 AW2 passengers (metro-6car)
- Peak frequency: 12 trains/hour/direction (5-min headway)
- Peak capacity per line per direction: $660 \times 12 = 7,920$ pphpd
- Network peak throughput (all lines, both directions): $9 \text{ lines} \times 2 \text{ directions} \times 7,920 = 142,560$ passengers/hour
- Daily theoretical capacity (peak $\times$ 10): $\approx 1,425,600$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 439,141 - 658,711$ trips/day

Catchment

- City population: 9,780,429
- Anchor-weighted coverage: 44.9%
- Catchment population: $\approx 4,391,412$ (within ~ 800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	37	500 kW	3000 kWh
Major	53	400 kW	2500 kWh
Standard	99	300 kW	2000 kWh
Terminal	15	500 kW	3000 kWh
Total installed	205	81,900 kW	526,500 kWh

Aggregate station-rail charging power: 113,750 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 720 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	24.0 kWh/km	6 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	1,359 kWh	56.6 km average line length
Onboard battery coverage	0.5 $\times$ average line run	720 kWh usable pack
Average 60 s dwell charge	8.7 kWh/stop	524 kW average charger across stops
Stops to refill one trainset pack	82 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	410 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	526 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the `[costs]` block in `design.toml` — emitted by the `osr-design` Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. `country-costs.toml` applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (476.2 km @ €3.5 M/km)	€1.67 bn
Elevated (31.4 km @ €18 M/km)	€565 M
Elevated-interchange premium (19 sites @ €20 M)	€380 M
Civil subtotal	€2.61 bn

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
halt	13	€0.4 M	€5.2 M
standard	99	€1.5 M	€148 M
major	53	€3.0 M	€159 M
terminal	15	€2.5 M	€38 M
depot-terminal	1	€3.0 M	€3.0 M
interchange-elevated	37	€4.5 M	€166 M
Stations subtotal			€520 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	15	€3.0 M	€45 M
Depots subtotal			€70 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
metro-6car (revenue + spare + cold reserve)	408	€6.0 M	€2.45 bn

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	509.5 km × €0.015 M/km	€7.6 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€77 M
EPC integration + project management (7%)	on subtotal	€401 M

Total

Bucket	Value
Civil works	€2.61 bn
Stations	€520 M
Depots	€70 M
Rolling stock	€2.45 bn
Residual train-control wayside + charging microgrids	€84 M
EPC overhead (7%)	€401 M
CAPEX total	€6.14 bn
Per-route-km	€12 M / km

Bucket	Value
Per-capita (city pop)	€627 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](https://github.com/OpenSourceRail/country-finance.toml). Pure function of the [costs] block above + the country code — regenerate by re-running `scripts/regenerate-city.sh` baghdad.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–5 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 6 and runs for 20 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–5)	€462 M / yr	€47
Steady-state, low-ridership (year 6+)	€509 M / yr	€52
Steady-state, high-ridership (year 6+)	€433 M / yr	€44
Lifecycle envelope (yr 1–25, low scenario)	€12.49 bn cumulative	€1,277
Lifecycle envelope (yr 1–25, high scenario)	€10.97 bn cumulative	€1,121

Population basis: 9,780,429 (catchment per [lib/city-batches/world-sample.toml](https://github.com/OpenSourceRail/city-batches/world-sample.toml)). After year 25, debt service drops to zero and only the OPEX shortfall remains — ~€76 M / yr (low) €0 k / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€3.68 bn	4.0%	25 y, 5 y grace	€271 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€1.53 bn	8.5%	25 y, 5 y grace	€162 M / yr
Government equity (no debt service)	15%	€920 M	—	—	—
Total	100%	€6.14 bn			€433 M / yr

During the 5-year grace period the operator pays interest only — multilateral €147 M / yr + bonds €130 M / yr = €278 M / yr total — plus the equity tranche amortised across construction (€184 M / yr × 5 yr). Principal repayment begins in year 6 on a 20-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€98 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€64 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€381 k
Traction energy (1715.9 GWh / yr)	trackside PV + Na-ion (RFC 0002) — self-generated, €0 / yr	€0 k
Labour (3,069 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€18 M
OPEX subtotal		€180 M / yr

Annual fleet utilisation: 364 revenue trainsets × 20.5 h/day × 365 d/yr × 35 km/h commercial × 75% revenue factor = 71.5 M train-km / yr (~196 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$380 USD (per [lib/templates/country-finance.toml](https://github.com/OpenSourceRail/country-finance.toml)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass — a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.58 (~\$0.63 USD)
Day pass (3 trips)	€1.49 (15 % bulk discount)
Monthly unlimited pass	€17.48 (~5 % of median monthly income)
Annual pass	€192.28 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	178.5 M	357.0 M
Farebox revenue	€104 M / yr	€208 M / yr
Farebox / OPEX recovery	58%	115%
Country policy-target recovery (diagnostic)	45%	45%
Operating shortfall (gov subsidy required)	€76 M / yr	€0 k / yr
Operating surplus (operator retained capex sinking fund)	€0 k / yr	€28 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€509 M / yr	€433 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
baghdad.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
baghdad-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
baghdad.corridor.geojson	Line polylines + stations (GeoJSON)
baghdad.stations.json	Machine-readable station list
baghdad.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug baghdad

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug baghdad \
  --sidecar .cache/osr-pipeline/rasters/baghdad.grid.json \
  --out-dir designs/.../Baghdad

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../baghdad.toml \
  --out designs/.../README.md
```

`scripts/regenerate-baghdad.sh` chains steps 3 + drift tests into a single command.

west-asia/Iraq/Baqubah

designs/west-asia/Iraq/Baqubah/README.md

Baqubah — Urban Rail Network

Country: IQ · Population: 470,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Baqubah rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [baqubah.corridor.geojson](#).

At a glance

Metric	Value
Lines	3
Unique stations	37
Interchange stations	3
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	50.0%
Route length (double track)	59.9 km
Revenue fleet	53 × 3-car trainsets
Spare + cold-reserve	7 × 3-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	19.1 km	13	19	W Outer E Mid
line-2	24.2 km	13	24	S Outer NE Outer
line-3	16.7 km	11	17	NW Outer SE Mid
Total	59.9 km	37 unique	60	

Rolling stock

Property	Value
Consist	3-car, 57 m
Max speed	90 km/h
Onboard battery	360 kWh per trainset
Seats	60 longitudinal seats
Nominal capacity (AW2)	330 pax (seated + standing, light-metro-3car per RFC 0008 §1)
Crush capacity (AW3)	420 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 330 AW2 passengers (light-metro-3car)
- Peak frequency: 12 trains/hour/direction (5-min headway)

- Peak capacity per line per direction: $330 \times 12 = 3,960$ pphpd
- Network peak throughput (all lines, both directions): $3 \text{ lines} \times 2 \text{ directions} \times 3,960 = 23,760$ passengers/hour
- Daily theoretical capacity (peak $\times 10$): $\approx 237,600$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 23,500 - 35,250$ trips/day

Catchment

- City population: 470,000
- Anchor-weighted coverage: 50.0%
- Catchment population: $\approx 235,000$ (within ~ 800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	3	500 kW	3000 kWh
Major	13	400 kW	2500 kWh
Standard	15	300 kW	2000 kWh
Terminal	5	500 kW	3000 kWh
Total installed	37	18,700 kW	126,500 kWh

Aggregate station-rail charging power: 21,500 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 360 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	12.0 kWh/km	3 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	240 kWh	20.0 km average line length
Onboard battery coverage	1.5 $\times$ average line run	360 kWh usable pack
Average 60 s dwell charge	9.7 kWh/stop	581 kW average charger across stops
Stops to refill one trainset pack	37 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	94 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	126 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in design.toml — emitted by the osr-design Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. country-costs.toml applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (56.5 km @ €3.5 M/km)	€198 M
Elevated (3.0 km @ €18 M/km)	€55 M
Elevated-interchange premium (1 sites @ €20 M)	€20 M
Civil subtotal	€272 M

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~ 11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
standard	15	€1.5 M	€22 M
major	13	€3.0 M	€39 M
terminal	5	€2.5 M	€12 M
depot-terminal	1	€3.0 M	€3.0 M
interchange-elevated	3	€4.5 M	€14 M
Stations subtotal			€90 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	5	€3.0 M	€15 M
Depots subtotal			€40 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
light-metro-3car (revenue + spare + cold reserve)	60	€3.0 M	€180 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	59.9 km × €0.015 M/km	€0.9 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€14 M
EPC integration + project management (7%)	on subtotal	€42 M

Total

Bucket	Value
Civil works	€272 M
Stations	€90 M
Depots	€40 M
Rolling stock	€180 M
Residual train-control wayside + charging microgrids	€14 M
EPC overhead (7%)	€42 M
CAPEX total	€639 M
Per-route-km	€11 M / km
Per-capita (city pop)	€1,360 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running scripts/regenerate-city.sh baqubah.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–5 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 6 and runs for 20 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–5)	€48 M / yr	€102
Steady-state, low-ridership (year 6+)	€58 M / yr	€123
Steady-state, high-ridership (year 6+)	€53 M / yr	€112
Lifecycle envelope (yr 1–25, low scenario)	€1.39 bn cumulative	€2,962
Lifecycle envelope (yr 1–25, high scenario)	€1.29 bn cumulative	€2,749

Population basis: 470,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 25, debt service drops to zero and only the OPEX shortfall remains – ~€12 M / yr (low) €7.5 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€383 M	4.0%	25 y, 5 y grace	€28 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€160 M	8.5%	25 y, 5 y grace	€17 M / yr
Government equity (no debt service)	15%	€96 M	—	—	—
Total	100%	€639 M			€45 M / yr

During the 5-year grace period the operator pays interest only – multilateral €15 M / yr + bonds €14 M / yr = €29 M / yr total – plus the equity tranche amortised across construction (€19 M / yr × 5 yr). Principal repayment begins in year 6 on a 20-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€7.2 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€8.1 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€45 k
Traction energy (107.1 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (371 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€2.2 M
OPEX subtotal		€17 M / yr

Annual fleet utilisation: 53 revenue trainsets × 20.5 h/day × 365 d/yr × 30 km/h commercial × 75% revenue factor = 8.9 M train-km / yr (~168 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$380 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.58 (~\$0.63 USD)
Day pass (3 trips)	€1.49 (15 % bulk discount)
Monthly unlimited pass	€17.48 (~5 % of median monthly income)
Annual pass	€192.28 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	8.6 M	17.2 M

	Low scenario	High scenario
Farebox revenue	€5.0 M / yr	€10.0 M / yr
Farebox / OPEX recovery	29%	57%
Country policy-target recovery (diagnostic)	45%	45%
Operating shortfall (gov subsidy required)	€12 M / yr	€7.5 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€58 M / yr	€53 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
baqubah.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
baqubah-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
baqubah.corridor.geojson	Line polylines + stations (GeoJSON)
baqubah.stations.json	Machine-readable station list
baqubah.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug baqubah

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug baqubah \
  --sidecar .cache/osr-pipeline/rasters/baqubah.grid.json \
  --out-dir designs/.../Baqubah

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../baqubah.toml \
  --out designs/.../README.md
```

`scripts/regenerate-baqubah.sh` chains steps 3 + drift tests into a single command.

west-asia/Iraq/Basra

designs/west-asia/Iraq/Basra/README.md

Basra — Urban Rail Network

Country: IQ · Population: 3,955,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Basra rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [basra.corridor.geojson](#).

At a glance

Metric	Value
Lines	7
Unique stations	118
Interchange stations	18
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	54.2%
Route length (double track)	289.0 km
Revenue fleet	211 $\times$ 6-car trainsets
Spare + cold-reserve	25 $\times$ 6-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 ($\approx$ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	46.5 km	20	38	N Outer SE Mid
line-2	35.4 km	17	29	SE Outer NW Inner
line-3	30.8 km	16	26	NE Mid SW Mid
line-4	34.4 km	13	28	E Inner SW Outer
line-5	27.7 km	14	24	S Inner N Mid
line-6	33.8 km	13	28	SW Outer NE Inner
line-7	80.4 km	26	63	NW Mid NW Mid
Total	289.0 km	118 unique	236	

Rolling stock

Property	Value
Consist	6-car, 111 m
Max speed	100 km/h
Onboard battery	720 kWh per trainset
Seats	120 longitudinal seats
Nominal capacity (AW2)	660 pax (seated + standing, metro-6car per RFC 0008 §1)
Crush capacity (AW3)	840 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 660 AW2 passengers (metro - 6 car)
- Peak frequency: 12 trains/hour/direction (5-min headway)
- Peak capacity per line per direction: $660 \times 12 = 7,920$ pphpd
- Network peak throughput (all lines, both directions): $7 \text{ lines} \times 2 \text{ directions} \times 7,920 = 110,880$ passengers/hour
- Daily theoretical capacity (peak $\times$ 10): $\approx 1,108,800$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 214,361 - 321,541$ trips/day

Catchment

- City population: 3,955,000
- Anchor-weighted coverage: 54.2%
- Catchment population: $\approx 2,143,610$ (within ~ 800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	18	500 kW	3000 kWh
Major	36	400 kW	2500 kWh
Standard	34	300 kW	2000 kWh
Terminal	11	500 kW	3000 kWh
Total installed	100	44,100 kW	285,000 kWh

Aggregate station-rail charging power: 60,750 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 720 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	24.0 kWh/km	6 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	991 kWh	41.3 km average line length
Onboard battery coverage	0.7 $\times$ average line run	720 kWh usable pack
Average 60 s dwell charge	8.6 kWh/stop	515 kW average charger across stops
Stops to refill one trainset pack	84 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	220 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	285 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the `[costs]` block in `design.toml` — emitted by the `osr-design` Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. `country-costs.toml` applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (250.2 km @ €3.5 M/km)	€876 M
Elevated (37.2 km @ €18 M/km)	€669 M
Elevated-interchange premium (8 sites @ €20 M)	€160 M
Civil subtotal	€1.71 bn

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
halt	19	€0.4 M	€7.6 M
standard	34	€1.5 M	€51 M
major	36	€3.0 M	€108 M
terminal	11	€2.5 M	€28 M
depot-terminal	1	€3.0 M	€3.0 M
interchange	4	€4.5 M	€18 M
interchange-elevated	14	€4.5 M	€63 M
Stations subtotal			€278 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	11	€3.0 M	€33 M
Depots subtotal			€58 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
metro-6car (revenue + spare + cold reserve)	236	€6.0 M	€1.42 bn

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	289.0 km × €0.015 M/km	€4.3 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€41 M
EPC integration + project management (7%)	on subtotal	€245 M

Total

Bucket	Value
Civil works	€1.71 bn
Stations	€278 M
Depots	€58 M
Rolling stock	€1.42 bn
Residual train-control wayside + charging microgrids	€46 M
EPC overhead (7%)	€245 M
CAPEX total	€3.75 bn

Bucket	Value
Per-route-km	€13 M / km
Per-capita (city pop)	€948 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code — regenerate by re-running `scripts/regenerate-city.sh` `basra`.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–5 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 6 and runs for 20 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–5)	€282 M / yr	€71
Steady-state, low-ridership (year 6+)	€330 M / yr	€84
Steady-state, high-ridership (year 6+)	€288 M / yr	€73
Lifecycle envelope (yr 1–25, low scenario)	€8.02 bn cumulative	€2,027
Lifecycle envelope (yr 1–25, high scenario)	€7.18 bn cumulative	€1,815

Population basis: 3,955,000 (catchment per `lib/city-batches/world-sample.toml`). After year 25, debt service drops to zero and only the OPEX shortfall remains — ~€66 M / yr (low) €24 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€2.25 bn	4.0%	25 y, 5 y grace	€165 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€937 M	8.5%	25 y, 5 y grace	€99 M / yr
Government equity (no debt service)	15%	€562 M	—	—	—
Total	100%	€3.75 bn			€264 M / yr

During the 5-year grace period the operator pays interest only — multilateral €90 M / yr + bonds €80 M / yr = €170 M / yr total — plus the equity tranche amortised across construction (€112 M / yr × 5 yr). Principal repayment begins in year 6 on a 20-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€57 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€41 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€216 k
Traction energy (994.6 GWh / yr)	trackside PV + Na-ion (RFC 0002) — self-generated, €0 / yr	€0 k
Labour (1,746 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€10 M
OPEX subtotal		€108 M / yr

Annual fleet utilisation: 211 revenue trainsets × 20.5 h/day × 365 d/yr × 35 km/h commercial × 75% revenue factor = 41.4 M train-km / yr (~196 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$380 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass — a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.58 (~\$0.63 USD)
Day pass (3 trips)	€1.49 (15 % bulk discount)
Monthly unlimited pass	€17.48 (~5 % of median monthly income)
Annual pass	€192.28 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	72.2 M	144.4 M
Farebox revenue	€42 M / yr	€84 M / yr
Farebox / OPEX recovery	39%	78%
Country policy-target recovery (diagnostic)	45%	45%
Operating shortfall (gov subsidy required)	€66 M / yr	€24 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€330 M / yr	€288 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
basra.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
basra-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
basra.corridor.geojson	Line polylines + stations (GeoJSON)
basra.stations.json	Machine-readable station list
basra.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug basra

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug basra \
  --sidecar .cache/osr-pipeline/rasters/basra.grid.json \
  --out-dir designs/.../Basra

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../basra.toml \
  --out designs/.../README.md
```

`scripts/regenerate-basra.sh` chains steps 3 + drift tests into a single command.

west-asia/Iraq/Diwaniyah

designs/west-asia/Iraq/Diwaniyah/README.md

Diwaniyah — Urban Rail Network

Country: IQ · Population: 440,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Diwaniyah rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [diwaniyah.corridor.geojson](#).

At a glance

Metric	Value
Lines	3
Unique stations	36
Interchange stations	3
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	43.1%
Route length (double track)	53.7 km
Revenue fleet	48 × 3-car trainsets
Spare + cold-reserve	7 × 3-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	25.9 km	15	25	SW Outer E Outer
line-2	16.2 km	12	17	SE Outer NW Mid
line-3	11.6 km	9	13	N Mid W Mid
Total	53.7 km	36 unique	55	

Rolling stock

Property	Value
Consist	3-car, 57 m
Max speed	90 km/h
Onboard battery	360 kWh per trainset
Seats	60 longitudinal seats
Nominal capacity (AW2)	330 pax (seated + standing, light-metro-3car per RFC 0008 §1)
Crush capacity (AW3)	420 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 330 AW2 passengers (light-metro-3car)
- Peak frequency: 12 trains/hour/direction (5-min headway)

- Peak capacity per line per direction: $330 \times 12 = 3,960$ pphpd
- Network peak throughput (all lines, both directions): $3 \text{ lines} \times 2 \text{ directions} \times 3,960 = 23,760$ passengers/hour
- Daily theoretical capacity (peak $\times 10$): $\approx 237,600$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 18,964 - 28,446$ trips/day

Catchment

- City population: 440,000
- Anchor-weighted coverage: 43.1%
- Catchment population: $\approx 189,640$ (within ~ 800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	3	500 kW	3000 kWh
Major	19	400 kW	2500 kWh
Standard	7	300 kW	2000 kWh
Terminal	5	500 kW	3000 kWh
Total installed	35	18,700 kW	125,500 kWh

Aggregate station-rail charging power: 20,750 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 360 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	12.0 kWh/km	3 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	215 kWh	17.9 km average line length
Onboard battery coverage	1.7 $\times$ average line run	360 kWh usable pack
Average 60 s dwell charge	9.6 kWh/stop	576 kW average charger across stops
Stops to refill one trainset pack	37 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	94 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	126 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in design.toml — emitted by the osr-design Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. country-costs.toml applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (47.6 km @ €3.5 M/km)	€166 M
Elevated (6.0 km @ €18 M/km)	€107 M
Elevated-interchange premium (2 sites @ €20 M)	€40 M
Civil subtotal	€314 M

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~ 11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
halt	1	€0.4 M	€0.4 M
standard	7	€1.5 M	€10 M
major	19	€3.0 M	€57 M
terminal	5	€2.5 M	€12 M
depot-terminal	1	€3.0 M	€3.0 M
interchange-elevated	3	€4.5 M	€14 M
Stations subtotal			€97 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	5	€3.0 M	€15 M
Depots subtotal			€40 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
light-metro-3car (revenue + spare + cold reserve)	55	€3.0 M	€165 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	53.7 km × €0.015 M/km	€0.8 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€14 M
EPC integration + project management (7%)	on subtotal	€44 M

Total

Bucket	Value
Civil works	€314 M
Stations	€97 M
Depots	€40 M
Rolling stock	€165 M
Residual train-control wayside + charging microgrids	€15 M
EPC overhead (7%)	€44 M
CAPEX total	€674 M
Per-route-km	€13 M / km
Per-capita (city pop)	€1,533 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running scripts/regenerate-city.sh diwaniyah.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–5 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 6 and runs for 20 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–5)	€51 M / yr	€115
Steady-state, low-ridership (year 6+)	€61 M / yr	€138
Steady-state, high-ridership (year 6+)	€56 M / yr	€127
Lifecycle envelope (yr 1–25, low scenario)	€1.46 bn cumulative	€3,328
Lifecycle envelope (yr 1–25, high scenario)	€1.37 bn cumulative	€3,115

Population basis: 440,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 25, debt service drops to zero and only the OPEX shortfall remains – ~€13 M / yr (low) €8.3 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€405 M	4.0%	25 y, 5 y grace	€30 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€169 M	8.5%	25 y, 5 y grace	€18 M / yr
Government equity (no debt service)	15%	€101 M	—	—	—
Total	100%	€674 M			€48 M / yr

During the 5-year grace period the operator pays interest only – multilateral €16 M / yr + bonds €14 M / yr = €31 M / yr total – plus the equity tranche amortised across construction (€20 M / yr × 5 yr). Principal repayment begins in year 6 on a 20-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€6.6 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€9.0 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€40 k
Traction energy (97.0 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (334 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€2.0 M
OPEX subtotal		€18 M / yr

Annual fleet utilisation: 48 revenue trainsets × 20.5 h/day × 365 d/yr × 30 km/h commercial × 75% revenue factor = 8.1 M train-km / yr (~168 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$380 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.58 (~\$0.63 USD)
Day pass (3 trips)	€1.49 (15 % bulk discount)
Monthly unlimited pass	€17.48 (~5 % of median monthly income)
Annual pass	€192.28 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	8.0 M	16.1 M

	Low scenario	High scenario
Farebox revenue	€4.7 M / yr	€9.4 M / yr
Farebox / OPEX recovery	27%	53%
Country policy-target recovery (diagnostic)	45%	45%
Operating shortfall (gov subsidy required)	€13 M / yr	€8.3 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€61 M / yr	€56 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
diwaniyah.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
diwaniyah-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
diwaniyah.corridor.geojson	Line polylines + stations (GeoJSON)
diwaniyah.stations.json	Machine-readable station list
diwaniyah.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug diwaniyah

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug diwaniyah \
  --sidecar .cache/osr-pipeline/rasters/diwaniyah.grid.json \
  --out-dir designs/.../Diwaniyah

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../diwaniyah.toml \
  --out designs/.../README.md
```

`scripts/regenerate-diwaniyah.sh` chains steps 3 + drift tests into a single command.

west-asia/Iraq/Duhok

designs/west-asia/Iraq/Duhok/README.md

Duhok — Urban Rail Network

Country: IQ · Population: 360,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Duhok rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [duhok.corridor.geojson](#).

At a glance

Metric	Value
Lines	3
Unique stations	32
Interchange stations	5
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	53.4%
Route length (double track)	53.2 km
Revenue fleet	48 × 3-car trainsets
Spare + cold-reserve	6 × 3-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	18.2 km	12	18	W Outer E Outer
line-2	21.0 km	12	21	E Outer NW Mid
line-3	14.0 km	8	15	N Inner SW Outer
Total	53.2 km	32 unique	54	

Rolling stock

Property	Value
Consist	3-car, 57 m
Max speed	90 km/h
Onboard battery	360 kWh per trainset
Seats	60 longitudinal seats
Nominal capacity (AW2)	330 pax (seated + standing, light-metro-3car per RFC 0008 §1)
Crush capacity (AW3)	420 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 330 AW2 passengers (light-metro-3car)
- Peak frequency: 12 trains/hour/direction (5-min headway)

- Peak capacity per line per direction: $330 \times 12 = 3,960$ pphpd
- Network peak throughput (all lines, both directions): $3 \text{ lines} \times 2 \text{ directions} \times 3,960 = 23,760$ passengers/hour
- Daily theoretical capacity (peak $\times 10$): $\approx 237,600$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 19,224 - 28,836$ trips/day

Catchment

- City population: 360,000
- Anchor-weighted coverage: 53.4%
- Catchment population: $\approx 192,240$ (within ~ 800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	5	500 kW	3000 kWh
Major	12	400 kW	2500 kWh
Standard	7	300 kW	2000 kWh
Terminal	5	500 kW	3000 kWh
Total installed	30	16,900 kW	114,000 kWh

Aggregate station-rail charging power: 18,500 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 360 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	12.0 kWh/km	3 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	213 kWh	17.7 km average line length
Onboard battery coverage	1.7 $\times$ average line run	360 kWh usable pack
Average 60 s dwell charge	9.6 kWh/stop	578 kW average charger across stops
Stops to refill one trainset pack	37 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	84 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	114 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the `[costs]` block in `design.toml` — emitted by the `osr-design` Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. `country-costs.toml` applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (51.0 km @ €3.5 M/km)	€179 M
Elevated (2.1 km @ €18 M/km)	€38 M
Elevated-interchange premium (2 sites @ €20 M)	€40 M
Civil subtotal	€256 M

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~ 11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
halt	2	€0.4 M	€0.8 M
standard	7	€1.5 M	€10 M
major	12	€3.0 M	€36 M
terminal	5	€2.5 M	€12 M
depot-terminal	1	€3.0 M	€3.0 M
interchange-elevated	5	€4.5 M	€22 M
Stations subtotal			€85 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	5	€3.0 M	€15 M
Depots subtotal			€40 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
light-metro-3car (revenue + spare + cold reserve)	54	€3.0 M	€162 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	53.2 km × €0.015 M/km	€0.8 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€13 M
EPC integration + project management (7%)	on subtotal	€39 M

Total

Bucket	Value
Civil works	€256 M
Stations	€85 M
Depots	€40 M
Rolling stock	€162 M
Residual train-control wayside + charging microgrids	€13 M
EPC overhead (7%)	€39 M
CAPEX total	€596 M
Per-route-km	€11 M / km
Per-capita (city pop)	€1,656 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running scripts/regenerate-city.sh duhok.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–5 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 6 and runs for 20 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–5)	€45 M / yr	€125
Steady-state, low-ridership (year 6+)	€54 M / yr	€151
Steady-state, high-ridership (year 6+)	€51 M / yr	€140
Lifecycle envelope (yr 1–25, low scenario)	€1.31 bn cumulative	€3,642
Lifecycle envelope (yr 1–25, high scenario)	€1.23 bn cumulative	€3,429

Population basis: 360,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 25, debt service drops to zero and only the OPEX shortfall remains – ~€12 M / yr (low) €8.4 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€358 M	4.0%	25 y, 5 y grace	€26 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€149 M	8.5%	25 y, 5 y grace	€16 M / yr
Government equity (no debt service)	15%	€89 M	—	—	—
Total	100%	€596 M			€42 M / yr

During the 5-year grace period the operator pays interest only – multilateral €14 M / yr + bonds €13 M / yr = €27 M / yr total – plus the equity tranche amortised across construction (€18 M / yr × 5 yr). Principal repayment begins in year 6 on a 20-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€6.5 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€7.6 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€40 k
Traction energy (97.0 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (331 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€1.9 M
OPEX subtotal		€16 M / yr

Annual fleet utilisation: 48 revenue trainsets × 20.5 h/day × 365 d/yr × 30 km/h commercial × 75% revenue factor = 8.1 M train-km / yr (~168 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$380 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.58 (~\$0.63 USD)
Day pass (3 trips)	€1.49 (15 % bulk discount)
Monthly unlimited pass	€17.48 (~5 % of median monthly income)
Annual pass	€192.28 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	6.6 M	13.1 M

	Low scenario	High scenario
Farebox revenue	€3.8 M / yr	€7.7 M / yr
Farebox / OPEX recovery	24%	48%
Country policy-target recovery (diagnostic)	45%	45%
Operating shortfall (gov subsidy required)	€12 M / yr	€8.4 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€54 M / yr	€51 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
duhok.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
duhok-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
duhok.corridor.geojson	Line polylines + stations (GeoJSON)
duhok.stations.json	Machine-readable station list
duhok.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug duhok

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug duhok \
  --sidecar .cache/osr-pipeline/rasters/duhok.grid.json \
  --out-dir designs/.../Duhok

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../duhok.toml \
  --out designs/.../README.md
```

`scripts/regenerate-duhok.sh` chains steps 3 + drift tests into a single command.

west-asia/Iraq/Erbil

designs/west-asia/Iraq/Erbil/README.md

Erbil — Urban Rail Network

Country: IQ · Population: 1,952,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Erbil rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [erbil.corridor.geojson](#).

At a glance

Metric	Value
Lines	6
Unique stations	96
Interchange stations	20
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	41.9%
Route length (double track)	199.4 km
Revenue fleet	148 × 4-car trainsets
Spare + cold-reserve	18 × 4-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	28.9 km	17	25	SW Outer NE Outer
line-2	26.5 km	16	23	SE Outer NW Outer
line-3	24.0 km	12	20	NW Outer E Outer
line-4	26.4 km	11	23	N Outer SE Mid
line-5	25.3 km	12	21	NE Outer S Outer
line-6	68.3 km	29	54	NW Mid NW Mid
Total	199.4 km	96 unique	166	

Rolling stock

Property	Value
Consist	4-car, 75 m
Max speed	90 km/h
Onboard battery	480 kWh per trainset
Seats	80 longitudinal seats
Nominal capacity (AW2)	440 pax (seated + standing, metro-4car per RFC 0008 §1)
Crush capacity (AW3)	560 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 440 AW2 passengers (metro - 4 car)
- Peak frequency: 12 trains/hour/direction (5-min headway)
- Peak capacity per line per direction: $440 \times 12 = 5,280$ pphpd
- Network peak throughput (all lines, both directions): $6 \text{ lines} \times 2 \text{ directions} \times 5,280 = 63,360$ passengers/hour
- Daily theoretical capacity (peak $\times$ 10): $\approx 633,600$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 81,788 - 122,683$ trips/day

Catchment

- City population: 1,952,000
- Anchor-weighted coverage: 41.9%
- Catchment population: $\approx 817,888$ (within ~800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	20	500 kW	3000 kWh
Major	33	400 kW	2500 kWh
Standard	32	300 kW	2000 kWh
Terminal	9	500 kW	3000 kWh
Total installed	95	42,300 kW	273,500 kWh

Aggregate station-rail charging power: 53,000 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 480 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	16.0 kWh/km	4 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	532 kWh	33.2 km average line length
Onboard battery coverage	0.9 $\times$ average line run	480 kWh usable pack
Average 60 s dwell charge	9.2 kWh/stop	552 kW average charger across stops
Stops to refill one trainset pack	52 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	212 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	274 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in design.toml — emitted by the osr-design Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. country-costs.toml applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (191.4 km @ €3.5 M/km)	€670 M
Elevated (7.4 km @ €18 M/km)	€134 M
Elevated-interchange premium (6 sites @ €20 M)	€120 M
Civil subtotal	€924 M

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
halt	2	€0.4 M	€0.8 M
standard	32	€1.5 M	€48 M
major	33	€3.0 M	€99 M
terminal	9	€2.5 M	€22 M
depot-terminal	1	€3.0 M	€3.0 M
interchange	4	€4.5 M	€18 M
interchange-elevated	16	€4.5 M	€72 M
Stations subtotal			€263 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	9	€3.0 M	€27 M
Depots subtotal			€52 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
metro - 4car (revenue + spare + cold reserve)	166	€4.0 M	€664 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	199.4 km × €0.015 M/km	€3.0 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€38 M
EPC integration + project management (7%)	on subtotal	€136 M

Total

Bucket	Value
Civil works	€924 M
Stations	€263 M
Depots	€52 M
Rolling stock	€664 M
Residual train-control wayside + charging microgrids	€41 M
EPC overhead (7%)	€136 M
CAPEX total	€2.08 bn
Per-route-km	€10 M / km
Per-capita (city pop)	€1,066 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running scripts/regenerate-city.sh erbil.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–5 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 6 and runs for 20 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–5)	€157 M / yr	€80
Steady-state, low-ridership (year 6+)	€185 M / yr	€95
Steady-state, high-ridership (year 6+)	€164 M / yr	€84
Lifecycle envelope (yr 1–25, low scenario)	€4.47 bn cumulative	€2,292
Lifecycle envelope (yr 1–25, high scenario)	€4.06 bn cumulative	€2,080

Population basis: 1,952,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 25, debt service drops to zero and only the OPEX shortfall remains – ~€38 M / yr (low) €17 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€1.25 bn	4.0%	25 y, 5 y grace	€92 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€520 M	8.5%	25 y, 5 y grace	€55 M / yr
Government equity (no debt service)	15%	€312 M	—	—	—
Total	100%	€2.08 bn			€147 M / yr

During the 5-year grace period the operator pays interest only – multilateral €50 M / yr + bonds €44 M / yr = €94 M / yr total – plus the equity tranche amortised across construction (€62 M / yr × 5 yr). Principal repayment begins in year 6 on a 20-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€27 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€25 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€149 k
Traction energy (465.1 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (1,208 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€7.1 M
OPEX subtotal		€59 M / yr

Annual fleet utilisation: 148 revenue trainsets × 20.5 h/day × 365 d/yr × 35 km/h commercial × 75% revenue factor = 29.1 M train-km / yr (~196 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$380 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.58 (~\$0.63 USD)
Day pass (3 trips)	€1.49 (15 % bulk discount)
Monthly unlimited pass	€17.48 (~5 % of median monthly income)
Annual pass	€192.28 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	35.6 M	71.2 M
Farebox revenue	€21 M / yr	€42 M / yr
Farebox / OPEX recovery	35%	71%
Country policy-target recovery (diagnostic)	45%	45%
Operating shortfall (gov subsidy required)	€38 M / yr	€17 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€185 M / yr	€164 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
erbil.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
erbil-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
erbil.corridor.geojson	Line polylines + stations (GeoJSON)
erbil.stations.json	Machine-readable station list
erbil.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug erbil

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug erbil \
  --sidecar .cache/osr-pipeline/rasters/erbil.grid.json \
  --out-dir designs/.../Erbil

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../erbil.toml \
  --out designs/.../README.md
```

`scripts/regenerate-erbil.sh` chains steps 3 + drift tests into a single command.

west-asia/Iraq/Fallujah

designs/west-asia/Iraq/Fallujah/README.md

Fallujah — Urban Rail Network

Country: IQ · Population: 360,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Fallujah rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [fallujah.corridor.geojson](#).

At a glance

Metric	Value
Lines	3
Unique stations	28
Interchange stations	3
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	57.0%
Route length (double track)	45.9 km
Revenue fleet	42 × 3-car trainsets
Spare + cold-reserve	6 × 3-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	20.5 km	10	20	SE Mid NW Outer
line-2	15.0 km	10	16	E Mid W Mid
line-3	10.4 km	8	12	W Inner SE Mid
Total	45.9 km	28 unique	48	

Rolling stock

Property	Value
Consist	3-car, 57 m
Max speed	90 km/h
Onboard battery	360 kWh per trainset
Seats	60 longitudinal seats
Nominal capacity (AW2)	330 pax (seated + standing, light-metro-3car per RFC 0008 §1)
Crush capacity (AW3)	420 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 330 AW2 passengers (light-metro-3car)
- Peak frequency: 12 trains/hour/direction (5-min headway)

- Peak capacity per line per direction: $330 \times 12 = 3,960$ pphpd
- Network peak throughput (all lines, both directions): $3 \text{ lines} \times 2 \text{ directions} \times 3,960 = 23,760$ passengers/hour
- Daily theoretical capacity (peak $\times 10$): $\approx 237,600$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 20,519 - 30,779$ trips/day

Catchment

- City population: 360,000
- Anchor-weighted coverage: 57.0%
- Catchment population: $\approx 205,199$ (within ~ 800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	3	500 kW	3000 kWh
Major	12	400 kW	2500 kWh
Standard	6	300 kW	2000 kWh
Terminal	5	500 kW	3000 kWh
Total installed	27	15,600 kW	106,000 kWh

Aggregate station-rail charging power: 16,750 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 360 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	12.0 kWh/km	3 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	184 kWh	15.3 km average line length
Onboard battery coverage	2.0 $\times$ average line run	360 kWh usable pack
Average 60 s dwell charge	10.0 kWh/stop	598 kW average charger across stops
Stops to refill one trainset pack	36 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	78 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	106 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in design.toml — emitted by the osr-design Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. country-costs.toml applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (42.5 km @ €3.5 M/km)	€149 M
Elevated (3.3 km @ €18 M/km)	€60 M
Elevated-interchange premium (2 sites @ €20 M)	€40 M
Civil subtotal	€249 M

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~ 11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
halt	1	€0.4 M	€0.4 M
standard	6	€1.5 M	€9.0 M
major	12	€3.0 M	€36 M
terminal	5	€2.5 M	€12 M
depot-terminal	1	€3.0 M	€3.0 M
interchange-elevated	3	€4.5 M	€14 M
Stations subtotal			€74 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	5	€3.0 M	€15 M
Depots subtotal			€40 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
light-metro-3car (revenue + spare + cold reserve)	48	€3.0 M	€144 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	45.9 km × €0.015 M/km	€0.7 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€11 M
EPC integration + project management (7%)	on subtotal	€36 M

Total

Bucket	Value
Civil works	€249 M
Stations	€74 M
Depots	€40 M
Rolling stock	€144 M
Residual train-control wayside + charging microgrids	€12 M
EPC overhead (7%)	€36 M
CAPEX total	€555 M
Per-route-km	€12 M / km
Per-capita (city pop)	€1,542 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running scripts/regenerate-city.sh fallujah.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–5 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 6 and runs for 20 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–5)	€42 M / yr	€116
Steady-state, low-ridership (year 6+)	€50 M / yr	€139
Steady-state, high-ridership (year 6+)	€46 M / yr	€128
Lifecycle envelope (yr 1–25, low scenario)	€1.21 bn cumulative	€3,363
Lifecycle envelope (yr 1–25, high scenario)	€1.13 bn cumulative	€3,150

Population basis: 360,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 25, debt service drops to zero and only the OPEX shortfall remains – ~€11 M / yr (low) €7.1 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€333 M	4.0%	25 y, 5 y grace	€25 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€139 M	8.5%	25 y, 5 y grace	€15 M / yr
Government equity (no debt service)	15%	€83 M	—	—	—
Total	100%	€555 M			€39 M / yr

During the 5-year grace period the operator pays interest only – multilateral €13 M / yr + bonds €12 M / yr = €25 M / yr total – plus the equity tranche amortised across construction (€17 M / yr × 5 yr). Principal repayment begins in year 6 on a 20-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€5.8 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€7.3 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€34 k
Traction energy (84.9 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (287 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€1.7 M
OPEX subtotal		€15 M / yr

Annual fleet utilisation: 42 revenue trainsets × 20.5 h/day × 365 d/yr × 30 km/h commercial × 75% revenue factor = 7.1 M train-km / yr (~168 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$380 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.58 (~\$0.63 USD)
Day pass (3 trips)	€1.49 (15 % bulk discount)
Monthly unlimited pass	€17.48 (~5 % of median monthly income)
Annual pass	€192.28 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	6.6 M	13.1 M

	Low scenario	High scenario
Farebox revenue	€3.8 M / yr	€7.7 M / yr
Farebox / OPEX recovery	26%	52%
Country policy-target recovery (diagnostic)	45%	45%
Operating shortfall (gov subsidy required)	€11 M / yr	€7.1 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€50 M / yr	€46 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
fallujah.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
fallujah-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
fallujah.corridor.geojson	Line polylines + stations (GeoJSON)
fallujah.stations.json	Machine-readable station list
fallujah.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug fallujah

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug fallujah \
  --sidecar .cache/osr-pipeline/rasters/fallujah.grid.json \
  --out-dir designs/.../Fallujah

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../fallujah.toml \
  --out designs/.../README.md
```

`scripts/regenerate-fallujah.sh` chains steps 3 + drift tests into a single command.

Hillah — Urban Rail Network

Country: IQ · Population: 700,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Hillah rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [hillah.corridor.geojson](#).

At a glance

Metric	Value
Lines	3
Unique stations	42
Interchange stations	3
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	42.8%
Route length (double track)	69.0 km
Revenue fleet	60 × 3-car trainsets
Spare + cold-reserve	8 × 3-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	23.4 km	16	23	NE Outer S Outer
line-2	27.0 km	13	26	NW Outer SE Outer
line-3	18.6 km	13	19	W Outer NE Mid
Total	69.0 km	42 unique	68	

Rolling stock

Property	Value
Consist	3-car, 57 m
Max speed	90 km/h
Onboard battery	360 kWh per trainset
Seats	60 longitudinal seats
Nominal capacity (AW2)	330 pax (seated + standing, light-metro-3car per RFC 0008 §1)
Crush capacity (AW3)	420 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 330 AW2 passengers (light-metro-3car)
- Peak frequency: 12 trains/hour/direction (5-min headway)

- Peak capacity per line per direction: $330 \times 12 = 3,960$ pphpd
- Network peak throughput (all lines, both directions): $3 \text{ lines} \times 2 \text{ directions} \times 3,960 = 23,760$ passengers/hour
- Daily theoretical capacity (peak $\times 10$): $\approx 237,600$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 29,960 - 44,940$ trips/day

Catchment

- City population: 700,000
- Anchor-weighted coverage: 42.8%
- Catchment population: $\approx 299,600$ (within ~ 800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	3	500 kW	3000 kWh
Major	16	400 kW	2500 kWh
Standard	17	300 kW	2000 kWh
Terminal	5	500 kW	3000 kWh
Total installed	42	20,500 kW	138,000 kWh

Aggregate station-rail charging power: 24,000 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 360 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	12.0 kWh/km	3 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	276 kWh	23.0 km average line length
Onboard battery coverage	1.3 $\times$ average line run	360 kWh usable pack
Average 60 s dwell charge	9.5 kWh/stop	571 kW average charger across stops
Stops to refill one trainset pack	38 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	102 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	138 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the `[costs]` block in `design.toml` — emitted by the `osr-design` Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. `country-costs.toml` applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (66.8 km @ €3.5 M/km)	€234 M
Elevated (2.0 km @ €18 M/km)	€36 M
Elevated-interchange premium (1 sites @ €20 M)	€20 M
Civil subtotal	€290 M

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~ 11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
standard	17	€1.5 M	€26 M
major	16	€3.0 M	€48 M
terminal	5	€2.5 M	€12 M
depot-terminal	1	€3.0 M	€3.0 M
interchange-elevated	3	€4.5 M	€14 M
Stations subtotal			€102 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	5	€3.0 M	€15 M
Depots subtotal			€40 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
light-metro-3car (revenue + spare + cold reserve)	68	€3.0 M	€204 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	69.0 km × €0.015 M/km	€1.0 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€15 M
EPC integration + project management (7%)	on subtotal	€46 M

Total

Bucket	Value
Civil works	€290 M
Stations	€102 M
Depots	€40 M
Rolling stock	€204 M
Residual train-control wayside + charging microgrids	€16 M
EPC overhead (7%)	€46 M
CAPEX total	€699 M
Per-route-km	€10 M / km
Per-capita (city pop)	€998 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running scripts/regenerate-city.sh hillah.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–5 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 6 and runs for 20 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–5)	€53 M / yr	€75
Steady-state, low-ridership (year 6+)	€61 M / yr	€87
Steady-state, high-ridership (year 6+)	€54 M / yr	€77
Lifecycle envelope (yr 1–25, low scenario)	€1.49 bn cumulative	€2,124
Lifecycle envelope (yr 1–25, high scenario)	€1.34 bn cumulative	€1,912

Population basis: 700,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 25, debt service drops to zero and only the OPEX shortfall remains – ~€12 M / yr (low) €4.5 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€419 M	4.0%	25 y, 5 y grace	€31 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€175 M	8.5%	25 y, 5 y grace	€18 M / yr
Government equity (no debt service)	15%	€105 M	—	—	—
Total	100%	€699 M			€49 M / yr

During the 5-year grace period the operator pays interest only – multilateral €17 M / yr + bonds €15 M / yr = €32 M / yr total – plus the equity tranche amortised across construction (€21 M / yr × 5 yr). Principal repayment begins in year 6 on a 20-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€8.2 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€8.7 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€52 k
Traction energy (121.2 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (426 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€2.5 M
OPEX subtotal		€19 M / yr

Annual fleet utilisation: 60 revenue trainsets × 20.5 h/day × 365 d/yr × 30 km/h commercial × 75% revenue factor = 10.1 M train-km / yr (~168 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$380 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.58 (~\$0.63 USD)
Day pass (3 trips)	€1.49 (15 % bulk discount)
Monthly unlimited pass	€17.48 (~5 % of median monthly income)
Annual pass	€192.28 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	12.8 M	25.6 M

	Low scenario	High scenario
Farebox revenue	€7.4 M / yr	€15 M / yr
Farebox / OPEX recovery	38%	77%
Country policy-target recovery (diagnostic)	45%	45%
Operating shortfall (gov subsidy required)	€12 M / yr	€4.5 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€61 M / yr	€54 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
hillah.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
hillah-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
hillah.corridor.geojson	Line polylines + stations (GeoJSON)
hillah.stations.json	Machine-readable station list
hillah.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug hillah

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug hillah \
  --sidecar .cache/osr-pipeline/rasters/hillah.grid.json \
  --out-dir designs/.../Hillah

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../hillah.toml \
  --out designs/.../README.md
```

`scripts/regenerate-hillah.sh` chains steps 3 + drift tests into a single command.

west-asia/Iraq/Karbala

designs/west-asia/Iraq/Karbala/README.md

Karbala — Urban Rail Network

Country: IQ · Population: 1,390,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Karbala rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [karbala.corridor.geojson](#).

At a glance

Metric	Value
Lines	6
Unique stations	89
Interchange stations	23
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	67.2%
Route length (double track)	169.9 km
Revenue fleet	128 × 4-car trainsets
Spare + cold-reserve	16 × 4-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	27.7 km	15	24	E Outer NW Outer
line-2	22.5 km	14	19	N Outer SW Mid
line-3	15.8 km	11	15	W Mid SE Mid
line-4	31.9 km	15	27	W Outer NE Outer
line-5	21.7 km	10	19	E Outer SW Mid
line-6	50.3 km	24	40	NW Mid NW Mid
Total	169.9 km	89 unique	144	

Rolling stock

Property	Value
Consist	4-car, 75 m
Max speed	90 km/h
Onboard battery	480 kWh per trainset
Seats	80 longitudinal seats
Nominal capacity (AW2)	440 pax (seated + standing, metro-4car per RFC 0008 §1)
Crush capacity (AW3)	560 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 440 AW2 passengers (metro - 4 car)
- Peak frequency: 12 trains/hour/direction (5-min headway)
- Peak capacity per line per direction: $440 \times 12 = 5,280$ pphpd
- Network peak throughput (all lines, both directions): $6 \text{ lines} \times 2 \text{ directions} \times 5,280 = 63,360$ passengers/hour
- Daily theoretical capacity (peak $\times$ 10): $\approx 633,600$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 93,408 - 140,112$ trips/day

Catchment

- City population: 1,390,000
- Anchor-weighted coverage: 67.2%
- Catchment population: $\approx 934,080$ (within ~800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	23	500 kW	3000 kWh
Major	28	400 kW	2500 kWh
Standard	22	300 kW	2000 kWh
Terminal	9	500 kW	3000 kWh
Total installed	83	38,800 kW	250,000 kWh

Aggregate station-rail charging power: 48,000 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 480 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	16.0 kWh/km	4 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	453 kWh	28.3 km average line length
Onboard battery coverage	1.1 $\times$ average line run	480 kWh usable pack
Average 60 s dwell charge	9.0 kWh/stop	539 kW average charger across stops
Stops to refill one trainset pack	53 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	194 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	250 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the `[costs]` block in `design.toml` — emitted by the `osr-design` Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. `country-costs.toml` applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (154.8 km @ €3.5 M/km)	€542 M
Elevated (14.5 km @ €18 M/km)	€261 M
Elevated-interchange premium (13 sites @ €20 M)	€260 M
Civil subtotal	€1.06 bn

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
halt	6	€0.4 M	€2.4 M
standard	22	€1.5 M	€33 M
major	28	€3.0 M	€84 M
terminal	9	€2.5 M	€22 M
depot-terminal	1	€3.0 M	€3.0 M
interchange-elevated	23	€4.5 M	€104 M
Stations subtotal			€248 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	9	€3.0 M	€27 M
Depots subtotal			€52 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
metro-4car (revenue + spare + cold reserve)	144	€4.0 M	€576 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	169.9 km × €0.015 M/km	€2.5 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€36 M
EPC integration + project management (7%)	on subtotal	€138 M

Total

Bucket	Value
Civil works	€1.06 bn
Stations	€248 M
Depots	€52 M
Rolling stock	€576 M
Residual train-control wayside + charging microgrids	€38 M
EPC overhead (7%)	€138 M
CAPEX total	€2.12 bn
Per-route-km	€12 M / km
Per-capita (city pop)	€1,522 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running `scripts/regenerate-city.sh karbala`.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–5 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 6 and runs for 20 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–5)	€159 M / yr	€115
Steady-state, low-ridership (year 6+)	€191 M / yr	€137
Steady-state, high-ridership (year 6+)	€176 M / yr	€127
Lifecycle envelope (yr 1–25, low scenario)	€4.62 bn cumulative	€3,321
Lifecycle envelope (yr 1–25, high scenario)	€4.32 bn cumulative	€3,108

Population basis: 1,390,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 25, debt service drops to zero and only the OPEX shortfall remains – ~€42 M / yr (low) €27 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€1.27 bn	4.0%	25 y, 5 y grace	€93 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€529 M	8.5%	25 y, 5 y grace	€56 M / yr
Government equity (no debt service)	15%	€317 M	—	—	—
Total	100%	€2.12 bn			€149 M / yr

During the 5-year grace period the operator pays interest only – multilateral €51 M / yr + bonds €45 M / yr = €96 M / yr total – plus the equity tranche amortised across construction (€63 M / yr × 5 yr). Principal repayment begins in year 6 on a 20-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€23 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€27 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€127 k
Traction energy (402.3 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (1,031 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€6.1 M
OPEX subtotal		€56 M / yr

Annual fleet utilisation: 128 revenue trainsets × 20.5 h/day × 365 d/yr × 35 km/h commercial × 75% revenue factor = 25.1 M train-km / yr (~196 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$380 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.58 (~\$0.63 USD)
Day pass (3 trips)	€1.49 (15 % bulk discount)
Monthly unlimited pass	€17.48 (~5 % of median monthly income)
Annual pass	€192.28 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	25.4 M	50.7 M
Farebox revenue	€15 M / yr	€30 M / yr
Farebox / OPEX recovery	26%	52%
Country policy-target recovery (diagnostic)	45%	45%
Operating shortfall (gov subsidy required)	€42 M / yr	€27 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€191 M / yr	€176 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
karbala.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
karbala-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
karbala.corridor.geojson	Line polylines + stations (GeoJSON)
karbala.stations.json	Machine-readable station list
karbala.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug karbala

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug karbala \
  --sidecar .cache/osr-pipeline/rasters/karbala.grid.json \
  --out-dir designs/.../Karbala

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../karbala.toml \
  --out designs/.../README.md
```

`scripts/regenerate-karbala.sh` chains steps 3 + drift tests into a single command.

west-asia/Iraq/Kirkuk

designs/west-asia/Iraq/Kirkuk/README.md

Kirkuk — Urban Rail Network

Country: IQ · Population: 1,780,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Kirkuk rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [kirkuk.corridor.geojson](#).

At a glance

Metric	Value
Lines	6
Unique stations	91
Interchange stations	19
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	59.8%
Route length (double track)	169.8 km
Revenue fleet	127 × 4-car trainsets
Spare + cold-reserve	15 × 4-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	27.6 km	15	24	S Outer N Outer
line-2	25.4 km	13	21	SW Outer E Mid
line-3	18.2 km	13	16	SE Mid N Mid
line-4	21.8 km	13	19	S Mid NW Outer
line-5	22.7 km	10	19	W Outer NE Mid
line-6	54.0 km	28	43	NW Mid NW Mid
Total	169.8 km	91 unique	142	

Rolling stock

Property	Value
Consist	4-car, 75 m
Max speed	90 km/h
Onboard battery	480 kWh per trainset
Seats	80 longitudinal seats
Nominal capacity (AW2)	440 pax (seated + standing, metro-4car per RFC 0008 §1)
Crush capacity (AW3)	560 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 440 AW2 passengers (metro - 4 car)
- Peak frequency: 12 trains/hour/direction (5-min headway)
- Peak capacity per line per direction: $440 \times 12 = 5,280$ pphpd
- Network peak throughput (all lines, both directions): $6 \text{ lines} \times 2 \text{ directions} \times 5,280 = 63,360$ passengers/hour
- Daily theoretical capacity (peak $\times$ 10): $\approx 633,600$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 106,444 - 159,666$ trips/day

Catchment

- City population: 1,780,000
- Anchor-weighted coverage: 59.8%
- Catchment population: $\approx 1,064,440$ (within ~800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	19	500 kW	3000 kWh
Major	40	400 kW	2500 kWh
Standard	19	300 kW	2000 kWh
Terminal	9	500 kW	3000 kWh
Total installed	88	40,700 kW	262,000 kWh

Aggregate station-rail charging power: 50,000 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 480 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	16.0 kWh/km	4 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	453 kWh	28.3 km average line length
Onboard battery coverage	1.1 $\times$ average line run	480 kWh usable pack
Average 60 s dwell charge	9.2 kWh/stop	549 kW average charger across stops
Stops to refill one trainset pack	52 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	204 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	262 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in design.toml — emitted by the osr-design Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. country-costs.toml applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (154.3 km @ €3.5 M/km)	€540 M
Elevated (14.8 km @ €18 M/km)	€266 M
Elevated-interchange premium (12 sites @ €20 M)	€240 M
Civil subtotal	€1.05 bn

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
halt	4	€0.4 M	€1.6 M
standard	19	€1.5 M	€28 M
major	40	€3.0 M	€120 M
terminal	9	€2.5 M	€22 M
depot-terminal	1	€3.0 M	€3.0 M
interchange-elevated	19	€4.5 M	€86 M
Stations subtotal			€261 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	9	€3.0 M	€27 M
Depots subtotal			€52 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
metro-4car (revenue + spare + cold reserve)	142	€4.0 M	€568 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	169.8 km × €0.015 M/km	€2.5 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€37 M
EPC integration + project management (7%)	on subtotal	€138 M

Total

Bucket	Value
Civil works	€1.05 bn
Stations	€261 M
Depots	€52 M
Rolling stock	€568 M
Residual train-control wayside + charging microgrids	€40 M
EPC overhead (7%)	€138 M
CAPEX total	€2.10 bn
Per-route-km	€12 M / km
Per-capita (city pop)	€1,183 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running scripts/regenerate-city.sh kirkuk.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–5 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 6 and runs for 20 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–5)	€158 M / yr	€89
Steady-state, low-ridership (year 6+)	€186 M / yr	€104
Steady-state, high-ridership (year 6+)	€167 M / yr	€94
Lifecycle envelope (yr 1–25, low scenario)	€4.51 bn cumulative	€2,532
Lifecycle envelope (yr 1–25, high scenario)	€4.13 bn cumulative	€2,319

Population basis: 1,780,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 25, debt service drops to zero and only the OPEX shortfall remains – ~€37 M / yr (low) €18 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€1.26 bn	4.0%	25 y, 5 y grace	€93 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€526 M	8.5%	25 y, 5 y grace	€56 M / yr
Government equity (no debt service)	15%	€316 M	—	—	—
Total	100%	€2.10 bn			€149 M / yr

During the 5-year grace period the operator pays interest only – multilateral €51 M / yr + bonds €45 M / yr = €95 M / yr total – plus the equity tranche amortised across construction (€63 M / yr × 5 yr). Principal repayment begins in year 6 on a 20-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€23 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€27 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€127 k
Traction energy (399.1 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (1,031 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€6.1 M
OPEX subtotal		€56 M / yr

Annual fleet utilisation: 127 revenue trainsets × 20.5 h/day × 365 d/yr × 35 km/h commercial × 75% revenue factor = 24.9 M train-km / yr (~196 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$380 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.58 (~\$0.63 USD)
Day pass (3 trips)	€1.49 (15 % bulk discount)
Monthly unlimited pass	€17.48 (~5 % of median monthly income)
Annual pass	€192.28 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	32.5 M	65.0 M
Farebox revenue	€19 M / yr	€38 M / yr
Farebox / OPEX recovery	34%	67%
Country policy-target recovery (diagnostic)	45%	45%
Operating shortfall (gov subsidy required)	€37 M / yr	€18 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€186 M / yr	€167 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
kirkuk.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
kirkuk-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
kirkuk.corridor.geojson	Line polylines + stations (GeoJSON)
kirkuk.stations.json	Machine-readable station list
kirkuk.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug kirkuk

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug kirkuk \
  --sidecar .cache/osr-pipeline/rasters/kirkuk.grid.json \
  --out-dir designs/.../Kirkuk

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../kirkuk.toml \
  --out designs/.../README.md
```

`scripts/regenerate-kirkuk.sh` chains steps 3 + drift tests into a single command.

west-asia/Iraq/Kut

designs/west-asia/Iraq/Kut/README.md

Kut — Urban Rail Network

Country: IQ · Population: 410,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Kut rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [kut.corridor.geojson](#).

At a glance

Metric	Value
Lines	3
Unique stations	32
Interchange stations	3
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	36.7%
Route length (double track)	55.7 km
Revenue fleet	49 × 3-car trainsets
Spare + cold-reserve	6 × 3-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	20.0 km	10	20	NE Mid W Outer
line-2	14.7 km	11	15	SW Mid NE Mid
line-3	20.9 km	11	20	SE Outer NW Mid
Total	55.7 km	32 unique	55	

Rolling stock

Property	Value
Consist	3-car, 57 m
Max speed	90 km/h
Onboard battery	360 kWh per trainset
Seats	60 longitudinal seats
Nominal capacity (AW2)	330 pax (seated + standing, light-metro-3car per RFC 0008 §1)
Crush capacity (AW3)	420 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 330 AW2 passengers (light-metro-3car)
- Peak frequency: 12 trains/hour/direction (5-min headway)

- Peak capacity per line per direction: $330 \times 12 = 3,960$ pphpd
- Network peak throughput (all lines, both directions): $3 \text{ lines} \times 2 \text{ directions} \times 3,960 = 23,760$ passengers/hour
- Daily theoretical capacity (peak $\times 10$): $\approx 237,600$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 15,047 - 22,570$ trips/day

Catchment

- City population: 410,000
- Anchor-weighted coverage: 36.7%
- Catchment population: $\approx 150,470$ (within ~ 800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	3	500 kW	3000 kWh
Major	14	400 kW	2500 kWh
Standard	7	300 kW	2000 kWh
Terminal	5	500 kW	3000 kWh
Total installed	30	16,700 kW	113,000 kWh

Aggregate station-rail charging power: 18,500 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 360 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	12.0 kWh/km	3 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	223 kWh	18.6 km average line length
Onboard battery coverage	1.6 $\times$ average line run	360 kWh usable pack
Average 60 s dwell charge	9.6 kWh/stop	578 kW average charger across stops
Stops to refill one trainset pack	37 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	84 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	113 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the `[costs]` block in `design.toml` — emitted by the `osr-design` Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. `country-costs.toml` applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (37.4 km @ €3.5 M/km)	€131 M
Elevated (18.0 km @ €18 M/km)	€323 M
Elevated-interchange premium (1 sites @ €20 M)	€20 M
Civil subtotal	€474 M

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~ 11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
halt	2	€0.4 M	€0.8 M
standard	7	€1.5 M	€10 M
major	14	€3.0 M	€42 M
terminal	5	€2.5 M	€12 M
depot-terminal	1	€3.0 M	€3.0 M
interchange-elevated	3	€4.5 M	€14 M
Stations subtotal			€82 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	5	€3.0 M	€15 M
Depots subtotal			€40 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
light-metro-3car (revenue + spare + cold reserve)	55	€3.0 M	€165 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	55.7 km × €0.015 M/km	€0.8 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€12 M
EPC integration + project management (7%)	on subtotal	€54 M

Total

Bucket	Value
Civil works	€474 M
Stations	€82 M
Depots	€40 M
Rolling stock	€165 M
Residual train-control wayside + charging microgrids	€13 M
EPC overhead (7%)	€54 M
CAPEX total	€829 M
Per-route-km	€15 M / km
Per-capita (city pop)	€2,022 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running scripts/regenerate-city.sh kut.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–5 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 6 and runs for 20 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–5)	€62 M / yr	€152
Steady-state, low-ridership (year 6+)	€75 M / yr	€182
Steady-state, high-ridership (year 6+)	€70 M / yr	€172
Lifecycle envelope (yr 1–25, low scenario)	€1.81 bn cumulative	€4,407
Lifecycle envelope (yr 1–25, high scenario)	€1.72 bn cumulative	€4,194

Population basis: 410,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 25, debt service drops to zero and only the OPEX shortfall remains – ~€16 M / yr (low) €12 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€497 M	4.0%	25 y, 5 y grace	€37 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€207 M	8.5%	25 y, 5 y grace	€22 M / yr
Government equity (no debt service)	15%	€124 M	—	—	—
Total	100%	€829 M			€58 M / yr

During the 5-year grace period the operator pays interest only – multilateral €20 M / yr + bonds €18 M / yr = €38 M / yr total – plus the equity tranche amortised across construction (€25 M / yr × 5 yr). Principal repayment begins in year 6 on a 20-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€6.6 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€12 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€42 k
Traction energy (99.0 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (346 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€2.0 M
OPEX subtotal		€21 M / yr

Annual fleet utilisation: 49 revenue trainsets × 20.5 h/day × 365 d/yr × 30 km/h commercial × 75% revenue factor = 8.2 M train-km / yr (~168 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$380 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.58 (~\$0.63 USD)
Day pass (3 trips)	€1.49 (15 % bulk discount)
Monthly unlimited pass	€17.48 (~5 % of median monthly income)
Annual pass	€192.28 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	7.5 M	15.0 M

	Low scenario	High scenario
Farebox revenue	€4.4 M / yr	€8.7 M / yr
Farebox / OPEX recovery	21%	42%
Country policy-target recovery (diagnostic)	45%	45%
Operating shortfall (gov subsidy required)	€16 M / yr	€12 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€75 M / yr	€70 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
kut.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
kut-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
kut.corridor.geojson	Line polylines + stations (GeoJSON)
kut.stations.json	Machine-readable station list
kut.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug kut

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug kut \
  --sidecar .cache/osr-pipeline/rasters/kut.grid.json \
  --out-dir designs/.../Kut

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../kut.toml \
  --out designs/.../README.md
```

`scripts/regenerate-kut.sh` chains steps 3 + drift tests into a single command.

west-asia/Iraq/Mosul

designs/west-asia/Iraq/Mosul/README.md

Mosul — Urban Rail Network

Country: IQ · Population: 1,940,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Mosul rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [mosul.corridor.geojson](#).

At a glance

Metric	Value
Lines	5
Unique stations	60
Interchange stations	4
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	38.2%
Route length (double track)	144.6 km
Revenue fleet	108 × 4-car trainsets
Spare + cold-reserve	14 × 4-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	29.0 km	14	25	SW Mid E Outer
line-2	31.5 km	12	26	SE Outer NW Outer
line-3	32.3 km	11	27	NW Outer E Mid
line-4	27.6 km	12	24	N Mid S Outer
line-5	24.2 km	11	20	NE Outer W Mid
Total	144.6 km	60 unique	122	

Rolling stock

Property	Value
Consist	4-car, 75 m
Max speed	90 km/h
Onboard battery	480 kWh per trainset
Seats	80 longitudinal seats
Nominal capacity (AW2)	440 pax (seated + standing, met ro - 4 car per RFC 0008 §1)
Crush capacity (AW3)	560 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 440 AW2 passengers (met ro - 4 car)

- Peak frequency: 12 trains/hour/direction (5-min headway)
- Peak capacity per line per direction: $440 \times 12 = 5,280$ pphpd
- Network peak throughput (all lines, both directions): $5 \text{ lines} \times 2 \text{ directions} \times 5,280 = 52,800$ passengers/hour
- Daily theoretical capacity (peak $\times$ 10): $\approx 528,000$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 74,108 - 111,162$ trips/day

Catchment

- City population: 1,940,000
- Anchor-weighted coverage: 38.2%
- Catchment population: $\approx 741,080$ (within ~800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	4	500 kW	3000 kWh
Major	18	400 kW	2500 kWh
Standard	24	300 kW	2000 kWh
Terminal	9	500 kW	3000 kWh
Total installed	56	25,900 kW	172,000 kWh

Aggregate station-rail charging power: 34,000 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 480 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	16.0 kWh/km	4 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	463 kWh	28.9 km average line length
Onboard battery coverage	1.0 $\times$ average line run	480 kWh usable pack
Average 60 s dwell charge	9.4 kWh/stop	567 kW average charger across stops
Stops to refill one trainset pack	51 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	130 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	172 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in design.toml — emitted by the osr-design Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. country-costs.toml applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (117.3 km @ €3.5 M/km)	€411 M
Elevated (26.5 km @ €18 M/km)	€477 M
Elevated-interchange premium (5 sites @ €20 M)	€100 M
Civil subtotal	€988 M

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
halt	4	€0.4 M	€1.6 M
standard	24	€1.5 M	€36 M
major	18	€3.0 M	€54 M
terminal	9	€2.5 M	€22 M
depot-terminal	1	€3.0 M	€3.0 M
interchange-elevated	4	€4.5 M	€18 M
Stations subtotal			€135 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	9	€3.0 M	€27 M
Depots subtotal			€52 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
metro-4car (revenue + spare + cold reserve)	122	€4.0 M	€488 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	144.6 km × €0.015 M/km	€2.2 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€20 M
EPC integration + project management (7%)	on subtotal	€118 M

Total

Bucket	Value
Civil works	€988 M
Stations	€135 M
Depots	€52 M
Rolling stock	€488 M
Residual train-control wayside + charging microgrids	€23 M
EPC overhead (7%)	€118 M
CAPEX total	€1.80 bn
Per-route-km	€12 M / km
Per-capita (city pop)	€930 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](https://github.com/OpenSourceRail/lib/templates/country-finance.toml). Pure function of the [costs] block above + the country code – regenerate by re-running `scripts/regenerate-city.sh` `mosul`.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–5 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 6 and runs for 20 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–5)	€136 M / yr	€70
Steady-state, low-ridership (year 6+)	€155 M / yr	€80
Steady-state, high-ridership (year 6+)	€134 M / yr	€69
Lifecycle envelope (yr 1–25, low scenario)	€3.78 bn cumulative	€1,947
Lifecycle envelope (yr 1–25, high scenario)	€3.36 bn cumulative	€1,734

Population basis: 1,940,000 (catchment per [lib/city-batches/world-sample.toml](https://github.com/OpenSourceRail/lib/city-batches/world-sample.toml)). After year 25, debt service drops to zero and only the OPEX shortfall remains – ~€28 M / yr (low) €7.0 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€1.08 bn	4.0%	25 y, 5 y grace	€80 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€451 M	8.5%	25 y, 5 y grace	€48 M / yr
Government equity (no debt service)	15%	€271 M	—	—	—
Total	100%	€1.80 bn			€127 M / yr

During the 5-year grace period the operator pays interest only – multilateral €43 M / yr + bonds €38 M / yr = €82 M / yr total – plus the equity tranche amortised across construction (€54 M / yr × 5 yr). Principal repayment begins in year 6 on a 20-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€20 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€23 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€108 k
Traction energy (339.4 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (880 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€5.2 M
OPEX subtotal		€48 M / yr

Annual fleet utilisation: 108 revenue trainsets × 20.5 h/day × 365 d/yr × 35 km/h commercial × 75% revenue factor = 21.2 M train-km / yr (~196 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$380 USD (per [lib/templates/country-finance.toml](https://github.com/OpenSourceRail/lib/templates/country-finance.toml)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.58 (~\$0.63 USD)
Day pass (3 trips)	€1.49 (15 % bulk discount)
Monthly unlimited pass	€17.48 (~5 % of median monthly income)
Annual pass	€192.28 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	35.4 M	70.8 M
Farebox revenue	€21 M / yr	€41 M / yr
Farebox / OPEX recovery	43%	85%
Country policy-target recovery (diagnostic)	45%	45%
Operating shortfall (gov subsidy required)	€28 M / yr	€7.0 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€155 M / yr	€134 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
mosul.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
mosul-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
mosul.corridor.geojson	Line polylines + stations (GeoJSON)
mosul.stations.json	Machine-readable station list
mosul.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug mosul

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug mosul \
  --sidecar .cache/osr-pipeline/rasters/mosul.grid.json \
  --out-dir designs/.../Mosul

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../mosul.toml \
  --out designs/.../README.md
```

`scripts/regenerate-mosul.sh` chains steps 3 + drift tests into a single command.

west-asia/Iraq/Najaf

designs/west-asia/Iraq/Najaf/README.md

Najaf — Urban Rail Network

Country: IQ · Population: 1,540,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Najaf rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [najaf.corridor.geojson](#).

At a glance

Metric	Value
Lines	6
Unique stations	90
Interchange stations	25
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	59.5%
Route length (double track)	172.3 km
Revenue fleet	129 × 4-car trainsets
Spare + cold-reserve	15 × 4-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	39.3 km	19	32	NW Outer SE Outer
line-2	20.3 km	9	18	NW Mid E Mid
line-3	25.3 km	12	21	N Mid SW Mid
line-4	21.9 km	13	19	SW Inner NE Mid
line-5	20.1 km	8	17	S Outer SE Inner
line-6	45.4 km	30	37	NW Mid NW Mid
Total	172.3 km	90 unique	144	

Rolling stock

Property	Value
Consist	4-car, 75 m
Max speed	90 km/h
Onboard battery	480 kWh per trainset
Seats	80 longitudinal seats
Nominal capacity (AW2)	440 pax (seated + standing, metro-4car per RFC 0008 §1)
Crush capacity (AW3)	560 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 440 AW2 passengers (metro - 4 car)
- Peak frequency: 12 trains/hour/direction (5-min headway)
- Peak capacity per line per direction: $440 \times 12 = 5,280$ pphpd
- Network peak throughput (all lines, both directions): $6 \text{ lines} \times 2 \text{ directions} \times 5,280 = 63,360$ passengers/hour
- Daily theoretical capacity (peak $\times$ 10): $\approx 633,600$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 91,630 - 137,445$ trips/day

Catchment

- City population: 1,540,000
- Anchor-weighted coverage: 59.5%
- Catchment population: $\approx 916,300$ (within ~800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	25	500 kW	3000 kWh
Major	29	400 kW	2500 kWh
Standard	22	300 kW	2000 kWh
Terminal	9	500 kW	3000 kWh
Total installed	86	40,200 kW	258,500 kWh

Aggregate station-rail charging power: 49,250 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 480 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	16.0 kWh/km	4 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	459 kWh	28.7 km average line length
Onboard battery coverage	1.0 $\times$ average line run	480 kWh usable pack
Average 60 s dwell charge	9.1 kWh/stop	547 kW average charger across stops
Stops to refill one trainset pack	53 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	201 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	258 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in design.toml — emitted by the osr-design Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. country-costs.toml applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (140.6 km @ €3.5 M/km)	€492 M
Elevated (30.9 km @ €18 M/km)	€556 M
Elevated-interchange premium (11 sites @ €20 M)	€220 M
Civil subtotal	€1.27 bn

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
halt	5	€0.4 M	€2.0 M
standard	22	€1.5 M	€33 M
major	29	€3.0 M	€87 M
terminal	9	€2.5 M	€22 M
depot-terminal	1	€3.0 M	€3.0 M
interchange-elevated	25	€4.5 M	€112 M
Stations subtotal			€260 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	9	€3.0 M	€27 M
Depots subtotal			€52 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
metro-4car (revenue + spare + cold reserve)	144	€4.0 M	€576 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	172.3 km × €0.015 M/km	€2.6 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€37 M
EPC integration + project management (7%)	on subtotal	€154 M

Total

Bucket	Value
Civil works	€1.27 bn
Stations	€260 M
Depots	€52 M
Rolling stock	€576 M
Residual train-control wayside + charging microgrids	€40 M
EPC overhead (7%)	€154 M
CAPEX total	€2.35 bn
Per-route-km	€14 M / km
Per-capita (city pop)	€1,526 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running `scripts/regenerate-city.sh` najaf.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–5 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 6 and runs for 20 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–5)	€177 M / yr	€115
Steady-state, low-ridership (year 6+)	€210 M / yr	€137
Steady-state, high-ridership (year 6+)	€194 M / yr	€126
Lifecycle envelope (yr 1–25, low scenario)	€5.09 bn cumulative	€3,306
Lifecycle envelope (yr 1–25, high scenario)	€4.76 bn cumulative	€3,093

Population basis: 1,540,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 25, debt service drops to zero and only the OPEX shortfall remains – ~€45 M / yr (low) €28 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€1.41 bn	4.0%	25 y, 5 y grace	€104 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€587 M	8.5%	25 y, 5 y grace	€62 M / yr
Government equity (no debt service)	15%	€352 M	—	—	—
Total	100%	€2.35 bn			€166 M / yr

During the 5-year grace period the operator pays interest only – multilateral €56 M / yr + bonds €50 M / yr = €106 M / yr total – plus the equity tranche amortised across construction (€70 M / yr × 5 yr). Principal repayment begins in year 6 on a 20-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€23 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€32 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€129 k
Traction energy (405.4 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (1,046 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€6.1 M
OPEX subtotal		€61 M / yr

Annual fleet utilisation: 129 revenue trainsets × 20.5 h/day × 365 d/yr × 35 km/h commercial × 75% revenue factor = 25.3 M train-km / yr (~196 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$380 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.58 (~\$0.63 USD)
Day pass (3 trips)	€1.49 (15 % bulk discount)
Monthly unlimited pass	€17.48 (~5 % of median monthly income)
Annual pass	€192.28 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	28.1 M	56.2 M
Farebox revenue	€16 M / yr	€33 M / yr
Farebox / OPEX recovery	27%	54%
Country policy-target recovery (diagnostic)	45%	45%
Operating shortfall (gov subsidy required)	€45 M / yr	€28 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€210 M / yr	€194 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
najaf.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
najaf-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
najaf.corridor.geojson	Line polylines + stations (GeoJSON)
najaf.stations.json	Machine-readable station list
najaf.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug najaf

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug najaf \
  --sidecar .cache/osr-pipeline/rasters/najaf.grid.json \
  --out-dir designs/.../Najaf

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../najaf.toml \
  --out designs/.../README.md
```

`scripts/regenerate-najaf.sh` chains steps 3 + drift tests into a single command.

west-asia/Iraq/Nasiriyah

designs/west-asia/Iraq/Nasiriyah/README.md

Nasiriyah — Urban Rail Network

Country: IQ · Population: 705,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Nasiriyah rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [nasiriyah.corridor.geojson](#).

At a glance

Metric	Value
Lines	3
Unique stations	33
Interchange stations	4
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	40.4%
Route length (double track)	56.4 km
Revenue fleet	50 × 3-car trainsets
Spare + cold-reserve	6 × 3-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	20.1 km	13	20	SE Mid N Outer
line-2	16.6 km	11	17	NE Mid SW Mid
line-3	19.6 km	9	19	W Inner SE Outer
Total	56.4 km	33 unique	56	

Rolling stock

Property	Value
Consist	3-car, 57 m
Max speed	90 km/h
Onboard battery	360 kWh per trainset
Seats	60 longitudinal seats
Nominal capacity (AW2)	330 pax (seated + standing, light-metro-3car per RFC 0008 §1)
Crush capacity (AW3)	420 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 330 AW2 passengers (light-metro-3car)
- Peak frequency: 12 trains/hour/direction (5-min headway)

- Peak capacity per line per direction: $330 \times 12 = 3,960$ pphpd
- Network peak throughput (all lines, both directions): $3 \text{ lines} \times 2 \text{ directions} \times 3,960 = 23,760$ passengers/hour
- Daily theoretical capacity (peak $\times 10$): $\approx 237,600$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 28,482 - 42,723$ trips/day

Catchment

- City population: 705,000
- Anchor-weighted coverage: 40.4%
- Catchment population: $\approx 284,820$ (within ~ 800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	4	500 kW	3000 kWh
Major	15	400 kW	2500 kWh
Standard	5	300 kW	2000 kWh
Terminal	5	500 kW	3000 kWh
Total installed	30	17,000 kW	114,500 kWh

Aggregate station-rail charging power: 18,750 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 360 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	12.0 kWh/km	3 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	226 kWh	18.8 km average line length
Onboard battery coverage	1.6 $\times$ average line run	360 kWh usable pack
Average 60 s dwell charge	9.5 kWh/stop	568 kW average charger across stops
Stops to refill one trainset pack	38 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	85 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	114 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in design.toml — emitted by the osr-design Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. country-costs.toml applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (37.8 km @ €3.5 M/km)	€132 M
Elevated (18.4 km @ €18 M/km)	€331 M
Elevated-interchange premium (1 sites @ €20 M)	€20 M
Civil subtotal	€483 M

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~ 11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
halt	3	€0.4 M	€1.2 M
standard	5	€1.5 M	€7.5 M
major	15	€3.0 M	€45 M
terminal	5	€2.5 M	€12 M
depot-terminal	1	€3.0 M	€3.0 M
interchange	2	€4.5 M	€9.0 M
interchange-elevated	2	€4.5 M	€9.0 M
Stations subtotal			€87 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	5	€3.0 M	€15 M
Depots subtotal			€40 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
light-metro-3car (revenue + spare + cold reserve)	56	€3.0 M	€168 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	56.4 km × €0.015 M/km	€0.8 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€13 M
EPC integration + project management (7%)	on subtotal	€55 M

Total

Bucket	Value
Civil works	€483 M
Stations	€87 M
Depots	€40 M
Rolling stock	€168 M
Residual train-control wayside + charging microgrids	€14 M
EPC overhead (7%)	€55 M
CAPEX total	€847 M
Per-route-km	€15 M / km
Per-capita (city pop)	€1,202 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running scripts/regenerate-city.sh nasiriyah.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–5 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 6 and runs for 20 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–5)	€64 M / yr	€90
Steady-state, low-ridership (year 6+)	€73 M / yr	€104
Steady-state, high-ridership (year 6+)	€66 M / yr	€93
Lifecycle envelope (yr 1–25, low scenario)	€1.78 bn cumulative	€2,532
Lifecycle envelope (yr 1–25, high scenario)	€1.64 bn cumulative	€2,319

Population basis: 705,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 25, debt service drops to zero and only the OPEX shortfall remains – ~€14 M / yr (low) €6.0 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€508 M	4.0%	25 y, 5 y grace	€37 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€212 M	8.5%	25 y, 5 y grace	€22 M / yr
Government equity (no debt service)	15%	€127 M	—	—	—
Total	100%	€847 M			€60 M / yr

During the 5-year grace period the operator pays interest only – multilateral €20 M / yr + bonds €18 M / yr = €38 M / yr total – plus the equity tranche amortised across construction (€25 M / yr × 5 yr). Principal repayment begins in year 6 on a 20-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€6.7 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€12 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€42 k
Traction energy (101.0 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (350 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€2.1 M
OPEX subtotal		€21 M / yr

Annual fleet utilisation: 50 revenue trainsets × 20.5 h/day × 365 d/yr × 30 km/h commercial × 75% revenue factor = 8.4 M train-km / yr (~168 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$380 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.58 (~\$0.63 USD)
Day pass (3 trips)	€1.49 (15 % bulk discount)
Monthly unlimited pass	€17.48 (~5 % of median monthly income)
Annual pass	€192.28 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	12.9 M	25.7 M

	Low scenario	High scenario
Farebox revenue	€7.5 M / yr	€15 M / yr
Farebox / OPEX recovery	36%	71%
Country policy-target recovery (diagnostic)	45%	45%
Operating shortfall (gov subsidy required)	€14 M / yr	€6.0 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€73 M / yr	€66 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
nasiriyah.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
nasiriyah-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
nasiriyah.corridor.geojson	Line polylines + stations (GeoJSON)
nasiriyah.stations.json	Machine-readable station list
nasiriyah.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug nasiriyah

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug nasiriyah \
  --sidecar .cache/osr-pipeline/rasters/nasiriyah.grid.json \
  --out-dir designs/.../Nasiriyah

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../nasiriyah.toml \
  --out designs/.../README.md
```

`scripts/regenerate-nasiriyah.sh` chains steps 3 + drift tests into a single command.

west-asia/Iraq/Ramadi

designs/west-asia/Iraq/Ramadi/README.md

Ramadi — Urban Rail Network

Country: IQ · Population: 525,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Ramadi rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [ramadi.corridor.geojson](#).

At a glance

Metric	Value
Lines	3
Unique stations	35
Interchange stations	3
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	39.2%
Route length (double track)	46.8 km
Revenue fleet	43 × 3-car trainsets
Spare + cold-reserve	6 × 3-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	18.6 km	13	19	W Outer E Outer
line-2	13.8 km	11	15	SE Mid W Outer
line-3	14.4 km	11	15	NW Mid E Outer
Total	46.8 km	35 unique	49	

Rolling stock

Property	Value
Consist	3-car, 57 m
Max speed	90 km/h
Onboard battery	360 kWh per trainset
Seats	60 longitudinal seats
Nominal capacity (AW2)	330 pax (seated + standing, light-metro-3car per RFC 0008 §1)
Crush capacity (AW3)	420 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 330 AW2 passengers (light-metro-3car)
- Peak frequency: 12 trains/hour/direction (5-min headway)

- Peak capacity per line per direction: $330 \times 12 = 3,960$ pphpd
- Network peak throughput (all lines, both directions): $3 \text{ lines} \times 2 \text{ directions} \times 3,960 = 23,760$ passengers/hour
- Daily theoretical capacity (peak $\times 10$): $\approx 237,600$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 20,580 - 30,870$ trips/day

Catchment

- City population: 525,000
- Anchor-weighted coverage: 39.2%
- Catchment population: $\approx 205,800$ (within ~ 800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	3	500 kW	3000 kWh
Major	20	400 kW	2500 kWh
Standard	6	300 kW	2000 kWh
Terminal	5	500 kW	3000 kWh
Total installed	35	18,800 kW	126,000 kWh

Aggregate station-rail charging power: 20,500 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 360 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	12.0 kWh/km	3 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	187 kWh	15.6 km average line length
Onboard battery coverage	1.9 $\times$ average line run	360 kWh usable pack
Average 60 s dwell charge	9.8 kWh/stop	586 kW average charger across stops
Stops to refill one trainset pack	37 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	94 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	126 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the `[costs]` block in `design.toml` — emitted by the `osr-design` Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. `country-costs.toml` applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (44.5 km @ €3.5 M/km)	€156 M
Elevated (2.2 km @ €18 M/km)	€39 M
Elevated-interchange premium (1 sites @ €20 M)	€20 M
Civil subtotal	€215 M

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~ 11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
standard	6	€1.5 M	€9.0 M
major	20	€3.0 M	€60 M
terminal	5	€2.5 M	€12 M
depot-terminal	1	€3.0 M	€3.0 M
interchange-elevated	3	€4.5 M	€14 M
Stations subtotal			€98 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	5	€3.0 M	€15 M
Depots subtotal			€40 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
light-metro-3car (revenue + spare + cold reserve)	49	€3.0 M	€147 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	46.8 km × €0.015 M/km	€0.7 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€14 M
EPC integration + project management (7%)	on subtotal	€36 M

Total

Bucket	Value
Civil works	€215 M
Stations	€98 M
Depots	€40 M
Rolling stock	€147 M
Residual train-control wayside + charging microgrids	€15 M
EPC overhead (7%)	€36 M
CAPEX total	€551 M
Per-route-km	€12 M / km
Per-capita (city pop)	€1,049 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running scripts/regenerate-city.sh ramadi.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–5 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 6 and runs for 20 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–5)	€41 M / yr	€79
Steady-state, low-ridership (year 6+)	€48 M / yr	€91
Steady-state, high-ridership (year 6+)	€42 M / yr	€81
Lifecycle envelope (yr 1–25, low scenario)	€1.17 bn cumulative	€2,221
Lifecycle envelope (yr 1–25, high scenario)	€1.05 bn cumulative	€2,009

Population basis: 525,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 25, debt service drops to zero and only the OPEX shortfall remains – ~€9.1 M / yr (low) €3.5 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€330 M	4.0%	25 y, 5 y grace	€24 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€138 M	8.5%	25 y, 5 y grace	€15 M / yr
Government equity (no debt service)	15%	€83 M	—	—	—
Total	100%	€551 M			€39 M / yr

During the 5-year grace period the operator pays interest only – multilateral €13 M / yr + bonds €12 M / yr = €25 M / yr total – plus the equity tranche amortised across construction (€17 M / yr × 5 yr). Principal repayment begins in year 6 on a 20-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€5.9 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€7.1 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€35 k
Traction energy (86.9 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (293 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€1.7 M
OPEX subtotal		€15 M / yr

Annual fleet utilisation: 43 revenue trainsets × 20.5 h/day × 365 d/yr × 30 km/h commercial × 75% revenue factor = 7.2 M train-km / yr (~168 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$380 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.58 (~\$0.63 USD)
Day pass (3 trips)	€1.49 (15 % bulk discount)
Monthly unlimited pass	€17.48 (~5 % of median monthly income)
Annual pass	€192.28 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	9.6 M	19.2 M

	Low scenario	High scenario
Farebox revenue	€5.6 M / yr	€11 M / yr
Farebox / OPEX recovery	38%	76%
Country policy-target recovery (diagnostic)	45%	45%
Operating shortfall (gov subsidy required)	€9.1 M / yr	€3.5 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€48 M / yr	€42 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
ramadi.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
ramadi-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
ramadi.corridor.geojson	Line polylines + stations (GeoJSON)
ramadi.stations.json	Machine-readable station list
ramadi.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug ramadi

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug ramadi \
  --sidecar .cache/osr-pipeline/rasters/ramadi.grid.json \
  --out-dir designs/.../Ramadi

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../ramadi.toml \
  --out designs/.../README.md
```

`scripts/regenerate-ramadi.sh` chains steps 3 + drift tests into a single command.

west-asia/Iraq/Samawah

designs/west-asia/Iraq/Samawah/README.md

Samawah — Urban Rail Network

Country: IQ · Population: 373,770

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Samawah rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [samawah.corridor.geojson](#).

At a glance

Metric	Value
Lines	3
Unique stations	33
Interchange stations	3
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	56.5%
Route length (double track)	54.9 km
Revenue fleet	49 × 3-car trainsets
Spare + cold-reserve	6 × 3-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	21.0 km	13	21	S Outer N Outer
line-2	22.2 km	12	21	SW Outer N Mid
line-3	11.7 km	8	13	E Mid NW Mid
Total	54.9 km	33 unique	55	

Rolling stock

Property	Value
Consist	3-car, 57 m
Max speed	90 km/h
Onboard battery	360 kWh per trainset
Seats	60 longitudinal seats
Nominal capacity (AW2)	330 pax (seated + standing, light-metro-3car per RFC 0008 §1)
Crush capacity (AW3)	420 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 330 AW2 passengers (light-metro-3car)
- Peak frequency: 12 trains/hour/direction (5-min headway)

- Peak capacity per line per direction: $330 \times 12 = 3,960$ pphpd
- Network peak throughput (all lines, both directions): $3 \text{ lines} \times 2 \text{ directions} \times 3,960 = 23,760$ passengers/hour
- Daily theoretical capacity (peak $\times 10$): $\approx 237,600$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 21,118 - 31,677$ trips/day

Catchment

- City population: 373,770
- Anchor-weighted coverage: 56.5%
- Catchment population: $\approx 211,180$ (within ~ 800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	3	500 kW	3000 kWh
Major	15	400 kW	2500 kWh
Standard	6	300 kW	2000 kWh
Terminal	5	500 kW	3000 kWh
Total installed	30	16,800 kW	113,500 kWh

Aggregate station-rail charging power: 18,750 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 360 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	12.0 kWh/km	3 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	220 kWh	18.3 km average line length
Onboard battery coverage	1.6 $\times$ average line run	360 kWh usable pack
Average 60 s dwell charge	9.5 kWh/stop	568 kW average charger across stops
Stops to refill one trainset pack	38 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	84 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	114 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the `[costs]` block in `design.toml` — emitted by the `osr-design` Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. `country-costs.toml` applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (53.1 km @ €3.5 M/km)	€186 M
Elevated (1.6 km @ €18 M/km)	€29 M
Elevated-interchange premium (1 sites @ €20 M)	€20 M
Civil subtotal	€235 M

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~ 11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
halt	3	€0.4 M	€1.2 M
standard	6	€1.5 M	€9.0 M
major	15	€3.0 M	€45 M
terminal	5	€2.5 M	€12 M
depot-terminal	1	€3.0 M	€3.0 M
interchange-elevated	3	€4.5 M	€14 M
Stations subtotal			€84 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	5	€3.0 M	€15 M
Depots subtotal			€40 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
light-metro-3car (revenue + spare + cold reserve)	55	€3.0 M	€165 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	54.9 km × €0.015 M/km	€0.8 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€12 M
EPC integration + project management (7%)	on subtotal	€38 M

Total

Bucket	Value
Civil works	€235 M
Stations	€84 M
Depots	€40 M
Rolling stock	€165 M
Residual train-control wayside + charging microgrids	€13 M
EPC overhead (7%)	€38 M
CAPEX total	€575 M
Per-route-km	€10 M / km
Per-capita (city pop)	€1,538 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running scripts/regenerate-city.sh samawah.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–5 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 6 and runs for 20 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–5)	€43 M / yr	€116
Steady-state, low-ridership (year 6+)	€52 M / yr	€140
Steady-state, high-ridership (year 6+)	€48 M / yr	€130
Lifecycle envelope (yr 1–25, low scenario)	€1.26 bn cumulative	€3,384
Lifecycle envelope (yr 1–25, high scenario)	€1.19 bn cumulative	€3,171

Population basis: 373,770 (catchment per [lib/city-batches/world-sample.toml](#)). After year 25, debt service drops to zero and only the OPEX shortfall remains – ~€12 M / yr (low) €7.9 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€345 M	4.0%	25 y, 5 y grace	€25 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€144 M	8.5%	25 y, 5 y grace	€15 M / yr
Government equity (no debt service)	15%	€86 M	—	—	—
Total	100%	€575 M			€41 M / yr

During the 5-year grace period the operator pays interest only – multilateral €14 M / yr + bonds €12 M / yr = €26 M / yr total – plus the equity tranche amortised across construction (€17 M / yr × 5 yr). Principal repayment begins in year 6 on a 20-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€6.6 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€7.2 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€41 k
Traction energy (99.0 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (341 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€2.0 M
OPEX subtotal		€16 M / yr

Annual fleet utilisation: 49 revenue trainsets × 20.5 h/day × 365 d/yr × 30 km/h commercial × 75% revenue factor = 8.2 M train-km / yr (~168 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$380 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.58 (~\$0.63 USD)
Day pass (3 trips)	€1.49 (15 % bulk discount)
Monthly unlimited pass	€17.48 (~5 % of median monthly income)
Annual pass	€192.28 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	6.8 M	13.6 M

	Low scenario	High scenario
Farebox revenue	€4.0 M / yr	€7.9 M / yr
Farebox / OPEX recovery	25%	50%
Country policy-target recovery (diagnostic)	45%	45%
Operating shortfall (gov subsidy required)	€12 M / yr	€7.9 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€52 M / yr	€48 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
samawah.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
samawah-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
samawah.corridor.geojson	Line polylines + stations (GeoJSON)
samawah.stations.json	Machine-readable station list
samawah.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug samawah

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug samawah \
  --sidecar .cache/osr-pipeline/rasters/samawah.grid.json \
  --out-dir designs/.../Samawah

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../samawah.toml \
  --out designs/.../README.md
```

`scripts/regenerate-samawah.sh` chains steps 3 + drift tests into a single command.

west-asia/Iraq/Sulaymaniyah

designs/west-asia/Iraq/Sulaymaniyah/README.md

Sulaymaniyah — Urban Rail Network

Country: IQ · Population: 2,150,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Sulaymaniyah rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [sulaymaniyah.corridor.geojson](#).

At a glance

Metric	Value
Lines	4
Unique stations	58
Interchange stations	8
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	49.0%
Route length (double track)	127.0 km
Revenue fleet	94 × 4-car trainsets
Spare + cold-reserve	12 × 4-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	37.1 km	14	30	SE Outer W Mid
line-2	16.9 km	11	15	E Inner NW Mid
line-3	27.7 km	11	24	W Inner NE Outer
line-4	45.2 km	23	37	W Mid W Mid
Total	127.0 km	58 unique	106	

Rolling stock

Property	Value
Consist	4-car, 75 m
Max speed	90 km/h
Onboard battery	480 kWh per trainset
Seats	80 longitudinal seats
Nominal capacity (AW2)	440 pax (seated + standing, met ro - 4car per RFC 0008 §1)
Crush capacity (AW3)	560 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 440 AW2 passengers (met ro - 4car)

- Peak frequency: 12 trains/hour/direction (5-min headway)
- Peak capacity per line per direction: $440 \times 12 = 5,280$ pphpd
- Network peak throughput (all lines, both directions): $4 \text{ lines} \times 2 \text{ directions} \times 5,280 = 42,240$ passengers/hour
- Daily theoretical capacity (peak $\times$ 10): $\approx 422,400$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 105,350 - 158,025$ trips/day

Catchment

- City population: 2,150,000
- Anchor-weighted coverage: 49.0%
- Catchment population: $\approx 1,053,500$ (within ~ 800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	8	500 kW	3000 kWh
Major	17	400 kW	2500 kWh
Standard	25	300 kW	2000 kWh
Terminal	5	500 kW	3000 kWh
Total installed	56	25,800 kW	171,500 kWh

Aggregate station-rail charging power: 31,750 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 480 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	16.0 kWh/km	4 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	508 kWh	31.8 km average line length
Onboard battery coverage	0.9 $\times$ average line run	480 kWh usable pack
Average 60 s dwell charge	9.1 kWh/stop	547 kW average charger across stops
Stops to refill one trainset pack	53 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	129 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	172 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in design.toml — emitted by the osr-design Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. country-costs.toml applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (115.3 km @ €3.5 M/km)	€403 M
Elevated (11.3 km @ €18 M/km)	€203 M
Elevated-interchange premium (4 sites @ €20 M)	€80 M
Civil subtotal	€686 M

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
halt	3	€0.4 M	€1.2 M
standard	25	€1.5 M	€38 M
major	17	€3.0 M	€51 M
terminal	5	€2.5 M	€12 M
depot-terminal	1	€3.0 M	€3.0 M
interchange-elevated	8	€4.5 M	€36 M
Stations subtotal			€141 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	5	€3.0 M	€15 M
Depots subtotal			€40 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
metro-4car (revenue + spare + cold reserve)	106	€4.0 M	€424 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	127.0 km × €0.015 M/km	€1.9 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€21 M
EPC integration + project management (7%)	on subtotal	€92 M

Total

Bucket	Value
Civil works	€686 M
Stations	€141 M
Depots	€40 M
Rolling stock	€424 M
Residual train-control wayside + charging microgrids	€23 M
EPC overhead (7%)	€92 M
CAPEX total	€1.41 bn
Per-route-km	€11 M / km
Per-capita (city pop)	€654 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running `scripts/regenerate-city.sh sulaymaniyah`.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–5 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 6 and runs for 20 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–5)	€106 M / yr	€49
Steady-state, low-ridership (year 6+)	€115 M / yr	€54
Steady-state, high-ridership (year 6+)	€99 M / yr	€46
Lifecycle envelope (yr 1–25, low scenario)	€2.84 bn cumulative	€1,319
Lifecycle envelope (yr 1–25, high scenario)	€2.51 bn cumulative	€1,169

Population basis: 2,150,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 25, debt service drops to zero and only the OPEX shortfall remains – ~€16 M / yr (low) €0 k / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€844 M	4.0%	25 y, 5 y grace	€62 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€352 M	8.5%	25 y, 5 y grace	€37 M / yr
Government equity (no debt service)	15%	€211 M	—	—	—
Total	100%	€1.41 bn			€99 M / yr

During the 5-year grace period the operator pays interest only – multilateral €34 M / yr + bonds €30 M / yr = €64 M / yr total – plus the equity tranche amortised across construction (€42 M / yr × 5 yr). Principal repayment begins in year 6 on a 20-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€17 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€17 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€95 k
Traction energy (295.4 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (774 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€4.5 M
OPEX subtotal		€39 M / yr

Annual fleet utilisation: 94 revenue trainsets × 20.5 h/day × 365 d/yr × 35 km/h commercial × 75% revenue factor = 18.5 M train-km / yr (~196 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$380 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.58 (~\$0.63 USD)
Day pass (3 trips)	€1.49 (15 % bulk discount)
Monthly unlimited pass	€17.48 (~5 % of median monthly income)
Annual pass	€192.28 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	39.2 M	78.5 M
Farebox revenue	€23 M / yr	€46 M / yr
Farebox / OPEX recovery	59%	117%
Country policy-target recovery (diagnostic)	45%	45%
Operating shortfall (gov subsidy required)	€16 M / yr	€0 k / yr
Operating surplus (operator retained capex sinking fund)	€0 k / yr	€6.8 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€115 M / yr	€99 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
sulaymaniyah.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
sulaymaniyah-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
sulaymaniyah.corridor.geojson	Line polylines + stations (GeoJSON)
sulaymaniyah.stations.json	Machine-readable station list
sulaymaniyah.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug sulaymaniyah

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug sulaymaniyah \
  --sidecar .cache/osr-pipeline/rasters/sulaymaniyah.grid.json \
  --out-dir designs/.../Sulaymaniyah

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../sulaymaniyah.toml \
  --out designs/.../README.md
```

`scripts/regenerate-sulaymaniyah.sh` chains steps 3 + drift tests into a single command.

west-asia/Jordan/Amman

designs/west-asia/Jordan/Amman/README.md

Amman — Urban Rail Network

Country: JO · Population: 4,007,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Amman rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [amman.corridor.geojson](#).

At a glance

Metric	Value
Lines	8
Unique stations	171
Interchange stations	29
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	50.9%
Route length (double track)	354.0 km
Revenue fleet	256 × 6-car trainsets
Spare + cold-reserve	30 × 6-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	52.5 km	24	42	NE Outer SW Outer
line-2	40.1 km	16	32	S Mid NE Outer
line-3	41.6 km	19	34	NW Outer S Mid
line-4	33.0 km	18	27	N Mid SE Mid
line-5	36.6 km	20	30	SE Mid W Outer
line-6	31.5 km	17	26	E Mid NW Outer
line-7	29.4 km	16	25	W Outer NE Mid
line-8	89.2 km	42	70	W Mid W Mid
Total	354.0 km	171 unique	286	

Rolling stock

Property	Value
Consist	6-car, 111 m
Max speed	100 km/h
Onboard battery	720 kWh per trainset
Seats	120 longitudinal seats
Nominal capacity (AW2)	660 pax (seated + standing, metro-6car per RFC 0008 §1)
Crush capacity (AW3)	840 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 660 AW2 passengers (metro - 6 car)
- Peak frequency: 12 trains/hour/direction (5-min headway)
- Peak capacity per line per direction: $660 \times 12 = 7,920$ pphpd
- Network peak throughput (all lines, both directions): $8 \text{ lines} \times 2 \text{ directions} \times 7,920 = 126,720$ passengers/hour
- Daily theoretical capacity (peak $\times$ 10): $\approx 1,267,200$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 203,956 - 305,934$ trips/day

Catchment

- City population: 4,007,000
- Anchor-weighted coverage: 50.9%
- Catchment population: $\approx 2,039,563$ (within ~800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	29	500 kW	3000 kWh
Major	62	400 kW	2500 kWh
Standard	64	300 kW	2000 kWh
Terminal	13	500 kW	3000 kWh
Total installed	169	70,000 kW	449,000 kWh

Aggregate station-rail charging power: 92,250 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 720 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	24.0 kWh/km	6 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	1,062 kWh	44.2 km average line length
Onboard battery coverage	0.7 $\times$ average line run	720 kWh usable pack
Average 60 s dwell charge	9.0 kWh/stop	539 kW average charger across stops
Stops to refill one trainset pack	80 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	350 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	449 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in design.toml — emitted by the osr-design Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. country-costs.toml applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (323.9 km @ €3.5 M/km)	€1.13 bn
Elevated (29.2 km @ €18 M/km)	€525 M
Elevated-interchange premium (14 sites @ €20 M)	€280 M
Civil subtotal	€1.94 bn

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
halt	3	€0.4 M	€1.2 M
standard	64	€1.5 M	€96 M
major	62	€3.0 M	€186 M
terminal	13	€2.5 M	€32 M
depot-terminal	1	€3.0 M	€3.0 M
interchange	2	€4.5 M	€9.0 M
interchange-elevated	27	€4.5 M	€122 M
Stations subtotal			€449 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	13	€3.0 M	€39 M
Depots subtotal			€64 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
metro-6car (revenue + spare + cold reserve)	286	€6.0 M	€1.72 bn

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	354.0 km × €0.015 M/km	€5.3 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€65 M
EPC integration + project management (7%)	on subtotal	€297 M

Total

Bucket	Value
Civil works	€1.94 bn
Stations	€449 M
Depots	€64 M
Rolling stock	€1.72 bn
Residual train-control wayside + charging microgrids	€70 M
EPC overhead (7%)	€297 M
CAPEX total	€4.53 bn

Bucket	Value
Per-route-km	€13 M / km
Per-capita (city pop)	€1,132 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code — regenerate by re-running scripts/regenerate-city.sh amman.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–5 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 6 and runs for 20 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–5)	€330 M / yr	€82
Steady-state, low-ridership (year 6+)	€383 M / yr	€96
Steady-state, high-ridership (year 6+)	€318 M / yr	€79
Lifecycle envelope (yr 1–25, low scenario)	€9.32 bn cumulative	€2,326
Lifecycle envelope (yr 1–25, high scenario)	€8.02 bn cumulative	€2,001

Population basis: 4,007,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 25, debt service drops to zero and only the OPEX shortfall remains — ~€72 M / yr (low) €7.0 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€2.72 bn	4.0%	25 y, 5 y grace	€200 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€1.13 bn	7.5%	25 y, 5 y grace	€111 M / yr
Government equity (no debt service)	15%	€680 M	—	—	—
Total	100%	€4.53 bn			€311 M / yr

During the 5-year grace period the operator pays interest only — multilateral €109 M / yr + bonds €85 M / yr = €194 M / yr total — plus the equity tranche amortised across construction (€136 M / yr × 5 yr). Principal repayment begins in year 6 on a 20-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€69 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€49 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€265 k
Traction energy (1206.8 GWh / yr)	trackside PV + Na-ion (RFC 0002) — self-generated, €0 / yr	€0 k
Labour (2,136 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€19 M
OPEX subtotal		€137 M / yr

Annual fleet utilisation: 256 revenue trainsets × 20.5 h/day × 365 d/yr × 35 km/h commercial × 75% revenue factor = 50.3 M train-km / yr (~196 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$580 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass — a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.89 (~\$0.97 USD)
Day pass (3 trips)	€2.27 (15 % bulk discount)
Monthly unlimited pass	€26.68 (~5 % of median monthly income)
Annual pass	€293.48 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	73.1 M	146.3 M
Farebox revenue	€65 M / yr	€130 M / yr
Farebox / OPEX recovery	47%	95%
Country policy-target recovery (diagnostic)	55%	55%
Operating shortfall (gov subsidy required)	€72 M / yr	€7.0 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€383 M / yr	€318 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
amman.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
amman-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
amman.corridor.geojson	Line polylines + stations (GeoJSON)
amman.stations.json	Machine-readable station list
amman.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug amman

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug amman \
  --sidecar .cache/osr-pipeline/rasters/amman.grid.json \
  --out-dir designs/.../Amman

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../amman.toml \
  --out designs/.../README.md
```

`scripts/regenerate-amman.sh` chains steps 3 + drift tests into a single command.

Aqaba — Urban Rail Network

Country: JO · Population: 250,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Aqaba rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [aqaba.corridor.geojson](#).

At a glance

Metric	Value
Lines	3
Unique stations	23
Interchange stations	3
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	56.7%
Route length (double track)	34.4 km
Revenue fleet	42 × 2-car trainsets
Spare + cold-reserve	6 × 2-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	15.2 km	10	20	E Outer W Outer
line-2	8.7 km	7	13	NE Outer SE Mid
line-3	10.4 km	6	15	SE Mid W Outer
Total	34.4 km	23 unique	48	

Rolling stock

Property	Value
Consist	2-car, 39 m
Max speed	70 km/h
Onboard battery	240 kWh per trainset
Seats	40 longitudinal seats
Nominal capacity (AW2)	210 pax (seated + standing, tram-2car per RFC 0008 §1)
Crush capacity (AW3)	260 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 210 AW2 passengers (tram-2car)
- Peak frequency: 12 trains/hour/direction (5-min headway)

- Peak capacity per line per direction: $210 \times 12 = 2,520$ pphpd
- Network peak throughput (all lines, both directions): $3 \text{ lines} \times 2 \text{ directions} \times 2,520 = 15,120$ passengers/hour
- Daily theoretical capacity (peak $\times 10$): $\approx 151,200$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 14,175 - 21,262$ trips/day

Catchment

- City population: 250,000
- Anchor-weighted coverage: 56.7%
- Catchment population: $\approx 141,750$ (within ~ 800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	3	500 kW	3000 kWh
Major	6	400 kW	2500 kWh
Standard	8	300 kW	2000 kWh
Terminal	5	500 kW	3000 kWh
Total installed	23	13,800 kW	95,000 kWh

Aggregate station-rail charging power: 14,500 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 240 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	8.0 kWh/km	2 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	92 kWh	11.5 km average line length
Onboard battery coverage	2.6 $\times$ average line run	240 kWh usable pack
Average 60 s dwell charge	10.5 kWh/stop	630 kW average charger across stops
Stops to refill one trainset pack	23 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	69 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	95 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in design.toml — emitted by the osr-design Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. country-costs.toml applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (34.2 km @ €3.5 M/km)	€120 M
Civil subtotal	€120 M

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~ 11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
standard	8	€1.5 M	€12 M

Archetype	Count	Unit	Subtotal
major	6	€3.0 M	€18 M
terminal	5	€2.5 M	€12 M
depot-terminal	1	€3.0 M	€3.0 M
interchange	3	€4.5 M	€14 M
Stations subtotal			€59 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	5	€3.0 M	€15 M
Depots subtotal			€40 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
tram-2car (revenue + spare + cold reserve)	48	€2.0 M	€96 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	34.4 km × €0.015 M/km	€0.5 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€8.9 M
EPC integration + project management (7%)	on subtotal	€23 M

Total

Bucket	Value
Civil works	€120 M
Stations	€59 M
Depots	€40 M
Rolling stock	€96 M
Residual train-control wayside + charging microgrids	€9.5 M
EPC overhead (7%)	€23 M
CAPEX total	€347 M
Per-route-km	€10 M / km
Per-capita (city pop)	€1,388 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code — regenerate by

re-running scripts/regenerate-city.sh aqaba.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–5 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 6 and runs for 20 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–5)	€25 M / yr	€101
Steady-state, low-ridership (year 6+)	€30 M / yr	€120
Steady-state, high-ridership (year 6+)	€26 M / yr	€104
Lifecycle envelope (yr 1–25, low scenario)	€725 M cumulative	€2,902
Lifecycle envelope (yr 1–25, high scenario)	€644 M cumulative	€2,577

Population basis: 250,000 (catchment per `lib/city-batches/world-sample.toml`). After year 25, debt service drops to zero and only the OPEX shortfall remains — ~€6.1 M / yr (low) €2.1 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€208 M	4.0%	25 y, 5 y grace	€15 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€87 M	7.5%	25 y, 5 y grace	€8.5 M / yr
Government equity (no debt service)	15%	€52 M	—	—	—
Total	100%	€347 M			€24 M / yr

During the 5-year grace period the operator pays interest only — multilateral €8.3 M / yr + bonds €6.5 M / yr = €15 M / yr total — plus the equity tranche amortised across construction (€10 M / yr × 5 yr). Principal repayment begins in year 6 on a 20-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€3.8 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€4.4 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€26 k
Traction energy (41.5 GWh / yr)	trackside PV + Na-ion (RFC 0002) — self-generated, €0 / yr	€0 k
Labour (218 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€2.0 M
OPEX subtotal		€10 M / yr

Annual fleet utilisation: 42 revenue trainsets × 20.5 h/day × 365 d/yr × 22 km/h commercial × 75% revenue factor = 5.2 M train-km / yr (~123 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$580 USD (per `lib/templates/country-finance.toml`). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass — a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.89 (~\$0.97 USD)
Day pass (3 trips)	€2.27 (15 % bulk discount)
Monthly unlimited pass	€26.68 (~5 % of median monthly income)
Annual pass	€293.48 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	4.6 M	9.1 M
Farebox revenue	€4.1 M / yr	€8.1 M / yr
Farebox / OPEX recovery	40%	80%

	Low scenario	High scenario
Country policy-target recovery (diagnostic)	55%	55%
Operating shortfall (gov subsidy required)	€6.1 M / yr	€2.1 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€30 M / yr	€26 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
aqaba.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
aqaba-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
aqaba.corridor.geojson	Line polylines + stations (GeoJSON)
aqaba.stations.json	Machine-readable station list
aqaba.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug aqaba

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug aqaba \
  --sidecar .cache/osr-pipeline/rasters/aqaba.grid.json \
  --out-dir designs/.../Aqaba

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../aqaba.toml \
  --out designs/.../README.md
```

`scripts/regenerate-aqaba.sh` chains steps 3 + drift tests into a single command.

west-asia/Jordan/Irbid

designs/west-asia/Jordan/Irbid/README.md

Irbid — Urban Rail Network

Country: JO · Population: 600,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Irbid rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [irbid.corridor.geojson](#).

At a glance

Metric	Value
Lines	3
Unique stations	33
Interchange stations	3
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	41.7%
Route length (double track)	56.1 km
Revenue fleet	50 × 3-car trainsets
Spare + cold-reserve	6 × 3-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	20.8 km	12	20	SE Outer N Mid
line-2	16.1 km	10	17	NE Outer W Outer
line-3	19.1 km	11	19	SW Outer NE Mid
Total	56.1 km	33 unique	56	

Rolling stock

Property	Value
Consist	3-car, 57 m
Max speed	90 km/h
Onboard battery	360 kWh per trainset
Seats	60 longitudinal seats
Nominal capacity (AW2)	330 pax (seated + standing, light-metro-3car per RFC 0008 §1)
Crush capacity (AW3)	420 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 330 AW2 passengers (light-metro-3car)
- Peak frequency: 12 trains/hour/direction (5-min headway)

- Peak capacity per line per direction: $330 \times 12 = 3,960$ pphpd
- Network peak throughput (all lines, both directions): $3 \text{ lines} \times 2 \text{ directions} \times 3,960 = 23,760$ passengers/hour
- Daily theoretical capacity (peak $\times 10$): $\approx 237,600$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 25,020 - 37,530$ trips/day

Catchment

- City population: 600,000
- Anchor-weighted coverage: 41.7%
- Catchment population: $\approx 250,200$ (within ~ 800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	3	500 kW	3000 kWh
Major	15	400 kW	2500 kWh
Standard	8	300 kW	2000 kWh
Terminal	5	500 kW	3000 kWh
Total installed	32	17,400 kW	117,500 kWh

Aggregate station-rail charging power: 19,250 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 360 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	12.0 kWh/km	3 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	224 kWh	18.7 km average line length
Onboard battery coverage	1.6 $\times$ average line run	360 kWh usable pack
Average 60 s dwell charge	9.7 kWh/stop	583 kW average charger across stops
Stops to refill one trainset pack	37 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	87 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	118 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in design.toml — emitted by the osr-design Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. country-costs.toml applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (54.7 km @ €3.5 M/km)	€192 M
Elevated (1.1 km @ €18 M/km)	€19 M
Elevated-interchange premium (1 sites @ €20 M)	€20 M
Civil subtotal	€231 M

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~ 11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
halt	1	€0.4 M	€0.4 M
standard	8	€1.5 M	€12 M
major	15	€3.0 M	€45 M
terminal	5	€2.5 M	€12 M
depot-terminal	1	€3.0 M	€3.0 M
interchange-elevated	3	€4.5 M	€14 M
Stations subtotal			€86 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	5	€3.0 M	€15 M
Depots subtotal			€40 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
light-metro-3car (revenue + spare + cold reserve)	56	€3.0 M	€168 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	56.1 km × €0.015 M/km	€0.8 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€13 M
EPC integration + project management (7%)	on subtotal	€38 M

Total

Bucket	Value
Civil works	€231 M
Stations	€86 M
Depots	€40 M
Rolling stock	€168 M
Residual train-control wayside + charging microgrids	€14 M
EPC overhead (7%)	€38 M
CAPEX total	€577 M
Per-route-km	€10 M / km
Per-capita (city pop)	€961 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running scripts/regenerate-city.sh irbid.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–5 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 6 and runs for 20 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–5)	€42 M / yr	€70
Steady-state, low-ridership (year 6+)	€47 M / yr	€78
Steady-state, high-ridership (year 6+)	€40 M / yr	€66
Lifecycle envelope (yr 1–25, low scenario)	€1.15 bn cumulative	€1,913
Lifecycle envelope (yr 1–25, high scenario)	€1.00 bn cumulative	€1,670

Population basis: 600,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 25, debt service drops to zero and only the OPEX shortfall remains – ~€7.3 M / yr (low) €0 k / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€346 M	4.0%	25 y, 5 y grace	€25 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€144 M	7.5%	25 y, 5 y grace	€14 M / yr
Government equity (no debt service)	15%	€87 M	—	—	—
Total	100%	€577 M			€40 M / yr

During the 5-year grace period the operator pays interest only – multilateral €14 M / yr + bonds €11 M / yr = €25 M / yr total – plus the equity tranche amortised across construction (€17 M / yr × 5 yr). Principal repayment begins in year 6 on a 20-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€6.7 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€7.2 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€42 k
Traction energy (101.0 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (349 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€3.1 M
OPEX subtotal		€17 M / yr

Annual fleet utilisation: 50 revenue trainsets × 20.5 h/day × 365 d/yr × 30 km/h commercial × 75% revenue factor = 8.4 M train-km / yr (~168 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$580 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.89 (~\$0.97 USD)
Day pass (3 trips)	€2.27 (15 % bulk discount)
Monthly unlimited pass	€26.68 (~5 % of median monthly income)
Annual pass	€293.48 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	10.9 M	21.9 M

	Low scenario	High scenario
Farebox revenue	€9.7 M / yr	€19 M / yr
Farebox / OPEX recovery	57%	114%
Country policy-target recovery (diagnostic)	55%	55%
Operating shortfall (gov subsidy required)	€7.3 M / yr	€0 k / yr
Operating surplus (operator retained capex sinking fund)	€0 k / yr	€2.4 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€47 M / yr	€40 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
irbid.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
irbid-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
irbid.corridor.geojson	Line polylines + stations (GeoJSON)
irbid.stations.json	Machine-readable station list
irbid.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug irbid

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug irbid \
  --sidecar .cache/osr-pipeline/rasters/irbid.grid.json \
  --out-dir designs/.../Irbid

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../irbid.toml \
  --out designs/.../README.md
```

`scripts/regenerate-irbid.sh` chains steps 3 + drift tests into a single command.

west-asia/Jordan/Zarqa

designs/west-asia/Jordan/Zarqa/README.md

Zarqa — Urban Rail Network

Country: JO · Population: 700,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Zarqa rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [zarqa.corridor.geojson](#).

At a glance

Metric	Value
Lines	3
Unique stations	57
Interchange stations	3
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	47.5%
Route length (double track)	83.2 km
Revenue fleet	71 × 3-car trainsets
Spare + cold-reserve	9 × 3-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	29.7 km	21	28	SW Outer NE Outer
line-2	27.0 km	19	26	NE Outer SW Outer
line-3	26.4 km	17	26	SW Outer NE Outer
Total	83.2 km	57 unique	80	

Rolling stock

Property	Value
Consist	3-car, 57 m
Max speed	90 km/h
Onboard battery	360 kWh per trainset
Seats	60 longitudinal seats
Nominal capacity (AW2)	330 pax (seated + standing, light-metro-3car per RFC 0008 §1)
Crush capacity (AW3)	420 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 330 AW2 passengers (light-metro-3car)
- Peak frequency: 12 trains/hour/direction (5-min headway)

- Peak capacity per line per direction: $330 \times 12 = 3,960$ pphpd
- Network peak throughput (all lines, both directions): $3 \text{ lines} \times 2 \text{ directions} \times 3,960 = 23,760$ passengers/hour
- Daily theoretical capacity (peak $\times 10$): $\approx 237,600$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 33,250 - 49,875$ trips/day

Catchment

- City population: 700,000
- Anchor-weighted coverage: 47.5%
- Catchment population: $\approx 332,500$ (within ~ 800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	3	500 kW	3000 kWh
Major	32	400 kW	2500 kWh
Standard	16	300 kW	2000 kWh
Terminal	5	500 kW	3000 kWh
Total installed	57	26,600 kW	176,000 kWh

Aggregate station-rail charging power: 31,500 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 360 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	12.0 kWh/km	3 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	333 kWh	27.7 km average line length
Onboard battery coverage	1.1 $\times$ average line run	360 kWh usable pack
Average 60 s dwell charge	9.2 kWh/stop	553 kW average charger across stops
Stops to refill one trainset pack	39 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	133 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	176 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in design.toml — emitted by the osr-design Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. country-costs.toml applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (62.9 km @ €3.5 M/km)	€220 M
Elevated (20.0 km @ €18 M/km)	€360 M
Elevated-interchange premium (1 sites @ €20 M)	€20 M
Civil subtotal	€600 M

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~ 11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
standard	16	€1.5 M	€24 M
major	32	€3.0 M	€96 M
terminal	5	€2.5 M	€12 M
depot-terminal	1	€3.0 M	€3.0 M
interchange	3	€4.5 M	€14 M
Stations subtotal			€149 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	5	€3.0 M	€15 M
Depots subtotal			€40 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
light-metro-3car (revenue + spare + cold reserve)	80	€3.0 M	€240 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	83.2 km × €0.015 M/km	€1.2 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€21 M
EPC integration + project management (7%)	on subtotal	€74 M

Total

Bucket	Value
Civil works	€600 M
Stations	€149 M
Depots	€40 M
Rolling stock	€240 M
Residual train-control wayside + charging microgrids	€23 M
EPC overhead (7%)	€74 M
CAPEX total	€1.13 bn
Per-route-km	€14 M / km
Per-capita (city pop)	€1,607 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running scripts/regenerate-city.sh zarqa.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–5 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 6 and runs for 20 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–5)	€82 M / yr	€117
Steady-state, low-ridership (year 6+)	€96 M / yr	€137
Steady-state, high-ridership (year 6+)	€85 M / yr	€121
Lifecycle envelope (yr 1–25, low scenario)	€2.33 bn cumulative	€3,325
Lifecycle envelope (yr 1–25, high scenario)	€2.10 bn cumulative	€3,001

Population basis: 700,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 25, debt service drops to zero and only the OPEX shortfall remains – ~€19 M / yr (low) €7.3 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€675 M	4.0%	25 y, 5 y grace	€50 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€281 M	7.5%	25 y, 5 y grace	€28 M / yr
Government equity (no debt service)	15%	€169 M	—	—	—
Total	100%	€1.13 bn			€77 M / yr

During the 5-year grace period the operator pays interest only – multilateral €27 M / yr + bonds €21 M / yr = €48 M / yr total – plus the equity tranche amortised across construction (€34 M / yr × 5 yr). Principal repayment begins in year 6 on a 20-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€9.6 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€16 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€62 k
Traction energy (143.4 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (511 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€4.6 M
OPEX subtotal		€30 M / yr

Annual fleet utilisation: 71 revenue trainsets × 20.5 h/day × 365 d/yr × 30 km/h commercial × 75% revenue factor = 12.0 M train-km / yr (~168 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$580 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.89 (~\$0.97 USD)
Day pass (3 trips)	€2.27 (15 % bulk discount)
Monthly unlimited pass	€26.68 (~5 % of median monthly income)
Annual pass	€293.48 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	12.8 M	25.6 M

	Low scenario	High scenario
Farebox revenue	€11 M / yr	€23 M / yr
Farebox / OPEX recovery	38%	76%
Country policy-target recovery (diagnostic)	55%	55%
Operating shortfall (gov subsidy required)	€19 M / yr	€7.3 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€96 M / yr	€85 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
zarqa.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
zarqa-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
zarqa.corridor.geojson	Line polylines + stations (GeoJSON)
zarqa.stations.json	Machine-readable station list
zarqa.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug zarqa

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug zarqa \
  --sidecar .cache/osr-pipeline/rasters/zarqa.grid.json \
  --out-dir designs/.../Zarqa

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../zarqa.toml \
  --out designs/.../README.md
```

`scripts/regenerate-zarqa.sh` chains steps 3 + drift tests into a single command.

west-asia/Lebanon/Beirut

designs/west-asia/Lebanon/Beirut/README.md

Beirut — Urban Rail Network

Country: LB · Population: 2,200,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Beirut rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [beirut.corridor.geojson](#).

At a glance

Metric	Value
Lines	6
Unique stations	82
Interchange stations	17
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	69.6%
Route length (double track)	159.5 km
Revenue fleet	119 × 4-car trainsets
Spare + cold-reserve	15 × 4-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	30.1 km	14	25	NE Outer S Mid
line-2	20.2 km	11	18	NE Outer W Mid
line-3	19.6 km	12	17	E Mid SW Mid
line-4	19.8 km	11	17	NW Mid S Outer
line-5	17.9 km	9	16	NW Inner SE Mid
line-6	52.0 km	26	41	W Mid W Mid
Total	159.5 km	82 unique	134	

Rolling stock

Property	Value
Consist	4-car, 75 m
Max speed	90 km/h
Onboard battery	480 kWh per trainset
Seats	80 longitudinal seats
Nominal capacity (AW2)	440 pax (seated + standing, metro-4car per RFC 0008 §1)
Crush capacity (AW3)	560 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 440 AW2 passengers (metro-4car)
- Peak frequency: 12 trains/hour/direction (5-min headway)
- Peak capacity per line per direction: $440 \times 12 = 5,280$ pphpd
- Network peak throughput (all lines, both directions): $6 \text{ lines} \times 2 \text{ directions} \times 5,280 = 63,360$ passengers/hour
- Daily theoretical capacity (peak $\times$ 10): $\approx 633,600$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 153,120 - 229,680$ trips/day

Catchment

- City population: 2,200,000
- Anchor-weighted coverage: 69.6%
- Catchment population: $\approx 1,531,200$ (within ~ 800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../.../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	17	500 kW	3000 kWh
Major	28	400 kW	2500 kWh
Standard	28	300 kW	2000 kWh
Terminal	9	500 kW	3000 kWh
Total installed	83	37,600 kW	244,000 kWh

Aggregate station-rail charging power: 46,500 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 480 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	16.0 kWh/km	4 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	425 kWh	26.6 km average line length
Onboard battery coverage	1.1 $\times$ average line run	480 kWh usable pack
Average 60 s dwell charge	9.5 kWh/stop	567 kW average charger across stops
Stops to refill one trainset pack	51 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	188 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	244 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in design.toml — emitted by the osr-design Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. country-costs.toml applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (146.8 km @ €3.5 M/km)	€514 M
Elevated (11.5 km @ €18 M/km)	€207 M
Elevated-interchange premium (10 sites @ €20 M)	€200 M
Civil subtotal	€920 M

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
standard	28	€1.5 M	€42 M
major	28	€3.0 M	€84 M
terminal	9	€2.5 M	€22 M
depot-terminal	1	€3.0 M	€3.0 M
interchange-elevated	17	€4.5 M	€76 M
Stations subtotal			€228 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	9	€3.0 M	€27 M
Depots subtotal			€52 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
metro-4car (revenue + spare + cold reserve)	134	€4.0 M	€536 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	159.5 km × €0.015 M/km	€2.4 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€33 M
EPC integration + project management (7%)	on subtotal	€124 M

Total

Bucket	Value
Civil works	€920 M
Stations	€228 M
Depots	€52 M
Rolling stock	€536 M
Residual train-control wayside + charging microgrids	€35 M
EPC overhead (7%)	€124 M
CAPEX total	€1.90 bn
Per-route-km	€12 M / km
Per-capita (city pop)	€862 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running scripts/regenerate-city.sh beirut.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–8 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 9 and runs for 22 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–8)	€205 M / yr	€93
Steady-state, low-ridership (year 9+)	€234 M / yr	€107
Steady-state, high-ridership (year 9+)	€217 M / yr	€99
Lifecycle envelope (yr 1–30, low scenario)	€6.80 bn cumulative	€3,090
Lifecycle envelope (yr 1–30, high scenario)	€6.42 bn cumulative	€2,918

Population basis: 2,200,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 30, debt service drops to zero and only the OPEX shortfall remains – ~€33 M / yr (low) €15 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€1.14 bn	4.5%	30 y, 8 y grace	€83 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€474 M	25.0%	30 y, 8 y grace	€119 M / yr
Government equity (no debt service)	15%	€284 M	—	—	—
Total	100%	€1.90 bn			€202 M / yr

During the 8-year grace period the operator pays interest only – multilateral €51 M / yr + bonds €118 M / yr = €170 M / yr total – plus the equity tranche amortised across construction (€36 M / yr × 8 yr). Principal repayment begins in year 9 on a 22-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€21 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€24 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€119 k
Traction energy (374.0 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (969 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€4.2 M
OPEX subtotal		€50 M / yr

Annual fleet utilisation: 119 revenue trainsets × 20.5 h/day × 365 d/yr × 35 km/h commercial × 75% revenue factor = 23.4 M train-km / yr (~196 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$280 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.43 (~\$0.47 USD)
Day pass (3 trips)	€1.09 (15 % bulk discount)
Monthly unlimited pass	€12.88 (~5 % of median monthly income)
Annual pass	€141.68 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	40.1 M	80.3 M
Farebox revenue	€17 M / yr	€34 M / yr
Farebox / OPEX recovery	35%	69%
Country policy-target recovery (diagnostic)	40%	40%
Operating shortfall (gov subsidy required)	€33 M / yr	€15 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€234 M / yr	€217 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
beirut.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
beirut-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
beirut.corridor.geojson	Line polylines + stations (GeoJSON)
beirut.stations.json	Machine-readable station list
beirut.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug beirut

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug beirut \
  --sidecar .cache/osr-pipeline/rasters/beirut.grid.json \
  --out-dir designs/.../Beirut

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../beirut.toml \
  --out designs/.../README.md
```

`scripts/regenerate-beirut.sh` chains steps 3 + drift tests into a single command.

west-asia/Lebanon/Sidon

designs/west-asia/Lebanon/Sidon/README.md

Sidon — Urban Rail Network

Country: LB · Population: 300,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Sidon rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [sidon.corridor.geojson](#).

At a glance

Metric	Value
Lines	3
Unique stations	23
Interchange stations	3
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	71.4%
Route length (double track)	37.2 km
Revenue fleet	46 × 2-car trainsets
Spare + cold-reserve	6 × 2-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	14.3 km	8	19	S Mid N Outer
line-2	14.5 km	8	20	SW Mid NE Outer
line-3	8.4 km	7	13	S Mid N Inner
Total	37.2 km	23 unique	52	

Rolling stock

Property	Value
Consist	2-car, 39 m
Max speed	70 km/h
Onboard battery	240 kWh per trainset
Seats	40 longitudinal seats
Nominal capacity (AW2)	210 pax (seated + standing, tram-2car per RFC 0008 §1)
Crush capacity (AW3)	260 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 210 AW2 passengers (tram-2car)
- Peak frequency: 12 trains/hour/direction (5-min headway)

- Peak capacity per line per direction: $210 \times 12 = 2,520$ pphpd
- Network peak throughput (all lines, both directions): $3 \text{ lines} \times 2 \text{ directions} \times 2,520 = 15,120$ passengers/hour
- Daily theoretical capacity (peak $\times 10$): $\approx 151,200$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 21,420 - 32,130$ trips/day

Catchment

- City population: 300,000
- Anchor-weighted coverage: 71.4%
- Catchment population: $\approx 214,200$ (within ~ 800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	3	500 kW	3000 kWh
Major	8	400 kW	2500 kWh
Standard	4	300 kW	2000 kWh
Terminal	5	500 kW	3000 kWh
Total installed	21	13,400 kW	92,000 kWh

Aggregate station-rail charging power: 14,000 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 240 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	8.0 kWh/km	2 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	99 kWh	12.4 km average line length
Onboard battery coverage	2.4 $\times$ average line run	240 kWh usable pack
Average 60 s dwell charge	10.1 kWh/stop	609 kW average charger across stops
Stops to refill one trainset pack	24 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	67 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	92 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in design.toml — emitted by the osr-design Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. country-costs.toml applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (35.1 km @ €3.5 M/km)	€123 M
Elevated (1.9 km @ €18 M/km)	€33 M
Elevated-interchange premium (2 sites @ €20 M)	€40 M
Civil subtotal	€196 M

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~ 11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
halt	2	€0.4 M	€0.8 M
standard	4	€1.5 M	€6.0 M
major	8	€3.0 M	€24 M
terminal	5	€2.5 M	€12 M
depot-terminal	1	€3.0 M	€3.0 M
interchange-elevated	3	€4.5 M	€14 M
Stations subtotal			€60 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	5	€3.0 M	€15 M
Depots subtotal			€40 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
tram-2car (revenue + spare + cold reserve)	52	€2.0 M	€104 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	37.2 km × €0.015 M/km	€0.6 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€9.0 M
EPC integration + project management (7%)	on subtotal	€29 M

Total

Bucket	Value
Civil works	€196 M
Stations	€60 M
Depots	€40 M
Rolling stock	€104 M
Residual train-control wayside + charging microgrids	€9.6 M
EPC overhead (7%)	€29 M
CAPEX total	€438 M
Per-route-km	€12 M / km
Per-capita (city pop)	€1,460 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running scripts/regenerate-city.sh sidon.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–8 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 9 and runs for 22 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–8)	€47 M / yr	€158
Steady-state, low-ridership (year 9+)	€55 M / yr	€185
Steady-state, high-ridership (year 9+)	€53 M / yr	€177
Lifecycle envelope (yr 1–30, low scenario)	€1.60 bn cumulative	€5,329
Lifecycle envelope (yr 1–30, high scenario)	€1.55 bn cumulative	€5,157

Population basis: 300,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 30, debt service drops to zero and only the OPEX shortfall remains – ~€8.8 M / yr (low) €6.4 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€263 M	4.5%	30 y, 8 y grace	€19 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€110 M	25.0%	30 y, 8 y grace	€28 M / yr
Government equity (no debt service)	15%	€66 M	—	—	—
Total	100%	€438 M			€47 M / yr

During the 8-year grace period the operator pays interest only – multilateral €12 M / yr + bonds €27 M / yr = €39 M / yr total – plus the equity tranche amortised across construction (€8.2 M / yr × 8 yr). Principal repayment begins in year 9 on a 22-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€4.2 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€5.9 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€28 k
Traction energy (45.4 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (235 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€1.0 M
OPEX subtotal		€11 M / yr

Annual fleet utilisation: 46 revenue trainsets × 20.5 h/day × 365 d/yr × 22 km/h commercial × 75% revenue factor = 5.7 M train-km / yr (~123 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$280 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.43 (~\$0.47 USD)
Day pass (3 trips)	€1.09 (15 % bulk discount)
Monthly unlimited pass	€12.88 (~5 % of median monthly income)
Annual pass	€141.68 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	5.5 M	10.9 M

	Low scenario	High scenario
Farebox revenue	€2.4 M / yr	€4.7 M / yr
Farebox / OPEX recovery	21%	42%
Country policy-target recovery (diagnostic)	40%	40%
Operating shortfall (gov subsidy required)	€8.8 M / yr	€6.4 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€55 M / yr	€53 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
sidon.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
sidon-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
sidon.corridor.geojson	Line polylines + stations (GeoJSON)
sidon.stations.json	Machine-readable station list
sidon.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug sidon

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug sidon \
  --sidecar .cache/osr-pipeline/rasters/sidon.grid.json \
  --out-dir designs/.../Sidon

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../sidon.toml \
  --out designs/.../README.md
```

`scripts/regenerate-sidon.sh` chains steps 3 + drift tests into a single command.

west-asia/Lebanon/Tripoli-Lb

designs/west-asia/Lebanon/Tripoli-Lb/README.md

Tripoli-Lb — Urban Rail Network

Country: LB · Population: 730,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Tripoli-Lb rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [tripoli-lb.corridor.geojson](#).

At a glance

Metric	Value
Lines	3
Unique stations	30
Interchange stations	2
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	49.9%
Route length (double track)	53.1 km
Revenue fleet	48 × 3-car trainsets
Spare + cold-reserve	7 × 3-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	24.8 km	14	25	NE Outer SW Outer
line-2	12.7 km	8	14	SE Mid NW Mid
line-3	15.6 km	8	16	S Outer NW Mid
Total	53.1 km	30 unique	55	

Rolling stock

Property	Value
Consist	3-car, 57 m
Max speed	90 km/h
Onboard battery	360 kWh per trainset
Seats	60 longitudinal seats
Nominal capacity (AW2)	330 pax (seated + standing, light-metro-3car per RFC 0008 §1)
Crush capacity (AW3)	420 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 330 AW2 passengers (light-metro-3car)
- Peak frequency: 12 trains/hour/direction (5-min headway)

- Peak capacity per line per direction: $330 \times 12 = 3,960$ pphpd
- Network peak throughput (all lines, both directions): $3 \text{ lines} \times 2 \text{ directions} \times 3,960 = 23,760$ passengers/hour
- Daily theoretical capacity (peak $\times 10$): $\approx 237,600$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 36,427 - 54,640$ trips/day

Catchment

- City population: 730,000
- Anchor-weighted coverage: 49.9%
- Catchment population: $\approx 364,270$ (within ~ 800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	2	500 kW	3000 kWh
Major	5	400 kW	2500 kWh
Standard	17	300 kW	2000 kWh
Terminal	5	500 kW	3000 kWh
Total installed	30	15,600 kW	107,500 kWh

Aggregate station-rail charging power: 18,000 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 360 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	12.0 kWh/km	3 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	212 kWh	17.7 km average line length
Onboard battery coverage	1.7 $\times$ average line run	360 kWh usable pack
Average 60 s dwell charge	10.0 kWh/stop	600 kW average charger across stops
Stops to refill one trainset pack	36 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	78 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	108 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in design.toml — emitted by the osr-design Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. country-costs.toml applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (51.6 km @ €3.5 M/km)	€181 M
Elevated (1.2 km @ €18 M/km)	€22 M
Elevated-interchange premium (1 sites @ €20 M)	€20 M
Civil subtotal	€223 M

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~ 11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
standard	17	€1.5 M	€26 M
major	5	€3.0 M	€15 M
terminal	5	€2.5 M	€12 M
depot-terminal	1	€3.0 M	€3.0 M
interchange-elevated	2	€4.5 M	€9.0 M
Stations subtotal			€65 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	5	€3.0 M	€15 M
Depots subtotal			€40 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
light-metro-3car (revenue + spare + cold reserve)	55	€3.0 M	€165 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	53.1 km × €0.015 M/km	€0.8 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€10 M
EPC integration + project management (7%)	on subtotal	€35 M

Total

Bucket	Value
Civil works	€223 M
Stations	€65 M
Depots	€40 M
Rolling stock	€165 M
Residual train-control wayside + charging microgrids	€11 M
EPC overhead (7%)	€35 M
CAPEX total	€539 M
Per-route-km	€10 M / km
Per-capita (city pop)	€738 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running scripts/regenerate-city.sh tripoli-lb.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–8 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 9 and runs for 22 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–8)	€58 M / yr	€80
Steady-state, low-ridership (year 9+)	€66 M / yr	€91
Steady-state, high-ridership (year 9+)	€61 M / yr	€83
Lifecycle envelope (yr 1–30, low scenario)	€1.93 bn cumulative	€2,638
Lifecycle envelope (yr 1–30, high scenario)	€1.80 bn cumulative	€2,465

Population basis: 730,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 30, debt service drops to zero and only the OPEX shortfall remains – ~€8.9 M / yr (low) €3.2 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€323 M	4.5%	30 y, 8 y grace	€23 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€135 M	25.0%	30 y, 8 y grace	€34 M / yr
Government equity (no debt service)	15%	€81 M	—	—	—
Total	100%	€539 M			€57 M / yr

During the 8-year grace period the operator pays interest only – multilateral €15 M / yr + bonds €34 M / yr = €48 M / yr total – plus the equity tranche amortised across construction (€10 M / yr × 8 yr). Principal repayment begins in year 9 on a 22-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€6.6 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€6.6 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€40 k
Traction energy (97.0 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (331 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€1.4 M
OPEX subtotal		€15 M / yr

Annual fleet utilisation: 48 revenue trainsets × 20.5 h/day × 365 d/yr × 30 km/h commercial × 75% revenue factor = 8.1 M train-km / yr (~168 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$280 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.43 (~\$0.47 USD)
Day pass (3 trips)	€1.09 (15 % bulk discount)
Monthly unlimited pass	€12.88 (~5 % of median monthly income)
Annual pass	€141.68 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	13.3 M	26.6 M

	Low scenario	High scenario
Farebox revenue	€5.7 M / yr	€11 M / yr
Farebox / OPEX recovery	39%	78%
Country policy-target recovery (diagnostic)	40%	40%
Operating shortfall (gov subsidy required)	€8.9 M / yr	€3.2 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€66 M / yr	€61 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
tripoli-lb.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
tripoli-lb-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
tripoli-lb.corridor.geojson	Line polylines + stations (GeoJSON)
tripoli-lb.stations.json	Machine-readable station list
tripoli-lb.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug tripoli-lb

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug tripoli-lb \
  --sidecar .cache/osr-pipeline/rasters/tripoli-lb.grid.json \
  --out-dir designs/.../Tripoli-Lb

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../tripoli-lb.toml \
  --out designs/.../README.md
```

`scripts/regenerate-tripoli-lb.sh` chains steps 3 + drift tests into a single command.

west-asia/Palestine/Gaza-City

designs/west-asia/Palestine/Gaza-City/README.md

Gaza-City — Urban Rail Network

Country: PS · Population: 600,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Gaza-City rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [gaza-city.corridor.geojson](#).

At a glance

Metric	Value
Lines	1
Unique stations	15
Interchange stations	0
Multi-line transfer reachability	100% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	29.5%
Route length (double track)	23.9 km
Revenue fleet	21 × 3-car trainsets
Spare + cold-reserve	3 × 3-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	23.9 km	15	24	SW Outer NE Outer
Total	23.9 km	15 unique	24	

Rolling stock

Property	Value
Consist	3-car, 57 m
Max speed	90 km/h
Onboard battery	360 kWh per trainset
Seats	60 longitudinal seats
Nominal capacity (AW2)	330 pax (seated + standing, light-metro-3car per RFC 0008 §1)
Crush capacity (AW3)	420 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 330 AW2 passengers (light-metro-3car)
- Peak frequency: 12 trains/hour/direction (5-min headway)
- Peak capacity per line per direction: 330 × 12 = 3,960 pphpd

- Network peak throughput (all lines, both directions): 1 lines × 2 directions × 3,960 = 7,920 passengers/hour
- Daily theoretical capacity (peak × 10): ≈ 79,200 passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): ≈ 17,700 – 26,550 trips/day

Catchment

- City population: 600,000
- Anchor-weighted coverage: 29.5%
- Catchment population: ≈ 177,000 (within ~800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Major	7	400 kW	2500 kWh
Standard	6	300 kW	2000 kWh
Terminal	1	500 kW	3000 kWh
Total installed	15	10,100 kW	72,500 kWh

Aggregate station-rail charging power: 8,500 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 360 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	12.0 kWh/km	3 cars × 4 kWh/car-km planning basis
Average one-way line energy	287 kWh	23.9 km average line length
Onboard battery coverage	1.3× average line run	360 kWh usable pack
Average 60 s dwell charge	9.4 kWh/stop	567 kW average charger across stops
Stops to refill one trainset pack	38 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	50 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	72 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in design.toml — emitted by the osr-design Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3× higher because of the line items OSR has architected away. country-costs.toml applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (23.4 km @ €3.5 M/km)	€82 M
Elevated (0.0 km @ €18 M/km)	€0.4 M
Civil subtotal	€82 M

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
standard	6	€1.5 M	€9.0 M
major	7	€3.0 M	€21 M

Archetype	Count	Unit	Subtotal
terminal	1	€2.5 M	€2.5 M
depot-terminal	1	€3.0 M	€3.0 M
Stations subtotal			€36 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	1	€3.0 M	€3.0 M
Depots subtotal			€28 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
light-metro-3car (revenue + spare + cold reserve)	24	€3.0 M	€72 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	23.9 km × €0.015 M/km	€0.4 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€5.5 M
EPC integration + project management (7%)	on subtotal	€16 M

Total

Bucket	Value
Civil works	€82 M
Stations	€36 M
Depots	€28 M
Rolling stock	€72 M
Residual train-control wayside + charging microgrids	€5.8 M
EPC overhead (7%)	€16 M
CAPEX total	€239 M
Per-route-km	€10 M / km
Per-capita (city pop)	€398 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code — regenerate by re-running scripts/regenerate-city.sh `gaza-city`.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–7 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 8 and runs for 23 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–7)	€15 M / yr	€24
Steady-state, low-ridership (year 8+)	€15 M / yr	€26
Steady-state, high-ridership (year 8+)	€15 M / yr	€25
Lifecycle envelope (yr 1–30, low scenario)	€454 M cumulative	€757
Lifecycle envelope (yr 1–30, high scenario)	€440 M cumulative	€733

Population basis: 600,000 (catchment per lib/city-batches/world-sample.toml). After year 30, debt service drops to zero and only the OPEX shortfall remains — ~€627 k / yr (low) €0 k / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€143 M	3.0%	30 y, 7 y grace	€8.7 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€60 M	8.5%	30 y, 7 y grace	€6.0 M / yr
Government equity (no debt service)	15%	€36 M	—	—	—
Total	100%	€239 M			€15 M / yr

During the 7-year grace period the operator pays interest only — multilateral €4.3 M / yr + bonds €5.1 M / yr = €9.4 M / yr total — plus the equity tranche amortised across construction (€5.1 M / yr × 7 yr). Principal repayment begins in year 8 on a 23-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€2.9 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€2.9 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€18 k
Traction energy (42.4 GWh / yr)	trackside PV + Na-ion (RFC 0002) — self-generated, €0 / yr	€0 k
Labour (155 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€862 k
OPEX subtotal		€6.7 M / yr

Annual fleet utilisation: 21 revenue trainsets × 20.5 h/day × 365 d/yr × 30 km/h commercial × 75% revenue factor = 3.5 M train-km / yr (~168 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$360 USD (per lib/templates/country-finance.toml). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass — a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.55 (~\$0.60 USD)
Day pass (3 trips)	€1.41 (15 % bulk discount)
Monthly unlimited pass	€16.56 (~5 % of median monthly income)
Annual pass	€182.16 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	10.9 M	21.9 M
Farebox revenue	€6.0 M / yr	€12 M / yr
Farebox / OPEX recovery	91%	181%
Country policy-target recovery (diagnostic)	45%	45%
Operating shortfall (gov subsidy required)	€627 k / yr	€0 k / yr

	Low scenario	High scenario
Operating surplus (operator retained capex sinking fund)	€0 k / yr	€5.4 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€15 M / yr	€15 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
gaza-city.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
gaza-city-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
gaza-city.corridor.geojson	Line polylines + stations (GeoJSON)
gaza-city.stations.json	Machine-readable station list
gaza-city.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug gaza-city

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug gaza-city \
  --sidecar .cache/osr-pipeline/rasters/gaza-city.grid.json \
  --out-dir designs/.../Gaza-City

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../gaza-city.toml \
  --out designs/.../README.md
```

`scripts/regenerate-gaza-city.sh` chains steps 3 + drift tests into a single command.

west-asia/Palestine/Hebron

designs/west-asia/Palestine/Hebron/README.md

Hebron — Urban Rail Network

Country: PS · Population: 800,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Hebron rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [hebron.corridor.geojson](#).

At a glance

Metric	Value
Lines	3
Unique stations	40
Interchange stations	3
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	61.3%
Route length (double track)	69.1 km
Revenue fleet	60 × 3-car trainsets
Spare + cold-reserve	8 × 3-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	25.7 km	15	25	SW Outer NE Outer
line-2	24.3 km	13	24	N Outer S Outer
line-3	19.2 km	12	19	E Mid W Mid
Total	69.1 km	40 unique	68	

Rolling stock

Property	Value
Consist	3-car, 57 m
Max speed	90 km/h
Onboard battery	360 kWh per trainset
Seats	60 longitudinal seats
Nominal capacity (AW2)	330 pax (seated + standing, light-metro-3car per RFC 0008 §1)
Crush capacity (AW3)	420 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 330 AW2 passengers (light-metro-3car)
- Peak frequency: 12 trains/hour/direction (5-min headway)

- Peak capacity per line per direction: $330 \times 12 = 3,960$ pphpd
- Network peak throughput (all lines, both directions): $3 \text{ lines} \times 2 \text{ directions} \times 3,960 = 23,760$ passengers/hour
- Daily theoretical capacity (peak $\times 10$): $\approx 237,600$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 49,040 - 73,560$ trips/day

Catchment

- City population: 800,000
- Anchor-weighted coverage: 61.3%
- Catchment population: $\approx 490,400$ (within ~ 800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	3	500 kW	3000 kWh
Major	13	400 kW	2500 kWh
Standard	16	300 kW	2000 kWh
Terminal	5	500 kW	3000 kWh
Total installed	38	19,000 kW	128,500 kWh

Aggregate station-rail charging power: 22,500 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 360 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	12.0 kWh/km	3 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	276 kWh	23.0 km average line length
Onboard battery coverage	1.3 $\times$ average line run	360 kWh usable pack
Average 60 s dwell charge	9.4 kWh/stop	562 kW average charger across stops
Stops to refill one trainset pack	38 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	95 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	128 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in design.toml — emitted by the osr-design Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. country-costs.toml applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (67.9 km @ €3.5 M/km)	€237 M
Elevated (1.2 km @ €18 M/km)	€22 M
Elevated-interchange premium (1 sites @ €20 M)	€20 M
Civil subtotal	€279 M

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~ 11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
halt	2	€0.4 M	€0.8 M
standard	16	€1.5 M	€24 M
major	13	€3.0 M	€39 M
terminal	5	€2.5 M	€12 M
depot-terminal	1	€3.0 M	€3.0 M
interchange-elevated	3	€4.5 M	€14 M
Stations subtotal			€93 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	5	€3.0 M	€15 M
Depots subtotal			€40 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
light-metro-3car (revenue + spare + cold reserve)	68	€3.0 M	€204 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	69.1 km × €0.015 M/km	€1.0 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€14 M
EPC integration + project management (7%)	on subtotal	€44 M

Total

Bucket	Value
Civil works	€279 M
Stations	€93 M
Depots	€40 M
Rolling stock	€204 M
Residual train-control wayside + charging microgrids	€15 M
EPC overhead (7%)	€44 M
CAPEX total	€675 M
Per-route-km	€9.8 M / km
Per-capita (city pop)	€844 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running scripts/regenerate-city.sh hebron.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–7 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 8 and runs for 23 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–7)	€41 M / yr	€51
Steady-state, low-ridership (year 8+)	€52 M / yr	€65
Steady-state, high-ridership (year 8+)	€44 M / yr	€55
Lifecycle envelope (yr 1–30, low scenario)	€1.49 bn cumulative	€1,864
Lifecycle envelope (yr 1–30, high scenario)	€1.31 bn cumulative	€1,632

Population basis: 800,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 30, debt service drops to zero and only the OPEX shortfall remains – ~€11 M / yr (low) €2.7 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€405 M	3.0%	30 y, 7 y grace	€25 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€169 M	8.5%	30 y, 7 y grace	€17 M / yr
Government equity (no debt service)	15%	€101 M	—	—	—
Total	100%	€675 M			€42 M / yr

During the 7-year grace period the operator pays interest only – multilateral €12 M / yr + bonds €14 M / yr = €27 M / yr total – plus the equity tranche amortised across construction (€14 M / yr × 7 yr). Principal repayment begins in year 8 on a 23-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€8.2 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€8.2 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€52 k
Traction energy (121.2 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (427 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€2.4 M
OPEX subtotal		€19 M / yr

Annual fleet utilisation: 60 revenue trainsets × 20.5 h/day × 365 d/yr × 30 km/h commercial × 75% revenue factor = 10.1 M train-km / yr (~168 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$360 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.55 (~\$0.60 USD)
Day pass (3 trips)	€1.41 (15 % bulk discount)
Monthly unlimited pass	€16.56 (~5 % of median monthly income)
Annual pass	€182.16 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	14.6 M	29.2 M

	Low scenario	High scenario
Farebox revenue	€8.1 M / yr	€16 M / yr
Farebox / OPEX recovery	43%	86%
Country policy-target recovery (diagnostic)	45%	45%
Operating shortfall (gov subsidy required)	€11 M / yr	€2.7 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€52 M / yr	€44 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
hebron.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
hebron-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
hebron.corridor.geojson	Line polylines + stations (GeoJSON)
hebron.stations.json	Machine-readable station list
hebron.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug hebron

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug hebron \
  --sidecar .cache/osr-pipeline/rasters/hebron.grid.json \
  --out-dir designs/.../Hebron

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../hebron.toml \
  --out designs/.../README.md
```

`scripts/regenerate-hebron.sh` chains steps 3 + drift tests into a single command.

west-asia/Palestine/Nablus

designs/west-asia/Palestine/Nablus/README.md

Nablus — Urban Rail Network

Country: PS · Population: 450,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Nablus rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [nablus.corridor.geojson](#).

At a glance

Metric	Value
Lines	3
Unique stations	34
Interchange stations	3
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	84.1%
Route length (double track)	64.6 km
Revenue fleet	57 × 3-car trainsets
Spare + cold-reserve	8 × 3-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	19.1 km	12	19	NW Mid SE Mid
line-2	22.9 km	12	23	W Mid E Mid
line-3	22.7 km	10	23	NE Outer SW Inner
Total	64.6 km	34 unique	65	

Rolling stock

Property	Value
Consist	3-car, 57 m
Max speed	90 km/h
Onboard battery	360 kWh per trainset
Seats	60 longitudinal seats
Nominal capacity (AW2)	330 pax (seated + standing, light-metro-3car per RFC 0008 §1)
Crush capacity (AW3)	420 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 330 AW2 passengers (light-metro-3car)
- Peak frequency: 12 trains/hour/direction (5-min headway)

- Peak capacity per line per direction: $330 \times 12 = 3,960$ pphpd
- Network peak throughput (all lines, both directions): $3 \text{ lines} \times 2 \text{ directions} \times 3,960 = 23,760$ passengers/hour
- Daily theoretical capacity (peak $\times 10$): $\approx 237,600$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 37,845 - 56,767$ trips/day

Catchment

- City population: 450,000
- Anchor-weighted coverage: 84.1%
- Catchment population: $\approx 378,450$ (within ~ 800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	3	500 kW	3000 kWh
Major	7	400 kW	2500 kWh
Standard	14	300 kW	2000 kWh
Terminal	5	500 kW	3000 kWh
Total installed	30	16,000 kW	109,500 kWh

Aggregate station-rail charging power: 19,000 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 360 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	12.0 kWh/km	3 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	258 kWh	21.5 km average line length
Onboard battery coverage	1.4 $\times$ average line run	360 kWh usable pack
Average 60 s dwell charge	9.3 kWh/stop	559 kW average charger across stops
Stops to refill one trainset pack	39 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	80 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	110 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in design.toml — emitted by the osr-design Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. country-costs.toml applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (63.3 km @ €3.5 M/km)	€222 M
Elevated (1.2 km @ €18 M/km)	€22 M
Elevated-interchange premium (1 sites @ €20 M)	€20 M
Civil subtotal	€264 M

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~ 11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
halt	4	€0.4 M	€1.6 M
standard	14	€1.5 M	€21 M
major	7	€3.0 M	€21 M
terminal	5	€2.5 M	€12 M
depot-terminal	1	€3.0 M	€3.0 M
interchange-elevated	3	€4.5 M	€14 M
Stations subtotal			€73 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	5	€3.0 M	€15 M
Depots subtotal			€40 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
light-metro-3car (revenue + spare + cold reserve)	65	€3.0 M	€195 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	64.6 km × €0.015 M/km	€1.0 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€11 M
EPC integration + project management (7%)	on subtotal	€41 M

Total

Bucket	Value
Civil works	€264 M
Stations	€73 M
Depots	€40 M
Rolling stock	€195 M
Residual train-control wayside + charging microgrids	€12 M
EPC overhead (7%)	€41 M
CAPEX total	€625 M
Per-route-km	€9.7 M / km
Per-capita (city pop)	€1,388 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running scripts/regenerate-city.sh nablus.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–7 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 8 and runs for 23 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–7)	€38 M / yr	€84
Steady-state, low-ridership (year 8+)	€52 M / yr	€115
Steady-state, high-ridership (year 8+)	€47 M / yr	€104
Lifecycle envelope (yr 1–30, low scenario)	€1.45 bn cumulative	€3,224
Lifecycle envelope (yr 1–30, high scenario)	€1.35 bn cumulative	€2,992

Population basis: 450,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 30, debt service drops to zero and only the OPEX shortfall remains – ~€13 M / yr (low) €8.5 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€375 M	3.0%	30 y, 7 y grace	€23 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€156 M	8.5%	30 y, 7 y grace	€16 M / yr
Government equity (no debt service)	15%	€94 M	—	—	—
Total	100%	€625 M			€38 M / yr

During the 7-year grace period the operator pays interest only – multilateral €11 M / yr + bonds €13 M / yr = €25 M / yr total – plus the equity tranche amortised across construction (€13 M / yr × 7 yr). Principal repayment begins in year 8 on a 23-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€7.8 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€7.5 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€48 k
Traction energy (115.2 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (400 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€2.2 M
OPEX subtotal		€18 M / yr

Annual fleet utilisation: 57 revenue trainsets × 20.5 h/day × 365 d/yr × 30 km/h commercial × 75% revenue factor = 9.6 M train-km / yr (~168 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$360 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.55 (~\$0.60 USD)
Day pass (3 trips)	€1.41 (15 % bulk discount)
Monthly unlimited pass	€16.56 (~5 % of median monthly income)
Annual pass	€182.16 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	8.2 M	16.4 M

	Low scenario	High scenario
Farebox revenue	€4.5 M / yr	€9.1 M / yr
Farebox / OPEX recovery	26%	52%
Country policy-target recovery (diagnostic)	45%	45%
Operating shortfall (gov subsidy required)	€13 M / yr	€8.5 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€52 M / yr	€47 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
nablus.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
nablus-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
nablus.corridor.geojson	Line polylines + stations (GeoJSON)
nablus.stations.json	Machine-readable station list
nablus.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug nablus

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug nablus \
  --sidecar .cache/osr-pipeline/rasters/nablus.grid.json \
  --out-dir designs/.../Nablus

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../nablus.toml \
  --out designs/.../README.md
```

`scripts/regenerate-nablus.sh` chains steps 3 + drift tests into a single command.

west-asia/Saudi Arabia/Abha

designs/west-asia/Saudi Arabia/Abha/README.md

Abha — Urban Rail Network

Country: SA · Population: 450,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Abha rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [abha.corridor.geojson](#).

At a glance

Metric	Value
Lines	3
Unique stations	45
Interchange stations	2
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	33.5%
Route length (double track)	69.8 km
Revenue fleet	61 × 3-car trainsets
Spare + cold-reserve	7 × 3-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	27.8 km	17	27	E Outer W Outer
line-2	20.0 km	13	20	W Mid SE Outer
line-3	22.0 km	15	21	SW Mid NE Outer
Total	69.8 km	45 unique	68	

Rolling stock

Property	Value
Consist	3-car, 57 m
Max speed	90 km/h
Onboard battery	360 kWh per trainset
Seats	60 longitudinal seats
Nominal capacity (AW2)	330 pax (seated + standing, light-metro-3car per RFC 0008 §1)
Crush capacity (AW3)	420 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 330 AW2 passengers (light-metro-3car)
- Peak frequency: 12 trains/hour/direction (5-min headway)

- Peak capacity per line per direction: $330 \times 12 = 3,960$ pphpd
- Network peak throughput (all lines, both directions): $3 \text{ lines} \times 2 \text{ directions} \times 3,960 = 23,760$ passengers/hour
- Daily theoretical capacity (peak $\times 10$): $\approx 237,600$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 15,075 - 22,612$ trips/day

Catchment

- City population: 450,000
- Anchor-weighted coverage: 33.5%
- Catchment population: $\approx 150,750$ (within ~ 800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	2	500 kW	3000 kWh
Major	17	400 kW	2500 kWh
Standard	20	300 kW	2000 kWh
Terminal	5	500 kW	3000 kWh
Total installed	45	21,300 kW	143,500 kWh

Aggregate station-rail charging power: 25,500 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 360 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	12.0 kWh/km	3 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	279 kWh	23.3 km average line length
Onboard battery coverage	1.3 $\times$ average line run	360 kWh usable pack
Average 60 s dwell charge	9.4 kWh/stop	567 kW average charger across stops
Stops to refill one trainset pack	38 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	106 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	144 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in design.toml — emitted by the osr-design Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. country-costs.toml applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (66.3 km @ €3.5 M/km)	€232 M
Elevated (3.2 km @ €18 M/km)	€58 M
Elevated-interchange premium (2 sites @ €20 M)	€40 M
Civil subtotal	€330 M

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~ 11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
standard	20	€1.5 M	€30 M
major	17	€3.0 M	€51 M
terminal	5	€2.5 M	€12 M
depot-terminal	1	€3.0 M	€3.0 M
interchange-elevated	2	€4.5 M	€9.0 M
Stations subtotal			€106 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	5	€3.0 M	€15 M
Depots subtotal			€40 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
light-metro-3car (revenue + spare + cold reserve)	68	€3.0 M	€204 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	69.8 km × €0.015 M/km	€1.0 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€16 M
EPC integration + project management (7%)	on subtotal	€49 M

Total

Bucket	Value
Civil works	€330 M
Stations	€106 M
Depots	€40 M
Rolling stock	€204 M
Residual train-control wayside + charging microgrids	€17 M
EPC overhead (7%)	€49 M
CAPEX total	€745 M
Per-route-km	€11 M / km
Per-capita (city pop)	€1,656 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running scripts/regenerate-city.sh abha.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–5 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 6 and runs for 20 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–5)	€48 M / yr	€106
Steady-state, low-ridership (year 6+)	€54 M / yr	€121
Steady-state, high-ridership (year 6+)	€47 M / yr	€104
Lifecycle envelope (yr 1–25, low scenario)	€1.32 bn cumulative	€2,943
Lifecycle envelope (yr 1–25, high scenario)	€1.17 bn cumulative	€2,603

Population basis: 450,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 25, debt service drops to zero and only the OPEX shortfall remains – ~€7.6 M / yr (low) €0 k / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€447 M	3.8%	25 y, 5 y grace	€32 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€186 M	4.5%	25 y, 5 y grace	€14 M / yr
Government equity (no debt service)	15%	€112 M	—	—	—
Total	100%	€745 M			€47 M / yr

During the 5-year grace period the operator pays interest only – multilateral €17 M / yr + bonds €8.4 M / yr = €25 M / yr total – plus the equity tranche amortised across construction (€22 M / yr × 5 yr). Principal repayment begins in year 6 on a 20-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€8.2 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€9.5 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€52 k
Traction energy (123.2 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (431 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€11 M
OPEX subtotal		€29 M / yr

Annual fleet utilisation: 61 revenue trainsets × 20.5 h/day × 365 d/yr × 30 km/h commercial × 75% revenue factor = 10.3 M train-km / yr (~168 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$1,700 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€2.61 (~\$2.83 USD)
Day pass (3 trips)	€6.65 (15 % bulk discount)
Monthly unlimited pass	€78.20 (~5 % of median monthly income)
Annual pass	€860.20 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	8.2 M	16.4 M

	Low scenario	High scenario
Farebox revenue	€21 M / yr	€43 M / yr
Farebox / OPEX recovery	74%	147%
Country policy-target recovery (diagnostic)	85%	85%
Operating shortfall (gov subsidy required)	€7.6 M / yr	€0 k / yr
Operating surplus (operator retained capex sinking fund)	€0 k / yr	€14 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€54 M / yr	€47 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
abha.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
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abha.corridor.geojson	Line polylines + stations (GeoJSON)
abha.stations.json	Machine-readable station list
abha.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug abha

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug abha \
  --sidecar .cache/osr-pipeline/rasters/abha.grid.json \
  --out-dir designs/.../Abha

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../abha.toml \
  --out designs/.../README.md
```

`scripts/regenerate-abha.sh` chains steps 3 + drift tests into a single command.

west-asia/Saudi Arabia/Buraidah

designs/west-asia/Saudi Arabia/Buraidah/README.md

Buraidah — Urban Rail Network

Country: SA · Population: 700,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Buraidah rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [buraidah.corridor.geojson](#).

At a glance

Metric	Value
Lines	3
Unique stations	54
Interchange stations	3
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	35.2%
Route length (double track)	79.6 km
Revenue fleet	69 × 3-car trainsets
Spare + cold-reserve	9 × 3-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	29.7 km	20	28	E Outer NW Outer
line-2	25.0 km	17	25	NW Outer SE Outer
line-3	25.0 km	17	25	SW Outer E Mid
Total	79.6 km	54 unique	78	

Rolling stock

Property	Value
Consist	3-car, 57 m
Max speed	90 km/h
Onboard battery	360 kWh per trainset
Seats	60 longitudinal seats
Nominal capacity (AW2)	330 pax (seated + standing, light-metro-3car per RFC 0008 §1)
Crush capacity (AW3)	420 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 330 AW2 passengers (light-metro-3car)
- Peak frequency: 12 trains/hour/direction (5-min headway)

- Peak capacity per line per direction: $330 \times 12 = 3,960$ pphpd
- Network peak throughput (all lines, both directions): $3 \text{ lines} \times 2 \text{ directions} \times 3,960 = 23,760$ passengers/hour
- Daily theoretical capacity (peak $\times 10$): $\approx 237,600$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 24,640 - 36,960$ trips/day

Catchment

- City population: 700,000
- Anchor-weighted coverage: 35.2%
- Catchment population: $\approx 246,400$ (within ~ 800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	3	500 kW	3000 kWh
Major	32	400 kW	2500 kWh
Standard	13	300 kW	2000 kWh
Terminal	5	500 kW	3000 kWh
Total installed	54	25,700 kW	170,000 kWh

Aggregate station-rail charging power: 30,000 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 360 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	12.0 kWh/km	3 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	318 kWh	26.5 km average line length
Onboard battery coverage	1.1 $\times$ average line run	360 kWh usable pack
Average 60 s dwell charge	9.3 kWh/stop	556 kW average charger across stops
Stops to refill one trainset pack	39 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	128 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	170 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in design.toml — emitted by the osr-design Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. country-costs.toml applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (78.2 km @ €3.5 M/km)	€274 M
Elevated (1.3 km @ €18 M/km)	€24 M
Elevated-interchange premium (1 sites @ €20 M)	€20 M
Civil subtotal	€318 M

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~ 11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
standard	13	€1.5 M	€20 M
major	32	€3.0 M	€96 M
terminal	5	€2.5 M	€12 M
depot-terminal	1	€3.0 M	€3.0 M
interchange-elevated	3	€4.5 M	€14 M
Stations subtotal			€144 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	5	€3.0 M	€15 M
Depots subtotal			€40 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
light-metro-3car (revenue + spare + cold reserve)	78	€3.0 M	€234 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	79.6 km × €0.015 M/km	€1.2 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€21 M
EPC integration + project management (7%)	on subtotal	€53 M

Total

Bucket	Value
Civil works	€318 M
Stations	€144 M
Depots	€40 M
Rolling stock	€234 M
Residual train-control wayside + charging microgrids	€22 M
EPC overhead (7%)	€53 M
CAPEX total	€811 M
Per-route-km	€10 M / km
Per-capita (city pop)	€1,159 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running scripts/regenerate-city.sh buraidah.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–5 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 6 and runs for 20 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–5)	€52 M / yr	€74
Steady-state, low-ridership (year 6+)	€51 M / yr	€73
Steady-state, high-ridership (year 6+)	€51 M / yr	€73
Lifecycle envelope (yr 1–25, low scenario)	€1.27 bn cumulative	€1,821
Lifecycle envelope (yr 1–25, high scenario)	€1.27 bn cumulative	€1,821

Population basis: 700,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 25, debt service drops to zero and only the OPEX shortfall remains – ~€0 k / yr (low) €0 k / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€487 M	3.8%	25 y, 5 y grace	€35 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€203 M	4.5%	25 y, 5 y grace	€16 M / yr
Government equity (no debt service)	15%	€122 M	—	—	—
Total	100%	€811 M			€51 M / yr

During the 5-year grace period the operator pays interest only – multilateral €18 M / yr + bonds €9.1 M / yr = €28 M / yr total – plus the equity tranche amortised across construction (€24 M / yr × 5 yr). Principal repayment begins in year 6 on a 20-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€9.4 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€10 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€60 k
Traction energy (139.4 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (490 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€13 M
OPEX subtotal		€32 M / yr

Annual fleet utilisation: 69 revenue trainsets × 20.5 h/day × 365 d/yr × 30 km/h commercial × 75% revenue factor = 11.6 M train-km / yr (~168 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$1,700 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€2.61 (~\$2.83 USD)
Day pass (3 trips)	€6.65 (15 % bulk discount)
Monthly unlimited pass	€78.20 (~5 % of median monthly income)
Annual pass	€860.20 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	12.8 M	25.6 M

	Low scenario	High scenario
Farebox revenue	€33 M / yr	€67 M / yr
Farebox / OPEX recovery	103%	206%
Country policy-target recovery (diagnostic)	85%	85%
Operating shortfall (gov subsidy required)	€0 k / yr	€0 k / yr
Operating surplus (operator retained capex sinking fund)	€962 k / yr	€34 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€51 M / yr	€51 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
buraidah.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
buraidah-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
buraidah.corridor.geojson	Line polylines + stations (GeoJSON)
buraidah.stations.json	Machine-readable station list
buraidah.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug buraidah

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug buraidah \
  --sidecar .cache/osr-pipeline/rasters/buraidah.grid.json \
  --out-dir designs/.../Buraidah

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../buraidah.toml \
  --out designs/.../README.md
```

`scripts/regenerate-buraidah.sh` chains steps 3 + drift tests into a single command.

west-asia/Saudi Arabia/Dammam

designs/west-asia/Saudi Arabia/Dammam/README.md

Dammam — Urban Rail Network

Country: SA · Population: 1,500,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Dammam rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [dammam.corridor.geojson](#).

At a glance

Metric	Value
Lines	4
Unique stations	94
Interchange stations	10
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	28.3%
Route length (double track)	209.9 km
Revenue fleet	151 × 4-car trainsets
Spare + cold-reserve	17 × 4-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	53.7 km	27	43	NW Outer SE Outer
line-2	37.2 km	15	30	SE Outer NW Outer
line-3	26.2 km	14	23	SE Mid N Mid
line-4	92.8 km	39	72	NW Mid NW Mid
Total	209.9 km	94 unique	168	

Rolling stock

Property	Value
Consist	4-car, 75 m
Max speed	90 km/h
Onboard battery	480 kWh per trainset
Seats	80 longitudinal seats
Nominal capacity (AW2)	440 pax (seated + standing, met ro - 4car per RFC 0008 §1)
Crush capacity (AW3)	560 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 440 AW2 passengers (met ro - 4car)

- Peak frequency: 12 trains/hour/direction (5-min headway)
- Peak capacity per line per direction: $440 \times 12 = 5,280$ pphpd
- Network peak throughput (all lines, both directions): $4 \text{ lines} \times 2 \text{ directions} \times 5,280 = 42,240$ passengers/hour
- Daily theoretical capacity (peak $\times$ 10): $\approx 422,400$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 42,449 - 63,674$ trips/day

Catchment

- City population: 1,500,000
- Anchor-weighted coverage: 28.3%
- Catchment population: $\approx 424,499$ (within ~800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	10	500 kW	3000 kWh
Major	41	400 kW	2500 kWh
Standard	34	300 kW	2000 kWh
Terminal	5	500 kW	3000 kWh
Total installed	91	39,100 kW	255,500 kWh

Aggregate station-rail charging power: 49,500 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 480 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	16.0 kWh/km	4 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	840 kWh	52.5 km average line length
Onboard battery coverage	0.6 $\times$ average line run	480 kWh usable pack
Average 60 s dwell charge	8.8 kWh/stop	527 kW average charger across stops
Stops to refill one trainset pack	55 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	196 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	256 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the `[costs]` block in `design.toml` — emitted by the `osr-design` Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. `country-costs.toml` applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (202.4 km @ €3.5 M/km)	€709 M
Elevated (7.0 km @ €18 M/km)	€125 M
Elevated-interchange premium (6 sites @ €20 M)	€120 M
Civil subtotal	€954 M

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
halt	4	€0.4 M	€1.6 M
standard	34	€1.5 M	€51 M
major	41	€3.0 M	€123 M
terminal	5	€2.5 M	€12 M
depot-terminal	1	€3.0 M	€3.0 M
interchange-elevated	10	€4.5 M	€45 M
Stations subtotal			€236 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	5	€3.0 M	€15 M
Depots subtotal			€40 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
metro-4car (revenue + spare + cold reserve)	168	€4.0 M	€672 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	209.9 km × €0.015 M/km	€3.1 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€34 M
EPC integration + project management (7%)	on subtotal	€136 M

Total

Bucket	Value
Civil works	€954 M
Stations	€236 M
Depots	€40 M
Rolling stock	€672 M
Residual train-control wayside + charging microgrids	€37 M
EPC overhead (7%)	€136 M
CAPEX total	€2.07 bn
Per-route-km	€9.9 M / km
Per-capita (city pop)	€1,383 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](https://github.com/dammam/lib/templates/country-finance.toml). Pure function of the [costs] block above + the country code – regenerate by re-running `scripts/regenerate-city.sh dammam`.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–5 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 6 and runs for 20 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–5)	€133 M / yr	€89
Steady-state, low-ridership (year 6+)	€144 M / yr	€96
Steady-state, high-ridership (year 6+)	€130 M / yr	€87
Lifecycle envelope (yr 1–25, low scenario)	€3.54 bn cumulative	€2,357
Lifecycle envelope (yr 1–25, high scenario)	€3.26 bn cumulative	€2,174

Population basis: 1,500,000 (catchment per [lib/city-batches/world-sample.toml](https://github.com/dammam/lib/city-batches/world-sample.toml)). After year 25, debt service drops to zero and only the OPEX shortfall remains – ~€14 M / yr (low) €0 k / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€1.24 bn	3.8%	25 y, 5 y grace	€90 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€519 M	4.5%	25 y, 5 y grace	€40 M / yr
Government equity (no debt service)	15%	€311 M	—	—	—
Total	100%	€2.07 bn			€130 M / yr

During the 5-year grace period the operator pays interest only – multilateral €47 M / yr + bonds €23 M / yr = €71 M / yr total – plus the equity tranche amortised across construction (€62 M / yr × 5 yr). Principal repayment begins in year 6 on a 20-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€27 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€25 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€157 k
Traction energy (474.5 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (1,271 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€33 M
OPEX subtotal		€85 M / yr

Annual fleet utilisation: 151 revenue trainsets × 20.5 h/day × 365 d/yr × 35 km/h commercial × 75% revenue factor = 29.7 M train-km / yr (~196 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$1,700 USD (per [lib/templates/country-finance.toml](https://github.com/dammam/lib/templates/country-finance.toml)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€2.61 (~\$2.83 USD)
Day pass (3 trips)	€6.65 (15 % bulk discount)
Monthly unlimited pass	€78.20 (~5 % of median monthly income)
Annual pass	€860.20 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	27.4 M	54.8 M
Farebox revenue	€71 M / yr	€143 M / yr
Farebox / OPEX recovery	84%	168%
Country policy-target recovery (diagnostic)	85%	85%
Operating shortfall (gov subsidy required)	€14 M / yr	€0 k / yr
Operating surplus (operator retained capex sinking fund)	€0 k / yr	€58 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€144 M / yr	€130 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
dammam.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
dammam-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
dammam.corridor.geojson	Line polylines + stations (GeoJSON)
dammam.stations.json	Machine-readable station list
dammam.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug dammam

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug dammam \
  --sidecar .cache/osr-pipeline/rasters/dammam.grid.json \
  --out-dir designs/.../Dammam

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../dammam.toml \
  --out designs/.../README.md
```

`scripts/regenerate-dammam.sh` chains steps 3 + drift tests into a single command.

west-asia/Saudi Arabia/Jeddah

designs/west-asia/Saudi Arabia/Jeddah/README.md

Jeddah — Urban Rail Network

Country: SA · Population: 4,700,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Jeddah rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [jeddah.corridor.geojson](#).

At a glance

Metric	Value
Lines	8
Unique stations	201
Interchange stations	35
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	45.2%
Route length (double track)	406.0 km
Revenue fleet	292 × 6-car trainsets
Spare + cold-reserve	34 × 6-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	53.1 km	27	42	NW Outer S Outer
line-2	48.3 km	25	39	SE Mid N Outer
line-3	50.1 km	26	40	NW Mid SE Outer
line-4	42.7 km	25	35	NW Mid SE Outer
line-5	54.0 km	21	43	N Outer S Outer
line-6	41.7 km	18	34	SW Outer E Mid
line-7	29.7 km	16	25	NE Outer SW Inner
line-8	86.5 km	44	68	NW Mid NW Mid
Total	406.0 km	201 unique	326	

Rolling stock

Property	Value
Consist	6-car, 111 m
Max speed	100 km/h
Onboard battery	720 kWh per trainset
Seats	120 longitudinal seats
Nominal capacity (AW2)	660 pax (seated + standing, metro-6car per RFC 0008 §1)
Crush capacity (AW3)	840 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 660 AW2 passengers (metro - 6 car)
- Peak frequency: 12 trains/hour/direction (5-min headway)
- Peak capacity per line per direction: $660 \times 12 = 7,920$ pphpd
- Network peak throughput (all lines, both directions): $8 \text{ lines} \times 2 \text{ directions} \times 7,920 = 126,720$ passengers/hour
- Daily theoretical capacity (peak $\times$ 10): $\approx 1,267,200$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 212,440 - 318,660$ trips/day

Catchment

- City population: 4,700,000
- Anchor-weighted coverage: 45.2%
- Catchment population: $\approx 2,124,400$ (within ~ 800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	35	500 kW	3000 kWh
Major	92	400 kW	2500 kWh
Standard	56	300 kW	2000 kWh
Terminal	13	500 kW	3000 kWh
Total installed	197	82,600 kW	526,000 kWh

Aggregate station-rail charging power: 106,750 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 720 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	24.0 kWh/km	6 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	1,218 kWh	50.8 km average line length
Onboard battery coverage	0.6 $\times$ average line run	720 kWh usable pack
Average 60 s dwell charge	8.9 kWh/stop	531 kW average charger across stops
Stops to refill one trainset pack	81 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	413 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	526 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in `design.toml` — emitted by the `osr-design` Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. `country-costs.toml` applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (385.6 km @ €3.5 M/km)	€1.35 bn
Elevated (19.4 km @ €18 M/km)	€350 M
Elevated-interchange premium (16 sites @ €20 M)	€320 M
Civil subtotal	€2.02 bn

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
halt	5	€0.4 M	€2.0 M
standard	56	€1.5 M	€84 M
major	92	€3.0 M	€276 M
terminal	13	€2.5 M	€32 M
depot-terminal	1	€3.0 M	€3.0 M
interchange-elevated	35	€4.5 M	€158 M
Stations subtotal			€555 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	13	€3.0 M	€39 M
Depots subtotal			€64 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
metro-6car (revenue + spare + cold reserve)	326	€6.0 M	€1.96 bn

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	406.0 km × €0.015 M/km	€6.1 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€78 M
EPC integration + project management (7%)	on subtotal	€328 M

Total

Bucket	Value
Civil works	€2.02 bn
Stations	€555 M
Depots	€64 M
Rolling stock	€1.96 bn
Residual train-control wayside + charging microgrids	€84 M
EPC overhead (7%)	€328 M
CAPEX total	€5.01 bn
Per-route-km	€12 M / km

Bucket	Value
Per-capita (city pop)	€1,065 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](https://github.com/OpenSourceRail/lib/templates/country-finance.toml). Pure function of the [costs] block above + the country code — regenerate by re-running `scripts/regenerate-city.sh jeddah`.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–5 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 6 and runs for 20 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–5)	€321 M / yr	€68
Steady-state, low-ridership (year 6+)	€313 M / yr	€67
Steady-state, high-ridership (year 6+)	€313 M / yr	€67
Lifecycle envelope (yr 1–25, low scenario)	€7.87 bn cumulative	€1,675
Lifecycle envelope (yr 1–25, high scenario)	€7.87 bn cumulative	€1,675

Population basis: 4,700,000 (catchment per [lib/city-batches/world-sample.toml](https://github.com/OpenSourceRail/lib/city-batches/world-sample.toml)). After year 25, debt service drops to zero and only the OPEX shortfall remains — ~€0 k / yr (low) €0 k / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€3.00 bn	3.8%	25 y, 5 y grace	€217 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€1.25 bn	4.5%	25 y, 5 y grace	€96 M / yr
Government equity (no debt service)	15%	€751 M	—	—	—
Total	100%	€5.01 bn			€313 M / yr

During the 5-year grace period the operator pays interest only — multilateral €114 M / yr + bonds €56 M / yr = €170 M / yr total — plus the equity tranche amortised across construction (€150 M / yr × 5 yr). Principal repayment begins in year 6 on a 20-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€78 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€53 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€304 k
Traction energy (1376.5 GWh / yr)	trackside PV + Na-ion (RFC 0002) — self-generated, €0 / yr	€0 k
Labour (2,448 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€64 M
OPEX subtotal		€196 M / yr

Annual fleet utilisation: 292 revenue trainsets × 20.5 h/day × 365 d/yr × 35 km/h commercial × 75% revenue factor = 57.4 M train-km / yr (~196 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$1,700 USD (per [lib/templates/country-finance.toml](https://github.com/OpenSourceRail/lib/templates/country-finance.toml)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass — a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€2.61 (~\$2.83 USD)
Day pass (3 trips)	€6.65 (15 % bulk discount)
Monthly unlimited pass	€78.20 (~5 % of median monthly income)
Annual pass	€860.20 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	85.8 M	171.6 M
Farebox revenue	€224 M / yr	€447 M / yr
Farebox / OPEX recovery	114%	229%
Country policy-target recovery (diagnostic)	85%	85%
Operating shortfall (gov subsidy required)	€0 k / yr	€0 k / yr
Operating surplus (operator retained capex sinking fund)	€28 M / yr	€252 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€313 M / yr	€313 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
jeddah.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
jeddah-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
jeddah.corridor.geojson	Line polylines + stations (GeoJSON)
jeddah.stations.json	Machine-readable station list
jeddah.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug jeddah

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug jeddah \
  --sidecar .cache/osr-pipeline/rasters/jeddah.grid.json \
  --out-dir designs/.../Jeddah

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../jeddah.toml \
  --out designs/.../README.md
```

`scripts/regenerate-jeddah.sh` chains steps 3 + drift tests into a single command.

west-asia/Saudi Arabia/Mecca

designs/west-asia/Saudi Arabia/Mecca/README.md

Mecca — Urban Rail Network

Country: SA · Population: 2,200,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Mecca rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [mecca.corridor.geojson](#).

At a glance

Metric	Value
Lines	6
Unique stations	113
Interchange stations	21
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	45.8%
Route length (double track)	251.5 km
Revenue fleet	183 × 4-car trainsets
Spare + cold-reserve	22 × 4-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	41.0 km	19	34	NE Outer SW Outer
line-2	33.4 km	17	28	NW Mid SE Outer
line-3	34.2 km	15	28	SE Outer NW Mid
line-4	28.6 km	14	24	W Outer N Mid
line-5	35.2 km	17	29	W Outer E Outer
line-6	79.0 km	32	62	W Mid W Mid
Total	251.5 km	113 unique	205	

Rolling stock

Property	Value
Consist	4-car, 75 m
Max speed	90 km/h
Onboard battery	480 kWh per trainset
Seats	80 longitudinal seats
Nominal capacity (AW2)	440 pax (seated + standing, metro-4car per RFC 0008 §1)
Crush capacity (AW3)	560 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 440 AW2 passengers (metro - 4 car)
- Peak frequency: 12 trains/hour/direction (5-min headway)
- Peak capacity per line per direction: $440 \times 12 = 5,280$ pphpd
- Network peak throughput (all lines, both directions): $6 \text{ lines} \times 2 \text{ directions} \times 5,280 = 63,360$ passengers/hour
- Daily theoretical capacity (peak $\times$ 10): $\approx 633,600$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 100,760 - 151,140$ trips/day

Catchment

- City population: 2,200,000
- Anchor-weighted coverage: 45.8%
- Catchment population: $\approx 1,007,600$ (within ~800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	21	500 kW	3000 kWh
Major	27	400 kW	2500 kWh
Standard	50	300 kW	2000 kWh
Terminal	9	500 kW	3000 kWh
Total installed	108	45,800 kW	297,500 kWh

Aggregate station-rail charging power: 60,500 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 480 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	16.0 kWh/km	4 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	671 kWh	41.9 km average line length
Onboard battery coverage	0.7 $\times$ average line run	480 kWh usable pack
Average 60 s dwell charge	8.9 kWh/stop	535 kW average charger across stops
Stops to refill one trainset pack	54 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	229 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	298 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in design.toml — emitted by the osr-design Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. country-costs.toml applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (236.7 km @ €3.5 M/km)	€828 M
Elevated (14.1 km @ €18 M/km)	€255 M
Elevated-interchange premium (11 sites @ €20 M)	€220 M
Civil subtotal	€1.30 bn

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
halt	6	€0.4 M	€2.4 M
standard	50	€1.5 M	€75 M
major	27	€3.0 M	€81 M
terminal	9	€2.5 M	€22 M
depot-terminal	1	€3.0 M	€3.0 M
interchange-elevated	21	€4.5 M	€94 M
Stations subtotal			€278 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	9	€3.0 M	€27 M
Depots subtotal			€52 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
metro-4car (revenue + spare + cold reserve)	205	€4.0 M	€820 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	251.5 km × €0.015 M/km	€3.8 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€41 M
EPC integration + project management (7%)	on subtotal	€175 M

Total

Bucket	Value
Civil works	€1.30 bn
Stations	€278 M
Depots	€52 M
Rolling stock	€820 M
Residual train-control wayside + charging microgrids	€45 M
EPC overhead (7%)	€175 M
CAPEX total	€2.67 bn
Per-route-km	€11 M / km
Per-capita (city pop)	€1,215 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running `scripts/regenerate-city.sh mecca`.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–5 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 6 and runs for 20 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–5)	€171 M / yr	€78
Steady-state, low-ridership (year 6+)	€168 M / yr	€76
Steady-state, high-ridership (year 6+)	€167 M / yr	€76
Lifecycle envelope (yr 1–25, low scenario)	€4.22 bn cumulative	€1,919
Lifecycle envelope (yr 1–25, high scenario)	€4.20 bn cumulative	€1,910

Population basis: 2,200,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 25, debt service drops to zero and only the OPEX shortfall remains – ~€964 k / yr (low) €0 k / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€1.60 bn	3.8%	25 y, 5 y grace	€116 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€668 M	4.5%	25 y, 5 y grace	€51 M / yr
Government equity (no debt service)	15%	€401 M	—	—	—
Total	100%	€2.67 bn			€167 M / yr

During the 5-year grace period the operator pays interest only – multilateral €61 M / yr + bonds €30 M / yr = €91 M / yr total – plus the equity tranche amortised across construction (€80 M / yr × 5 yr). Principal repayment begins in year 6 on a 20-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€33 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€33 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€188 k
Traction energy (575.1 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (1,521 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€40 M
OPEX subtotal		€106 M / yr

Annual fleet utilisation: 183 revenue trainsets × 20.5 h/day × 365 d/yr × 35 km/h commercial × 75% revenue factor = 35.9 M train-km / yr (~196 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$1,700 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€2.61 (~\$2.83 USD)
Day pass (3 trips)	€6.65 (15 % bulk discount)
Monthly unlimited pass	€78.20 (~5 % of median monthly income)
Annual pass	€860.20 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	40.1 M	80.3 M
Farebox revenue	€105 M / yr	€209 M / yr
Farebox / OPEX recovery	99%	198%
Country policy-target recovery (diagnostic)	85%	85%
Operating shortfall (gov subsidy required)	€964 k / yr	€0 k / yr
Operating surplus (operator retained capex sinking fund)	€0 k / yr	€104 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€168 M / yr	€167 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
mecca.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
mecca-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
mecca.corridor.geojson	Line polylines + stations (GeoJSON)
mecca.stations.json	Machine-readable station list
mecca.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug mecca

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug mecca \
  --sidecar .cache/osr-pipeline/rasters/mecca.grid.json \
  --out-dir designs/.../Mecca

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../mecca.toml \
  --out designs/.../README.md
```

`scripts/regenerate-mecca.sh` chains steps 3 + drift tests into a single command.

west-asia/Saudi Arabia/Medina

designs/west-asia/Saudi Arabia/Medina/README.md

Medina — Urban Rail Network

Country: SA · Population: 1,500,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Medina rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [medina.corridor.geojson](#).

At a glance

Metric	Value
Lines	5
Unique stations	103
Interchange stations	14
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	47.0%
Route length (double track)	211.5 km
Revenue fleet	153 × 4-car trainsets
Spare + cold-reserve	18 × 4-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	34.4 km	19	28	NE Outer SW Outer
line-2	31.4 km	18	26	W Outer E Outer
line-3	27.7 km	15	24	N Mid SE Outer
line-4	40.1 km	20	32	E Outer W Outer
line-5	77.9 km	32	61	W Outer W Outer
Total	211.5 km	103 unique	171	

Rolling stock

Property	Value
Consist	4-car, 75 m
Max speed	90 km/h
Onboard battery	480 kWh per trainset
Seats	80 longitudinal seats
Nominal capacity (AW2)	440 pax (seated + standing, met ro - 4 car per RFC 0008 §1)
Crush capacity (AW3)	560 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 440 AW2 passengers (met ro - 4 car)

- Peak frequency: 12 trains/hour/direction (5-min headway)
- Peak capacity per line per direction: $440 \times 12 = 5,280$ pphpd
- Network peak throughput (all lines, both directions): $5 \text{ lines} \times 2 \text{ directions} \times 5,280 = 52,800$ passengers/hour
- Daily theoretical capacity (peak $\times$ 10): $\approx 528,000$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 70,500 - 105,750$ trips/day

Catchment

- City population: 1,500,000
- Anchor-weighted coverage: 47.0%
- Catchment population: $\approx 705,000$ (within ~800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	14	500 kW	3000 kWh
Major	42	400 kW	2500 kWh
Standard	33	300 kW	2000 kWh
Terminal	7	500 kW	3000 kWh
Total installed	97	42,200 kW	274,000 kWh

Aggregate station-rail charging power: 54,250 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 480 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	16.0 kWh/km	4 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	677 kWh	42.3 km average line length
Onboard battery coverage	0.7 $\times$ average line run	480 kWh usable pack
Average 60 s dwell charge	8.8 kWh/stop	527 kW average charger across stops
Stops to refill one trainset pack	55 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	211 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	274 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the `[costs]` block in `design.toml` — emitted by the `osr-design` Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. `country-costs.toml` applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (196.9 km @ €3.5 M/km)	€689 M
Elevated (13.8 km @ €18 M/km)	€248 M
Elevated-interchange premium (11 sites @ €20 M)	€220 M
Civil subtotal	€1.16 bn

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
halt	7	€0.4 M	€2.8 M
standard	33	€1.5 M	€50 M
major	42	€3.0 M	€126 M
terminal	7	€2.5 M	€18 M
depot-terminal	1	€3.0 M	€3.0 M
interchange-elevated	14	€4.5 M	€63 M
Stations subtotal			€262 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	7	€3.0 M	€21 M
Depots subtotal			€46 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
metro-4car (revenue + spare + cold reserve)	171	€4.0 M	€684 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	211.5 km × €0.015 M/km	€3.2 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€38 M
EPC integration + project management (7%)	on subtotal	€153 M

Total

Bucket	Value
Civil works	€1.16 bn
Stations	€262 M
Depots	€46 M
Rolling stock	€684 M
Residual train-control wayside + charging microgrids	€41 M
EPC overhead (7%)	€153 M
CAPEX total	€2.34 bn
Per-route-km	€11 M / km
Per-capita (city pop)	€1,562 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](https://github.com/medina/lib/templates/country-finance.toml). Pure function of the [costs] block above + the country code – regenerate by re-running `scripts/regenerate-city.sh medina`.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–5 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 6 and runs for 20 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–5)	€150 M / yr	€100
Steady-state, low-ridership (year 6+)	€166 M / yr	€111
Steady-state, high-ridership (year 6+)	€147 M / yr	€98
Lifecycle envelope (yr 1–25, low scenario)	€4.07 bn cumulative	€2,711
Lifecycle envelope (yr 1–25, high scenario)	€3.68 bn cumulative	€2,456

Population basis: 1,500,000 (catchment per [lib/city-batches/world-sample.toml](https://github.com/medina/lib/city-batches/world-sample.toml)). After year 25, debt service drops to zero and only the OPEX shortfall remains – ~€19 M / yr (low) €0 k / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€1.41 bn	3.8%	25 y, 5 y grace	€102 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€586 M	4.5%	25 y, 5 y grace	€45 M / yr
Government equity (no debt service)	15%	€352 M	—	—	—
Total	100%	€2.34 bn			€147 M / yr

During the 5-year grace period the operator pays interest only – multilateral €53 M / yr + bonds €26 M / yr = €80 M / yr total – plus the equity tranche amortised across construction (€70 M / yr × 5 yr). Principal repayment begins in year 6 on a 20-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€27 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€29 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€158 k
Traction energy (480.8 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (1,281 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€34 M
OPEX subtotal		€90 M / yr

Annual fleet utilisation: 153 revenue trainsets × 20.5 h/day × 365 d/yr × 35 km/h commercial × 75% revenue factor = 30.1 M train-km / yr (~196 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$1,700 USD (per [lib/templates/country-finance.toml](https://github.com/medina/lib/templates/country-finance.toml)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€2.61 (~\$2.83 USD)
Day pass (3 trips)	€6.65 (15 % bulk discount)
Monthly unlimited pass	€78.20 (~5 % of median monthly income)
Annual pass	€860.20 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	27.4 M	54.8 M
Farebox revenue	€71 M / yr	€143 M / yr
Farebox / OPEX recovery	79%	158%
Country policy-target recovery (diagnostic)	85%	85%
Operating shortfall (gov subsidy required)	€19 M / yr	€0 k / yr
Operating surplus (operator retained capex sinking fund)	€0 k / yr	€52 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€166 M / yr	€147 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
medina.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
medina-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
medina.corridor.geojson	Line polylines + stations (GeoJSON)
medina.stations.json	Machine-readable station list
medina.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug medina

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug medina \
  --sidecar .cache/osr-pipeline/rasters/medina.grid.json \
  --out-dir designs/.../Medina

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../medina.toml \
  --out designs/.../README.md
```

`scripts/regenerate-medina.sh` chains steps 3 + drift tests into a single command.

west-asia/Saudi Arabia/Tabuk

designs/west-asia/Saudi Arabia/Tabuk/README.md

Tabuk — Urban Rail Network

Country: SA · Population: 650,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Tabuk rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [tabuk.corridor.geojson](#).

At a glance

Metric	Value
Lines	3
Unique stations	45
Interchange stations	2
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	27.2%
Route length (double track)	62.9 km
Revenue fleet	56 × 3-car trainsets
Spare + cold-reserve	7 × 3-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	25.3 km	17	25	E Outer W Outer
line-2	20.2 km	15	20	S Outer NW Outer
line-3	17.3 km	13	18	S Mid N Mid
Total	62.9 km	45 unique	63	

Rolling stock

Property	Value
Consist	3-car, 57 m
Max speed	90 km/h
Onboard battery	360 kWh per trainset
Seats	60 longitudinal seats
Nominal capacity (AW2)	330 pax (seated + standing, light-metro-3car per RFC 0008 §1)
Crush capacity (AW3)	420 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 330 AW2 passengers (light-metro-3car)
- Peak frequency: 12 trains/hour/direction (5-min headway)

- Peak capacity per line per direction: $330 \times 12 = 3,960$ pphpd
- Network peak throughput (all lines, both directions): $3 \text{ lines} \times 2 \text{ directions} \times 3,960 = 23,760$ passengers/hour
- Daily theoretical capacity (peak $\times 10$): $\approx 237,600$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 17,680 - 26,520$ trips/day

Catchment

- City population: 650,000
- Anchor-weighted coverage: 27.2%
- Catchment population: $\approx 176,800$ (within ~ 800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	2	500 kW	3000 kWh
Major	27	400 kW	2500 kWh
Standard	10	300 kW	2000 kWh
Terminal	5	500 kW	3000 kWh
Total installed	45	22,300 kW	148,500 kWh

Aggregate station-rail charging power: 25,500 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 360 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	12.0 kWh/km	3 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	252 kWh	21.0 km average line length
Onboard battery coverage	1.4 $\times$ average line run	360 kWh usable pack
Average 60 s dwell charge	9.4 kWh/stop	567 kW average charger across stops
Stops to refill one trainset pack	38 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	112 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	148 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in design.toml — emitted by the osr-design Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. country-costs.toml applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (56.6 km @ €3.5 M/km)	€198 M
Elevated (6.0 km @ €18 M/km)	€108 M
Elevated-interchange premium (1 sites @ €20 M)	€20 M
Civil subtotal	€326 M

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~ 11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
standard	10	€1.5 M	€15 M
major	27	€3.0 M	€81 M
terminal	5	€2.5 M	€12 M
depot-terminal	1	€3.0 M	€3.0 M
interchange-elevated	2	€4.5 M	€9.0 M
Stations subtotal			€120 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	5	€3.0 M	€15 M
Depots subtotal			€40 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
light-metro-3car (revenue + spare + cold reserve)	63	€3.0 M	€189 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	62.9 km × €0.015 M/km	€0.9 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€17 M
EPC integration + project management (7%)	on subtotal	€49 M

Total

Bucket	Value
Civil works	€326 M
Stations	€120 M
Depots	€40 M
Rolling stock	€189 M
Residual train-control wayside + charging microgrids	€18 M
EPC overhead (7%)	€49 M
CAPEX total	€743 M
Per-route-km	€12 M / km
Per-capita (city pop)	€1,143 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running scripts/regenerate-city.sh tabuk.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–5 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 6 and runs for 20 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–5)	€48 M / yr	€73
Steady-state, low-ridership (year 6+)	€46 M / yr	€72
Steady-state, high-ridership (year 6+)	€46 M / yr	€72
Lifecycle envelope (yr 1–25, low scenario)	€1.17 bn cumulative	€1,796
Lifecycle envelope (yr 1–25, high scenario)	€1.17 bn cumulative	€1,796

Population basis: 650,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 25, debt service drops to zero and only the OPEX shortfall remains – ~€0 k / yr (low) €0 k / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€446 M	3.8%	25 y, 5 y grace	€32 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€186 M	4.5%	25 y, 5 y grace	€14 M / yr
Government equity (no debt service)	15%	€111 M	—	—	—
Total	100%	€743 M			€46 M / yr

During the 5-year grace period the operator pays interest only – multilateral €17 M / yr + bonds €8.4 M / yr = €25 M / yr total – plus the equity tranche amortised across construction (€22 M / yr × 5 yr). Principal repayment begins in year 6 on a 20-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€7.6 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€9.7 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€47 k
Traction energy (113.1 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (389 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€10 M
OPEX subtotal		€28 M / yr

Annual fleet utilisation: 56 revenue trainsets × 20.5 h/day × 365 d/yr × 30 km/h commercial × 75% revenue factor = 9.4 M train-km / yr (~168 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$1,700 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€2.61 (~\$2.83 USD)
Day pass (3 trips)	€6.65 (15 % bulk discount)
Monthly unlimited pass	€78.20 (~5 % of median monthly income)
Annual pass	€860.20 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	11.9 M	23.7 M

	Low scenario	High scenario
Farebox revenue	€31 M / yr	€62 M / yr
Farebox / OPEX recovery	112%	224%
Country policy-target recovery (diagnostic)	85%	85%
Operating shortfall (gov subsidy required)	€0 k / yr	€0 k / yr
Operating surplus (operator retained capex sinking fund)	€3.4 M / yr	€34 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€46 M / yr	€46 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
tabuk.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
tabuk-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
tabuk.corridor.geojson	Line polylines + stations (GeoJSON)
tabuk.stations.json	Machine-readable station list
tabuk.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug tabuk

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug tabuk \
  --sidecar .cache/osr-pipeline/rasters/tabuk.grid.json \
  --out-dir designs/.../Tabuk

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../tabuk.toml \
  --out designs/.../README.md
```

`scripts/regenerate-tabuk.sh` chains steps 3 + drift tests into a single command.

west-asia/Saudi Arabia/Taif

designs/west-asia/Saudi Arabia/Taif/README.md

Taif — Urban Rail Network

Country: SA · Population: 700,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Taif rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [taif.corridor.geojson](#).

At a glance

Metric	Value
Lines	3
Unique stations	42
Interchange stations	3
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	40.8%
Route length (double track)	73.5 km
Revenue fleet	64 × 3-car trainsets
Spare + cold-reserve	9 × 3-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	24.4 km	14	24	SW Outer N Outer
line-2	25.1 km	15	25	SE Outer NW Outer
line-3	23.9 km	13	24	W Mid NE Outer
Total	73.5 km	42 unique	73	

Rolling stock

Property	Value
Consist	3-car, 57 m
Max speed	90 km/h
Onboard battery	360 kWh per trainset
Seats	60 longitudinal seats
Nominal capacity (AW2)	330 pax (seated + standing, light-metro-3car per RFC 0008 §1)
Crush capacity (AW3)	420 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 330 AW2 passengers (light-metro-3car)
- Peak frequency: 12 trains/hour/direction (5-min headway)

- Peak capacity per line per direction: $330 \times 12 = 3,960$ pphpd
- Network peak throughput (all lines, both directions): $3 \text{ lines} \times 2 \text{ directions} \times 3,960 = 23,760$ passengers/hour
- Daily theoretical capacity (peak $\times 10$): $\approx 237,600$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 28,560 - 42,840$ trips/day

Catchment

- City population: 700,000
- Anchor-weighted coverage: 40.8%
- Catchment population: $\approx 285,600$ (within ~ 800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../.../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	3	500 kW	3000 kWh
Major	12	400 kW	2500 kWh
Standard	20	300 kW	2000 kWh
Terminal	5	500 kW	3000 kWh
Total installed	41	19,800 kW	134,000 kWh

Aggregate station-rail charging power: 23,750 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 360 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	12.0 kWh/km	3 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	294 kWh	24.5 km average line length
Onboard battery coverage	1.2 $\times$ average line run	360 kWh usable pack
Average 60 s dwell charge	9.4 kWh/stop	565 kW average charger across stops
Stops to refill one trainset pack	38 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	99 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	134 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in design.toml — emitted by the osr-design Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. country-costs.toml applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (71.1 km @ €3.5 M/km)	€249 M
Elevated (2.2 km @ €18 M/km)	€40 M
Elevated-interchange premium (1 sites @ €20 M)	€20 M
Civil subtotal	€308 M

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~ 11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
halt	1	€0.4 M	€0.4 M
standard	20	€1.5 M	€30 M
major	12	€3.0 M	€36 M
terminal	5	€2.5 M	€12 M
depot-terminal	1	€3.0 M	€3.0 M
interchange-elevated	3	€4.5 M	€14 M
Stations subtotal			€95 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	5	€3.0 M	€15 M
Depots subtotal			€40 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
light-metro-3car (revenue + spare + cold reserve)	73	€3.0 M	€219 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	73.5 km × €0.015 M/km	€1.1 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€14 M
EPC integration + project management (7%)	on subtotal	€47 M

Total

Bucket	Value
Civil works	€308 M
Stations	€95 M
Depots	€40 M
Rolling stock	€219 M
Residual train-control wayside + charging microgrids	€16 M
EPC overhead (7%)	€47 M
CAPEX total	€726 M
Per-route-km	€9.9 M / km
Per-capita (city pop)	€1,037 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running scripts/regenerate-city.sh taif.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–5 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 6 and runs for 20 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–5)	€46 M / yr	€66
Steady-state, low-ridership (year 6+)	€45 M / yr	€65
Steady-state, high-ridership (year 6+)	€45 M / yr	€65
Lifecycle envelope (yr 1–25, low scenario)	€1.14 bn cumulative	€1,630
Lifecycle envelope (yr 1–25, high scenario)	€1.14 bn cumulative	€1,630

Population basis: 700,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 25, debt service drops to zero and only the OPEX shortfall remains – ~€0 k / yr (low) €0 k / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€436 M	3.8%	25 y, 5 y grace	€31 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€181 M	4.5%	25 y, 5 y grace	€14 M / yr
Government equity (no debt service)	15%	€109 M	—	—	—
Total	100%	€726 M			€45 M / yr

During the 5-year grace period the operator pays interest only – multilateral €17 M / yr + bonds €8.2 M / yr = €25 M / yr total – plus the equity tranche amortised across construction (€22 M / yr × 5 yr). Principal repayment begins in year 6 on a 20-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€8.8 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€8.9 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€55 k
Traction energy (129.3 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (453 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€12 M
OPEX subtotal		€30 M / yr

Annual fleet utilisation: 64 revenue trainsets × 20.5 h/day × 365 d/yr × 30 km/h commercial × 75% revenue factor = 10.8 M train-km / yr (~168 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$1,700 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€2.61 (~\$2.83 USD)
Day pass (3 trips)	€6.65 (15 % bulk discount)
Monthly unlimited pass	€78.20 (~5 % of median monthly income)
Annual pass	€860.20 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	12.8 M	25.6 M

	Low scenario	High scenario
Farebox revenue	€33 M / yr	€67 M / yr
Farebox / OPEX recovery	113%	225%
Country policy-target recovery (diagnostic)	85%	85%
Operating shortfall (gov subsidy required)	€0 k / yr	€0 k / yr
Operating surplus (operator retained capex sinking fund)	€3.7 M / yr	€37 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€45 M / yr	€45 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
taif.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
taif-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
taif.corridor.geojson	Line polylines + stations (GeoJSON)
taif.stations.json	Machine-readable station list
taif.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug taif

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug taif \
  --sidecar .cache/osr-pipeline/rasters/taif.grid.json \
  --out-dir designs/.../Taif

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../taif.toml \
  --out designs/.../README.md
```

`scripts/regenerate-taif.sh` chains steps 3 + drift tests into a single command.

west-asia/Syria/Aleppo

designs/west-asia/Syria/Aleppo/README.md

Aleppo — Urban Rail Network

Country: SY · Population: 1,639,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Aleppo rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [aleppo.corridor.geojson](#).

At a glance

Metric	Value
Lines	5
Unique stations	88
Interchange stations	16
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	46.2%
Route length (double track)	175.7 km
Revenue fleet	129 × 4-car trainsets
Spare + cold-reserve	16 × 4-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	29.8 km	16	25	W Outer E Outer
line-2	32.4 km	17	27	SE Outer NW Outer
line-3	35.4 km	17	29	NE Outer SW Outer
line-4	20.7 km	13	18	W Outer E Mid
line-5	57.4 km	26	46	NW Mid NW Mid
Total	175.7 km	88 unique	145	

Rolling stock

Property	Value
Consist	4-car, 75 m
Max speed	90 km/h
Onboard battery	480 kWh per trainset
Seats	80 longitudinal seats
Nominal capacity (AW2)	440 pax (seated + standing, met ro - 4 car per RFC 0008 §1)
Crush capacity (AW3)	560 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 440 AW2 passengers (met ro - 4 car)

- Peak frequency: 12 trains/hour/direction (5-min headway)
- Peak capacity per line per direction: $440 \times 12 = 5,280$ pphpd
- Network peak throughput (all lines, both directions): $5 \text{ lines} \times 2 \text{ directions} \times 5,280 = 52,800$ passengers/hour
- Daily theoretical capacity (peak $\times$ 10): $\approx 528,000$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 75,721 - 113,582$ trips/day

Catchment

- City population: 1,639,000
- Anchor-weighted coverage: 46.2%
- Catchment population: $\approx 757,218$ (within ~ 800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	16	500 kW	3000 kWh
Major	30	400 kW	2500 kWh
Standard	32	300 kW	2000 kWh
Terminal	7	500 kW	3000 kWh
Total installed	86	38,100 kW	248,000 kWh

Aggregate station-rail charging power: 47,750 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 480 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	16.0 kWh/km	4 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	562 kWh	35.1 km average line length
Onboard battery coverage	0.9 $\times$ average line run	480 kWh usable pack
Average 60 s dwell charge	9.0 kWh/stop	543 kW average charger across stops
Stops to refill one trainset pack	53 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	190 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	248 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in design.toml — emitted by the osr-design Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. country-costs.toml applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (168.1 km @ €3.5 M/km)	€588 M
Elevated (6.9 km @ €18 M/km)	€124 M
Elevated-interchange premium (6 sites @ €20 M)	€120 M
Civil subtotal	€833 M

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
halt	3	€0.4 M	€1.2 M
standard	32	€1.5 M	€48 M
major	30	€3.0 M	€90 M
terminal	7	€2.5 M	€18 M
depot-terminal	1	€3.0 M	€3.0 M
interchange	2	€4.5 M	€9.0 M
interchange-elevated	14	€4.5 M	€63 M
Stations subtotal			€232 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	7	€3.0 M	€21 M
Depots subtotal			€46 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
metro-4car (revenue + spare + cold reserve)	145	€4.0 M	€580 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	175.7 km × €0.015 M/km	€2.6 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€34 M
EPC integration + project management (7%)	on subtotal	€121 M

Total

Bucket	Value
Civil works	€833 M
Stations	€232 M
Depots	€46 M
Rolling stock	€580 M
Residual train-control wayside + charging microgrids	€36 M
EPC overhead (7%)	€121 M
CAPEX total	€1.85 bn
Per-route-km	€11 M / km
Per-capita (city pop)	€1,127 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running `scripts/regenerate-city.sh aleppo`.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–10 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 11 and runs for 25 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–10)	€170 M / yr	€104
Steady-state, low-ridership (year 11+)	€212 M / yr	€129
Steady-state, high-ridership (year 11+)	€208 M / yr	€127
Lifecycle envelope (yr 1–35, low scenario)	€6.99 bn cumulative	€4,265
Lifecycle envelope (yr 1–35, high scenario)	€6.91 bn cumulative	€4,216

Population basis: 1,639,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 35, debt service drops to zero and only the OPEX shortfall remains – ~€43 M / yr (low) €40 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€1.11 bn	4.5%	35 y, 10 y grace	€75 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€462 M	20.0%	35 y, 10 y grace	€93 M / yr
Government equity (no debt service)	15%	€277 M	—	—	—
Total	100%	€1.85 bn			€168 M / yr

During the 10-year grace period the operator pays interest only — multilateral €50 M / yr + bonds €92 M / yr = €142 M / yr total — plus the equity tranche amortised across construction (€28 M / yr × 10 yr). Principal repayment begins in year 11 on a 25-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€23 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€22 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€131 k
Traction energy (405.4 GWh / yr)	trackside PV + Na-ion (RFC 0002) — self-generated, €0 / yr	€0 k
Labour (1,066 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€1.2 M
OPEX subtotal		€47 M / yr

Annual fleet utilisation: 129 revenue trainsets × 20.5 h/day × 365 d/yr × 35 km/h commercial × 75% revenue factor = 25.3 M train-km / yr (~196 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$70 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass — a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.11 (~\$0.12 USD)
Day pass (3 trips)	€0.27 (15 % bulk discount)
Monthly unlimited pass	€3.22 (~5 % of median monthly income)
Annual pass	€35.42 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	29.9 M	59.8 M
Farebox revenue	€3.2 M / yr	€6.4 M / yr
Farebox / OPEX recovery	7%	14%
Country policy-target recovery (diagnostic)	30%	30%
Operating shortfall (gov subsidy required)	€43 M / yr	€40 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€212 M / yr	€208 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
aleppo.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
aleppo-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
aleppo.corridor.geojson	Line polylines + stations (GeoJSON)
aleppo.stations.json	Machine-readable station list
aleppo.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug aleppo

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug aleppo \
  --sidecar .cache/osr-pipeline/rasters/aleppo.grid.json \
  --out-dir designs/.../Aleppo

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../aleppo.toml \
  --out designs/.../README.md
```

`scripts/regenerate-aleppo.sh` chains steps 3 + drift tests into a single command.

west-asia/Syria/Damascus

designs/west-asia/Syria/Damascus/README.md

Damascus — Urban Rail Network

Country: SY · Population: 2,503,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Damascus rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [damascus.corridor.geojson](#).

At a glance

Metric	Value
Lines	6
Unique stations	112
Interchange stations	25
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	45.1%
Route length (double track)	233.1 km
Revenue fleet	171 × 4-car trainsets
Spare + cold-reserve	21 × 4-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	38.5 km	18	31	SW Outer NE Outer
line-2	30.9 km	15	26	SE Mid NW Outer
line-3	27.8 km	15	24	S Mid N Outer
line-4	30.5 km	17	26	NE Mid SW Outer
line-5	26.0 km	13	23	W Mid E Outer
line-6	79.4 km	35	62	NW Mid NW Mid
Total	233.1 km	112 unique	192	

Rolling stock

Property	Value
Consist	4-car, 75 m
Max speed	90 km/h
Onboard battery	480 kWh per trainset
Seats	80 longitudinal seats
Nominal capacity (AW2)	440 pax (seated + standing, metro-4car per RFC 0008 §1)
Crush capacity (AW3)	560 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 440 AW2 passengers (metro - 4 car)
- Peak frequency: 12 trains/hour/direction (5-min headway)
- Peak capacity per line per direction: $440 \times 12 = 5,280$ pphpd
- Network peak throughput (all lines, both directions): $6 \text{ lines} \times 2 \text{ directions} \times 5,280 = 63,360$ passengers/hour
- Daily theoretical capacity (peak $\times$ 10): $\approx 633,600$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 112,885 - 169,327$ trips/day

Catchment

- City population: 2,503,000
- Anchor-weighted coverage: 45.1%
- Catchment population: $\approx 1,128,853$ (within ~ 800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../.../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	25	500 kW	3000 kWh
Major	28	400 kW	2500 kWh
Standard	48	300 kW	2000 kWh
Terminal	9	500 kW	3000 kWh
Total installed	111	47,600 kW	308,000 kWh

Aggregate station-rail charging power: 61,000 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 480 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	16.0 kWh/km	4 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	622 kWh	38.9 km average line length
Onboard battery coverage	0.8 $\times$ average line run	480 kWh usable pack
Average 60 s dwell charge	9.1 kWh/stop	545 kW average charger across stops
Stops to refill one trainset pack	53 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	238 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	308 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in design.toml — emitted by the osr-design Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. country-costs.toml applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (217.4 km @ €3.5 M/km)	€761 M
Elevated (14.3 km @ €18 M/km)	€258 M
Elevated-interchange premium (12 sites @ €20 M)	€240 M
Civil subtotal	€1.26 bn

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
halt	2	€0.4 M	€0.8 M
standard	48	€1.5 M	€72 M
major	28	€3.0 M	€84 M
terminal	9	€2.5 M	€22 M
depot-terminal	1	€3.0 M	€3.0 M
interchange	3	€4.5 M	€14 M
interchange-elevated	22	€4.5 M	€99 M
Stations subtotal			€295 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	9	€3.0 M	€27 M
Depots subtotal			€52 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
metro-4car (revenue + spare + cold reserve)	192	€4.0 M	€768 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	233.1 km × €0.015 M/km	€3.5 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€43 M
EPC integration + project management (7%)	on subtotal	€169 M

Total

Bucket	Value
Civil works	€1.26 bn
Stations	€295 M
Depots	€52 M
Rolling stock	€768 M
Residual train-control wayside + charging microgrids	€46 M
EPC overhead (7%)	€169 M
CAPEX total	€2.59 bn
Per-route-km	€11 M / km
Per-capita (city pop)	€1,034 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running `scripts/regenerate-city.sh damascus`.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–10 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 11 and runs for 25 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–10)	€238 M / yr	€95
Steady-state, low-ridership (year 11+)	€295 M / yr	€118
Steady-state, high-ridership (year 11+)	€290 M / yr	€116
Lifecycle envelope (yr 1–35, low scenario)	€9.76 bn cumulative	€3,900
Lifecycle envelope (yr 1–35, high scenario)	€9.64 bn cumulative	€3,851

Population basis: 2,503,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 35, debt service drops to zero and only the OPEX shortfall remains – ~€60 M / yr (low) €55 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€1.55 bn	4.5%	35 y, 10 y grace	€105 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€647 M	20.0%	35 y, 10 y grace	€131 M / yr
Government equity (no debt service)	15%	€388 M	—	—	—
Total	100%	€2.59 bn			€236 M / yr

During the 10-year grace period the operator pays interest only — multilateral €70 M / yr + bonds €129 M / yr = €199 M / yr total — plus the equity tranche amortised across construction (€39 M / yr × 10 yr). Principal repayment begins in year 11 on a 25-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€31 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€32 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€174 k
Traction energy (537.4 GWh / yr)	trackside PV + Na-ion (RFC 0002) — self-generated, €0 / yr	€0 k
Labour (1,411 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€1.5 M
OPEX subtotal		€65 M / yr

Annual fleet utilisation: 171 revenue trainsets × 20.5 h/day × 365 d/yr × 35 km/h commercial × 75% revenue factor = 33.6 M train-km / yr (~196 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$70 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass — a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.11 (~\$0.12 USD)
Day pass (3 trips)	€0.27 (15 % bulk discount)
Monthly unlimited pass	€3.22 (~5 % of median monthly income)
Annual pass	€35.42 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	45.7 M	91.4 M
Farebox revenue	€4.9 M / yr	€9.8 M / yr
Farebox / OPEX recovery	8%	15%
Country policy-target recovery (diagnostic)	30%	30%
Operating shortfall (gov subsidy required)	€60 M / yr	€55 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€295 M / yr	€290 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
damascus.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
damascus-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
damascus.corridor.geojson	Line polylines + stations (GeoJSON)
damascus.stations.json	Machine-readable station list
damascus.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug damascus

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug damascus \
  --sidecar .cache/osr-pipeline/rasters/damascus.grid.json \
  --out-dir designs/.../Damascus

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../damascus.toml \
  --out designs/.../README.md
```

`scripts/regenerate-damascus.sh` chains steps 3 + drift tests into a single command.

west-asia/Syria/Deir-Ez-Zor

designs/west-asia/Syria/Deir-Ez-Zor/README.md

Deir-Ez-Zor — Urban Rail Network

Country: SY · Population: 500,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Deir-Ez-Zor rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [deir-ez-zor.corridor.geojson](#).

At a glance

Metric	Value
Lines	3
Unique stations	32
Interchange stations	3
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	62.5%
Route length (double track)	56.7 km
Revenue fleet	51 × 3-car trainsets
Spare + cold-reserve	6 × 3-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	16.1 km	10	17	N Outer S Mid
line-2	18.9 km	10	19	SW Mid NE Outer
line-3	21.6 km	12	21	SE Outer NW Mid
Total	56.7 km	32 unique	57	

Rolling stock

Property	Value
Consist	3-car, 57 m
Max speed	90 km/h
Onboard battery	360 kWh per trainset
Seats	60 longitudinal seats
Nominal capacity (AW2)	330 pax (seated + standing, light-metro-3car per RFC 0008 §1)
Crush capacity (AW3)	420 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 330 AW2 passengers (light-metro-3car)
- Peak frequency: 12 trains/hour/direction (5-min headway)

- Peak capacity per line per direction: $330 \times 12 = 3,960$ pphpd
- Network peak throughput (all lines, both directions): $3 \text{ lines} \times 2 \text{ directions} \times 3,960 = 23,760$ passengers/hour
- Daily theoretical capacity (peak $\times 10$): $\approx 237,600$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 31,250 - 46,875$ trips/day

Catchment

- City population: 500,000
- Anchor-weighted coverage: 62.5%
- Catchment population: $\approx 312,500$ (within ~ 800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	3	500 kW	3000 kWh
Major	10	400 kW	2500 kWh
Standard	12	300 kW	2000 kWh
Terminal	5	500 kW	3000 kWh
Total installed	31	16,600 kW	113,000 kWh

Aggregate station-rail charging power: 18,750 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 360 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	12.0 kWh/km	3 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	227 kWh	18.9 km average line length
Onboard battery coverage	1.6 $\times$ average line run	360 kWh usable pack
Average 60 s dwell charge	9.8 kWh/stop	586 kW average charger across stops
Stops to refill one trainset pack	37 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	83 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	113 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in design.toml — emitted by the osr-design Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. country-costs.toml applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (47.7 km @ €3.5 M/km)	€167 M
Elevated (8.6 km @ €18 M/km)	€156 M
Elevated-interchange premium (1 sites @ €20 M)	€20 M
Civil subtotal	€343 M

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~ 11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
halt	1	€0.4 M	€0.4 M
standard	12	€1.5 M	€18 M
major	10	€3.0 M	€30 M
terminal	5	€2.5 M	€12 M
depot-terminal	1	€3.0 M	€3.0 M
interchange-elevated	3	€4.5 M	€14 M
Stations subtotal			€77 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	5	€3.0 M	€15 M
Depots subtotal			€40 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
light-metro-3car (revenue + spare + cold reserve)	57	€3.0 M	€171 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	56.7 km × €0.015 M/km	€0.8 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€12 M
EPC integration + project management (7%)	on subtotal	€45 M

Total

Bucket	Value
Civil works	€343 M
Stations	€77 M
Depots	€40 M
Rolling stock	€171 M
Residual train-control wayside + charging microgrids	€13 M
EPC overhead (7%)	€45 M
CAPEX total	€688 M
Per-route-km	€12 M / km
Per-capita (city pop)	€1,377 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running scripts/regenerate-city.sh deir-ez-zor.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–10 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 11 and runs for 25 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–10)	€63 M / yr	€127
Steady-state, low-ridership (year 11+)	€78 M / yr	€156
Steady-state, high-ridership (year 11+)	€77 M / yr	€154
Lifecycle envelope (yr 1–35, low scenario)	€2.59 bn cumulative	€5,173
Lifecycle envelope (yr 1–35, high scenario)	€2.56 bn cumulative	€5,124

Population basis: 500,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 35, debt service drops to zero and only the OPEX shortfall remains – ~€15 M / yr (low) €15 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€413 M	4.5%	35 y, 10 y grace	€28 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€172 M	20.0%	35 y, 10 y grace	€35 M / yr
Government equity (no debt service)	15%	€103 M	—	—	—
Total	100%	€688 M			€63 M / yr

During the 10-year grace period the operator pays interest only – multilateral €19 M / yr + bonds €34 M / yr = €53 M / yr total – plus the equity tranche amortised across construction (€10 M / yr × 10 yr). Principal repayment begins in year 11 on a 25-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€6.8 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€9.2 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€42 k
Traction energy (103.0 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (352 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€381 k
OPEX subtotal		€16 M / yr

Annual fleet utilisation: 51 revenue trainsets × 20.5 h/day × 365 d/yr × 30 km/h commercial × 75% revenue factor = 8.6 M train-km / yr (~168 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$70 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.11 (~\$0.12 USD)
Day pass (3 trips)	€0.27 (15 % bulk discount)
Monthly unlimited pass	€3.22 (~5 % of median monthly income)
Annual pass	€35.42 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	9.1 M	18.2 M

	Low scenario	High scenario
Farebox revenue	€979 k / yr	€2.0 M / yr
Farebox / OPEX recovery	6%	12%
Country policy-target recovery (diagnostic)	30%	30%
Operating shortfall (gov subsidy required)	€15 M / yr	€15 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€78 M / yr	€77 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
deir-ez-zor.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
deir-ez-zor-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
deir-ez-zor.corridor.geojson	Line polylines + stations (GeoJSON)
deir-ez-zor.stations.json	Machine-readable station list
deir-ez-zor.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug deir-ez-zor

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug deir-ez-zor \
  --sidecar .cache/osr-pipeline/rasters/deir-ez-zor.grid.json \
  --out-dir designs/.../Deir-Ez-Zor

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../deir-ez-zor.toml \
  --out designs/.../README.md
```

`scripts/regenerate-deir-ez-zor.sh` chains steps 3 + drift tests into a single command.

west-asia/Syria/Hama

designs/west-asia/Syria/Hama/README.md

Hama — Urban Rail Network

Country: SY · Population: 600,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Hama rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [hama.corridor.geojson](#).

At a glance

Metric	Value
Lines	3
Unique stations	31
Interchange stations	3
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	52.7%
Route length (double track)	55.3 km
Revenue fleet	49 × 3-car trainsets
Spare + cold-reserve	6 × 3-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	17.8 km	11	18	N Outer S Outer
line-2	20.3 km	10	20	SE Mid NW Outer
line-3	17.2 km	10	17	NE Mid SW Outer
Total	55.3 km	31 unique	55	

Rolling stock

Property	Value
Consist	3-car, 57 m
Max speed	90 km/h
Onboard battery	360 kWh per trainset
Seats	60 longitudinal seats
Nominal capacity (AW2)	330 pax (seated + standing, light-metro-3car per RFC 0008 §1)
Crush capacity (AW3)	420 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 330 AW2 passengers (light-metro-3car)
- Peak frequency: 12 trains/hour/direction (5-min headway)

- Peak capacity per line per direction: $330 \times 12 = 3,960$ pphpd
- Network peak throughput (all lines, both directions): $3 \text{ lines} \times 2 \text{ directions} \times 3,960 = 23,760$ passengers/hour
- Daily theoretical capacity (peak $\times 10$): $\approx 237,600$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 31,620 - 47,430$ trips/day

Catchment

- City population: 600,000
- Anchor-weighted coverage: 52.7%
- Catchment population: $\approx 316,200$ (within ~ 800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	3	500 kW	3000 kWh
Major	9	400 kW	2500 kWh
Standard	11	300 kW	2000 kWh
Terminal	5	500 kW	3000 kWh
Total installed	29	15,900 kW	108,500 kWh

Aggregate station-rail charging power: 18,000 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 360 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	12.0 kWh/km	3 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	221 kWh	18.4 km average line length
Onboard battery coverage	1.6 $\times$ average line run	360 kWh usable pack
Average 60 s dwell charge	9.7 kWh/stop	581 kW average charger across stops
Stops to refill one trainset pack	37 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	80 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	108 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in design.toml — emitted by the osr-design Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. country-costs.toml applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (53.6 km @ €3.5 M/km)	€187 M
Elevated (1.5 km @ €18 M/km)	€27 M
Civil subtotal	€214 M

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~ 11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
halt	2	€0.4 M	€0.8 M
standard	11	€1.5 M	€16 M
major	9	€3.0 M	€27 M
terminal	5	€2.5 M	€12 M
depot-terminal	1	€3.0 M	€3.0 M
interchange	3	€4.5 M	€14 M
Stations subtotal			€73 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	5	€3.0 M	€15 M
Depots subtotal			€40 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
light-metro-3car (revenue + spare + cold reserve)	55	€3.0 M	€165 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	55.3 km × €0.015 M/km	€0.8 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€11 M
EPC integration + project management (7%)	on subtotal	€35 M

Total

Bucket	Value
Civil works	€214 M
Stations	€73 M
Depots	€40 M
Rolling stock	€165 M
Residual train-control wayside + charging microgrids	€12 M
EPC overhead (7%)	€35 M
CAPEX total	€540 M
Per-route-km	€9.8 M / km
Per-capita (city pop)	€900 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running scripts/regenerate-city.sh hama.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–10 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 11 and runs for 25 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–10)	€50 M / yr	€83
Steady-state, low-ridership (year 11+)	€62 M / yr	€103
Steady-state, high-ridership (year 11+)	€60 M / yr	€101
Lifecycle envelope (yr 1–35, low scenario)	€2.03 bn cumulative	€3,391
Lifecycle envelope (yr 1–35, high scenario)	€2.00 bn cumulative	€3,342

Population basis: 600,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 35, debt service drops to zero and only the OPEX shortfall remains – ~€12 M / yr (low) €11 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€324 M	4.5%	35 y, 10 y grace	€22 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€135 M	20.0%	35 y, 10 y grace	€27 M / yr
Government equity (no debt service)	15%	€81 M	—	—	—
Total	100%	€540 M			€49 M / yr

During the 10-year grace period the operator pays interest only – multilateral €15 M / yr + bonds €27 M / yr = €42 M / yr total – plus the equity tranche amortised across construction (€8.1 M / yr × 10 yr). Principal repayment begins in year 11 on a 25-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€6.6 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€6.6 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€41 k
Traction energy (99.0 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (344 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€372 k
OPEX subtotal		€14 M / yr

Annual fleet utilisation: 49 revenue trainsets × 20.5 h/day × 365 d/yr × 30 km/h commercial × 75% revenue factor = 8.2 M train-km / yr (~168 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$70 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.11 (~\$0.12 USD)
Day pass (3 trips)	€0.27 (15 % bulk discount)
Monthly unlimited pass	€3.22 (~5 % of median monthly income)
Annual pass	€35.42 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	10.9 M	21.9 M

	Low scenario	High scenario
Farebox revenue	€1.2 M / yr	€2.4 M / yr
Farebox / OPEX recovery	9%	17%
Country policy-target recovery (diagnostic)	30%	30%
Operating shortfall (gov subsidy required)	€12 M / yr	€11 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€62 M / yr	€60 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
hama.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
hama-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
hama.corridor.geojson	Line polylines + stations (GeoJSON)
hama.stations.json	Machine-readable station list
hama.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug hama

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug hama \
  --sidecar .cache/osr-pipeline/rasters/hama.grid.json \
  --out-dir designs/.../Hama

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../hama.toml \
  --out designs/.../README.md
```

`scripts/regenerate-hama.sh` chains steps 3 + drift tests into a single command.

west-asia/Syria/Homs

designs/west-asia/Syria/Homs/README.md

Homs — Urban Rail Network

Country: SY · Population: 775,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Homs rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [homs.corridor.geojson](#).

At a glance

Metric	Value
Lines	3
Unique stations	33
Interchange stations	2
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	41.6%
Route length (double track)	51.3 km
Revenue fleet	46 × 3-car trainsets
Spare + cold-reserve	7 × 3-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	16.4 km	12	17	SW Mid N Outer
line-2	23.0 km	12	23	E Mid SW Outer
line-3	11.9 km	9	13	NE Mid NW Mid
Total	51.3 km	33 unique	53	

Rolling stock

Property	Value
Consist	3-car, 57 m
Max speed	90 km/h
Onboard battery	360 kWh per trainset
Seats	60 longitudinal seats
Nominal capacity (AW2)	330 pax (seated + standing, light-metro-3car per RFC 0008 §1)
Crush capacity (AW3)	420 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 330 AW2 passengers (light-metro-3car)
- Peak frequency: 12 trains/hour/direction (5-min headway)

- Peak capacity per line per direction: $330 \times 12 = 3,960$ pphpd
- Network peak throughput (all lines, both directions): $3 \text{ lines} \times 2 \text{ directions} \times 3,960 = 23,760$ passengers/hour
- Daily theoretical capacity (peak $\times 10$): $\approx 237,600$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 32,240 - 48,360$ trips/day

Catchment

- City population: 775,000
- Anchor-weighted coverage: 41.6%
- Catchment population: $\approx 322,400$ (within ~ 800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	2	500 kW	3000 kWh
Major	15	400 kW	2500 kWh
Standard	9	300 kW	2000 kWh
Terminal	5	500 kW	3000 kWh
Total installed	32	17,200 kW	116,500 kWh

Aggregate station-rail charging power: 19,250 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 360 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	12.0 kWh/km	3 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	205 kWh	17.1 km average line length
Onboard battery coverage	1.8 $\times$ average line run	360 kWh usable pack
Average 60 s dwell charge	9.7 kWh/stop	583 kW average charger across stops
Stops to refill one trainset pack	37 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	86 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	116 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in design.toml — emitted by the osr-design Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. country-costs.toml applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (49.2 km @ €3.5 M/km)	€172 M
Elevated (2.0 km @ €18 M/km)	€35 M
Elevated-interchange premium (1 sites @ €20 M)	€20 M
Civil subtotal	€228 M

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~ 11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
halt	1	€0.4 M	€0.4 M
standard	9	€1.5 M	€14 M
major	15	€3.0 M	€45 M
terminal	5	€2.5 M	€12 M
depot-terminal	1	€3.0 M	€3.0 M
interchange-elevated	2	€4.5 M	€9.0 M
Stations subtotal			€83 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	5	€3.0 M	€15 M
Depots subtotal			€40 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
light-metro-3car (revenue + spare + cold reserve)	53	€3.0 M	€159 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	51.3 km × €0.015 M/km	€0.8 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€12 M
EPC integration + project management (7%)	on subtotal	€37 M

Total

Bucket	Value
Civil works	€228 M
Stations	€83 M
Depots	€40 M
Rolling stock	€159 M
Residual train-control wayside + charging microgrids	€13 M
EPC overhead (7%)	€37 M
CAPEX total	€560 M
Per-route-km	€11 M / km
Per-capita (city pop)	€722 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running scripts/regenerate-city.sh

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–10 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 11 and runs for 25 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–10)	€52 M / yr	€66
Steady-state, low-ridership (year 11+)	€63 M / yr	€82
Steady-state, high-ridership (year 11+)	€62 M / yr	€80
Lifecycle envelope (yr 1–35, low scenario)	€2.09 bn cumulative	€2,703
Lifecycle envelope (yr 1–35, high scenario)	€2.06 bn cumulative	€2,654

Population basis: 775,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 35, debt service drops to zero and only the OPEX shortfall remains – ~€12 M / yr (low) €11 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€336 M	4.5%	35 y, 10 y grace	€23 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€140 M	20.0%	35 y, 10 y grace	€28 M / yr
Government equity (no debt service)	15%	€84 M	—	—	—
Total	100%	€560 M			€51 M / yr

During the 10-year grace period the operator pays interest only – multilateral €15 M / yr + bonds €28 M / yr = €43 M / yr total – plus the equity tranche amortised across construction (€8.4 M / yr × 10 yr). Principal repayment begins in year 11 on a 25-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€6.4 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€7.0 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€38 k
Traction energy (92.9 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (320 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€346 k
OPEX subtotal		€14 M / yr

Annual fleet utilisation: 46 revenue trainsets × 20.5 h/day × 365 d/yr × 30 km/h commercial × 75% revenue factor = 7.7 M train-km / yr (~168 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$70 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.11 (~\$0.12 USD)
Day pass (3 trips)	€0.27 (15 % bulk discount)
Monthly unlimited pass	€3.22 (~5 % of median monthly income)
Annual pass	€35.42 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	14.1 M	28.3 M

	Low scenario	High scenario
Farebox revenue	€1.5 M / yr	€3.0 M / yr
Farebox / OPEX recovery	11%	22%
Country policy-target recovery (diagnostic)	30%	30%
Operating shortfall (gov subsidy required)	€12 M / yr	€11 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€63 M / yr	€62 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
homs.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
homs-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
homs.corridor.geojson	Line polylines + stations (GeoJSON)
homs.stations.json	Machine-readable station list
homs.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug homs

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug homs \
  --sidecar .cache/osr-pipeline/rasters/homs.grid.json \
  --out-dir designs/.../Homs

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../homs.toml \
  --out designs/.../README.md
```

`scripts/regenerate-homs.sh` chains steps 3 + drift tests into a single command.

west-asia/Syria/Latakia

designs/west-asia/Syria/Latakia/README.md

Latakia — Urban Rail Network

Country: SY · Population: 700,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Latakia rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [latakia.corridor.geojson](#).

At a glance

Metric	Value
Lines	3
Unique stations	26
Interchange stations	2
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	54.1%
Route length (double track)	41.0 km
Revenue fleet	37 × 3-car trainsets
Spare + cold-reserve	6 × 3-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	15.9 km	10	16	NW Outer E Mid
line-2	17.1 km	11	17	E Outer SW Mid
line-3	8.0 km	5	10	SW Mid NW Mid
Total	41.0 km	26 unique	43	

Rolling stock

Property	Value
Consist	3-car, 57 m
Max speed	90 km/h
Onboard battery	360 kWh per trainset
Seats	60 longitudinal seats
Nominal capacity (AW2)	330 pax (seated + standing, light-metro-3car per RFC 0008 §1)
Crush capacity (AW3)	420 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 330 AW2 passengers (light-metro-3car)
- Peak frequency: 12 trains/hour/direction (5-min headway)

- Peak capacity per line per direction: $330 \times 12 = 3,960$ pphpd
- Network peak throughput (all lines, both directions): $3 \text{ lines} \times 2 \text{ directions} \times 3,960 = 23,760$ passengers/hour
- Daily theoretical capacity (peak $\times 10$): $\approx 237,600$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 37,870 - 56,805$ trips/day

Catchment

- City population: 700,000
- Anchor-weighted coverage: 54.1%
- Catchment population: $\approx 378,700$ (within ~ 800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	2	500 kW	3000 kWh
Major	10	400 kW	2500 kWh
Standard	8	300 kW	2000 kWh
Terminal	5	500 kW	3000 kWh
Total installed	26	14,900 kW	102,000 kWh

Aggregate station-rail charging power: 16,000 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 360 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	12.0 kWh/km	3 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	164 kWh	13.7 km average line length
Onboard battery coverage	2.2 $\times$ average line run	360 kWh usable pack
Average 60 s dwell charge	10.3 kWh/stop	615 kW average charger across stops
Stops to refill one trainset pack	35 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	74 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	102 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in design.toml — emitted by the osr-design Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. country-costs.toml applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (40.8 km @ €3.5 M/km)	€143 M
Civil subtotal	€143 M

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~ 11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
standard	8	€1.5 M	€12 M

Archetype	Count	Unit	Subtotal
major	10	€3.0 M	€30 M
terminal	5	€2.5 M	€12 M
depot-terminal	1	€3.0 M	€3.0 M
interchange	2	€4.5 M	€9.0 M
Stations subtotal			€66 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	5	€3.0 M	€15 M
Depots subtotal			€40 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
light-metro-3car (revenue + spare + cold reserve)	43	€3.0 M	€129 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	41.0 km × €0.015 M/km	€0.6 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€9.9 M
EPC integration + project management (7%)	on subtotal	€27 M

Total

Bucket	Value
Civil works	€143 M
Stations	€66 M
Depots	€40 M
Rolling stock	€129 M
Residual train-control wayside + charging microgrids	€11 M
EPC overhead (7%)	€27 M
CAPEX total	€416 M
Per-route-km	€10 M / km
Per-capita (city pop)	€595 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code — regenerate by

re-running scripts/regenerate-city.sh latakia.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–10 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 11 and runs for 25 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–10)	€38 M / yr	€55
Steady-state, low-ridership (year 11+)	€47 M / yr	€67
Steady-state, high-ridership (year 11+)	€46 M / yr	€65
Lifecycle envelope (yr 1–35, low scenario)	€1.56 bn cumulative	€2,224
Lifecycle envelope (yr 1–35, high scenario)	€1.52 bn cumulative	€2,175

Population basis: 700,000 (catchment per `lib/city-batches/world-sample.toml`). After year 35, debt service drops to zero and only the OPEX shortfall remains — ~€9.1 M / yr (low) €7.7 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€250 M	4.5%	35 y, 10 y grace	€17 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€104 M	20.0%	35 y, 10 y grace	€21 M / yr
Government equity (no debt service)	15%	€62 M	—	—	—
Total	100%	€416 M			€38 M / yr

During the 10-year grace period the operator pays interest only — multilateral €11 M / yr + bonds €21 M / yr = €32 M / yr total — plus the equity tranche amortised across construction (€6.2 M / yr × 10 yr). Principal repayment begins in year 11 on a 25-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€5.2 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€5.0 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€31 k
Traction energy (74.8 GWh / yr)	trackside PV + Na-ion (RFC 0002) — self-generated, €0 / yr	€0 k
Labour (258 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€279 k
OPEX subtotal		€10 M / yr

Annual fleet utilisation: 37 revenue trainsets × 20.5 h/day × 365 d/yr × 30 km/h commercial × 75% revenue factor = 6.2 M train-km / yr (~168 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$70 USD (per `lib/templates/country-finance.toml`). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass — a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.11 (~\$0.12 USD)
Day pass (3 trips)	€0.27 (15 % bulk discount)
Monthly unlimited pass	€3.22 (~5 % of median monthly income)
Annual pass	€35.42 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	12.8 M	25.6 M
Farebox revenue	€1.4 M / yr	€2.7 M / yr
Farebox / OPEX recovery	13%	26%

	Low scenario	High scenario
Country policy-target recovery (diagnostic)	30%	30%
Operating shortfall (gov subsidy required)	€9.1 M / yr	€7.7 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€47 M / yr	€46 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
latakia.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
latakia-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
latakia.corridor.geojson	Line polylines + stations (GeoJSON)
latakia.stations.json	Machine-readable station list
latakia.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug latakia

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug latakia \
  --sidecar .cache/osr-pipeline/rasters/latakia.grid.json \
  --out-dir designs/.../Latakia

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../latakia.toml \
  --out designs/.../README.md
```

`scripts/regenerate-latakia.sh` chains steps 3 + drift tests into a single command.

west-asia/Syria/Tartus

designs/west-asia/Syria/Tartus/README.md

Tartus — Urban Rail Network

Country: SY · Population: 250,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Tartus rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [tartus.corridor.geojson](#).

At a glance

Metric	Value
Lines	3
Unique stations	23
Interchange stations	3
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	73.4%
Route length (double track)	35.0 km
Revenue fleet	43 × 2-car trainsets
Spare + cold-reserve	6 × 2-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	15.1 km	9	20	N Outer S Outer
line-2	10.7 km	7	15	SE Outer NW Mid
line-3	9.2 km	7	14	S Outer W Inner
Total	35.0 km	23 unique	49	

Rolling stock

Property	Value
Consist	2-car, 39 m
Max speed	70 km/h
Onboard battery	240 kWh per trainset
Seats	40 longitudinal seats
Nominal capacity (AW2)	210 pax (seated + standing, tram-2car per RFC 0008 §1)
Crush capacity (AW3)	260 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 210 AW2 passengers (tram-2car)
- Peak frequency: 12 trains/hour/direction (5-min headway)

- Peak capacity per line per direction: $210 \times 12 = 2,520$ pphpd
- Network peak throughput (all lines, both directions): $3 \text{ lines} \times 2 \text{ directions} \times 2,520 = 15,120$ passengers/hour
- Daily theoretical capacity (peak $\times 10$): $\approx 151,200$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 18,350 - 27,525$ trips/day

Catchment

- City population: 250,000
- Anchor-weighted coverage: 73.4%
- Catchment population: $\approx 183,500$ (within ~ 800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	3	500 kW	3000 kWh
Major	9	400 kW	2500 kWh
Standard	5	300 kW	2000 kWh
Terminal	5	500 kW	3000 kWh
Total installed	23	14,100 kW	96,500 kWh

Aggregate station-rail charging power: 14,500 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 240 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	8.0 kWh/km	2 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	93 kWh	11.7 km average line length
Onboard battery coverage	2.6 $\times$ average line run	240 kWh usable pack
Average 60 s dwell charge	10.5 kWh/stop	630 kW average charger across stops
Stops to refill one trainset pack	23 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	70 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	96 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the `[costs]` block in `design.toml` — emitted by the `osr-design` Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. `country-costs.toml` applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (31.6 km @ €3.5 M/km)	€111 M
Elevated (3.2 km @ €18 M/km)	€58 M
Elevated-interchange premium (3 sites @ €20 M)	€60 M
Civil subtotal	€229 M

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~ 11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
standard	5	€1.5 M	€7.5 M
major	9	€3.0 M	€27 M
terminal	5	€2.5 M	€12 M
depot-terminal	1	€3.0 M	€3.0 M
interchange-elevated	3	€4.5 M	€14 M
Stations subtotal			€64 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	5	€3.0 M	€15 M
Depots subtotal			€40 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
tram-2car (revenue + spare + cold reserve)	49	€2.0 M	€98 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	35.0 km × €0.015 M/km	€0.5 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€9.4 M
EPC integration + project management (7%)	on subtotal	€31 M

Total

Bucket	Value
Civil works	€229 M
Stations	€64 M
Depots	€40 M
Rolling stock	€98 M
Residual train-control wayside + charging microgrids	€9.9 M
EPC overhead (7%)	€31 M
CAPEX total	€471 M
Per-route-km	€13 M / km
Per-capita (city pop)	€1,883 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running scripts/regenerate-city.sh tartus.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–10 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 11 and runs for 25 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–10)	€43 M / yr	€173
Steady-state, low-ridership (year 11+)	€53 M / yr	€213
Steady-state, high-ridership (year 11+)	€53 M / yr	€211
Lifecycle envelope (yr 1–35, low scenario)	€1.76 bn cumulative	€7,050
Lifecycle envelope (yr 1–35, high scenario)	€1.75 bn cumulative	€7,001

Population basis: 250,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 35, debt service drops to zero and only the OPEX shortfall remains – ~€10 M / yr (low) €9.8 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€282 M	4.5%	35 y, 10 y grace	€19 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€118 M	20.0%	35 y, 10 y grace	€24 M / yr
Government equity (no debt service)	15%	€71 M	—	—	—
Total	100%	€471 M			€43 M / yr

During the 10-year grace period the operator pays interest only – multilateral €13 M / yr + bonds €24 M / yr = €36 M / yr total – plus the equity tranche amortised across construction (€7.1 M / yr × 10 yr). Principal repayment begins in year 11 on a 25-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€3.9 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€6.6 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€26 k
Traction energy (42.5 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (222 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€240 k
OPEX subtotal		€11 M / yr

Annual fleet utilisation: 43 revenue trainsets × 20.5 h/day × 365 d/yr × 22 km/h commercial × 75% revenue factor = 5.3 M train-km / yr (~123 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$70 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.11 (~\$0.12 USD)
Day pass (3 trips)	€0.27 (15 % bulk discount)
Monthly unlimited pass	€3.22 (~5 % of median monthly income)
Annual pass	€35.42 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	4.6 M	9.1 M

	Low scenario	High scenario
Farebox revenue	€490 k / yr	€979 k / yr
Farebox / OPEX recovery	5%	9%
Country policy-target recovery (diagnostic)	30%	30%
Operating shortfall (gov subsidy required)	€10 M / yr	€9.8 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€53 M / yr	€53 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
tartus.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
tartus-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
tartus.corridor.geojson	Line polylines + stations (GeoJSON)
tartus.stations.json	Machine-readable station list
tartus.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug tartus

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug tartus \
  --sidecar .cache/osr-pipeline/rasters/tartus.grid.json \
  --out-dir designs/.../Tartus

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../tartus.toml \
  --out designs/.../README.md
```

`scripts/regenerate-tartus.sh` chains steps 3 + drift tests into a single command.

west-asia/Yemen/Aden

designs/west-asia/Yemen/Aden/README.md

Aden — Urban Rail Network

Country: YE · Population: 985,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Aden rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [aden.corridor.geojson](#).

At a glance

Metric	Value
Lines	3
Unique stations	28
Interchange stations	2
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	42.9%
Route length (double track)	44.4 km
Revenue fleet	40 × 3-car trainsets
Spare + cold-reserve	6 × 3-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	19.4 km	12	19	N Outer SE Outer
line-2	12.2 km	7	13	E Mid NW Outer
line-3	12.8 km	9	14	E Mid SW Outer
Total	44.4 km	28 unique	46	

Rolling stock

Property	Value
Consist	3-car, 57 m
Max speed	90 km/h
Onboard battery	360 kWh per trainset
Seats	60 longitudinal seats
Nominal capacity (AW2)	330 pax (seated + standing, light-metro-3car per RFC 0008 §1)
Crush capacity (AW3)	420 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 330 AW2 passengers (light-metro-3car)
- Peak frequency: 12 trains/hour/direction (5-min headway)

- Peak capacity per line per direction: $330 \times 12 = 3,960$ pphpd
- Network peak throughput (all lines, both directions): $3 \text{ lines} \times 2 \text{ directions} \times 3,960 = 23,760$ passengers/hour
- Daily theoretical capacity (peak $\times 10$): $\approx 237,600$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 42,256 - 63,384$ trips/day

Catchment

- City population: 985,000
- Anchor-weighted coverage: 42.9%
- Catchment population: $\approx 422,565$ (within ~ 800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	2	500 kW	3000 kWh
Major	10	400 kW	2500 kWh
Standard	9	300 kW	2000 kWh
Terminal	5	500 kW	3000 kWh
Total installed	27	15,200 kW	104,000 kWh

Aggregate station-rail charging power: 16,750 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 360 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	12.0 kWh/km	3 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	178 kWh	14.8 km average line length
Onboard battery coverage	2.0 $\times$ average line run	360 kWh usable pack
Average 60 s dwell charge	10.0 kWh/stop	598 kW average charger across stops
Stops to refill one trainset pack	36 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	76 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	104 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in design.toml — emitted by the osr-design Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. country-costs.toml applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (41.9 km @ €3.5 M/km)	€147 M
Elevated (2.2 km @ €18 M/km)	€40 M
Elevated-interchange premium (2 sites @ €20 M)	€40 M
Civil subtotal	€227 M

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~ 11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
halt	1	€0.4 M	€0.4 M
standard	9	€1.5 M	€14 M
major	10	€3.0 M	€30 M
terminal	5	€2.5 M	€12 M
depot-terminal	1	€3.0 M	€3.0 M
interchange-elevated	2	€4.5 M	€9.0 M
Stations subtotal			€68 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	5	€3.0 M	€15 M
Depots subtotal			€40 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
light-metro-3car (revenue + spare + cold reserve)	46	€3.0 M	€138 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	44.4 km × €0.015 M/km	€0.7 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€10 M
EPC integration + project management (7%)	on subtotal	€34 M

Total

Bucket	Value
Civil works	€227 M
Stations	€68 M
Depots	€40 M
Rolling stock	€138 M
Residual train-control wayside + charging microgrids	€11 M
EPC overhead (7%)	€34 M
CAPEX total	€518 M
Per-route-km	€12 M / km
Per-capita (city pop)	€526 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running scripts/regenerate-city.sh aden.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–10 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 11 and runs for 30 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–10)	€40 M / yr	€41
Steady-state, low-ridership (year 11+)	€50 M / yr	€51
Steady-state, high-ridership (year 11+)	€48 M / yr	€48
Lifecycle envelope (yr 1–40, low scenario)	€1.90 bn cumulative	€1,926
Lifecycle envelope (yr 1–40, high scenario)	€1.83 bn cumulative	€1,859

Population basis: 985,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 40, debt service drops to zero and only the OPEX shortfall remains – ~€10 M / yr (low) €8.2 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€311 M	3.0%	40 y, 10 y grace	€16 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€130 M	18.0%	40 y, 10 y grace	€23 M / yr
Government equity (no debt service)	15%	€78 M	—	—	—
Total	100%	€518 M			€39 M / yr

During the 10-year grace period the operator pays interest only – multilateral €9.3 M / yr + bonds €23 M / yr = €33 M / yr total – plus the equity tranche amortised across construction (€7.8 M / yr × 10 yr). Principal repayment begins in year 11 on a 30-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€5.5 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€6.7 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€33 k
Traction energy (80.8 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (278 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€344 k
OPEX subtotal		€13 M / yr

Annual fleet utilisation: 40 revenue trainsets × 20.5 h/day × 365 d/yr × 30 km/h commercial × 75% revenue factor = 6.7 M train-km / yr (~168 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$80 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.12 (~\$0.13 USD)
Day pass (3 trips)	€0.31 (15 % bulk discount)
Monthly unlimited pass	€3.68 (~5 % of median monthly income)
Annual pass	€40.48 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	18.0 M	36.0 M

	Low scenario	High scenario
Farebox revenue	€2.2 M / yr	€4.4 M / yr
Farebox / OPEX recovery	17%	35%
Country policy-target recovery (diagnostic)	25%	25%
Operating shortfall (gov subsidy required)	€10 M / yr	€8.2 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€50 M / yr	€48 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
aden.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
aden-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
aden.corridor.geojson	Line polylines + stations (GeoJSON)
aden.stations.json	Machine-readable station list
aden.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug aden

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug aden \
  --sidecar .cache/osr-pipeline/rasters/aden.grid.json \
  --out-dir designs/.../Aden

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../aden.toml \
  --out designs/.../README.md
```

`scripts/regenerate-aden.sh` chains steps 3 + drift tests into a single command.

west-asia/Yemen/Hodeidah

designs/west-asia/Yemen/Hodeidah/README.md

Hodeidah — Urban Rail Network

Country: YE · Population: 750,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Hodeidah rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [hodeidah.corridor.geojson](#).

At a glance

Metric	Value
Lines	3
Unique stations	27
Interchange stations	3
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	61.4%
Route length (double track)	36.2 km
Revenue fleet	34 × 3-car trainsets
Spare + cold-reserve	6 × 3-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	10.2 km	9	12	NW Mid S Mid
line-2	10.2 km	8	12	SE Outer N Mid
line-3	15.8 km	10	16	NW Outer S Outer
Total	36.2 km	27 unique	40	

Rolling stock

Property	Value
Consist	3-car, 57 m
Max speed	90 km/h
Onboard battery	360 kWh per trainset
Seats	60 longitudinal seats
Nominal capacity (AW2)	330 pax (seated + standing, light-metro-3car per RFC 0008 §1)
Crush capacity (AW3)	420 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 330 AW2 passengers (light-metro-3car)
- Peak frequency: 12 trains/hour/direction (5-min headway)

- Peak capacity per line per direction: $330 \times 12 = 3,960$ pphpd
- Network peak throughput (all lines, both directions): $3 \text{ lines} \times 2 \text{ directions} \times 3,960 = 23,760$ passengers/hour
- Daily theoretical capacity (peak $\times 10$): $\approx 237,600$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 46,050 - 69,075$ trips/day

Catchment

- City population: 750,000
- Anchor-weighted coverage: 61.4%
- Catchment population: $\approx 460,500$ (within ~ 800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	3	500 kW	3000 kWh
Major	14	400 kW	2500 kWh
Standard	4	300 kW	2000 kWh
Terminal	5	500 kW	3000 kWh
Total installed	27	15,800 kW	107,000 kWh

Aggregate station-rail charging power: 16,500 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 360 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	12.0 kWh/km	3 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	145 kWh	12.1 km average line length
Onboard battery coverage	2.5 $\times$ average line run	360 kWh usable pack
Average 60 s dwell charge	10.2 kWh/stop	611 kW average charger across stops
Stops to refill one trainset pack	35 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	79 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	107 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in design.toml — emitted by the osr-design Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. country-costs.toml applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (34.2 km @ €3.5 M/km)	€120 M
Elevated (1.9 km @ €18 M/km)	€34 M
Elevated-interchange premium (2 sites @ €20 M)	€40 M
Civil subtotal	€194 M

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~ 11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
standard	4	€1.5 M	€6.0 M
major	14	€3.0 M	€42 M
terminal	5	€2.5 M	€12 M
depot-terminal	1	€3.0 M	€3.0 M
interchange-elevated	3	€4.5 M	€14 M
Stations subtotal			€77 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	5	€3.0 M	€15 M
Depots subtotal			€40 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
light-metro-3car (revenue + spare + cold reserve)	40	€3.0 M	€120 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	36.2 km × €0.015 M/km	€0.5 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€11 M
EPC integration + project management (7%)	on subtotal	€31 M

Total

Bucket	Value
Civil works	€194 M
Stations	€77 M
Depots	€40 M
Rolling stock	€120 M
Residual train-control wayside + charging microgrids	€12 M
EPC overhead (7%)	€31 M
CAPEX total	€474 M
Per-route-km	€13 M / km
Per-capita (city pop)	€632 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running scripts/regenerate-city.sh hodeidah.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–10 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 11 and runs for 30 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–10)	€37 M / yr	€49
Steady-state, low-ridership (year 11+)	€46 M / yr	€61
Steady-state, high-ridership (year 11+)	€44 M / yr	€59
Lifecycle envelope (yr 1–40, low scenario)	€1.74 bn cumulative	€2,317
Lifecycle envelope (yr 1–40, high scenario)	€1.69 bn cumulative	€2,250

Population basis: 750,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 40, debt service drops to zero and only the OPEX shortfall remains – ~€9.7 M / yr (low) €8.0 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€284 M	3.0%	40 y, 10 y grace	€15 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€118 M	18.0%	40 y, 10 y grace	€21 M / yr
Government equity (no debt service)	15%	€71 M	—	—	—
Total	100%	€474 M			€36 M / yr

During the 10-year grace period the operator pays interest only – multilateral €8.5 M / yr + bonds €21 M / yr = €30 M / yr total – plus the equity tranche amortised across construction (€7.1 M / yr × 10 yr). Principal repayment begins in year 11 on a 30-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€4.8 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€6.2 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€27 k
Traction energy (68.7 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (229 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€283 k
OPEX subtotal		€11 M / yr

Annual fleet utilisation: 34 revenue trainsets × 20.5 h/day × 365 d/yr × 30 km/h commercial × 75% revenue factor = 5.7 M train-km / yr (~168 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$80 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.12 (~\$0.13 USD)
Day pass (3 trips)	€0.31 (15 % bulk discount)
Monthly unlimited pass	€3.68 (~5 % of median monthly income)
Annual pass	€40.48 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	13.7 M	27.4 M

	Low scenario	High scenario
Farebox revenue	€1.7 M / yr	€3.4 M / yr
Farebox / OPEX recovery	15%	30%
Country policy-target recovery (diagnostic)	25%	25%
Operating shortfall (gov subsidy required)	€9.7 M / yr	€8.0 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€46 M / yr	€44 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
hodeidah.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
hodeidah-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
hodeidah.corridor.geojson	Line polylines + stations (GeoJSON)
hodeidah.stations.json	Machine-readable station list
hodeidah.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug hodeidah

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug hodeidah \
  --sidecar .cache/osr-pipeline/rasters/hodeidah.grid.json \
  --out-dir designs/.../Hodeidah

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../hodeidah.toml \
  --out designs/.../README.md
```

`scripts/regenerate-hodeidah.sh` chains steps 3 + drift tests into a single command.

west-asia/Yemen/Ibb

designs/west-asia/Yemen/Ibb/README.md

Ibb — Urban Rail Network

Country: YE · Population: 750,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Ibb rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [ibb.corridor.geojson](#).

At a glance

Metric	Value
Lines	3
Unique stations	40
Interchange stations	5
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	58.2%
Route length (double track)	71.6 km
Revenue fleet	63 × 3-car trainsets
Spare + cold-reserve	8 × 3-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	20.8 km	14	20	N Mid S Mid
line-2	26.0 km	12	26	SW Mid NE Outer
line-3	24.8 km	14	25	NW Mid SE Outer
Total	71.6 km	40 unique	71	

Rolling stock

Property	Value
Consist	3-car, 57 m
Max speed	90 km/h
Onboard battery	360 kWh per trainset
Seats	60 longitudinal seats
Nominal capacity (AW2)	330 pax (seated + standing, light-metro-3car per RFC 0008 §1)
Crush capacity (AW3)	420 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 330 AW2 passengers (light-metro-3car)
- Peak frequency: 12 trains/hour/direction (5-min headway)

- Peak capacity per line per direction: $330 \times 12 = 3,960$ pphpd
- Network peak throughput (all lines, both directions): $3 \text{ lines} \times 2 \text{ directions} \times 3,960 = 23,760$ passengers/hour
- Daily theoretical capacity (peak $\times 10$): $\approx 237,600$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 43,650 - 65,475$ trips/day

Catchment

- City population: 750,000
- Anchor-weighted coverage: 58.2%
- Catchment population: $\approx 436,500$ (within ~ 800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	5	500 kW	3000 kWh
Major	10	400 kW	2500 kWh
Standard	17	300 kW	2000 kWh
Terminal	5	500 kW	3000 kWh
Total installed	38	19,100 kW	129,000 kWh

Aggregate station-rail charging power: 22,500 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 360 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	12.0 kWh/km	3 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	286 kWh	23.9 km average line length
Onboard battery coverage	1.3 $\times$ average line run	360 kWh usable pack
Average 60 s dwell charge	9.4 kWh/stop	562 kW average charger across stops
Stops to refill one trainset pack	38 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	96 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	129 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in design.toml — emitted by the osr-design Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. country-costs.toml applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (69.2 km @ €3.5 M/km)	€242 M
Elevated (2.2 km @ €18 M/km)	€39 M
Elevated-interchange premium (2 sites @ €20 M)	€40 M
Civil subtotal	€321 M

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~ 11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
halt	2	€0.4 M	€0.8 M
standard	17	€1.5 M	€26 M
major	10	€3.0 M	€30 M
terminal	5	€2.5 M	€12 M
depot-terminal	1	€3.0 M	€3.0 M
interchange-elevated	5	€4.5 M	€22 M
Stations subtotal			€94 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	5	€3.0 M	€15 M
Depots subtotal			€40 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
light-metro-3car (revenue + spare + cold reserve)	71	€3.0 M	€213 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	71.6 km × €0.015 M/km	€1.1 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€14 M
EPC integration + project management (7%)	on subtotal	€48 M

Total

Bucket	Value
Civil works	€321 M
Stations	€94 M
Depots	€40 M
Rolling stock	€213 M
Residual train-control wayside + charging microgrids	€15 M
EPC overhead (7%)	€48 M
CAPEX total	€732 M
Per-route-km	€10 M / km
Per-capita (city pop)	€976 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running scripts/regenerate-city.sh ibb.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–10 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 11 and runs for 30 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–10)	€57 M / yr	€76
Steady-state, low-ridership (year 11+)	€72 M / yr	€96
Steady-state, high-ridership (year 11+)	€70 M / yr	€94
Lifecycle envelope (yr 1–40, low scenario)	€2.73 bn cumulative	€3,645
Lifecycle envelope (yr 1–40, high scenario)	€2.68 bn cumulative	€3,578

Population basis: 750,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 40, debt service drops to zero and only the OPEX shortfall remains – ~€17 M / yr (low) €15 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€439 M	3.0%	40 y, 10 y grace	€22 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€183 M	18.0%	40 y, 10 y grace	€33 M / yr
Government equity (no debt service)	15%	€110 M	—	—	—
Total	100%	€732 M			€56 M / yr

During the 10-year grace period the operator pays interest only – multilateral €13 M / yr + bonds €33 M / yr = €46 M / yr total – plus the equity tranche amortised across construction (€11 M / yr × 10 yr). Principal repayment begins in year 11 on a 30-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€8.5 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€9.1 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€54 k
Traction energy (127.3 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (442 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€547 k
OPEX subtotal		€18 M / yr

Annual fleet utilisation: 63 revenue trainsets × 20.5 h/day × 365 d/yr × 30 km/h commercial × 75% revenue factor = 10.6 M train-km / yr (~168 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$80 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.12 (~\$0.13 USD)
Day pass (3 trips)	€0.31 (15 % bulk discount)
Monthly unlimited pass	€3.68 (~5 % of median monthly income)
Annual pass	€40.48 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	13.7 M	27.4 M

	Low scenario	High scenario
Farebox revenue	€1.7 M / yr	€3.4 M / yr
Farebox / OPEX recovery	9%	18%
Country policy-target recovery (diagnostic)	25%	25%
Operating shortfall (gov subsidy required)	€17 M / yr	€15 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€72 M / yr	€70 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
ibb.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
ibb-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
ibb.corridor.geojson	Line polylines + stations (GeoJSON)
ibb.stations.json	Machine-readable station list
ibb.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug ibb

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug ibb \
  --sidecar .cache/osr-pipeline/rasters/ibb.grid.json \
  --out-dir designs/.../Ibb

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../ibb.toml \
  --out designs/.../README.md
```

`scripts/regenerate-ibb.sh` chains steps 3 + drift tests into a single command.

west-asia/Yemen/Mukalla

designs/west-asia/Yemen/Mukalla/README.md

Mukalla — Urban Rail Network

Country: YE · Population: 550,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Mukalla rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [mukalla.corridor.geojson](#).

At a glance

Metric	Value
Lines	3
Unique stations	33
Interchange stations	4
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	72.0%
Route length (double track)	60.9 km
Revenue fleet	54 × 3-car trainsets
Spare + cold-reserve	7 × 3-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	18.6 km	11	19	NE Mid SW Outer
line-2	19.5 km	11	19	E Inner SW Outer
line-3	22.8 km	11	23	E Outer SW Mid
Total	60.9 km	33 unique	61	

Rolling stock

Property	Value
Consist	3-car, 57 m
Max speed	90 km/h
Onboard battery	360 kWh per trainset
Seats	60 longitudinal seats
Nominal capacity (AW2)	330 pax (seated + standing, light-metro-3car per RFC 0008 §1)
Crush capacity (AW3)	420 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 330 AW2 passengers (light-metro-3car)
- Peak frequency: 12 trains/hour/direction (5-min headway)

- Peak capacity per line per direction: $330 \times 12 = 3,960$ pphpd
- Network peak throughput (all lines, both directions): $3 \text{ lines} \times 2 \text{ directions} \times 3,960 = 23,760$ passengers/hour
- Daily theoretical capacity (peak $\times 10$): $\approx 237,600$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 39,600 - 59,400$ trips/day

Catchment

- City population: 550,000
- Anchor-weighted coverage: 72.0%
- Catchment population: $\approx 396,000$ (within ~ 800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	4	500 kW	3000 kWh
Major	7	400 kW	2500 kWh
Standard	15	300 kW	2000 kWh
Terminal	5	500 kW	3000 kWh
Total installed	32	16,800 kW	114,500 kWh

Aggregate station-rail charging power: 19,250 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 360 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	12.0 kWh/km	3 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	244 kWh	20.3 km average line length
Onboard battery coverage	1.5 $\times$ average line run	360 kWh usable pack
Average 60 s dwell charge	9.7 kWh/stop	583 kW average charger across stops
Stops to refill one trainset pack	37 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	84 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	114 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in design.toml — emitted by the osr-design Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. country-costs.toml applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (58.2 km @ €3.5 M/km)	€204 M
Elevated (1.9 km @ €18 M/km)	€34 M
Elevated-interchange premium (2 sites @ €20 M)	€40 M
Civil subtotal	€278 M

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~ 11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
halt	1	€0.4 M	€0.4 M
standard	15	€1.5 M	€22 M
major	7	€3.0 M	€21 M
terminal	5	€2.5 M	€12 M
depot-terminal	1	€3.0 M	€3.0 M
interchange-elevated	4	€4.5 M	€18 M
Stations subtotal			€77 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	5	€3.0 M	€15 M
Depots subtotal			€40 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
light-metro-3car (revenue + spare + cold reserve)	61	€3.0 M	€183 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	60.9 km × €0.015 M/km	€0.9 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€12 M
EPC integration + project management (7%)	on subtotal	€41 M

Total

Bucket	Value
Civil works	€278 M
Stations	€77 M
Depots	€40 M
Rolling stock	€183 M
Residual train-control wayside + charging microgrids	€13 M
EPC overhead (7%)	€41 M
CAPEX total	€632 M
Per-route-km	€10 M / km
Per-capita (city pop)	€1,150 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running scripts/regenerate-city.sh mukalla.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–10 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 11 and runs for 30 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–10)	€49 M / yr	€90
Steady-state, low-ridership (year 11+)	€63 M / yr	€114
Steady-state, high-ridership (year 11+)	€61 M / yr	€111
Lifecycle envelope (yr 1–40, low scenario)	€2.37 bn cumulative	€4,307
Lifecycle envelope (yr 1–40, high scenario)	€2.33 bn cumulative	€4,240

Population basis: 550,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 40, debt service drops to zero and only the OPEX shortfall remains – ~€15 M / yr (low) €13 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€379 M	3.0%	40 y, 10 y grace	€19 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€158 M	18.0%	40 y, 10 y grace	€29 M / yr
Government equity (no debt service)	15%	€95 M	—	—	—
Total	100%	€632 M			€48 M / yr

During the 10-year grace period the operator pays interest only – multilateral €11 M / yr + bonds €28 M / yr = €40 M / yr total – plus the equity tranche amortised across construction (€9.5 M / yr × 10 yr). Principal repayment begins in year 11 on a 30-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€7.3 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€7.9 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€45 k
Traction energy (109.1 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (377 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€466 k
OPEX subtotal		€16 M / yr

Annual fleet utilisation: 54 revenue trainsets × 20.5 h/day × 365 d/yr × 30 km/h commercial × 75% revenue factor = 9.1 M train-km / yr (~168 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$80 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.12 (~\$0.13 USD)
Day pass (3 trips)	€0.31 (15 % bulk discount)
Monthly unlimited pass	€3.68 (~5 % of median monthly income)
Annual pass	€40.48 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	10.0 M	20.1 M

	Low scenario	High scenario
Farebox revenue	€1.2 M / yr	€2.5 M / yr
Farebox / OPEX recovery	8%	16%
Country policy-target recovery (diagnostic)	25%	25%
Operating shortfall (gov subsidy required)	€15 M / yr	€13 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€63 M / yr	€61 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
mukalla.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
mukalla-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
mukalla.corridor.geojson	Line polylines + stations (GeoJSON)
mukalla.stations.json	Machine-readable station list
mukalla.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug mukalla

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug mukalla \
  --sidecar .cache/osr-pipeline/rasters/mukalla.grid.json \
  --out-dir designs/.../Mukalla

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../mukalla.toml \
  --out designs/.../README.md
```

`scripts/regenerate-mukalla.sh` chains steps 3 + drift tests into a single command.

west-asia/Yemen/Sanaa

designs/west-asia/Yemen/Sanaa/README.md

Sanaa — Urban Rail Network

Country: YE · Population: 3,937,500

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Sanaa rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [sanaa.corridor.geojson](#).

At a glance

Metric	Value
Lines	9
Unique stations	126
Interchange stations	32
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	78.0%
Route length (double track)	260.8 km
Revenue fleet	193 × 6-car trainsets
Spare + cold-reserve	25 × 6-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	41.2 km	17	34	SE Outer NW Outer
line-2	28.8 km	15	24	S Mid N Outer
line-3	22.1 km	10	19	N Mid SW Mid
line-4	27.1 km	13	23	N Mid S Mid
line-5	27.1 km	10	23	SE Outer SW Inner
line-6	24.5 km	12	20	E Inner W Outer
line-7	20.6 km	9	18	NW Mid E Mid
line-8	17.2 km	11	15	SW Mid NE Inner
line-9	52.3 km	29	42	NW Mid W Inner
Total	260.8 km	126 unique	218	

Rolling stock

Property	Value
Consist	6-car, 111 m
Max speed	100 km/h
Onboard battery	720 kWh per trainset
Seats	120 longitudinal seats
Nominal capacity (AW2)	660 pax (seated + standing, metro-6car per RFC 0008 §1)
Crush capacity (AW3)	840 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 660 AW2 passengers (metro - 6 car)
- Peak frequency: 12 trains/hour/direction (5-min headway)
- Peak capacity per line per direction: $660 \times 12 = 7,920$ pphpd
- Network peak throughput (all lines, both directions): $9 \text{ lines} \times 2 \text{ directions} \times 7,920 = 142,560$ passengers/hour
- Daily theoretical capacity (peak $\times$ 10): $\approx 1,425,600$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 307,125 - 460,687$ trips/day

Catchment

- City population: 3,937,500
- Anchor-weighted coverage: 78.0%
- Catchment population: $\approx 3,071,250$ (within ~ 800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	32	500 kW	3000 kWh
Major	34	400 kW	2500 kWh
Standard	36	300 kW	2000 kWh
Terminal	15	500 kW	3000 kWh
Total installed	118	52,900 kW	338,000 kWh

Aggregate station-rail charging power: 69,000 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 720 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	24.0 kWh/km	6 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	695 kWh	29.0 km average line length
Onboard battery coverage	1.0 $\times$ average line run	720 kWh usable pack
Average 60 s dwell charge	9.1 kWh/stop	548 kW average charger across stops
Stops to refill one trainset pack	79 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	264 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	338 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the `[costs]` block in `design.toml` — emitted by the `osr-design` Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. `country-costs.toml` applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (239.4 km @ €3.5 M/km)	€838 M
Elevated (20.6 km @ €18 M/km)	€370 M
Elevated-interchange premium (19 sites @ €20 M)	€380 M
Civil subtotal	€1.59 bn

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
halt	8	€0.4 M	€3.2 M
standard	36	€1.5 M	€54 M
major	34	€3.0 M	€102 M
terminal	15	€2.5 M	€38 M
depot-terminal	1	€3.0 M	€3.0 M
interchange-elevated	32	€4.5 M	€144 M
Stations subtotal			€344 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	15	€3.0 M	€45 M
Depots subtotal			€70 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
metro-6car (revenue + spare + cold reserve)	218	€6.0 M	€1.31 bn

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	260.8 km × €0.015 M/km	€3.9 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€50 M
EPC integration + project management (7%)	on subtotal	€235 M

Total

Bucket	Value
Civil works	€1.59 bn
Stations	€344 M
Depots	€70 M
Rolling stock	€1.31 bn
Residual train-control wayside + charging microgrids	€53 M
EPC overhead (7%)	€235 M
CAPEX total	€3.60 bn
Per-route-km	€14 M / km

Bucket	Value
Per-capita (city pop)	€914 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from lib/templates/country-finance.toml. Pure function of the [costs] block above + the country code — regenerate by re-running `scripts/regenerate-city.sh` `sanaa`.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–10 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 11 and runs for 30 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–10)	€281 M / yr	€71
Steady-state, low-ridership (year 11+)	€359 M / yr	€91
Steady-state, high-ridership (year 11+)	€350 M / yr	€89
Lifecycle envelope (yr 1–40, low scenario)	€13.58 bn cumulative	€3,448
Lifecycle envelope (yr 1–40, high scenario)	€13.31 bn cumulative	€3,381

Population basis: 3,937,500 (catchment per `lib/city-batches/world-sample.toml`). After year 40, debt service drops to zero and only the OPEX shortfall remains — ~€86 M / yr (low) €77 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€2.16 bn	3.0%	40 y, 10 y grace	€110 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€900 M	18.0%	40 y, 10 y grace	€163 M / yr
Government equity (no debt service)	15%	€540 M	—	—	—
Total	100%	€3.60 bn			€273 M / yr

During the 10-year grace period the operator pays interest only — multilateral €65 M / yr + bonds €162 M / yr = €227 M / yr total — plus the equity tranche amortised across construction (€54 M / yr × 10 yr). Principal repayment begins in year 11 on a 30-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€52 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€40 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€195 k
Traction energy (909.8 GWh / yr)	trackside PV + Na-ion (RFC 0002) — self-generated, €0 / yr	€0 k
Labour (1,577 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€1.9 M
OPEX subtotal		€95 M / yr

Annual fleet utilisation: 193 revenue trainsets × 20.5 h/day × 365 d/yr × 35 km/h commercial × 75% revenue factor = 37.9 M train-km / yr (~196 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$80 USD (per lib/templates/country-finance.toml). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass — a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.12 (~\$0.13 USD)
Day pass (3 trips)	€0.31 (15 % bulk discount)
Monthly unlimited pass	€3.68 (~5 % of median monthly income)
Annual pass	€40.48 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	71.9 M	143.7 M
Farebox revenue	€8.8 M / yr	€18 M / yr
Farebox / OPEX recovery	9%	19%
Country policy-target recovery (diagnostic)	25%	25%
Operating shortfall (gov subsidy required)	€86 M / yr	€77 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€359 M / yr	€350 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
sanaa.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
sanaa-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
sanaa.corridor.geojson	Line polylines + stations (GeoJSON)
sanaa.stations.json	Machine-readable station list
sanaa.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug sanaa

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug sanaa \
  --sidecar .cache/osr-pipeline/rasters/sanaa.grid.json \
  --out-dir designs/.../Sanaa

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../sanaa.toml \
  --out designs/.../README.md
```

`scripts/regenerate-sanaa.sh` chains steps 3 + drift tests into a single command.

west-asia/Yemen/Sayun

designs/west-asia/Yemen/Sayun/README.md

Sayun — Urban Rail Network

Country: YE · Population: 200,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Sayun rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [sayun.corridor.geojson](#).

At a glance

Metric	Value
Lines	2
Unique stations	15
Interchange stations	2
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	57.6%
Route length (double track)	21.5 km
Revenue fleet	27 × 2-car trainsets
Spare + cold-reserve	4 × 2-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	12.5 km	8	17	W Outer E Mid
line-2	9.1 km	7	14	S Inner E Outer
Total	21.5 km	15 unique	31	

Rolling stock

Property	Value
Consist	2-car, 39 m
Max speed	70 km/h
Onboard battery	240 kWh per trainset
Seats	40 longitudinal seats
Nominal capacity (AW2)	210 pax (seated + standing, t ram - 2 car per RFC 0008 §1)
Crush capacity (AW3)	260 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 210 AW2 passengers (t ram - 2 car)
- Peak frequency: 12 trains/hour/direction (5-min headway)
- Peak capacity per line per direction: 210 × 12 = 2,520 pphpd

- Network peak throughput (all lines, both directions): $2 \text{ lines} \times 2 \text{ directions} \times 2,520 = 10,080 \text{ passengers/hour}$
- Daily theoretical capacity (peak $\times 10$): $\approx 100,800 \text{ passenger-trips/day}$
- Practical daily ridership estimate (10–15 % of catchment): $\approx 11,519 - 17,279 \text{ trips/day}$

Catchment

- City population: 200,000
- Anchor-weighted coverage: 57.6%
- Catchment population: $\approx 115,199$ (within ~800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [.././././././lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	2	500 kW	3000 kWh
Major	5	400 kW	2500 kWh
Standard	4	300 kW	2000 kWh
Terminal	3	500 kW	3000 kWh
Total installed	15	10,700 kW	75,500 kWh

Aggregate station-rail charging power: 9,500 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 240 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	8.0 kWh/km	2 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	86 kWh	10.8 km average line length
Onboard battery coverage	2.8 $\times$ average line run	240 kWh usable pack
Average 60 s dwell charge	10.6 kWh/stop	633 kW average charger across stops
Stops to refill one trainset pack	23 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	54 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	76 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the [costs] block in design.toml — emitted by the osr-design Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. country-costs.toml applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (20.4 km @ €3.5 M/km)	€71 M
Elevated (1.1 km @ €18 M/km)	€21 M
Elevated-interchange premium (1 sites @ €20 M)	€20 M
Civil subtotal	€112 M

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
standard	4	€1.5 M	€6.0 M
major	5	€3.0 M	€15 M
terminal	3	€2.5 M	€7.5 M
depot-terminal	1	€3.0 M	€3.0 M
interchange-elevated	2	€4.5 M	€9.0 M
Stations subtotal			€40 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	3	€3.0 M	€9.0 M
Depots subtotal			€34 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
tram-2car (revenue + spare + cold reserve)	31	€2.0 M	€62 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	21.5 km × €0.015 M/km	€0.3 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€6.2 M
EPC integration + project management (7%)	on subtotal	€18 M

Total

Bucket	Value
Civil works	€112 M
Stations	€40 M
Depots	€34 M
Rolling stock	€62 M
Residual train-control wayside + charging microgrids	€6.5 M
EPC overhead (7%)	€18 M
CAPEX total	€273 M
Per-route-km	€13 M / km
Per-capita (city pop)	€1,363 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running scripts/regenerate-city.sh sayun.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–10 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 11 and runs for 30 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–10)	€21 M / yr	€106
Steady-state, low-ridership (year 11+)	€27 M / yr	€133
Steady-state, high-ridership (year 11+)	€26 M / yr	€131
Lifecycle envelope (yr 1–40, low scenario)	€1.01 bn cumulative	€5,060
Lifecycle envelope (yr 1–40, high scenario)	€999 M cumulative	€4,993

Population basis: 200,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 40, debt service drops to zero and only the OPEX shortfall remains – ~€5.9 M / yr (low) €5.5 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€164 M	3.0%	40 y, 10 y grace	€8.3 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€68 M	18.0%	40 y, 10 y grace	€12 M / yr
Government equity (no debt service)	15%	€41 M	—	—	—
Total	100%	€273 M			€21 M / yr

During the 10-year grace period the operator pays interest only – multilateral €4.9 M / yr + bonds €12 M / yr = €17 M / yr total – plus the equity tranche amortised across construction (€4.1 M / yr × 10 yr). Principal repayment begins in year 11 on a 30-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€2.5 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€3.7 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€16 k
Traction energy (26.7 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (141 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€174 k
OPEX subtotal		€6.4 M / yr

Annual fleet utilisation: 27 revenue trainsets × 20.5 h/day × 365 d/yr × 22 km/h commercial × 75% revenue factor = 3.3 M train-km / yr (~123 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$80 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.12 (~\$0.13 USD)
Day pass (3 trips)	€0.31 (15 % bulk discount)
Monthly unlimited pass	€3.68 (~5 % of median monthly income)
Annual pass	€40.48 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	3.6 M	7.3 M

	Low scenario	High scenario
Farebox revenue	€448 k / yr	€895 k / yr
Farebox / OPEX recovery	7%	14%
Country policy-target recovery (diagnostic)	25%	25%
Operating shortfall (gov subsidy required)	€5.9 M / yr	€5.5 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€27 M / yr	€26 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
sayun.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
sayun-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
sayun.corridor.geojson	Line polylines + stations (GeoJSON)
sayun.stations.json	Machine-readable station list
sayun.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug sayun

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug sayun \
  --sidecar .cache/osr-pipeline/rasters/sayun.grid.json \
  --out-dir designs/.../Sayun

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../sayun.toml \
  --out designs/.../README.md
```

`scripts/regenerate-sayun.sh` chains steps 3 + drift tests into a single command.

Taiz — Urban Rail Network

Country: YE · Population: 615,000

Auto-planned by the OpenSourceRail design pipeline: [osr_geo](#) rasterises Overpass-verified OpenStreetMap features (arterial road graph, buildings, water, protected land, demand-anchor POIs) onto a 20 m cost / demand / buildability grid; [osr-design](#) (rust) runs a demand-rewarded Dijkstra on that grid to synthesise corridors, places stations against the demand surface, and classifies every segment (at-grade / elevated / bridge — no tunnels per [RFC 0011](#)). Population, country, and bbox are read from the canonical city catalog at [lib/city-batches/world-sample.toml](#).

Network map

Taiz rail network on OpenStreetMap

Every line visible end-to-end — radials out to the city edge, forced-coverage suburbs, and the ring line if present. Auto-fit zoom based on the network's actual bounding box.

Corridor polylines + stations as GeoJSON for GIS / alignment tooling: [taiz.corridor.geojson](#).

At a glance

Metric	Value
Lines	3
Unique stations	33
Interchange stations	3
Multi-line transfer reachability	0% (line-pairs sharing ≥ 1 station)
Anchor-weighted coverage	54.8%
Route length (double track)	49.0 km
Revenue fleet	45 × 3-car trainsets
Spare + cold-reserve	6 × 3-car trainsets
Peak headway	5 min
Service hours	05:30 – 02:00 (≈ 20 h/day)

Lines

Termini are tagged by compass quadrant + radial band (Inner < 0.33 R, Mid 0.33–0.67 R, Outer > 0.67 R, where R is the network's outermost station-to-centre distance).

Line	Length	Stations	Trainsets	Termini
line-1	21.7 km	14	21	SW Outer NE Mid
line-2	17.4 km	11	18	W Mid NE Outer
line-3	10.0 km	8	12	SE Inner NW Mid
Total	49.0 km	33 unique	51	

Rolling stock

Property	Value
Consist	3-car, 57 m
Max speed	90 km/h
Onboard battery	360 kWh per trainset
Seats	60 longitudinal seats
Nominal capacity (AW2)	330 pax (seated + standing, light-metro-3car per RFC 0008 §1)
Crush capacity (AW3)	420 pax, short-duration structural/egress reference

Ridership capacity

- Per-train planning capacity: 330 AW2 passengers (light-metro-3car)
- Peak frequency: 12 trains/hour/direction (5-min headway)

- Peak capacity per line per direction: $330 \times 12 = 3,960$ pphpd
- Network peak throughput (all lines, both directions): $3 \text{ lines} \times 2 \text{ directions} \times 3,960 = 23,760$ passengers/hour
- Daily theoretical capacity (peak $\times 10$): $\approx 237,600$ passenger-trips/day
- Practical daily ridership estimate (10–15 % of catchment): $\approx 33,702 - 50,553$ trips/day

Catchment

- City population: 615,000
- Anchor-weighted coverage: 54.8%
- Catchment population: $\approx 337,020$ (within ~ 800 m walk of a station)

Energy infrastructure (solar + battery)

On-site trackside + depot PV and battery storage. Per-tier sizing (from [../../../../lib/templates/energy-sites.toml](#)):

Tier	Sites	PV each	Battery each
Depot-Main	1	5000 kW	40000 kWh
Interchange	3	500 kW	3000 kWh
Major	13	400 kW	2500 kWh
Standard	11	300 kW	2000 kWh
Terminal	5	500 kW	3000 kWh
Total installed	33	17,500 kW	118,500 kWh

Aggregate station-rail charging power: 19,500 kW. Trains opportunity-charge during station dwell per RFC 0002; onboard 360 kWh battery covers running.

Energy Feasibility Check

Check	Value	Interpretation
Trainset line-haul intensity	12.0 kWh/km	3 cars $\times$ 4 kWh/car-km planning basis
Average one-way line energy	196 kWh	16.3 km average line length
Onboard battery coverage	1.8 $\times$ average line run	360 kWh usable pack
Average 60 s dwell charge	9.8 kWh/stop	591 kW average charger across stops
Stops to refill one trainset pack	37 stops	Opportunity charging supplements, not replaces, onboard reserve
PV daily yield proxy	88 MWh/day	5 peak-sun-hour planning proxy before local derates
Station/depot stationary storage	118 MWh	Distributed Na-ion buffer for charging peaks and grid outages

CAPEX (planning grade)

All figures come from the `[costs]` block in `design.toml` — emitted by the `osr-design` Rust planner per RFC 0011 §9. OSR-discipline unit costs: prefab portal-frame canopies (no bespoke architectural cladding), at-grade depots without overhead bridge cranes, €1.0 M per self-contained car rolling stock, commodity Na-ion cells + tier-2 PMSM motors + DIY SiC inverters, onboard-first train control with only residual wayside (no trackside fibre backbone, no proprietary CBTC vendor stack, no trackside computer interlockings — the function moves into the trainset, already counted in rolling-stock CAPEX), no overhead catenary, and self-EPC overhead. Conventional metro budgets land 2–3 $\times$ higher because of the line items OSR has architected away. `country-costs.toml` applies the per-country labour/material multiplier downstream.

Civil works

Bucket	Value
At-grade (46.8 km @ €3.5 M/km)	€164 M
Elevated (2.1 km @ €18 M/km)	€38 M
Elevated-interchange premium (2 sites @ €20 M)	€40 M
Civil subtotal	€242 M

Stations

Prefab portal-frame canopy + factory-bonded PV sandwich panel (RFC 0010 §3, ~ 11 t / 13-bay canopy delivered on two lorries, 3–5 day erection). Precast L-unit platform edge. Vertical circulation per archetype.

Archetype	Count	Unit	Subtotal
standard	11	€1.5 M	€16 M
major	13	€3.0 M	€39 M
terminal	5	€2.5 M	€12 M
depot-terminal	1	€3.0 M	€3.0 M
interchange-elevated	3	€4.5 M	€14 M
Stations subtotal			€84 M

Depots

At-grade portal-frame workshop sheds; pit tracks with stinger + portable wheel lathe (no overhead bridge crane); on-site PV array; Na-ion stationary storage; no traction substation.

Archetype	Count	Unit	Subtotal
main-heavy	1	€25 M	€25 M
layup-minimal	5	€3.0 M	€15 M
Depots subtotal			€40 M

Rolling stock

Rolling stock is costed at €1.0 M per self-contained car (wagon). Each car carries one powered bogie, one trailer bogie, under-seat Na-ion battery, traction inverter, onboard sensor/control stack, doors, HVAC, interior, and aluminium body. Motors, sensors, train-control computers, and onboard batteries appear here ONLY — never re-billed elsewhere in the cost stack.

Per-car cost bucket	Basis	Cost
Body shell + interior + doors	Aluminium extrusion body, glazing, seats, PRM zone, plug doors	€300 k
Bogies + brakes	One powered bogie + one trailer bogie, wheelsets, suspension, discs	€220 k
Traction package	PMSM motors, gearbox, SiC inverter, cooling, HV contactors	€180 k
Battery + BMS	120 kWh usable under-seat Na-ion pack, BMS, fire containment	€120 k
Driverless onboard stack	T-ECU/S, T-ECU/A, T-OBS sensors, radios, cameras, event recorder	€90 k
HVAC, auxiliaries, fit-out margin	HVAC, lighting, PIS, wiring, assembly QA	€90 k
Total per car		€1.0 M

Item	Count	Unit	Subtotal
light-metro-3car (revenue + spare + cold reserve)	51	€3.0 M	€153 M

Systems

Item	Basis	Subtotal
Residual signalling / train-control wayside (onboard ATP/ATO + T-OBS carries the function; W-Nodes, balises, LoRa gateways, OCC interfaces remain)	49.0 km × €0.015 M/km	€0.7 M
Station/depot charging microgrids (conductive charger, switchgear, inverter interface, local PV/battery tie-in; no continuous wayside supply)	per-stop allowance by station archetype	€12 M
EPC integration + project management (7%)	on subtotal	€37 M

Total

Bucket	Value
Civil works	€242 M
Stations	€84 M
Depots	€40 M
Rolling stock	€153 M
Residual train-control wayside + charging microgrids	€13 M
EPC overhead (7%)	€37 M
CAPEX total	€570 M
Per-route-km	€12 M / km
Per-capita (city pop)	€927 / person

Funding & affordability

Planning-grade financing model anchored to country financial parameters from [lib/templates/country-finance.toml](#). Pure function of the [costs] block above + the country code – regenerate by re-running scripts/regenerate-city.sh taiz.

Government commitment summary (budgetable)

Bottom line for next year's budget submission. Construction phase runs years 1–10 (equity drawdown + interest-only grace on multilateral + bonds); steady-state operation begins year 11 and runs for 30 years until the loans amortise.

Phase	Annual gov / municipal commitment	Per resident / yr
Construction (years 1–10)	€44 M / yr	€72
Steady-state, low-ridership (year 11+)	€56 M / yr	€91
Steady-state, high-ridership (year 11+)	€54 M / yr	€88
Lifecycle envelope (yr 1–40, low scenario)	€2.12 bn cumulative	€3,442
Lifecycle envelope (yr 1–40, high scenario)	€2.08 bn cumulative	€3,375

Population basis: 615,000 (catchment per [lib/city-batches/world-sample.toml](#)). After year 40, debt service drops to zero and only the OPEX shortfall remains – ~€12 M / yr (low) €11 M / yr (high).

CAPEX funding stack

Tranche	Share	Principal	Rate	Tenor	Annual debt service (post-grace)
Multilateral concessional loan (IBRD / AfDB / ADB class)	60%	€342 M	3.0%	40 y, 10 y grace	€17 M / yr
Sovereign bonds (10-y benchmark + project)	25%	€142 M	18.0%	40 y, 10 y grace	€26 M / yr
Government equity (no debt service)	15%	€85 M	—	—	—
Total	100%	€570 M			€43 M / yr

During the 10-year grace period the operator pays interest only – multilateral €10 M / yr + bonds €26 M / yr = €36 M / yr total – plus the equity tranche amortised across construction (€8.5 M / yr × 10 yr). Principal repayment begins in year 11 on a 30-year amortisation schedule.

Annual OPEX (steady state)

Component	Basis	Annual cost
Rolling-stock maintenance	4 % of rolling-stock CAPEX	€6.1 M
Civil + station + depot maintenance	2 % of fixed-asset CAPEX	€7.3 M
Residual train-control wayside maintenance	5 % of residual signalling CAPEX	€37 k
Traction energy (90.9 GWh / yr)	trackside PV + Na-ion (RFC 0002) – self-generated, €0 / yr	€0 k
Labour (306 FTE)	~6 FTE/route-km + 12 admin core × country median × 12 × engineer-premium 1.4	€378 k
OPEX subtotal		€14 M / yr

Annual fleet utilisation: 45 revenue trainsets × 20.5 h/day × 365 d/yr × 30 km/h commercial × 75% revenue factor = 7.6 M train-km / yr (~168 k km / trainset / yr).

Ticket pricing anchored to median income

Country median monthly income: \$80 USD (per [lib/templates/country-finance.toml](#)). Affordability target: a monthly unlimited-ride pass costs 5 % of median monthly income. Single-trip fare set so that 30 single trips equal one monthly pass – a frequent commuter averaging ~50 trips / month then pays an effective ~40 % bulk discount on the pass, matching the structure used by Delhi Metro, Cairo Metro, and STIB.

Product	Price target
Single-trip fare	€0.12 (~\$0.13 USD)
Day pass (3 trips)	€0.31 (15 % bulk discount)
Monthly unlimited pass	€3.68 (~5 % of median monthly income)
Annual pass	€40.48 (10 × monthly = ~1 free month)

Farebox & operating subsidy

Practical-ridership bracket = 5–10 % of urban population × 365 service-days. At the affordability-anchored fare:

	Low scenario	High scenario
Annual paid trips	11.2 M	22.4 M

	Low scenario	High scenario
Farebox revenue	€1.4 M / yr	€2.8 M / yr
Farebox / OPEX recovery	10%	20%
Country policy-target recovery (diagnostic)	25%	25%
Operating shortfall (gov subsidy required)	€12 M / yr	€11 M / yr
Steady-state government commitment (debt service + OPEX shortfall)	€56 M / yr	€54 M / yr

Caveats: The funding-stack 60/25/15 split, the 5 % income-share affordability target, and the 5–10 % daily-pax bracket are project-level defaults. Real deployments will negotiate the share with the financing institutions and will tune fares iteratively from boarding data. Treat the numbers above as a first-iteration sanity check, not as a bid-ready financial close.

Files

File	Role
design.toml	Authoritative design
taiz.toml	Expanded simulation scenario (input to <code>osr-sim</code>)
taiz-network-map.png	Auto-fit network map (rendered by <code>osr_scenario.render_map</code>)
taiz.corridor.geojson	Line polylines + stations (GeoJSON)
taiz.stations.json	Machine-readable station list
taiz.design-quality.yaml	Coverage / anchor-hit / civil-mix metrics + auto-gate result

Reproducibility

```
# 1. raster bundle from OpenStreetMap (cached by query hash)
python -m osr_geo.cli --slug taiz

# 2. design.toml + corridor.geojson + design-quality.yaml
# (population + country pulled from lib/city-batches/world-sample.toml)
cargo run --release --bin osr-design -- --slug taiz \
  --sidecar .cache/osr-pipeline/rasters/taiz.grid.json \
  --out-dir designs/.../Taiz

# 3. scenario.toml + map PNGs + this README
python -m osr_scenario --design designs/.../design.toml
python -m osr_scenario.render_map --design designs/.../design.toml
python -m osr_scenario.network_readme \
  --design designs/.../design.toml \
  --scenario designs/.../taiz.toml \
  --out designs/.../README.md
```

`scripts/regenerate-taiz.sh` chains steps 3 + drift tests into a single command.